

AMC 10

§1 AMC 10 10B 2020

1. (AMC 10 10B 2020) What is the value of

$$1 - (-2) - 3 - (-4) - 5 - (-6)?$$

(A) -20 (B) -3 (C) 3 (D) 5 (E) 21

Answer: D

Solution 1

We know that when we subtract negative numbers, $a - (-b) = a + b$.

The equation becomes $1 + 2 - 3 + 4 - 5 + 6 = \boxed{\text{(D)} 5}$.

Solution 2

Like Solution 1, we know that when we subtract $a - (-b)$, that will equal $a + b$ as the opposite/negative of a negative is a positive. Thus, $1 - (-2) - 3 - (-4) - 5 - (-6) = 1 + 2 - 3 + 4 - 5 + 6$. We can group together a few terms to make our computation a bit simpler. $1 + (2 - 3) + 4 + (-5 + 6) = 1 + (-1) + 4 + 1 = \boxed{\text{(D)} 5}$.

Solution 3

Notice $1 - (-2) - 3 - (-4) - 5 - (-6)$ has three groups: $1 - (-2)$, $-3 - (-4)$, $-5 - (-6)$

The first group, $1 - (-2)$, can be expressed as $a - [-(a + 1)]$.

Simplify,

$$a - [-(a + 1)] \text{ is } 2a + 1$$

The second and third groups can be expressed as $-c - [-(c + 1)]$.

Simplify,

$$-c - [-(c + 1)] \text{ is } 1$$

Thus, the sum of the three groups is $2 \cdot 1 + 1 + 1 + 1 = 5$ or $\boxed{\text{(D)} 5}$

2. (AMC 10 10B 2020) Carl has 5 cubes each having side length 1, and Kate has 5 cubes each having side length 2. What is the total volume of these 10 cubes?

(A) 24 (B) 25 (C) 28 (D) 40 (E) 45

Answer: E

Solution 1

A cube with side length 1 has volume $1^3 = 1$, so 5 of these will have a total volume of $5 \cdot 1 = 5$.

A cube with side length 2 has volume $2^3 = 8$, so 5 of these will have a total volume of $5 \cdot 8 = 40$.

$$5 + 40 = \boxed{\text{(E)} 45}.$$

Solution 2 (more in depth about Solution 1)

The total volume of Carl's cubes is 5. This is because to find the volume of a cube or a rectangular prism, you have to multiply the height by the length by the width. So in this question, it would be $1 \times 1 \times 1$. This is equal to 1. Since Carl has 5 cubes, you will have to multiply 1 by 5, to account for all the 5 cubes.

Next, to find the total volume of Kate's cubes you have to do the same thing. Except, this time, the height, the width, and the length, are all 2, so it will be $2 \times 2 \times 2 = 8$. Now you have to multiply by 5 to account for all the 5 blocks. This is 40. So the total volume of Kate's cubes is 40.

Lastly, to find the total of Carl's and Kate's cubes, you must add the total volume of their cubes together. This is going to be $5 + 40 = \boxed{\text{(E)} 45}$.

3. (AMC 10 10B 2020) The ratio of w to x is $4 : 3$, the ratio of y to z is $3 : 2$, and the ratio of z to x is $1 : 6$. What is the ratio of w to y ? (A) $4 : 3$ (B) $3 : 2$ (C) $8 : 3$ (D) $4 : 1$ (E) $16 : 3$

Answer: E

Solution 1 (One Sentence)

We have

$$\frac{w}{y} = \frac{w}{x} \cdot \frac{x}{z} \cdot \frac{z}{y} = \frac{4}{3} \cdot \frac{6}{1} \cdot \frac{2}{3} = \frac{16}{3},$$

from which $w : y = \boxed{\text{(E)} 16 : 3}$.

Solution 2

We need to somehow link all three of the ratios together. We can start by connecting the last two ratios together by multiplying the last ratio by two.

$z : x = 1 : 6 = 2 : 12$, and since $y : z = 3 : 2$, we can link them together to get $y : z : x = 3 : 2 : 12$.

Finally, since $x : w = 3 : 4 = 12 : 16$, we can link this again to get: $y : z : x : w = 3 : 2 : 12 : 16$, so $w : y = \boxed{\text{(E)} 16 : 3}$.

Solution 3

We have the equations $\frac{w}{x} = \frac{4}{3}$, $\frac{y}{z} = \frac{3}{2}$, and $\frac{z}{x} = \frac{1}{6}$. Clearing denominators, we have $3w = 4x$, $2y = 3z$, and $6z = x$. Since we want $\frac{w}{y}$, we look to find y in terms of x since we know the relationship between x and w . We begin by multiplying both sides of $2y = 3z$ by two, obtaining $4y = 6z$. We then substitute that into $6z = x$ to get $4y = x$. Now, to be able to substitute this into our first equation, we need to have $4x$ on the RHS. Multiplying both sides by 4, we have $16y = 4x$. Substituting this into our first equation, we have $3w = 16y$, or $\frac{w}{y} = \frac{16}{3}$, so our answer is $\boxed{\text{(E)} 16 : 3}$.

Solution 4

WLOG, let $w = 4$ and $x = 3$.

Since the ratio of z to x is $1 : 6$, we can substitute in the value of x to get $\frac{z}{3} = \frac{1}{6} \implies z = \frac{1}{2}$.

The ratio of y to z is $3 : 2$, so $\frac{y}{\frac{1}{2}} = \frac{3}{2} \implies y = \frac{3}{4}$.

The ratio of w to y is then $\frac{4}{3} = \frac{16}{12}$ so our answer is **(E) 16 : 3**.

Solution 5

We have $\frac{w}{x} = \frac{4}{3}$, $\frac{y}{z} = \frac{3}{2}$, $\frac{z}{x} = \frac{1}{6}$. Find LCD: $\frac{w}{x} = \frac{8}{6}$, $\frac{z}{x} = \frac{1}{6}$, we have $\frac{w}{z} = \frac{8}{1}$. Similarly, $\frac{w}{z} = \frac{16}{2}$, $\frac{y}{z} = \frac{3}{2}$, so $\frac{w}{y} = \frac{16}{3}$, **(E) 16 : 3**.

4. (AMC 10 10B 2020) The acute angles of a right triangle are a° and b° , where $a > b$ and both a and b are prime numbers. What is the least possible value of b ?
- (A) 2 (B) 3 (C) 5 (D) 7 (E) 11

Answer: D

Solution 1

Since the three angles of a triangle add up to 180° and one of the angles is 90° because it's a right triangle, $a^\circ + b^\circ = 90^\circ$.

The greatest prime number less than 90 is 89. If $a = 89^\circ$, then $b = 90^\circ - 89^\circ = 1^\circ$, which is not prime.

The next greatest prime number less than 90 is 83. If $a = 83^\circ$, then $b = 7^\circ$, which IS prime, so we have our answer **(D) 7** quacker88

Solution 2

Looking at the answer choices, only 7 and 11 are coprime to 90. Testing 7, the smaller angle, makes the other angle 83 which is prime, therefore our answer is **(D) 7**

Solution 3 (Euclidean Algorithm)

It is clear that $\gcd(a, b) = 1$. By the Euclidean Algorithm, we have

$$\gcd(a, b) = \gcd(a + b, b) = \gcd(90, b) = 1,$$

so 90 and b are relatively prime.

The least such prime number b is 7, from which $a = 90 - b = 83$ is also a prime number. Therefore, the answer is **(D) 7**.

5. (AMC 10 10B 2020) How many distinguishable arrangements are there of 1 brown tile, 1 purple tile, 2 green tiles, and 3 yellow tiles in a row from left to right? (Tiles of the same color are indistinguishable.)
- (A) 210 (B) 420 (C) 630 (D) 840 (E) 1050

Answer: B

Solution 1

Let's first find how many possibilities there would be if they were all distinguishable, then divide out the ones we overcounted.

There are $7!$ ways to order 7 objects. However, since there's $3! = 6$ ways to switch the yellow tiles around without changing anything (since they're indistinguishable) and $2! = 2$ ways to order the green tiles, we have to divide out these possibilities: $\frac{7!}{3! \cdot 2} = \mathbf{(B) 420}$.

Solution 2

There are $\binom{7}{3}$ ways to arrange the yellow tiles which is 35. Then from the remaining tiles there are $\binom{4}{2} = 6$ ways to arrange the green tiles. And now from the remaining

two tiles and two slots we can see there are two ways to arrange the purple and brown tiles, giving us an answer of $35 \cdot 6 \cdot 2 = 420$.

Solution 3

We can choose a different frame to solve this problem. Our tile combination can be written as Y, Y, Y, G, G, B, P . We can focus on Y, Y, Y, G, G first, which gives us $\binom{5}{3} = 10$. Now we can insert our brown tile into this which only has 6 choices (like Y, B, Y, Y, G, G and Y, Y, Y, G, G, B etc.), then insert purple tile which only has 7 choices (like B, Y, P, Y, Y, G, G and B, Y, Y, Y, G, G, P etc.). Multiply them together we get $10 \cdot 6 \cdot 7 = \boxed{\text{(B)} 420}$.

Solution 4 (Concise)

Let B be brown, P be purple, G be green, and Y be yellow. Then, we are just ordering Y, Y, Y, G, G, B , and P . Hence, $\frac{7!}{3!2!} = \boxed{\text{(B)} 420}$.

Solution 5 (Reverse Order)

There are $1 + 1 + 2 + 3 = 7$ ways you can arrange the brown tile, $1 + 2 + 3 = 6$ ways you can arrange the purple tile, $\binom{5}{2} = 10$ ways to arrange the green tiles, and only 1 possible arrangement of the yellow tiles after all of this. Therefore, $7 * 6 * 10 * 1 = \boxed{\text{(B)} 420}$.

6. (AMC 10 10B 2020) Driving along a highway, Megan noticed that her odometer showed 15951 (miles). This number is a palindrome—it reads the same forward and backward. Then 2 hours later, the odometer displayed the next higher palindrome. What was her average speed, in miles per hour, during this 2-hour period?

(A) 50 (B) 55 (C) 60 (D) 65 (E) 70

Answer: B

Solution

In order to get the smallest palindrome greater than 15951, we need to raise the middle digit. If we were to raise any of the digits after the middle, we would be forced to also raise a digit before the middle to keep it a palindrome, making it unnecessarily larger.

So we raise 9 to the next largest value, 10, but obviously, that's not how place value works, so we're in the 16000s now. To keep this a palindrome, our number is now 16061.

So Megan drove $16061 - 15951 = 110$ miles. Since this happened over 2 hours, she drove at $\frac{110}{2} = \boxed{\text{(B)} 55}$ mph. quacker88

7. (AMC 10 10B 2020) How many positive even multiples of 3 less than 2020 are perfect squares?

(A) 7 (B) 8 (C) 9 (D) 10 (E) 12

Answer: A

Solution 1

Any even multiple of 3 is a multiple of 6, so we need to find multiples of 6 that are perfect squares and less than 2020. Any solution that we want will be in the form $(6n)^2$, where n is a positive integer. The smallest possible value is at $n = 1$, and the largest is at $n = 7$ (where the expression equals 1764). Therefore, there are a total of $\boxed{\text{(A)} 7}$ possible numbers. -PCChess

Solution 2

A even multiple square of 3 can be represented by $3^2 \cdot 2^2 \cdot x^2$, where 3^2 is the multiple of 3 and 2^2 makes it even. Simplifying we have $36 \cdot x^2$. We can divide 2020 by 36 (floor) and get 56 as the result. We can then see that there are 7 different values for x . It cannot be larger than 7 or else $x^2 > 56$. And thus **(A) 7**

Solution 3

It can be seen that the problem is just asking for squares that are multiples of six. Thus, all squares of multiples of six can be listed out: $6^2, 12^2, 18^2, 24^2, 30^2, 36^2$, and 42^2 . Since $48^2 = 2304 > 2020$ and $42^2 = 1764 < 2020$, there are **(A) 7** valid answers. airbus-a321, November 2023

8. (AMC 10 10B 2020) Points P and Q lie in a plane with $PQ = 8$. How many locations for point R in this plane are there such that the triangle with vertices P , Q , and R is a right triangle with area 12 square units?

(A) 2 (B) 4 (C) 6 (D) 8 (E) 12

Answer: D

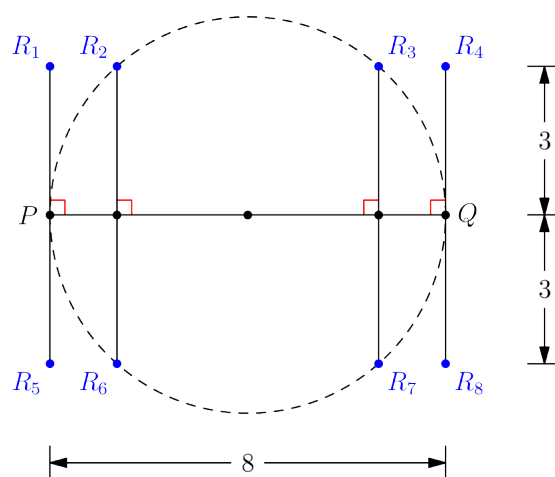
Solution 1 (Geometry)

Let the brackets denote areas. We are given that

$$[PQR] = \frac{1}{2} \cdot PQ \cdot h_R = 12.$$

Since $PQ = 8$, it follows that $h_R = 3$.

We construct a circle with diameter $\overline{PQ}$. All such locations for R are shown below:



We apply casework to the right angle of $\triangle PQR$:

- If $\angle P = 90^\circ$, then $R \in \{R_1, R_5\}$ by the tangent.
- If $\angle Q = 90^\circ$, then $R \in \{R_4, R_8\}$ by the tangent.
- If $\angle R = 90^\circ$, then $R \in \{R_2, R_3, R_6, R_7\}$ by the Inscribed Angle Theorem.

Together, there are **(D) 8** such locations for R . **Remarks**

- The reflections of R_1, R_2, R_3, R_4 about $\overleftrightarrow{PQ}$ are R_5, R_6, R_7, R_8 , respectively.

- b) The reflections of R_1, R_2, R_5, R_6 about the perpendicular bisector of $\overline{PQ}$ are R_4, R_3, R_8, R_7 , respectively.

Solution 2 (Algebra)

Let the brackets denote areas. We are given that

$$[PQR] = \frac{1}{2} \cdot PQ \cdot h_R = 12.$$

Since $PQ = 8$, it follows that $h_R = 3$.

Without loss of generality, let $P = (-4, 0)$ and $Q = (4, 0)$. We conclude that the y -coordinate of R must be ± 3 .

We apply casework to the right angle of $\triangle PQR$:

- a) $\angle P = 90^\circ$. The x -coordinate of R must be -4 , so we have $R = (-4, \pm 3)$.

In this case, there are 2 such locations for R .

- b) $\angle Q = 90^\circ$. The x -coordinate of R must be 4 , so we have $R = (4, \pm 3)$.

In this case, there are 2 such locations for R .

- c) $\angle R = 90^\circ$. For $R = (x, 3)$, the Pythagorean Theorem $PR^2 + QR^2 = PQ^2$ gives

$$[(x+4)^2 + 3^2] + [(x-4)^2 + 3^2] = 8^2.$$

Solving this equation, we have $x = \pm\sqrt{7}$, or $R = (\pm\sqrt{7}, 3)$.

For $R = (x, -3)$, we have $R = (\pm\sqrt{7}, -3)$ by a similar process.

In this case, there are 4 such locations for R .

Together, there are $2 + 2 + 4 = \boxed{\text{(D)} 8}$ such locations for R .

9. (AMC 10 10B 2020) How many ordered pairs of integers (x, y) satisfy the equation

$$x^{2020} + y^2 = 2y?$$

- (A) 1 (B) 2 (C) 3 (D) 4 (E) infinitely many

Answer: D

Solution 1

Rearranging the terms and completing the square for y yields the result $x^{2020} + (y-1)^2 = 1$. Then, notice that x can only be 0, 1, or -1 because any value of x^{2020} that is greater than 1 will cause the term $(y-1)^2$ to be less than 0, which is impossible as y must be real. Therefore, plugging in the above values for x gives the ordered pairs $(0, 0)$, $(1, 1)$, $(-1, 1)$, and $(0, 2)$ gives a total of $\boxed{\text{(D)} 4}$ ordered pairs.

Solution 2

Bringing all of the terms to the LHS, we see a quadratic equation

$$y^2 - 2y + x^{2020} = 0$$

in terms of y . Applying the quadratic formula, we get

$$y = \frac{2 \pm \sqrt{4 - 4 \cdot 1 \cdot x^{2020}}}{2} = \frac{2 \pm \sqrt{4(1 - x^{2020})}}{2}.$$

In order for y to be real, which it must be given the stipulation that we are seeking integral answers, we know that the discriminant, $4(1 - x^{2020})$ must be nonnegative. Therefore,

$$4(1 - x^{2020}) \geq 0 \implies x^{2020} \leq 1.$$

Here, we see that we must split the inequality into a compound inequality, resulting in $-1 \leq x \leq 1$.

The only integers that satisfy this are $x \in \{-1, 0, 1\}$. Plugging these values back into the quadratic equation, we see that $x = \{-1, 1\}$ both produce a discriminant of 0, meaning that there is only 1 solution for y . If $x = \{0\}$, then the discriminant is nonzero, resulting in two solutions for y .

Thus, the answer is $2 \cdot 1 + 1 \cdot 2 = \boxed{\text{(D)} 4}$.

Solution 3: Solve for x first

Set it up as a quadratic in terms of y :

$$y^2 - 2y + x^{2020} = 0$$

Then the discriminant is

$$\Delta = 4 - 4x^{2020}$$

This will clearly only yield real solutions when $|x^{2020}| \leq 1$, because the discriminant must be positive. Then $x = -1, 0, 1$. Checking each one: -1 and 1 are the same when raised to the 2020th power:

$$y^2 - 2y + 1 = (y - 1)^2 = 0$$

This has only has solutions 1, so $(\pm 1, 1)$ are solutions. Next, if $x = 0$:

$$y^2 - 2y = 0 \implies y(y - 2) = 0$$

Which has 2 solutions, so $(0, 2)$ and $(0, 0)$.

These are the only 4 solutions, so our answer is $\boxed{\text{(D)} 4}$.

Solution 4: Solve for y first

Move the y^2 term to the other side to get $x^{2020} = 2y - y^2 = y(2 - y)$.

Because $x^{2020} \geq 0$ for all x , then $y(2 - y) \geq 0 \implies y = 0, 1, 2$.

If $y = 0$ or $y = 2$, the right side is 0 and therefore $x = 0$.

When $y = 1$, the right side become 1, therefore $x = 1, -1$.

Our solutions are $(0, 2)$, $(0, 0)$, $(1, 1)$, $(-1, 1)$. These are the only 4 solutions, so the answer is $\boxed{\text{(D)} 4}$

- wwt7535

Solution 5: Similar to solution 4

Since x^{2020} and y^2 are perfect squares, they are both nonnegative. That means y^2 plus a nonnegative number equals $2y$, which means $y^2 \leq 2y$. The only possible integer values for y are 0, 1, 2.

For $y = 0$, x can only be 0.

For $y = 1$, $x^2 = 1$ so $x = 1, -1$.

For $y = 2$, x can only be 0 as well.

This gives us the solutions $(0, 0)$, $(1, 1)$, $(-1, 1)$, and $(0, 2)$. These are the only solutions, so there is a total of $\boxed{\text{(D)} 4}$ ordered pairs.

- kc1374

Solution 6: (Casework)

We see that x has to be 1, 0, or -1 , as any other integer would make this value too large. We also know that because 2020 is even, both -1 , and 1 for x will yield the same x^{2020} value of 1.

Case 1: $x = 0$. This gives us that $y^2 = 2y$. Dividing both sides by y gives us $y = 2$. Additionally, we know intuitively that $y = 0$ is also a case, which gives us 2 possibilities for this case.

Case 2: $x = 1$ or -1 . This gives us that $1 + y^2 = 2y$. Bringing the $2y$ to the other side, we have a simple quadratic. $y^2 - 2y + 1 = 0$. Factor to get $(y - 1)^2 = 0$ so $y = 1$. Because this works for x as -1 and 1, there are 2 possibilities for this case.

Adding the cases gets us our final answer of $\boxed{\text{(D)} 4}$ ordered pairs. iluvme

Solution 7: (Casework)

We can move the y^2 term to the other side to obtain $x^{2020} = 2y - y^2$, which can be easily factored into $x^{2020} = (2 - y)y$. If we raise each side to the power $1/2020$, we get $x = ((2 - y)(y))^{1/2020}$. Since we're dealing with integer values, the inside of the square root and thus $(2 - y)y$ must be greater than 0. For this to happen, y must be between 0 and 2 inclusive. Testing each integer value, we get the cases:

Case 1: When $y = 0$, then $y(2 - y)$ is also 0, and since the square root of 0 is 0, x must also be 0. So our first pair of solutions is $(0, 0)$.

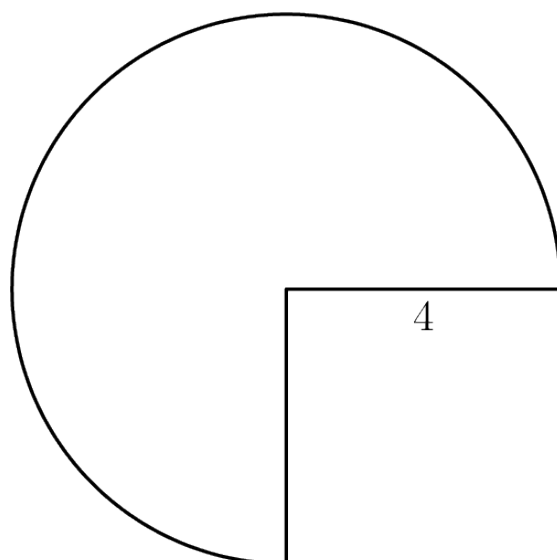
Case 2: When $y = 1$, then $y(2 - y)$ is also 1. The possible values for x are 1 and -1 , so we get 2 more pairs of solutions: $(1, -1)$ and $(1, 1)$.

Case 3: When $y = 2$, then $y(2 - y)$ is 0, and since the square root of 0 is 0, x must also be 0 - so we get one more pair of solutions, $(2, 0)$

If we count all these pairs $((0, 0), (1, -1), (1, 1), (2, 0))$, we have 4 pairs of solutions, so our answer is $\boxed{\text{(D)} 4}$ ordered pairs.

- youtube.com/indianmathguy

10. (AMC 10 10B 2020) A three-quarter sector of a circle of radius 4 inches together with its interior can be rolled up to form the lateral surface area of a right circular cone by taping together along the two radii shown. What is the volume of the cone in cubic inches?



- (A) $3\pi\sqrt{5}$ (B) $4\pi\sqrt{3}$ (C) $3\pi\sqrt{7}$ (D) $6\pi\sqrt{3}$ (E) $6\pi\sqrt{7}$

Answer: C

Solution 1

Notice that when the cone is formed, the 2 shown radii, when joined, will form the slant height of the cone, and the intact circumference of the circle will become the circumference of the base of the cone.

We can calculate that the intact circumference of the circle is $8\pi \cdot \frac{3}{4} = 6\pi$. Since that is also equal to the cone's circumference, the radius of the cone is 3. We also have that the slant height of the cone is 4. Therefore, we use the Pythagorean Theorem to calculate that the height of the cone is $\sqrt{4^2 - 3^2} = \sqrt{7}$. The volume of the cone is $\frac{1}{3} \cdot \pi \cdot 3^2 \cdot \sqrt{7} = \boxed{\text{(C)} 3\pi\sqrt{7}}$

-PCChess minor edits by WrenMath

Solution 2 (Last Resort/Cheap)

Using a ruler, measure a circle of radius 4 and cut out the circle and then the quarter missing. Then, fold it into a cone and measure the diameter to be 6 cm $\implies r = 3$. You can form a right triangle with sides 3, 4, and then through the Pythagorean theorem the height h is found to be $h^2 = 4^2 - 3^2 \implies h = \sqrt{7}$. The volume of a cone is $\frac{1}{3}\pi r^2 h$. Plugging in we find $V = 3\pi\sqrt{7} \implies \boxed{\text{(C)}}$

- DBlack2021

Solution 3

The radius of the given $\frac{3}{4}$ - circle will end up being the slant height of the cone. Thus, the radius and height of the cone are legs of a right triangle with hypotenuse 4. The volume of a cone is $\frac{1}{3}\pi r^2 h$. Using this with the options, we can take out the π and multiply the coefficient of the radical by 3 to get $r^2 h$. We can then use the r and h values to see that the only option that satisfies $r^2 + h^2 = 4^2 = 16$ is $\boxed{\text{(C)}}$

- ColtsFan10

11. (AMC 10 10B 2020) Ms. Carr asks her students to read any 5 of the 10 books on a reading list. Harold randomly selects 5 books from this list, and Betty does the same. What is the probability that there are exactly 2 books that they both select?

- (A) $\frac{1}{8}$ (B) $\frac{5}{36}$ (C) $\frac{14}{45}$ (D) $\frac{25}{63}$ (E) $\frac{1}{2}$

Answer: D

Solution 1

We don't care about which books Harold selects. We just care that Betty picks 2 books from Harold's list and 3 that aren't on Harold's list.

The total amount of combinations of books that Betty can select is $\binom{10}{5} = 252$.

There are $\binom{5}{2} = 10$ ways for Betty to choose 2 of the books that are on Harold's list.

From the remaining 5 books that aren't on Harold's list, there are $\binom{5}{3} = 10$ ways to choose 3 of them.

$$\frac{10 \cdot 10}{252} = \boxed{\text{(D)} \frac{25}{63}} \text{ quacker88}$$

Solution 2

We can analyze this as two containers with 10 balls each, with the two people grabbing 5 balls each. First, we need to find the probability of two of the balls being the same among five: $\frac{1}{3} \cdot \frac{4}{9} \cdot \frac{3}{8} \cdot \frac{5}{7} \cdot \frac{2}{4}$. After that we must multiply this probability by $\binom{5}{2}$, for choosing the 2 balls that are the same chosen among 5 balls. The answer

$$\text{will be } \frac{5}{126} * 10 = \frac{25}{63}. \quad \boxed{\text{(D)} \frac{25}{63}}$$

Solution 3

Firstly, we know that the denominator will be $\binom{10}{5} \cdot \binom{10}{5}$. To calculate the numerator, or successful events, we first find the number of ways both Betty and Harold can choose the same 2 books. Then we find the number of ways for Betty to choose 3 different other books and the number of ways for Harold to choose 3 different other books. That is $\frac{\binom{10}{2} \cdot \binom{8}{3} \cdot \binom{5}{3}}{\binom{10}{5} \cdot \binom{10}{5}} = \frac{45 \cdot 56 \cdot 10}{252 \cdot 252}$. From here, do not multiply, instead cancel common factors, then simplify. In fact, to make this expression even more manageable, leave the combinations in simplest factored form (for example, $\binom{10}{2}$ is $5 \cdot 3 \cdot 3$). After doing so, we get, $\boxed{\text{(D)} \frac{25}{63}}$.

Solution 4

Say that there is a set S $a, b, c, \dots, j$ which represent the 10 books. We know that intersection of the subsets that Bob and Alice choose has 2 elements. Thus there are $\binom{10}{2}$ ways for that to happen. Out of the 8 remaining in the set S , Bob and

Alice both need to choose 3 different elements $= \binom{8}{3} * \binom{5}{3}$. Thus the number of ways this works is $\binom{10}{2} * \binom{8}{3} * \binom{5}{3}$. The denominator for the fraction is just $\binom{10}{5} * \binom{10}{5}$. Simplifying, you get $\boxed{\frac{25}{63}}$.

Solution 5 (similar to 3 but way faster & cheese)

Following the same method as solution 3, you end up getting $\frac{\binom{10}{2} \cdot \binom{8}{3} \cdot \binom{5}{3}}{\binom{10}{5} \cdot \binom{10}{5}} =$

$\frac{45 \cdot 56 \cdot 10}{252 \cdot 252}$. However you notice that there are two 5s in the numerator which can't cancel out. The only answer with two 5 in the numerator is (D) $\frac{25}{63}$.

12. (AMC 10 10B 2020) The decimal representation of

$$\frac{1}{20^{20}}$$

consists of a string of zeros after the decimal point, followed by a 9 and then several more digits. How many zeros are in that initial string of zeros after the decimal point?

- (A) 23 (B) 24 (C) 25 (D) 26 (E) 27

Answer: D

Solution 1

We have

$$\frac{1}{20^{20}} = \frac{1}{(10 \cdot 2)^{20}} = \frac{1}{10^{20} \cdot 2^{20}}$$

Now we do some estimation. Notice that $2^{20} = 1024^2$, which means that 2^{20} is a little more than $1000^2 = 1,000,000$. Multiplying it with 10^{20} , we get that the denominator is about $1 \underbrace{00 \dots 0}_{26 \text{ zeros}}$. Notice that when we divide 1 by an n digit number as long as n is not a power of 10, there are $n - 1$ zeros before the first nonzero digit. This means that when we divide 1 by the 27 digit integer $1 \underbrace{00 \dots 0}_{26 \text{ zeros}}$, there are

(D) 26 zeros in the initial string after the decimal point. -PCCchess

Solution 2

First rewrite $\frac{1}{20^{20}}$ as $\frac{5^{20}}{10^{40}}$. Then, we know that when we write this in decimal form, there will be 40 digits after the decimal point. Therefore, we just have to find the number of digits in 5^{20} .

$\log 5^{20} = 20 \log 5$ and memming $\log 5 \approx 0.69$ (alternatively use the fact that $\log 5 = 1 - \log 2$), $\lfloor 20 \log 5 \rfloor + 1 = \lfloor 20 \cdot 0.69 \rfloor + 1 = 13 + 1 = 14$ digits.

Our answer is (D) 26.

Solution 3 (Brute Force)

Just as in Solution 2, we rewrite $\frac{1}{20^{20}}$ as $\frac{5^{20}}{10^{40}}$. We then calculate 5^{20} entirely by hand, first doing $5^5 \cdot 5^5$, then multiplying that product by itself, resulting in 95,367,431,640,625. Because this is 14 digits, after dividing this number by 10 fourteen times, the decimal point is before the 9. Dividing the number again by 10 twenty-six more times allows a string of (D) 26 zeroes to be formed. -OreoChocolate

Solution 4 (Alternate Brute Force)

Just as in Solutions 2 and 3, we rewrite $\frac{1}{20^{20}}$ as $\frac{5^{20}}{10^{40}}$. We can then look at the number of digits in powers of 5. $5^1 = 5$, $5^2 = 25$, $5^3 = 125$, $5^4 = 625$, $5^5 = 3125$, $5^6 = 15625$, $5^7 = 78125$ and so on. We notice after a few iterations that every

power of five with an exponent of $1 \pmod 3$, the number of digits doesn't increase. This means 5^{20} should have $20 - 6$ digits since there are 6 numbers which are $1 \pmod 3$ from 0 to 20, or 14 digits total. This means our expression can be written as $\frac{k \cdot 10^{14}}{10^{40}}$, where k is in the range $[1, 10)$. Canceling gives $\frac{k}{10^{26}}$, or 26 zeroes before the k since the number k should start on where the one would be in 10^{26} . aop2014

Solution 5 (Scientific Notation)

We see that $\frac{1}{20^{20}} = 9.5367432 \dots \times 10^{-27}$. We see that this has $27 - 1 = 26$ zeros after the decimal point before coming to 9.

Therefore, the answer is (D) 26

but how would this be possible without the use of a calculator?

(Brute Force)

Solution 6 (Logarithms Without Words)

$$\begin{aligned} \lceil \log \frac{1}{20^{20}} \rceil &= \lceil \log 20^{-20} \rceil \\ &= \lceil -20 \log(20) \rceil \\ &= \lceil -20(\log 10 + \log 2) \rceil \\ &= \lceil -20(1 + 0.301) \rceil \\ &= \lceil -26.02 \rceil \\ &= -26 \\ &= \boxed{\text{(D) } 26} \end{aligned}$$

Solution 7 (Algebra)

Define n as the number of consecutive 0s after the decimal point in the decimal representation of $\frac{1}{20^{20}}$. Thus, $\frac{1}{10^n}$ has $n - 1$ zeroes followed by a 1 when expressed as a decimal, and is greater than $\frac{1}{20^{20}}$. So, we have that

$$\frac{1}{10^n} > \frac{1}{20^{20}}.$$

Taking the reciprocal and switching signs grants

$$20^{20} > 10^n.$$

We can express 20^{20} as $10^{20} \cdot 1024^2$. We know that

$$\begin{aligned} (1000 + 24)^2 &= 1024^2 \\ 1000000 + 48000 + 576 &= 1048576. \end{aligned}$$

Multiplying this with 10^{20} adds 20 more zeroes at the end, which means the left side of our inequality has 27 digits. Then, the 10 on the right side must have an exponent less than 27, meaning that $n < 27$.

We are also given that the initial string of zeroes in the decimal representation of $\frac{1}{20^{20}}$ is followed by a 9. If we consider just these $n + 1$ digits after the decimal point, we can express it as the fraction $\frac{9}{10^{n+1}}$. Since there are more digits after this 9 in the decimal representation of $\frac{1}{20^{20}}$, it is true that

$$\frac{1}{20^{20}} > \frac{9}{10^{n+1}}.$$

Rinse and repeat and we have

$$20^{20} < \frac{10^{n+1}}{9}.$$

Multiplying both sides by $\frac{9}{10^{20}}$ gives

$$9 \cdot 2^{20} < 10^{n-19}.$$

We previously found that $2^{20} = 1048576$, which has 7 digits. Multiplying this by 9 will keep the number of digits the same (make sure to see why). Thus,

$$n - 19 > 6 \quad \text{or} \quad n > 25.$$

If $n < 27$ and $n > 25$, then $n = \boxed{\text{(D)} 26}$.

Solution 8 (reworded Solution 1)

We split $\frac{1}{20^{20}}$ into $\frac{1}{2^{20}} \cdot \frac{1}{10^{20}}$. Memorizing (or just multiplying $1024 \cdot 1024$) $2^{20} = 1048576$, we know that

$$\frac{1}{1000000} < \frac{1}{2^{20}} = \frac{1}{1048576}.$$

Taking notice of the fact that $\frac{1}{1000000} = 10^{-6}$ and has 5 zeroes following the decimal point, we know that $\frac{1}{1048576}$ must be a bit smaller than that and therefore have 6 of them. So, multiplying that up with $\frac{1}{10^{20}}$ would give another 20 zeroes from 10^{-20} , and therefore there are $20 + 6 = \boxed{26}$ zeroes following the decimal point in the original fraction.

by DearPrince

13. (AMC 10 10B 2020) Andy the Ant lives on a coordinate plane and is currently at $(-20, 20)$ facing east (that is, in the positive x -direction). Andy moves 1 unit and then turns 90° left. From there, Andy moves 2 units (north) and then turns 90° left. He then moves 3 units (west) and again turns 90° left. Andy continues his progress, increasing his distance each time by 1 unit and always turning left. What is the location of the point which Andy makes the 2020th left turn?

- (A) $(-1030, -994)$ (B) $(-1030, -990)$ (C) $(-1026, -994)$ (D) $(-1026, -990)$
 (E) $(-1022, -994)$

Answer: B

Solution 1 (Associative Property)

Andy makes a total of 2020 moves: 1010 horizontal (505 left and 505 right) and 1010 vertical (505 up and 505 down).

The x -coordinate of Andy's final position is

$$-20 + \overbrace{\underbrace{1-3}_{-2} + \underbrace{5-7}_{-2} + \underbrace{9-11}_{-2} + \cdots + \underbrace{2017-2019}_{-2}}^{1010 \text{ terms, } 505 \text{ pairs}} = -20 - 2 \cdot 505 = -1030.$$

The y -coordinate of Andy's final position is

$$20 + \overbrace{\underbrace{2-4}_{-2} + \underbrace{6-8}_{-2} + \underbrace{10-12}_{-2} + \cdots + \underbrace{2018-2020}_{-2}}^{1010 \text{ terms, } 505 \text{ pairs}} = 20 - 2 \cdot 505 = -990.$$

Together, we have $(x, y) = \boxed{\text{(B)} (-1030, -990)}$.

Solution 2 (Pattern)

You can find that every four moves both coordinates decrease by 2. Therefore, both coordinates need to decrease by two 505 times. You subtract, giving you the answer of $\boxed{\text{(B)} (-1030, -990)}$.

Solution 3 (Pattern)

Let's first mark the first few points Andy will arrive at:

Starting Point (0th move): $(-20, 20)$ 1st move: $(-19, 20)$ 2nd move: $(-19, 22)$ 3rd move: $(-22, 22)$ 4th move: $(-22, 18)$ 5th move: $(-17, 18)$ 6th move: $(-17, 24)$ 7th move: $(-24, 24)$

In the 3rd move Andy lands on $(-22, 22)$, in the 7th move, Andy lands on $(-24, 24)$.

There is a pattern, for every 4 moves (starting from the 3rd move), Andy will arrive on a coordinate in the form of $(-2n, 2n)$. From this we can deduce:

(1) 3rd move: $(-22, 22)$ (2) 7th move: $(-24, 24)$ (3) 11th move: $(-26, 26)$ (4) 15th move: $(-28, 28)$ (n) ($4n - 1$)th move: $(-20 - 2n, 20 + 2n)$

We have $2019 = 4 \cdot 505 - 1$, for which $n = 505$ and $20 + 2 \cdot 505 = 1030$. So, on the 2019th move Andy is at $(-1030, 1030)$.

Because the problem asks for the 2020th move, $1030 - 2020 = -990$, on the 2020th move, Andy will be on $\boxed{\text{(B)} (-1030, -990)}$.

Solution 4 (Answer Choices)

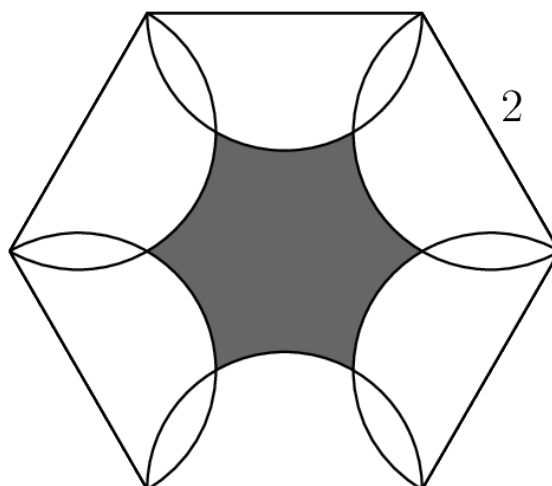
Drawing the spiral out, you obtain that every 4th turn occurs on the bottom left corner, whose coordinates difference stay the same(40):

$(-20, 20)(-22, 18)(-24, 16)...$

So the X and Y coordinates will always have a difference of 40.

Now looking at the answer choices, only answer choice B's X and Y coordinates have a difference of 40, there (B) is the answer. :)

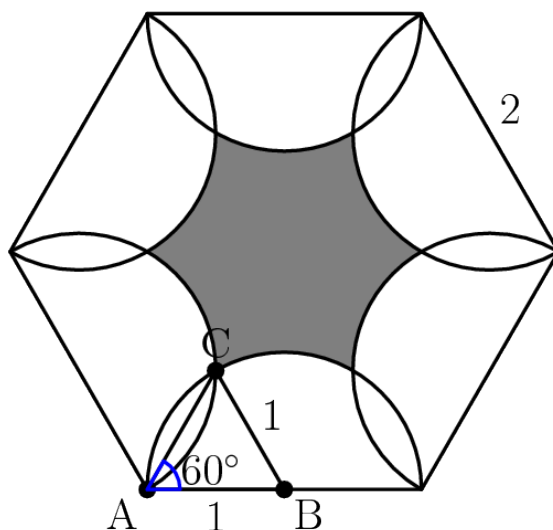
14. (AMC 10 10B 2020) As shown in the figure below, six semicircles lie in the interior of a regular hexagon with side length 2 so that the diameters of the semicircles coincide with the sides of the hexagon. What is the area of the shaded region — inside the hexagon but outside all of the semicircles?



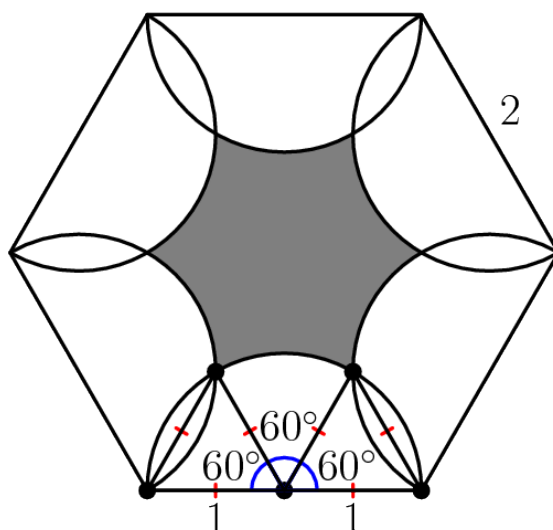
- (A) $6\sqrt{3}-3\pi$ (B) $\frac{9\sqrt{3}}{2}-2\pi$ (C) $\frac{3\sqrt{3}}{2}-\frac{\pi}{3}$ (D) $3\sqrt{3}-\pi$ (E) $\frac{9\sqrt{3}}{2}-\pi$

Answer: D

Solution 1



Let point A be a vertex of the regular hexagon, let point B be the midpoint of the line connecting point A and a neighboring vertex, and let point C be the second intersection of the two semicircles that pass through point A . Then, $BC = 1$, since B is the center of the semicircle with radius 1 that C lies on, $AB = 1$, since B is the center of the semicircle with radius 1 that A lies on, and $\angle BAC = 60^\circ$, as a regular hexagon has angles of 120° , and $\angle BAC$ is half of any angle in this hexagon. Now, since $AB = BC$, $\angle ACB = \angle BAC = 60^\circ$. Since the angles in a triangle sum to 180° , $\angle ABC$ is also 60° . Therefore, $\triangle ABC$ is an equilateral triangle with side lengths of 1.

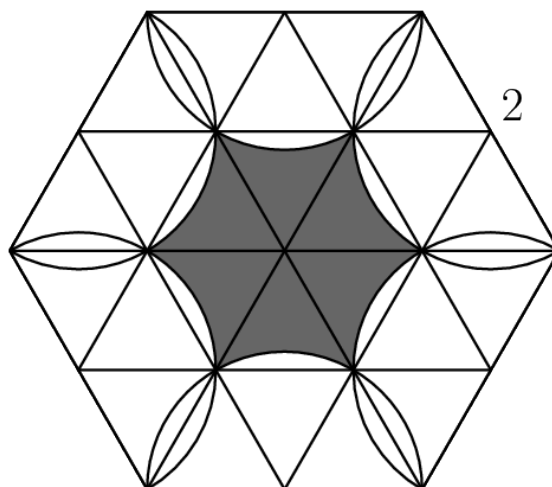


Since the area of a regular hexagon can be found with the formula $\frac{3\sqrt{3}s^2}{2}$, where s is the side length of the hexagon, the area of this hexagon is $\frac{3\sqrt{3}(2^2)}{2} = 6\sqrt{3}$. Since the area of an equilateral triangle can be found with the formula $\frac{\sqrt{3}}{4}s^2$, where s is the side length of the equilateral triangle, the area of an equilateral triangle with

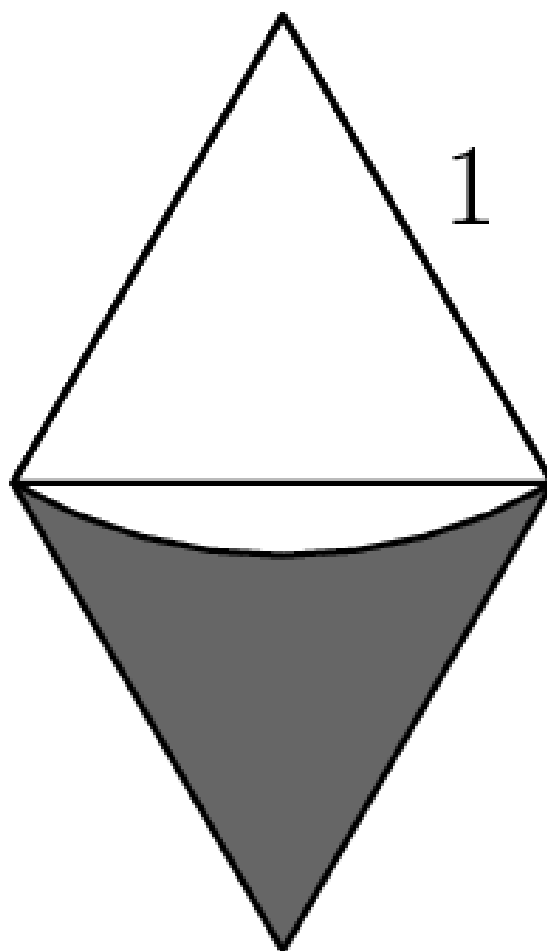
side lengths of 1 is $\frac{\sqrt{3}}{4} (1^2) = \frac{\sqrt{3}}{4}$. Since the area of a circle can be found with the formula πr^2 , the area of a sixth of a circle with radius 1 is $\frac{\pi(1^2)}{6} = \frac{\pi}{6}$. In each sixth of the hexagon, there are two equilateral triangles colored white, each with an area of $\frac{\sqrt{3}}{4}$, and one-sixth of a circle with radius 1 colored white, with an area of $\frac{\pi}{6}$. The rest of the sixth is colored gray. Therefore, the total area that is colored white in each sixth of the hexagon is $2 \left(\frac{\sqrt{3}}{4} \right) + \frac{\pi}{6}$, which equals $\frac{\sqrt{3}}{2} + \frac{\pi}{6}$, and the total area colored white is $6 \left(\frac{\sqrt{3}}{2} + \frac{\pi}{6} \right)$, which equals $3\sqrt{3} + \pi$. Since the area colored gray equals the total area of the hexagon minus the area colored white, the area colored gray is $6\sqrt{3} - (3\sqrt{3} + \pi)$, which equals **(D)** $3\sqrt{3} - \pi$.

Solution 2

First, subdivide the hexagon into 24 equilateral triangles with side length 1:



Now note that the entire shaded region is just 6 times this part:



The entire rhombus is just 2 equilateral triangles with side lengths of 1, so it has an area of:

$$2 \cdot \frac{\sqrt{3}}{4} = \frac{\sqrt{3}}{2}$$

The sector that is not included has an area of:

$$\frac{1}{6} \cdot \pi \cdot 1^2 = \frac{\pi}{6}$$

Hence, the area of the shaded region in that section is

$$\frac{\sqrt{3}}{2} - \frac{\pi}{6}$$

For a final area of:

$$6 \left(\frac{\sqrt{3}}{2} - \frac{\pi}{6} \right) = 3\sqrt{3} - \pi \Rightarrow \boxed{(D)}$$

N828335

Solution 3

We can see there are six congruent "leaves" on the original figure, and let us say the total area of them is S . If we make a circle with radius one in the circle (the dashed one), we can get another six congruent "leaves", and they are the same as the old six ones. So the area of the six new leaves can also be represented by S . Then the area of the shaded region can be expressed in the following two ways:

[1]. **Area** of hexagon - (**Area** of six semicircles - **Area** of six **old** "leaves") = $\frac{\sqrt{3}}{4} \cdot 4 \cdot 6 - (3\pi - S)$; [2]. **Area** of dashed circle - **Area** of six **new** "leaves" = $\pi - S$;

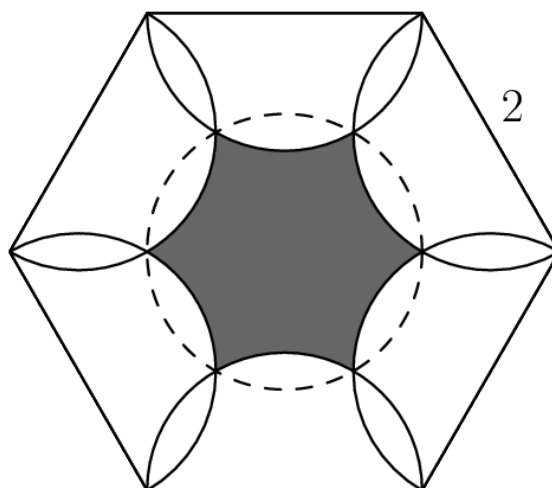
Then we have $\frac{\sqrt{3}}{4} \cdot 4 \cdot 6 - (3\pi - S) = \pi - S$ $S = 2\pi - 3\sqrt{3}$

Plugging S back in to either [1] or [2], we can solve for the shaded area = $3\sqrt{3} - \pi$

→ (D)

–RyanZ@BRS

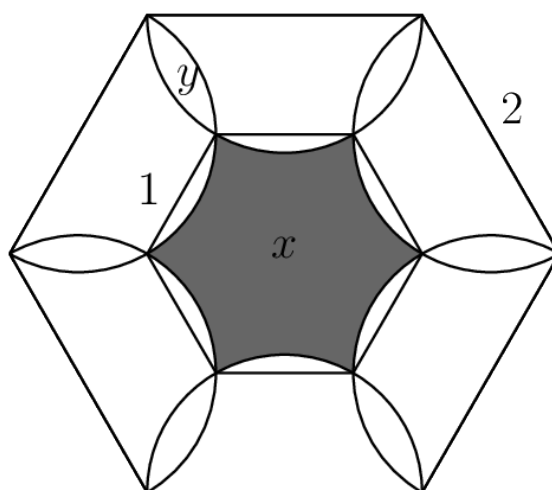
–Extremelysupercooldude (Minor latex and grammar edits)



Solution 4 (Variables)

We can set several variables.

Let x be the area of the shaded region, and let y be the area of one of the 'leaf-shaped' regions.



We then can connect the vertices of x , and we end up with another hexagon, with side length 1. Since each of the regions that are outside of x but inside of the smaller hexagon is half of y , their total area is $\frac{6y}{2} = 3y$. The smaller hexagon minus this region would be x . So,

$$x = \frac{3\sqrt{3}}{2} \cdot 1^2 - 3y$$

Also, the area of the larger hexagon minus the semicircles *then add* back $6y$ would also be x . (Note that by just subtracting the semicircles from the larger hexagon would result in an overlap of $6y$.) So,

$$x = \frac{3\sqrt{3}}{2} \cdot 2^2 - 3\pi + 6y$$

Equating the equations, we have

$$\begin{aligned} \frac{3\sqrt{3}}{2} \cdot 1^2 - 3y &= \frac{3\sqrt{3}}{2} \cdot 2^2 - 3\pi + 6y \\ \frac{3\sqrt{3}}{2} - 3y &= 6\sqrt{3} - 3\pi + 6y \end{aligned}$$

Solving for y , we have

$$\begin{aligned} 9y &= \frac{3\sqrt{3}}{2} - 6\sqrt{3} + 3\pi \\ &= -\frac{9\sqrt{3}}{2} + 3\pi \end{aligned}$$

Then, we can plug this back into

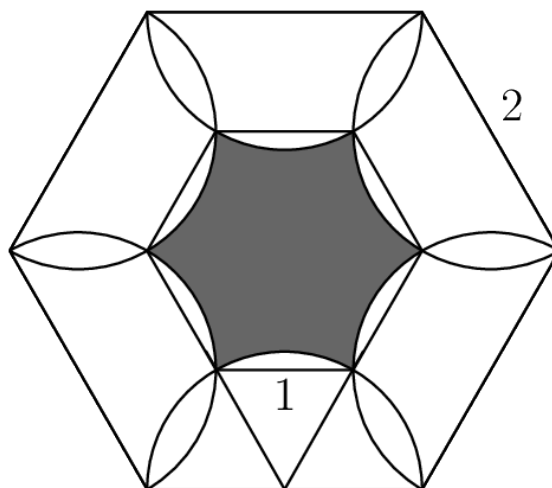
$$x = \frac{3\sqrt{3}}{2} \cdot 1^2 - 3y$$

Since $9y = -\frac{9\sqrt{3}}{2} + 3\pi$, dividing by 3 gives

$$\begin{aligned} 3y &= -\frac{9\sqrt{3}}{2} \times \frac{1}{3} + \pi \\ &= -\frac{3\sqrt{3}}{2} + \pi \end{aligned}$$

Therefore, $x = \frac{3\sqrt{3}}{2} - (-\frac{3\sqrt{3}}{2} + \pi) \implies x = \boxed{\text{(D)} 3\sqrt{3} - \pi}$.

Solution 5



First we connect every vertex of the shaded portion to another, forming an equilateral hexagon. Each interior angle of this hexagon is 120° , the interior angles of the new triangle are all 60° . As the radius of the semicircle is 1, we know that this hexagon has side length 1, so its area is given by the combined area of six equilateral triangles with side length 1, or $\frac{\sqrt{3}}{4} \cdot 6 = \frac{3\sqrt{3}}{2}$.

Now, we have to subtract off the six mini-sectors between the triangle and the shaded area. The area of one of these mini-sectors is the difference between a 60° sector of the semicircle and the equilateral we just drew. The area of a sector is just $\frac{\pi}{6}$, and the area of a triangle is $\frac{\sqrt{3}}{4}$. Therefore our combined area of the mini-sectors is

$$6 \left(\frac{\pi}{6} - \frac{\sqrt{3}}{4} \right) = \pi - \frac{3\sqrt{3}}{2}.$$

Subtracting this from the area of the (smaller) hexagon, the desired area is

$$\frac{3\sqrt{3}}{2} - \left(\pi - \frac{3\sqrt{3}}{2} \right) = \boxed{\text{(D)} 3\sqrt{3} - \pi}$$

-Benedict T (countmath1)

15. (AMC 10 10B 2020) Steve wrote the digits 1, 2, 3, 4, and 5 in order repeatedly from left to right, forming a list of 10,000 digits, beginning 123451234512... He then erased every third digit from his list (that is, the 3rd, 6th, 9th, ... digits from the left), then erased every fourth digit from the resulting list (that is, the 4th, 8th, 12th, ... digits from the left in what remained), and then erased every fifth digit from what remained at that point. What is the sum of the three digits that were then in the positions 2019, 2020, 2021?

(A) 7 (B) 9 (C) 10 (D) 11 (E) 12

Answer: D

Solution 1 (Simulation)

Note that cycles exist initially and after each round of erasing.

Let the parentheses denote cycles. It follows that:

a) Initially, the list has cycles of length 5 :

$$(12345) = 12345123451234512345 \dots$$

- b) To find one cycle after the first round of erasing, we need one cycle of length $\text{lcm}(3, 5) = 15$ before erasing. So, we first group $\frac{15}{5} = 3$ copies of the current cycle into one, then erase:

$$\begin{aligned} (12345) &\longrightarrow (123451234512345) \\ &\longrightarrow (\cancel{123451234512345}) \\ &\longrightarrow (1245235134). \end{aligned}$$

As a quick confirmation, one cycle should have length $15 \cdot \left(1 - \frac{1}{3}\right) = 10$ at this point.

- c) To find one cycle after the second round of erasing, we need one cycle of length $\text{lcm}(4, 10) = 20$ before erasing. So, we first group $\frac{20}{10} = 2$ copies of the current cycle into one, then erase:

$$\begin{aligned} (1245235134) &\longrightarrow (12452351341245235134) \\ &\longrightarrow (\cancel{12452351341245235134}) \\ &\longrightarrow (124235341452513). \end{aligned}$$

As a quick confirmation, one cycle should have length $20 \cdot \left(1 - \frac{1}{4}\right) = 15$ at this point.

- d) To find one cycle after the third round of erasing, we need one cycle of length $\text{lcm}(5, 15) = 15$ before erasing. We already have it here, so we erase:

$$\begin{aligned} (124235341452513) &\longrightarrow (\cancel{124235341452513}) \\ &\longrightarrow (124253415251). \end{aligned}$$

As a quick confirmation, one cycle should have length $15 \cdot \left(1 - \frac{1}{5}\right) = 12$ at this point.

Since 2019, 2020, 2021 are congruent to 3, 4, 5 modulo 12, respectively, the three digits in the final positions 2019, 2020, 2021 are 4, 2, 5, respectively:

$$(124253415251).$$

Therefore, the answer is $4 + 2 + 5 = \boxed{\text{(D) } 11}$.

Solution 2 (Simulation)

After erasing every third digit, the list becomes 1245235134... repeated. After erasing every fourth digit from this list, the list becomes 124235341452513... repeated. Finally, after erasing every fifth digit from this list, the list becomes 124253415251... repeated. Since this list repeats every 12 digits and 2019, 2020, 2021 are 3, 4, 5 respectively in $(\text{mod } 12)$, we have that the 2019th, 2020th, and 2021st digits are the 3rd, 4th, and 5th digits respectively. It follows that the answer is $4 + 2 + 5 = \boxed{\text{(D) } 11}$.

Solution 3 (Analysis)

Note that cycles exist initially and after each round of erasing.

We will consider one cycle after all three rounds of erasing. Suppose this cycle has length L before any round of erasing. It follows that:

- a) Initially, one cycle has length 5, from which L must be divisible by 5.

- b) After the first round of erasing, one cycle has length $L \left(1 - \frac{1}{3}\right) = \frac{2}{3}L$, from which L must be divisible by 3.
- c) After the second round of erasing, one cycle has length $L \left(1 - \frac{1}{3}\right) \left(1 - \frac{1}{4}\right) = \frac{1}{2}L$, from which L must be divisible by 2.
- d) After the third round of erasing, one cycle has length $L \left(1 - \frac{1}{3}\right) \left(1 - \frac{1}{4}\right) \left(1 - \frac{1}{5}\right) = \frac{2}{5}L$, from which L must be divisible by 5.

The least such positive integer L is $\text{lcm}(5, 3, 2) = 30$. So, there is a repeating pattern for every 30 digits on the original list. As shown below, the digits erased in the first, second, and third rounds are colored in red, yellow, and green, respectively:

1	2	3	4	5	2	3	4	5	1	2	3	4	5	1	2	3	4	5	1	2	3	4	5	1	2	3	4	5	1	2	3	4	5
1	2	3	4	5	2	3	4	5	1	2	3	4	5	1	2	3	4	5	1	2	3	4	5	1	2	3	4	5	1	2	3	4	5
1	2	3	4	5	2	3	4	5	1	2	3	4	5	1	2	3	4	5	1	2	3	4	5	1	2	3	4	5	1	2	3	4	5
1	2	3	4	5	2	3	4	5	1	2	3	4	5	1	2	3	4	5	1	2	3	4	5	1	2	3	4	5	1	2	3	4	5

As indicated by the white squares, each group of 30 digits on the original list has $\frac{2}{5} \cdot 30 = 12$ digits remaining on the final list.

Since 2019, 2020, 2021 are congruent to 3, 4, 5 modulo 12, respectively, the three digits in the final positions 2019, 2020, 2021 are 4, 2, 5, respectively:

1	2	3	4	5	2	3	4	5	1	2	3	4	5	1	2	3	4	5	1	2	3	4	5	1	2	3	4	5	1	2	3	4	5

Therefore, the answer is $4 + 2 + 5 = \boxed{\text{(D)} 11}$.

Solution 4 (Analysis)

As the LCM of 3, 4, and 5 is 60, let us look at a 60-digit block of original numbers (many will be erased by Steve). After he erases every third number $\left(\frac{1}{3}\right)$, then every fourth number of what remains $\left(\frac{1}{4}\right)$, then every fifth number of what remains $\left(\frac{1}{5}\right)$, we are left with $\frac{2}{3} \cdot \frac{3}{4} \cdot \frac{4}{5} \cdot 60 = 24$ digits from this 60-digit block. $2019 \equiv 3 \pmod{24}$, $2020 \equiv 4 \pmod{24}$, $2021 \equiv 5 \pmod{24}$. Writing out the first few digits of this sequence, we have $\underbrace{1}_{\#1}, \underbrace{2}_{\#2}, \cancel{3}, \underbrace{4}_{\#3}, \cancel{5}, \cancel{1}, \underbrace{2}_{\#4}, \cancel{3}, \cancel{4}, \underbrace{5}_{\#5}, \dots$. Therefore, our

answer is $4 + 2 + 5 = \boxed{\text{(D)} 11}$.

Solution 5

Lemma: Given a sequence $a_1, a_2, a_3, \dots$, and an positive integer $k > 2$. If we erase every k th item in this sequence, and we name $b_1, b_2, b_3, \dots$ as the remaining sequence. Then we have

$$b_{(k-1)m+1} = a_{km+1}, b_{(k-1)m+2} = a_{km+2}, \dots, b_{(k-1)m+k-1} = a_{km+k-1}.$$

Proof: For a_{km+j} with some $j, 1 \leq j \leq k-1$, we will have m items removed before this item, so it becomes $b_{(k-1)m+j}$ in the new sequence. Hence, we have $b_{(k-1)m+j} = a_{km+j}$.

If we start with $a_1, a_2, a_3, \dots$, and let $b_1, b_2, \dots$ be the sequence after removing all the indexes that are multiples of 3. Then, we have $b_{2n+1} = a_{3n+1}, b_{2n+2} = a_{3n+2}$.

Similarly, if we start with $b_1, b_2, b_3, \dots$, and remove all multiples of 4, and get $c_1, c_2, \dots$. We have $c_{3n+1} = b_{4n+1}, c_{3n+2} = b_{4n+2}, c_{3n+3} = b_{4n+3}$.

If we start with $c_1, c_2, \dots$ and $d_1, d_2, \dots$ are remove all multiples of 5, we get

$$d_{4n+1} = c_{5n+1}, d_{4n+2} = c_{5n+2}, d_{4n+3} = c_{5n+3}, d_{4n+4} = c_{5n+4}.$$

Therefore,

$$\begin{aligned} d_{2019} &= d_{4 \cdot 504 + 3} = c_{5 \cdot 504 + 3} = c_{2523} = c_{3 \cdot 840 + 3} = b_{4 \cdot 840 + 3} = b_{3363} = b_{2 \cdot 1681 + 1} = a_{3 \cdot 1681 + 1} = a_{5044} = a_4 = 4 \\ d_{2020} &= d_{4 \cdot 504 + 4} = c_{5 \cdot 504 + 4} = c_{2524} = c_{3 \cdot 841 + 1} = b_{4 \cdot 841 + 1} = b_{3365} = b_{2 \cdot 1682 + 1} = a_{3 \cdot 1682 + 1} = a_{5047} = a_2 = 2 \\ d_{2021} &= d_{4 \cdot 505 + 1} = c_{5 \cdot 505 + 1} = c_{2526} = c_{3 \cdot 841 + 3} = b_{4 \cdot 841 + 3} = b_{3367} = b_{2 \cdot 1683 + 1} = a_{3 \cdot 1683 + 1} = a_{5050} = a_5 = 5 \end{aligned}$$

and the answer is $4 + 2 + 5 = \boxed{\text{(D)} 11}$.

16. (AMC 10 10B 2020) Bela and Jenn play the following game on the closed interval $[0, n]$ of the real number line, where n is a fixed integer greater than 4. They take turns playing, with Bela going first. At his first turn, Bela chooses any real number in the interval $[0, n]$. Thereafter, the player whose turn it is chooses a real number that is more than one unit away from all numbers previously chosen by either player. A player unable to choose such a number loses. Using optimal strategy, which player will win the game?

(A) Bela will always win. (B) Jenn will always win. (C) Bela will win if and only if n is odd.
(D) Jenn will win if and only if n is odd. (E) Jenn will win if and only if $n > 8$.

Answer: A

Solution 1 (Game Theory)

We use game theory.

Notice how they say Optimal Strategy. This means that each player plays the best move in any scene.

We then solve through cases.

We set $n > 4$, n is even. Say $n = 10$. Then, we see that Bela could pick either 3 or 4, and then Jenn would be forced to choose either 1(2), 5, or 6. But notice how you can only make one move, and 1 and $[5, 6]$ are on opposite sides. So therefore, Bela will always win.

We set $n > 4$, n is odd. Say $n = 9$. This is even easier! Bela could just choose the middle, and then again, Jenn would be forced to pick a move that Bela could just replicate and win. So, Bela will also always win.

Through deduction, our answer is (A) Bela will always win. *But that's just a theory, a game theory!*

Solution 2 (Symmetry)

If Bela selects the middle number in the range $[0, n]$ and then mirror whatever number Jenn selects, then if Jenn can select a number within the range, so can Bela. Jenn will always be the first person to run out of a number to choose, so the answer is (A) Bela will always win.

Solution 3 (Educated Guess)

First of all, realize that the parity of n likely has no effect on the strategy, since Bela obviously wins when $n = 1$, and it's not hard to find a winning strategy for

$n = 2$ or $n = 3$ (start with $n/2$). Similarly, there is no reason the strategy would change when $n > 8$.

So we are left with (A) and (B). From here it is best to try out random numbers and try to find the strategy that will let Bela win, but if you can't find it, realize that it is more likely the answer is (A) Bela will always win since Bela has the first move and thus has more control.

Solution 4 (Simulation)

We can just play the game. We can draw a number line to 9 and assume Bela and Jenn will only play the integer values. From now after playing one round of the game, Bela will win so answers (B), (D), and (E) are eliminated. Now we want to test if Bela will win if n is even so we can draw another number line this time up to 10. Now after playing the game, we can find that Bela won yet again so the answer is (A) Bela will always win.

-minor edits by jjbros

Solution 5 (Answer Choices)

Play out $n = 5$ and $n = 6$. In the first case, Bela can choose 0 and win, and in the second case, Bela can choose 1 and win (try it out). This rules out (B), (C), and (D).

Next, consider (E). Assume for the sake of contradiction that this is true; an alternate way of stating this is that for $n > 8$, the second player always wins (we ignore the "only if" statement). Thus, $n = 9$ is a win for the second player. Consider $n = 11$. If Bela chooses 11, then the game becomes an $n = 9$ case where the first player is instead Jenn. However, by the claim that the second player should win $n > 8$, this game should be a win for Bela, but we have already stated that any games with $n > 8$ should be a win for Jenn, which is a contradiction. Thus (E) is incorrect, and so the answer is (A) Bela will always win.

Solution 6 (Similar to Solution 2)

Bela can choose the exact midpoint of the number line. Whatever Jenn does, Bela can copy her moves in the opposite side half of the number line. Jenn will always run out of space first.

17. (AMC 10 10B 2020) There are 10 people standing equally spaced around a circle. Each person knows exactly 3 of the other 9 people: the 2 people standing next to him or her, as well as the person directly across the circle. How many ways are there for the 10 people to split up into 5 pairs so that the members of each pair know each other?

(A) 11 (B) 12 (C) 13 (D) 14 (E) 15

Answer: C

Solution

Consider the 10 people to be standing in a circle, where two people opposite of each other form a diameter of the circle.

Let us use casework on the number of pairs that form a diameter of the circle.

Case 1: 0 diameters

There are 2 ways: either 1 pairs with 2, 3 pairs with 4, and so on or 10 pairs with 1, 2 pairs with 3, etc. So 2 cases.

Case 2: 1 diameter

There are 5 possible diameters to draw (everyone else pairs with the person next to them). So $\boxed{5}$ cases.

Note that there cannot be 2 diameters since there would be one person on either side that will not have a pair adjacent to them. The only scenario forced is when the two people on either side would be paired up across a diameter. Thus, a contradiction will arise.

Case 3: 3 diameters

There are 5 possible sets of 3 diameters to draw. Notice we are technically choosing the number of ways to choose a pair of two diameters that are neighbors to each other. This means we can choose the first diameter in the pair, and have only two diameters to choose from for the second in the pair. This means we have $5 \cdot 2 = 10$ possibilities for choosing 5 neighboring diameters. However, notice that there are duplicates, so we divide the 10 possibilities by 2 to get $\boxed{5}$.

Note that there cannot be a case with 4 diameters because then there would have to be 5 diameters for the two remaining people as they have to be connected with a diameter. A contradiction arises.

Case 4: 5 diameters

There is only $\boxed{1}$ way to do this.

Thus, in total there are $2 + 5 + 5 + 1 = \boxed{(C) 13}$ possible ways.

18. (AMC 10 10B 2020) An urn contains one red ball and one blue ball. A box of extra red and blue balls lies nearby. George performs the following operation four times: he draws a ball from the urn at random and then takes a ball of the same color from the box and returns those two matching balls to the urn. After the four iterations the urn contains six balls. What is the probability that the urn contains three balls of each color?

(A) $\frac{1}{6}$ (B) $\frac{1}{5}$ (C) $\frac{1}{4}$ (D) $\frac{1}{3}$ (E) $\frac{1}{2}$

Answer: B

Solution 1 (The Shortcut Passed Every Barrier)

Suppose that we have a deck, currently containing just one black card. We then insert n red cards one-by-one into the deck at random positions. It is easy to see using induction, that the black card is randomly situated in the deck.

Now, suppose that we have this deck again, with only one black card. Each time we pick a red ball, we place a card above the black card, and each time we pick a blue ball, we place a card below the black card. It is easy to see that the probability that the card is inserted into the top part of the deck is simply equal to the number of red balls divided by the total number of balls, and the probability that the card is inserted into the bottom part of the deck is equal to the number of blue balls divided by the total number of balls. Therefore, this is equivalent to inserting the card randomly into the deck.

Finally, four more red cards will be inserted into the deck, and so the black card can be in five possible positions. Only one corresponds to having three balls of each type. Our probability is thus $\frac{1}{5}$, and so the answer is $\boxed{B}$.

Solution 2

Let R denote the action where George selects a red ball and B denote the action where he selects a blue one. Now, in order to get 3 balls of each color, he needs 2 more of both R and B .

There are 6 cases: $RRBB, RBRB, RBBR, BBRR, BRBR, BRRB$ (we can confirm that there are only 6 since $\binom{4}{2} = 6$). However we can clump $RRBB + BBRR$, $RBRB + BRBR$, and $RBBR + BRRB$ together since they are equivalent by symmetry.

CASE 1: $RRBB$ and $BBRR$

Let's find the probability that he picks the balls in the order of $RRBB$.

The probability that the first ball he picks is red is $\frac{1}{2}$.

Now there are 2 reds and 1 blue in the urn. The probability that he picks red again is now $\frac{2}{3}$.

There are 3 reds and 1 blue now. The probability that he picks a blue is $\frac{1}{4}$.

Finally, there are 3 reds and 2 blues. The probability that he picks a blue is $\frac{2}{5}$.

So the probability that the $RRBB$ case happens is $\frac{1}{2} \cdot \frac{2}{3} \cdot \frac{1}{4} \cdot \frac{2}{5} = \frac{1}{30}$. However, since the $BBRR$ case is the exact same by symmetry, case 1 has a probability of $\frac{1}{30} \cdot 2 = \frac{1}{15}$ chance of happening.

CASE 2: $RBRB$ and $BRBR$

Let's find the probability that he picks the balls in the order of $RBRB$.

The probability that the first ball he picks is red is $\frac{1}{2}$.

Now there are 2 reds and 1 blue in the urn. The probability that he picks blue is $\frac{1}{3}$.

There are 2 reds and 2 blues now. The probability that he picks a red is $\frac{1}{2}$.

Finally, there are 3 reds and 2 blues. The probability that he picks a blue is $\frac{2}{5}$.

So the probability that the $RBRB$ case happens is $\frac{1}{2} \cdot \frac{1}{3} \cdot \frac{1}{2} \cdot \frac{2}{5} = \frac{1}{30}$. However, since the $BRBR$ case is the exact same by symmetry, case 2 has a probability of $\frac{1}{30} \cdot 2 = \frac{1}{15}$ chance of happening.

CASE 3: $RBBR$ and $BRRB$

Let's find the probability that he picks the balls in the order of $RBBR$.

The probability that the first ball he picks is red is $\frac{1}{2}$.

Now there are 2 reds and 1 blue in the urn. The probability that he picks blue is $\frac{1}{3}$.

There are 2 reds and 2 blues now. The probability that he picks a blue is $\frac{1}{2}$.

Finally, there are 2 reds and 3 blues. The probability that he picks a red is $\frac{2}{5}$.

So the probability that the $RBBR$ case happens is $\frac{1}{2} \cdot \frac{1}{3} \cdot \frac{1}{2} \cdot \frac{2}{5} = \frac{1}{30}$. However, since the $BRRB$ case is the exact same by symmetry, case 3 has a probability of $\frac{1}{30} \cdot 2 = \frac{1}{15}$ chance of happening.

Adding up the cases, we have $\frac{1}{15} + \frac{1}{15} + \frac{1}{15} = \boxed{\text{(B)} \frac{1}{5}}$ quacker88

Solution 3

We know that we need to find the probability of adding 2 red and 2 blue balls in some order. There are 6 ways to do this, since there are $\binom{4}{2} = 6$ ways to arrange $RRBB$ in some order. We will show that the probability for each of these 6 ways is the same.

We first note that the denominators should be counted by the same number. This number is $2 \cdot 3 \cdot 4 \cdot 5 = 120$. This is because 2, 3, 4, and 5 represent how many choices there are for the four steps. No matter what the k -th step involves $k+1$ numbers to choose from.

The numerators are the number of successful operations. No matter the order, the first time a red is added will come from 1 choice and the second time will come from 2 choices, since that is how many reds are in the urn originally. The same goes for the blue ones. The numerator must equal $(1 \cdot 2)^2$.

Therefore, the probability for each of the orderings of $RRBB$ is $\frac{4}{120} = \frac{1}{30}$. There are 6 of these, so the total probability is $\boxed{(B) \frac{1}{5}}$.

Solution 4

First, notice that when George chooses a ball he just adds another ball of the same color. On George's first move, he either chooses the red or the blue with a $\frac{1}{2}$ chance each. We can assume he chooses Red (chance $\frac{1}{2}$), and then multiply the final answer by two for symmetry. Now, there are two red balls and one blue ball in the urn. Then, he can either choose another Red (chance $\frac{2}{3}$), in which case he must choose two blues to get three of each, with probability $\frac{1}{4} \cdot \frac{2}{5} = \frac{1}{10}$ or a blue for two blue and two red in the urn, with chance $\frac{1}{3}$. If he chooses blue next, he can either choose a red then a blue, or a blue then a red. Each of these has a $\frac{1}{2} \cdot \frac{2}{5}$ for total of $2 \cdot \frac{1}{5} = \frac{2}{5}$. The total probability that he ends up with three red and three blue is $2 \cdot \frac{1}{2} \left(\frac{2}{3} \cdot \frac{1}{10} + \frac{1}{3} \cdot \frac{2}{5} \right) = \frac{1}{15} + \frac{2}{15} = \boxed{(B) \frac{1}{5}}$.

Solution 4 Alternate

A solution pretty similar to the above is as follows.

Your only two cases of having 3R and 3B after 4 iterations are (without regard to order; if you do order, you multiply by 2 on both the top and the bottom and yield the same result):

$1|1 \rightarrow 2|1 \rightarrow 3|1 \rightarrow 3|2 \rightarrow 3|3$, and this occurs with probability $(1)(2/3)(1/4)(2/5) = 1/15$.
 $1|1 \rightarrow 2|1 \rightarrow 2|2 \rightarrow 3|2 \rightarrow 3|3$, and this occurs with probability $(1)(1/3)(1)(2/5) = 2/15$.
 Probability of $2|2 \rightarrow 3|2$ is 1 because it doesn't matter what you do. We don't count for order here so either ball you put from the box into of the urn is going to get you a $3|2$ distribution. Same argument goes for the iteration from $1|1$ to $2|1$.

Summing these two up we get $\frac{1}{15} + \frac{2}{15} = \boxed{(B) \frac{1}{5}}$.

mathboy282

Solution 5

Let the probability that the urn ends up with more red balls be denoted $P(R)$. Since this is equal to the probability there are more blue balls, the probability there are equal amounts is $1 - 2P(R)$. $P(R)$ = the probability no more blues are chosen plus the probability only 1 more blue is chosen. The first case, $P(\text{no more blues}) = \frac{1}{2} \cdot \frac{2}{3} \cdot \frac{3}{4} \cdot \frac{4}{5} = \frac{1}{5}$.

The second case, $P(1 \text{ more blue}) = 4 \cdot \frac{1 \cdot 1 \cdot 2 \cdot 3}{2 \cdot 3 \cdot 4 \cdot 5} = \frac{1}{5}$. Thus, the answer is $1 - 2\left(\frac{1}{5} + \frac{1}{5}\right) = 1 - \frac{4}{5} = \boxed{(B) \frac{1}{5}}$.

Solution 6

Here X stands for R or B, and Y for the remaining color. After 3 rounds one can either have a $4 + 1$ configuration ($XXXXY$), or $3 + 2$ configuration ($XXXY$). The probability of getting to $XXXXYY$ from $XXXY$ is $\frac{2}{5}$. Observe that the probability of arriving to $4 + 1$ configuration is

$$\frac{2}{3} \cdot \frac{3}{4} = \frac{1}{2}$$

($\frac{2}{3}$ to get from XXY to $XXXY$, $\frac{3}{4}$ to get from $XXXY$ to $XXXXY$). Thus the probability of arriving to $3 + 2$ configuration is also $\frac{1}{2}$, and the answer is

$$\frac{1}{2} \cdot \frac{2}{5} = \boxed{\text{(B)} \frac{1}{5}}.$$

Solution 7

We can use dynamic programming to solve this problem.

We let $dp[i][j]$ be the probability that we end up with i red balls and j blue balls. Notice that there are only two ways that we can end up with i red balls and j blue balls: one is by fetching a red ball from the urn when we have $i - 1$ red balls and j blue balls and the other is by fetching a blue ball from the urn when we have i red balls and $j - 1$ blue balls.

Then we have $dp[i][j] = \frac{i-1}{i-1+j} dp[i-1][j] + \frac{j-1}{i-1+j} dp[i][j-1]$

Then we start can with $dp[1][1] = 1$ and try to compute $dp[3][3]$.

$i \setminus j$	1	2	3
1	1	1/2	1/3
2	1/2	1/3	1/4
3	1/3	1/4	1/5

The answer is $\boxed{\text{(B)} \frac{1}{5}}$.

(Solution by CircleOO)

Solution 8 (Easier Solution)

Basically, George is picking a ball and adding another ball of that color to the urn. Basically, he needs to pick a red ball twice and a blue ball twice to get 3 red and 3 blue balls in the urn. The number of ways to choose a red ball twice is $1 \cdot 2$ (at the start there's 1 red ball, then he needs to choose it again when there's 2, and then there is 3 red balls so he doesn't need to choose it again). Similarly the number of ways to choose blue balls twice is $2 \cdot 2 = 4$ ways to choose 2 red balls and 2 blue balls (not considering order). The total number of ways is $2 \cdot 3 \cdot 4 \cdot 5$. Adding order to the 4 ways to choose 2 red and 2 blue balls, it is $4C2$ because out of the 4 ball choosings, 2 of them must be red, and we automatically make the other ones blue. $\frac{\binom{4}{2} \cdot 2 \cdot 2}{2 \cdot 3 \cdot 4 \cdot 5} = \frac{1}{5}$.

Solution 9 (Desperate Guess)

Notice that the answer can either be (1R, 5B) (2R, 4B) (3R, 3B) (4R, 2B) or (5R, 1B). Also, the answer choices are all really basic. Just guess that each of these

cases are equal by symmetry so the probability of (3R, 3B) is $\boxed{\text{(B)} \frac{1}{5}}$.

-unhappyfarmer

19. (AMC 10 10B 2020) In a certain card game, a player is dealt a hand of 10 cards from a deck of 52 distinct cards. The number of distinct (unordered) hands that can be dealt to the player can be written as $158A00A4AA0$. What is the digit A ?
 (A) 2 (B) 3 (C) 4 (D) 6 (E) 7

Answer: A

Solution 1 (Mod 9)

$$158A00A4AA0 \equiv 1 + 5 + 8 + A + 0 + 0 + A + 4 + A + A + 0 \equiv 4A \pmod{9}$$

We're looking for the amount of ways we can get 10 cards from a deck of 52, which is represented by $\binom{52}{10}$.

$$\binom{52}{10} = \frac{52 \cdot 51 \cdot 50 \cdot 49 \cdot 48 \cdot 47 \cdot 46 \cdot 45 \cdot 44 \cdot 43}{10 \cdot 9 \cdot 8 \cdot 7 \cdot 6 \cdot 5 \cdot 4 \cdot 3 \cdot 2 \cdot 1}$$

We need to get rid of the multiples of 3, which will subsequently get rid of the multiples of 9 (if we didn't, the zeroes would mess with the equation since you can't divide by 0)

$$9 \cdot 5 = 45, 8 \cdot 6 = 48, \frac{51}{3} \text{ leaves us with } 17.$$

$$\frac{52 \cdot 51^{17} \cdot 50 \cdot 49 \cdot 48 \cdot 47 \cdot 46 \cdot 45 \cdot 44 \cdot 43}{10 \cdot 9 \cdot 8 \cdot 7 \cdot 6 \cdot 5 \cdot 4 \cdot 3 \cdot 2 \cdot 1}$$

Converting these into (mod 9), we have

$$\binom{52}{10} \equiv \frac{(-2) \cdot (-1) \cdot (-4) \cdot 4 \cdot 2 \cdot 1 \cdot (-1) \cdot (-2)}{1 \cdot (-2) \cdot 4 \cdot 2 \cdot 1} \equiv (-1) \cdot (-4) \cdot (-1) \cdot (-2) \equiv 8 \pmod{9}$$

$$4A \equiv 8 \pmod{9} \implies A = \boxed{\text{(A) } 2} \text{ quacker88}$$

Solution 2 (Ten's digit)

We're looking for the amount of ways we can get 10 cards from a deck of 52, which is represented by $\binom{52}{10}$.

$$\binom{52}{10} = \frac{52 \cdot 51 \cdot 50 \cdot 49 \cdot 48 \cdot 47 \cdot 46 \cdot 45 \cdot 44 \cdot 43}{10 \cdot 9 \cdot 8 \cdot 7 \cdot 6 \cdot 5 \cdot 4 \cdot 3 \cdot 2 \cdot 1}$$

And after simplifying, we get $26 \cdot 17 \cdot 7 \cdot 47 \cdot 46 \cdot 5 \cdot 11 \cdot 43$. Now, if we examine the number $158A00A4AA0$, we can notice that it is equal to some number n times 10. Therefore, we can divide 10 from the aforementioned expression and find the units digit, which will be A .

Now, after dividing ten, we will have $26 \cdot 17 \cdot 7 \cdot 47 \cdot 23 \cdot 11 \cdot 43$. We can then use modulo 10 and find that the unit digit of the expression is $\boxed{\text{(A) } 2}$ lucaswujc
 i_equal_tan_90 emerald_block

Solution 3 (Factors of 2)

$$\binom{52}{10} = \frac{52 \cdot 51 \cdot 50 \cdot 49 \cdot 48 \cdot 47 \cdot 46 \cdot 45 \cdot 44 \cdot 43}{10 \cdot 9 \cdot 8 \cdot 7 \cdot 6 \cdot 5 \cdot 4 \cdot 3 \cdot 2 \cdot 1} = 26 \cdot 17 \cdot 5 \cdot 7 \cdot 47 \cdot 46 \cdot 11 \cdot 43$$

Since this number is divisible by 4 but not 8, the last 2 digits must be divisible by 4 but the last 3 digits cannot be divisible by 8. This narrows the options down to 2 and 6.

Also, the number cannot be divisible by 3. Adding up the digits, we get $18 + 4A$. If $A = 6$, then the expression equals 42, a multiple of 3. This would mean that the entire number would be divisible by 3, which is not what we want. Therefore, the only option is $\boxed{\text{(A) } 2}$

Solution 4 (Mod 13)

It is not hard to check that 13 divides the number,

$$\binom{52}{10} = \frac{52 \cdot 51 \cdot 50 \cdot 49 \cdot 48 \cdot 47 \cdot 46 \cdot 45 \cdot 44 \cdot 43}{10 \cdot 9 \cdot 8 \cdot 7 \cdot 6 \cdot 5 \cdot 4 \cdot 3 \cdot 2 \cdot 1} = 26 \cdot 17 \cdot 5 \cdot 7 \cdot 47 \cdot 46 \cdot 11 \cdot 43.$$

As $10^3 \equiv -1 \pmod{13}$, using $\pmod{13}$ we have $13 \mid \overline{AA0} - \overline{0AA} + \overline{8A0} - \overline{15} = 110A + 781$. Thus $6A + 1 \equiv 0 \pmod{13}$, implying $A \equiv 2 \pmod{13}$ so the answer is **(A) 2**.

Solution 5 (Factors of 3, 7)

As in Solution 2, we see that

$$\binom{52}{10} = 26 \cdot 17 \cdot 5 \cdot 7 \cdot 47 \cdot 46 \cdot 11 \cdot 43,$$

which contains no factors of 3. Therefore, the sum of the digits must not be a multiple of 3. This sum is

$$1 + 5 + 8 + A + 0 + 0 + A + 4 + A + A + 0 = 18 + 4A.$$

It follows that $4A$ cannot be a multiple of 3, ruling out choices (B) and (D). Therefore, our possibilities are $A = 2, 4$, and 7. Now, notice that $\binom{52}{10}$ is divisible by 7. Therefore, we can plug each possible value of A into $158A00A4AA0$ and test for divisibility by 7. Conveniently, we see that the first value, $A = 2$, works. Thus, the answer is **(A) 2**. (To make our argument more rigorous, we can also test divisibility by 7 for $A = 4$ and 7 to show that these values do not work.)

Solution 6 (Very Factor Bashy CRT)

We note that:

$$\frac{(52)(51)(50)(49)(48)(47)(46)(45)(44)(43)}{(10)(9)(8)(7)(6)(5)(4)(3)(2)(1)} = (13)(17)(7)(47)(46)(5)(22)(43).$$

Let $K = (13)(17)(7)(47)(46)(5)(22)(43)$. This will help us find the last two digits modulo 4 and modulo 25. It is obvious that $K \equiv 0 \pmod{4}$. Also (although this not so obvious),

$$\begin{aligned} K &\equiv (13)(17)(7)(47)(46)(5)(22)(43) \\ &\equiv (13)(-8)(7)(-3)(-4)(5)(-3)(-7) \\ &\equiv (13)(-96)(21)(35) \\ &\equiv (13)(4)(-4)(10) \\ &\equiv (13)(-16)(10) \\ &\equiv (13)(9)(10) \\ &\equiv (117)(10) \\ &\equiv (-8)(10) \\ &\equiv 20 \pmod{25}. \end{aligned}$$

Therefore, $K \equiv 20 \pmod{100}$. Thus $K = 20$, implying that **(A) 2**.

Solution 7 (STRAIGHT BASH!!!)

Compute $\frac{52!}{10!42!} = 15820024220$. Therefore our answer is **(A) 2**.

Note: This is a very impractical solution for the AMC 10, here only for fun.

20. (AMC 10 10B 2020) Let B be a right rectangular prism (box) with edges lengths 1, 3, and 4, together with its interior. For real $r \geq 0$, let $S(r)$ be the set of points in 3-dimensional space that lie within a distance r of some point in B . The volume of $S(r)$ can be expressed as $ar^3 + br^2 + cr + d$, where a , b , c , and d are positive real numbers. What is $\frac{bc}{ad}$? (A) 6 (B) 19 (C) 24 (D) 26 (E) 38

Answer: B

Solution 1

Split $S(r)$ into 4 regions:

1. The rectangular prism itself
2. The extensions of the faces of B
3. The quarter cylinders at each edge of B
4. The one-eighth spheres at each corner of B

(To better visualize $S(r)$, visualize every point in or on B being a center of a sphere of radius r)

Region 1: Firstly, notice that $S(0) = d$, thus the set of all points a distance of 0 (the point itself) from any point in B is equal to d . This is all the points in B . The volume of B is $1 \cdot 3 \cdot 4 = 12$, so $d = 12$.

Region 2: This volume is equal to the surface area of B times r (these "extensions" are just more boxes). The volume is then $SA \cdot r = 2(1 \cdot 3 + 1 \cdot 4 + 3 \cdot 4)r = 38r$ to get $c = 38$. (Note that since we are asked to find a ratio, what r equals does not matter)

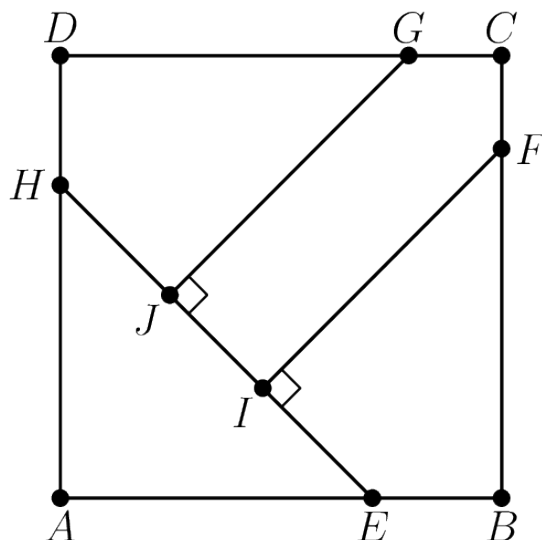
Region 3: We see that there are 12 quarter-cylinders, 4 of each type. We have 4 quarter-cylinders of height 1, 4 quarter-cylinders of height 3, 4 quarter-cylinders of height 4. Since 4 quarter-cylinders make a full cylinder, the total volume is $4 \cdot \frac{1\pi r^2}{4} + 4 \cdot \frac{3\pi r^2}{4} + 4 \cdot \frac{4\pi r^2}{4} = 8\pi r^2$. Therefore, $b = 8\pi$.

Region 4: There is an eighth-sphere of radius r at each corner of B . Since there are 8 corners, these eighth-spheres add up to 1 full sphere of radius r . The volume of this sphere is then $\frac{4}{3}\pi \cdot r^3$, so $a = \frac{4\pi}{3}$.

Using these values, $\frac{bc}{ad} = \frac{(8\pi)(38)}{(\frac{4\pi}{3})(12)} = \boxed{\text{(B) } 19}$.

To see a diagram of $S(r)$, view TheBeautyofMath's explanation video (Video Solution 2).

21. (AMC 10 10B 2020) In square $ABCD$, points E and H lie on $\overline{AB}$ and $\overline{DA}$, respectively, so that $AE = AH$. Points F and G lie on $\overline{BC}$ and $\overline{CD}$, respectively, and points I and J lie on $\overline{EH}$ so that $\overline{FI} \perp \overline{EH}$ and $\overline{GJ} \perp \overline{EH}$. See the figure below. Triangle AEH , quadrilateral $BFIE$, quadrilateral $DHJG$, and pentagon $FCGJI$ each has area 1. What is FI^2 ?



- (A) $\frac{7}{3}$ (B) $8 - 4\sqrt{2}$ (C) $1 + \sqrt{2}$ (D) $\frac{7}{4}\sqrt{2}$ (E) $2\sqrt{2}$

Answer: B

Solution 1

Since the total area is 4, the side length of square $ABCD$ is 2. We see that since triangle HAE is a right isosceles triangle with area 1, we can determine sides HA and AE both to be $\sqrt{2}$. Now, consider extending FB and IE until they intersect. Let the point of intersection be K . We note that EBK is also a right isosceles triangle with side $2 - \sqrt{2}$ and find its area to be $3 - 2\sqrt{2}$. Now, we notice that FIK is also a right isosceles triangle (because $\angle EKB = 45^\circ$) and find its area to be $\frac{1}{2}FI^2$. This is also equal to $1 + 3 - 2\sqrt{2}$ or $4 - 2\sqrt{2}$. Since we are looking for FI^2 , we want two times this. That gives **(B)** $8 - 4\sqrt{2}$. TLiu

Solution 2

Draw the auxiliary line AC . Denote by M the point it intersects with HE , and by N the point it intersects with GF . Last, denote by x the segment FN , and by y the segment FI . We will find two equations for x and y , and then solve for y^2 .

Since the overall area of $ABCD$ is 4 $\implies AB = 2$, and $AC = 2\sqrt{2}$. In addition, the area of $\triangle AME = \frac{1}{2} \implies AM = 1$.

The two equations for x and y are then:

- Length of AC : $1 + y + x = 2\sqrt{2} \implies x = (2\sqrt{2} - 1) - y$
- Area of CMIF: $\frac{1}{2}x^2 + xy = \frac{1}{2} \implies x(x + 2y) = 1$.

Substituting the first into the second, yields $[(2\sqrt{2} - 1) - y] \cdot [(2\sqrt{2} - 1) + y] = 1$

Solving for y^2 gives **(B)** $8 - 4\sqrt{2}$ DrB

Solution 3

Plot a point F' such that $F'I$ and AB are parallel and extend line FB to point B' such that $FIB'F'$ forms a square. Extend line AE to meet line $F'B'$ and point E' is the intersection of the two. The area of this square is equivalent to FI^2 . We see that the area of square $ABCD$ is 4, meaning each side is of length 2. The area of the pentagon $EIFF'E'$ is 2. Length $AE = \sqrt{2}$, thus $EB = 2 - \sqrt{2}$. Triangle

$EB'E'$ is isosceles, and the area of this triangle is $\frac{1}{2}(4 - 2\sqrt{2})(2 - \sqrt{2}) = 6 - 4\sqrt{2}$. Adding these two areas, we get

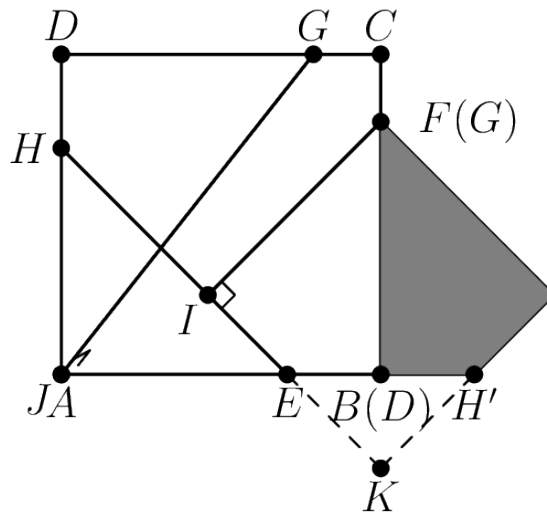
$$2 + 6 - 4\sqrt{2} = 8 - 4\sqrt{2} \rightarrow \boxed{\text{(B)} 8 - 4\sqrt{2}}$$

. -OGBooger

Solution 4 (HARD Calculation)

We can easily observe that the area of square $ABCD$ is 4 and its side length is 2 since all four regions that build up the square has area 1. Extend FI and let the intersection with AB be K . Connect AC , and let the intersection of AC and HE be L . Notice that since the area of triangle AEH is 1 and $AE = AH$, $AE = AH = \sqrt{2}$, therefore $BE = HD = 2 - \sqrt{2}$. Let $CG = CF = m$, then $BF = DG = 2 - m$. Also notice that $KB = 2 - m$, thus $KE = KB - BE = 2 - m - (2 - \sqrt{2}) = \sqrt{2} - m$. Now use the condition that the area of quadrilateral $BFIE$ is 1, we can set up the following equation: $\frac{1}{2}(2 - m)^2 - \frac{1}{4}(\sqrt{2} - m)^2 = 1$. We solve the equation and yield $m = \frac{8 - 2\sqrt{2} - \sqrt{64 - 32\sqrt{2}}}{2}$. Now notice that $FI = AC - AL - \frac{m}{\sqrt{2}} = 2\sqrt{2} - 1 - \frac{\sqrt{2}}{2} \cdot \frac{8 - 2\sqrt{2} - \sqrt{64 - 32\sqrt{2}}}{2} = 2\sqrt{2} - 1 - \frac{8\sqrt{2} - 4 - \sqrt{128 - 64\sqrt{2}}}{4} = \frac{\sqrt{128 - 64\sqrt{2}}}{4}$. Hence $FI^2 = \frac{128 - 64\sqrt{2}}{16} = 8 - 4\sqrt{2}$. -HarryW

Solution 5 (Basically Same as Solution 3)

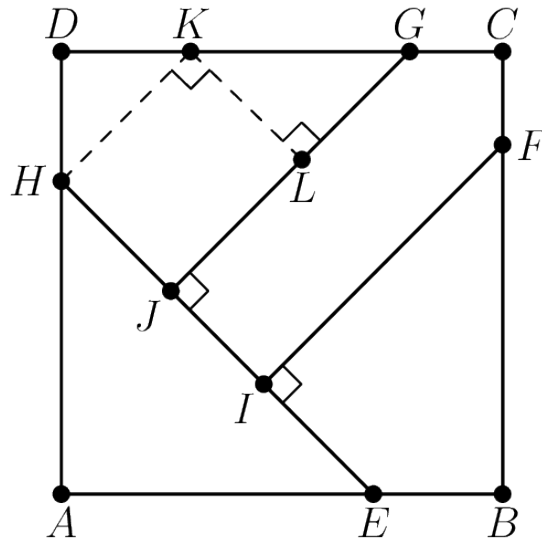


Easily, we can find that: quadrilateral $BFIE$ and $DHJG$ are congruent with each other, so we can move $DHJG$ to the shaded area (F and G , B and D overlapping) to form a square $FIKJ'$ ($DG = FB$, $CG = FC$, $\angle CGF = \angle CFG = 45^\circ$ so $\angle IFJ' = 90^\circ$). Then we can solve $AH = AE = \sqrt{2}$, $EB = 2 - \sqrt{2}$, $EK = 2\sqrt{2} - 2$.

$$FI^2 = \text{area of } BFIE + \text{area of } FJ'H'B + \text{area of } EH'K \\ = 1 + 1 + \frac{1}{2}(2\sqrt{2} - 2)^2 = 8 - 4\sqrt{2} \rightarrow \boxed{\text{(B)} 8 - 4\sqrt{2}}$$

-Ryan Zhang @BRS-Minor edit by WhySean38

Solution 6



$$[ABCD] = 4, AB = 2, [AHE] = 1, AH = AE = \sqrt{2}, DH = 2 - \sqrt{2}, JL = HK = \sqrt{2} \cdot DH = 2\sqrt{2} - 2$$

Because $ABCD$ is a square and $AH = AE$, AC is the line of symmetry of pentagon $CDHEB$. Because $[DHJG] = [BFIE]$, $DHJG$ is the reflection of $BFIE$ about line AC

$$\begin{aligned} \text{Let } FI = GJ = x, KL = LG = GJ - LJ = x - 2\sqrt{2} + 2 \quad [DHK] &= \frac{(2-\sqrt{2})^2}{2} = 3 - 2\sqrt{2} \\ [GKL] &= \frac{(x-2\sqrt{2}+2)^2}{2} = \frac{x^2}{2} + 2x - 2x\sqrt{2} - 4\sqrt{2} + 6 \quad [HKJL] = (x - 2\sqrt{2} + 2) \cdot (2\sqrt{2} - 2) \\ &= 2x\sqrt{2} - 2x + 8\sqrt{2} - 12 \end{aligned}$$

$$[DHK] + [GKL] + [HKLJ] = [DHJG]$$

$$3 - 2\sqrt{2} + \frac{x^2}{2} + 2x - 2x\sqrt{2} - 4\sqrt{2} + 6 + 2x\sqrt{2} - 2x + 8\sqrt{2} - 12 = 1$$

$$\frac{x^2}{2} + 2\sqrt{2} - 4 = 0$$

$$x^2 = 8 - 4\sqrt{2}$$

$$FI^2 = \boxed{\text{(B)} \ 8 - 4\sqrt{2}}$$

Solution 7 (Easy to See)

Note that the side length of $ABCD$ is 2 and thus the diagonal is of length $2\sqrt{2}$. However, the height to side HE in triangle HAE is 1, implying that $CM = 2\sqrt{2} - 1$ where M is the midpoint of JI . From here suppose that N is the midpoint of $\overline{FG}$ and let $x = NC$, which means $FG = 2x$. The area of the pentagon is then

$$[FIJG] + [GCF] = GF \cdot FI + x^2 = (2x)(2\sqrt{2} - 1 - x) + x^2 = 1$$

Solving this quadratic for x yields $x = 2\sqrt{2} - 1 \pm \sqrt{8 - 4\sqrt{2}}$ (technically the smaller value is the correct one but it doesn't matter for our purposes). We can then calculate $FI^2 = (2\sqrt{2} - 1 - x)^2 = \boxed{\mathbf{(B)} \ 8 - 4\sqrt{2}}$.

Solution 8

We extend $\overline{FB}$ and $\overline{IE}$ to meet at a point X . Since $\angle AEI = \angle BEX$ and FX is parallel to DA , we know that $\triangle BEX \sim \triangle AEH$, and because $\angle BXE = \angle IXF$ and $\angle XBE = \angle XIF$, we can conclude that $\triangle BEX \sim \triangle AEH \sim \triangle IFX$.

Now, because $\triangle AEH$ is isosceles, right, and has an area of 1, we can conclude that $AE = AH = \sqrt{2}$ and that $BE = 2 - \sqrt{2}$. Armed with this knowledge, and setting $IF = a$ and the area of $\triangle BEX = b$, we can use similarity to say that

$$\left(\frac{a}{2 - \sqrt{2}}\right)^2 = \frac{1 + b}{b}$$

Since we know the side lengths of $\triangle BEX$ due to the fact that it is also an isosceles right triangle, we know that the area is $\frac{(2 - \sqrt{2})^2}{2}$. Simplifying further and plugging in values, we have

$$\frac{a^2}{(2 - \sqrt{2})^2} = 1 + \frac{2}{(2 - \sqrt{2})^2}$$

Multiplying by $(2 - \sqrt{2})^2$ on both sides, we get

$$a^2 = (2 - \sqrt{2})^2 + 2 = \boxed{\text{(B)} \ 8 - 4\sqrt{2}}$$

yingkai_0_

22. (AMC 10 10B 2020) What is the remainder when $2^{202} + 202$ is divided by $2^{101} + 2^{51} + 1$?

(A) 100 (B) 101 (C) 200 (D) 201 (E) 202

Answer: D

Solution 1 (MAA Original Solution)

Completing the square, then difference of squares:

$$\begin{aligned} 2^{202} + 202 &= (2^{101})^2 + 2 \cdot 2^{101} + 1 - 2 \cdot 2^{101} + 201 \\ &= (2^{101} + 1)^2 - 2^{102} + 201 \\ &= (2^{101} - 2^{51} + 1)(2^{101} + 2^{51} + 1) + 201. \end{aligned}$$

Thus, we see that the remainder is $\boxed{\text{(D)} \ 201}$

(Source: <https://artofproblemsolving.com/community/c5h2001950p14000817>)

Solution 2

Let $x = 2^{50}$. We are now looking for the remainder of $\frac{4x^4 + 202}{2x^2 + 2x + 1}$.

We could proceed with polynomial division, but the numerator looks awfully similar to the **Sophie Germain Identity**, which states that

$$a^4 + 4b^4 = (a^2 + 2b^2 + 2ab)(a^2 + 2b^2 - 2ab)$$

Let's use the identity, with $a = 1$ and $b = x$, so we have

$$1 + 4x^4 = (1 + 2x^2 + 2x)(1 + 2x^2 - 2x)$$

Rearranging, we can see that this is exactly what we need:

$$\frac{4x^4 + 1}{2x^2 + 2x + 1} = 2x^2 - 2x + 1$$

So

$$\frac{4x^4 + 202}{2x^2 + 2x + 1} = \frac{4x^4 + 1}{2x^2 + 2x + 1} + \frac{201}{2x^2 + 2x + 1}$$

Since the first half divides cleanly as shown earlier, the remainder must be (D) 201

Solution 3 (Same As Solution 2)

We let

$$x = 2^{50}$$

and

$$2^{202} + 202 = 4x^4 + 202$$

.Next we write

$$2^{101} + 2^{51} + 1 = 2x^2 + 2x + 1$$

.We know that

$$4x^4 + 1 = (2x^2 + 2x + 1)(2x^2 - 2x + 1)$$

by the Sophie Germain identity so to find

$$4x^4 + 202,$$

we find that

$$4x^4 + 202 = 4x^4 + 201 + 1$$

which shows that the remainder is (D) 201

Solution 4

We let $x = 2^{50.5}$. That means $2^{202} + 202 = x^4 + 202$ and $2^{101} + 2^{51} + 1 = x^2 + 2^{0.5}x + 1$. Then, we simply do polynomial division, and find that the remainder is (D) 201.

The long division is essentially the same if you work with $x = 2$, or do repeated multiplication and subtraction using the original expression.

Solution 5 (Modular Arithmetic)

Let $n = 2^{101} + 2^{51} + 1$. Then, mod n :

$$2^{202} + 202 \equiv (-2^{51} - 1)^2 + 202 \equiv 2^{102} + 2^{52} + 203 = 2(n - 1) + 203 \equiv 201 \pmod{n}.$$

Thus, the remainder is (D) 201.

Solution 6 (by Shiva Kumar Kannan - Least insightful very straightforward + Manipulation)

We can repeatedly manipulate the numerator to make parts of it divisible by the denominator:

$$\begin{aligned} & \frac{2^{202} + 202}{2^{101} + 2^{51} + 1} \\ &= \frac{2^{202} + 2^{152} + 2^{101}}{2^{101} + 2^{51} + 1} - \frac{2^{152} + 2^{101} - 202}{2^{101} + 2^{51} + 1} \end{aligned}$$

$$\begin{aligned}
&= 2^{101} - \frac{2^{152} + 2^{101} - 202}{2^{101} + 2^{51} + 1} \\
&= 2^{101} - \frac{2^{152} + 2^{101} + 2^{101} + 2^{51} - 2^{101} - 2^{51} - 202}{2^{101} + 2^{51} + 1} \\
&= 2^{101} - 2^{51} + \frac{2^{101} + 2^{51} + 202}{2^{101} + 2^{51} + 1} \\
&= 2^{101} - 2^{51} + \frac{2^{101} + 2^{51} + 1 + 201}{2^{101} + 2^{51} + 1} \\
&= 2^{101} - 2^{51} + 1 + \frac{201}{2^{101} + 2^{51} + 1}.
\end{aligned}$$

Clearly, $201 < 2^{201} + 2^{51} + 1$, hence, we cannot manipulate the numerator further to make the denominator divide into one of its parts. This concludes, that the remainder is **(D) 201**.

23. (AMC 10 10B 2020) Square $ABCD$ in the coordinate plane has vertices at the points $A(1,1)$, $B(-1,1)$, $C(-1,-1)$, and $D(1,-1)$. Consider the following four transformations:

- L , a rotation of 90° counterclockwise around the origin;
- R , a rotation of 90° clockwise around the origin;
- H , a reflection across the x -axis; and
- V , a reflection across the y -axis.

Each of these transformations maps the squares onto itself, but the positions of the labeled vertices will change. For example, applying R and then V would send the vertex A at $(1,1)$ to $(-1,-1)$ and would send the vertex B at $(-1,1)$ to itself. How many sequences of 20 transformations chosen from $\{L, R, H, V\}$ will send all of the labeled vertices back to their original positions? (For example, R, R, V, H is one sequence of 4 transformations that will send the vertices back to their original positions.)

- (A) 2^{37} (B) $3 \cdot 2^{36}$ (C) 2^{38} (D) $3 \cdot 2^{37}$ (E) 2^{39}

Answer: C

Solution 1

For each transformation:

- a) Each labeled vertex will move to an adjacent position.
- b) The labeled vertices will maintain the consecutive order $ABCD$ in either direction (clockwise or counterclockwise).
- c) L and R will retain the direction of the labeled vertices, but H and V will alter the direction of the labeled vertices.

After the 19th transformation, vertex A will be at either $(1,-1)$ or $(-1,1)$. All possible configurations of the labeled vertices are shown below:

Configuration	$C \ B$ $D \ A$	$A \ D$ $B \ C$	$C \ D$ $B \ A$	$A \ B$ $D \ C$
The 20th Transformation	L	R	H	V

Each sequence of 19 transformations generates one valid sequence of 20 transformations. Therefore, the answer is $4^{19} = \boxed{(C) 2^{38}}$.

Solution 2

Let $(+)$ denote counterclockwise/starting orientation and $(-)$ denote clockwise orientation. Let 1, 2, 3, and 4 denote which quadrant A is in.

Realize that from any odd quadrant and any orientation, the 4 transformations result in some permutation of $(2+, 2-, 4+, 4-)$.

The same goes that from any even quadrant and any orientation, the 4 transformations result in some permutation of $(1+, 1-, 3+, 3-)$.

We start our first 19 moves by doing whatever we want, 4 choices each time. Since 19 is odd, we must end up on an even quadrant.

As said above, we know that exactly one of the four transformations will give us $(1+)$, and we must use that transformation.

Thus, the answer is $4^{19} = \boxed{(C) 2^{38}}$.

IGNORE THIS SOLUTION IT IS WRONG Notice that any **pair** of transformations either swaps the x and y -coordinates, negates the x and y -coordinates, swaps and negates the x and y -coordinates, or leaves the original unchanged. Furthermore, notice that for each of these results, if we apply another pair of transformations, one of these results will happen again, and with equal probability. Therefore, no matter what state we are in after we apply the first 9 pairs of transformations, there is a $\frac{1}{4}$ chance the last pair of transformations will return the figure to its original position. Therefore, the answer is $\frac{4^{20}}{8} = 2^{37} = \boxed{(A) 2^{37}}$.

note that this solution a few flaws and the answer is wrong- Happymaths

Solution 4

The total number of sequences is $4^{20} = 2^{40}$.

Note that there can only be an even number of reflections since they result in the same anti-clockwise orientation of the vertices A, B, C, D . Therefore, the probability of having the same anti-clockwise orientation with the original arrangement after the transformation is $\frac{1}{2}$.

Next, the even number of reflections means that there must be an even number of rotations since their sum is even. Even rotations result in only the original position or a 180° rotation of it.

Since rotation R and rotation L cancel each other out, the difference between the numbers of them define the final position. The probability of the transformation returning the vertices to the original position, given that there are even number of rotations, is equivalent to the probability that

$$|n(R) - n(L)| \equiv 0 \pmod{4} \text{ when } |n(H) - n(V)| \equiv 0 \pmod{4}$$

or

$$|n(R) - n(L)| \equiv 2 \pmod{4} \text{ when } |n(H) - n(V)| \equiv 2 \pmod{4}$$

which is again, $\frac{1}{2}$.

Therefore, the answer is $2^{40} \cdot \frac{1}{2} \cdot \frac{1}{2} = \boxed{(C) 2^{38}}$.

Solution 5 (Group Theory)

This problem is a **Dihedral Group** problem, D_4 , in **Group Theory**.

The transformation has associativity, for $x, y, z \in \{L, R, H, V\}$, $(x \circ y) \circ z = x \circ (y \circ z)$.

Let I be the initial state of the square $A(1, 1), B(-1, 1), C(-1, -1)$, and $D(1, -1)$.

$$R \circ L = L \circ R = V \circ V = H \circ H = I \quad (1)$$

$$V \circ H = R \circ R = H \circ V = L \circ L \quad (2)$$

$$H \circ R = L \circ H = V \circ L = R \circ V \quad (3)$$

$$V \circ R = L \circ V = H \circ L = R \circ H \quad (4)$$

It's not hard to see that after a series of transformations from initial state I to initial state I , the number of transformations must be even. Denote $f(2n)$ be the number of sequences of $2n$ transformations from initial state I to initial state I . We are going to prove $f(2n) = 4^{2n-1}$.

For each transformation composite operator, there are 4 replacements.

For example, when $n = 2$:

$$R \circ R \circ R \circ R = (R \circ R) \circ R \circ R = I.$$

From (2), $R \circ R = V \circ H = H \circ V = L \circ L$, so $R \circ R$ can be replaced with $V \circ H$, $H \circ V$, $L \circ L$ without changing the result. Suppose we choose $V \circ H$, then

$$(R \circ R) \circ R \circ R = (V \circ H) \circ R \circ R = V \circ (H \circ R) \circ R.$$

From (3), $H \circ R = L \circ H = V \circ L = R \circ V$, so $H \circ R$ can be replaced with $L \circ H$, $V \circ L$, $R \circ V$ without changing the result. Suppose we choose $R \circ V$, then

$$V \circ (H \circ R) \circ R = V \circ (R \circ V) \circ R = V \circ R \circ (V \circ R).$$

From (4), $V \circ R = L \circ V = H \circ L = R \circ H$, so $V \circ R$ can be replaced with $L \circ V$, $H \circ L$, $R \circ H$ without changing the result. Suppose we choose $H \circ L$, then

$$V \circ R \circ (V \circ R) = V \circ R \circ (H \circ L) = V \circ R \circ H \circ L = I.$$

So, we have $f(4) = 4^{4-1}$:

$$\underbrace{R \circ R \circ R \circ R \circ \dots \circ R \circ R \circ R}_{2n}.$$

With $2n$ R transformations, it will go from initial state to initial state. There are $2n - 1$ transformation composite operators $\circ$ between the transformations, and each pair of transformations surrounding the transformations composite operator $\circ$ have 4 options. So, we have $f(2n) = 4^{2n-1}$, from which $f(20) = 4^{20-1} = \boxed{(C) 2^{38}}$.

Side Note

Equations (2), (3), (4) are equivalent. Here I will prove that (2) is equivalent to (3).

From (1), $R \circ L = V \circ V$. We have

$$H \circ (R \circ L) = H \circ (V \circ V) = (H \circ V) \circ V.$$

From (2), $H \circ V = V \circ H = L \circ L$. We have

$$(H \circ V) \circ V = (V \circ H) \circ V = V \circ (H \circ V) = V \circ (L \circ L).$$

So,

$$(H \circ R) \circ L = H \circ (R \circ L) = V \circ (L \circ L) = (V \circ L) \circ L.$$

It follows that $H \circ R = V \circ L$, which is (3).

Solution 6 (Quick)

The number of possible ways to orient a square using this method 20 times would be 4^{20} , as for each of the transformations we can only use the four transformations L , V , H , or R .

After 20 transformations, there are only 4 possible outcomes, as parity prohibits the other 4 outcomes. By symmetry, all 4 of these possibilities are equally likely. Thus, our answer is $\frac{4^{20}}{4} = 4^{19} = \boxed{(C) 2^{38}}$.

Solution 7 (Generating Functions)

Let x represent rotating $\overline{AC}$ 90° counterclockwise about the origin and let y represent rotating $\overline{BD}$ 90° counterclockwise about the origin. Then, note that the transformations L, V, H , and R can be encoded by $xy, \frac{1}{xy}, \frac{y}{x}$, and $\frac{x}{y}$, respectively. Thus, we wish to compute the sum of the coefficients of the terms of the form $cx^{4a}y^{4b}$ for $a, b, c \geq 0$ in the polynomial

$$(xy + \frac{1}{xy} + \frac{y}{x} + \frac{x}{y})^{20}.$$

We can factor the polynomial to get

$$((x + \frac{1}{x})(y + \frac{1}{y}))^{20} = (x + \frac{1}{x})^{20}(y + \frac{1}{y})^{20}.$$

We can multiply the polynomial by $x^{20}y^{20}$ and it wouldn't affect the sum of the coefficients of the terms of the form $cx^{4a}y^{4b}$, so we can now work with

$$(x^2 + 1)^{20}(y^2 + 1)^{20}.$$

Now, it is very evident that we will get a term of the form $cx^{4a}y^{4b}$ exactly when the first part of the polynomial contributes an even number of x^2 's and the second part of the polynomial contributes an even number of y^2 's, so the answer is

$$\begin{aligned} \left(\binom{20}{0} + \binom{20}{2} + \dots + \binom{20}{20} \right) \left(\binom{20}{0} + \binom{20}{2} + \dots + \binom{20}{20} \right) &= 2^{19} \cdot 2^{19} \\ &= \boxed{(C) 2^{38}}. \end{aligned}$$

□

24. (AMC 10 10B 2020) How many positive integers n satisfy

$$\frac{n + 1000}{70} = \lfloor \sqrt{n} \rfloor?$$

(Recall that $\lfloor x \rfloor$ is the greatest integer not exceeding x .)

- (A) 2 (B) 4 (C) 6 (D) 30 (E) 32

Answer: C

Solution 1 (Fakesolve)

We can first consider the equation without a floor function:

$$\frac{n + 1000}{70} = \sqrt{n}$$

Multiplying both sides by 70 and then squaring:

$$n^2 + 2000n + 1000000 = 4900n$$

Moving all terms to the left:

$$n^2 - 2900n + 1000000 = 0$$

Now we can determine the factors:

$$(n - 400)(n - 2500) = 0$$

This means that for $n = 400$ and $n = 2500$, the equation will hold without the floor function.

Now we can simply check the multiples of 70 around 400 and 2500 in the original equation, which we abbreviate as $L = R$.

For $n = 330$, $L = 19$ but $18^2 < 330 < 19^2$ so $R = 18$

For $n = 400$, $L = 20$ and $R = 20$

For $n = 470$, $L = 21$, $R = 21$

For $n = 540$, $L = 22$ but $540 > 23^2$ so $R = 23$

Now we move to $n = 2500$

For $n = 2430$, $L = 49$ and $49^2 < 2430 < 50^2$ so $R = 49$

For $n = 2360$, $L = 48$ and $48^2 < 2360 < 49^2$ so $R = 48$

For $n = 2290$, $L = 47$ and $47^2 < 2360 < 48^2$ so $R = 47$

For $n = 2220$, $L = 46$ but $47^2 < 2220$ so $R = 47$

For $n = 2500$, $L = 50$ and $R = 50$

For $n = 2570$, $L = 51$ but $2570 < 51^2$ so $R = 50$

Therefore we have 6 total solutions, $n = 400, 470, 2290, 2360, 2430, 2500 = \boxed{\text{(C) } 6}$

Solution 2

This is my first solution here, so please forgive me for any errors.

We are given that

$$\frac{n + 1000}{70} = \lfloor \sqrt{n} \rfloor.$$

We know $\lfloor \sqrt{n} \rfloor$ must be an integer, which means that $n + 1000$ is divisible by 70. As $1000 \equiv 20 \pmod{70}$, this means that $n \equiv 50 \pmod{70}$, so we can write $n = 70k + 50$ for $k \in \mathbb{Z}$. Note that n has to be nonnegative for $\sqrt{n}$ to be defined. Thus, $n = 70k + 50 \geq 0$. Simplifying this, we have $k \geq -5/7$. In other words, k also has to be nonnegative.

Therefore,

$$\frac{n + 1000}{70} = \frac{70k + 1050}{70} = k + 15 = \lfloor \sqrt{70k + 50} \rfloor.$$

Also, we can say that $\sqrt{70k+50} - 1 < k + 15$ and $k + 15 \leq \sqrt{70k+50}$.

Since k is nonnegative, both sides of the second inequality are nonnegative, so we can square them to get $k^2 + 30k + 225 \leq 70k + 50 \implies k^2 - 40k + 175 \leq 0 \implies (k-5)(k-35) \leq 0 \implies 5 \leq k \leq 35$.

Similarly, solving the first inequality gives us $k < 19 - \sqrt{155}$ or $k > 19 + \sqrt{155}$.

We know that $\sqrt{155}$ is larger than 12 and smaller than 13, so instead, we can say $k \leq 6$ or $k \geq 32$.

Combining this with $5 \leq k \leq 35$, we get $k = 5, 6, 32, 33, 34, 35$ are all solutions for k that give a valid solution for n , meaning that our answer is **(C) 6**.-Solution By Qqqwerw

Solution 3

We start with the given equation

$$\frac{n+1000}{70} = \lfloor \sqrt{n} \rfloor$$

From there, we can start with the general inequality that $\lfloor \sqrt{n} \rfloor \leq \sqrt{n} < \lfloor \sqrt{n} \rfloor + 1$. This means that

$$\frac{n+1000}{70} \leq \sqrt{n} < \frac{n+1070}{70}$$

Solving each inequality separately gives us two inequalities:

$$n - 70\sqrt{n} + 1000 \leq 0 \rightarrow (\sqrt{n} - 50)(\sqrt{n} - 20) \leq 0 \rightarrow 20 \leq \sqrt{n} \leq 50$$

$$n - 70\sqrt{n} + 1070 > 0 \rightarrow \sqrt{n} < 35 - \sqrt{155}, \sqrt{n} > 35 + \sqrt{155}$$

Simplifying and approximating decimals yields 2 solutions for one inequality and 4 for the other. Hence, the answer is $2 + 4 = \mathbf{(C) 6}$.

Solution 4

Let n be uniquely of the form $n = k^2 + r$ where $0 \leq r \leq 2k$ ★. Then,

$$\frac{k^2 + r + 1000}{70} = k$$

Rearranging and completeing the square gives

$$\begin{aligned} (k-35)^2 + r &= 225 \\ \Rightarrow r &= (k-20)(50-k) \end{aligned}$$

This gives us

$$(k-35)^2 \leq (k-35)^2 + r = 225 \leq (k-35)^2 + 2k$$

Solving the left inequality shows that $20 \leq k \leq 50$. Combing this with the right inequality gives that

$$(k-35)^2 + r = 225 \leq (k-35)^2 + 2k \leq (k-35)^2 + 100$$

which implies either $k \geq 47$ or $k \leq 23$. By directly computing the cases for $k = 20, 21, 22, 23, 47, 48, 49, 50$ using ★, it follows that only $k = 22, 23$ yield and invalid r from ★. Since each k corresponds to one r and thus to one n (from the original form), there must be 6 such n .

Solution 5

Since the right-hand-side is an integer, so must be the left-hand-side. Therefore, we must have $n \equiv -20 \pmod{70}$; let $n = 70j - 20$. The given equation becomes

$$j + 14 = \lfloor \sqrt{70j - 20} \rfloor$$

Since $\lfloor x \rfloor \leq x < \lfloor x \rfloor + 1$ for all real x , we can take $x = \sqrt{70j - 20}$ with $\lfloor x \rfloor = j + 14$ to get

$$j + 14 \leq \sqrt{70j - 20} < j + 15$$

We can square the inequality to get

$$196 + 28j + j^2 \leq 70j - 20 < 225 + 30j + j^2$$

The left inequality simplifies to $(j - 36)(j - 6) \leq 0$, which yields

$$6 \leq j \leq 36.$$

The right inequality simplifies to $(j - 20)^2 - 155 > 0$, which yields

$$j < 20 - \sqrt{155} < 8 \quad \text{or} \quad j > 20 + \sqrt{155} > 32$$

Solving $j < 8$, and $6 \leq j \leq 36$, we get $6 \leq j < 8$, for 2 values $j \in \{6, 7\}$.

Solving $j > 32$, and $6 \leq j \leq 36$, we get $32 < j \leq 36$, for 4 values $k \in \{33, \dots, 36\}$.

Thus, our answer is $2 + 4 = \boxed{\text{(C)} 6}$

Solution 6

Set $x = \sqrt{n}$ in the given equation and solve for x to get $x^2 = 70 \cdot \lfloor x \rfloor - 1000$. Set $k = \lfloor x \rfloor \geq 0$; since $\lfloor x \rfloor^2 \leq x^2 < (\lfloor x \rfloor + 1)^2$, we get

$$k^2 \leq 70k - 1000 < k^2 + 2k + 1.$$

The left inequality simplifies to $(k - 20)(k - 50) \leq 0$, which yields

$$20 \leq k \leq 50.$$

The right inequality simplifies to $(k - 34)^2 > 155$, which yields

$$k < 34 - \sqrt{155} < 22 \quad \text{or} \quad k > 34 + \sqrt{155} > 46$$

Solving $k < 22$, and $20 \leq k \leq 50$, we get $20 \leq k < 22$, for 2 values $k \in \{20, 21\}$.

Solving $k > 46$, and $20 \leq k \leq 50$, we get $46 < k \leq 50$, for 4 values $k \in \{47, \dots, 50\}$.

Thus, our answer is $2 + 4 = \boxed{\text{(C)} 6}$

Solution 7

If n is a perfect square, we can write $n = k^2$ for a positive integer k , so $\lfloor \sqrt{n} \rfloor = \sqrt{n} = k$. The given equation turns into

$$\begin{aligned} \frac{k^2 + 1000}{70} &= k \\ k^2 - 70k + 1000 &= 0 \end{aligned}$$

$$(k - 20)(k - 50) = 0,$$

so $k = 20$ or $k = 50$, so $n = 400, 2500$.

If n is not square, then we can say that, for a positive integer k , we have

$$\begin{aligned} k^2 &< n < (k + 1)^2 \\ k^2 + 1000 &< n + 1000 = 70\lfloor\sqrt{n}\rfloor = 70k < (k + 1)^2 + 1000 \\ k^2 + 1000 &< 70k < (k + 1)^2 + 1000. \end{aligned}$$

To solve this inequality, we take the intersection of the two solution sets to each of the two inequalities $k^2 + 1000 < 70k$ and $70k < (k + 1)^2 + 1000$. To solve the first one, we have

$$\begin{aligned} k^2 - 70k + 1000 &< 0 \\ (k - 20)(k - 50) &< 0 \end{aligned}$$

$k \in (20, 50)$, because the portion of the parabola between its two roots will be negative.

The second inequality yields

$$\begin{aligned} 70k &< k^2 + 2k + 1 + 1000 \\ 0 &< k^2 - 68k + 1001. \end{aligned}$$

This time, the inequality will hold for all portions of the parabola that are not on or between the its two roots, which are $34 + \sqrt{155} > 46$ and $34 - \sqrt{155} < 22$ (they are roughly equal, but this is to ensure that we do not miss any solutions).

Notation wise, we need all integers k such that

$$k \in (20, 50) \cap (-\infty, 34 - \sqrt{155})$$

or

$$k \in (20, 50) \cap (34 + \sqrt{155}, \infty).$$

For the first one, since our upper bound is a little less than 22, the k that works is 21. For the second, our lower bound is a little more than 46, so the k that work are 47, 48, and 49.

(C) 6 total solutions for n , since each value of k corresponds to exactly one value of n .

-Benedict T (countmath1)

Solution 8

First, express $\lfloor\sqrt{n}\rfloor = \sqrt{n} - \{\sqrt{n}\}$ and simplify the equation out:

$$n + 1000 = 70\sqrt{n} - 70\{\sqrt{n}\}$$

For simplicity, define $a = \sqrt{n}$. Then, $a^2 + 1000 = 70a - 70\{a\}$. Rearranging and factoring: $(a - 20)(a - 50) = -70\{a\}$. We know that $0 \leq \{a\} < 1$, so we can set $\{a\} = 0$ to obtain $a = 20, 50 \implies n = 400, 2500$. For the next solutions, we can use the quadratic formula on the equation $a^2 - 70a + (1000 + 70\{a\}) = 0$:

$$a = 35 \pm \sqrt{1225 - (1000 + 70\{a\})} = 35 \pm \sqrt{225 - 70\{a\}}$$

Since $a = 35 \pm \sqrt{225 - 70\{a\}}$ must produce real values of a that satisfy $\lfloor a \rfloor = a - \{a\}$, the expression inside the square root must correspond to a value of a whose integer part stays consistent. Therefore, we only consider values of $\{a\}$ that make a land near an integer, which happens when $\sqrt{225 - 70\{a\}}$ is close to an integer.

From earlier, we can use the bounds of $\{a\}$ to find out that $225 - 70\{a\}$ will approach a minimum of $225 - 70 = 155$ but never reach it. Now, we can just list out the perfect squares between 155 and 225, as setting $225 - 70\{a\}$ equal to that perfect square will result in an integer value of a corresponding to an integer value of n .

The two perfect squares that satisfy these restrictions are $k = 13$ and $k = 14$. Then, $a = 35 \pm 14 \implies a = 49, 21$ and $a = 35 \pm 13 \implies a = 48, 22$.

We know that each value of $a = \sqrt{n}$ corresponds to exactly one value of n , so we have a total of $2 + 4 = \boxed{\text{(C)} 6}$ total solutions for n .

25. (AMC 10 10B 2020) Let $D(n)$ denote the number of ways of writing the positive integer n as a product

$$n = f_1 \cdot f_2 \cdots f_k,$$

where $k \geq 1$, the f_i are integers strictly greater than 1, and the order in which the factors are listed matters (that is, two representations that differ only in the order of the factors are counted as distinct). For example, the number 6 can be written as 6, $2 \cdot 3$, and $3 \cdot 2$, so $D(6) = 3$. What is $D(96)$?

- (A) 112 (B) 128 (C) 144 (D) 172 (E) 184

Answer: A

Solution 1

Note that $96 = 2^5 \cdot 3$. Since there are at most six not necessarily distinct factors > 1 multiplying to 96, we have six cases: $k = 1, 2, \dots, 6$. Now we look at each of the six cases.

$k = 1$: We see that there is 1 way, merely 96.

$k = 2$: This way, we have the 3 in one slot and 2 in another, and symmetry. The four other 2's leave us with 5 ways and symmetry doubles us so we have 10.

$k = 3$: We have 3, 2, 2 as our baseline. We need to multiply by 2 in 3 places, and see that we can split the remaining three powers of 2 in a manner that is $3 - 0 - 0$, $2 - 1 - 0$ or $1 - 1 - 1$. A $3 - 0 - 0$ split has $6 + 3 = 9$ ways of happening ($24 - 2 - 2$ and symmetry; $2 - 3 - 16$ and symmetry), a $2 - 1 - 0$ split has $6 \cdot 3 = 18$ ways of happening (due to all being distinct) and a $1 - 1 - 1$ split has 3 ways of happening ($6 - 4 - 4$ and symmetry) so in this case we have $9 + 18 + 3 = 30$ ways.

$k = 4$: We have 3, 2, 2, 2 as our baseline, and for the two other 2's, we have a $2 - 0 - 0 - 0$ or $1 - 1 - 0 - 0$ split. The former grants us $4 + 12 = 16$ ways ($12 - 2 - 2 - 2$ and symmetry and $3 - 8 - 2 - 2$ and symmetry) and the latter

grants us also $12 + 12 = 24$ ways ($6 - 4 - 2 - 2$ and symmetry and $3 - 4 - 4 - 2$ and symmetry) for a total of $16 + 24 = 40$ ways.

$k = 5$: We have $3, 2, 2, 2, 2$ as our baseline and one place to put the last two: on another two or on the three. On the three gives us 5 ways due to symmetry and on another two gives us $5 \cdot 4 = 20$ ways due to symmetry. Thus, we have $5 + 20 = 25$ ways.

$k = 6$: We have $3, 2, 2, 2, 2, 2$ and symmetry and no more twos to multiply, so by symmetry, we have 6 ways.

Thus, adding, we have $1 + 10 + 30 + 40 + 25 + 6 = \boxed{\text{(A)} 112}$.

Solution 2

As before, note that $96 = 2^5 \cdot 3$, and we need to consider 6 different cases, one for each possible value of k , the number of factors in our factorization. However, instead of looking at each individually, find a general form for the number of possible factorizations with k factors. First, the factorization needs to contain one factor that is itself a multiple of 3. There are k to choose from. That leaves $k - 1$ slots left to fill, each of which must contain at least one factor of 2. Once we have filled in a 2 to each of the remaining slots, we're left with $5 - (k - 1) = 6 - k$ twos.

Consider the remaining $6 - k$ factors of 2 left to assign to the k factors. Using stars and bars, the number of ways to do this is:

$$\binom{(6 - k) + k - 1}{6 - k} = \binom{5}{6 - k}$$

This makes $k \binom{5}{6 - k}$ possibilities for each k .

To obtain the total number of factorizations, add all possible values for k :

$$\sum_{k=1}^6 k \binom{5}{6 - k} = 1 + 10 + 30 + 40 + 25 + 6 = \boxed{\text{(A)} 112}.$$

Solution 3

Begin by examining f_1 . f_1 can take on any value that is a factor of 96 except 1. For each choice of f_1 , the resulting $f_2 \dots f_k$ must have a product of $96/f_1$. This means the number of ways the rest f_a , $1 < a \leq k$ can be written by the scheme stated in the problem for each f_1 is equal to $D(96/f_1)$, since the product of $f_2 \cdot f_3 \dots f_k = x$ is counted as one valid product if and only if $f_1 \cdot x = 96$, the product x has the properties that factors are greater than 1, and differently ordered products are counted separately.

For example, say the first factor is 2. Then, the remaining numbers must multiply to 48, so the number of ways the product can be written beginning with 2 is $D(48)$. To add up all the number of solutions for every possible starting factor, $D(96/f_1)$ must be calculated and summed for all possible f_1 , except 96 and 1, since a single 1 is not counted according to the problem statement. The 96 however, is counted, but only results in 1 possibility, the first and only factor being 96. This means

$$D(96) = D(48) + D(32) + D(24) + D(16) + D(12) + D(8) + D(6) + D(4) + D(3) + D(2) + 1.$$

Instead of calculating D for the larger factors first, reduce $D(48)$, $D(32)$, and $D(24)$ into sums of $D(m)$ where $m \leq 16$ to ease calculation. Following the recursive

definition $D(n) = (\text{sums of } D(c)) + 1$ where c takes on every divisor of n except for 1 and itself, the sum simplifies to

$D(96) = (D(24) + D(16) + D(12) + D(8) + D(6) + D(4) + D(3) + D(2) + 1) + (D(16) + D(8) + D(4) + D(2) + 1) + D(24) + D(16) + D(12) + D(8) + D(6) + D(4) + D(3) + D(2) + 1$.
 $D(24) = D(12) + D(8) + D(6) + D(4) + D(3) + D(2) + 1$, so the sum further simplifies to

$D(96) = 3D(16) + 4D(12) + 5D(8) + 4D(6) + 5D(4) + 4D(3) + 5D(2) + 5$, after combining terms. From quick casework,

$D(16) = 8, D(12) = 8, D(8) = 4, D(6) = 3, D(4) = 2, D(3) = 1$ and $D(2) = 1$. Substituting these values into the expression above,

$$D(96) = 3 \cdot 8 + 4 \cdot 8 + 5 \cdot 4 + 4 \cdot 3 + 5 \cdot 2 + 4 \cdot 1 + 5 \cdot 1 + 5 = \boxed{(A) 112}.$$

Solution 4

Note that $96 = 3 \cdot 2^5$, and that D of a perfect power of a prime is relatively easy to calculate. Also note that you can find $D(96)$ from $D(32)$ by simply totaling the number of ways there are to insert a 3 into a set of numbers that multiply to 32.

First, calculate $D(32)$. Since $32 = 2 \cdot 2 \cdot 2 \cdot 2 \cdot 2$, all you have to do was find the number of ways to divide the 2's into groups, such that each group has at least one 2. By stars and bars, this results in 1 way with five terms, 4 ways with four terms, 6 ways with three terms, 4 ways with two terms, and 1 way with one term. (The total, 16, is not needed for the remaining calculations.)

Then, to get $D(96)$, in each possible $D(32)$ sequence, insert a 3 somewhere in it, either by placing it somewhere next to the original numbers (in one of $n + 1$ ways, where n is the number of terms in the $D(32)$ sequence), or by multiplying one of the numbers by 3 (in one of n ways). There are $2 + 1 = 3$ ways to do this with one term, $3 + 2 = 5$ with two, 7 with three, 9 with four, and 11 with five.

The resulting number of possible sequences is $3 \cdot 1 + 5 \cdot 4 + 7 \cdot 6 + 9 \cdot 4 + 11 \cdot 1 = \boxed{(A) 112}$.

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Solution 5 (Minimal Casework)

Consider the arrangement of the prime factors of 96 in a line $(2, 2, 2, 2, 2, 3)$. An arrangement of factors can be created by placing "dividers" to group primes. For example, $(2, 2, |, 2, 2, 2, |, 3)$ is equivalent to the arrangement $4 \cdot 8 \cdot 3$. Because there are 6 ways to order the prime factors, and 2^5 ways to place dividers, this gives us an initial $6 \cdot 2^5$ ways to arrange dividers.

However, through this method, we overcount cases where 3 is combined with another factor. For example, the arrangement $4 \cdot 6 \cdot 4$ can be written as $(2, 2, |, 2, 3, |, 2, 2)$ or $(2, 2, |, 3, 2, |, 2, 2)$. Precisely, we double count any case with 6 as a factor, triple count any case with 12, quadruple count any case with 24, etc.

Now, consider all cases where 3 must be grouped with at least one 2. This can be expressed in the same "line" format as $(2, 2, 2, 2, 6)$, where dividers can again be placed to group divisors. In this case, there are 5 ways to order divisors, and 2^4 ways to place dividers, so we have an $5 \cdot 2^4$ possible sequences for this case. Notice that in this format, we double count cases where 12 is a factor, we triple count cases where 24 is a factor, etc. Precisely, for any case counted n times in the first step, it is counted $n - 1$ times in this step. Thus, if we subtract, we count each case exactly once.

So, we get:

$$6 \cdot 2^5 - 5 \cdot 2^4 = \boxed{(A) 112}. \quad \text{hdail122}$$

Solution 6 (Another Fast Way)

First we factor 32 into m numbers $g_1, \dots, g_m$ where $g_i > 1, i = 1, \dots, m$. By applying stars and bars there are $\binom{5-1}{m-1}$ ways. Then we can either insert 3 into each of the $m+1$ spaces between (or beyond) g_i 's, or multiply it to one of the g_i 's, a total of $2m+1$ ways. Hence the answer to the problem is

$$\sum_{m=1}^5 (2m+1) \binom{5-1}{m-1} = \sum_{n=0}^4 (2n+3) \binom{4}{n} = 8 \sum_{n=0}^4 \frac{n}{4} \binom{4}{n} + 3 \sum_{n=0}^4 \binom{4}{n} = 8 \sum_{n=0}^4 \binom{3}{n-1} + 3 \sum_{n=0}^4 \binom{4}{n} = 8 \cdot 2^3 + 3 \cdot 2^4 = 8 \cdot 8 + 3 \cdot 16 = 64 + 48 = 112$$

Solution 7 (Integer Partition)

Note that $96 = 2^5 \cdot 3$. $D(n)$ depends on dividing 2^5 into different terms, which is the integer partition of 5.

Divide 96 into 1 term:

There is only one way. 1

Divide 96 into 2 terms:

Case 1\$: 3\$ is alone $(2^5, 3)$ has $\text{\textbf{2}}$ different arrangements.

Case 2\$: 3\$ is with 2^n $5 = 1 + 4 = 2 + 3$. For $(2, 2^4)$ and $(2^2, 2^3)$, 3\$ can be with 2 or 2^2 .
 $2 + 8 = \underline{10}$

Divide 96 into 3 terms:

Case 1\$: 3\$ is alone $5 = 2 + 3 = 1 + 4$. For $(2^2, 2^3, 3)$ and $(2, 2^4, 3)$, the 3\$ can be with 2 or 2^2 or 2^3 .
 $3 + 12 = 15$

Case 2\$: 3\$ is with 2^n $5 = 1 + 1 + 3 = 1 + 2 + 2$. For $(2, 2, 2^3)$ and $(2, 2^2, 2^2)$, 3\$ can be with 2 or 2^2 .
 $12 + 18 = \underline{30}$

Divide 96 into 4 terms:

Case 1\$: 3\$ is alone $5 = 1 + 1 + 3 = 1 + 2 + 2$. For $(2, 2, 2^3, 3)$ and $(2, 2^2, 2^2, 3)$, the 3\$ can be with 2 or 2^2 .
 $15 + 15 = 30$

Case 2\$: 3\$ is with 2^n $5 = 1 + 1 + 1 + 2$. For $(2, 2, 2, 2^2)$, 3\$ can be with 2 or 2^2 .
 $24 + 16 = \underline{40}$

Divide 96 into 5 terms:

When dividing 5 into 5 parts there are 2 cases.

Case 1\$: 3\$ is alone $5 = 1 + 1 + 1 + 2$. For $(2, 2, 2, 2^2, 3)$, there are $\frac{6!}{5!} = 6$ arrangements.

Case 2\$: 3\$ is with 2^n $5 = 1 + 1 + 1 + 1 + 1$. For $(2, 2, 2, 2, 2)$, 3\$ can be with 2 or 2^2 .
 $20 + 5 = \underline{25}$

Divide 96 into 6 terms:

$5 = 1 + 1 + 1 + 1 + 1$. The number of arrangements of $(2, 2, 2, 2, 2, 3)$ is $\frac{6!}{5!} = 6$
 $1 + 10 + 30 + 40 + 25 + 6 = \boxed{(A) 112}$

Solution 8

Ignore the 3 first and first count $2^{x_1+x_2+\dots+x_n} = 32$ which $x_1 + x_2 + \dots + x_n = 5$. This implies that n is less than or equal to 5. Now, we can see that 3 can lie between

the two x_i, x_{i+1} , or contribute to one of them. This gives $2k + 1$ if $x_1 + \dots + x_k = 5$. Now, just sum up gives $\binom{5-1}{5-1} \cdot 11 + \binom{5-1}{4-1} \cdot 9 + \binom{5-1}{3-1} \cdot 7 + \binom{5-1}{2-1} \cdot 5 + \binom{5-1}{1-1} \cdot 3 = \boxed{112}$.

Solution 9 (Cleanest Algebra)

Consider how we partition the factors of $96 = 2^5 \cdot 3$. For each k , there are two cases. Either we can put the 2s into k nonzero parts, so that the 3 shares a partition with some 2s, which can be done in $k \binom{5-(k-1)+(k-1)-1}{k-2} = k \binom{4}{k-2}$ ways, or we can put the 2s into $k - 1$ nonzero parts and put the 3 in its own partition, which can be done in $k \binom{5-k+k-1}{k-1} = k \binom{4}{k-1}$ ways. Summing over all k , we have

$$\sum_{k=1}^6 k \binom{4}{k-2} + \sum_{k=1}^6 k \binom{4}{k-1}.$$

But $\sum_{k=1}^6 k \binom{4}{k-2} = 2 \binom{4}{0} + 3 \binom{4}{1} + \dots + 6 \binom{4}{4} = 4 \cdot \sum_{i=0}^4 \binom{4}{i} = 4 \cdot 2^4 = 64$. Similarly, $\sum_{k=1}^6 k \binom{4}{k-1} = 3 \cdot 2^4 = 48$. So our answer is $64 + 48 = \boxed{(A) 112}$.

Solution 10 (Word Arrangements and Brute Forcing)

After careful inspection we notice that for each respect value of k , there are a certain number of unique set of numbers that form total word arrangements. Using case work we can deduce that for each respective k value:

$k = 1: 96$

$k = 2: 2 \ 48, 3 \ 32, 4 \ 24, 6 \ 16, 8 \ 12$

$k = 3: 2 \ 6 \ 8, 2 \ 3 \ 16, 3 \ 4 \ 8, 2 \ 4 \ 12, 2 \ 2 \ 24, 4 \ 4 \ 6$

$k = 4: 2 \ 2 \ 3 \ 8, 2 \ 3 \ 4 \ 4, 2 \ 2 \ 2 \ 12, 2 \ 2 \ 4 \ 6$

$k = 5: 2 \ 2 \ 2 \ 3 \ 4, 2 \ 2 \ 2 \ 2 \ 6$

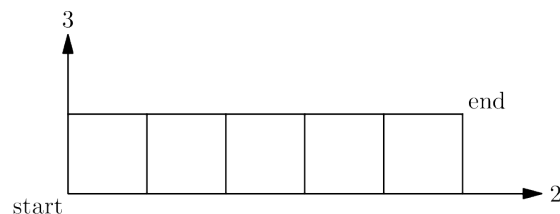
$k = 6: 2 \ 2 \ 2 \ 2 \ 2 \ 3$

Using the word arrangement formula $\left(\frac{n!}{f_1! f_2! \dots f_n!} \right)$, where n is the total number of letters in the word, and f is the frequency of each respective letter for instance in "aabc" the letter "a" has a frequency of 2, b has 1, and c has 1) for each respective set of unique integers we obtain the sum $1 + 10 + 30 + 40 + 25 + 6 = 112 = \boxed{(A) 112}$

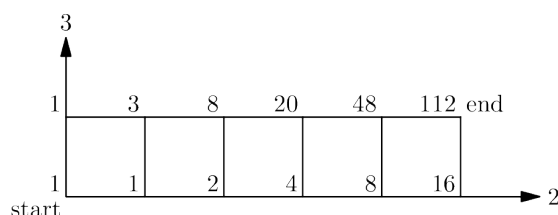
Solution 11 (Grid with Labels)

This question is like [2010 AMC 8 problem 25](#), if we treat each prime factor of $96 = 2^5 \cdot 3$ as a step, this is a recursive problem of climbing stairs.

The stair can be designed like



In this grid, we can only take any number of steps to the right or up, so the number of each point is the sum of all numbers to its left and below it. We label the points to obtain number of paths.



The result is **(A) 112**

Solution 12 (The simplest solution)

There are $2^4 = 16$ ways of placing (or not placing) $\times$'s between five 2's. These placements can be grouped into 8 pairs such that in each pair, the total number of $\times$'s is 4.

In each pair, there are $4 + 2 = 6$ powers of 2 and $6 + 2 = 8$ spaces between/beyond these powers. We can either multiply 3 with one of the powers, or place 3 in the spaces between/beyond the powers (so that we have a single 3 in the product), giving a total of $6 + 8 = 14$ ways of decomposing 96 into products greater than 1.

The answer is then $8 \cdot 14 = 112$.

§2 AMC 10 10A 2020

1. (AMC 10 10A 2020) What value of x satisfies

$$x - \frac{3}{4} = \frac{5}{12} - \frac{1}{3}?$$

- (A) $-\frac{2}{3}$ (B) $\frac{7}{36}$ (C) $\frac{7}{12}$ (D) $\frac{2}{3}$ (E) $\frac{5}{6}$

Answer: E

Solution 1

Adding $\frac{3}{4}$ to both sides, $x = \frac{5}{12} - \frac{1}{3} + \frac{3}{4} = \frac{5}{12} - \frac{4}{12} + \frac{9}{12} = \textbf{(E) } \frac{5}{6}$.

Solution 2

Multiplying 12 on both sides gets us $12x - 9 = 1 \Rightarrow 12x = 10$, therefore $x = \textbf{(E) } \frac{5}{6}$.

2. (AMC 10 10A 2020) The numbers 3, 5, 7, a , and b have an average (arithmetic mean) of 15. What is the average of a and b ?

- (A) 0 (B) 15 (C) 30 (D) 45 (E) 60

Answer: C

Solution

The arithmetic mean of the numbers 3, 5, 7, a , and b is equal to $\frac{3+5+7+a+b}{5} = \frac{15+a+b}{5} = 15$. Solving for $a + b$, we get $a + b = 60$. Dividing by 2 to find the average of the two numbers a and b gives $\frac{60}{2} = \textbf{(C) } 30$.

Solution (two solutions - balancing and summing)

We know the average is 15. 3, 5, and 7 are, respectively, 12, 10, and 8 below the average. So far then we are $-12 - 10 - 8$ or 30 below the average. We have to make

this up with a and b so on average each is $30/2 = 15$ above the average. The average is 15 so each is on average $15+15 = 30$, or answer C.

Incidentally we can immediately rule out answers A and B as being too small - with these we'd never catch up to get our average of 15. An alternative solution is that to get an average of 15 we need a sum of $15 \cdot 5 = 75$. We have 3, 5, and 7 which adds up to 15, so a and b must make up the remaining $75-15 = 60$. $60/2 = 30$ so the average of a and b is 30.

~Dilip

3. (AMC 10 10A 2020) Assuming $a \neq 3$, $b \neq 4$, and $c \neq 5$, what is the value in simplest form of the following expression?

$$\frac{a-3}{5-c} \cdot \frac{b-4}{3-a} \cdot \frac{c-5}{4-b}$$

- (A) -1 (B) 1 (C) $\frac{abc}{60}$ (D) $\frac{1}{abc} - \frac{1}{60}$ (E) $\frac{1}{60} - \frac{1}{abc}$

Answer: A

Solution 1 (Negatives) (Fastest)

If $x \neq y$, then $\frac{x-y}{y-x} = -1$. We use this fact to simplify the original expression:

$$\frac{\overset{-1}{\cancel{a-3}}}{\underset{1}{\cancel{5-c}}} \cdot \frac{\overset{-1}{\cancel{b-4}}}{\underset{1}{\cancel{3-a}}} \cdot \frac{\overset{-1}{\cancel{c-5}}}{\underset{1}{\cancel{4-b}}} = (-1)(-1)(-1) = \boxed{\text{(A)} -1}.$$

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Solution 2 (Answer Choices)

At $(a, b, c) = (4, 5, 6)$, the answer choices become

- (A) -1 (B) 1 (C) 2 (D) $-\frac{1}{120}$ (E) $\frac{1}{120}$

and the original expression becomes

$$\frac{-1}{1} \cdot \frac{-1}{1} \cdot \frac{-1}{1} = \boxed{\text{(A)} -1}.$$

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Solution 3

We can simply set $x = a - 3$, $y = b - 4$, and $z = 5 - c$. Now, the problem simplifies to

$$\frac{x}{z} \cdot \frac{-y}{x} \cdot \frac{z}{y} = \frac{-xyz}{xyz} = \boxed{\text{(A)} -1}.$$

Explanation: After substituting x , y , and z , the opposites (for example $5 - c$ and $c - 5$) can just be written as the negative of each. With the same example, this can be shown by: $c - 5 = -(5 - c)$.

4. (AMC 10 10A 2020) A driver travels for 2 hours at 60 miles per hour, during which her car gets 30 miles per gallon of gasoline. She is paid \$0.50 per mile, and her only expense is gasoline at \$2.00 per gallon. What is her net rate of pay, in dollars per hour, after this expense?

- (A) 20 (B) 22 (C) 24 (D) 25 (E) 26

Answer: E

Solution 1

Since the driver travels 60 miles per hour and each hour she uses 2 gallons of gasoline, she spends \$4 per hour on gas. If she gets \$0.50 per mile, then she gets \$30 per hour of driving. Subtracting the gas cost, her net rate of money earned per hour is **(E) 26**. mathsmiley

Solution 2 (longer)

The driver is driving for 2 hours at 60 miles per hour, she drives 120 miles. Therefore, she uses $\frac{120}{30} = 4$ gallons of gasoline. So, total she has $\$0.50 \cdot 120 - \$2.00 \cdot 4 = \$60 - \$8 = \$52$. So, her rate is $\frac{52}{2} = \mathbf{(E) 26}$. sosiaops

5. (AMC 10 10A 2020) What is the sum of all real numbers x for which $|x^2 - 12x + 34| = 2$? (A) 12 (B) 15 (C) 18 (D) 21 (E) 25

Answer: C

Solution 1 (Casework and Factoring)

Split the equation into two cases, where the value inside the absolute value is positive and nonpositive.

Case 1:

The equation yields $x^2 - 12x + 34 = 2$, which is equal to $(x - 4)(x - 8) = 0$. Therefore, the two values for the positive case is 4 and 8.

Case 2:

Similarly, taking the nonpositive case for the value inside the absolute value notation yields $x^2 - 12x + 34 = -2$. Factoring and simplifying gives $(x - 6)^2 = 0$, so the only value for this case is 6.

Summing all the values results in $4 + 8 + 6 = \mathbf{(C) 18}$.

Solution 2 (Casework and Vieta)

We have the equations $x^2 - 12x + 32 = 0$ and $x^2 - 12x + 36 = 0$.

Notice that the second is a perfect square with a double root at $x = 6$, and the first has two distinct real roots. By Vieta's, the sum of the roots of the first equation is $-(-12)$ or 12. $12 + 6 = \mathbf{(C) 18}$.

Solution 3 (Casework and Graphing)

Completing the square gives

$$\begin{aligned} |(x - 6)^2 - 2| &= 2 \\ (x - 6)^2 - 2 &= \pm 2. \end{aligned} \quad (\star)$$

Note that the graph of $y = (x - 6)^2 - 2$ is an upward parabola with the vertex $(6, -2)$ and the axis of symmetry $x = 6$; the graphs of $y = \pm 2$ are horizontal lines.

We apply casework to $(\star)$:

- a) $(x - 6)^2 - 2 = 2$ The line $y = 2$ intersects the parabola $y = (x - 6)^2 - 2$ at two points that are symmetric about the line $x = 6$.

In this case, the average of the solutions is 6, so the sum of the solutions is 12.

- b) $(x - 6)^2 - 2 = -2$ The line $y = -2$ intersects the parabola $y = (x - 6)^2 - 2$ at one point: the vertex of the parabola.

In this case, the only solution is $x = 6$.

Finally, the sum of all solutions is $12 + 6 = \boxed{\text{(C)} 18}$.

Solution 4 (Time Saving)

We have $x^2 - 12x - 34 = 2$ or $x^2 - 12x - 34 = -2$. Solving the latter yields $x = 6$. Now, from here, we know that the first equation either has 2 solutions or no solution since it isn't a perfect square. By Vieta's, this would imply the sum to be either 0 or 12, making the whole sum 6 or 18. Since 6 isn't an answer choice, the answer must be 18.

6. (AMC 10 10A 2020) How many 4-digit positive integers (that is, integers between 1000 and 9999, inclusive) having only even digits are divisible by 5? (A) 80 (B) 100 (C) 125

Answer: B

Solution

The units digit, for all numbers divisible by 5, must be either 0 or 5. However, since all digits are even, the units digit must be 0. The middle two digits can be 0, 2, 4, 6, or 8, but the thousands digit can only be 2, 4, 6, or 8 since it cannot be zero. There is one choice for the units digit, 5 choices for each of the middle 2 digits, and 4 choices for the thousands digit, we have a total of $4 \cdot 5 \cdot 5 \cdot 1 = \boxed{\text{(B)} 100}$ numbers.

7. (AMC 10 10A 2020) The 25 integers from -10 to 14 , inclusive, can be arranged to form a 5-by-5 square in which the sum of the numbers in each row, the sum of the numbers in each column, and the sum of the numbers along each of the main diagonals are all the same. What is the value of this common sum?
(A) 2 (B) 5 (C) 10 (D) 25 (E) 50

Answer: C

Solution 1

Without loss of generality, consider the five rows in the square. Each row must have the same sum of numbers, meaning that the sum of all the numbers in the square divided by 5 is the total value per row. The sum of the 25 integers is $-10 + -9 + \dots + 14 = 11 + 12 + 13 + 14 = 50$, and the common sum is $\frac{50}{5} = \boxed{\text{(C)} 10}$.

Solution 2

Take the sum of the middle 5 values of the set (they will turn out to be the mean of each row). We get $0 + 1 + 2 + 3 + 4 = \boxed{\text{(C)} 10}$ as our answer. Baolan

Solution 3

Taking the average of the first and last terms, -10 and 14 , we have that the mean of the set is 2. There are 5 values in each row, column or diagonal, so the value of the common sum is $5 \cdot 2$, or $\boxed{\text{(C)} 10}$. Arctic_Bunny, edited by KINGLOGIC

Solution 4

Let us consider the horizontal rows. Since there are five of them, each with constant sum x , we can add up the 25 numbers in 5 rows for a sum of $5x$. Since the sum of the 25 numbers used is $-10 - 9 - 8 - \dots + 12 + 13 + 14 + 15 = 11 + 12 + 13 + 14 + 15 = 50$, $5x = 50$ and $x = \boxed{\text{(C)} 10}$. cw357

Solution 5

The mean of the set of numbers is $(14 - 10) \div 2 = 2$. The numbers around it must be equal (i.e. if the mean of 1, 2, 3, 4, and 5 is 3, then $2 + 4 = 1 + 5$.) One row of the square would be

$$\square\square 2\square\square$$

Adding the numbers would be

$$0, 1, 2, 3, 4$$

with a sum of $\boxed{(C) 10}$.

Solution 6

If the sum of each row, column, and diagonal is x , then we have a total of $12x$ for the sum. The sum of the rows and columns is the sum of all the numbers doubled, which is $50 \cdot 2 = 100$. Therefore $100 + 2x = 12x$, $100 = 10x$, and $x = \boxed{(C) 10}$. MC413551

Solution 7

A cool thing to find is that with magic squares, the smallest and largest number are either in the same row, column and diagonal to balance the equation out. To continue doing this, the other numbers in that same line form an arithmetic sequence. With our A term generally being 1, and our last term being n^2 . The $d = n + 1$, by working out the $a + (n - 1)d$ formula. For a 5 by 5, $d = 6$. To apply this we start with A being -10 , and our last term being 14. We increase by increments by 6 to get the 5 numbers in our line is $-10, -4, 2, 8, 14$. This adds up to $\boxed{(C) 10}$.

(Note the goal of this solution was just to point out something I found that I thought was cool, I don't advise using it in actual competitions)

-CubiksRube

Solution 8

We can solve this by counting how many times each square is included when summing all possible lines. The number of possible lines that can be drawn is 12 (5 rows + 5 columns + 2 main diagonals).

First, the sum of all 25 integers in the set $\{-10, -9, \dots, 14\}$ is 50. Let S be the common sum of a single line. The sum of all 12 lines combined is $12S$.

However, not all squares are used an equal amount of times in these 12 lines. Let's categorize them:

- The center square is used in 4 lines. Let its value be a .
- The remaining 8 squares on the main diagonals are used in 3 lines each. Let the sum of these 8 squares be b .
- The remaining 16 squares are used in 2 lines each. Let the sum of these 16 squares be c .

We can set up a system of equations based on this information:

1. The sum of all squares is 50:

$$a + b + c = 50$$

2. The sum of all 12 lines counts each square multiple times based on its position:

$$12S = 4a + 3b + 2c$$

This means $4a + 3b + 2c \equiv 0 \pmod{12}$.

We just need to find a combination of a , b , and c such that $a + b + c$ is a factor of 50, and then scale it up. A simple combination that works is $a = 0$, $b = 2$, and $c = 3$. The sum here is $0 + 2 + 3 = 5$. Since we need the sum to be 50, we can multiply our combination by 10 to get $a = 0$, $b = 20$, and $c = 30$. (Note: $0 + 20 + 30 = 50$, satisfying our first equation).

Now, plug this combination back into our line sum equation:

$$12S = 4(0) + 3(20) + 2(30)$$

$$12S = 0 + 60 + 60$$

$$12S = 120$$

$$S = 10$$

The common sum is thus (C) 10.

-PandaGod7

8. (AMC 10 10A 2020) What is the value of

$$1 + 2 + 3 - 4 + 5 + 6 + 7 - 8 + \cdots + 197 + 198 + 199 - 200?$$

- (A) 9,800 (B) 9,900 (C) 10,000 (D) 10,100 (E) 10,200

Answer: B

Solution 1

Looking at the numbers, you see that every set of 4 has 3 positive numbers and 1 negative number. Calculating the sum of the first couple sets gives us $2 + 10 + 18 \dots + 394$. Clearly, this pattern is an arithmetic sequence. By using the formula we get $\frac{2+394}{2} \cdot 50 = \text{span style="border: 1px solid black; padding: 0 5px;">(B) 9900.$

-ViratKohli2018

Solution 2

Split the even numbers and the odd numbers apart. If we group every 2 even numbers together and add them, we get a total of $50 \cdot (-2) = -100$. Summing the odd numbers is equivalent to summing the first 100 odd numbers, which is equal to $100^2 = 10000$. Adding these two, we obtain the answer of (B) 9900.

Solution 3 (bashing)

We can break this entire sum down into 4 integer bits, in which the sum is $2x$, where x is the first integer in this bit. We can find that the first sum of every sequence is $4x - 3$, which we plug in for the

9. (AMC 10 10A 2020) A single bench section at a school event can hold either 7 adults or 11 children. When N bench sections are connected end to end, an equal number of adults and children seated together will occupy all the bench space. What is the least possible positive integer value of N ? (A) 9 (B) 18 (C) 27 (D) 36 (E) 77

Answer: B

Solution 1

The least common multiple of 7 and 11 is 77. Therefore, there must be 77 adults and 77 children. The total number of benches is $\frac{77}{7} + \frac{77}{11} = 11 + 7 = \boxed{\text{(B) } 18}$. taarun-ganesh

Solution 2

Let x denote how many adults there are. Since the number of adults is equal to the number of children we can write N as $\frac{x}{7} + \frac{x}{11} = N$. Simplifying we get $\frac{18x}{77} = N$. Since both n and x have to be positive integers, x has to equal 77. Therefore, $N = \boxed{\text{(B) } 18}$ is our final answer.

Solution 3

We have 1 Bench = 7 Adults = 11 Children, or $A = \frac{11}{7}C$.

The minimum number of benches is $N = x(A + C)$.

This gives $\frac{77}{18}N = x$, so that the minimum value for x is the integer 18, or (B).

Solution 4 (Fast)

The ratio of adults and children is 7:11 and the total amount of benches is the amount of adult and children combined which gives $7 + 11 = \boxed{\text{(B) } 18}$. We see that this is the smallest integer value.

10. (AMC 10 10A 2020) Seven cubes, whose volumes are 1, 8, 27, 64, 125, 216, and 343 cubic units, are stacked vertically to form a tower in which the volumes of the cubes decrease from bottom to top. Except for the bottom cube, the bottom face of each cube lies completely on top of the cube below it. What is the total surface area of the tower (including the bottom) in square units?
(A) 644 (B) 658 (C) 664 (D) 720 (E) 749

Answer: B

Solution 1

The volume of each cube follows the pattern of n^3 , for n is between 1 and 7.

We see that the total surface area can be comprised of three parts: the sides of the cubes, the tops of the cubes, and the bottom of the $7 \times 7 \times 7$ cube (which is just $7 \times 7 = 49$). The sides areas can be measured as the sum $4 \sum_{n=1}^7 n^2$, giving us 560. Structurally, if we examine the tower from the top, we see that it really just forms a 7×7 square of area 49. Therefore, we can say that the total surface area is $560 + 49 + 49 = \boxed{\text{(B) } 658}$. Alternatively, for the area of the tops, we could have found the sum $\sum_{n=2}^7 ((n)^2 - (n-1)^2)$, giving us 49 as well.

Note: The area on top and bottom are 49 because the largest area is 49, and the other cubes are "inscribed" in it.

Solution 2

It can quickly be seen that the side lengths of the cubes are the integers from 1 to 7 inclusive.

First, we will calculate the total surface area of the cubes, ignoring overlap. This value is $6(1^2 + 2^2 + \cdots + 7^2) = 6 \sum_{n=1}^7 n^2 = 6 \left(\frac{7(7+1)(2 \cdot 7+1)}{6} \right) = 7 \cdot 8 \cdot 15 = 840$. Then, we need to subtract out the overlapped parts of the cubes. Between each consecutive pair of cubes, one of the smaller cube's faces is completely covered, along with an equal area of one of the larger cube's faces. The total area of the overlapped parts of the cubes is thus equal to $2 \sum_{n=1}^6 n^2 = 182$. Subtracting the overlapped surface area from the total surface area, we get $840 - 182 = \boxed{\text{(B)} 658}$.

emerald_block

Solution 3 (a bit more tedious than other solutions)

It can be seen that the side lengths of the cubes using cube roots are all integers from 1 to 7, inclusive.

Only the cubes with side length 1 and 7 have 5 faces in the surface area and the rest have 4. Also, since the cubes are stacked, we have to find the difference between each n^2 and $(n-1)^2$ side length as n ranges from 7 to 2.

We then come up with this: $5(49) + 13 + 4(36) + 11 + 4(25) + 9 + 4(16) + 7 + 4(9) + 5 + 4(4) + 3 + 5(1)$.

We then add all of this and get $\boxed{\text{(B)} 658}$.

Solution 4

Notice that the surface area of the top cube is $6s^2$ and the others are $4s^2$. Then we can directly compute. The edge length for the first cube is 7 and has a surface area of 294. The surface area of the next cube is 144. The surface area of the next cube is 100. The surface area of the next cube is 64. The surface area of the next cube is 36. The surface area of the next cube is 16. The surface area of the next cube is 4. We then sum up $294 + 144 + 100 + 64 + 36 + 16 + 4$ to get $\boxed{\text{(B)} 658}$.

Solution 5

First of all, compute the area of the sides, excluding the top and bottoms, of the cubes. The side lengths (cube root the volumes) are 1, 2, 3, 4, 5, 6, 7. Each cube's area of the sides can be calculated with $4(\text{area of one side}) = 4(l^2)$ so in total that is $4(1 + 4 + 16 + \dots + 49)$ so $4(140) = 560$ the area of all the sides of the cubes is 560. Then, calculate the bottom face of the largest cube, $7 * 7 = 49$. Now, notice that if you stack the cubes up on top of each other, and look directly down on them, the tops of the cubes showing add up to the area of the bottom cube, the 7×7 . Therefore, the sum of the area of the tops of the cubes is $7 * 7 = 49$.

Now add them all up: $49 + 49 + 560 = 658$. Therefore, the answer is $\boxed{\text{(B)} 658}$.

Solution 6 (Similar to Solution 2, without notation)

Firstly, calculate the total surface area for all the cubes. Knowing that the area of each square of each cube are $2^2, 3^2, 4^2, 5^2, 6^2, 7^2$, we have $6(1 + 4 + 9 + 16 + 25 + 36 + 49) = 6(140) = 840$. When you stack these cubes, the top square of each are subtracted by the square area of the cube above it. Meanwhile, the bottom square of each (except the $7 \times 7 \times 7$ cube) won't be counted because of the stacking. So, we have to subtract $2(1 + 4 + 9 + 16 + 25 + 36) = 2(91) = 182$ from 840, getting $\boxed{\text{(B)} 658}$.

Solution 7 (Going from Bottom to Top)

The surface area of the bottom cube is the bottom face, the 4 side faces, and the top face minus the bottom face of the next cube. Therefore, the surface area of

the bottom cube is 6 faces minus the bottom face of the next cube. That will be $6 \cdot 49 - 36$. For the next cube, we take the 4 side faces and the top face, and subtract the bottom face of the next cube. Therefore, the surface area of the next cube will be 5 of its faces minus the bottom face of the next cube. That will be $5 \cdot 36 - 25$. We can do the next cube the same way. It will end up as $5 \cdot 25 - 16$. From here we start noticing a pattern. The next will be $5 \cdot 16 - 9$, and then $5 \cdot 9 - 4$, next $5 \cdot 4 - 1$, and to include the top cube, we have to add $5 \cdot 1$, which is basically $5 \cdot 1 - 0$. Now, putting it all together, we get $6 \cdot 49 - 36 + 5 \cdot 36 - 25 + 5 \cdot 25 - 16 + 5 \cdot 16 - 9 + 5 \cdot 9 - 4 + 5 \cdot 4 - 1 + 5 \cdot 1 - 0 = 6 \cdot 49 + 5(36 + 25 + 16 + 9 + 4 + 1) - (36 + 25 + 16 + 9 + 4 + 1) = 6 \cdot 49 + 4(36 + 25 + 16 + 9 + 4 + 1) = 294 + 4(91) = 294 + 364 = \boxed{\text{(B)} 658}$.

11. (AMC 10 10A 2020) What is the median of the following list of 4040 numbers?

$$1, 2, 3, \dots, 2020, 1^2, 2^2, 3^2, \dots, 2020^2$$

- (A) 1974.5 (B) 1975.5 (C) 1976.5 (D) 1977.5 (E) 1978.5

Answer: C

Solution 1

We can see that $44^2 = 1936$ which is less than 2020. Therefore, there are $2020 - 44 = 1976$ of the 4040 numbers greater than 2020. Also, there are $2020 + 44 = 2064$ numbers that are less than or equal to 2020.

Since there are 44 duplicates/extras, it will shift up our median's placement down 44. Had the list of numbers been $1, 2, 3, \dots, 4040$, the median of the whole set would be $\frac{1 + 4040}{2} = 2020.5$.

Thus, our answer is $2020.5 - 44 = \boxed{\text{(C)} 1976.5}$.

Solution 2

As we are trying to find the median of a 4040-term set, we must find the average of the 2020th and 2021st terms.

Since $45^2 = 2025$ is slightly greater than 2020, we know that the 44 perfect squares 1^2 through 44^2 are less than 2020, and the rest are greater. Thus, from the number 1 to the number 2020, there are $2020 + 44 = 2064$ terms. Since 44^2 is $44 + 45 = 89$ less than $45^2 = 2025$ and 84 less than 2020, we will only need to consider the perfect square terms going down from the 2064th term, 2020, after going down 84 terms. Since the 2020th and 2021st terms are only 44 and 43 terms away from the 2064th term, we can simply subtract 44 from 2020 and 43 from 2020 to get the two terms, which are 1976 and 1977. Averaging the two, we get $\boxed{\text{(C)} 1976.5}$.

Solution 3

We want to know the 2020th term and the 2021st term to get the median.

We know that $44^2 = 1936$. So, numbers $1^2, 2^2, \dots, 44^2$ are in between 1 and 1936. So, the sum of 44 and 1936 will result in 1980, which means that 1936 is the 1980th number.

Also, notice that $45^2 = 2025$, which is larger than 2021.

Then the 2020th term will be $1936 + 40 = 1976$, and similarly the 2021st term will be 1977.

Solving for the median of the two numbers, we get $\boxed{\text{(C)} 1976.5}$

Solution 4

We note that $44^2 = 1936$, which is the largest square less than 2020, which means that there are 44 additional terms before 2020. This makes 2020 the 2064th term. To find the median, we need the 2020th and 2021st term. We note that every term before 2020 is one less than the previous term (That is, we subtract 1 to get the previous term.). If 2020 is the 2064th term, then $2020 - 44$ is the $(2064 - 44)$ th term. So, the 2020th term is 1976, and the 2021st term is 1977, and the average of these two terms is the median, or **(C) 1976.5**.

Solution 5 (Decreasing Order)

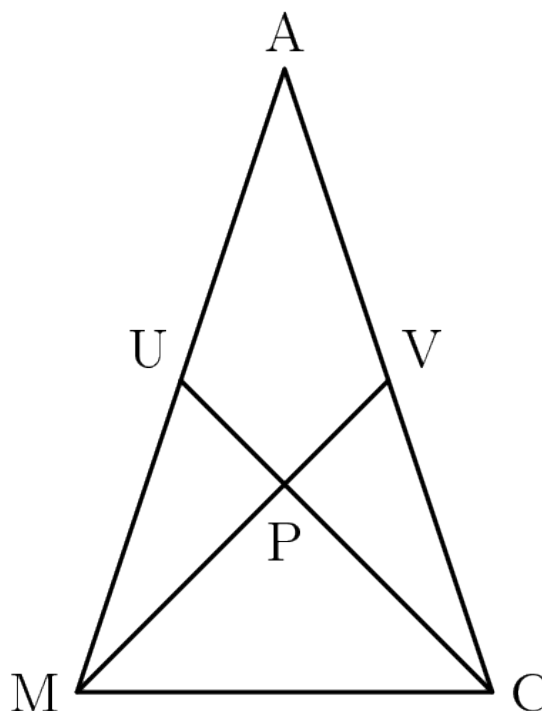
To find the median, we sort the 4040 numbers in decreasing order, then average the 2020th and the 2021st numbers of the sorted list.

Since $45^2 = 2025$ and $44^2 = 1936$, the first 2021 numbers of the sorted list are

$$\underbrace{2020^2, 2019^2, 2018^2, \dots, 46^2, 45^2}_{1976 \text{ numbers}}, \underbrace{2020, 2019, 2018, \dots, 1977, 1976}_{45 \text{ numbers}},$$

from which the answer is $\frac{1977+1976}{2} = \mathbf{(C) 1976.5}$.

12. (*AMC 10 10A 2020*) Triangle AMC is isosceles with $AM = AC$. Medians $\overline{MV}$ and $\overline{CU}$ are perpendicular to each other, and $MV = CU = 12$. What is the area of $\triangle AMC$?



- (A) 48 (B) 72 (C) 96 (D) 144 (E) 192

Answer: C

Solution 1

Since quadrilateral $UVC M$ has perpendicular diagonals, its area can be found as half of the product of the length of the diagonals. Also note that $\triangle AUV$ has $\frac{1}{4}$ the

area of triangle AMC by similarity, so $[UVC M] = \frac{3}{4} \cdot [AMC]$. Thus,

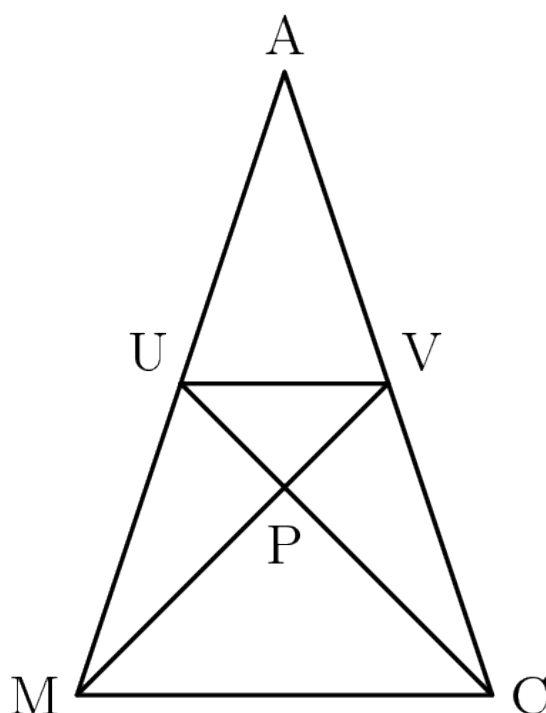
$$\frac{1}{2} \cdot 12 \cdot 12 = \frac{3}{4} \cdot [AMC]$$

$$72 = \frac{3}{4} \cdot [AMC]$$

$$[AMC] = 96 \rightarrow \boxed{(C)}.$$

Note: We know that since AU is the median AU divided by AM is $1/2$. Since triangle AUV is similar to triangle AMC , and by a factor of $1/2$, then every sidelength of AMC must also be scaled by a factor of $1/2$ to get AUV . The base and height are all scaled by $1/2$, so then the ratio is $1/2 \cdot 1/2$.

Solution 2 (Trapezoid)



We know that $\triangle AUV \sim \triangle AMC$, and since the ratios of its sides are $\frac{1}{2}$, the ratio of of their areas is $\left(\frac{1}{2}\right)^2 = \frac{1}{4}$.

If $\triangle AUV$ is $\frac{1}{4}$ the area of $\triangle AMC$, then trapezoid $MUVC$ is $\frac{3}{4}$ the area of $\triangle AMC$.

Let's call the intersection of $\overline{UC}$ and $\overline{MV}$ P . Let $\overline{UP} = x$. Then $\overline{PC} = 12 - x$. Since $\overline{UC} \perp \overline{MV}$, $\overline{UP}$ and $\overline{CP}$ are heights of triangles $\triangle MUV$ and $\triangle MCV$, respectively. Both of these triangles have base 12.

$$\text{Area of } \triangle MUV = \frac{x \cdot 12}{2} = 6x$$

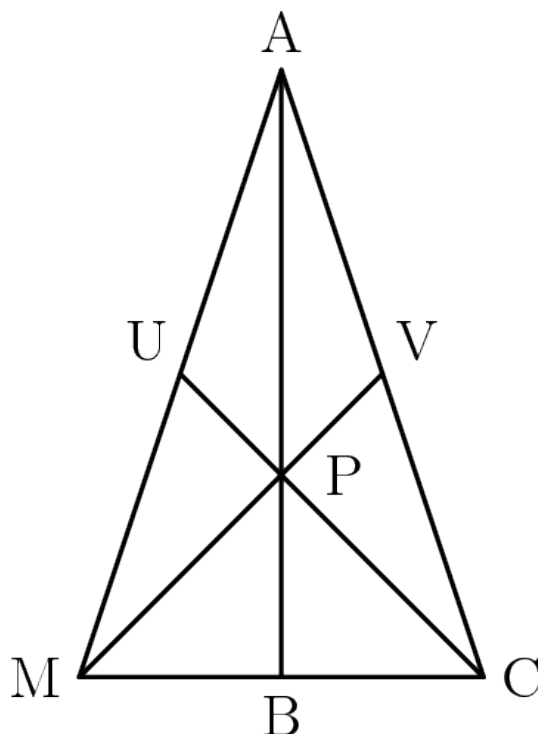
$$\text{Area of } \triangle MCV = \frac{(12-x) \cdot 12}{2} = 72 - 6x$$

Adding these two gives us the area of trapezoid $MUVC$, which is $6x + (72 - 6x) = 72$.

This is $\frac{3}{4}$ of the triangle, so the area of the triangle is $\frac{4}{3} \cdot 72 = \boxed{(C) \ 96}$ quacker88,
diagram by programjames1

Solution 3 (Medians)

Draw median $\overline{AB}$.

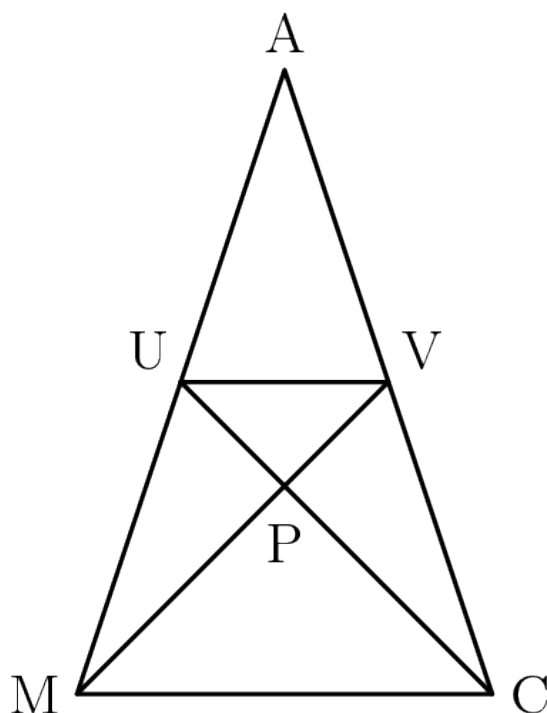


Since we know that all medians of a triangle intersect at the centroid, we know that $\overline{AB}$ passes through point P . We also know that medians of a triangle divide each other into segments of ratio $2 : 1$. Knowing this, we can see that $\overline{PC} : \overline{UP} = 2 : 1$, and since the two segments sum to 12, $\overline{PC}$ and $\overline{UP}$ are 8 and 4, respectively.

Finally knowing that the medians divide the triangle into 6 sections of equal area, finding the area of $\triangle PUM$ is enough. $\overline{PC} = \overline{MP} = 8$.

The area of $\triangle PUM = \frac{4 \cdot 8}{2} = 16$. Multiplying this by 6 gives us $6 \cdot 16 = \boxed{\text{(C)} 96}$

Solution 4 (Triangles)



We know that $AU = UM$, $AV = VC$, so $UV = \frac{1}{2}MC$.

As $\angle UPM = \angle VPC = 90^\circ$, we can see that $\triangle UPM \cong \triangle VPC$ and $\triangle UVP \sim \triangle MPC$ with a side ratio of $1 : 2$.

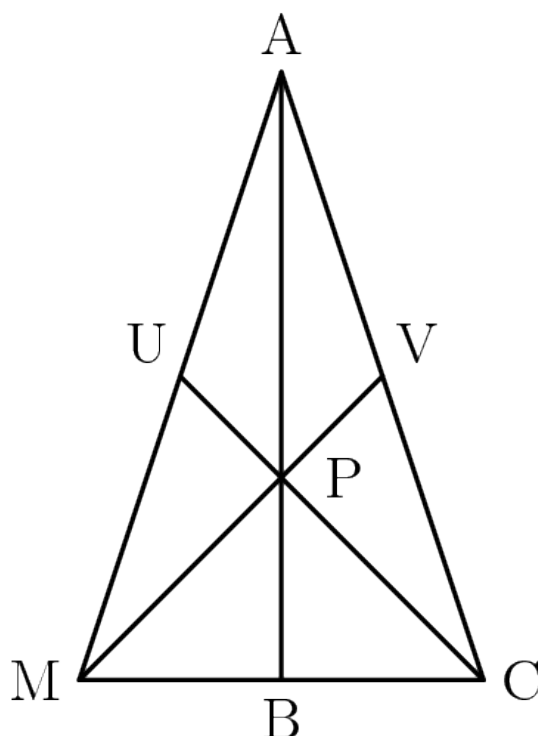
So $UP = VP = 4$, $MP = PC = 8$.

With that, we can see that $[\triangle UPM] = 16$, and the area of trapezoid $MUCV$ is 72.

As said in solution 1, $[\triangle AMC] = 72/\frac{3}{4} = \boxed{\text{(C) } 96}$.

-QuadraticFunctions, solution 1 by ???

Solution 5 (Only Pythagorean Theorem)



Let AB be the height. Since medians divide each other into a $2 : 1$ ratio, and the medians have length 12, we have $PC = MP = 8$ and $UP = VP = 4$. From right triangle $\triangle MUP$,

$$MU^2 = MP^2 + UP^2 = 8^2 + 4^2 = 80,$$

so $MU = \sqrt{80} = 4\sqrt{5}$. Since CU is a median, $AM = 8\sqrt{5}$. From right triangle $\triangle MPC$,

$$MC^2 = MP^2 + PC^2 = 8^2 + 8^2 = 128,$$

which implies $MC = \sqrt{128} = 8\sqrt{2}$. By symmetry $MB = \frac{8\sqrt{2}}{2} = 4\sqrt{2}$.

Applying the Pythagorean Theorem to right triangle $\triangle MAB$ gives $AB^2 = AM^2 - MB^2 = 8^2 \cdot 5 - 4^2 \cdot 2 = 288$, so $AB = \sqrt{288} = 12\sqrt{2}$. Then the area of $\triangle AMC$ is

$$\frac{AB \cdot MC}{2} = \frac{8\sqrt{2} \cdot 12\sqrt{2}}{2} = \frac{96 \cdot 2}{2} = \boxed{(C) 96}$$

Solution 6 (Drawing)

By similarity, the area of AUV is equal to $\frac{1}{4}$.

The area of $UVC M$ is equal to 72.

Assuming the total area of the triangle is S , the equation will be : $\frac{3}{4}S = 72$.

$$S = \boxed{(C) 96}$$

Solution 7 (fastest)

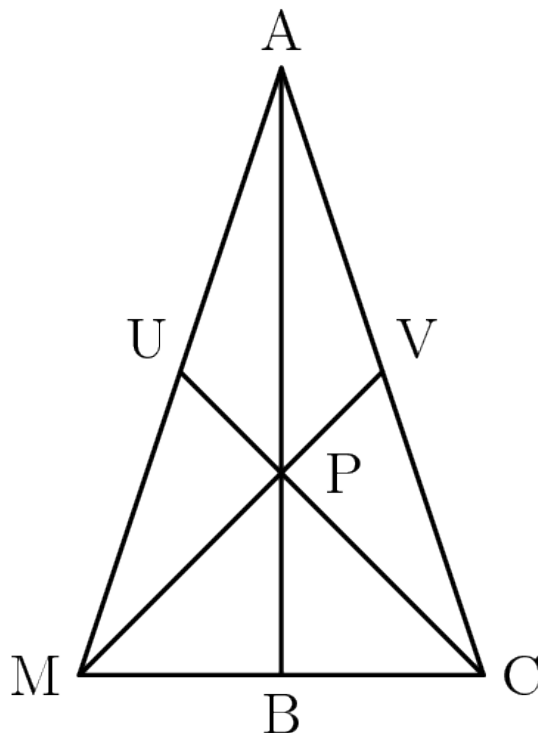
Given a triangle with perpendicular medians with lengths x and y , the area will be $\frac{2xy}{3} = \boxed{(C) 96}$.

Solution 8

Connect the line segment UV and it's easy to see quadrilateral $UVMC$ has an area of the product of its diagonals divided by 2 which is 72. Now, solving for triangle AUV could be an option, but the drawing shows the area of AUV will be less than the quadrilateral meaning the the area of AMC is less than $72 * 2$ but greater than 72, leaving only one possible answer choice, **(C) 96**.

-Rohan S.

Solution 9



Connect AP , and let B be the point where AP intersects MC . $MB = CB$ because all medians of a triangle intersect at one point, which in this case is P . $MP : PV = 2 : 1$ because the point at which all medians intersect divides the medians into segments of ratio $2 : 1$, so $MP = 8$ and similarly $CP = 8$. We apply the Pythagorean Theorem to triangle MPC and get $MC = \sqrt{128} = 8\sqrt{2}$. The area of triangle MPC is $\frac{MP \cdot CP}{2} = 32$, and that must equal to $\frac{MC \cdot BP}{2}$, so $BP = \frac{8}{\sqrt{2}} = 4 \cdot \sqrt{2}$. $BP = \frac{1}{3}BA$, so $BA = 12\sqrt{2}$. The area of triangle AMC is equal to $\frac{MC \cdot BA}{2} = \frac{8 \cdot \sqrt{2} \cdot 12 \cdot \sqrt{2}}{2} = \boxed{\text{(C) } 96}$.

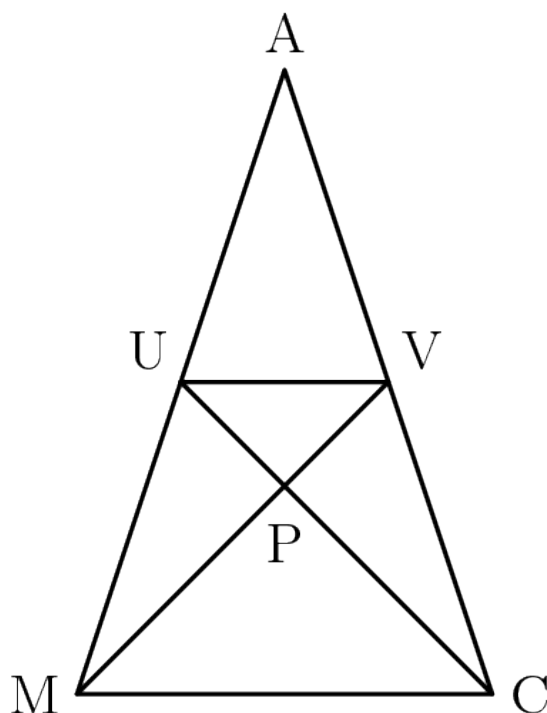
-SmileKat32

Solution 10 (High IQ)

$[\square MUV C] = 72$. Let intersection of line AP and base MC be B

$$[AUV] = [MUB] = [UVB] = [BVC] \implies \left[\frac{\triangle AMC}{4} \right] = \left[\frac{\square MUV C}{3} \right] \implies [AMC] = \boxed{\text{(C) } 96}$$

Solution 11 (Kite)



Since $\overline{MV}$ and $\overline{CU}$ intersect at a right angle, this means $MUV C$ is a kite. Hence, the area of this kite is $\frac{12 \cdot 12}{2} = 72$.

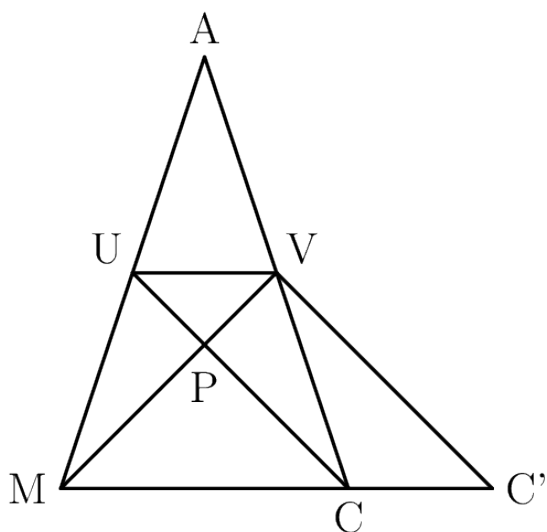
Also, notice that $\triangle AUV \sim \triangle AMC$ by AAA Similarity. Since the ratios of its sides is $\frac{1}{2}$, the ratios of the area is $\left(\frac{1}{2}\right)^2 = \frac{1}{4}$. Therefore,

$$[AMC] = [MUV C] + \frac{1}{4} \cdot [AMC]$$

Simplifying gives us $\frac{3}{4} \cdot [AMC] = 72$, so $[AMC] = 72 \cdot \frac{4}{3} = \boxed{\text{(C) } 96}$

Solution 12 (Educated guessing)

Horizontally translate line $\overline{UC}$ until point U is at point V , with C subsequently at C' , and then connect up C' and C to create $\triangle MVC'$, which is a right triangle.



Notice that $\triangle MVC' = 12 \cdot 12 \cdot \frac{1}{2} = 72$, and $\triangle MVC' = \triangle MVC + \triangle MUV$ (since the latter has the same base and height as the sub-triangle $\triangle CVC'$ inside $\triangle MVC'$).

From this, we can deduce that **(B)** cannot be true, since an incomplete part of $\triangle AMC$ is equal to it. We can also deduce that **(D)** also cannot be true, since the unknown triangle $\triangle AUV = \triangle MUV$, and $\triangle MUV = \triangle CVC' < \triangle MVC'$. Therefore, the answer must be between 72 and 144, leaving **(C) 96** as the only possible correct answer.

13. (AMC 10 10A 2020) A frog sitting at the point $(1, 2)$ begins a sequence of jumps, where each jump is parallel to one of the coordinate axes and has length 1, and the direction of each jump (up, down, right, or left) is chosen independently at random. The sequence ends when the frog reaches a side of the square with vertices $(0, 0)$, $(0, 4)$, $(4, 4)$, and $(4, 0)$. What is the probability that the sequence of jumps ends on a vertical side of the square?

(A) $\frac{1}{2}$ (B) $\frac{5}{8}$ (C) $\frac{2}{3}$ (D) $\frac{3}{4}$ (E) $\frac{7}{8}$

Answer: B

Solution 1

Drawing out the square, it's easy to see that if the frog goes to the left, it will immediately hit a vertical end of the square. Therefore, the probability of this happening is $\frac{1}{4} \cdot 1 = \frac{1}{4}$. If the frog goes to the right, it will be in the center of the square at $(2, 2)$, and by symmetry (since the frog is equidistant from all sides of the square), the chance it will hit a vertical side of a square is $\frac{1}{2}$. The probability of this happening is $\frac{1}{4} \cdot \frac{1}{2} = \frac{1}{8}$.

If the frog goes either up or down, it will hit a line of symmetry along the corner it is closest to and furthest to, and again, is equidistant relating to the two closer sides and also equidistant relating the two further sides. The probability for it to hit a vertical wall is $\frac{1}{2}$. Because there's a $\frac{1}{2}$ chance of the frog going up or down, the total probability for this case is $\frac{1}{2} \cdot \frac{1}{2} = \frac{1}{4}$ and summing up all the cases,

$$\frac{1}{4} + \frac{1}{8} + \frac{1}{4} = \frac{5}{8} \implies \boxed{\text{(B)} \frac{5}{8}}.$$

Solution 2

Let's say we have our five by five grid and we work this out by casework. A is where the frog is, while B and C are possible locations for his second jump, while O is everything else. If we land on a C, we have reached the vertical side. However, if we land on a B, the probability of landing on a C is $1/4$, while B is $3/4$. Since C means that we have "succeeded", while B means that we have a half chance, we compute $1 \cdot C + \frac{1}{2} \cdot B$.

$$1 \cdot \frac{1}{4} + \frac{1}{2} \cdot \frac{3}{4} \\ \frac{1}{4} + \frac{3}{8}$$

We get $\frac{5}{8}$, or B

O	O	O	O	O
O	B	O	O	O
C	A	B	O	O

O B O O O
O O O O O

-yeskay

Solution 3

If the frog is on one of the 2 diagonals, the chance of landing on vertical or horizontal each becomes $\frac{1}{2}$. Since it starts on $(1, 2)$, there is a $\frac{3}{4}$ chance (up, down, or right) it will reach a diagonal on the first jump and $\frac{1}{4}$ chance (left) it will reach the vertical side. The probability of landing on a vertical is $\frac{1}{4} + \frac{3}{4} \cdot \frac{1}{2} = \boxed{\text{(B)} \frac{5}{8}}$.- Lingjun

Solution 4 (Complete States)

Let $P_{(x,y)}$ denote the probability of the frog's sequence of jumps ends with it hitting a vertical edge when it is at (x, y) . Note that $P_{(1,2)} = P_{(3,2)}$ by reflective symmetry over the line $x = 2$. Similarly, $P_{(1,1)} = P_{(1,3)} = P_{(3,1)} = P_{(3,3)}$, and $P_{(2,1)} = P_{(2,3)}$. Now we create equations for the probabilities at each of these points/states by considering the probability of going either up, down, left, or right from that point:

$$P_{(1,2)} = \frac{1}{4} + \frac{1}{2}P_{(1,1)} + \frac{1}{4}P_{(2,2)}$$

$$P_{(2,2)} = \frac{1}{2}P_{(1,2)} + \frac{1}{2}P_{(2,1)}$$

$$P_{(1,1)} = \frac{1}{4} + \frac{1}{4}P_{(1,2)} + \frac{1}{4}P_{(2,1)}$$

$$P_{(2,1)} = \frac{1}{2}P_{(1,1)} + \frac{1}{4}P_{(2,2)}$$

We have a system of 4 equations in 4 variables, so we can solve for each of these probabilities. Plugging the second equation into the fourth equation gives

$$P_{(2,1)} = \frac{1}{2}P_{(1,1)} + \frac{1}{4} \left(\frac{1}{2}P_{(1,2)} + \frac{1}{2}P_{(2,1)} \right)$$

$$P_{(2,1)} = \frac{8}{7} \left(\frac{1}{2}P_{(1,1)} + \frac{1}{8}P_{(1,2)} \right) = \frac{4}{7}P_{(1,1)} + \frac{1}{7}P_{(1,2)}$$

Plugging in the third equation into this gives

$$P_{(2,1)} = \frac{4}{7} \left(\frac{1}{4} + \frac{1}{4}P_{(1,2)} + \frac{1}{4}P_{(2,1)} \right) + \frac{1}{7}P_{(1,2)}$$

$$P_{(2,1)} = \frac{7}{6} \left(\frac{1}{7} + \frac{2}{7}P_{(1,2)} \right) = \frac{1}{6} + \frac{1}{3}P_{(1,2)} \quad (*)$$

Next, plugging in the second and third equation into the first equation yields

$$P_{(1,2)} = \frac{1}{4} + \frac{1}{2} \left(\frac{1}{4} + \frac{1}{4}P_{(1,2)} + \frac{1}{4}P_{(2,1)} \right) + \frac{1}{4} \left(\frac{1}{2}P_{(1,2)} + \frac{1}{2}P_{(2,1)} \right)$$

$$P_{(1,2)} = \frac{3}{8} + \frac{1}{4}P_{(1,2)} + \frac{1}{4}P_{(2,1)}$$

Now plugging in $(*)$ into this, we get

$$P_{(1,2)} = \frac{3}{8} + \frac{1}{4}P_{(1,2)} + \frac{1}{4} \left(\frac{1}{6} + \frac{1}{3}P_{(1,2)} \right)$$

$$P_{(1,2)} = \frac{3}{2} \cdot \frac{5}{12} = \boxed{\text{(B)} \frac{5}{8}}$$

-mathisawesome2169

Note: We can also think of each state $P_{(x,y)}$ as the probability the frog will reach a vertical side given that it is x away from a vertical side and y away from a horizontal side. This is just another way to think about the reflective symmetry.

-GardenRock10

Solution 5 (States)

this is basically another version of solution 4; shoutout to mathisawesome2169 :D

First, we note the different places the frog can go at certain locations in the square:

If the frog is at a border vertical point $((1, 2), (3, 2))$, it moves with probability $\frac{1}{4}$ to a vertical side of the square, probability $\frac{1}{4}$ to the center of the square, and probability $\frac{1}{2}$ to a corner square.

If the frog is at a border horizontal point $((2, 1), (2, 3))$, it moves with probability $\frac{1}{4}$ to a horizontal side of the square, probability $\frac{1}{4}$ to the center of the square, and probability $\frac{1}{2}$ to a corner square.

If the frog is at a center square $((2, 2))$, it moves with probability $\frac{1}{2}$ to a border horizontal point and probability $\frac{1}{2}$ to a border vertical point.

If the frog is at a corner $((1, 1), (1, 3), (3, 3), (3, 1))$, it moves with probability $\frac{1}{4}$ to a vertical side of the square, probability $\frac{1}{4}$ to a horizontal side, probability $\frac{1}{4}$ to a border horizontal point, and probability $\frac{1}{4}$ to a border vertical point.

Now, let x denote the probability of the frog reaching a vertical side when it is at a border vertical point. Similarly, let y denote the probability of the frog reaching a vertical side when it is at a border horizontal point. Now, the probability of the frog reaching a vertical side of the square at any location inside the square can be expressed in terms of x and y .

First, the two easier ones: $P_{center} = \frac{1}{2}x + \frac{1}{2}y$, and $P_{corner} = \frac{1}{4} + \frac{1}{4}x + \frac{1}{4}y$. Now, we can write x and y in terms of x and y , allowing us to solve a system of two variables:

$$x = \frac{1}{4} + \frac{1}{4}P_{center} + \frac{1}{2}P_{corner} = \frac{1}{4} + \frac{1}{4}\left(\frac{1}{2}x + \frac{1}{2}y\right) + \frac{1}{2}\left(\frac{1}{4} + \frac{1}{4}x + \frac{1}{4}y\right)$$

and

$$y = \frac{1}{4}P_{center} + \frac{1}{2}P_{corner} = \frac{1}{4}\left(\frac{1}{2}x + \frac{1}{2}y\right) + \frac{1}{2}\left(\frac{1}{4} + \frac{1}{4}x + \frac{1}{4}y\right).$$

From these two equations, it is apparent that $y = x - \frac{1}{4}$. We can then substitute this value for y back into any of the two equations above to get

$$x = \frac{1}{4} + \frac{1}{4}\left(\frac{1}{2}x + \frac{1}{2}\left(x - \frac{1}{4}\right)\right) + \frac{1}{2}\left(\frac{1}{4} + \frac{1}{4}x + \frac{1}{4}\left(x - \frac{1}{4}\right)\right).$$

Although this certainly looks intimidating, we can expand the parentheses and multiply both sides by 16 to eliminate the fractions, which upon simplification yields the equation

$$16x = 5 + 8x,$$

giving us the desired probability $x = \frac{5}{8}$. The answer is then $\boxed{\text{(B)} \frac{5}{8}}$.

- curiousmind888 TGSN

14. (AMC 10 10A 2020) Real numbers x and y satisfy $x + y = 4$ and $x \cdot y = -2$. What is the value of

$$x + \frac{x^3}{y^2} + \frac{y^3}{x^2} + y?$$

(A) 360 (B) 400 (C) 420 (D) 440 (E) 480

Answer: D

Solution 1

$$x + \frac{x^3}{y^2} + \frac{y^3}{x^2} + y = x + \frac{x^3}{y^2} + y + \frac{y^3}{x^2} = \frac{x^3}{x^2} + \frac{y^3}{x^2} + \frac{y^3}{y^2} + \frac{x^3}{y^2}$$

Continuing to combine

$$\frac{x^3 + y^3}{x^2} + \frac{x^3 + y^3}{y^2} = \frac{(x^2 + y^2)(x^3 + y^3)}{x^2 y^2} = \frac{(x^2 + y^2)(x + y)(x^2 - xy + y^2)}{x^2 y^2}$$

From the givens, it can be concluded that $x^2 y^2 = 4$. Also,

$$(x + y)^2 = x^2 + 2xy + y^2 = 16$$

This means that $x^2 + y^2 = 20$. Substituting this information into $\frac{(x^2 + y^2)(x + y)(x^2 - xy + y^2)}{x^2 y^2}$, we have $\frac{(20)(4)(22)}{4} = 20 \cdot 22 = \boxed{\text{(D) 440}}$. PCCchess

Solution 2

As above, we need to calculate $\frac{(x^2 + y^2)(x^3 + y^3)}{x^2 y^2}$. Note that x, y , are the roots of $x^2 - 4x - 2$ and so $x^3 = 4x^2 + 2x$ and $y^3 = 4y^2 + 2y$. Thus $x^3 + y^3 = 4(x^2 + y^2) + 2(x + y) = 4(20) + 2(4) = 88$ where $x^2 + y^2 = 20$ and $x^2 y^2 = 4$ as in the previous solution. Thus the answer is $\frac{(20)(88)}{4} = \boxed{\text{(D) 440}}$. Note $(x^2 + y^2 = (x + y)^2 - 2xy = 20$, and $x^2 y^2 = (xy)^2 = 4$)

Solution 3

Note that $(x^3 + y^3)(\frac{1}{y^2} + \frac{1}{x^2}) = x + \frac{x^3}{y^2} + \frac{y^3}{x^2} + y$. Now, we only need to find the values of $x^3 + y^3$ and $\frac{1}{y^2} + \frac{1}{x^2}$.

Recall that $x^3 + y^3 = (x + y)(x^2 - xy + y^2)$, and that $x^2 - xy + y^2 = (x + y)^2 - 3xy$. We are able to solve the second equation, and doing so gets us $4^2 - 3(-2) = 22$. Plugging this into the first equation, we get $x^3 + y^3 = 4(22) = 88$.

In order to find the value of $\frac{1}{y^2} + \frac{1}{x^2}$, we find a common denominator so that we can add them together. This gets us $\frac{x^2}{x^2 y^2} + \frac{y^2}{x^2 y^2} = \frac{x^2 + y^2}{(xy)^2}$. Recalling that $x^2 + y^2 = (x + y)^2 - 2xy$ and solving this equation, we get $4^2 - 2(-2) = 20$. Plugging this into the first equation, we get $\frac{1}{y^2} + \frac{1}{x^2} = \frac{20}{(-2)^2} = 5$.

Solving the original equation, we get $x + \frac{x^3}{y^2} + \frac{y^3}{x^2} + y = (88)(5) = \boxed{\text{(D) 440}}$.

[emerald_block](#)

Solution 3.5 (Binomial Theorem and Sum of Cubes, not that messy)

The expression can be rewritten as:

$$x + y + \frac{x^3}{y^2} + \frac{y^3}{x^2} = \frac{x^5 + y^5}{(xy)^2} + (x + y)$$

Plug in what we know:

$$\frac{x^5 + y^5}{4} + 4$$

Use Binomial Theorem to expand and simplify the numerator of the fraction:

$$\begin{aligned} x^5 + y^5 &= (x + y)^5 - 5(x^4y + xy^4 + 2x^3y^2 + 2x^2y^3) \\ &= (x + y)^5 - 5xy((x + y)^3 - xy(x + y)) \\ &= 4^5 + 10(4^3 + 8). \end{aligned}$$

Plug this back into the expression to finally get:

$$\frac{4^5 + 10(4^3 + 8)}{4} + 4 = 4^4 + 10(4^2 + 2) + 4 = 256 + 180 + 4 = \boxed{440}.$$

Solution 4 (Bashing)

This is basically bashing using Vieta's formulas to find x and y (which I highly do not recommend, I only wrote this solution for fun).

We use Vieta's to find a quadratic relating x and y . We set x and y to be the roots of the quadratic $Q(n) = n^2 - 4n - 2$ (because $x + y = 4$, and $xy = -2$). We can solve the quadratic to get the roots $2 + \sqrt{6}$ and $2 - \sqrt{6}$. x and y are "interchangeable" (WLOG, without loss of generality, we could have x equal one and y equal the other), meaning that it doesn't matter which solution x or y is, because it'll return the same result when plugged in. So we plug in $2 + \sqrt{6}$ for x and $2 - \sqrt{6}$ and get $\boxed{\text{(D) 440}}$ as our answer.

Solution 5 (Bashing Part 2)

This usually wouldn't work for most problems like this, but we're lucky that we can quickly expand and factor this expression in this question.

We first change the original expression to $4 + \frac{x^5 + y^5}{x^2y^2}$, because $x + y = 4$. This is equal to $4 + \frac{(x+y)(x^4 - x^3y + x^2y^2 - xy^3 + y^4)}{x^2y^2} = x^4 + y^4 - x^3y - xy^3 + 8$. We can factor and reduce $x^4 + y^4$ to $(x^2 + y^2)^2 - 2x^2y^2 = ((x + y)^2 - 2xy)^2 - 8 = 400 - 8 = 392$. Now our expression is just $400 - (x^3y + xy^3)$. We factor $x^3y + xy^3$ to get $(xy)(x^2 + y^2) = -40$. So the answer would be $400 - (-40) = \boxed{\text{(D)440}}$.

Solution 6 (Complete Binomial Theorem)

We first simplify the expression to

$$x + y + \frac{x^5 + y^5}{x^2y^2}.$$

Then, we can solve for x and y given the system of equations in the problem. Since $xy = -2$, we can substitute $\frac{-2}{x}$ for y . Thus, this becomes the equation

$$x - \frac{2}{x} = 4.$$

Multiplying both sides by x , we obtain $x^2 - 2 = 4x$, or

$$x^2 - 4x - 2 = 0.$$

By the quadratic formula we obtain $x = 2 \pm \sqrt{6}$. We also easily find that given $x = 2 \pm \sqrt{6}$, y equals the conjugate of x . Thus, plugging our values in for x and y , our expression equals

$$4 + \frac{(2 - \sqrt{6})^5 + (2 + \sqrt{6})^5}{(2 - \sqrt{6})^2(2 + \sqrt{6})^2}$$

By the binomial theorem, we observe that every second terms of the expansions x^5 and y^5 will cancel out (since a positive plus a negative of the same absolute value equals zero). We also observe that the other terms not canceling out are doubled when summing the expansions of $x^5 + y^5$. Thus, our expression equals

$$4 + \frac{2(2^5 + \binom{5}{2}2^3 \times 6 + \binom{5}{4}2 \times 36)}{(2 - \sqrt{6})^2(2 + \sqrt{6})^2}.$$

which equals

$$4 + \frac{2(872)}{4}$$

which equals (D)440.

Solution 7

As before, simplify the expression to

$$x + y + \frac{x^5 + y^5}{x^2y^2}.$$

Since $x + y = 4$ and $x^2y^2 = 4$, we substitute that in to obtain

$$4 + \frac{x^5 + y^5}{4}.$$

Now, we must solve for $x^5 + y^5$. Start by squaring $x + y$, to obtain

$$x^2 + 2xy + y^2 = 16$$

Simplifying, $x^2 + y^2 = 20$. Squaring once more, we obtain

$$x^4 + y^4 + 2x^2y^2 = 400$$

Once again simplifying, $x^4 + y^4 = 392$. Now, to obtain the fifth powers of x and y , we multiply both sides by $x + y$. We now have

$$x^5 + x^4y + xy^4 + y^5 = 1568$$

, or

$$x^5 + y^5 + xy(x^3 + y^3) = 1568$$

We now solve for $x^3 + y^3$. $(x + y)^3 = x^3 + y^3 + 3xy(x + y) = 64$, so $x^3 + y^3 = 88$. Plugging this back into $x^5 + x^4y + xy^4 + y^5 = 1568$, we find that $x^5 + y^5 = 1744$, so we have

$$4 + \frac{1744}{4}.$$

. This equals 440, so our answer is (D)440.

Solution 8

We can use Newton Sums to solve this problem. We start by noticing that we can rewrite the equation as $\frac{x^3}{y^2} + \frac{y^3}{x^2} + x + y$. Then, we know that $x + y = 4$, so we have $\frac{x^3}{y^2} + \frac{y^3}{x^2} + 4$. We can use the equation $x \cdot y = -2$ to write $x = \frac{-2}{y}$ and $y = \frac{-2}{x}$. Next, we can plug in these values of x and y to get $\frac{x^3}{y^2} + \frac{y^3}{x^2} = \frac{x^5}{4} + \frac{y^5}{4}$, which is the same as

$$\frac{x^3}{y^2} + \frac{y^3}{x^2} = \frac{x^5 + y^5}{4}.$$

Then, we use Newton sums where S_n is the elementary symmetric sum of the sequence and P_n is the power sum ($x^n + y^n$). Using this, we can make the following Newton sums:

$$P_1 = S_1$$

$$P_2 = P_1 S_1 - 2S_2$$

$$P_3 = P_2 S_1 - P_1 S_2$$

$$P_4 = P_3 S_1 - P_2 S_2$$

$$P_5 = P_4 S_1 - P_3 S_2.$$

We also know that S_1 is 4 because $x + y$ is four, and we know that S_2 is -2 because $x \cdot y$ is -2 as well. Then, we can plug in values! We have

$$P_1 = S_1 = 4$$

$$P_2 = P_1 S_1 - 2S_2 = 16 - (-4) = 20$$

$$P_3 = P_2 S_1 - P_1 S_2 = 80 - (-8) = 88$$

$$P_4 = P_3 S_1 - P_2 S_2 = 88 \cdot 4 - (-40) = 392$$

$$P_5 = P_4 S_1 - P_3 S_2 = 392 \cdot 4 - (-2) \cdot 88 = 1744.$$

We earlier noted that $\frac{x^3}{y^2} + \frac{y^3}{x^2} = \frac{x^5 + y^5}{4}$, so we have that this equals $\frac{1744}{4}$, or 436. Then, plugging this back into the original equation, this is $436 + 4$ or 440, so our answer is (D) 440.

Solution 9

As in the first solution, we get the expression to be $\frac{x^3 + y^3}{x^2} + \frac{x^3 + y^3}{y^2}$.

Then, since the numerators are the same, we can put the two fractions as a common denominator and multiply the numerator by $x^2 y^2$. This gets us $\frac{(x^2 + y^2)(x^3 + y^3)}{x^2 y^2}$.

Now, since we know $x + y = 4$ and $xy = -2$, instead of solving for x and y , we will try to manipulate the above expression them into a manner that we can substitute the sum and product that we know. Also, another form of $x^3 + y^3$ is $(x + y)(x^2 - xy + y^2)$.

Thus, we can convert the current expression to $\frac{(x^2 + y^2)(x + y)(x^2 - xy + y^2)}{x^2 y^2}$.

Doing some algebraic multiplications, we get $\frac{((x + y)^2 - 2xy)(x + y)((x + y)^2 - 3xy)}{(xy)^2}$.

Since we know $x + y = 4$ and $xy = -2$, we have $\frac{(16 - (-4))(4)(16 - (-6))}{4} = \frac{20 \cdot 4 \cdot 22}{4} = 20 \cdot 22 = 440$.

Therefore the answer is (D) 440.

Solution 10 (Algebra Bash)

We give all of the terms in this expression a common denominator. $x + \frac{x^3}{y^2} + \frac{y^3}{x^2} + y = \frac{x^3y^2 + x^5 + y^5 + x^2y^3}{x^2y^2}$. We can find $x^2y^2 = (xy)^2 = (-2)^2 = 4$ and we can find $x^3y^2 + x^2y^3 = x^2y^2(x + y) = 16$. Our expression $\frac{xy^2 + x^5 + y^5 + x^2y}{x^2y^2}$ is now $\frac{16 + x^5 + y^5}{4}$. Now all we need to find is $x^5 + y^5$. Using the binomial theorem, $(x + y)^5 = x^5 + 5x^4y + 10x^3y^2 + 10x^2y^3 + 5xy^4 + y^5 = 4^5 = 1024$. The extra terms that we don't need is $5x^4y + 10x^3y^2 + 10x^2y^3 + 5xy^4 = 5xy(x^3 + y^3) + 10x^2y^2(x + y) = -10(x^3 + y^3) + 160$. What's $x^3 + y^3$? We use the same method. Using the binomial theorem, $(x + y)^3 = x^3 + 3x^2y + 3xy^2 + y^3 = x^3 + y^3 + 3xy(x + y) = x^3 + y^3 - 24 = 4^3 = 64$. Now we know that $x^3 + y^3 = 64 + 24 = 88$, and plugging that into $-10(x^3 + y^3) + 160$ gives $-10(88) + 160 = -880 + 160 = -720$. Now we see that those extra terms have a sum of -720 . Thus $(x + y)^5 = x^5 + y^5 - 720 = 1024$ so $x^5 + y^5 = 1024 + 720 = 1744$. Remember our goal: we want to find $\frac{-8 + x^5 + y^5}{4}$. Using $x^5 + y^5 = 1744$, $\frac{16 + x^5 + y^5}{4} = \frac{16 + 1744}{4} = \frac{1760}{4} = \boxed{\text{(D) 440}}$.

Solution 11

Since we know $x + y = 4$, we can simplify to $4 + \frac{x^3}{y^2} + \frac{y^3}{x^2}$. Then, when you add the fractions, you get $\frac{x^5 + y^5}{x^2y^2} = \frac{x^5 + y^5}{4}$. To find $x^5 + y^5$, we expand $(x + y)^5 = 4^5$ and then simplify. $x^5 + y^5 + 5x^4y + 5xy^4 + 10x^3y^2 + 10x^2y^3 = 1024$. We can use the fact that $xy = -2$ to simplify to $x^5 + y^5 - 10x^3 - 10y^3 + 40x + 40y = 1024$. We can factor to get $x^5 + y^5 - 10(x^3 + y^3) + 40(x + y)$. To find $x^3 + y^3$, we simplify $(x + y)^3 = 4^3$. $x^3 + y^3 + 3x^2y + 3y^2x = 64$. We can use the same process as we did before and get $x^3 + y^3 - 6(x + y) = 64$ (using the fact that $xy = -2$). Therefore, $x^3 + y^3 = 88$. Now, plugging it back in to our other expansion, we get $x^5 + y^5 + 160 - 880 = 1024$. $x^5 + y^5 = 1744$. Now to get the final answer, all we need to do is to plug it back into the original equation and get $\frac{1744}{4} + 4 = \boxed{\text{(D) 440}}$.

Solution 12

We have $x + y = 4$ and $xy = -2$. We solve for x to get either $2 + \sqrt{6}$ or $2 - \sqrt{6}$, and WLOG, we say $x = 2 + \sqrt{6}$ and $y = 2 - \sqrt{6}$. $\sqrt{6}$ is about 2.44, We then substitute $\sqrt{6} = 2.44$ for x and y , yielding $x = 4.44$ and $y = -0.44$. We now have $x + \frac{x^3}{y^2} + \frac{y^3}{x^2} + y$. $x + y = 4$, so we now have $4 + \frac{x^3}{y^2} + \frac{y^3}{x^2}$. Substituting the respective values of x and y yields $4 + 456 + \text{a negative number}$, and we get $460 - n$, where n is a positive number. We look at our answer choices and see that $460 - n$ lies between $\boxed{\text{(D) 440}}$ and $\boxed{\text{(E) 480}}$, but because we have a negative number, $\boxed{\text{(E) 480}}$ is not possible. So our answer is $\boxed{\text{(D) 440}}$, and we're done.

(Note from Pinotation: Not the most ideal solution, but it works).

15. (AMC 10 10A 2020) A positive integer divisor of $12!$ is chosen at random. The probability that the divisor chosen is a perfect square can be expressed as $\frac{m}{n}$, where m and n are relatively prime positive integers. What is $m + n$?

(A) 3 (B) 5 (C) 12 (D) 18 (E) 23

Answer: E

Solution

The prime factorization of $12!$ is $2^{10} \cdot 3^5 \cdot 5^2 \cdot 7 \cdot 11$. This yields a total of $11 \cdot 6 \cdot 3 \cdot 2 \cdot 2$ divisors of $12!$. In order to produce a perfect square divisor, there must be an even exponent for each number in the prime factorization. Note that the divisor can't have any factors of 7 and 11 in the prime factorization because there is only one of

each in $12!$. Thus, there are $6 \cdot 3 \cdot 2$ perfect squares. (For 2, you can have 0, 2, 4, 6, 8, or 10 2s, etc.) The probability that the divisor chosen is a perfect square is

$$\frac{6 \cdot 3 \cdot 2}{11 \cdot 6 \cdot 3 \cdot 2 \cdot 2} = \frac{1}{22} \implies \frac{m}{n} = \frac{1}{22} \implies m + n = 1 + 22 = \boxed{\text{(E)} 23}$$

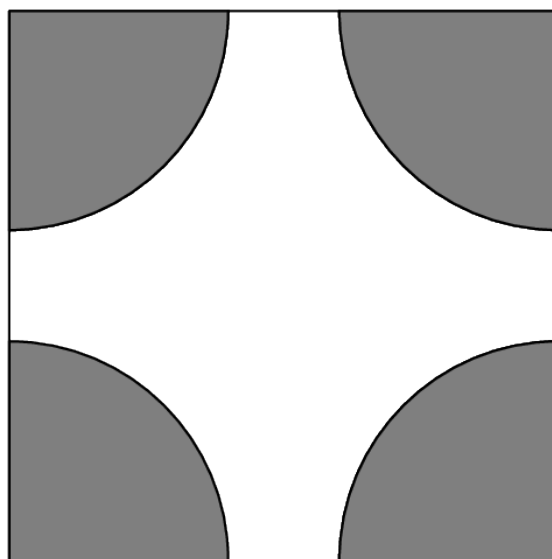
16. (AMC 10 10A 2020) A point is chosen at random within the square in the coordinate plane whose vertices are $(0, 0)$, $(2020, 0)$, $(2020, 2020)$, and $(0, 2020)$. The probability that the point is within d units of a lattice point is $\frac{1}{2}$. (A point (x, y) is a lattice point if x and y are both integers.) What is d to the nearest tenth?

(A) 0.3 (B) 0.4 (C) 0.5 (D) 0.6 (E) 0.7

Answer: B

Solutions

Diagram



The diagram represents each unit square of the given 2020×2020 square.

Solution 1

We consider an individual one-by-one block.

If we draw a quarter of a circle from each corner (where the lattice points are located), each with radius d , the area covered by the circles should be 0.5. Because of this, and the fact that there are four circles, we write

$$4 \cdot \frac{1}{4} \cdot \pi d^2 = \frac{1}{2}$$

Solving for d , we obtain $d = \frac{1}{\sqrt{2\pi}}$, where with $\pi \approx 3$, we get $d \approx \frac{1}{\sqrt{6}} \approx \frac{1}{2.5} = \frac{10}{25} = \frac{2}{5}$, and from here, we see that $d \approx 0.4 \implies \boxed{\text{(B)} 0.4}$. **Note:** To be more rigorous, note that $d < 0.5$ since if $d \geq 0.5$ then clearly the probability is greater than $\frac{1}{2}$. This would make sure the above solution works, as if $d \geq 0.5$ there is overlap with the quartercircles. - **Emathmaster**

Solution 2

As in the previous solution, we obtain the equation $4 \cdot \frac{1}{4} \cdot \pi d^2 = \frac{1}{2}$, which simplifies to $\pi d^2 = \frac{1}{2} = 0.5$. Since π is slightly more than 3, d^2 is slightly less than $\frac{0.5}{3} = 0.1\bar{6}$. We notice that $0.1\bar{6}$ is slightly more than $0.4^2 = 0.16$, so d is roughly (B) 0.4.

emerald_block

Solution 3 (Estimating)

As above, we find that we need to estimate $d = \frac{1}{\sqrt{2\pi}}$.

Note that we can approximate $2\pi \approx 6.28318 \approx 6.25$ and so $\frac{1}{\sqrt{2\pi}} \approx \frac{1}{\sqrt{6.25}} = \frac{1}{2.5} = 0.4$.

And so our answer is (B) 0.4.

Solution 4 (Estimating but a bit different)

We only need to figure out the probability for a unit square, as it will scale up to the 2020×2020 square. Since we want to find the probability that a point inside a unit square that is d units away from a lattice point (a corner of the square) is $\frac{1}{2}$, we can find which answer will come the closest to covering $\frac{1}{2}$ of the area.

Since the closest is 0.4 which turns out to be $(0.4)^2 \times \pi = 0.16 \times \pi$ which is about 0.502, we find that the answer rounded to the nearest tenth is 0.4 or (B).

Solution 5 (Estimating but differently again)

As per the above diagram, realize that $\pi d^2 = \frac{1}{2}$, so $d = \frac{1}{(\sqrt{2})(\sqrt{\pi})}$.

$$\sqrt{2} \approx 1.4 = \frac{7}{5}.$$

$\sqrt{\pi}$ is between 1.7 and 1.8 ($(1.7)^2 = 2.89$ and $(1.8)^2 = 3.24$), so we can say $\sqrt{\pi} \approx 1.75 = \frac{7}{4}$.

So $d \approx \frac{1}{(\frac{7}{5})(\frac{7}{4})} = \frac{1}{\frac{49}{20}} = \frac{20}{49}$. This is slightly above (B) 0.4, since $\frac{20}{49} \approx \frac{2}{5}$.

-Solution by JoeYa

Solution 6 (Estimating but differently again, again)

As above, we have the equation $\pi d^2 = \frac{1}{2}$, and we want to find the most accurate value of d . We resort to the answer choices and can plug those values of d in and see which value of d will lead to the most accurate value of π .

Starting off in the middle, we try option C with $d = 0.5$. Plugging this in, we get $\pi \left(\frac{1}{2}\right)^2 = \frac{1}{2}$, and after simplifying we get $\pi = \frac{1}{2} \cdot 4 = 2$. That's not very good. We know $\pi \approx 3.14$.

Let's see if we can do better. Trying option A with $d = 0.3$, we get $\pi = \frac{1}{2} \cdot \frac{100}{9} = \frac{50}{9} = 5\frac{5}{9}$.

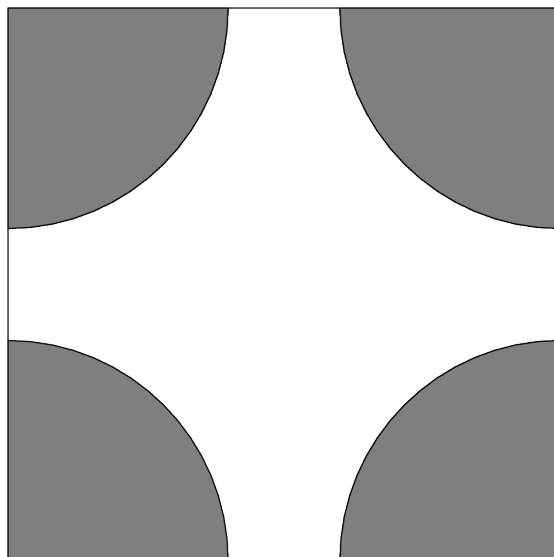
Hm, let's try option B with $d = 0.4$. We get $\pi = \frac{1}{2} \cdot \frac{25}{4} = \frac{25}{8} = 3\frac{1}{8}$. This is very close to π and is the best estimate for π of the 5 options.

Therefore, the answer is (B) 0.4. epiconan

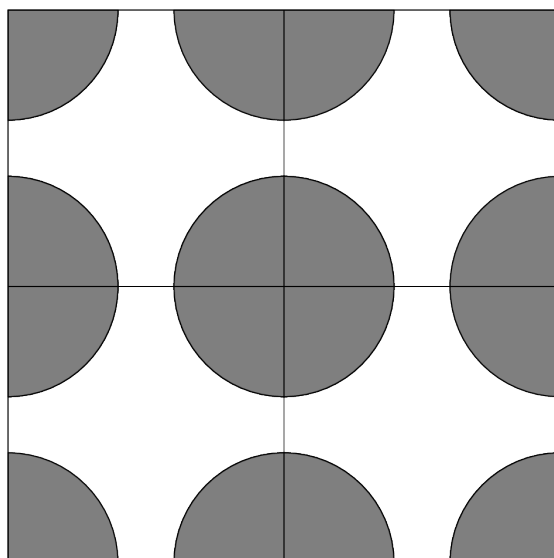
Solution 7 (Sol. 1, but rigorous (and excessive))

PLEASE NOTE: Solution 1 IS rigorous. Say there are k unit squares (it doesn't matter how many). There is a $\frac{1}{k}$ probability the point is in some unit square. There is a πd^2 probability the point is in the shaded region. So, there is a $\frac{1}{k} \pi d^2 \cdot k = \pi d^2$ probability the point is in any shaded region (since there are k unit squares).

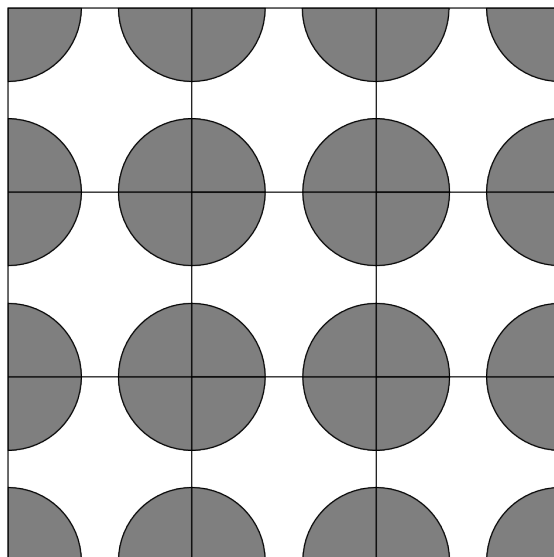
Let n be the side length of a square. When $n = 1$, the shaded areas represent half of the total area:



When $n = 2$:



For $n = 3$:



We can calculate the total number of shaded circles given some n . There are $(n-1)^2$ full circles on the inside, $4(n-1)$ semicircles on the sides, and 4 quarter circles for the corners.

Full circles are, of course, worth one circle. Semicircles are worth half a circle each, and quarter circles are worth $\frac{1}{4}$ of a circle. Thus, weighing our sum gives

$$(n-1)^2 + \frac{4(n-1)}{2} + \frac{4}{4} = n^2 - 2n + 1 + 2(n-1) + 1 = n^2 - 2n + 2n + 2 - 2 = n^2.$$

Thus, there is $n^2 \cdot \pi r^2$ worth of the shaded area for any n , and since the area of each circle is πr^2 if r is the radius of each.

We want the ratio of this shaded area to the entire to be $\frac{1}{2}$. The area of the entire square is n^2 , so dividing, we see that $\frac{n^2 \cdot \pi r^2}{n^2} = \pi r^2 = \frac{1}{2}$.

The rest is the same as solution 1.

Solution 8 (most simple, easy, estimation)

First, let's test what happens if $d = 0.5$. If $d = 0.5$, the area of the circle around each lattice point is $\pi(0.5)^2 = 0.25\pi$. Since each lattice point corresponds to a 1×1 square of area, the probability of the point falling within 0.5 units is exactly $\frac{0.25\pi}{1} \approx 0.785$. Because this strictly exceeds our target probability of $\frac{1}{2}$, we know that $d < 0.5$. This conveniently eliminates 0.5, 0.6, and 0.7, and proves that the circles centered at the lattice points do not overlap.

Since the circles do not overlap, we can simply find the combined favorable area of the circles (and partial circles) inside the 2020×2020 square.

We can break this down into three types of lattice points: There are 2019^2 interior lattice points, each contributing a full circle of area πd^2 . There are $4 \cdot 2019$ edge lattice points (excluding corners), each contributing a semicircle of area $\frac{1}{2}\pi d^2$. There are 4 corner lattice points, each contributing a quarter-circle of area $\frac{1}{4}\pi d^2$.

Summing these gives the total favorable area: $2019^2 \pi d^2 + 4(2019) \left(\frac{1}{2} \pi d^2 \right) + 4 \left(\frac{1}{4} \pi d^2 \right)$

Factoring out the πd^2 , we get a perfect square: $(2019^2 + 2(2019) + 1)\pi d^2 = (2019 + 1)^2 \pi d^2 = 2020^2 \pi d^2$.

The total area of the bounding 2020×2020 square is 2020^2 . Therefore, the probability is the ratio of the favorable area to the total area: $\frac{2020^2 \pi d^2}{2020^2} = \pi d^2$.

We are given that the probability is $\frac{1}{2}$, so we equate and solve: $\pi d^2 = \frac{1}{2} \implies d^2 = \frac{1}{2\pi}$.

Solving for d gives $d = \sqrt{\frac{1}{2\pi}} \approx \sqrt{0.159}$. Since $0.4^2 = 0.16$, we can clearly see that $d \approx 0.4$.

by chundoowan

17. (AMC 10 10A 2020) Define

$$P(x) = (x - 1^2)(x - 2^2) \cdots (x - 100^2).$$

How many integers n are there such that $P(n) \leq 0$?

- (A) 4900 (B) 4950 (C) 5000 (D) 5050 (E) 5100

Answer: E

Solution 1 (Casework)

We perform casework on $P(n) \leq 0$:

- a) $P(n) = 0$ In this case, there are 100 such integers n :

$$1^2, 2^2, 3^2, \dots, 100^2.$$

- b) $P(n) < 0$ There are 100 factors in $P(x)$, and we need an odd number of them to be negative. We construct the table below:

Interval of x	# of Negative Factors	Valid?
$(-\infty, 1^2)$	100	
$(1^2, 2^2)$	99	✓
$(2^2, 3^2)$	98	
$(3^2, 4^2)$	97	✓
$(4^2, 5^2)$	96	
$(5^2, 6^2)$	95	✓
$(6^2, 7^2)$	94	
$\vdots$	$\vdots$	$\vdots$
$(99^2, 100^2)$	1	✓
$(100^2, \infty)$	0	

Note that there are 50 valid intervals of x . We count the integers in these intervals:

$$(2^2 - 1^2 - 1) + (4^2 - 3^2 - 1) + (6^2 - 5^2 - 1) + \cdots + (100^2 - 99^2 - 1) = \underbrace{(2^2 - 1^2)}_{(2+1)(2-1)} + \underbrace{(4^2 - 3^2)}_{(4+3)(4-3)} + \cdots + \underbrace{(100^2 - 99^2)}_{(100+99)(100-99)}$$

$$\begin{aligned}
 &= \underbrace{(2+1) + (4+3) + (6+2) + \dots + (100+1)}_{1+2+3+4+\dots+100} \\
 &= \frac{101(100)}{2} - 50 \\
 &= 5000.
 \end{aligned}$$

In this case, there are 5000 such integers n .

Together, the answer is $100 + 5000 = \boxed{\text{(E)} 5100}$.

Solution 2 (Casework)

Notice that $P(x)$ is nonpositive when x is between 100^2 and 99^2 , 98^2 and 97^2 , $\dots$, 2^2 and 1^2 (inclusive), because there are an odd number of negatives, which means that the number of values equals

$$((100+99)(100-99)+1) + ((98+97)(98-97)+1) + \dots + ((2+1)(2-1)+1).$$

This reduces to

$$200 + 196 + 192 + \dots + 4 = 4(1 + 2 + \dots + 50) = 4 \cdot \frac{50 \cdot 51}{2} = \boxed{\text{(E)} 5100}.$$

Zeric

Solution 3 (End Behavior)

We know that $P(x)$ is a 100-degree function with a positive leading coefficient. That is, $P(x) = x^{100} + ax^{99} + bx^{98} + \dots + (\text{constant})$.

Since the degree of $P(x)$ is even, its end behaviors match. And since the leading coefficient is positive, we know that both ends approach ∞ as x goes in either direction, from which

$$\lim_{x \rightarrow -\infty} P(x) = \lim_{x \rightarrow \infty} P(x) = \infty.$$

So the first time $P(x)$ is going to be negative is when it intersects the x -axis at an x -intercept and it's going to dip below. This happens at 1^2 , which is the smallest intercept.

However, when it hits the next intercept, it's going to go back up again into positive territory, we know this happens at 2^2 . And when it hits 3^2 , it's going to dip back into negative territory. Clearly, this is going to continue to snake around the intercepts until 100^2 .

To get the amount of integers below and/or on the x -axis, we simply need to count the integers. For example, the amount of integers in between the $[1^2, 2^2]$ interval we got earlier, we subtract and add one. $(2^2 - 1^2 + 1) = 4$ integers, so there are four integers in this interval that produce a negative result.

Doing this with all of the other intervals, we have

$$(2^2 - 1^2 + 1) + (4^2 - 3^2 + 1) + \dots + (100^2 - 99^2 + 1) = \boxed{\text{(E)} 5100}$$

from Solution 2's result.

Solution 4 (Fast)

We know $P(x) \leq 0$ when an odd number of its factors are positive and negative. For example, to make the first factor positive, $x \in [1^2, 2^2]$. then there will be a even

number of positive factors. We would do $2^2 - 1^2 + 1$ (inclusive) to find all integers that work. In short we can generalize too:

$$\begin{aligned}x^2 - (x - 1)^2 + 1 &= 2x \\x^2 - (x^2 - 2x + 1) + 1 &= 2x \\x^2 - x^2 + 2x - 1 + 1 &= 2x.\end{aligned}$$

But remember this only works when $x \in \{2, 4, 6, 8 \dots 98, 100\}$ because only then will there be an odd amount of positive and negative factors. So we can set $x = 2k$, for $k \in \{1, 2, 3, 4, \dots 49, 50\}$ Now we only have to solve:

$$\sum_{k=1}^{50} 2(2k) = 2 \sum_{k=1}^{50} 2k = 4 \sum_{k=1}^{50} k = 4 \cdot \frac{(50)(51)}{2} = 2 \cdot (50)(51) = \boxed{\text{(E) } 5100}.$$

Wiselion

Solution 5 (Rigorous, No Bashing)

When looking at the question, it should immediately jump out that we need an odd number of terms to be negative. This means that we need to find the number of n such that $(2k + 1)^2 \leq n \leq (2k + 2)^2$ for some $0 \leq k \leq 49$ such that $k \in \mathbb{Z}/\mathbb{Z}^-$. The number of integers n s.t. $n \in [a, b]$ is $b - a + 1$. We want $n \in [(2k + 1)^2, (2k + 2)^2]$.

Thus, it is obvious we want to find

$$\sum_{k=0}^{49} ((2k + 2)^2 - (2k + 1)^2 + 1)$$

If we do some investigation on the term inside the summation, we find that:

$$(2k + 2)^2 - (2k + 1)^2 + 1 = (4k^2 + 8k + 4) - (4k^2 + 4k + 1) + 1 = 4k + 4$$

thus:

$$\sum_{k=0}^{49} ((2k + 2)^2 - (2k + 1)^2 + 1) = \sum_{k=0}^{49} 4(k + 1) = 4 \sum_{k=0}^{49} (k + 1) = 4 \sum_{k=1}^{50} k$$

The reason the last step works is that adding 1 inside the summation effectively shifts the bounds up by 1, as it turns $k = 0$ into $k = 1$, and so on.

$$4 \sum_{k=1}^{50} k = 4 \frac{(50)(51)}{2} = 2(50)(51) = 100(51) = \boxed{\text{(E) } 5100}$$

Solution 6

Since $P(x)$ starts off at infinity before $x = 1^2$ and goes back to infinity after $x = 100^2$, we can draw a sign chart, and notice that every interval that starts where x is the square of an odd number and ends where x is the square of an even number, $P(x)$ is nonpositive. We notice a pattern - from $x = 1$ to $x = 4$, there are 4 integers. From $x = 9$ to $x = 16$, there are 8 integers. So, the number of integers in each interval where $P(x)$ is negative is part of an arithmetic sequence that starts at 4 and has common difference 4. Since there are 50 such intervals (since every

other interval has $P(x)$ be nonpositive), the final number is 200. So, the sum of this arithmetic series is $\frac{4+200}{2} \cdot 50 = \boxed{\text{(E) } 5100}$.

Solution Visual Aid

Visualization that makes it easier to see solutions:



Note Clearly, there are a finite amount of integers that result in $P(x)$ being nonpositive. This is because there are an even number of factors of $P(x)$. From here, you can read other solutions that count the amount of negative numbers of $P(x)$.

Solution 7 (General)

Let

$$p_k(x) = \prod_{j=1}^{2k} (x - j^2).$$

where k is a positive integer. We see that $P(x) = p_{50}(x)$. Then $\lim_{x \rightarrow \pm\infty} p_k(x) = \infty$ for all $k \in \mathbb{Z}^+$ so you can think of the graph hitting $(1, 0)$ from above then oscillating then hitting $(4k^2, 0)$ from below and shooting back up. With this out of the way, we see that all integers z_k such that $p_k(x) \leq 0$ is formed by $[1, 4] \cup [9, 16] \cup \dots \cup [(2k-1)^2, (2k)^2] \cap \mathbb{Z}$. Since the number of integers between $(2j)^2$ and $(2j-1)^2$ inclusive is $(2j)^2 - (2j-1)^2 + 1 = 4j^2 - (4j^2 - 4j + 1) + 1 = 4j$ we have that $z_k = 4 \sum_{j=1}^k j = 2k(k+1)$. Hence $k = 50 \implies z_{50} = n = \boxed{\text{(E) } 5100}$.

Note: To solve fast I just did small casework and so $z_1 = 4$, $z_2 = 4 + 8 = 12$, $z_3 = 4 + 8 + 12 = 24$ and guessed $n = 4(1 + 2 + \dots + 50) = 5100$. Took me no more than like a minute or two with a crappy graph that looked like some squiggles with two lines going way up attached :D

Solution 8

We can find a pattern for the number of values that satisfy the given. We see that 0 will make the function positive; the integers 1 through 4 will make it negative; the next 4 integers will make it positive, and this pattern continues with a common difference of 4 for each set of integers that satisfies the property we are looking for. The last interval of favorable integers will be in the interval from $(99)^2$ to $(100)^2$; we subtract to find that there are 200 integers in this interval. We now sum the arithmetic sequence to find our answer. $\frac{4+204}{2} \cdot 50$. This simplifies to **(E)** $\boxed{5100}$.

18. (AMC 10 10A 2020) Let (a, b, c, d) be an ordered quadruple of not necessarily distinct integers, each one of them in the set $\{0, 1, 2, 3\}$. For how many such quadruples is it true that $a \cdot d - b \cdot c$ is odd? (For example, $(0, 3, 1, 1)$ is one such quadruple, because $0 \cdot 1 - 3 \cdot 1 = -3$ is odd.)

(A) 48 (B) 64 (C) 96 (D) 128 (E) 192

Answer: C

Solution 1 (Parity)

In order for $a \cdot d - b \cdot c$ to be odd, consider parity. We must have (even)-(odd) or (odd)-(even). There are $2(2+4) = 12$ ways to pick numbers to obtain an even

product. There are $2 \cdot 2 = 4$ ways to obtain an odd product. Therefore, the total amount of ways to make $a \cdot d - b \cdot c$ odd is $2 \cdot (12 \cdot 4) = \boxed{\text{(C) } 96}$.

Solution 2 (Solution 1 but more in-depth)

Consider parity. We need exactly one term to be odd, one term to be even. Because of symmetry, we can set ad to be odd and bc to be even, then multiply by 2. If ad is odd, both a and d must be odd, therefore there are $2 \cdot 2 = 4$ possibilities for ad . Consider bc . Let us say that b is even. Then there are $2 \cdot 4 = 8$ possibilities for bc . However, b can be odd, in which case we have $2 \cdot 2 = 4$ more possibilities for bc . Thus there are 12 ways for us to choose bc and 4 ways for us to choose ad . Therefore, also considering symmetry, we have $2 \cdot 4 \cdot 12 = \boxed{\text{(C) } 96}$ total values of $ad - bc$.

Solution 3 (Complementary Counting)

There are 4 ways to choose any number independently and 2 ways to choose any odd number independently. To get an even products, we count: $P(\text{any number}) \cdot P(\text{any number}) - P(\text{odd}) \cdot P(\text{odd})$, which is $4 \cdot 4 - 2 \cdot 2 = 12$. The number of ways to get an odd product can be counted like so: $P(\text{odd}) \cdot P(\text{odd})$, which is $2 \cdot 2$, or 4. So, for one product to be odd the other to be even: $2 \cdot 4 \cdot 12 = \boxed{\text{(C) } 96}$ (order matters).

Solution 4 (Solution 3 but more in-depth)

We use complementary counting: If the difference is even, then we can subtract those cases. There are a total of $4^4 = 256$ cases.

For an even difference, we have (even)-(even) or (odd-odd).

From Solution 3:

"There are 4 ways to choose any number independently and 2 ways to choose any odd number independently. even products: $(\text{number}) \cdot (\text{number}) - (\text{odd}) \cdot (\text{odd})$: $4 \cdot 4 - 2 \cdot 2 = 12$. odd products: $(\text{odd}) \cdot (\text{odd})$: $2 \cdot 2 = 4$."

With this, we easily calculate $256 - 12^2 - 4^2 = \boxed{\text{(C) } 96}$.

Solution 5 (Casework)

As in solution 1, we must have (even)-(odd) or (odd)-(even). We see that there are two cases, if ad is even and bc is odd and if ad is odd and bc is even. Because of symmetry, we can multiply by two for when ad is odd and bc is even. Let e denote an even number and let o denote an odd number.

If ad is even and bc is odd, there are three cases:

$$(e, o, o, e)$$

$$(e, o, o, o)$$

$$(o, o, o, e)$$

For each of these cases, there are 2^4 ways to choose from the set $(0, 1, 2, 3)$ as there are 2 even's and 2 odd's; because there are three cases, we multiply this by 3. Also, because of there are 2 cases (ad is even and bc is odd and if ad is odd and bc is even), we multiply this by 2. This gives us:

$$2^4 \cdot 3 \cdot 2 = \boxed{\text{(C) } 96}$$

Solution 6

For parity reasons, if $ad - bc$ is to be odd, we must have ad odd and bc even or ad even and bc odd. By symmetry, these cases are identical, so we consider the first one and multiply by two at the end. For ad to be odd, we must have both a and d odd, and there are $2 \cdot 2$ ways to do so. To count the cases where bc is odd, we use PIE. there are $2 \cdot 4 = 8$ ways for b to be odd and $4 \cdot 2 = 8$ ways for c to be odd, and there are $2 \cdot 2 = 4$ ways for both to be odd. Thus, there are $8 + 8 - 4 = 12$ ways for bc to be even. Multiplying out, there are $2 \cdot 4 \cdot 12$ ways to have $ad - bc$ odd for a total of **(C) 96**.

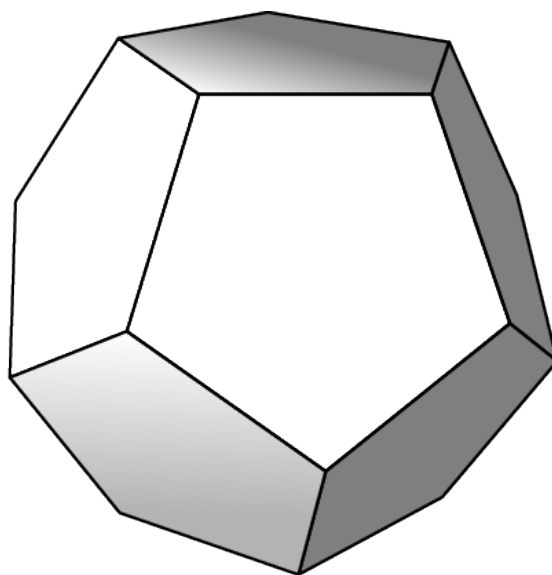
Solution 7(Probability)

Consider the scenario where each number in the quadruple is randomly picked uniformly from the set $\{0, 1, 2, 3\}$. In that case the probability for ad and bc to be odd is $\frac{1}{4}$. Hence, the probability that $ad - bc$ is odd is

$$\frac{1}{4} \cdot \frac{3}{4} + \frac{3}{4} \cdot \frac{1}{4} = \frac{3}{8}$$

Clearly, there are 4^4 total outcomes from the uniform sample space, and thus the amount of ways is $4^4 \cdot \frac{3}{8} = \mathbf{(C) 96}$.

19. (AMC 10 10A 2020) As shown in the figure below, a regular dodecahedron (the polyhedron consisting of 12 congruent regular pentagonal faces) floats in empty space with two horizontal faces. Note that there is a ring of five slanted faces adjacent to the top face, and a ring of five slanted faces adjacent to the bottom face. How many ways are there to move from the top face to the bottom face via a sequence of adjacent faces so that each face is visited at most once and moves are not permitted from the bottom ring to the top ring?



- (A) 125 (B) 250 (C) 405 (D) 640 (E) 810

Answer: E

Solution 1

Since we start at the top face and end at the bottom face without moving from the lower ring to the upper ring or revisiting a face, our journey must consist of the top face, a series of faces in the upper ring, a series of faces in the lower ring, and the bottom face, in that order.

We have 5 choices for which face we visit first on the top ring. From there, we have 9 choices for how far around the top ring we go before moving down: 1, 2, 3, or 4 faces around clockwise, 1, 2, 3, or 4 faces around counterclockwise, or immediately going down to the lower ring without visiting any other faces in the upper ring.

We then have 2 choices for which lower ring face to visit first (since every upper-ring face is adjacent to exactly 2 lower-ring faces) and then once again 9 choices for how to travel around the lower ring. We then proceed to the bottom face, completing the trip.

Multiplying together all the numbers of choices we have, we get $5 \cdot 9 \cdot 2 \cdot 9 = \boxed{(E) \ 810}$.

Solution 2 (Similar to Solution 1)

From the top, we can go down in five different ways to the five faces underneath the first face. From here we can go down or go to the adjacent faces. From the face you went down from the top face, you can either go clockwise or counterclockwise 1, 2, 3, or 4 times, or you can go straight down. Then from there, you go down into the lower row, which you have two choices, left or right down. From here we have $5 \cdot 9 \cdot 2$ ways multiplied by the ways you can move from the bottom ring to the bottom face, but we don't need to know that since from here we can see that $\boxed{(E)}$ is the only answer choice divisible by 90, so our answer is $\boxed{(E)}$. Terribleteeth

Solution 3

There are five ways to get to the top ring. Casework:

Case 1: directly go to bottom ring For each of the 5 initial top ring faces, we have two ways of directly going to the bottom ring, as each face on the top is adjacent to two faces on the bottom. There are $5 \cdot 2 = 10$ paths here.

Case 2: traversing sideways After entering the top ring, we can also choose to travel sideways along the top ring before going to the bottom ring. For each of the 5 initial top ring faces, we can choose 2 directions to go (clockwise or counterclockwise). Once we choose a direction, we can go 1, 2, 3, or 4 faces in that direction. Notice we cannot travel 5 faces because then we would land on the original top ring face, but the problem specifies that we can only travel each face once. Once again, top ring faces are each adjacent to two bottom ring faces, so after traversing sideways we can choose 2 paths to exit. There are $5 \cdot 2 \cdot 4 \cdot 2 = 80$ paths here.

Adding the two cases together, there are 90 ways to get to the bottom ring from the top.

Next, from the bottom ring to the bottom we do casework again:

Case 1: directly go to the bottom For each of the 90 ways we can get to faces on the bottom ring, there is only 1 way to get to the bottom. So there are 90 ways here.

Case 2: traversing sideways For each of the 90 ways we can get to faces on the bottom ring, there are once again 2 ways to choose the direction and 4 ways to choose the amount of faces traveled. However, after traveling the faces there is only 1 way to go to the bottom face. There are $90 \cdot 2 \cdot 4 \cdot 1 = 720$ paths here.

Adding together the cases, there are 810 paths, or answer choice E. JH. L

Solution 4

Like Solution 3, we can use casework, but in a different way. We can split the caseworks into the amount of moves we spend on each row (top and bottom, where both rows have 5 faces).

Case 1: 1 move on the top row; There are 5 ways to make 1 move on the top row from the starting point (just moving on each of the five faces). Now we want to look at how many ways we can move towards the bottom. Notice how we have two faces on the bottom row that are adjacent to each of the top faces. If we spend one move on the bottom, there are 2 ways. If we spend two moves on the bottom, there are 4 ways (since order matters, we can flip the direction of these moves). If we spend three moves on the bottom, there are also 4 ways (similar reasoning). If we spend four moves on the bottom, there are 4 ways. And if we spend five moves on the bottom, there are again, 4 ways. In total there are $2 + 4 + 4 + 4 + 4 = 18$ ways to move from the bottom after the top moves are chosen. And since there are 5 ways to make 1 move on the top row, there are $5 \cdot 18 = 90$ ways.

Case 2: 2 moves on the top row; From just counting, we can see that there are five ways to move once on the top, and after that, two options for the second move (either left or right), so there are $5 \cdot 2 = 10$ ways. And from using what we obtained in Case 1, there are $10 \cdot 18 = 180$ ways for this case.

Case 3: three moves on the top row; There are, again, 10 ways to do this. And so, there are $10 \cdot 18 = 180$ ways for this case to happen.

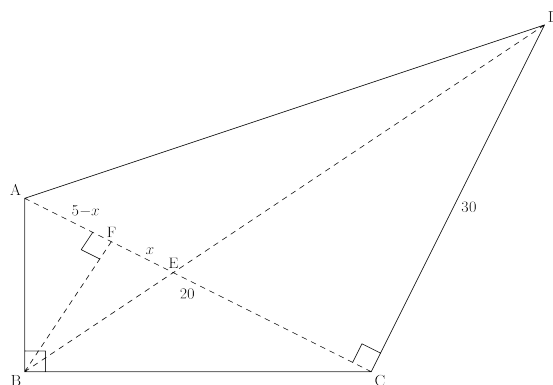
Now, we can see that there will be 10 ways for the remaining cases of the first move for the top row (because 5 ways to choose the first, and then you can go either left or right, so $5 \cdot 2 = 10$). So for Case 4 and Case 5, there will both also be $10 \cdot 18 = 180$ ways.

Finally, adding all these cases up, we obtain $90 + 180 + 180 + 180 + 180 = 810$, or
 (E) Mislicked

20. (AMC 10 10A 2020) Quadrilateral $ABCD$ satisfies $\angle ABC = \angle ACD = 90^\circ$, $AC = 20$, and $CD = 30$. Diagonals $\overline{AC}$ and $\overline{BD}$ intersect at point E , and $AE = 5$. What is the area of quadrilateral $ABCD$? (A) 330 (B) 340 (C) 350 (D) 360 (E) 370

Answer: D

Solution 1



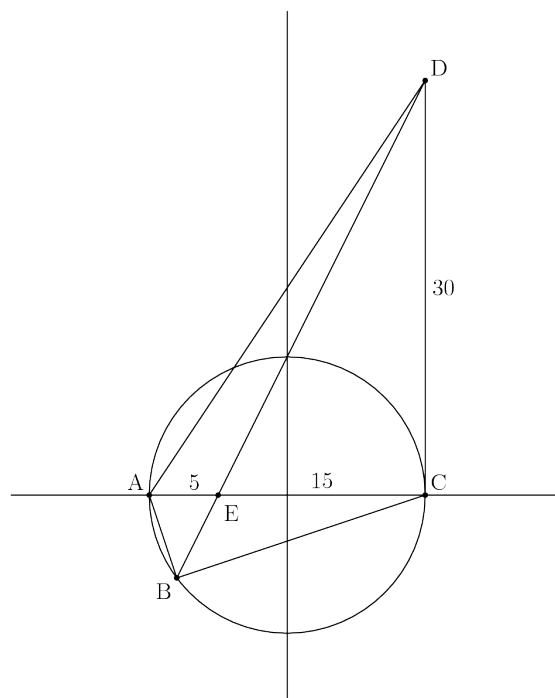
It's crucial to draw a good diagram for this one. Since $AC = 20$ and $CD = 30$, we get $[ACD] = 300$. Now we need to find $[ABC]$ to get the area of the whole quadrilateral. Drop an altitude from B to AC and call the point of intersection F . Let $FE = x$. Since $AE = 5$, then $AF = 5 - x$.

By dropping this altitude, we can also see two similar triangles, $\triangle BFE \sim \triangle DCE$. Since EC is $20 - 5 = 15$, and $DC = 30$, we get that $BF = 2x$.

Now, if we redraw another diagram just of ABC , we get that $(2x)^2 = (5-x)(15+x)$ because of the altitude geometric mean theorem which states that in any right triangle, the altitude squared is equal to the product of the two lengths that it divides the base into.

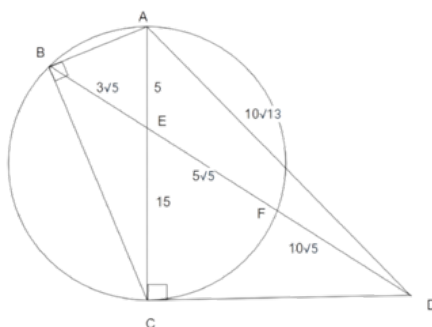
Expanding, simplifying, and dividing by the GCF, we get $x^2 + 2x - 15 = 0$. This factors to $(x + 5)(x - 3)$, which has roots of $x = -5, 3$. Since lengths cannot be negative, $x = 3$. Since $x = 3$, that means the altitude $BF = 2 \cdot 3 = 6$, or $[ABC] = 60$. Thus $[ABCD] = [ACD] + [ABC] = 300 + 60 = \boxed{\text{(D)} 360}$

Solution 2 (Coordinates)



Let the points be $A(-10, 0)$, $B(x, y)$, $C(10, 0)$, $D(10, 30)$, and $E(-5, 0)$, respectively. Since B lies on line DE , we know that $y = 2x + 10$. Furthermore, since $\angle ABC = 90^\circ$, B lies on the circle with diameter AC , so $x^2 + y^2 = 100$. Solving for x and y with these equations, we get the solutions $(0, 10)$ and $(-8, -6)$. We immediately discard the $(0, 10)$ solution as y should be negative. Thus, we conclude that $[ABCD] = [ACD] + [ABC] = \frac{20 \cdot 30}{2} + \frac{20 \cdot 6}{2} = \boxed{\text{(D)} 360}$.

Solution 3 (Power of A Point)



Draw the circumcircle of $\triangle ABC$. Let $BE = x$, and let F be the point where the circumcircle of $\triangle ABC$ meets BD . By the Pythagorean Theorem, $ED = \sqrt{1125}$, so $\frac{75}{x} + DF = \sqrt{1125} \implies DF = \sqrt{1125} - \frac{75}{x}$. ($EF = \frac{75}{x}$ by the Intersecting Chords Theorem.) From Power of A Point we have $CD^2 = 30^2 = FD \cdot BD = FD^2 + FD \cdot BF = (\sqrt{1125} - \frac{75}{x})^2 + (\sqrt{1125} - \frac{75}{x})(x + \frac{75}{x}) = (\sqrt{1125} - \frac{75}{x})(\sqrt{1125} + x)$. Solving we get $x = 3\sqrt{5}$. $\sin \angle CED = \frac{30}{15\sqrt{5}}$. The area of a quadrilateral is half the product of its diagonals times the sine of the angle between them, so the answer is $18\sqrt{5} \cdot 20 \cdot \frac{30}{15\sqrt{5}} \cdot \frac{1}{2} = 180 \cdot 2 = \boxed{360}$.

Solution 4 (Trigonometry)

Let $\angle C = \angle ACB$ and $\angle B = \angle CBE$. Using Law of Sines on $\triangle BCE$ we get

$$\frac{BE}{\sin C} = \frac{CE}{\sin B} = \frac{15}{\sin B}$$

and LoS on $\triangle ABE$ yields

$$\frac{BE}{\sin(90 - C)} = \frac{5}{\sin(90 - B)} = \frac{BE}{\cos C} = \frac{5}{\cos B}.$$

Divide the two to get $\tan B = 3 \tan C$. Now,

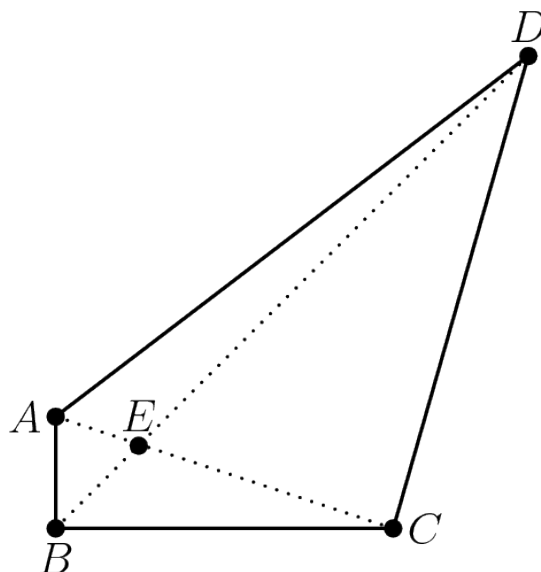
$$\tan \angle CED = 2 = \tan \angle B + \angle C = \frac{4 \tan C}{1 - 3 \tan^2 C}$$

and solve the quadratic, taking the positive solution (C is acute) to get $\tan C = \frac{1}{3}$. So if $AB = a$, then $BC = 3a$ and $[ABC] = \frac{3a^2}{2}$. By Pythagorean Theorem, $10a^2 = 400 \iff \frac{3a^2}{2} = 60$ and the answer is $300 + 60 \iff \boxed{(D)}$.

(This solution is incomplete, can someone complete it please) Edit by kc5170

We could use the famous m-n rule in trigonometry in $\triangle ABC$ with Point E [Unable to write it here. Could anybody write the expression]. We will find that $\overrightarrow{BD}$ is an angle bisector of $\triangle ABC$ (because we will get $\tan(x) = 1$). Therefore by converse of angle bisector theorem $AB : BC = 1 : 3$. By using Pythagorean theorem, we have values of AB and AC . Computing $AB \cdot AC = 120$. Adding the areas of ABC and ACD , hence the answer is $\boxed{(D) 360}$.

Solution 5 (Law of Cosines)



Denote EB as x . By the Law of Cosines:

$$AB^2 = 25 + x^2 - 10x \cos(\angle DEC)$$

$$BC^2 = 225 + x^2 + 30x \cos(\angle DEC)$$

Adding these up yields:

$$400 = 250 + 2x^2 + 20x \cos(\angle DEC) \implies x^2 + \frac{10x}{\sqrt{5}} - 75 = 0$$

By the quadratic formula, $x = 3\sqrt{5}$.

Observe:

$$[AEB] + [BEC] = \frac{1}{2}(x)(5) \sin(\angle DEC) + \frac{1}{2}(x)(15) \sin(180 - \angle DEC) = (3)(20) = 60$$

.

Thus the desired area is $\frac{1}{2}(30)(20) + 60 = \boxed{\text{(D) } 360}$

Solution 6 (Vectors / Coordinates)

Let $C = (0, 0)$ and $D = (0, 30)$. Then $E = (-15, 0)$, $A = (-20, 0)$, and B lies on the line $y = 2x + 30$. So the coordinates of B are

$$(x, 2x + 30).$$

We can make this a vector problem. $\vec{B} = \begin{pmatrix} x \\ 2x + 30 \end{pmatrix}$. We notice that point B forms a right angle, meaning vectors $\vec{BC}$ and $\vec{BA}$ are orthogonal, and their dot-product is 0.

We determine $\vec{BC}$ and $\vec{BA}$ to be $\begin{pmatrix} -x \\ -2x - 30 \end{pmatrix}$ and $\begin{pmatrix} -20 - x \\ -2x - 30 \end{pmatrix}$, respectively. (To get this, we use the fact that $\vec{BC} = \vec{C} - \vec{B}$ and similarly, $\vec{BA} = \vec{A} - \vec{B}$.)

Equating the cross-product to 0 gets us the quadratic $-x(-20 - x) + (-2x - 30)(-2x - 30) = 0$. The solutions are $x = -18, -10$. Since B clearly has a more negative x-coordinate than E , we take $x = -18$. So $B = (-18, -6)$.

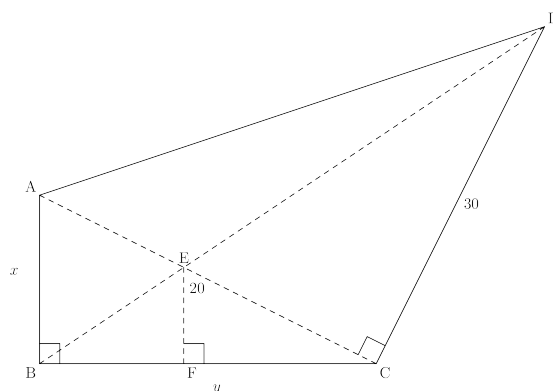
From here, there are multiple ways to get the area of $\triangle ABC$ to be 60, and since the area of $\triangle ACD$ is 300, we get our final answer to be

$$60 + 300 = \boxed{\text{(D) } 360}.$$

-PureSwag

Solution 7 (Power of a Point)

Solution 8 (Solving Equations)



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Because $\triangle ABC$ and $\triangle ACD$ share the same base, $\frac{[ABC]}{[ACD]} = \frac{BE}{DE}$ $[ABC] = [ACD]$.

$$\frac{BE}{DE} = 300 \cdot \frac{\frac{\sqrt{2x^2+100}}{2}}{15\sqrt{5}} \cdot \frac{1}{2} \cdot x \cdot y = 20 \cdot \frac{\frac{\sqrt{2x^2+100}}{2}}{\sqrt{5}} \quad xy = 4\sqrt{10x^2+500}$$

By $x^2 + y^2 = 400$, $y = \sqrt{400 - x^2}$. So, $x \cdot \sqrt{400 - x^2} = 4\sqrt{10x^2 + 500}$ $x^2(400 - x^2) = 16(10x^2 + 500)$

Let $x^2 = a$, $a(400 - a) = 16(10a + 500)$, $400a - a^2 = 160a + 8000$, $a^2 - 240a + 8000 = 0$, $(a - 200)(a - 40) = 0$

Because $x < 20$, a can only equal 40. $a = 40$, $x = 2\sqrt{10}$, $y = 6\sqrt{10}$ $[ABC] = \frac{1}{2} \cdot 2\sqrt{10} \cdot 6\sqrt{10} = 60$ $[ABCD] = [ABC] + [ACD] = 60 + 300 = \boxed{(D) 360}$

Solution 9

Drop perpendiculars $\overline{AF}$ and $\overline{CG}$ to $\overline{BD}$. Notice that since $\angle AEF = \angle CEG$ (since they are vertical angles) and $\angle AFE = \angle CGE = 90^\circ$, triangles AEF and CEG are similar. Therefore, we have

$$x/EF = CE/AE = 15/5 = 3,$$

where $EG = x$. Therefore, $EF = x/3$.

Additionally, angle chasing shows that triangles CEG and DCG are also similar. This gives $CG/x = DC/CE = 30/15 = 2$, so $CG = 2x$. Thus, applying the Pythagorean Theorem to triangle CEG gives

$$x^2 + (2x)^2 = 15^2,$$

so $EG = x = 3\sqrt{5}$. Our pairs of similar triangles then allow us to fill in the following lengths (in this order):

$$EF = x/3 = \sqrt{5}, CG = 2x = 6\sqrt{5}, AF = CG/3 = 2\sqrt{5}, DG = 2 \cdot CG = 12\sqrt{5}.$$

Now, let $BF = y$. Angle chasing shows that triangle ABF and BCG are similar, so $BG/AF = CG/BF$. Plugging in known lengths gives

$$\frac{y + 4\sqrt{5}}{2\sqrt{5}} = \frac{6\sqrt{5}}{y}.$$

This gives $y = 2\sqrt{5}$. Now we know all the lengths that make up BD , which allows us to find

$$BD = 2\sqrt{5} + \sqrt{5} + 3\sqrt{5} + 12\sqrt{5} = 18\sqrt{5}.$$

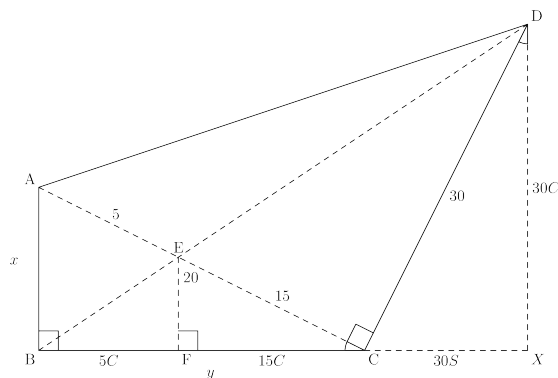
Therefore,

$$\begin{aligned} [ABCD] &= [ABD] + [CBD] \\ &= (BD)(AF)/2 + (BD)(CG)/2 \\ &= (18\sqrt{5})(2\sqrt{5})/2 + (18\sqrt{5})(6\sqrt{5})/2 \end{aligned}$$

$$= \boxed{(\text{D}) } 360.$$

–vaporwave

Solution 10 (Trigonometry)



set

$$\angle ACB = \theta, C = \cos(\theta), S = \sin(\theta)$$

$$\frac{E_y}{E_x} = \frac{30C}{20C + 30S} = \frac{15S}{20C - 15C}$$

$$2SC = \frac{3}{5}$$

$$\begin{aligned}[ABCD] &= [ABD] + [CBD] \\ &= \frac{1}{2} \cdot 20C \cdot 30C + \frac{1}{2} \cdot 20S(20C + 30S) \\ &= 100 \cdot 2SC + 300 \\ &= \boxed{\text{(D) } 360}.\end{aligned}$$

21. (*AMC 10 10A 2020*) There exists a unique strictly increasing sequence of nonnegative integers $a_1 < a_2 < \dots < a_k$ such that

$$\frac{2^{289} + 1}{2^{17} + 1} = 2^{a_1} + 2^{a_2} + \dots + 2^{a_k}.$$

What is k ? (A) 117 (B) 136 (C) 137 (D) 273 (E) 306

Answer: C

Solution 1 (Simplest)

If we begin with a common identity or just try polynomial long division on $\frac{2^{289}+1}{2^{17}+1}$, we quickly find it is equal to $(2^{272} - 2^{255} \pm \dots + 1)$. We recognize that we want all of these differences to turn into sums, and so we express this as $(2^{255} + 2^{221} + \dots + 2^{17})(2^{17} - 1) + 1$ (we grouped every two terms and factored out $2^{17} - 1$). If we keep this product in its current form, the -1 term on the second factor will create subtraction which we don't want, so we express this as the sum of a geometric sequence, which is $2^{16} + 2^{15} + \dots + 2^0$, so we have

- SumDumNerd

First, substitute 2^{17} with x . Then, the given equation becomes $\frac{x^{17}+1}{x+1} = x^{16} - x^{15} + x^{14} \dots - x^1 + x^0$ by sum of powers factorization. Now consider only $x^{16} - x^{15}$. This equals $x^{15}(x - 1) = x^{15} \cdot (2^{17} - 1)$. Note that $2^{17} - 1$ equals $2^{16} + 2^{15} + \dots + 1$, by difference of powers factorization (or by considering the expansion of $2^{17} = 2^{16} + 2^{15} + \dots + 2^2 + 2 + 2$). Thus, we can see that $x^{16} - x^{15}$ forms the sum of 17 different powers of 2. Applying the same method to each of $x^{14} - x^{13}$, $x^{12} - x^{11}$, .., $x^2 - x^1$, we can see that each of the pairs forms the sum of 17 different powers of 2. This gives us $17 \cdot 8 = 136$. But we must count also the x^0 term. Thus, Our answer is $136 + 1 = \boxed{\text{(C)} \ 137}$.

(There is actually an error...) Multiply both sides by $2^{17} + 1$ to get

Notice that $a_1 = 0$, since there is a 1 on the LHS. However, now we have an extra term of 2^{18} on the right from 2^{a_1+17} . To cancel it, we let $a_2 = 18$. The two 2^{18} 's now combine into a term of 2^{19} , so we let $a_3 = 19$. And so on, until we get to $a_{18} = 34$. Now everything we don't want telescopes into 2^{35} . We already have that term since we let $a_2 = 18 \implies a_2 + 17 = 35$. Everything from now on will automatically telescope to 2^{52} . So we let a_{19} be 52.

As you can see, we will have to add 17 a_n 's at a time, then "wait" for the sum to automatically telescope for the next 17 numbers, etc, until we get to 2^{289} . We only need to add a_n 's between odd multiples of 17 and even multiples. The largest even multiple of 17 below 289 is $17 \cdot 16$, so we will have to add a total of $17 \cdot 8$ a_n 's. However, we must not forget we let $a_1 = 0$ at the beginning, so our answer is $17 \cdot 8 + 1 = \boxed{\text{(C)} \ 137}$.

In order to shorten expressions, # will represent 16 consecutive 0s when expressing numbers.

[illegible]

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$2^{n+16} + 2^{n+15} + 2^{n+14} + \dots + 2^n$ This means that each pair has 17 terms of the form 2^n .

Since there are 8 of these pairs, there are a total of $8 \cdot 17 = 136$ terms. Accounting for the 2^0 term, which was not in the pair, we have a total of $136 + 1 = \boxed{\text{(C) } 137}$ terms. emerald_block

Solution 5 (Similar to Solution 4)

We can first look at smaller similar cases in binary. We can treat the initial problem as $\frac{2^{n^2}+1}{2^{n+1}}$ where $n = 17$. We can first look at the case $n = 3$ or $\frac{2^9+1}{2^3+1}$, which is equivalent to $\frac{1000000001_2}{1001_2}$. If we do long division we find that this equals 111001_2 . Then we can also look at the case $n = 5$ or $\frac{2^{25}+1}{2^5+1}$. Doing long division on this in binary gives us a quotient that is a repeating pattern of 5 zeros and 5 ones. This pattern does hold, as $111\dots n \text{ ones } \dots 1_2 \cdot 10\dots n-1 \text{ zeroes } \dots 01_2 = (10_2)^{2n} - 1$. Then at the end, there is a remainder of $10\dots n-1 \text{ zeroes } \dots 01_2$, which is the same as the denominator of the original fraction. Thus, for the original problem, there are $\frac{17-1}{2}$ repeats of 17 zeroes and 17 ones, giving $8 \cdot 17 + 1 = \boxed{\text{(C) } 137}$. Hi2937

Solution 6 (Psychology)

The two answers that are closest to each other are 136 and 137. Since it seems more likely to forget an edge case than to overcount something in this context, the right answer is most likely **(C) 137**.-AtomicBoy was here

22. (AMC 10 10A 2020) For how many positive integers $n \leq 1000$ is

$$\left\lfloor \frac{998}{n} \right\rfloor + \left\lfloor \frac{999}{n} \right\rfloor + \left\lfloor \frac{1000}{n} \right\rfloor$$

not divisible by 3? (Recall that $\lfloor x \rfloor$ is the greatest integer less than or equal to x .)

- (A) 22 (B) 23 (C) 24 (D) 25 (E) 26

Answer: A

Solution 1

If n is a factor of 999 (and $n \neq 1$), then let $999 = an$ for some positive integer a . Then we get $\lfloor 998/n \rfloor = a - 1$, $\lfloor 999/n \rfloor = a$, $\lfloor 1000/n \rfloor = a$. The sum is $3a - 1$, which isn't a multiple of 3. If n is a factor of 1000 (and $n \neq 1$), then let $1000 = an$. Then we get $\lfloor 998/n \rfloor = a - 1$, $\lfloor 999/n \rfloor = a - 1$, and $\lfloor 1000/n \rfloor = a$. The sum is $3a - 2$, which also isn't a multiple of 3.

We find that the only way for the problem's condition to hold true is if n is a factor of 999 or 1000 and $n \neq 1$.

- $1000 = 5^3 \cdot 2^3 \implies (3 + 1)(3 + 1) = 16$ factors.
- $999 = 3^3 \cdot 37 \implies (3 + 1)(1 + 1) = 8$ factors.

We counted 1 twice. So the answer is $16 + 8 - 2 = \boxed{22}$.

Solution 2

Clearly, $n = 1$ fails. Except for the special case of $n = 1$,

$$\left\lfloor \frac{1000}{n} \right\rfloor - \left\lfloor \frac{998}{n} \right\rfloor$$

equals either 0 or 1. If it equals 0, this implies that $\left\lfloor \frac{998}{n} \right\rfloor = \left\lfloor \frac{999}{n} \right\rfloor = \left\lfloor \frac{1000}{n} \right\rfloor$, so their sum is clearly a multiple of 3, so this will always fail. If it equals 1, the sum of the three floor terms is $3 \left\lfloor \frac{999}{n} \right\rfloor \pm 1$, so it is never a multiple of 3. Thus, we are looking for all $n \neq 1$ such that

$$\left\lfloor \frac{1000}{n} \right\rfloor - \left\lfloor \frac{998}{n} \right\rfloor = 1.$$

This implies that either

$$\left\lfloor \frac{998}{n} \right\rfloor + 1 = \left\lfloor \frac{999}{n} \right\rfloor,$$

or

$$\left\lfloor \frac{999}{n} \right\rfloor + 1 = \left\lfloor \frac{1000}{n} \right\rfloor.$$

Let's analyze the first equation of these two. This equation is equivalent to the statement that there is a positive integer a such that

$$\frac{998}{n} < a \leq \frac{999}{n} \implies 998 < an \leq 999 \implies an = 999 \implies a = \frac{999}{n} \implies n|999.*$$

Analogously, the second equation implies that

$$n|1000.$$

So our only n that satisfy this condition are $n \neq 1$ that divide 999 or 1000. Using the method to find the number of divisors of a number, we see that 999 has 8 divisors and 1000 has 16 divisors. Their only common factor is 1, so there are $8 + 16 - 1 = 23$ positive integers that divide either 999 or 1000. Since the integer 1 is a special case and does not count, we must subtract this from our 23, so our final answer is $23 - 1 = \boxed{\text{(A) } 22}$.

*While this observation may seem strange, it is actually "trivial by intuition" to g

Solution 3

Counting down n from 1000, 999, 998... we notice that $\left\lfloor \frac{998}{n} \right\rfloor + \left\lfloor \frac{999}{n} \right\rfloor + \left\lfloor \frac{1000}{n} \right\rfloor$ is only not divisible by 3 when n is a divisor of *only* 1000 or 999 (1000, 999, 500, 333...). Notice how the factors of 998 : 1, 2, 499, and 998, do not work.

The prime factorization of 999 is $3^3 \cdot 37$, so $4 \cdot 2 = 8$ factors in total.

The prime factorization of 1000 is $2^3 \cdot 5^3$, so $4 \cdot 4 = 16$ factors in total.

However, 1 obviously does not work, so we have to subtract 2 (1 is counted twice) from the total. $8 + 16 - 2 = \boxed{\text{(A) } 22}$.

Solution 4

First, we notice the following lemma:

Lemma: For $N, n \in \mathbb{N}$, $\left\lfloor \frac{N}{n} \right\rfloor = \left\lfloor \frac{N-1}{n} \right\rfloor + 1$ if $n \mid N$; and $\left\lfloor \frac{N}{n} \right\rfloor = \left\lfloor \frac{N-1}{n} \right\rfloor$ if $n \nmid N$.

Proof: Let $A = kn + r$, with $0 \leq r < n$. If $n \mid N$, then $r = 0$. Hence $\left\lfloor \frac{N}{n} \right\rfloor = k$, $\left\lfloor \frac{N-1}{n} \right\rfloor = \left\lfloor \frac{(k-1)n + n-1}{n} \right\rfloor = k - 1 + \left\lfloor \frac{n-1}{n} \right\rfloor = k - 1$, and $\left\lfloor \frac{N}{n} \right\rfloor = \left\lfloor \frac{N-1}{n} \right\rfloor + 1$.

If $n \nmid N$, then $1 \leq r < n$. Hence $\left\lfloor \frac{N}{n} \right\rfloor = k$, $\left\lfloor \frac{N-1}{n} \right\rfloor = k + \left\lfloor \frac{r-1}{n} \right\rfloor = k$, and $\left\lfloor \frac{N}{n} \right\rfloor = \left\lfloor \frac{N-1}{n} \right\rfloor$.

From the lemma and the given equation, we have four possible cases:

$$\left\lfloor \frac{998}{n} \right\rfloor + 1 = \left\lfloor \frac{999}{n} \right\rfloor = \left\lfloor \frac{1000}{n} \right\rfloor - 1 \quad (1)$$

$$\left\lfloor \frac{998}{n} \right\rfloor + 1 = \left\lfloor \frac{999}{n} \right\rfloor + 1 = \left\lfloor \frac{1000}{n} \right\rfloor \quad (2)$$

$$\left\lfloor \frac{998}{n} \right\rfloor + 1 = \left\lfloor \frac{999}{n} \right\rfloor = \left\lfloor \frac{1000}{n} \right\rfloor \quad (3)$$

$$\left\lfloor \frac{998}{n} \right\rfloor = \left\lfloor \frac{999}{n} \right\rfloor = \left\lfloor \frac{1000}{n} \right\rfloor \quad (4)$$

Note that cases (2) and (3) are the cases in which the term, $\left\lfloor \frac{998}{n} \right\rfloor + \left\lfloor \frac{999}{n} \right\rfloor + \left\lfloor \frac{1000}{n} \right\rfloor$, is not divisible by 3. So we only need to count the number of n 's for which cases (2) and (3) stand.

Case (2): By the lemma, we have $n \mid 1000$ and $n \nmid 999$. Hence n can be any factor of 1000 except for $n = 1$. Since $1000 = 2^3 * 5^3$, there are $(3 + 1)(3 + 1) - 1 = 15$ possible values of n for this case.

Case (3): By the lemma, we have $n \mid 999$ and $n \nmid 998$. Hence n can be any factor of 999 except for $n = 1$. Since $999 = 3^3 * 37^1$, there are $(3 + 1)(1 + 1) - 1 = 7$ possible values of n for this case.

So in total, we have total of $15 + 7 = \boxed{\text{(A)}22}$ possible n 's.

Solution 5 (Casework)

Note that $\left\lfloor \frac{998}{n} \right\rfloor + \left\lfloor \frac{999}{n} \right\rfloor + \left\lfloor \frac{1000}{n} \right\rfloor$ is a multiple of 3 if $\left\lfloor \frac{998}{n} \right\rfloor + \left\lfloor \frac{999}{n} \right\rfloor + \left\lfloor \frac{1000}{n} \right\rfloor$ lies between two consecutive multiples of n .

Explanation:

Let's assume that the above expression does indeed lie between two consecutive multiples of n . This implies that n does not divide either 998, 999 or 1000, meaning that when divided by n , none of the quotients are whole. In turn, this also means that they will all have the same whole number part.

If $\frac{998}{n}$, $\frac{999}{n}$, and $\frac{1000}{n}$ were to have a different whole number part, then n would have to lie within 998 and 1000. If this is confusing, think of why $\frac{6}{5}$, $\frac{7}{5}$ and $\frac{8}{5}$ have the same whole number part (1 in this case). Here, $n = 5$. What happens to the whole number part when $n = 7$?

Since $\frac{998}{n}$, $\frac{999}{n}$, and $\frac{1000}{n}$ have identical whole number parts, their floors are identical, since the floor of a number is equal to its whole number part, discarding its fractional component.

Let k be the number we get when we floor $\frac{998}{n}$, $\frac{999}{n}$, or $\frac{1000}{n}$ if the three of those numbers lie between two consecutive multiples of n . Adding them up, we get $3k$ (due to their floors being the same), which is a multiple of 3. So no matter what, we cannot have $\frac{998}{n}$, $\frac{999}{n}$, and $\frac{1000}{n}$ lie between two consecutive multiples of n .

What does this mean? It means that there must be some multiple of n within this expression (with some restrictions, as we'll see), in order to prevent the violation of the main restriction. We can proceed with casework now.

Case 1

Let's assume that the multiple of n is located at $\frac{998}{n}$. Luckily, the only prime factors of 998 are 2 and 499.

We can observe that 499 doesn't work, since $\frac{998}{499}$ and $\frac{1000}{499}$ will both round down to 2 when divided by it. However, 2 does work, since it divides both 998 and 1000. The floor of $\frac{999}{2}$ is 499, meaning that when we evaluate the given expression for $n = 2$, we will get $2 \cdot 499 + 500$ which isn't a multiple of 3.

This case works because we have a multiple of n at the end of the expression, as well as the beginning, so the whole number parts of 998, 999 and 1000 when divided by n are not all the same, due to n dividing two of these numbers.

The restriction of n dividing more than one number within the expression is only valid when we're testing the first number in the given expression (otherwise, the other floors wouldn't round down to the same value as the first).

Case 1 is complete, and we've found that only 2 works.

Case 2

Now, let's assume that the multiple of n is located at $\frac{999}{n}$. In this case, if n divides 999, it doesn't divide 998 (since two multiples of a number greater than 1 are never consecutive), nor does it divide 1000, for the same reason.

The prime factorization of 999 is $3^3 \cdot 37$, and thus has $(3 + 1)(1 + 1) = 8$ divisors.

When testing a few values of n initially, we observed that $n = 1$ causes the expression to be divisible by 3. Subtracting 1, we see that there are 7 values of n that work for this case.

Case 3

Finally, let's assume that the multiple of n is located at $\frac{1000}{n}$.

Our goal is to have the other multiple of n below whatever $\frac{998}{n}$ is, so we won't have identical floors throughout the expression.

Once again, any factor of 1000, except for 2 is relatively prime to both 998 and 999, so when we floor those two numbers, we get an integer that isn't 500.

1000 factors as $2^3 \cdot 5^3$, meaning that it has $(3 + 1)(3 + 1) = 16$ divisors. 1 and 2 either don't work or have already been counted, so there are 14 valid values of n for this case.

Ending

Adding these three cases, we get $1 + 7 + 14 = \boxed{\text{A)} 22}$ values of n .

-Benedict T (countmath1)

Solution 6 (simple, fast)

Let $\lfloor \frac{998}{n} \rfloor + \lfloor \frac{999}{n} \rfloor + \lfloor \frac{1000}{n} \rfloor$ be (1). Claim : If n is a factor of at least one of 998, 999, 1000, then (1) is NOT divisible by 3. Proof of the claim :

If you don't understand this, let's test out some possible values. Evaluating the expression using $n = 1, 2, 3, 4, 5$ show that the expression isn't divisible by 3. And when $n = 6$, we have

$$\lfloor \frac{998}{6} \rfloor = \lfloor \frac{999}{6} \rfloor = \lfloor \frac{1000}{6} \rfloor = 166$$

This justifies that the expression is divisible by 3.

Therefore, let S be the set of numbers (not incl. 1) for which this is divisible by 3, which is $S = \{2, 3, 4, 5, 8, 9, 10, 20, 27, 37, 40, 50, 100, 111, 200, 333, 400, 499, 500, 999, 1000\}$. Adding 1 to this gives $21 + 1 = 22$ values. (A) 22 Q. E. D. elpianista227

23. (AMC 10 10A 2020) Let T be the triangle in the coordinate plane with vertices $(0, 0)$, $(4, 0)$, and $(0, 3)$. Consider the following five isometries (rigid transformations) of the plane: rotations of 90° , 180° , and 270° counterclockwise around the origin, reflection across the x -axis, and reflection across the y -axis. How many of the 125 sequences of three of these transformations (not necessarily distinct) will return T to its original position? (For example, a 180° rotation, followed by a reflection across the x -axis, followed by a reflection across the y -axis will return T to its original position, but a 90° rotation, followed by a reflection across the x -axis, followed by another reflection across the x -axis will not return T to its original position.)
- (A) 12 (B) 15 (C) 17 (D) 20 (E) 25

Answer: A

Solution 1 (Simplest)

Label each rotation A, B, C, D , and E respectively.

One can notice that any rotation on any figure is formed by the transformation on its vertices. Therefore, instead of thinking about the entire triangle, just think of the point

24. (AMC 10 10A 2020) Let n be the least positive integer greater than 1000 for which

$$\gcd(63, n + 120) = 21 \quad \text{and} \quad \gcd(n + 63, 120) = 60.$$

What is the sum of the digits of n ?

- (A) 12 (B) 15 (C) 18 (D) 21 (E) 24

Answer: C

Solution 1

When looking at $38 = 7\alpha - 20\gamma$, instead of attempting to take α modulo 40, it may be more intuitively obvious to take α modulo 20:

As 7 and 20 are coprime, 7 has an inverse modulo 20. Recall that $7 \cdot 3 \equiv 1 \pmod{20}$, so $7^{-1} \equiv 3 \pmod{20}$. Thus, $\alpha \equiv 54 \equiv 14 \pmod{20}$. Furthermore, consider that $21\alpha - 57 > 1000$, so $\alpha \geq 51$. $\alpha = 54$ fails, as $\gcd(\alpha, 3) = 1$ from $\gcd(63, n + 57) = 21$. 74 fails as well; this causes γ to be even from $114 = 21\alpha - 60\gamma$. 94 is the first number not violating any restrictions, leaving $94 \cdot 21 - 57 = 1917$, which has digital sum $1 + 9 + 1 + 7 = 18$, so we choose (C).

Alternatively, we could have said $\alpha = 40k + 14 \equiv 0 \pmod{3}$ for $k \equiv 1 \pmod{3}$ only, so $k \equiv 0, 2 \pmod{3}$, giving us our answer. Since the problem asks for the sum of the digits of n , our answer is $1 + 9 + 1 + 7 = \span style="border: 1px solid black; padding: 0 2px;">(C) 18.$

Supplement When looking at $38 = 7\alpha - 20\gamma$, instead of attempting to take α modulo 40, it may be more intuitively obvious to take α modulo 20:

As 7 and 20 are coprime, 7 has an inverse modulo 20. Recall that $7 \cdot 3 \equiv 1 \pmod{20}$, so $7^{-1} \equiv 3 \pmod{20}$. Thus, $\alpha \equiv 54 \equiv 14 \pmod{20}$. Furthermore, consider that $21\alpha - 57 > 1000$, so $\alpha \geq 51$. $\alpha = 54$ fails, as $\gcd(\alpha, 3) = 1$ from $\gcd(63, n + 57) = 21$. 74 fails as well; this causes γ to be even from $114 = 21\alpha - 60\gamma$. 94 is the first

number not violating any restrictions, leaving $94 \cdot 21 - 57 = 1917$, which has digital sum $1 + 9 + 1 + 7 = 18$, so we choose $\boxed{(C)}$.

In the end, though, it is just the same steps in a different order. LeonQS

Solution 2

The conditions of the problem reduce to the following: $n + 120 = 21k$ and $n + 63 = 60\ell$, where $\gcd(k, 3) = 1$ and $\gcd(\ell, 2) = 1$. From these equations, we see that $21k - 60\ell = 57$. Solving this Diophantine equation gives us that $k = 20a + 17$ and $\ell = 7a + 5$. Since, $n > 1000$, we can do some bounding and get that $k > 53$ and $\ell > 17$. Now we start bashing by plugging in numbers that satisfy these conditions: $a = 4$ is the first number that works so we get $\ell = 33$, $k = 97$. Therefore, we have $n = 21(97) - 120 = 60(33) - 63 = 1917$, and our answer is $1 + 9 + 1 + 7 = \boxed{(C) 18}$.

Solution 3

We first find that $n \equiv 6 \pmod{21}$ and $n \equiv 57 \pmod{60}$, then we get $n = 21x + 6$ and $n = 60y + 57$ by definitions, where x and y are integers. It follows that y must be odd, since the GCD will be 120 instead of 60 if y is even. Also, the unit digit of n must be 7, since the unit digit of $60y$ is always 0 and the unit digit of 57 is 7. Therefore, we find that x must end in 1 to satisfy n having a unit digit of 7. Also, we find that x must not be a multiple of 3 or else the GCD will be 63. Therefore, we test values for x and find that $x = 91$ satisfies all these conditions. Therefore, $n = 1917$ and $1 + 9 + 1 + 7 = \boxed{(C) 18}$.

Solution 4

After applying the **Euclidean Algorithm** we get $\gcd(n - 6, 63) = 21$ and $\gcd(n - 57, 120) = 60$. So we have $n - 6 = 21a$ and $n - 57 = 60b$, where a is not a multiple of 3 and b is not a multiple of 2. Setting these equations equal to each other, we get $21a = 60b + 51 \implies 20b = 7a - 17$. We note that a must be ≥ 48 (since n has to be greater than 1000) and the last digit of a must be 1 so that $7a - 17$ is a multiple of 10 and has a possibility of also being a multiple of 20.

25. (AMC 10 10A 2020) Jason rolls three fair standard six-sided dice. Then he looks at the rolls and chooses a subset of the dice (possibly empty, possibly all three dice) to reroll. After rerolling, he wins if and only if the sum of the numbers face up on the three dice is exactly 7. Jason always plays to optimize his chances of winning. What is the probability that he chooses to reroll exactly two of the dice?

(A) $\frac{7}{36}$ (B) $\frac{5}{24}$ (C) $\frac{2}{9}$ (D) $\frac{17}{72}$ (E) $\frac{1}{4}$

Answer: A

Solution 1

Consider the probability that rolling two dice gives a sum of s , where $s \leq 7$. There are $s - 1$ pairs that satisfy this, namely $(1, s - 1), (2, s - 2), \dots, (s - 1, 1)$, out of $6^2 = 36$ possible pairs. The probability is $\frac{s-1}{36}$.

Therefore, if one die has a value of a and Jason rerolls the other two dice, then the probability of winning is $\frac{7-a-1}{36} = \frac{6-a}{36}$.

In order to maximize the probability of winning, a must be minimized. This means that if Jason rerolls two dice, he must choose the two dice with the maximum values.

Thus, we can let $a \leq b \leq c$ be the values of the three dice. Consider the case when $a + b < 7$. If $a + b + c = 7$, then we do not need to reroll any dice. Otherwise, if we

reroll one die, we can reroll any one die in the hope that we get the value that makes the sum of the three dice 7. This happens with probability $\frac{1}{6}$. If we reroll two dice, we will roll the two maximum dice, and the probability of winning is $\frac{6-a}{36}$, as stated above. However, $\frac{1}{6} > \frac{6-a}{36}$, so rolling one die is always better than rolling two dice if $a + b < 7$.

Now consider the case where $a + b \geq 7$. Rerolling one die will not help us win since the sum of the three dice will always be greater than 7. If we reroll two dice, the probability of winning is, once again, $\frac{6-a}{36}$. To find the probability of winning if we reroll all three dice, we can let each die have 1 dot and find the number of ways to distribute the remaining 4 dots. By stars and bars, there are $\binom{6}{2} = 15$ ways to do this, making the probability of winning $\frac{15}{6^3} = \frac{5}{72}$.

In order for rolling two dice to be more favorable than rolling three dice, $\frac{6-a}{36} > \frac{5}{72} \rightarrow a \leq 3$.

Thus, rerolling two dice is optimal if and only if $a \leq 3$ and $a + b \geq 7$. The possible triplets (a, b, c) that satisfy these conditions, and the number of ways they can be permuted, are

$(3, 4, 4) \rightarrow 3$ ways.

$(3, 4, 5) \rightarrow 6$ ways.

$(3, 4, 6) \rightarrow 6$ ways.

$(3, 5, 5) \rightarrow 3$ ways.

$(3, 5, 6) \rightarrow 6$ ways.

$(3, 6, 6) \rightarrow 3$ ways.

$(2, 5, 5) \rightarrow 3$ ways.

$(2, 5, 6) \rightarrow 6$ ways.

$(2, 6, 6) \rightarrow 3$ ways.

$(1, 6, 6) \rightarrow 3$ ways.

There are $3 + 6 + 6 + 3 + 6 + 3 + 3 + 6 + 3 + 3 = 42$ ways in which rerolling two dice is optimal, out of $6^3 = 216$ possibilities. Therefore, the probability that Jason will reroll two dice is $\frac{42}{216} = \boxed{\text{(A)} \frac{7}{36}}$.

Solution 2 (Illustration)

We conclude all of the following after the initial roll:

- Jason rerolls exactly zero dice if and only if the sum of the three dice is 7, in which the probability of winning is always 1.
- If Jason rerolls exactly one die, then the sum of the two other dice must be 2, 3, 4, 5, or 6. The probability of winning is always $\frac{1}{6}$, as exactly 1 of the 6 possible outcomes of the die rerolled results in a win.
- If Jason rerolls exactly two dice, then the outcome of the remaining die must be 1, 2, 3, 4, or 5. Applying casework to the remaining die produces the following

table:

Remaining Die	Sum Needed for the Two Dice Rerolled	Probability of Winning
1	6	$5/36$
2	5	$4/36$
3	4	$3/36$
4	3	$2/36$
5	2	$1/36$

The probability of winning is at most $\frac{5}{36}$.

- d) If Jason (re)rolls all three dice, then the probability of winning is always $\frac{\binom{6}{2}}{6^3} = \frac{15}{216} = \frac{5}{72}$. For the denominator, rolling three dice gives a total of $6^3 = 216$ possible outcomes.

For the numerator, this is the same as counting the ordered triples of positive integers (a, b, c) for which $a + b + c = 7$. Suppose that 7 balls are lined up in a row. There are 6 gaps between the balls, and placing dividers in 2 of the gaps separates the balls into 3 piles. From left to right, the numbers of balls in the piles correspond to a, b , and c , respectively. There are $\binom{6}{2} = 15$ ways to place the dividers. Note that the dividers' positions and the ordered triples have one-to-one correspondence, and $1 \leq a, b, c \leq 6$ holds for all such ordered triples.

The optimal strategy is that:

- If Jason needs to reroll at least zero dice to win, then he rerolls exactly zero dice.
- If Jason needs to reroll at least one die to win, then he rerolls exactly one die.
- If Jason needs to reroll at least two dice to win, then he rerolls exactly two dice if and only if the probability of winning is greater than $\frac{5}{72}$, the probability of winning for rerolling all three dice. The first three cases in the table above satisfy this requirement. We will analyze these cases by considering the initial outcomes of the two dice rerolled. Note that any of the three dice can be the remaining die, so we need a factor of 3 for all counts in the third column:

Remaining Die	Initial Outcomes of the Two Dice Rerolled	# of Ways
1	Both are in $\{6\}$, not necessarily distinct.	$3 \cdot 1^2 = 3$
2	Both are in $\{5, 6\}$, not necessarily distinct.	$3 \cdot 2^2 = 12$
3	Both are in $\{4, 5, 6\}$, not necessarily distinct.	$3 \cdot 3^2 = 27$

Finally, the requested probability is

$$\frac{3 + 12 + 27}{6^3} = \frac{42}{216} = \boxed{\text{(A)} \frac{7}{36}}.$$

Solution 3 (Similar to Solution 2)

We count the numerator. Jason will pick up no dice if he already has a 7 as a sum. We need to assume he does not have a 7 to begin with. If Jason decides to pick up all the dice to re-roll, by the stars and bars rule ways to distribute, $\binom{n+k-1}{k-1}$, there will be 2 bars and 4 stars (3 of them need to be guaranteed because a roll is at least

1) for a probability of $\frac{15}{216} = \frac{2.5}{36}$. If Jason picks up 2 dice and leaves a die showing k , he will need the other two to sum to $7 - k$. This happens with probability

$$\frac{6 - k}{36}$$

for integers $1 \leq k \leq 6$. If the roll is not 7, Jason will pick up exactly one die to re-roll if there can remain two other dice with sum less than 7, since this will give him a $\frac{1}{6}$ chance which is a larger probability than all the cases unless he has a 7 to begin with. We have

$$\frac{1}{6} > \frac{5, 4, 3}{36} > \frac{2.5}{36} > \frac{2, 1, 0}{36}.$$

We count the underlined part's frequency for the numerator without upsetting the probability greater than it. Let a be the roll we keep. We know $a \leq 3$ since $a = 4$ would cause Jason to pick up all the dice. When $a = 1$, there are 3 choices for whether it is rolled 1st, 2nd, or 3rd, and in this case the other two rolls have to be at least 6 (or he would have only picked up 1). This gives $3 \cdot 1^2 = 3$ ways. Similarly, $a = 2$ gives $3 \cdot 2^2 = 12$ because the 2 can be rolled in 3 places and the other two rolls are at least 5. $a = 3$ gives $3 \cdot 3^2 = 27$. Summing together gives the numerator of 42. The denominator is $6^3 = 216$, so we have $\frac{42}{216} = \boxed{\text{(A)} \frac{7}{36}}$.

Solution 4

We can quickly write out all possible rolls that result in a total of 7, to get that there is a $\frac{5}{72}$ chance that rolling three dice will result in a total of 7.

Claim 1: If all of Jason's initial outcomes are 4, 5 or 6, Jason is best off rerolling everything.

Proof of if part: If Jason rerolls two dice, he would like for the sum of the two outcomes to be 7 - (The initial outcome of the untouched dice). Since in this case, every dice's initial outcome is 4, 5 or 6, this value is 1, 2 or 3. By basic dice probability, the probability that they add to one is 0; the probability that they add to two is $\frac{1}{36}$; the probability that they add to three is $\frac{1}{18}$. Each of these probabilities is less than $\frac{5}{72}$, (the probability of winning by rerolling everything) so Jason should not reroll one die in this scenario. Rolling one die is also nonsense in this situation, since the other two dice will already add up to something greater than 7.

Proof of only-if part: If Jason rolls a 1, 2 or 3, even just once, then, the probability that rerolling the other two dice will result in win is $\frac{5}{36}$, $\frac{1}{9}$ and $\frac{1}{12}$, by basic die probability. Each of these is greater than $\frac{5}{72}$, so if Jason rolls a 1, 2 or 3, he should not reroll everything.

Claim 2: If the sum of any two of the initial dice is less than 7, Jason is best off rerolling one or zero die.

Proof of if part: If the sum of any two of the initial dice is less than 7, when the other die is rolled, there is always a fixed $\frac{1}{6}$ chance that this results in a win for Jason (or, he has his initial roll already won, so he doesn't have to roll anything). Rerolling that one die is better than rerolling all three dice, since, as we have seen there is a $\frac{5}{72}$ chance that Jason wins from that. Rerolling that one die is also better than rerolling two dice: by basic die probability, if Jason wishes for the sum of two rerolled dice to equal one value, the maximum probability this happens is $\frac{1}{6}$, and this is when he wishes for the sum to be 7; however, since he wishes for the sum of

all three dice to be 7, this will never happen, so the probability that rerolling two dice results in a win is always less than $\frac{1}{6}$.

Proof of only-if part: If the sum of each pair of the three dice is at least 7, it would make no sense to reroll one die, since the final sum of all three dice will always be greater than 7. Also, if the sum of each pair of the three dice is at least 7, the total initial sum must also be greater than 7, so Jason should not reroll zero dice.

From having shown these two claims, we know that Jason should reroll two dice if at least one of them is a 1, 2 or 3, and if the sum of any two of the initial dice is at least 7. We can write out all the possibilities of initial rolls, account for their permutations, and add everything together to see that there are 42 total outcomes, where Jason should reroll two dice. Therefore, our final answer is $\frac{42}{216} = \boxed{\text{(A)} \frac{7}{36}}$.

Solution 5

If the sum of the dice is 7, we reroll no dice. If the sum of any two dice is ≤ 6 , we reroll 1 die because rolling 1 die will give Jason a $\frac{1}{6}$ chance of winning. Thus, we consider the scenario where the sum of any two dice is 7 or greater.

If the smallest die is 1, the other two dice must be 6, which has 3 possible permutations.

If the smallest die is 2, the other two dice can be 5 or 6. There are 3 ways to choose the position of the smallest die, 2 ways each to choose the value of the other dice, so there are $2 \cdot 2 \cdot 3 = 12$ possible permutations.

If the smallest die is 3, the other two dice can be 4, 5, or 6. Similarly, there are 3 ways to choose the position of the smallest die, and 3 ways each to choose the value of the other dice, so there are $3 \cdot 3 \cdot 3 = 27$ possible permutations.

If all die are 4 or above, then we reroll 3 dice. When we reroll all 3 dice, the winning rolls are 1,1,5; 1,2,4; 1,3,3; 2,2,3 in some arrangement, for a total of 15 permutations. Thus, the probability of winning after rerolling 3 dice is $\frac{15}{6^3} = \frac{5}{72}$. If we reroll only 2 dice with the smallest die being 4, we would need to roll 1,2 or 2,1 to win, which gives a $\frac{2}{36} = \frac{1}{18}$ chance of winning, so this is worse.

If the smallest die is 5 or 6, rerolling 2 dice gives us even worse odds.

Therefore, the cases where 1, 2, or 3 are the smallest die covers all cases where 2 rerolls is optimal. There are $3 + 12 + 27 = 42$ permutations out of 216 total permutations, giving us a probability of $\frac{42}{216} = \boxed{\frac{7}{36}}$.

§3 AMC 10 10B 2021

1. (AMC 10 10B 2021) How many integer values of x satisfy $|x| < 3\pi$?
 (A) 9 (B) 10 (C) 18 (D) 19 (E) 20

Answer: D

Solution 1

Since $3\pi \approx 9.42$, we multiply 9 by 2 for the integers from 1 to 9 and the integers from -1 to -9 and add 1 to account for 0 to get $\boxed{\text{(D)} 19}$.

Solution 2

$|x| < 3\pi \iff -3\pi < x < 3\pi$. Since π is approximately 3.14, 3π is approximately 9.42. We are trying to solve for $-9.42 < x < 9.42$, where $x \in \mathbb{Z}$. Hence, $-9.42 < x < 9.42 \implies -9 \leq x \leq 9$, for $x \in \mathbb{Z}$. The number of integer values of x is $9 - (-9) + 1 = 19$. Therefore, the answer is **(D) 19**.

Solution 3

$3\pi \approx 9.4$. There are two cases here.

When $x > 0$, $|x| > 0$, and $x = |x|$. So then $x < 9.4$

When $x < 0$, $|x| > 0$, and $x = -|x|$. So then $-x < 9.4$. Dividing by -1 and flipping the sign, we get $x > -9.4$.

From case 1 and 2, we know that $-9.4 < x < 9.4$. Since x is an integer, we must have x between -9 and 9 . There are a total of

$$9 - (-9) + 1 = \boxed{\text{(D) } 19} \text{ integers.}$$

PureSwag

Solution 4

Looking at the problem, we see that instead of directly saying x , we see that it is $|x|$. That means all the possible values of x in this case are positive and negative. Rounding π to 3 we get $3(3) = 9$. There are 9 positive solutions and 9 negative solutions: $9 + 9 = 18$. But what about zero? Even though zero is neither negative nor positive, but we still need to add it into the solution. Hence, the answer is $9 + 9 + 1 = 18 + 1 = \boxed{\text{(D) } 19}$.

Solution 5

There are an odd number of integer solutions x to this inequality since if any non-zero integer x satisfies this inequality, then so does $-x$, and we must also account for 0, which gives us the desired. Then, the answer is either **(A)** or **(D)**, and since $3\pi > 3 \cdot 3 > 9$, the answer is at least $9 \cdot 2 + 1 = 19$, yielding **(D) 19**.

2. (AMC 10 10B 2021) What is the value of $\sqrt{(3 - 2\sqrt{3})^2} + \sqrt{(3 + 2\sqrt{3})^2}$?
- (A)** 0 **(B)** $4\sqrt{3} - 6$ **(C)** 6 **(D)** $4\sqrt{3}$ **(E)** $4\sqrt{3} + 6$

Answer: D

Solution 1

Note that the square root of any number squared is always the absolute value of the squared number because the square root function will only return a nonnegative number. By squaring both 3 and $2\sqrt{3}$, we see that $2\sqrt{3} > 3$, thus $3 - 2\sqrt{3}$ is negative, so we must take the absolute value of $3 - 2\sqrt{3}$, which is just $2\sqrt{3} - 3$. Knowing this, the first term in the expression equals $2\sqrt{3} - 3$ and the second term is $3 + 2\sqrt{3}$, and summing the two gives **(D) $4\sqrt{3}$** .

Solution 2

Let $x = \sqrt{(3 - 2\sqrt{3})^2} + \sqrt{(3 + 2\sqrt{3})^2}$, then $x^2 = (3 - 2\sqrt{3})^2 + 2\sqrt{(-3)^2} + (3 + 2\sqrt{3})^2$. The $2\sqrt{(-3)^2}$ term is there due to difference of squares when you simplify $2ab$ from $(a + b)^2$. Simplifying the expression gives us $x^2 = 48$, so $x = \boxed{\text{(D) } 4\sqrt{3}}$

shrungpatel

Solution 3 (Memorizing Roots)

Memorizing your square roots from 1 - 10 are really important for cheesing some AMC problems, so try to memorize them.

$\sqrt{3}$ is about 1.7.

We then substitute $\sqrt{3}$ for 1.7 to solve this.

We get

$$\begin{aligned} & \sqrt{(3 - 2 \cdot 1.7)^2} + \sqrt{(3 + 2 \cdot 1.7)^2} \\ &= \sqrt{(-0.4)^2} + \sqrt{(6.4)^2} \\ &= \sqrt{0.16} + \sqrt{6.4^2} \\ &= 0.4 + 6.4 \\ &= 6.8 \end{aligned}$$

Looking at the answer choices, we see that (D) $4\sqrt{3}$ gives 6.8 for when $\sqrt{3} = 1.7$.

3. (*AMC 10 10B 2021*) In an after-school program for juniors and seniors, there is a debate team with an equal number of students from each class on the team. Among the 28 students in the program, 25% of the juniors and 10% of the seniors are on the debate team. How many juniors are in the program?
- (A) 5 (B) 6 (C) 8 (D) 11 (E) 20

Answer: C

Solution 1

Say there are j juniors and s seniors in the program. Converting percentages to fractions, $\frac{j}{4}$ and $\frac{s}{10}$ are on the debate team, and since an equal number of juniors and seniors are on the debate team, $\frac{j}{4} = \frac{s}{10}$.

Cross-multiplying and simplifying we get $5j = 2s$. Additionally, since there are 28 students in the program, $j + s = 28$. It is now a matter of solving the system of equations

$$\begin{aligned} 5j &= 2s \\ j + s &= 28, \end{aligned}$$

and the solution is $j = 8, s = 20$. Since we want the number of juniors, the answer is

$$\boxed{(C) \ 8}.$$

-PureSwag

Solution 2 (Logic)

Clearly, the number of seniors must be a number divisible by 10, since 10% points to a whole number, so the number of seniors must be either 10 or 20. If there are ten seniors, there is 1 senior on the debate team, which would mean that there are

4 juniors, since $10\%s = 25\%j$. This is impossible, so the number of seniors is 20, and the number of juniors is $28 - 20 = 8 \rightarrow \boxed{C}$.

Note that by using the answer choices, we could automatically rule out 10.

Solution 3

Since there are an equal number of juniors and seniors on the debate team, suppose there are x juniors and x seniors. This number represents $25\% = \frac{1}{4}$ of the juniors and $10\% = \frac{1}{10}$ of the seniors, which tells us that there are $4x$ juniors and $10x$ seniors. There are 28 juniors and seniors in the program altogether, so we get

$$10x + 4x = 28,$$

$$14x = 28,$$

$$x = 2.$$

Which means there are $4x = 8$ juniors on the debate team, $\boxed{(C) 8}$.

Solution 4 (Elimination)

The amount of juniors must be a multiple of 4, since exactly $\frac{1}{4}$ of the students are on the debate team. Thus, we can immediately see that $\boxed{C}$ and $\boxed{E}$ are the only possibilities for the number of juniors. However, if there are 20 juniors, then there are 8 seniors, so it is not true that $1/10$ of the seniors are on the debate team, since $\frac{1}{10} \cdot 8 = \frac{4}{5}$, which is not an integer. Thus, we conclude that there are 8 juniors, so the answer is $\boxed{C}$.

4. (AMC 10 10B 2021) At a math contest, 57 students are wearing blue shirts, and another 75 students are wearing yellow shirts. The 132 students are assigned into 66 pairs. In exactly 23 of these pairs, both students are wearing blue shirts. In how many pairs are both students wearing yellow shirts?

(A) 23 (B) 32 (C) 37 (D) 41 (E) 64

Answer: B

Solution

There are 46 blue students paired with a blue partner. The other 11 students wearing blue shirts must each be paired with a partner wearing a shirt of the opposite color. There are 64 yellow students remaining. Therefore the requested number of pairs is $\frac{64}{2} = \boxed{(B) 32}$

5. (AMC 10 10B 2021) The ages of Jonie's four cousins are distinct single-digit positive integers. Two of the cousins' ages multiplied together give 24, while the other two multiply to 30. What is the sum of the ages of Jonie's four cousins?

(A) 21 (B) 22 (C) 23 (D) 24 (E) 25

Answer: B

Solution

First look at the two cousins' ages that multiply to 24. Since the ages must be single-digit, the ages must either be 3 and 8 or 4 and 6.

Next, look at the two cousins' ages that multiply to 30. Since the ages must be single-digit, the only ages that work are 5 and 6. Remembering that all the ages must all be distinct, the only solution that works is when the ages are 3, 8 and 5, 6.

We are required to find the sum of the ages, which is

$$3 + 8 + 5 + 6 = \boxed{\text{(B)} 22}.$$

-PureSwag

6. (AMC 10 10B 2021) Ms. Blackwell gives an exam to two classes. The mean of the scores of the students in the morning class is 84, and the afternoon class's mean score is 70. The ratio of the number of students in the morning class to the number of students in the afternoon class is $\frac{3}{4}$. What is the mean of the scores of all the students?

(A) 74 (B) 75 (C) 76 (D) 77 (E) 78

Answer: C

Solution 1 (One Variable)

Let there be $3x$ students in the morning class and $4x$ students in the afternoon class. The total number of students is $3x + 4x = 7x$. The average is $\frac{3x \cdot 84 + 4x \cdot 70}{7x} = 76$. Therefore, the answer is $\boxed{\text{(C)} 76}$.

Solution 2 (Two Variables)

Suppose the morning class has m students and the afternoon class has a students. We have the following table:

	# of Students	Mean	Total
Morning	m	84	$84m$
Afternoon	a	70	$70a$

We are also given that $\frac{m}{a} = \frac{3}{4}$, which rearranges as $m = \frac{3}{4}a$.

The mean of the scores of all the students is

$$\frac{84m + 70a}{m + a} = \frac{84\left(\frac{3}{4}a\right) + 70a}{\frac{3}{4}a + a} = \frac{133a}{\frac{7}{4}a} = 133 \cdot \frac{4}{7} = \boxed{\text{(C)} 76}.$$

MRENTHUSIASM

Solution 3 (Ratio)

Of the average, $\frac{3}{3+4} = \frac{3}{7}$ of the scores came from the morning class and $\frac{4}{7}$ came from the afternoon class. The average is $\frac{3}{7} \cdot 84 + \frac{4}{7} \cdot 70 = \boxed{\text{(C)} 76}$.

Solution 4 (Convenient Values)

WLOG, assume there are 3 students in the morning class and 4 in the afternoon class. Then the average is $\frac{3 \cdot 84 + 4 \cdot 70}{7} = \boxed{\text{(C)} 76}$.

Solution 5 (Basic manipulation)

Let S_1 be the sum of the morning class's scores, and S_2 be the sum of the afternoon class's scores. Let $3x$ be the number of students in the morning class, and $4x$ be the number of students in the afternoon class. We can write $\frac{S_1}{3x} = 84$ and $\frac{S_2}{4x} = 70$, so $S_2 = 70 \cdot 4x$ and $S_1 = 84 \cdot 3x$. Adding these, we get $S_1 + S_2 = x(70 \cdot 4 + 84 \cdot 3)$. We want to find the mean, which is $\frac{S_1 + S_2}{7x}$, so dividing by $7x$ on both sides of the equation, we get the mean to be $10 \cdot 4 + 12 \cdot 3 = \boxed{\text{(C)} 76}$.

7. (AMC 10 10B 2021) In a plane, four circles with radii 1, 3, 5, and 7 are tangent to line ℓ at the same point A , but they may be on either side of ℓ . Region S consists of all the points that lie inside exactly one of the four circles. What is the maximum possible area of region S ?

(A) 24π (B) 32π (C) 64π (D) 65π (E) 84π

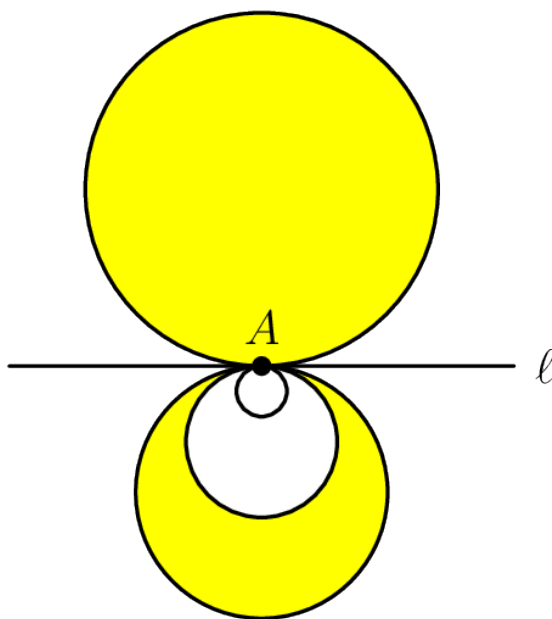
Answer: D

Solution

Suppose that line ℓ is horizontal, and each circle lies either north or south to ℓ . We construct the circles one by one:

- Without the loss of generality, we draw the circle with radius 7 north to ℓ .
- To maximize the area of region S , we draw the circle with radius 5 south to ℓ .
- Now, we need to subtract the circle with radius 3 **at least**. The optimal situation is that the circle with radius 3 encompasses the circle with radius 1, in which we do not need to subtract more. That is, the two smallest circles are on the same side of ℓ , but can be on either side.

The diagram below shows one possible configuration of the four circles:



Together, the answer is $\pi \cdot 7^2 + \pi \cdot 5^2 - \pi \cdot 3^2 = \boxed{\text{(D)} 65\pi}$.

8. (AMC 10 10B 2021) Mr. Zhou places all the integers from 1 to 225 into a 15 by 15 grid. He places 1 in the middle square (eighth row and eighth column) and places other numbers one by one clockwise, as shown in part in the diagram below. What is the sum of the greatest number and the least number that appear in the second row from the top?

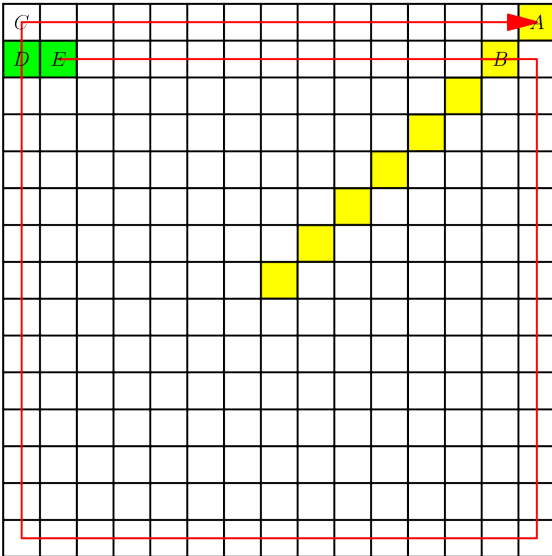
...	...	...	...	...	...	...
...	21	22	23	24	25	...
...	20	7	8	9	10	...
...	19	6	1	2	11	...
...	18	5	4	3	12	...
...	17	16	15	14	13	...
...	...	...	...	...	...	...

- (A) 367 (B) 368 (C) 369 (D) 379 (E) 380

Answer: A

Solution 1 (Observations and Patterns: Considers Only 5 Squares)

In the diagram below, the red arrows indicate the progression of numbers. In the second row from the top, the greatest number and the least number are D and E , respectively. Note that the numbers in the yellow cells are consecutive odd perfect squares, as we can prove by induction.



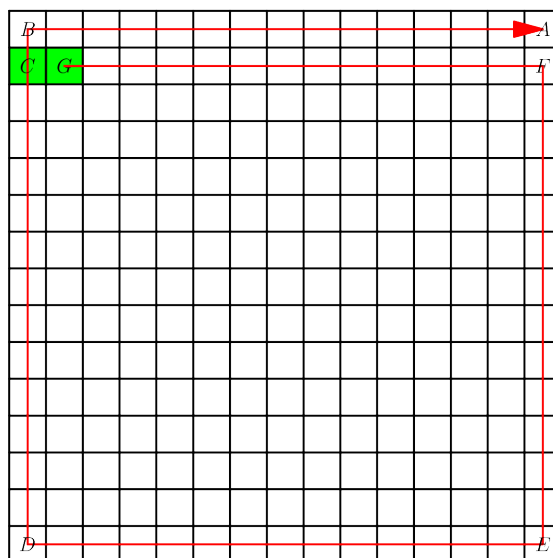
By observations, we proceed as follows:

$$\begin{aligned} A = 15^2 = 225, \ B = 13^2 = 169 &\implies C = A - 14 = 211 \\ &\implies D = C - 1 = 210 \\ &\implies E = B - 12 = 157. \end{aligned}$$

Therefore, the answer is $D + E = \boxed{\text{(A)} 367}$.

Solution 2 (Observations and Patterns: Considers Only 7 Squares)

In the diagram below, the red arrows indicate the progression of numbers. In the second row from the top, the greatest number and the least number are C and G , respectively.



By observations, we proceed as follows:

$$\begin{aligned}
 A = 15^2 = 225 &\implies B = A - 14 = 211 \\
 &\implies C = B - 1 = 210 \\
 &\implies D = C - 13 = 197 \\
 &\implies E = D - 14 = 183 \\
 &\implies F = E - 13 = 170 \\
 &\implies G = F - 13 = 157.
 \end{aligned}$$

Therefore, the answer is $C + G = \boxed{\text{(A)} 367}$.

Solution 3 (Brute Force: Draws All 225 Squares Out)

From the full diagram below, the answer is $210 + 157 = \boxed{\text{(A)} 367}$.

211	212	213	214	215	216	217	218	219	220	221	222	223	224	225
210	157	158	159	160	161	162	163	164	165	166	167	168	169	170
209	156	111	112	113	114	115	116	117	118	119	120	121	122	171
208	155	110	73	74	75	76	77	78	79	80	81	82	123	172
207	154	109	72	43	44	45	46	47	48	49	50	83	124	173
206	153	108	71	42	21	22	23	24	25	26	51	84	125	174
205	152	107	70	41	20	7	8	9	10	27	52	85	126	175
204	151	106	69	40	19	6	1	2	11	28	53	86	127	176
203	150	105	68	39	18	5	4	3	12	29	54	87	128	177
202	149	104	67	38	17	16	15	14	13	30	55	88	129	178
201	148	103	66	37	36	35	34	33	32	31	56	89	130	179
200	147	102	65	64	63	62	61	60	59	58	57	90	131	180
199	146	101	100	99	98	97	96	95	94	93	92	91	132	181
198	145	144	143	142	141	140	139	138	137	136	135	134	133	182
197	196	195	194	193	192	191	190	189	188	187	186	185	184	183

This solution is extremely time-consuming and error-prone. Please do not try it in a real competition unless you have no other options.

9. (AMC 10 10B 2021) The point $P(a, b)$ in the xy -plane is first rotated counter-clockwise by 90° around the point $(1, 5)$ and then reflected about the line $y = -x$. The image of P after these two transformations is at $(-6, 3)$. What is $b - a$?
 (A) 1 (B) 3 (C) 5 (D) 7 (E) 9

Answer: D

Solution 1 (Transformation Rules)

The final image of P is $(-6, 3)$. We know the reflection rule for reflecting over $y = -x$ is $(x, y) \rightarrow (-y, -x)$. So before the reflection and after rotation the point is $(-3, 6)$.

By definition of rotation, the slope between $(-3, 6)$ and $(1, 5)$ must be perpendicular to the slope between (a, b) and $(1, 5)$. The first slope is $\frac{5-6}{1-(-3)} = \frac{-1}{4}$. This means the slope of P and $(1, 5)$ is 4.

Rotations also preserve distance to the center of rotation, and since we only "travelled" up and down by the slope once to get from $(-3, 6)$ to $(1, 5)$ it follows we shall only use the slope once to travel from $(1, 5)$ to P .

Therefore point P is located at $(1 + 1, 5 + 4) = (2, 9)$. The answer is $9 - 2 = 7 =$
(D) 7.

-abhinavg0627

Solution 2 (Complex Numbers)

Let us reconstruct that coordinate plane as the complex plane. Then, the point $P(a, b)$ becomes $a + b \cdot i$. A 90° rotation around the point $(1, 5)$ can be done by translating the point $(1, 5)$ to the origin, rotating around the origin by 90° , and then translating the origin back to the point $(1, 5)$.

$$a + b \cdot i \implies (a-1) + (b-5) \cdot i \implies ((a-1) + (b-5) \cdot i) \cdot i = 5 - b + (a-1)i \implies 5 + 1 - b + (a-1+5)i = 6 - b + (a+4)i$$

By basis reflection rules, the reflection of $(-6, 3)$ about the line $y = -x$ is $(-3, 6)$. Hence, we have

$$6 - b + (a + 4)i = -3 + 6i \implies b = 9, a = 2,$$

from which $b - a = 9 - 2 =$ (D) 7.

Solution 3 (Reverso)

The problem gives a series of transformations and proceeds to give the resultant point, being $(-6, 3) = P$. Therefore, all we must do is reverse it. First, we reverse the last transformation by computing the distance from the point $(-6, 3)$ to $y = -x$ by using the formula $d = \frac{|Ax + By + C|}{\sqrt{A^2 + B^2}}$. Where $Ax + By + C$ is the standard form of a line. Computing, we obtain that $d = \frac{3\sqrt{2}}{2}$. We also know that this is magnitude is at an angle of $\frac{\pi}{4}$. Therefore, to do the transformation, we double that vector and add it to the point. We get:

$$P' = P + 2\vec{v} \implies P' = (-6, 3) + \left\langle 3\sqrt{2} \cdot \frac{\sqrt{2}}{2}, 3\sqrt{2} \cdot \frac{\sqrt{2}}{2} \right\rangle \implies P' = (-3, 6).$$

Now, we must reverse the second transformation. To do so, realize that $P' - (1, 5) \Leftrightarrow \vec{v}_2 = \langle -4, 1 \rangle$. Simply make this vector perpendicular by switching the x and y

components and switching the sign of the initial y component. Therefore, we get $\overrightarrow{v_{2\perp}} = \langle 1, 4 \rangle$. Therefore, adding this vector to $(1, 5)$ yields $(2, 9)$, which leads us to $9 - 2 = \boxed{\text{(D)} 7}$.

Solution 4 (Vector Dot Product)

Using the same method as in Solution 1, we can obtain that the point before the reflection is $(-3, 6)$. If we let the original point be (x, y) , then we can use that the starting point is $(1, 5)$ to obtain two vectors $\langle -4, 1 \rangle$ and $\langle x - 1, y - 5 \rangle$. We know that two vectors are perpendicular if their dot product is equal to 0, and that both points are the same distance $(\sqrt{17})$ from $(1, 5)$.

Therefore, we can write two equations using these vectors: $(x - 1)^2 + (y - 5)^2 = 17$ (from distance and pythagorean theorem) and $-4x + y - 1 = 0$ (from dot product)

Solving, we simplify the second equation to $y = 4x + 1$, and plug it into the first equation. We obtain $(x - 1)^2 + (4x - 4)^2 = 17$. We can simplify this to the quadratic $17x^2 - 34x = 0$. When we factor out $17x$, we find that $x = 2$ or $x = 0$. However, x cannot equal 0. Therefore, $x = 2$, and plugging this into the second equation gives us that $y = 9$. Since the point is $(9, 2)$, we compute $9 - 2 = \boxed{\text{(D)} 7}$.

Solution 5 (Vector Dot Product scuffed version)

Using the same method as in Solution 1 reflecting $(-6, 3)$ about the line $y = -x$ gives us $(-3, 6)$.

Let the original point be $\langle x, y \rangle$. From point $(1, 5)$, we form the vectors $\langle -4, 1 \rangle$ and $\langle x - 1, y - 5 \rangle$ that extend out from the initial point. If they are perpendicular, we know that their dot product has to equal zero. Therefore,

$$\langle -4, 1 \rangle \cdot \langle x - 1, y - 5 \rangle = 0 \implies -4x + y - 1 = 0.$$

Now, we have to do some guess and check from the multiple choices. Let $y - x = A$ where A is one of the answer choices. Then, $A - 3x = 1$. By intuition and logical reasoning we deduce that A must be 1 (mod 3) so that brings our potential answers down to **(A)** and **(C)**. If $A = 1$ from **(A)**, then $x = 0$, which we can quickly rule out since we know that P rotated counterclockwise not clockwise. Hence, **(D) 7** is the answer.

Solution 6 (Working Backwards Approach, Super Easy!)

The last thing that happened was the reflection over the line $y = -x$, so we reverse that by reflecting $(-6, 3)$ over $y = -x$ to get $(-3, 6)$. Now, we have to rotate it **clockwise**, to reverse the first step. To rotate $(-3, 6)$ over the point $(1, 5)$ clockwise 90 degrees, we can subtract 1 from each x-coordinate and subtract 5 from each y-coordinate. That way, we only have to rotate $(-4, 1)$ clockwise 90 degrees over the origin $(0, 0)$. This is pretty easy to do in your head, so you get $(1, 4)$. But we are not done, we still have to add 1 to each x-coordinate and add 5 to each y-coordinate, to get $(2, 9)$, so our final answer is $9 - 2 = 7$, which is **(D) 7**.

10. (AMC 10 10B 2021) An inverted cone with base radius 12cm and height 18cm is full of water. The water is poured into a tall cylinder whose horizontal base has a radius of 24cm. What is the height in centimeters of the water in the cylinder?

(A) 1.5 (B) 3 (C) 4 (D) 4.5 (E) 6

Answer: A

Solution 1

The volume of a cone is $\frac{1}{3} \cdot \pi \cdot r^2 \cdot h$ where r is the base radius and h is the height. The water completely fills up the cone so the volume of the water is $\frac{1}{3} \cdot 18 \cdot 144\pi = 6 \cdot 144\pi$. The volume of a cylinder is $\pi \cdot r^2 \cdot h$ so the volume of the water in the cylinder would be $24 \cdot 24 \cdot \pi \cdot h$.

We can equate these two expressions because the water volume stays the same like this $24 \cdot 24 \cdot \pi \cdot h = 6 \cdot 144\pi$. We get $4h = 6$ and $h = \frac{6}{4}$.

So the answer is (A) 1.5.

Solution 2

The water completely fills up the cone. For now, assume the radius of both cone and cylinder are the same. Then the cone has $\frac{1}{3}$ of the volume of the cylinder, and so the height is divided by 3. Then, from the problem statement, the radius is doubled, meaning the area of the base is quadrupled (since $2^2 = 4$).

Therefore, the height is divided by 3 and divided by 4, which is $18 \div 3 \div 4 = \span style="border: 1px solid black; padding: 2px;">(A) 1.5.$

11. (AMC 10 10B 2021) Grandma has just finished baking a large rectangular pan of brownies. She is planning to make rectangular pieces of equal size and shape, with straight cuts parallel to the sides of the pan. Each cut must be made entirely across the pan. Grandma wants to make the same number of interior pieces as pieces along the perimeter of the pan. What is the greatest possible number of brownies she can produce?

(A) 24 (B) 30 (C) 48 (D) 60 (E) 64

Answer: D

Solution 1

Let the side lengths of the rectangular pan be m and n . It follows that $(m - 2)(n - 2) = \frac{mn}{2}$, since half of the brownie pieces are in the interior. This gives $2(m - 2)(n - 2) = mn \iff mn - 4m - 4n + 8 = 0$. Adding 8 to both sides and applying **Simon's Favorite Factoring Trick**, we obtain $(m - 4)(n - 4) = 8$. Since m and n are both positive, we obtain $(m, n) = (5, 12), (6, 8)$ (up to ordering). By inspection, $5 \cdot 12 = \span style="border: 1px solid black; padding: 2px;">(D) 60 maximizes the number of brownies.$

Solution 2

Obviously, no side of the rectangular pan can have less than 5 brownies beside it. We let one side of the pan have 5 brownies, and let the number of brownies on its adjacent side be x . Therefore, $5x = 2 \cdot 3(x - 2)$, and solving yields $x = 12$ and there are $5 \cdot 12 = 60$ brownies in the pan. 64 is the only choice larger than 60, but it cannot be the answer since the only way to fit 64 brownies in a pan without letting a side of it have less than 5 brownies beside it is by forming a square of 8 brownies on each side, which does not meet the requirement. Thus the answer is (D) 60.

-SmileKat32

Solution 3 (rational function)

Let the **interior** of the pan have n vertical cuts and m horizontal cuts. We see that $mn = 2(m + n) + 4$. We want the greatest possible value of $(m + 2)(n + 2) = 2mn$. Expanding the first equation we get

$$mn = 2m + 2n + 4 \implies (m - 2)n = 2m + 4 \implies n = \frac{2m + 4}{m - 2}.$$

In the first quadrant, this is a rational function which decreases as m approaches infinity. So we want the smallest value for m for the largest value of n . We know $m \geq 3$. If $m = 3, n = 10$. And we know since the function only decreases from here, these must be the greatest possible values. So the answer is $2mn = 2 \cdot 30 = \boxed{60}$.

12. (AMC 10 10B 2021) Let $N = 34 \cdot 34 \cdot 63 \cdot 270$. What is the ratio of the sum of the odd divisors of N to the sum of the even divisors of N ?

(A) 1 : 16 (B) 1 : 15 (C) 1 : 14 (D) 1 : 8 (E) 1 : 3

Answer: C

Solution 1

Prime factorize N to get $N = 2^3 \cdot 3^5 \cdot 5 \cdot 7 \cdot 17^2$. For each odd divisor n of N , there exist even divisors $2n, 4n, 8n$ of N , therefore the ratio is $1 : (2+4+8) = \boxed{\text{(C) } 1 : 14}$

Solution 2

Prime factorizing N , we see $N = 2^3 \cdot 3^5 \cdot 5 \cdot 7 \cdot 17^2$. The sum of N 's odd divisors are the sum of the factors of N without 2, and the sum of the even divisors is the sum of the odds subtracted by the total sum of divisors.

BY SUM OF FACTORS FORMULA (search it up if you don't know): The formula is actually for all factors, but we can just take out the 2^3 , so we have:

The sum of odd divisors is given by

$$a = (1 + 3 + 3^2 + 3^3 + 3^4 + 3^5)(1 + 5)(1 + 7)(1 + 17 + 17^2)$$

and the total sum of divisors is

$$(1 + 2 + 4 + 8)(1 + 3 + 3^2 + 3^3 + 3^4 + 3^5)(1 + 5)(1 + 7)(1 + 17 + 17^2) = 15a.$$

Thus, our ratio is

$$\frac{a}{15a - a} = \frac{a}{14a} = \boxed{\text{(C) } 1 : 14}.$$

Solution 3

Prime factorizing N , we have that there is 2^3 in our factorization. Now, call the sum of the odd divisors k . We know that if we multiply k by 2, we will have even divisors. So, we can multiply k by 2, $2^2, 2^3$ respectively to get $14k$ as the sum of the even divisors. Therefore, the answer is

$$\frac{k}{14k} = \boxed{\text{(C) } 1 : 14}.$$

MC

13. (AMC 10 10B 2021) Let n be a positive integer and d be a digit such that the value of the numeral $\underline{32d}$ in base n equals 263, and the value of the numeral $\underline{324}$ in base n equals the value of the numeral $\underline{11d1}$ in base six. What is $n + d$?
(A) 10 (B) 11 (C) 13 (D) 15 (E) 16

Answer: B

Solution 1

We can start by setting up an equation to convert $\underline{32d}$ base n to base 10. To convert this to base 10, it would be $3n^2 + 2n + d$. Because it is equal to 263, we can set this equation to 263. Finally, subtract d from both sides to get $3n^2 + 2n = 263 - d$.

We can also set up equations to convert $\underline{324}$ base n and $\underline{11d1}$ base 6 to base 10. The equation to convert $\underline{324}$ base n to base 10 is $3n^2 + 2n + 4$. The equation to convert $\underline{11d1}$ base 6 to base 10 is $6^3 + 6^2 + 6d + 1$.

Simplify $6^3 + 6^2 + 6d + 1$ so it becomes $6d + 253$. Setting the above equations equal to each other, we have

$$3n^2 + 2n + 4 = 6d + 253.$$

Subtracting 4 from both sides gets $3n^2 + 2n = 6d + 249$.

We can then use equations

$$3n^2 + 2n = 263 - d$$

$$3n^2 + 2n = 6d + 249$$

to solve for d . Set $263 - d$ equal to $6d + 249$ and solve to find that $d = 2$.

Plug $d = 2$ back into the equation $3n^2 + 2n = 263 - d$. Subtract 261 from both sides to get your final equation of $3n^2 + 2n - 261 = 0$. We solve using the quadratic formula to find that the solutions are 9 and $-29/3$. Because the base must be positive, $n = 9$.

Adding 2 to 9 gets (B) 11

-Zeusthemoose (edited for readability)-solution corrected by Billowingsweater

Solution 2

$32d$ is greater than 263 when both are interpreted in base 10, so n is less than 10. Some trial and error gives $n = 9$. 263 in base 9 is 322 , so the answer is $9 + 2 = \text{span style="border: 1px solid black; padding: 2px;">(B) 11.$

-SmileKat32

Solution 3

We have

$$3n^2 + 2n + d = 263$$

$$3n^2 + 2n + 4 = 6^3 + 6^2 + 6d + 1$$

Subtracting the 2nd from the 1st equation we get

$$\begin{aligned} d - 4 &= 263 - (216 + 36 + 6d + 1) \\ &= 263 - 253 - 6d \\ &= 10 - 6d \end{aligned}$$

Thus we have $d = 2$. Substituting into the first, we have $3n^2 + 2n + 2 = 263 \Rightarrow 3n^2 + 2n - 261 = 0$. Factoring, we have $(n - 9)(3n + 29) = 0$. A digit cannot be negative, so we have $n = 9$. Thus, $d + n = 2 + 9 = \text{span style="border: 1px solid black; padding: 2px;">(B) 11$

mathboy282 signing off $\text{\LaTeX}$ fixed by Lamboreghini

Solution 4

Find that $d = 2$ using one of the methods above. Then we have that $3n^2 + 2n = 261$. We know that n is an integer, so we can solve the equation $n(3n + 2) = 261$ (this is guaranteed to have a solution if we did this correctly). The prime factorization of 261 is $3^2 \cdot 29$, so the corresponding factor pairs are $(1, 261)$, $(3, 87)$, and $(9, 29)$. If $n = 9$, then the equation is true, so $n + d = 9 + 2 = \text{span style="border: 1px solid black; padding: 2px;">(B) 11.$

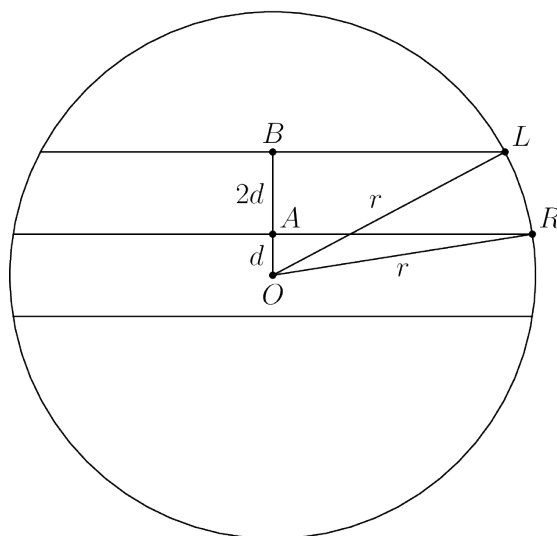
Note: This solution provides an alternate way to complete the final part of the problem if, like me, you really dislike quadratics. cxsmi

14. (AMC 10 10B 2021) Three equally spaced parallel lines intersect a circle, creating three chords of lengths 38, 38, and 34. What is the distance between two adjacent parallel lines?

(A) $5\frac{1}{2}$ (B) 6 (C) $6\frac{1}{2}$ (D) 7 (E) $7\frac{1}{2}$

Answer: B

Solution 1 (Pythagorean Theorem)



Since two parallel chords have the same length (38), they must be equidistant from the center of the circle. Let the perpendicular distance of each chord from the center of the circle be d . Thus, the distance from the center of the circle to the chord of length 34 is

$$2d + d = 3d$$

and the distance between each of the chords is just $2d$. Let the radius of the circle be r . Drawing radii to the points where the lines intersect the circle, we create two different right triangles:

- One with base $\frac{38}{2} = 19$, height d , and hypotenuse r ($\triangle RAO$ on the diagram)
- Another with base $\frac{34}{2} = 17$, height $3d$, and hypotenuse r ($\triangle LBO$ on the diagram)

By the Pythagorean theorem, we can create the following system of equations:

$$19^2 + d^2 = r^2$$

$$17^2 + (2d + d)^2 = r^2$$

Solving, we find $d = 3$, so $2d = \boxed{\text{(B) } 6}$.

Solution 2 (Coordinates)

Because we know that the equation of a circle is $(x - a)^2 + (y - b)^2 = r^2$ where the center of the circle is (a, b) and the radius is r , we can find the equation of this circle by centering it on the origin. Doing this, we get that the equation is

$x^2 + y^2 = r^2$. Now, we can set the distance between the chords as $2d$ so the distance from the chord with length 38 to the diameter is d .

Therefore, the following points are on the circle as the y -axis splits the chord in half, that is where we get our x value:

$(19, d)$ $(19, -d)$ $(17, -3d)$

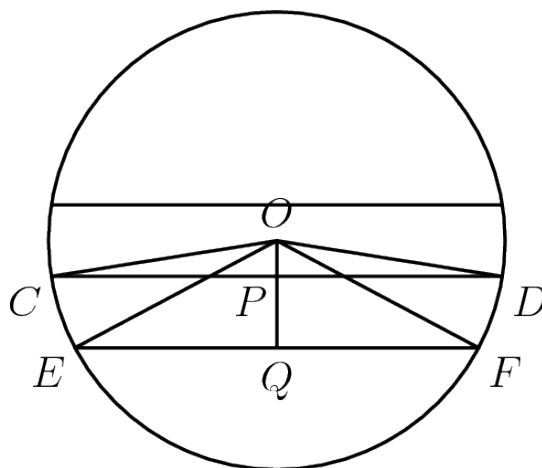
Now, we can plug one of the first two value in as well as the last one to get the following equations:

$$19^2 + d^2 = r^2$$

$$17^2 + (3d)^2 = r^2$$

Subtracting these two equations, we get $19^2 - 17^2 = 8d^2$ - therefore, we get $72 = 8d^2 \rightarrow d^2 = 9 \rightarrow d = 3$. We want to find $2d = 6$ because that's the distance between two chords. So, our answer is $\boxed{B}$.

Solution 3 (Stewart's Theorem)



If d is the requested distance, and r is the radius of the circle, Stewart's Theorem applied to $\triangle OCD$ with cevian $\overrightarrow{OP}$ gives

$$19 \cdot 38 \cdot 19 + \frac{1}{2}d \cdot 38 \cdot \frac{1}{2}d = 19r^2 + 19r^2.$$

This simplifies to $13718 + \frac{19}{2}d^2 = 38r^2$. Similarly, another round of Stewart's Theorem applied to $\triangle OEF$ with cevian $\overrightarrow{OQ}$ gives

$$17 \cdot 34 \cdot 17 + \frac{3}{2}d \cdot 34 \cdot \frac{3}{2}d = 17r^2 + 17r^2.$$

This simplifies to $9826 + \frac{153}{2}d^2 = 34r^2$. Dividing the top equation by 38 and the bottom equation by 34 results in the system of equations

$$\begin{aligned} 361 + \frac{1}{4}d^2 &= r^2 \\ 289 + \frac{9}{4}d^2 &= r^2 \end{aligned}$$

By transitive, $361 + \frac{1}{4}d^2 = 289 + \frac{9}{4}d^2$. Therefore $(\frac{9}{4} - \frac{1}{4})d^2 = 361 - 289 \rightarrow 2d^2 = 72 \rightarrow d^2 = 36 \rightarrow d = \boxed{\text{(B)} 6}$.

15. (AMC 10 10B 2021) The real number x satisfies the equation $x + \frac{1}{x} = \sqrt{5}$. What is the value of $x^{11} - 7x^7 + x^3$? (A) -1 (B) 0 (C) 1 (D) 2 (E) $\sqrt{5}$

Answer: B

Solution 1

We square $x + \frac{1}{x} = \sqrt{5}$ to get $x^2 + 2 + \frac{1}{x^2} = 5$. We subtract 2 on both sides for $x^2 + \frac{1}{x^2} = 3$ and square again, and see that $x^4 + 2 + \frac{1}{x^4} = 9$ so $x^4 + \frac{1}{x^4} = 7$. We can factor out x^7 from our original expression of $x^{11} - 7x^7 + x^3$ to get that it is equal to $x^7(x^4 - 7 + \frac{1}{x^4})$. Therefore because $x^4 + \frac{1}{x^4}$ is 7, it is equal to $x^7(0) = \boxed{\text{(B)} 0}$.

Solution 2

Multiplying both sides by x and using the quadratic formula, we get

16. (AMC 10 10B 2021) Call a positive integer an uphill integer if every digit is strictly greater than the previous digit. For example, 1357, 89, and 5 are all uphill integers, but 32, 1240, and 466 are not. How many uphill integers are divisible by 15?
(A) 4 (B) 5 (C) 6 (D) 7 (E) 8

Answer: C

Solution 1

The divisibility rule of 15 is that the number must be congruent to 0 mod 3 and congruent to 0 mod 5. Being divisible by 5 means that it must end with a 5 or a 0. We can rule out the case when the number ends with a 0 immediately because the only integer that is uphill and ends with a 0 is 0 which is not positive. So now we know that the number ends with a 5. Looking at the answer choices, the answer choices are all pretty small, so we can generate all of the numbers that are uphill and are divisible by 3. These numbers are 15, 45, 135, 345, 1245, 12345, or $\boxed{\text{(C)} 6}$ numbers.

Solution 2

First, note how the number must end in either 5 or 0 in order to satisfying being divisible by 15. However, the number can't end in 0 because it's not strictly greater than the previous digits. Thus, our number must end in 5. We do casework on the number of digits.

Case 1: 1 digit. No numbers work, so 0 numbers.

Case 2: 2 digits. We have the numbers 15, 45, and 75, but 75 isn't an uphill number, so 2 numbers

Case 3: 3 digits. We have the numbers 135, 345, so 2 numbers.

Case 4: 4 digits. We have the numbers 1235, 1245 and 2345, but only 1245 satisfies this condition, so 1 number.

Case 5: 5 digits. We have only 12345, so 1 number.

Adding these up, we have $2 + 2 + 1 + 1 = \boxed{\text{(C)} 6}$.

Solution 3

Like solution 2, we can proceed by using casework. A number is divisible by 15 if it is divisible by 3 and 5. In this case, the units digit must be 5, otherwise no number can be formed.

Case 1: sum of digits = 6

There is only one number, 15.

Case 2: sum of digits = 9

There are two numbers: 45 and 135.

Case 3: sum of digits = 12

There are two numbers: 345 and 1245.

Case 4: sum of digits = 15

There is only one number, 12345.

We can see that we have exhausted all cases, because in order to have a larger sum of digits, then a number greater than 5 needs to be used, breaking the conditions of the problem. The answer is $\boxed{(C) 6}$.

Solution 4

An integer is divisible by 15 if it is divisible by 3 and 5. Divisibility by 5 means ending in 0 or 5, but since no digit is less than 0, the only uphill integer ending in 0 could be 0, which is not positive. This means the integer must end in 5.

All uphill integers ending in 5 are formed by picking any subset of the sequence (1, 2, 3, 4) of digits (keeping their order), then appending a 5. Divisibility by 3 means the sum of the digits is a multiple of 3, so our choice of digits must add to 0 modulo 3.

$5 \equiv -1 \pmod{3}$, so the other digits we pick must add to 1 modulo 3. Since $(1, 2, 3, 4) \equiv (1, -1, 0, 1) \pmod{3}$, we can pick either nothing, or one residue 1 (from 1 or 4) and one residue -1 (from 2), and we can then optionally add a residue 0 (from 3). This gives $(1 + 2 \cdot 1) \cdot 2 = \boxed{(C) 6}$ possibilities.

Solution 5 (Solution 2 but more in depth)

First, by intuition note how the number must end in either 5 or 0 in order to satisfying being divisible by 15. However, the number can't end in 0 because it's not greater than the previous digits and it has to be positive. Thus, our number must end in 5. Now we can do casework on the number of digits.

Case 1: 1 digit. Only 0 is a 1 digit number that is divisible by 15 but it must be positive, so 0 numbers.

Case 2: 2 digits. We have to find numbers that end in 5 and is divisible by 3, so we get 15, 45, and 75, but 75 isn't an uphill number, so we only have 2 numbers.

Case 3: 3 digits. Finding 3 digits is the same as removing 2 digits that add to a multiple of 3 from 12345, so we can remove 24 and 15 becoming 135 and 234 respectively, so we have 2 numbers.

Case 4: 4 digits. This is the same as removing one multiple of 3 from 12345 so removing 3 becomes 1245, so we have 1 number.

Case 5: 5 digits. We have only 12345, so 1 number.

Adding these up, we have $2 + 2 + 1 + 1 = \boxed{(C) 6}$.

Note: This is my first solution so please forgive me for any errors, thank you.

Note The number of "uphill" positive integers as specified in the problem is 511.

17. (AMC 10 10B 2021) Ravon, Oscar, Aditi, Tyrone, and Kim play a card game. Each person is given 2 cards out of a set of 10 cards numbered $1, 2, 3, \dots, 10$. The score of a player is the sum of the numbers of their cards. The scores of the players are as follows: Ravon–11, Oscar–4, Aditi–7, Tyrone–16, Kim–17. Which of the following statements is true?
- (A) Ravon was given card 3. (B) Aditi was given card 3. (C) Ravon was given card 4. (D) Aditi was given card 4. (E) Tyrone was given card 7.

Answer: C

Solution 1 (Logical Deduction)

By logical deduction, we consider the scores from lowest to highest:

Oscar's score is 4. $\implies$ Oscar is given cards 1 and 3.
 $\implies$ Aditi is given cards 2 and 5.
 $\implies$ Ravon is given cards 4 and 7. (★)
 $\implies$ Tyrone is given cards 6 and 10.
 $\implies$ Kim is given cards 8 and 9.

Therefore, the answer is (C) Ravon was given card 4.

Certainly, if we read the answer choices sooner, then we can stop at (★) and pick (C).

Solution 2 (Elimination)

Rather than figuring out each person's cards individually, we can use elimination. Let's eliminate answer choices (B) and (D) since if one is true, then the other also must (Aditi's cards add up to 7). However, since there is only one solution, it holds true that neither of these can be true. We also know that Oscar's cards added up to 4, and since he can't have two of the same, he must have a 1 and a 3, eliminating (A). Furthermore, if Tyrone got a 7, his other card must be a 9, which makes it impossible for Kim to have 17 points with the remaining cards. Thus, the only option remaining is (C) Ravon was given card 4.

18. (AMC 10 10B 2021) A fair 6-sided die is repeatedly rolled until an odd number appears. What is the probability that every even number appears at least once before the first occurrence of an odd number?
- (A) $\frac{1}{120}$ (B) $\frac{1}{32}$ (C) $\frac{1}{20}$ (D) $\frac{3}{20}$ (E) $\frac{1}{6}$

Answer: C

Solution 1

Since 3 out of 6 of the numbers are even, there is a $\frac{3}{6}$ chance that the first number we choose is even.

Since the number rolled first is irrelevant (see note on bottom), we don't have to consider it. Therefore there are 2 even numbers out of the 5 choices left. There is a $\frac{2}{5}$ chance that the next number that is distinct from the first is even.

There is a $\frac{1}{4}$ chance that the next number distinct from the first two is even. (There is only one even integer left.) With all the even integers taken, the next integer rolled must be odd.

$\frac{3}{6} \cdot \frac{2}{5} \cdot \frac{1}{4} = \frac{1}{20}$, so the answer is $\boxed{(C) \frac{1}{20}}$.

NOTE: The first number is irrelevant because if it is a odd number, then the probability is automatically 0, not adding or subtracting from the total probability.

Solution 2

Every set of three numbers chosen from $\{1, 2, 3, 4, 5, 6\}$ has an equal chance of being the first 3 distinct numbers rolled.

Therefore, the probability that the first 3 distinct numbers are $\{2, 4, 6\}$ is $\frac{1}{\binom{6}{3}} =$

$$\boxed{(C) \frac{1}{20}}$$

Solution 3

Note that the problem is basically asking us to find the probability that in some permutation of 1, 2, 3, 4, 5, 6 that we get the three even numbers in the first three spots.

There are $6!$ ways to order the 6 numbers and $3!(3!)$ ways to order the evens in the first three spots and the odds in the next three spots.

Therefore the probability is $\frac{3!(3!)}{6!} = \frac{1}{20} = \boxed{(C)}$.

Solution 4

Let P_n denote the probability that the first odd number appears on roll n and all our conditions are met. We now proceed with complementary counting.

For $n \leq 3$, it's impossible to have all 3 evens appear before an odd. Note that for $n \geq 4$,

$$P_n = \frac{1}{2^n} - \frac{1}{2^n} \left(\frac{\binom{3}{2}(2^{n-1} - 2) + \binom{3}{1}}{3^{n-1}} \right)$$

since there's a $\frac{1}{2^n}$ chance that the first odd appears on roll n (disregarding the other conditions) and the other term is subtracting the probability that less than 3 of the evens show up before the first odd roll. Simplifying, we arrive at

$$P_n = \frac{1}{2^n} - \left(\frac{3(2^{n-1}) - 3}{2^n \cdot 3^{n-1}} \right) = \frac{1}{2^n} - \frac{1}{2 \cdot 3^{n-2}} + \frac{1}{2^n \cdot 3^{n-2}}.$$

Summing for all n , we get our answer of

$$\left(\frac{1}{2^4} + \frac{1}{2^5} + \dots \right) - \left(\frac{1}{2 \cdot 3^2} + \frac{1}{2 \cdot 3^3} + \dots \right) + \left(\frac{1}{2^4 \cdot 3^2} + \frac{1}{2^5 \cdot 3^3} + \dots \right) = \left(\frac{1}{8} \right) - \left(\frac{\frac{1}{18}}{\frac{2}{3}} \right) + \left(\frac{\frac{1}{144}}{\frac{5}{6}} \right) = \frac{1}{8} - \frac{1}{12} + \frac{1}{24} = \frac{1}{24}$$

Solution 5 (States)

Let E_n be that probability that the condition in the problem is satisfied given that we need n more distinct even numbers. Then,

$$E_1 = \frac{1}{6} + \frac{1}{3} \cdot E_1 + \frac{1}{2} \cdot 0,$$

since there is a $\frac{1}{3}$ probability that we will roll an even number we already have rolled and will be in the same position again. Solving, we find that $E_1 = \frac{1}{4}$.

We can apply the same concept for E_2 and E_3 . We find that

$$E_2 = \frac{1}{3} \cdot E_1 + \frac{1}{6} \cdot E_2 + \frac{1}{2} \cdot 0,$$

and so $E_2 = \frac{1}{10}$. Also,

$$E_3 = \frac{1}{2} \cdot E_2 + \frac{1}{2} \cdot 0,$$

so $E_3 = \frac{1}{20}$. Since the problem is asking for E_3 , our answer is (C) $\frac{1}{20}$. -BorealBear

Solution 6 (Infinite Geometric Sequence Method)

Let's say that our even integers are found in the first n numbers where n must be greater than or equal to 3. Then, we can form an argument based on this. There are 3^n total ways to assign even numbers being (2,4,6) to each space. Furthermore, we must subtract the cases where we have only a single even integer present in the total n spaces and the case where there are only 2 distinct even integers present. There is 1 way we can have a single even integer in the entire n spaces, therefore giving us just 1 option for each of the three integers, so we have 3 total cases for this. Moreover, the amount of cases with just 2 distinct even integers is 2^n but subtract the cases where all of the n spaces is either a single integer giving us $2^n - 2$, but we multiply by 3C2 because of the ways to choose 2 distinct even integers that are used in the sequence of n . Finally, we have $\sum_{n=3}^{\infty} \frac{(3^n - 3 - 3(2^n - 2))}{6^n}$ note: we must divide all of this by 6^n for probability. Additionally, over the entire summation, we multiply by $1/2$ because of the $1/2$ probability of selecting an odd at the end of all the evens. Therefore, if you compute this using infinite geometric sequences, you get $1/20$. (C) $\frac{1}{20}$.

Solution 7 (Complementary + PIE)

We know that the sequence with odd ending will eventually add up to probability 1.

For every even number appears at least once, let's look at the complementary, i.e. without some even numbers 2, 4, 6. $A = \{\text{without 2}\}$, $B = \{\text{without 4}\}$, $C = \{\text{without 6}\}$.

$$1 - A \cup B \cup C = 1 - (A + B + C - A \cap B - B \cap C - A \cap C + A \cap B \cap C)$$

$$\text{Let } f(x) = \frac{1}{2} + x\frac{1}{2} + x^2\frac{1}{2} + \dots = \frac{1}{2} \frac{1}{1-x}$$

$$2 \text{ even numbers left } |A| = |B| = |C| : f\left(\frac{2}{6}\right) = \frac{1}{2} \frac{1}{1-\frac{1}{3}} = \frac{3}{4},$$

$$1 \text{ even number left } |A \cap B| = |B \cap C| = |A \cap C| : f\left(\frac{1}{6}\right) = \frac{1}{2} \frac{1}{1-\frac{1}{6}} = \frac{3}{5}$$

$$0 \text{ even number left } |A \cap B \cap C| : f(0) = \frac{1}{2}$$

$$\text{ans} = 1 - A \cup B \cup C = 1 - \left(\frac{3}{4} * 3 - \frac{3}{5} * 3 + \frac{1}{2}\right) = \frac{1}{20}$$

Solution 8

We don't care what happens after we pick the first three distinct even numbers. This can be done in $3! = 6$ ways. Totally there are $6 \cdot 5 \cdot 4$ ways to pick three distinct numbers. So the answer is $\frac{6}{6 \cdot 5 \cdot 4} = \frac{1}{20}$.

Solution 9

We can use complementary counting for this problem by eliminating the chances that we get 0, 1, and 2 even numbers before the odd number. To get 0 even numbers

before the odd number has a probability of $3/6$ or $1/2$. To get 1 even number before the odd number has a probability of $3/6 \cdot 3/5$ which is $3/10$ (0.3). To get the probability of 2 even numbers before the odd number it is $3/6 \cdot 2/5 \cdot 3/4$ which is $3/20$ (0.15). Now we eliminate all of these cases to get the answer to the question by doing $1 - (0.5 + 0.3 + 0.15)$ which is equal to 0.05 or $1/20$ so the answer is C $1/20$

Solution 10 (Infinite Geometric Sequence 2)

We can first try to find an expression for, given n dice rolls, the probability that it contains only all the even digits. Firstly, we know that it must be $\frac{1}{2^n}$ (the probability that it contains only even digits). We can do complementary counting to find an expression for the probability that it contains all digits. For all the cases where the sequence does not contain all digits, we know that it must be a sequence of two digits, each one being one or the other. This works out to 2^n , but it has to be multiplied by 3 (for the 3 possible choices of 2 digits) and have 3 subtracted from it (because we will double count the cases where it's only one digit repeated). Then, we can put it over the total number of cases, 3^n .

This gives $1 - \frac{3 \cdot 2^n - 3}{3^n} = 1 - \frac{2^n - 1}{3^{n-1}} = \frac{3^{n-1} - 2^n + 1}{3^{n-1}}$. We can then multiply this by the probability that it contains only even digits, giving $\frac{3^{n-1} - 2^n + 1}{3^{n-1} \cdot 2^n}$. We still need to make sure we don't double count, as we don't explicitly state that the next digit will be odd. So, the final answer will be the result of $\sum_{n=3}^{\infty} \frac{3^{n-1} - 2^n + 1}{3^{n-1} \cdot 2^n}$.

We can continue to simplify the expression to $\sum_{n=3}^{\infty} \frac{3^n \cdot \frac{1}{3} - 2^n + 1}{3^n \cdot 2^n \cdot \frac{2}{3}} = \frac{3}{2} \cdot \sum_{n=3}^{\infty} \frac{\frac{1}{3} \cdot 3^n - 2^n + 1}{6^n}$.

This becomes the sum $\frac{3}{2} \cdot \sum_{n=3}^{\infty} \frac{1}{3} \cdot \frac{1}{2^n} - \frac{1}{3^n} + \frac{1}{6^n}$, which is $\frac{3}{2} \cdot (\frac{1}{12} - \frac{1}{18} + \frac{1}{180}) = \frac{3}{2} \cdot \frac{1}{30} = \boxed{(C) \frac{1}{20}}$

Solution 11 (Quick Easy to Understand Infinite Geometric Sequence)

We need to have every even number from 1 to 6 at least once before we get an odd number. We find that the probability of the first number being even is $\frac{3}{6} = \frac{1}{2}$. Regardless of what the first number is, the probability of the second number being distinct and even is $\frac{2}{5}$, because now only 2 even numbers are left. Next, the probability of the third number being even and distinct is $\frac{1}{4}$. The final number must be odd, and it can be any of the 3 odd numbers, thus $\frac{1}{2}$. Multiplying these all out yields $\frac{1}{2} \cdot \frac{2}{5} \cdot \frac{1}{4} \cdot \frac{1}{2} = \frac{1}{40}$. Now, this is the probability that all even numbers appear exactly once before the odd number, and we haven't accounted for the fact that an even number could appear more than once. Since any even number can appear more than once, we multiply $\frac{1}{40}$ by $\frac{1}{2}$. We could keep multiplying by $\frac{1}{2}$ infinitely before we get an odd number, thus we use the infinite geometric sequence sum formula (we could have 3 even numbers, all the way to ∞ even numbers before we get our first odd number, so that's why we're summing the probabilities). Our first term, a_1 is $\frac{1}{40}$, and the common ratio is $\frac{1}{2}$. Plugging it into the formula $\frac{a_1}{1 - r}$,

we get $\frac{\frac{1}{40}}{1 - \frac{1}{2}} = \frac{\frac{1}{40}}{\frac{1}{2}} = \frac{1}{40} \cdot 2 = \frac{1}{20}$, and we are done. Therefore, our answer is

$$\boxed{(C) \frac{1}{20}}$$

Solution 12 (Unnecessarily Complex Infinite Geometric Sequence)

We need to collect all 3 distinct even numbers ($\{2, 4, 6\}$) before rolling our first odd number. Since repeated evens can occur at any point in the sequence, we evaluate the infinite geometric series for transitioning successfully through each stage. Stage 1: Getting the 1st distinct even. Before seeing any even numbers, a roll either gives a new even (probability $\frac{3}{6}$) or an odd (probability $\frac{3}{6}$). Since no evens have been seen yet, there are no repeated evens to re-roll. The probability of getting the first distinct even before an odd is:

$$S_1 = \frac{3}{6} = \frac{1}{2}$$

Stage 2: Getting the 2nd distinct even. Now 1 even has been seen. A roll can give a new distinct even ($\frac{2}{6}$), an odd ($\frac{3}{6}$), or repeat the 1st even ($\frac{1}{6}$). Rolling the repeated even extends the sequence without failing or progressing. The probability of getting a second distinct even before an odd is the infinite geometric series:

$$S_2 = \frac{2}{6} + \frac{2}{6} \left(\frac{1}{6} \right) + \frac{2}{6} \left(\frac{1}{6} \right)^2 + \dots$$

Using the infinite geometric series sum formula $\frac{a}{1-r}$ with first term $a_1 = \frac{2}{6}$ and common ratio $r = \frac{1}{6}$:

$$S_2 = \frac{\frac{2}{6}}{1 - \frac{1}{6}} = \frac{\frac{2}{6}}{\frac{5}{6}} = \frac{2}{5}$$

Stage 3: Getting the 3rd distinct even. Now 2 evens have been seen. A roll can give the final distinct even ($\frac{1}{6}$), an odd ($\frac{3}{6}$), or repeat either of the first 2 evens ($\frac{2}{6}$). Rolling either repeated even extends the sequence. The probability of getting the final distinct even before an odd is the infinite geometric series:

$$S_3 = \frac{1}{6} + \frac{1}{6} \left(\frac{2}{6} \right) + \frac{1}{6} \left(\frac{2}{6} \right)^2 + \dots$$

Using $\frac{a}{1-r}$ with first term $a_1 = \frac{1}{6}$ and common ratio $r = \frac{2}{6}$:

$$S_3 = \frac{\frac{1}{6}}{1 - \frac{2}{6}} = \frac{\frac{1}{6}}{\frac{4}{6}} = \frac{1}{4}$$

Overall Probability. Multiplying the probabilities of successfully completing each infinite series in order yields:

$$(ErrorcompilingLaTeX.Unknownerror_msg) \text{ Total Probability} = S_1 \cdot S_2 \cdot S_3 = \frac{1}{2} \cdot \frac{2}{5} \cdot \frac{1}{4} = \frac{2}{40} =$$

(C) $\frac{1}{20}$.

19. (AMC 10 10B 2021) Suppose that S is a finite set of positive integers. If the greatest integer in S is removed from S , then the average value (arithmetic mean) of the integers remaining is 32. If the least integer in S is also removed, then the average value of the integers remaining is 35. If the greatest integer is then returned to the set, the average value of the integers rises to 40. The greatest integer in the original set S is 72 greater than the least integer in S . What is the average value of all the integers in the set S ?
- (A) 36.2 (B) 36.4 (C) 36.6 (D) 36.8 (E) 37

Answer: D

Solution 1

We can then say that $\frac{A+S(n)}{n+1} = 32$, $\frac{S(n)}{n} = 35$, and $\frac{B+S(n)}{n+1} = 40$.

Expanding gives us $A + S(n) = 32n + 32$, $S(n) = 35n$, and $B + S(n) = 40n + 40$.

Substituting $S(n) = 35n$ to all gives us $A + 35n = 32n + 32$ and $B + 35n = 40n + 40$.

Solving for A and B gives $A = -3n + 32$ and $B = 5n + 40$.

We now need to find $\frac{S(n)+A+B}{n+2}$. We substitute everything to get $\frac{35n+(-3n+32)+(5n+40)}{n+2}$, or $\frac{37n+72}{n+2}$.

Say that the answer to this is Z . Then, Z needs to be a number that makes n a positive integer. The only options that work is **(C) 36.6** and **(D) 36.8**.

However, if 36.6 is an option, we get $n = 3$. So that means that A is 23 and B is 55, and $S(n) = 105$. But if there is 3 terms, then the middle number is 105, but we said that B is the largest number in the set, so therefore our answer cannot be **(C) 36.6** and is instead **(D) 36.8** and now, we're finished!

Solution 1 Alternative Ending

If A is smaller than B by 72 therefore from the equation on the top you can find out that $N = 8$ using substitution the plug it in to the equation $\frac{37n+72}{n+2}$ then you will get that $Z = \mathbf{(D) 36.8}$.

This was more efficient then the last part.

Solution 2

Let the lowest value be L and the highest G , and let the sum be Z and the amount of numbers n . We have $\frac{Z-G}{n-1} = 32$, $\frac{Z-L-G}{n-2} = 35$, $\frac{Z-L}{n-1} = 40$, and $G = L + 72$. Clearing denominators gives $Z - G = 32n - 32$, $Z - L - G = 35n - 70$, and $Z - L = 40n - 40$. We use $G = L + 72$ to turn the first equation into $Z - L = 32n + 40$. Since $Z - L = 40(n - 1)$ we substitute it into the equation which gives $n = 10$. Turning the second into $Z - 2L = 35n + 2$ using $G = L + 72$ we see $L = 8$ and $Z = 368$ so the average is $\frac{Z}{n} = \mathbf{(D) 36.8}$
aop2014

Solution 3

Let x be the greatest integer, y be the smallest, z be the sum of the numbers in S excluding x and y , and k be the number of elements in S .

Then, $S = x + y + z$

First, when the greatest integer is removed, $\frac{S-x}{k-1} = 32$

When the smallest integer is also removed, $\frac{S-x-y}{k-2} = 35$

When the greatest integer is added back, $\frac{S-y}{k-1} = 40$

We are given that $x = y + 72$

After you substitute $x = y + 72$, you have 3 equations with 3 unknowns S , y and k .

$$S - y - 72 = 32k - 32 \quad S - 2y - 72 = 35k - 70 \quad S - y = 40k - 40$$

This can be easily solved to yield $k = 10$, $y = 8$, $S = 368$.

$$\therefore \text{average value of all integers in the set} = S/k = 368/10 = \mathbf{(D) 36.8}$$

Solution 4

We should plug in 36.2 and assume everything is true except the 35 part. We then calculate that part and end up with 35.75. We also see with the formulas we used with the plug in that when you increase by 0.2 the 35.75 part decreases by 0.25. The answer is then **(D) 36.8**. You can work backwards because it is multiple choice and you don't have to do critical thinking. Lopkiloim

Solution 5

Let $S = \{a_1, a_2, a_3, \dots, a_n\}$ with $a_1 < a_2 < a_3 < \dots < a_n$. We are given the following:

$$\begin{cases} \sum_{i=1}^{n-1} a_i = 32(n-1) = 32n - 32, \\ \sum_{i=2}^n a_i = 40(n-1) = 40n - 40, \\ \sum_{i=2}^{n-1} a_i = 35(n-2) = 35n - 70, \\ a_n - a_1 = 72 \implies a_1 + 72 = a_n. \end{cases}$$

Subtracting the third equation from the sum of the first two, we find that

$$\sum_{i=1}^n a_i = (32n - 32) + (40n - 40) - (35n - 70) = 37n - 2.$$

Furthermore, from the fourth equation, we have

$$\sum_{i=2}^n a_i - \sum_{i=1}^{n-1} a_i = \left[(a_1 + 72) + \sum_{i=2}^{n-1} a_i \right] - \left[(a_1) + \sum_{i=2}^{n-1} a_i \right] = (40n - 40) - (32n - 32).$$

Combining like terms and simplifying, we have

$$72 = 8n - 8 \implies 8n = 80 \implies n = 10.$$

Thus, the sum of the elements in S is $37 \cdot 10 - 2 = 368$, and since there are 10 elements in S , the average of the elements in S is $\frac{368}{10} = \mathbf{(D) 36.8}$

Solution 6

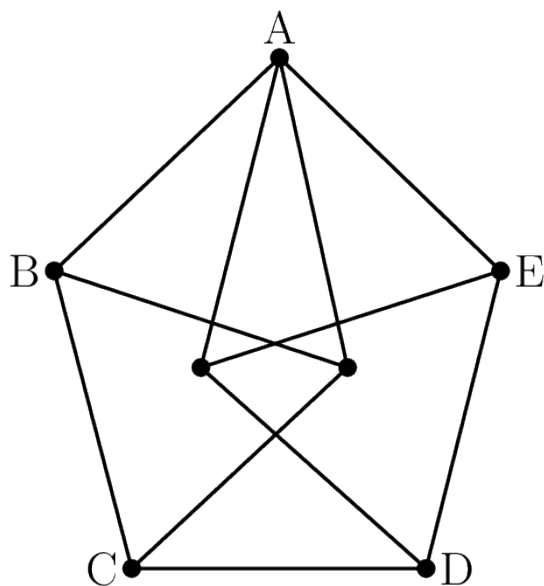
Let n be the number of elements in S , $m_l = 32$, $\Sigma_l = m_l \cdot (n-1)$, $m_g = 40$, $\Sigma_g = m_g \cdot (n-1)$, $m_{lg} = 35$, $\Sigma_{lg} = m_{lg} \cdot (n-2)$.

$$\Sigma_g - \Sigma_l = 72 = (n-1) \cdot (m_g - m_l) \implies n = 1 + \frac{72}{m_g - m_l} = 1 + \frac{72}{40 - 32} = 10.$$

$$m = \frac{\Sigma_g + \Sigma_l - \Sigma_{lg}}{n} = \frac{40 \cdot 9 + 32 \cdot 9 - 35 \cdot 8}{10} = 36.8.$$

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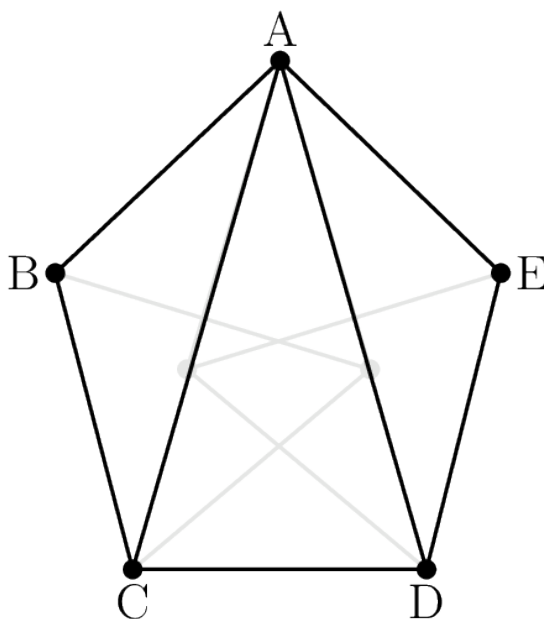
20. (AMC 10 10B 2021) The figure is constructed from 11 line segments, each of which has length 2. The area of pentagon $ABCDE$ can be written as $\sqrt{m} + \sqrt{n}$, where m and n are positive integers. What is $m + n$?



- (A) 20 (B) 21 (C) 22 (D) 23 (E) 24

Answer: D

Solution 1



Draw diagonals AC and AD to split the pentagon into three parts. We can compute the area for each triangle and sum them up at the end. For triangles ABC and ADE , they each have area $2 \cdot \frac{1}{2} \cdot \frac{4\sqrt{3}}{4} = \sqrt{3}$.^{*} For triangle ACD , we can see that $AC = AD = 2\sqrt{3}$ and $CD = 2$. Using Pythagorean Theorem, the altitude for this triangle is $\sqrt{11}$, so the area is $\sqrt{11}$. Adding each part up, we get $2\sqrt{3} + \sqrt{11} = \sqrt{12} + \sqrt{11} \Rightarrow \boxed{\text{(D) } 23}$.

Note: Another way to find the areas would be to use the formula $A = \frac{1}{2}ab \sin C$ - erringbubble

^{*} A more intuitive way to find this: We know $\angle B = \angle E = 120^\circ$ because of equilateral triangles, so the triangles can each be partitioned into 2 30-60-90 right triangles with side

-

Answer: A

We can set the point on CD where the fold occurs as point F . Then, we can set FD as x , and CF as $1 - x$ because of symmetry due to the fold. It can be recognized that this is a right triangle, and solving for x , we get,

We know this is a 3-4-5 triangle because the side lengths are $\frac{3}{9}, \frac{4}{9}, \frac{5}{9}$. We also know that EAC' is similar to $C'DF$ because angle $EC'F$ is a right angle. Now, we can use similarity to find out that the perimeter is just the perimeter of $C'DF \times \frac{AC'}{DF}$. That's just $\frac{4}{3} \times \frac{\frac{2}{4}}{\frac{5}{9}} = \frac{4}{3} \times \frac{3}{2} = 2$. Therefore, the final answer is **(A) 2**

Let the line we're reflecting over be ℓ , and let the points where it hits AB and CD , be M and N , respectively. Notice, to reflect over a line we find the perpendicular passing through the midpoint of the two points (which are the reflected and the original). So, we first find the equation of the line ℓ . The segment CC' has slope $\frac{0-1}{1-\frac{2}{3}} = -3$, implying line ℓ has a slope of $\frac{1}{3}$. Also, the midpoint of segment CC' is $(\frac{5}{6}, \frac{1}{2})$, so line ℓ passes through this point. Then, we get the equation of line ℓ is simply $y = \frac{1}{3}x + \frac{2}{9}$. Then, if the point where B is reflected over line ℓ is B' , then we get BB' is the line $y = -3x$. The intersection of ℓ and segment BB' is $(-\frac{1}{15}, \frac{1}{5})$. So, we get $B' = (-\frac{2}{15}, \frac{2}{5})$. Then, line segment $B'C'$ has equation $y = \frac{3}{4}x + \frac{1}{2}$, so the point E is the y -intercept, or $(0, \frac{1}{2})$.

This implies that $AE = \frac{1}{2}$, $AC' = \frac{2}{3}$, and by the Pythagorean Theorem, $EC' = \frac{5}{6}$ (or you could notice $\triangle AEC'$ is a 3-4-5 right triangle). Then, the perimeter is $\frac{1}{2} + \frac{2}{3} + \frac{5}{6} = 2$, so our answer is (A) 2. rocketsri

Solution 3 (Fakesolve):

Assume that E is the midpoint of $\overline{AB}$. Then, $\overline{AE} = \frac{1}{2}$ and since $C'D = \frac{1}{3}$, $\overline{AC'} = \frac{2}{3}$. By the Pythagorean Theorem, $\overline{EC'} = \frac{5}{6}$. It easily follows that our desired perimeter is (A) 2 samrocksnature

Note: This method is only correct because it turns out that AE does in fact equal $\frac{1}{2}$. However, since it is already given that $C'D$ is $\frac{1}{3}$, and that all of the points are fixed we thus cannot make any assumptions on AE 's length.

Solution 4

As described in Solution 1, we can find that $DF = \frac{4}{9}$, and $C'F = \frac{5}{9}$.

Then, we can find we can find the length of $\overline{AE}$ by expressing the length of $\overline{EF}$ in two different ways, in terms of AE . If let $AE = a$, by the Pythagorean Theorem we have that $EC' = \sqrt{a^2 + \left(\frac{2}{3}\right)^2} = \sqrt{a^2 + \frac{4}{9}}$. Therefore, since we know that $\angle EC'F$ is right, by

Pythagoras again we have that $EF = \sqrt{\left(\sqrt{a^2 + \frac{4}{9}}\right)^2 + \left(\frac{5}{9}\right)^2} = \sqrt{a^2 + \frac{61}{81}}$.

Another way we can express EF is by using Pythagoras on $\triangle XEF$, where X is the foot of the perpendicular from F to $\overline{AE}$. We see that $ADFX$ is a rectangle, so we know that $AD = 1 = FX$. Secondly, since $FD = \frac{4}{9}$, $EX = a - \frac{4}{9}$. Therefore, through the Pythagorean Theorem, we find that $EF = \sqrt{\left(a - \frac{4}{9}\right)^2 + 1^2} = \sqrt{a^2 - \frac{8}{9}a + \frac{97}{81}}$.

Since we have found two expressions for the same length, we have the equation $\sqrt{a^2 + \frac{61}{81}} = \sqrt{a^2 - \frac{8}{9}a + \frac{97}{81}}$. Solving this, we find that $a = \frac{1}{2}$.

Finally, we see that the perimeter of $\triangle AEC'$ is $\frac{1}{2} + \frac{2}{3} + \sqrt{\left(\frac{1}{2}\right)^2 + \frac{4}{9}}$, which we can simplify to be 2. Thus, the answer is (A) 2. laffytaffy

Solution 5 (Trig)

Draw a perpendicular line from $\overline{AB}$ at E , and let it intersect $\overline{DC}$ at E' . The angle between $\overline{AB}$ and $\overline{EE'}$ is 2θ , where θ is the angle between the fold and a line perpendicular to $\overline{AD}$. The slope of the fold is $\frac{1}{3}$ because it is perpendicular to $\overline{CC'}$ ($\overline{CC'}$ has a slope of -3 using points C' and C , and perpendicular lines have slopes negative inverses of each other). Using tangent double angle formula, the slope of $\overline{EC'}$ is $\frac{3}{4}$, which implies $\overline{AE} = \frac{1}{2}$. By the Pythagorean Theorem, $\overline{EC'} = \frac{5}{6}$. It easily follows that our desired perimeter is (A) 2 forrestc

Solution 6

It is easy to prove that the ratio of the sum of the larger leg and hypotenuse to the smaller leg depends monotonically on the angle of a right triangle, which means:

$$C'F + DF = CF + DF = CD = AD = 3C'D \implies C'D : DF : C'F = 3 : 4 : 5.$$

For a similar triangle, the ratio of the perimeter to the larger leg is $\frac{3+4+5}{4} = 3$. $\triangle AEC' \sim \triangle DC'F \implies$ the perimeter of triangle $\triangle AEC'$ is $3AC' = \span style="border: 1px solid black; padding: 2px;">(A) 2.$

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Solution 7 (Super Easy Trig)

Basic Idea: Use Pythagorean Theorem to solve for DF and $C'F$ and then use the fact that $\angle AC'E = 90 - \angle FC'D$ to find the sine of $\angle AC'E$, and then use a basic substitution to solve for AE and EC' .

Solution: Let F be the point on $\overline{CD}$ where the paper folds. Then, we can set $FD = x$ and $FC' = 1 - x$, since if FC' was folded down, it would be $FC = 1 - x$. Since $\triangle FDC'$ is a right triangle, we can use the Pythagorean Theorem to state that $\left(\frac{1}{3}\right)^2 + x^2 = (1 - x)^2$. Solving, we get $x = \frac{4}{9}$, so $FD = \frac{4}{9}$ and $FC' = \frac{5}{9}$. Now, we can let $\angle DC'F = \theta$. We know that $\angle EC'F = 90^\circ$ and that $\angle AC'E$, $\angle EC'F$ and $\angle DC'F$ are supplementary, so $\theta + 90 + \angle AC'E = 180$. Therefore, $\angle AC'E = 90 - \theta$. We know $\sin(\angle AC'E) = \sin(90 - \theta) = \sin(\angle DFC')$. We know $\sin(\angle DFC') = \frac{3}{5}$, so $\sin(\angle AC'E) = \frac{3}{5}$. Therefore, we can set $AE = 3x$ and $EC' = 5x$. Then, using the Pythagorean Theorem, we get $25x^2 = 9x^2 + \frac{4}{9}$. We isolate x : $x^2 = \frac{4}{9 \cdot 16}$ and then take the square root: $x = \frac{2}{12} = \frac{1}{6}$. We need the perimeter of $AC'E$, which is $3x + 5x + \frac{2}{3} = 8x + \frac{2}{3} = \frac{8}{6} + \frac{2}{3} = \boxed{\text{(A) } 2}$.

-NumberNinja11

22. (AMC 10 10B 2021) Ang, Ben, and Jasmin each have 5 blocks, colored red, blue, yellow, white, and green; and there are 5 empty boxes. Each of the people randomly and independently of the other two people places one of their blocks into each box. The probability that at least one box receives 3 blocks all of the same color is $\frac{m}{n}$, where m and n are relatively prime positive integers. What is $m + n$?
 (A) 47 (B) 94 (C) 227 (D) 471 (E) 542

Answer: D

Solution 1 (Principle of Inclusion-Exclusion)

Let our denominator be $(5!)^3$, so we consider all possible distributions.

We use PIE (Principle of Inclusion and Exclusion) to count the successful ones.

When we have at 1 box with all 3 blocks the same color in that box, there are ${}_5C_1 \cdot {}_5P_1 \cdot (4!)^3$ ways for the distributions to occur (${}_5C_1$ for selecting one of the five boxes for a uniform color, ${}_5P_1$ for choosing the color for that box, $4!$ for each of the three people to place their remaining items).

However, we overcounted those distributions where two boxes had uniform color, and there are ${}_5C_2 \cdot {}_5P_2 \cdot (3!)^3$ ways for the distributions to occur (${}_5C_2$ for selecting two of the five boxes for a uniform color, ${}_5P_2$ for choosing the color for those boxes, $3!$ for each of the three people to place their remaining items).

Again, we need to re-add back in the distributions with three boxes of uniform color... and so on so forth.

Our success by PIE is

$${}_5C_1 \cdot {}_5P_1 \cdot (4!)^3 - {}_5C_2 \cdot {}_5P_2 \cdot (3!)^3 + {}_5C_3 \cdot {}_5P_3 \cdot (2!)^3 - {}_5C_4 \cdot {}_5P_4 \cdot (1!)^3 + {}_5C_5 \cdot {}_5P_5 \cdot (0!)^3 = 120 \cdot 2556.$$

$$\frac{120 \cdot 2556}{120^3} = \frac{71}{400},$$

yielding an answer of $\boxed{\text{(D) } 471}$.

Alternate Simplification As In Solution 1, the probability is

$$\frac{\binom{5}{1} \cdot 5 \cdot (4!)^3 - \binom{5}{2} \cdot 5 \cdot 4 \cdot (3!)^3 + \binom{5}{3} \cdot 5 \cdot 4 \cdot 3 \cdot (2!)^3 - \binom{5}{4} \cdot 5 \cdot 4 \cdot 3 \cdot 2 \cdot (1!)^3 + \binom{5}{5} \cdot 5 \cdot 4 \cdot 3 \cdot 2 \cdot 1}{(5!)^3}$$

$$= \frac{5 \cdot 5 \cdot (4!)^3 - 10 \cdot 5 \cdot 4 \cdot (3!)^3 + 10 \cdot 5 \cdot 4 \cdot 3 \cdot (2!)^3 - 5 \cdot 5! + 5!}{(5!)^3}.$$

Dividing by $5!$, we get

$$\frac{5 \cdot (4!)^2 - 10 \cdot (3!)^2 + 10 \cdot (2!)^2 - 5 + 1}{(5!)^2}.$$

Dividing by 4, we get

$$\frac{5 \cdot 6 \cdot 24 - 10 \cdot 9 + 10 - 1}{30 \cdot 120}.$$

Dividing by 9, we get

$$\frac{5 \cdot 2 \cdot 8 - 10 + 1}{10 \cdot 40} = \frac{71}{400} \implies \boxed{\text{(D) } 471}.$$

Solution 2 (Complementary Counting)

Use complementary counting. Denote T_n as the total number of ways to put n colors to n boxes by 3 people out of which f_n ways are such that no box has uniform color. Notice $T_n = (n!)^3$. From this setup we see the question is asking for $1 - \frac{f_5}{(5!)^3}$. To find f_5 we want to exclude the cases of a) one box of the same color, b) 2 boxes of the same color, c) 3 boxes of same color, d) 4 boxes of the same color, and e) 5 boxes of the same color. Cases d) and e) coincide. From this, we have

$$f_5 = T_5 - \binom{5}{1} \binom{5}{1} \cdot f_4 - \binom{5}{2} \binom{5}{2} \cdot 2! \cdot f_3 - \binom{5}{3} \binom{5}{3} \cdot 3! \cdot f_2 - 5!$$

In case b), there are $\binom{5}{2}$ ways to choose 2 boxes that have the same color, $\binom{5}{2}$ ways to choose the two colors, $2!$ ways to permute the 2 chosen colors, and f_3 ways so that the remaining 3 boxes don't have the same color. The same goes for cases a) and c). In case e), the total number of ways to permute 5 colors is $5!$. Now, we just need to calculate f_2 , f_3 and f_4 .

We have $f_2 = T_2 - 2 = (2!)^3 - 2 = 6$, since we subtract the number of cases where the boxes contain uniform colors, which is 2.

In the same way, $f_3 = T_3 - \left[3! + \binom{3}{1} \binom{3}{1} \cdot f_2 \right] = 156$ - again, we must subtract the number of ways at least 1 box has uniform color. There are $3!$ ways if 3 boxes each contains uniform color. Two boxes each contains uniform color is the same as previous. If one box has the same color, there are $\binom{3}{1}$ ways to pick a box, and $\binom{3}{1}$ ways to pick a color for that box, $1!$ ways to permute the chosen color, and f_2 ways for the remaining 2 boxes to have non-uniform colors. Similarly, $f_4 = (4!)^3 - \left[4! + \binom{4}{2} \binom{4}{2} \cdot 2! \cdot f_2 + \binom{4}{1} \binom{4}{1} \cdot f_3 \right] = 10,872$

Thus, $f_5 = f_5 = (5!)^3 - \left[\binom{5}{1} \binom{5}{1} \cdot f_4 + \binom{5}{2} \binom{5}{2} \cdot 2! \cdot f_3 + \binom{5}{3} \binom{5}{3} \cdot 3! \cdot f_2 + 5! \right] = (5!)^3 - 306,720$

Thus, the probability is $\frac{306,720}{(5!)^3} = 71/400$ and the answer is $\boxed{\text{(D) } 471}$

-angelinasheeen

Solution 3 (WLOG and PIE)

WLOG fix which block Ang places into each box. (This can also be thought of as labeling each box by the color of Ang's block.) There are then $(5!)^2$ total possibilities.

As in [Solution 1](#), we use PIE. With 1 box of uniform color, there are ${}_5C_1 \cdot (4!)^2$ possibilities (${}_5C_1$ for selecting one of the five boxes (whose color is fixed by Ang), $4!$ for each of Ben

and Jasmin to place their remaining items). We overcounted cases with 2 boxes, of which there are ${}_5C_2 \cdot (3!)^2$ possibilities, and so on.

The probability is thus

$$\frac{{}_5C_1 \cdot (4!)^2 - {}_5C_2 \cdot (3!)^2 + {}_5C_3 \cdot (2!)^2 - {}_5C_4 \cdot (1!)^2 + {}_5C_5 \cdot (0!)^2}{(5!)^2}$$

$$= \frac{5 \cdot (4!)^2 - 10 \cdot (3!)^2 + 10 \cdot (2!)^2 - 5 + 1}{(5!)^2}$$

at which point we can proceed as in the **Alternate simplification** to simplify to $\frac{71}{400} \implies$

(D) 471.

Solution 4 (Derangements)

$!n$ denotes the number of derangements of n elements, i.e. the number of permutations where no element appears in its original position. Recall the recursive formula $!0 = 1, !1 = 0, !n = (n-1)(!(n-1) + !(n-2))$.

We will consider the number of candidate boxes (ones that currently remain uniform-color) after each person places their blocks.

After Ang, all 5 boxes are candidates, and each has a fixed color its blocks must be to remain uniform-color.

Ben has $5!$ total ways to place blocks. There are $\binom{5}{k} \cdot !k$ ways to disqualify k candidates, $0 \leq k \leq 5$. (There are $\binom{5}{k}$ ways to choose the candidates to disqualify, and $!k$ ways to arrange the disqualified candidates' same-color blocks.)

Jasmin has $5!$ total ways to place blocks. Currently, k boxes are no longer candidates. Consider the number of ways for Jasmin to place blocks such that no box remains a candidate, where n is the number of boxes and k is the number of non-candidates, and call it $D(n, k)$ (not to be confused with partial derangements). We wish to find $5! - D(5, k)$ for each k , $0 \leq k \leq 5$.

Notice that $D(n, 0) = !n$, since all boxes are candidates and no block can be placed in the same-colored box. Also notice the recursive formula $D(n, k) = D(n-1, k-1) + D(n, k-1)$, since the first non-candidate box can either contain its same-color block (in which case it can be ignored), or a different-color block (in which case it can be treated as a candidate).

We can make a table (n horizontal, k vertical):

$D(n, k)$	0	1	2	3	4	5
0	1	0	1	2	9	44
1		1	1	3	11	53
2			2	4	14	64
3				6	18	78
4					24	96
5						120

(Note that $D(n, n) = n!$ since there are no candidates to begin with, which checks out with the diagonal.)

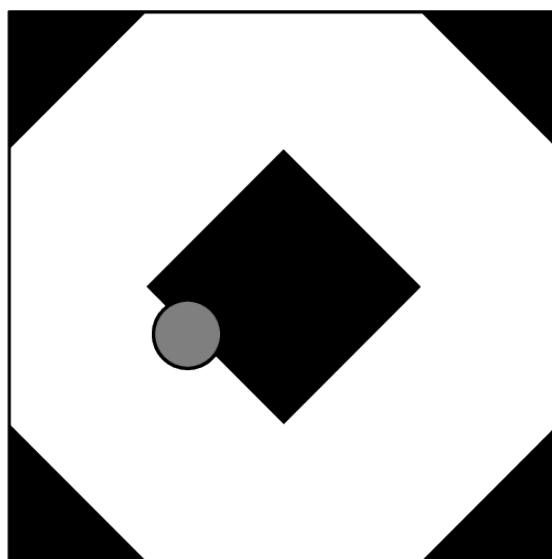
The desired values of $D(5, k)$ are in the rightmost column.

The probability that at least one box is of uniform color is thus

$$\frac{\sum_{k=0}^5 \left(\binom{5}{k} \cdot !k \right) (5! - D(5, k))}{(5!)^2}$$

$$\begin{aligned}
 &= \frac{(1 \cdot 1)(120 - 44) + (5 \cdot 0)(120 - 53) + (10 \cdot 1)(120 - 64) + (10 \cdot 2)(120 - 78) + (5 \cdot 9)(120 - 96) + (1 \cdot 44)}{(5!)^2} \\
 &= \frac{76 + 0 + 560 + 840 + 1080 + 0}{(5!)^2} \\
 &= \frac{19 + 140 + 210 + 270}{(5!)(5 \cdot 3 \cdot 2)} \\
 &= \frac{639}{(5!)(5 \cdot 3 \cdot 2)} \\
 &= \frac{71}{(5 \cdot 4 \cdot 2)(5 \cdot 2)} \\
 &= \frac{71}{400} \implies \boxed{\text{(D) } 471}.
 \end{aligned}$$

23. (AMC 10 10B 2021) A square with side length 8 is colored white except for 4 black isosceles right triangular regions with legs of length 2 in each corner of the square and a black diamond with side length $2\sqrt{2}$ in the center of the square, as shown in the diagram. A circular coin with diameter 1 is dropped onto the square and lands in a random location where the coin is completely contained within the square. The probability that the coin will cover part of the black region of the square can be written as $\frac{1}{196} (a + b\sqrt{2} + \pi)$, where a and b are positive integers. What is $a + b$?



- (A) 64 (B) 66 (C) 68 (D) 70 (E) 72

Answer: C

Solution

To find the probability, we look at the $\frac{\text{success region}}{\text{total possible region}}$. For the coin to be completely contained within the square, we must have the distance from the center of the coin to a side of the square to be at least $\frac{1}{2}$, as it's the radius of the coin. This implies the total possible region is a square with side length $8 - \frac{1}{2} - \frac{1}{2} = 7$, with an area of 49. Now, we consider cases where needs to land to partially cover a black region.

Near The Center Square

We can have the center of the coin land within $\frac{1}{2}$ outside of the center square, or inside of the center square. So, we have a region with $\frac{1}{2}$ emanating from every point on the exterior of the square, forming four quarter circles and four rectangles. The four quarter circles combine to make a full circle of radius $\frac{1}{2}$, so the area is $\frac{\pi}{4}$. The area of a rectangle is $2\sqrt{2} \cdot \frac{1}{2} = \sqrt{2}$, so 4 of them combine to an area of $4\sqrt{2}$. The area of the black square is simply $(2\sqrt{2})^2 = 8$. So, for this case, we have a combined total of $8 + 4\sqrt{2} + \frac{\pi}{4}$. Onto the second (and last) case.

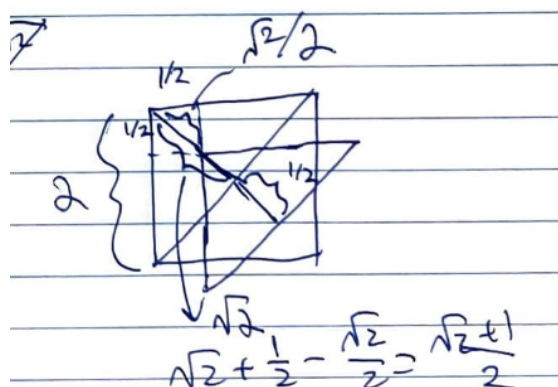
Near A Triangle

We can also have the coin land within $\frac{1}{2}$ outside of one of the triangles. By symmetry, we can just find the successful region for one of them, then multiply by 4. Consider the above diagram. We can draw an altitude from the bottom corner of the square to hit the hypotenuse of the green triangle. The length of this when passing through the black region is $\sqrt{2}$, and when passing through the white region (while being contained in the green triangle) is $\frac{1}{2}$. However, we have to subtract off when it doesn't pass through the red square. Then, it's the hypotenuse of a small isosceles right triangle with side lengths of $\frac{1}{2}$ which is $\frac{\sqrt{2}}{2}$. So, the altitude of the green triangle is $\sqrt{2} + \frac{1}{2} - \frac{\sqrt{2}}{2} = \frac{\sqrt{2}+1}{2}$. Then, recall, the area of an isosceles right triangle is $\frac{h^2}{2}$, where h is the altitude from the right angle. So, squaring this, we get $\frac{3+2\sqrt{2}}{4}$. Now, we have to multiply this by 4 to account for all of the black triangles, to get $3 + 2\sqrt{2}$ as the final area for this case.

Finishing

Then, to have the coin touching a black region, we add up the area of our successful regions, or $8 + 4\sqrt{2} + \frac{\pi}{4} + 3 + 2\sqrt{2} = 11 + 6\sqrt{2} + \frac{\pi}{4} = \frac{44+24\sqrt{2}+\pi}{4}$. The total region is 49, so our probability is $\frac{\frac{44+24\sqrt{2}+\pi}{4}}{49} = \frac{44+24\sqrt{2}+\pi}{196}$, which implies $a + b = 44 + 24 = 68$. This corresponds to answer choice **(C) 68**.

Image



24. (AMC 10 10B 2021) Arjun and Beth play a game in which they take turns removing one brick or two adjacent bricks from one "wall" among a set of several walls of bricks, with gaps possibly creating new walls. The walls are one brick tall. For example, a set of walls of sizes 4 and 2 can be changed into any of the following by one move: (3, 2), (2, 1, 2), (4), (4, 1), (2, 2), or (1, 1, 2).



Arjun plays first, and the player who removes the last brick wins. For which starting configuration is there a strategy that guarantees a win for Beth?

- (A) (6, 1, 1) (B) (6, 2, 1) (C) (6, 2, 2) (D) (6, 3, 1) (E) (6, 3, 2)

Answer: B

Solution 1

We say that a game state is an N-position if it is winning for the next player (the player to move), and a P-position if it is winning for the other player. We are trying to find which of the given states is a P-position.

First we note that symmetrical positions are P-positions, as the second player can win by mirroring the first player's moves. It follows that (6, 1, 1) is an N-position, since we can win by moving to (2, 2, 1, 1); this rules out (A). We next look at (6, 2, 1). The possible next states are

(6, 2), (6, 1, 1), (6, 1), (5, 2, 1), (4, 2, 1, 1), (4, 2, 1), (3, 2, 2, 1), (3, 2, 1, 1), (2, 2, 2, 1).

None of these are symmetrical, so we might reasonably suspect that they are all N-positions. Indeed, it just so happens that for all of these states except (6, 2) and (6, 1), we can win by moving to (2, 2, 1, 1); it remains to check that (6, 2) and (6, 1) are N-positions.

To save ourselves work, it would be nice if we could find a single P-position directly reachable from both (6, 2) and (6, 1). We notice that (3, 2, 1) is directly reachable from both states, so it would suffice to show that (3, 2, 1) is a P-position. Indeed, the possible next states are

(3, 2), (3, 1, 1), (3, 1), (2, 2, 1), (2, 1, 1, 1), (2, 1, 1),

which allow for the following refutations:

$$\begin{array}{lll} (3, 2) \rightarrow (2, 2), & (3, 1, 1) \rightarrow (1, 1, 1, 1), & (3, 1) \rightarrow (1, 1), \\ (2, 2, 1) \rightarrow (2, 2), & (2, 1, 1, 1) \rightarrow (1, 1, 1, 1), & (2, 1, 1) \rightarrow (1, 1). \end{array}$$

Hence, (3, 2, 1) is a P-position, so (6, 2) and (6, 1) are both N-positions, along with all other possible next states from (6, 2, 1) as noted before. Thus, (6, 2, 1) is a P-position, so our answer is (B). (For completeness, we could also rule out (C), (D) and (E) as in Solution 2.)

-Note: In general, this game is very complicated. For example (8, 7, 5, 3, 2) is winning for the first player but good luck showing that.

Solution 2 (Elimination)

(6, 1, 1) can be turned into (2, 2, 1, 1) by Arjun, which is symmetric, so Beth will lose.

(6, 3, 1) can be turned into (3, 1, 3, 1) by Arjun, which is symmetric, so Beth will lose.

(6, 2, 2) can be turned into (2, 2, 2, 2) by Arjun, which is symmetric, so Beth will lose.

(6, 3, 2) can be turned into (3, 2, 3, 2) by Arjun, which is symmetric, so Beth will lose.

That leaves (6, 2, 1) or (B).

Solution 3 (Nim-values)

Let the nim-value of the ending game state, where someone has just removed the final brick, be 0. Then, any game state with a nim-value of 0 is losing. It is well-known

that the nim-value of a supergame (a combination of two or more individual games) is the binary xor function on the nim-values of the individual games that compose the supergame. Therefore, we calculate the nim-values of the states with a single wall up to 6 bricks long (since the answer choices only go up to 6).

First, the game with 1 brick has a nim-value of 1.

Similarly, the game with 2 bricks has a nim-value of 2.

Next, we consider a 3 brick wall. After the next move, the possible resulting game states are 1 brick, a 2 brick wall, or 2 separate bricks. The first two options have nim-values of 1 and 2. The final option has a nim-value of $1 \oplus 1 = 0$, so the nim-value of this game state is 3.

Next, the 4 brick wall. The possible states are a 2 brick wall, a 3 brick wall, a 2 brick wall and a 1 brick wall, or a 1 brick wall and a 1 brick wall. The nim-values of these states are 2, 3, 3, and 0, respectively, and hence the nim-value of this game state is 1.

[Why is the nim-value of it 1? - awesomediabrine

[https://en.wikipedia.org/wiki/Mex_\(mathematics\)](https://en.wikipedia.org/wiki/Mex_(mathematics))]

The possible game states after the 5 brick wall are the following: a 3 brick wall, a 4 brick wall, a 3 brick wall and a 1 brick wall, a two 2 brick walls, and a 2 brick wall plus a 1 brick wall. The nim-values of these are 3, 1, 2, 0, and 3, respectively, meaning the nim-value of a 5 brick wall is 4.

Finally, we find the nim-value of a 6 brick wall. The possible states are a 5 brick wall, a 4 brick wall and a 1 brick wall, a 3 brick wall and a 2 brick wall, a 4 brick wall, a 3 brick wall and a 1 brick wall, and finally two 2 brick walls. The nim-values of these game states are 4, 0, 1, 1, 2, and 0, respectively. This means the 6 brick wall has a nim-value of 3.

The problem is asking which of the answer choices is losing, or has a nim-value of 0. We see that option **(A)** has a nim-value of $3 \oplus 1 \oplus 1 = 3$, option **(B)** has a nim-value of $3 \oplus 2 \oplus 1 = 0$, option **(C)** has a nim-value of $3 \oplus 2 \oplus 2 = 3$, option **(D)** has a nim-value of $3 \oplus 3 \oplus 1 = 1$, and option **(E)** has a nim-value of $3 \oplus 3 \oplus 2 = 2$, so the answer is **(B)** (6, 2, 1).

This method can also be extended to solve the note after the first solution. The nim-values of the 7 brick wall and the 8 brick wall are 2 and 1, using the same method as above. The nim-value of (8, 7, 5, 3, 2) is therefore $1 \oplus 2 \oplus 4 \oplus 3 \oplus 2 = 6$, which is winning.

Solution 4

Consider the following, much simpler game.

Arjun and Beth can each either take 1 or 2 bricks from the right-hand-side of a continuous row of initially n bricks. It is easy to see that for n a multiple of 3, Beth can "mirror" whatever Arjun plays: if he takes 1, she takes 2, and if he takes 2, she takes 1. With this strategy, Beth always takes the last brick. If n is not a multiple of 3, then Arjun takes whichever amount puts Beth in the losing position.

The total number of bricks in the initial states given by the answer choices is 8, 9, 10, 10, 11. Thus, answer choice **(B)** appears promising as a winning position for Beth. The difference between this game and the simplified game is that in certain positions, namely those consisting of fragments of size only 1, taking 2 bricks is not allowed. We can assume that for the starting position (6, 2, 1), Beth always has a move to ensure that she can continue to mirror Arjun throughout the game. (This could be proven rigorously with lots of casework. In particular, she must avoid providing Arjun with a position of 3 continuous

bricks, because then he could take the middle block and force a win.) The assumption seems reasonable, so the answer is **(B)**.

Solution 5

We can start by guessing and checking our solutions. Let's start with A , $(6, 1, 1)$. Arjun goes first, and he has a winning strategy. His strategy is to cut the 6 block in half, so whatever Beth does, Arjun will copy. If we see this in practice, Arjun has made the block in to $(2, 2, 1, 1)$. If Beth takes 1, he will take one. If Beth takes 2, he will take 2 as well.

Then, we can guess solution B , $(6, 2, 1)$. If Arjun starts by taking 1 of the blocks, he will either have $(6, 1, 1)$, $(6, 2)$, $(x, y, 2, 1)$ (Where $x + y = 5$). If he has $(6, 1, 1)$, Beth can take 2 out of the middle of $(6, 1, 1)$ to get $(2, 2, 1, 1)$ which we know Arjun will lose because we know by symmetry from answer choice A . If he has $(6, 2)$, Beth will take out one from the edge, and we get $(5, 2)$. Beth can now copy whatever Arjun does, so she will win.

Next, if he takes one out to get $(x, y, 2, 1)$, we can either have $(4, 1, 2, 1)$ or $(3, 2, 2, 1)$. If it is the first case, Beth can take the edge 2 out of the 4, which gives us $(2, 1, 2, 1)$ and whatever Arjun does, Beth can do, so she has established a win. If it is the 2nd case, Beth can take out 2 from the 3 to get $(1, 2, 2, 1)$, in which, Beth can copy whatever Arjun does, so Beth will win. So in all 4 cases, Beth wins. So our answer is **(B)**.

Solution 6 (Mirroring)

In many game problems, the winning strategy is to simply copy all of your opponent's moves. Since you will always have a valid move if you mirror a valid move, you must move last and win.

Then, if Arjun can make the game "mirrorable" in his first move, then he can just copy all of Beth's moves after that.

$(6, 1, 1)$ can be turned into $(2, 2, 1, 1)$ by Arjun. Any move Beth makes, Arjun can just copy on the other block with same value. For example, if Beth took one block off the first 2, Arjun takes one block off the second 2.

$(6, 2, 2)$ turns into $(2, 2, 2, 2)$.

$(6, 3, 1)$ turns into $(3, 1, 3, 1)$.

$(6, 3, 2)$ turns into $(3, 2, 3, 2)$.

There are definitely other ways to split, but Arjun must win these games. This leaves answer **$B(6, 2, 1)$** .

Solution 7 (Side Games)

We will call the 6-wall the main game and the remaining walls the side game. It is clear that if there is no side game, Arjun can win by taking the middle two walls out. However, if Beth is ever able to force Arjun to take two turns in a row on the main game, their roles will swap and she will win. To do so, she must force Arjun to take two turns in a row, which means she needs to take the last wall in the side game. In other words, we can reduce the problem into Beth winning a game with starting configuration ignoring the 6-wall.

From here, it's clear that Beth loses $(1, 1)$, $(2, 2)$, and $(3, 2)$. However, Beth does win both $(3, 1)$ and $(2, 1)$. However, with the $(3, 1)$ case, Arjun can instead start the game by splitting the $(3, 1)$ into $(1, 1, 1)$, which would swap their roles. Since Arjun now wins the side game, Beth will lose the $(3, 1)$ case, making the answer **$B(6, 2, 1)$** .

25. (AMC 10 10B 2021) Let S be the set of lattice points in the coordinate plane, both of whose coordinates are integers between 1 and 30, inclusive. Exactly 300 points in S lie on or below a line with equation $y = mx$. The possible values of m lie in an interval of length $\frac{a}{b}$, where a and b are relatively prime positive integers. What is $a + b$?
 (A) 31 (B) 47 (C) 62 (D) 72 (E) 85

Answer: E

Solution 1

First, we find a numerical representation for the number of lattice points in S that are under the line $y = mx$. For any value of x , the highest lattice point under $y = mx$ is $\lfloor mx \rfloor$. Because every lattice point from $(x, 1)$ to $(x, \lfloor mx \rfloor)$ is under the line, the total number of lattice points under the line is $\sum_{x=1}^{30} (\lfloor mx \rfloor)$.

Now, we proceed by finding lower and upper bounds for m . To find the lower bound, we start with an approximation. If 300 lattice points are below the line, then around $\frac{1}{3}$ of the area formed by S is under the line. By using the formula for a triangle's area, we find that when $x = 30, y \approx 20$. Solving for m assuming that $(30, 20)$ is a point on the line, we get $m = \frac{2}{3}$. Plugging in m to $\sum_{x=1}^{30} (\lfloor mx \rfloor)$, we get:

$$\sum_{x=1}^{30} (\lfloor \frac{2}{3}x \rfloor) = 0 + 1 + 2 + 2 + 3 + \cdots + 18 + 18 + 19 + 20$$

We have a repeat every 3 values (every time $y = \frac{2}{3}x$ goes through a lattice point). Thus, we can use arithmetic sequences to calculate the value above:

$$\begin{aligned} \sum_{x=1}^{30} (\lfloor \frac{2}{3}x \rfloor) &= 0 + 1 + 2 + 2 + 3 + \cdots + 18 + 18 + 19 + 20 \\ &= \frac{20(21)}{2} + 2 + 4 + 6 + \cdots + 18 \\ &= 210 + \frac{20}{2} \cdot 9 \\ &= 300 \end{aligned}$$

This means that $\frac{2}{3}$ is a possible value of m . Furthermore, it is the lower bound for m . This is because $y = \frac{2}{3}x$ goes through many points (such as $(21, 14)$). If m was lower, $y = mx$ would no longer go through some of these points, and there would be less than 300 lattice points under it.

Now, we find an upper bound for m . Imagine increasing m slowly and rotating the line $y = mx$, starting from the lower bound of $m = \frac{2}{3}$. The upper bound for m occurs when $y = mx$ intersects a lattice point again (look at this problem to get a better idea of what's happening: https://artofproblemsolving.com/wiki/index.php/2011_AMC_10B_Problems/Problem_24).

In other words, we are looking for the first $m > \frac{2}{3}$ that is expressible as a ratio of positive integers $\frac{p}{q}$ with $q \leq 30$. For each $q = 1, \dots, 30$, the smallest multiple of $\frac{1}{q}$ which exceeds $\frac{2}{3}$ is $1, \frac{2}{2}, \frac{3}{3}, \frac{3}{4}, \frac{4}{5}, \dots, \frac{19}{27}, \frac{19}{28}, \frac{20}{29}, \frac{21}{30}$ respectively, and the smallest of these is $\frac{19}{28}$.

Alternatively, see the method of finding upper bounds in solution 2.

The lower bound is $\frac{2}{3}$ and the upper bound is $\frac{19}{28}$. Their difference is $\frac{1}{84}$, so the answer is $1 + 84 = \boxed{85}$.

An alternative would be using Farey fractions and the median theorem to find the upper bound. $\frac{2}{3}$ and $\frac{7}{10}$ gives $\frac{9}{13}$, and so on using Farey addition.

The above alternative can be made quicker by using a different Farey sequence rule. The Farey sequence rule states that if $\frac{a}{b}$ and $\frac{c}{d}$ are adjacent in a Farey sequence, we have $bc - ad = 1$ (see https://en.wikipedia.org/wiki/Farey_sequence#Farey_neighbours). We use the rule with $a = 2$, $b = 3$, so we have $3c - 2d = 1$. Starting with $d = 30$ (since $d \leq 30$), we see that c would be noninteger and the same can be shown for $d = 29$. When $d = 28$, $c = 19$ so the range is $[\frac{2}{3}, \frac{19}{28}] \Rightarrow \frac{1}{84} \Rightarrow \boxed{85}$

An alternative approach with the same methodology can be done using Pick's Theorem. Wikipedia page: https://en.wikipedia.org/wiki/Pick%27s_theorem. It's a formula to find the number of lattice points strictly inside a polygon. Approximation of the lower bound is still necessary.

Solution 2

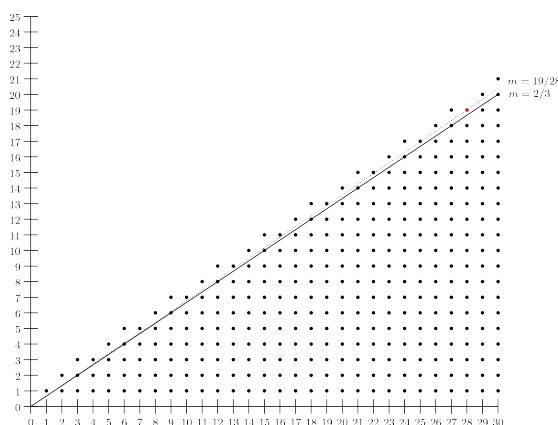
Based on area ratios between a square of side length 30 and a triangle with base 30, we estimate that the slope m of the line we want is approximately $\frac{2}{3}$. Following this estimate, we see if there are approximately $30 \cdot 30 - 300 = 600$ lattice points above the line $y = \frac{2}{3}x$.

Counting the number of lattice points with $x = 1$ above the line, the number of lattice points with $x = 2$ above the line, and so on, we find that the total number of lattice points above the line is $30 + 29 + 28 + 28 + 27 + 26 + 26 \dots + 10$, with the even integers repeating every third term. We see that the average of the 30 terms is 20, which means that exactly $20 \cdot 30 = 600$ lattice points above the line as desired. This gives a lower bound because any decrease in the slope of the line would cause points that were already on the line to shift to being above it.

To find the upper bound, notice that each lattice point less than 1 unit above the line is either $\frac{1}{3}$ or $\frac{2}{3}$ above. Since the slope through a point is the y-coordinate divided by the x-coordinate, a shift in the slope will increase the y-value of the higher x-coordinates more than the y-value of the lower x-coordinates. So, we turn our attention to $x = 28, 29, 30$ for which the line $y = \frac{2}{3}x$ intersects at $y = \frac{56}{3}, \frac{58}{3}, 20$. The point $(30, 20)$ is already counted, and we can clearly see that if we slowly increase the slope of the line, we will first hit the point $(28, 19)$ since $(28, \frac{56}{3})$ is the closest to it. The equation of the line which goes through both the origin and $(28, 19)$ is $y = \frac{19}{28}x$. This gives an upper bound of $m = \frac{19}{28}$.

Taking the upper bound of m and subtracting the lower bound yields $\frac{19}{28} - \frac{2}{3} = \frac{1}{84}$. The answer is therefore $1 + 84 = \boxed{\text{(E)} 85}$.

Diagram



Solution 3

As the procedure shown in the Solution 1, the lower bound of m is $\frac{2}{3}$. Here I give a more logic way to show how to find the upper bound of m . Denote $N = \sum_{x=1}^{30} \lfloor mx \rfloor$ as the number of lattice points in S .

$$N = \lfloor m \rfloor + \lfloor 2m \rfloor + \lfloor 3m \rfloor + \cdots + \lfloor 30m \rfloor = 300.$$

Let $m = \frac{2}{3} + k$. for $\forall x_i \leq 30, x \in N^*, \lfloor mx_i \rfloor = \lfloor \frac{2}{3}x + xk \rfloor$.

Our target is finding the minimum value of k which can increase one unit of $\lfloor mx_i \rfloor$.

Notice that:

When $x_i = 3n, \lfloor mx_i \rfloor = \lfloor 2n + (3n)k \rfloor$ We don't have to discuss this case.

When $x_i = 3n + 1, \lfloor mx_i \rfloor = \lfloor 2n + \frac{2}{3} + (3n + 1)k \rfloor, k_{min1} = \frac{1}{3(3n+1)}.$

When $x_i = 3n + 2, \lfloor mx_i \rfloor = \lfloor 2n + 1 + \frac{1}{3} + (3n + 2)k \rfloor, k_{min2} = \frac{2}{3(3n+2)}.$

Here $n \in N^*, n \leq 9$.

Denote $k_{min} = \min \{k_{min1}, k_{min2}\}.$

Obviously k_{min1} and k_{min2} are decreasing. Let's considering the situation when $n = 9$.

$$k_{min} = \min \left\{ \frac{1}{84}, \frac{2}{87} \right\} = \frac{1}{84}.$$

This means that the answer is just $\frac{1}{84}$, so $a + b = 85$. Choose $\boxed{E}$.

Solution 4

It's easier to calculate the number of lattice points inside a rectangle with vertices $(0, 0), (p, 0), (p, q), (0, q)$. Those lattice points are divided by the diagonal $y = \frac{p}{q} \cdot x$ into 2 halves. In this problem the number of lattice points on or below the diagonal and $x \geq 1$ is

$$\frac{1}{2} [(p+1)(q+1) - d] + d - (p+1), \quad d \text{ is the number of lattice points on the diagonal}$$

$(p+1)(q+1)$ is the total number of lattice points inside the rectangle. Subtract the number of lattice points on the diagonal, divided by 2 is the number of lattice points below the diagonal, add the number of lattice points on the diagonal, and subtract the lattice points on the x axis, then we get the total number of lattice points on or below the diagonal and $x \geq 1$.

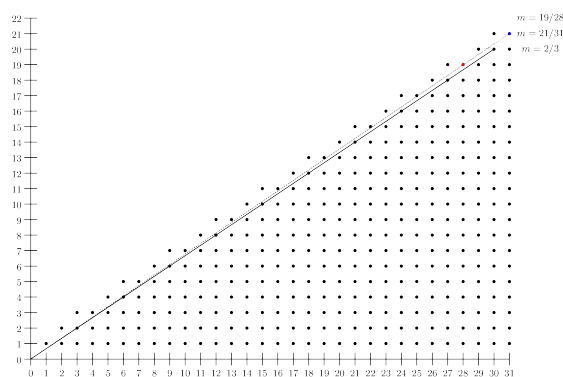
There are 900 lattice points in total. 300 is $\frac{1}{3}$ of 900. The x coordinate of the top-right vertex of the rectangle is 30, $\frac{1}{2} \cdot 30 \cdot 20 = 300$. I guess the y coordinate of the top-right vertex of the rectangle is 20. Now I am going to verify that. The slope of the diagonal is $\frac{20}{30} = \frac{2}{3}$, there are 11 lattice points on the diagonal. Substitute $(p, q) = (30, 20), d = 11$ to the above formula:

$$\frac{1}{2} [(30+1)(10+1) - 11] + 11 - (30+1) = 300$$

Because there are 11 lattice points on line $y = \frac{2}{3}x$, if $m < \frac{2}{3}$, then the number of lattice points on or below the line is less than 300. So $m = \frac{2}{3}$ is the lower bound.

Now I am going to calculate the upper bound. From $\frac{b}{a} < \frac{b+1}{a+1}$,

$$\frac{2}{3} = \frac{18}{27} < \frac{19}{28} \quad \frac{2}{3} = \frac{20}{30} < \frac{21}{31} < \frac{19}{28}$$



If $m = \frac{21}{31}$, I will calculate by using the rectangle with blue vertex $(p, q) = (31, 21)$, then subtract lattice points on line $x = 31$, which is 21. There are 2 lattice points on the diagonal, $d = 2$.

$$\frac{1}{2} [(31+1)(21+1) - 2] + 2 - (31+1) - 21 = 300, \text{ same as that of } m = \frac{19}{28}$$

If $m = \frac{19}{28}$, I will calculate by using the rectangle with red vertex $(p, q) = (28, 19)$, then add lattice points on line $x = 29$ and $x = 30$, which is $19 + 20 = 39$. There are 2 lattice points on the diagonal, $d = 2$.

$$\frac{1}{2} [(28+1)(19+1) - 2] + 2 - (28+1) + 39 = 301, \$1\$ \text{ more than that of } m = \frac{19}{28}$$

When m increases, more lattice points falls below the line $y = mx$. Any value larger than $\frac{19}{28}$ has more than 301 lattice points on or below $y = \frac{19}{28}x$. So the upper bound is $\frac{19}{28}$.

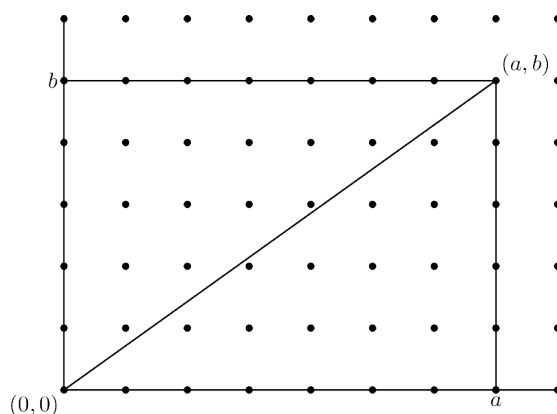
$$\frac{19}{28} - \frac{2}{3} = \frac{1}{84}, \quad \boxed{\text{(E) } 85}.$$

Solution 5

The lower bound of m is $\frac{2}{3} = \frac{20}{30}$. Inside the rectangle with vertices $(0, 0)$, $(30, 0)$, $(30, 20)$, $(0, 20)$ and diagonal $y = \frac{2}{3}x$, there are $(30 - 1)(20 - 1) = 551$ lattice points inside, not including the edges. There are 9 lattice points on diagonal $y = \frac{2}{3}x$ inside the rectangle, $551 + 9 = 560$. Half of the 560 lattice points are below diagonal $y = \frac{2}{3}x$, $560 \cdot \frac{1}{2} = 280$. There are 20 lattice points on edge $x = 30$, $280 + 20 = 300$. Once $m < \frac{2}{3}$, the 9 lattice points on diagonal $y = \frac{2}{3}x$ will be above the new diagonal, making the number of lattice points on and below the diagonal less than 300.

Now we are going to calculate the upper bound by the following formula:

The number of lattice points inside rectangle $(0, 0)$, $(a, 0)$, (a, b) , $(0, b)$ is



$$\frac{2}{3} = \frac{20}{30} < \frac{21}{31} < \frac{19}{28}$$

When $a = 31$, $b = 21$, $\frac{(31-1)(21-1)}{2} = 300$.

When $a = 28$, $b = 19$, $\frac{(28-1)(19-1)}{2} = 243$. Below the line $y = \frac{19}{28}x$, there are 19 lattice points on line $x = 28$, 19 lattice points on line $x = 29$, 20 lattice points on line $x = 30$, $243 + 19 + 19 + 20 = 301$.

More lattice points fall below the line $y = mx$ as m increases. There are more than 301 lattice points on and below the line for any m greater than $\frac{19}{28}$. Therefore, the upper bound is $\frac{19}{28}$.

$$\frac{19}{28} - \frac{2}{3} = \frac{1}{84}, \text{ so } 1 + 84 = \boxed{\text{(E)} 85}.$$

Remark $\lfloor \frac{b}{a}k \rfloor$ is the number of lattice points on line $x = k$, below line $y = \frac{b}{a}x$ and above the x axis, where k is an integer and $0 < k < a$. Therefore, $\sum_{k=1}^{a-1} \lfloor \frac{b}{a}k \rfloor$ is the number of lattice points inside the rectangle $(0, 0)$, $(a, 0)$, (a, b) , $(0, b)$, below diagonal $y = \frac{b}{a}x$. If a and b are relatively prime, $\sum_{k=1}^{a-1} \lfloor \frac{b}{a}k \rfloor = \frac{(a-1)(b-1)}{2}$, as explained in solution 5. This problem is about finding the upper and lower bound of $\frac{b}{a}$, given $\sum_{k=1}^{30} \lfloor \frac{b}{a}k \rfloor = 300$. The same problem can have geometric representation as stated in the original problem, or algebraic representation as stated here.

Solution 6

As the procedure shown in the Solution 1, the lower bound of m is $\frac{2}{3}$, whose equation is $2x - 3y = 0$.

Now, we are going to find the upper bound of m .

The signed distance between a point (x_0, y_0) and the line $2x - 3y = 0$ is

$$d = \frac{2x_0 - 3y_0}{\sqrt{2^2 + 3^2}}$$

If (x_0, y_0) is the lowest lattice point above the line, then it should have the largest values of coordinates among all the solutions of $2x_0 - 3y_0 = 1$, because among the points with the shortest distance from the line, the slope of the farthest point to the origin is the smallest. Solve the inequality:

$$2x - 3y = 1, \quad 1 \leq x \leq 30$$

We obtain $(x_0, y_0) = (28, 19)$, so the upper bound of $m = \frac{19}{28}$.

$$\frac{19}{28} - \frac{2}{3} = \frac{1}{84}, \quad \boxed{\text{(E)} 85}.$$

§4 AMC 10 10A 2021

1. (AMC 10 10A 2021) What is the value of

$$(2^2 - 2) - (3^2 - 3) + (4^2 - 4)?$$

- (A) 1 (B) 2 (C) 5 (D) 8 (E) 12

Answer: D

Solution 1

$$(4 - 2) - (9 - 3) + (16 - 4) = 2 - 6 + 12 = 8. \text{ This corresponds to } \boxed{\text{(D)} 8}.$$

-happykeeper

Solution 2

We have

$$\begin{aligned}(2^2 - 2) - (3^2 - 3) + (4^2 - 4) &= 2(2 - 1) - 3(3 - 1) + 4(4 - 1) \\ &= 2(1) - 3(2) + 4(3) \\ &= 2 - 6 + 12 \\ &= \boxed{\text{(D)} 8}.\end{aligned}$$

MRENTHUSIASM

Solution 3 (Overkill: Just for Fun)

We have

$$\begin{aligned}(2^2 - 2) - (3^2 - 3) + (4^2 - 4) &= 2^2 + 4^2 - 3^2 - 2 + 3 - 4 \\ &= 2^2 + (4 - 3)(4 + 3) - 3 \\ &= 2^2 + 7 - 3 = 2^2 + 4 \\ &= 4 \cdot 2 \\ &= \boxed{\text{(D)} 8}.\end{aligned}$$

-PureSwag

Solution 4 (When you have too much time)

Using the difference of squares, we have

$$(2 - \sqrt{2})(2 + \sqrt{2}) - (3 - \sqrt{3})(3 + \sqrt{3}) + (4 - 2)(4 + 2).$$

Knowing that $\sqrt{2} \approx 1.41$ and $\sqrt{3} \approx 1.73$, we get

$$(2 - \sqrt{2})(2 + \sqrt{2}) - (3 - \sqrt{3})(3 + \sqrt{3}) + (4 - 2)(4 + 2) \approx 0.59 \cdot 3.41 - 1.26 \cdot 4.73 + 2 \cdot 6 = 8.0521,$$

which is closest to $\boxed{\text{(D)} 8}$.

2. (*AMC 10 10A 2021*) Portia's high school has 3 times as many students as Lara's high school. The two high schools have a total of 2600 students. How many students does Portia's high school have?

(A) 600 (B) 650 (C) 1950 (D) 2000 (E) 2050

Answer: C

Solution 1 (Two Variables)

The following system of equations can be formed with P representing the number of students in Portia's high school and L representing the number of students in Lara's high school:

$$\begin{aligned}P &= 3L, \\ P + L &= 2600.\end{aligned}$$

Substituting $P = 3L$ gives $4L = 2600$. Solving for L gives $L = 650$. Since we need to find P , we multiply 650 by 3 to get $P = \boxed{\text{(C)} 1950}$.

Solution 2 (One Variable)

Suppose Lara's high school has x students, so Portia's high school has $3x$ students. We have $x + 3x = 2600$, or $4x = 2600$. The answer is

$$3x = 2600 \cdot \frac{3}{4} = 650 \cdot 3 = \boxed{\text{(C)} 1950}.$$

Solution 3 (Arithmetic)

Clearly, 2600 is 4 times the number of students in Lara's high school. Therefore, Lara's high school has $2600 \div 4 = 650$ students, and Portia's high school has $650 \cdot 3 = \boxed{\text{(C)} 1950}$ students.

Solution 4 (Observations)

The number of students in Portia's high school must be a multiple of 3. This eliminates (B), (D), and (E). Since (A) is too small (as it is clear that $600 + \frac{600}{3} < 2600$), we are left with $\boxed{\text{(C)} 1950}$.

3. (AMC 10 10A 2021) The sum of two natural numbers is 17,402. One of the two numbers is divisible by 10. If the units digit of that number is erased, the other number is obtained. What is the difference of these two numbers?

(A) 10,272 (B) 11,700 (C) 13,362 (D) 14,238 (E) 15,426

Answer: D

Solution 1 (Algebra)

The units digit of a multiple of 10 will always be 0. We add a 0 whenever we multiply by 10. So, removing the units digit is equal to dividing by 10.

Let the smaller number (the one we get after removing the units digit) be a . This means the bigger number would be $10a$.

We know the sum is $10a + a = 11a$ so $11a = 17402$. So $a = 1582$. The difference is $10a - a = 9a$. So, the answer is $9(1582) = \boxed{\text{(D)} 14,238}$.

Solution 2 (Arithmetic)

Since the unit's place of a multiple of 10 is 0, the other integer must end with a 2, for both integers sum up to a number ending in a 2. Thus, the unit's place of the difference must be $10 - 2 = 8$, and the only answer choice that ends with an 8 is $\boxed{\text{(D)} 14,238}$.

Another quick solution is to realize that the sum represents a number n added to $10n$. The difference is $9n$, which is $\frac{9}{11}$ of the given sum.

Solution 3 (Vertical Addition and Logic)

Let the larger number be $\overline{AB,CD0}$. It follows that the smaller number is $\overline{A,BCD}$. Adding vertically, we have

$$\begin{array}{r} A \ B \ C \ D \ 0 \\ + \quad \quad A \ B \ C \ D \\ \hline 1 \ 7 \ 4 \ 0 \ 2 \end{array}$$

Working from right to left, we get

$$D = 2 \implies C = 8 \implies B = 5 \implies A = 1.$$

The larger number is 15,820 and the smaller number is 1,582. Their difference is $15,820 - 1,582 = \boxed{\text{(D)} 14,238}$.

Solution 4 (Logic)

We know that the larger number has a units digit of 0 since it is divisible by 10. If D is the ten's digit of the larger number, then D is the units digit of the smaller number. Since the sum of the natural numbers has a unit's digit of 2, $D = 2$.

The units digit of the larger number is 0 and the units digit of the smaller number is 2, so the positive difference between the numbers is 8. There is only one answer choice that has this units digit, and that is $\boxed{\text{(D)} 14,238}$.

(Similar to MRENTHUSIASM's solution)

4. (AMC 10 10A 2021) A cart rolls down a hill, travelling 5 inches the first second and accelerating so that during each successive 1-second time interval, it travels 7 inches more than during the previous 1-second interval. The cart takes 30 seconds to reach the bottom of the hill. How far, in inches, does it travel?

(A) 215 (B) 360 (C) 2992 (D) 3195 (E) 3242

Answer: D

Solution 1 (Arithmetic Series)

Since

$$\text{Distance} = \text{Speed} \cdot \text{Time},$$

we seek the sum

$$5 \cdot 1 + 12 \cdot 1 + 19 \cdot 1 + 26 \cdot 1 + \cdots = 5 + 12 + 19 + 26 + \cdots,$$

in which there are 30 terms.

The last term is $5 + 7 \cdot (30 - 1) = 208$. Therefore, the requested sum is

$$5 + 12 + 19 + 26 + \cdots + 208 = \frac{5 + 208}{2} \cdot 30 = \boxed{\text{(D)} 3195}.$$

Recall that to find the sum of an arithmetic series, we take the average of the first and last terms and then multiply by the number of terms:

$$\text{Sum} = \frac{\text{First} + \text{Last}}{2} \cdot \text{Count}.$$

MRENTHUSIASM

Solution 2 (Arithmetic Series)

The distance (in inches) traveled within each 1-second interval is:

$$5, 5 + 1(7), 5 + 2(7), \dots, 5 + 29(7).$$

This is an arithmetic sequence so the total distance travelled, found by summing them up is:

$$\text{number of terms} \cdot \text{average of terms} = \text{number of terms} \cdot \frac{\text{first term} + \text{last term}}{2}.$$

Or,

$$30 \cdot \frac{5 + 5 + 29(7)}{2} = 15 \cdot 213 = \boxed{\text{(D)} 3195}.$$

BakedPotato66

Solution 3 (Answer Choices and Modular Arithmetic)

From the 30-term sum

$$5 + 12 + 19 + 26 + \cdots$$

in Solution 1, taking modulo 10 gives

$$5 + 12 + 19 + 26 + \cdots \equiv 3 \cdot (5 + 2 + 9 + 6 + 3 + 0 + 7 + 4 + 1 + 8) = 3 \cdot 45 \equiv 5 \pmod{10}.$$

The only answer choices congruent to 5 modulo 10 are (A) and (D). By a quick estimation, (A) is too small, leaving us with (D) 3195.

Solution 4 (Motion With Constant Acceleration)

This problem can be solved by physics method. This method is perhaps the quickest too and shows the beauty of the problem. The average speed increases 7 in/s per second. So, the acceleration $a = 7 \text{ in/s}^2$. The average speed of the first second is 5 in/s. We can know the initial velocity $v_0 = 5 - 0.5 \cdot 7 = 1.5$. (Because of the formula for the distance traveled in the k th one-second interval being $\Delta x_k = v_0 + a \left(k - \frac{1}{2}\right)$). The displacement at $t = 30$ is

$$s = \frac{1}{2}at^2 + v_0t = \frac{1}{2} \cdot 7 \cdot 30^2 + 1.5 \cdot 30 = \span style="border: 1px solid black; padding: 2px;">(D) 3195.$$

Bran_Qin

5. (AMC 10 10A 2021) The quiz scores of a class with $k > 12$ students have a mean of 8. The mean of a collection of 12 of these quiz scores is 14. What is the mean of the remaining quiz scores in terms of k ?

(A) $\frac{14-8}{k-12}$ (B) $\frac{8k-168}{k-12}$ (C) $\frac{14}{12} - \frac{8}{k}$ (D) $\frac{14(k-12)}{k^2}$ (E) $\frac{14(k-12)}{8k}$

Answer: B**Solution 1 (Generalized Value of k)**

The total score of the class is $8k$, and the total score of the 12 quizzes is $12 \cdot 14 = 168$. Therefore, for the remaining quizzes ($k - 12$ of them), the total score is $8k - 168$.

Their mean score is (B) $\frac{8k - 168}{k - 12}$.

Solution 2 (Specified Values of k)

Set $k = 13$. The answer is the same as the last student's quiz score, which is

$8 \cdot 13 - 14 \cdot 12 < 0$. From the answer choices, only (B) $\frac{8k - 168}{k - 12}$ is negative at $k = 13$.

Solution 3

You know that the mean of the first 12 students is 14, so that means all of them combined had a score of $12 \cdot 14 = 168$. Set the mean of the remaining students (in other words the value you are trying to solve for), to a . The total number of remaining students in a class of size k can be written as $k - 12$. The total score $k - 12$ students got combined can be written as $a(k - 12)$, and the total score all of the students in the class got was $168 + a(k - 12)$ (the first twelve students, plus the remaining students). The mean of the whole class can be written as $\frac{168 + a(k - 12)}{k}$. The mean of the class has already been given as 8, so by just writing the equation

$\frac{168+a(k-12)}{k} = 8$, and solving for a (the mean of $k - 12$ students) will give you the

answer in terms of k , which is (B) $\frac{8k - 168}{k - 12}$.

6. (AMC 10 10A 2021) Chantal and Jean start hiking from a trailhead toward a fire tower. Jean is wearing a heavy backpack and walks slower. Chantal starts walking at 4 miles per hour. Halfway to the tower, the trail becomes really steep, and Chantal slows down to 2 miles per hour. After reaching the tower, she immediately turns around and descends the steep part of the trail at 3 miles per hour. She meets Jean at the halfway point. What was Jean's average speed, in miles per hour, until they meet?

(A) $\frac{12}{13}$ (B) 1 (C) $\frac{13}{12}$ (D) $\frac{24}{13}$ (E) 2

Answer: A

Solution 1 (Generalized Distance)

Let $2d$ miles be the distance from the trailhead to the fire tower, where $d > 0$. When Chantal meets Jean, the two have traveled for

$$\frac{d}{4} + \frac{d}{2} + \frac{d}{3} = d \left(\frac{1}{4} + \frac{1}{2} + \frac{1}{3} \right) = d \left(\frac{3}{12} + \frac{6}{12} + \frac{4}{12} \right) = \frac{13}{12}d$$

hours. At that point, Jean has traveled for d miles, so his average speed is

$$\frac{d}{\frac{13}{12}d} = \text{(A) } \frac{12}{13} \text{ miles per hour.}$$

Solution 2 (Specified Distance)

<i>We will follow the same template as shown in Solution 1, except that we will replace d with a convenient constant.</i>

Let 24 miles be the distance from the trailhead to the fire tower. When Chantal meets Jean, the two have traveled for

$$\frac{12}{4} + \frac{12}{2} + \frac{12}{3} = 3 + 6 + 4 = 13$$

hours. At that point, Jean has traveled for 12 miles, so his average speed is (A) $\frac{12}{13}$ miles per hour.

Solution 3 (d=st)

<i>Shortened Solution</i> $d = st$, so Jean's speed can be represented as $s = \frac{d}{t}$. We know the time is $\left(\frac{d}{4} + \frac{d}{2} + \frac{d}{3}\right)$ (if the total distance is $2d$), so Jean's speed is

$$\frac{d}{\left(\frac{d}{4} + \frac{d}{2} + \frac{d}{3}\right)} = \text{(A) } \frac{12}{13}.$$

<i>Full Solution</i>

We know that distance traveled is equal to the speed multiplied with the time. So, $d = st$ and $t = \frac{d}{s}$. Let $2d$ be equal to the distance from the trailhead to the tower. Then, originally, Chantal travels a d distance with at 4 miles per hour. So, Chantal's time is $\frac{d}{4}$. From the midpoint to the tower, Chantal takes $\frac{d}{2}$ hours (since Chantal has speed of 2 miles per hour.) Similarly, the time it takes for Chantal to return to the midpoint is $\frac{d}{3}$. Therefore, the total time is $\left(\frac{d}{4} + \frac{d}{2} + \frac{d}{3}\right)$. We can can

substitute this time into the original equation of $d = st$ to obtain $d = s \left(\frac{d}{4} + \frac{d}{2} + \frac{d}{3} \right)$,

$$\text{so } \frac{d}{\left(\frac{d}{4} + \frac{d}{2} + \frac{d}{3}\right)} = s \implies \frac{1}{\frac{13}{12}} \implies \boxed{\text{(A)} \frac{12}{13}}.$$

7. (AMC 10 10A 2021) Tom has a collection of 13 snakes, 4 of which are purple and 5 of which are happy. He observes that

- all of his happy snakes can add,
- none of his purple snakes can subtract, and
- all of his snakes that can't subtract also can't add.

Which of these conclusions can be drawn about Tom's snakes?

- (A) Purple snakes can add.
 (B) Purple snakes are happy.
 (C) Snakes that can add are purple.
 (D) Happy snakes are not purple.
 (E) Happy snakes can't subtract.

Answer: D

Solution 1 (Comprehensive Explanation of Logic)

We are given that

$$\text{happy} \implies \text{can add}, \quad (1)$$

$$\text{purple} \implies \text{cannot subtract}, \quad (2)$$

$$\text{cannot subtract} \implies \text{cannot add}. \quad (3)$$

Two solutions follow from here:

Solution 1.1 (Intuitive)

Combining (2) and (3) gives

$$\text{happy} \implies \text{can add}, \quad (1)$$

$$\underbrace{\text{purple} \implies \text{cannot subtract}}_{(2)} \overset{(3)}{\implies} \text{cannot add}. \quad (*)$$

Clearly, the answer is $\boxed{\text{(D)}}$.

Solution 1.2 (Rigorous)

Recall that every conditional statement $p \implies q$ is always logically equivalent to its contrapositive $\neg q \implies \neg p$.

Combining (1), (2) and (3) gives

$$\underbrace{\text{purple} \implies \text{cannot subtract}}_{(2)} \overset{(3)}{\implies} \text{cannot add} \overset{\text{Contrapositive of (1)}}{\implies} \text{not happy}. \quad (**)$$

Applying the hypothetical syllogism to (**), we conclude that

$$\text{purple} \implies \text{not happy},$$

whose contrapositive is

$$\text{happy} \implies \text{not purple.}$$

Therefore, the answer is **(D)**. Remark

The conclusions in the other choices do not follow from (**): **(A)** purple $\implies$ can add **(B)** purple $\implies$ happy **(C)** can add $\implies$ purple **(E)** happy $\implies$ cannot subtract

Solution 2 (Process of Elimination)

From Solution 1.1, we can also see this through the process of elimination. Statement *A* is false because purple snakes cannot add. *B* is false as well because since happy snakes can add and purple snakes can not add, purple snakes are not happy snakes. *E* is false using the same reasoning, purple snakes are not happy snakes so happy snakes can subtract since purple snakes cannot subtract. *C* is false since snakes that can add are happy, not purple. That leaves statement *D*. **(D)** is the only correct statement.

Solution 3 (Rigorous)

We first convert each statement to "If X, then Y" form:

- If a snake is happy, then it can add.
- If a snake is purple, then it can't subtract.
- If a snake can't subtract, then it can't add.

Now, we simply check the truth value for each statement:

- a) Combining the last two propositions, we have

If a snake is purple, then it can't add.

Thus, **(A)** is never true.

- b) From the last part, we found that

If a snake is purple, then it can't add.

Also, since the contrapositive of a proposition has the same truth value as the proposition itself, we know, from the first statement, that

If a snake can't add, then it isn't happy.

Combining these two propositions, we find that

If a snake is purple, then it isn't happy. Purple snakes are not happy.

Thus, **(B)** is never true.

- c) From part **(A)**, we found that "If a snake is purple, then it can't add." This implies its contrapositive, "If a snake can add, then it is not purple." is true, meaning **(C)** is NEVER true. [Thanks again to MRENTHUSIASM for pointing this out!]

- d) From the first statement, we have

If a snake is happy, then it can add.

From the contrapositive of the third statement, we have

If a snake can add, then it can subtract.

Then, from the contrapositive of the second statement, we have

If a snake can subtract, then it is not purple.

Combining all of these yields

If a snake is happy, then it is not purple.

Thus, (D) is always true.

e) From the first proposition, we have

If a snake is happy, then it can add.

From the contrapositive of the third proposition, we have

If a snake can add, then it can subtract.

Combining these two propositions gives

If a snake is happy, then it can subtract.

Thus, (E) is never true.

Therefore, (D) is our answer.

8. (AMC 10 10A 2021) When a student multiplied the number 66 by the repeating decimal,

$$\underline{1.\underline{a}\underline{b}\underline{a}\underline{b}\dots} = \underline{1.\underline{a}\underline{b}},$$

where a and b are digits, he did not notice the notation and just multiplied 66 times $\underline{1.\underline{a}\underline{b}}$. Later he found that his answer is 0.5 less than the correct answer. What is the 2-digit number $\underline{a}\underline{b}$? (A) 15 (B) 30 (C) 45 (D) 60 (E) 75

Answer: E

Solution 1

We are given that $66(\underline{1.\underline{a}\underline{b}}) - 0.5 = 66(\underline{1.\underline{a}\underline{b}})$, from which

$$66(\underline{1.\underline{a}\underline{b}}) - 66(\underline{1.\underline{a}\underline{b}}) = 0.5$$

$$66(\underline{1.\underline{a}\underline{b}} - \underline{1.\underline{a}\underline{b}}) = 0.5$$

$$66(\underline{0.\underline{0}\underline{0}\underline{a}\underline{b}}) = 0.5$$

$$66\left(\frac{1}{100} \cdot \underline{0.\underline{a}\underline{b}}\right) = \frac{1}{2}$$

$$\underline{0.\underline{a}\underline{b}} = \frac{25}{33}$$

$$\underline{0.\underline{a}\underline{b}} = 0.\overline{75}$$

$$\underline{a}\underline{b} = \boxed{\text{(E) } 75}.$$

MRENTHUSIASM

Solution 2

It is known that $\underline{0.\underline{a}\underline{b}} = \frac{a}{99} + \frac{b}{990}$ and $\underline{0.\underline{a}\underline{b}} = \frac{a}{100} + \frac{b}{1000}$.

Let $x = \underline{a}\underline{b}$. We have

$$66\left(1 + \frac{x}{99}\right) - 66\left(1 + \frac{x}{100}\right) = 0.5.$$

Expanding and simplifying give $\frac{x}{150} = 0.5$, so $x = \boxed{\text{(E) } 75}$.

Solution 3 (Similar to Solution 2)

We have

$$66 \cdot \left(1 + \frac{10a + b}{100}\right) + \frac{1}{2} = 66 \cdot \left(1 + \frac{10a + b}{99}\right).$$

Expanding both sides, we have

$$66 + \frac{33(10a + b)}{50} + \frac{1}{2} = 66 + \frac{2(10a + b)}{3}.$$

Subtracting 66 from both sides, we have

$$\frac{33(10a + b)}{50} + \frac{1}{2} = \frac{2(10a + b)}{3}.$$

Multiplying both sides by $50 \cdot 3 = 150$, we have

$$99(10a + b) + 75 = 100(10a + b).$$

Thus, the answer is $10a + b = \boxed{\text{(E)} 75}$.

By letting $x = \underline{a} \underline{b} = 10a + b$, this solution is similar to Solution 2. In this solution, we solve for $10a + b$ as a whole.

-mathboy282 (Solution)

Solution 4 (Answer Choices & Modular)

Let $\underline{a} \underline{b}$ represent the two-digit number ab , not the product of the digits a and b . We can construct fractions for the values $1.\underline{a} \underline{b}$, and $1.\overline{\underline{a} \underline{b}}$, which are $\frac{1a \ b}{100}$ and $\frac{1a \ b}{99}$ respectively. Multiplying by 66 on both sides and adding $1/2$ to $\frac{1a \ b}{100}$ and simplifying, we get this:

$$\frac{33 * \underline{ab} + 25}{50} = \frac{2ab}{3}.$$

Looking at the answer choices, we notice that all of them are divisible by 3. This means that since the right-hand side will result in an integer, the left-hand side needs to as well. This means that the numerator of the left-hand side fraction has to be divisible by 50. So, we get this expression:

$$33 * \underline{ab} + 25 \equiv 0 \pmod{50}. \quad 33 * \underline{ab} \equiv -25 \pmod{50}. \quad 33 * \underline{ab} \equiv 25 \pmod{50}.$$

This means that the product of 33 and $\underline{ab}$ must have a remainder of 25 when divided by 50. Since it must have a remainder of 25, the product should have a units digit of 5, which eliminates **(B)** and **(D)**. Multiplying 33 to the rest of the answer choices, the only one which fills this requirement is 75, which is $\boxed{\text{(E)} 75}$.

Solution 5 (A different way to calculate)

$$1.\overline{ab} = 1 + \frac{\overline{ab}_{10}}{10^2 - 1} = 1 + \frac{10a + b}{99} \text{ and } 1.ab = 1 + \frac{a}{10} + \frac{b}{100}, \text{ so our equation is}$$

$$66 \times \left(1 + \frac{10a + b}{99}\right) = 66 \times \left(1 + \frac{a}{10} + \frac{b}{100}\right) + 0.5,$$

so now let's simplify!

$$\begin{aligned} 66 \times \left(1 + \frac{10a + b}{99}\right) &= 66 \times \left(1 + \frac{a}{10} + \frac{b}{100}\right) + 0.5 \\ 1 + \frac{10a + b}{99} &= 1 + \frac{a}{10} + \frac{b}{100} + \frac{1}{132} \end{aligned}$$

$$\frac{10a + b}{99} = \frac{10a + b}{100} + \frac{1}{132}.$$

Notice that the 2-digit number $\overline{ab}$ is just $10a + b$. Therefore, let's just find the value of $10a + b$.

$$\begin{aligned}\left(\frac{1}{99} - \frac{1}{100}\right)(10a + b) &= \frac{1}{132} \\ 10a + b &= \frac{\frac{1}{132}}{\frac{1}{99} - \frac{1}{100}} \\ 10a + b &= \frac{\frac{1}{132}}{\frac{100 - 99}{99 \times 100}} \\ 10a + b &= \frac{9900}{132} \\ 10a + b &= 75.\end{aligned}$$

We just got $10a + b = 75$, and 75 is indeed a 2-digit number, so the two-digit number $\overline{ab}$ is $\boxed{75}$.

9. (AMC 10 10A 2021) What is the least possible value of $(xy - 1)^2 + (x + y)^2$ for real numbers x and y ?
- (A) 0 (B) $\frac{1}{4}$ (C) $\frac{1}{2}$ (D) 1 (E) 2

Answer: D

Solution 1 (Expand)

Expanding, we get that the expression is $x^2 + 2xy + y^2 + x^2y^2 - 2xy + 1$ or $x^2 + y^2 + x^2y^2 + 1$. By the Trivial Inequality (all squares are nonnegative) the minimum value for this is $\boxed{\text{(D) } 1}$, which can be achieved at $x = y = 0$.

Solution 2 (Expand and then Factor)

We expand the original expression, then factor the result by grouping:

$$\begin{aligned}(xy - 1)^2 + (x + y)^2 &= (x^2y^2 - 2xy + 1) + (x^2 + 2xy + y^2) \\ &= x^2y^2 + x^2 + y^2 + 1 \\ &= x^2(y^2 + 1) + (y^2 + 1) \\ &= (x^2 + 1)(y^2 + 1).\end{aligned}$$

Clearly, both factors are positive. By the Trivial Inequality, we have

$$(x^2 + 1)(y^2 + 1) \geq (0 + 1)(0 + 1) = \boxed{\text{(D) } 1}.$$

Note that the least possible value of $(xy - 1)^2 + (x + y)^2$ occurs at $x = y = 0$.

Solution 3 (Beyond Overkill)

Like solution 1, expand and simplify the original equation to $x^2 + y^2 + x^2y^2 + 1$ and let $f(x, y) = x^2 + y^2 + x^2y^2 + 1$. To find local extrema, find where $\nabla f(x, y) = \mathbf{0}$. First, find the first partial derivative with respect to x and y and find where they are 0:

$$\begin{aligned}\frac{\partial f}{\partial x} &= 2x + 2xy^2 = 2x(1 + y^2) = 0 \implies x = 0 \\ \frac{\partial f}{\partial y} &= 2y + 2yx^2 = 2y(1 + x^2) = 0 \implies y = 0\end{aligned}$$

Thus, there is a local extremum at $(0, 0)$. Because this is the only extremum, we can assume that this is a minimum because the problem asks for the minimum (though this can also be proven using the partial second derivative test) and the global minimum since it's the only minimum, meaning $f(0, 0)$ is the minimum of $f(x, y)$. Plugging $(0, 0)$ into $f(x, y)$, we find $1 \implies \boxed{\text{(D)} 1}$.

Remark: Honestly, this isn't even that overkill. I did it this way, and it was not very bashy. vaishnav

10. (AMC 10 10A 2021) Which of the following is equivalent to

$$(2 + 3)(2^2 + 3^2)(2^4 + 3^4)(2^8 + 3^8)(2^{16} + 3^{16})(2^{32} + 3^{32})(2^{64} + 3^{64})?$$

- (A) $3^{127} + 2^{127}$ (B) $3^{127} + 2^{127} + 2 \cdot 3^{63} + 3 \cdot 2^{63}$ (C) $3^{128} - 2^{128}$ (D) $3^{128} + 2^{128}$
(E) 5^{127}

Answer: C

Solution 1

By multiplying the entire equation by $3 - 2 = 1$, all the terms will simplify by difference of squares, and the final answer is $\boxed{\text{(C)} 3^{128} - 2^{128}}$.

Additionally, we could also multiply the entire equation (we can let it be equal to x) by $2 - 3 = -1$. The terms again simplify by difference of squares. This time, we get $-x = 2^{128} - 3^{128} \implies x = 3^{128} - 2^{128}$. Both solutions yield the same answer.

Note: Also notice when you multiply it by the first pair $(2 + 3)$, it immediately factors. Notice how it "domino" effects to the others, ultimately collapsing into $\boxed{\text{(C)} 3^{128} - 2^{128}}$.

Solution 2

If you weren't able to come up with the $(3 - 2)$ insight, then you could just notice that the answer is divisible by $2 + 3 = 5$, and $2^2 + 3^2 = 13$. We can then use Fermat's Little Theorem for $p = 5, 13$ on the answer choices to determine which of the answer choices are divisible by both 5 and 13. This is $\boxed{\text{(C)} 3^{128} - 2^{128}}$.

-mewto

Solution 3 (Units Digit and Multiples)

Notice that simplifying the first term gives 5, which means that the units digit of the entire expression must be 5. Also, note how the numbers appear to form multiples of 2, so therefore our answer should contain a power of 2. The only one that satisfies the above properties is $\boxed{\text{(C)} 3^{128} - 2^{128}}$.

Solution 4

After expanding the first few terms, the result after each term appears to be $2^{2^n-1} + 2^{2^n-2} \cdot 3^1 + 2^{2^n-3} \cdot 3^2 + \dots + 2^1 \cdot 3^{2^n-2} + 3^{2^n-1}$ where n is the number of terms expanded. We can prove this using mathematical induction. The base step is trivial. When expanding another term, all of the previous terms multiplied by $2^{2^{n-1}}$ would give $2^{2^n-1} + 2^{2^n-2} \cdot 3^1 + 2^{2^n-3} \cdot 3^2 + \dots + 2^{2^{n-1}+1} \cdot 3^{2^{n-1}-1} + 2^{2^{n-1}} \cdot 3^{2^{n-1}}$, and all the previous terms multiplied by $3^{2^{n-1}}$ would give $3^{2^n-1} + 3^{2^n-2} \cdot 2^1 + 3^{2^n-3} \cdot 2^2 + \dots + 3^{2^{n-1}+1} \cdot 2^{2^{n-1}-1} + 3^{2^{n-1}} \cdot 2^{2^{n-1}}$. Their sum is equal to $2^{2^n-1} + 2^{2^n-2} \cdot 3^1 + 2^{2^n-3} \cdot 3^2 + \dots + 2^1 \cdot 3^{2^n-2} + 3^{2^n-1}$, so the proof is complete. Since $\frac{3^{2^n}-2^{2^n}}{3-2}$ is equal to $2^{2^n-1} + 2^{2^n-2} \cdot 3^1 + 2^{2^n-3} \cdot 3^2 + \dots + 2^1 \cdot 3^{2^n-2} + 3^{2^n-1}$, the answer is $\frac{3^{2^7}-2^{2^7}}{3-2} = \boxed{\text{(C)} 3^{128} - 2^{128}}$.

Solution 5 (Engineer's Induction)

We can compute some of the first few partial products, and notice that $\prod_{k=0}^{2^n} (2^{2^k} + 3^{2^k}) = 3^{2^{n+1}} - 2^{2^{n+1}}$. As we don't have to prove this, we get the product is $3^{2^7} - 2^{2^7} = 3^{128} - 2^{128}$, and click (C) $3^{128} - 2^{128}$.

Solution 6 (Difference of Squares)

We notice that the first term is equal to $3^2 - 2^2$. If we multiply this by the second term, then we will get $(3^2 - 2^2)(3^2 + 2^2)$, and we can simplify by using difference of squares to obtain $3^4 - 2^4$. If we multiply this by the third term and simplify using difference of squares again, we get $3^8 - 2^8$. We can continue down the line until we multiply by the last term, $3^{64} + 2^{64}$, and get (C) $3^{128} - 2^{128}$.

Solution 7 (Generalization)

Let's generalize to $x = 3$ and $y = 2$. Then we get:

$$(y + x)(y^2 + x^2)(y^4 + x^4)(y^8 + x^8)(y^{16} + x^{16})(y^{32} + x^{32})(y^{64} + x^{64}).$$

We see that the first term is $y + x$, and the next is $y^2 + x^2$. We realize that if we multiply the first term by $y - x$, the result by difference of squares will be $y^2 - x^2$: we can proceed to use difference of squares on that. In other words, the equation has the domino effect, and all we need to get started is to multiply the whole equation by $y - x$ (keeping in mind to divide $y - x$ at the end.)

Now we get:

$$\begin{aligned} & (y - x)(y + x)(y^2 + x^2)(y^4 + x^4)(y^8 + x^8)(y^{16} + x^{16})(y^{32} + x^{32})(y^{64} + x^{64}) \\ &= (y^2 - x^2)(y^2 + x^2)(y^4 + x^4)(y^8 + x^8)(y^{16} + x^{16})(y^{32} + x^{32})(y^{64} + x^{64}) \\ &= (y^4 - x^4)(y^4 + x^4)(y^8 + x^8)(y^{16} + x^{16})(y^{32} + x^{32})(y^{64} + x^{64}) \\ &\vdots \\ &= (y^{64} - x^{64})(y^{64} + x^{64}) \\ &= y^{128} - x^{128}. \end{aligned}$$

Now we can plug in $y = 2$ and $x = 3$:

$$(2 - 3)(2 + 3)(2^2 + 3^2)(2^4 + 3^4)(2^8 + 3^8)(2^{16} + 3^{16})(2^{32} + 3^{32})(2^{64} + 3^{64}) = 2^{128} - 3^{128}.$$

However, we must not forget to divide by $2 - 3 = -1$ at the end! Dividing, we get the answer of

$$3^{128} - 2^{128} = \boxed{\text{(C) } 3^{128} - 2^{128}}.$$

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Solution 8 (Generalization)

More generally, we have

$$(a^n - b^n)(a^n + b^n) = a^{2n} - b^{2n} \quad (\star)$$

by the difference of squares.

Multiplying the original expression by $1 = 3 - 2$ and then applying $(\star)$ repeatedly, we get

$$(2 + 3)(2^2 + 3^2)(2^4 + 3^4)(2^8 + 3^8)(2^{16} + 3^{16})(2^{32} + 3^{32})(2^{64} + 3^{64})$$

$$\begin{aligned}
 &= \underbrace{\binom{1}{3-2}}_{\text{Apply the Commutative Property of Addition to all factors.}} (2+3)(2^2+3^2)(2^4+3^4)(2^8+3^8)(2^{16}+3^{16})(2^{32}+3^{32})(2^{64}+3^{64}) \\
 &= \underbrace{(3-2)(3+2)}_{\text{Apply } (\star) \text{ to get } 3^2-2^2.} (3^2+2^2)(3^4+2^4)(3^8+2^8)(3^{16}+2^{16})(3^{32}+2^{32})(3^{64}+2^{64}) \\
 &= \underbrace{(3^2-2^2)(3^2+2^2)}_{\text{Apply } (\star) \text{ to get } 3^4-2^4.} (3^4+2^4)(3^8+2^8)(3^{16}+2^{16})(3^{32}+2^{32})(3^{64}+2^{64}) \\
 &= \underbrace{(3^4-2^4)(3^4+2^4)}_{\text{Apply } (\star) \text{ to get } 3^8-2^8.} (3^8+2^8)(3^{16}+2^{16})(3^{32}+2^{32})(3^{64}+2^{64}) \\
 &\vdots \\
 &= \boxed{(\text{C}) 3^{128} - 2^{128}}.
 \end{aligned}$$

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Solution 9 (Desperate Cheese, Similar to Solution 5)

One would probably assume that the factorization that this is equal to can be generalized for the pattern

$$(2+3)(2^2+3^2)(2^4+3^4)\cdots(2^n+3^n).$$

So we can take a smaller portion and apply the answer choices there. We get

$$(2+3)(2^2+3^2) = (5)(13) = 65.$$

We notice that the patterns have a relationship where its either the sum of the powers or the next power not included that is incorporated. So we change the numbers from 127 to 3, 128 to 4, and 63 to 1 (since 63 is the sum of the powers, divided by 2, rounded down). Checking the values of the other patterns, the only one that gives 65 is $3^4 - 2^4$, which means the answer is $\boxed{(\text{C}) 3^{128} - 2^{128}}$.

11. (AMC 10 10A 2021) For which of the following integers b is the base- b number $2021_b - 221_b$ not divisible by 3?

(A) 3 (B) 4 (C) 6 (D) 7 (E) 8

Answer: E

Solution 1 (Factor)

We have

$$\begin{aligned}
 2021_b - 221_b &= (2021_b - 21_b) - (221_b - 21_b) \\
 &= 2000_b - 200_b \\
 &= 2b^3 - 2b^2 \\
 &= 2b^2(b-1),
 \end{aligned}$$

which is divisible by 3 *unless* $b \equiv 2 \pmod{3}$. The only choice congruent to 2 modulo 3 is $\boxed{(\text{E}) 8}$.

Solution 2 (Vertical Subtraction)

Vertically subtracting $2021_b - 221_b$, we see that the ones place becomes 0, and so does the b^1 place. Then, we perform a carry (make sure the carry is in base b). Let $b - 2 = A$. Then, we have our final number as

$$1A00_b.$$

Now, when expanding, we see that this number is simply $b^3 - (b - 2)^2$.

Now, notice that the final number will only be congruent to

$$b^3 - (b - 2)^2 \equiv 0 \pmod{3}.$$

If either $b \equiv 0 \pmod{3}$, or if $b \equiv 1 \pmod{3}$ (because note that $(b - 2)^2$ would become $\equiv 1 \pmod{3}$, and b^3 would become $\equiv 1 \pmod{3}$ as well, and therefore the final expression would become $1 - 1 \equiv 0 \pmod{3}$). Therefore, b must be $\equiv 2 \pmod{3}$. Among the answers, only 8 is $\equiv 2 \pmod{3}$, and therefore our answer is (E) 8.

Solution 3 (Answer Choices)

By the definition of bases, we have

$$2021_b - 221_b = (2b^3 + 2b + 1) - (2b^2 + 2b + 1).$$

For values b_1 and b_2 such that $b_1 \equiv b_2 \pmod{3}$, we get

$$(2b_1^3 + 2b_1 + 1) - (2b_1^2 + 2b_1 + 1) \equiv (2b_2^3 + 2b_2 + 1) - (2b_2^2 + 2b_2 + 1) \pmod{3}.$$

Note that answer choices (A), (B), (C), (D), (E) are congruent to 0, 1, 0, 1, 2 modulo 3, respectively. So, (A) and (C) are either both correct or both incorrect. Since there is only one correct answer, (A) and (C) are both incorrect. Similarly, (B) and (D) are both incorrect. This leaves us with (E) 8, the answer choice with a unique residue modulo 3.

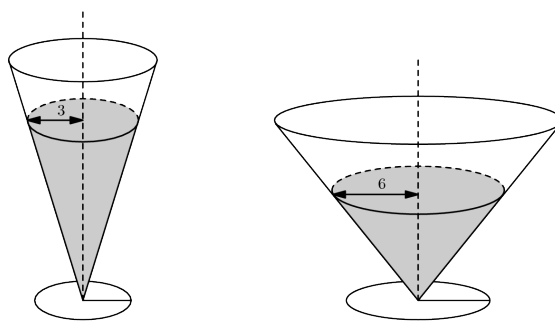
Solution 4 (like Solution 1)

As with the beginning of Solution 1, we get that 3 does not divide $2b^2(b - 1)$. We can just test each answer choice (doing short circuit evaluation for 3 and 6), and finding that only (E) 8 works. Honestly, Solution 1 is easier, but during an actual test when your brain is fried, I'd just check each option.

Solution 5

Test all the answer choices and see that 8 (E) is the answer!

12. (AMC 10 10A 2021) Two right circular cones with vertices facing down as shown in the figure below contain the same amount of liquid. The radii of the tops of the liquid surfaces are 3 cm and 6 cm. Into each cone is dropped a spherical marble of radius 1 cm, which sinks to the bottom and is completely submerged without spilling any liquid. What is the ratio of the rise of the liquid level in the narrow cone to the rise of the liquid level in the wide cone?



- (A) 1 : 1 (B) 47 : 43 (C) 2 : 1 (D) 40 : 13 (E) 4 : 1

Answer: E

Solution 1 (Algebra)

Initial Scenario

Let the heights of the narrow cone and the wide cone be h_1 and h_2 , respectively. We have the following table:

	Base Radius	Height	Volume
Narrow Cone	3	h_1	$\frac{1}{3}\pi(3)^2h_1 = 3\pi h_1$
Wide Cone	6	h_2	$\frac{1}{3}\pi(6)^2h_2 = 12\pi h_2$

Equating the volumes gives $3\pi h_1 = 12\pi h_2$, which simplifies to $\frac{h_1}{h_2} = 4$.

Furthermore, by similar triangles:

- For the narrow cone, the ratio of the base radius to the height is $\frac{3}{h_1}$, which always remains constant.
- For the wide cone, the ratio of the base radius to the height is $\frac{6}{h_2}$, which always remains constant.

Two solutions follow from here:

Solution 1.1 (Properties of Fractions)

Final Scenario

For the narrow cone and the wide cone, let their base radii be $3x$ and $6y$ (for some $x, y > 1$), respectively. By the similar triangles discussed above, their heights must be h_1x and h_2y , respectively. We have the following table:

	Base Radius	Height	Volume
Narrow Cone	$3x$	h_1x	$\frac{1}{3}\pi(3x)^2(h_1x) = 3\pi h_1x^3$
Wide Cone	$6y$	h_2y	$\frac{1}{3}\pi(6y)^2(h_2y) = 12\pi h_2y^3$

Recall that $\frac{h_1}{h_2} = 4$. Equating the volumes gives $3\pi h_1x^3 = 12\pi h_2y^3$, which simplifies to $x^3 = y^3$, or $x = y$.

Finally, the requested ratio is

$$\frac{h_1x - h_1}{h_2y - h_2} = \frac{h_1(x - 1)}{h_2(y - 1)} = \frac{h_1}{h_2} = \boxed{\text{(E)} 4 : 1}.$$

Remarks

- a) This solution uses the following property of fractions: For unequal positive numbers a, b, c and d , if $\frac{a}{b} = \frac{c}{d} = k$, then $\frac{a \pm c}{b \pm d} = \frac{bk \pm dk}{b \pm d} = \frac{(b \pm d)k}{b \pm d} = k$.
- b) This solution shows that, regardless of the shape or the volume of the solid dropped into each cone, the requested ratio stays the same as long as the solid sinks to the bottom and is completely submerged without spilling any liquid.

Solution 1.2 (Bash)

<u>Final Scenario</u>

For the narrow cone and the wide cone, let their base radii be r_1 and r_2 , respectively; let their rises of the liquid levels be Δh_1 and Δh_2 , respectively. We have the following table:

	Base Radius	Height	Volume
Narrow Cone	r_1	$h_1 + \Delta h_1$	$\frac{1}{3}\pi r_1^2(h_1 + \Delta h_1)$
Wide Cone	r_2	$h_2 + \Delta h_2$	$\frac{1}{3}\pi r_2^2(h_2 + \Delta h_2)$

By the similar triangles discussed above, we get

$$\frac{3}{h_1} = \frac{r_1}{h_1 + \Delta h_1} \implies r_1 = \frac{3}{h_1}(h_1 + \Delta h_1), \quad (1)$$

$$\frac{6}{h_2} = \frac{r_2}{h_2 + \Delta h_2} \implies r_2 = \frac{6}{h_2}(h_2 + \Delta h_2). \quad (2)$$

The volume of the marble dropped into each cone is $\frac{4}{3}\pi(1)^3 = \frac{4}{3}\pi$.

Now, we set up an equation for the volume of the narrow cone, then express Δh_1 in terms of h_1 :

$$\begin{aligned} \frac{1}{3}\pi r_1^2(h_1 + \Delta h_1) &= 3\pi h_1 + \frac{4}{3}\pi \\ \frac{1}{3}r_1^2(h_1 + \Delta h_1) &= 3h_1 + \frac{4}{3} \\ \frac{1}{3}\left(\frac{3}{h_1}(h_1 + \Delta h_1)\right)^2(h_1 + \Delta h_1) &= 3h_1 + \frac{4}{3} && \text{by (1)} \\ \frac{3}{h_1^2}(h_1 + \Delta h_1)^3 &= 3h_1 + \frac{4}{3} \\ (h_1 + \Delta h_1)^3 &= h_1^3 + \frac{4h_1^2}{9} \\ \Delta h_1 &= \sqrt[3]{h_1^3 + \frac{4h_1^2}{9}} - h_1. \end{aligned}$$

Next, we set up an equation for the volume of the wide cone, then express Δh_2 in terms of h_2 :

$$\frac{1}{3}\pi r_2^2(h_2 + \Delta h_2) = 12\pi h_2 + \frac{4}{3}\pi.$$

Using a similar process from above, we get

$$\Delta h_2 = \sqrt[3]{h_2^3 + \frac{h_2^2}{9}} - h_2.$$

Recall that $\frac{h_1}{h_2} = 4$. Therefore, the requested ratio is

$$\begin{aligned}\frac{\Delta h_1}{\Delta h_2} &= \frac{\sqrt[3]{h_1^3 + \frac{4h_1^2}{9}} - h_1}{\sqrt[3]{h_2^3 + \frac{h_2^2}{9}} - h_2} \\ &= \frac{\sqrt[3]{(4h_2)^3 + \frac{4(4h_2)^2}{9}} - 4h_2}{\sqrt[3]{h_2^3 + \frac{h_2^2}{9}} - h_2} \\ &= \frac{\sqrt[3]{4^3 \left(h_2^3 + \frac{h_2^2}{9}\right)} - 4h_2}{\sqrt[3]{h_2^3 + \frac{h_2^2}{9}} - h_2} \\ &= \frac{4\sqrt[3]{h_2^3 + \frac{h_2^2}{9}} - 4h_2}{\sqrt[3]{h_2^3 + \frac{h_2^2}{9}} - h_2} \\ &= \boxed{\text{(E)} \ 4 : 1}.\end{aligned}$$

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Solution 2 (Approximate Cones with Cylinders)

The heights of the cones are not given, so suppose the heights are very large (i.e. tending towards infinity) in order to approximate the cones as cylinders with base radii 3 and 6 and infinitely large height. Then the base area of the wide cylinder is 4 times that of the narrow cylinder. Since we are dropping a ball of the same volume into each cylinder, the water level in the narrow cone/cylinder should rise

$\boxed{\text{(E)} \ 4}$ times as much.

Solution 3 (Calculus)

The volume of the shorter cone can be expressed as $V_1 = 3\pi h_1$, and the volume of the larger cone can be expressed as $V_2 = 12\pi h_2$. We also know that the volume changes by $\frac{4\pi}{3}$, because the volume of the 1cm sphere is $\frac{4\pi}{3}$.

Taking the derivative of the equations, we get: $dV_1 = 3\pi(dh_1) = \frac{4\pi}{3}$ and $dV_2 = 12\pi(dh_2) = \frac{4\pi}{3}$.

Therefore, $dh_1 = \frac{4\pi}{3} \cdot \frac{1}{3\pi} = \frac{4}{9}$ and $dh_2 = \frac{4\pi}{3} \cdot \frac{1}{12\pi} = \frac{1}{9}$. The ratio is $\frac{4}{9} : \frac{1}{9}$ giving us the answer of $\boxed{\text{(E)} \ 4 : 1}$.

Solution 4 (Extremely Quick Observation)

Note that as long as the volumes are equal, the ratio of the heights of Cone 2 to Cone 1 is always 1 : 4. We can prove this as follows:

If we fix Cone 2 to have a certain height, then the volume of the cone is $\frac{1}{3} \cdot 36\pi h_1 = 12\pi h_1$. Now if Cone 1 has the same volume, its height would be $\frac{12\pi h_1}{\frac{1}{3} \cdot 9\pi} = \frac{12\pi h_1}{3\pi} = 4h_1$.

So the answer is $\boxed{\text{(E)} \ 4 : 1}$

Minor edits by: T3CHN0B14D3

Remark You can cheese this by letting r_1 and r_2 be some small integers. I recommend 4 and 1.

13. (AMC 10 10A 2021) What is the volume of tetrahedron $ABCD$ with edge lengths $AB = 2$, $AC = 3$, $AD = 4$, $BC = \sqrt{13}$, $BD = 2\sqrt{5}$, and $CD = 5$?

- (A) 3 (B) $2\sqrt{3}$ (C) 4 (D) $3\sqrt{3}$ (E) 6

Answer: C

Solution 1 (Three Right Triangles)

Drawing the tetrahedron out and testing side lengths, we realize that the $\triangle ACD$, $\triangle ABC$, and $\triangle ABD$ are right triangles by the Converse of the Pythagorean Theorem. It is now easy to calculate the volume of the tetrahedron using the formula for the volume of a pyramid. If we take $\triangle ADC$ as the base, then $\overline{AB}$ must be the altitude. The volume of tetrahedron $ABCD$ is $\frac{1}{3} \cdot \frac{3 \cdot 4}{2} \cdot 2 = \boxed{\text{(C) } 4}$. This solution does not have a reasoning for using triangle ADC as base
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Solution 2 (One Right Triangle)

We will place tetrahedron $ABCD$ in the xyz -plane. By the Converse of the Pythagorean Theorem, we know that $\triangle ACD$ is a right triangle. Without the loss of generality, let $A = (0, 0, 0)$, $C = (3, 0, 0)$, $D = (0, 4, 0)$, and $B = (x, y, z)$.

We apply the Distance Formula to $\overline{BA}$, $\overline{BC}$, and $\overline{BD}$, respectively:

$$x^2 + y^2 + z^2 = 2^2, \quad (1)$$

$$(x - 3)^2 + y^2 + z^2 = \sqrt{13}^2, \quad (2)$$

$$x^2 + (y - 4)^2 + z^2 = (2\sqrt{5})^2. \quad (3)$$

Subtracting (1) from (2) gives $-6x + 9 = 9$, from which $x = 0$.

Subtracting (1) from (3) gives $-8y + 16 = 16$, from which $y = 0$.

Substituting $(x, y) = (0, 0)$ into (1) produces $z^2 = 4$, or $|z| = 2$.

Let the brackets denote areas. Finally, we find the volume of tetrahedron $ABCD$ using $\triangle ACD$ as the base:

$$\begin{aligned} V_{ABCD} &= \frac{1}{3} \cdot [ACD] \cdot h_B \\ &= \frac{1}{3} \cdot \left(\frac{1}{2} \cdot AC \cdot AD \right) \cdot |z| \\ &= \boxed{\text{(C) } 4}. \end{aligned}$$

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Solution 3 (Trirectangular Tetrahedron)

<https://mathworld.wolfram.com/TrirectangularTetrahedron.html>

Given the observations from Solution 1, where $\triangle ACD$, $\triangle ABC$, and $\triangle ABD$ are right triangles, the base is $\triangle ABD$. We can apply the information about a trirectangular tetrahedron (all of the face angles are right angles), which states that the volume is

$$\begin{aligned} V &= \frac{1}{6} \cdot AB \cdot AC \cdot BD \\ &= \frac{1}{6} \cdot 2 \cdot 4 \cdot 3 \\ &= \boxed{\text{(C) } 4}. \end{aligned}$$

AMC60 (Solution)

Remark Here is a similar problem from another AMC test: [2015 AMC 10A Problem 21](#).

14. (AMC 10 10A 2021) All the roots of the polynomial $z^6 - 10z^5 + Az^4 + Bz^3 + Cz^2 + Dz + 16$ are positive integers, possibly repeated. What is the value of B ?

(A) -88 (B) -80 (C) -64 (D) -41 (E) -40

Answer: A

Solution 1

By Vieta's formulas, the sum of the six roots is 10 and the product of the six roots is 16. By inspection, we see the roots are 1, 1, 2, 2, 2, and 2, so the function is $(z - 1)^2(z - 2)^4 = (z^2 - 2z + 1)(z^4 - 8z^3 + 24z^2 - 32z + 16)$. Therefore, calculating just the z^3 terms, we get $B = -32 - 48 - 8 = \boxed{\text{(A)} -88}$.

Solution 2

Using the same method as Solution 1, we find that the roots are 2, 2, 2, 2, 1, and 1. Note that B is the negation of the 3rd symmetric sum of the roots. Using casework on the number of 1's in each of the $\binom{6}{3} = 20$ products $r_a \cdot r_b \cdot r_c$, we obtain

$$B = - \left(\binom{4}{3} \binom{2}{0} \cdot 2^3 + \binom{4}{2} \binom{2}{1} \cdot 2^2 \cdot 1 + \binom{4}{1} \binom{2}{2} \cdot 2 \right) = -(32 + 48 + 8) = \boxed{\text{(A)} -88}.$$

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Solution 3 (brute force)

Same as solution 1, we find the roots are 1, 1, 2, 2, 2, 2 by Vieta's formula. This next part is brute force expansion. $-B =$

$$r_1 r_2 r_3 + r_1 r_2 r_4 + r_1 r_2 r_5 + r_1 r_2 r_6 +$$

$$r_1 r_3 r_4 + r_1 r_3 r_5 + r_1 r_3 r_6 +$$

$$r_1 r_4 r_5 + r_1 r_4 r_6 +$$

$$r_1 r_5 r_6 +$$

$$r_2 r_3 r_4 + r_2 r_3 r_5 + r_2 r_3 r_6 +$$

$$r_2 r_4 r_5 + r_2 r_4 r_6 +$$

$$r_2 r_5 r_6 +$$

$$r_3 r_4 r_5 + r_3 r_4 r_6 +$$

$$r_3 r_5 r_6 +$$

$$r_4 r_5 r_6$$

$$= 2 \times 4 + 4 \times 12 + 8 \times 4$$

$$= 8 + 48 + 32$$

$$B = \boxed{\text{(A)} -88}$$

15. (AMC 10 10A 2021) Values for A, B, C , and D are to be selected from $\{1, 2, 3, 4, 5, 6\}$ without replacement (i.e. no two letters have the same value). How many ways are there to make such choices so that the two curves $y = Ax^2 + B$ and $y = Cx^2 + D$ intersect? (The order in which the curves are listed does not matter; for example, the choices $A = 3, B = 2, C = 4, D = 1$ is considered the same as the choices $A = 4, B = 1, C = 3, D = 2$.)

(A) 30 (B) 60 (C) 90 (D) 180 (E) 360

Answer: C

Solution 1 (Intuition):

Visualizing the two curves, we realize they are both parabolas with the same axis of symmetry. WLOG the first equation is above the second, since order doesn't matter. Then $C > A$ and $B > D$. Therefore the number of ways to choose the four integers is $\binom{6}{2}\binom{4}{2} = 90$, and the answer is (C) 90.

Solution 2 (Algebra):

Setting $y = Ax^2 + B = Cx^2 + D$, we find that $Ax^2 - Cx^2 = x^2(A - C) = D - B$, so $x^2 = \frac{D-B}{A-C} \geq 0$ by the trivial inequality. This implies that $D - B$ and $A - C$ must both be positive or negative. If two distinct values are chosen for (A, C) and (B, D) respectively, there are 2 ways to order them so that both the numerator and denominator are positive/negative (increasing and decreasing). We must divide by 2 at the end, however, since the 2 curves aren't considered distinct. Calculating, we get

$$\frac{1}{2} \cdot \binom{6}{2} \binom{4}{2} \cdot 2 = \span style="border: 1px solid black; padding: 2px;">(C) 90.$$

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Solution 3 (Symmetry):

Like in Solution 2, we find $\frac{D-B}{A-C} \geq 0$. Notice that, since $D \neq B$, this expression can never equal 0, and since $A \neq C$, there won't be a divide-by-0. This means that every choice results in either a positive or a negative value.

For every choice of (A, B, C, D) that results in a positive value, we can flip B and D to obtain a corresponding negative value. This is a bijection (we could flip B and D again to obtain the original choice (injectivity) and we could flip B and D from any negative choice to obtain the corresponding positive choice (surjectivity)), so half of the choices are positive (where the curves intersect) and half are negative (where they don't).

This means that of the $\frac{6 \cdot 5 \cdot 4 \cdot 3}{2} = 180$ total choices (dividing by 2 because the order of the curves does not matter), half of them, or $\frac{180}{2} = \span style="border: 1px solid black; padding: 2px;">(C) 90, lead to intersecting curves.$

Solution 4 (Simple)

Notice that one of the parabolas must be the wider one and the other one must be the thinner one. There are only two options: the wider one is above the thinner one, or the thinner one is above the wider one. Only the first option works. Therefore, out of the $\frac{1}{2} \cdot 6 \cdot 5 \cdot 4 \cdot 3 = 180$ ways to pick A, B, C , and D without regard to order, only $\frac{1}{2} \cdot 180 = 90$ of these work.

Solution 5

We notice that the only way this will work is if the wider parabola is lower and the thinner one is higher. We have 15 ways to pick values for one parabola to be thinner than the other; we multiply this by 2 to get 30 because there are 2 choices for which parabola is thinner. After picking these 2 numbers, there are 3 ways to choose values for the thinner parabola to have a higher origin than the wider one. We multiply 30 by 3 to get the answer of (C) 90.

16. (AMC 10 10A 2021) In the following list of numbers, the integer n appears n times in the list for $1 \leq n \leq 200$.

1, 2, 2, 3, 3, 3, 4, 4, 4, 4, $\dots$, 200, 200, $\dots$, 200

What is the median of the numbers in this list?

- (A) 100.5 (B) 134 (C) 142 (D) 150.5 (E) 167

Answer: C

Solution 1

There are $1 + 2 + \dots + 199 + 200 = \frac{(200)(201)}{2} = 20100$ numbers in total. Let the median be k . We want to find the median k such that

$$\frac{k(k+1)}{2} = 20100/2,$$

or

$$k(k+1) = 20100.$$

Note that $\sqrt{20100} \approx 142$. Plugging this value in as k gives

$$\frac{1}{2}(142)(143) = 10153.$$

$10153 - 142 < 10050$, so 142 is the 152nd and 153rd numbers, and hence, our desired answer. (C) 142.

Note that we can derive $\sqrt{20100} \approx 142$ through the formula

$$\sqrt{n} = \sqrt{a+b} \approx \sqrt{a} + \frac{b}{2\sqrt{a}+1},$$

where a is a perfect square less than or equal to n . We set a to 19600, so $\sqrt{a} = 140$, and $b = 500$. We then have $n \approx 140 + \frac{500}{2(140)+1} \approx 142$. approximation by ciceronii

Note by Fasolinka (use answer choices): Once you know that the answer is in the 140s range (200,000 is around 14^2 times 10^2) by the approximation, it is highly improbable for the answer to be anything but C.

Solution 2

The x th number of this sequence is $\left\lceil \frac{-1 \pm \sqrt{1+8x}}{2} \right\rceil$ via the quadratic formula. We can see that if we halve x we end up getting $\left\lceil \frac{-1 \pm \sqrt{1+4x}}{2} \right\rceil$. This is approximately the number divided by $\sqrt{2}$. $\frac{200}{\sqrt{2}} = 141.4$ and since 142 looks like the only number close to it, it is answer (C)142 Lopkiloinm

Solution 3 (Answer Choices)

We can look at answer choice (C), which is 142 first. That means that the number of numbers from 1 to 142 is roughly the number of numbers from 143 to 200.

The number of numbers from 1 to 142 is $\frac{142(142+1)}{2}$ which is approximately 10000.

The number of numbers from 143 to 200 is $\frac{200(200+1)}{2} - \frac{142(142+1)}{2}$ which is approximately 10000 as well. Therefore, we can be relatively sure the answer choice is

(C) 142.

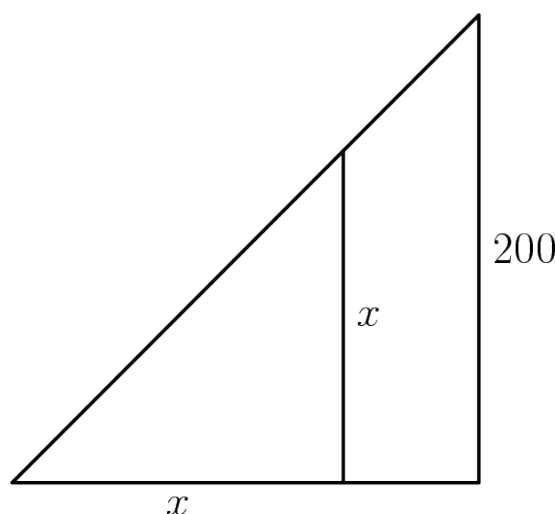
-PureSwag

Solution 4 (Geometry)

We can arrange the numbers in the following pattern:

$$\begin{array}{ccccccc} & & & & & 200 & \\ & & & & & 199 & 200 \\ & & & & & \vdots & \\ & & & & & \vdots & \\ & & & & & \vdots & \\ & & & & & 2 & \dots & 199 & 200 \\ 1 & 2 & \dots & 199 & 200 \end{array}$$

We can see this as a isosceles right triangle, with legs of length 200.



Let x be the side length such that both sides of the triangle have the same area. The desired answer is then around x because about half of the numbers in the list fall on each side.

Solving for x yields:

$$\begin{aligned}\frac{x^2}{2} &= \frac{1}{2} \cdot \frac{200^2}{2} \\ x^2 &= \frac{1}{2} \cdot 200^2 \\ x &= \frac{200}{\sqrt{2}} = 100\sqrt{2} \approx 141.\end{aligned}$$

We see that $\boxed{(C) 142}$ is the closest to x by far, and thus, can be relatively certain this is the answer.

17. (AMC 10 10A 2021) Trapezoid $ABCD$ has $\overline{AB} \parallel \overline{CD}$, $BC = CD = 43$, and $\overline{AD} \perp \overline{BD}$. Let O be the intersection of the diagonals $\overline{AC}$ and $\overline{BD}$, and let P be the midpoint of $\overline{BD}$. Given that $OP = 11$, the length AD can be written in the form $m\sqrt{n}$, where m and n are positive integers and n is not divisible by the square of any prime. What is $m + n$?

(A) 65 (B) 132 (C) 157 (D) 194 (E) 215

Answer: D

Solution 1 (Similar Triangles and Pythagorean Theorem)

Angle chasing* reveals that $\triangle BPC \sim \triangle BDA$, therefore

$$2 = \frac{BD}{BP} = \frac{AB}{BC} = \frac{AB}{43},$$

or $AB = 86$.

Additional angle chasing shows that $\triangle ABO \sim \triangle CDO$, therefore

$$2 = \frac{AB}{CD} = \frac{BO}{OD} = \frac{BP + 11}{BP - 11},$$

or $BP = 33$ and $BD = 66$.

Since $\triangle ADB$ is right, the Pythagorean theorem implies that

$$AD = \sqrt{86^2 - 66^2} = 4\sqrt{190}.$$

The answer is $4 + 190 = \boxed{\text{(D)} 194}$.

- Angle Chasing: If we set $\angle DBC = \alpha$, then we know that $\angle DCB = 180^\circ - 2\alpha$ because $\triangle DBC$ is isosceles. Then, $\angle BCP = 90^\circ - \alpha$ (since P is the midpoint of $\overline{BD}$, and therefore line $\overline{CP}$ is a median, which then means $\angle BCP$ bisects $\angle DBC$), so $\angle BPC$ is a right angle. Because $\angle BDC = \alpha$ and $\overline{AB} \parallel \overline{DC}$, we conclude that $\angle ABD = \alpha$ too. Lastly, because $\triangle BPC$ and $\triangle BDA$ are both right triangles, they are similar by AA.

=Another way to angle chase It is easy to show that triangles CDO and ODA are similar through AA. From there, we already know that through isosceles triangles CPD and CPB are congruent. Hence we can simply see that angle PCO and angle OAD are congruent through similar triangles. Also, angle OAB and angle OCD are congruent through parallelism. Thus, angle BAD and angle DCP are congruent. Then, through AA, $\triangle BCP \cong \triangle DCP \sim \triangle BDA$.

This skips the unnecessary step of angle chasing via similar angles of an isosceles triangle and skips the step of manually counting each angle's measure in terms of α . However, no discredit to the original angle chasing comment, it has its merits as well (using the angle measures in the above angle chasing technique proves useful for showing other statements [and cooler facts] about the problem).

Solution 2 (Similar Triangles, Areas, Pythagorean Theorem)

Since $\triangle BCD$ is isosceles with base $\overline{BD}$, it follows that median $\overline{CP}$ is also an altitude. Let $OD = x$ and $CP = h$, so $PB = x + 11$.

Since $\angle AOD = \angle COP$ by vertical angles, we conclude that $\triangle AOD \sim \triangle COP$ by AA, from which $\frac{AD}{CP} = \frac{OD}{OP}$, or

$$AD = CP \cdot \frac{OD}{OP} = h \cdot \frac{x}{11}.$$

Let the brackets denote areas. Notice that $[AOD] = [BOC]$ (By the same base and height, we deduce that $[ACD] = [BDC]$. Subtracting $[OCD]$ from both sides gives $[AOD] = [BOC]$.). Doubling both sides produces

$$\begin{aligned} 2[AOD] &= 2[BOC] \\ OD \cdot AD &= OB \cdot CP \\ x \left(\frac{hx}{11} \right) &= (x + 22)h \\ x^2 &= 11(x + 22). \end{aligned}$$

Rearranging and factoring result in $(x - 22)(x + 11) = 0$, from which $x = 22$.

Applying the Pythagorean Theorem to right $\triangle CPB$, we have

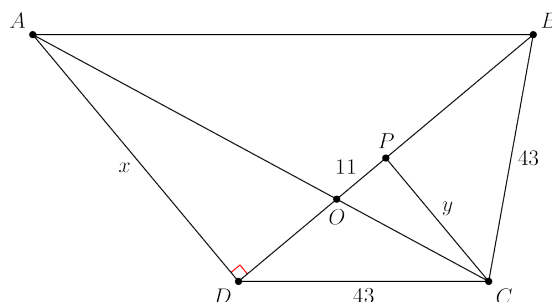
$$h = \sqrt{43^2 - 33^2} = \sqrt{(43 + 33)(43 - 33)} = \sqrt{760} = 2\sqrt{190}.$$

Finally, we get

$$AD = h \cdot \frac{x}{11} = 4\sqrt{190},$$

so the answer is $4 + 190 = \boxed{\text{(D)} 194}$.

Solution 3 (Short)



Let $CP = y$. CP is a perpendicular bisector of DB . Then, let $DO = x$, thus $DP = PB = 11 + x$.

(1) $\triangle CPO \sim \triangle ADO$, so we get $\frac{AD}{x} = \frac{y}{11}$, or $AD = \frac{xy}{11}$.

(2) Applying Pythagorean Theorem on $\triangle CDP$ gives $(11 + x)^2 + y^2 = 43^2$.

(3) $\triangle BPC \sim \triangle BDA$ with ratio $1 : 2$, so $AD = 2y$ using the fact that P is the midpoint of BD .

Thus, $\frac{xy}{11} = 2y$, or $x = 22$. And $y = \sqrt{43^2 - 33^2} = 2\sqrt{190}$, so $AD = 4\sqrt{190}$ and the answer is $4 + 190 = \boxed{\text{(D)} 194}$.

Solution 4 (Law of Sines and Similar Triangles)

This solution will refer to the diagram section at the top of this page.

We know $\angle ABD = \angle BDC = \angle DBC$. Let $PB = 11 + x$. Notice that $\triangle ABD$ is similar to $\triangle DOC$, so $\frac{AB}{43} = \frac{22+x}{x} \implies AB = \frac{43(22+x)}{x}$. Now we apply the Law of Sines to isosceles triangle $\triangle DCB$:

$$\frac{\sin(\angle ABD)}{43} = \frac{\sin(180^\circ - 2\angle ABD)}{22 + 2x} = \frac{2\sin(\angle ABD)\cos(\angle ABD)}{22 + 2x}$$

$$\frac{1}{43} = \frac{\cos(\angle ABD)}{11 + x} \implies \cos(\angle ABD) = \frac{11 + x}{43}$$

From right triangle $\triangle BAD$ we already know that $\cos(\angle ABD) = \frac{BD}{AB} = \frac{(22+2x)x}{43(22+x)}$.

Setting these equations equal to each other gives us:

$$\frac{(22 + 2x)x}{43(22 + x)} = \frac{11 + x}{43}$$

$$x = 22 \implies BD = 2(11 + x) = 66, AB = \frac{43 \cdot 44}{22} = 86.$$

So $AD = \sqrt{86^2 - 66^2} = 4\sqrt{190}$. Ans = $\boxed{194}$.

Solution 5 (Extending the Line)

Observe that $\triangle BPC$ is congruent to $\triangle DPC$; both are similar to $\triangle BDA$. Let's extend $\overline{AD}$ and $\overline{BC}$ past points D and C respectively, such that they intersect at a point E . Observe that $\angle BDE$ is 90 degrees, and that $\angle DBE \cong \angle PBC \cong$

$\angle DBA \implies \angle DBE \cong \angle DBA$. Thus, by ASA, we know that $\triangle ABD \cong \triangle EBD$, thus, $AD = ED$, meaning D is the midpoint of AE . Let M be the midpoint of $\overline{DE}$. Note that $\triangle CME$ is congruent to $\triangle BPC$, thus $BC = CE$, meaning C is the midpoint of $\overline{BE}$.

Therefore, $\overline{AC}$ and $\overline{BD}$ are both medians of $\triangle ABE$. This means that O is the centroid of $\triangle ABE$; therefore, because the centroid divides the median in a 2:1 ratio, $\frac{BO}{OD} = \frac{AO}{OE} = \frac{BD}{DO}$. Recall that P is the midpoint of BD ; $DP = \frac{BD}{2}$. The question tells us that $OP = 11$; $DP - DO = 11$; we can write this in terms of DB ; $\frac{DB}{2} - \frac{DB}{3} = \frac{DB}{6} = 11 \implies DB = 66$.

We are almost finished. Each side length of $\triangle ABD$ is twice as long as the corresponding side length $\triangle CBP$ or $\triangle CPD$, since those triangles are similar; this means that $AB = 2 \cdot 43 = 86$. Now, by Pythagorean theorem on $\triangle ABD$, $AB^2 - BD^2 = AD^2 \implies 86^2 - 66^2 = AD^2 \implies AD = \sqrt{3040} \implies AD = 4\sqrt{190}$.

The answer is $4 + 190 = \boxed{\text{(D)} 194}$.

Solution 6

Since P is the midpoint of isosceles triangle BCD , it would be pretty easy to see that $CP \perp BD$. Since $AD \perp BD$ as well, $AD \parallel CP$. Connecting AP , it's obvious that $[ADC] = [ADP]$. Since $DP = BP$, $[APB] = [ADC]$.

Since P is the midpoint of BD , the height of $\triangle APB$ on side AB is half that of $\triangle ADC$ on CD . Since $[APB] = [ADC]$, $AB = 2CD$.

As a basic property of a trapezoid, $\triangle AOB \sim \triangle COD$, so $\frac{OB}{OD} = \frac{AB}{CD} = 2$, or $OB = 2OD$. Letting $OD = x$, then $PB = DP = 11 + x$, and $OB = 22 + x$. Hence $22 + x = 2x$ and $x = 22$.

Since $\triangle AOD \sim \triangle COP$, $\frac{AD}{PC} = \frac{OD}{OP} = 2$. Since $PD = 11 + 22 = 33$, $PC = \sqrt{43^2 - 33^2} = \sqrt{760}$.

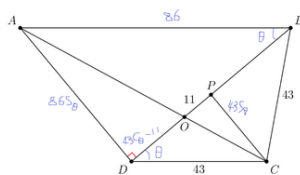
So, $AD = 2\sqrt{760} = 4\sqrt{190}$. The correct answer is $\boxed{\text{(D)} 194}$.

Solution 7 (Coordinate Geometry)

Let D be the origin of the cartesian coordinate plane, B lie on the positive x -axis, and A lie on the negative y -axis. Then let the coordinates of $B = (2a, 0)$, $A = (0, -2b)$. Then the slope of AB is $\frac{b}{a}$. Since $AB \parallel CD$ the slope of CD is the same. Note that as $\triangle DCB$ is isosceles C lies on $x = a$. Thus since CD has equation $y = \frac{b}{a}x$ (D is the origin), $C = (a, b)$. Therefore AC has equation $y = \frac{3b}{a}x - 2b$ and intersects BD (x -axis) at $O = (\frac{2}{3}a, 0)$. The midpoint of BD is $P = (a, 0)$, so $OP = \frac{a}{3} = 11$, from which $a = 33$. Then by Pythagorean theorem on $\triangle DPC$ ($\triangle DBC$ is isosceles), we have $b = \sqrt{43^2 - 33^2} = 2\sqrt{190}$, so $2b = 4\sqrt{190}$.

Finally, the answer is $4 + 190 = \boxed{\text{(D)} 194}$.

Solution 8 (Trigonometry)



set $\angle BDC = \theta$ $BD = 2 \cdot DP = 2 \cdot 43 \cdot \cos \theta$ $AB = \frac{BD}{\cos \angle DBA} = \frac{BD}{\cos \theta} = 2 \cdot 43 \cdot \frac{\cos \theta}{\cos \theta} = 86$
 $\frac{OP}{DO} = \frac{CP}{AD} \cdot \frac{11}{43 \cdot \cos \theta - 11} = \frac{43 \cdot \sin \theta}{86 \cdot \sin \theta} \cos \theta = \frac{33}{43}$ $AD = 86 \cdot \sin \theta = 2\sqrt{760} = 4\sqrt{190} =$
(D) 194.

18. (AMC 10 10A 2021) Let f be a function defined on the set of positive rational numbers with the property that $f(a \cdot b) = f(a) + f(b)$ for all positive rational numbers a and b . Furthermore, suppose that f also has the property that $f(p) = p$ for every prime number p . For which of the following numbers x is $f(x) < 0$?

(A) $\frac{17}{32}$ (B) $\frac{11}{16}$ (C) $\frac{7}{9}$ (D) $\frac{7}{6}$ (E) $\frac{25}{11}$

Answer: E

Solution 1 (Intuitive)

From the answer choices, note that

$$\begin{aligned} f(25) &= f\left(\frac{25}{11} \cdot 11\right) \\ &= f\left(\frac{25}{11}\right) + f(11) \\ &= f\left(\frac{25}{11}\right) + 11. \end{aligned}$$

On the other hand, we have

$$\begin{aligned} f(25) &= f(5 \cdot 5) \\ &= f(5) + f(5) \\ &= 5 + 5 \\ &= 10. \end{aligned}$$

Equating the expressions for $f(25)$ produces

$$f\left(\frac{25}{11}\right) + 11 = 10,$$

from which $f\left(\frac{25}{11}\right) = -1$. Therefore, the answer is **(E)** $\frac{25}{11}$. **Remark**

Similarly, we can find the outputs of f at the inputs of the other answer choices:

$$\begin{aligned} \text{(A)} \quad f\left(\frac{17}{32}\right) &= 7 \\ \text{(B)} \quad f\left(\frac{11}{16}\right) &= 3 \\ \text{(C)} \quad f\left(\frac{7}{9}\right) &= 1 \\ \text{(D)} \quad f\left(\frac{7}{6}\right) &= 2 \end{aligned}$$

Alternatively, refer to Solutions 2 and 4 for the full processes.

Solution 2 (Specific)

We know that $f(p) = f(p \cdot 1) = f(p) + f(1)$. By transitive, we have

$$f(p) = f(p) + f(1).$$

Subtracting $f(p)$ from both sides gives $0 = f(1)$. Also

$$f(2) + f\left(\frac{1}{2}\right) = f(1) = 0 \implies 2 + f\left(\frac{1}{2}\right) = 0 \implies f\left(\frac{1}{2}\right) = -2$$

$$f(3) + f\left(\frac{1}{3}\right) = f(1) = 0 \implies 3 + f\left(\frac{1}{3}\right) = 0 \implies f\left(\frac{1}{3}\right) = -3$$

$$f(11) + f\left(\frac{1}{11}\right) = f(1) = 0 \implies 11 + f\left(\frac{1}{11}\right) = 0 \implies f\left(\frac{1}{11}\right) = -11$$

In **(A)** we have $f\left(\frac{17}{32}\right) = 17 + 5f\left(\frac{1}{2}\right) = 17 - 5(2) = 7$.

In **(B)** we have $f\left(\frac{11}{16}\right) = 11 + 4f\left(\frac{1}{2}\right) = 11 - 4(2) = 3$.

In **(C)** we have $f\left(\frac{7}{9}\right) = 7 + 2f\left(\frac{1}{3}\right) = 7 - 2(3) = 1$.

In **(D)** we have $f\left(\frac{7}{6}\right) = 7 + f\left(\frac{1}{2}\right) + f\left(\frac{1}{3}\right) = 7 - 2 - 3 = 2$.

In **(E)** we have $f\left(\frac{25}{11}\right) = 10 + f\left(\frac{1}{11}\right) = 10 - 11 = -1$.

Thus, our answer is **(E)** $\frac{25}{11}$.

Solution 3 (Generalized)

Consider the rational $\frac{a}{b}$, for a, b integers. We have $f(a) = f\left(\frac{a}{b} \cdot b\right) = f\left(\frac{a}{b}\right) + f(b)$. So $f\left(\frac{a}{b}\right) = f(a) - f(b)$. Let p be a prime. Notice that $f(p^k) = kf(p)$. And $f(p) = p$. So if $a = p_1^{a_1} p_2^{a_2} \cdots p_k^{a_k}$, $f(a) = a_1 p_1 + a_2 p_2 + \cdots + a_k p_k$. We simply need this to be greater than what we have for $f(b)$. Notice that for answer choices **(A)**, **(B)**, **(C)**, and **(D)**, the numerator has fewer prime factors than the denominator, and so they are less likely to work. We check **(E)** first, and it works, therefore the answer is

(E) $\frac{25}{11}$

Solution 4 (Generalized)

We derive the following properties of f :

a) By induction, we have

$$f\left(\prod_{k=1}^n a_k\right) = \sum_{k=1}^n f(a_k)$$

for all positive rational numbers a_k and positive integers n . Since positive powers are just repeated multiplication of the base, it follows that

$$f(a^n) = f\left(\prod_{k=1}^n a\right) = \sum_{k=1}^n f(a) = nf(a)$$

for all positive rational numbers a and positive integers n .

b) For all positive rational numbers a , we have

$$f(a) = f(a \cdot 1) = f(a) + f(1),$$

from which $f(1) = 0$.

c) For all positive rational numbers a , we have

$$f(a) + f\left(\frac{1}{a}\right) = f\left(a \cdot \frac{1}{a}\right) = f(1) = 0,$$

from which $f\left(\frac{1}{a}\right) = -f(a)$.

For all positive integers x and y , suppose $\prod_{k=1}^m p_k^{d_k}$ and $\prod_{k=1}^n q_k^{e_k}$ are their respective prime factorizations. We get

$$\begin{aligned} f\left(\frac{x}{y}\right) &= f(x) + f\left(\frac{1}{y}\right) \\ &= f(x) - f(y) && \text{by Property 3} \\ &= f\left(\prod_{k=1}^m p_k^{d_k}\right) - f\left(\prod_{k=1}^n q_k^{e_k}\right) \\ &= \left[\sum_{k=1}^m f(p_k^{d_k})\right] - \left[\sum_{k=1}^n f(q_k^{e_k})\right] && \text{by Property 1} \\ &= \left[\sum_{k=1}^m d_k f(p_k)\right] - \left[\sum_{k=1}^n e_k f(q_k)\right] && \text{by Property 1} \\ &= \left[\sum_{k=1}^m d_k p_k\right] - \left[\sum_{k=1}^n e_k q_k\right]. \end{aligned}$$

We apply f to each fraction in the answer choices:

$$\begin{aligned} \text{(A)} \quad f\left(\frac{17}{32}\right) &= f\left(\frac{17^1}{2^5}\right) &= [1(17)] - [5(2)] &= 7 \\ \text{(B)} \quad f\left(\frac{11}{16}\right) &= f\left(\frac{11^1}{2^4}\right) &= [1(11)] - [4(2)] &= 3 \\ \text{(C)} \quad f\left(\frac{7}{9}\right) &= f\left(\frac{7^1}{3^2}\right) &= [1(7)] - [2(3)] &= 1 \\ \text{(D)} \quad f\left(\frac{7}{6}\right) &= f\left(\frac{7^1}{2^1 \cdot 3^1}\right) &= [1(7)] - [1(2) + 1(3)] &= 2 \\ \text{(E)} \quad f\left(\frac{25}{11}\right) &= f\left(\frac{5^2}{11^1}\right) &= [2(5)] - [1(11)] &= -1 \end{aligned}$$

Therefore, the answer is **(E)** $\frac{25}{11}$.

Solution 5 (Quick, Dirty, and Frantic Last Hope)

Note that answer choices **(A)** through **(D)** are $\frac{\text{prime}}{\text{composite}}$, whereas **(E)** is $\frac{\text{composite}}{\text{prime}}$. Because the functional equation is related to primes, we hope that the uniqueness of answer choice **(E)** $\frac{25}{11}$ is enough.

Solution 6 (Rushed Generalization)

If $f(a \cdot b) = f(a) + f(b)$, and if $f(p) = p$, then $f(p \cdot p) = 2p$. You can do this multiple times (Ex: $f(p^3) = 3p$). You can quickly assume then, that $f(p^n) = np$. Thus the answer choices can then be rewritten as the product of a prime and another prime to the negative power. Answer choices A-C are straightforward. For

D, you can rewrite $\frac{1}{6}$ as $\frac{1}{2} \cdot \frac{1}{3}$. When you get to E, you get $f(25) + f(\frac{1}{11})$, which is $10 - 11$, which is -1 . So the answer is (E) $\frac{25}{11}$

Solution 7 (Generalized)

Note that for each of the answer choices we can multiply the fractions by their denominators to be left with only the numerator and also have the prime factorization of the denominators, then set the function equal to the numerator. This will be shown throughout the solution.

For answer choice (A) $\frac{17}{32}$, we have $f\left(\frac{17}{32}\right)$. Now, we can set up an equation by multiplying $\frac{17}{32}$ by 32 and setting it equal to 17. This will give $f\left(32 \cdot \frac{17}{32}\right) = f(17) \rightarrow f(32) + f\left(\frac{17}{32}\right) = f(17)$. Our goal is to find $f\left(\frac{17}{32}\right)$, therefore we must find $f(32)$ and $f(17)$. Since $f(p) = p$ for any prime p , $f(17) = 17$. Taking the prime factorization of 32 gives 2^5 , so $f(32) = f(2 \cdot 2 \cdot 2 \cdot 2 \cdot 2) = f(2) + f(2) + f(2) + f(2) + f(2) = 5f(2) = 5 \cdot 2 = 10$. Therefore, $10 + f\left(\frac{17}{32}\right) = 17$ and $f\left(\frac{17}{32}\right) = 7$ (B) $\frac{11}{16}$: $f\left(\frac{11}{16}\right) \rightarrow f\left(16 \cdot \frac{11}{16}\right) = f(11) \rightarrow f(16) + f\left(\frac{11}{16}\right) = 11 \rightarrow 4f(2) + f\left(\frac{11}{16}\right) = 11 \rightarrow f\left(\frac{11}{16}\right) = 3$ (C) $\frac{7}{9}$: $f\left(\frac{7}{9}\right) \rightarrow f\left(9 \cdot \frac{7}{9}\right) = f(7) \rightarrow f(9) + f\left(\frac{7}{9}\right) = 7 \rightarrow 2f(3) + f\left(\frac{7}{9}\right) = 7 \rightarrow f\left(\frac{7}{9}\right) = 1$ (D) $\frac{7}{6}$: $f\left(\frac{7}{6}\right) \rightarrow f\left(6 \cdot \frac{7}{6}\right) = f(7) \rightarrow f(6) + f\left(\frac{7}{6}\right) = 7 \rightarrow f(2) + f(3) + f\left(\frac{7}{6}\right) = 7 \rightarrow f\left(\frac{7}{6}\right) = 2$ (E) $\frac{25}{11}$: $f\left(\frac{25}{11}\right) \rightarrow f\left(11 \cdot \frac{25}{11}\right) = f(25) \rightarrow f(11) + f\left(\frac{25}{11}\right) = 2f(5) \rightarrow 11 + f\left(\frac{25}{11}\right) = 10 \rightarrow f\left(\frac{25}{11}\right) = -1$

Therefore, the answer is (E) $\frac{25}{11}$

Note: The general strategy here was the setting up of equations to find $f(x)$. By setting it equal to $f(a)$ where a was an integer, we could take the prime factorization to find the value of $f(a)$ and also set up an equation involving $f(\text{denominator})$ because the denominator was also an integer, therefore we could take the prime factorization and find its value

Solution 8 (Generalized But Fast)

Let p be a prime number. We have that $f(p) = p$. Now, consider $f\left(\frac{1}{p^a}\right)$ where a is a positive integer. Notice that

$$f\left(p^{a+1} \cdot \frac{1}{p^a}\right) = f(p) = p \quad \text{and} \quad f\left(p^{a+1} \cdot \frac{1}{p^a}\right) = f(p^{a+1}) + f\left(\frac{1}{p^a}\right) = (a+1)p + f\left(\frac{1}{p^a}\right)$$

and hence

$$(a+1)p + f\left(\frac{1}{p^a}\right) = p \implies f\left(\frac{1}{p^a}\right) = -ap.$$

Thus, we see that

$$f\left(\frac{25}{11}\right) = 2f(5) + f\left(\frac{1}{11}\right) = 10 - 11 = -1 < 0 \implies \boxed{\text{(E)} \frac{25}{11}}$$

shockfront99

19. (AMC 10 10A 2021) The area of the region bounded by the graph of

$$x^2 + y^2 = 3|x - y| + 3|x + y|$$

is $m + n\pi$, where m and n are integers. What is $m + n$?

- (A) 18 (B) 27 (C) 36 (D) 45 (E) 54

Answer: E

Solution 1

In order to attack this problem, we can use casework on the sign of $|x - y|$ and $|x + y|$.

Case 1: $|x - y| = x - y, |x + y| = x + y$

Substituting and simplifying, we have $x^2 - 6x + y^2 = 0$, i.e. $(x - 3)^2 + y^2 = 3^2$, which gives us a circle of radius 3 centered at $(3, 0)$.

Case 2: $|x - y| = y - x, |x + y| = x + y$

Substituting and simplifying again, we have $x^2 + y^2 - 6y = 0$, i.e. $x^2 + (y - 3)^2 = 3^2$. This gives us a circle of radius 3 centered at $(0, 3)$.

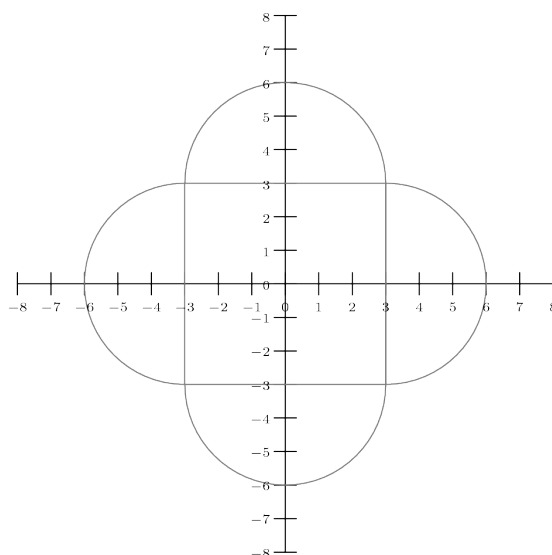
Case 3: $|x - y| = x - y, |x + y| = -x - y$

Doing the same process as before, we have $x^2 + y^2 + 6y = 0$, i.e. $x^2 + (y + 3)^2 = 3^2$. This gives us a circle of radius 3 centered at $(0, -3)$.

Case 4: $|x - y| = y - x, |x + y| = -x - y$

One last time: we have $x^2 + y^2 + 6x = 0$, i.e. $(x + 3)^2 + y^2 = 3^2$. This gives us a circle of radius 3 centered at $(-3, 0)$.

After combining all the cases and drawing them on the Cartesian Plane, this is what the diagram looks like:



Now, the area of the shaded region is just a square with side length 6 with four semicircles of radius 3. The area is $6 \cdot 6 + 4 \cdot \frac{9\pi}{2} = 36 + 18\pi$. The answer is $36 + 18\pi$ which is **(E) 54**

Solution 2

A somewhat faster variant of solution 1 is to use a bit of symmetry in order to show that the remaining three cases are identical to Case 1 in the above solution, up to rotations by 90° about the origin. This allows us to quickly sketch the region after solving Case 1.

Upon simplifying Case 1, we obtain $(x - 3)^2 + y^2 = 3^2$ which is a circle of radius 3 centered at $(3, 0)$. We remark that only the points on the semicircle where $x \geq 3$ work here, since Case 1 assumes $x - y \geq 0$ and $x + y \geq 0$. Next, we observe that an ordered pair is a solution to the given equation if and only if any of its 90° rotations about the origin is a solution. This follows as the value of $x^2 + y^2 - 3(|x - y| + |x + y|)$ is invariant to 90° rotations, since $x^2 + y^2$ simply represents the square of the distance to the origin (which is unchanged upon rotation), and $|x - y| + |x + y|$ is the sum of the distances to the lines $y = x$ and $y = -x$, multiplied by $\sqrt{2}$ (also unchanged upon 90° rotation).

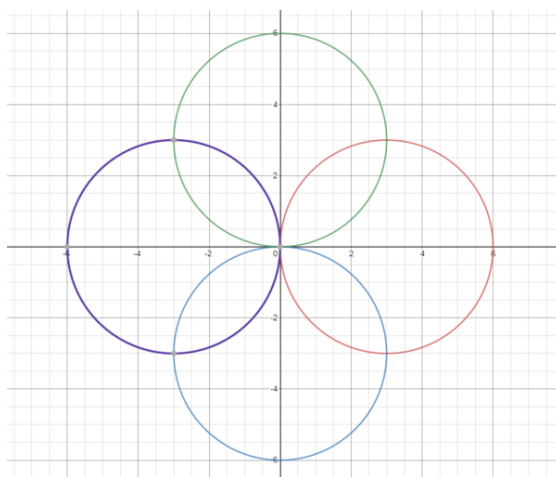
By the above observation, we can quickly sketch the remainder of the region, and the area is (E) 54 as above.

Solution 3 (Guessing)

Assume $y = 0$. We get that $x = 6$. That means that this figure must contain the points $(0, 6), (6, 0), (0, -6), (-6, 0)$. Now, assume that $x = y$. We get that $x = 3\sqrt{3}$. We get the points $(3, 3), (3, -3), (-3, 3), (-3, -3)$.

Since this contains $x^2 + y^2$, assume that there are circles. Therefore, we can guess that there is a center square with area $6 \cdot 6 = 36$ and 4 semicircles with radius 3. We get 4 semicircles with area 4.5π , and therefore the answer is $36 + 18 =$ (E)54

Remark This problem asks for the area of **the union of these four circles**:



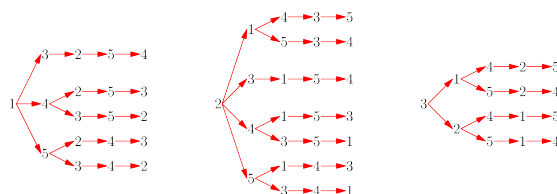
Rotation by 45 degrees with the help of rotation <https://www.wolframalpha.com/input/?i=rotate+45> we can simplify the equation to $a^2 + b^2 = 3\sqrt{2}|a| + 3\sqrt{2}|b|$. Then follow the previous question https://artofproblemsolving.com/wiki/index.php/2016_AMC_10B_Problems/Problem_21

20. (AMC 10 10A 2021) In how many ways can the sequence 1, 2, 3, 4, 5 be rearranged so that no three consecutive terms are increasing and no three consecutive terms are decreasing?
- (A) 10 (B) 18 (C) 24 (D) 32 (E) 44

Answer: D

Solution 1 (Enumeration by Symmetry)

By symmetry with respect to 3, note that $(x_1, x_2, x_3, x_4, x_5)$ is a valid sequence if and only if $(6 - x_1, 6 - x_2, 6 - x_3, 6 - x_4, 6 - x_5)$ is a valid sequence. We enumerate the valid sequences that start with 1, 2, 31, or 32, as shown below:



There are 16 valid sequences that start with 1, 2, 31, or 32. By symmetry, there are 16 valid sequences that start with 5, 4, 35, or 34. So, the answer is $16 + 16 = \boxed{\text{(D) } 32}$.

Solution 2 (Casework on the Consecutive Digits)

Reading the terms from left to right, we have two cases for the consecutive digits, where $+$ means increase and $-$ means decrease:

Case #1: $+, -, +, -$ **Case #2:** $-, +, -, +$

For Case #1, note that for the second and fourth terms, one term must be 5, and the other term must be either 3 or 4. We have four subcases:

(1) $_3_5_$ (2) $_5_3_$ (3) $_4_5_$ (4) $_5_4_$

For (1), the first two blanks must be 1 and 2 in some order, and the last blank must be 4. So, we get 2 possibilities. Similarly, (2) also has 2 possibilities.

For (3), there are no restrictions for the numbers 1, 2, and 3. So, we get $3! = 6$ possibilities. Similarly, (4) also has 6 possibilities.

Together, Case #1 has $2 + 2 + 6 + 6 = 16$ possibilities. By symmetry, Case #2 also has 16 possibilities.

Finally, the answer is $16 + 16 = \boxed{\text{(D) } 32}$. **Remark**

This problem is somewhat similar to [2004 AIME I Problem 6](#).

Solution 3 (Casework Similar to Solution 2)

Like Solution 2, we have two cases. Due to symmetry, we just need to count one of the cases. For the purpose of this solution, we will be doing $-, +, -, +$. Instead of starting with 5, we start with 1.

There are two ways to place it:

$_1_ _ _$
 $_ _ _1_$

Now we place 2, it can either be next to 1 and on the outside, or is place in where 1 would go in the other case. So now we have another two "sub case":

$_1_2_$ (case 1)

$21_ _ _$ (case 2)

There are $3!$ ways to arrange the rest for case 1, since there is no restriction.

For case 2, we need to consider how many ways to arrange 3,4,5 in a $a > b < c$ fashion. It should seem pretty obvious that b has to be 3, so there will be $2!$ way to put 4 and 5.

Now we find our result, times 2 for symmetry, times 2 for placement of 1 and times $(3!+2!)$ for the two different cases for placement of 2. This give us $2 * 2 * (3! + 2!) = 4 * (6 + 2) = \boxed{\text{(D)} 32}$.

Solution 4 (Casework on the Position of 5)

We only need to find the # of rearrangements when 5 is the 4th digit and 5th digit. Find the total, and multiply by 2. Then we can get the answer by adding the case when 5 is the third digit.

Case 1: 5 is the 5th digit. $___ ___ ___ ___ 5$

Then 4 can only be either 1st digit or the 3rd digit.

4 $___ ___ ___ 5$, then the only way is that 3 is the 3rd digit, so it can be either 231 or 132, give us 2 results.

$___ ___ 4 ___ 5$, then the 1st digit must be 2 or 3, 2 gives us 1 way, and 3 gives us 2 ways. (Can't be 1 because the first digit would increasing). Therefore, 4 in the middle and 5 in the last would result in 3 ways.

Case 2: 5 is the fourth digit. $___ ___ ___ 5 ___$

Then the last digit can be all of the 4 numbers 1, 2, 3, and 4. Let's say if the last digit is 4, then the 2nd digit would be the largest for the remaining digits to prevent increasing order or decreasing order. Then the remaining two are interchangeable, give us $2!$ ways. All of the 4 can work, so case 2 would result in $2! + 2! + 2! + 2! = 8$ ways.

Case 3: 5 is in the middle. $___ ___ 5 ___ ___$

Then there are only two cases: 1. 42513, then 4 and 3 are interchangeable, which results in $2! * 2!$. Or it can be 43512, then 4 and 2 are interchangeable, but it can not be 23514, so there can only be 2 possible ways: 43512, 21534.

Therefore, case 3 would result in $4 + 2 = 6$ ways.

$8 + 3 + 2 = 13$, so the total ways for case 1 and case 2 with both increasing and decreasing would be $13 * 2 = 26$.

Finally, we have $26 + 6 = \boxed{\text{(D)} 32}$.

Solution 5 (Overcounting)

First, we list the triples that are invalid:

543, 542, 541, 532, 531, 521, 432, 431, 321

By symmetry, there are the same amount of increasing triplets as there are decreasing ones. This yields 18 invalid 3 digit permutations in total.

Suppose the triplet is ABC and the other 2 digits are X and Y. We then have 3 ways to arrange a triplet with 2 other digits.

ABCXY, XABCY, XYABC

X and Y can be arranged 2 ways.

XY, YX

This produces $18 * 3 * 2 = 108$ permutations of invalid results. We have $5!$ ways to arrange 5 numbers so $120 - 108 = 12$.

Now, we must account for overcounting. For example, when 543 is counted, it only registers as one invalid permutation but in fact, it is 3 whole invalid permutations. We then complete this for the rest of the list:

54321 has 543, 432, and 321

54213 has 542 and 421

54123 has 541 and 123

53214 has 532 and 321

53124 has 531 and 124

52134 has 521 and 134

43215 has 432 and 321

43125 has 431 and 125

32145 has 321 and 145

This produces 19 values that we have overcounted but this value itself is also overcounted. We already counted 9 of the terms. This brings the final value of overcounted terms down to 10 for the decreasing triplets. By symmetry, 10 increasing triplets were overcounted.

This gives us $120 - 108 + 20 = \boxed{(D) 32}$.

Solution 6 (Using a Table)

It is easier to consider the complement of the desired cases, so try to find the cases that DO have three integers in increasing order. First, write down the sets of three numbers that feature the numbers in increasing order. They are 123, 124, 125, 134, 135, 145, 234, 245, 345. Each of these can be in three positions: the three are in the front with two more numbers in the back, the three are in the middle with two on each side, and two in the front and the set of three in the back. Now, count the number of combinations of 5 numbers that each of the set of three can form that have not been previously accounted for. Also, if the set features both 3 increasing and 3 decreasing, then do not count it because we will separately count them.

Three Increasing Numbers	Front	Middle	Back
123	2	2	2
124	2	2	2
125	1	2	2
134	2	2	2
135	1	2	2
145	1	2	2
234	2	1	1
245	1	1	1
345	1	0	1

This gives us a total of 41 possibilities. These account for the possibilities with ONLY increasing numbers. Mirror over to the other side to get the set of combinations with either at least 1 set of 3 increasing numbers in a row or only at least 1 set of 3 decreasing numbers, but not both. We can count the ones with both separately. 14532, 12543, 13542, 54123, 53124, 52134. Total of 6.

$41 \cdot 2 + 6 = 82 + 6 = 88$. This is the compliment. There are a total of $5! = 120$. $120 - 88 = 32$.

Thus, the answer is (D) 32.

Solution 7 (Symmetry)

First, note that there is a symmetry from $+, -, +, -$ to $-, +, -, +$ as follows: $a, b, c, d, e \leftrightarrow 6 - a, 6 - b, 6 - c, 6 - d, 6 - e$. Now, consider the placement of 5 in the $+, -, +, -$ case. Clearly, 5 is the maximum value, so it must be placed in the 2nd position or the 4th position, but we also have symmetry $a, b, c, d, e \leftrightarrow e, d, c, b, a$. We will find the number of sequences with 5 in the second position by using casework on the position of the 4.

Case 1:

4 5 ____: 3 is the maximum value, so it must go into the 4th position, leaving us 2 ways to fill in 1 and 2.

Case 2:

____ 5 ____ 4 ____: 1, 2, and 3 are all less than 4 or 5, so they can be filled in any manner: $3! = 6$.

Overall, we have 8 ways to fill in the sequences. By our two types of symmetry, there are $8 \cdot 2 \cdot 2 =$ (D) 32.

Solution 8 (PIE: No Casework)

We use complementary counting.

Let A be the set of all permutations $(a_1, a_2, a_3, a_4, a_5)$ of $(1, 2, 3, 4, 5)$ in which there exists at least one set of three consecutive increasing terms, and let B be the set of all permutations of $(1, 2, 3, 4, 5)$ in which there exists at least one set of three consecutive decreasing terms. The complement is $|A \cup B| = |A| + |B| - |A \cap B|$.

First, we calculate $|A|$. Let S_i be the set of permutations in which (a_i, a_{i+1}, a_{i+2}) is increasing for $i = 1, 2, 3$. We have

$$|A| = |S_1 \cup S_2 \cup S_3| = |S_1| + |S_2| + |S_3| - |S_1 \cap S_2| - |S_2 \cap S_3| - |S_3 \cap S_1| + |S_1 \cap S_2 \cap S_3|.$$

Since there are $\binom{5}{3}$ ways to pick (a_1, a_2, a_3) (there's only one way to arrange them in increasing order) and $2!$ ways to order a_4, a_5 ,

$$|S_1| = \binom{5}{3} \cdot 2! = 20.$$

By symmetry, we find $|S_2| = |S_3| = 20$.

Since we require $a_1 < a_2 < a_3 < a_4$ and there are $\binom{5}{4}$ ways to pick the numbers (again, only one way to arrange them in increasing order),

$$|S_1 \cap S_2| = \binom{5}{4} \cdot 1! = 5.$$

By symmetry, $|S_2 \cap S_3| = 5$.

Notice that $S_3 \cap S_1 = S_1 \cap S_2 \cap S_3$ since both translate to $a_1 < a_2 < a_3 < a_4 < a_5$, so we get

$$|A| = 20 + 20 + 20 - 5 - 5 - |S_3 \cap S_1| + |S_1 \cap S_3| = 60 - 10 = 50.$$

By symmetry, $|B| = 50$, and it remains to compute $|A \cap B|$.

We have two cases, either $a_1 < a_2 < a_3 > a_4 > a_5$ or $a_1 > a_2 > a_3 < a_4 < a_5$ (it is easy to verify that these are the only ones that work). WLOG consider the first. We must have $a_3 = 5$ since it's the largest, and for every choice of a_1, a_2 , there is exactly one way to arrange them in order and to choose/arrange a_3, a_4 . So, there are $\binom{4}{2}$ ways to pick a_1, a_2 . Multiplying by 2 gives

$$|A \cap B| = 2 \cdot 6 = 12 \implies |A \cup B| = 50 + 50 - 12 = 88.$$

Since there are $5! = 120$ permutations in total, the answer is $120 - 88 = \boxed{\text{(D) } 32}$.

Solution 9 (Quick)

Notice that the ways to arrange the numbers is $n, n + 2, n + 3, n + 2$, and n , for when it starts with 1, 2, 3, 4, and 5, respectively. Now that we have found a pattern, we can just solve for n , so we can choose to find the ways if it starts with 1 or 5. Solving, we get $n = 5$. So, the answer is $5 + 7 + 8 + 7 + 5 = \boxed{\text{(D) } 32}$.

Solution 10 (Brute Force)

We write out the $5! = 120$ cases, then filter out the valid ones:

13254, 14253, 14352, 15243, 15342, 21435, 21534, 23154, 24153, 24351, 25143, 25341, 31425, 31524, 32415, 32514, 34152, 34251, 35142, 35241, 41325, 41523, 42315, 42513, 43512, 45132, 45231, 51324, 51423, 52314, 52413, 53412.

We count these out and get $\boxed{\text{(D) } 32}$ permutations that work.

Note: We are NOT going to write our 120 cases and factor them out. I don't know why this is the first solution to pop up.

Note: Moved to number ten due to the sheer unnecessary nature of this solution :)

This solution is (kind of) realistic though, as there is enough space between the solutions where you could be off by a few and still be correct.

For anyone reading this, you shouldn't do this during the competition unless you have a lot of time and nothing else to do. Ning2020

Solution 11

We take our 120 cases and subtract the cases with 3 increasing or 3 decreasing. By symmetry, we can find the amount for one and multiply the number by 2.

Originally, we find that there are 10 choices for the numbers we will choose to be increasing, 3 choices for where to place them, and 2 choices for where to place the remaining numbers, for a total of 60 ways. However, we have overcounted all of the cases with multiple triplets of increasing integers in a row. We write these out and find that we have overcounted by 10 ways, for an actual total of 50 ways. Be careful at this step, as for certain cases like 12345 we actually overcounted by 2.

We multiply this by 2 to find that there are also 50 ways for the decreasing.

However, we have now overcounted the cases in which there are increasing and decreasing triplets.

We break these cases into 2 different groups: the first group is where the first 3 integers are decreasing, and the last triplet will be increasing. This forces the middle number to be 1; we quickly write out the cases and find 6.

The second case is where the first 3 are increasing and the second 3 are decreasing. This means the middle number can be a 5. We write out our cases for these and

find another 6 cases, for a grand total of 12 overcounted cases. This means there are actually a total of $50 + 50 - 12 = 88$ distinct cases where there are 3 increasing in a row or 3 decreasing.

We now subtract these 88 cases from 120 to get our answer of $\boxed{32}$.

21. (AMC 10 10A 2021) Let $ABCDEF$ be an equiangular hexagon. The lines AB, CD , and EF determine a triangle with area $192\sqrt{3}$, and the lines BC, DE , and FA determine a triangle with area $324\sqrt{3}$. The perimeter of hexagon $ABCDEF$ can be expressed as $m + n\sqrt{p}$, where m, n , and p are positive integers and p is not divisible by the square of any prime. What is $m + n + p$?

(A) 47 (B) 52 (C) 55 (D) 58 (E) 63

Answer: C

Solution 1

Let P, Q, R, X, Y , and Z be the intersections $\overleftrightarrow{AB} \cap \overleftrightarrow{CD}, \overleftrightarrow{CD} \cap \overleftrightarrow{EF}, \overleftrightarrow{EF} \cap \overleftrightarrow{AB}, \overleftrightarrow{BC} \cap \overleftrightarrow{DE}, \overleftrightarrow{DE} \cap \overleftrightarrow{FA}$, and $\overleftrightarrow{FA} \cap \overleftrightarrow{BC}$, respectively.

The sum of the interior angles of any hexagon is 720° . Since hexagon $ABCDEF$ is equiangular, each of its interior angles is $720^\circ \div 6 = 120^\circ$. By angle chasing, we conclude that the interior angles of $\triangle PQR, \triangle QDE, \triangle RFA, \triangle XCD, \triangle YEF$, and $\triangle ZAB$ are all 60° . Therefore, these triangles are all equilateral triangles, from which $\triangle PQR$ and $\triangle XYZ$ are both equilateral triangles.

We are given that

$$[PQR] = \frac{\sqrt{3}}{4} \cdot PQ^2 = 192\sqrt{3},$$

$$[XYZ] = \frac{\sqrt{3}}{4} \cdot YZ^2 = 324\sqrt{3},$$

so we get $PQ = 16\sqrt{3}$ and $YZ = 36$, respectively.

By equilateral triangles and segment addition, we find the perimeter of hexagon $ABCDEF$:

$$\begin{aligned} AB + BC + CD + DE + EF + FA &= AZ + PC + CD + DQ + YF + FA \\ &= (YF + FA + AZ) + (PC + CD + DQ) \\ &= YZ + PQ \\ &= 36 + 16\sqrt{3}. \end{aligned}$$

Finally, the answer is $36 + 16 + 3 = \boxed{(C) 55}$.

Solution 2

Let the length $AB = x, BC = y$. Then, we have

$$(y + 2x)^2 \cdot \frac{\sqrt{3}}{4} = 324\sqrt{3},$$

$$(x + 2y)^2 \cdot \frac{\sqrt{3}}{4} = 192\sqrt{3}.$$

We get

$$y + 2x = 36,$$

$$x + 2y = 16\sqrt{3}.$$

We want $3x + 3y$, and it follows that

$$3x + 3y = (y + 2x) + (x + 2y) = 36 + 16\sqrt{3}.$$

Finally, the answer is $36 + 16 + 3 = \boxed{\text{(C)} 55}$.

Note: This solution is kind of a fakesolve because it assumes the hexagon is rotationally symmetric, which we can see it is not by the diagram in Solution 1.

Solution 3 (Solution 1 But More Condensed)

Since it is an equiangular hexagon, each interior angle measures 120 degrees, and by angle chasing you can conclude that all the triangles are equilateral. Since the area formula for an equiangular triangle is $s^2 \frac{\sqrt{3}}{4}$, the side lengths of the triangles are 36 and $16\sqrt{3}$. In the above diagram the sum of the lengths of AF , AB , and FE is 36 and the sum of the lengths of BC , CD , and DE is $16\sqrt{3}$. Since these are also the side lengths of the hexagon, the answer is $36 + 16 + 3 = 55(C)$.

22. (AMC 10 10A 2021) Hiram's algebra notes are 50 pages long and are printed on 25 sheets of paper; the first sheet contains pages 1 and 2, the second sheet contains pages 3 and 4, and so on. One day he leaves his notes on the table before leaving for lunch, and his roommate decides to borrow some pages from the middle of the notes. When Hiram comes back, he discovers that his roommate has taken a consecutive set of sheets from the notes and that the average (mean) of the page numbers on all remaining sheets is exactly 19. How many sheets were borrowed?

(A) 10 (B) 13 (C) 15 (D) 17 (E) 20

Answer: B

Solution 1

Suppose the roommate took sheets a through b , or equivalently, page numbers $2a - 1$ through $2b$. Because there are $(2b - 2a + 2)$ numbers taken,

$$\frac{(2a - 1 + 2b)(2b - 2a + 2)}{2} + 19(50 - (2b - 2a + 2)) = \frac{50 \cdot 51}{2} \implies (2a + 2b - 39)(b - a + 1) = \frac{50 \cdot 13}{2} = 25 \cdot 13.$$

The first possible solution that comes to mind is if $2a + 2b - 39 = 25$, $b - a + 1 = 13 \implies a + b = 32$, $b - a = 12$, which indeed works, giving $b = 22$ and $a = 10$. The answer is $22 - 10 + 1 = \boxed{\text{(B)} 13}$.

Solution 2

Suppose the smallest page number borrowed is k , and n pages are borrowed. It follows that the largest page number borrowed is $k + n - 1$.

We have the following preconditions:

- n pages are borrowed means that $\frac{n}{2}$ sheets are borrowed, from which n must be even and $n < 50$.
- k must be odd, as the smallest page number borrowed is on the right side (odd-numbered).
- $1 + 2 + 3 + \dots + 50 = \frac{51(50)}{2} = 1275$.
- The sum of the page numbers borrowed is $\frac{(2k + n - 1)n}{2}$.

Together, we have

$$\begin{aligned}\frac{1275 - \frac{(2k+n-1)n}{2}}{50-n} &= 19 \\ 1275 - \frac{(2k+n-1)n}{2} &= 19(50-n) \\ 2550 - (2k+n-1)n &= 38(50-n) \\ 2550 - (2k+n-1)n &= 1900 - 38n \\ 650 &= (2k+n-39)n.\end{aligned}$$

The factors of 650 are

$$1, 2, 5, 10, 13, 25, 26, 50, 65, 130, 325, 650.$$

Since n is even and $n < 50$, we only have a few cases to consider:

n	$2k + n - 39$	k
2	325	181
10	65	47
26	25	19

Since $1 \leq k \leq 49$, only $k = 47$ or $k = 19$ is possible. If $k = 47$, then there will not be enough pages when we take 10 pages out starting from page 47. Therefore, the only possibility is $k = 19$. We conclude that $n = 26$ pages, or $\frac{n}{2} = \boxed{\text{(B)} 13}$ sheets, are borrowed.

Solution 2a (Basically the same thing as Solution 2)

Suppose the smallest page number borrowed is k , and the number of sheets is n . It follows that the largest page number borrowed is $k + 2n - 1$.

Then the total sum of pages he took is $(2k + 2n - 1)2n/2 = (2k + 2n - 1)n$.

The total sum of all the pages is $\frac{50 \cdot 51}{2} = 1275$.

With the mean being 19, we can write $\frac{1275 - (2k + 2n - 1)n}{50 - 2n} = 19$.

Simplifying, $1275 - (2k + 2n - 1)n = 950 - 38n$ and $325 = (2k + 2n - 39)n$.

Since n (the number of sheets) is a positive integer, it must be a factor of $325 = 25 \cdot 13$.

From the answer choices, $\boxed{\text{(B)} 13}$ is the only divisor of 325.

-unhappyfarmer

Solution 3

Let n be the number of sheets borrowed, with an average page number $k + 25.5$. The remaining $25 - n$ sheets have an average page number of 19 which is less than 25.5, the average page number of all 50 pages, therefore $k > 0$. Since the borrowed sheets start with an odd page number and end with an even page number we have $k \in \mathbb{N}$. We notice that $n < 25$ and $k \leq (49 + 50)/2 - 25.5 = 24 < 25$.

The weighted increase of average page number from 25.5 to $k + 25.5$ should be equal to the weighted decrease of average page number from 25.5 to 19, where the weights are the page number in each group (borrowed vs. remained), therefore

$$2nk = 2(25 - n)(25.5 - 19) = 13(25 - n) \implies 13|n \text{ or } 13|k$$

Since $n, k < 25$ we have either $n = 13$ or $k = 13$. If $n = 13$ then $k = 6$. If $k = 13$ then $2n = 25 - n$ which is impossible. Therefore the answer should be $n = \boxed{\text{(B)} 13}$.

Solution 4

Let $(2k - 1) - 2n$ be pages be borrowed, the sum of the page numbers on those pages is $(2n + 2k + 1)(n - k)$ while the sum of the rest pages is $1275 - (2n + 2k + 1)(n - k)$ and we know the average of the rest is $\frac{1275 - (2n + 2k + 1)(n - k)}{50 - 2n + 2k}$ which equals to 19; multiply this out we got $950 - 38(n - k) = 1275 - (2n + 2k + 1)(n - k)$ and we got $(2n + 2k - 37)(n - k) = 325$. As $325 = 25 \cdot 13$, we can see $n - k = 13$ and that is desired $\boxed{\text{(B)} 13}$.

Solution 5

Let c be the number of consecutive sheets Hiram's roommate borrows, and let b be the number of sheets preceding the c borrowed sheets (i.e. if the friend borrows sheets 3, 4, and 5, then $c = 3$ and $b = 2$).

The sum of the page numbers up till b sheets is $1 + 2 + 3 + \dots + 2b = \frac{2b \cdot (2b + 1)}{2} = b(2b + 1)$. The last page number of the borrowed sheets would be $2(b + c)$. Therefore, the sum of the remaining page numbers of the sheets after the c borrowed sheets would be $2(b + c) + 1 + 2(b + c) + 2 + \dots + 50$.

The total number of page numbers after the borrow would be $50 - 2c$.

Thus the average of the page numbers after the borrow would be:

$$\frac{b(2b + 1) + 2(b + c) + 1 + 2(b + c) + 2 + \dots + 50}{50 - 2c} = 19.$$

By the arithmetic series formula, this turns out to be:

$$\frac{b(2b + 1) + \frac{(2(b + c) + 1 + 50) \cdot (50 - 2c - 2b)}{2}}{50 - 2c} = 19$$

because in the changed sum, there are 50 numbers minus $2c$ borrowed numbers and $2b$ numbers from the first b sheets.

This simplifies to

$$\frac{b(2b + 1) + (2(b + c) + 51) \cdot (25 - c - b)}{50 - 2c} = 19 \implies 19(50 - 2c) = b(2b + 1) + (2b + 2c + 51) \cdot (25 - c - b).$$

Noticing that some terms will cancel, we expand, leading to:

$$\begin{aligned} 950 - 38c &= 2b^2 + b + 50b - 2bc - 2b^2 + 50c - 2c^2 - 2bc + 1275 - 51c - 51b \\ \implies 950 - 38c &= 1275 - 4bc - c - 2c^2 \implies 2c^2 + 4b - 37c = 325. \end{aligned}$$

Factoring, we get

$$c(2c + 4b - 37) = 325.$$

The prime factorization of 325 is $5^2 \cdot 13$. Recall that $0 \leq c \leq 25$, so c could be 1, 5, 13, or 25.

We can rule out $c = 25$ since Hiram would have no paper left over, so the average of the page numbers he has would be 0. We can now plug in the other answers for c and we if we get a valid answer for b .

- $c = 1 \implies 4b - 35 = 325 \implies 4b = 360 \implies b = 90$. Since Hiram only has 25 sheets, this is clearly wrong and we can rule out $c = 1$.

- $c = 5 \implies 10 + 4b - 37 = 65 \implies 4b = 92 \implies b = 23$. Since $b + c$ is 28 but we have only 25 sheets, this is also implausible so we can rule out $c = 5$.

Finally, just to make sure, we test $c = 13 \implies 26 + 4b - 37 = 25 \implies 4b = 36 \implies b = 9$.

b is an integer and $b + c = 22$, so everything checks out. The number of consecutive sheets borrowed by Hiram's friend is $c = \boxed{\text{(B) } 13}$.

Solution 6

The sum of all the page numbers is

$$1 + 2 + 3 + \cdots + 50 = 1275.$$

If we add the page numbers on each sheet, we get this sequence:

$$3, 7, 11, \dots, 99.$$

So we can write the sum of the numbers on the first sheet that the roommate borrowed as $4n + 3$ for some nonnegative integer, n . If the roommate borrowed k sheets, he borrowed sheets

$$4n + 3, 4n + 7, \dots, 4n + 4k - 1.$$

The sum of the numbers in this sequence is

$$\frac{k(8n + 4k + 2)}{2} = 4nk + 2k^2 + k.$$

Since there are 2 pages per sheet, there are $50 - 2k$ pages remaining, so the average page number of the remaining sheets is

$$\frac{1275 - (4nk + 2k^2 + k)}{50 - 2k}.$$

Therefore,

$$\frac{1275 - (4nk + 2k^2 + k)}{50 - 2k} = 19,$$

which simplifies to

$$2k^2 - 37k - 325 = -4nk.$$

Factoring the left-hand side,

$$(2k + 13)(k - 25) = -4nk.$$

Since the right-hand side of this equation is divisible by k , the left-hand side must also be divisible by k .

In order for $(2k + 13)(k - 25)$ to be divisible by k , either $2k + 13$ or $k - 25$ must be divisible by k .

$$\frac{2k + 13}{k} = 2 + \frac{13}{k},$$

so $2k + 13$ is divisible by k only if k is a factor of 13.

$$\frac{k - 25}{k} = 1 - \frac{25}{k},$$

so $k - 25$ is divisible by k only if k is a factor of 25.

None of the answer choices are factors of 25, but answer choice B is a factor of 13. Hence, the answer is **(B) 13**.

Solution 7

We see the sum of the first 50 page numbers is 1275. We can then write an equation to solve for k , the number of pages removed. Let the average of the page numbers removed be a . We write $\frac{1275 - ka}{50 - k} = 19$.

We simplify this equation to get $325 = k(a - 19)$.

We now test multiples of 325.

We know that a can be at max 49, so $k = 1$ and $k = 5$ won't work.

We test $k = 13$, and this value works. $k = 25$ also works, but this is not an answer choice, so we see that the answer must be **13**.

23. (AMC 10 10A 2021) Frieda the frog begins a sequence of hops on a 3×3 grid of squares, moving one square on each hop and choosing at random the direction of each hop-up, down, left, or right. She does not hop diagonally. When the direction of a hop would take Frieda off the grid, she "wraps around" and jumps to the opposite edge. For example if Frieda begins in the center square and makes two hops "up", the first hop would place her in the top row middle square, and the second hop would cause Frieda to jump to the opposite edge, landing in the bottom row middle square. Suppose Frieda starts from the center square, makes at most four hops at random, and stops hopping if she lands on a corner square. What is the probability that she reaches a corner square on one of the four hops?

(A) $\frac{9}{16}$ (B) $\frac{5}{8}$ (C) $\frac{3}{4}$ (D) $\frac{25}{32}$ (E) $\frac{13}{16}$

Answer: D

Solution 1 (Complementary Counting)

We will use complementary counting. First, the frog can go left with probability $\frac{1}{4}$. We observe symmetry, so our final answer will be multiplied by 4 for the 4 directions, and since $4 \cdot \frac{1}{4} = 1$, we will ignore the leading probability.

From the left, she either goes left to another edge ($\frac{1}{4}$) or back to the center ($\frac{1}{4}$). Time for some casework.

Case 1: She goes back to the center.

Now, she can go in any 4 directions, and then has 2 options from that edge. This gives $\frac{1}{2}$. –End case 1

Case 2: She goes to another edge (rightmost).

Subcase 1: She goes back to the left edge. She now has 2 places to go, giving $\frac{1}{2}$

Subcase 2: She goes to the center. Now any move works.

$\frac{1}{4} \cdot \frac{1}{2} + \frac{1}{4} \cdot 1 = \frac{1}{8} + \frac{1}{4} = \frac{3}{8}$ for this case. –End case 2

She goes back to the center in Case 1 with probability $\frac{1}{4}$, and to the right edge with probability $\frac{1}{4}$

So, our answer is $\frac{1}{4} \cdot \frac{1}{2} + \frac{1}{4} \cdot \frac{3}{8} = \frac{1}{4}(\frac{1}{2} + \frac{3}{8}) = \frac{1}{4} \cdot \frac{7}{8} = \frac{7}{32}$

But, don't forget complementary counting. So, we get $1 - \frac{7}{32} = \frac{25}{32} \implies \boxed{D}$.

Solution 2 (Direct Counting and Probability States)

We can draw a state diagram with three states: center, edge, and corner. Denote center by M, edge by E, and corner by C. There are a few ways Frieda can reach a corner in four or less moves: EC, EEC, EEEC, EMEC. Then, calculating the probabilities of each of these cases happening, we have $1 \cdot \frac{1}{2} + 1 \cdot \frac{1}{4} \cdot \frac{1}{2} + 1 \cdot \frac{1}{4} \cdot \frac{1}{4} \cdot \frac{1}{2} + 1 \cdot \frac{1}{4} \cdot 1 \cdot \frac{1}{2} = \frac{25}{32}$, so the answer is $\boxed{D}$. IceWolf10

Solution 3 (Finds Numerator and Denominator Separately)

Suppose Frieda makes four independent hops without stopping so that each outcome is equally likely.

Denominator

There are $4^4 = 256$ ways for Frieda to make four hops without restrictions.

Numerator

We perform casework on which hop Frieda reaches a corner for the first time:

- a) The second hop (The third and the fourth hops have no restrictions.) The four hops have 4, 2, 4, 4 options, respectively. So, this case has $4 \cdot 2 \cdot 4 \cdot 4 = 128$ ways.
 - b) The third hop (The fourth hop has no restrictions.) No matter which direction the first hop takes, the second hop must "wrap around".
The four hops have 4, 1, 2, 4 options, respectively. So, this case has $4 \cdot 1 \cdot 2 \cdot 4 = 32$ ways.
 - c) The fourth hop (There are two subcases based on the second hop.) <ol style="margin-left: 1.5em;" type="A">
 - d) The second hop "wraps around". It follows that the third hop must also "wrap around". The four hops have 4, 1, 1, 2 options, respectively. So, this subcase has $4 \cdot 1 \cdot 1 \cdot 2 = 8$ ways.
 - e) The second hop returns to the center. The four hops have 4, 1, 4, 2 options, respectively. So, this subcase has $4 \cdot 1 \cdot 4 \cdot 2 = 32$ ways.
- Together, this case has $8 + 32 = 40$ ways.

The numerator is $128 + 32 + 40 = 200$. **Probability**

The requested probability is $\frac{200}{256} = \boxed{(D) \frac{25}{32}}$. **Remark**

This problem is quite similar to [1995 AIME Problem 3](#).

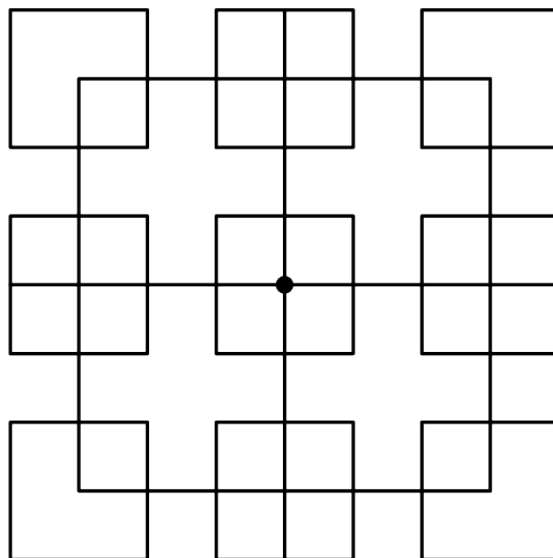
Solution 4

Let C_n be the probability that Frieda is on the central square after n moves, E_n be the probability that Frieda is on one of the four squares on the middle of the edges after n moves, and V_n (V for vertex) be the probability that Frieda is on a corner after n moves. The only way to reach the center is by moving in 1 specific direction out of 4 total directions from the middle of an edge, so $C_{n+1} = \frac{E_n}{4}$. The ways to reach the middle of an edge are by moving in any direction from the center or by moving in 1 specific direction from the middle of an edge, so $E_{n+1} = C_n + \frac{E_n}{4}$. The ways to reach a corner are by simply staying there after reaching there in a previous move or by moving in 2 specific directions from the middle of an edge, so $V_{n+1} = V_n + \frac{E_n}{2}$. Since Frieda always start from the center, $C_0 = 1$, $E_0 = 0$, and $V_0 = 0$. We use the previous formulas to work out V_4 and find it to be $\boxed{(D) \frac{25}{32}}$.

-SmileKat32

Solution 5

Imagine an infinite grid of 2 by 2 squares such that there is a 2 by 2 square centered at $(3x, 3y)$ for all ordered pairs of integers (x, y) .



It is easy to see that the problem is equivalent to Frieda moving left, right, up, or down on this infinite grid starting at $(0, 0)$ (minus the teleportations). Since counting the complement set is easier, we'll count the number of 4-step paths such that Frieda never reaches a corner point.

In other words, since the reachable corner points are $(\pm 1, \pm 1)$, $(\pm 1, \pm 2)$, $(\pm 2, \pm 1)$, and $(\pm 2, \pm 2)$, Frieda can only travel along the collection of points included in S , where S is all points on $x = 0$ and $y = 0$ such that $|y| < 4$ and $|x| < 4$, respectively, plus all points on the big square with side length 6 centered at $(0, 0)$. We then can proceed with casework:

Case 1: Frieda never reaches $(0, \pm 3)$ nor $(\pm 3, 0)$.

When Frieda only moves horizontally or vertically for her four moves, she can do so in $2^4 - 4 = 12$ ways for each case. Thus, $12 \cdot 2$ total paths for the subcase of staying in one direction. (For instance, all length 4 combinations of F and B except $FFFF$, $BBBB$, $FFFB$, and $BBBF$ for the horizontal direction.)

There is another subcase where she changes directions during her path. There are four symmetric cases for this subcase depending on which of the four quadrants Frieda hugs. For the first quadrant, the possible paths are $FBUD$, $FBUU$, $UDFB$, and $UDFF$. Thus, a total of $4 \cdot 4 = 16$ ways for this subcase.

Total for Case 1: $24 + 16 = 40$

Case 2: Frieda reaches $(0, \pm 3)$ or $(\pm 3, 0)$.

Once Frieda reaches one of the points listed above (by using three moves), she has four choices for her last move. Thus, a total of $4 \cdot 4 = 16$ paths for this case.

Our total number of paths never reaching coroners is thus $16 + 40 = 56$, making for an answer of

$$\frac{4^4 - 56}{4^4} = \boxed{\text{(D)} \frac{25}{32}}.$$

-fidgetboss_4000

Solution 6 (Casework)

We take cases on the number of hops needed to reach a corner. For simplicity, denote E as a move that takes Frieda to an edge, W as wrap-around move and C as a corner move. Also, denote O as a move that takes us to the center.

2 Hops

Then, Frieda will have to (E, C) as her set of moves. There are 4 ways to move to an edge, and 2 corners to move to, for a total of $4 \cdot 2 = 8$ cases here. Then, there are 4 choices for each move, for a probability of $\frac{8}{4 \cdot 4} = \frac{1}{2}$.

3 Hops

In this case, Frieda must wrap-around. There's only one possible combination, just (E, W, C) . There are 4 ways to move to an edge, 1 way to wrap-around (you must continue in the same direction) and 2 corners, for a total of $4 \cdot 1 \cdot 2 = 8$ cases here. Then, there are 4 choices for each move, for a probability of $\frac{8}{4 \cdot 4 \cdot 4} = \frac{1}{8}$.

4 Hops

Lastly, there are two cases we must consider here. The first case is (E, O, E, C) , and the second is (E, W, W, C) . For the first case, there are 4 ways to move to an edge, 1 way to return to the center, 4 ways to move to an edge once again, and 2 ways to move to a corner. Hence, there is a total of $4 \cdot 1 \cdot 4 \cdot 2 = 32$ cases here. Then, for the second case, there are 4 ways to move to a corner, 1 way to wrap-around, 1 way to wrap-around the way Frieda just came from, and 2 ways to move to a corner. This implies there are $4 \cdot 1 \cdot 1 \cdot 2 = 8$ cases here. Then, there is a total of $8 + 32 = 40$ cases, out of a total of $4^4 = 256$ cases, for a probability of $\frac{40}{256} = \frac{5}{32}$.

Then, the total probability that Frieda ends up on a corner is $\frac{1}{2} + \frac{1}{8} + \frac{5}{32} = \frac{25}{32}$, corresponding to choice

(D) $\frac{25}{32}$

.

Solution 7

I denote 3x3 grid by

- HOME square (x1)
- CORN squares (x4)
- SIDE squares (x4)

Transitions:

- HOME always move to SIDE
- CORN is DONE
- SIDE move to HOME with $p = 1/4$, move to SIDE with $p = 1/4$, and move to CORN with $p = 1/2$.

After one move, will be on SIDE square

After two moves, will be $1/2 + 1/4(\text{HOME}) + 1/4(\text{SIDE})$

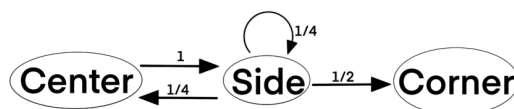
After three moves, will be $1/2 + 1/4(\text{SIDE}) + 1/4(1/2 + 1/4(\text{SIDE}) + 1/4(\text{HOME}))$

After four moves, probability on CORN will be $1/2 + 1/4(1/2) + 1/4(1/2 + 1/4(1/2)) = 1/2 + 1/8 + 5/32 = 25/32$.

Solution 8 (Markov Chain)

(i) Main article: [Markov Chains](#)

The grid is symmetrical. Denote the center cell as state Center, the 4 middle side cells as state Side, the 4 corner cells as state Corner. We can draw the following State Transition Diagram with [Markov Chain](#). The numbers on the transition arc are the transition probability.



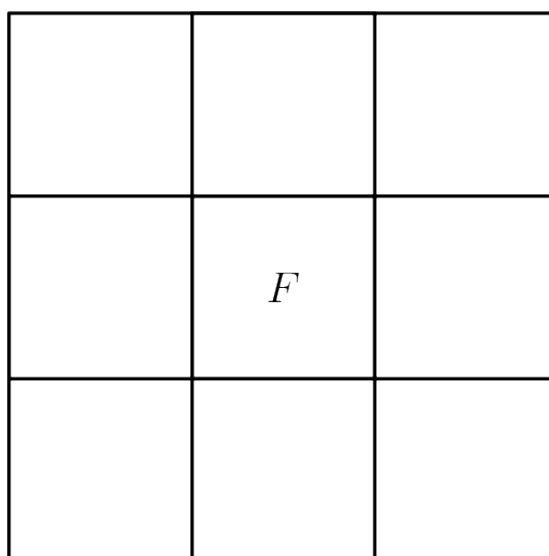
We calculate the probability of different states in each round in the following table by iterating a few more times:

Round	Center	Side	Corner
1	0	1	0
2	$\frac{1}{4}$	$\frac{1}{4}$	$\frac{1}{2}$
3	$\frac{1}{4} \cdot \frac{1}{4} = \frac{1}{16}$	$\frac{1}{4} + \frac{1}{4} \cdot \frac{1}{4} = \frac{5}{16}$	$\frac{1}{4} \cdot \frac{1}{2} + \frac{1}{2} = \frac{5}{8}$
4	$\frac{5}{16} \cdot \frac{1}{4} = \frac{5}{64}$	$\frac{1}{16} + \frac{5}{16} \cdot \frac{1}{4} = \frac{9}{64}$	$\frac{5}{16} \cdot \frac{1}{2} + \frac{5}{8} = \frac{25}{32}$

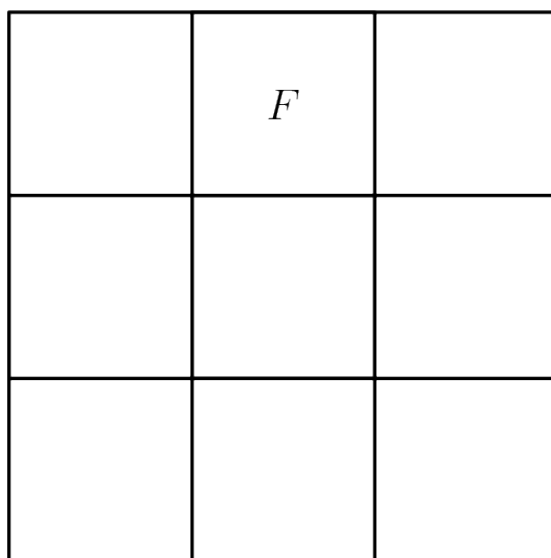
Therefore, the answer is (D) $\frac{25}{32}$.

Solution 9

Let us go through the frog's jumps step by step to find the probability that it will land in a corner. Denote the frog's position as F :

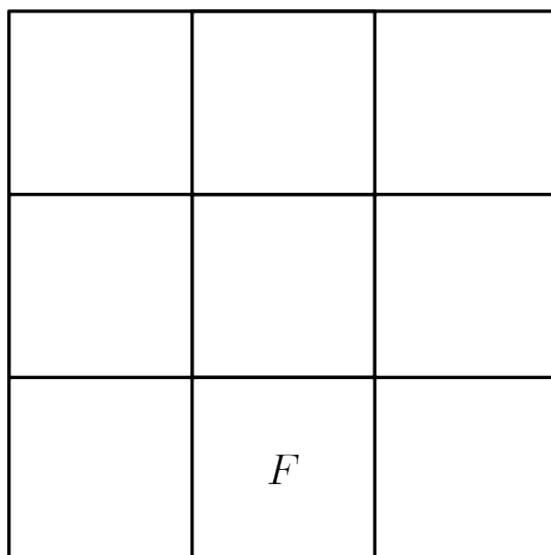


First notice that since the frog only moves up, down, left or right, it's original hop will not matter by symmetry, because all first hops will bring the frog at the edge next to two corners. WLOG we let the frog hop up:



Notice that in its second hop, there are two ways for it to move to a corner (left and right) out of four, and thus the probability of Frieda reaching a corner on the second hop is $\frac{1}{2}$.

There are two other cases for Frieda's second hop: it could either wrap around to another edge square or move back to the middle. We analyze each of these cases individually, noting that each has probability $\frac{1}{4}$:



In this case, there are again 2 choices to go to a corner on the third hop, go back to the previous square, or go to the middle. Here there is a $\frac{1}{2}$ probability that Frieda will reach a corner on the third hop. If Frieda hops back with a $\frac{1}{4}$ probability, she will again have a $\frac{1}{2}$ chance of reaching a corner on her fourth hop, but after four hops she stops hopping, so there are no more chances for her to reach a corner if

she hops back. If she hops to the middle, then her fourth hop cannot reach a corner and that case yields 0 probability of reaching a corner.

To sum up this case, the total probability of her reaching a corner is $\frac{1}{4} \left(\frac{1}{2} + \frac{1}{4} \cdot \frac{1}{2} \right)$.

Finally suppose Frieda goes back to her initial position after 2 steps. Then her third step again doesn't matter because whatever she does, she will be next to 2 corners, so the probability of her reaching a corner on her fourth hop in this case is $\frac{1}{4} \cdot \frac{1}{2}$.

Summing up all the cases, we get

$$\frac{1}{2} + \frac{1}{4} \left(\frac{1}{2} + \frac{1}{4} \cdot \frac{1}{2} \right) + \frac{1}{4} \cdot \frac{1}{2} = \frac{1}{2} + \frac{1}{4} \left(\frac{1}{2} + \frac{1}{8} \right) + \frac{1}{8} = \frac{5}{4} \cdot \left(\frac{1}{2} + \frac{1}{8} \right) = \frac{5}{4} \cdot \frac{5}{8} = \boxed{\text{(D)} \frac{25}{32}}.$$

KingRavi

Solution 10 (Similar to the States Solution)

Denote the center square as C , the edge squares as E , and the corner squares as Co . On the first move, the probability of Frieda on the C is 0 because she's moving away from the center square. The probability she's on E is 1 because no matter which direction she goes, she's always going to end up on an edge square. Probability she's on Co equals $1 - (0 + 1) = 0$ by complementary counting. Notice that the probability Frieda will be on the C can be expressed as: $P(C) = \frac{1}{4} \cdot P(E \text{ from last move})$ because one-fourth of the moves Frieda takes from an edge square sends her back to C . Also, she can't reach C from a Co square. Now $P(E) = P(C \text{ from last move}) + \frac{1}{4} \cdot P(E \text{ from last move})$ simply from the same logic. We can find $P(Co)$ using complementary counting. On the second move, $P(C) = \frac{1}{4} \cdot 1 = \frac{1}{4}$ and $P(E) = 0 + \frac{1}{4} \cdot 1 = \frac{1}{4}$. Now $P(Co) = 1 - \frac{1}{4} - \frac{1}{4} = \frac{1}{2}$. On the third move, $P(C) = \frac{1}{4} \cdot \frac{1}{4} = \frac{1}{16}$ and $P(E) = \frac{1}{4} + \frac{1}{4} \cdot \frac{1}{4} = \frac{5}{16}$. $P(Co) = 1 - \frac{1}{16} - \frac{5}{16} = \frac{5}{8}$. Now on the fourth and last move, $P(C) = \frac{1}{4} \cdot \frac{5}{16} = \frac{5}{64}$ and $P(E) = \frac{1}{16} + \frac{1}{4} \cdot \frac{5}{16} = \frac{9}{64}$. Finally, $P(Co) = 1 - \frac{5}{64} - \frac{9}{64} = 1 - \frac{14}{64} = \frac{50}{64} = \frac{25}{32}$.

24. (AMC 10 10A 2021) The interior of a quadrilateral is bounded by the graphs of $(x + ay)^2 = 4a^2$ and $(ax - y)^2 = a^2$, where a is a positive real number. What is the area of this region in terms of a , valid for all $a > 0$?

(A) $\frac{8a^2}{(a+1)^2}$ (B) $\frac{4a}{a+1}$ (C) $\frac{8a}{a+1}$ (D) $\frac{8a^2}{a^2+1}$ (E) $\frac{8a}{a^2+1}$

Answer: D

Solution 1 (Generalized Value of a)

The cases for $(x + ay)^2 = 4a^2$ are $x + ay = \pm 2a$, or two parallel lines. We rearrange each case and construct the table below:

Case	Line's Equation	x -Intercept	y -Intercept	Slope
1	$x + ay - 2a = 0$	$2a$	2	$-\frac{1}{a}$
2	$x + ay + 2a = 0$	$-2a$	-2	$-\frac{1}{a}$

The cases for $(ax - y)^2 = a^2$ are $ax - y = \pm a$, or two parallel lines. We rearrange each case and construct the table below:

Case	Line's Equation	x -Intercept	y -Intercept	Slope
1*	$ax - y - a = 0$	1	$-a$	a
2*	$ax - y + a = 0$	-1	a	a

Since the slopes of intersecting lines $(1) \cap (1^*)$, $(1) \cap (2^*)$, $(2) \cap (1^*)$, and $(2) \cap (2^*)$ are negative reciprocals, we get four right angles, from which the quadrilateral is a rectangle.

Two solutions follow from here:

Solution 1.1 (Distance Between Parallel Lines)

Recall that for constants A, B, C_1 and C_2 , the distance d between parallel lines $\begin{cases} Ax + By + C_1 = 0 \\ Ax + By + C_2 = 0 \end{cases}$ is

$$d = \frac{|C_2 - C_1|}{\sqrt{A^2 + B^2}}.$$

From this formula:

- The distance between lines (1) and (2) is $\frac{4a}{\sqrt{1+a^2}}$, the length of this rectangle.
- The distance between lines (1^*) and (2^*) is $\frac{2a}{\sqrt{a^2+1}}$, the width of this rectangle.

The area we seek is

$$\frac{4a}{\sqrt{1+a^2}} \cdot \frac{2a}{\sqrt{a^2+1}} = \boxed{\text{(D)} \frac{8a^2}{a^2+1}}.$$

MRENTHUSIASM

Solution 1.2 (Distance Between Points)

The solutions to systems of equations $(1) \cap (1^*)$, $(1) \cap (2^*)$, $(2) \cap (2^*)$, $(2) \cap (1^*)$ are

$$(x, y) = \left(\frac{a(a+2)}{a^2+1}, \frac{a(2a-1)}{a^2+1} \right), \left(-\frac{a(a-2)}{a^2+1}, \frac{a(2a+1)}{a^2+1} \right), \left(-\frac{a(a+2)}{a^2+1}, -\frac{a(2a-1)}{a^2+1} \right), \left(\frac{a(a-2)}{a^2+1}, -\frac{a(2a+1)}{a^2+1} \right),$$

respectively, which are the consecutive vertices of this rectangle.

By the Distance Formula, the length and width of this rectangle are $\frac{4a\sqrt{a^2+1}}{a^2+1}$ and $\frac{2a\sqrt{a^2+1}}{a^2+1}$, respectively.

The area we seek is

$$\frac{4a\sqrt{a^2+1}}{a^2+1} \cdot \frac{2a\sqrt{a^2+1}}{a^2+1} = \boxed{\text{(D)} \frac{8a^2}{a^2+1}}.$$

MRENTHUSIASM

Solution 2 (Specified Value of a)

In this solution, we will refer to equations (1) , (2) , (1^*) , and (2^*) from Solution 1.

Substituting $a = 2$ into the answer choices gives

$$\text{(A)} \frac{32}{9} \quad \text{(B)} \frac{8}{3} \quad \text{(C)} \frac{16}{3} \quad \text{(D)} \frac{32}{5} \quad \text{(E)} \frac{16}{5}$$

At $a = 2$, the solutions to systems of equations $(1) \cap (1^*)$, $(1) \cap (2^*)$, $(2) \cap (2^*)$, $(2) \cap (1^*)$ are

$$(x, y) = \left(\frac{8}{5}, \frac{6}{5} \right), (0, 2), \left(-\frac{8}{5}, -\frac{6}{5} \right), (0, -2),$$

respectively, which are the consecutive vertices of the quadrilateral.

Two solutions follow from here:

Solution 2.1 (Area of a Rectangle)

From the tables in Solution 1, we conclude that the quadrilateral is a rectangle.

By the Distance Formula, the length and width of this rectangle are $\frac{8\sqrt{5}}{5}$ and $\frac{4\sqrt{5}}{5}$, respectively.

The area we seek is

$$\frac{8\sqrt{5}}{5} \cdot \frac{4\sqrt{5}}{5} = \frac{32}{5},$$

from which the answer is **(D)** $\frac{8a^2}{a^2 + 1}$.

Solution 2.2 (Area of a General Quadrilateral)

Even if we do not recognize that the quadrilateral is a rectangle, we can apply the Shoelace Theorem to its *consecutive* vertices

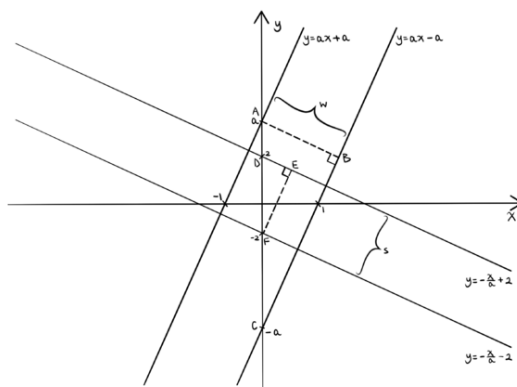
$$\begin{aligned}(x_1, y_1) &= \left(\frac{8}{5}, \frac{6}{5}\right), \\(x_2, y_2) &= (0, 2), \\(x_3, y_3) &= \left(-\frac{8}{5}, -\frac{6}{5}\right), \\(x_4, y_4) &= (0, -2).\end{aligned}$$

The area we seek is

$$\frac{1}{2} |(x_1y_2 + x_2y_3 + x_3y_4 + x_4y_1) - (y_1x_2 + y_2x_3 + y_3x_4 + y_4x_1)| = \frac{32}{5}.$$

from which the answer is **(D)** $\frac{8a^2}{a^2 + 1}$.

Solution 3 (Slopes and Intercepts)



The quadrilateral is enclosed by four lines. Similar to Solution 1, we will use the equations from the four cases:

- $x + ay = 2a$. This is a line with x -intercept $2a$, y -intercept 2 , and slope $-\frac{1}{a}$.
- $x + ay = -2a$. This is a line with x -intercept $-2a$, y -intercept -2 , and slope $-\frac{1}{a}$.
- $ax - y = a$. This is a line with x -intercept 1 , y -intercept $-a$, and slope a .
- $ax - y = -a$. This is a line with x -intercept -1 , y -intercept a , and slope a .

It follows that $DF = 4$ and $DE = \sqrt{4^2 - s^2}$.

Because the slope of line $y = -\frac{x}{a} + 2$ is $-\frac{1}{a}$, $\frac{1}{a} = \frac{DE}{EF} = \frac{\sqrt{16-s^2}}{s}$, $s^2(a^2 + 1) = 16a^2$, $s = \frac{4a}{\sqrt{a^2+1}}$.

It follows that $AC = 2a$ and $BC = \sqrt{(2a)^2 - w^2}$.

Because the slope of line $y = ax - a$ is a , $a = \frac{BC}{AB} = \frac{\sqrt{4a^2-w^2}}{w}$, $w^2(a^2 + 1) = 4a^2$, $w = \frac{2a}{\sqrt{a^2+1}}$.

Therefore, the answer is

$$\text{Area} = s \cdot w = \frac{4a}{\sqrt{a^2+1}} \cdot \frac{2a}{\sqrt{a^2+1}} = \boxed{\text{(D)} \frac{8a^2}{a^2+1}}.$$

Solution 4 (Trigonometry)

Similar to Solution 1, we will use the equations from the four cases:

- a) $x + ay = 2a$. This is a line with x -intercept $2a$, y -intercept 2 , and slope $-\frac{1}{a}$.
- b) $x + ay = -2a$. This is a line with x -intercept $-2a$, y -intercept -2 , and slope $-\frac{1}{a}$.
- c) $ax - y = a$. This is a line with x -intercept 1 , y -intercept $-a$, and slope a .
- d) $ax - y = -a$. This is a line with x -intercept -1 , y -intercept a , and slope a .

Let $\tan A = a$. The area of the rectangle created by the four equations can be written as

$$\begin{aligned} 2a \cdot \cos A \cdot 4 \sin A &= 8a \cos A \cdot \sin A \\ &= 8a \cdot \frac{1}{\sqrt{a^2+1}} \cdot \frac{a}{\sqrt{a^2+1}} \\ &= \boxed{\text{(D)} \frac{8a^2}{a^2+1}}. \end{aligned}$$

fnothing4994 (Solution)

Solution 5 (Observations)

The conditions $(x + ay)^2 = 4a^2$ and $(ax - y)^2 = a^2$ give $|x + ay| = |2a|$ and $|ax - y| = |a|$ or $x + ay = \pm 2a$ and $ax - y = \pm a$. The slopes here are perpendicular, so the quadrilateral is a rectangle. Plug in $a = 1$ and graph it. We quickly see that the area is $2\sqrt{2} \cdot \sqrt{2} = 4$, so the answer can't be (A) or (B) by testing the values they give (test it!). Now plug in $a = 2$. We see using a ruler that the sides of the rectangle are about $\frac{7}{4}$ and $\frac{7}{2}$. So the area is about $\frac{49}{8} = 6.125$. Testing (C), we get $\frac{16}{3}$ which is clearly less than 6, so it is out. Testing (D), we get $\frac{32}{5}$ which is near our answer, so we leave it. Testing (E), we get $\frac{16}{5}$, way less than 6, so it is out. So,

the only plausible answer is $\boxed{\text{(D)} \frac{8a^2}{a^2+1}}$.

Solution 6 (Observations)

Trying $a = 1$ narrows down the choices to options (C), (D) and (E). Trying $a = 2$

and $a = 3$ eliminates (C) and (E), to obtain $\boxed{\text{(D)} \frac{8a^2}{a^2+1}}$ as our answer. Refer to

Solution 2 for a detailed explanation.

Solution 7 (Observations: Cheap)

Note that $a = 2$ yields different values for all answer choices. If we put in $a = 2$, we find that the area of the quadrilateral is $\frac{32}{5}$. This means that the answer must

be **(D)** $\frac{8a^2}{a^2 + 1}$. Refer to Solution 2 for a detailed explanation.

25. (AMC 10 10A 2021) How many ways are there to place 3 indistinguishable red chips, 3 indistinguishable blue chips, and 3 indistinguishable green chips in the squares of a 3×3 grid so that no two chips of the same color are directly adjacent to each other, either vertically or horizontally?

(A) 12 (B) 18 (C) 24 (D) 30 (E) 36

Answer: E

Solution 1 (Casework on the Center's Color Chip's Configurations)

Call the different colors A,B,C. There are $3! = 6$ ways to rearrange these colors to these three letters, so 6 must be multiplied after the letters are permuted in the grid. WLOG assume that A is in the center.

$$\begin{array}{ccc} ? & ? & ? \\ ? & A & ? \\ ? & ? & ? \end{array}$$

In this configuration, there are two cases, either all the A's lie on the same diagonal:

$$\begin{array}{ccc} ? & ? & A \\ ? & A & ? \\ A & ? & ? \end{array}$$

or all the other two A's are on adjacent corners:

$$\begin{array}{ccc} A & ? & A \\ ? & A & ? \\ ? & ? & ? \end{array}$$

In the first case there are two ways to order them since there are two diagonals, and in the second case there are four ways to order them since there are four pairs of adjacent corners.

In each case there is only one way to put the three B's and the three C's as shown in the diagrams.

$$\begin{array}{ccc} C & B & A \\ B & A & C \\ A & C & B \end{array}$$

$$\begin{array}{ccc} A & B & A \\ C & A & C \\ B & C & B \end{array}$$

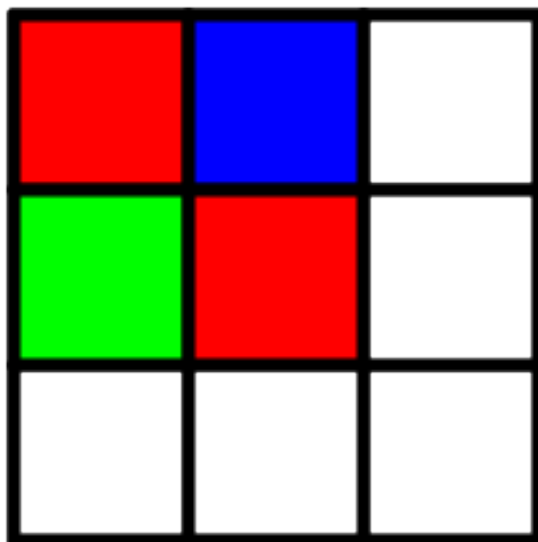
This means that there are $4 + 2 = 6$ ways to arrange A,B, and C in the grid, and there are 6 ways to rearrange the colors. Therefore, there are $6 \cdot 6 = 36$ ways in total, which is **(E)** 36.

-happykeeper

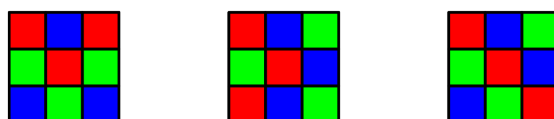
Solution 2 (Casework on the Top-Center and Center-Left Chips)

Without the loss of generality, we fix the top-left square with a red chip. We apply casework to its two adjacent chips:

Case (1): The top-center and center-left chips have different colors.

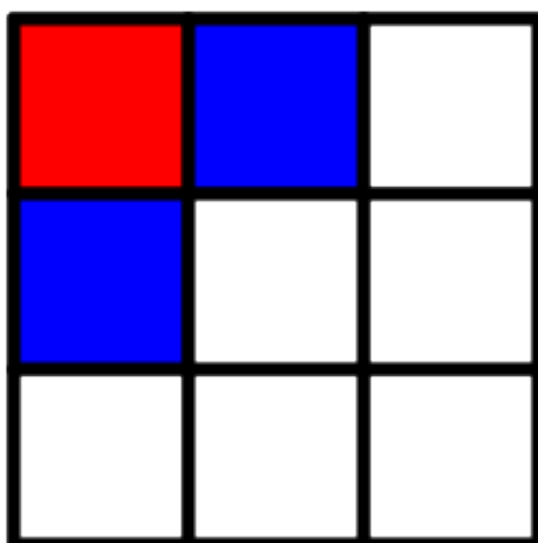


There are three subcases for Case (1):

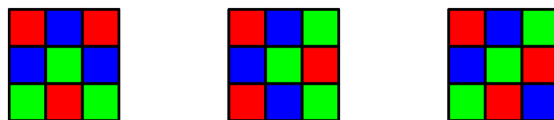


As there are $3! = 6$ permutations of the three colors, each subcase has 6 ways. So, Case (1) has $3 \cdot 6 = 18$ ways in total.

Case (2): The top-center and center-left chips have the same color.



There are three subcases for Case (2):



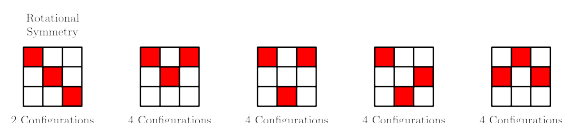
As there are $3! = 6$ permutations of the three colors, each subcase has 6 ways. So, Case (2) has $3 \cdot 6 = 18$ ways in total.

Answer

Together, the answer is $18 + 18 = \boxed{(E) 36}$.

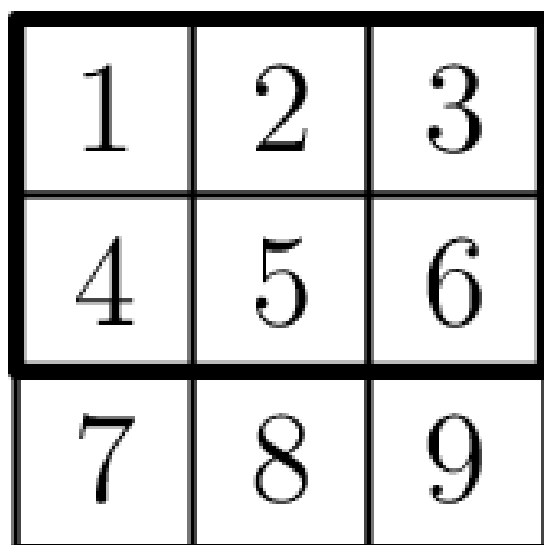
Solution 3 (Casework on the Red Chips' Configurations)

We consider all possible configurations of the red chips for which rotations matter:



As there are $2! = 2$ permutations of blue and green for each configuration, the answer is $2 \cdot (2 + 4 + 4 + 4 + 4) = \boxed{(E) 36}$.

Solution 4 (Focusing On Top 2 Rows)



We will first count the colorings where if the top two rows are filled out, the grid is uniquely determined. This will only happen if there are 3 cells of one color, 2 cells of another color, and 1 cell of the remaining color in the top 3×2 grid made up of cells 1 – 6. With this, we are left with two tokens of one color and one token of another color for the bottom row, so there must only be 1 way to fill it up since the two tokens of the same color can't be next to each other. This makes the grid uniquely determined.

The 3 cells with the same color can either be cells 1, 5, 3 or 2, 4, 6. If we choose 1, 3, 5, the two cells with the same color must be cells 4 and 6 and cell 2 must be the cell of the other color. If we choose 2, 4, 6, the two cells with the same color must be cells 1 and 3, and cell 5 must be the cell of the other color. So there are 2

ways to choose which cells share 3 colors and 1 way to choose which cells share 2 and 1 color. There are $3! = 6$ ways to assign colors to the cells. We have a total of $2 \cdot 1 \cdot 3! = 12$ colorings for this case. An example coloring is shown below with cells 1, 3, 5 being chosen for the 3 cells of the same color.

1	2	3
4	5	6
7	8	9

Only 1 configuration for last row, grid is uniquely determined from top 2 rows

Now, what about the colorings where even if the top two rows are filled out, the grid is still not uniquely determined? In other words, what about the colorings where even if we fill in the top two rows, there is still more than one possible color combination for the last row?

In this case, the 3×2 grid made up of the top two rows would have to use two tokens of each color, and the second row would be some permutation of the colors *blue*, *red*, *green*. Then for the last row, we would be left with one token of each color.

There are $3! = 6$ ways to permute the second row. WLOG, say we fill the second row of the grid so that cell 4 is red, cell 5 is blue, and cell 6 is green. Then cells 1, 2, 3 can be *green*, *red*, *blue* or *blue*, *green*, *red*, respectively. Finally, cells 7, 8, 9 can also either be *green*, *red*, *blue* or *blue*, *green*, *red*, respectively. This gives us $6 \cdot 4 = 24$ colorings.

1	2	3	1	2	3	1	2	3	1	2	3
4	5	6	4	5	6	4	5	6	4	5	6
7	8	9	7	8	9	7	8	9	7	8	9

Our final answer is $12 + 24 = \boxed{\text{(E)} 36}$.

Solution 5 (Casework and Symmetry)

$$\begin{array}{ccccccc}
 & R & G & ? & & R & G & B & & R & G & R \\
 (1) & B & R & ? & \implies & B & R & G & \implies & B & R & B & 3 \cdot 2 \cdot 2 = 12 \\
 & ? & ? & ? & & R & G & B & & G & B & G
 \end{array}$$

There are 3 choices for R , 2 choices for G . R on the down left corner can be switched with B on the upper right corner.

$$\begin{array}{ccccccc}
 & R & G & ? & & R & G & B \\
 (2) & B & R & ? & \implies & B & R & G & 3 \cdot 2 = 6 \\
 & ? & ? & ? & & G & B & R
 \end{array}$$

There are 3 choices for R , 2 choices for G .

$$\begin{array}{ccccccc}
 & G & R & ? & & G & R & B & & G & R & G \\
 (3) & R & B & ? & \implies & R & B & G & \implies & R & B & R & 3 \cdot 2 \cdot 2 = 12 \\
 & ? & ? & ? & & G & R & B & & B & G & B
 \end{array}$$

Note that (3) is a 180° rotation of (1).

$$\begin{array}{ccccccc}
 & G & R & ? & & G & R & B \\
 (4) & R & B & ? & \implies & R & B & G & 3 \cdot 2 = 6 \\
 & ? & ? & ? & & B & G & R
 \end{array}$$

Note that (4) is a 90° rotation of (2).

Therefore, the answer is $2 \cdot (12 + 6) = \boxed{\text{(E)} 36}$.

Solution 6 (Casework and Derangements)

Case (1): We have a permutation of R, B, and G as all of the rows. There are $3!$ ways to rearrange these three colors. After finishing the first row, we move onto the second. Notice how the second row must be a derangement of the first one. By the derangement formula, $\frac{3!}{e} \approx 2$, so there are two possible permutations of the second row. (Note: You could have also found the number of derangements of PIE). Finally, there are 2 possible permutations for the last row. Thus, there are $3! \cdot 2 \cdot 2 = 24$ possibilities.

Case (2): All of the rows have two chips that are the same color and one that is different. There are obviously 3 possible configurations for the first row, 2 for the second, and 2 for the third. Thus, there are $3 \cdot 2 \cdot 2 = 12$ possibilities.

Therefore, our answer is $24 + 12 = \boxed{\text{(E)} 36}$.

Solution 7 (Rush, Use if you have less than 5 minutes)

Ignore the center piece. Notice that when you place 3 of the chips, there are 6 ways, making it inevitable for 2 ways left for the other 2, so 6 multiply 2 is 12. Now, notice there are 3 ways to place a center piece, so our final statement is $12 \cdot 3 = \boxed{\text{(E)} 36}$.

Solution 8 (Casework Using the Center and Surrounding)

There are 3 indistinguishable red chips, 3 indistinguishable blue chips, and 3 indistinguishable green chips. Let's label the red chips R, the blue chips B, and the green chips G.

WLOG, let's assume that the chip in the middle of the array is R.

?	?	?
?	R	?
?	?	?

For the convenience of our proof, we will replace the question marks as follows.

1	2	3
4	R	6
7	8	9

Doing this helps us easily identify where to place a certain chip. For example, if I want to place a green chip in the top leftmost square, I'd say, "Place a green chip in position 1."

Now that we've assumed the center square is occupied by a red (R) chip, notice that the chips in positions 2, 4, 6, and 8 can only be blue or green. With the information we now know, we can break the problem into cases based off of the number of a certain color of chip in the positions 2, 4, 6, 8.

Case 1 (2 consecutive blue, 2 consecutive green): An example case to show you what this means is shown below.

1	G	3
B	R	G
7	B	9

There are 4 ways to figure out where the two blue and two green chips are placed. Notice that after I've placed the two blue, two green, and one red chip, the rest of the chips are in fixed places (try it!). Now we only need to worry about what color the center chip is.

There are 3 ways to choose the color for the center chip, and afterwards, there are only two colors left for the two kinds of chips in positions 2, 4, 6, and 8, and there are two of each, so there is no need to worry about the color of those chips.

In conclusion, there are $4 * 3 = 12$ ways to arrange the chips in this case.

Case 2 (3 blue, 1 green OR 3 green, 1 blue): An example case to show you what this means is shown below.

1	G	3
B	R	G
7	G	9

WLOG, let's first assume that there are 3 green chips in such a position, and only 1 blue chip. There are 4 ways to figure out where they are placed. Notice that after I've placed the three green, one blue, and one red chip, the rest of the chips are yet again in fixed places (try it!). Now we only need to worry about what color the center chip is and which color of chip has 3 in the positions of 2, 4, 6, 8.

There are 3 ways to choose the color of the center chip. After that is chosen, there are only two colors left for the two kinds of chips needed in the positions 2, 4, 6, and 8. However, since this time there are a different number of each color of chip, there are now 2 ways to assign colors to the chips.

In conclusion, there are $4 * 3 * 2 = 24$ ways to arrange the chips in this case.

Notice how there should be a **Case 3** with cases that look like some variation of this:

1	G	3
B	R	B
7	G	9

There isn't, because if such an arrangement were to happen, we wouldn't be able to arrange the grid following the problem's conditions. There would either be a blue chip next to a blue chip, and/or a green chip next to a green chip.

$12 + 24 + 0 = 36$, so the answer is (E) 36.

Solution 9

Imagine a 3×3 grid as stated in the problem. Next, since this grid is extremely symmetric, let's assume the center square is R (red). We realize that since R is the center, the remaining 2 R 's can either go in the top left square and the top right square or in the top left square and the bottom right square. Let's look at the first case. Note that the remaining 2 R 's don't have in the top right and top left squares. We can rotate the board 4 times to find the same case. So, let's multiply the final value in this case by 4. Anyways, let's find the number of ways to color the rest of the board. Upon examination we can conclude that the top center square, the bottom left square, and the bottom right square all have to be the same color, so it can either be B (blue) or G (green), so we have 2 choices. Thus, for this case, we have $4 \times 2 = 8$ ways to color the grid. Now, it is time for case 2. Similarly, we need to multiply this case by 2 because we can rotate the board twice to get the same case. So, upon some examination, we find that the top right square, the

bottom center square, and left center square must be all the same color, so we've got 2 choices. So in total we have $2 \times 2 = 4$ ways to color the grid this way. So in total, we have 12 ways, but remember, we have to multiply by 3 because we let the center square be R , so our answer is $12 \times 3 = 36 \Rightarrow \boxed{\text{(E)}}$.

-jib2015007

§5 AMC 10 10B Fall 2021

1. (AMC 10 10B Fall 2021) What is the value of $1234 + 2341 + 3412 + 4123$?
 (A) 10,000 (B) 10,010 (C) 10,110 (D) 11,000 (E) 11,110

Answer: E

Solution 1

We see that 1, 2, 3, and 4 each appear in the ones, tens, hundreds, and thousands digit exactly once. Since $1 + 2 + 3 + 4 = 10$, we find that the sum is equal to

$$10 \cdot (1 + 10 + 100 + 1000) = \boxed{\text{(E) } 11,110}.$$

Note that it is equally valid to manually add all four numbers together to get the answer.

Solution 2

We have

$$1234 + 2341 + 3412 + 4123 = 1111(1 + 2 + 3 + 4) = \boxed{\text{(E) } 11,110}.$$

Steven Chen (www.professorchen.edu.com)

Solution 3

We see that the units digit must be 0, since $4 + 3 + 2 + 1$ is 0. But every digit from there, will be a 1 since we have that each time afterwards, we must carry the 1 from the previous sum. The answer choice that satisfies these conditions is $\boxed{\text{(E) } 11,110}$.

Solution 4 (Brute Force)

We can simply add the numbers. $1234 + 2341 + 3412 + 4123 = 11110 \Rightarrow \boxed{\text{(E)}}$.

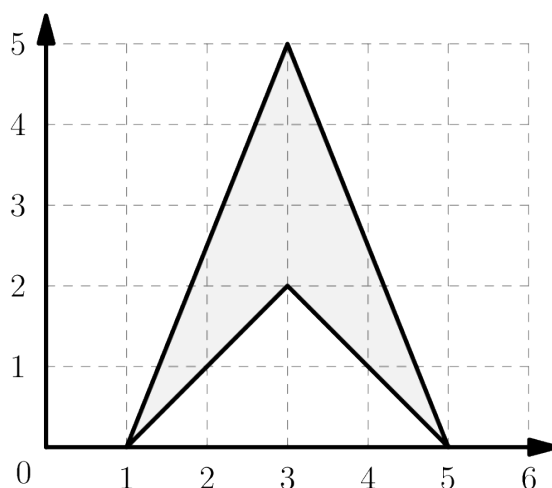
Note: Although this would not take terribly long, it is not recommended to do this in a real contest.

Solution 5 (Primary School But Also Similar to Solution 4)

We can simply just add each place value.

$$1000 + 2000 + 3000 + 4000 = 10,000 + 200 + 300 + 400 + 100 = 1,000 + 30 + 40 + 10 + 20 = 100 + 4 + 1 + 2 + 3 = 10 \quad 10,000 + 1,000 + 100 + 10 = \boxed{\text{(E) } 11,110}.$$

2. (AMC 10 10B Fall 2021) What is the area of the shaded figure shown below?



- (A) 4 (B) 6 (C) 8 (D) 10 (E) 12

Answer: B

Solution 1 (Area Addition)

The line of symmetry divides the shaded figure into two congruent triangles, each with base 3 and height 2.

Therefore, the area of the shaded figure is

$$2 \cdot \left(\frac{1}{2} \cdot 3 \cdot 2 \right) = 2 \cdot 3 = \boxed{\text{(B)} 6}.$$

MRENTHUSIASM Wilhelm Z

Solution 2 (Area Subtraction)

To find the area of the shaded figure, we subtract the area of the smaller triangle (base 4 and height 2) from the area of the larger triangle (base 4 and height 5):

$$\frac{1}{2} \cdot 4 \cdot 5 - \frac{1}{2} \cdot 4 \cdot 2 = 10 - 4 = \boxed{\text{(B)} 6}.$$

MRENTHUSIASM Steven Chen (www.professorchen.edu.com)

Solution 3 (Shoelace Theorem)

The consecutive vertices of the shaded figure are $(1, 0)$, $(3, 2)$, $(5, 0)$, and $(3, 5)$. By the **Shoelace Theorem**, the area is

$$\frac{1}{2} \cdot |(1 \cdot 2 + 3 \cdot 0 + 5 \cdot 5 + 3 \cdot 0) - (0 \cdot 3 + 2 \cdot 5 + 0 \cdot 3 + 5 \cdot 1)| = \frac{1}{2} \cdot 12 = \boxed{\text{(B)} 6}.$$

Taco12 I-AM-DA-KING

Solution 4 (Pick's Theorem)

We have 4 lattice points in the interior and 6 lattice points on the boundary. By **Pick's Theorem**, the area of the shaded figure is

$$4 + \frac{6}{2} - 1 = 4 + 3 - 1 = \boxed{\text{(B)} 6}.$$

danprathab

3. (AMC 10 10B Fall 2021) The expression $\frac{2021}{2020} - \frac{2020}{2021}$ is equal to the fraction $\frac{p}{q}$ in which p and q are positive integers whose greatest common divisor is 1. What is p ?
 (A) 1 (B) 9 (C) 2020 (D) 2021 (E) 4041

Answer: E

Solution 1

We write the given expression as a single fraction:

$$\frac{2021}{2020} - \frac{2020}{2021} = \frac{2021 \cdot 2021 - 2020 \cdot 2020}{2020 \cdot 2021}$$

by cross multiplication. Then by factoring the numerator, we get

$$\frac{2021 \cdot 2021 - 2020 \cdot 2020}{2020 \cdot 2021} = \frac{(2021 - 2020)(2021 + 2020)}{2020 \cdot 2021}.$$

The question is asking for the numerator, so our answer is $2021 + 2020 = 4041$, giving **(E) 4041**.

Solution 2

Denote $a = 2020$. Hence,

$$\begin{aligned} \frac{2021}{2020} - \frac{2020}{2021} &= \frac{a+1}{a} - \frac{a}{a+1} \\ &= \frac{(a+1)^2 - a^2}{a(a+1)} \\ &= \frac{2a+1}{a(a+1)}. \end{aligned}$$

We observe that $\gcd(2a+1, a) = 1$ and $\gcd(2a+1, a+1) = 1$.

Hence, $\gcd(2a+1, a(a+1)) = 1$.

Therefore, $p = 2a+1 = 4041$.

Therefore, the answer is **(E) 4041**.

Solution 3

Turning term 1 to a fraction: $\frac{2021}{2020} - \frac{2020}{2021} = 1 + \frac{1}{2020} - \frac{2020}{2021}$

Subtracting the last term from the first term gives us: $\frac{1}{2021} + \frac{1}{2020}$

Doing some simple cross multiplication, you get $\frac{4041}{2020 \cdot 2021}$, here you can see the numerator is **(E) 4041**.

4. (AMC 10 10B Fall 2021) At noon on a certain day, Minneapolis is N degrees warmer than St. Louis. At 4:00 the temperature in Minneapolis has fallen by 5 degrees while the temperature in St. Louis has risen by 3 degrees, at which time the temperatures in the two cities differ by 2 degrees. What is the product of all possible values of N ? (A) 10 (B) 30 (C) 60 (D) 100 (E) 120

Answer: C

Solution 1 (Two Variables)

At noon on a certain day, let M and L be the temperatures (in degrees) in Minneapolis and St. Louis, respectively. It follows that $M = L + N$.

At 4:00, we get

$$\begin{aligned} |(M - 5) - (L + 3)| &= 2 \\ |M - L - 8| &= 2 \\ |N - 8| &= 2. \end{aligned}$$

We have two cases:

- a) If $N - 8 = 2$, then $N = 10$.
- b) If $N - 8 = -2$, then $N = 6$.

Together, the product of all possible values of N is $10 \cdot 6 = \boxed{\text{(C)} 60}$.

Solution 2 (One Variable)

At noon on a certain day, the difference of temperatures in Minneapolis and St. Louis is N degrees.

At 4:00, the difference of temperatures in Minneapolis and St. Louis is $N - 8$ degrees.

It follows that

$$|N - 8| = 2.$$

We continue with the casework in Solution 1 to get the answer $\boxed{\text{(C)} 60}$.

Solution 3

Let the temperature in Minneapolis be N . Then St. Louis is 0 degrees. $N - 5 = 3 \pm 2$, $N = -10, -6, -10 \cdot -6 = \boxed{\text{(C)} 60}$.

5. (AMC 10 10B Fall 2021) Let $n = 8^{2022}$. Which of the following is equal to $\frac{n}{4}$?
 (A) 4^{1010} (B) 2^{2022} (C) 8^{2018} (D) 4^{3031} (E) 4^{3032}

Answer: E

Solution 1

We have

$$n = 8^{2022} = \left(8^{\frac{2}{3}}\right)^{3033} = 4^{3033}.$$

Therefore,

$$\frac{n}{4} = \boxed{\text{(E)} 4^{3032}}.$$

kingofpineapplz

Solution 2

The requested value is

$$\frac{8^{2022}}{4} = \frac{2^{6066}}{4} = \frac{2^{6066}}{2^2} = 2^{6064} = \boxed{\text{(E)} 4^{3032}}.$$

NH14

Solution 3

If we rewrite everything in powers of 2, we get: $n = 8^{2022} = (2^3)^{2022} = 2^{6064} = (4^{\frac{1}{2}})^{6064} = 4^{\frac{6064}{2}} = \boxed{\text{(E)} 4^{3032}}$.

- abed_nadir (youtube.com/@indianmathguy)

Solution 4

$$n = 8^{2022} \cdot 8^{2022} / 4 = 2 \cdot 8^{2021} = 2 \cdot 2^{6063} = 2^{6064} = \boxed{\text{(E)} 4^{3032}}.$$

-sphericalfox

6. (AMC 10 10B Fall 2021) The least positive integer with exactly 2021 distinct positive divisors can be written in the form $m \cdot 6^k$, where m and k are integers and 6 is not a divisor of m . What is $m+k$? (A) 47 (B) 58 (C) 59 (D) 88 (E) 90

Answer: B

Solution 1

Let this positive integer be written as $p_1^{e_1} \cdot p_2^{e_2}$. The number of factors of this number is therefore $(e_1 + 1) \cdot (e_2 + 1)$, and this must equal 2021. The prime factorization of 2021 is $43 \cdot 47$, so $e_1 + 1 = 43 \implies e_1 = 42$ and $e_2 + 1 = 47 \implies e_2 = 46$. To minimize this integer, we set $p_1 = 3$ and $p_2 = 2$. Then this integer is $3^{42} \cdot 2^{46} = 2^4 \cdot 2^{42} \cdot 3^{42} = 16 \cdot 6^{42}$. Now $m = 16$ and $k = 42$ so $m + k = 16 + 42 = \boxed{\text{(B)} 58}$.

Solution 2

Recall that 6^k can be written as $2^k \cdot 3^k$. Since we want the integer to have 2021 divisors, we must have it in the form $p_1^{42} \cdot p_2^{46}$, where p_1 and p_2 are prime numbers. Therefore, we want p_1 to be 3 and p_2 to be 2. To make up the remaining 2^4 , we multiply $2^{42} \cdot 3^{42}$ by m , which is 2^4 which is 16. Therefore, we have $42+16 = \boxed{\text{(B)} 58}$.

Solution 3

If a number has prime factorization $p_1^{k_1} p_2^{k_2} \cdots p_m^{k_m}$, then the number of distinct positive divisors of this number is $(k_1 + 1)(k_2 + 1) \cdots (k_m + 1)$.

We have $2021 = 43 \cdot 47$. Hence, if a number N has 2021 distinct positive divisors, then N takes one of the following forms: $p_1^{2020}, p_1^{42} p_2^{46}$.

Therefore, the smallest N is $3^{42} 2^{46} = 2^4 \cdot 6^{42} = 16 \cdot 6^{42}$.

Therefore, the answer is $\boxed{\text{(B)} 58}$.

7. (AMC 10 10B Fall 2021) Call a fraction $\frac{a}{b}$, not necessarily in the simplest form, *special*, if a and b are positive integers whose sum is 15. How many distinct integers can be written as the sum of two, not necessarily different, special fractions?
(A) 9 (B) 10 (C) 11 (D) 12 (E) 13

Answer: C

Solution 1

The special fractions are

$$\frac{1}{14}, \frac{2}{13}, \frac{3}{12}, \frac{4}{11}, \frac{5}{10}, \frac{6}{9}, \frac{7}{8}, \frac{8}{7}, \frac{9}{6}, \frac{10}{5}, \frac{11}{4}, \frac{12}{3}, \frac{13}{2}, \frac{14}{1}.$$

We rewrite them in the simplest form:

$$\frac{1}{14}, \frac{2}{13}, \frac{1}{4}, \frac{4}{11}, \frac{1}{2}, \frac{2}{3}, \frac{7}{8}, 1\frac{1}{7}, 1\frac{1}{2}, 2, 2\frac{3}{4}, 4, 6\frac{1}{2}, 14.$$

Note that two unlike fractions in the simplest form cannot sum to an integer. So, we only consider the fractions whose denominators appear more than once:

$$\frac{1}{4}, \frac{1}{2}, 1\frac{1}{2}, 2, 2\frac{3}{4}, 4, 6\frac{1}{2}, 14.$$

For the set $\{2, 4, 14\}$, two elements (not necessarily different) can sum to 4, 6, 8, 16, 18, 28.

For the set $\{\frac{1}{2}, 1\frac{1}{2}, 6\frac{1}{2}\}$, two elements (not necessarily different) can sum to 1, 2, 3, 7, 8, 13.

For the set $\{\frac{1}{4}, 2\frac{3}{4}\}$, two elements (not necessarily different) can sum to 3.

Together, there are **(C) 11** distinct integers that can be written as the sum of two, not necessarily different, special fractions:

$$1, 2, 3, 4, 6, 7, 8, 13, 16, 18, 28.$$

KingRavi samrocksnature Wilhelm Z MRENTHUSIASM Steven Chen (www.professorchen.edu.com)

Solution 2

Let $a = 15 - b$, so the special fraction is

$$\frac{a}{b} = \frac{15 - b}{b} = \frac{15}{b} - 1.$$

We can ignore the -1 part and only focus on $\frac{15}{b}$.

The integers are $\frac{15}{1}, \frac{15}{3}, \frac{15}{5}$, which are 15, 5, 3, respectively. We get 30, 20, 18, 10, 8, 6 from this group of numbers.

The halves are $\frac{15}{2}, \frac{15}{6}, \frac{15}{10}$, which are $7\frac{1}{2}, 2\frac{1}{2}, 1\frac{1}{2}$, respectively. We get 15, 10, 9, 5, 4, 3 from this group of numbers.

The quarters are $\frac{15}{4}, \frac{15}{12}$, which are $3\frac{3}{4}, 1\frac{1}{4}$, respectively. We get 5 from this group of numbers.

Note that 10 and 5 each appear twice. Therefore, the answer is **(C) 11**.

Solution 3 (Casework)

We split this up into two cases:

Case 1: integer + integer

The whole numbers we have are $\frac{10}{5}$ (or 2), $\frac{12}{3}$ (or 4), and $\frac{14}{1}$ (or 14). There are $\binom{3}{2} = 3$ ways to choose different-numbered pairs and 3 ways to choose the same-numbered pairs. So, $3 + 3 = 6$.

Case 2: fraction + fraction

The fractions we have are $\frac{5}{10}$ (or $\frac{1}{2}$), $\frac{9}{6}$ (or $\frac{3}{2}$), and $\frac{13}{2}$. Similarly, there are $\binom{3}{2} = 3$ ways to choose different-numbered pairs and 3 ways to choose the same-numbered pairs. So, $3 + 3 = 6$.

Thus, $6 + 6 = 12$.

So now you would just go ahead and innocently choose **(D) 12**, right? No! We overcounted 8, as $\frac{13}{2} + \frac{9}{6} = \frac{12}{3} + \frac{12}{3} = 8$. Therefore, the correct answer is actually $12 - 1 = \mathbf{(C) 11}$.

8. (AMC 10 10B Fall 2021) The greatest prime number that is a divisor of 16,384 is 2 because $16,384 = 2^{14}$. What is the sum of the digits of the greatest prime number that is a divisor of 16,383?

(A) 3 (B) 7 (C) 10 (D) 16 (E) 22

Answer: C

Solution

We have

$$\begin{aligned} 16383 &= 2^{14} - 1 \\ &= (2^7 + 1)(2^7 - 1) \\ &= 129 \cdot 127 \end{aligned}$$

Since 129 is composite, 127 is the largest prime which can divide 16383. The sum of 127's digits is $1 + 2 + 7 = \boxed{\text{(C)} 10}$.

Note Note that you can quickly tell that $2^7 - 1$ is prime because it is a **Mersenne prime**.

9. (AMC 10 10B Fall 2021) The knights in a certain kingdom come in two colors. $\frac{2}{7}$ of them are red, and the rest are blue. Furthermore, $\frac{1}{6}$ of the knights are magical, and the fraction of red knights who are magical is 2 times the fraction of blue knights who are magical. What fraction of red knights are magical?

(A) $\frac{2}{9}$ (B) $\frac{3}{13}$ (C) $\frac{7}{27}$ (D) $\frac{2}{7}$ (E) $\frac{1}{3}$

Answer: C

Solution 1

Let k be the number of knights: then the number of red knights is $\frac{2}{7}k$ and the number of blue knights is $\frac{5}{7}k$.

Let b be the fraction of blue knights that are magical - then $2b$ is the fraction of red knights that are magical. Thus we can write the equation $b \cdot \frac{5}{7}k + 2b \cdot \frac{2}{7}k = \frac{k}{6} \implies \frac{5}{7}b + \frac{4}{7}b = \frac{1}{6} \implies \frac{9}{7}b = \frac{1}{6} \implies b = \frac{7}{54}$

We want to find the fraction of red knights that are magical, which is $2b = \frac{7}{27} =$

$\boxed{\text{(C)} \frac{7}{27}}$

Solution 2

We denote by p the fraction of red knights who are magical.

Hence,

$$\frac{1}{6} = \frac{2}{7}p + \left(1 - \frac{2}{7}\right)\frac{p}{2}.$$

By solving this equation, we get $p = \frac{7}{27}$.

Therefore, the answer is $\boxed{\text{(C)} \frac{7}{27}}$.

Solution 3

Suppose there are 42 knights. Then there are $\frac{2}{7} \cdot 42 = 12$ red knights and $42 - 12 = 30$ blue knights.

There are $\frac{1}{6} \cdot 42 = 7$ magical knights. Let x be the number of red magical knights.

Then we have the equation $\frac{x}{12} = 2 \cdot \frac{7-x}{30} = \frac{7-x}{15}$. Solving, we get $x = 84/27$.

Plugging x back gives us the fraction of red magical knights, which is $\frac{\frac{84}{27}}{12} = \frac{84}{27} \cdot \frac{1}{12} =$

$$\boxed{\text{(C)} \frac{7}{27}}.$$

Solution 4

Similar to Solution 3, let the total number of knights be 42 (We can say this because the fractions we know have denominators of 7 and 6 which have a LCM of 42)

This means that there are $\frac{2}{7} \times 42 = 12$ red knights meaning there are 30 blue knights

This also means that there are $\frac{1}{6} \times 42 = 7$ magical knights. This is important information for our next step.

We are given that the fraction of magical red knights is twice the fraction of magical blue knights. As this problem is dealing with the fraction being twice as large and is asking for the fraction, let's just deal with the variable being a fraction. Let $\frac{p}{q}$ be the fraction of red magical knights. This means that $\frac{p}{2q}$ is the fraction of blue magical knights.

We know that total number of magical knights from what we figured out before is 7, so let's use the fraction we figured and make the equation:

$$(12 \times \frac{p}{q}) + (30 \times \frac{p}{2q}) = 7$$

Solving this gives us $\frac{p}{q} = \frac{7}{27}$ or the fraction of red magical knights.

Hence, the answer is $\boxed{\frac{7}{27}}$

Solution 5

This solution is similar to Solution 2.

Suppose x is the fraction of red knights that are magical. Then $\frac{x}{2}$ is the fraction of blue knights that are magical. We can then create an equation in terms of x .

$$\frac{2}{7}x + \frac{5}{7}(\frac{x}{2}) = \frac{1}{6}$$

Solving, we get that $x = \frac{7}{27}$.

The answer is $\boxed{\text{(C)} \frac{7}{27}}$.

10. (AMC 10 10B Fall 2021) Forty slips of paper numbered 1 to 40 are placed in a hat. Alice and Bob each draw one number from the hat without replacement, keeping their numbers hidden from each other. Alice says, "I can't tell who has the larger number." Then Bob says, "I know who has the larger number." Alice says, "You do? Is your number prime?" Bob replies, "Yes." Alice says, "In that case, if I multiply your number by 100 and add my number, the result is a perfect square. " What is the sum of the two numbers drawn from the hat?

(A) 27 (B) 37 (C) 47 (D) 57 (E) 67

Answer: A

Solution 1

Let Alice have the number A , Bob B . When Alice says that she can't tell who has the larger number, it means that A cannot equal 1. Therefore, it makes sense that Bob has 2 because he now knows that Alice has the larger number. 2 is also prime. The last statement means that $200 + A$ is a perfect square. The three squares in the range $200 - 300$ are 225, 256, and 289. So, A could equal 25, 56, or 89, so $A + B$ is 27, 58, or 91, of only $\boxed{\text{(A)} 27}$ is below 40, therefore it is the correct answer since the maximum number one can have is 40.

–stjwyl

Minor edit SIGMAMATHEMATICIAN

Solution 2

Denote by A and B the numbers drawn by Alice and Bob, respectively.

Alice's sentence "I can't tell who has the larger number." implies $A \in \{2, \dots, 39\}$.

Bob's sentence "I know who has the larger number." implies $B \in \{1, 2, 39, 40\}$.

Their subsequent conversation that B is prime implies $B = 2$.

Then, Alice's next sentence "In that case, if I multiply your number by 100 and add my number, the result is a perfect square." implies $200 + A$ is a perfect square. Hence, $A = 25$.

Therefore, the answer is $\boxed{\text{(A)} 27}$.

Solution 3 - Guessing those Squares

We see that 225 is one such square that works. Bob gets 2 and Alice gets 25 which is valid. Thus, $2 + 25 = 27$. So $\boxed{\text{(A)} 27}$ is our answer.

-D1r

Solution 4 (Ambiguity, Sanity Checks)

The MAA knows this problem is ambiguous, as the interpretation of what might be viewed as the "middle" of the set could be different for all. Then, they would have specifically designed the answer choices such that one choice is guaranteed to be correct while all others are unobtainable.

So, for a sanity check and our general satisfaction, we will use the set $\mathcal{A} \in \{5, 6, 7, \dots, 34, 35\}$ as all the possible values of $\mathcal{A}$ and $\mathcal{B} \in \{1, 2, 3, 4, 36, 37, 38, 39, 40\}$. Alice notes that Bob's number is prime. Therefore, $\mathcal{B} \in \{2, 3, 4, 37\}$. Now, say Alice's number is A . Then, each of $200 + A$, $300 + A$, and $3700 + A$ is prime. Note that in each case, $A = 25, 24, 21$ respectively. Then, $A + B = 27, 27, 58$ respectively. Only $\boxed{27}$ is in the answer choices, so that would be our answer.

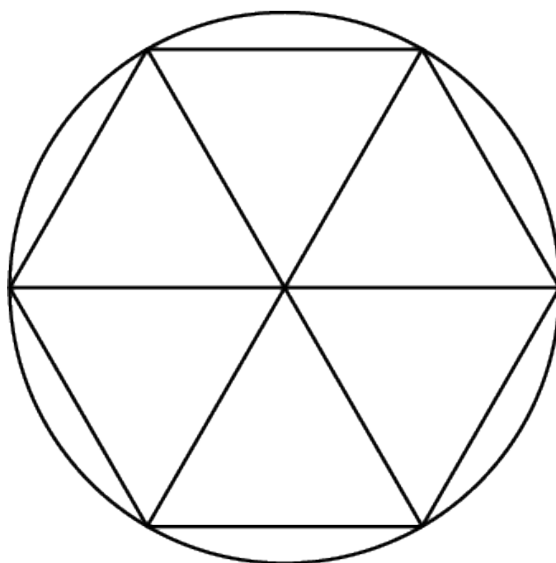
Sidenote Note that Bob's statement (that he knows who won/lost) comes after Alice tells him that she doesn't know who won. Since Alice doesn't know who won, Bob knows that she didn't get draw a 1 (or a 40), which tells him that his 2 must be lower than Alice's number.

11. (AMC 10 10B Fall 2021) A regular hexagon of side length 1 is inscribed in a circle. Each minor arc of the circle determined by a side of the hexagon is reflected over that side. What is the area of the region bounded by these 6 reflected arcs?

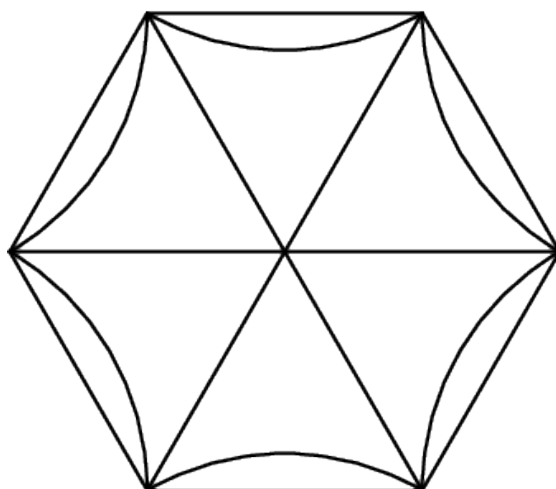
(A) $\frac{5\sqrt{3}}{2} - \pi$ (B) $3\sqrt{3} - \pi$ (C) $4\sqrt{3} - \frac{3\pi}{2}$ (D) $\pi - \frac{\sqrt{3}}{2}$ (E) $\frac{\pi + \sqrt{3}}{2}$

Answer: B

Solution 1



This is the graph of the original Hexagon. After reflecting each minor arc over the sides of the hexagon it will look like this;



This bounded region is the same as the area of the hexagon minus the area of each of the reflect arcs. From the first diagram, the area of each arc is the area of the 60° sector minus the area of the equilateral triangle, so each arc has an area of $\frac{\pi r^2}{6} - \frac{s^2\sqrt{3}}{4} \implies \frac{\pi}{6} - \frac{\sqrt{3}}{4}$.

There are 6 total arcs, so the total area of the arcs is $6 \cdot (\frac{\pi}{6} - \frac{\sqrt{3}}{4}) = \pi - \frac{3\sqrt{3}}{2}$.

The area of the hexagon is $6 \cdot \frac{\sqrt{3}}{4} = \frac{3\sqrt{3}}{2}$, so the area of the bounded region is: $\frac{3\sqrt{3}}{2} - (\pi - \frac{3\sqrt{3}}{2}) = 3\sqrt{3} - \pi = \boxed{B}$

Solution 2

Let the hexagon described be of area H and let the circle's area be C . Let the area we want to aim for be A . Thus, we have that $C - H = H - A$, or $A = 2H - C$. By some formulas, $C = \pi r^2 = \pi$ and $H = 6 \cdot \frac{1}{2} \cdot 1 \cdot (\frac{1}{2} \cdot \sqrt{3}) = \frac{3\sqrt{3}}{2}$. Thus, $A = 3\sqrt{3} - \pi$ or $\boxed{(B)}$.

Solution 3

Denote by O the center of this circle. Hence, the radius of this circle is 1. Denote this hexagon as $ABCDEF$.

We have $\angle AOB = 60^\circ$. Hence, the area of the region formed between segment AB and the minor arc formed by A and B , denoted as M , is

$$\begin{aligned} M &= \pi 1^2 \frac{60}{360} - \frac{\sqrt{3}}{4} 1^2 \\ &= \frac{\pi}{6} - \frac{\sqrt{3}}{4}. \end{aligned}$$

Therefore, the area of the region that this problem asks us to compute is

$$\pi 1^2 - 12M = 3\sqrt{3} - \pi.$$

Therefore, the answer is **(B)** $3\sqrt{3} - \pi$.

Solution 4

The area of the desired shape is equal to the $A(\text{whole hexagon}) - A(\text{Six arcs})$.

Since the arcs are reflected upon the sides of the hexagon, we can see that $A(\text{Six arcs}) = A(\text{Whole circle}) - A(\text{Whole hexagon})$.

The area of the circle is $\pi \cdot 1^2$, since the shape has side length 1 and is inscribed within the circle (so its diameter is 2).

Combining these two, we see that

$$A(\text{Desired shape}) = 2A(\text{Hexagon}) - A(\text{Circle}) \rightarrow$$

$$A(\text{Desired shape}) = 2A(\text{Hexagon}) - \pi.$$

From here, three solutions can progress:

Solution 4.1 (General polygons)

The area of a regular polygon is equal to $\frac{p \cdot a}{2}$, where p is the perimeter of the polygon, and a is the length of its apothem.

The apothem is the distance from the center of the polygon to the midpoint of two adjacent vertices. If we were to create an equilateral triangle, whose base is at the side of the polygon, and its two sides meeting at the center, its height would be the apothem.

In this case, the side length of the hexagon is 1. We can now split this equilateral triangle into two congruent right triangles, who both have the apothem as a side. Each triangle has side lengths of $\frac{1}{2}$, half of the base, 1, the radius, and the apothem. By the pythagorean theorem, $1^2 = (\frac{1}{2})^2 + a^2$, for the apothem a . Solving yields $a = \frac{\sqrt{3}}{2}$.

Since $p = 6$, our formula becomes

$$A = 6\left(\frac{\sqrt{3}}{2}\right) \cdot \frac{1}{2} = \frac{3\sqrt{3}}{2}.$$

Recall that the area of the figure is $2A(\text{Hexagon}) - \pi$. $2A = 3\sqrt{3}$, so the sought area is $3\sqrt{3} - \pi$, so our answer is **(B)** $3\sqrt{3} - \pi$

Alternatively, we could find the length of the apothem by the formula $\frac{s}{2\tan(\frac{180}{n})}$, where s is the side length and n is the number of sides.

Solution 4.2 (Quick)

The area of a regular hexagon is given by the formula $\frac{3\sqrt{3}}{2} \cdot s$, for the side length s , which in this case, is 1, so the area of this hexagon is $\frac{3\sqrt{3}}{2}$.

We seek $2A(\text{Hexagon}) - \pi$, which is $\boxed{\text{(B)} \ 3\sqrt{3} - \pi}$

Solution 5

Use the diagrams provided in Solution 1.

The problem wants the area of the region bounded by the 6 reflected arcs, a.k.a the area of the hexagon minus the area of the arcs drawn in the first figure. The area of those arcs can be calculated by taking the total area of the circle minus the total area of the hexagon.

In the end, we are looking for the value of the (area of the hexagon)-((area of the circle)-(area of the hexagon)).

The area of a hexagon with side length 1 is $\frac{3\sqrt{3}}{2}$. The area of a circle with radius 1 is π .

$$\frac{3\sqrt{3}}{2} - (\pi - \frac{3\sqrt{3}}{2}) = 3\sqrt{3} - \pi.$$

The answer is $\boxed{\text{(B)} \ 3\sqrt{3} - \pi}$.

12. (AMC 10 10B Fall 2021) Which of the following conditions is sufficient to guarantee that integers x , y , and z satisfy the equation

$$x(x - y) + y(y - z) + z(z - x) = 1?$$

- (A) $x > y$ and $y = z$ (B) $x = y - 1$ and $y = z - 1$ (C) $x = z + 1$ and $y = x + 1$
(D) $x = z$ and $y - 1 = x$ (E) $x + y + z = 1$

Answer: D

Solution 1 (Completing the Square)

It is obvious x , y , and z are symmetrical. We are going to solve the problem by Completing the Square.

$$x^2 + y^2 + z^2 - xy - yz - zx = 1 \quad 2x^2 + 2y^2 + 2z^2 - 2xy - 2yz - 2zx = 2 \quad (x - y)^2 + (y - z)^2 + (z - x)^2 = 2$$

Because x, y, z are integers, $(x - y)^2$, $(y - z)^2$, and $(z - x)^2$ can only equal 0, 1, 1. So one variable must equal another, and the third variable is 1 different from those 2 equal variables. So the answer is $\boxed{D}$.

Solution 2

Plugging in every choice, we see that choice (D) works.

We have $y = x + 1, z = x$, so

$$x(x - y) + y(y - z) + z(z - x) = x(x - (x + 1)) + (x + 1)((x + 1) - x) + x(x - x) = x(-1) + (x + 1)(1) = 1.$$

Our answer is (D).

Solution 3 (Bash)

Just plug in all these options one by one, and one sees that all but D fails to satisfy the equation.

For D , substitute $z = x$ and $y = x + 1$:

$$LHS = x(x - (x + 1)) + (x + 1)(x + 1 - x) + x(x - x) = (-x) + (x + 1) = 1 = RHS$$

Hence the answer is $\boxed{(D)}$.

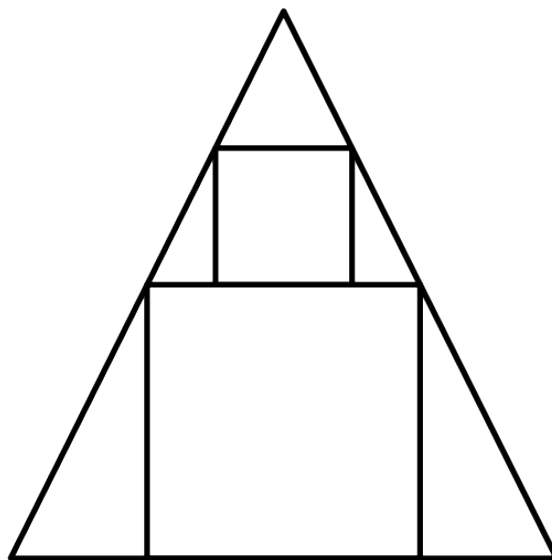
Solution 4 (Strategy)

Looking at the answer choices and the question, the simplest ones to plug in would be equalities because it would make one term of the equation become zero. We see that answer choices A and D have the simplest equalities in them. However, A has an inequality too, so it would be simpler to plug in D which has another equality. We see that $x = z$ and $y - 1 = x$ means the equation becomes $x(x - (x + 1)) + (x + 1)(x + 1 - x) = 1 \implies -x + x + 1 = 1 \implies 1 = 1$, which is always true, so the answer is $\boxed{D}$.

Solution 5 (Logic and some Guesswork)

We know that x , y , and z are symmetric, which means that answer choices B and C are the same, so it can't be either of them. Answer choice E doesn't make sense because it is only 1 equation, which probably won't result in only 1 possible answer. This leaves us with choices A and D. Choice A isn't right because if x is really big and $y = z = 0$, then the equation becomes $x^2 - x = 1$ which doesn't work with a large integer x . Therefore, the only remaining answer choice is $\boxed{D}$.

13. (AMC 10 10B Fall 2021) A square with side length 3 is inscribed in an isosceles triangle with one side of the square along the base of the triangle. A square with side length 2 has two vertices on the other square and the other two on sides of the triangle, as shown. What is the area of the triangle?

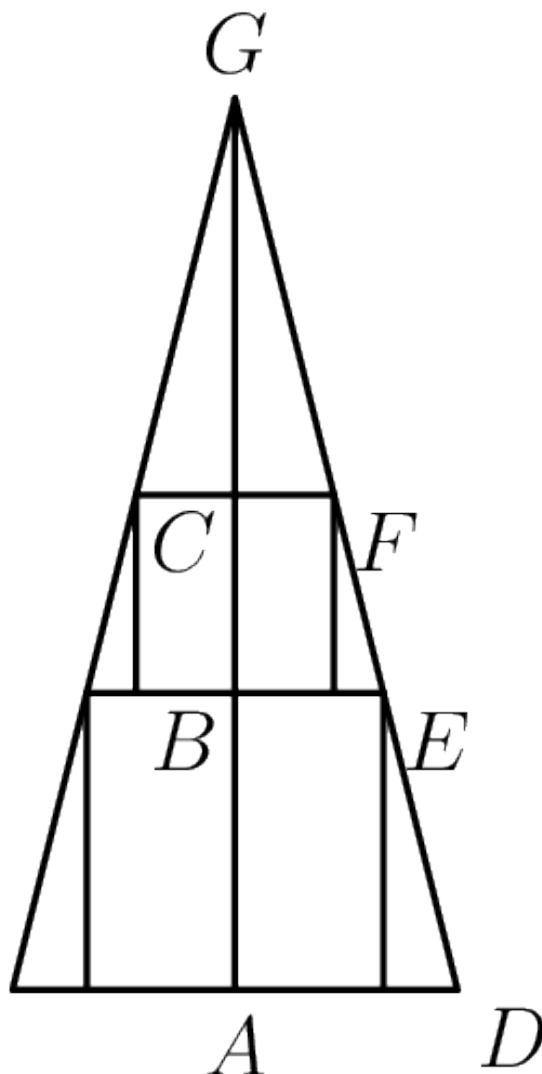


- (A) $19\frac{1}{4}$ (B) $20\frac{1}{4}$ (C) $21\frac{3}{4}$ (D) $22\frac{1}{2}$ (E) $23\frac{3}{4}$

Answer: B

Solution 1

Let's split the triangle down the middle and label it:



We see that $\triangle ADG \sim \triangle BEG \sim \triangle CFG$ by AA similarity. $BE = \frac{3}{2}$ because AG cuts the side length of the square in half; similarly, $CF = 1$. Let $CG = h$: then by side ratios,

$$\frac{h+2}{h} = \frac{\frac{3}{2}}{1} \implies 2(h+2) = 3h \implies h = 4$$

.

Now the height of the triangle is $AG = 4 + 2 + 3 = 9$. By side ratios,

$$\frac{9}{4} = \frac{AD}{1} \implies AD = \frac{9}{4}$$

.

The area of the triangle is $AG \cdot AD = 9 \cdot \frac{9}{4} = \frac{81}{4} = \boxed{\text{(B)} 20\frac{1}{4}}$

Solution 2

By similarity, the height is $3 + \frac{3}{1} \cdot 2 = 9$ and the base is $\frac{9}{2} \cdot 1 = 4.5$. Thus the area is $\frac{9 \cdot 4.5}{2} = 20.25 = \boxed{\text{(B)} 20\frac{1}{4}}$.

Solution 3 (With two different endings)

This solution is based on this figure: [Image:2021_AMC_10B_\(Nov\)_Problem_13,_sol.png](#)

Denote by O the midpoint of AB .

Because $FG = 3$, $JK = 2$, $FJ = KG$, we have $FJ = \frac{1}{2}$.

We observe $\triangle ADF \sim \triangle FJH$. Hence, $\frac{AD}{FJ} = \frac{FD}{HJ}$. Hence, $AD = \frac{3}{4}$. By symmetry, $BE = AD = \frac{3}{4}$.

Therefore, $AB = AD + DE + BE = \frac{9}{2}$.

Because O is the midpoint of AB , $AO = \frac{9}{4}$.

We observe $\triangle AOC \sim \triangle ADF$. Hence, $\frac{OC}{DF} = \frac{AO}{AD}$. Hence, $OC = 9$.

Therefore, $\text{Area } \triangle ABC = \frac{1}{2}AB \cdot OC = \frac{81}{4} = 20\frac{1}{4}$.

Therefore, the answer is **(B)** $20\frac{1}{4}$.

Alternatively, we can find the height in a slightly different way.

Following from our finding that the base of the large triangle $AB = \frac{9}{2}$, we can label the length of the altitude of $\triangle CHI$ as x . Notice that $\triangle CHI \sim \triangle CAB$. Hence, $\frac{HI}{AB} = \frac{x}{CO}$. Substituting and simplifying, $\frac{HI}{AB} = \frac{x}{CO} \Rightarrow \frac{\frac{2}{9}}{\frac{9}{2}} = \frac{x}{x+5} \Rightarrow \frac{x}{x+5} = \frac{4}{9} \Rightarrow x =$

$4 \Rightarrow CO = 4 + 5 = 9$. Therefore, the area of the triangle is $\frac{\frac{9}{2} \cdot 9}{2} = \frac{81}{4} =$ **(B)** $20\frac{1}{4}$.

Solution 4 (Coordinates)

For convenience, we will use the image provided in the third solution.

We can set O as the origin.

We know that $FG = 3$ and $JK = 2$.

We subtract JK from FG and divide by 2 to get $KG = FJ = \frac{1}{2}$.

Since $HIKJ$ is a square, we know that $IK = 2$.

Using rise over run, we find that the slope of CB is $\frac{-2}{0.5} = -4$.

The coordinates of I are $(1, 5)$. We plug this in to get the equation of the line that CB runs along:

$$y = -4x + 9$$

We know that the x -value of C is 0. Using this, we find that the y -value is 9. So the coordinates of C are $(0, 9)$.

This gives us the height of $\triangle ACB$: $CO = 9$.

Now we need to find the coordinates of B .

We know that the y -value is 0. Plugging this in, we find $0 = -4x + 9$, or $\frac{9}{4} = x$.

The coordinates of B are $(\frac{9}{4}, 0)$.

Since $\triangle ACB$ is symmetrical along CO , we can multiply CO by OB to get

$$9 \cdot \frac{9}{4} = \frac{81}{4}$$

Simplifying, we get **(B)** $20\frac{1}{4}$ for the area.

Solution 5 (interesting)

Call the right triangle to the right of the square with side length 2 triangle a_1 . Similarly, call the triangle to the right of the square with side length 3 a_2 . Label the ENTIRE triangle triangle ABC. We notice AA similarity between a_1 and a_2 . Additionally, the base of a_1 must have a length of 0.5, because the middle square takes up a length of 2 and the other length of 1 must be evenly split between the two congruent triangles next to the square, being all situated on a length of 3. We then see that the ratio of 2 : 1 must hold for the base of triangle ABC. $3 : x = 2 : 1$, and we find that $x = 1.5$. The length of the base is then $1.5 + 3 = 9/2$.

We can then find the height of the triangle by thinking about a geometrical sequence. It must be possible that you could continue inscribing squares in the triangles that are made, as every triangle created is similar to the one before it. Using this, we have the geometrical sequence of $3 + 2 + 4/3 + \dots$ (for each side length is the height that is being added) Using the geometrical sequence formula, we find $\frac{3}{1-\frac{2}{3}} = \frac{3}{\frac{1}{3}} = 3 \cdot 3 = 9$. The height is 9 whilst the base length is $\frac{9}{2}$, and multiplying

both values and dividing by 2 yields $\frac{9(9)}{2(2)} = 81/4 = \boxed{\text{(B)} 20\frac{1}{4}}$

14. (AMC 10 10B Fall 2021) Una rolls 6 standard 6-sided dice simultaneously and calculates the product of the 6 numbers obtained. What is the probability that the product is divisible by 4? (A) $\frac{3}{4}$ (B) $\frac{57}{64}$ (C) $\frac{59}{64}$ (D) $\frac{187}{192}$ (E) $\frac{63}{64}$

Answer: C

Solution

We will use complementary counting to find the probability that the product is not divisible by 4. Then, we can find the probability that we want by subtracting this from 1. We split this into 2 cases.

Case 1: The product is not divisible by 2.

We need every number to be odd, and since the chance we roll an odd number is $\frac{1}{2}$, our probability is $\left(\frac{1}{2}\right)^6 = \frac{1}{64}$.

Case 2: The product is divisible by 2, but not by 4.

We need 5 numbers to be odd, and one to be divisible by 2, but not by 4. There is a $\frac{1}{2}$ chance that an odd number is rolled, a $\frac{1}{3}$ chance that we roll a number satisfying the second condition (only 2 and 6 work), and 6 ways to choose the order in which the even number appears.

Our probability is $\left(\frac{1}{2}\right)^5 \left(\frac{1}{3}\right) \cdot 6 = \frac{1}{16}$.

Therefore, the probability the product is not divisible by 4 is $\frac{1}{64} + \frac{1}{16} = \frac{5}{64}$.

Our answer is $1 - \frac{5}{64} = \boxed{\text{(C)} \frac{59}{64}}$.

Solution 2(Pretty much same as above)

Complementary counting implies that we can have either 0 factors of 2 or just 1 factor of 2.

Case 1: 0 factors of 2

The only odd numbers are 1, 3, 5 on a die so we have these 3 odd numbers with 6 positions so 3^6 ways here.

Case 2: 1 factor of 2

The only even numbers that have only one factor of 2 on a die are 2, 6. On the 6 positions, there are 6 ways to place one of these even numbers and the rest of the 5 positions must be one of the three odd numbers. Since we could have either 2 or 6, we have

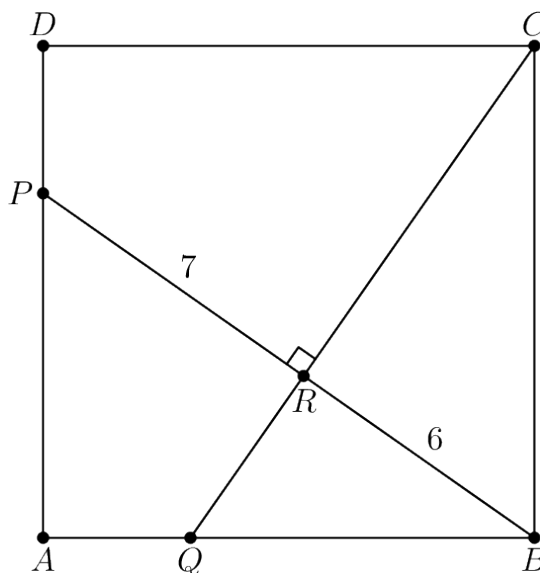
$$2 \cdot 6 \cdot 3^5 = 4 \cdot 3^6.$$

Thus, we have $3^6 + 4 \cdot 3^6 = 5 \cdot 3^6$ cases that don't work.

By complementary counting we get

$$1 - \frac{5 \cdot 3^6}{6^6} = 1 - \frac{5}{2^6} = \boxed{\frac{59}{64}}$$

15. (AMC 10 10B Fall 2021) In square $ABCD$, points P and Q lie on $\overline{AD}$ and $\overline{AB}$, respectively. Segments $\overline{BP}$ and $\overline{CQ}$ intersect at right angles at R , with $BR = 6$ and $PR = 7$. What is the area of the square?



- (A) 85 (B) 93 (C) 100 (D) 117 (E) 125

Answer: D

Solution 1

Note that $\triangle APB \cong \triangle BQC$ by ASA. ($\angle PAB = \angle QBC = 90^\circ$, $AB = CB$, and $\angle PBA = \angle QCB$.) Then, it follows that $\overline{PB} \cong \overline{QC}$. Thus, $QC = PB = PR + RB = 7 + 6 = 13$. Define x to be the length of side CR , then $RQ = 13 - x$. Because $\overline{BR}$ is the altitude of the triangle, we can use the property that $QR \cdot RC = BR^2$. Substituting the given lengths, we have

$$(13 - x) \cdot x = 36.$$

Solving, gives $RQ = 4$ and $RC = 9$. We eliminate the possibility of $x = 4$ because $RC > QR$. Thus, the side length of the square, by Pythagorean Theorem, is

$$\sqrt{9^2 + 6^2} = \sqrt{81 + 36} = \sqrt{117}.$$

Thus, the area of the square is $(\sqrt{117})^2 = 117$, so the answer is **(D) 117**.

Please note that there is another way to prove that $CR = 4$ is impossible. If $CR = 4$, then the side length would be $\sqrt{4^2 + 6^2} = \sqrt{52}$, and the area would be 52, but that isn't in the answer choices. Thus, CR must be 9.

Extra Note: Another way to prove 4 is impossible. The side length of the square, S , is equal to $\sqrt{4^2 + 6^2} = \sqrt{52}$. Because $x = 4$, $RQ = 9$. Because $QB = \sqrt{RB^2 + RQ^2} = \sqrt{6^2 + 9^2} = \sqrt{117}$ and $QB < S$ but $\sqrt{117} > \sqrt{52}$, we have proof by contradiction. And so $x = 9$.

Solution 2 (Similarity, Pythagorean Theorem, and Systems of Equations)

As above, note that $\triangle BPA \cong \triangle CQB$, which means that $QC = 13$. In addition, note that BR is the altitude of a right triangle to its hypotenuse, so $\triangle BQR \sim \triangle CBR \sim \triangle CQB$. Let the side length of the square be x ; using similarity side ratios of $\triangle BQR$ to $\triangle CQB$, we get

$$\frac{6}{x} = \frac{QB}{13} \implies QB \cdot x = 78$$

Note that $QB^2 + x^2 = 13^2 = 169$ by the Pythagorean theorem, so we can use the expansion $(a + b)^2 = a^2 + 2ab + b^2$ to produce two equations and two variables;

$$(QB+x)^2 = QB^2 + 2QB \cdot x + x^2 \implies (QB+x)^2 = 169 + 2 \cdot 78 \implies QB+x = \sqrt{13(13) + 13(12)} = \sqrt{13 \cdot 25}$$

$$(QB-x)^2 = QB^2 - 2QB \cdot x + x^2 \implies (QB-x)^2 = 169 - 2 \cdot 78 \implies \pm(QB-x) = \sqrt{13(13) - 13(12)}$$

Since $QB - x$ is negative, it doesn't make sense in the context of this problem, so we go with

$$x - QB = \sqrt{13(13) - 13(12)} = \sqrt{13 \cdot 1} = \sqrt{13}$$

We want x^2 , so we want to find x . Adding the first equation to the second, we get

$$2x = 6\sqrt{13} \implies x = 3\sqrt{13}$$

$$\text{Then } x^2 = (3\sqrt{13})^2 = 9 \cdot 13 = 117 = \boxed{D}$$

-yingkai_0_ (Minor Edits)

Solution 3

We have that $\triangle CRB \sim \triangle BAP$. Thus, $\frac{CB}{CR} = \frac{PB}{AB}$. Now, let the side length of the square be s . Then, by the Pythagorean theorem, $CR = \sqrt{x^2 - 36}$. Plugging all of this information in, we get

$$\frac{s}{\sqrt{s^2 - 36}} = \frac{13}{s}.$$

Simplifying gives

$$s^2 = 13\sqrt{s^2 - 36},$$

Squaring both sides gives

$$s^4 = 169s^2 - 169 \cdot 36 \implies s^4 - 169s^2 + 169 \cdot 36 = 0.$$

We now set $s^2 = t$, and get the equation $t^2 - 169t + 169 \cdot 36 = 0$. From here, notice we want to solve for t , as it is precisely s^2 , or the area of the square. So we use the

Quadratic formula, and though it may seem bashy, we hope for a nice cancellation of terms.

$$t = \frac{169 \pm \sqrt{169^2 - 4 \cdot 36 \cdot 169}}{2}.$$

It seems scary, but factoring 169 from the square root gives us

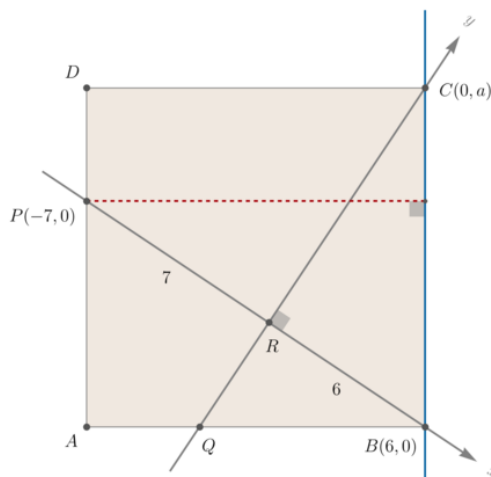
$$t = \frac{169 \pm \sqrt{169 \cdot (169 - 144)}}{2} = \frac{169 \pm \sqrt{169 \cdot 25}}{2} = \frac{169 \pm 13 \cdot 5}{2} = \frac{169 \pm 65}{2},$$

giving us the solutions

$$t = 52, 117.$$

We instantly see that $t = 52$ is way too small to be an area of this square (52 isn't even an answer choice, so you can skip this step if out of time) because then the side length would be $2\sqrt{13}$ and then, even the largest line you can draw inside the square (the diagonal) is $2\sqrt{26}$, which is less than 13 (line PB). And thus, t must be 117, and our answer is **(D)**. ■

Solution 4 (Point-line distance formula)



Denote $a = RC$. Now tilt your head to the right and view $R, \overrightarrow{RB}$ and $\overrightarrow{RC}$ as the origin, x -axis and y -axis, respectively. In particular, we have points $B(6,0), C(0,a), P(-7,0)$. Note that side length of the square $ABCD$ is $BC = \sqrt{a^2 + 36}$. Also equation of line BC is

$$\underbrace{\frac{x}{6} + \frac{y}{a} = 1}_{\text{intercepts form}} \implies ax + 6y - 6a = 0.$$

Because the distance from $P(-7,0)$ to line $BC : ax + 6y - 6a = 0$ is also the side length $\sqrt{a^2 + 36}$, we can apply the point-line distance formula to get

$$\frac{|a \cdot (-7) + 6 \cdot 0 - 6a|}{\sqrt{a^2 + 36}} = \sqrt{a^2 + 36}$$

which reduces to $|13a| = a^2 + 36$. Since a is positive, the last equations factors $asa^2 - 13a + 36 = (a - 4)(a - 9) = 0$. Now judging from the figure, we learn

that $a > RB = 6$. So $a = 9$. Therefore, the area of the square $ABCD$ is $BC^2 = RC^2 + RB^2 = a^2 + 6^2 = \boxed{\text{(D) 117}}$.

Solution 5

Denote $\angle PBA = \alpha$. Because $\angle QRB = \angle QBC = 90^\circ$, $\angle BCQ = \alpha$.

Hence, $AB = BP \cos \angle PBA = 13 \cos \alpha$, $BC = \frac{BR}{\sin \angle BCQ} = \frac{6}{\sin \alpha}$.

Because $ABCD$ is a square, $AB = BC$. Hence, $13 \cos \alpha = \frac{6}{\sin \alpha}$.

Therefore,

$$\begin{aligned} \sin 2\alpha &= 2 \sin \alpha \cos \alpha \\ &= \frac{12}{13}. \end{aligned}$$

Thus, $\cos 2\alpha = \pm \frac{5}{13}$.

Case 1: $\cos 2\alpha = \frac{5}{13}$.

Thus, $\cos \alpha = \sqrt{\frac{1+\cos 2\alpha}{2}} = \frac{3}{\sqrt{13}}$.

Hence, $AB = 13 \cos \alpha = 3\sqrt{13}$.

Therefore, $\text{Area } ABCD = AB^2 = 117$.

Case 2: $\cos 2\alpha = -\frac{5}{13}$.

Thus, $\cos \alpha = \sqrt{\frac{1+\cos 2\alpha}{2}} = \frac{2}{\sqrt{13}}$.

Hence, $AB = 13 \cos \alpha = 2\sqrt{13}$.

However, we observe $BQ = \frac{BR}{\cos \alpha} = 3\sqrt{13} > AB$. Therefore, in this case, point Q is not on the segment AB .

Therefore, this case is infeasible.

Putting all cases together, the answer is $\boxed{\text{(D) 117}}$.

Solution 6 (Answer choices and areas)

Note that if we connect points P and C , we get a triangle with height RC and length 13. This triangle has an area of $\frac{1}{2}$ the square. We can now use answer choices to our advantage!

Answer choice A: If BC was $\sqrt{85}$, RC would be 7. The triangle would therefore have an area of $\frac{91}{2}$ which is not half of the area of the square. Therefore, A is wrong.

Answer choice B: If BC was $\sqrt{93}$, RC would be $\sqrt{57}$. This is obviously wrong.

Answer choice C: If BC was 10, we would have that RC is 8. The area of the triangle would be 52, which is not half the area of the square. Therefore, C is wrong.

Answer choice D: If BC was $\sqrt{117}$, that would mean that RC is 9. The area of the triangle would therefore be $\frac{117}{2}$ which IS half the area of the square. Therefore, our answer is $\boxed{\text{(D) 117}}$.

Solution 7 (Power of a Point)

Note that $PRQA$ is a cyclic quadrilateral (opposite angles add to 180°). Call the circumcircle of quadrilateral $PRQA$ O . Then the power of B to O is $6 \cdot (6 + 7) = 78$.

Let a be the length of BQ and s the side length of the square, then we have $a \cdot s = 78$, and we also have $a^2 + s^2 = 13^2$ (see note), solving the two equations will give us $s^2 = 117$.

- Note: This is from the fact that $\triangle QCB$ and $\triangle PBA$ are congruent. A short proof: We can prove that $\triangle PBA$ and $\triangle QBR$ are similar by ASA similarity. Likewise, we can prove that $\triangle QCB$ and $\triangle QBR$ are similar. Therefore, $\triangle QCB$ and $\triangle PBA$ are similar. Since they both have the side of a square as a hypotenuse, their hypotenuses are equal, so they are congruent. Therefore, $\overline{CQ} = \overline{PB} = 13$.

Solution 8 (Pythagorean Theorem)

Let's start by connecting the line CP and calling the side length of the square x .

By the Pythagorean Theorem, we will have $CR^2 + RB^2 = CB^2$, so $CR^2 + 6^2 = x^2$, so $CR = \sqrt{x^2 - 36}$.

Now, let's find CP by using the Pythagorean theorem with the equation $CP^2 = PR^2 + CR^2$. Because $PR = 7$ and $CR = \sqrt{x^2 - 36}$, we have $CP^2 = 7^2 + (\sqrt{x^2 - 36})^2$, so $CP = \sqrt{x^2 + 13}$. Because CD is a side of square $ABCD$, $CD = x$, and so now we can have $DP^2 + CD^2 = CP^2$, which will give us $DP = \sqrt{13}$.

Because $AB = x$, $AP = AD - DP = x - \sqrt{13}$, and $PB = 6 + 7 = 13$, we now have $AB^2 + AP^2 = PB^2$ by the Pythagorean theorem, and the equation will be $x^2 + (x - \sqrt{13})^2 = 169$. After solving this equation for x , we will get $x = -2\sqrt{13}$ or $3\sqrt{13}$, and because side lengths can't be negative, we will take the solution $x = 3\sqrt{13}$, which means the area of the square will be $(3\sqrt{13})^2 = 117$.

Therefore, our answer is (D) 117.

16. (AMC 10 10B Fall 2021) Five balls are arranged around a circle. Chris chooses two adjacent balls at random and interchanges them. Then Silva does the same, with her choice of adjacent balls to interchange being independent of Chris's. What is the expected number of balls that occupy their original positions after these two successive transpositions?

(A) 1.6 (B) 1.8 (C) 2.0 (D) 2.2 (E) 2.4

Answer: D

Solution 1

After the first swap, we do casework on the next swap.

Case 1: Silva swaps the two balls that were just swapped

There is only one way for Silva to do this, and it leaves 5 balls occupying their original position.

Case 2: Silva swaps one ball that has just been swapped with one that hasn't been swapped

There are two ways for Silva to do this, and it leaves 2 balls occupying their original positions.

Case 3: Silva swaps two balls that have not been swapped

There are two ways for Silva to do this, and it leaves 1 ball occupying their original position.

Our answer is the average of all 5 possible swaps, so we get

$$\frac{5 + 2 \cdot 2 + 2 \cdot 1}{5} = \frac{11}{5} = \boxed{\text{(D)} 2.2}.$$

Solution 2 (Linearity of Expectation)

The "expected value" in the question tips us off to this technique. Consider any ball. The probability it returns to the same position is the probability of being swapped twice plus the probability of never being swapped:

$$\frac{2}{5} \cdot \frac{1}{5} + \left(\frac{3}{5}\right)^2 = \frac{11}{25}.$$

Multiply by 5 for 5 balls to get $\boxed{\text{(D)} 2.2}$.

Solution 3 (Calculating-Mean)

In probability and statistics, you learn that the mean and expected value are fundamentally equivalent values.

The total number of ways to pick the balls is $\binom{5}{2} = 10$.

Now there are three cases either Chris and Silva choose the same two balls (C1), they choose two different balls (C2), or they choose one overlapping ball and three nonoverlapping (C3).

C1: There are $\binom{5}{2} = 10$ possible placements of choosing 2 balls. This gives a remaining 5 unarranged balls. This gives a weighted mean pair of $10(5) = 50$.

C2: There are 5 ways of choosing 4 balls adjacently, and due to Chris and Silva each choosing 2 gives 10 ways due to symmetry. This gives a weighted mean pair of $10(1) = 10$.

C3: There are now $100 - 10 - 10 = 80$ ways for C3 to occur. This gives 2 balls that remain in their original positions for a weighted mean pair of $80(2) = 160$.

Calculating the mean gives $\frac{50+10+160}{100} = \boxed{\text{(D)} 2.2}$.

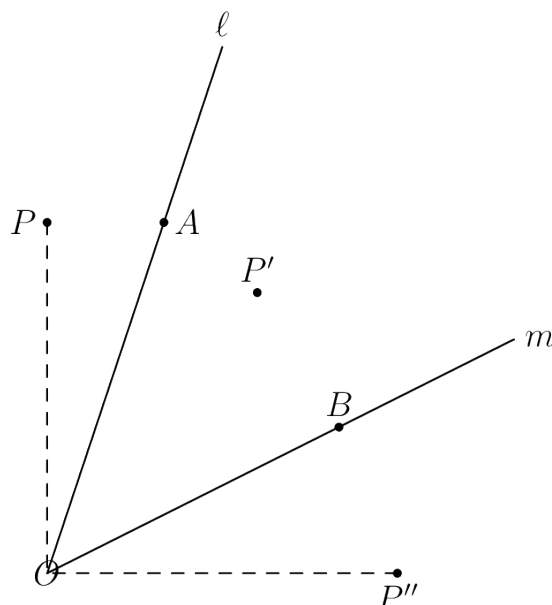
17. (AMC 10 10B Fall 2021) Distinct lines ℓ and m lie in the xy -plane. They intersect at the origin. Point $P(-1, 4)$ is reflected about line ℓ to point P' , and then P' is reflected about line m to point P'' . The equation of line ℓ is $5x - y = 0$, and the coordinates of P'' are $(4, 1)$. What is the equation of line m ? (A) $5x + 2y = 0$ (B) $3x + 2y = 0$ (C) $x - 3y = 0$ (D) $2x - 3y = 0$ (E) $5x - 3y = 0$

Answer: D

Solution 1

Denote O as the origin.

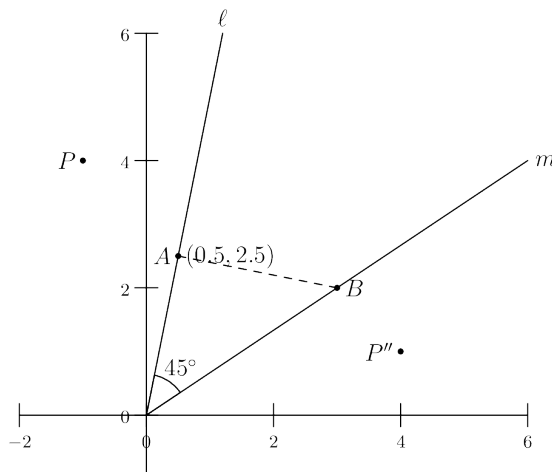
Even though the problem is phrased as a coordinate bash, that looks disgusting. Instead, let's try to phrase this problem in terms of Euclidean geometry, using the observation that $\angle POP'' = 90^\circ$, and that both ℓ and m must pass through O in order to preserve the distance from P to the origin.



(A and B are just defined as points on lines ℓ and m .) Because of how reflections work, we have that $\angle AOP' = \angle POA$ and $\angle P'OB = \angle BOP''$; adding these two equations together and using angle addition, we have that $\angle AOB = \angle POA + \angle BOP''$. Since the sum of both sides combined must be 90° by angle addition,

$$\angle AOB = 45^\circ.$$

This is helpful! We can now return to using coordinates, with this piece of information in mind:



The 45° angle is a little bit unwieldy in the coordinate plane, so we should try to make a $45 - 45 - 90$ triangle. Let A be a point on ℓ ; to make A fit nicely in the diagram, let it be $(0.5, 2.5)$. Now, let's draw a perpendicular to ℓ through point A , intersecting m at point B . OAB is a $45 - 45 - 90$ triangle, so B is a 90 degree counterclockwise rotation from O about A . Therefore, the coordinates of B are

$$(0.5 + 2.5, 2.5 - 0.5) = (3, 2).$$

So, $(3, 2)$ is a point on line m , which we already know passes through the origin; therefore, m 's equation is $y = \frac{2x}{3} \implies \boxed{\text{(D)} \ 2x - 3y = 0}$.

(We never actually had to use the information of the exact coordinates of P ; as long as $\angle POP'' = 90^\circ$, when we move P around, this will not affect m 's equation.)

Supplement In case you are confused about the coordinates of B, first transform O, A, and B such that A is the origin $A'(0,0)$ in a new coordinate system. From there it is not too hard to see that O now has the coordinates $O'(-0.5, -2.5)$. Thus point B, from a 90 deg CCW rotation around origin A, will have a coordinate of $B'(2.5, -0.5)$.

Now in order to go from this new coord system $A'(0,0)$ to its original point $A(0.5, 2.5)$, we have to $+0.5, +2.5$ to the x and y coordinates respectively. Doing this with B' to B, we have $(2.5+0.5, -0.5+2.5) = B(3, 2)$ as desired.

Solution 2

It is well known that the composition of 2 reflections, one after another, about two lines l and m , respectively, that meet at an angle θ is a rotation by 2θ around the intersection of l and m .

Now, we note that $(4, 1)$ is a 90 degree rotation clockwise of $(-1, 4)$ about the origin, which is also where l and m intersect. So m is a 45 degree rotation of l about the origin clockwise.

To rotate l 90 degrees clockwise, we build a square with adjacent vertices $(0, 0)$ and $(1, 5)$. The other two vertices are at $(5, -1)$ and $(6, 4)$. The center of the square is at $(3, 2)$, which is the midpoint of $(1, 5)$ and $(5, -1)$. The line m passes through the origin and the center of the square we built, namely at $(0, 0)$ and $(3, 2)$. Thus the line is $y = \frac{2}{3}x$. The answer is (D) $2x - 3y = 0$.

Solution 3

We know that the equation of line ℓ is $y = 5x$. This means that P' is $(-1, 4)$ reflected over the line $y = 5x$. This means that the line with P and P' is perpendicular to ℓ , so it has slope $-\frac{1}{5}$. Then the equation of this perpendicular line is $y = -\frac{1}{5}x + c$, and plugging in $(-1, 4)$ for x and y yields $c = \frac{19}{5}$.

The midpoint of P' and P lies at the intersection of $y = 5x$ and $y = -\frac{1}{5}x + \frac{19}{5}$. Solving, we get the x-value of the intersection is $\frac{19}{26}$ and the y-value is $\frac{95}{26}$. Let the x-value of P' be x' - then by the midpoint formula, $\frac{x'-1}{2} = \frac{19}{26} \implies x' = \frac{32}{13}$. We can find the y-value of P' the same way, so $P' = (\frac{32}{13}, \frac{43}{13})$.

Now we have to reflect P' over m to get to $(4, 1)$. The midpoint of P' and P'' will lie on m , and this midpoint is, by the midpoint formula, $(\frac{42}{13}, \frac{28}{13})$. $y = mx$ must satisfy this point, so $m = \frac{\frac{28}{13}}{\frac{42}{13}} = \frac{28}{42} = \frac{2}{3}$.

Now the equation of line m is $y = \frac{2}{3}x \implies 2x - 3y = 0 = \boxed{D}$

Solution 4

First, use Solution 1's method to get $\angle POP'' = 90^\circ$ and that the angle between lines ℓ and m is 45° . From here, note that the slope of line m is less than that of line ℓ as otherwise P'' wouldn't even be close to $(4, 1)$. Thus, line m is a 45° clockwise rotation of line ℓ . Line ℓ makes an angle of $\tan^{-1}(5)$ with the positive x axis. Thus, line m makes an angle of $\tan^{-1}(5) - 45^\circ$ with the positive x axis. Thus, the slope of line m is

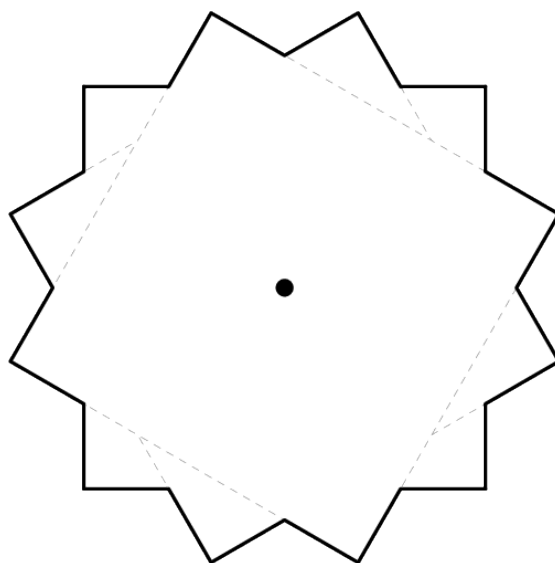
$$\tan(\tan^{-1}(5) - 45^\circ) = \frac{5 - 1}{1 + 5 \cdot 1} = \frac{2}{3},$$

by the tangent addition formula. Since the slope of line m is $\frac{2}{3}$, its equation is $y = \frac{2}{3}x \implies 2x - 3y = 0$, which is choice **D**.

Solution 5(cheese)

When we graph all the lines and points with a ruler, you can see that a slope of $\frac{5}{3}$ is too big while $\frac{1}{3}$ is too small. We also see that the slope cannot be negative, therefore the answer is **D**. agentdabber

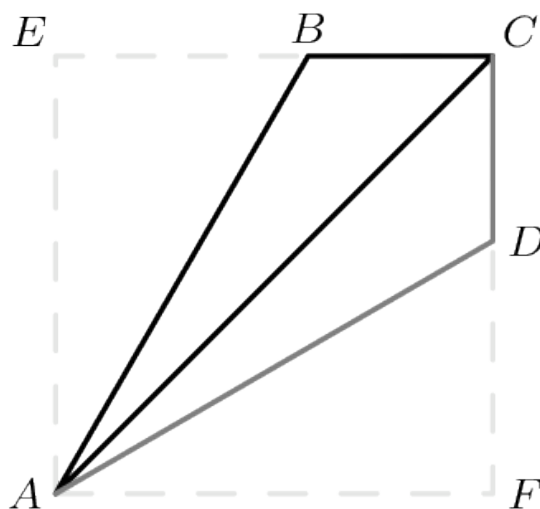
18. (*AMC 10 10B Fall 2021*) Three identical square sheets of paper each with side length 6 are stacked on top of each other. The middle sheet is rotated clockwise 30° about its center and the top sheet is rotated clockwise 60° about its center, resulting in the 24-sided polygon shown in the figure below. The area of this polygon can be expressed in the form $a - b\sqrt{c}$, where a , b , and c are positive integers, and c is not divisible by the square of any prime. What is $a + b + c$?



- (A) 75 (B) 93 (C) 96 (D) 129 (E) 147

Answer: E

Solution 1 (Trigonometry)

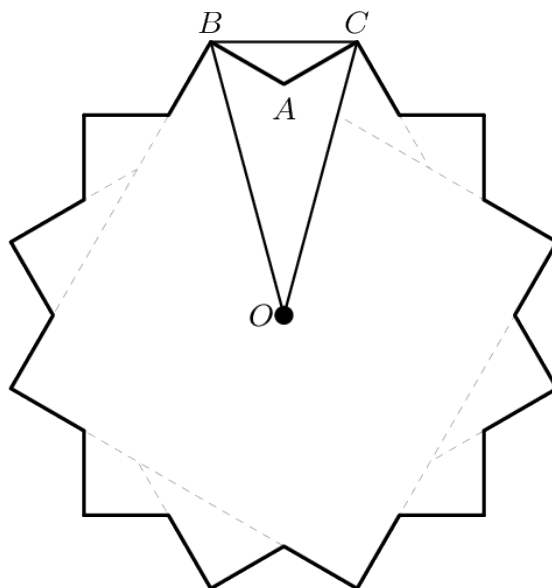


The 24-sided polygon is made out of 24 shapes like $\triangle ABC$. Then $\angle BAC = 360^\circ/24 = 15^\circ$, and $\angle EAC = 45^\circ$, so $\angle EAB = 30^\circ$. Then $EB = AE \tan 30^\circ = \sqrt{3}$; therefore $BC = EC - EB = 3 - \sqrt{3}$. Thus

$$[ABC] = \frac{BC}{EC} \cdot [ACE] = \frac{3 - \sqrt{3}}{3} \cdot \frac{9}{2}$$

and the required area is $24 \cdot [ABC] = 108 - 36\sqrt{3}$. Finally $108 + 36 + 3 = \boxed{\text{(E)} 147}$.

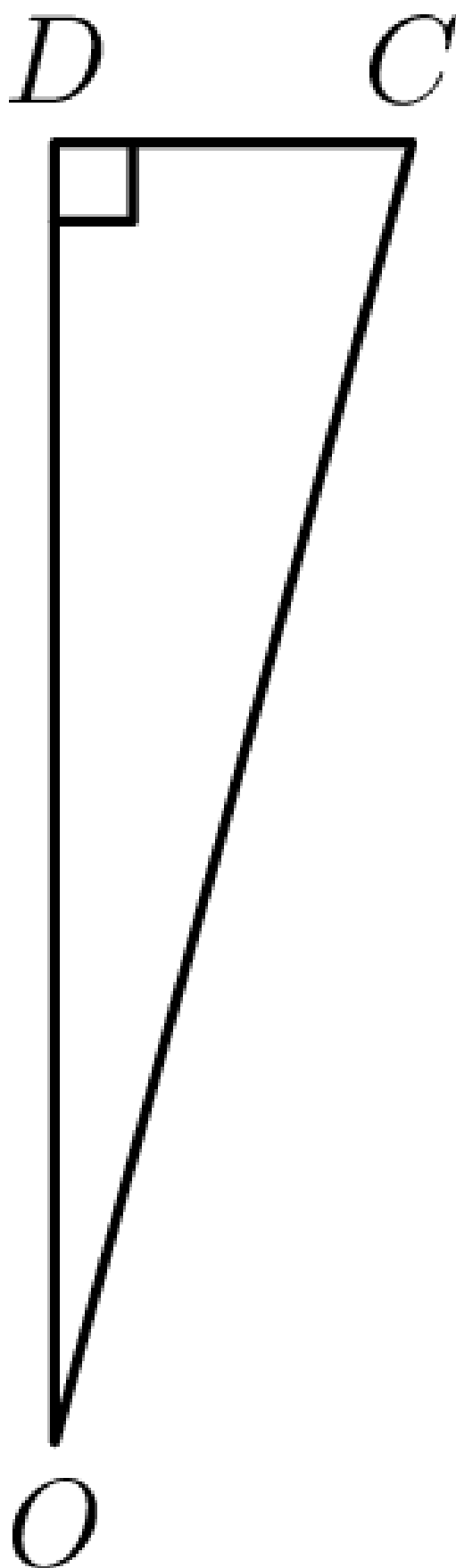
Solution 2 (15-75-90 triangles)



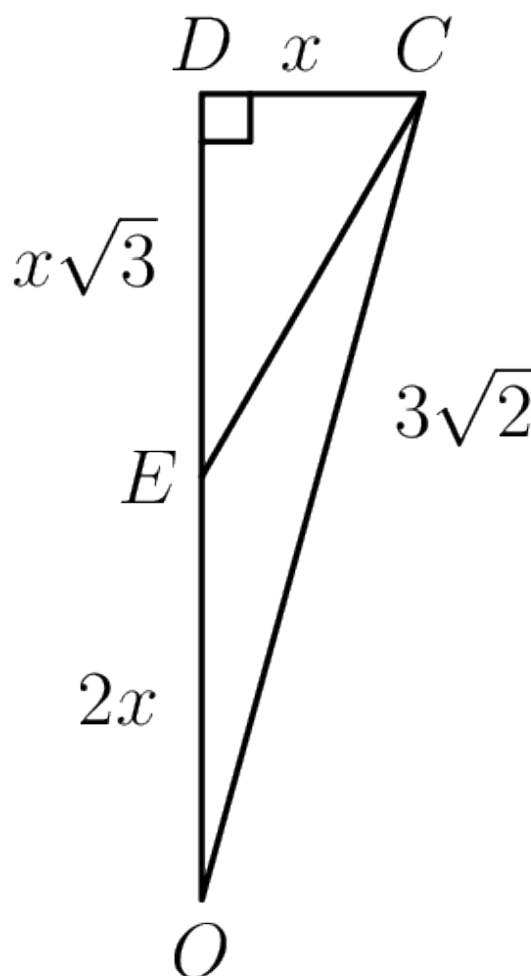
Break the 24-gon as shown so that there are 12 copies of quadrilateral $OBAC$. We can find the area of this quadrilateral by finding the area of triangle OBC and subtracting the area of triangle ABC .

The angle from O to one of the vertices of the original square is 45° , and this point rotates 30° , so the angle that OC makes with the horizontal is 75° .

Let D be the intersection of OA with BC . Since ABC and OBC are both isosceles, OD forms a right angle with BC . Furthermore, $\angle DOC = 90^\circ - 75^\circ = 15^\circ$.



Note that OC is equal to $3\sqrt{2}$ because it is half the diagonal of the square with side length 6. We can now split up the $15-75-90$ into $30-60-90$ and $15-15-150$ triangles to find the height and length of the triangle:



Now we have use Pythagorean: $x^2 + (x \cdot (2 + \sqrt{3}))^2 = 3\sqrt{2}^2 \implies x^2 + x^2 \cdot (7 + 4\sqrt{3}) = 18 \implies x^2 = \frac{18}{8 + 4\sqrt{3}} = \frac{9}{4 + 2\sqrt{3}} \implies x^2 = \frac{36 - 18\sqrt{3}}{4}$

Now note that x must be in the form of $\frac{a\sqrt{3}-b}{2}$. Therefore, $-2ab = -18 \implies ab = 9$. We can guess that $a = b = 3$. Sure enough, $3\sqrt{3}^2 + 3^2 = 36$, so $x = \frac{3\sqrt{3}-3}{2}$. Then the height of the triangle is $x \cdot (2 + \sqrt{3}) = \frac{3\sqrt{3}+3}{2}$.

The area of triangle OBC is equal to the length times the height of triangle ODC since this triangle has half the area of the full triangle. Therefore the area of triangle $OBC = \frac{3\sqrt{3}-3}{2} \cdot \frac{3\sqrt{3}+3}{2} = \frac{18}{4}$.

Now we look at triangle ABC . $\angle OCA = 45^\circ$ because OC is the diagonal of a square, and $\angle COA$ is 15° as we saw earlier. Therefore, $\angle OAC = \angle OAB = 120^\circ$. Because both these angles are 120 , $\angle BAC = 360 - 240 = 120$.

We can now split triangle ABC into two congruent $30-60-90$ triangles, and we know the base of each since $BC = 3\sqrt{3} - 3$.

The height of these triangles is $\frac{\frac{3\sqrt{3}-3}{2}}{\sqrt{3}} = \frac{3-\sqrt{3}}{2}$. Therefore the area of this triangle is $\frac{3\sqrt{3}-3}{2} \cdot \frac{3-\sqrt{3}}{2} = \frac{12\sqrt{3}-18}{4}$

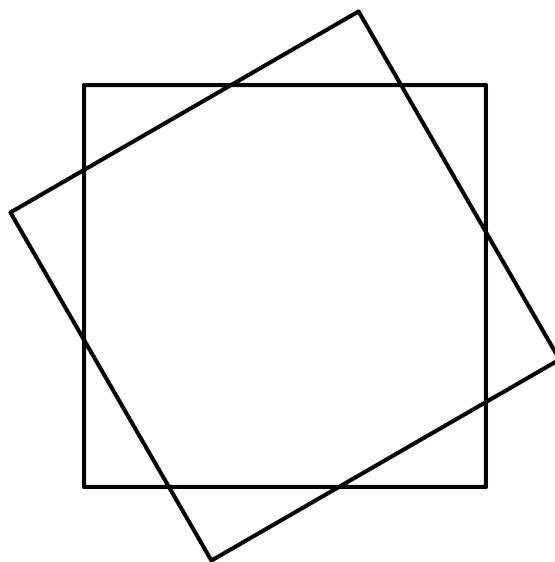
Therefore the total area of this segment is $\frac{18}{4} - (\frac{12\sqrt{3}-18}{4}) = \frac{36-12\sqrt{3}}{4} = 9 - 3\sqrt{3}$. Multiplying by 12 to find the entire area, the area of the figure is $108 - 36\sqrt{3}$ and the answer is $108 + 36 + 3 = 147 = \boxed{E}$

Solution 3

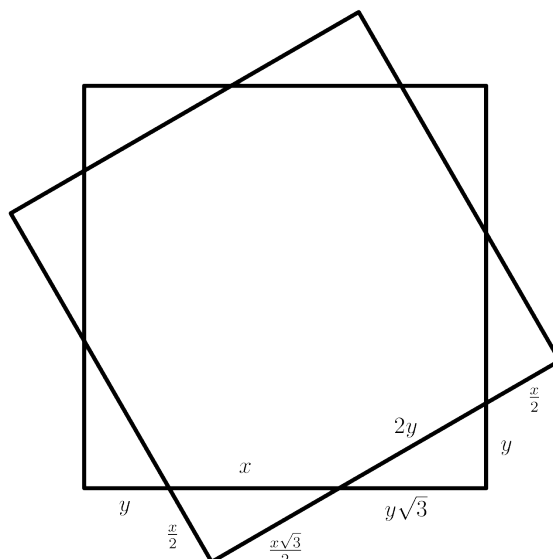
First note the useful fact that if R is the circumradius of a dodecagon (12-gon) the area of the figure is $3R^2$. If we connect the vertices of the 3 squares we get a dodecagon. The radius of circumcircle of the dodecagon is simply half the diagonal of the square, which is $3\sqrt{2}$. Thus the area of the dodecagon is $3 \cdot (3\sqrt{2})^2 = 3 \cdot 18 = 54$. But, the problem asked for the area of the combined figure which was made from the rotated squares. This area is the area of the dodecagon, which was found, subtracting the 12 isosceles triangles, which are formed when connecting the vertices of the squares to created the dodecagon. To find this area, we need to know the base of the isosceles triangle, call this x . Then, we can use the Law of Cosines on the triangle that is formed from the two vertices of the square and the center of the square. After computing, we get that $x = 3\sqrt{3} - 3$. Realize that the 12 isosceles are congruent with an angle measure of 120° , this means that we can create 4 congruent equilateral triangles with side length of $3\sqrt{3} - 3$. The area of the equilateral triangle is $\frac{\sqrt{3}}{4} \cdot (3\sqrt{3} - 3)^2 = \frac{\sqrt{3}}{4} \cdot (36 - 18\sqrt{3}) = \frac{36\sqrt{3}-54}{4}$. Thus, the area of all the twelve small equilateral triangles are $4 \cdot \frac{36\sqrt{3}-54}{4} = 36\sqrt{3} - 54$. Thus, the requested area is $54 - (36\sqrt{3} - 54) = 108 - 36\sqrt{3}$. Thus, $a + b + c = 108 + 36 + 3 = 147$, so the answer is $\boxed{(E)147}$.

Solution 4 (30-60-90 Triangles)

To make things simpler, let's take only the original sheet and the 30 degree rotated sheet. Then the diagram is this;



The area of this diagram is the original square plus the area of the four triangles that 'jut' out of the square. Because the square is rotated 30° , each triangle is a 30-60-90 triangle. Similarly, the triangles that are bounded on the inside of the original square outside of the rotated square are also congruent 30-60-90 triangles. Noting this, we can do some labelling:



Since the side lengths of the squares must be the same, and they are both 6, we have a system of equations;

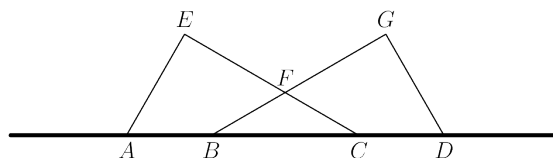
$$\begin{aligned} y + x + y\sqrt{3} &= 6 \\ \frac{x\sqrt{3}}{2} + 2y + \frac{x}{2} &= 6 \end{aligned}$$

We solve this to get $x = 6 - 2\sqrt{3}$ and $y = 3 - \sqrt{3}$.

The area of each triangle is $\frac{x}{2} \cdot \frac{x\sqrt{3}}{2} \cdot \frac{1}{2} = 6\sqrt{3} - 9$ by plugging in x .

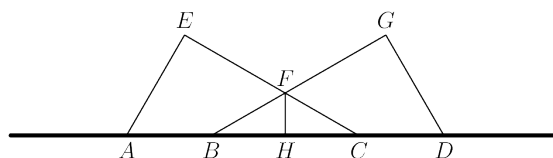
The rotated 60 degree square is the same thing as rotating it 30 degrees counterclockwise, so it's triangles that jut out of the square will be congruent to the triangles we have found, and therefore they will have the same area.

Unfortunately, when drawing all three squares, we see the two triangles overlap; take the very top for example.



The area of this shape is twice the area of each of the triangles that we have already found minus the area of the small triangle that is overlapped by the two by PIE. Now we only need to find the area of $\triangle BCF$.

$\angle GBD \cong \angle ECA \cong 30^\circ$ and by symmetry $\triangle BCF$ is isosceles, so it is a 30-30-120 triangle. If we draw a perpendicular, we split it into two 30-60-90 triangles;



By symmetry, the distance from A to the edge of the square is equal to the distance from D to the edge of the square is equal to y . $AC = BD = x$, and the side length of the square is 6, so we use PIE to obtain

$$x + x - BC = 6 - y - y \implies BC = 12 - 6\sqrt{3}$$

To find the height of $\triangle BFC$, we see that $HC = \frac{BC}{2} = 6 - 3\sqrt{3}$. Then by 30-60-90 triangles, $HF = \frac{HC}{\sqrt{3}} = 2\sqrt{3} - 3$. Finally, the area of $\triangle BFC = \frac{BC \cdot HF}{2} = 21\sqrt{3} - 36$.

Putting it all together, the area of the entire diagram is the area of the square plus four of these triangle-triangle intersections. The area of these intersections by PIE is $2 \cdot [ACE] - [BFC] = 12\sqrt{3} - 18 - (21\sqrt{3} - 36) = 18 - 9\sqrt{3}$. Therefore the total area is $36 + 4 \cdot (18 - 9\sqrt{3}) = 36 + 72 - 36\sqrt{3} = 108 - 36\sqrt{3}$.

Thus $a + b + c = 108 + 36 + 3 = 147 = \boxed{\text{(E)} \ 147}$.

Solution 5

As shown in [Image:2021_AMC_12B_\(Nov\)_Problem_15_sol.png](#), all 12 vertices of three squares form a regular dodecagon (12-gon). Denote by O the center of this dodecagon.

Hence, $\angle AOB = \frac{360^\circ}{12} = 30^\circ$.

Because the length of a side of a square is 6, $AO = 3\sqrt{2}$.

Hence, $AB = 2AO \sin \frac{\angle AOB}{2} = 3(\sqrt{3} - 1)$.

We notice that $\angle MAB = \angle MBA = 30^\circ$. Hence, $AM = \frac{AB}{2 \cos \angle MAB} = 3 - \sqrt{3}$.

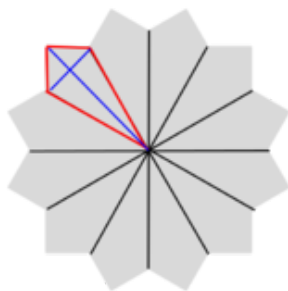
Therefore, the area of the region that three squares cover is

$$\begin{aligned} & [ABCDEFGH IJ KL] - 12[MAB] \\ &= 12[OAB] - 12[MAB] \\ &= 12 \cdot \frac{1}{2} OA \cdot OB \sin \angle AOB - 12 \cdot \frac{1}{2} MA \cdot MB \sin \angle AMB \\ &= 6OA^2 \sin \angle AOB - 6MA^2 \sin \angle AMB \\ &= 108 - 36\sqrt{3} \end{aligned}$$

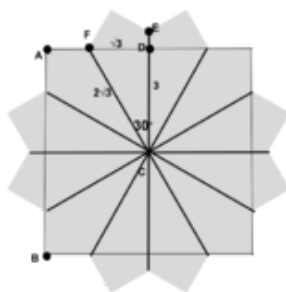
Therefore, the answer is $\boxed{\text{(E)} \ 147}$.

Note: If you have square brackets around a specified polygon (i.e. $[ABCD]$), it means the writer is referring to the area of that polygon.

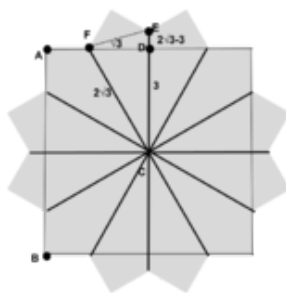
Solution 6



First, we can separate the shape into 12 congruent kites. The area of the figure can be determined by finding the area of one kite and multiplying it by 12. In order to get the area of one kite, we need to find its diagonals, shown in blue.



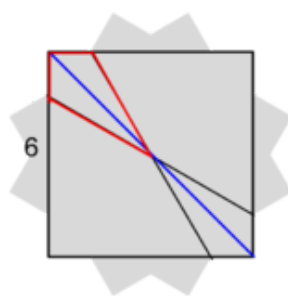
We notice that angle FCE is $\frac{360^\circ}{12} = 30^\circ$. Also, we know that CD is half of AB, so it has a length of 3. Now, we can find the lengths of FC and FD using the 30-60-90 triangle. We find that FC is $2\sqrt{3}$ and FD is $\sqrt{3}$. Since FC is congruent to CE, CE is also $2\sqrt{3}$. Using this information, we can conclude that ED is $2\sqrt{3} - 3$.



Now, we can find the shorter diagonal by using the Pythagorean theorem:

$$FC^2 = \sqrt{3}^2 + (2\sqrt{3} - 3)^2$$

$$FC = \sqrt{24 - 12\sqrt{3}}$$



We can find the longer diagonal of the kite by looking at one of the square sheets of paper. We know that the side of the square has a length of 6, so the diagonal of the square must be $6\sqrt{2}$. The longer diagonal of the kite is half of this length, so it has a length of $3\sqrt{2}$.

The area of the entire figure is

$$\begin{aligned} &= 12 \cdot \frac{d_1 \cdot d_2}{2} \\ &= 12 \cdot \frac{3\sqrt{2} \cdot \sqrt{24 - 12\sqrt{3}}}{2} \\ &= 12 \cdot \frac{6\sqrt{12 - 6\sqrt{3}}}{2} \\ &= 36\sqrt{12 - 6\sqrt{3}} \end{aligned}$$

Now we can use algebra to make our answer look a little nicer.

$$\begin{aligned} a - \sqrt{b} &= \sqrt{12 - 6\sqrt{3}} \\ (a - \sqrt{b})^2 &= (\sqrt{12 - 6\sqrt{3}})^2 \\ a^2 - 2a\sqrt{b} + b &= 12 - 6\sqrt{3} \end{aligned}$$

$$\begin{aligned} a^2 + b &= 12 \\ 2a\sqrt{b} &= 6\sqrt{3} \\ a = 3, b &= 3 \end{aligned}$$

$$a - \sqrt{b} = \sqrt{12 - 6\sqrt{3}} = 3 - \sqrt{3}$$

The area of the entire region is $36(3 - \sqrt{3})$, or $108 - 36\sqrt{3}$.

Therefore, $a + b + c = 108 + 36 + 3 = 147 = \boxed{E}$.

Solution 7

Let O be the center of the polygon, A be the bottom right corner of the first square, C be the next vertex to the left of A , and M be the midpoint between A and B , where B is the bottom left corner of the first square. Note that because there are

three 90° squares separated by $\frac{90^\circ}{3} = 30^\circ$, each side of the 24-sided polygon is equal in length, meaning to calculate the area of the whole polygon, we find the area of $\triangle OAC$ and multiply by 24.

To find $[\triangle OAC]$, we already know the height $\overline{OM}$ is the sidelength of the square over 2, or $\frac{6}{2} = 3$, so we just need the length of the base $\overline{AC}$. Notice that $\triangle OCM$ is a $30 - 60 - 90$ triangle since $\angle COM = \frac{360^\circ}{12} = 30^\circ$, so $\overline{CM} = \frac{\overline{OM}}{\sqrt{3}} = \frac{3}{\sqrt{3}} = \sqrt{3}$. Then $\overline{AC} = \overline{AM} - \overline{CM} = \frac{6}{2} - \sqrt{3} = 3 - \sqrt{3}$, so

$$\begin{aligned} [\triangle AOC] &= \frac{1}{2} \cdot \overline{OM} \cdot \overline{AC} \\ &= \frac{1}{2}(3)(3 - \sqrt{3}) \\ &= \frac{9 - 3\sqrt{3}}{2} \end{aligned}$$

Then the whole area of the polygon is $\frac{9-3\sqrt{3}}{2} \cdot 24 = 108 - 36\sqrt{3}$. The desired solution is then $108 + 36 + 3 = 147$, so the answer is **(E) 147**.

Solution 8

Using the diagram from Solution 2, we observe two isosceles triangles. The larger triangle has angles of 75° , 75° , and 30° .

To find the angle measures of the smaller triangle BCO , note that the congruent sides of $\triangle BCO$ are each half of the diagonals of two squares. Since a diagonal of a square bisects the right angle (45°), we have $75 - 45 = 30^\circ$ for each base angle, making the remaining vertex angle 120° .

Because both triangles are isosceles, we have $BO = CO$ and $AB = AC$.

Using the formula for the area of a square, $A = \frac{d^2}{2}$, we find

$$72 = \frac{d^2}{2} \Rightarrow d = 6\sqrt{2},$$

so $BO = 3\sqrt{2}$.

Let $[BCO]$ denote the area of $\triangle BCO$. Using the area formula $\frac{1}{2}ab \sin C$, we find

$$[BCO] = \frac{1}{2}(3\sqrt{2})(3\sqrt{2}) \sin(30^\circ) = \frac{1}{2}(18) \left(\frac{1}{2}\right) = \frac{9}{2}.$$

Next, to find the side lengths of $\triangle ABC$, we apply the Law of Cosines to $\triangle BCO$:

$$BC^2 = (3\sqrt{2})^2 + (3\sqrt{2})^2 - 2(3\sqrt{2})(3\sqrt{2}) \cos(120^\circ) = 36(1 - \frac{\sqrt{3}}{2}) = 18(2 - \sqrt{3}).$$

Now, let $AB = x$. Applying the Law of Cosines to $\triangle ABC$:

$$2x^2 - 2x^2 \cos(120^\circ) = 18(2 - \sqrt{3}).$$

Simplifying,

$$3x^2 = 18(2 - \sqrt{3}) \Rightarrow x^2 = 6(2 - \sqrt{3}).$$

We can now find the area of $\triangle ABC$ using the same area formula:

$$[ABC] = \frac{1}{2}x^2 \sin(120^\circ) = \frac{1}{2}(6(2 - \sqrt{3})) \left(\frac{\sqrt{3}}{2}\right) = 3\sqrt{3} - \frac{9}{2}.$$

Subtracting the two areas gives the area of the quadrilateral $OBAC$:

$$[OBAC] = 9 - 3\sqrt{3}.$$

Since there are 12 such regions in the figure, the total area is

$$12(9 - 3\sqrt{3}) = 108 - 36\sqrt{3}.$$

The problem asks for $a + b + c$, so

$$108 + 36 + 3 = \boxed{\text{(E)} 147}$$

19. (AMC 10 10B Fall 2021) Let N be the positive integer $7777 \dots 777$, a 313-digit number where each digit is a 7. Let $f(r)$ be the leading digit of the r th root of N . What is

$$f(2) + f(3) + f(4) + f(5) + f(6)?$$

- (A) 8 (B) 9 (C) 11 (D) 22 (E) 29

Answer: A

Solution 1

We can rewrite N as $\frac{7}{9} \cdot 9999 \dots 999 = \frac{7}{9} \cdot (10^{313} - 1)$. When approximating values, as we will shortly do, the minus one will become negligible so we can ignore it. When we take the power of ten out of the square root, we'll be multiplying by another power of ten, so the leading digit will not change. Thus the leading digit of $f(r)$ will be equal to the leading digit of $\sqrt[r]{\frac{7}{9} \cdot 10^{313(\bmod r)}}$.

Then $f(2)$ is the first digit of $\sqrt{\frac{7}{9} \cdot (10)} = \sqrt{\frac{70}{9}} = \sqrt{7.7\ldots} \approx 2$ $f(3) = \sqrt[3]{\frac{7}{9} \cdot 10} = \sqrt[3]{\frac{70}{9}} = \sqrt[3]{7.7\ldots} \approx 1$.

$$f(4) = \sqrt[4]{\frac{7}{9} \cdot 10} = \sqrt[4]{\frac{70}{9}} = \sqrt[4]{7.7\ldots} \approx 1.$$

$$f(5) = \sqrt[5]{\frac{7}{9} \cdot 1000} = \sqrt[5]{\frac{7000}{9}} = \sqrt[5]{777.7\ldots} \approx 3.$$

$$f(6) = \sqrt[6]{\frac{7}{9} \cdot 10} = \sqrt[6]{\frac{70}{9}} = \sqrt[6]{7.7\ldots} \approx 1.$$

The final answer is therefore $2 + 1 + 1 + 3 + 1 = \boxed{\text{(A)} 8}$.

Note: in all of the divisions, we omitted the decimal places, because they are irrelevant to finding the leading digit.

Solution 2

For notation purposes, let x be the number $777 \dots 777$ with 313 digits, and let $B(n)$ be the leading digit of n . As an example, $B(x) = 7$, because $x = 777 \dots 777$, and the first digit of that is 7.

Notice that

$$B\left(\sqrt{\frac{n}{100}}\right) = B(\sqrt{n})$$

for all numbers $n \geq 100$; this is because $\sqrt{\frac{n}{100}} = \frac{\sqrt{n}}{10}$, and dividing by 10 does not affect the leading digit of a number. Similarly,

$$B\left(\sqrt[3]{\frac{n}{1000}}\right) = B(\sqrt[3]{n}).$$

In general, for positive integers k and real numbers $n > 10^k$, it is true that

$$B\left(\sqrt[k]{\frac{n}{10^k}}\right) = B(\sqrt[k]{n}).$$

Behind all this complex notation, all that we're really saying is that the first digit of something like $\sqrt[3]{123456789}$ has the same first digit as $\sqrt[3]{123456.789}$ and $\sqrt[3]{123.456789}$.

The problem asks for

$$B(\sqrt[2]{x}) + B(\sqrt[3]{x}) + B(\sqrt[4]{x}) + B(\sqrt[5]{x}) + B(\sqrt[6]{x}).$$

From our previous observation, we know that

$$B(\sqrt[2]{x}) = B(\sqrt[2]{\frac{x}{100}}) = B(\sqrt[2]{\frac{x}{10,000}}) = B(\sqrt[2]{\frac{x}{1,000,000}}) = \dots$$

Therefore, $B(\sqrt[2]{x}) = B(\sqrt[2]{7.777\dots})$. We can evaluate $B(\sqrt[2]{7.777\dots})$, the leading digit of $\sqrt[2]{7.777\dots}$, to be 2. Therefore, $f(2) = 2$.

Similarly, we have

$$B(\sqrt[3]{x}) = B(\sqrt[3]{\frac{x}{1,000}}) = B(\sqrt[3]{\frac{x}{1,000,000}}) = B(\sqrt[3]{\frac{x}{1,000,000,000}}) = \dots$$

Therefore, $B(\sqrt[3]{x}) = B(\sqrt[3]{7.777\dots})$. We know $B(\sqrt[3]{7.777\dots}) = 1$, so $f(3) = 1$.

Next,

$$B(\sqrt[4]{x}) = B(\sqrt[4]{7.777\dots})$$

and $B(\sqrt[4]{7.777\dots}) = 1$, so $f(4) = 1$.

We also have

$$B(\sqrt[5]{x}) = B(\sqrt[5]{777.777\dots})$$

and $B(\sqrt[5]{777.777\dots}) = 3$, so $f(5) = 3$.

Finally,

$$B(\sqrt[6]{x}) = B(\sqrt[6]{7.777\dots})$$

and $B(\sqrt[6]{7.777\dots}) = 1$, so $f(6) = 1$.

We have that $f(2) + f(3) + f(4) + f(5) + f(6) = 2 + 1 + 1 + 3 + 1 = \boxed{\text{(A)} 8}$.

Solution 3 (Condensed Solution 1)

Since 7777.7 is a 313 digit number and $\sqrt{7}$ is around 2.5, we have $f(2)$ is 2. $f(3)$ is the same story, so $f(3)$ is 1. It is the same as $f(4)$ as well, so $f(4)$ is also 1. However, 313 is $3 \pmod{5}$, so we need to take the 5th root of 777, which is between 3 and 4, and therefore, $f(5)$ is 3. $f(6)$ is the same as $f(4)$, since it is 1 more than a multiple of 6. Therefore, we have $2 + 1 + 1 + 3 + 1$ which is $\boxed{\text{(A)} 8}$.

Solution 4

First, we compute $f(2)$.

Because $N > 4 \cdot 10^{312}$, $\sqrt{N} > 2 \cdot 10^{156}$. Because $N < 9 \cdot 10^{312}$, $\sqrt{N} < 3 \cdot 10^{156}$.

Therefore, $f(2) = 2$.

Second, we compute $f(3)$.

Because $N > 1 \cdot 10^{312}$, $\sqrt[3]{N} > 1 \cdot 10^{104}$. Because $N < 8 \cdot 10^{312}$, $\sqrt[3]{N} < 2 \cdot 10^{104}$.

Therefore, $f(3) = 1$.

Third, we compute $f(4)$.

Because $N > 1 \cdot 10^{312}$, $\sqrt[4]{N} > 1 \cdot 10^{78}$. Because $N < 16 \cdot 10^{312}$, $\sqrt[4]{N} < 2 \cdot 10^{78}$.

Therefore, $f(4) = 1$.

Fourth, we compute $f(5)$.

Because $N > 3^5 \cdot 10^{310}$, $\sqrt[5]{N} > 3 \cdot 10^{62}$. Because $N < 4^5 \cdot 10^{310}$, $\sqrt[5]{N} < 4 \cdot 10^{62}$.

Therefore, $f(5) = 3$.

Fifth, we compute $f(6)$.

Because $N > 1 \cdot 10^{312}$, $\sqrt[6]{N} > 1 \cdot 10^{52}$. Because $N < 2^6 \cdot 10^{312}$, $\sqrt[6]{N} < 2 \cdot 10^{52}$.

Therefore, $f(6) = 1$.

Therefore,

$$\begin{aligned} f(2) + f(3) + f(4) + f(5) + f(6) &= 2 + 1 + 1 + 3 + 1 \\ &= 8. \end{aligned}$$

Therefore, the answer is (A) 8.

Solution 5 (Guessing)

Benford's Law states that in random numbers, the leading digit is more likely to be 1, 2, or 3 rather than 8 or 9. From here, we can eliminate C, D, E. It is better to guess between A and B than not guess at all since your expected score from doing this is 3 points.

Solution 6 (Intuition)

To begin with, you can test some numbers and find a pattern. For $f(2)$, we get the leading digit of $\sqrt{7}$ is 2, and the leading digit of $\sqrt{77}$ is 8. With each 7 afterward, it repeats in a loop: 2 for 7, 8 for 77, 2 for 777, 8 for 777, and so on. Since the problem states that N is a 313-digit number, we get the leading digit as 2 for $f(2)$. For $f(3)$, it repeats in a loop of 3 (3rd root means it repeats every 3 times). In this case, you would get the leading digit for the 3rd root of 7 to be 1, 77 it will be 4, and for 777 it will be 9. Since N is 1 more than a multiple of 3, the leading digit for $f(3)$ is 1. Likewise, we can find the leading digits for $f(4)$ to be 1, $f(5)$ to be 3, and $f(6)$ to be 1:

For $f(2)$: 7 : 2, 77 : 8...

For $f(3)$: 7 : 1, 77 : 4, 777 : 9...

For $f(4)$: 7 : 1, 77 : 2, 777 : 5, 7777 : 9...

For $f(5)$: 7 : 1, 77 : 2, 777 : 3, 7777 : 6, 77777 : 9...

For $f(6)$: 7 : 1, 77 : 2, 777 : 3, 7777 : 4, 77777 : 6, 777777 : 9...

Remember that it loops every r times; the boxed numbers are where N falls. Thus, the desired sum is $2 + 1 + 1 + 3 + 1 =$ (A) 8

Solution 7 (Logs)

Let's express N as a geometric series:

$$7 + 70 + 700 + \dots + 7 \cdot 10^{313} = N$$

We can now apply the geometric series formula, thus:

$$N = 7 \cdot \frac{10^{313} - 1}{9}$$

Since we are dealing with large values, we can discard -1 , as the value will virtually remain the same. Thus,

$$N = 7 \cdot \frac{10^{313}}{9}$$

Next, we must take the base 10 logarithm of N , as we are trying to express the r th roots of N in the form $a \cdot 10^b$, thus

$$\begin{aligned}\log \frac{7}{9} \cdot 10^{313} &= \\ 313 + \log \frac{7}{9} & \\ \approx 312.9 &\end{aligned}$$

So now to find the leading digits of the r th roots of N , simply take $\log \sqrt[r]{N} = \frac{\log N}{r} = \frac{312.9}{r}$. This calculates the power of 10 when $f(r)$ is taken in scientific notation ($a \cdot 10^b$). a here in the notation corresponds with the fractional part of $\frac{312.9}{r}$ raised to the 10th power. We must find a as it indicates the leading digit. Thus, for $f(2)$,

$$\begin{aligned}\frac{312.9}{2} &= 156.45 \\ a &= 10^{0.45}\end{aligned}$$

Since $2 < a < 3$, $f(2) = 2$ For $f(3)$,

$$\begin{aligned}\frac{312.9}{3} &= 104.3 \\ a &= 10^{0.3}\end{aligned}$$

Since $1 < a < 2$, $f(3) = 1$ For $f(4)$,

$$\begin{aligned}\frac{312.9}{4} &= 78.225 \\ a &= 10^{0.225}\end{aligned}$$

Since $1 < a < 2$, $f(4) = 1$ For $f(5)$,

$$\begin{aligned}\frac{312.9}{5} &= 62.58 \\ a &= 10^{0.58}\end{aligned}$$

Since $3 < a < 4$, $f(5) = 3$ For $f(6)$,

$$\begin{aligned}\frac{312.9}{6} &= 52.15 \\ a &= 10^{0.15}\end{aligned}$$

,Thus, $f(6) = 1$ So, $2 + 1 + 1 + 3 + 1 = \boxed{8}$ (Note that this is very bashy, would not recommend on real test, as logs basically require a calculator)

20. (AMC 10 10B Fall 2021) In a particular game, each of 4 players rolls a standard 6-sided die. The winner is the player who rolls the highest number. If there is a tie for the highest roll, those involved in the tie will roll again and this process will continue until one player wins. Hugo is one of the players in this game. What is the probability that Hugo's first roll was a 5, given that he won the game?

(A) $\frac{61}{216}$ (B) $\frac{367}{1296}$ (C) $\frac{41}{144}$ (D) $\frac{185}{648}$ (E) $\frac{11}{36}$

Answer: C

Solution 1

Define the events H , A , and B as follows:

H - Hugo wins.

A - Hugo's first roll is a 5.

B - The highest number any player rolls on their first turn is 5.

The desired probability is $P(A|H)$. In all games Hugo wins, the number Hugo rolls on his first turn must equal the highest number any player rolls on their first turn, so $P(A|H) = P(B|H)$. By symmetry, B and H are independent events and $P(B|H) = P(B)$.

Define the random variable X as the highest number any player rolls on their first turn.

$$\text{Then } P(A|H) = P(B) = P(X = 5) = P(X \leq 5) - P(X \leq 4) = \left(\frac{5}{6}\right)^4 - \left(\frac{4}{6}\right)^4 = \frac{5^4 - 4^4}{6^4} = \frac{369}{1296} = \boxed{(C) \frac{41}{144}}.$$

Solution 2 (Conditional Probability)

The conditional probability formula states that $P(A|B) = \frac{P(A \cap B)}{P(B)}$, where $A|B$ means A given B and $A \cap B$ means A and B. Therefore the probability that Hugo rolls a five given he won is $\frac{P(A \cap B)}{P(B)}$, where A is the probability that he rolls a five and B is the probability that he wins. In written form,

$$P(\text{Hugo rolled a 5 given he won}) = \frac{P(\text{Hugo rolls a 5 and wins})}{P(\text{Hugo wins})}.$$

The probability that Hugo wins is $\frac{1}{4}$ by symmetry since there are four people playing and there is no bias for any one player. The probability that he gets a 5 and wins is more difficult; we will have to consider cases on how many players tie with Hugo. We consider the cases in which Hugo wins first, find the probabilities of those, and then account for the probability that Hugo rolled a 5, at the end.

Case 1: No Players Tie With Hugo

In this case, all other players must have numbers from 1 through four, and Hugo had a probability of 1 to win after the first round.

There is a $\left(\frac{4}{6}\right)^3 \cdot 1 = \frac{8}{27}$ chance of this happening.

Case 2: One Player Ties With Hugo

In this case, there are $\binom{3}{1} = 3$ ways to choose which other player ties with Hugo, and the probability that this happens is $\frac{1}{6} \cdot \left(\frac{4}{6}\right)^2$. The probability that Hugo wins the game is then $\frac{1}{2}$ because there are now two players rolling the die.

Therefore the total probability in this case is $3 \cdot \frac{1}{2} \cdot \frac{1}{6} \cdot \left(\frac{4}{6}\right)^2 = \frac{1}{9}$.

Case 3: Two Players Tie With Hugo

In this case, there are $\binom{3}{2} = 3$ ways to choose which other players tie with Hugo, and the probability that this happens is $\left(\frac{1}{6}\right)^2 \cdot \frac{4}{6}$. The probability that Hugo wins the game is then $\frac{1}{3}$ because there are now three players rolling the die.

Therefore the total probability in this case is $3 \cdot \frac{1}{3} \cdot \left(\frac{1}{6}\right)^2 \cdot \frac{4}{6} = \frac{1}{54}$.

Case 4: All Three Players Tie With Hugo

In this case, the probability that all three players tie with Hugo is $\left(\frac{1}{6}\right)^3$. The probability that Hugo wins the game is $\frac{1}{4}$, so the total probability is $\frac{1}{4} \cdot \left(\frac{1}{6}\right)^3 = \frac{1}{864}$. Hugo has a $\frac{1}{6}$ probability of rolling a five himself, so the total probability that Hugo rolls a 5 and wins is

$$\frac{1}{6} \left(\frac{8}{27} + \frac{1}{9} + \frac{1}{54} + \frac{1}{864} \right) = \frac{1}{6} \left(\frac{369}{864} \right) = \frac{1}{6} \left(\frac{41}{96} \right).$$

Finally, the total probability is this probability divided by $\frac{1}{4}$ which is this probability times four; the final answer is

$$4 \cdot \frac{1}{6} \left(\frac{41}{96} \right) = \frac{2}{3} \cdot \frac{41}{96} = \frac{41}{48 \cdot 3} = \frac{41}{144} = \boxed{C}.$$

Solution 3

We use H to refer to Hugo. We use H_1 to denote the outcome of Hugo's t th toss. We denote by A, B, C the other three players. We denote by N the number of players among A, B, C whose first tosses are 5. We use W to denote the winner.

We have

$$\begin{aligned} P(H_1 = 5 | W = H) &= \frac{P(H_1 = 5, W = H)}{P(W = H)} \\ &= \frac{P(H_1 = 5) P(W = H | H_1 = 5)}{P(W = H)} \\ &= \frac{\frac{1}{6} P(W = H | H_1 = 5)}{\frac{1}{4}} \\ &= \frac{2}{3} P(W = H | H_1 = 5). \end{aligned}$$

Now, we compute $P(W = H | H_1 = 5)$.

We have

$$\begin{aligned} P(W = H | H_1 = 5) &= P(W = H | H_1 = 5, \max\{A_1, B_1, C_1\} \leq 4) P(\max\{A_1, B_1, C_1\} \leq 4 | H_1 = 5) \\ &\quad + P(W = H | H_1 = 5, \max\{A_1, B_1, C_1\} = 6) P(\max\{A_1, B_1, C_1\} = 6 | H_1 = 5) \\ &\quad + \sum_{N=1}^3 P(W = H | H_1 = 5, \max\{A_1, B_1, C_1\} = 5, N) P(\max\{A_1, B_1, C_1\} = 5, N | H_1 = 5) \\ &= P(W = H | H_1 = 5, \max\{A_1, B_1, C_1\} \leq 4) P(\max\{A_1, B_1, C_1\} \leq 4) \\ &\quad + P(W = H | H_1 = 5, \max\{A_1, B_1, C_1\} = 6) P(\max\{A_1, B_1, C_1\} = 6) \\ &\quad + \sum_{N=1}^3 P(W = H | H_1 = 5, \max\{A_1, B_1, C_1\} = 5, N) P(\max\{A_1, B_1, C_1\} = 5, N) \\ &= 1 \cdot P(\max\{A_1, B_1, C_1\} \leq 4) + 0 \cdot P(\max\{A_1, B_1, C_1\} = 6) \\ &\quad + \sum_{N=1}^3 P(W = H | H_1 = 5, \max\{A_1, B_1, C_1\} = 5, N) P(\max\{A_1, B_1, C_1\} = 5, N) \end{aligned}$$

$$\begin{aligned}
 &= P(\max\{A_1, B_1, C_1\} \leq 4) \\
 &\quad + \sum_{N=1}^3 P(W = H | H_1 = 5, \max\{A_1, B_1, C_1\} = 5, N) P(\max\{A_1, B_1, C_1\} = 5, N) \\
 &= \left(\frac{4}{6}\right)^3 + \sum_{N=1}^3 \frac{1}{N+1} \cdot \binom{3}{N} \left(\frac{1}{6}\right)^N \left(\frac{4}{6}\right)^{3-N} \\
 &= \frac{41}{96}.
 \end{aligned}$$

The first equality follows from the law of total probability. The second equality follows from the property that Hugo's outcome is independent from other players' outcomes.

Therefore,

$$\begin{aligned}
 P(H_1 = 5 | W = H) &= \frac{2}{3} P(W = H | H_1 = 5) \\
 &= \frac{2}{3} \frac{41}{96} \\
 &= \frac{41}{144},
 \end{aligned}$$

so the answer is (C) $\frac{41}{144}$.

21. (AMC 10 10B Fall 2021) Regular polygons with 5, 6, 7, and 8 sides are inscribed in the same circle. No two of the polygons share a vertex, and no three of their sides intersect at a common point. At how many points inside the circle do two of their sides intersect?
- (A) 52 (B) 56 (C) 60 (D) 64 (E) 68

Answer: E

Solution

Imagine we have 2 regular polygons with m and n sides and $m > n$ inscribed in a circle without sharing a vertex. We see that each side of the polygon with n sides (the polygon with fewer sides) will be intersected twice. (We can see this because to have a vertex of the m -gon on an arc subtended by a side of the n -gon, there will be one intersection to "enter" the arc and one to "exit" the arc. KingRavi)

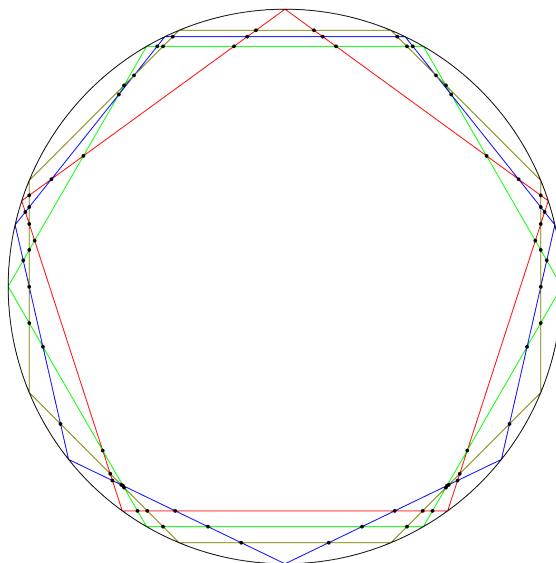
This means that we will end up with 2 times the number of sides in the polygon with fewer sides.

If we have polygons with 5, 6, 7, and 8 sides, we need to consider each possible pair of polygons and count their intersections.

Throughout 6 of these pairs, the 5-sided polygon has the least number of sides 3 times, the 6-sided polygon has the least number of sides 2 times, and the 7-sided polygon has the least number of sides 1 time.

Therefore the number of intersections is $2 \cdot (3 \cdot 5 + 2 \cdot 6 + 1 \cdot 7) = \boxed{\text{(E) } 68}$.

Remark For regular polygons with 5, 6, 7, and 8 sides, the 68 points of intersection inside the circle are shown below:



MRENTHUSIASM

22. (AMC 10 10B Fall 2021) For each integer $n \geq 2$, let S_n be the sum of all products jk , where j and k are integers and $1 \leq j < k \leq n$. What is the sum of the 10 least values of n such that S_n is divisible by 3?
- (A) 196 (B) 197 (C) 198 (D) 199 (E) 200

Answer: B

Solution 1

To get from S_n to S_{n+1} , we add $1(n+1) + 2(n+1) + \cdots + n(n+1) = (1+2+\cdots+n)(n+1) = \frac{n(n+1)^2}{2}$.

Now, we can look at the different values of $n \pmod 3$. For $n \equiv 0 \pmod 3$ and $n \equiv 2 \pmod 3$, then we have $\frac{n(n+1)^2}{2} \equiv 0 \pmod 3$. However, for $n \equiv 1 \pmod 3$, we have

$$\frac{1 \cdot 2^2}{2} \equiv 2 \pmod 3.$$

Clearly, $S_2 \equiv 2 \pmod 3$. Using the above result, we have $S_5 \equiv 1 \pmod 3$, and S_8, S_9 , and S_{10} are all divisible by 3. After $3 \cdot 3 = 9$, we have S_{17}, S_{18} , and S_{19} all divisible by 3, as well as S_{26}, S_{27}, S_{28} , and S_{35} . Thus, our answer is $8+9+10+17+18+19+26+27+28+35 = 27+54+81+35 = 162+35 = \boxed{\text{(B) } 197}$.

-BorealBear

-minor edit by Yiyj1

Solution 2 (bash)

Since we have a wonky function, we start by trying out some small cases and see what happens. If j is 1 and k is 2, then there is one case. We have $2 \pmod 3$ for this case. If N is 3, we have $1 \cdot 2 + 1 \cdot 3 + 2 \cdot 3$ which is still $2 \pmod 3$. If N is 4, we have to add $1 \cdot 4 + 2 \cdot 4 + 3 \cdot 4$ which is a multiple of 3, meaning that we are still at $2 \pmod 3$. If we try a few more cases, we find that when N is 8, we get a multiple of 3. When N is 9, we are adding $0 \pmod 3$, and therefore, we are still at a multiple of 3.

When N is 10, then we get $0 \pmod 3 + 10(1+2+3+\dots+9)$ which is 10 times a multiple of 3. Therefore, we have another multiple of 3. When N is 11, so we have

2 mod 3. So, every time we have $-1 \bmod 9$, $0 \bmod 9$, and $1 \bmod 9$, we always have a multiple of 3. Think about it: When N is 1, it will have to be $0 \cdot 1$, so it is a multiple of 3. Therefore, our numbers are 8, 9, 10, 17, 18, 19, 26, 27, 28, 35. Adding the numbers up, we get **(B) 197**

Solution 3

Denote $A_{n,<} = \{(j, k) : 1 \leq j < k \leq n\}$, $A_{n,>} = \{(j, k) : 1 \leq k < j \leq n\}$ and $A_{n,=} = \{(j, k) : 1 \leq j = k \leq n\}$.

Hence, $\sum_{(j,k) \in A_{n,<}} jk = \sum_{(j,k) \in A_{n,>}} jk = S_n$.

Therefore,

$$\begin{aligned} S_n &= \frac{1}{2} \left(\sum_{(j,k) \in A_{n,<}} jk = \sum_{(j,k) \in A_{n,>}} jk \right) \\ &= \frac{1}{2} \left(\sum_{1 \leq j, k \leq n} jk - \sum_{(j,k) \in A_{n,=}} jk \right) \\ &= \frac{1}{2} \left(\sum_{j=1}^n \sum_{k=1}^n jk - \sum_{j=1}^n j^2 \right) \\ &= \frac{1}{2} \left(\frac{n^2(n+1)^2}{4} - \frac{n(n+1)(2n+1)}{6} \right) \\ &= \frac{(n-1)n(n+1)(3n+2)}{24}. \end{aligned}$$

Hence, S_n is divisible by 3 if and only if $(n-1)n(n+1)(3n+2)$ is divisible by $24 \cdot 3 = 8 \cdot 9$.

First, $(n-1)n(n+1)(3n+2)$ is always divisible by 8. Otherwise, S_n is not even an integer.

Second, we find conditions for n , such that $(n-1)n(n+1)(3n+2)$ is divisible by 9.

Because $3n+2$ is not divisible by 3, it cannot be divisible by 9.

Hence, we need to find conditions for n , such that $(n-1)n(n+1)$ is divisible by 9. This holds of $n \equiv 0, \pm 1 \pmod{9}$.

Therefore, the 10 least values of n such that $(n-1)n(n+1)$ is divisible by 9 (equivalently, S_n is divisible by 3) are 8, 9, 10, 17, 18, 19, 26, 27, 28, 35. Their sum is 197.

Therefore, the answer is **(B) 197**.

23. (AMC 10 10B Fall 2021) Each of the 5 sides and the 5 diagonals of a regular pentagon are randomly and independently colored red or blue with equal probability. What is the probability that there will be a triangle whose vertices are among the vertices of the pentagon such that all of its sides have the same color?

(A) $\frac{2}{3}$ (B) $\frac{105}{128}$ (C) $\frac{125}{128}$ (D) $\frac{253}{256}$ (E) 1

Answer: D

Solution 1

Instead of finding the probability of a same-colored triangle appearing, let us find the probability that one does not appear. After drawing the regular pentagon out,

note the topmost vertex; it has 4 sides/diagonals emanating outward from it. We do casework on the color distribution of these sides/diagonals.

Case 1: all 4 are colored one color. In that case, all of the remaining sides must be of the other color to not have a triangle where all three sides are of the same color. We can correspondingly fill out each color based on this constraint, but in this case you will always end up with a triangle where all three sides have the same color by inspection.

Case 2: 3 are one color and one is the other. Following the steps from the previous case, you can try filling out the colors, but will always arrive at a contradiction so this case does not work either.

Case 3: 2 are one color and 2 are of the other color. Using the same logic as previously, we can color the pentagon 2 different ways by inspection to satisfy the requirements. There are $\binom{4}{2}$ ways to color the original sides/diagonals and 2 ways after that to color the remaining ones for a total of $6 \cdot 2 = 12$ ways to color the pentagon so that no such triangle has the same color for all of its sides.

These are all the cases, and there are a total of 2^{10} ways to color the pentagon. Therefore the answer is $1 - \frac{12}{1024} = 1 - \frac{3}{256} = \frac{253}{256} = \boxed{D}$

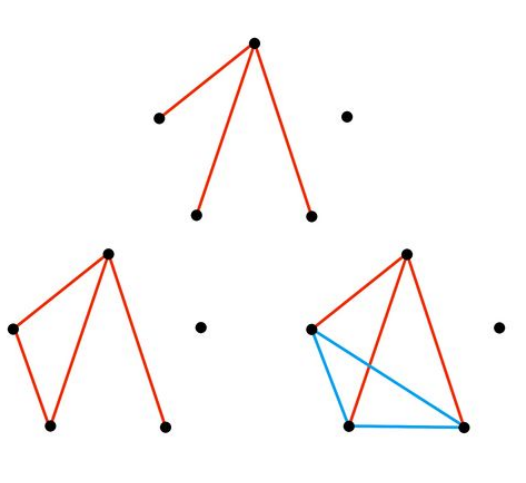
Solution 2 (Ramsey's Theorem)

This problem is related to a special case of **Ramsey's Theorem**, $R(3, 3) = 6$. Suppose we color every edge of a 6 vertex complete graph (K_6) with 2 colors, there must exist a 3 vertex complete graph (K_3) with all it's edges in the same color. That is, K_6 with edges in 2 colors contains a monochromatic K_3 . For K_5 with edges in 2 colors, a monochromatic K_3 does not always exist.

This is a problem about the probability of a monochromatic K_3 exist in a 5 vertex complete graph K_5 with edges in 2 colors.

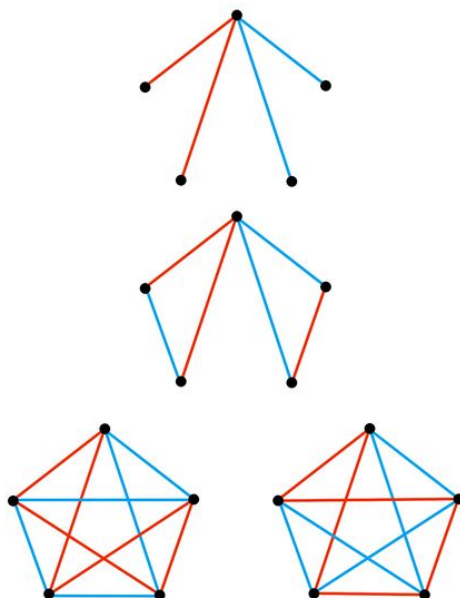
Choose a vertex, it has 4 edges.

Case 1: When 3 or more edges are the same color, there must exist a monochromatic K_3 . Suppose the color is red, as shown below.



There is only 1 way to color all the edges in the same color. There is $\binom{4}{3} = 4$ ways to color 3 edges in the same color. There are 2 colors. The probability of 3 or more edges the same color is $\frac{(1+4) \cdot 2}{2^4} = \frac{5}{8}$. So the probability of containing a monochromatic K_3 is $\frac{5}{8}$.

Case 2: When 2 edges are the same color, graphs that does not contain a monochromatic K_3 can exist. The following diagram shows steps to obtain graphs that does not contain a monochromatic K_3 .



There are $\binom{4}{2} = 6$ ways to choose 2 edges with the same color. For the other 4 vertices there are $\binom{4}{2} = 6$ edges among them, there are $2^6 = 64$ ways to color the edges. There are only 2 cases without a monochromatic K_3 .

So the probability without monochromatic K_3 is $\frac{2}{64} = \frac{1}{32}$.

The probability with monochromatic K_3 is $1 - \frac{1}{32} = \frac{31}{32}$.

From case 1 and case 2, the probability with monochromatic K_3 is $\frac{5}{8} + \left(1 - \frac{5}{8}\right) \cdot \frac{31}{32} =$

$$\boxed{(D) \frac{253}{256}}$$

Solution 3 (Elementary Casework)

We can break this problem down into cases based on the distribution of the two colors throughout the 5 sides(not diagonals) of the pentagon. This idea comes from the fact that if you play around with the sides for a while and their colors, you will see that the interior diagonals actually depend largely on the colors of the five exterior sides. The total number of combinations is $2^{10} = 1024$. We will be counting the number of arrangements in which there are no triangles with all 3 sides the same color, and we can subtract from the total(complementary counting).

- Exterior sides are the five sides of the regular pentagon
- Interior Diagonals are the five diagonals inside the pentagon made by the connection of two non-adjacent vertices

Case 1: The five exterior sides are all blue or all red.

There are 2 options for the color given in the title of this case, and for each color, only one option is given for the exterior (All 5 sides the same color; either Red or Blue). Once you color the five sides a single color, it is simple to notice that all the 5 interior diagonals must be in the other color which the sides are not. To make it more clear, if all of the 5 exterior sides are blue, all of the 5 interior diagonals must be red, and if all of the 5 exterior sides are red, all of the 5 interior diagonals must be blue. This gives us a total of 2 (Choices of Colors) $\cdot$ 1 (Configuration for particular color existing on all 5 exterior edges) = 2 ways.

Case 2: Four of the five exterior sides are in one color, while the remaining one exterior side is in the other color.

There are 2 ways to choose which color occupies four exterior sides and which color occupies the remaining one. Either we can have four red and one blue, or four blue and one red, which are the 2 ways mentioned above. When we calculated the total number of combinations, we took into account that each segment could either be red or blue which gives 2 choices for each of 10 segments yielding 2^{10} . What you must understand is that when we calculated this total, we did not account for any rotations or reflections or any other symmetry. For the same reason, we must not account for any symmetry such as rotations or reflections when calculating the number of arrangements where there is not a triangle consisting of three sides of the same color. There are 5 ways to arrange the one blue and four reds or one red and four blues on the 5 exterior sides (draw it out to see), and once you start playing with our condition, that no triangle may have all three sides in the same color, you will see that this case actually yields zero solutions as there comes a triangle in the middle consisting of all three sides of the same color. Hence this case yields 0 ways.

Case 3: Three of the five exterior sides in one color and the remaining two in another color.

There are two sub-cases, one in which the two exterior sides which are colored differently from the other three are adjacent, and the other case in which they are separated by one other exterior side.

Subcase 1: The case in which the two exterior sides colored differently from the other three are adjacent.

If you draw the five exterior sides with two colored in one color and the other three colored in a different color, you will see, that there are absolutely no ways to color the interior diagonals such that no triangle with all three sides the same color exists. This subcase yields 0 ways.

Subcase 2: The case in which the two exterior sides colored differently from the other three are separated by another exterior side. There are 5 arrangements for the exterior (draw and see for yourself) and the colors can be swapped with each other and since each of the exterior configurations force a particular interior configuration, this yields $2 \cdot 5 = 10$ ways.

In total, there are $10 + 2 = 12$ ways, such that no triangle has all three sides in the same color. This yields $1024 - 12 = 1012$ ways such that there is a triangle such that it has all three sides in the same color. Thus, the probability is:

$$\frac{1012}{2^{10}} = \boxed{(D) \frac{253}{256}} \text{ Shiva Kannan}$$

24. (AMC 10 10B Fall 2021) A cube is constructed from 4 white unit cubes and 4 blue unit cubes. How many different ways are there to construct the $2 \times 2 \times 2$ cube

using these smaller cubes? (Two constructions are considered the same if one can be rotated to match the other.)

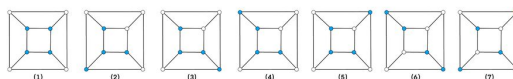
- (A) 7 (B) 8 (C) 9 (D) 10 (E) 11

Answer: A

Solution 1 (Graph Theory)

This problem is about the relationships between the white unit cubes and the blue unit cubes, which can be solved by **Graph Theory**. We use a **Planar Graph** to represent the larger cube. Each vertex of the planar graph represents a unit cube. Each edge of the planar graph represents a shared face between 2 neighboring unit cubes. Each face of the planar graph represents a face of the larger cube.

Now the problem becomes a **Graph Coloring** problem of how many ways to assign 4 vertices blue and 4 vertices white with **Topological Equivalence**. For example, in Figure (1), as long as the 4 blue vertices belong to the same planar graph face, the different planar graphs are considered to be topological equivalent by rotating the larger cube.



Here is how the 4 blue unit cubes are arranged:

In Figure (1): 4 blue unit cubes are on the same layer (horizontal or vertical).

In Figure (2): 4 blue unit cubes are in T shape.

In Figure (3) and (4): 4 blue unit cubes are in S shape.

In Figure (5): 3 blue unit cubes are in L shape, and the other is isolated without a shared face.

In Figure (6): 2 pairs of neighboring blue unit cubes are isolated from each other without a shared face.

In Figure (7): 4 blue unit cubes are isolated from each other without a shared face.

So the answer is **(A) 7**

Solution 2 (Casework, Counting Down)

Let's split the cube into two layers; a bottom and top. Note that there must be four of each color, so however many number of one color are in the bottom, there will be four minus that number of the color on the top. We do casework on the color distribution of the bottom layer.

Case 1: 4, 0

In this case, there is only one possibility for the top layer - all of the other color - $\binom{4}{4}$. Therefore there is 1 construction from this case.

Case 2: 3, 1

In this case, the top layer has four possibilities, because there are four different ways to arrange it so that it also has a 3, 1 color distribution - $\binom{4}{3}$. Therefore there are 4 constructions from this case.

Case 3: 2, 2

In this case, the top layer has six possibilities of arrangement - $\binom{4}{2}$. However, having adjacent colors one way can be rotated to having adjacent colors any other way, so there is only one construction for the adjacent colors subcase and similarly, only one for the diagonal color subcase. Therefore the total number of constructions for this case is 2.

The total number of constructions for the cube is thus $1 + 4 + 2 = 7 = \boxed{A}$

Solution 3 (Casework, Counting Up)

Divide the $2 \times 2 \times 2$ cube into two layers, say, front and back. Any possible construction can be rotated such that the front layer has the same or greater number of white cubes than blue cubes, so we only need to count the number of cases given that is true.

- a) **Case 1: Each layer contains 2 cubes of each color.** Note that we only need to consider the configuration of the white cubes because all the other cubes will be blue cubes. There are 2 ways that the 2 white cubes in each layer can be arranged: adjacent or diagonal to each other.
 - Case 1.1: Both layers have 2 white cubes adjacent to each other.** Rotate the cube such that there are white cubes along the top edge of the front layer. Now, the white cubes in the back layer can be along the top, bottom, right, or left edges. So, case 1.1 results in 4 constructions.
 - Case 1.2: One layer has 2 white cubes adjacent to each other, and the other has 2 white cubes diagonal from each other.** Rotate the cube such that there are white cubes along the top of the front layer. The white cubes in the back layer can be at the top-left and bottom-right or at the top-right and bottom-left. If we rotate the former case by 90 degrees clockwise into the page, it becomes the same as the latter case. So, case 1.2 results in 1 additional construction.
 - Case 1.3: Both layers have white cubes diagonal from each other.** Rotate the cube such that there is a white cube at the top-left and bottom-right of the front layer. The back layer could also have white cubes at the top-left and bottom-right, but this is the same as case 1.1 with the white cubes in the back layer along the bottom edge. Alternatively, the back layer could have white cubes at the top-right and the bottom-left. This is a distinct case. So, case 1.3 results in 1 additional construction.
 So, case 1 results in $4 + 1 + 1 = 6$ distinct constructions.
- b) **Case 2: The front layer contains 3 white cubes, and the back layer contains one white cube.** In this case, unless the sole white and sole blue cubes in the front and back layers are on opposite corners of the $2 \times 2 \times 2$ cube, then that cube can be split into left and right layers with 2 cubes of each color in each layer. Those constructions were counted in case 1, so case 2 results in 1 additional construction.
- c) **Case 3: The front layer contains 4 white cubes, and the back layer contains no white cubes.** If we split this construction into its left and right layers, then each layer will have 2 cubes of each color. So, this construction was counted in case 1, and case 3 results in 0 additional constructions.

Therefore, our answer is $6 + 1 + 0 = \boxed{(A) 7}$.

Solution 4 (Burnside Lemma)

Burnside lemma is used to counting number of orbit where the element on the same orbit can be achieved by the defined operator, naming rotation, reflection and etc.

The facts for Burnside lemma are

1. the sum of stabilizer on the same orbit equals to the # of operators;
2. the sum of stabilizer can be counted as $\text{fix}(g)$
3. the sum of the $\text{fix}(g)/|G|$ equals the # of orbit.

Let's start with defining the operator for a cube,

1. **e (identity)**

For identity, there are $\frac{8!}{4!4!} = 70$

2. $\mathbf{r}^1, \mathbf{r}^2, \mathbf{r}^3$ to be the rotation axis along three pair of opposite face,

each contains $r_{90}^i, r_{180}^i, r_{270}^i$ where $i = 1, 2, 3$ $\text{fix}(r_{90}^i) = \text{fix}(r_{270}^i) = 2 \cdot 1 = 2$
 $\text{fix}(r_{180}^i) = \frac{4!}{2!2!} = 6$

therefore $\text{fix}(\mathbf{r}^i) = 2 + 2 + 6 = 10$, and $\text{fix}(\mathbf{r}^1) + \text{fix}(\mathbf{r}^2) + \text{fix}(\mathbf{r}^3) = 30$

3. $\mathbf{r}^4, \mathbf{r}^5, \mathbf{r}^6, \mathbf{r}^7$ to the rotation axis along four cube diagonals.

each contains r_{120}^i, r_{240}^i where $i = 4, 5, 6, 7$ $\text{fix}(r_{120}^i) = \text{fix}(r_{240}^i) = 2 \cdot 1 \cdot 2 \cdot 1 = 4$

therefore $\text{fix}(\mathbf{r}^i) = 4 + 4 = 8$, and $\text{fix}(\mathbf{r}^4) + \text{fix}(\mathbf{r}^5) + \text{fix}(\mathbf{r}^6) + \text{fix}(\mathbf{r}^7) = 32$

4. $\mathbf{r}^8, \mathbf{r}^9, \mathbf{r}^{10}, \mathbf{r}^{11}, \mathbf{r}^{12}, \mathbf{r}^{13}$ to be the rotation axis along 6 pairs of diagonally opposite sides

each contains r_{180}^i where $i = 8, 9, 10, 11, 12, 13$ $\text{fix}(r_{180}^i) = \frac{4!}{2!2!} = 6$

therefore $\text{fix}(\mathbf{r}^8) + \text{fix}(\mathbf{r}^9) + \text{fix}(\mathbf{r}^{10}) + \text{fix}(\mathbf{r}^{11}) + \text{fix}(\mathbf{r}^{12}) + \text{fix}(\mathbf{r}^{13}) = 36$

5. The total number of operators are

$$|G| = 1 + 3 \cdot 3 + 4 \cdot 2 + 6 \cdot 1 = 24$$

Based on 1, 2, 3, 4 the total number of stabilizers is $70 + 30 + 32 + 36 = 168$

therefore the number of orbits = $\frac{168}{G=24} = \boxed{7}$

Solution 5 (Quick 'n Easy but not rigorous)

Since rotations of a single pattern are considered indistinguishable, we can assume that the forward upper right corner of the 2-by-2-by-2 cube is a blue cube (since we can always rotate the big cube to place a blue cube in that spot).

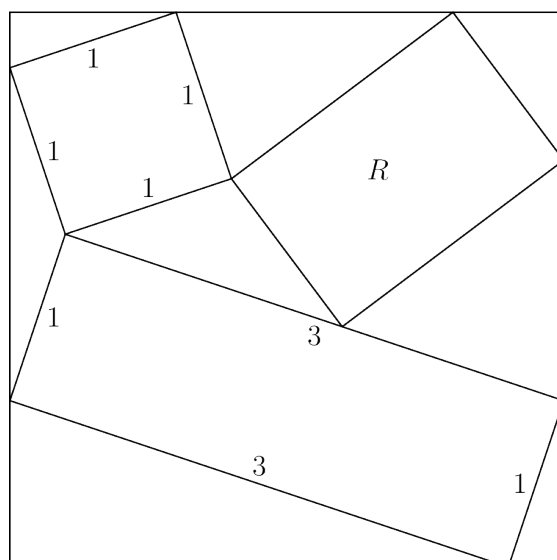
Once we've assigned this cube to be blue, we note that 3 1-by-1-by-1 cubes share a side with it, 3 1-by-1-by-1 cubes share a corner with it, and 1 1-by-1-by-1 cube does not touch the assigned cube at all, from the perspective of someone who can only see the cube's faces. We'll call the first 3 "adjacent", the second 3 "cornering", and the last one "opposite."

We can use a little bit of intuition to confirm that due to the rotation condition, we should treat all adjacents as indistinguishable, all cornerings as indistinguishable, and of course the opposite one is unique from all the others. Thus, we can list out like so (keeping in mind that there are 3 adjacents, 3 cornerings, and 1 opposite, and that we're choosing the positions of the remaining 3 blue cubes):

OAA, OAC, OCC, CCC, CAA, CCA, AAA.

This gives the answer to be $\boxed{(\mathbf{A}) 7}$.

25. (AMC 10 10B Fall 2021) A rectangle with side lengths 1 and 3, a square with side length 1, and a rectangle R are inscribed inside a larger square as shown. The sum of all possible values for the area of R can be written in the form $\frac{m}{n}$, where m and n are relatively prime positive integers. What is $m + n$?

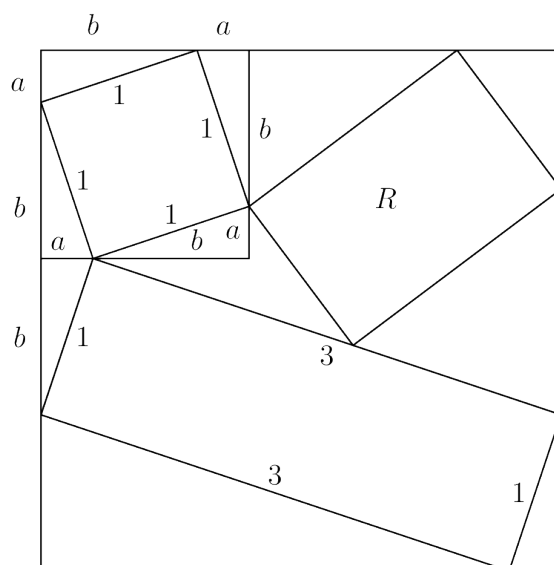


- (A) 14 (B) 23 (C) 46 (D) 59 (E) 67

Answer: E

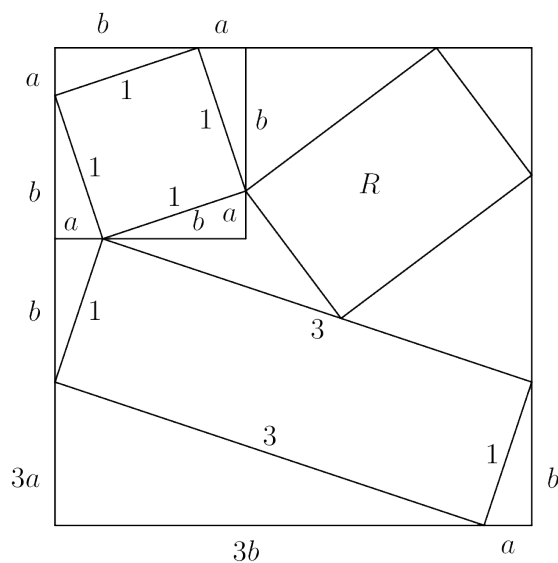
Solution 1

We see that the polygon bounded by the small square, large square, and rectangle of known lengths is an isosceles triangle. Let's draw a perpendicular from the vertex of this triangle to its opposing side;

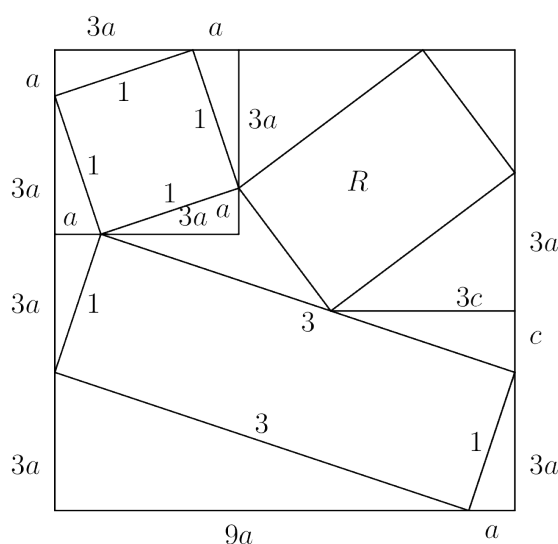


We see that this creates two congruent triangles. Let the smaller side of the triangle have length a and let the larger side of the triangle have length b . Now we see by AAS congruency that if we draw perpendiculars that surround the smaller square, each outer triangle will be congruent to these two triangles.

Now notice that these small triangles are also similar to the large triangle bounded by the bigger square and the rectangle by AA, and the ratio of the sides are 1:3, so we can fill in the lengths of that triangle. Similarly, the small triangle on the right bounded by the rectangle and the square is also congruent to the other small triangles by AAS, so we can fill in those sides;



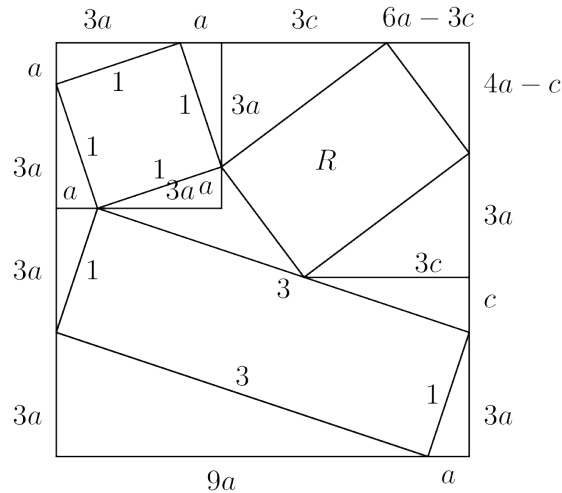
Since the larger square by definition has all equal sides, we can set the sum of the lengths of the sides equal to each other. $3a + b + b + a = 3b + a \implies 3a = b$. Now let's draw some more perpendiculars and rename the side lengths.



By AA similarity, when we draw a perpendicular from the intersection of the two rectangles to the large square, we create a triangle below that is similar to the small congruent triangles with length $a, 3a$. Since we don't know its scale, we'll label its sides $c, 3c$.

The triangle that is created above the perpendicular is congruent to the triangle on the opposite of the rectangle with unknown dimensions because they share the same hypotenuse and have two angles in common. Thus we can label these two triangles accordingly.

The side length of the big square is $10a$, so we can find the remaining dimensions of the triangle bounded by the rectangle with unknown dimensions and the large square in terms of a and c :



This triangle with side lengths $4a - c$ and $6a - 3c$ is similar to the triangle directly below it with side lengths $3a$ and $3c$ by AA similarity, so we can set up a ratio equation: $\frac{3a}{3c} = \frac{6a-3c}{4a-c} \implies 4a^2 - ac = -3c^2 + 6ac \implies 4a^2 - 7ac + 3c^2 = 0 \implies (4a - 3c)(a - c) = 0$. There are two solutions to this equation; $c = \frac{4}{3}a$ and $c = a$. For the first solution, the triangle in the corner has sides $2a$ and $\frac{8}{3}a$. Using Pythagorean theorem on that triangle, the hypotenuse has length $\frac{10}{3}a$. The triangle directly below has side lengths $3a$ and $4a$ in this case, so special right triangle yields the hypotenuse to be $5a$. The area of the rectangle is thus $5a \cdot \frac{10}{3}a = \frac{50}{3}a^2$. For the second solution, the side lengths of the corner triangle are $3a$ and $3a$, so the hypotenuse of the triangle is $3\sqrt{2}a$. The triangle below that also has side lengths $3a$ and $3a$, so its hypotenuse is the same. Then the area of the rectangle is $(3\sqrt{2}a)^2 = 18a^2$.

The sum of the possible areas of the rectangle is therefore $18a^2 + \frac{50}{3}a^2 = \frac{104}{3}a^2$.

Using Pythagorean theorem on the original small congruent triangles, $a^2 + 9a^2 = 1$ or $a^2 = \frac{1}{10}$. Therefore the sum of the possible areas of the rectangle is $\frac{104}{3} \cdot \frac{1}{10} = \frac{52}{15}$.

Therefore $m = 52$, $n = 15$, and $m + n = 67 = \boxed{E}$

Solution 2

We use [Image:2021_AMC_10B_\(Nov\)_Problem_25,_sol.png](#) to facilitate our analysis.

Denote $\angle AFE = \theta$. Thus, $\angle FIB = \angle CEF = \angle EKG = \angle KLC = \theta$.

Hence,

$$\begin{aligned} AB &= AF + FB \\ &= EF \cos \angle EFA + IF \sin \angle FIB \\ &= 3 \cos \theta + \sin \theta, \end{aligned}$$

and

$$\begin{aligned} AC &= AE + EK + KC \\ &= EF \sin \angle EFA + EG \cos \angle CEG + KG \cos \angle EKG + KL \sin \angle CLK \\ &= 3 \sin \theta + \cos \theta + \cos \theta + \sin \theta \\ &= 2 \cos \theta + 4 \sin \theta. \end{aligned}$$

Because $AB = AC$, $3 \cos \theta + \sin \theta = 2 \cos \theta + 4 \sin \theta$. Hence, $\tan \theta = \frac{1}{3}$. Hence, $\sin \theta = \frac{1}{\sqrt{10}}$ and $\cos \theta = \frac{3}{\sqrt{10}}$.

Hence, $AB = AC = BD = CD = \sqrt{10}$.

Now, we put the graph to a coordinate plane by setting point A as the origin, putting AB in the x -axis and AC on the y -axis.

Hence, $A = (0, 0)$, $B = (\sqrt{10}, 0)$, $C = (0, \sqrt{10})$, $D = (\sqrt{10}, \sqrt{10})$, $E = (0, \frac{3}{\sqrt{10}})$, $F = (\frac{9}{\sqrt{10}}, 0)$, $G = (\frac{1}{\sqrt{10}}, \frac{6}{\sqrt{10}})$, $H = (\frac{4}{\sqrt{10}}, \frac{7}{\sqrt{10}})$, $I = (\sqrt{10}, \frac{3}{\sqrt{10}})$.

Denote $P = (\frac{10-u}{\sqrt{10}}, \sqrt{10})$, $Q = (\sqrt{10}, \frac{10-v}{\sqrt{10}})$.

Because $HPQJ$ is a rectangle, $HP \perp PQ$. Hence, $m_{HP}m_{PQ} = -1$. We have $m_{HP} = \frac{3}{6-u}$ and $m_{PQ} = -\frac{v}{u}$. Hence,

$$\frac{3}{6-u} \cdot \left(-\frac{v}{u}\right) = -1. \quad (1)$$

Because $HPQJ$ is a rectangle, $x_J + x_P = x_H + x_Q$ and $y_J + y_P = y_H + y_Q$. Hence, $J = (\frac{4+u}{\sqrt{10}}, \frac{7-v}{\sqrt{10}})$.

The equation of line GI is

$$\begin{aligned} y &= \frac{\frac{3}{\sqrt{10}} - \frac{6}{\sqrt{10}}}{\sqrt{10} - \frac{1}{\sqrt{10}}} \left(x - \frac{1}{\sqrt{10}}\right) + \frac{6}{\sqrt{10}} \\ &= -\frac{x}{3} + \frac{19}{3\sqrt{10}}. \end{aligned}$$

Because point J is on line GI , plugging the coordinates of J into the equation of line GI , we get

$$\frac{7-v}{\sqrt{10}} = -\frac{\frac{4+u}{\sqrt{10}}}{3} + \frac{19}{3\sqrt{10}}. \quad (2)$$

By solving Equations (1) and (2), we get $(u, v) = (2, \frac{8}{3})$ or $(3, 3)$.

Case 1: $(u, v) = (2, \frac{8}{3})$.

We have $P = (\frac{8}{\sqrt{10}}, \sqrt{10})$ and $Q = (\sqrt{10}, \frac{22}{3\sqrt{10}})$. Thus, $HP = \frac{5}{\sqrt{10}}$ and $PQ = \frac{10}{3\sqrt{10}}$.

Therefore, $\text{Area } R = HP \cdot PQ = \frac{5}{3}$.

Case 2: $(u, v) = (3, 3)$.

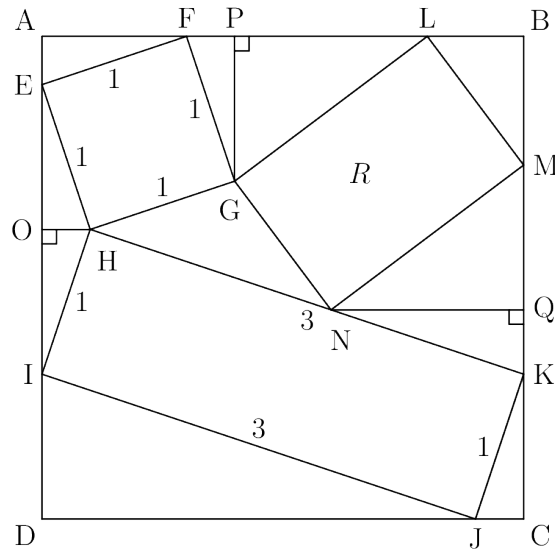
We have $P = (\frac{7}{\sqrt{10}}, \sqrt{10})$ and $Q = (\sqrt{10}, \frac{7}{\sqrt{10}})$. Thus, $HP = \frac{3\sqrt{2}}{\sqrt{10}}$ and $PQ = \frac{3\sqrt{2}}{\sqrt{10}}$.

Therefore, $\text{Area } R = HP \cdot PQ = \frac{9}{5}$.

Putting these two cases together, the sum of all possible values of the area of R is $\frac{5}{3} + \frac{9}{5} = \frac{52}{15}$.

Therefore, the answer is (E) 67.

Solution 3



We will scale every number up by a factor of $\sqrt{10}$. This implies our final area will be $\frac{1}{10}$ of the answer we receive.

We have

$$FAE \sim EOH \sim IOH \sim JDI \sim KCJ \sim NQK \sim GPF.$$

Let $AE = a$ and $FA = b$. We have

$$FP = AE = OH = JC = \frac{1}{3}ID = a$$

and

$$PG = AF = EO = OI = KC = \frac{1}{3}DJ = b$$

As $ABCD$ is a square, we have $AD = DC$ or

$$a + 2b + 3a = 3b + a \Rightarrow 3a = b.$$

Since $a^2 + b^2 = 10$, we have

$$a = 1, b = 3.$$

We have $\triangle GPL \cong \triangle MQN$ which implies

$$MQ = GP = 3.$$

Denote $QK = x$. As $\triangle NQK \sim \triangle JDI$, we have $NQ = 3x$.

We have

$$\begin{aligned} BM &= BC - (CK + QK + MQ) \\ &= 4 - x. \end{aligned}$$

In addition,

$$\begin{aligned} LB &= AB - (AF + FP + PL) \\ &= 6 - 3x. \end{aligned}$$

Since $\triangle LBM \sim \triangle MQN$, we have

$$\frac{LB}{BM} = \frac{MQ}{QN} \Rightarrow \frac{6-3x}{4-x} = \frac{3}{3x} = \frac{1}{x}.$$

Simplifying we have

$$3x^2 - 7x + 4 = 0 \Rightarrow x = \frac{4}{3}, 1.$$

We have

$$\begin{aligned} [GLMN] &= MN \cdot LM \\ &= 3\sqrt{x^2 + 1} \cdot \sqrt{10x^2 - 44x + 52}. \end{aligned}$$

Plugging in $x = 1$, we have $[GLMN] = 18$.

Plugging in $x = \frac{4}{3}$, we have $[GLMN] = \frac{50}{3}$.

The sum of the two possible R s is

$$\frac{1}{10} \cdot \frac{104}{3} = \frac{52}{15}.$$

Hence, $52 + 15 = \boxed{\text{(E) } 67}$.

Solution 4

We use the same diagram as solution 3, without scaling by $\sqrt{10}$. Denote S the foot from G to HN . Using the same method, obtain $JC = \frac{1}{\sqrt{10}}$ and $KC = \frac{3}{\sqrt{10}}$.

We have $\angle GHS = 2\angle HIO \Rightarrow \sin \angle GHS = 2\sin \angle HIO \cos \angle HIO = \frac{3}{5}$, so $GS = \frac{3}{5}$, and $HS = \frac{4}{5}$.

Denote $x = KN$, then $SN = HK - HS - NK = \frac{11-5x}{5}$, and $NQ = \frac{3x}{\sqrt{10}}$.

Also, $\triangle NQM \cong \triangle LPG$ because they're similar by AA, and then $LG = NM$, so $MQ = PG = \frac{3}{\sqrt{10}}$.

Thus $\tan \angle KNQ = \frac{1}{3}$ and $\tan \angle QNM = \frac{1}{x}$. Then $\cot \angle GNS = \tan(90^\circ - \angle GNS) = \tan \angle KNM = \tan(\angle KNQ + \angle QNM) = \frac{\tan \angle KNQ + \tan \angle QNM}{1 - \tan \angle KNQ \tan \angle QNM}$

which simplifies to

$$\frac{11-5x}{3} = \frac{3+x}{3x-1}$$

Cross multiplying gives

$$3x^2 - 7x + 4 = 0, \text{ with solutions } x = 1, \frac{4}{3}.$$

Plugging these back into the setup gives areas of $\frac{9}{5}$ and $\frac{5}{3}$, respectively, which have sum $\frac{52}{15}$, and thus the answer is $52 + 15 = \boxed{\text{(E) } 67}$.

§6 AMC 10 10A Fall 2021

- (AMC 10 10A Fall 2021) What is the value of $\frac{(2112-2021)^2}{169}$?
(A) 7 (B) 21 (C) 49 (D) 64 (E) 91

Answer: C

Solution 1 (Laws of Exponents)

We have

$$\frac{(2112 - 2021)^2}{169} = \frac{91^2}{169} = \frac{91^2}{13^2} = \left(\frac{91}{13}\right)^2 = 7^2 = \boxed{\text{(C) } 49}.$$

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Solution 2 (Difference of Squares)

We have

$$\frac{(2112 - 2021)^2}{169} = \frac{91^2}{169} = \frac{(10^2 - 3^2)^2}{13^2} = \frac{((10 + 3)(10 - 3))^2}{13^2} = \frac{(13 \cdot 7)^2}{13^2} = \frac{13^2 \cdot 7^2}{13^2} = 7^2 = \boxed{\text{(C) } 49}.$$

Solution 3 (Estimate)

We know that $2112 - 2021 = 91$. Approximate this as 100 as it is pretty close to it. Also, approximate 169 to 170. We then have

$$\frac{(2112 - 2021)^2}{169} \approx \frac{100^2}{170} \approx \frac{1000}{17} \approx 58.$$

Now check the answer choices. The two closest answers are 49 and 64. As the numerator is actually bigger than it should be, it should be the smaller answer, or $\boxed{\text{(C) } 49}$.

2. (*AMC 10 10A Fall 2021*) Menkara has a 4×6 index card. If she shortens the length of one side of this card by 1 inch, the card would have area 18 square inches. What would the area of the card be in square inches if instead she shortens the length of the other side by 1 inch?
- (A) 16 (B) 17 (C) 18 (D) 19 (E) 20

Answer: E

Solution

We construct the following table:

Scenario	Length	Width	Area
Initial	4	6	24
Menkara shortens one side.	3	6	18
Menkara shortens other side instead.	4	5	20

Therefore, the answer is $\boxed{\text{(E) } 20}$.

3. (*AMC 10 10A Fall 2021*) What is the maximum number of balls of clay of radius 2 that can completely fit inside a cube of side length 6 assuming the balls can be reshaped but not compressed before they are packed in the cube?
- (A) 3 (B) 4 (C) 5 (D) 6 (E) 7

Answer: D

Solution 1 (Inequality)

The volume of the cube is $V_{\text{cube}} = 6^3 = 216$, and the volume of a clay ball is $V_{\text{ball}} = \frac{4}{3} \cdot \pi \cdot 2^3 = \frac{32}{3}\pi$.

Since the balls can be reshaped but not compressed, the maximum number of balls that can completely fit inside a cube is

$$\left\lfloor \frac{V_{\text{cube}}}{V_{\text{ball}}} \right\rfloor = \left\lfloor \frac{81}{4\pi} \right\rfloor.$$

Approximating with $\pi \approx 3.14$, we have $12 < 4\pi < 13$, or $\left\lfloor \frac{81}{13} \right\rfloor \leq \left\lfloor \frac{81}{4\pi} \right\rfloor \leq \left\lfloor \frac{81}{12} \right\rfloor$. We simplify to get

$$6 \leq \left\lfloor \frac{81}{4\pi} \right\rfloor \leq 6,$$

from which $\left\lfloor \frac{81}{4\pi} \right\rfloor = \boxed{\text{(D)} 6}$.

Solution 2 (Inequality)

As shown in Solution 1, we conclude that the maximum number of balls that can completely fit inside a cube is $\left\lfloor \frac{81}{4\pi} \right\rfloor$.

By an underestimation $\pi \approx 3$, we have $4\pi > 12$, or $\frac{81}{4\pi} < 6\frac{3}{4}$.

By an overestimation $\pi \approx \frac{22}{7}$, we have $4\pi < \frac{88}{7}$, or $\frac{81}{4\pi} > 6\frac{39}{88}$.

Together, we get

$$6 < 6\frac{39}{88} < \frac{81}{4\pi} < 6\frac{3}{4} < 7,$$

from which $\left\lfloor \frac{81}{4\pi} \right\rfloor = \boxed{\text{(D)} 6}$.

Solution 3 (Approximation)

As shown in Solution 1, we conclude that the maximum number of balls that can completely fit inside a cube is $\left\lfloor \frac{81}{4\pi} \right\rfloor$.

Approximating with $\pi \approx 3$, we have $\frac{81}{4\pi} \approx 6\frac{3}{4}$. Since π is about 5% greater than 3, it is safe to claim that $\left\lfloor \frac{81}{4\pi} \right\rfloor = \boxed{\text{(D)} 6}$.

4. (AMC 10 10A Fall 2021) Mr. Lopez has a choice of two routes to get to work. Route A is 6 miles long, and his average speed along this route is 30 miles per hour. Route B is 5 miles long, and his average speed along this route is 40 miles per hour, except for a $\frac{1}{2}$ -mile stretch in a school zone where his average speed is 20 miles per hour. By how many minutes is Route B quicker than Route A?

(A) $2\frac{3}{4}$ (B) $3\frac{3}{4}$ (C) $4\frac{1}{2}$ (D) $5\frac{1}{2}$ (E) $6\frac{3}{4}$

Answer: B

Solution 1

If Mr. Lopez chooses Route A, then he will spend $\frac{6}{30} = \frac{1}{5}$ hour, or $\frac{1}{5} \cdot 60 = 12$ minutes.

If Mr. Lopez chooses Route B, then he will spend $\frac{9/2}{40} + \frac{1/2}{20} = \frac{11}{80}$ hour, or $\frac{11}{80} \cdot 60 = 8\frac{1}{4}$ minutes.

Therefore, Route B is quicker than Route A by $12 - 8\frac{1}{4} = \boxed{\text{(B)} 3\frac{3}{4}}$ minutes.

Solution 2

We use the equation $d = st$ to solve this problem. Recall that 1 mile per hour is equal to $\frac{1}{60}$ mile per minute.

For Route A, the distance is 6 miles and the speed to travel this distance is $\frac{1}{2}$ mile per minute. Thus, the time it takes on Route A is 12 minutes.

For Route B, we have to use the equation twice: once for the distance of $5 - \frac{1}{2} = \frac{9}{2}$ miles with a speed of $\frac{2}{3}$ mile per minute and a distance of $\frac{1}{2}$ miles at a speed of $\frac{1}{3}$ mile per minute. Thus, the time it takes to go on Route B is $\frac{9}{2} \cdot \frac{3}{2} + \frac{1}{2} \cdot 3 = \frac{27}{4} + \frac{3}{2} = \frac{33}{4}$ minutes. Thus, Route B is $12 - \frac{33}{4} = \frac{15}{4} = 3\frac{3}{4}$ faster than Route A. Thus, the

answer is (B) $3\frac{3}{4}$.

5. (AMC 10 10A Fall 2021) The six-digit number 20210A is prime for only one digit A. What is A? (A) 1 (B) 3 (C) 5 (D) 7 (E) 9

Answer: E

Solution 1

First, modulo 2 or 5, $20210A \equiv A$. Hence, $A \neq 0, 2, 4, 5, 6, 8$.

Second modulo 3, $20210A \equiv 2 + 0 + 2 + 1 + 0 + A \equiv 5 + A$. Hence, $A \neq 1, 4, 7$.

Third, modulo 11, $20210A \equiv A + 1 + 0 - 0 - 2 - 2 \equiv A - 3$. Hence, $A \neq 3$.

Therefore, the answer is (E) 9.

Solution 2 (Elimination)

Any number ending in 5 is divisible by 5. So we can eliminate option (C).

If the sum of the digits of a number is divisible by 3, the number is divisible by 3. The sum of the digits of this number is $2 + 0 + 2 + 1 + 0 + A = 5 + A$. If $5 + A$ is divisible by 3, the number is divisible by 3. Thus we can eliminate options (A) and (D).

So the correct option is either (B) or (E). Let's try dividing the number with some integers.

$20210A/7 = 2887x$, where x is $1A/7$. Since 13 and 19 are both indivisible by 7, this does not help us narrow the choices down.

$20210A/11 = 1837x$, where x is $3A/11$. Since $33/11 = 3$, option (B) would make $20210A$ divisible by 11. Thus, by elimination, the correct choice must be option (E) 9.

Solution 3

$202100 \implies$ divisible by 2.

$202101 \implies$ divisible by 3.

$202102 \implies$ divisible by 2.

$202103 \implies$ divisible by 11.

$202104 \implies$ divisible by 2.

$202105 \implies$ divisible by 5.

$202106 \implies$ divisible by 2.

$202107 \implies$ divisible by 3.

$202108 \implies$ divisible by 2.

This leaves only $A =$ (E) 9.

Sidenote The divisibility test for 11 is if the difference between the sum of the alternating digits is 0 or 11, then that number is divisible by 11. For 202103, we have $2+2-1-3=0$.

For 7, the divisibility test can be slow. Take the last digit, double it, and subtract it from the rest of the number. If it is divisible by 7, then the original number is divisible by 7. If the number is too large to determine whether or not it is a multiple of 7, then we repeat the process until we can determine whether or not the number is divisible by 7.

6. (AMC 10 10A Fall 2021) Elmer the emu takes 44 equal strides to walk between consecutive telephone poles on a rural road. Oscar the ostrich can cover the same distance in 12 equal leaps. The telephone poles are evenly spaced, and the 41st pole along this road is exactly one mile (5280 feet) from the first pole. How much longer, in feet, is Oscar's leap than Elmer's stride?
- (A) 6 (B) 8 (C) 10 (D) 11 (E) 15

Answer: B

Solution 1

There are $41 - 1 = 40$ gaps between the 41 telephone poles, so the distance of each gap is $5280 \div 40 = 132$ feet.

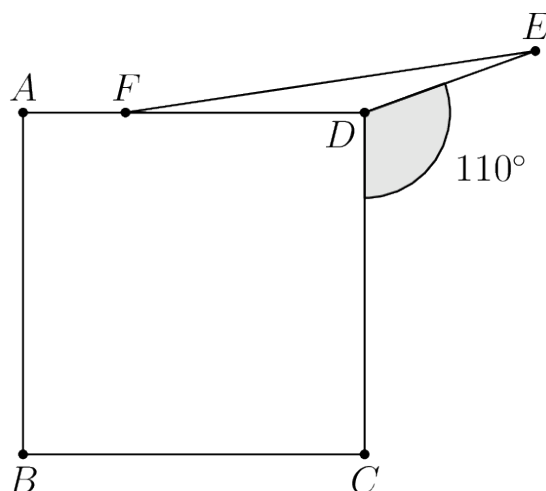
Each of Oscar's leaps covers $132 \div 12 = 11$ feet, and each of Elmer's strides covers $132 \div 44 = 3$ feet.

Therefore, Oscar's leap is $11 - 3 = \boxed{\text{(B) } 8}$ feet longer than Elmer's stride.

Solution 2

There are $41 - 1 = 40$ gaps between the 41 telephone poles, so Elmer takes $44 \cdot 40 = 1760$ strides in total, and Oscar takes $12 \cdot 40 = 480$ leaps in total. Therefore, the answer is $(5280 \div 480) - (5280 \div 1760) = 11 - 3 = \boxed{\text{(B) } 8}$.

7. (AMC 10 10A Fall 2021) As shown in the figure below, point E lies on the opposite half-plane determined by line CD from point A so that $\angle CDE = 110^\circ$. Point F lies on $\overline{AD}$ so that $DE = DF$, and $ABCD$ is a square. What is the degree measure of $\angle AFE$?



- (A) 160 (B) 164 (C) 166 (D) 170 (E) 174

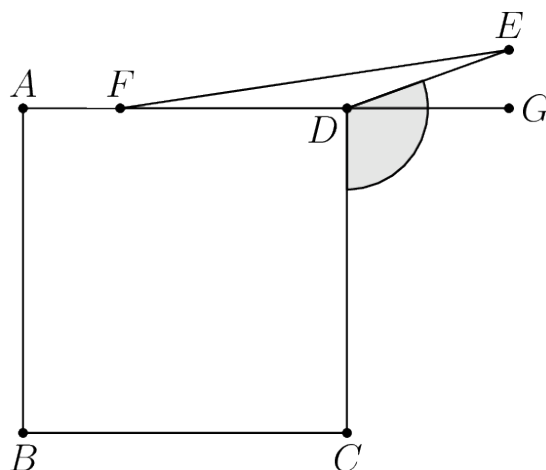
Answer: D

Solution 1

By angle subtraction, we have $\angle ADE = 360^\circ - \angle ADC - \angle CDE = 160^\circ$. Note that $\triangle DEF$ is isosceles, so $\angle DFE = \frac{180^\circ - \angle ADE}{2} = 10^\circ$. Finally, we get $\angle AFE = 180^\circ - \angle DFE = \boxed{\text{(D) } 170}$ degrees.

Solution 2 (Extension)

We can extend $\overline{AD}$ to G , making $\angle CDG$ a right angle. It follows that $\angle GDE$ is $110^\circ - 90^\circ = 20^\circ$, as shown below.



Since $\angle DFE = \angle DEF$, we see that $\angle DFE = \angle DEF = \frac{20^\circ}{2} = 10^\circ$. Thus, $\angle AFE = 180^\circ - 10^\circ = \boxed{\text{(D) } 170}$ degrees.

8. (AMC 10 10A Fall 2021) A two-digit positive integer is said to be *cuddly* if it is equal to the sum of its nonzero tens digit and the square of its units digit. How many two-digit positive integers are cuddly?

(A) 0 (B) 1 (C) 2 (D) 3 (E) 4

Answer: B

Solution 1

Note that the number $\underline{xy} = 10x + y$. By the problem statement,

$$10x + y = x + y^2 \implies 9x = y^2 - y \implies 9x = y(y - 1).$$

From this we see that $y(y - 1)$ must be divisible by 9. This only happens when $y = 9$. Then, $x = 8$. Thus, there is only $\boxed{\text{(B) } 1}$ cuddly number, which is 89.

Solution 2

If the tens digit is a and the ones digit is b then the number is $10a + b$ so we have the equation $10a + b = a + b^2$. We can guess and check after narrowing the possible cuddly numbers down to 13, 14, 24, 25, 35, 36, 46, 47, 57, 68, 78, 89, and 99. (We can narrow it down to these by just thinking about how a 's value affects b 's value and then check all the possibilities.) Checking all of these we get that there is only $\boxed{\text{(B) } 1}$ 2-digit cuddly number, and it is 89. Yay!!!

9. (*AMC 10 10A Fall 2021*) When a certain unfair die is rolled, an even number is 3 times as likely to appear as an odd number. The die is rolled twice. What is the probability that the sum of the numbers rolled is even?

(A) $\frac{3}{8}$ (B) $\frac{4}{9}$ (C) $\frac{5}{9}$ (D) $\frac{9}{16}$ (E) $\frac{5}{8}$

Answer: E

Solution 1

Since an even number is 3 times more likely to appear than an odd number, the probability of an even number appearing is $\frac{3}{4}$. Since the problem states that the sum of the two die must be even, the numbers must both be even or both be odd. We either have EE or OO, so we have

$$\frac{3}{4} \cdot \frac{3}{4} + \frac{1}{4} \cdot \frac{1}{4} = \frac{1}{16} + \frac{9}{16} = \frac{10}{16} = \boxed{\text{(E)} \frac{5}{8}}.$$

Solution 2 (Complementary Counting)

As explained in the above solution, the probability of an even number appearing is $\frac{3}{4}$, while the probability of an odd number appearing is $\frac{1}{4}$. Then the probability of getting an odd and an even (to make an odd number) is $\frac{3}{4} \cdot \frac{1}{4} \cdot 2 = \frac{3}{8}$. Then the probability of getting an even number is $1 - \frac{3}{8} = \boxed{\text{(E)} \frac{5}{8}}$.

Solution 3 (Answer Choices)

As explained in the above solutions, the probability of an even number appearing is $\frac{3}{4}$. Getting two even numbers in a row would result in an even sum, and the probability of this happening is $\frac{3}{4} \cdot \frac{3}{4} = \frac{9}{16}$. As an even sum can also be a result of the sum of two odd numbers, the probability of an even sum is slightly greater than $\frac{9}{16}$. The only answer choice with a probability greater than $\frac{9}{16}$ is $\boxed{\text{(E)} \frac{5}{8}}$.

10. (*AMC 10 10A Fall 2021*) A school has 100 students and 5 teachers. In the first period, each student is taking one class, and each teacher is teaching one class. The enrollments in the classes are 50, 20, 20, 5, and 5. Let t be the average value obtained if a teacher is picked at random and the number of students in their class is noted. Let s be the average value obtained if a student was picked at random and the number of students in their class, including the student, is noted. What is $t - s$?

(A) -18.5 (B) -13.5 (C) 0 (D) 13.5 (E) 18.5

Answer: B

Solution

The formula for expected values is

$$\text{Expected Value} = \sum (\text{Outcome} \cdot \text{Probability}).$$

We have

$$\begin{aligned} t &= 50 \cdot \frac{1}{5} + 20 \cdot \frac{1}{5} + 20 \cdot \frac{1}{5} + 5 \cdot \frac{1}{5} + 5 \cdot \frac{1}{5} \\ &= (50 + 20 + 20 + 5 + 5) \cdot \frac{1}{5} \end{aligned}$$

$$\begin{aligned}
&= 100 \cdot \frac{1}{5} \\
&= 20, \\
s &= 50 \cdot \frac{50}{100} + 20 \cdot \frac{20}{100} + 20 \cdot \frac{20}{100} + 5 \cdot \frac{5}{100} + 5 \cdot \frac{5}{100} \\
&= 25 + 4 + 4 + 0.25 + 0.25 \\
&= 33.5.
\end{aligned}$$

Therefore, the answer is $t - s = \boxed{\text{(B)} -13.5}$.

11. (AMC 10 10A Fall 2021) Emily sees a ship traveling at a constant speed along a straight section of a river. She walks parallel to the riverbank at a uniform rate faster than the ship. She counts 210 equal steps walking from the back of the ship to the front. Walking in the opposite direction, she counts 42 steps of the same size from the front of the ship to the back. In terms of Emily's equal steps, what is the length of the ship?

(A) 70 (B) 84 (C) 98 (D) 105 (E) 126

Answer: A

Solution 1 (One Variable)

Let x be the length of the ship. Then, in the time that Emily walks 210 steps, the ship moves $210 - x$ steps. Also, in the time that Emily walks 42 steps, the ship moves $x - 42$ steps. Since the ship and Emily have the same ratio of absolute speeds in either direction, $\frac{210}{210-x} = \frac{42}{x-42}$. Dividing both sides by 42 and cross multiplying, we get $5(x - 42) = 210 - x$, so $6x = 420$, and $x = \boxed{\text{(A)} 70}$.

Solution 2 (Two Variables)

Let the speed at which Emily walks be 42 steps per hour. Let the speed at which the ship is moving be s . Walking in the direction of the ship, it takes her 210 steps, or $\frac{210}{42} = 5$ hours, to travel. We can create an equation:

$$d = 5(42 - s),$$

where d is the length of the ship. Walking in the opposite direction of the ship, it takes her 42 steps, or $42/42 = 1$ hour. We can create a similar equation:

$$d = 1(42 + s).$$

Now we have two variables and two equations. We can equate the expressions for d and solve for s :

$$\begin{aligned}
210 - 5s &= 42 + s \\
s &= 28.
\end{aligned}$$

Therefore, we have $d = 42 + s = \boxed{\text{(A)} 70}$.

Solution 3 (Three Variables)

Suppose that Emily and the ship take steps simultaneously such that Emily's steps cover a greater length than the ship's steps.

Let L be the length of the ship, E be Emily's step length, and S be the ship's step length. We wish to find $\frac{L}{E}$.

When Emily walks from the back of the ship to the front, she walks a distance of $210E$ and the front of the ship moves a distance of $210S$. We have $210E = L + 210S$ for this scenario, which rearranges to

$$210E - 210S = L. \quad (1)$$

When Emily walks in the opposite direction, she walks a distance of $42E$ and the back of the ship moves a distance of $42S$. We have $42E = L - 42S$ for this scenario, which rearranges to

$$42E + 42S = L. \quad (2)$$

We multiply (2) by 5 and then add (1) to get $420E = 6L$, from which $\frac{L}{E} = \boxed{\text{(A) } 70}$.

Solution 4 (Using the Boat's "Step")

Every time Emily takes a step, the boat also "takes a step". Call the length of the boat's step s . Call the length of the boat x .

When Emily is walking in the same direction as the boat, every time she takes a step the boat moves an additional distance of s . This means that she travels a total distance of $x + 210s$ to reach the other end of the boat.

When Emily is walking in the opposite direction of the boat, every time she takes a step the distance till the end of the boat reduces by s (since the boat is coming towards her and moves a distance of s). This means that she travels a total distance of $x - 42s$ to reach the other end of the boat.

Taking Emily's step as a unit of distance, we now have two equations

$$\begin{aligned} 210 &= x + 210s, \\ 42 &= x - 42s. \end{aligned}$$

Solving for x you get $\boxed{\text{(A) } 70}$.

Solution 5 (Relative Speeds)

Call the speed of the boat v_s and the speed of Emily v_e .

Consider the scenario when Emily is walking along with the boat. Relative to an observer on the boat, her speed is $v_e - v_s$.

Consider the scenario when Emily is walking in the opposite direction. Relative to an observer on the boat, her speed is $v_e + v_s$.

Since Emily takes 210 steps to walk along with the boat and 42 steps to walk opposite the boat, that means it takes her 5 times longer to walk the length of a stationary boat at $v_e - v_s$ compared to $v_e + v_s$.

This means that $5(v_e - v_s) = v_e + v_s$, so $v_s = \frac{2v_e}{3}$.

As Emily takes 210 steps to walk the length of the boat at a speed of $v_e - \frac{2v_e}{3} = \frac{v_e}{3}$, she must take $\frac{1}{3}$ of the time to walk the length of the boat at a speed of v_e , so our answer is $\frac{210}{3} = \boxed{\text{(A) } 70}$.

12. (AMC 10 10A Fall 2021) The base-nine representation of the number N is $27,006,000,052_{\text{nine}}$. What is the remainder when N is divided by 5? (A) 0 (B) 1 (C) 2 (D)

Answer: D

Solution 1 (Modular Arithmetic)

Recall that $9 \equiv -1 \pmod{5}$. We expand N by the definition of bases:

$$\begin{aligned} N &= 27,006,000,052_9 \\ &= 2 \cdot 9^{10} + 7 \cdot 9^9 + 6 \cdot 9^6 + 5 \cdot 9 + 2 \\ &\equiv 2 \cdot (-1)^{10} + 7 \cdot (-1)^9 + 6 \cdot (-1)^6 + 2 \pmod{5} \\ &\equiv 2 - 7 + 6 + 2 \pmod{5} \\ &\equiv \boxed{\text{(D)} 3} \pmod{5}. \end{aligned}$$

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Solution 2 (Powers of 9)

We need to first convert N into a regular base-10 number:

$$N = 27,006,000,052_9 = 2 \cdot 9^{10} + 7 \cdot 9^9 + 6 \cdot 9^6 + 5 \cdot 9 + 2.$$

Now, consider how the last digit of 9 changes with changes of the power of 9 :

$$\begin{aligned} 9^0 &= 1, \\ 9^1 &= 9, \\ 9^2 &= \dots 1, \\ 9^3 &= \dots 9, \\ 9^4 &= \dots 1, \\ &\vdots \end{aligned}$$

Note that if x is odd, then $9^x \equiv 4 \pmod{5}$. On the other hand, if x is even, then $9^x \equiv 1 \pmod{5}$.

Therefore, we have

$$\begin{aligned} N &\equiv 2 \cdot (1) + 7 \cdot (4) + 6 \cdot (1) + 5 \cdot (4) + 2 \cdot (1) \pmod{5} \\ &\equiv 2 + 28 + 6 + 20 + 2 \pmod{5} \\ &\equiv 58 \pmod{5} \\ &\equiv \boxed{\text{(D)} 3} \pmod{5}. \end{aligned}$$

Note that for the odd case, $9^x \equiv -1 \pmod{5}$ may simplify the process further, as given by Solution 1.

Remark By the long division algorithm, you can work from left to right accumulating the answer, and don't need to count the digits to get the power. You only need to take the current leading digit modulo 5, negate it, and add it to the next digit, and repeat the process until you reach the units digit.

13. (AMC 10 10A Fall 2021) Each of 6 balls is randomly and independently painted either black or white with equal probability. What is the probability that every ball is different in color from more than half of the other 5 balls?
- (A) $\frac{1}{64}$ (B) $\frac{1}{6}$ (C) $\frac{1}{4}$ (D) $\frac{5}{16}$ (E) $\frac{1}{2}$

Answer: D

Solution 1

Note that for this restriction to be true, there must be 3 balls of each color. There are a total of $2^6 = 64$ ways to color the balls, and there are $\binom{6}{3} = 20$ ways for three balls chosen to be painted white. Thus, the answer is $\frac{20}{64} = \boxed{\text{(D)} \frac{5}{16}}$.

-Aidensharp

Solution 2

For this restriction to be upheld, there must be three black and three white balls. One such way for this to occur is the arrangement $BBBWWW$, which has a $\frac{1}{2^6}$ probability of occurring. However, there are $\frac{6!}{3!3!}$ ways to arrange the three black and three white balls, meaning that the answer is, $\frac{1}{64} \cdot \frac{6!}{3!3!} = \boxed{\text{(D)} \frac{5}{16}}$.

Solution 3

To get every ball different in color from more than half of the other 5 balls, we must have 3 black balls and 3 white balls.

Following from the binomial theorem, this happens with probability

$$\binom{6}{3} \left(\frac{1}{2}\right)^3 \left(1 - \frac{1}{2}\right)^{6-3} = \frac{5}{16}.$$

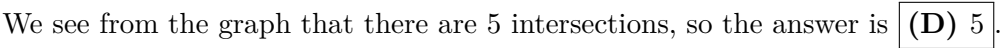
Therefore, the answer is $\boxed{\text{(D)} \frac{5}{16}}$.

14. (AMC 10 10A Fall 2021) How many ordered pairs (x, y) of real numbers satisfy the following system of equations?
- $$\begin{cases} x^2 + y^2 = 1 \\ x^2 - y^2 = 0 \end{cases}$$

Answer: D

The second equation is $(|x| + |y| - 4)^2 = 1$. We know that the graph of $|x| + |y|$ is a very simple diamond shape, so let's see if we can reduce this equation to that form:

We now have two separate graphs for this equation and one graph for the first equation, so let's put it on the coordinate plane:



Solution 2 (Casework)

From the first equation, we can express y in terms of x :

$$3y = 9 - x^2 \text{ which is } y = (9 - x^2)/3$$

The second equation can be rewritten as:

15. (AMC 10 10A Fall 2021) Isosceles triangle ABC has $AB = AC = 3\sqrt{6}$, and a circle with radius $5\sqrt{2}$ is tangent to line AB at B and to line AC at C . What is the area of the circle that passes through vertices A , B , and C ? (A) 24π (B) 25π (C) 26π (D) 27π

Answer: C

Solution 1 (Cyclic Quadrilateral)

Let $\odot O_1$ be the circle with radius $5\sqrt{2}$ that is tangent to $\overleftrightarrow{AB}$ at B and to $\overleftrightarrow{AC}$ at C . Note that $\angle ABO_1 = \angle ACO_1 = 90^\circ$. Since the opposite angles of quadrilateral ABO_1C are supplementary, quadrilateral ABO_1C is cyclic.

Let $\odot O_2$ be the circumcircle of quadrilateral ABO_1C . It follows that $\odot O_2$

16. (AMC 10 10A Fall 2021) The graph of

$$f(x) = ||x|| - ||1 - x||$$

is symmetric about which of the following? (Here $\lfloor x \rfloor$ is the greatest integer not exceeding x .)

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Answer: D

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Note that

$$f(1-x) = ||1-x|| - ||x|| = -f(x),$$

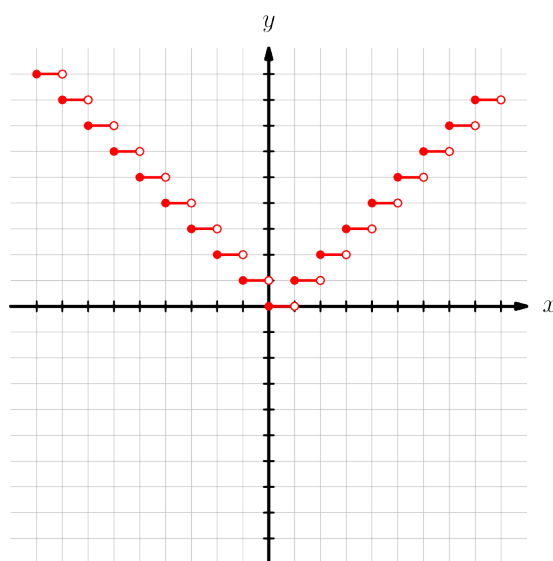
$$\text{so } f\left(\frac{1}{2}+x\right) = -f\left(\frac{1}{2}-x\right).$$

This means that the graph is symmetric about **(D)** the point $\left(\frac{1}{2}, 0\right)$.

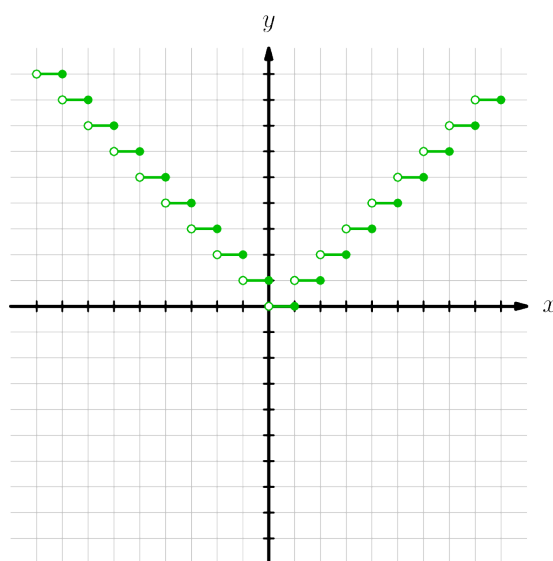
Solution 2 (Graphing)

Let $y_1 = ||x||$ and $y_2 = ||1-x|| = ||-(x-1)||$. Note that the graph of y_2 is a reflection of the graph of y_1 about the y -axis, followed by a translation 1 unit to the right.

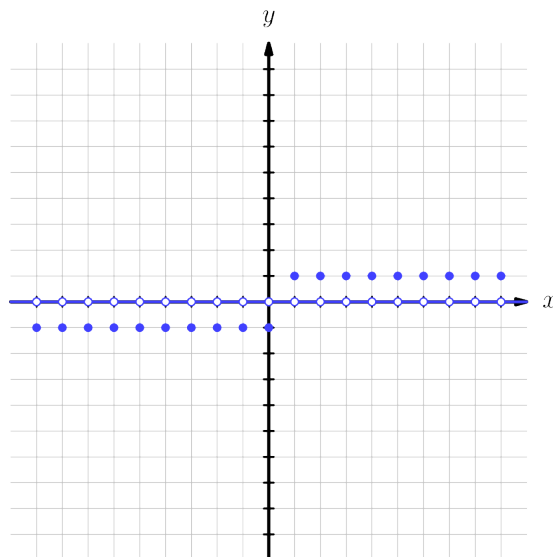
The graph of y_1 is shown below:



The graph of y_2 is shown below:



The graph of $f(x) = y_1 - y_2$ is shown below:



Therefore, the graph of $f(x)$ is symmetric about **(D)** the point $\left(\frac{1}{2}, 0\right)$.

Solution 3 (Casework)

For all $x \in \mathbb{R}$ and $n \in \mathbb{Z}$, note that:

a) $\lfloor x + n \rfloor = \lfloor x \rfloor + n$ and $\lceil x + n \rceil = \lceil x \rceil + n$

b) $\lfloor -x \rfloor = -\lceil x \rceil$

c) $\lceil x \rceil - \lfloor x \rfloor = \begin{cases} 0 & \text{if } x \in \mathbb{Z} \\ 1 & \text{if } x \notin \mathbb{Z} \end{cases}$

We rewrite $f(x)$ as

$$\begin{aligned} f(x) &= |\lfloor x \rfloor| - |\lceil 1 - x \rceil| \\ &= |\lfloor x \rfloor| - |-\lceil x - 1 \rceil| \\ &= |\lfloor x \rfloor| - |-\lceil x \rceil + 1|. \end{aligned}$$

We apply casework to the value of x :

a) $x \in \mathbb{Z}^-$ It follows that $f(x) = -x - (-x + 1) = -1$.

b) $x = 0$ It follows that $f(x) = 0 - 1 = -1$.

c) $x \in \mathbb{Z}^+$ It follows that $f(x) = x - (x - 1) = 1$.

d) $x \notin \mathbb{Z}$ and $x < 0$ It follows that $f(x) = -\lfloor x \rfloor - (-\lceil x \rceil + 1) = (\lceil x \rceil - \lfloor x \rfloor) - 1 = 0$.

e) $x \notin \mathbb{Z}$ and $0 < x < 1$ It follows that $f(x) = 0 - 0 = 0$.

f) $x \notin \mathbb{Z}$ and $x > 1$ It follows that $f(x) = \lfloor x \rfloor - (\lceil x \rceil - 1) = (\lfloor x \rfloor - \lceil x \rceil) + 1 = 0$.

Together, we have

$$f(x) = \begin{cases} -1 & \text{if } x \in \mathbb{Z}^- \cup \{0\} \\ 1 & \text{if } x \in \mathbb{Z}^+ \\ 0 & \text{if } x \notin \mathbb{Z} \end{cases},$$

so the graph of $f(x)$ is symmetric about **(D)** the point $\left(\frac{1}{2}, 0\right)$.

Alternatively, we can eliminate (A), (B), (C), and (E) once we finish with Case 3. This leaves us with (D).

Solution 4 (Casework)

Denote $x = a + b$, where $a \in \mathbb{Z}$ and $b \in [0, 1)$. Hence, a is the integer part of x and b is the decimal part of x .

Case 1: $b = 0$.

We have

$$\begin{aligned} f(x) &= ||x|| - ||1 - x|| \\ &= |a| - |1 - a| \\ &= \begin{cases} a - (a - 1) & \text{if } a \in \mathbb{Z} \text{ and } a \geq 1 \\ -1 & \text{if } a = 0 \\ -a - (1 - a) & \text{if } a \in \mathbb{Z} \text{ and } a \leq -1 \end{cases} \\ &= \begin{cases} 1 & \text{if } a \in \mathbb{Z} \text{ and } a \geq 1 \\ -1 & \text{if } a = 0 \\ -1 & \text{if } a \in \mathbb{Z} \text{ and } a \leq -1 \end{cases} \\ &= \begin{cases} 1 & \text{if } a \in \mathbb{Z} \text{ and } a \geq 1 \\ -1 & \text{if } a \in \mathbb{Z} \text{ and } a \leq 0 \end{cases} \end{aligned}$$

Case 2: $b \neq 0$.

We have

$$\begin{aligned} f(x) &= ||x|| - ||1 - x|| \\ &= |a| - ||1 - a - b|| \\ &= |a| - ||-a + (1 - b)|| \\ &= |a| - |-a| \\ &= 0. \end{aligned}$$

Therefore, the graph of $f(x)$ is symmetric through the point $(\frac{1}{2}, 0)$.

Therefore, the answer is (D) the point $(\frac{1}{2}, 0)$.

Solution 5 (Semi-Fakesolve)

Suppose $x \in \mathbb{Z}$, making the equation equivalent to $f(x) = |x| - |1 - x|$. We consider the cases when $x \in (-\infty, 0), 0, 1, (1, \infty)$.

If $x \in (-\infty, 0)$, we have $|x| = -x$ and $|1 - x| = 1 - x$, so $f(x) = -x - (1 - x) = -1$.

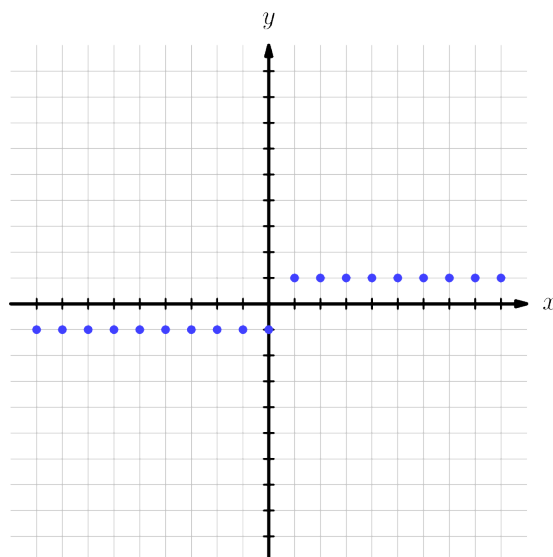
If $x = 0$ or $x = 1$, we trivially get $f(x) = -1$ and 1 respectively.

If $x \in (1, \infty)$, we have $|x| = x$ and $|1 - x| = x - 1$, giving $f(x) = x - (x - 1) = 1$.

Since, for all $x \in \mathbb{Z} \leq 0$, $f(x) = -1$ and $x \in \mathbb{Z} \geq 1$, $f(x) = 1$, we can conclude that it is symmetric across the coordinate pair

$$\left(\frac{0+1}{2}, \frac{-1+1}{2}\right) = \left(\frac{1}{2}, 0\right) = \text{(D) } \left(\frac{1}{2}, 0\right),$$

the midpoint of the "endpoints" of these line segments.



Just considering the integers is never a good idea when dealing with any function, especially one with floor functions. However, after dealing with the case "when $x \in \mathbb{Z}$ ", it becomes apparent that the graph of $f(x)$ is symmetric about $x = \frac{1}{2}$, or more specifically, the point $(\frac{1}{2}, 0)$.

-Benedict T (countmath1)

Solution 6 (Ultra Fast if you have like 20 sec left)

Take some simple values of x . Here we can take $-2, -1, 0, 1, 2$ and get $-1, -1, -1, 1, 1$

respectively. We easily see that choices A, B, C, E is already false, so **(D)** the point $(\frac{1}{2}, 0)$.

17. (AMC 10 10A Fall 2021) An architect is building a structure that will place vertical pillars at the vertices of regular hexagon $ABCDEF$, which is lying horizontally on the ground. The six pillars will hold up a flat solar panel that will not be parallel to the ground. The heights of pillars at A, B , and C are 12, 9, and 10 meters, respectively. What is the height, in meters, of the pillar at E ?

(A) 9 (B) $6\sqrt{3}$ (C) $8\sqrt{3}$ (D) 17 (E) $12\sqrt{3}$

Answer: D

Solution 1 (Height From the Center)

The pillar at B has height 9 and the pillar at A has height 12. Since the solar panel is flat, the inclination from pillar B to pillar A is 3. Call the center of the hexagon G . Since $\overrightarrow{CG} \parallel \overrightarrow{BA}$, it follows that the solar panel has height 13 at G . Since the solar panel is flat, the heights of the solar panel at B, G , and E are collinear. Therefore, the pillar at E has height $9 + 4 + 4 = \mathbf{(D) 17}$.

Solution 2 (Height From Each Vertex)

Let the height of the pillar at D be x . Notice that the difference between the heights of pillar C and pillar D is equal to the difference between the heights of pillar A and pillar F . So, the height at F is $x + 2$. Now, doing the same thing for pillar E we get the height is $x + 3$. Therefore, we can see the difference between the heights at pillar C and pillar D is half the difference between the heights at B and E , so

$$x + 3 - 9 = 2 \cdot (x - 10)$$

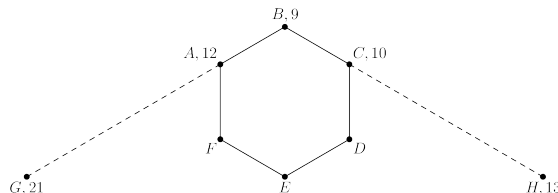
$$x - 6 = 2 \cdot (x - 10)$$

$$x = 14.$$

The answer is $x + 3 = \boxed{\text{(D)} 17}$.

Solution 3 (Extend the Sides)

We can extend BA and BC to G and H , respectively, such that $AG = CH$ and E lies on $\overline{GH}$:



Because of hexagon proportions, $\frac{BA}{AG} = \frac{1}{3}$ and $\frac{BC}{CH} = \frac{1}{3}$. Let g be the height of G . Because A , B and G lie on the same line, $\frac{12-9}{g-12} = \frac{1}{3}$, so $g - 12 = 9$ and $g = 21$. Similarly, the height of H is 13. E is the midpoint of GH , so we can take the average of these heights to get our answer, $\boxed{\text{(D)} 17}$.

Solution 4 (Averages of Heights)

Denote by h_X the height of any point X .

Denote by M the midpoint of A and C . Hence,

$$h_M = \frac{h_A + h_C}{2} = 11.$$

Denote by O the center of $ABCDEF$. Because $ABCDEF$ is a regular hexagon, O is the midpoint of B and E . Hence,

$$h_O = \frac{h_E + h_B}{2} = \frac{h_E + 9}{2}.$$

Because $ABCDEF$ is a regular hexagon, M is the midpoint of B and O . Hence,

$$h_M = \frac{h_B + h_O}{2} = \frac{9 + h_O}{2}.$$

Solving these equations, we get $h_E = \boxed{\text{(D)} 17}$.

Solution 5 (Vectors)

In this solution, we define **rise** as the change of height (in meters) from the solar panel to the ground. It follows that the rise from B to A is $12 - 9 = 3$, and the rise from B to C is $10 - 9 = 1$. Note that $\vec{BE} = 2\vec{BA} + 2\vec{BC}$, so the rise from B to E is $2 \cdot 3 + 2 \cdot 1 = 8$.

Together, the height of the pillar at E is $9 + 8 = \boxed{\text{(D)} 17}$ meters.

Solution 6 (Vectors)

WLOG, let the side length of the hexagon be 6.

Establish a 3D coordinate system, in which $A = (0, 0, 0)$. Let the coordinates of B and C be $(6, 0, 0)$, $(9, -3\sqrt{3}, 0)$, respectively. Then, the solar panel passes through $P = (0, 0, 12)$, $Q = (6, 0, 9)$, $R = (9, -3\sqrt{3}, 10)$.

The vector $\vec{PQ} = \langle 6, 0, -3 \rangle$ and $\vec{PR} = \langle 9, -3\sqrt{3}, -2 \rangle$. Computing $\vec{PQ} \times \vec{PR}$ by the matrix

$$\begin{bmatrix} i & j & k \\ 6 & 0 & -3 \\ 9 & -3\sqrt{3} & -2 \end{bmatrix}$$

gives the result $-9\sqrt{3}i - 15j - 18\sqrt{3}k$. Therefore, a normal vector of the plane of the solar panel is $\langle -9\sqrt{3}, -15, -18\sqrt{3} \rangle$, and the equation of the plane is $-9\sqrt{3}x - 15y - 18\sqrt{3}z = k$. Substituting $(x, y, z) = (0, 0, 12)$, we find that $k = -216\sqrt{3}$.

Since $E = (0, -6\sqrt{3})$, we substitute $(x, y) = (0, -6\sqrt{3})$ into $-9\sqrt{3}x - 15y - 18\sqrt{3}z = -216\sqrt{3}$, which gives $z = \boxed{\text{(D)} 17}$.

Solution 7 (3D Slopes)

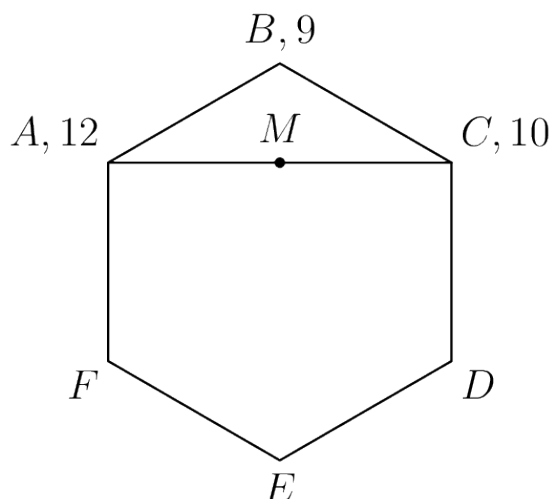
Let the pillars be $AA', BB', \dots, FF'$. Since solar panel $A'B'C'D'E'F'$ is a hexagon, the line $B'E'$ hits the midpoint M of $A'C'$. So, the 3D slope (change in x : change in y : change in z) of BE is same as BD . If a is side of the hexagonal solar panel,

$$B'M' = \frac{1}{2}a, B'E' = a + 2 \cdot \frac{1}{2}a = 2a$$

. So, $B'M : B'E' = 1 : 4$. Since the height of M to ground $ABCDEF$ is $(10 + 12)/2 = 11$, the rise (in z) from B' to M is 2 meaning the rise from B' to E' is 8. Thus, $EE' = 8 + BB' = \boxed{\text{(D)} 17}$.

Solution 8 (Midpoints, SIMPLE)

Set the midpoint of $\overline{AC}$ as M :



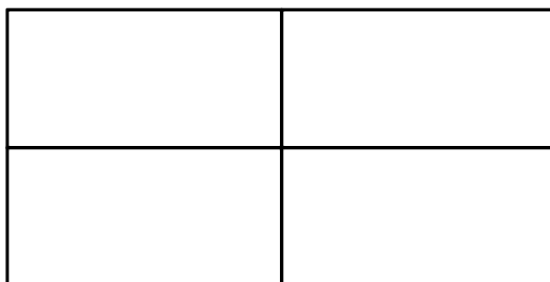
We know that the height of M is 11 as it is the midpoint of $\overline{AC}$, so the height is the average of A and C , which is $\frac{10+12}{2} = 11$. Since $ABCDEF$ is a regular hexagon, $BE = 4 \cdot BM$. Because the increase in height is proportional to the length of the line segments, and the increase in height from B to M is 2, the increase in height from B to E is $2 \cdot 4 = 8$. Adding to the height of B , we get $8 + 9 = \boxed{\text{(D)} 17}$.

Solution 9 (Educated Guess)

Because the three points given are integers, it is likely that the answer is also an integer. This leaves us with 9 or 17. Because both A and C are greater than 9

and closer to E than B , we can assume that the height increases as the point gets closer to E . Thus, we know the answer is greater than 9. The only choice that satisfies both these criteria is **(D) 17**.

18. (*AMC 10 10A Fall 2021*) A farmer's rectangular field is partitioned into 2 by 2 grid of 4 rectangular sections as shown in the figure. In each section the farmer will plant one crop: corn, wheat, soybeans, or potatoes. The farmer does not want to grow corn and wheat in any two sections that share a border, and the farmer does not want to grow soybeans and potatoes in any two sections that share a border. Given these restrictions, in how many ways can the farmer choose crops to plant in each of the four sections of the field?



- (A) 12 (B) 64 (C) 84 (D) 90 (E) 144

Answer: C

Solution 1 (Casework)

There are 4 possibilities for the top-left section. It follows that the top-right and bottom-left sections each have 3 possibilities, so they have $3^2 = 9$ combinations. We have two cases:

- a) **The top-right and bottom-left sections have the same crop.** Note that 3 of the 9 combinations of the top-right and bottom-left sections satisfy this case, from which the bottom-right section has 3 possibilities. Therefore, there are $4 \cdot 3 \cdot 3 = 36$ ways in this case.
- b) **The top-right and bottom-left sections have different crops.** Note that 6 of the 9 combinations of the top-right and bottom-left sections satisfy this case, from which the bottom-right section has 2 possibilities. Therefore, there are $4 \cdot 6 \cdot 2 = 48$ ways in this case.

Together, the answer is $36 + 48 = \mathbf{(C) 84}$.

Solution 2 (Casework)

We will do casework on the type of crops in the field.

Case 1: all of a kind.

If all four sections have the same type of crop, there are simply 4 ways to choose crops for the sections.

Case 2: 3 of a kind, 1 of another kind.

Since the one of another kind must be adjacent to two of the other crops, when choosing the type of crops in this case, we cannot choose soybeans and potatoes, or corn and wheat. Therefore, there are $4 \cdot 3 - 2 \cdot 2 = 8$ choices for the two crops we choose for the section (notice we did not choose by 2, since the crop we pick

first will be the unique one), and 4 ways to choose which section the unique crop is planted on. This gives us a total of $4 \cdot 8 = \underline{32}$ ways to choose crops for the sections.

Case 3: 2 of a kind, 2 of another kind.

We cannot choose corn and wheat, or soybeans and potatoes, once again, because if we do, the two would have to be adjacent in some way, which the problem disallows. So, there are $\binom{4}{2} - 2 = 4$ ways to choose our two crops (notice that we did choose by 2, since there are two of both crops). There are $\binom{4}{2} = 6$ ways to choose where one of the crops go, so there are $4 \cdot 6 = \underline{24}$ ways to choose crops for the sections.

Case 4: 2 of a kind, 1 of another kind, 1 of another kind.

In cases 2 and 3, we excluded the possibility of choosing bad pairs for our crops (i.e. soybeans and potatoes, or corn and wheat). In this case, it is inevitable that we choose a bad pair, because we are choosing 3 crops this time. The two sections of the same kind must contain the crop that is not part of the bad pair in the trio: for example, if we choose corn, soybeans and potatoes as our three crop types, nor soybeans and potatoes can be the type which occupies two sections in this case; corn must be the one to do so. There are 4 ways to choose the crop that is not part of the bad pair, and then 1 way to choose the bad pair, giving us $4 \cdot 1 = 4$ ways to choose the crops. To separate the bad pair of crops, the two of a kind must be diagonally placed. There are 2 ways to choose where the two of a kind go, and 2 ways to choose which of the bad pair goes where, giving us $2 \cdot 2 = 4$ ways to choose the positions for the crops. In total, there are $4 \cdot 4 = \underline{16}$ ways to choose crops for the sections.

Case 5: every single crop.

Bad pairs must be on the same diagonal, so there are 2 ways to choose which pair gets which diagonal, and $2 \cdot 2 = 4$ ways to choose which of each pair goes where on the diagonal, giving us $2 \cdot 4 = \underline{8}$ ways to choose crops for the sections.

Adding up all our values, we get our final answer of $4 + 32 + 24 + 16 + 8 = \boxed{\text{(C)} 84}$.

Solution 3 (Casework)

To lighten notation, we use C, W, S, P to denote corn, wheat, soybeans, and potatoes, respectively.

We use I, II, III, IV to denote four quadrants, respectively.

We determine an arrangement in the following steps.

Step 1: Determine the crop planted in I.

The number of ways is 4.

Step 2: Determine the crops planted in II, III, IV.

To find the number of arrangements in this step, without loss of generality, we assume that we plant C in I.

We do the following casework analysis.

Case 1: Both II and IV are planted with C.

In this case, the number of ways to plant in III is 3.

Case 2: In II and IV, only one quadrant is planted with C, and another quadrant is planted with either S or P.

In this case, we determine an arrangement in the following steps.

Step 2.1: Determine whether C is planted in II or IV.

The number of ways is 2.

Step 2.2: In II or IV not planted with C, determine whether it is planted with S or P.

The number of ways is 2.

Step 2.3: Determine the crop planted in III.

The number of ways is 2.

Following from the rule of product, the number of ways in this case is $2 \cdot 2 \cdot 2 = 8$.

Case 3: II and IV are both planted with S or W.

In this case, we determine an arrangement in the following steps.

Step 2.1: Determine whether S or W is planted in II and IV.

The number of ways is 2.

Step 2.2: Determine the crop planted in III.

The number of ways is 3.

Following from the rule of product, the number of ways in this case is $2 \cdot 3 = 6$.

Case 4: In II and IV, exactly one quadrant is planted with S and another one is planted with W.

Step 2.1: Determine which quadrant in II and IV is planted with S.

The number of ways is 2.

Step 2.2: Determine the crop planted in III.

The number of ways is 2.

Following from the rule of product, the number of ways in this case is $2 \cdot 2 = 4$.

Putting all cases together, the total number of arrangements in Step 2 is $3+8+6+4 = 21$.

Following from the rule of product, the total number of arrangements is $4 \cdot 21 = 84$.

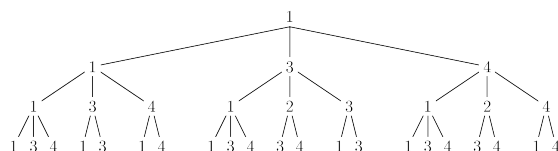
Therefore, the answer is $\boxed{(C) 84}$.

Solution 4 (Principle of Inclusion-Exclusion)

The number of cases with at least one pair of corn and wheat adjacent is $4 \cdot 2 \cdot 4^2 - 4 \cdot 2 \cdot 4 + 0 - 2 = 94$ possible fields (You can easily see this for yourself using PIE.), and WLOG, the same goes for soybean and potatoes. Now, applying PIE on both sets (number of cases with at least one pair of corn and wheat and the number of cases with at least one pair of soybeans and potatoes) yields $2 \cdot 94 - 16 = 172$. We want the number of cases **without** adjacent pairs of soybeans and potatoes or wheat and corn, so we subtract $256 - 172$, yielding an answer of $\boxed{(C) 84}$.

Solution 5 (Tree Diagram)

We can create a tree representing an arbitrary box we start with and the possibilities for other boxes around the grid. We can just designate the crop we start with as 1, and the other crops as 2, 3, and 4, where 1 cannot be next to 2 and 3 cannot be next to 4 in the grid.



As we can see, there are 21 possibilities. We multiply this by the 4 (the number of possibilities there are for the crop we started with), so our answer is **(C) 84**.

Solution 6 (Educated Guess)

The top right box has 4 choices and the top left box has 3 choices. Thus, it is reasonable to assume that the answer is a multiple of 12. We know that the answer will not be too small or too large, so the answer is **(C) 84**.

Solution 7 (Metasolve)

Estimation suggests B is too low, and clearly 90 is incorrect as it has a factor of 5 which is impossible to achieve with combinations less than 5. Thus, the answer is **(C) 84**.

19. (AMC 10 10A Fall 2021) A disk of radius 1 rolls all the way around the inside of a square of side length $s > 4$ and sweeps out a region of area A . A second disk of radius 1 rolls all the way around the outside of the same square and sweeps out a region of area $2A$. The value of s can be written as $a + \frac{b\pi}{c}$, where a, b , and c are positive integers and b and c are relatively prime. What is $a + b + c$?
- (A) 10 (B) 11 (C) 12 (D) 13 (E) 14

Answer: A

Solution 1

The side length of the inner square traced out by the disk with radius 1 is $s - 4$. However, there is a piece at each corner (bounded by two line segments and one 90° arc) where the disk never sweeps out. The combined area of these four pieces is $(1 + 1)^2 - \pi \cdot 1^2 = 4 - \pi$. As a result, we have

$$A = s^2 - (s - 4)^2 - (4 - \pi) = 8s - 20 + \pi.$$

Now, we consider the second disk. The part it sweeps is comprised of four quarter circles with radius 2 and four rectangles with side lengths of 2 and s . When we add it all together, we have $2A = 8s + 4\pi$, or

$$A = 4s + 2\pi.$$

We equate the expressions for A , and then solve for s :

$$8s - 20 + \pi = 4s + 2\pi.$$

We get $s = 5 + \frac{\pi}{4}$, so the answer is $5 + 1 + 4 = \mathbf{(A) 10}$.

20. (AMC 10 10A Fall 2021) For how many ordered pairs (b, c) of positive integers does neither $x^2 + bx + c = 0$ nor $x^2 + cx + b = 0$ have two distinct real solutions?
- (A) 4 (B) 6 (C) 8 (D) 12 (E) 16

Answer: B

Solution 1 (Casework)

A quadratic equation does not have two distinct real solutions if and only if the **discriminant** is nonpositive. We conclude that:

- a) Since $x^2 + bx + c = 0$ does not have real solutions, we have $b^2 \leq 4c$.
 b) Since $x^2 + cx + b = 0$ does not have real solutions, we have $c^2 \leq 4b$.

Squaring the first inequality, we get $b^4 \leq 16c^2$. Multiplying the second inequality by 16, we get $16c^2 \leq 64b$. Combining these results, we get

$$b^4 \leq 16c^2 \leq 64b.$$

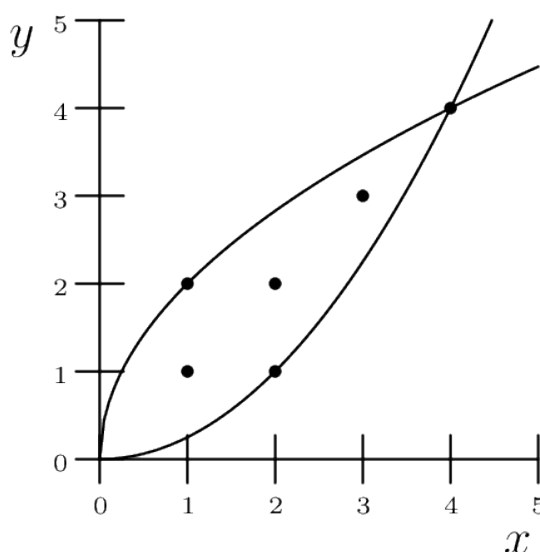
Since $b^4 \leq 64b$, it follows that $b \leq 4$. We apply casework to the value of b :

- If $b = 1$, then $1 \leq 16c^2 \leq 64$, from which $c = 1, 2$.
- If $b = 2$, then $16 \leq 16c^2 \leq 128$, from which $c = 1, 2$.
- If $b = 3$, then $81 \leq 16c^2 \leq 192$, from which $c = 3$.
- If $b = 4$, then $256 \leq 16c^2 \leq 256$, from which $c = 4$.

Together, there are (B) 6 ordered pairs (b, c) , namely $(1, 1), (1, 2), (2, 1), (2, 2), (3, 3)$, and $(4, 4)$.

Solution 2 (Graphing)

Similar to Solution 1, use the discriminant to get $b^2 \leq 4c$ and $c^2 \leq 4b$. These can be rearranged to $c \geq \frac{1}{4}b^2$ and $b \geq \frac{1}{4}c^2$. Now, we can roughly graph these two inequalities, letting one of them be the x axis and the other be y . The graph of solutions should be above the parabola and under its inverse, meaning we want points on the graph or in the first area enclosed by the two graphs:



We are looking for lattice points (since b and c are positive integers), of which we can count (B) 6.

Solution 3 (Graphing)

We need to solve the following system of inequalities:

$$\begin{cases} b^2 - 4c \leq 0 \\ c^2 - 4b \leq 0 \end{cases}.$$

Feasible solutions are in the region formed between two parabolas $b^2 - 4c = 0$ and $c^2 - 4b = 0$.

Define $f(b) = \frac{b^2}{4}$ and $g(b) = 2\sqrt{b}$. Therefore, all feasible solutions are in the region formed between the graphs of these two functions.

For $b = 1$, we have $f(b) = \frac{1}{4}$ and $g(b) = 2$. Hence, the feasible c are 1, 2.

For $b = 2$, we have $f(b) = 1$ and $g(b) = 2\sqrt{2}$. Hence, the feasible c are 1, 2.

For $b = 3$, we have $f(b) = \frac{9}{4}$ and $g(b) = 2\sqrt{3}$. Hence, the feasible c is 3.

For $b = 4$, we have $f(b) = 4$ and $g(b) = 4$. Hence, the feasible c is 4.

For $b > 4$, we have $f(b) > g(b)$. Hence, there is no feasible c .

Putting all cases together, the correct answer is **(B) 6**.

Solution 4 (Oversimplified but Risky)

A quadratic equation $Ax^2 + Bx + C = 0$ has one real solution if and only if $B^2 - 4AC = 0$. Similarly, it has imaginary solutions if and only if $B^2 - 4AC < 0$. We proceed as following:

We want both $x^2 + bx + c$ to be 1 value or imaginary and $x^2 + cx + b$ to be 1 value or imaginary. $x^2 + 4x + 4$ is one such case since $\sqrt{b^2 - 4ac}$ is 0. Also, $x^2 + 3x + 3$, $x^2 + 2x + 2$, $x^2 + x + 1$ are always imaginary for both b and c . We also have $x^2 + x + 2$ along with $x^2 + 2x + 1$ since the latter has one solution, while the first one is imaginary. Therefore, we have **(B) 6** total ordered pairs of integers.

Solution 5 (Quick and Easy)

We see that $b^2 \leq 4c$ and $c^2 \leq 4b$. WLOG, assume that $b \geq c$. Then we have that $b^2 \leq 4c \leq 4b$, so $b^2 \leq 4b$ and therefore $b \leq 4$, also meaning that $c \leq 4$. This means that we only need to try 16 cases. Now we can get rid of the assumption that $b \geq c$, because we want ordered pairs. For $b = 1$ and $b = 2$, $c = 1$ and $c = 2$ work. When $b = 3$, c can only be 3, and when $b = 4$, only $c = 4$ works, for a total of **(B) 6** ordered pairs of integers.

Solution 6 (Fastest)

We need both $b^2 \leq 4c$ and $c^2 \leq 4b$.

If $b = c$ then the above become $b^2 \leq 4b \iff b \leq 4$, so we have four solutions (k, k) , where $k = 1, 2, 3, 4$.

If $b < c$ then we only need $c^2 \leq 4b$ since it implies $b^2 < 4c$. Now $c^2 \leq 4b \leq 4(c-1) \implies (c-2)^2 \leq 0 \implies c = 2$, so $b = 1$. We plug $b = 1$, $c = 2$ back into $c^2 \leq 4b$ and it works. So there is another solution $(1, 2)$.

By symmetry, if $b > c$ then $(b, c) = (2, 1)$.

Therefore the total number of solutions is **(B) 6**.

Solution 7 (Shortest)

Since $b^2 - 4c \leq 0$ and $c^2 - 4b \leq 0$, adding the two together yields $b^2 + c^2 \leq 4(c + b)$. Obviously, this is not true if either b or c get too large, and they are equal when $b = c = 4$, so the greatest pair is $(4, 4)$ and both numbers must be lesser for further pairs. For there to be two distinct real solutions, we can test all these pairs where (b, c) are less than 4 (except for the already valid solution) on the original quadratics, and we find the working pairs are $(1, 1)$, $(2, 1)$, $(1, 2)$, $(2, 2)$, $(3, 3)$, $(4, 4)$ meaning there are **(B) 6** pairs.

- youtube.com/indianmathguy

21. (AMC 10 10A Fall 2021) Each of the 20 balls is tossed independently and at random into one of the 5 bins. Let p be the probability that some bin ends up with 3 balls, another with 5 balls, and the other three with 4 balls each. Let q be the probability that every bin ends up with 4 balls. What is $\frac{p}{q}$?
- (A) 1 (B) 4 (C) 8 (D) 12 (E) 16

Answer: E

Solution 1 (Multinomial Coefficients)

For simplicity purposes, we assume that the balls and the bins are both distinguishable.

Recall that there are 5^{20} ways to distribute 20 balls into 5 bins. For p , we choose one of the 5 bins to have 3 balls and another one of the 4 bins to have 5 balls. We get

$$p = \frac{5 \cdot 4 \cdot \binom{20}{3,5,4,4,4}}{5^{20}} \text{ and } q = \frac{\binom{20}{4,4,4,4,4}}{5^{20}}.$$

Therefore, the answer is

$$\frac{p}{q} = \frac{5 \cdot 4 \cdot \binom{20}{3,5,4,4,4}}{\binom{20}{4,4,4,4,4}} = \frac{5 \cdot 4 \cdot \frac{20!}{3! \cdot 5! \cdot 4! \cdot 4! \cdot 4!}}{\frac{20!}{4! \cdot 4! \cdot 4! \cdot 4! \cdot 4!}} = \frac{5 \cdot 4 \cdot (4! \cdot 4! \cdot 4! \cdot 4! \cdot 4!)}{3! \cdot 5! \cdot 4! \cdot 4! \cdot 4!} = \frac{5 \cdot 4 \cdot 4}{5} = \boxed{\text{(E)} 16}.$$

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Solution 2

For simplicity purposes, we assume that the balls and the bins are both distinguishable.

Let $q = \frac{x}{a}$, where a is the total number of combinations and x is the number of cases where every bin ends up with 4 balls.

We can take 1 ball from one bin and place it in another bin so that some bin ends up with 3 balls, another with 5 balls, and the other three with 4 balls each. Note that one configuration of 4-4-4-4-4 corresponds to $5 \cdot 4 \cdot 4 = 80$ configurations of 3-5-4-4-4. On the other hand, one configuration of 3-5-4-4-4 corresponds to 5 configurations of 4-4-4-4-4.

Therefore, we have

$$p = \frac{80}{5} \cdot \frac{x}{a} = 16 \cdot \frac{x}{a},$$

from which $\frac{p}{q} = \boxed{\text{(E)} 16}$.

Solution 3 (Binomial Coefficients)

Since both of the cases will have 3 bins with 4 balls in them, we can leave those out. There are $2 \cdot \binom{5}{2} = 20$ ways to choose where to place the 3 and the 5. After that, there are $\binom{8}{3} = 56$ ways to put the 3 and 5 balls being put into the bins. For the 4, 4, 4, 4, 4 case, after we canceled the 4, 4, 4 out, we have $\binom{8}{4} = 70$ ways to put the 4 balls inside the bins. Therefore, we have $\frac{56 \cdot 20}{70}$ which is equal to $8 \cdot 2 = \boxed{\text{(E)} 16}$.

Solution 4 (Set Theory / Graph Theory)

Construct the set A consisting of all possible 3–5–4–4–4 bin configurations, and construct set B consisting of all possible 4–4–4–4–4 configurations. If we let N be the total number of configurations possible, it's clear we want to solve for $\frac{p}{q} = \frac{\frac{|A|}{N}}{\frac{|B|}{N}} = \frac{|A|}{|B|}$.

Consider drawing an edge between an element in A and an element in B if it is possible to reach one configuration from the other by moving a single ball (Note this process is reversible.). Let us consider the total number of edges drawn.

For any element in A , we may choose one of the 5 balls in the 5-bin and move it to the 3-bin to get a valid element in B . This implies the number of edges is $5|A|$.

On the other hand, for any element in B , we may choose one of the 20 balls and move it to one of the other 4-bins to get a valid element in A . This implies the number of edges is $80|B|$.

We equate the expressions to get $5|A| = 80|B|$, from which $\frac{|A|}{|B|} = \frac{80}{5} = \boxed{\text{(E)} 16}$.

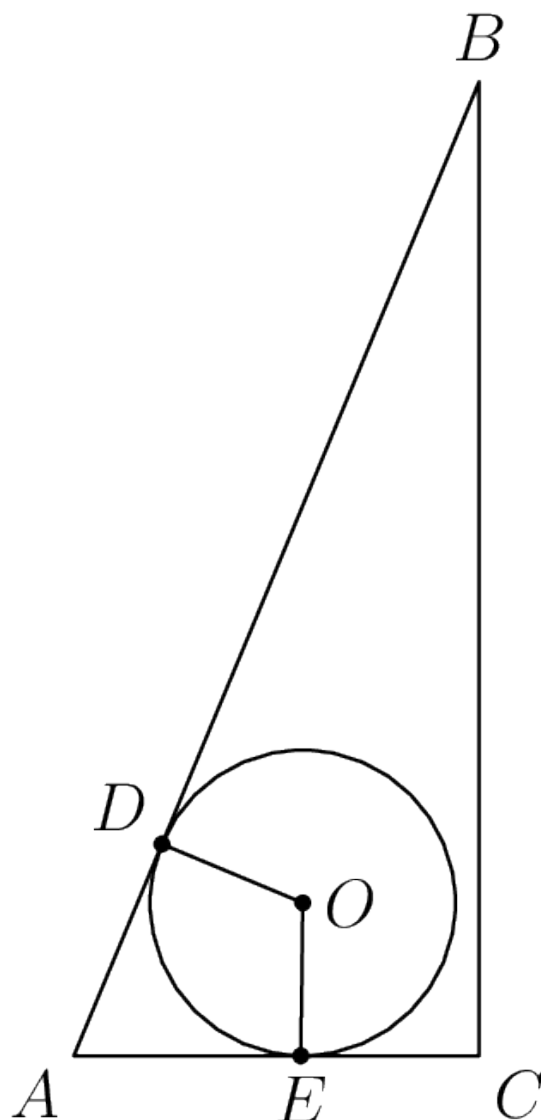
22. (AMC 10 10A Fall 2021) Inside a right circular cone with base radius 5 and height 12 are three congruent spheres with radius r . Each sphere is tangent to the other two spheres and also tangent to the base and side of the cone. What is r ?

(A) $\frac{3}{2}$ (B) $\frac{90-40\sqrt{3}}{11}$ (C) 2 (D) $\frac{144-25\sqrt{3}}{44}$ (E) $\frac{5}{2}$

Answer: B

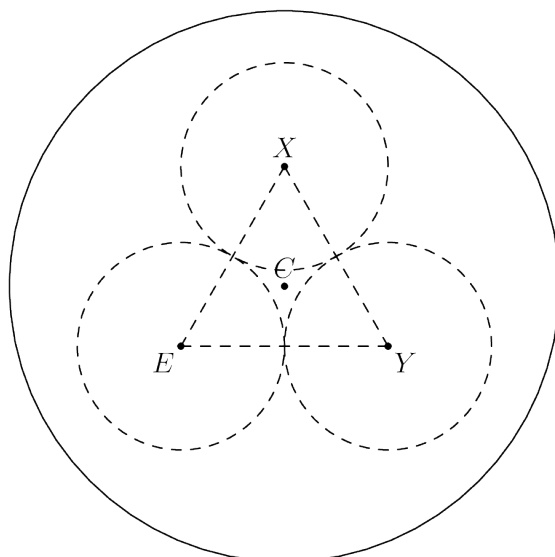
Solution 1 (Cross Sections and Angle Bisectors)

We can take half of a cross section of the sphere, as such:

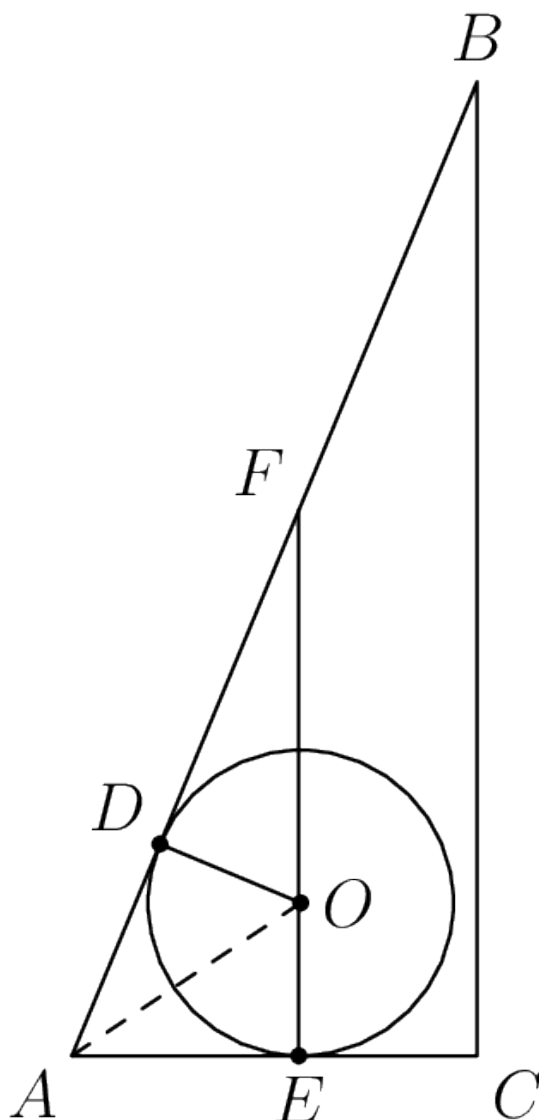


Notice that we chose a cross section where one of the spheres was tangent to the lateral surface of the cone at D .

To evaluate r , we will find AE and EC in terms of r ; we also know that $AE + EC = 5$, so with this, we can solve r . Firstly, to find EC , we can take a bird's eye view of the cone:



Note that C is the centroid of equilateral triangle EXY . Also, since all of the medians of an equilateral triangle are also altitudes, we want to find two-thirds of the altitude from E to XY ; this is because medians cut each other into a 2 to 1 ratio. This equilateral triangle has a side length of $2r$, therefore it has an altitude of length $r\sqrt{3}$; two thirds of this is $\frac{2r\sqrt{3}}{3}$, so $EC = \frac{2r\sqrt{3}}{3}$.



To evaluate AE in terms of r , we will extend $\overline{OE}$ past point O to $\overline{AB}$ at point F . $\triangle AEF$ is similar to $\triangle ACB$. Also, AO is the angle bisector of $\angle EAB$. Therefore, by the angle bisector theorem, $\frac{OE}{OF} = \frac{AE}{AF} = \frac{5}{13}$. Also, $OE = r$, so $\frac{r}{OF} = \frac{5}{13}$, so $OF = \frac{13r}{5}$. This means that

$$AE = \frac{5 \cdot EF}{12} = \frac{5 \cdot (OE + OF)}{12} = \frac{5 \cdot (r + \frac{13r}{5})}{12} = \frac{18r}{12} = \frac{3r}{2}.$$

We have that $EC = \frac{2r\sqrt{3}}{3}$ and that $AE = \frac{3r}{2}$, so $AC = EC + AE = \frac{2r\sqrt{3}}{3} + \frac{3r}{2} = \frac{4r\sqrt{3}+9r}{6}$. We also were given that $AC = 5$. Therefore, we have

$$\frac{4r\sqrt{3} + 9r}{6} = 5.$$

This is a simple linear equation in terms of r . We can solve for r to get $r =$

(B) $\frac{90 - 40\sqrt{3}}{11}$.

Solution 2 (Cross Sections and Areas)

Denote by O_1, O_2, O_3 the centers of three spheres.

Because three congruent spheres are tangent to the base of the cone, the plane formed by O_1, O_2, O_3 (denoted as α) is parallel to the base, with the distance r .

Let D be the point that the sphere with center O_1 meets the base of the cone at. Hence, $O_1D = r$.

Because three congruent spheres are mutually externally tangent to each other, $\triangle O_1O_2O_3$ is equilateral, with side length $2r$.

Let O be the center of the base, V be the vertex of the cone. Let line OV and plane α intersect at point E . By symmetry, E is the center $\triangle O_1O_2O_3$. Hence, $O_1E = \frac{\sqrt{3}}{3}O_1O_2 = \frac{2\sqrt{3}}{3}r$.

Let F be the point that the sphere with center O_1 meets the side of the cone at. Hence, $O_1F = r$.

Let line VF and the base intersect at point A .

Hence, we only need to analyze the following 2-d geometry problem: In $\triangle VOA$ with $\angle O = 90^\circ$, $VO = 12$, $OA = 5$, there is an interior point O_1 whose distances to OA , OV , VA , are r , $\frac{2\sqrt{3}}{3}r$, and r , respectively. What is r ?

Now, we solve this problem.

We compute the area of $\triangle VOA$ in two ways.

First, we have

$$\text{Area } \triangle VOA = \frac{1}{2}OA \cdot OV = 30.$$

Second, we have

$$\begin{aligned} \text{Area } \triangle VOA &= \text{Area } \triangle O_1OA + \text{Area } \triangle O_1OV + \text{Area } \triangle O_1VA \\ &= \frac{1}{2} \cdot OA \cdot O_1D + \frac{1}{2} \cdot OV \cdot O_1E + \frac{1}{2} \cdot VA \cdot O_1F \\ &= \frac{1}{2} \cdot 5 \cdot r + \frac{1}{2} \cdot 12 \cdot \frac{2\sqrt{3}}{3} \cdot r + \frac{1}{2} \cdot 13 \cdot r \\ &= (9 + 4\sqrt{3})r. \end{aligned}$$

These two approaches to compute $\text{Area } \triangle VOA$ should give me the same number. Hence,

$$r = \frac{30}{9 + 4\sqrt{3}} = \boxed{\text{(B)} \frac{90 - 40\sqrt{3}}{11}}.$$

Solution 3 (Coordinate Geometry)

We will use coordinates. WLOG, let the coordinates of the center of the base of the cone be the origin. Then, let the center of one of the spheres be $(0, \frac{2r}{\sqrt{3}}, r)$. Note that the distance between this point and the plane given by $12y + 5z = 60$ is r . Thus, by the point-to-plane distance formula, we have

$$\frac{\left| 12 \cdot \frac{2r}{\sqrt{3}} + 5r - 60 \right|}{\sqrt{0^2 + 5^2 + 12^2}} = r.$$

$$\text{Solving for } r \text{ yields } r = \boxed{\text{(B)} \frac{90 - 40\sqrt{3}}{11}}.$$

Solution 4 (Very Cheesy and Risky)

Clearly the answer must have square roots, leaving (B) and (D). D has a lot of perfect squares, so we suspect it is a trap. This leads us to choose (B).

23. (AMC 10 10A Fall 2021) For each positive integer n , let $f_1(n)$ be twice the number of positive integer divisors of n , and for $j \geq 2$, let $f_j(n) = f_1(f_{j-1}(n))$. For how many values of $n \leq 50$ is $f_{50}(n) = 12$? (A) 7 (B) 8 (C) 9 (D) 10 (E) 11

Answer: D

Solution 1

First, we can test values that would make $f(x) = 12$ true. For this to happen x must have 6 divisors, which means its prime factorization is in the form pq^2 or p^5 , where p and q are prime numbers. Listing out values less than 50 which have these prime factorizations, we find 12, 18, 20, 28, 44, 45, 50 for pq^2 , and just 32 for p^5 . Here 12 especially catches our eyes, as this means if one of $f_i(n) = 12$, each of $f_{i+1}(n), f_{i+2}(n), \dots$ will all be 12. This is because $f_{i+1}(n) = f(f_i(n))$ (as given in the problem statement), so were $f_i(n) = 12$, plugging this in we get $f_{i+1}(n) = f(12) = 12$, and thus the pattern repeats. Hence, as long as for a i , such that $i \leq 50$ and $f_i(n) = 12$, $f_{50}(n) = 12$ must be true, which also immediately makes all our previously listed numbers, where $f_1(x) = 12$, possible values of n .

We also know that if $f_1(x)$ were to be any of these numbers, x would satisfy $f_{50}(n)$ as well. Looking through each of the possibilities aside from 12, we see that $f_1(x)$ could only possibly be equal to 20 and 18, and still have x less than or equal to 50. This is because if $f_1(x) = 20$ or 18, $f_2(x) = 12$. This would mean x must have 10, or 9 divisors, and testing out, we see that x will then be of the form p^4q , or p^2q^2 . The only two values less than or equal to 50 would be 48 and 36 respectively. From here there are no more possible values, so tallying our possibilities we count (D) 10 values (Namely 12, 18, 20, 28, 32, 36, 44, 45, 48, 50).

Solution 2 (Rigorous reasoning on why there cannot be any other solutions)

First, take note that the maximum possible value of $f_1(n)$ for $1 \leq n \leq k$ increases as k increases. When we increase k , we either include a new value of n that has more divisors than any previous values of n , increasing the maximum of $f_1(x)$, or none of the new n have more divisors, which would leave the maximum of $f_1(x)$ the same as before. In other words, $f_1(x)$ is increasing. This means that $f_1(x)$ decreases when we decrease k .

Also, let $f_1(n) = 2d(n)$ where $d(n)$ is the number of divisors of n .

Since $n \leq 50$, $f_1(n) \leq 20$. This maximum occurs when $d(n) = 10 \implies n = 2^4 \cdot 3 = 48$. Next, since $f_1(n) \leq 20$, $f_1(f_1(n)) \leq 12 \implies f_2(n) \leq 12$. This maximum occurs when $d(f_1(n)) = 6 \implies n = 2 \cdot 3^2 = 18, n = 2^2 \cdot 3 = 12$. Since $f_2(n) \leq 12$, $f_1(f_2(n)) \leq 12 \implies f_3(n) \leq 12$, once again. This maximum again occurs when $d(f_2(n)) = 6 \implies f_2(n) = 2^2 \cdot 3 = 12$. Notice that, from this point on, for $k \geq 3$, $f_k(n) \leq 12$ by a straightforward induction argument.

On a hunch, we investigate what happens when $f_2(n) < 12$, as this is the first time when $f_k(n)$ is bounded to a maximum of 12. Suppose for the sake of contradiction that $f_2(n) < 12$. Then, $f_3(n) < 12$, as $f_2(n) = 12$ is the only number that would maximize $f_3(n)$ for $f_2(n) \leq 12$. As a result, since $f_1(n)$ is increasing, and because 12 is the only value for which f_1 is maximized, f_1 "steps down" from a maximum of

$6 \cdot 2 = 12$ if $n < 12$. This means we must have that $f_1(f_3(n)) < f_1(12) = 12 \implies f_4(n) < 12$. We continue applying f_1 on both sides (which is possible since f_1 is increasing) until we reach f_{50} , giving us that $f_{50}(n) < 12$. However, $f_{50}(n) = 12$, which is a contradiction. Thus, $f_2(n) = 12$.

Now, we can finally solve for the solutions with a clean conscience - we proceed similarly to Solution 1. $f_2(n) = 12 \implies f_1(f_1(n)) = 12 \implies d(f_1(n)) = 6$. $d(f_1(n)) = 6$. This gives us two cases. First, we have the case where $f_1(n) = p^2 \cdot q$ where p and q are primes. Second, we have the case where $f_1(n) = p^5$ where p is a prime. For both cases, since $f_1(n) \leq 20$, $f_1(n)$ can only be 12, 18, or 20.

If $f_1(n) = 12$, then $d(n) = 6 \implies n = p^5, p^2 \cdot q \implies n \in \{12, 18, 20, 28, 32, 44, 45, 50\}$, resulting in 8 solutions.

If $f_1(n) = 18$, then $d(n) = 9 \implies n = p^8, p^2 \cdot q^2 \implies n = 36$, giving us one more solution. Finally, $f_1(n) = 20 \implies d(n) = 10 \implies n = p^9, p^4 \cdot q \implies n = 48$.

Thus, in total, we have (D)10 solutions.

Solution 3

Observation 1: $f_1(12) = 12$.

Hence, if n has the property that $f_j(n) = 12$ for some j , then $f_k(n) = 12$ for all $k > j$.

Observation 2: $f_1(8) = 8$.

Hence, if n has the property that $f_j(n) = 8$ for some j , then $f_k(n) = 8$ for all $k > j$.

Case 1: $n = 1$.

We have $f_1(n) = 2$, $f_2(n) = f_1(2) = 4$, $f_3(n) = f_1(4) = 6$, $f_4(n) = f_1(6) = 8$. Hence, Observation 2 implies $f_{50}(n) = 8$.

Case 2: n is prime.

We have $f_1(n) = 4$, $f_2(n) = f_1(4) = 6$, $f_3(n) = f_1(6) = 8$. Hence, Observation 2 implies $f_{50}(n) = 8$.

Case 3: The prime factorization of n takes the form p_1^2 .

We have $f_1(n) = 6$, $f_2(n) = f_1(6) = 8$. Hence, Observation 2 implies $f_{50}(n) = 8$.

Case 4: The prime factorization of n takes the form p_1^3 .

We have $f_1(n) = 8$. Hence, Observation 2 implies $f_{50}(n) = 8$.

Case 5: The prime factorization of n takes the form p_1^4 .

We have $f_1(n) = 10$, $f_2(n) = f_1(10) = 8$. Hence, Observation 2 implies $f_{50}(n) = 8$.

Case 6: The prime factorization of n takes the form p_1^5 .

We have $f_1(n) = 12$. Hence, Observation 1 implies $f_{50}(n) = 12$.

In this case the only n is $2^5 = 32$.

Case 7: The prime factorization of n takes the form $p_1 p_2$.

We have $f_1(n) = 8$. Hence, Observation 2 implies $f_{50}(n) = 8$.

Case 8: The prime factorization of n takes the form $p_1 p_2^2$.

We have $f_1(n) = 12$. Hence, Observation 1 implies $f_{50}(n) = 12$.

In this case, all n are 12, 18, 20, 28, 44, 45, and 50.

Case 9: The prime factorization of n takes the form $p_1 p_2^3$.

We have $f_1(n) = 16$, $f_2(n) = f_1(16) = 10$, $f_3(n) = f_1(10) = 8$. Hence, Observation 2 implies $f_{50}(n) = 8$.

Case 10: The prime factorization of n takes the form $p_1 p_2^4$.

We have $f_1(n) = 20$, $f_2(n) = f_1(20) = 12$. Hence, Observation 1 implies $f_{50}(n) = 12$.

In this case, the only n is 48.

Case 11: The prime factorization of n takes the form $p_1^2 p_2^2$.

We have $f_1(n) = 18$, $f_2(n) = f_1(18) = 12$. Hence, Observation 1 implies $f_{50}(n) = 12$.

In this case, the only n is 36.

Case 12: The prime factorization of n takes the form $p_1 p_2 p_3$.

We have $f_1(n) = 16$, $f_2(n) = f_1(16) = 10$, $f_3(n) = f_2(10) = 8$. Hence, Observation 2 implies $f_{50}(n) = 8$.

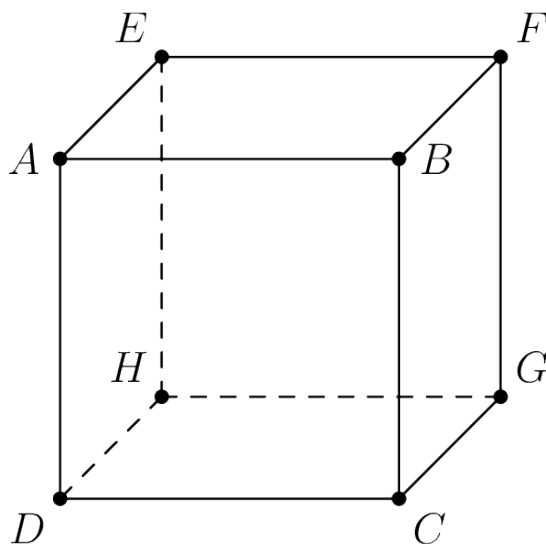
Putting all cases together, the number of feasible $n \leq 50$ is **(D) 10**.

24. (AMC 10 10A Fall 2021) Each of the 12 edges of a cube is labeled 0 or 1. Two labelings are considered different even if one can be obtained from the other by a sequence of one or more rotations and/or reflections. For how many such labelings is the sum of the labels on the edges of each of the 6 faces of the cube equal to 2?
- (A) 8 (B) 10 (C) 12 (D) 16 (E) 20

Answer: E

Solution 1

For simplicity, we will name this cube $ABCDEFGH$ by vertices, as shown below.

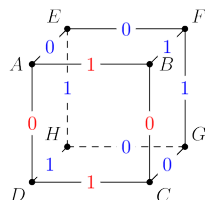


Note that for each face of this cube, two edges are labeled 0, and two edges are labeled 1. For all twelve edges of this cube, we conclude that six edges are labeled 0, and six edges are labeled 1.

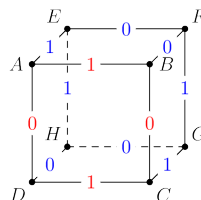
We apply casework to face $ABCD$. Recall that there are $\binom{4}{2} = 6$ ways to label its edges:

- a) **Opposite edges have the same label.** There are 2 ways to label the edges of $ABCD$. We will consider one of the ways, then multiply the count by 2. Without loss of generality, we assume that $\overline{AB}, \overline{BC}, \overline{CD}, \overline{DA}$ are labeled 1, 0, 1, 0, respectively:

We apply casework to the label of $\overline{AE}$, as shown below.



The label of $\overline{AE}$ is 0.

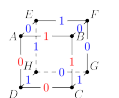


The label of $\overline{AE}$ is 1.

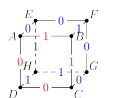
We have $2 \cdot 2 = 4$ such labelings for this case.

- b) **Opposite edges have different labels.** There are 4 ways to label the edges of $ABCD$. We will consider one of the ways, then multiply the count by 4. Without loss of generality, we assume that $\overline{AB}, \overline{BC}, \overline{CD}, \overline{DA}$ are labeled 1, 1, 0, 0, respectively:

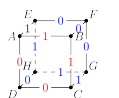
We apply casework to the labels of $\overline{AE}$ and $\overline{BF}$, as shown below.



The label of $\overline{AE}$ is 0.
The label of $\overline{BF}$ is 0.



The label of $\overline{AE}$ is 0.
The label of $\overline{BF}$ is 1.



The label of $\overline{AE}$ is 1.
The label of $\overline{BF}$ is 0.

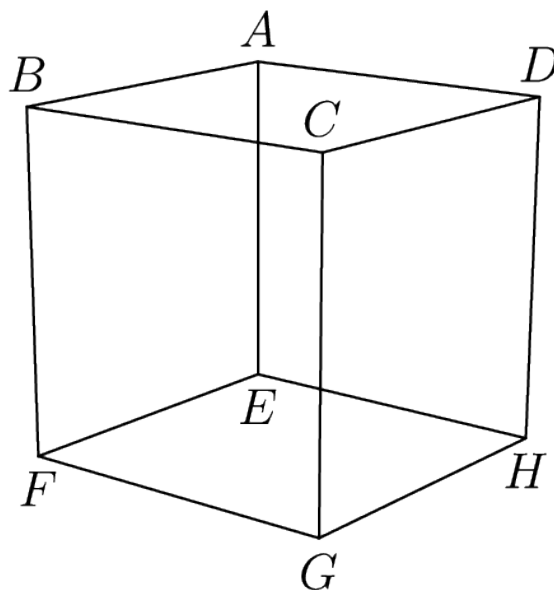


The label of $\overline{AE}$ is 1.
The label of $\overline{BF}$ is 1.

We have $4 \cdot 4 = 16$ such labelings for this case.

Therefore, we have $4 + 16 = \boxed{\text{(E)} 20}$ such labelings in total.

Solution 1.5



We will do casework on face $ABCD$.

Case 1: Two adjacent edges on face $ABCD$ are both labeled 1.

There are 4 ways to choose two adjacent edges on the top face. WLOG say we choose edges BC and CD to be labeled 1. Then AB and AD must be labeled 0.

We have three subcases for face $BCGF$:

SC1: FG is labeled 1

Then the rest of the cube becomes uniquely determined.

SC2: CG is labeled 1

The cube still becomes uniquely determined.

SC3: BF is labeled 1

If BF is labeled 1, we can choose either DH or GH to be labeled 1 on face $CDHG$. Then the rest of the cube is uniquely determined.

In total we get $4(1 + 1 + 2) = 16$ ways for this case.

Case 2: Two opposite edges on face $ABCD$ are both labeled 1

There are 2 ways to choose the opposite edges. WLOG say we choose edges BC and AD . Then for face $BCGF$ we can choose either CG or BF to be labeled 1 (we can't choose FG because that would force an invalid labeling).

This case gives us $2 \cdot 2 = 4$ ways.

So the answer is $16 + 4 = \boxed{20}$.

Solution 2

Since we want the sum of the edges of each face to be 2, we need there to be two 1s and two 0s on each face. Through experimentation, we find that either 2, 4, or all of them have 1s adjacent to 1s and 0s adjacent to 0 on each face. WLOG, let the first face (counterclockwise) be 0, 0, 1, 1. In this case we are trying to have all of them be adjacent to each other. First face: 0, 0, 1, 1. Second face: 2 choices: 1, 0, 0, 1 or 0, 0, 1, 1. After that, it is basically forced and everything will fall in to place. Since we assumed WLOG, we need to multiply 2 by 4 to get a total of 8 different arrangements.

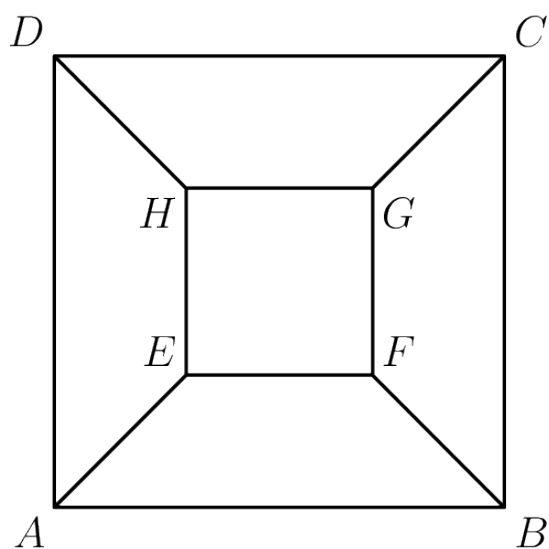
Secondly, 4 of the faces have all of them adjacent and 2 of the faces do not: WLOG counting counterclockwise, we have 0, 0, 1, 1. Then, we choose the other face next to it. There are two cases, which are 0, 1, 0, 1 and 1, 0, 1, 0. Therefore, this subcase has 4 different arrangements. Then, we can choose the face at front to be 1, 0, 1, 0. This has 4 cases. The sides can either be 0, 1, 1, 0 or 1, 1, 0, 0. Therefore, we have another 8 cases.

Summing these up, we have $8 + 4 + 8 = 20$. Therefore, our answer is $\boxed{(E) 20}$.

<u>Remark</u>

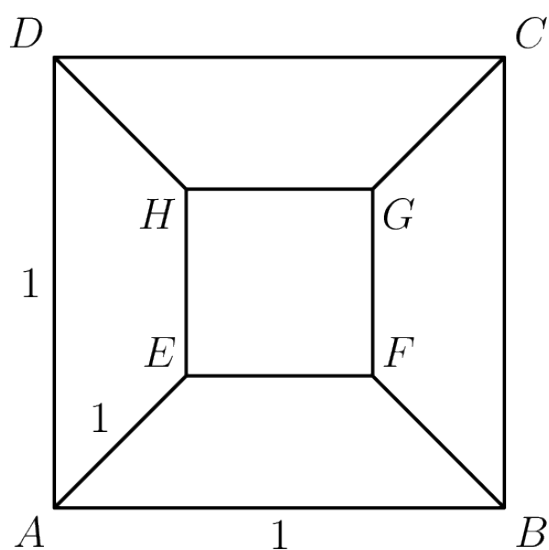
It is very easy to get disorganized when counting, resulting in incorrect calculations, so when doing this problem, make sure to draw a diagram of the cube. Labeling is a bit harder, since we often confuse one side with another. Try doing the problem by labeling sides on the lines (literally letting the lines pass through your 0s and 1s.) I found that to be very helpful when solving this problem.

Solution 3

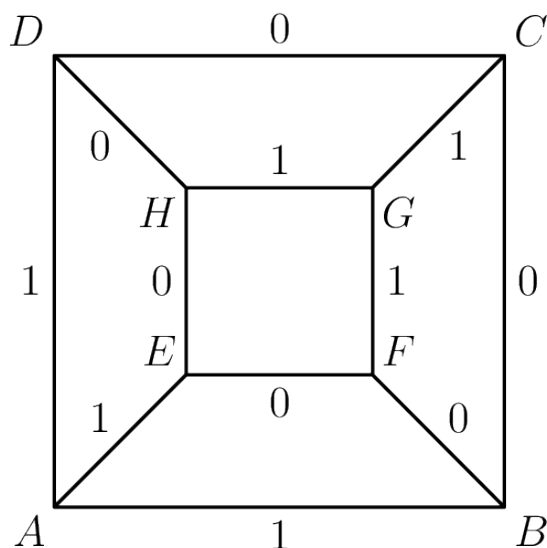


We see that each face has to have 2 1's and 2 0's. We can start with edges connecting to A.

Case 1



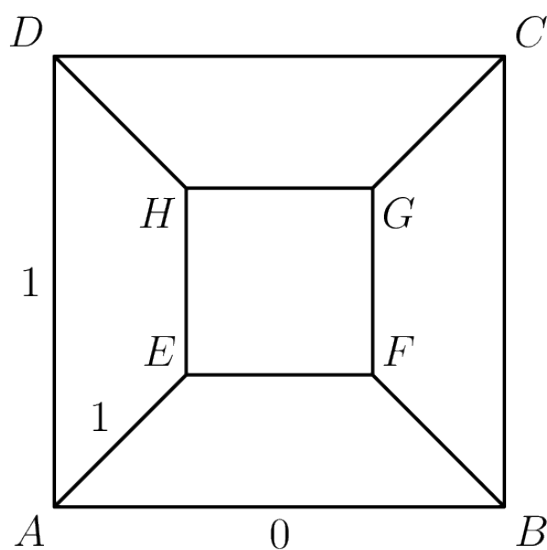
This goes to:



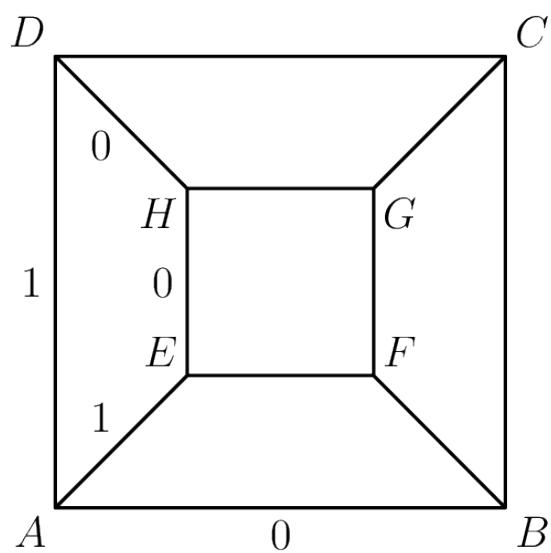
We can see that we choose 2 diametrically opposite vertices to put 3 1's on the connecting edges. As a result, this case has $\frac{8}{2} = 4$ orientations.

An alternate way to count would be to realize that you can rotate the configuration four times yielding four different configurations, but when you reflect horizontally and diagonally, you don't get another unique configuration, rather you get the same configuration as if you rotated it. mathboy282

Case 2

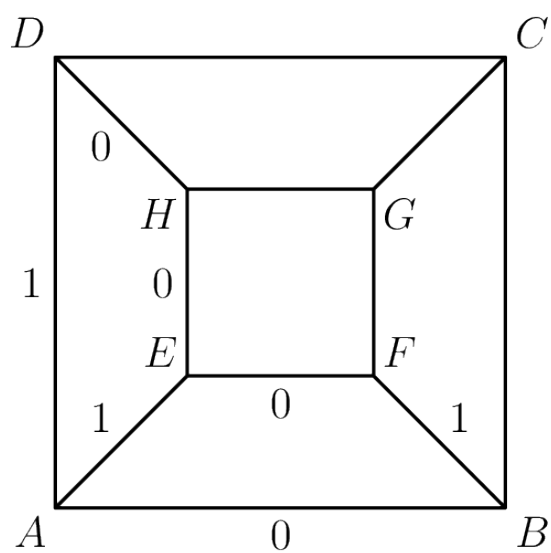


Filling out a bit more, we have:

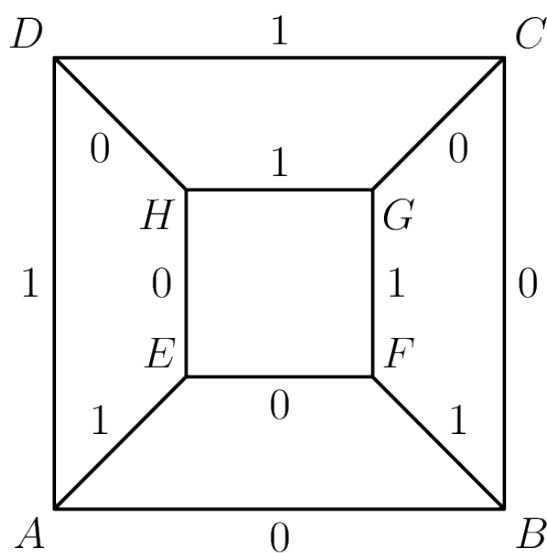


Let's try filling out BC and CD first.

Case 2.1

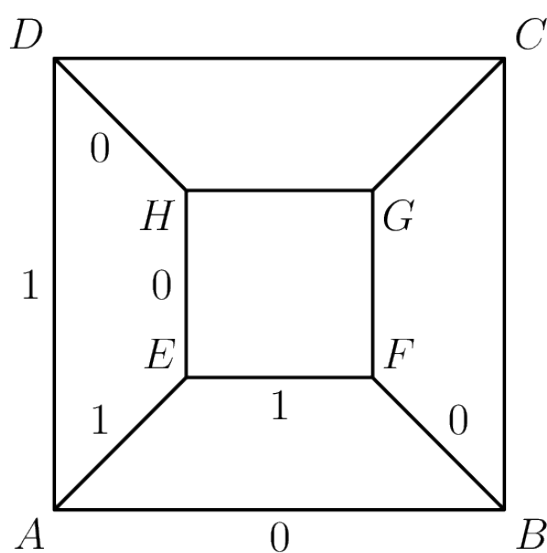


This goes to:



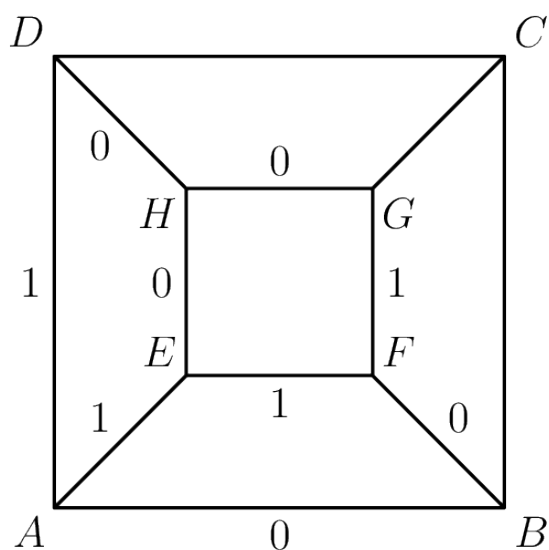
We can see that it consists of chains of three 1's, with the middle of each chain being opposite edges. As a result, this case has $\frac{12}{2} = 6$ orientations.

Case 2.2

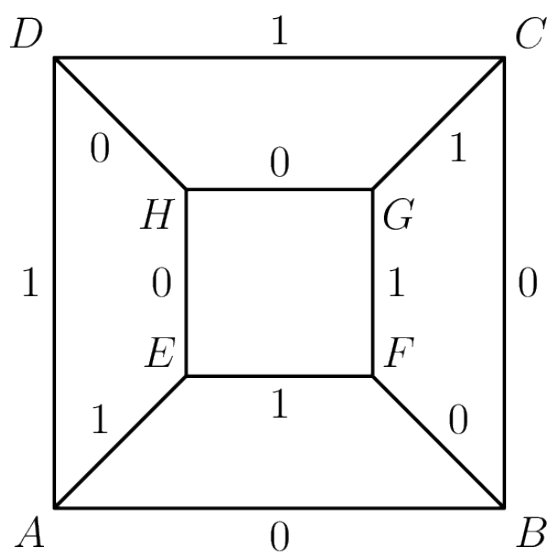


Oh no... We have different ways of filling out FG and GH . More casework!

Case 2.2.1

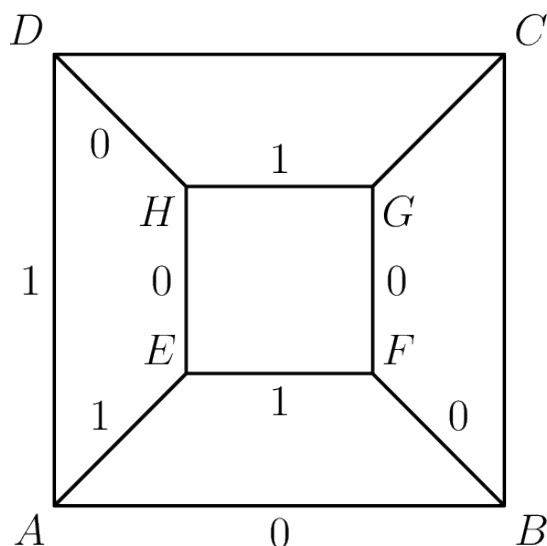


This goes to:

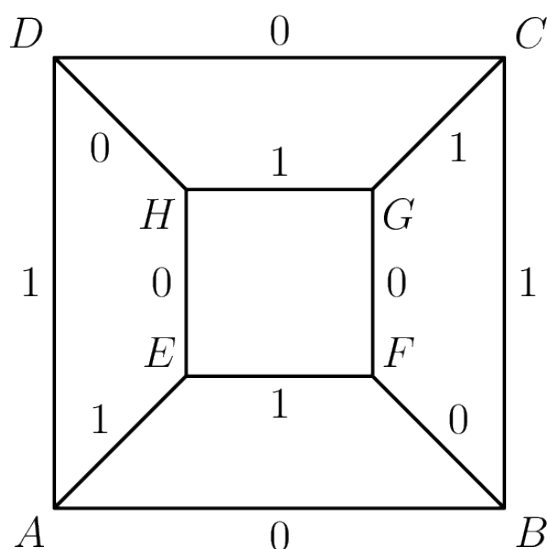


We can see that this is the inverse of case 1 (Define inverse to mean swapping 1's for 0's and 0's for 1's). Therefore, this should also have 4 orientations.

Case 2.2.2



This goes to:



This is the inverse of case 2.1, so this will also have 6 orientations.

Putting Them All Together We see that if the 3 edges connecting to A has two 0's, and one 1, it would have the same solutions as if it had two 1's, and one 0. The solutions would just be inverted. As case 2.1 and case 2.2.2 are inverses, and case 2.2.1 has case 1 as an inverse, there would not be any additional solutions.

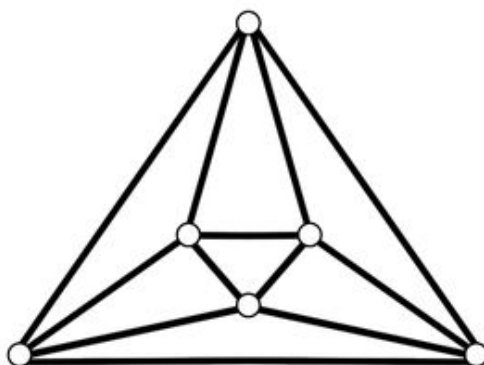
Similarly, if the 3 edges connecting to A has three 0's, it would be the same as the inverse of case 1, or case 2.2.1, resulting in no new solutions.

Putting all the cases together, we have $4 + 6 + 4 + 6 = \boxed{\text{(E)} 20}$ solutions.

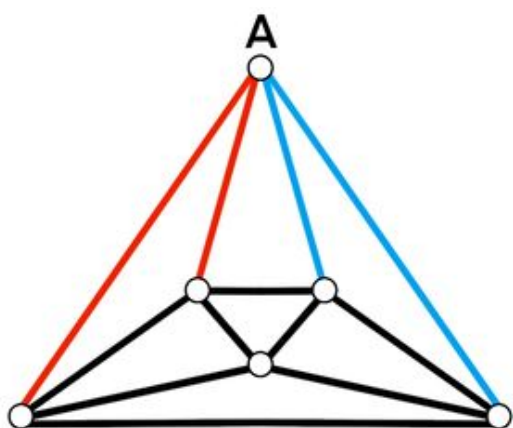
Solution 4

The problem states the sum of the labels on the edges of each of the 6 faces of the cube equal to 2. That is, the sum of the labels on the 4 edges of a face is equal to 2. The labels can only be 0 or 1, meaning 2 edges are labeled 1, the other 2 are labeled 0.

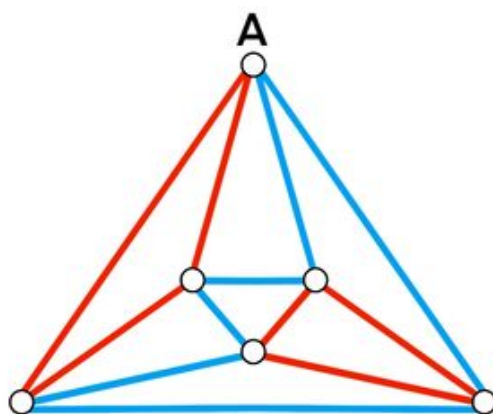
This problem can be approached by **Graph Coloring** or **Graph Theory**. Note that each face of the cube connects to 4 other faces, each with a shared edge. We use the following graph to represent the problem. Each vertex represents a face, each edge represent the cube's edge. Each vertex has 4 edges connecting to 4 other vertices. The edges can be colored red or blue, with red as label 0, and blue as label 1. Each vertex must have 2 red edges and 2 blue edges.



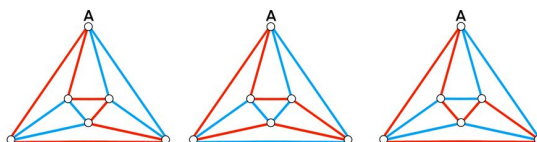
Case 1: 2 adjacent red edges from vertex A. There are 4 ways to choose 2 red edges adjacent to each other and connect to 2 vertices with an edge between them as shown below.



Case 1.1: 2 adjacent red edges from vertex A form a closed loop with a third red edge. There is only 1 case as shown below.

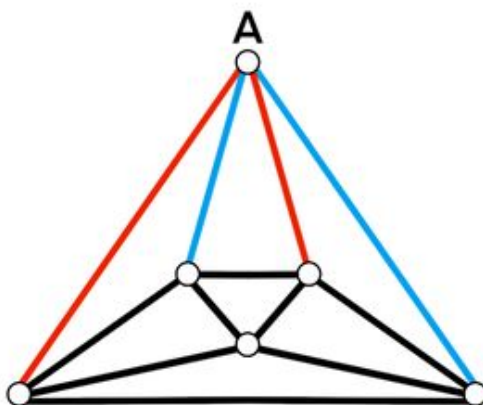


Case 1.2: 2 adjacent red edges from vertex A does not form a closed loop with a third red edge. There are 3 cases as shown below.

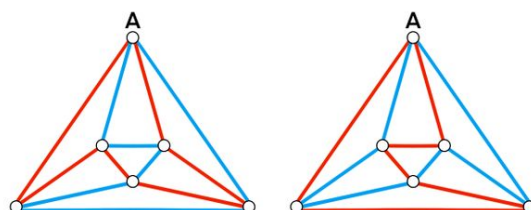


In case 1, there are total $4 \cdot (1 + 3) = 16$ ways.

Case 2: 2 red edges from vertex A with 1 blue edge in between. There are 2 ways to choose 2 red edges with 1 blue edge in between.



There are only 2 cases as shown below.



In case 2, there are total $2 \cdot 2 = 4$ ways.

From both case 1 and case 2, there are $16 + 4 = \boxed{\text{(E)} 20}$ ways in total.

25. (AMC 10 10A Fall 2021) A quadratic polynomial with real coefficients and leading coefficient 1 is called *disrespectful* if the equation $p(p(x)) = 0$ is satisfied by exactly three real numbers. Among all the disrespectful quadratic polynomials, there is a unique such polynomial $\tilde{p}(x)$ for which the sum of the roots is maximized. What is $\tilde{p}(1)$?

(A) $\frac{5}{16}$ (B) $\frac{1}{2}$ (C) $\frac{5}{8}$ (D) 1 (E) $\frac{9}{8}$

Answer: A

Solution 1 (Standard Form)

Let r_1 and r_2 be the roots of $\tilde{p}(x)$. Then, $\tilde{p}(x) = (x - r_1)(x - r_2) = x^2 - (r_1 + r_2)x + r_1r_2$. The solutions to $\tilde{p}(\tilde{p}(x)) = 0$ is the union of the solutions to

$$\tilde{p}(x) - r_1 = x^2 - (r_1 + r_2)x + (r_1r_2 - r_1) = 0$$

and

$$\tilde{p}(x) - r_2 = x^2 - (r_1 + r_2)x + (r_1r_2 - r_2) = 0.$$

Note that one of these two quadratics has one solution (a double root) and the other has two as there are exactly three solutions. WLOG, assume that the quadratic with one root is $x^2 - (r_1 + r_2)x + (r_1r_2 - r_1) = 0$. Then, the discriminant is 0, so $(r_1 + r_2)^2 = 4r_1r_2 - 4r_1$. Thus, $r_1 - r_2 = \pm 2\sqrt{-r_1}$, but for $x^2 - (r_1 + r_2)x + (r_1r_2 - r_2) = 0$ to have two solutions, it must be the case that $r_1 - r_2 = -2\sqrt{-r_1}$ (*). It follows that the sum of the roots of $\tilde{p}(x)$ is $2r_1 + 2\sqrt{-r_1}$, whose maximum value occurs when $r_1 = -\frac{1}{4}$ (*). Solving for r_2 yields $r_2 = \frac{3}{4}$. Therefore, $\tilde{p}(x) = x^2 - \frac{1}{2}x - \frac{3}{16}$, so

$$\tilde{p}(1) = \boxed{\text{(A)} \frac{5}{16}}.$$

<u>**Remarks**</u>

- For $x^2 - (r_1 + r_2)x + (r_1r_2 - r_2) = 0$ to have two solutions, the discriminant $(r_1 + r_2)^2 - 4r_1r_2 + 4r_2$ must be positive. From here, we get that $(r_1 - r_2)^2 > -4r_2$, so $-4r_1 > -4r_2 \implies r_1 < r_2$. Hence, $r_1 - r_2$ is negative, so $r_1 - r_2 = -2\sqrt{-r_1}$.
- Set $\sqrt{-r_1} = x$. Now $r_1 + \sqrt{-r_1} = -x^2 + x$, for which the maximum occurs when $x = \frac{1}{2} \rightarrow r_1 = -\frac{1}{4}$.

Solution 2 (Vertex Form)

Let $p(x) = (x - h)^2 + k$ for some real constants h and k . Suppose that $p(x)$ has real roots r and s .

Since $p(p(x)) = 0$, we conclude that $p(x) = r$ or $p(x) = s$. Without loss of generality, we assume that $p(x) = r$ has two real solutions and $p(x) = s$ has one real solution. Therefore, we have $k = s$, from which

$$p(x) = (x - h)^2 + s.$$

As $p(s) = 0$, we expand the left side to obtain $(s - h)^2 + s = 0$, or

$$s^2 - (2h - 1)s + h^2 = 0. \quad (\star)$$

Since $(\star)$ has real solutions for s , the discriminant is nonnegative: $(2h - 1)^2 - 4h^2 \geq 0$. We solve this inequality to get $h \leq \frac{1}{4}$.

Either by the axis of symmetry or Vieta's formulas, note that $r + s = 2h$. As we wish to maximize $2h$, we maximize h . Substituting $h = \frac{1}{4}$ into (★), we obtain $s^2 + \frac{1}{2}s + \frac{1}{16} = 0$. We factor the left side to get $\left(s + \frac{1}{4}\right)^2 = 0$, or $s = -\frac{1}{4}$.

Finally, the unique such polynomial is

$$\tilde{p}(x) = \left(x - \frac{1}{4}\right)^2 - \frac{1}{4},$$

from which $\tilde{p}(1) = \boxed{\text{(A)} \frac{5}{16}}$.

Solution 2.5 (Vertex Form but Done Differently)

Continuing from the previous solution's reasoning, we know that the y -coordinate of the vertex must also be a root of the polynomial. Thus, $\tilde{p}(x) = \left(x - \frac{s+r}{2}\right)^2 + s$, where r and s are roots of the polynomial.

We also know that in order for $\tilde{p}(x) = r$ to have two solutions, $y = r$ must intersect $\tilde{p}(x)$ above its minimum/vertex. Thus, $r > s$. Let $r = s + a$. Then, $\tilde{p}(x) = \left(x - \frac{2s+a}{2}\right)^2 + s$.

Since s is a root, $\tilde{p}(s) = 0$ so $\left(s - \frac{2s+a}{2}\right)^2 + s = 0 \implies \frac{a^2}{4} + s = 0 \implies s = -\frac{1}{4}a^2$. We wish to maximize $s + s + a = 2s + a = -\frac{1}{2}a^2 + a$.

This quadratic is maximized at $a = 1$ (when converting the quadratic to vertex form). Then, since $\tilde{p}(s+a) = 0 \implies \tilde{p}(s+1) = 0$, $\left(s+1 - s - \frac{1}{2}\right)^2 + s = 0 \implies \frac{1}{4} + s = 0 \implies s = -\frac{1}{4}$.

Thus, $\tilde{p}(x) = \left(x - \frac{1}{4}\right)^2 - \frac{1}{4}$. So, $\tilde{p}(1) = \boxed{\text{(A)} \frac{5}{16}}$.

Solution 3 (Symmetry)

Let $\tilde{p}(x) = (x - h)^2 + k$. We seek to maximize the sum of the roots $2h$, so we maximize h .

Let $P(x) = \tilde{p}(\tilde{p}(x)) = ((x - h)^2 + k - h)^2 + k$. Note that $P(x)$ is symmetric about $x = h$.

$P(x) = 0$ has 3 real solutions. Due to the complex conjugate theorem, $P(x)$ must have 4 real roots. Therefore, $P(x)$ must have exactly 1 double root.

This root cannot be to the left or to the right of $x = h$, as the symmetry of the function would mean that there would be another double root reflected across the $x = h$. It follows that the double root could only be situated at $x = h$.

$0 = P(h) = ((h - h)^2 + k - h)^2 + k = (k - h)^2 + k$. Expanding and writing this out in terms of k , $k^2 + (1 - 2h)k + h^2 = 0$.

In order for this to have a solution, the discriminant has to be non-negative. In other words, $(1 - 2h)^2 - 4h^2 \geq 0$.

This simplifies to $1 - 4h \geq 0$, or $h \leq \frac{1}{4}$.

As we seek to maximize h , we set $h = \frac{1}{4}$ and see that $k = -\frac{1}{4}$.

Therefore, $\tilde{p}(x) = \left(x - \frac{1}{4}\right)^2 - \frac{1}{4}$, and $\tilde{p}(1) = \left(1 - \frac{1}{4}\right)^2 - \frac{1}{4} = \frac{9}{16} - \frac{1}{4} = \boxed{\text{(A)} \frac{5}{16}}$.

Solution 4 (Discriminant)

Equation $p(p(x)) = 0$ is equivalent to the following system of equations:

$$\begin{aligned} p(u) &= 0, \\ p(x) - u &= 0. \end{aligned}$$

Denote $p(x) = x^2 + px + q$.

Denote by r_1 and r_2 two roots of $p(x) = 0$.

Because $p(p(x)) = 0$ has three real roots, we must have the properties that r_1 and r_2 are real with $r_1 \neq r_2$. Without loss of generality, we assume $r_1 < r_2$.

We notice that all roots of $p(p(x)) = 0$ are the collection of all roots of $p(x) - r_1 = 0$ and all roots of $p(x) - r_2 = 0$.

Because each of these two equations is quadratic, it has two roots (may be identical). To get a total number of three roots, one equation must have two identical roots.

Because $r_1 < r_2$, equation $p(x) - r_1 = 0$ has two identical roots. Hence, the discriminant of this equation satisfies

$$p^2 - 4(q - r_1) = 0.$$

Because $r_1 < r_2$, $r_1 = \frac{-p - \sqrt{p^2 - 4q}}{2}$.

Hence, the above equation can be written as

$$p^2 - 4\left(q - \frac{-p - \sqrt{p^2 - 4q}}{2}\right) = 0.$$

This can be reorganized as

$$(p^2 - 4q) - 2\sqrt{p^2 - 4q} - 2p = 0. \quad (1)$$

Define $x = \sqrt{p^2 - 4q}$. Hence, the value of p should ensure that equation $x^2 - 2x - 2p = 0$ has at least one real nonnegative root.

This condition can be satisfied if the discriminant of this equation is nonnegative: $(-2)^2 - 4 \cdot 1 \cdot (-2p) \geq 0$. Hence, $p \geq -\frac{1}{2}$.

Now, we are ready to find $\tilde{p}(x)$.

Following from Vieta's formula, $r_1 + r_2 = -p$. Hence, to get $r_1 + r_2$ maximized, we need to find smallest p .

Because $p \geq -\frac{1}{2}$, the smallest p is $-\frac{1}{2}$. Plugging this value into Equation (1), we get $\sqrt{p^2 - 4q} = 1$. Hence, $q = -\frac{3}{16}$.

Thus, $\tilde{p}(x) = x^2 - \frac{1}{2}x - \frac{3}{16}$. Therefore, $\tilde{p}(1) = \boxed{\text{(A)} \frac{5}{16}}$.

Solution 5 (Factored Form)

The disrespectful function $p(x)$ has leading coefficient 1, so it can be written in factored form as $(x - r)(x - s)$. Now the problem states that all $p(x)$ must satisfy $p(p(x)) = 0$. Plugging our form in, we get

$$((x - r)(x - s) - r)((x - r)(x - s) - s) = 0.$$

The roots of this equation are $(x - r)(x - s) = r$ and $(x - r)(x - s) = s$. By the fundamental theorem of algebra, each root must have two roots for a total of four possible values of x yet the problem states that this equation is satisfied by three values of x . Therefore one equation must give a double root. Without loss of generality, let the equation $(x - r)(x - s) = r$ be the equation that produces the double root. Expanding gives $x^2 - (r + s)x + rs - r = 0$. We know that if there is a double root to this equation, the discriminant must be equal to zero, so $(r + s)^2 - 4(rs - r) = 0 \implies r^2 + 2rs + s^2 - 4rs + 4r = 0 \implies r^2 - 2rs + s^2 + 4r = 0$.

From here two solutions can progress.

Solution 5.1 (Quadratic Formula)

We can rewrite $r^2 - 2rs + s^2 + 4r = 0$ as $(r - s)^2 + 4r = 0$. Let's keep our eyes on the ball; we want to find the disrespectful quadratic that maximizes the sum of the roots, which is $r + s$. Let this be equal to a new variable, m , so that our problem is reduced to maximizing this variable. We can rewrite our equation in terms of m as $(2r - m)^2 + 4r = 0 \implies m^2 - 4rm + 4r^2 + 4r = 0$.

This is a quadratic in m , so we can use the quadratic formula:

$$m = \frac{4r \pm \sqrt{16r^2 - 4(4r^2 + 4r)}}{2} = 2r \pm \sqrt{-4r} = 2(r \pm \sqrt{-r}).$$

It will be easier to think without the square root, so let $q = \sqrt{-r}$. We can rewrite the equation as $m = 2(-q^2 \pm q)$. We want to maximize m , so we take the plus value of the right-hand-side of the equation. Then,

$$m = 2(-q^2 + q) \implies m = -2q(q - 1).$$

To maximize m , we find the vertex of the right-hand side of the equation. The vertex of $-2q(q - 1)$ is the average of the roots of the equation which is $\frac{0+1}{2} = \frac{1}{2}$. This means that $r = -q^2 \implies r = -\frac{1}{4}$ and $m = -2q(q - 1) \implies m = \frac{1}{2}$. Therefore, $m - r = s \implies s = \frac{3}{4}$.

Solution 5.2 (Derivation-Rotated Conics)

We see that the equation $r^2 - 2rs + s^2 + 4r = 0$ is in the form of the general equation of a rotated conic

$$Ax^2 + Bxy + Cy^2 + Dx + Ey + F = 0.$$

Because $B^2 - 4AC = (-2)^2 - 4(1)(1) = 0$, this rotated conic is a parabola.

The definition of a parabola is the locus of all points that are equidistant from a point (focus) and line (directrix). Let the focus and directrix of this particular parabola be (a, b) and $y = mx + c$. Then we can try to find the general form of a rotated parabola in terms of a, b, m , and c .

The distance between two points (x, y) and (a, b) is $\sqrt{(x - a)^2 + (y - b)^2}$. Therefore this is the distance from any point on the parabola to the focus.

The distance from a point (x, y) to a line $y = mx + c \implies mx - y + c = 0$ is $\frac{|mx - y + c|}{\sqrt{m^2 + 1}}$.

We can set these two equal to each other and we get

$$\sqrt{(x - a)^2 + (y - b)^2} = \frac{|mx - y + c|}{\sqrt{m^2 + 1}}.$$

Squaring both sides of the equation, we get

$$(x - a)^2 + (y - b)^2 = \frac{(mx - y + c)^2}{m^2 + 1}.$$

Expanding both sides of the equation gives

$$x^2 - 2ax + a^2 + y^2 - 2by + b^2 = \frac{m^2x^2 + y^2 + c^2 + 2mxc - 2yc - 2mxy}{m^2 + 1}.$$

Multiplying both sides of the equation by $m^2 + 1$ and rearranging gives

$$x^2 + 2mx + m^2y^2 - 2((m^2 + 1)a + mc)x - 2((m^2 + 1)b - c)y + (m^2 + 1)(a^2 + b^2) - c^2.$$

Now we can compare to our rotated parabola, $r^2 - 2rs + s^2 + 4r = 0$. From this, $-2 = 2m$ or $m = -1$. From here we have a system of three equations:

$$\begin{aligned} -2((m^2 + 1)a + mc) &= 4, \\ -2((m^2 + 1)b - c) &= 0, \\ (m^2 + 1)(a^2 + b^2) - c^2 &= 0. \end{aligned}$$

Plugging in $m = -1$ we get

$$\begin{aligned} -2(2a - c) &= 4, \\ -2(2b - c) &= 0, \\ 2(a^2 + b^2) - c^2 &= 0. \end{aligned}$$

Solving for the first equation, $c = 2 + 2a$.

Subtracting the first two equations, $-4a + 4b = 4 \implies b = a + 1$.

Plugging into the third equation, $2a^2 + 2a^2 + 4a + 2 = c^2 \implies 4a^2 + 4a + 2 = c^2$.

Substituting c in, we get $4a^2 + 4a + 2 = 4a^2 + 8a + 4 \implies 4a + 2 = 0 \implies a = -\frac{1}{2}$.

Now $b = a + 1 = \frac{1}{2}$ and $c = 2 + 2a = 1$.

This means that the focus of the parabola is $\left(-\frac{1}{2}, \frac{1}{2}\right)$ and the directrix is $y = -x + 1$. The maximum value of $r + s$ would lie at the vertex of the parabola, which is the midpoint of the focus and the foot of the focus at the directrix. The line that the vertex and focus lie on is perpendicular to the directrix, so it has slope 1. It can be written as $y = x + d$ and must go through $\left(-\frac{1}{2}, \frac{1}{2}\right)$ so $d = 1$. This perpendicular line intersects the directrix, so to find the point at which this foot occurs, we set the equation of the lines equal to each other:

$$\begin{aligned} y &= x + 1, \\ y &= -x + 1. \end{aligned}$$

Adding, we get $2y = 2$ or $y = 1$ and $x = 0$. The vertex of the parabola is now at the midpoint of $\left(-\frac{1}{2}, \frac{1}{2}\right)$ and $(0, 1)$ which is $\left(-\frac{1}{4}, \frac{3}{4}\right)$.

Therefore, we have $r = -\frac{1}{4}$ and $s = \frac{3}{4}$.

Solutions 5.1 and 5.2 Rejoined

Now that we know the roots of $\tilde{p}(1)$, we can plug in our equation:

$$(x - r)(x - s) = \left(1 - \left(-\frac{1}{4}\right)\right)\left(1 - \frac{3}{4}\right) = \frac{5}{4} \cdot \frac{1}{4} = \frac{5}{16} = \boxed{\text{(A)} \frac{5}{16}}.$$

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Solution 6 (Factored Form)

Let $p(x) = (x - p)(x - q)$. Then,

$$p(p(x)) = ((x - p)(x - q) - p)((x - p)(x - q) - q) = 0,$$

which means that either $(x - p)(x - q) - p = 0$ or $(x - p)(x - q) - q = 0$. Both of these equations are quadratics, so $p(p(x))$ has four roots, unless there's a double root.

Without loss of generality, let $(x - p)(x - q) - p$ be the expression that produces the double root, so its discriminant is zero. When expanded,

$$(x - p)(x - q) - p = x^2 - (p + q)x + pq - p = 0.$$

The value of x is irrelevant to the discriminant, which is $-(p + q)^2 - 4(pq - p)$. Setting this equal to zero and simplifying, this equation becomes $p^2 - 2pq + q^2 + 4p = 0$, which is a quadratic in p .

Now, we seek the value of p . The previous quadratic is equivalent to $p^2 - (2q - 4)p + q^2 = 0$. Using the quadratic formula by having $a = 1$, $b = -(2q - 4)$, and $c = q^2$, we have

$$p = \frac{2q - 4 \pm \sqrt{-(2q - 4)^2 - 4(1)(q^2)}}{2(1)} = \frac{2q - 4 \pm 4\sqrt{1 - q}}{2} = q - 2 + 2\sqrt{1 - q}.$$

Our focus is on maximizing $p + q$, so we need the maximum values of p and q respectively (by taking the positive square root). Adding q , we see that

$$p + q = 2(q - 1 + \sqrt{1 - q}).$$

Let $k = \sqrt{1 - q}$. Then, $-k^2 = q - 1$, so $p + q = 2(-k^2 + k)$.

The graph of $-k^2 + k$ is a parabola that opens up downwards, and has its maximum value at y -value of the vertex. The x coordinate of the vertex is the average of the two roots, which in this case, are 0 and 1, so the average of these two is $\frac{1}{2}$. This means that the maximum value of $-k^2 + k$ occurs at $k = \frac{1}{2}$.

Substituting $k = \sqrt{1 - q}$, we see that $q = \frac{3}{4}$. Since $p = q - 2 + 2\sqrt{1 - q}$, we can plug $q = \frac{3}{4}$ in, to see that $p = -\frac{1}{4}$.

Because the roots of this quadratic are $\frac{3}{4}$ and $-\frac{1}{4}$, our quadratic is $(x - \frac{3}{4})(x + \frac{1}{4})$.

$\tilde{p}(1) = \frac{5}{16}$, so the answer is

(A) $\frac{5}{16}$

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Solution 7 (Vieta's Formulas)

Since $\tilde{p}(x)$ is a disrespectful polynomial, it must satisfy $\tilde{p}(\tilde{p}(x))$. Let $\tilde{p}(x)$ have roots x_1, x_2 . So, the equation $\tilde{p}(\tilde{p}(x))$ becomes two separate equations:

$$x^2 + bx + c = x_1,$$

$$x^2 + bx + c = x_2.$$

Let r_1, r_2, r_3, r_4 be the roots of $\tilde{p}(\tilde{p}(x))$. Since there are 3 real roots, then either one of them is complex, or two of them are the same. It is known that if complex number w is a root to a certain polynomial, then so is $\bar{w}$. So, if one root to $\tilde{p}(\tilde{p}(x))$

is complex, then another root should also be complex — we can't have that. So, two roots are equal; WLOG, assume it is $r_3 = r_4$. That means the roots to $\tilde{p}(\tilde{p}(x))$ are r_1, r_2, r_3, r_3 .

Now, if the roots of $x^2 + bx + c = x_1$ are r_1 and r_3 and the roots of $x^2 + bx + c = x_2$ are r_2 and r_3 , then we have $-b = r_1 + r_3 = r_2 + r_3$ giving $r_1 = r_2$ by **Vieta's Formula**. That can't happen either. So, $x^2 + bx + c = x_1$ must have roots r_1, r_2 and $x^2 + bx + c = x_2$ must have a double root r_3 . Again by Vieta's, we have

$$\begin{aligned} r_1 + r_2 &= -b, \\ 2r_3 &= -b. \end{aligned}$$

That implies $r_2 - r_3 = r_3 - r_1$. So, let $r_1 = r_3 - k$ and $r_2 = r_3 + k$. Also, from $2r_3 = -b$, we have $r_3 = -\frac{b}{2}$. By Vieta's again, we have

$$\begin{aligned} r_1 \cdot r_2 &= c - x_1 = r_1 \cdot r_2 = (r_3 - k)(r_3 + k) = r_3^2 - k^2, \\ r_3^2 &= c - x_2. \end{aligned}$$

Subtracting yields $x_1 - x_2 = k^2$. Adding that with $x_1 + x_2 = -b$ brings us to

$$\begin{aligned} x_1 &= \frac{k^2 - b}{2}, \\ x_2 &= \frac{-k^2 - b}{2}. \end{aligned}$$

Thus, $c = x_1 \cdot x_2 = \frac{b^2 - k^4}{4}$. Now, $c - x_2 = r_3^2$ by Vieta's. Plugging in $c, x_2, r_3 = -\frac{b}{2}$ gives

$$\frac{b^2 - k^4}{4} + \frac{k^2 + b}{2} = \frac{b^2 - k^4 + 2k^2 - 2b}{4} = \frac{b^2}{4}.$$

Simplifying yields $2b = k^4 - 2k^2 = (k^2 - 1)^2 - 1 \geq -1$ by **Trivial Inequality**. Thus, $\min(b) = -\frac{1}{2}$. Since $\tilde{p}(x)$ have the maximum value to the sum of the roots, which is $-b$, we want to minimize b . That means that b for $\tilde{p}(x)$ is $\frac{1}{2}$. That means $k^2 = 1$

and $c = -\frac{3}{16}$, and thus giving us $\tilde{p}(1) = 1 + b + c = 1 - \frac{1}{2} - \frac{3}{16} = \boxed{\text{(A)} \frac{5}{16}}$.

Solution 8 (Calculus)

Let $p(x) = (x - r_1)(x - r_2)$. Then

$$p(p(x)) = ((x - r_1)(x - r_2) - r_1)((x - r_1)(x - r_2) - r_2)$$

As noted in other solutions, in order for $p(p(x))$ to have three (distinct) real roots, one of the roots must be repeated twice.

For a general polynomial $q(x)$ with an n -times repeated root s , write $q(x) = (x - s)^n u(x)$, where $u(x)$ is another polynomial. Taking the derivative,

$$\begin{aligned} q'(x) &= n(x - s)^{n-1}u(x) + (x - s)^n u'(x) \\ &= (x - s)^{n-1} [nu(x) + (x - s)u'(x)] \end{aligned}$$

This shows that a repeated root / linear factor will appear in the derivative as well.

Returning to the problem, consider the derivative of $p(p(x))$, which is

$$[p(p(x))]' = p'(p(x))p'(x) = [2(x - r_1)(x - r_2) - r_1 - r_2] [2x - r_1 - r_2]$$

By the observation above, the repeated root of $p(p(x))$ must be a root of this derivative. Therefore, a search through its roots will eventually give information to resolve r_1 and r_2 . Setting the derivative equal to zero yields the candidate equations

$$2(x - r_1)(x - r_2) - r_1 - r_2 = 0 \quad \text{or} \quad 2x - r_1 - r_2 = 0$$

Case 1: Starting with the first equation, if x_0 is a solution, then rearrange to obtain

$$(x_0 - r_1)(x_0 - r_2) = \frac{r_1 + r_2}{2}$$

Then

$$\begin{aligned} p(p(x_0)) &= ((x_0 - r_1)(x_0 - r_2) - r_1)((x_0 - r_1)(x_0 - r_2) - r_2) \\ &= \left(\frac{r_1 + r_2}{2} - r_1 \right) \left(\frac{r_1 + r_2}{2} - r_2 \right) \\ &= \frac{1}{4}(r_1 - r_2)^2 = 0 \end{aligned}$$

This shows that $r_1 = r_2$. However, this gives $p(x) = (x - r_1)^2$ and $p(p(x)) = [(x - r_1)^2 - r_1]^2$, which only has at most two distinct real roots. This contradicts the three-root condition given by the problem.

Case 2: The failure of Case 1 shows that the second equation must contain the repeated root. The solution for the second equation is readily $x_0 = (r_1 + r_2)/2$. Then,

$$p(p(x_0)) = \left[\left(\frac{r_1 + r_1}{2} - r_1 \right) \left(\frac{r_1 + r_2}{2} - r_2 \right) - r_1 \right] \left[\left(\frac{r_1 + r_2}{2} - r_1 \right) \left(\frac{r_1 + r_2}{2} - r_2 \right) - r_2 \right] = 0$$

Solving for r_1 yields four candidate equations:

$$r_1 = r_2 \pm 2\sqrt{-r_2} \quad \text{or} \quad r_1 = r_2 \pm 2\sqrt{1 - r_2} - 2$$

Since it is desired to maximise the sum of roots, add r_2 to both sides to consider

$$r_1 + r_2 = 2r_2 \pm 2\sqrt{-r_2} \quad \text{or} \quad r_1 + r_2 = 2r_2 \pm 2\sqrt{1 - r_2} - 2$$

For the first family of solutions, the "plus" version is immediately larger, so maximise

$$f(r_2) = 2r_2 + 2\sqrt{-r_2}$$

Taking the derivative and setting equal to zero,

$$f'(r_2) = 2 - \frac{1}{\sqrt{-r_2}} = 0$$

yielding $r_2 = -1/4$ and $f(-1/4) = 1/2$, which can be verified to be a maximum. Since $f(r_2) = r_1 + r_2$, $r_1 = 3/4$. A similar calculation on the second family of solutions yields the complementary version $r_1 = -1/4$ and $r_2 = 3/4$.

In summary, this calculation shows that $\tilde{p}(x) = \left(x + \frac{1}{4}\right) \left(x - \frac{3}{4}\right)$. Finally, $\tilde{p}(1) =$

$$\left(1 + \frac{1}{4}\right) \left(1 - \frac{3}{4}\right) = \boxed{\text{(A)} \frac{5}{16}}.$$

Solution 9 (Lagrange Multipliers)

First note that $p(x)$ must have two distinct roots, else $p(p(x))$ will have fewer than 3 roots. Let these be a, b . Since $p(x)$ has 3 real roots, and complex roots come in pairs, there must be a double root. So there must be a_1, a_2 s.t $p(a_1) = p(a_2) = a$ and b_1, b_2 s.t $p(b_1) = p(b_2) = b$ but two of a_1, a_2, b_1, b_2 must be the same. WLOG either $a_1 = a_2$ or $a_1 = b_1$. In the latter, case, then,

$$\begin{aligned} p(x) &= a + (x - a_1)(x - a_2) \\ p(x) &= b + (x - b_1)(x - b_2) \\ &= b + (x - a_1)(x - b_2) \\ \implies -(a_1 + a_2)x &= -(a_1 + b_2)x \\ \implies a_2 &= b_2 \end{aligned}$$

which implies $a = b$ which is not possible. Hence assume WLOG $a_1 = a_2$. Then note that

$$\begin{aligned} p(x) &= (x - a)(x - b) \\ &= a + (x - a_1)^2 \\ -(a + b)x &= -2a_1x, \end{aligned}$$

so

$$\begin{aligned} p(x) &= a + \left(x - \frac{a+b}{2}\right)^2 \\ x^2 - (a+b)x + ab &= x^2 - (a+b)x + \left(a + \frac{1}{4}(a^2 + 2ab + b^2)\right) \\ 0 &= a^2 - 2ab + b^2 + 4a. \end{aligned}$$

The problem wants us to maximize the sum of the roots, $a + b$. Thus define $f(a, b) = a + b, g(a, b) = a^2 - 2ab + b^2 + 4a$. By the method of Lagrange Multipliers, we need to solve,

$$\begin{aligned} \vec{\nabla} f &= \lambda \vec{\nabla} g \\ g &= 0. \end{aligned}$$

In the first equation, note,

$$\begin{aligned} \vec{\nabla} f &= \lambda \vec{\nabla} g \\ \langle 1, 1 \rangle &= \lambda \langle 2a - 2b + 4, -2a + 2b \rangle \\ 2a - 2b + 4 &= -2a + 2b \\ a - b &= -1. \end{aligned}$$

Substituting into the second equation,

$$\begin{aligned} g(a, b) &= (a - b)^2 + 4a \\ 0 &= 1 + 4a \\ a &= -\frac{1}{4}, b = \frac{3}{4} \\ \tilde{p}(1) &= \left(1 + \frac{1}{4}\right) \left(1 - \frac{3}{4}\right) = \boxed{\text{(A)} \frac{5}{16}}. \end{aligned}$$

Aaryabhata1

Solution 10 (Solution 8 but Simpler)

Let $p(x) = (x - r)(x - s)$, having roots r, s . WLOG, set $r < s$. The graph of $p(x)$ will have a vertex at $(\frac{r+s}{2}, p(\frac{r+s}{2}))$. Since $p(p(x)) = 0 \iff p(x) = r$ or $p(x) = s$ has exactly 3 distinct solutions, we know that $r = p(\frac{r+s}{2})$ and $s > p(\frac{r+s}{2})$ providing 1 and 2 solutions respectively for a total of 3 (if both $r, s > p(\frac{r+s}{2})$ there will be 4 solutions and if either $r, s < p(\frac{r+s}{2})$ there will be at most 2 solutions). Then,

$$r = p\left(\frac{r+s}{2}\right) = \left(\frac{s-r}{2}\right)\left(\frac{r-s}{2}\right) = -\frac{(s-r)^2}{4}$$

$$(s-r)^2 = -4r$$

$$s+r = 2r \pm \sqrt{-4r}$$

The negative case clearly doesn't work (giving max $s+r = 0$ where $r, s = 0$), leaving us with $s+r = 2r + \sqrt{-4r}$. Finding the derivative we see that $s+r$ is maximized when $r = -\frac{1}{4}$, which gives $s = \frac{3}{4}$. Plugging these back into $p(x)$ we see

that $\tilde{p}(x) = (x + \frac{1}{4})(x - \frac{3}{4})$. Then $\tilde{p}(1) = \boxed{\text{(A)} \frac{5}{16}}$.

Solution 11 (AM-GM Inequality)

Because $p(x)$ is a quadratic equation with a leading coefficient of 1, let $p(x) = x^2 + ax + b$. Moreover, let r_1, r_2, r_3 be the roots of the equation $p(p(x)) = 0$.

$$p(r_1) = a$$

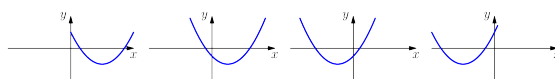
$$p(r_2) = b$$

$$p(r_3) = c$$

$$p(a) = p(b) = p(c) = 0$$

$$p(x) = (x - \alpha)(x - \beta)$$

The following cases are the possible graphs for $p(x)$.



The second graph is the case that could provide the maximum sum of the roots.

$$\begin{aligned} \therefore \left(x - \frac{\alpha + \beta}{2}\right)^2 + \alpha &= (x - \alpha)(x - \beta) \\ x^2 - (\alpha + \beta)x + \frac{(\alpha + \beta)^2}{4} + \alpha &= x^2 - (\alpha + \beta)x + \alpha\beta \\ \frac{(\alpha + \beta)^2}{4} + \alpha &= \alpha\beta \\ (\alpha + \beta)^2 &= 4\alpha\beta - 4\alpha = 4\alpha(\beta - 1) \end{aligned}$$

$$\sqrt{\alpha(\beta - 1)} \leq \frac{\alpha + (\beta - 1)}{2}$$

leads to

$$\alpha = \beta - 1.$$

$$(\alpha + \beta)^2 = 4\alpha^2$$

$$\alpha = \beta \text{ or } 3\alpha + \beta = 0$$

$$\therefore \alpha = -\frac{1}{4}, \beta = \frac{3}{4}$$

$$p(1) = (1 - \alpha)(1 - \beta) = \frac{5}{4} \cdot \frac{1}{4} = \boxed{\text{(A)} \frac{5}{16}}$$

Solution 12: 2 Ways

In this solution are really 2 solutions: my first one, that I did really fast, but was basically solution 1 except I showed that $\tilde{p}(\tilde{p}(x))$ can only have real roots and neither quadratic factor can share any roots. This is too simple, however, so I had to come up with something overkill and unnecessary. So in my second solution, I explicitly list root cases, use derivatives when the optimization could be done with algebra, etc.

Solution 12.1: The Sane Way

Let a and b be the roots of $\tilde{p}(x)$. We have that

$$\tilde{p}(x) = (x - a)(x - b) \implies \tilde{p}(\tilde{p}(x)) = ((x - a)(x - b) - a)((x - a)(x - b) - b).$$

For this to have 3 real solutions, one quadratic factor must have a discriminant of 0 and the other discriminant must be positive. From this, it can quickly be shown that a and b must be real: Assume a and b are nonreal such that $a + b$ and ab are real; then neither quadratic factor can have any real roots because $\tilde{p}(x) = x^2 - (a + b)x + ab$ must be real for all $x \in \mathbb{R}$, so for all $x \in \mathbb{R}$ the equations $\tilde{p}(x) = a$ and $\tilde{p}(x) = b$ cannot be satisfied. Then, it quite immediately follows that for the discriminants to differ, $a \neq b$ so $p(x) = a$ and $p(x) = b$ cannot be satisfied simultaneously, which ensures neither quadratic factor shares a root. Thus, without loss of generality:

$$x^2 - (a + b)x + ab - a = 0 \text{ for 2 distinct } x \in \mathbb{R} \implies (a + b)^2 - 4(ab - a) = a^2 + b^2 - 2ab + 4a = 0 \implies (a - b)^2 = 4a$$

and

$$x^2 - (a + b)x + ab - b = 0 \text{ for 1 } x \in \mathbb{R} \implies (a + b)^2 - 4(ab - b) = a^2 + b^2 - 2ab + 4b > 0 \implies (a - b)^2 = -4a > 0$$

Hence,

$$b - a = 2\sqrt{-a} \implies a + b = 2(a + \sqrt{-a}).$$

We want to maximize $a + \sqrt{-a}$ for $a \leq 0$. Substitute $x = \sqrt{-a}$ so we maximize $-x^2 + x$ for $x \geq 0$. We have $-x^2 + x = -x(x - 1)$ which is maximized at $x = \frac{1}{2} \implies a = -\frac{1}{4} \implies \max(a + \sqrt{-a}) = \frac{1}{4}$. Then $\max(a + b) = \frac{1}{2}$, $a = -\frac{1}{4} \implies b = \frac{3}{4}$. Hence

$$\tilde{p}(x) = x^2 - \frac{1}{2}x - \frac{3}{16} \implies \tilde{p}(1) = \boxed{\text{(A)} \frac{5}{16}}$$

shockfront99

Solution 12.2: Projectile Motion and General Inefficiency

Let $q(x)$ be a disrespectful quadratic. It follows that the graph of $q(x)$ is a convex parabola, and thus the graph of $-q(x)$ is a concave parabola. Suppose $q(x) = (x - a)(x - b)$ where a and b are real numbers. Then:

$$q(q(x)) = (q(x) - a)(q(x) - b).$$

The graph of $q(q(x))$ can be viewed as essentially graphing $q(t)$ where $t \in \text{ran}(q)$. The following cases yield different amounts of real numbers such that $q(q(x)) = 0$:

Case 1: 4 real roots We need the equations

$$q(x) = a \quad \text{and} \quad q(x) = b$$

to have 2 real roots each.

Case 2: 3 real roots We need the equations

$$q(x) = a \quad \text{and} \quad q(x) = b$$

to have 2 real roots and 1 real root respectively, or 1 real root and 2 real roots respectively.

Case 3: 2 real roots We need the equations

$$q(x) = a \quad \text{and} \quad q(x) = b$$

to have 1 real roots each.

Case 4: 1 real root We need the equations

$$q(x) = a \quad \text{and} \quad q(x) = b$$

to have 1 real root and no real roots respectively, or no real roots and 1 real root respectively.

=Case 5: no real roots We need the equations

$$q(x) = a \quad \text{and} \quad q(x) = b$$

to both have no real roots.

Case 2 is what we want. For the $q(x) = a$ and $q(x) = b$ to have different amounts of real solutions, it immediately follows that $a \neq b$. It can also be quickly shown that to have a disrespectful polynomial, a and b must be real: since any disrespectful quadratic has real coefficients, all real inputs yield real outputs and so the equations $q(x) = a$ and $q(x) = b$ can't have any real solutions unless a and b are real. Let

$$\tilde{p}(x) = (x - \alpha)(x - \beta),$$

and fix $\alpha < \beta$ for simplicity. Define the k -shift of $g(x)$ as $g(x - k)$. It follows that $-\tilde{p}(x)$ is concave parabola such that $\max(-\tilde{p}) > 0$, so $-\tilde{p}(x)$ is the α -shift of the path of a projectile that experiences negligible air resistance launched from a horizontal surface with constant downward acceleration. A projectile launched with speed u at an angle θ to the horizontal has initial horizontal velocity $u \cos(\theta)$, on which no relevant horizontal drag, friction, or gravitational force effects and thus the projectile has horizontal position

$$p_h(t) = ut \cos(\theta)$$

after t seconds. Then, the projectile has initial vertical velocity $u \sin(\theta)$ and has a vertical acceleration g downwards and thus a total vertical velocity of $u \sin(\theta) - gt$ and a vertical position of

$$p_v(t) = ut \sin(\theta) - \frac{1}{2}gt^2.$$

We can now see that setting $p'_v(t) = 0$ yields that $t = \frac{u \sin(\theta)}{g}$ so letting $H_{\max} = \max(p_v)$ denote the maximum height,

$$H_{\max} = \frac{u^2 \sin^2(\theta)}{g} - \frac{u^2 \sin^2(\theta)}{2g} = \frac{u^2 \sin^2(\theta)}{2g}.$$

Then, to find the range R (horizontal distance the projectile travels) we need $p_v(t) = 0$ such that $t \neq 0$ so

$$u \sin(\theta) - \frac{gt}{2} = 0 \implies t = \frac{2u \sin(\theta)}{g} \implies R = \frac{2u^2 \sin(\theta) \cos(\theta)}{g} = \frac{u^2 \sin(2\theta)}{g}.$$

Now, we can see that modeling $-\tilde{p}(x)$ as the α -shift of such a trajectory, $\beta - \alpha$ is the range of the projectile and so we are maximizing

$$f(\alpha, \theta, u) = \alpha + (\alpha + R) = 2\alpha + \frac{u^2 \sin(2\theta)}{g}$$

subject to some constraints where g is a positive real constant. Then, the equation $\tilde{p}(\tilde{p}(x)) = 0$ can be rewritten as

$$-\tilde{p}(\tilde{p}(x)) = -(\tilde{p}(x) - \alpha) \left(\tilde{p}(x) - \alpha - \frac{u^2 \sin(2\theta)}{g} \right).$$

We now use the constraints of case 2: we either have that $\tilde{p}(x) - \alpha$ has 2 real roots and $\tilde{p}(x) - \alpha - \frac{u^2 \sin(2\theta)}{g}$ has 1, or vice versa. It can be seen that

$$-\alpha > -\alpha - \frac{u^2 \sin(2\theta)}{g}$$

(which was immediate from $\alpha < \beta$ anyways) so the vertex of $\tilde{p}(x) - \alpha - \frac{u^2 \sin(2\theta)}{g}$ lies below the vertex of $\tilde{p}(x) - \alpha$. Thus $\tilde{p}(x) - \alpha$ has to have its vertex on the x -axis and $\tilde{p}(x) - \alpha - \frac{u^2 \sin(2\theta)}{g}$ has to have its vertex below the x -axis. Since $\tilde{p}(x) - \alpha$ has its vertex on the x -axis it must be expressible in the form $(x - h)^2 = x^2 - 2hx + h^2$ for some real h , and be zero at this h . Thus we'd have

$$0 = x^2 - \left(2\alpha + \frac{u^2 \sin(2\theta)}{g} \right) x + \alpha^2 + \frac{\alpha u^2 \sin(2\theta)}{g} - \alpha = x^2 - 2hx + h^2.$$

Comparing linear coefficients instantly yields

$$h = \alpha + \frac{u^2 \sin(2\theta)}{2g}.$$

Comparing our constant term and using our expression for h yields

$$h^2 = \alpha^2 + \frac{u^2 \sin(2\theta)}{g} + \frac{u^4 \sin^2(2\theta)}{4g^2} = \alpha^2 + \frac{u^2 \sin(2\theta)}{g} - \alpha \implies -4\alpha = \frac{u^2 \sin^2(2\theta)}{g^2} \implies 2\sqrt{-\alpha} = \frac{u \sin(2\theta)}{g}$$

Thus to maximize f we can just maximize

$$f_1(\alpha) = 2\alpha + 2\sqrt{-\alpha} \quad (\alpha \leq 0)$$

We used common derivatives/integrals for the projectile motion but we're not doing that here for no reason

$$\frac{df}{d\alpha} = \lim_{h \rightarrow 0} \frac{2(\alpha + h) + 2\sqrt{-\alpha - h} - 2\alpha - 2\sqrt{-\alpha}}{h} = 2 + 2 \lim_{h \rightarrow 0} \frac{\sqrt{-\alpha - h} - \sqrt{-\alpha}}{h} = 2 + 2 \lim_{h \rightarrow 0} \frac{-h}{h(\sqrt{-\alpha - h} + \sqrt{-\alpha})}$$

So

$$\frac{df}{d\alpha} = 0 \implies \frac{1}{\sqrt{-\alpha}} = 2 \implies \sqrt{-\alpha} = \frac{1}{2} \implies \alpha = -\frac{1}{4}.$$

We then have that $\max(f) = \alpha + \beta = -\frac{1}{2} + 1 = \frac{1}{2}$ so $\beta = \frac{3}{4}$. Thus

$$\tilde{p}(x) = \left(x + \frac{1}{4}\right) \left(x - \frac{3}{4}\right) \implies \tilde{p}(1) = \frac{5}{4} \cdot \frac{1}{4} = \boxed{\text{(A)} \frac{5}{16}}$$

shockfront99

§7 AMC 10 10B 2022

1. (*AMC 10 10B 2022*) Define $x \diamond y$ to be $|x - y|$ for all real numbers x and y . What is the value of

$$(1 \diamond (2 \diamond 3)) - ((1 \diamond 2) \diamond 3)?$$

(A) -2 (B) -1 (C) 0 (D) 1 (E) 2

Answer: A

Solution 1

We have

$$(1 \diamond (2 \diamond 3)) - ((1 \diamond 2) \diamond 3) = |1 - |2 - 3|| - ||1 - 2| - 3| \quad (5)$$

$$= |1 - 1| - |1 - 3| \quad (6)$$

$$= 0 - 2 \quad (7)$$

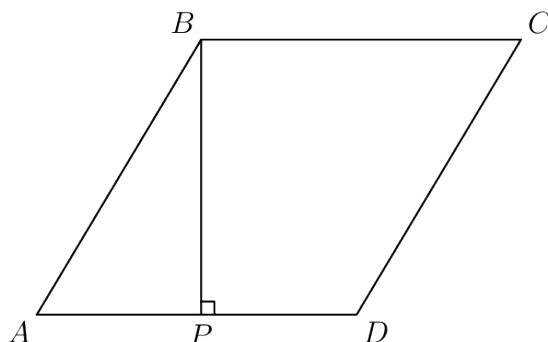
$$= \boxed{\text{(A)} -2}. \quad (8)$$

MRENTHUSIASM

Solution 2

Observe that the $\diamond$ function is simply the positive difference between two numbers. Thus, we evaluate: the difference between 2 and 3 is 1; the difference between 1 and 1 is 0; the difference between 1 and 2 is 1; the difference between 1 and 3 is 2; and finally, $0 - 2 = \boxed{\text{(A)} -2}$.

2. (*AMC 10 10B 2022*) In rhombus $ABCD$, point P lies on segment $\overline{AD}$ so that $\overline{BP} \perp \overline{AD}$, $AP = 3$, and $PD = 2$. What is the area of $ABCD$? (Note: The figure is not drawn to scale.)



- (A) $3\sqrt{5}$ (B) 10 (C) $6\sqrt{5}$ (D) 20 (E) 25

Answer: D

Solution

Notice that $BC = AD = AP + PD = 3 + 2 = 5$. Now, because the question tells us $ABCD$ is a rhombus, $BC = AB = 5$. Thus, by Pythagorean Theorem on APB , $BP = 4$. Therefore,

$$[ABCD] = AD \times BP = 5 \times 4 = \boxed{\text{D } 20}$$

DUODUOLUO (First time using $\LaTeX$, may have some minor mistakes)

3. (AMC 10 10B 2022) How many three-digit positive integers have an odd number of even digits?

- (A) 150 (B) 250 (C) 350 (D) 450 (E) 550

Answer: D

Solution 1

We use simple case work to solve this problem.

Case 1: even, even, even $= 4 \cdot 5 \cdot 5 = 100$

Case 2: even, odd, odd $= 4 \cdot 5 \cdot 5 = 100$

Case 3: odd, even, odd $= 5 \cdot 5 \cdot 5 = 125$

Case 4: odd, odd, even $= 5 \cdot 5 \cdot 5 = 125$

Simply sum up the cases to get your answer, $100 + 100 + 125 + 125 = \boxed{\text{(D) } 450}$.

- Wesseywes7254

Solution 2 (Bijection)

We will show that the answer is 450 by proving a bijection between the three digit integers that have an even number of even digits and the three digit integers that have an odd number of even digits. For every even number with an odd number of even digits, we increment the number's last digit by 1, unless the last digit is 9, in which case it becomes 0. It is very easy to show that every number with an even number of even digits is mapped to every number with an odd number of even digits, and vice versa. Thus, the answer is half the number of three digit numbers, or $\boxed{\text{(D) } 450}$

Solution 3

There are 9 possibilities for the first choice of a three-digit number in the hundreds place, excluding 0. Now we have a few cases:

Case 1: The first digit is odd

In this case, only 1 even digit is permitted, since 3 is impossible. The even digit can be either in the tens or the ones, so we have $5 \cdot 5 \cdot 2$ ways to place the even digit and odd digit.

Case 2: The first digit is even

In this case, either all digits can be even or the rest of the digits can be odd. That's $5 \cdot 5$ for both of them, yielding $5 \cdot 5 \cdot 2$ in total.

Thus, both cases result in multiplying the 9 by $5 \cdot 5 \cdot 2$, or **(D) 450**

-ichang3765

4. (AMC 10 10B 2022) A donkey suffers an attack of hiccups and the first hiccup happens at 4 : 00 one afternoon. Suppose that the donkey hiccups regularly every 5 seconds. At what time does the donkey's 700th hiccup occur?
- (A) 15 seconds after 4 : 58 (B) 20 seconds after 4 : 58 (C) 25 seconds after 4 : 58
(D) 30 seconds after 4 : 58 (E) 35 seconds after 4 : 58

Answer: A

Solution 1

Since the donkey hiccupped the 1st hiccup at 4 : 00, it hiccupped for $5 \cdot (700 - 1) = 3495$ seconds, which is 58 minutes and 15 seconds, so the answer is **(A) 15 seconds after 4 : 58**.

Solution 2 (Faster)

We see that the minute has already been determined. The donkey hiccups once every 5 seconds, or 12 times a minute. $700 \equiv 4 \pmod{12}$, so the 700th hiccup happened on the same second as the 4th, which occurred on the $5(4 - 1) = 15$ th second. **(A) 15 seconds after 4 : 58**.

Solution 3 (Another Fast Way)

We can add $7 \cdot 500 = 3500$ and then minus 5 seconds.

$$\begin{aligned} 4 : 00 + 3500 \text{ seconds} &= 4 : 00 + 3600 \text{ seconds} - 100 \text{ seconds} \\ &= 4 : 00 + 1 \text{ hour} - 100 \text{ seconds} \\ &= 5 : 00 - 100 \text{ seconds} \\ &= 20 \text{ seconds after } 4 : 58. \end{aligned}$$

Finally, we minus 5 seconds giving **(A) 15 seconds after 4 : 58**.

5. (AMC 10 10B 2022) What is the value of

$$\frac{\left(1 + \frac{1}{3}\right) \left(1 + \frac{1}{5}\right) \left(1 + \frac{1}{7}\right)}{\sqrt{\left(1 - \frac{1}{3^2}\right) \left(1 - \frac{1}{5^2}\right) \left(1 - \frac{1}{7^2}\right)}}?$$

- (A) $\sqrt{3}$ (B) 2 (C) $\sqrt{15}$ (D) 4 (E) $\sqrt{105}$

Answer: B

Solution 1 (Difference of Squares)

We apply the difference of squares to the denominator, and then regroup factors:

$$\begin{aligned}
 \frac{\left(1 + \frac{1}{3}\right)\left(1 + \frac{1}{5}\right)\left(1 + \frac{1}{7}\right)}{\sqrt{\left(1 - \frac{1}{3^2}\right)\left(1 - \frac{1}{5^2}\right)\left(1 - \frac{1}{7^2}\right)}} &= \frac{\left(1 + \frac{1}{3}\right)\left(1 + \frac{1}{5}\right)\left(1 + \frac{1}{7}\right)}{\sqrt{\left(1 + \frac{1}{3}\right)\left(1 + \frac{1}{5}\right)\left(1 + \frac{1}{7}\right) \cdot \sqrt{\left(1 - \frac{1}{3}\right)\left(1 - \frac{1}{5}\right)\left(1 - \frac{1}{7}\right)}}} \\
 &= \frac{\sqrt{\left(1 + \frac{1}{3}\right)\left(1 + \frac{1}{5}\right)\left(1 + \frac{1}{7}\right)}}{\sqrt{\left(1 - \frac{1}{3}\right)\left(1 - \frac{1}{5}\right)\left(1 - \frac{1}{7}\right)}} \\
 &= \frac{\sqrt{\frac{4}{3} \cdot \frac{6}{5} \cdot \frac{8}{7}}}{\sqrt{\frac{2}{3} \cdot \frac{4}{5} \cdot \frac{6}{7}}} \\
 &= \frac{\sqrt{4 \cdot 6 \cdot 8}}{\sqrt{2 \cdot 4 \cdot 6}} \\
 &= \frac{\sqrt{8}}{\sqrt{2}} \\
 &= \boxed{\text{(B)} 2}.
 \end{aligned}$$

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Solution 2 (Brute Force)

Since these numbers are fairly small, we can use brute force as follows:

$$\frac{\left(1 + \frac{1}{3}\right)\left(1 + \frac{1}{5}\right)\left(1 + \frac{1}{7}\right)}{\sqrt{\left(1 - \frac{1}{3^2}\right)\left(1 - \frac{1}{5^2}\right)\left(1 - \frac{1}{7^2}\right)}} = \frac{\frac{4}{3} \cdot \frac{6}{5} \cdot \frac{8}{7}}{\sqrt{\frac{8}{9} \cdot \frac{24}{25} \cdot \frac{48}{49}}} = \frac{\frac{4 \cdot 6 \cdot 8}{3 \cdot 5 \cdot 7}}{\sqrt{\frac{(2^3)(2^3 \cdot 3^1)(2^4 \cdot 3^1)}{(3^2)(5^2)(7^2)}}} = \frac{\frac{64}{35}}{\frac{96}{105}} = \frac{64}{35} \cdot \frac{105}{96} = \boxed{\text{(B)} 2}.$$

not_slay

Solution 3 (Brute Force)

This solution starts precisely like the one above. We simplify to get the following:

$$\frac{\left(1 + \frac{1}{3}\right)\left(1 + \frac{1}{5}\right)\left(1 + \frac{1}{7}\right)}{\sqrt{\left(1 - \frac{1}{3^2}\right)\left(1 - \frac{1}{5^2}\right)\left(1 - \frac{1}{7^2}\right)}} = \frac{\frac{4 \cdot 6 \cdot 8}{3 \cdot 5 \cdot 7}}{\sqrt{\frac{(2^3)(2^3 \cdot 3^1)(2^4 \cdot 3^1)}{(3^2)(5^2)(7^2)}}}$$

But now, we can get a nice simplification as shown:

$$\frac{\frac{4 \cdot 6 \cdot 8}{3 \cdot 5 \cdot 7}}{\sqrt{\frac{(2^3)(2^3 \cdot 3^1)(2^4 \cdot 3^1)}{(3^2)(5^2)(7^2)}}} = \frac{\frac{4 \cdot 6 \cdot 8}{3 \cdot 5 \cdot 7}}{\sqrt{\frac{2^{10} \cdot 3^2}{3^2 \cdot 5^2 \cdot 7^2}}} = \frac{\frac{4 \cdot 6 \cdot 8}{3 \cdot 5 \cdot 7}}{\frac{2^5 \cdot 3}{3 \cdot 5 \cdot 7}} = \frac{4 \cdot 6 \cdot 8}{3 \cdot 5 \cdot 7} \cdot \frac{3 \cdot 5 \cdot 7}{2^5 \cdot 3} = \frac{2^6 \cdot 3}{2^5 \cdot 3} = \boxed{\text{(B)} 2}.$$

6. (AMC 10 10B 2022) How many of the first ten numbers of the sequence 121, 11211, 1112111, ... are prime numbers?

(A) 0 (B) 1 (C) 2 (D) 3 (E) 4

Answer: A

Solution 1 (Generalization)

The n th term of this sequence is

$$\sum_{k=n}^{2n} 10^k + \sum_{k=0}^n 10^k = 10^n \sum_{k=0}^n 10^k + \sum_{k=0}^n 10^k = (10^n + 1) \sum_{k=0}^n 10^k.$$

It follows that the terms are

$$\begin{aligned} 121 &= 11 \cdot 11, \\ 11211 &= 101 \cdot 111, \\ 1112111 &= 1001 \cdot 1111, \\ &\vdots \end{aligned}$$

Therefore, there are $\boxed{(\mathbf{A}) \ 0}$ prime numbers in this sequence.

Solution 2 (Detailed Explanation of Solution 1)

Denote this sequence as a_n , then we can find that

$$\begin{aligned} a_1 &= 121 = 10^2 + 2 \cdot 10 + 1 = (10^2 + 10) + (10 + 1), \\ a_2 &= 11211 = (10^4 + 10^3 + 10^2) + (10^2 + 10 + 1), \\ a_3 &= 1112111 = (10^6 + 10^5 + 10^4 + 10^3) + (10^3 + 10^2 + 10 + 1), \\ &\vdots \end{aligned}$$

So, we can induct that the general term is

$$\begin{aligned} a_n &= (10^{2n} + 10^{2n-1} + \dots + 10^{n+1} + 10^n) + (10^n + 10^{n-1} + \dots + 10 + 1) \\ &= 10^n \cdot (10^n + 10^{n-1} + \dots + 10 + 1) + (10^n + 10^{n-1} + \dots + 10 + 1) \\ &= (10^n + 1) \sum_{k=0}^n 10^k. \end{aligned}$$

Therefore, there are $\boxed{(\mathbf{A}) \ 0}$ prime numbers in this sequence.

Solution 3 (Simple Sums)

Observe how

$$\begin{aligned} 121 &= 110 + 11, \\ 11211 &= 11100 + 111, \\ 1112111 &= 1111000 + 1111, \\ &\vdots \end{aligned}$$

all take the form of

$$\underbrace{111\dots}_{n+1} \underbrace{00\dots}_n + \underbrace{111\dots}_{n+1} = \underbrace{111\dots}_{n+1} (10^n + 1).$$

Factoring each of the sums, we have

$$11(10 + 1), 111(100 + 1), 1111(1000 + 1), \dots$$

respectively. With each number factored, there are $\boxed{(\mathbf{A}) \ 0}$ primes in the set.

Solution 4 (Educated Guess)

Note that 121 is divisible by 11 and 11211 is divisible by 3. Because this is Problem 6 of the AMC 10, we assume we do not need to check two-digit prime divisibility or use obscure theorems. Therefore, the answer is $\boxed{(\mathbf{A}) \ 0}$.

7. (AMC 10 10B 2022) For how many values of the constant k will the polynomial $x^2 + kx + 36$ have two distinct integer roots?

(A) 6 (B) 8 (C) 9 (D) 14 (E) 16

Answer: B

Solution 1

Let p and q be the roots of $x^2 + kx + 36$. By **Vieta's Formulas**, we have $p + q = -k$ and $pq = 36$.

It follows that p and q must be distinct factors of 36. The possibilities of $\{p, q\}$ are

$$\pm\{1, 36\}, \pm\{2, 18\}, \pm\{3, 12\}, \pm\{4, 9\}.$$

Each unordered pair gives a unique value of k . Therefore, there are **(B) 8** values of k , corresponding to $\mp 37, \mp 20, \mp 15, \mp 13$, respectively.

Solution 2

Note that k must be an integer. Using the **quadratic formula**, $x = \frac{-k \pm \sqrt{k^2 - 144}}{2}$. Since 4 divides 144 evenly, k and $k^2 - 144$ have the same parity, so x is an integer if and only if $k^2 - 144$ is a perfect square.

Let $k^2 - 144 = n^2$. Then, $(k + n)(k - n) = 144$. Since k is an integer and 144 is even, $k + n$ and $k - n$ must both be even. Assuming that k is positive, we get 5 possible values of $k + n$, namely 2, 4, 8, 6, 12, which will give distinct positive values of k , but $k + n = 12$ gives $k + n = k - n$ and $n = 0$, giving 2 identical integer roots. Therefore, there are 4 distinct positive values of k . Multiplying that by 2 to take the negative values into account, we get $4 \cdot 2 =$ **(B) 8** values of k .

Solution 3 (Pythagorean Triples)

Proceed similar to Solution 2 and deduce that the discriminant of $x^2 + kx + 36$ must be a perfect square greater than 0 to satisfy all given conditions. Seeing something like $k^2 - 144$ might remind us of a right triangle, where k is the hypotenuse, and 12 is a leg. There are four ways we could have this: a 9-12-15 triangle, a 12-16-20 triangle, a 5-12-13 triangle, and a 12-35-37 triangle.

Multiply by 2 to account for negative k values (since k is being squared), and our answer is **(B) 8**.

Solution 4

Since $36 = 2^2 \cdot 3^2$, that means there are $(2 + 1)(2 + 1) = 9$ possible factors of 36. Since $6 \cdot 6$ violates the distinct root condition, subtract 1 from 9 to get 8. Each sum is counted twice, and we count of those twice for negatives. This cancels out, so we get **(B) 8**.

Solution 5 (Discriminant and Factor Rainbow)

Since to have two solutions, the discriminant must be greater than 0, $k^2 - 144 > 0$ so k must be greater than 12 or less than -12 (since it's squared). Next, we know that when factoring, the two numbers must multiply to 36. So you can create a factor rainbow (remember from elementary school) which has the pairs of integers that multiply to 36 (1, 2, 3, 4, 6, 6, 9, 12, 18, 36) (you just look at the position and the one on the opposite side, for example 2 is the second value, and 18 is the second last value, and they multiply to 36). See that there are 4 pairs of values which add up to greater than 12 ($6 + 6$ is equal to 12, not greater), and we look at the negative values of them too (since k is squared) we get **(B) 8** total values.

8. (AMC 10 10B 2022) Consider the following 100 sets of 10 elements each:

$$\begin{aligned} &\{1, 2, 3, \dots, 10\}, \\ &\{11, 12, 13, \dots, 20\}, \\ &\{21, 22, 23, \dots, 30\}, \\ &\vdots \\ &\{991, 992, 993, \dots, 1000\}. \end{aligned}$$

How many of these sets contain exactly two multiples of 7?

- (A) 40 (B) 42 (C) 43 (D) 49 (E) 50

Answer: B

Solution 1 (Competition)

There are $\left\lfloor \frac{1000}{7} \right\rfloor = 142$ numbers divisible by 7. We split these into 100 sets containing 10 numbers each, giving us 1.42 multiples of 7 per set. After the first set, the numbers come evenly, and we multiply 100 by $1.42 - 1 = \boxed{\text{(B) } 42}$.

(Genuinely genius solution MW)

Solution 2 (Casework)

We apply casework to this problem. The only sets that contain two multiples of seven are those for which:

- The multiples of 7 are 1 (mod 10) and 8 (mod 10). That is, the first and eighth elements of such sets are multiples of 7. The first element is $1 + 10k$ for some integer $0 \leq k \leq 99$. It is a multiple of 7 when $k = 2, 9, 16, \dots, 93$.
- The multiples of 7 are 2 (mod 10) and 9 (mod 10). That is, the second and ninth elements of such sets are multiples of 7. The second element is $2 + 10k$ for some integer $0 \leq k \leq 99$. It is a multiple of 7 when $k = 4, 11, 18, \dots, 95$.
- The multiples of 7 are 3 (mod 10) and 0 (mod 10). That is, the third and tenth elements of such sets are multiples of 7. The third element is $3 + 10k$ for some integer $0 \leq k \leq 99$. It is a multiple of 7 when $k = 6, 13, 20, \dots, 97$.

Each case has $\left\lfloor \frac{100}{7} \right\rfloor = 14$ sets. Therefore, the answer is $14 \cdot 3 = \boxed{\text{(B) } 42}$.

Solution 3 (Find A Pattern)

We find a pattern.

$$\begin{aligned} &\{1, 2, 3, \dots, 10\}, \\ &\{11, 12, 13, \dots, 20\}, \\ &\{21, 22, 23, \dots, 30\}, \\ &\vdots \\ &\{991, 992, 993, \dots, 1000\}. \end{aligned}$$

We can figure out that the first set has 1 multiple of 7. The second set also has 1 multiple of 7. The third set has 2 multiples of 7. The fourth set has 1 multiple of 7. The fifth set has 2 multiples of 7. The sixth set has 1 multiple of 7. The

seventh set has 1 multiple of 7. The eighth set has 2 multiples of 7. Disregarding the first set and then calculating this pattern further, we can see (reasonably) that it repeats for each 1 sets. We see that the pattern for the number of multiples per 7 sets (again, disregarding the first set) goes: 1, 2, 1, 2, 1, 1, 2. So, for every 7 sets after the first, there are three sets with 2 multiples of 7. We calculate $\left\lfloor \frac{100-1}{7} \right\rfloor$ and multiply that by 3. (We also disregard the remainder of 1 since it doesn't add any extra sets with 2 multiples of 7.). We get $14 \cdot 3 = \boxed{\text{(B) } 42}$.

Solution 4 (Fastest)

Each set contains exactly 1 or 2 multiples of 7 (Because it will need at least $2 \cdot 7 = 14$ numbers in a group for 3 or more multiples of 7).

There are $\frac{1000}{10} = 100$ total sets and $\left\lfloor \frac{1000}{7} \right\rfloor = 142$ multiples of 7. Thus, there are $142 - 100 = \boxed{\text{(B) } 42}$ sets with 2 multiples of 7.

Solution 5 (Simple Counting, Similar to Solution 1)

Consecutive multiples of 7 must differ by 7. So, if a set $\{\dots 1, \dots 2, \dots 3, (\dots), \dots 8, \dots 9, \dots 0\}$ contains two multiples of 7, they must end with the digits 1 and 8, 2 and 9, or 3 and 0. This reduces the problem to counting the amount of multiples of 7 less than 1000 that end with 1, 2, and 3.

The first multiple of 7 that ends with 1 is 21. The next multiple that ends with 1 occurs 70 later, since that is the smallest multiple of 7 we can add to 21 without affecting the last digit. The greatest number of 70's we can add to 21 while keeping it less than 1000 is 13, because $21 + 70(13) = 931$. Therefore, the set of multiples of 7 less than 1000 ending with 1 is $\{21 + 70(0), 21 + 70(1), \dots, 21 + 70(13)\}$, meaning there are 14 of these particular multiples. We can use the same reasoning to count the multiples of 7 that end with 2 and 3.

The first multiple of 7 that ends with 2 is 42. The greatest number of 70's we can add to 42 here is also 13, since $42 + 70(13) = 952$. The set of multiples of 7 less than 1000 ending with 2 is $\{42 + 70(0), 42 + 70(1), \dots, 42 + 70(13)\}$, giving 14 multiples.

The first multiple of 7 that ends with 3 is 63. The greatest number of 70's we can add to 63 here is yet again 13, since $63 + 70(13) = 973$. The set of multiples of 7 less than 1000 ending with 3 is $\{63 + 70(0), 63 + 70(1), \dots, 63 + 70(13)\}$, giving another 14 multiples.

In total, there are $14 + 14 + 14 = 42$ of these multiples, and so $\boxed{\text{(B) } 42}$ sets with two multiples of 7.

Solution 6 (Very Fast System of Equations)

Let a be the number of sets with 1 multiple of 7 and b be the number of sets with 2 multiples of 7. Note that it is impossible for a set to have more than two multiples of 7.

Since there are a total of 100 sets, $a + b = 100$. Also, since there are $\left\lfloor \frac{1000}{7} \right\rfloor = 142$ multiples of 7 between 1 and 1000, we must have $a + 2b = 142$.

Solving the system of equations

$$a + b = 100$$

$$a + 2b = 142$$

for b gives us the answer of (B) 42.

Solution 7 (reasonable count)

(Similar to Solution 3, but a little more intuitive and less numbers.) Note that this system loops every cycle of length 70, or 7 such sets. From 1 to 70, the multiples of 7 are 7, 14, 21, 28, 35, 42, 49, 56, 63, and 70; note that (21, 28), (42, 49), and (63, 70) are in the same sets. Thus, for every 7 sets, we have 3 sets with exactly two multiples of 7. We have 100 sets, which is $98 + 2 = 7 \cdot 14 + 2$; the first 98 sets contain $\frac{3}{7}(7 \cdot 14) = 42$ desired sets. The last two sets comprise the integers from 981 to 999; the multiples of 7 here are 987 and 994. Neither of the two last sets contains two multiples of 7, so our answer is simply (D) 42.

Solution 8 (Pigeonhole Principle, very fast)

Consider any consecutive seven sets beginning on one more than a multiple of 70. There are guaranteed to be 10 multiples of 7 in these seven sets. Note that no set may have three multiples of 7. This means that some three of the ten sets have 2 multiples of seven. There are 14 such groups of seven sets, and then the sets $\{981, 982, \dots, 989\}$ and $\{991, 992, \dots, \}$. These last two sets are congruent to $\{1, 2, \dots, 9\}$ and $\{11, 12, \dots, 19\}$ modulo 7, thus making it easy to check that each only has one multiple of 7. Therefore, there are $14 \cdot 3 =$ (D) 42 sets with 2 multiples of 7.

9. (AMC 10 10B 2022) The sum

$$\frac{1}{2!} + \frac{2}{3!} + \frac{3}{4!} + \cdots + \frac{2021}{2022!}$$

can be expressed as $a - \frac{1}{b!}$, where a and b are positive integers. What is $a + b$?

- (A) 2020 (B) 2021 (C) 2022 (D) 2023 (E) 2024

Answer: D

Solution 1

For all positive integers n , we have

$$\frac{n}{(n+1)!} = \frac{(n+1) - 1}{(n+1)!} = \frac{1}{n!} - \frac{1}{(n+1)!}.$$

Note that the original sum is a telescoping series:

$$\begin{aligned} \frac{1}{2!} + \frac{2}{3!} + \frac{3}{4!} + \cdots + \frac{2021}{2022!} &= \left(\frac{1}{1!} - \frac{1}{2!}\right) + \left(\frac{1}{2!} - \frac{1}{3!}\right) + \left(\frac{1}{3!} - \frac{1}{4!}\right) + \cdots + \left(\frac{1}{2021!} - \frac{1}{2022!}\right) \\ &= \frac{1}{1!} + \left(-\frac{1}{2!} + \frac{1}{2!} - \frac{1}{3!} + \frac{1}{3!} - \frac{1}{4!} + \cdots + \frac{1}{2021!}\right) - \frac{1}{2022!} \\ &= 1 - \frac{1}{2022!}. \end{aligned}$$

Therefore, the answer is $1 + 2022 =$ (D) 2023.

Solution 2

We add $\frac{1}{2022!}$ to the original expression

$$\left(\frac{1}{2!} + \frac{2}{3!} + \frac{3}{4!} + \cdots + \frac{2021}{2022!}\right) + \frac{1}{2022!} = \left(\frac{1}{2!} + \frac{2}{3!} + \frac{3}{4!} + \cdots + \frac{2020}{2021!}\right) + \frac{1}{2021!}.$$

This sum clearly telescopes, thus we end up with $\left(\frac{1}{2!} + \frac{2}{3!}\right) + \frac{1}{3!} = \frac{2}{2!} = 1$

10. (AMC 10 10B 2022) Camila writes down five positive integers. The unique mode of these integers is 2 greater than their median, and the median is 2 greater than their arithmetic mean. What is the least possible value for the mode?

(A) 5 (B) 7 (C) 9 (D) 11 (E) 13

Answer: D

Solution 1 (Variables)

Let M be the median. It follows that the two largest integers are both $M + 2$.

Let a and b be the two smallest integers such that $a < b$. The sorted list is

$$a, b, M, M + 2, M + 2.$$

Since the median is 2 greater than their arithmetic mean, we have $\frac{a+b+M+(M+2)+(M+2)}{5} + 2 = M$, or

$$a + b + 14 = 2M.$$

Note that $a + b$ must be even. We minimize this sum so that the arithmetic mean, the median, and the unique mode are minimized. Let $a = 1$ and $b = 3$, from which $M = 9$ and $M + 2 = \boxed{\text{(D) } 11}$.

Solution 2 (Elimination)

We can also easily test all the answer choices. (This strategy is generally good to use for multiple-choice questions if you don't have a concrete method to proceed with!)

For answer choice (A), the mode is 5, the median is 3, and the arithmetic mean is 1. However, we can quickly see this doesn't work, as there are five integers, and they can't have an arithmetic mean of 1 while having a mode of 5.

Trying answer choice (B), the mode is 7, the median is 5, and the arithmetic mean is 3. From the arithmetic mean, we know that all the numbers have to sum to 15. We know three of the numbers: __, __, 5, 7, 7. This exceeds the sum of 15.

Now we try answer choice (C). The mode is 9, the median is 7, and the arithmetic mean is 5. From the arithmetic mean, we know that the list sums to 25. Three of the numbers are __, __, 7, 9, 9, which is exactly 25. However, our list needs positive integers, so this won't work.

Since we were really close on answer choice (C), we can intuitively feel that the answer is probably going to be (D). We can confirm this by creating a list that satisfies the problem and choose (D) : 1, 3, 9, 11, 11.

So, our answer is $\boxed{\text{(D) } 11}$.

11. (AMC 10 10B 2022) All the high schools in a large school district are involved in a fundraiser selling T-shirts. Which of the choices below is logically equivalent to the statement "No school bigger than Euclid HS sold more T-shirts than Euclid HS"?

(A) All schools smaller than Euclid HS sold fewer T-shirts than Euclid HS.
 (B) No school that sold more T-shirts than Euclid HS is bigger than Euclid HS.
 (C) All schools bigger than Euclid HS sold fewer T-shirts than Euclid HS.
 (D) All schools that sold fewer T-shirts than Euclid HS are smaller than Euclid HS.
 (E) All schools smaller than Euclid HS sold more T-shirts than Euclid HS.

Answer: B

Solution 1

Let B denote a school that is bigger than Euclid HS, and M denote a school that sold more T-shirts than Euclid HS.

It follows that $\neg B$ denotes a school that is **not bigger than** Euclid HS, and $\neg M$ denotes a school that **did not sell more** T-shirts than Euclid HS.

Converting everything to conditional statements (if-then form), the given statement becomes

$$B \implies \neg M.$$

Its contrapositive is $M \implies \neg B$, which is **(B)**.

Note that "not bigger than" does not mean "smaller than", and "not selling more" does not mean "selling fewer". There is an equality case. Therefore, none of the other answer choices is equivalent to $B \implies \neg M$.

Solution 2 (Elimination)

Suppose we have five schools: Euclid HS with 50 students and 10 T-shirts sold.

- School A with 51 students and 10 T-shirts sold.
- School B with 49 students and 10 T-shirts sold.
- School C with 49 students and 9 T-shirts sold.
- School D with 51 students and 9 T-shirts sold (This configuration is legal.)

Then, school B rules out **(A)**, school A rules out **(C)**, school D rules out **(D)**, and school C rules out **(E)**, leaving us with **(B)** as the correct answer.

Solution 3 (Logic/Common Sense to Eliminate)

First, you can eliminate A because the question doesn't say anything about smaller schools. Next, you can notice that B seems right at first glance, so you can keep it in contention. Next, you can eliminate C because it says they sold fewer, while the problem only states they didn't sell more than Euclid HS, so it can be fewer or equal to the number of shirts. Next, you can eliminate D because for the same reason as A. Finally, you can eliminate E for the same reason as A. Since everything except B is eliminated, **(B)** is the correct answer.

12. (AMC 10 10B 2022) A pair of fair 6-sided dice is rolled n times. What is the least value of n such that the probability that the sum of the numbers face up on a roll equals 7 at least once is greater than $\frac{1}{2}$?
- (A) 2 (B) 3 (C) 4 (D) 5 (E) 6

Answer: C

Solution 1 (Complement)

Rolling a pair of fair 6-sided dice, the probability of getting a sum of 7 is $\frac{1}{6}$. Regardless what the first die shows, the second die has exactly one outcome to make the sum 7. We consider the complement: The probability of not getting a sum of 7 is $1 - \frac{1}{6} = \frac{5}{6}$. Rolling the pair of dice n times, the probability of getting a sum of 7 at least once is $1 - \left(\frac{5}{6}\right)^n$.

Therefore, we have $1 - \left(\frac{5}{6}\right)^n > \frac{1}{2}$, or

$$\left(\frac{5}{6}\right)^n < \frac{1}{2}.$$

Since $\left(\frac{5}{6}\right)^4 < \frac{1}{2} < \left(\frac{5}{6}\right)^3$, the least integer n satisfying the inequality is **(C) 4**.

Solution 2 (Fakesolve)

Let's try the answer choices. We can quickly find that when we roll 3 dice, either the first and second sum to 7, the first and third sum to 7, or the second and third sum to 7. There are 6 ways for the first and second dice to sum to 7, 6 ways for the first and third to sum to 7, and 6 ways for the second and third dice to sum to 7. However, we overcounted (but not by much) so we can assume that the answer is **(C) 4**.

Solution 3 (Fakesolve)

We can start by figuring out what the probability is for each die to add up to 7 if there is only 1 roll. We can quickly see that the probability is $\frac{1}{6}$, as there are 6 ways to make 7 from 2 numbers on a die, and there are a total of 36 ways to add 2 numbers on a die. And since each time we roll the dice, we are adding to the probability, we can conclude that the total probability for rolling a sum of 7 in n rolls would be $\frac{1}{6}n$. The smallest number that satisfies this is **(C) 4**.

Solution 4 (Fakesolve)

On each roll, there is a $\frac{1}{6}$ chance of rolling a sum of 7. You would need 4 of these rolls to get $4 \cdot \frac{1}{6}$, which is larger than $\frac{1}{2}$. Therefore, the answer is **(C) 4**.

13. (AMC 10 10B 2022) The positive difference between a pair of primes is equal to 2, and the positive difference between the cubes of the two primes is 31106. What is the sum of the digits of the least prime that is greater than those two primes?
(A) 8 **(B) 10** **(C) 11** **(D) 13** **(E) 16**

Answer: E

Solution 1 (Simplest)

Denote the two primes as a and b . Then,

$$a - b = 2$$

$$a^3 - b^3 = 31106$$

$$\text{We see that } a = 2 + b$$

$$\text{Now, we have } (2 + b)^3 - b^3 = 31106$$

We apply the binomial theorem (or just expand) to $(2 + b)^3$, getting

$$8 + 12b + 6b^2 + b^3 - b^3 = 31106$$

$$\Rightarrow 8 + 12b + 6b^2 = 31106$$

Subtracting by 8 on both sides results in

$$6b^2 + 12b = 31098$$

Then dividing it all by 6 and pushing it all to one side results in

$$b^2 + 2b - 5183 = 0$$

$$(b - 71)(b + 73) = 0$$

We see that $b \in \{-73, 71\}$. We negate all negative values, and see that $b = 71$.

Therefore $a - b = 2$, $a = 2 + b$, $a = 2 + 71 = 73$.

The next prime number greater than both of these is 79, and therefore our answer is $7 + 9 = \boxed{\text{(E)} 16}$.

Solution 2

Let the two primes be a and b with $a > b$. We would have $a - b = 2$ and $a^3 - b^3 = 31106$. Using difference of cubes, we would have $(a - b)(a^2 + ab + b^2) = 31106$. Since we know $a - b$ is equal to 2, $(a - b)(a^2 + ab + b^2)$ would become $2(a^2 + ab + b^2) = 31106$. Simplifying more, we would get $a^2 + ab + b^2 = 15553$.

Now let's introduce another variable. Instead of using a and b , we can express the primes as $x + 2$ and x where a is $x + 2$ and b is x . Plugging x and $x + 2$ in, we would have $(x + 2)^2 + x(x + 2) + x^2$. When we expand the parenthesis, it would become $x^2 + 4x + 4 + x^2 + 2x + x^2$. Then we combine like terms to get $3x^2 + 6x + 4$ which equals 15553. Then we subtract 4 from both sides to get $3x^2 + 6x = 15549$. Since all three numbers are divisible by 3, we can divide by 3 to get $x^2 + 2x = 5183$.

Notice how if we add 1 to both sides, the left side would become a perfect square trinomial: $x^2 + 2x + 1 = 5184$ which is $(x + 1)^2 = 5184$. Since 2 is too small to be a valid number, the two primes must be odd, therefore $x + 1$ is the number in the middle of them. Conveniently enough, $5184 = 72^2$ so the two numbers are 71 and 73. The next prime number is 79, and $7 + 9 = 16$ so the answer is $\boxed{\text{(E)} 16}$.

A note: If you aren't entirely familiar with square trinomials, after we have $x^2 + 2x = 5183$, we can simply guess and check from here. We know that $70^2 = 4900$, and we notice that for $x^2 + 2x = 5183$, x must be a odd number. Thus we can guess that $x = 71$, which proves to be right. Continuing, we then know the larger prime is 73, and we know that 75 and 77 aren't primes, so thus our next bigger prime is 79, and $7 + 9 = 16$. Thus the answer is $\boxed{\text{(E)} 16}$.

Understanding the "Completing a Square" formula: Adding 1 to $x^2 + 2x$ is called completing a square. In the expression $x^2 + bx$, where b is a constant, you can add $b/2$ to make the square $(x + b/2)^2$.

Solution 3

Let the two primes be a and b , with a being the larger prime. We have $a - b = 2$, and $a^3 - b^3 = 31106$. Using difference of cubes, we obtain $a^2 + ab + b^2 = 15553$. Now, we use the equation $a - b = 2$ to obtain $a^2 - 2ab + b^2 = 4$. Hence,

$$a^2 + ab + b^2 - (a^2 - 2ab + b^2) = 3ab = 15553 - 4 = 15549$$

$$ab = 5183.$$

Because we have $a = b + 2$, $ab = (b + 1)^2 - (1)^2$. Thus, $(b + 1)^2 = 5183 + 1 = 5184$, so $b + 1 = 72$. This implies $a = 73$, $b = 71$, and thus the next biggest prime is 79, so our answer is $7 + 9 = \boxed{\text{(E)} 16}$.

Solution 4 (Estimation)

Let the two primes be p and q such that $p - q = 2$ and $p^3 - q^3 = 31106$

By the difference of cubes formula, $p^3 - q^3 = (p - q)(p^2 + pq + q^2)$

Plugging in $p - q = 2$ and $p^3 - q^3 = 31106$,

$$31106 = 2(p^2 + pq + q^2)$$

Through the givens, we can see that $p \approx q$.

$$\begin{aligned}\text{Thus, } 31106 &= 2(p^2 + pq + q^2) \approx 6p^2 \\ p^2 &\approx \frac{31106}{6} \approx 5200\end{aligned}$$

Recall that $70^2 = 4900$ and $80^2 = 6400$. It follows that our primes must be only marginally larger than 70, where we conveniently find $p = 73, q = 71$

The least prime greater than these two primes is 79 $\implies 7 + 9 \implies \boxed{\text{(E)} 16}$

Solution 5

Let the two primes be $x + 1$ and $x - 1$. Then, plugging it into the second condition, we get $(x + 1)^3 - (x - 1)^3 = 31106$. Expanding the left side,

$$6x^2 + 2 = 31106 \implies x^2 = 5184.$$

Taking the square root of both sides, we get that $x = 72$ and the larger prime is 73. The smallest prime larger than 73 is 79, which has a digit sum of $7 + 9 = \boxed{16}$.

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Solution 6

Let the two primes be x and y . We write $x^3 - y^3 = 31106$. So $(x - y)(x^2 + xy + y^2) = 31106$.

We manipulate the equation by subtracting $3xy$ to get a perfect square.

$$\begin{aligned}(x^2 - 2xy + y^2) &= 15553 - 3xy \quad (x - y)^2 = 15553 - 3xy \\ 4 &= 15553 - 3xy \\ 3xy &= 15549 \\ xy &= 5183.\end{aligned}$$

We write out the factors of 5183 and find 71 and 73.

We can then list out primes greater than 73 to find 79, which has a digit sum of $\boxed{16}$.

14. (AMC 10 10B 2022) Suppose that S is a subset of $\{1, 2, 3, \dots, 25\}$ such that the sum of any two (not necessarily distinct) elements of S is never an element of S . What is the maximum number of elements S may contain?

(A) 12 (B) 13 (C) 14 (D) 15 (E) 16

Answer: B

Solution 1 (Pigeonhole Principle)

Let M be the largest number in S . We categorize numbers $\{1, 2, \dots, M - 1\}$ (except $\frac{M}{2}$ if M is even) into $\left\lfloor \frac{M-1}{2} \right\rfloor$ groups, such that the i th group contains two numbers i and $M - i$.

Recall that $M \in S$ and the sum of two numbers in S cannot be equal to M , and the sum of numbers in each group above is equal to S . Thus, each of the above $\left\lfloor \frac{M-1}{2} \right\rfloor$ groups can have at most one number in S . Therefore,

$$\begin{aligned}|S| &\leq 1 + \left\lfloor \frac{M-1}{2} \right\rfloor \\ &\leq 1 + \left\lfloor \frac{25}{2} \right\rfloor \\ &= 13.\end{aligned}$$

Next, we construct an instance of S with $|S| = 13$. Let $S = \{13, 14, \dots, 25\}$. Thus, this set is feasible. Therefore, the most number of elements in S is $\boxed{\text{(B)} 13}$.

Solution 2 (Pigeonhole v2)

We construct a possible subset S with 13 elements by including all odd integers from 1 to 25, inclusive. $S = \{1, 3, 5, \dots, 25\}$. The sum of any 2 elements is even, and thus cannot be an element of S .

To show that S cannot have more than 13 elements, assume for sake of contradiction that $|S| \geq 14$. Let $S = \{x_1, x_2, \dots, x_n\}$ where $n \geq 14$ and $x_1 < x_2 < \dots < x_n$. Because the sums of any 2 (not necessarily distinct) elements do not appear in S , $x_1 + x_i$ is not an element of S for all $1 \leq i \leq n$. So, $x_1, x_2, \dots, x_n, x_1 + x_1, x_1 + x_2, \dots, x_1 + x_n$ are all distinct integers. Let these integers be elements of the set T . $|T| = 2n$, and because $n \geq 14$, $|T| \geq 28$. But all elements of T must be $\geq x_1$ and $\leq x_1 + x_n \leq x_1 + 25$, leaving only 26 possible values for the elements in T . By the Pigeonhole Principle, the elements cannot be distinct, and we have a contradiction.

Thus, $\boxed{\text{(B)} 13}$ is the maximum possible size of S .

Solution 3 (Intuitive and Fast)

Notice how $24 + 25 > 25$, $23 + 24 > 25$, and this continues backward until $n + n + 1 > 25$. We then see that $2n + 1 > 25 \rightarrow 2n > 24 \rightarrow n > 12$.

Thus the elements in the set must have the property $n > 12 : n \in \mathbb{Z}^+$ and therefore our set is $\{13, 14, 15, \dots, 25\}$ in which there is $25 - 13 + 1 \Rightarrow \boxed{\text{(B)} 13}$ elements.

15. (AMC 10 10B 2022) Let S_n be the sum of the first n terms of an arithmetic sequence that has a common difference of 2. The quotient $\frac{S_{3n}}{S_n}$ does not depend on n . What is S_{20} ?

(A) 340 (B) 360 (C) 380 (D) 400 (E) 420

Answer: D

Solution 1

Let's say that our sequence is

$$a, a + 2, a + 4, a + 6, a + 8, a + 10, \dots$$

Then, since the value of n doesn't matter in the quotient $\frac{S_{3n}}{S_n}$, we can say that

$$\frac{S_3}{S_1} = \frac{S_6}{S_2}.$$

Simplifying, we get $\frac{3a+6}{a} = \frac{6a+30}{2a+2}$, from which

$$\frac{3a+6}{a} = \frac{3a+15}{a+1}.$$

$$3a^2 + 9a + 6 = 3a^2 + 15a$$

$$6a = 6$$

Solving for a , we get that $a = 1$.

Since the sum of the first n odd numbers is n^2 , $S_{20} = 20^2 = \boxed{\text{(D)} 400}$.

Note: you could also plug in the formulas for $\frac{S_{3n}}{S_n}$ and simplify, getting $3 + \frac{6n}{a+n-1}$. You would then find $a = 1$ megacleverstarfish15

(Optimal competition solution if you're short on time. Lunar Paradox)

Solution 2 (Quick Insight)

Recall that the sum of the first n odd numbers is n^2 .

Since $\frac{S_{3n}}{S_n} = \frac{9n^2}{n^2} = 9$, we have $S_{20} = 20^2 = \boxed{\text{(D) } 400}$.

Solution 3 (I didn't know Solution 1!)

We have a slightly challenging problem :-(, but that's okay!

If you didn't come up with the Solution 1 intuition, then here is a more direct approach.

We want $\frac{S_{3n}}{S_n}$ to be a natural ($\frac{S_{3n}}{S_n} > 0$; basically a whole number $\neq 0$) number. Then only can the progression be incremental by 2.

Assume that $a_1 = 1$. Then, $a_2 = 3, a_3 = 5$, etc. We take the first term $n = 1$, which is 1. Then $3n$; the next 3 terms will sum to 9. $\frac{9}{1} = 9$, and this is an arithmetic sequence, and therefore works!

What about $a_1 = 0$. We immediately see that this would be undefined, so we cannot have $a_1 = 0$. So we say $a_1 = 2$. Then the sum for $n = 1$ is 2, and for $3n$; it is simply 6. This is $6/2 = 3$, so it works!

Then, what about $a_2 = 3$? Then this means that the $n = 2$ sum is 4, and the $3n$ sum is 36. This is $36/4 = 9$, and this is the same as the difference before, proving that $a_1 = 1$ is indeed correct.

And then? What about $a_2 = 4$? This means $n = 2$ sum is 6, and the $3n$ sum is 42, but uh oh, the progression breaks! Therefore, the evens cannot work.

Therefore, $a_1 = 1, a_{20}$ is just the sum of the first 20 odd numbers, which is $\boxed{\text{(D) } 400}$

Remark 3.0.1 Hi! It me (Pinotation) again!

If you are confused about why $a_1 \neq 3$ or $a_1 \neq 4$ or something along those lines, we can prove by induction.

We have

$$S_n = \frac{n}{2}(2a_1 + (n-1) \cdot 2) = n(a_1 + n - 1).$$

Then

$$S_{3n} = 3n(a_1 + 3n - 1),$$

so

$$\frac{S_{3n}}{S_n} = \frac{3(a_1 + 3n - 1)}{a_1 + n - 1}.$$

Base case $n = 1$:

$$\frac{S_3}{S_1} = \frac{3(a_1 + 2)}{a_1}.$$

Suppose for some n the ratio is independent of n and equal to a constant k . Then

$$\frac{S_{3n}}{S_n} = \frac{3(a_1 + 3n - 1)}{a_1 + n - 1} = k.$$

For $n + 1$ we must also have

$$\frac{S_{3(n+1)}}{S_{n+1}} = \frac{3(a_1 + 3n + 2)}{a_1 + n} = k.$$

So

$$\frac{3(a_1 + 3n - 1)}{a_1 + n - 1} = \frac{3(a_1 + 3n + 2)}{a_1 + n}.$$

Cross multiplying gives

$$(a_1 + 3n - 1)(a_1 + n) = (a_1 + 3n + 2)(a_1 + n - 1).$$

Expanding,

$$a_1^2 + 4na_1 + a_1 - n - 3 = a_1^2 + 4na_1 + a_1 + 2n - 2.$$

Simplifying,

$$-n - 3 = 2n - 2,$$

so

$$3n = -1.$$

This is impossible for integer n . The only way the equality can hold for all n is if the proportionality factor between numerator and denominator is fixed. Comparing coefficients in

$$3(a_1 + 3n - 1) = C(a_1 + n - 1),$$

we get $C = 9$ and hence

$$3a_1 - 3 = 9a_1 - 9 \implies a_1 = 1.$$

I hope this helped! If not, then you can check Solution 1, as it is practically the friendlier version LOL!

Solution 4 (Answer Choices)

Let a be the first element. Then, $S_{20} = 20(a + 19)$. Let it equal to all the answer choices respectively, we get $a = -2, -1, 0, 1, 2$. We test $\frac{S_{3n}}{S_n}$ for each a , using $n = 1, 2$. We can see that for $a = -2, -1, 0$ there exists n to let $S_n = 0$, which implies they are invalid. For $a = 2$ we can also see it's invalid, since $\frac{S_3}{S_1} = 6 \neq 7 = \frac{S_6}{S_2}$. Therefore, $a = 1$, and **(D) 400** is the correct choice.

Solution 5 (No usage of initial term or difference)

Let a and d be the initial term and difference respectively. (These won't be used it later)

Notice that the terms $n+1, n+2, \dots, 2n = (a+d+nd), (a+2d+nd), \dots, (a+nd+nd) = S_n + n(nd) = S_n + n^2d$. In a similar fashion, we have $S_{3n} = 3S_n + 2n^2 + 4n^2 = 3S_n + 6n^2$.

We are given that the quotient $\frac{S_{3n}}{S_n}$ does not depend on n . With the result above, we have

$$\frac{3S_n + 6n^2}{S_n}$$

where we get

$$3 + \frac{6n^2}{S_n}$$

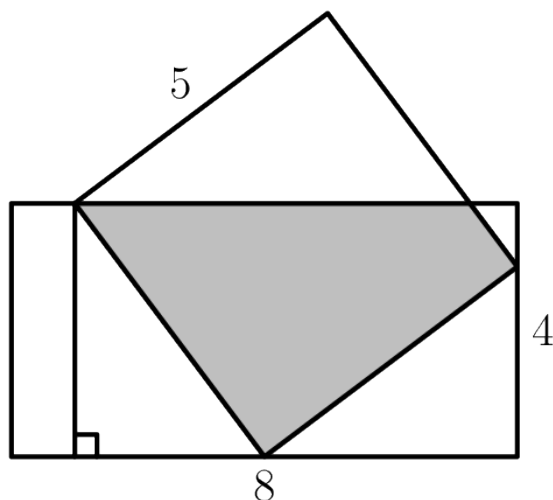
With the information above, we must have

$$S_n = kn^2$$

If $k \neq 1$, then $d \neq 2$, so we must have $S_n = n^2$.

Hence, $S_{20} = 20^2 = \mathbf{(D) 400}$

16. (AMC 10 10B 2022) The diagram below shows a rectangle with side lengths 4 and 8 and a square with side length 5. Three vertices of the square lie on three different sides of the rectangle, as shown. What is the area of the region inside both the square and the rectangle?

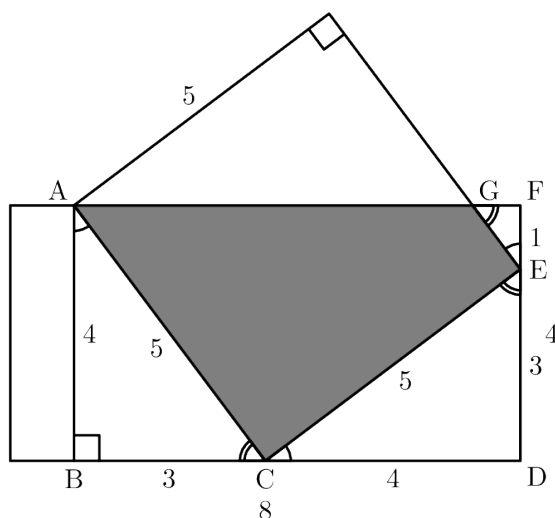


- (A) $15\frac{1}{8}$ (B) $15\frac{3}{8}$ (C) $15\frac{1}{2}$ (D) $15\frac{5}{8}$ (E) $15\frac{7}{8}$

Answer: D

Solution 1

Let us label the points on the diagram.



By doing some angle chasing using the fact that $\angle ACE$ and $\angle CEG$ are right angles, we find that $\angle BAC = \angle DCE = \angle FEG$. Similarly, $\angle ACB = \angle CED = \angle EGF$. Therefore, $\triangle ABC \sim \triangle CDE \sim \triangle EFG$.

As we are given a rectangle and a square, $AB = 4$ and $AC = 5$. Therefore, $\triangle ABC$ is a 3-4-5 right triangle and $BC = 3$.

CE is also 5. So, using the similar triangles, $CD = 4$ and $DE = 3$.

$EF = DF - DE = 4 - 3 = 1$. Using the similar triangles again, EF is $\frac{1}{4}$ of the corresponding AB . So,

$$\begin{aligned}
 [\triangle EFG] &= \left(\frac{1}{4}\right)^2 \cdot [\triangle ABC] \\
 &= \frac{1}{16} \cdot 6 \\
 &= \frac{3}{8}.
 \end{aligned}$$

Finally, we have

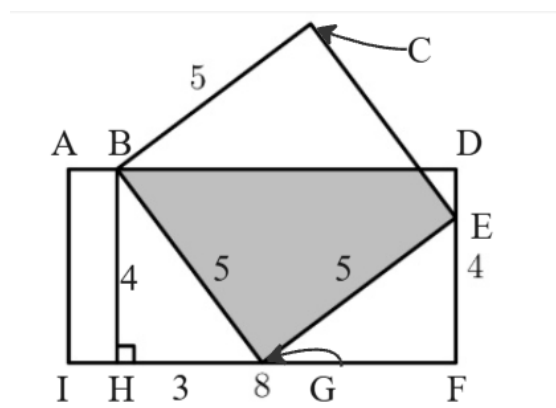
$$\begin{aligned}
 [ACEG] &= [ABDF] - [\triangle ABC] - [\triangle CDE] - [\triangle EFG] \\
 &= 7 \cdot 4 - \frac{1}{2} \cdot 3 \cdot 4 - \frac{1}{2} \cdot 3 \cdot 4 - \frac{3}{8} \\
 &= 28 - 6 - 6 - \frac{3}{8} \\
 &= \boxed{\text{(D)} 15\frac{5}{8}}.
 \end{aligned}$$

Solution 2 (Clever)

(Refer to the diagram above) Proceed the same way as Solution 1 until you get all of the side lengths. Then, it is clear that due to the answer choices, we only need to find the fractional part of the shaded area. The area of the whole rectangle is integral, as is the area of $\triangle ABC$, $\triangle CDE$, and the rectangle to the far left of the diagram. The area of EFG is $\frac{3}{8}$ and thus the fractional part of the answer is $\frac{5}{8}$.

The only answer choice that has $\frac{5}{8}$ in it is **(D)** $15\frac{5}{8}$

Solution 3 (Intuitive Risking)



We can quickly find out through Pythagorean Triples that $GH = 3$. Now, we make an educated guess that G is the midpoint of FI , and therefore $HI = 1$.

Noice! So, now what? Well, FG is then 4, and EF would then be 3.

Given that info, we also see that $DE = 1$.

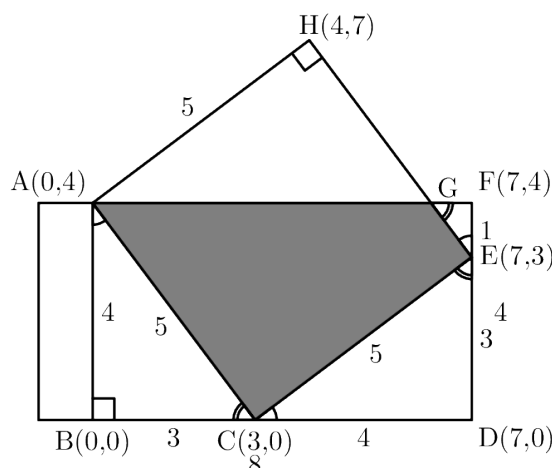
Our area of the rectangle $ADFI$ is $8 \times 4 = 32$. We then see that we must subtract rectangle $ABHI = 4 \times 1 = 4$, triangle $BHG = \frac{1}{2} \times 3 \times 4 = 6$, and triangle $GFE = \frac{1}{2} \times 4 \times 3 = 6$. This gives $32 - (4 + 6 + 6) = 16$.

We make a guess that the intersection between B and D , call it J , to E is exactly $5/4$. Then, through the Pythagorean theorem, $JD = 1/2$, and the area is $\frac{1}{2} \times \frac{1}{2} \times 1 = \frac{1}{4}$. So, the area of that mini triangle is $\frac{1}{4}$, in reality we know it is bigger. We subtract this from 16 to get $15\frac{3}{4}$ and take the closest answer choice that is lower than the result we got, which is **(D)** $15\frac{5}{8}$.

Take risks everyone! You might do great things like solve an AMC problem :D!

Solution 4 (Coordinate Geometry)

Same diagram as Solution 1, but added point H , which is $(4, 7)$. I also renamed all the points to form coordinates using B as the origin.



In order to find the area, point G 's coordinates must be found. Notice how EH and AG intercept at point G . This means that we need to find the equations for EH and AG and make a system of linear equations.

Using the slope formula $m = \frac{y_2 - y_1}{x_2 - x_1}$, we get the slope for EH , which means $m = \frac{3-7}{7-4} = -\frac{4}{3}$

Then, by using point slope form. $y - y_1 = m(x - x_1)$. We can say that the equation for EH is $y - 7 = -\frac{4}{3}(x - 4)$ or in this case, $y = -\frac{4}{3}x + 12\frac{1}{3}$.

And it is easy to figure out that the equation for AG is $y = 4$.

The best way to solve the system of linear equations is to substitute the y for the 4 in equation EH . $4 = -\frac{4}{3}x + 12\frac{1}{3}$, so $x = 6\frac{1}{4}$ and $y = 4$ This would mean $G(6\frac{1}{4}, 4)$.

Since we have our G coordinate, we can continue with Solution 3, with the area of the trapezoid $\left(\frac{EG+AC}{2}\right)(CE)$, where $EG = \frac{5}{4}$ (using distance formula for E to G), $AC = 5$, and $CE = 5$.

By substitution, we get $\left(\frac{\frac{5}{4}+5}{2}\right)(5) = \mathbf{(D)} \ 15\frac{5}{8}$.

Solution 5 (FAST)

Notice the small triangle in the upper right corner is a $3 - 4 - 5$ triangle. Then that triangle is similar to the big triangle by AA similarity. From that, do similar triangles and you find that the longer leg of the small triangle is 1. Then you find that the triangle below is $3 - 4 - 5$, so the side length of the rectangle (without the outer rectangle) is 7. afterwards you just add the half of the square + the

remaining triangle which can be found by multiplying base and height (in which we already know)

I found the same solution when solving this, but essentially, we know all triangles in the picture are similar, and we know the lower left hand triangle is a 3-4-5 triangle, and it is similar to the top triangle, where 4 corresponds to 5, therefore the unknown side length of that triangle is $15/4$. Continuing on, the area of the white part of the square is $5 * 15/4 * 1/2 = 75/8$, $25 - 75/8 = \boxed{(D)}$

-kaiser

Solution 6 (Fastest Similar Triangles)

For reference, use the points labelled in the diagram of Solution 3. Let the point one unit to the right of A be A' (so that A' is one of the vertices of the square). The square means $A'C = 5$, so we get a 3-4-5 triangle $A'BC$.

$$m\angle BA'C = 90^\circ - m\angle CA'G = m\angle GA'H.$$

Therefore $\triangle GA'H$ is proportional to a 3-4-5 triangle, with HG corresponding to 3 and $A'H$ corresponding to 4. By similar triangles, we find

$$HG = A'H \cdot \frac{3}{4} = \frac{15}{4}.$$

$$\text{then, finally, } [A'CEG] = [A'CEH] - [\triangle GA'H] = 5^2 - \frac{1}{2} \cdot \frac{15}{4} \cdot 5 =$$

$$\boxed{(D) \ 15\frac{5}{8}}.$$

17. (AMC 10 10B 2022) One of the following numbers is not divisible by any prime number less than 10. Which is it?

(A) $2^{606} - 1$ (B) $2^{606} + 1$ (C) $2^{607} - 1$ (D) $2^{607} + 1$ (E) $2^{607} + 3^{607}$

Answer: C

Solution 1 (Modular Arithmetic)

For (A) modulo 3,

$$\begin{aligned} 2^{606} - 1 &\equiv (-1)^{606} - 1 \\ &\equiv 1 - 1 \\ &\equiv 0. \end{aligned}$$

Thus, $2^{606} - 1$ is divisible by 3.

For (B) modulo 5,

$$\begin{aligned} 2^{606} + 1 &\equiv 2^{\text{Rem}(606, \phi(5))} + 1 \\ &\equiv 2^{\text{Rem}(606, 4)} + 1 \\ &\equiv 2^2 + 1 \\ &\equiv 0. \end{aligned}$$

Thus, $2^{606} + 1$ is divisible by 5.

For (D) modulo 3,

$$\begin{aligned} 2^{607} + 1 &\equiv (-1)^{607} + 1 \\ &\equiv -1 + 1 \\ &\equiv 0. \end{aligned}$$

Thus, $2^{607} + 1$ is divisible by 3.

For (E) modulo 5,

$$\begin{aligned} 2^{607} + 3^{607} &\equiv 2^{607} + (-2)^{607} \\ &\equiv 2^{607} - 2^{607} \\ &\equiv 0. \end{aligned}$$

Thus, $2^{607} + 3^{607}$ is divisible by 5.

Therefore, the answer is (C) $2^{607} - 1$.

Solution 2 (Factoring)

We have

$$\begin{aligned} 2^{606} - 1 &= 4^{303} - 1 = (4 - 1)(4^{302} + 4^{301} + 4^{300} + \cdots + 4^0), \\ 2^{606} + 1 &= 4^{303} + 1 = (4 + 1)(4^{302} - 4^{301} + 4^{300} - \cdots + 4^0), \\ 2^{607} + 1 &= (2 + 1)(2^{606} - 2^{605} + 2^{604} - \cdots + 2^0), \\ 2^{607} + 3^{607} &= (2 + 3)(2^{606} \cdot 3^0 - 2^{605} \cdot 3^1 + 2^{604} \cdot 3^2 - \cdots + 2^0 \cdot 3^{606}). \end{aligned}$$

We conclude that (A) is divisible by 3, (B) is divisible by 5, (D) is divisible by 3, and (E) is divisible by 5.

Since all of the other choices have been eliminated, we are left with (C) $2^{607} - 1$.

Solution 3 (Elimination)

Mersenne Primes are primes of the form $2^n - 1$, where n is prime. Using the process of elimination, we can eliminate every option except for (A) and (C). Clearly, 606 isn't prime, so the answer must be (C) $2^{607} - 1$.

Note: A organization of computer scientist and mathematicians named Great Internet Mersenne Prime Search (GIMPS) search for the worlds biggest prime numbers. More information can be found on <https://www.mersenne.org/primes/> hash-brown2009

Solution 3a (Elimination)

We examine option E first. 2^{607} has a units digit of 8 (Taking the units digit of the first few powers of two gives a pattern of 2, 4, 8, 6, 2, 4, 8, 6, 2, 4, 8, 6, ...) and 3^{607} has a units digit of 7 (Taking the units digit of the first few powers of three gives a pattern of 3, 9, 7, 1, 3, 9, 7, 1, 3, 9, 7, 1, ...). Adding 7 and 8 together, we get 15, which is a multiple of 5, meaning that $2^{607} + 3^{607}$ is divisible by 5.

Next, we examine option D. We take the first few powers of 2 added with 1:

$$\begin{aligned} 2^1 + 1 &= 3 \\ 2^2 + 1 &= 5 \\ 2^3 + 1 &= 9 \\ 2^4 + 1 &= 17 \\ 2^5 + 1 &= 33 \\ 2^6 + 1 &= 65 \\ 2^7 + 1 &= 129 \end{aligned}$$

We see that the odd powers of 2 added with 1 are multiples of three. If we continue this pattern, $2^{607} + 1$ will be divisible by 3. (The reason why this pattern works: When you multiply $2 \equiv 2 \pmod{3}$ by 2, you obtain $4 \equiv 1 \pmod{3}$. Multiplying by 2 again, we get $1 \cdot 2 \equiv 2 \pmod{3}$. We see that in every cycle of two powers of 2, it goes from $2 \pmod{3}$ to $1 \pmod{3}$ and back to $2 \pmod{3}$.)

Next, we examine option B. We see that 2^{606} has a units digit of 4 (Taking the units digit of the first few powers of 2 gives a pattern of 2, 4, 8, 6, 2, 4, 8, 6, 2, 4, 8, 6, $\dots$). Adding 1 to 4, we get 5. Since $2^{606} + 1$ has a units digit of 5, it is divisible by 5.

Lastly, we examine option A. Using the difference of cubes factorization $a^3 - b^3 = (a - b)(a^2 + ab + b^2)$, we have $2^{606} - 1^3 = (2^{202} - 1)(2^{404} + 2^{202} + 1)$. Since $2^{404} + 2^{202} + 1 \equiv 0 \pmod{3}$ (Every term in the sequence is equivalent to 1 $\pmod{3}$), $2^{606} - 1$ is divisible by 3.

Since we have eliminated every option except C, $(C) \ 2^{607} - 1$ is not divisible by any prime less than 10.

Solution 3b (Elimination + Number Theory)

We know that the prime numbers less than 10 are 2, 3, 5 and 7. We can start by testing if any of the answer choices are divisible by 2. We see that they are all sums of one even number and one odd number, which is simply odd. So, we cannot exclude any answer choices so far. Now, let's check divisibility by 3. We can use the fact that $2 \equiv -1 \pmod{3}$ to our advantage:

(A) $2^{606} - 1 \equiv (-1)^{606} - 1 \equiv 1 - 1 \equiv 0 \pmod{3}$ (B) $2^{606} + 1 \equiv (-1)^{606} + 1 \equiv 1 + 1 \equiv 2 \pmod{3}$ (C) $2^{607} - 1 \equiv (-1)^{607} - 1 \equiv -1 - 1 \equiv -2 \pmod{3}$ (D) $2^{607} + 1 \equiv (-1)^{607} + 1 \equiv -1 + 1 \equiv 0 \pmod{3}$ (E) $2^{607} + 3^{607} \equiv 2^{607} \equiv (-1)^{607} \equiv -1 \pmod{3}$

So, we eliminate choices A and D from divisibility by 3. Now, we move onto divisibility by 5. We can use cycling of powers to find useful remainders. Let's start with choice B.

(B) $2^{606} + 1$

We see that the remainders of powers of 2 when divided by 5 cycle in a pattern: 2, 4, 3, 1, 2, 4, 3, 1 $\dots$. Since the pattern cycles every 4 terms, we use modulo 4 to simplify 606, getting that $606 \equiv 2 \pmod{4}$. So, we get that 2^{606} is congruent modulo 5 to the second term of our pattern, so $2^{606} \equiv 4 \pmod{5}$. Thus, $2^{606} + 1 \equiv 4 + 1 \equiv 5 \equiv 0 \pmod{5}$. We now eliminate choice B and compare choices C and E.

(E) $2^{607} + 3^{607}$

Looking at choice E, we see that we have to do similar cycling for powers of 3. We get a pattern of 3, 4, 2, 1, 3, 4, 2, 1 $\dots$. Since this pattern also cycles every 4 terms, we use modulo 4 to simplify 607, getting that $607 \equiv 3 \pmod{4}$. Both the power of 3 and the power of 2 have an exponent of 607, so we use the third term (since we just found that $607 \equiv 3 \pmod{4}$) in each corresponding 4 term pattern to get that $2^{607} + 3^{607} \equiv 3 + 2 \equiv 5 \equiv 0 \pmod{5}$. We eliminate choice E, and we are left with the correct answer: choice $(C) \ 2^{607} - 1$

Solution 3c (Elimination, Number Theory, and Function Analysis)

Let's start by calculating the unit's digit of each answer choice.

2^{606} has a unit's digit of 4, so choices A and B have unit's digits of 3 and 5 respectively.

2^{607} has a unit's digit of 8, so choices C and D have unit's digits of 7 and 9 respectively.

3^{607} has a unit's digit of 7. Combining with the statement above, choice E has a unit's digit of 5.

So, none of the answer choices are divisible by 2. But, we can eliminate choices B and E because they both have unit's digit of 5, and therefore, are divisible by 5.

We are left now with choices A, C, and D. Let's now apply function analysis.

Look at choices A and C again. They share the form $2^n - 1$. Notice that if n is even, it follows that $2^n - 1$ is divisible by 3. Since 606 is even, choice A is divisible by 3. However, our answer cannot be divisible by 3, so eliminate A. This leaves us with choices C and D.

Look at choice D again. It has the form $2^n + 1$. Notice that if n is odd, it follows that $2^n + 1$ is divisible by 3. Since 607 is odd, choice D is divisible by 3. However, since the answer cannot be divisible by 3, we eliminate D.

At this point, you would choose choice C, but let's now test divisibility by 7. Notice that if n is a multiple of 3, it follows that $2^n - 1$ is divisible by 7. However, since 607 is not a multiple of 3, choice C is not divisible by 7.

Since it is not divisible by any prime less than 10, $\boxed{(C) \ 2^{607} - 1}$ is the correct answer.

Solution 4 (modulo 15 elimination)

We proceed by taking the solutions modulo 15, allowing us to eliminate both multiples of 3 and multiples of 5.

First, we must identify cycles of powers of 2 modulo 15. $2^0 \equiv 1$, $2^1 \equiv 2$, $2^2 \equiv 4$, $2^3 \equiv 8$, $2^4 \equiv 1 \pmod{15}$. Thus, the pattern repeats every 4 powers: $2^n \equiv \begin{cases} 1 & \text{if } n \equiv 0 \pmod{4}, \\ 2 & \text{if } n \equiv 1 \pmod{4}, \\ 4 & \text{if } n \equiv 2 \pmod{4}, \\ 8 & \text{if } n \equiv 3 \pmod{4}. \end{cases}$

Since $606 \equiv 2 \pmod{4}$ and $607 \equiv 3 \pmod{4}$, $2^{606} \equiv 4 \pmod{15}$, $2^{607} \equiv 8 \pmod{15}$.

Now let's look at cycles of powers of 3 mod 15. $3^0 \equiv 1$, $3^1 \equiv 3$, $3^2 \equiv 9$, $3^3 \equiv 12$, $3^4 \equiv 6$, $3^5 \equiv 3 \pmod{15}$. The cycle from here repeats every 4 terms for

$$n \geq 1: 3^n \equiv \begin{cases} 6 & \text{if } n \equiv 0 \pmod{4}, \\ 3 & \text{if } n \equiv 1 \pmod{4}, \\ 9 & \text{if } n \equiv 2 \pmod{4}, \\ 12 & \text{if } n \equiv 3 \pmod{4}. \end{cases}$$

Since $607 \equiv 3 \pmod{4}$, $3^{607} \equiv 12 \pmod{15}$.

We then evaluate each option modulo 15.

$$(A) : 2^{606} - 1 \equiv 4 - 1 \equiv 3 \pmod{15},$$

$$(B) : 2^{606} + 1 \equiv 4 + 1 \equiv 5 \pmod{15},$$

$$(C) : 2^{607} - 1 \equiv 8 - 1 \equiv 7 \pmod{15},$$

$$(D) : 2^{607} + 1 \equiv 8 + 1 \equiv 9 \pmod{15},$$

$$(E) : 2^{607} + 3^{607} \equiv 8 + 12 \equiv 20 \equiv 5 \pmod{15}.$$

The residues divisible by 3 or 5 modulo 15 are 0, 3, 5, 6, 9, 10, 12. Among our results, only 7 (from option (C)) is not among these, eliminating all the other answer choices.

(C) $2^{607} - 1$ is the number not divisible by any prime less than 10.

Loquacious_Autodidact

Solution 5 (Contradiction; Less than 1 minute if you're good at number theory)

This is like solution 1, but don't find the remainder, we assume it.

The primes less than 10 are 2, 3, 5, and 7.

Option choice A: Let $2^{606} \equiv 1 \pmod{3}$ as it clearly isn't divisible by 2. Then, because $2^2 \equiv 1 \pmod{3}$, it follows that 2^{606} satisfies the modulo. Option A is not possible.

Option choice B ends in 5, and thus not possible.

Option choice C: It is obviously not divisible by 2. We set $2^{607} \equiv 1 \pmod{3}$, which we see is not possible through Fermat's Little Theorem. We see 2^{607} doesn't end in 5. We set $2^{607} \equiv 1 \pmod{7}$, and through Fermat's Little Theorem $2^6 \equiv 1 \pmod{7}$. It follows that 607 isn't a factor of 2, and thus does not hold the desired claim, a contradiction.

At this point, for speed purposes, one would circle C and move on. However, given the directory that this is a solution, I shall show the last two cases are not possible.

Option choice D: It is obviously not divisible by 2. We set $2^{607} \equiv 2 \pmod{3}$. Through properties of modulo 3 we have that $(-1)^{607} \equiv -1 \pmod{3}$, which is true. Thus, D is not plausible.

Option choice E is divisible by 5, and thus not possible. $\square$

Additionally, for option D. One could notice that $2^{607} \equiv 2 \pmod{3}$ in the following way. We analyze the pattern. $2^1 \equiv 2 \pmod{3}$. $2^3 \equiv 2 \pmod{3}$. $2^5 \equiv 2 \pmod{3}$. $2^{2n+1} \pmod{3} \equiv 4^n \cdot 2^1 \pmod{3} \equiv 1^n \cdot 2 \pmod{3} \equiv 2 \pmod{3}$. 607 is odd, and thus $2^{607} \equiv 2 \pmod{3}$, as desired. $\square$

Solution 6 (Extremely Illegal Cheese)

All of the other solutions are extremely long, but this is how to cheese it using some methods that MIGHT be unreliable: First, turn 606 to 0 and 607 to 1. Next, Calculate each value, giving you $A = 0B = 2C = 1D = 3E = 5$. Now since 2, 3, and 5 are prime, they can be divided by a prime (themselves). 0 doesn't really have multiples, so you can literally ignore it, and since 1 is only divisible by 1, which isn't prime, you get the correct answer, (C).

18. (AMC 10 10B 2022) Consider systems of three linear equations with unknowns x , y , and z ,

$$\begin{aligned}a_1x + b_1y + c_1z &= 0 \\a_2x + b_2y + c_2z &= 0 \\a_3x + b_3y + c_3z &= 0\end{aligned}$$

where each of the coefficients is either 0 or 1 and the system has a solution other than $x = y = z = 0$. For example, one such system is

$$\{1x + 1y + 0z = 0, 0x + 1y + 1z = 0, 0x + 0y + 0z = 0\}$$

with a nonzero solution of $\{x, y, z\} = \{1, -1, 1\}$. How many such systems of equations are there? (The equations in a system need not be distinct, and two systems containing the same equations in a different order are considered different.)

- (A) 302 (B) 338 (C) 340 (D) 343 (E) 344

Answer: B

Solution 1 (Casework)

Let $M_1 = \begin{bmatrix} a_1 & b_1 & c_1 \end{bmatrix}$, $M_2 = \begin{bmatrix} a_2 & b_2 & c_2 \end{bmatrix}$, and $M_3 = \begin{bmatrix} a_3 & b_3 & c_3 \end{bmatrix}$.

We wish to count the ordered triples (M_1, M_2, M_3) of row matrices. We perform casework:

- a) $M_1 = M_2 = M_3$. There are $2^3 = 8$ options for M_1 . Once M_1 is chosen, M_2 and M_3 are uniquely determined.

In this case, we have 8 ordered triples (M_1, M_2, M_3) .

- b) Exactly two of M_1, M_2 , and M_3 are equal. For $M_1 = M_2 \neq M_3$, there are $2^3 = 8$ options for M_1 . Once M_1 is chosen, M_2 is uniquely determined, and M_3 has $2^3 - 1 = 7$ options. So, there are $8 \cdot 7 = 56$ ordered triples (M_1, M_2, M_3) .

Similarly, for each of $M_1 = M_3 \neq M_2$ and $M_2 = M_3 \neq M_1$, there are 56 ordered triples (M_1, M_2, M_3) .

In this case, we have $56 \cdot 3 = 168$ ordered triples (M_1, M_2, M_3) .

- c) All of M_1, M_2 , and M_3 are different. There are two subcases:

<ol style="margin-left: 1.5em;" type="A">

- d) Exactly one of M_1, M_2 , and M_3 is $\begin{bmatrix} 0 & 0 & 0 \end{bmatrix}$. For $M_1 = \begin{bmatrix} 0 & 0 & 0 \end{bmatrix}$, there are $2^3 - 1 = 7$ options for M_2 and $2^3 - 2 = 6$ options for M_3 . So, there are $7 \cdot 6 = 42$ ordered triples (M_1, M_2, M_3) .

Similarly, for each of $M_2 = \begin{bmatrix} 0 & 0 & 0 \end{bmatrix}$ and $M_3 = \begin{bmatrix} 0 & 0 & 0 \end{bmatrix}$, there are 42 ordered triples (M_1, M_2, M_3) .

In this subcase, we have $42 \cdot 3 = 126$ ordered triples (M_1, M_2, M_3) .

- e) The sum of two of M_1, M_2 , and M_3 is equal to the third matrix. For $M_1 + M_2 = M_3$:

- If $M_1 = \begin{bmatrix} 1 & 0 & 0 \end{bmatrix}$, then $M_2 \in \{\begin{bmatrix} 0 & 1 & 1 \end{bmatrix}, \begin{bmatrix} 0 & 1 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 0 & 1 \end{bmatrix}\}$. More generally, if M_1 consists of one 1 and two 0's, then M_2 has 3 options, and M_3 is uniquely determined. So, there are $\binom{3}{1} \cdot 3 = 9$ ordered triples (M_1, M_2, M_3) .

- If $M_1 = \begin{bmatrix} 1 & 1 & 0 \end{bmatrix}$, then $M_2 = \begin{bmatrix} 0 & 0 & 1 \end{bmatrix}$. More generally, if M_1 consists of two 1's and one 0, then M_2 and M_3 are uniquely determined. So, there are $\binom{3}{2} = 3$ ordered triples (M_1, M_2, M_3) .

There are $9 + 3 = 12$ ordered triples (M_1, M_2, M_3) .

Similarly, for each of $M_1 + M_3 = M_2$ and $M_2 + M_3 = M_1$, there are 12 ordered triples (M_1, M_2, M_3) .

In this subcase, we have $12 \cdot 3 = 36$ ordered triples (M_1, M_2, M_3) .

Together, the answer is $8 + 168 + 126 + 36 = \boxed{\text{(B)} 338}$.

Solution 2 (Intuitive, Straightforward)

Begin by adding the three equations.

$$(a_1 + a_2 + a_3)x + (b_1 + b_2 + b_3)y + (c_1 + c_2 + c_3)z = 0$$

Each of the coefficients is 0 or 1, and x, y, z cannot all equal 0. Notice that then, either $a_1 + a_2 + a_3 = 0, 1, 2, 3$, and the same for $b_1 + b_2 + b_3$, and $c_1 + c_2 + c_3$. Considering just $a_1 + a_2 + a_3$ for now, if equals to 0, then only 1 case exists. If equal to 1, then 3 cases exist. If equal to 2, then 3 cases still exist. If 3, then only 1 case exists. That is 8 total, so we have 24 total cases. Considering all 8 other viable systems (any permutation of (x, y, z) besides $x = y = z = 0$), given the possible coefficient cases here, we have $24 \cdot 8 = 192$ possible cases.

Now, subtract the three equations.

$$(a_1 - a_2 - a_3)x - (b_1 - b_2 - b_3)y - (c_1 - c_2 - c_3)z = 0$$

Now, notice that $a_1 - a_2 - a_3 = -2, -1, 0$, and the same for b_1, b_2, b_3 and c_1, c_2, c_3 . For when $a_1 - a_2 - a_3 = -2$, that can only happen when we have only $(a_1, a_2, a_3) = (0, 1, 1)$, which is 1 way. If when $a_1 - a_2 - a_3 = -1$, we have this can happen in two ways, some permutations (except for when $a_1 = 1$) of either $(a_1, a_2, a_3) = (1, 1, 1), (0, 0, 1), (0, 1, 0)$, which is 3 ways total. For $a_1 - a_2 - a_3 = 0$, there are 3 ways, those being $(a_1, a_2, a_3) = (0, 0, 0), (1, 0, 1), (1, 1, 0)$. Finally, for $a_1 - a_2 - a_3 = 1$, we have the case $(a_1, a_2, a_3) = (1, 0, 0)$. Again 8 ways total, so times 3 is 24. When we consider this system, we only consider 6 plausible cases that do not degenerate into $x = y = z = 0$. To see where 6 comes from; due to subtraction, if $x + y = x + z = y + z = 0$, then subtracting will yield $x = y = z = 0$ regardless, unlike the first case, where there could exist some x or y not necessarily equal to 0 but add to 0 that will still satisfy. Due to properties of subtraction, that is simply impossible for three distinct equations. So, all such equations must not contain any opposites of one another. Any subset from $\{-2, -1, -1, 0, 1, 1, 2\}$ of three elements not consisting of both -2 and 2 are viable roots. That is $\binom{7}{3} - 5 = 30$ cases. Only one fifth of these are viable as four fifths pertain values that will subtract to 0, a contradiction from the first claim (particularly, casework will show that, on $(-1, -1, z), (1, 1, z), (0, -1, z), (0, 1, z)$, which is pertained by $4/5$ of the total will not work), so only 6 systems are valid. Therefore, we have $24 \cdot 6 = 144$ possible solutions.

Now, the two forgotten solutions. When we add these two equations, we forget the case of redundancy (as we always count viable solutions, not infinite). When an

equation is redundant, we must be able to deduce a viable ordered pair, which is the same as saying we must have systems in which their coefficients are of different parity. That is 2 cases. This however never applies for subtraction, as even if two equations have different parity, when we subtract the equations, that parity becomes constant, and will always cancel to 0, yielding only one possible solution, $(0, 0, 0)$, which is not allowed.

So, exactly $192 + 144 + 2 = 336 + 2 = \boxed{338}$ solutions are viable

Solution 3 (Solution 8 Alternate, Slightly Easier)

We proceed with casework. The following will hold if

Case 1: At least one system has $0 = 0$, in which through the compliment, we have $512 - 343 = 169$ ways.

Case 2: Exactly two equations are equal, in which we have $\binom{3}{2} \cdot 7 \cdot 6 = 126$ ways.

Case 3: An equation is the result of the sum of the other two. The equation must have some coefficients between 0 and 1, so the possible cases for the coefficients of the third equation (excluding all 0 as that has already been counted) is $(0, 0, 1)$ or permutations, $(1, 1, 0)$ or permutations, and $(1, 1, 1)$. The first case has 0 ways. The second has 2 cases per each of the three cases, which is 6 ways total. The third has 6 cases in all, either the first equation is $(0, 0, 1)$ or some permutation (3 ways) which uniquely determines the second, or is $(0, 1, 1)$ or some permutation (3 ways) which uniquely determines the second. So, there are $6 \cdot 6 = 36$ total cases.

Case 4: All 3 are equal (except all 0, as already counted). That is 7 ways.

So, exactly $169 + 126 + 36 + 7 = \boxed{338}$ cases exist.

Solution 4

Let nV represent an equation in the system with n variables. There are $2^9 = 512$ total possible systems (each coefficient can either be 0 or 1). We will subtract off the invalid ones (the equations with solution $x = y = z = 0$) using casework.

We have 1 case of selecting a 3-variable, 2-variable, and 1-variable equation, 6 cases of selecting two n -variable equations and one different m -variable equation, and 3 cases of selecting 3 of the same m -variable equations. However, we can't pick two 3-variable equations in our system because then it's impossible to get to the solution $x = y = z = 0$. So there are 3 invalid cases. Then we need to look for a total of $1 + 6 + 3 - 3 = 7$ different cases. (Doing this beforehand makes sure we don't accidentally skip a case.)

Case 1: 3V-2V-2V: WLOG say we have $x + y + z = 0$, $y + z = 0$, and $x + y = 0$. We are choosing two variables out of the 3 to be in the second equation, and we have 1 choice for the third. There are 6 ways to rearrange them. In total we get $6 \times \binom{3}{2} = 18$ systems for this case.

Case 2: 1V-1V-1V: When we have $x = 0$, $y = 0$, and $z = 0$. There are 6 systems for this case.

Case 3: 2V-1V-1V: We have 3 choices for the first equation: $x + y = 0$, $y + z = 0$, or $x + z = 0$. WLOG if $x + z = 0$ is the first equation, the second equation can either be $x = 0$ or $z = 0$, and the third equation must be $y = 0$. So we have 2 choices for the second equation and 1 choice for the third. $6 \times 3 \times 2$ gives us 36 systems for this case.

Case 4: 3V-2V-1V: There is obviously one choice for the first equation ($x + y + z = 0$). For the second equation, we can choose any 2 of the 3 variables (in 3 ways).

For the third, we must choose one of the two variables we included in our second equation (in 2 ways). In total, we get $6 \times 3 \times 2 = 36$ systems for this case.

Case 5: 2V-2V-2V: There is only 1 combination of equations: $x + y = 0$, $y + z = 0$, and $x + z = 0$. These can be rearranged in 6 ways, giving us 6 systems.

Case 6: 3V-1V-1V: Again, there's only one choice for the first equation ($x + y + z = 0$). For the second and third, we can choose any two of the three variables in 3 ways, giving us $6 \times 3 = 18$ systems.

Case 7: 2V-2V-1V: We want to choose a variable to appear twice in both of the two variable equations. This can be done in 3 ways. The second equation will be the variable appearing twice added to the variable left out of the first (so only 1 choice), and 3 choices for the one variable equation. The system can be rearranged in 6 ways. This gives us $6 \times 3 \times 3 = 54$ systems.

Our final answer is $512 - (18 + 6 + 36 + 36 + 6 + 18 + 54) = \boxed{\text{(B)} 338}$.

Solution 5 (Complementary Counting)

We will use complementary counting and do casework on the equations.

There are 8 possible equations:

Equation 1: $0 = 0$

Equation 2: $x = 0$

Equation 3: $y = 0$

Equation 4: $z = 0$

Equation 5: $x + y = 0$

Equation 6: $x + z = 0$

Equation 7: $y + z = 0$

Equation 8: $x + y + z = 0$

We will continue to refer to the equations by their number on this list.

$8^3 = 512$ total systems. Note that no two equations by themselves can force $x = y = z = 0$. Therefore no system with Equation 1 or with repeated equations can force $x = y = z = 0$.

Case 1: Equation 8 ($x + y + z = 0$) is present.

Case 1a: Equation 8, and two equations from $\{5, 6, 7\}$.

There are $\binom{3}{2} = 3$ ways to choose two equations from $\{5, 6, 7\}$ and $3! = 6$ ways to arrange each case. The number of options that force $x = y = z = 0$ is $3 \cdot 3! = 18$.

Case 1b: Equation 8, one equation from $\{5, 6, 7\}$, and one equation from $\{2, 3, 4\}$.

There are $\binom{3}{1} = 3$ ways to choose one equation from $\{5, 6, 7\}$. WLOG let us choose Equation 7. Given $x + y + z = 0$ and $y + z = 0$, we conclude that $x = 0$. The third equation can be either $y = 0$ or $z = 0$. There are $3!$ ways to arrange each case. The number of options that force $x = y = z = 0$ is $3 \cdot 2 \cdot 3! = 36$.

Case 1c: Equation 8, and two equations from $\{2, 3, 4\}$.

There are $\binom{3}{2} = 3$ ways to choose two equations from $\{2, 3, 4\}$ and $3! = 6$ ways to arrange each case. Each of these cases forces $x = y = z = 0$. $3 \cdot 3! = 18$ total options.

Case 2: Equation 8 is **not** present, at least one equation from $\{5, 6, 7\}$ is present.

Case 2a: Equations $\{5, 6, 7\}$ are all present.

There are $3!$ ways to arrange the three equations. 6 options.

Case 2b: Two equations from $\{5, 6, 7\}$ are present. One equation from $\{2, 3, 4\}$ is present.

There are $\binom{3}{2}$ ways to choose two equations from $\{5, 6, 7\}$. WLOG let Equations 5 and 6 be in our system: $x + y = 0$ and $x + z = 0$. Any equation from $\{2, 3, 4\}$ will force $x = y = z = 0$. There are $3!$ ways to arrange the equations. The number of options that force $x = y = z = 0$ is $\binom{3}{2} \cdot \binom{3}{1} \cdot 3! = 54$.

Case 2c: One equation from $\{5, 6, 7\}$ is present. Two equations from $\{2, 3, 4\}$ are present.

There are $\binom{3}{1}$ ways to choose one equation from $\{5, 6, 7\}$. WLOG let Equation 5 ($x + y = 0$) be present. One of the two equations from $\{2, 3, 4\}$ must be Equation 4, $z = 0$, since it is the only equation that restricts z . The last equation can be either 2 or 3. There are $3!$ ways to arrange the equations. The number of options that force $x = y = z = 0$ is $\binom{3}{1} \cdot \binom{2}{1} \cdot 3! = 36$.

Case 3: Only equations $\{2, 3, 4\}$ are present.

There are $3!$ ways to arrange the three equations. 6 options.

We add up the cases: $18 + 36 + 18 + 6 + 54 + 36 + 6 = 174$ total systems force $x = y = z = 0$. Thus $512 - 174 = \boxed{\text{(B) } 338}$ do not.

Solution 6 (Complementary Counting)

The total number of possible systems is $2^9 = 512$, with 8 possible sets of coefficients per equation. We will use complementary counting to find the number of systems which only have the solution $(0, 0, 0)$ and subtract that from the total. Similar to what is observed in Solution 3, if any equation is repeated or $0x + 0y + 0z = 0$, there will only be two or fewer equations for three variables, making one unique solution impossible. Therefore, we must choose 3 different equations from 7 possible ones, giving $7 \cdot 6 \cdot 5 = 210$ systems. However, there are two exceptions to consider, which will have more than one solution. The first is of the form $x + y + z = 0$, $x + y = 0$, $z = 0$; the second is of the form $x = 0$, $y = 0$, $x + y = 0$. In both cases, there are 3 ways to choose the variables in the equations, and then 6 ways to arrange them, giving $2 \cdot 3 \cdot 6 = 36$ exceptions. Subtracting this gives $210 - 36 = 174$ systems with only one solution, and the answer is then $512 - 174 = \boxed{\text{(B) } 338}$.

Solution 7 (Complementary Counting, Linear Dependence, Vector Analysis)

Denote vector $\vec{i} = (i_1, i_2, i_3)^T$ for $i \in \{a, b, c\}$. Thus, we need to count how many vector tuples $(\vec{a}, \vec{b}, \vec{c})$ are linearly dependent.

We do complementary counting.

First, the total number of vector tuples $(\vec{a}, \vec{b}, \vec{c})$ is $(2^3)^3 = 512$.

Second, we count how many many vector tuples $(\vec{a}, \vec{b}, \vec{c})$ are linearly independent.

To meet this condition, no vector can be a zero vector $\vec{0} = (0, 0, 0)^T$.

Next, we do the casework analysis.

Case 1^c: Three vectors are all on axes.

In this case, the number of $(\vec{a}, \vec{b}, \vec{c})$ is $3!$.

Case 2^c: Two vectors are on axes and the third vector is not.

We construct such an instance in the following steps.

Step 1: We determine which two vectors lie on axes.

The number of ways is 3.

Step 2: For two vectors selected in Step 1, we determine which two axes they lie on.

The number of ways is $3 \cdot 2$.

Step 3: For the third unselected vector, we determine its value.

To make three vectors linear independent, the third vector cannot be on the plane formed by the first two vectors. So the number of ways is 3.

Following from the rule of product, the number of $(\vec{a}, \vec{b}, \vec{c})$ in this case is $3 \cdot 3 \cdot 2 \cdot 3$.

Case 3^c: One vector is on an axis and the other two are not.

We construct such an instance in the following steps.

Step 1: We determine which vector lies on an axis.

The number of ways is 3.

Step 2: For the selected vector, we determine which axis it lies on.

The number of ways is 3.

Step 3: We determine the values of the two unselected vectors.

First, to be linearly independent, these two vectors are distinct. Second, to be linearly independent, we cannot have one vector $(1, 1, 1)$ and another one that is a diagonal vector on the plane that is perpendicular to the first selected vector.

Thus, the number of ways in this step is $4 \cdot 3 - 2 = 10$.

Following from the rule of product, the number of $(\vec{a}, \vec{b}, \vec{c})$ in this case is $3 \cdot 3 \cdot 10$.

Case (4.4)^c: No vector is on any axis.

In this case, any three distinct vectors are linearly independent. So the number of $(\vec{a}, \vec{b}, \vec{c})$ in this case is $4 \cdot 3 \cdot 2$.

Putting all cases together, the number of vector tuples $(\vec{a}, \vec{b}, \vec{c})$ that are linearly independent is

$$8^3 - 3! - 3 \cdot 3 \cdot 2 \cdot 3 - 3 \cdot 3 \cdot 10 - 4 \cdot 3 \cdot 2 = \boxed{\text{(B) } 338}.$$

Steven Chen (Professor Chen Education Palace, www.professorchen.edu.com)

Solution 8 (Casework, but more intuitive)

Case 1: At least one of the equations is $0 = 0$. By complementary counting, there are $512 - 343 = \boxed{169}$ possible systems which satisfy this condition.

Case 2: Exactly 2 of the equations are equal. By directly counting, we have $\binom{3}{2} \cdot 7 \cdot 6 = \boxed{126}$, where there is $\binom{3}{2}$ places to put the two identical equations, and 7, 6 ways to choose the nonzero equations.

Case 3: The sum of two different equations is equal to the last equation. The equation which serves as the total sum of the other two can either be $x + y, x + z, y + z$, which each have 1 way respectively to pick a set of equations, and $x + y + z$ has 3

ways to pick a set of equations. Then, we have $6 \cdot 6 = \boxed{36}$ for this case accounting for permutations.

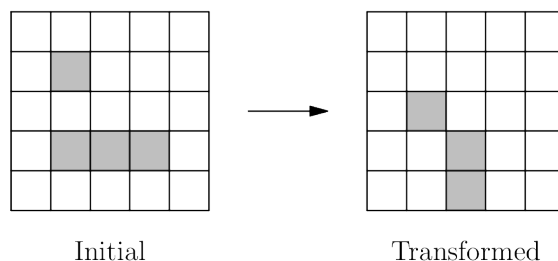
Case 4: All 3 of the equations are equal. There are $2^3 - 1 = \boxed{7}$ systems which satisfy this case (they cannot all be 0 because it's already counted in Case 1).

Thus, there are $169 + 126 + 36 + 7 = \boxed{\text{(B)} 338}$ total ways to pick such a system.

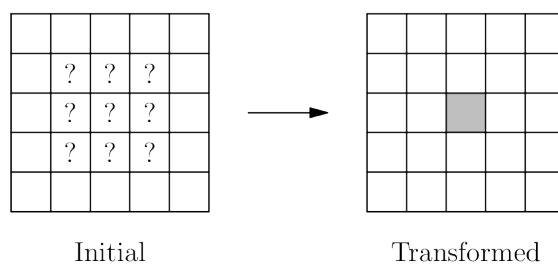
19. (AMC 10 10B 2022) Each square in a 5×5 grid is either filled or empty, and has up to eight adjacent neighboring squares, where neighboring squares share either a side or a corner. The grid is transformed by the following rules:

- Any filled square with two or three filled neighbors remains filled.
- Any empty square with exactly three filled neighbors becomes a filled square.
- All other squares remain empty or become empty.

A sample transformation is shown in the figure below.



Suppose the 5×5 grid has a border of empty squares surrounding a 3×3 subgrid. How many initial configurations will lead to a transformed grid consisting of a single filled square in the center after a single transformation? (Rotations and reflections of the same configuration are considered different.)



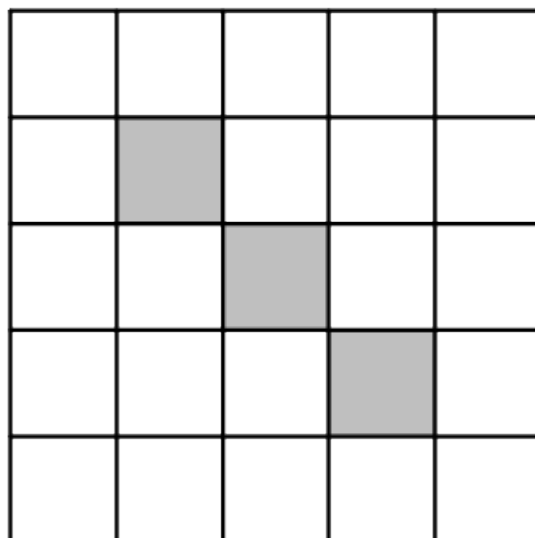
- (A) 14 (B) 18 (C) 22 (D) 26 (E) 30

Answer: C

Solution

There are two cases for the initial configuration:

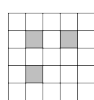
- a) The center square is filled. Exactly two of the eight adjacent neighboring squares of the center are filled. Clearly, the only possibility is that the squares along one diagonal are filled, as shown below:



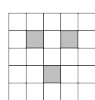
2 Configurations

In this case, there are 2 possible initial configurations. All rotations and reflections are considered.

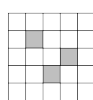
- b) The center square is empty. Exactly three of the eight adjacent neighboring squares of the center are filled. The possibilities are shown below:



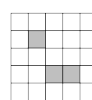
4 Configurations



4 Configurations



4 Configurations



8 Configurations

In this case, there are $4 + 4 + 4 + 8 = 20$ possible initial configurations. All rotations and reflections are considered.

Together, the answer is $2 + 20 = \boxed{(C) 22}$.

Solution Alternate

Key Idea:

Only the 3×3 inner grid can initially contain filled squares, since the outer border is empty. We want exactly the center square to be filled after one step.

Case 1: The center is initially filled.

A filled square remains filled precisely when it has 2 or 3 filled neighbors. We examine the possible choices of filled neighbors of the center and require that no other square becomes filled.

This gives 2 valid configurations, counting rotations/reflections.

Case 2: The center is initially empty.

For the center to become filled, it must have exactly 3 filled neighbors. There are several possible shapes for these three neighbors. Counting the configurations of each shape gives

$$2 + 4 + 4 + 4 + 8 = 22.$$

The total number of configurations is

$$2 + (2 + 4 + 4 + 4 + 8) = 2 + 22 \\ = \boxed{24}.$$

Therefore the answer is $\boxed{\text{(C)} 22}$.

Lunar paradox (talk) 18:06, 24 August 2026 (EDT)

Remark This problem is based off of Conway's Game of Life, a cellular automaton created by British mathematician John Conway in 1970. The game follows the exact same pattern of rules as this problem, but is played on an infinite grid of cells.

20. (AMC 10 10B 2022) Let $ABCD$ be a rhombus with $\angle ADC = 46^\circ$. Let E be the midpoint of $\overline{CD}$, and let F be the point on $\overline{BE}$ such that $\overline{AF}$ is perpendicular to $\overline{BE}$. What is the degree measure of $\angle BFC$?

(A) 110 (B) 111 (C) 112 (D) 113 (E) 114

Answer: D

Solution 1

Extend segments $\overline{AD}$ and $\overline{BE}$ until they meet at point G .

Because $\overline{AB} \parallel \overline{ED}$, we have $\angle ABG = \angle DEG$ and $\angle GDE = \angle GAB$, so $\triangle ABG \sim \triangle DEG$ by AA.

Because $ABCD$ is a rhombus, $AB = CD = 2DE$, so $AG = 2GD$, meaning that D is a midpoint of segment $\overline{AG}$.

Now, $\overline{AF} \perp \overline{BE}$, so $\triangle GFA$ is right and median $FD = AD$.

So now, because $ABCD$ is a rhombus, $FD = AD = CD$. This means that there exists a circle from D with radius AD that passes through F , A , and C .

AG is a diameter of this circle because $\angle AFG = 90^\circ$. This means that $\angle GFC = \angle GAC = \frac{1}{2}\angle GDC$, so $\angle GFC = \frac{1}{2}(180^\circ - 46^\circ) = 67^\circ$, which means that $\angle BFC =$

$\boxed{\text{(D)} 113}$

Solution 3

Let $\overline{AC}$ meet $\overline{BD}$ at O , then $AOFB$ is cyclic and $\angle FBO = \angle FAO$. Also, $AC \cdot BO = [ABCD] = 2 \cdot [ABE] = AF \cdot BE$, so $\frac{AF}{BO} = \frac{AC}{BE}$, thus $\triangle AFC \sim \triangle BOE$ by SAS, and $\angle OEB = \angle ACF$, then $\angle CFE = \angle EOC = \angle DAC = 67^\circ$, and $\angle BFC =$ $\boxed{\text{(D)} 113}$

A little bit faster: $AOFB$ is cyclic $\implies \angle OFE = \angle BAO$.

$AB \parallel CD \implies \angle BAO = \angle OCE$.

Therefore $\angle OFE = \angle OCE \implies OEFC$ is cyclic.

Hence $\angle CFE = \angle COE = \angle CAD = 67^\circ$.

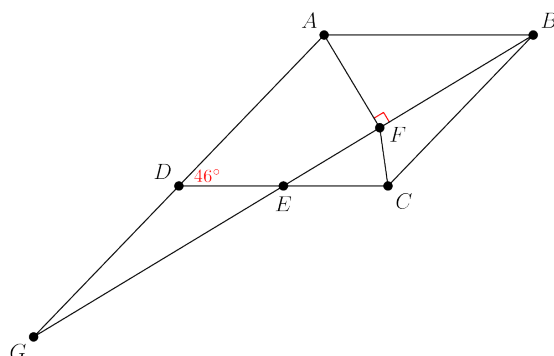
Solution 4

Observe that all answer choices are close to $112.5 = 90 + \frac{45}{2}$. A quick solve shows that having $\angle D = 90^\circ$ yields $\angle BFC = 135^\circ = 90 + \frac{90}{2}$, meaning that $\angle BFC$ increases with $\angle D$. Substituting, $\angle BFC = 90 + \frac{46}{2} = \boxed{\text{(D)} 113}$.

Solution 5 (Similarity and Circle Geometry)

This solution refers to the **Diagram** section.

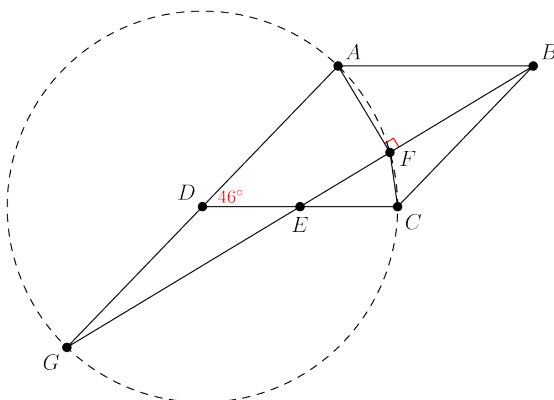
We extend AD and BE to point G , as shown below:



We know that $AB = AD = 2$ and $CE = DE = 1$.

By AA Similarity, $\triangle ABG \sim \triangle DEG$ with a ratio of $2 : 1$. This implies that $2AD = AG$ and $AD \cong DG$, so $AG = 2AD = 2 \cdot 2 = 4$. That is, D is the midpoint of AG .

Note that as $\angle AFG$ has an angle of 90 deg and $AG = 2DG$, we can redraw our previous diagram, but construct a circle with radius AD or 2 centered at D and by extending CD to point H , which is on the circle, as shown below:



Notice how F and C are on the circle and that $\angle CFE$ intercepts with $\widehat{CG}$.

Let's call $\angle CFE = \theta$.

Note that $\angle CDG$ also intercepts $\widehat{CG}$, So $\angle CDG = 2\angle CFE$.

Let $\angle CDG = 2\theta$. Notice how $\angle CDG$ and $\angle ADC$ are supplementary to each other. We conclude that

$$2\theta = 180 - \angle ADC$$

$$2\theta = 180 - 46$$

$$2\theta = 134$$

$$\theta = 67.$$

Since $\angle BFC = 180 - \theta$, we have $\angle BFC = 180 - 67 = \boxed{\text{(D) } 113}$.

Note: You can also find that $\angle CFE$ is half of $\angle CDG$ via circle theorems. We know $\angle CDG = 180 - 46 = 134$, therefore making $\angle CFE = 67$ and $\angle BFC = 113$.

Solution 6 (Simplification/Reduction)

If angle ADC was a right angle, it would be much easier. Thus, first pretend that ADC is a right angle. $ABCD$ is now a square. WLOG, let each of the side lengths be 1. We can use the Pythagorean Theorem to find the length of line AE , which is $\sqrt{5}/2$. We want the measure of angle BFC , so to work closer to it, we should try finding the length of line BF . Angle FAB and angle ABF are complementary. Angle ABF and angle FBC are also complementary. Thus, $\sin FAB = \cos ABF = \sin FBC$. $\sin FAB = \sin FBC = (1/2)/(\sqrt{5}/2) = 1/\sqrt{5}$. Since $\sin FAB = 1/\sqrt{5}$, and $AB = 1$, $FB = \sin FAB$. It follows now that $FE = 3 * \sqrt{5}/10$.

Now, zoom in on triangle BEC . To use the Law of Cosines on triangle FBC , we need the length of FC . Use the Law of Cosines on triangle EFC . $\cos E = 1/\sqrt{5}$. Thus, after using the Law of Cosines, $FC = \sqrt{2/5}$.

Since we now have SSS on BEC , we can get use the Law of Cosines. $\cos BFC = 1/(-\sqrt{2})$. $\arccos 1/(-\sqrt{2})$ is 45° , but if the cosine is negative that means that the angle is the supplement of the positive cosine value. $180 - 45 = 135$. Angle BFC is 135° .

Realize that, around point F , there will always be 3 right angles, regardless of what angle ADC is. There are only two angles that change when ADC changes. Break up angle BFC into angle BFB' , which is always 90 degrees, and angle $B'FC$, which we have discovered to be half of ADC . Thus, when angle ADC is 46 degrees, then $B'FC$ will be 23. $23 + 90 = 113$. Angle BFC is **(D) 113** degrees.

Solution 7

Draw an auxiliary line from D to the midpoint of AB . Label it G . Then quadrilateral $BGDE$ is a parallelogram. Hence AF and BD are perpendicular. Now, G being the midpoint of the hypotenuse of triangle ABF , it is the circumcenter of it. Thus, $GF = GA$ and so DG is the perpendicular bisector of AF . Therefore, triangle AFD is also an isosceles triangle. Since $AD = FD = CD$, triangle CFD is also an isosceles triangle. Their one distinct angles' sum being 46 degrees, angle $BFC = 113$.

Jeongha Cho

Solution 8(lots and lots of tedious angle chasing)

It is a well known fact that connecting the midpoints of the sides of a rhombus gives us a rectangle. We let the midpoint of side DC as E , the midpoint of side AD as F , the midpoint of side AB as G , and the midpoint of side BC as H . We can connect E, F, G, H to get rectangle $EFGH$. Note that the obtuse angles of the rhombus are each 134 degrees. We can perform a little bit of angle chasing following this diagram(I cannot draw diagrams in LaTeX so the rest of this solution will be diagram-free). Let the intersection point between line segment BE and line segment GH as I . Let angle FEB to be θ . Performing more angle chasing leads us to finding that angle GIB is angle θ and angle IBG is $113 - \theta$. Let point J be on BE such that AJ is perpendicular to BE . Then, by performing yet more angle chasing leads us to finding that angle JEC is $90 - \theta + 23 = 113 - \theta$. We can predict that triangle JEC is similar to triangle BIG . This is because since GB and EC both bisect the sides of the rhombus and one angle is common(the $113 - \theta$ angle). Therefore, we can safely say that all angles in these two triangles must be

the same and thus angle $CJE = IGB = 67$ degrees. Thus, our desired angle, CJB is simply just $180 - 67 = \boxed{113}$.

Solution 9 (educated guess)

The answer choices must have something to do with 46. We can do some playing around, like trying $180 - 46 = 134$. Since the answers are all around the 110s, we can start manipulating our 134. We could divide by 2, subtract 90, etc. Dividing by 2 gives 67, and we can see $180 - 67 = 113$. However, we can also see that $180 - ((180 - (90 - 46))/2) = 112$, but this seems much too complicated for a simple rhombus. Therefore, we can reasonably assume the answer is $\boxed{113}$.

21. (AMC 10 10B 2022) Let $P(x)$ be a polynomial with rational coefficients such that when $P(x)$ is divided by the polynomial $x^2 + x + 1$, the remainder is $x + 2$, and when $P(x)$ is divided by the polynomial $x^2 + 1$, the remainder is $2x + 1$. There is a unique polynomial of least degree with these two properties. What is the sum of the squares of the coefficients of that polynomial?

(A) 10 (B) 13 (C) 19 (D) 20 (E) 23

Answer: E

Solution 1 (Experimentation)

Given that all the answer choices and coefficients are integers, we hope that $P(x)$ has positive integer coefficients.

Throughout this solution, we will express all polynomials in base x . E.g. $x^2 + x + 1 = 111_x$.

We are given:

$$111a + 12 = 101b + 21 = P(x).$$

We add 111 and 101 to each side and balance respectively:

$$111(a - 1) + 123 = 101(b - 1) + 122 = P(x).$$

We make the unit's digits equal:

$$111(a - 1) + 123 = 101(b - 2) + 223 = P(x).$$

We now notice that:

$$111(a - 11) + 1233 = 101(b - 12) + 1233 = P(x).$$

Therefore $a = 11_x = x + 1$, $b = 12_x = x + 2$, and $P(x) = 1233_x = x^3 + 2x^2 + 3x + 3$. 3 is the minimal degree of $P(x)$ since there is no way to influence the x 's digit in $101b + 21$ when b is an integer. The desired sum is $1^2 + 2^2 + 3^2 + 3^2 = \boxed{\text{(E) } 23}$

P.S. The four computational steps can be deduced through quick experimentation.

Solution 2

Let $P(x) = Q(x)(x^2 + x + 1) + x + 2$, then $P(x) = Q(x)(x^2 + 1) + xQ(x) + x + 2$, therefore $xQ(x) + x + 2 \equiv 2x + 1 \pmod{x^2 + 1}$, or $xQ(x) \equiv x - 1 \pmod{x^2 + 1}$. The rule says we can change any x^2 into -1 . This is because $x^2 \equiv -1 \pmod{x^2 + 1}$ (since $x^2 - (-1) = x^2 + 1$, which is a multiple of $x^2 + 1$). To find the least possible degree $Q(x)$, it would be logical to first assume it's a constant. Let's say $Q(x) = c$, which makes $xQ(x) = cx$. Is cx congruent to $x - 1$? The difference is $cx - (x - 1) = (c - 1)x + 1$. This is a polynomial of degree 1, so it's not a multiple of $x^2 + 1$ (a degree 2

polynomial). So, a constant won't work. Could $Q(x)$ be a simple degree 1 polynomial? Let's try $Q(x) = x + 1$. Let's check if $Q(x) = x + 1$ works. Calculate $xQ(x)$: $x(x + 1) = x^2 + x$.

Now, apply our rule: we can replace x^2 with -1 . Therefore, $x^2 + x$ becomes $-1 + x$, which is $x - 1$. This shows that $xQ(x)$ is congruent to $x - 1$. Since we showed that no constant polynomial works, and we found a degree 1 polynomial that does work, the minimum degree must be 1. The simplest polynomial of degree 1 that satisfies the condition is $Q(x) = x + 1$, and expanding gives $P(x) = x^3 + 2x^2 + 3x + 3$. Summing the squares of coefficients gives **(E) 23**

Solution 3

Let $P(x) = (x^2 + x + 1)Q_1(x) + x + 2$, then $P(x) = (x^2 + 1)Q_1(x) + xQ_1(x) + x + 2$

Also, $P(x) = (x^2 + 1)Q_2(x) + 2x + 1$

We infer that $Q_1(x)$ and $Q_2(x)$ have same degree, we can assume $Q_1(x) = x + a$, and $Q_2(x) = x + b$, since $P(x)$ has least degree. If this cannot work, we will try quadratic, etc.

Then we get: $(x^2 + 1)(Q_1(x) - Q_2(x)) + xQ_1(x) - x + 1 = 0$

The constant term gives us: $(Q_1(x) - Q_2(x)) + 1 = 0$

So $Q_1(x) - Q_2(x) = -1$

Substituting this in gives: $-(x^2 + 1) + xQ_1(x) - x + 1 = 0$

Solving this equation, we get $Q_1(x) = x + 1$

Plugging this into our original equation we get $P(x) = x^3 + 2x^2 + 3x + 3$

Verify this works with $P(x) = (x^2 + 1)Q_2(x) + 2x + 1$

Therefore the answer is $1^2 + 2^2 + 3^2 + 3^2 = \mathbf{(E) 23}$

Solution 4 (Undetermined Coefficients)

Notice that we cannot have the quotients equal to some constants, since the same constant will yield different constant terms for $P(x)$ (which is bad) and different constants will yield different first coefficients (also bad). Thus, we try setting the quotients equal to linear terms (for minimizing degree).

Let $P(x) = (x^2 + x + 1)(ax + b) + (x + 2)$ and $P(x) = (x^2 + 1)(ax + c) + (2x + 1)$. The quotients have the same x coefficient, since $P(x)$ must have the same x^3 coefficient in both cases. Expanding, we get

$$P(x) = ax^3 + (a + b)x^2 + (a + b + 1)x + (b + 2)$$

and

$$P(x) = ax^3 + cx^2 + (a + 2)x + (c + 1).$$

Equating coefficients, we get $b + 2 = c + 1$, $a + b + 1 = a + 2$, and $a + b = c$. From the second equation, we get $b = 1$, then substituting into the first, $c = 2$. Finally, from $a + b = c$, we have $a = 1$. Now, $P(x) = (x^2 + x + 1)(ax + b) + (x + 2) = (x^2 + x + 1)(x + 1) + (x + 2) = x^3 + 2x^2 + 3x + 3$ and our answer is

$$1^2 + 2^2 + 3^2 + 3^2 = \mathbf{(E) 23}.$$

Solution 5 (Quick, but Not Quicker Than Solution 2)

We construct the following equations in terms of $P(x)$ and the information given by the problem:

$$(1) \quad P(x) = (x^2 + x + 1) \cdot Q(x) + x + 2$$

$$(2) \quad P(x) = (x^2 + 1) \cdot R(x) + 2x + 1$$

Upon inspection, $Q(x)$ and $R(x)$ cannot be constant, so the smallest possible degree of $P(x)$ is 3, and both $Q(x)$ and $R(x)$ are linear.

Let $Q(x) = x - q$ and $R(x) = x - r$. We know there will be values for q and r that make the below equation hold, so we can assume that $P(x)$ has a leading coefficient of 1.

Substituting these values in, and setting (1) and (2) equal to each other,

$$(x^2 + x + 1)(x - q) + x + 2 = (x^2 + 1)(x - r) + 2x + 1.$$

We plug in $x = 0$, yielding $r + 1 = q$. Substituting this value into the above equation,

$$(x^2 + x + 1)(x - r - 1) + x + 2 = (x^2 + 1)(x - r) + 2x + 1.$$

Letting $x = 1$, we conclude that $r = -2$, so $R(x) = x + 2$. Therefore,

$$P(x) = (x^2 + 1)(x + 2) + 2x + 1 = x^3 + 2x^2 + 3x + 3.$$

The requested sum is

$$1^2 + 2^2 + 3^2 + 3^2 = \boxed{(E) \ 23}$$

-Benedict T (countmath1)

Solution 6 (Similar to Solution 3)

By remainder theorem, the polynomial can be written as follows.

$$P(x) = (x^2 + x + 1)Q_1(x) + x + 2 = (x^2 + 1)Q_2(x) + 2x + 1$$

This is a timed exam, we can use the information given by answer choices. The answer choices tell us this is the polynomial with integer coefficients, and we need to find the polynomial with the least degree so we can assume both $Q_1(x)$ and $Q_2(x)$ are linear (the coefficient of x should be same).

Then we can write $P(x)$ as a cubic polynomial.

$$P(x) = (x^2 + x + 1)(ax + b) + x + 2 = (x^2 + 1)(ax + c) + 2x + 1$$

Substituting $x = 0, 1, -1$ to determine the value of a and b .

We have:

$$P(0) = b + 2 = c + 1$$

$$P(1) = 3a + 3b + 3 = 2a + 2c + 3$$

$$P(-1) = -a + b + 1 = -2a + 2c - 1$$

We can solve the simultaneous equations: $a = 1, b = 1, c = 2$

Hence, $P(x) = (x^2 + x + 1)(x + 1) + x + 2 = x^3 + 2x^2 + 3x + 3$. The answer is $1^2 + 2^2 + 3^2 + 3^2 = \boxed{(E) \ 23}$

Solution 7 (Bounding with Answer Choices)

We are given:

$$P(x) \equiv x + 2 \pmod{x^2 + x + 1} \quad P(x) \equiv 2x + 1 \pmod{x^2 + 1}$$

Since the problem asks us about the coefficients of the polynomial, we should evaluate $P(x)$ at 1.

Our congruences now become:

$$P(1) \equiv 1 + 2 \pmod{(1)^2 + (1) + 1} \implies P(1) \equiv 3 \pmod{3} \implies P(1) \equiv 0 \pmod{3}.$$

$$P(1) \equiv 2(1) + 1 \pmod{(1)^2 + 1} \implies P(1) \equiv 3 \pmod{2} \implies P(1) \equiv 1 \pmod{2}.$$

We now know that the coefficients of the polynomial sum to an odd multiple of 3.

We now begin bounding the value of the sum of the coefficients squared. For our answer, the sum of the coefficients squared must be between 10 and 23. Trying a quadratic polynomial, if $P(1) = 3$, then the maximum of the coefficients squared is 9 ($3^2 + 0^2 + 0^2$ is the maximum, any other permutation can be trivially shown to be lower). If $P(1) = 9$, the minimum of the coefficients squared is 27 ($3^2 + 3^2 + 3^2$, once again, any other permutation can be trivially shown to be higher). Any higher value of $P(1)$ obviously will not work, so $P(x)$ cannot be a quadratic.

Since a quadratic did not work, we now move up to a cubic. We start with $P(1) = 3$. Similarly to the quadratic, there is no way to get an answer within the range of the answer choices.

However, moving up to $P(1) = 9$, we get that the minimum of the coefficients squared is $81/4$ (which we get from all 4 coefficients being equal at $9/4$). Since this is the minimum, any other answers will be higher. We get that **(E) 23** is our answer, since it is the only answer choice greater than $81/4$.

-SwordOfJustice

Solution 8

Since the divisors $x^2 + x + 1$ and $x^2 + 1$ are both powers of 2, and the remainders $x + 2$ and $2x + 1$ are both linear, we can assume that $P(x)$ has a minimum degree of 3.

We can then set the coefficients of the polynomial as $P(x) = ax^3 + bx^2 + cx + d$.

Then, using long division, we divide $P(x) = ax^3 + bx^2 + cx + d$ by $x^2 + x + 1$ and obtain a remainder of $(c - b)x + (a + d - b)$, which is equal to $x + 2$

We also divide $P(x) = ax^3 + bx^2 + cx + d$ by $x^2 + 1$ and obtain a remainder of $(c - a)x + (d - b)$, which is equal to $2x + 1$.

From here we can set up a systems of equations:

$$c - b = 1 \quad a + d - b = 2 \quad c - a = 2 \quad d - b = 1$$

Thus giving us the solutions $a = 1$, $b = 2$, $c = 3$, $d = 3$ and hence $P(x) = x^3 + 2x^2 + 3x + 3$

Squaring the coefficients, then adding them, $1^2 + 2^2 + 3^2 + 3^2$ we get the final answer **(E) 23**

Solution 9

We assume $P(x)$ is a monic cubic polynomial. Now, we set $P(x) = x^3 + ax^2 + bx + c$, and set the equations:

$$x^3 + ax^2 + bx + c = Q_1(x)(x^2 + x + 1) + x + 2 \quad x^3 + ax^2 + bx + c = Q_2(x)(x^2 + 1) + 2x + 1$$

Plugging in $x = i$ for the second equation, we get $-i - a + bi + c = 2i + 1$, and matching coefficients gives $c - a = 1$ and $b - 1 = 2 \rightarrow b = 3$. Now, we can guess values for a and c based on the answer choices. We have that the sum of the squares of the current known coefficients is $1^2 + 3^2 = 10$, and guessing $c = 1, a = 0$ clearly doesn't match up with the answer choices, so we guess $c = 3, a = 2$ and find that the answer is **(E) 23**.

Solution 10 (EZ CRT)

If we let $x = 10$, we just get a CRT problem. Note: this only works because all the coefficients are integers between 0 and 9. Our problem now is to find n such that

$$n \equiv 12 \pmod{111}$$

$$n \equiv 21 \pmod{101}$$

After using some careful Euclidean Algorithm (or just eyeing it), $n = 1233$, so our answer is $1^2 + 2^2 + 3^2 + 3^2 = \mathbf{(E) 23}$.

Solution 11 (Polynomial CRT)

$x^2 + x + 1$ and $x^2 + 1$ are relatively prime according to Bézout's theorem.

$P(x) \equiv U(x) + V(x) \pmod{(x^2 + x + 1)(x^2 + 1)}$, where

$$U(x) \equiv x + 2 \pmod{x^2 + x + 1}, \quad U(x) \equiv 0 \pmod{x^2 + 1}$$

$$V(x) \equiv 2x + 1 \pmod{x^2 + 1}, \quad V(x) \equiv 0 \pmod{x^2 + x + 1}$$

Notice that

$$x^2 + 1 \equiv -x \pmod{x^2 + x + 1} \tag{9}$$

$$(x^2 + 1)(x + 1) \equiv -x(x + 1) \equiv 1 \pmod{x^2 + x + 1} \tag{10}$$

$$U(x) \equiv -(x^2 + 1) + 2(x^2 + 1)(x + 1) \pmod{(x^2 + x + 1)(x^2 + 1)} \tag{11}$$

$$U(x) \equiv 2x^3 + x^2 + 2x + 1 \pmod{(x^2 + x + 1)(x^2 + 1)} \tag{12}$$

And

$$x^2 + x + 1 \equiv x \pmod{x^2 + 1} \tag{13}$$

$$(x^2 + x + 1)x \equiv x^2 \equiv -1 \pmod{x^2 + 1} \tag{14}$$

$$V(x) \equiv 2(x^2 + x + 1) - (x^2 + x + 1)x \pmod{(x^2 + x + 1)(x^2 + 1)} \tag{15}$$

$$V(x) \equiv -x^3 + x^2 + x + 2 \pmod{(x^2 + x + 1)(x^2 + 1)} \tag{16}$$

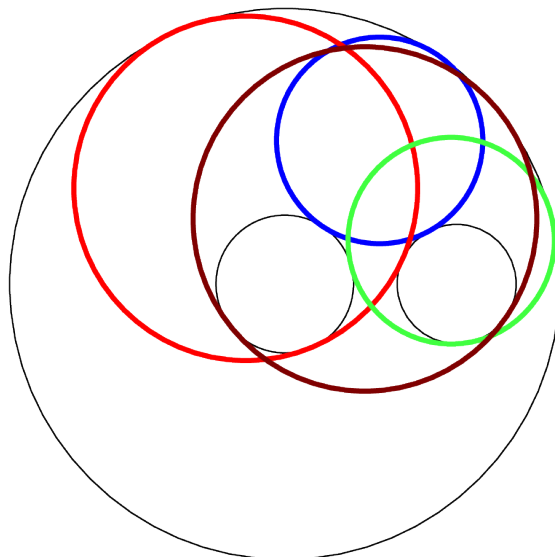
Therefore $P(x) \equiv U(x) + V(x) \equiv x^3 + 2x^2 + 3x + 3 \pmod{(x^2 + x + 1)(x^2 + 1)}$, the answer is $1^2 + 2^2 + 3^2 + 3^2 = \mathbf{(E) 23}$.

22. (AMC 10 10B 2022) Let S be the set of circles in the coordinate plane that are tangent to each of the three circles with equations $x^2 + y^2 = 4$, $x^2 + y^2 = 64$, and $(x - 5)^2 + y^2 = 3$. What is the sum of the areas of all circles in S ?

(A) 48π (B) 68π (C) 96π (D) 102π (E) 136π

Answer: E

Solution 1



The circles match up as follows: Case 1 is brown, Case 2 is blue, Case 3 is green, and Case 4 is red. Let $x^2 + y^2 = 64$ be circle O , $x^2 + y^2 = 4$ be circle P , and $(x - 5)^2 + y^2 = 3$ be circle Q . All the circles in S are internally tangent to circle O . There are four cases with two circles belonging to each:

- * P and Q are internally tangent to S .
- * P and Q are externally tangent to S .
- * P is externally and Circle Q is internally tangent to S .
- * P is internally and Circle Q is externally tangent to S .

Consider Cases 1 and 4 together. Since circles O and P have the same center, the line connecting the center of S and the center of O will pass through the tangency point of both S and O and the tangency point of S and P . This line will be the diameter of S and have length $r_P + r_O = 10$. Therefore the radius of S in these cases is 5.

Consider Cases 2 and 3 together. Similarly to Cases 1 and 4, the line connecting the center of S to the center of O will pass through the tangency points. This time, however, the diameter of S will have length $r_P - r_O = 6$. Therefore, the radius of S in these cases is 3.

The set of circles S consists of 8 circles - 4 of which have radius 5 and 4 of which have radius 3. The total area of all circles in S is $4(5^2\pi + 3^2\pi) = 136\pi \Rightarrow \boxed{\text{(E)}}$.

-naman12

Solution 2

We denote by C_1 the circle that has the equation $x^2 + y^2 = 4$. We denote by C_2 the circle that has the equation $x^2 + y^2 = 64$. We denote by C_3 the circle that has the equation $(x - 5)^2 + y^2 = 3$.

We denote by C_0 a circle that is tangent to C_1 , C_2 and C_3 . We denote by (u, v) the coordinates of circle C_0 , and r the radius of this circle.

From the graphs of circles C_1 , C_2 , C_3 , we observe that if C_0 is tangent to all of them, then C_0 must be internally tangent to C_2 . We have

$$u^2 + v^2 = (8 - r)^2. \quad (1)$$

We do the following casework analysis in terms of the whether C_0 is externally tangent to C_1 and C_3 .

Case 1: C_0 is externally tangent to C_1 and C_3 .

We have

$$u^2 + v^2 = (r + 2)^2 \quad (2)$$

and

$$(u - 5)^2 + v^2 = (r + \sqrt{3})^2. \quad (3)$$

Taking (2) – (1), we get $r + 2 = 8 - r$. Thus, $r = 3$. We can further compute (omitted here) that there exist feasible (u, v) with this given r .

Case 2: C_1 is internally tangent to C_0 and C_3 is externally tangent to C_0 .

We have

$$u^2 + v^2 = (r - 2)^2 \quad (2)$$

and

$$(u - 5)^2 + v^2 = (r + \sqrt{3})^2. \quad (3)$$

Taking (2) – (1), we get $r - 2 = 8 - r$. Thus, $r = 5$. We can further compute (omitted here) that there exist feasible (u, v) with this given r .

Case 3: C_1 is externally tangent to C_0 and C_3 is internally tangent to C_0 .

We have

$$u^2 + v^2 = (r + 2)^2 \quad (2)$$

and

$$(u - 5)^2 + v^2 = (r - \sqrt{3})^2. \quad (3)$$

Taking (2) – (1), we get $r + 2 = 8 - r$. Thus, $r = 3$. We can further compute (omitted here) that there exist feasible (u, v) with this given r .

Case 4: C_1 is internally tangent to C_0 and C_3 is internally tangent to C_0 .

We have

$$u^2 + v^2 = (r - 2)^2 \quad (2)$$

and

$$(u - 5)^2 + v^2 = (r - \sqrt{3})^2. \quad (3)$$

Taking (2) – (1), we get $r - 2 = 8 - r$. Thus, $r = 5$. We can further compute (omitted here) that there exist feasible (u, v) with this given r .

Because the graph is symmetric with the x -axis, and for each case above, the solution of v is not 0. Hence, in each case, there are two congruent circles whose centers are symmetric through the x -axis.

Therefore, the sum of the areas of all the circles in S is $2(3^2\pi + 5^2\pi + 3^2\pi + 5^2\pi) =$
(E) 136π .

Solution 3

The problem of counting the number of circles tangent to a given number of other circles, lines, or points is called Apollonius's Problem. Here, there are three circles; solutions to Apollonius's problem tell us that there are 8 possible tangent circles. When looking for all eight, we can notice that the diagram is symmetrical about

the x -axis, so we only need to find four different circles and then reflect them to the other side.

After drawing in those circles, we can use any of the above methods to find the radii of the circle. The important step is to realize that the segment between tangent circles' centers must pass through the point of tangency. From here, we can prove that the center of the circle with equation $x^2 + y^2 = 4$ and the center of the circle tangent to all three must both lie on the radius of $x^2 + y^2 = 64$, allowing us to quickly find the radius of the tangent circle.

For the tangent circles mostly above the x -axis, there are two of radius 5, and two of radius 3. The total area is $50\pi + 18\pi = 68\pi$. As due to symmetry there are four other circles below the x -axis, the total area is 136π , so we choose $\boxed{(E)}$.

23. (AMC 10 10B 2022) Ant Amelia starts on the number line at 0 and crawls in the following manner. For $n = 1, 2, 3$, Amelia chooses a time duration t_n and an increment x_n independently and uniformly at random from the interval $(0, 1)$. During the n th step of the process, Amelia moves x_n units in the positive direction, using up t_n minutes. If the total elapsed time has exceeded 1 minute during the n th step, she stops at the end of that step; otherwise, she continues with the next step, taking at most 3 steps in all. What is the probability that Amelia's position when she stops will be greater than 1?

(A) $\frac{1}{3}$ (B) $\frac{1}{2}$ (C) $\frac{2}{3}$ (D) $\frac{3}{4}$ (E) $\frac{5}{6}$

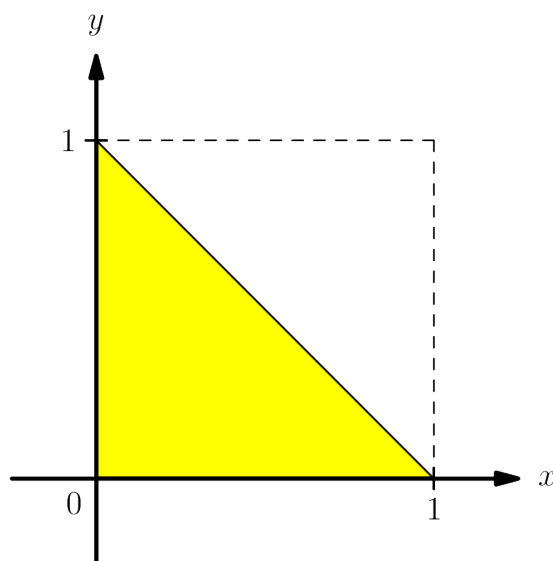
Answer: C

Solution 1 (Geometric Probability)

Let x and y be random variables that are independently and uniformly distributed in the interval $(0, 1)$. Note that

$$P(x + y \leq 1) = \frac{\frac{1}{2} \cdot 1^2}{1^2} = \frac{1}{2},$$

as shown below:

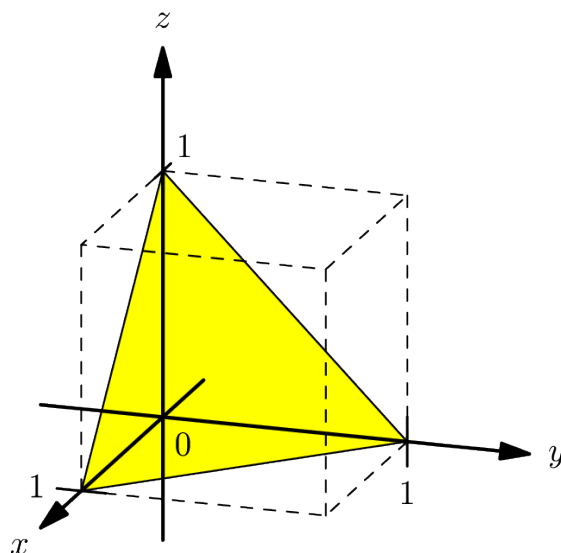


Let x , y , and z be random variables that are independently and uniformly distributed

in the interval $(0, 1)$. Note that

$$P(x + y + z \leq 1) = \frac{\frac{1}{3} \cdot \left(\frac{1}{2} \cdot 1^2\right) \cdot 1}{1^3} = \frac{1}{6},$$

as shown below:



We have two cases:

- a) Amelia takes exactly 2 steps. We need $x_1 + x_2 > 1$ and $t_1 + t_2 > 1$. So, the probability is

$$P(x_1 + x_2 > 1) \cdot P(t_1 + t_2 > 1) = \left(1 - \frac{1}{2}\right) \cdot \left(1 - \frac{1}{2}\right) = \frac{1}{4}.$$

- b) Amelia takes exactly 3 steps. We need $x_1 + x_2 + x_3 > 1$ and $t_1 + t_2 \leq 1$. So, the probability is

$$P(x_1 + x_2 + x_3 > 1) \cdot P(t_1 + t_2 \leq 1) = \left(1 - \frac{1}{6}\right) \cdot \frac{1}{2} = \frac{5}{12}.$$

Together, the answer is $\frac{1}{4} + \frac{5}{12} = \boxed{\text{(C)} \frac{2}{3}}.$

Solution 2 (Generalization and Induction)

We can in fact find the probability that any number of randomly distributed numbers on the interval $[0, 1]$ sum to more than 1 using geometric probability, as shown in the video below.

If we graph the points that satisfy $x + y < 1$, $0 < x, y < 1$, we get the triangle with points $(0, 0)$, $(1, 0)$, and $(0, 1)$. If we graph the points that satisfy $x + y + z < 1$, $0 < x, y, z < 1$, we get the tetrahedron with points $(0, 0, 0)$, $(1, 0, 0)$, $(0, 1, 0)$, and $(0, 0, 1)$.

Of course, the probability of either of these cases happening is simply the area/volume of the points we graphed divided by the total area of the graph, which is always 1 (this would be much simpler than my calculus proof above).

Thus, we can now solve for the probability that the sum is less than one for n numbers using induction.

Claim: The probability that the sum is less than one is $\frac{1}{n!}$.

Base Case: For just 1 number, the probability is 1.

Induction step: Suppose that the probability for n numbers is $\frac{1}{n!}$. We will prove that the probability for $n + 1$ numbers is $\frac{1}{(n+1)!}$. To prove this, we consider that the area of an $n + 1$ -dimensional tetrahedron is simply the area/volume of the base times the height divided by $n + 1$.

Of course, the area of the base is $\frac{1}{n!}$, and the height is 1, and thus, we obtain $\frac{1}{n! \cdot (n+1)} = \frac{1}{(n+1)!}$ as our volume (this may be hard to visualize for higher dimensions). The induction step is complete.

The probability of the sum being less than 1 is $\frac{1}{n!}$, and the probability of the sum being more than 1 is $\frac{n!-1}{n!}$. This trivializes the problem. The answer is

$$\frac{1}{2} \cdot \frac{2!-1}{2!} + \frac{1}{2} \cdot \frac{3!-1}{3!} = \boxed{(C) \frac{2}{3}}.$$

Solution 3 (Observations)

There are two cases: Amelia takes two steps or three steps.

The former case has a probability of $\frac{1}{2}$, as stated in Solution 1, and thus the latter also has a probability of $\frac{1}{2}$.

The probability that Amelia passes 1 after two steps is also $\frac{1}{2}$, as it is symmetric to the probability above.

Thus, if the probability that Amelia passes 1 after three steps is x , our total probability is $\frac{1}{2} \cdot \frac{1}{2} + \frac{1}{2} \cdot x$. We know that $0 < x < 1$, and it is relatively obvious that $x > \frac{1}{2}$ (because the probability that $x > \frac{3}{2}$ is $\frac{1}{2}$). This means that our total probability is between $\frac{1}{2}$ and $\frac{3}{4}$, non-inclusive, so the only answer choice that fits is

$$\boxed{(C) \frac{2}{3}}.$$

Remark (Calculus) It is not immediately clear why three random numbers between 0 and 1 have a probability of $\frac{5}{6}$ of summing to more than 1. Here is a proof:

Let us start by finding the probability that two random numbers between 0 and 1 have a sum of more than x , where $0 \leq x \leq 1$.

Suppose that our two numbers are y and z . Then, the probability that $y > x$ (which means that $y + z > x$) is $1 - x$, and the probability that $y < x$ is x .

If $y < x$, the probability that $y + z > x$ is $1 - x + y$. This is because the probability that $y + z < x$ is equal to the probability that $z < x - y$, which is $x - y$, so our total probability is $1 - (x - y) = 1 - x + y$.

Let us now find the average of the probability that $y + z > x$ when $y < x$. Since y is a random number between 0 and x , its average is $\frac{x}{2}$. Thus, our average is $1 - x + \frac{1}{2} = 1 - \frac{x}{2}$.

Hence, our total probability is equal to

$$1(1 - x) + \left(1 - \frac{x}{2}\right)(x) = 1 - \frac{1}{2}x^2.$$

Now, let us find the probability that three numbers uniformly distributed between 0 and 1 sum to more than 1.

Let our three numbers be a , b , and c . Then, the probability that $a + b + c > 1$ is equal to the probability that $b + c$ is greater than $1 - a$, which is equal to $1 - \frac{1}{2}(1 - a)^2$.

To find the total probability, we must average over all values of a . This average is simply equal to the area under the curve $1 - \frac{1}{2}(1 - x)^2$ from 0 to 1, all divided by 1. We can compute this value using integrals:

$$\begin{aligned} \frac{\int_0^1 1 - \frac{1}{2}(1 - x)^2 dx}{1} &= \int_0^1 1 - \frac{1}{2}(1 - x)^2 dx \\ &= 1 - \frac{1}{2} \int_0^1 (1 - x)^2 dx \\ &= 1 - \frac{1}{2} \int_0^1 x^2 dx \\ &= 1 - \frac{1}{2} \left(\frac{1}{3} \right) \\ &= \frac{5}{6}. \end{aligned}$$

For those who don't know calculus, $\int_m^n f(x) dx$ is the area under the curve $f(x)$ from m to n .

Remark (Rigorous Calculus) In the language of probability, we have three random variables X , Y , Z , each independent and uniformly distributed over the interval $(0, 1)$. We are interested in the probability that $X + Y + Z > 1$, or $P(X + Y + Z > 1)$.

It follows from the definition of the joint probability density function $f(x, y, z)$ that (we assume that $0 < x, y, z < 1$):

$$P(X + Y + Z > 1) = \iiint_{x+y+z>1} f(x, y, z) dx dy dz. \quad (17)$$

Since X, Y, Z are independent, we have $f(x, y, z) = f_X(x) * f_Y(y) * f_Z(z)$, where $f_X(x), f_Y(y), f_Z(z)$ represent the probability densities of X, Y, Z respectively. Recall that X, Y, Z are uniformly distributed over the interval $(0, 1)$. Hence, $f_X(x) = f_Y(y) = f_Z(z) = 1$.

It remains to evaluate the following triple integral

$$\iiint_{x+y+z>1} dx dy dz = \frac{5}{6}. \quad (18)$$

Therefore, $P(X + Y + Z > 1) = \frac{5}{6}$.

respect to whoever would do this during the comp

-guy_who_likes_frogs

24. (AMC 10 10B 2022) Consider functions f that satisfy

$$|f(x) - f(y)| \leq \frac{1}{2}|x - y|$$

for all real numbers x and y . Of all such functions that also satisfy the equation $f(300) = f(900)$, what is the greatest possible value of

$$f(f(800)) - f(f(400))?$$

(A) 25 (B) 50 (C) 100 (D) 150 (E) 200

Answer: B

Solution 1 (Absolute Values and Inequalities)

We have

$$\begin{aligned} |f(f(800)) - f(f(400))| &\leq \frac{1}{2} |f(800) - f(400)| & (\star) \\ &\leq \frac{1}{2} \left| \frac{1}{2} |800 - 400| \right| \\ &= 100, \end{aligned}$$

from which we eliminate answer choices (D) and (E).

Note that

$$\begin{aligned} |f(800) - f(300)| &\leq \frac{1}{2} |800 - 300| = 250, \\ |f(800) - f(900)| &\leq \frac{1}{2} |800 - 900| = 50, \\ |f(400) - f(300)| &\leq \frac{1}{2} |400 - 300| = 50, \\ |f(400) - f(900)| &\leq \frac{1}{2} |400 - 900| = 250. \end{aligned}$$

Let $a = f(300) = f(900)$. Together, it follows that

$$\begin{aligned} |f(800) - a| &\leq 50, \\ |f(400) - a| &\leq 50. \end{aligned}$$

We rewrite (★) as

$$\begin{aligned} |f(f(800)) - f(f(400))| &\leq \frac{1}{2} |f(800) - f(400)| \\ &= \frac{1}{2} |(f(800) - a) - (f(400) - a)| \\ &\leq \frac{1}{2} |50 - (-50)| \\ &= \boxed{\text{(B) } 50}. \end{aligned}$$

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Solution 2 (Lipschitz Condition)

Denote $f(900) - f(600) = a$. Because $f(300) = f(900)$, $f(300) - f(600) = a$.

Following from the Lipschitz condition given in this problem, $|a| \leq 150$ and

$$f(800) - f(600) \leq \min \{a + 50, 100\}$$

and

$$f(400) - f(600) \geq \max \{a - 50, -100\}.$$

Thus,

$$\begin{aligned} f(800) - f(400) &\leq \min \{a + 50, 100\} - \max \{a - 50, -100\} \\ &= 100 + \min \{a, 50\} - \max \{a, -50\} \\ &= 100 + \begin{cases} a + 50 & \text{if } a \leq -50 \\ 0 & \text{if } -50 < a < 50 \\ -a + 50 & \text{if } a \geq 50 \end{cases} . \end{aligned}$$

Thus, $f(800) - f(400)$ is maximized at $a = 0$, $f(800) - f(600) = 50$, $f(400) - f(600) = -50$, with the maximal value 100.

By symmetry, following from an analogous argument, we can show that $f(800) - f(400)$ is minimized at $a = 0$, $f(800) - f(600) = -50$, $f(400) - f(600) = 50$, with the minimal value -100 .

Following from the Lipschitz condition,

$$\begin{aligned} f(f(800)) - f(f(400)) &\leq \frac{1}{2} |f(800) - f(400)| \\ &\leq 50. \end{aligned}$$

We have already construct instances in which the second inequality above is augmented to an equality.

Now, we construct an instance in which the first inequality above is augmented to an equality.

Consider the following piecewise-linear function:

$$f(x) = \begin{cases} \frac{1}{2}(x - 300) & \text{if } x \leq 300 \\ -\frac{1}{2}(x - 300) & \text{if } 300 < x \leq 400 \\ \frac{1}{2}(x - 600) & \text{if } 400 < x \leq 800 \\ -\frac{1}{2}(x - 900) & \text{if } x > 800 \end{cases} .$$

Therefore, the maximum value of $f(f(800)) - f(f(400))$ is **(B)** 50.

Solution 3 (Slopes)

Divide both sides by $|x - y|$ to get $\frac{|f(x) - f(y)|}{|x - y|} \leq \frac{1}{2}$. This means that when we take any two points on f , the absolute value of the slope between the two points is at most $\frac{1}{2}$.

Let $f(300) = f(900) = c$, and since we want to find the maximum value of $|f(800) - f(400)|$, we can take the most extreme case and draw a line with slope $-\frac{1}{2}$ down from $f(300)$ to $f(400)$ and a line with slope $\frac{1}{2}$ up from $f(800)$ to $f(900)$. Then $f(400) = c - 50$ and $f(800) = c + 50$, so $|f(800) - f(400)| = |c + 50 - (c - 50)| = 100$, and this is attainable because the slope of the line connecting $f(400)$ and $f(800)$ still has absolute value less than $\frac{1}{2}$.

Therefore, $|f(f(800)) - f(f(400))| \leq \frac{1}{2} |f(800) - f(400)| = \frac{1}{2}(100) =$ **(B)** 50.

Solution 4 (Translation)

Consider $g(x) = f(x) - f(300)$. Then $g(x)$ satisfies all the conditions and $g(300) = g(900) = 0$. We would want $g(400)$ and $g(800)$ as distant from each other as possible. So assign $g(400) = -50$ and $g(800) = 50$, the possible lower and upper bounds respectively. It follows that one can obtain the upper bound for $|g(g(800)) - g(g(400))| = |g(50) - g(-50)| \leq \frac{1}{2}(100) =$ **(B)** 50 as the answer.

25. (AMC 10 10B 2022) Let $x_0, x_1, x_2, \dots$ be a sequence of numbers, where each x_k is either 0 or 1. For each positive integer n , define

$$S_n = \sum_{k=0}^{n-1} x_k 2^k$$

Suppose $7S_n \equiv 1 \pmod{2^n}$ for all $n \geq 1$. What is the value of the sum

$$x_{2019} + 2x_{2020} + 4x_{2021} + 8x_{2022}?$$

- (A) 6 (B) 7 (C) 12 (D) 14 (E) 15

Answer: A

Solution 1

In binary numbers, we have

$$S_n = (x_{n-1}x_{n-2}x_{n-3}x_{n-4} \dots x_2x_1x_0)_2.$$

It follows that

$$8S_n = (x_{n-1}x_{n-2}x_{n-3}x_{n-4} \dots x_2x_1x_0000)_2.$$

We obtain $7S_n$ by subtracting the equations:

$$\begin{array}{r} (x_{n-1} \ x_{n-2} \ x_{n-3} \ x_{n-4} \ \dots \ x_2 \ x_1 \ x_0 \ 0 \ 0 \ 0)_2 \\ - \quad (x_{n-1} \ \dots \ x_5 \ x_4 \ x_3 \ x_2 \ x_1 \ x_0)_2 \\ \hline (\ ? \ \quad ? \ \quad ? \ \quad 0 \ \quad \dots \ 0 \ 0 \ 0 \ 0 \ 0 \ 1)_2 \end{array}$$

We work from right to left:

$$\begin{aligned} x_{-0} = x_{-1} = x_{-2} = 1 &\implies x_{-3} = 0 \\ &\implies x_{-4} = 1 \\ &\implies x_{-5} = 1 \\ &\implies x_{-6} = 0 \\ &\implies x_{-7} = 1 \\ &\implies x_{-8} = 1 \\ &\vdots \end{aligned}$$

For all $n \geq 3$, we conclude that

- $x_n = 0$ if and only if $n \equiv 0 \pmod{3}$.
- $x_n = 1$ if and only if $n \not\equiv 0 \pmod{3}$.

Finally, we get $(x_{2019}, x_{2020}, x_{2021}, x_{2022}) = (0, 1, 1, 0)$, from which

$$x_{2019} + 2x_{2020} + 4x_{2021} + 8x_{2022} = \boxed{\text{(A) } 6}.$$

Steven Chen (Professor Chen Education Palace, www.professorchenedu.com)

Solution 2

First, notice that

$$x_{2019} + 2x_{2020} + 4x_{2021} + 8x_{2022} = \frac{S_{2023} - S_{2019}}{2^{2019}}.$$

Then since S_n is the modular inverse of 7 in $\mathbb{Z}_{2^n}$, we can perform the Euclidean Algorithm to find it for $n = 2019, 2023$.

Starting with 2019,

$$\begin{aligned} 7S_{2019} &\equiv 1 \pmod{2^{2019}} \\ 7S_{2019} &= 2^{2019}k + 1. \end{aligned}$$

Now, take both sides mod 7:

$$0 \equiv 2^{2019}k + 1 \pmod{7}.$$

Using Fermat's Little Theorem,

$$2^{2019} = (2^{336})^6 \cdot 2^3 \equiv 2^3 \equiv 1 \pmod{7}.$$

Thus,

$$0 \equiv k + 1 \pmod{7} \implies k \equiv 6 \pmod{7} \implies k = 7j + 6.$$

Therefore,

$$7S_{2019} = 2^{2019}(7j + 6) + 1 \implies S_{2019} = \frac{2^{2019}(7j + 6) + 1}{7}.$$

We may repeat this same calculation with S_{2023} to yield

$$S_{2023} = \frac{2^{2023}(7h + 3) + 1}{7}.$$

Now, we notice that S_n is basically an integer expressed in binary form with n bits. This gives rise to a simple inequality,

$$0 \leq S_n \leq 2^n.$$

Since the maximum possible number that can be generated with n bits is

$$\underbrace{11111 \dots 1}_n = \sum_{k=0}^{n-1} 2^k = 2^n - 1 \leq 2^n.$$

Looking at our calculations for S_{2019} and S_{2023} , we see that the only valid integers that satisfy that constraint are $j = h = 0$.

$$\frac{S_{2023} - S_{2019}}{2^{2019}} = \frac{\frac{2^{2023} \cdot 3 + 1}{7} - \frac{2^{2019} \cdot 6 + 1}{7}}{2^{2019}} = \frac{2^4 \cdot 3 - 6}{7} = \boxed{\text{(A)} 6}.$$

zoomanTV

Solution 3

As in Solution 2, we note that

$$x_{2019} + 2x_{2020} + 4x_{2021} + 8x_{2022} = \frac{S_{2023} - S_{2019}}{2^{2019}}.$$

We also know that $7S_{2023} \equiv 1 \pmod{2^{2023}}$ and $7S_{2019} \equiv 1 \pmod{2^{2019}}$, this implies:

$$(1) \quad 7S_{2023} = 2^{2023} \cdot x + 1,$$

$$(2) \quad 7S_{2019} = 2^{2019} \cdot y + 1.$$

Dividing by 7, we can isolate the previous sums:

$$(3) \quad S_{2023} = \frac{2^{2023} \cdot x + 1}{7},$$

$$(4) \quad S_{2019} = \frac{2^{2019} \cdot y + 1}{7}.$$

The maximum value of S_n occurs when every x_i is equal to 1. Even when this happens, the value of S_n is less than 2^n . Therefore, we can construct the following inequalities:

$$(3) \quad S_{2023} = \frac{2^{2023} \cdot x + 1}{7} < 2^{2023},$$

$$(4) \quad S_{2019} = \frac{2^{2019} \cdot y + 1}{7} < 2^{2019}.$$

From these two equations, we can deduce that both x and y are less than 7.

Reducing **1** and **2** (mod 7), we see that

$$2^{2023} \cdot x \equiv 6 \pmod{7},$$

and

$$2^{2019} \cdot y \equiv 6 \pmod{7}.$$

The powers of 2 repeat every 3, (mod 7).

Therefore, $2^{2023} \equiv 2 \pmod{7}$ and $2^{2019} \equiv 1 \pmod{7}$. Substituting this back into the above equations,

$$2x \equiv 6 \pmod{7}$$

and

$$y \equiv 6 \pmod{7}.$$

Since x and y are integers less than 7, the only values of x and y are 3 and 6 respectively.

The requested sum is

$$\begin{aligned} \frac{S_{2023} - S_{2019}}{2^{2019}} &= \frac{\frac{2^{2023} \cdot x + 1}{7} - \frac{2^{2019} \cdot y + 1}{7}}{2^{2019}} \\ &= \frac{1}{2^{2019}} \left(\frac{2^{2023} \cdot 3 + 1}{7} - \left(\frac{2^{2019} \cdot 6 + 1}{7} \right) \right) \\ &= \frac{3 \cdot 2^4 - 6}{7} \\ &= \boxed{(A) \ 6}. \end{aligned}$$

-Benedict T (countmath1)

Solution 4

Note that, as in Solution 2, we have

$$x_{2019} + 2x_{2020} + 4x_{2021} + 8x_{2022} = \frac{S_{2023} - S_{2019}}{2^{2019}}.$$

This is because

$$S_{2023} = x_0 2^0 + x_1 2^1 + \cdots + x_{2019} 2^{2019} + \cdots + x_{2022} 2^{2022}$$

and

$$S_{2019} = x_0 2^0 + x_1 2^1 + \cdots + x_{2018} 2^{2018}.$$

Note that

$$S_{2023} - S_{2019} = x_{2019} 2^{2019} + \cdots + x_{2022} 2^{2022} = 2^{2019} (x_{2019} + 2x_{2020} + 4x_{2021} + 8x_{2022}).$$

Therefore,

$$x_{2019} + 2x_{2020} + 4x_{2021} + 8x_{2022} = \frac{S_{2023} - S_{2019}}{2^{2019}}.$$

Multiplying both sides by 7 gives us

$$7(x_{2019} + 2x_{2020} + 4x_{2021} + 8x_{2022}) = \frac{7S_{2023} - 7S_{2019}}{2^{2019}}.$$

We can write

$$7S_{2023} = 1 \pmod{2^{2023}} = 1 + 2^{2023}a = 1 + 2^{2019} * 16a$$

and

$$7S_{2019} = 1 \pmod{2^{2019}} = 1 + 2^{2019}b$$

for some a and b. Substituting, we get

$$7(x_{2019} + 2x_{2020} + 4x_{2021} + 8x_{2022}) = \frac{(1 + 2^{2019} * 16a) - (1 + 2^{2019}b)}{2^{2019}} = 16a - b.$$

Therefore, our answer can be written as

$$\frac{16a - b}{7}.$$

Another thing to notice is that a and b are integers between 0 and 6. This is because

$$7(1 + 2 + 4 + 8 + \cdots + 2^{2022}) \geq 7S_{2023} = 1 + 2^{2023}a$$

which is

$$\begin{aligned} 7(2^{2023}) - 7 &\geq 1 + 2^{2023}a \\ (7 - a) &\geq \frac{8}{2^{2023}} \end{aligned}$$

which only holds when a is less than 7 because the right is very small positive number, so the left must be positive, too. Clearly, a is also non-negative, because otherwise,

$$7S_{2023} = 1 + 2^{2023}a < 0$$

which would mean

$$S_{2023} < 0$$

which cannot happen, so a is greater than 0. A similar explanation for b shows that b is an integer between 0 and 6 inclusive.

Going back to the solution, if our answer to the problem is n, then

$$16a - b = 7n$$

and

$$16a = 7n + b,$$

so we can try the five option choices and see which one, when multiplied by 7 and added to some whole number between 0 and 6 results in a multiple of 16. Trying all the option choices, we see that you need to add 7n to something more than 6 to equal a multiple of 16 other than for option A. Therefore, the answer is (A) 6.

§8 AMC 10 10A 2022

1. (AMC 10 10A 2022) What is the value of

$$3 + \frac{1}{3 + \frac{1}{3 + \frac{1}{3}}}$$

- (A) $\frac{31}{10}$ (B) $\frac{49}{15}$ (C) $\frac{33}{10}$ (D) $\frac{109}{33}$ (E) $\frac{15}{4}$

Answer: D

Solution 1

We have

$$\begin{aligned} 3 + \frac{1}{3 + \frac{1}{3 + \frac{1}{3}}} &= 3 + \frac{1}{3 + \frac{1}{\left(\frac{10}{3}\right)}} \\ &= 3 + \frac{1}{3 + \frac{3}{10}} \\ &= 3 + \frac{1}{\left(\frac{33}{10}\right)} \\ &= 3 + \frac{10}{33} \\ &= \boxed{\text{(D)} \frac{109}{33}}. \end{aligned}$$

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Solution 2

Continued fractions with integer parts q_i and numerators all 1 can be calculated as

$$\frac{[q_0, q_1, q_2, \dots, q_n]}{[q_1, q_2, \dots, q_n]}$$

where

$$\begin{aligned} [] &= 1 \\ [q_0, q_1, q_2, \dots, q_n] &= q_0[q_1, q_2, \dots, q_n] + [q_2, \dots, q_n] \end{aligned}$$

Solution 3

Finite continued fractions of form $n + \frac{1}{n + \frac{1}{n + \dots}}$ have linear combinations of x, y that solve Pell's Equation. Specifically, the denominator y and numerator x are solutions to the Diophantine equation $(n^2 + 4)\left(\frac{y}{2}\right)^2 - \left(x - \frac{ny}{2}\right)^2 = \pm 1$. So for this problem in particular, the denominator y and numerator x are solutions to the Diophantine equation $13\left(\frac{y}{2}\right)^2 - \left(x - \frac{3y}{2}\right)^2 = \pm 1$. That leaves two answers. Since the number of 1's in the continued fraction is odd, we further narrow it down to $13\left(\frac{y}{2}\right)^2 - \left(x - \frac{3y}{2}\right)^2 = -1$, which only leaves us with 1 answer and that is

$$(x, y) = (109, 33) \text{ which means } \boxed{\text{(D)} \frac{109}{33}}.$$

(Note: Integer solutions increase exponentially, so our next solution will have a numerator greater than $3^2(109)$. Therefore, when you don't see numerators greater than $3^2(109)$ in the answer choices, this method should be fine.)

2. (AMC 10 10A 2022) Mike cycled 15 laps in 57 minutes. Assume he cycled at a constant speed throughout. Approximately how many laps did he complete in the first 27 minutes?

(A) 5 (B) 7 (C) 9 (D) 11 (E) 13

Answer: B

Solution 1

Mike's speed is $\frac{15}{57} = \frac{5}{19}$ laps per minute.

In the first 27 minutes, he completed approximately $\frac{5}{19} \cdot 27 \approx \frac{1}{4} \cdot 28 = \boxed{\text{(B) } 7}$ laps.

Solution 2

Mike runs 1 lap in $\frac{57}{15} = \frac{19}{5}$ minutes. So, in 27 minutes, Mike ran about $\frac{27}{\frac{19}{5}} \approx \boxed{\text{(B) } 7}$ laps.

Solution 3

Mike's rate is

$$\frac{15}{57} = \frac{x}{27},$$

where x is the number of laps he can complete in 27 minutes. If you cross multiply, $57x = 405$.

So, $x = \frac{405}{57} \approx \boxed{\text{(B) } 7}$.

Solution 4 (Quick Estimate)

Note that 27 minutes is a little bit less than half of 57 minutes. Mike will therefore run a little bit less than $15/2 = 7.5$ laps, which is about $\boxed{\text{(B) } 7}$.

Solution 5 (Approximation)

Note that 57 minutes is almost equal to 1 hour. Running 15 laps in 1 hour is running approximately 1 lap every 4 minutes. This means that in 27 minutes, Mike will run approximately $\frac{27}{4}$ laps. This is very close to $\frac{28}{4} = \boxed{\text{(B) } 7}$.

3. (AMC 10 10A 2022) The sum of three numbers is 96. The first number is 6 times the third number, and the third number is 40 less than the second number. What is the absolute value of the difference between the first and second numbers?

(A) 1 (B) 2 (C) 3 (D) 4 (E) 5

Answer: E

Solution 1

Let x be the third number. It follows that the first number is $6x$, and the second number is $x + 40$.

We have

$$6x + (x + 40) + x = 8x + 40 = 96,$$

from which $x = 7$.

Therefore, the first number is 42, and the second number is 47. Their absolute value of the difference is $|42 - 47| = \boxed{\text{(E) } 5}$.

Solution 2

Solve this using a system of equations. Let x, y , and z be the three numbers, respectively. We get three equations:

$$\begin{aligned}x + y + z &= 96, \\x &= 6z, \\z &= y - 40.\end{aligned}$$

Rewriting the third equation gives us $y = z + 40$, so we can substitute x as $6z$ and y as $z + 40$.

Therefore, we get

$$\begin{aligned}6z + (z + 40) + z &= 96 \\8z + 40 &= 96 \\8z &= 56 \\z &= 7.\end{aligned}$$

Substituting 7 in for z gives us $x = 6z = 6(7) = 42$ and $y = z + 40 = 7 + 40 = 47$.

So, the answer is $|x - y| = |42 - 47| = \boxed{\text{(E)} 5}$.

Solution 3

In accordance with Solution 2,

$$y = z + 40, x = 6z \implies |x - y| = |6z - z - 40| = 5|z - 8| \implies \boxed{\text{(E)} 5}.$$

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4. (AMC 10 10A 2022) In some countries, automobile fuel efficiency is measured in liters per 100 kilometers while other countries use miles per gallon. Suppose that 1 kilometer equals m miles, and 1 gallon equals l liters. Which of the following gives the fuel efficiency in liters per 100 kilometers for a car that gets x miles per gallon?

(A) $\frac{x}{100lm}$ (B) $\frac{xlm}{100}$ (C) $\frac{lm}{100x}$ (D) $\frac{100}{xlm}$ (E) $\frac{100lm}{x}$

Answer: E

Solution 1

The formula for fuel efficiency is

$$\frac{\text{Distance}}{\text{Gas Consumption}}.$$

Note that 1 mile equals $\frac{1}{m}$ kilometers. We have

$$\frac{x \text{ miles}}{1 \text{ gallon}} = \frac{\frac{x}{m} \text{ kilometers}}{l \text{ liters}} = \frac{1 \text{ kilometer}}{\frac{lm}{x} \text{ liters}} = \frac{100 \text{ kilometers}}{\frac{100lm}{x} \text{ liters}}.$$

Therefore, the answer is $\boxed{\text{(E)} \frac{100lm}{x}}$.

Solution 2

Since it can be a bit odd to think of "liters per 100 km", this statement's numerical value is equivalent to 100 km per 1 liter:

1 km requires l liters, so the numerator is simply l . Since l liters is 1 gallon, and x miles is 1 gallon, we have $1 \text{ liter} = \frac{x}{l}$.

Therefore, the requested expression is

$$100 \cdot \frac{m}{\left(\frac{x}{l}\right)} = \boxed{\text{(E)} \frac{100lm}{x}}.$$

-Benedict T (countmath1)

Solution 3

A car that gets a gallon for x miles has a fuel efficiency of $\frac{1 \text{ gallon}}{x \text{ miles}}$. We want to convert these units to liters and kilometers. Since one gallon is l liters, that can be rewritten as $\frac{l \text{ liters}}{x \text{ miles}}$.

Now to convert the miles into kilometers. 1 km is m miles, so 1 mile is $\frac{1}{m}$ km. Plug that in to get $\frac{l \text{ liter}}{x \cdot \frac{1}{m} \text{ km}}$. Simplifying, we get $\frac{lm \text{ liters}}{x \text{ km}}$.

But the problem asked for the fuel efficiency of liters per 100 km, not just 1 km.

So instead of m , we put in $100m$, getting $\boxed{\text{(E)} \frac{100lm}{x}}$ as the answer.

5. (AMC 10 10A 2022) Square $ABCD$ has side length 1. Points P , Q , R , and S each lie on a side of $ABCD$ such that $APQCRS$ is an equilateral convex hexagon with side length s . What is s ?

(A) $\frac{\sqrt{2}}{3}$ (B) $\frac{1}{2}$ (C) $2 - \sqrt{2}$ (D) $1 - \frac{\sqrt{2}}{4}$ (E) $\frac{2}{3}$

Answer: C

Solution 1

Note that $BP = BQ = DR = DS = 1 - s$. It follows that $\triangle BPQ$ and $\triangle DRS$ are congruent isosceles right triangles.

In $\triangle BPQ$, we have $PQ = BP\sqrt{2}$, or

$$\begin{aligned} s &= (1 - s)\sqrt{2} \\ s &= \sqrt{2} - s\sqrt{2} \\ (\sqrt{2} + 1)s &= \sqrt{2} \\ s &= \frac{\sqrt{2}}{\sqrt{2} + 1}. \end{aligned}$$

Therefore, the answer is

$$s = \frac{\sqrt{2}}{\sqrt{2} + 1} \cdot \frac{\sqrt{2} - 1}{\sqrt{2} - 1} = \boxed{\text{(C)} 2 - \sqrt{2}}.$$

MRENTHUSIASM

Solution 2

Since it is an equilateral convex hexagon, all sides are the same, so we will call the side length x . Notice that $(1 - x)^2 + (1 - x)^2 = x^2$. We can solve this equation which gives us our answer.

$$\begin{aligned} 1 + x^2 - 2x + 1 + x^2 - 2x &= x^2 \\ 2x^2 - 4x + 2 &= x^2 \end{aligned}$$

$$x^2 - 4x + 2 = 0$$

We then use the quadratic formula which gives us:

$$\begin{aligned} x &= \frac{4 \pm \sqrt{4^2 - 4 \cdot 1 \cdot 2}}{2 \cdot 1} \\ &= \frac{4 \pm \sqrt{8}}{2} \\ &= \frac{4 \pm 2\sqrt{2}}{2} \end{aligned}$$

Then we simplify it by dividing and crossing out 2 which gives us $2 \pm \sqrt{2}$ and that gives us (C) $2 - \sqrt{2}$.

Solution 3 (Area)

We can find areas in terms of s . From the diagram, draw in segments SP and RQ . We then have two non-shaded triangles, two shaded triangles, and a rectangle.

The non-shaded triangles have leg lengths of $1 - s$, so they each have area $\frac{(1-s)^2}{2}$. Therefore, the total area of the two triangles is $(1 - s)^2$.

The shaded triangles have side lengths s , so they each have area $\frac{s^2}{2}$. Then, we get that their combined area is s^2 .

Looking at the rectangle, we find that $SP = RQ = s\sqrt{2}$, from 45-45-90 triangles APS and CRQ . Multiplying this with the other side length s , we see that the rectangle has area $s^2\sqrt{2}$.

These three expressions of area sum up to the big square, which has area 1. So, we add them up and solve:

$$\begin{aligned} (1 - s)^2 + s^2 + s^2\sqrt{2} &= 1 \\ s^2 - 2s + 1 + s^2 + s^2\sqrt{2} &= 1 \\ 2s^2 + s^2\sqrt{2} - 2s &= 0 \\ s((2 + \sqrt{2})s - 2) &= 0 \end{aligned}$$

s cannot be 0, since it represents a positive side length. This means that s satisfies $(2 + \sqrt{2})s - 2 = 0$. Solving, we see that

$$s = \frac{2}{2 + \sqrt{2}} = \frac{2}{2 + \sqrt{2}} \cdot \frac{2 - \sqrt{2}}{2 - \sqrt{2}} = \frac{4 - 2\sqrt{2}}{2} = \boxed{\text{(C)} \ 2 - \sqrt{2}}.$$

Solution 4 (Pythagorean Theorem)

A corner of the square gives a right triangle (e.g. $\triangle DRS$) with legs $1 - s$ and hypotenuse s . It follows that

$$\begin{aligned} 2(1-s)^2 &= s^2 \\ s^2 - 4s + 2 &= 0 \end{aligned}$$

Completing the square gives

$$\begin{aligned} s^2 - 4s + 4 &= 2 \\ (s-2)^2 &= 2 \\ s &= \pm\sqrt{2} + 2 \end{aligned}$$

The only answer choice that appears in our solution is $-\sqrt{2} + 2$, or **(C)** $2 - \sqrt{2}$.

Solution 5 (Similar to Solution 1)

$AS = s$ and $SD = \frac{\sqrt{2}}{2}s$ based on **45-45-90 Triangles** (refer to Solution 1 why) so $s = \frac{1}{1+\frac{\sqrt{2}}{2}}$. Simplify and get **(C)** $2 - \sqrt{2}$.

6. (AMC 10 10A 2022) Which expression is equal to

$$\left| a - 2 - \sqrt{(a-1)^2} \right|$$

for $a < 0$? **(A)** $3 - 2a$ **(B)** $1 - a$ **(C)** 1 **(D)** $a + 1$ **(E)** 3

Answer: A

Solution 1

We have

$$\begin{aligned} \left| a - 2 - \sqrt{(a-1)^2} \right| &= |a - 2 - |a - 1|| \\ &= |a - 2 - (1 - a)| \\ &= |2a - 3| \\ &= \boxed{\text{(A)} \ 3 - 2a}. \end{aligned}$$

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Solution 2

Assume that $a = -1$. Then, the given expression simplifies to 5:

$$\begin{aligned} \left| a - 2 - \sqrt{(a-1)^2} \right| &= \left| -1 - 2 - \sqrt{(-1-1)^2} \right| \\ &= \left| -1 - 2 - \sqrt{4} \right| \\ &= \left| -1 - 2 - 2 \right| \\ &= 5. \end{aligned}$$

Then, we test each of the answer choices to see which one is equal to 5:

(A) $3 - 2a = 3 - 2 \cdot (-1) = 3 + 2 = 5$. (B) $1 - a = 1 - (-1) = 2 \neq 5$. (C) $1 \neq 5$.
(D) $a + 1 = -1 + 1 = 0 \neq 5$. (E) $3 \neq 5$.

The only answer choice equal to 5 for $a = -1$ is (A) $3 - 2a$.

-MathWizard09

Solution 3

The given function is continuous, so assume that $a = 0$. Then, the given expression simplifies to 3.

We test each of the answer choices and get (A) $3 - 2a$ or (E) 3.

We test $x = -1000$ and get $|-1000 - 2 - \text{positive}| \neq 3 \implies$ (A) $3 - 2a$.

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Solution 4

We know that ,

$$|x| = x, \text{ if } x > 0$$

$$= -x, \text{ if } x < 0$$

$$\text{Now, } \left| a - 2 - \sqrt{(a - 1)^2} \right| = \left| a - 2 - \sqrt{(1 - a)^2} \right|, \text{ as } a < 0$$

$$= |a - 2 - (1 - a)|$$

$$= |2a - 3|$$

$$= -(2a - 3) \text{ as } a < 0$$

$$= 3 - 2a$$

So, the correct choice is option (A) $3 - 2a$.

Solution 5

Because the degree of the expression is 1, and all the solutions are of degree 1 or 0, we can assume the function is linear for $a < 0$ and try values to find what the function is.

We can first say from Solution 1 that:

$$\left| a - 2 - \sqrt{(a - 1)^2} \right| = |a - 2 - |a - 1||$$

Then, we can try values:

a	Plug in values	=	Result
-1	$ -1 - 2 - -1 - 1 $	=	$ -3 - 2 = 5$
-2	$ -2 - 2 - -2 - 1 $	=	$ -4 - 3 = 7$
-3	$ -3 - 2 - -3 - 1 $	=	$ -5 - 4 = 9$

We can see

that this is a linear function. We can find that the slope is $m = \frac{7-5}{-2-(-1)} = \frac{2}{-1} = -2$.

$$y = mx + b$$

$$5 = -2(-1) + b$$

Using the first pair: $5 = 2 + b$

$$3 = b$$

$$y = -2x + 3$$

We take that y equals the expression, and x is a . This gives $-2a + 3$, or $3 - 2a$. Testing to make sure the answer is correct (like that in Solution 2) is a good practice to have and shows that the correct choice must be (A) $3 - 2a$.

7. (AMC 10 10A 2022) The least common multiple of a positive integer n and 18 is 180, and the greatest common divisor of n and 45 is 15. What is the sum of the digits of n ?

(A) 3 (B) 6 (C) 8 (D) 9 (E) 12

Answer: B

Solution 1

Note that

$$\begin{aligned} 18 &= 2 \cdot 3^2, \\ 180 &= 2^2 \cdot 3^2 \cdot 5, \\ 45 &= 3^2 \cdot 5 \\ 15 &= 3 \cdot 5. \end{aligned}$$

Let $n = 2^a \cdot 3^b \cdot 5^c$. It follows that:

- a) From the least common multiple condition, we have

$$\text{lcm}(n, 18) = \text{lcm}(2^a \cdot 3^b \cdot 5^c, 2 \cdot 3^2) = 2^{\max(a,1)} \cdot 3^{\max(b,2)} \cdot 5^{\max(c,0)} = 2^2 \cdot 3^2 \cdot 5,$$

from which $a = 2, b \in \{0, 1, 2\}$, and $c = 1$.

- b) From the greatest common divisor condition, we have

$$\text{gcd}(n, 45) = \text{gcd}(2^2 \cdot 3^b \cdot 5, 3^2 \cdot 5) = 2^{\min(2,0)} \cdot 3^{\min(b,2)} \cdot 5^{\min(1,1)} = 3 \cdot 5,$$

from which $b = 1$.

Together, we conclude that $n = 2^2 \cdot 3 \cdot 5 = 60$. The sum of its digits is $6+0 = \boxed{\text{(B)} 6}$.

Solution 2

The options for $\text{lcm}(x, 18) = 180$ are 20, 60, and 180. The options for $\text{gcd}(y, 45) = 15$ are 15, 30, 60, 75, etc. We see that 60 appears in both lists; therefore, $6+0 = \boxed{\text{(B)} 6}$.

Remark If you ignore or mess up the LCM, and get $n = 15$, you'll still get the correct answer.

8. (AMC 10 10A 2022) A data set consists of 6 (not distinct) positive integers: 1, 7, 5, 2, 5, and X . The average (arithmetic mean) of the 6 numbers equals a value in the data set. What is the sum of all possible values of X ?

(A) 10 (B) 26 (C) 32 (D) 36 (E) 40

Answer: D

Solution (Casework)

First, note that $1 + 7 + 5 + 2 + 5 = 20$. There are 3 possible cases:

Case 1: the mean is 5.

$$X = 5 \cdot 6 - 20 = 10.$$

Case 2: the mean is 7.

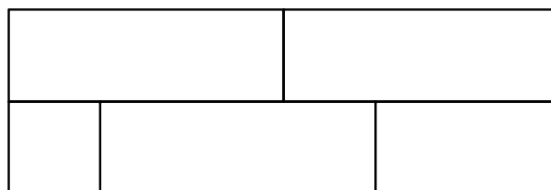
$$X = 7 \cdot 6 - 20 = 22.$$

Case 3: the mean is X .

$$X = \frac{20+X}{6} \Rightarrow X = 4.$$

Therefore, the answer is $10 + 22 + 4 = \boxed{\text{(D)} 36}$.

9. (AMC 10 10A 2022) A rectangle is partitioned into 5 regions as shown. Each region is to be painted a solid color - red, orange, yellow, blue, or green - so that regions that touch are painted different colors, and colors can be used more than once. How many different colorings are possible?



- (A) 120 (B) 270 (C) 360 (D) 540 (E) 720

Answer: D

Solution 1

The top left rectangle can be 5 possible colors. Then the bottom left region can only be 4 possible colors, and the bottom middle can only be 3 colors since it is next to the top left and bottom left. Similarly, we have 3 choices for the top right and 3 choices for the bottom right, which gives us a total of $5 \cdot 4 \cdot 3 \cdot 3 \cdot 3 = \boxed{\text{(D) } 540}$.

Note that the number of choices for a rectangle is $5 - n$ when there are n rectangles adjacent to it. GoldemGem4848

Solution 2 (Casework)

Case 1: All the rectangles are different colors. It would be $5! = 120$ choices.

Case 2: Two rectangles that are the same color. Grouping these two rectangles as one gives us $5 \cdot 4 \cdot 3 \cdot 2 = 120$. But, you need to multiply this number by three because the same-colored rectangles can be chosen at the top left and bottom right, the top right and bottom left, or the bottom right and bottom left, which gives us a grand total of 360.

Case 3: We have two sets of rectangles chosen from these choices (top right bottom left, top left bottom right) that have the same color. However, the choice of the bottom left and bottom right does not work for this case, as the second pair would be chosen from two touching rectangles. Again, grouping the same-colored rectangles gives us $5 \cdot 4 \cdot 3 = 60$.

Therefore, we have $120 + 360 + 60 = \boxed{\text{(D) } 540}$.

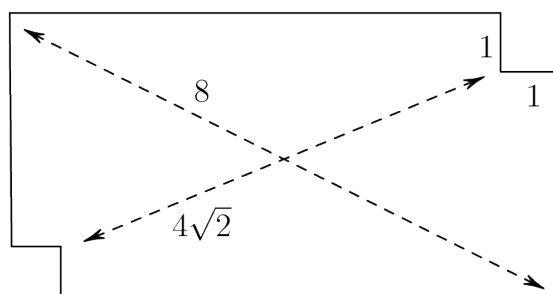
Solution 3 (Different Kind Of Casework)

Let us first choose the color of the bottom-middle rectangle, as it is touching all the other rectangles and has to be a different color than all of them. This rectangle has 5 possibilities. To make things easier for us, let's assume we chose the color red. Now, let's choose the color of the top-left rectangle. This has 4 options: orange, yellow, blue, or green (it can't be the same color as the middle rectangle, which we assumed was red). Let's assume this is orange. Now, we know that the bottom-left rectangle only has 3 choices: yellow, blue, or green. Let's assume this is colored yellow. Now, let's do casework on the color of the bottom right-rectangle. It either can be the same color as the top-left (orange), or different. CASE 1: (It's the same color) We then know that the bottom-right rectangle is also colored orange. We now know that the top-left rectangle only has 3 choices: yellow, blue, or green (it can't be orange or red). The sum of the possibilities of this case is $5 \cdot 4 \cdot 3 \cdot 3$.

CASE 2: (Its a different color) We then know that the bottom-right rectangle has 3 options (yellow, blue, or green, since it can't be red, and it can't be orange since we already covered that in the case 1). Lets assume it is colored yellow. We know have 3 choices for the top-right rectangle (blue or green). The sum of the possibilities of this case is $5 \cdot 4 \cdot 3 \cdot 3 \cdot 2$

We can then sum up the possibilities of our two cases, which is $5 \cdot 4 \cdot 3 \cdot 3 + 5 \cdot 4 \cdot 3 \cdot 3 \cdot 2 = 5 \cdot 4 \cdot 3(3 + 3 \cdot 2) = 60 \cdot 9 = \boxed{\text{(D)} 540}$

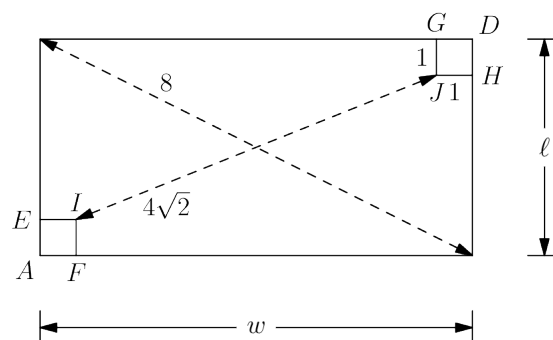
10. (AMC 10 10A 2022) Daniel finds a rectangular index card and measures its diagonal to be 8 centimeters. Daniel then cuts out equal squares of side 1 cm at two opposite corners of the index card and measures the distance between the two closest vertices of these squares to be $4\sqrt{2}$ centimeters, as shown below. What is the area of the original index card?



- (A) 14 (B) $10\sqrt{2}$ (C) 16 (D) $12\sqrt{2}$ (E) 18

Answer: E

Solution 1



Label the bottom left corner of the larger rectangle (without the square cut out) as A and the top right as D . w is the width of the rectangle and ℓ is the length. Now we have vertices E, F, G, H as vertices of the irregular octagon created by cutting out the squares. Let I, J be the two closest vertices formed by the squares. The distance between the two closest vertices of the squares is thus $IJ = 4\sqrt{2}$. Substituting, we get:

$$(IJ)^2 = (w - 2)^2 + (\ell - 2)^2 = (4\sqrt{2})^2 = 32 \implies w^2 + \ell^2 - 4w - 4\ell = 24.$$

Using the fact that the diagonal of the rectangle is 8, we get:

$$w^2 + \ell^2 = 64.$$

Subtracting the first equation from the second equation, we get:

$$4w + 4\ell = 40 \implies w + \ell = 10.$$

Squaring yields:

$$w^2 + 2w\ell + \ell^2 = 100.$$

Subtracting the second equation from this, we get: $2w\ell = 36$, and thus the area of the original rectangle is $w\ell = \boxed{\text{(E)} 18}$.

Edits and Diagram by KingRavi and MRENTHUSIASM

Minor edit by yanes04

11. (AMC 10 10A 2022) Ted mistakenly wrote $2^m \cdot \sqrt{\frac{1}{4096}}$ as $2 \cdot \sqrt[m]{\frac{1}{4096}}$. What is the sum of all real numbers m for which these two expressions have the same value?

(A) 5 (B) 6 (C) 7 (D) 8 (E) 9

Answer: C

Solution 1

We are given that

$$2^m \cdot \sqrt{\frac{1}{4096}} = 2 \cdot \sqrt[m]{\frac{1}{4096}}.$$

Converting everything into powers of 2 and equating exponents, we have

$$\begin{aligned} 2^m \cdot (2^{-12})^{\frac{1}{2}} &= 2 \cdot (2^{-12})^{\frac{1}{m}} \\ 2^{m-6} &= 2^{1-\frac{12}{m}} \\ m-6 &= 1 - \frac{12}{m}. \end{aligned}$$

We multiply both sides by m , then rearrange as

$$m^2 - 7m + 12 = 0.$$

By Vieta's Formulas, the sum of such values of m is $\boxed{\text{(C)} 7}$.

Note that $m = 3$ or $m = 4$ from the quadratic equation above.

Solution 2 (Logarithms)

We can rewrite the equation using fractional exponents and take logarithms of both sides:

$$\log_2(2^m \cdot 4096^{-1/2}) = \log_2(2 \cdot 4096^{-1/m}).$$

We can then use the additive properties of logarithms to split them up:

$$\log_2(2^m) + \log_2(4096^{-1/2}) = \log_2 2 + \log_2(4096^{-1/m}).$$

Using the power rule, the fact that $4096 = 2^{12}$, and bringing the exponents down, we get

$$\begin{aligned} m-6 &= 1 - \frac{12}{m} \\ m + \frac{12}{m} &= 7 \\ m^2 + 12 &= 7m \end{aligned}$$

$$m^2 - 7m + 12 = 0$$

$$(m - 3)(m - 4) = 0,$$

from which $m = 3$ or $m = 4$. Therefore, the answer is $3 + 4 = \boxed{\text{(C)} 7}$.

- youtube.com/indianmathguy

Solution 3

Since third roots are conventionally positive integers, assume m is an integer, so m can only be 1, 2, 3, 4, 6, and 12. $\sqrt[3]{\frac{1}{4096}} = \frac{1}{64}$. Testing out m , we see that only 3 and 4 work. Hence, $3 + 4 = \boxed{\text{(C)} 7}$.

Solution 4 (Beginner)

$2^m \cdot \sqrt[3]{\frac{1}{4096}}$ becomes $\frac{2^m}{64}$, and $2 \cdot \sqrt[m]{\frac{1}{4096}}$ becomes $2 \cdot \left(\frac{1}{2^{12}}\right)^{\frac{1}{m}}$, or $\frac{2}{2^{\frac{12}{m}}}$. Setting these two values equal and cross multiplying results in $128 = 2^m \cdot 2^{\frac{12}{m}}$, or $128 = 2^{m + \frac{12}{m}}$. $128 = 2^7$, and since all the numbers have the same base, the exponents become the new equation, resulting in $7 = m + \frac{12}{m}$. Testing values of m results in two solutions: $m = 3$ and $m = 4$. $3 + 4 = 7$, so the answer is $\boxed{\text{(C)} 7}$.

(note: this is my first time posting a solution, so things may not be formatted correctly. if the writing or LaTeX needs to be fixed, please fix it, thanks :D)

12. (AMC 10 10A 2022) On Halloween, 31 children walked into the principal's office asking for candy. They can be classified into three types: Some always lie; some always tell the truth; and some alternately lie and tell the truth. The alternaters arbitrarily choose their first response, either a lie or the truth, but each subsequent statement has the opposite truth value from its predecessor. The principal asked everyone the same three questions in this order.

"Are you a truth-teller?" The principal gave a piece of candy to each of the 22 children who answered yes.

"Are you an alternater?" The principal gave a piece of candy to each of the 15 children who answered yes.

"Are you a liar?" The principal gave a piece of candy to each of the 9 children who answered yes.

How many pieces of candy in all did the principal give to the children who always tell the truth?

- (A) 7 (B) 12 (C) 21 (D) 27 (E) 31

Answer: A

Solution 1

Note that:

- Truth-tellers would answer yes-no-no to the three questions in this order.
- Liars would answer yes-yes-no to the three questions in this order.
- Alternaters who responded truth-lie-truth would answer no-no-no to the three questions in this order.
- Alternaters who responded lie-truth-lie would answer yes-yes-yes to the three questions in this order.

Suppose that there are T truth-tellers, L liars, and A alternaters who responded lie-truth-lie.

The conditions of the first two questions imply that

$$\begin{aligned} T + L + A &= 22, \\ L + A &= 15. \end{aligned}$$

Subtracting the second equation from the first, we have $T = 22 - 15 = \boxed{(A) 7}$.

Remark

The condition "31 children" and the third question are extraneous. However, they imply $A = 9$ and $L = 6$, from which there are $31 - T - L - A = 9$ alternaters who responded truth-lie-truth.

Solution 2

Consider when the principal asks "Are you a liar?": The truth tellers truthfully say no, and the liars lie and say no. This leaves only alternaters who lie on this question to answer yes. Thus, all 9 children that answered yes are alternaters that falsely answer Questions 1 and 3, and truthfully answer Question 2. The rest of the alternaters, however many there are, have the opposite behavior.

Consider the second question, "Are you an alternater?": The truth tellers again answer no, the liars falsely answer yes, and alternaters that truthfully answer also say yes. From the previous part, we know that 9 alternaters truthfully answer here. Because only liars and 9 alternaters answer yes, we can deduce that there are $15 - 9 = 6$ liars.

Consider the first question, "Are you a truth teller?": Truth tellers say yes, liars also say yes, and alternaters that lie on this question also say yes. From the first part, we know that 9 alternaters lie here. From the previous part, we know that there are 6 liars. Because only the number of truth tellers is unknown here, we can deduce that there are $22 - 9 - 6 = 7$ truth tellers.

The final question is how many pieces of candy did the principal give to truth tellers. Because truth tellers answer yes on only the first question, we know that all 7 of them said yes once, resulting in $\boxed{(A) 7}$ pieces of candy.

Solution 3 (case analysis)

There are T truth-tellers, L liars, and A alternaters. We translate from logic to math via **formalization**.

From statement 3 we see that $A \geq 9$. From statement 2 we have that $A + L = 24$, and from statement 1 we have that $T + L = 13$.

We also know that $T + L + A = 31$. Having a system of three, we see that case 3 is redundant and unnecessary. Therefore, solving

$$T + L + A = 31$$

$$T + L = 13$$

$$A + L = 24$$

the solution $(T, A, L) = (\boxed{7}, 18, 6)$. This is exactly solution 1 and 2 except faster and mathematically intuitive rather than logically intuitive.

Solution 4 (logical deduction via answer choices)

Note that liars will always lie and truth-tellers will always tell the truth. Therefore, when he asks, "Are you a liar", only the alternators who answered "Yes" to question 1 will answer question 3 as "Yes". Therefore, we currently know 9 alternators exist.

Now when he asks, "Are you an alternator", we know the remaining alternators and the liars will both say "Yes". Therefore, we know that the remaining alternators and liars answer yes to that. In addition, when he asks "Are you a truth-teller", all the 9 alternators will say yes, so we have 13 people, some of which are truth-tellers, and some of which are liars. This reduces our answer to either 7 or 12.

Now, say we have 7 truth-tellers. Then, that must mean we have 6 liars. Therefore, we have $9 + 9 = 18$ alternators, for a total of 31, which is satisfactory. On the other hand, if we have 12 truth-tellers, that must mean we have 1 liar, meaning we have $9 + 14 = 23$ alternators. That count is 36, which is a contradiction. The answer is 7.

Remark (Fake Solve) This problem is broken in an interesting way that helps the test-taker. Since the true answers alternate for alternators, you can still get the correct answer if you misinterpret the problem as "alternators alternate their yes/no answers, but not their truth values". Suppose that A alternators answer yes-no-yes, and A' alternators answer no-yes-no. You still get $A = A' = 9$.

In addition, you can even get the correct answer with the misinterpretation "alternators have two types: alternators of the same type give the same arbitrary answers to the three questions, and alternators of different types give the opposite arbitrary answers to the three questions".

It's also notable that the misinterpretation makes the problem harder, so that the solution actually relies on all the information. This suggests that the question-writer may have been mistaken but got lucky.

13. (AMC 10 10A 2022) Let $\triangle ABC$ be a scalene triangle. Point P lies on $\overline{BC}$ so that $\overline{AP}$ bisects $\angle BAC$. The line through B perpendicular to $\overline{AP}$ intersects the line through A parallel to $\overline{BC}$ at point D . Suppose $BP = 2$ and $PC = 3$. What is AD ?
 (A) 8 (B) 9 (C) 10 (D) 11 (E) 12

Answer: C

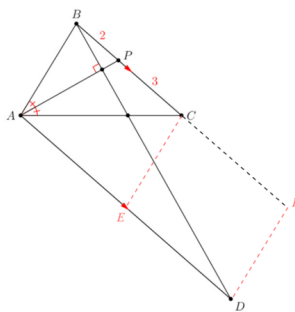
Solution 1 (Angle Bisector Theorem and Similar Triangles)

Suppose that $\overline{BD}$ intersects $\overline{AP}$ and $\overline{AC}$ at X and Y , respectively. By Angle-Side-Angle, we conclude that $\triangle ABX \cong \triangle AXY$.

Let $AB = AY = 2x$. By the Angle Bisector Theorem, we have $AC = 3x$, or $YC = x$.

By alternate interior angles, we get $\angle YAD = \angle YCB$ and $\angle YDA = \angle YBC$. Note that $\triangle ADY \sim \triangle CBY$ by the Angle-Angle Similarity, with the ratio of similitude $\frac{AY}{CY} = 2$. It follows that $AD = 2CB = 2(BP + PC) = \textbf{(C) } 10$.

Solution 2 (No Angle Bisectors!)



We can draw lines from C to a new point E , and extend the line BC to a point F such that the lines FD , CE and AB are parallel.

Notice how we have two parallelograms, but more specifically, these parallelograms are just made up of $2 \times \triangle ABC$. We see that the parallelograms $ABCE$ and $ECFD$ are congruent, and therefore $BC \cong AE \cong CF \cong ED$. $BC = 5$, so $AE = 5$, and because $AE \cong ED$, we have $ED = 5$.

Finally, we have $AE + ED = AD$, so $AD = 5 + 5 = \boxed{(C) 10}$.

And just like that! No angle bisectors needed!

Solution 3 (Auxiliary Lines)

Let the intersection of AC and BD be M , and the intersection of AP and BD be N . Draw a line from M to BC , and label the point of intersection O .

By adding this extra line, we now have many pairs of similar triangles. We have $\triangle BPN \sim \triangle BOM$, with a ratio of 2, so $BO = 4$ and $OC = 1$. We also have $\triangle COM \sim \triangle CAP$ with ratio 3. Additionally, $\triangle BPN \sim \triangle ADN$ (with an unknown ratio). It is also true that $\triangle BAN \cong \triangle MAN$.

Suppose the area of $\triangle COM$ is x . Then, $[\triangle CAP] = 9x$. Because $\triangle CAP$ and $\triangle BAP$ share the same height and have a base ratio of $3 : 2$, $[\triangle BAP] = 6x$. Because $\triangle BOM$ and $\triangle COM$ share the same height and have a base ratio of $4 : 1$, $[\triangle BOM] = 4x$, $[\triangle BPN] = x$, and thus $[OMNP] = 4x - x = 3x$. Thus, $[\triangle MAN] = [\triangle BAN] = 5x$.

Finally, we have $\frac{[\triangle BAN]}{[\triangle BPN]} = \frac{5x}{x} = 5$, and because these triangles share the same height $\frac{AN}{PN} = 5$. Notice that these side lengths are corresponding side lengths of the similar triangles BPN and ADN . This means that $AD = 5 \cdot BP = \boxed{(C) 10}$.

Solution 4 (Slopes)

Let point B be the origin, with C being on the positive x -axis and A being in the first quadrant.

By the Angle Bisector Theorem, $AB : AC = 2 : 3$. Thus, assume that $AB = 4$, and $AC = 6$.

Let the perpendicular from A to BC be AM .

Using Heron's formula,

$$[ABC] = \sqrt{\frac{15}{2} \left(\frac{15}{2} - 4 \right) \left(\frac{15}{2} - 5 \right) \left(\frac{15}{2} - 6 \right)} = \frac{15}{4} \sqrt{7}.$$

Hence,

$$AM = \frac{\frac{15}{4} \sqrt{7}}{\frac{5}{2}} = \frac{3}{2} \sqrt{7}.$$

Next, we have

$$BM^2 + AM^2 = AB^2$$

$$\therefore BM = \sqrt{16 - \frac{63}{4}} = \sqrt{\frac{1}{4}} = \frac{1}{2} \text{ and } PM = \frac{3}{2}.$$

The slope of line AP is thus

$$\frac{-\frac{3}{2}\sqrt{7}}{\frac{3}{2}} = -\sqrt{7}.$$

Therefore, since the slopes of perpendicular lines have a product of -1 , the slope of line BD is $\frac{1}{\sqrt{7}}$. This means that we can solve for the coordinates of D :

$$y = \frac{3}{2}\sqrt{7}$$

$$y = \frac{1}{\sqrt{7}}x$$

$$\frac{1}{\sqrt{7}}x = \frac{3}{2}\sqrt{7}$$

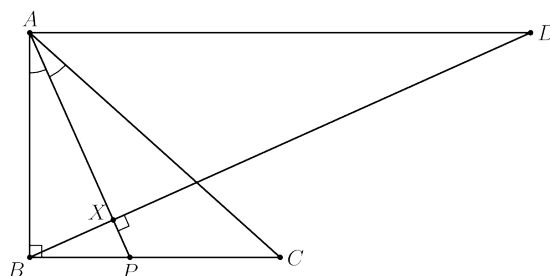
$$x = \frac{7 \cdot 3}{2} = \frac{21}{2}$$

$$D = \left(\frac{21}{2}, \frac{3}{2}\sqrt{7}\right).$$

We also know that the coordinates of A are $\left(\frac{1}{2}, \frac{3}{2}\sqrt{7}\right)$, because $BM = \frac{1}{2}$ and $AM = \frac{3}{2}\sqrt{7}$.

Since the y -coordinates of A and D are the same, and their x -coordinates differ by 10, the distance between them is 10. Our answer is **(C) 10**.

Solution 5 (Assumption)



Since there is only one possible value of AD , we assume $\angle B = 90^\circ$. By the angle bisector theorem, $\frac{AB}{AC} = \frac{2}{3}$, so $AB = 2\sqrt{5}$ and $AC = 3\sqrt{5}$. Now observe that $\angle BAD = 90^\circ$. Let the intersection of BD and AP be X . Then $\angle ABD = 90^\circ - \angle BAX = \angle APB$. Consequently,

$$\triangle DAB \sim \triangle ABP$$

and therefore $\frac{DA}{AB} = \frac{AB}{BP}$, so $AD = \mathbf{(C) 10}$, and we're done!

14. (AMC 10 10A 2022) How many ways are there to split the integers 1 through 14 into 7 pairs such that in each pair, the greater number is at least 2 times the lesser number?

(A) 108 (B) 120 (C) 126 (D) 132 (E) 144

Answer: E

Solution 1 (Multiplication Principle)

Clearly, the integers from 8 through 14 must be in different pairs, so are the integers from 1 through 7. Note that 7 must pair with 14.

We pair the numbers 1, 2, 3, 4, 5, 6 with the numbers 8, 9, 10, 11, 12, 13 systematically:

- 6 can pair with either 12 or 13.
- 5 can pair with any of the three remaining numbers from 10, 11, 12, 13.
- 1, 2, 3, 4 can pair with the other four remaining numbers from 8, 9, 10, 11, 12, 13 without restrictions.

Together, the answer is $2 \cdot 3 \cdot 4! = \boxed{\text{(E) } 144}$.

Solution 2 (Multiplication Principle)

As said in Solution 1, clearly, the integers from 8 through 14 must be in different pairs.

We know that 8 or 9 can pair with any integer from 1 to 4, 10 or 11 can pair with any integer from 1 to 5, and 12 or 13 can pair with any integer from 1 to 6. Thus, 8 will have 4 choices to pair with, 9 will then have 3 choices to pair with (9 cannot pair with the same number as the one 8 pairs with). 10 cannot pair with the numbers 8 and 9 has paired with but can also now pair with 5, so there are 3 choices. 11 cannot pair with 8's, 9's, or 10's paired numbers, so there will be 2 choices for 11. 12 can pair with an integer from 1 to 5 that hasn't been paired with already, or it can pair with 6. 13 will only have one choice left, and 7 must pair with 14.

So, the answer is $4 \cdot 3 \cdot 3 \cdot 2 \cdot 2 \cdot 1 \cdot 1 = \boxed{\text{(E) } 144}$.

Solution 3 (Generalization)

The integers $x \in \{8, \dots, 14\}$ must each be the larger elements of a distinct pair.

Assign partners in decreasing order for $x \in \{7, \dots, 1\}$:

Note that 7 must pair with 14: **1 choice**.

For $5 \leq x \leq 7$, the choices are $\{2x, \dots, 14\} - \{\text{previous choices}\}$. As x decreases by 1, The minuend increases by 2 elements, and the subtrahend increases by 1 element, so the difference increases by 1, yielding **3! combined choices**.

After assigning a partner to 5, there are no invalid pairings for yet-unpaired numbers, so there are **4! ways** to choose partners for $\{1, 2, 3, 4\}$.

The answer is $3! \cdot 4! = \boxed{\text{(E) } 144}$.

In general, for $1, \dots, 2n$, the same logic yields answer: $\left\lfloor \frac{n}{2} \right\rfloor! \cdot \left\lceil \frac{n}{2} \right\rceil!$

15. (AMC 10 10A 2022) Quadrilateral $ABCD$ with side lengths $AB = 7, BC = 24, CD = 20, DA = 15$ is inscribed in a circle. The area interior to the circle but exterior to the quadrilateral can be written in the form $\frac{a\pi - b}{c}$, where a, b , and c

are positive integers such that a and c have no common prime factor. What is $a + b + c$? (A) 260 (B) 855 (C) 1235 (D) 1565 (E) 1997

Answer: D

Solution 1 (Inscribed Angle Theorem)

Opposite angles of every cyclic quadrilateral are supplementary, so

16. (AMC 10 10A 2022) The roots of the polynomial $10x^3 - 39x^2 + 29x - 6$ are the height, length, and width of a rectangular box (right rectangular prism). A new rectangular box is formed by lengthening each edge of the original box by 2 units. What is the volume of the new box?

(A) $\frac{24}{5}$ (B) $\frac{42}{5}$ (C) $\frac{81}{5}$ (D) 30 (E) 48

Answer: D

Solution 1 (Vieta's Formulas)

Let a, b, c be the three roots of the polynomial. The lengthened prism's volume is

$$V = (a+2)(b+2)(c+2) = abc + 2ac + 2ab + 2bc + 4a + 4b + 4c + 8 = abc + 2(ab + ac + bc) + 4(a + b + c) + 8.$$

By Vieta's formulas, we know that a cubic polynomial $Ax^3 + Bx^2 + Cx + D$ with roots a, b, c satisfies:

$$\begin{aligned} a + b + c &= -\frac{B}{A} = \frac{39}{10}, \\ ab + ac + bc &= \frac{C}{A} = \frac{29}{10}, \\ abc &= -\frac{D}{A} = \frac{6}{10}. \end{aligned}$$

We can substitute these into the expression, obtaining

$$V = \frac{6}{10} + 2\left(\frac{29}{10}\right) + 4\left(\frac{39}{10}\right) + 8 = \boxed{\text{(D) } 30}.$$

- phuang1024

Solution 2 (Guessing Roots)

From the answer choices, we can assume the roots are rational numbers, and therefore this polynomial should be easily factorable. The coefficients of x must multiply to 10, so these coefficients must be 5, 2, 1 or 10, 1, in some order. We can try one at a time, and therefore write the factored form as follows:

$$(5x - p)(2x - q)(x - r).$$

Note that p, q, r have to multiply to 6, so they must be either 3, 2, 1 or 6, 1, 1 in some order. Again, we can try one at a time in different positions and see if they multiply correctly. We try $(5x - 2)(2x - 1)(x - 3)$ and multiply the x -terms, and sure enough they add up to 29. You can try to add up the x^2 terms and they add up to -39 . Therefore the roots are $\frac{2}{5}$, $\frac{1}{2}$ and 3. Now if you add 2 to each root, you get the volume is $\frac{12}{5} \cdot \frac{5}{2} \cdot 5 = 6 \cdot 5 = 30 = \boxed{\text{(D) } 30}$.

Solution 3 (Rational Root Theorem Bash)

We can find the roots of the cubic using the Rational Root Theorem, which tells us that the rational roots of the cubic must be in the form $\frac{p}{q}$, where p is a factor of

the constant (-6) and q is a factor of the leading coefficient (10) . Therefore, p is $\pm(1, 2, 3, 6)$ and q is $\pm(1, 2, 5, 10)$.

Doing Synthetic Division, we find that 3 is a root of the cubic:

$$\begin{array}{r|rrrr} 3 & 10 & -39 & 29 & -6 \\ & & 30 & -27 & 6 \\ \hline & 10 & -9 & 2 & 0 \end{array}$$

Then, we have a quadratic $10x^2 - 9x + 2$. Using the Quadratic Formula, we can find the other two roots:

$$x = \frac{9 \pm \sqrt{(-9)^2 - 4(10)(2)}}{2 \cdot 10},$$

which simplifies to $x = \frac{1}{2}, \frac{2}{5}$.

To find the new volume, we add 2 to each of the roots we found:

$$(3 + 2) \cdot \left(\frac{1}{2} + 2\right) \cdot \left(\frac{2}{5} + 2\right).$$

Simplifying, we find that the new volume is (D) 30.

-MathWizard09

Solution 4

Let $P(x) = 10x^3 - 39x^2 + 29x - 6$, and let a, b, c be the roots of $P(x)$. The roots of $P(x - 2)$ are then $a + 2, b + 2, c + 2$, so the product of the roots of $P(x - 2)$ is the area of the desired rectangular prism.

$P(x - 2)$ has leading coefficient 10 and constant term $P(0 - 2) = P(-2) = 10(-2)^3 - 39(-2)^2 + 29(-2) - 6 = -300$.

Thus, by Vieta's Formulas, the product of the roots of $P(x - 2)$ is $\frac{-(-300)}{10} =$
(D) 30.

-Orange_Quail_9

Solution 5

Let $P(x) = 10x^3 - 39x^2 + 29x - 6$. This can be factored as $P(x) = 10(x - a)(x - b)(x - c)$, where a, b , and c are the roots of $P(x)$. We want $V = (a + 2)(b + 2)(c + 2)$.

"Luckily" $P(-2) = 10(-2 - a)(-2 - b)(-2 - c) = -10V$. $P(-2) = -300$, giving $V =$
(D) 30.

-Oxymoronic15

(It's not just lucky. If $P(x)$ has roots $x \in \{r_i\}$, $Q(x) = P(x - 2)$ has roots $y \in \{r_i + 2\}$. By Vieta, the product V of the roots is the negation of the constant term divided by the leading coefficient 10, which is $-Q(0)/10$, which is $-P(0 - 2)/10$. -oinava)

Solution 6 (Very Nice Vieta Solve)

If we let the roots of the cubic polynomial be p, q, r , then we can write what we want in terms of p, q, r . So, we have

$$(p+2)(q+2)(r+2) = pqr + 2pq + 2qr + 4q + 2pr + 4p + 4r + 8 = pqr + 2(pq + qr + pr) + 4(q + p + r) + 8.$$

Now, we can just use Vieta's to finish up from here! We have $pqr = -(-\frac{6}{10}) = \frac{3}{5}$.
 Now, we can find the sum of the pairwise roots which is just $pq + qr + pr = \frac{29}{10}$
 and then multiply by 2 which gives $\frac{29}{5}$. Looking good so far! Now, we need to find
 the sum of the roots which is just $p + q + r = \frac{39}{10}$ and then multiply by 4 which
 gives $\frac{78}{5}$. Almost there! Now, we just add 8 to that sum, and then sum it all up to
 get (D) 30

-jb2015007

Note: This is the same as Solution 1.

17. (AMC 10 10A 2022) How many three-digit positive integers $\underline{a} \underline{b} \underline{c}$ are there whose nonzero digits a, b , and c satisfy

$$0.\overline{\underline{a} \underline{b} \underline{c}} = \frac{1}{3}(0.\overline{a} + 0.\overline{b} + 0.\overline{c})?$$

(The bar indicates repetition, thus $0.\overline{\underline{a} \underline{b} \underline{c}}$ is the infinite repeating decimal $0.\underline{a} \underline{b} \underline{c} \underline{a} \underline{b} \underline{c} \dots$)

(A) 9 (B) 10 (C) 11 (D) 13 (E) 14

Answer: D

Solution

We rewrite the given equation, then rearrange:

$$\begin{aligned} \frac{100a + 10b + c}{999} &= \frac{1}{3} \left(\frac{a}{9} + \frac{b}{9} + \frac{c}{9} \right) \\ 100a + 10b + c &= 37a + 37b + 37c \\ 63a &= 27b + 36c \\ 7a &= 3b + 4c. \end{aligned}$$

Now, this problem is equivalent to counting the ordered triples (a, b, c) that satisfies the equation.

Clearly, the 9 ordered triples $(a, b, c) = (1, 1, 1), (2, 2, 2), \dots, (9, 9, 9)$ are solutions to this equation.

The expression $3b + 4c$ has the same value when:

- b increases by 4 as c decreases by 3.
- b decreases by 4 as c increases by 3.

We find 4 more solutions from the 9 solutions above: $(a, b, c) = (4, 8, 1), (5, 1, 8), (5, 9, 2), (6, 2, 9)$. Note that all solutions are symmetric about $(a, b, c) = (5, 5, 5)$.

Together, we have $9 + 4 =$ (D) 13 ordered triples (a, b, c) .

Remark One way to solve the Diophantine Equation, $7a = 3b + 4c$ is by taking (mod 7), from which the equation becomes $0 \equiv 3b - 3c \pmod{7} \implies b \equiv c \pmod{7}$ so either $b = c$ or WLOG $b < c, b + 7 = c$.

18. (AMC 10 10A 2022) Let T_k be the transformation of the coordinate plane that first rotates the plane k degrees counterclockwise around the origin and then reflects the

plane across the y -axis. What is the least positive integer n such that performing the sequence of transformations $T_1, T_2, T_3, \dots, T_n$ returns the point $(1, 0)$ back to itself?

- (A) 359 (B) 360 (C) 719 (D) 720 (E) 721

Answer: A

Solution 1 (Polar Coords)

Let $P = (r, \theta)$ be a point in polar coordinates, where θ is in degrees.

Rotating P by k° counterclockwise around the origin gives the transformation $(r, \theta) \rightarrow (r, \theta + k^\circ)$. Reflecting P across the y -axis gives the transformation $(r, \theta) \rightarrow (r, 180^\circ - \theta)$. Note that

$$\begin{aligned} T_k(P) &= (r, 180^\circ - \theta - k^\circ), \\ T_{k+1}(T_k(P)) &= (r, \theta - 1^\circ). \end{aligned}$$

We start with $(1, 0^\circ)$ in polar coordinates. For the sequence of transformations $T_1, T_2, T_3, \dots, T_k$, it follows that

- After T_1 , we have $(1, 179^\circ)$.
- After T_2 , we have $(1, -1^\circ)$.
- After T_3 , we have $(1, 178^\circ)$.
- After T_4 , we have $(1, -2^\circ)$.
- After T_5 , we have $(1, 177^\circ)$.
- After T_6 , we have $(1, -3^\circ)$.
- ...
- After T_{2k-1} , we have $(1, 180^\circ - k^\circ)$.
- After T_{2k} , we have $(1, -k^\circ)$.

The least such positive integer k is 180. Therefore, the least such positive integer n is $2k - 1 = \boxed{\text{(A) } 359}$.

Solution 2

Note that since we're reflecting across the y -axis, if the point ever makes it to $(-1, 0)$ then it will flip back to the original point. Note that after T_1 the point will be 1 degree clockwise from the negative x -axis. Applying T_2 will rotate it to be 1 degree counterclockwise from the negative x -axis, and then flip it so that it is 1 degree clockwise from the positive x -axis. Therefore, after every 2 transformations, the point rotates 1 degree clockwise. To rotate it so that it will rotate 179 degrees clockwise will require $179 \cdot 2 = 358$ transformations. Then finally on the last transformation, it will rotate on to $(-1, 0)$ and then flip back to its original position. Therefore, the answer is $358 + 1 = 359 = \boxed{\text{(A) } 359}$.

Solution 3

So, you don't know polar coordinates (Solution 1) nor came up with Solution 2's logic, then, here we go!

The easy solution is to notice that, without reflections and taking $k = 1$, there are 360 turns that need to be made.

Now take $k = 2$. We see that there are 180 rotations, so the number of rotations is just $360/k$.

Now include the reflections over the y axis.

Say that each time we reflect a point over the y axis, we draw a line.

Each time we reflect over the y axis, we see more and more lines that generate. For $k = 1$, there are quite obviously 360 of these lines.

However, one can see that this works for every single case, because when we reflect it over the y axis and make a k° rotation, we fill up all the missed gaps until we reach the end of a sequence - I prove this on section 1.1.

But wait! We overcounted by one line! Notice how I said there were 360 lines, but the line $x = 1$ intersects $y = 0$ is already given to you, and that is when $T_k = 0$, which is the point $(1, 0)$! So our final answer is $360 - 1 = \boxed{\text{(A) 359}}$.

Proof A way you can prove such an assumption is by taking ideas like addition. We know that eventually there exists a number N such that $l \times N = m$, where m has a units digit of l . Utilizing the same approach, we can find that this actually works.

Solution 4

In degrees:

Starting with $n = 0$, the sequence goes $0 \rightarrow 179 \rightarrow 359 \rightarrow 178 \rightarrow 358 \rightarrow 177 \rightarrow 357 \rightarrow \dots$.

We see that it takes 2 steps to downgrade the point by 1° . Since the 1st point in the sequence is 179, the answer is $1 + 2(179) = \boxed{\text{(A) 359}}$.

Solution 5 (Simple)

We can consider the rotations and reflections separately. For the rotations, each rotation turns it by the next natural number. Thus the total number of degrees turned would be a triangle number. We test the smallest number, 359 first, and we get that it turns $\frac{(1+359)359}{2} = 180n$, where n is an integer. Thus, the point would be rotated to $(-1, 0)$. We may be tempted to dismiss this option but we haven't considered the reflections. Each reflection acts as a 180° rotation, so every two reflections cancel. However, 359 is odd so we have to reflect $(-1, 0)$, taking us to $(1, 0)$, which is what we want. Thus we get $\boxed{\text{(A) 359}}$.

Solution 6 (Complex)

Rotations and reflections in 2D are very nice to describe using complex numbers in polar form. Reflections and rotations also don't affect the length (modulus or absolute value) of a complex number. This motivates us to set $z = \exp(i\theta)$, starting with $z = 1 + 0i = \exp(i0)$. Then, we can describe rotations and reflections about the y -axis using these formulas:

$$\text{Rotation}(\exp(i\theta), k) : \exp(i\theta) \mapsto \exp(i(\theta + k))$$

$$\text{Reflection}(\exp(i\theta)) : \exp(i\theta) \mapsto \exp(i(180^\circ - \theta)) = -\exp(i(-\theta))$$

If we apply two successive iterations, we see a simplification:

$$T_k(\exp(i\theta)) = -\exp(-i(\theta + k))$$

$$\begin{aligned} T_{-k+1}(T_{-k}(\exp(i\theta))) &= T_{-k+1}(-\exp(i(-\theta - k))) \\ &= -\exp(-i(-\theta - k + (k + 1))) \\ &= \exp(i(\theta - 1)) \end{aligned}$$

We also can calculate $T_1(\exp i0) = \exp 179^\circ$. Thus, the point $(1, 0)$ gets sent back to when all double iterations after 1 cancel 179° . $179 - \left(\frac{n-1}{2}\right) = 0$ so $n = 1 + 2 \cdot 179 = \boxed{\text{(A) } 359}$.

Solution 7 (Matrices)

Similar to the above solution, one can use rotation matrices to compute it.

(i.e. $\begin{bmatrix} \cos(\theta) & -\sin(\theta) \\ \sin(\theta) & \cos(\theta) \end{bmatrix}$)

For the reflection, use the reflection matrix (i.e. $\begin{bmatrix} -1 & 0 \\ 0 & 1 \end{bmatrix}$)

Then proceed like Solution 5.

Solution 8 (Very simple and quick)

Let's test a few small cases for T_k . When $k = 1$, the point will move $(1 + 180)^\circ$. When $k = 2$, the point will move $(2 + 180)^\circ$. We can see that T_k will result in the point moving $(k + 180)^\circ$ from its previous location. Since we want the point to return to itself, the total number of degrees it has rotated must be a multiple of 360° . This means $T_1 + T_2 + T_3 + \dots T_k \equiv 0 \pmod{360}$. Using the formula for the sum of the first k integers, this can be expressed as

$$180k + \frac{k(k+1)}{2} \equiv 0 \pmod{360}$$

Testing the answer choices gives $\boxed{\text{(A) } 359}$.

19. (AMC 10 10A 2022) Define L_n as the least common multiple of all the integers from 1 to n inclusive. There is a unique integer h such that

$$\frac{1}{1} + \frac{1}{2} + \frac{1}{3} + \dots + \frac{1}{17} = \frac{h}{L_{17}}$$

What is the remainder when h is divided by 17?

- (A) 1 (B) 3 (C) 5 (D) 7 (E) 9

Answer: C

Solution 1

Notice that L_{17} contains the highest power of every prime below 17 since higher primes cannot divide L_{17} . Thus, $L_{17} = 16 \cdot 9 \cdot 5 \cdot 7 \cdot 11 \cdot 13 \cdot 17$.

When writing the sum under a common fraction, we multiply the denominators by L_{17} divided by each denominator. However, since L_{17} is a multiple of 17, all terms will be a multiple of 17 until we divide out 17, and the only term that will do this is $\frac{1}{17}$. Thus, the remainder of all other terms when divided by 17 will be 0, so the problem is essentially asking us what the remainder of $\frac{L_{17}}{17} = L_{16}$ divided by 17 is. This is equivalent to finding the remainder of $16 \cdot 9 \cdot 5 \cdot 7 \cdot 11 \cdot 13$ divided by 17.

We use modular arithmetic to simplify our answer:

This is congruent to $-1 \cdot 9 \cdot 5 \cdot 7 \cdot 11 \cdot 13 \pmod{17}$.

Evaluating, we get:

$$\begin{aligned} (-1) \cdot 9 \cdot 35 \cdot 11 \cdot 13 &\equiv (-1) \cdot 9 \cdot 1 \cdot 11 \cdot 13 \pmod{17} \\ &\equiv 9 \cdot 11 \cdot (-13) \pmod{17} \\ &\equiv 9 \cdot 11 \cdot 4 \pmod{17} \\ &\equiv 2 \cdot 11 \pmod{17} \\ &\equiv 5 \pmod{17} \end{aligned}$$

Therefore the remainder is $\boxed{\text{(C) } 5}$.

Solution 2

As in solution 1, we express the LHS as a sum under one common denominator. We note that

$$\frac{1}{1} + \frac{1}{2} + \cdots + \frac{1}{17} = \frac{17!}{17!} + \frac{17!}{2 \cdot 17!} + \frac{17!}{3 \cdot 17!} + \cdots + \frac{17!}{17!}$$

Now, we have $h = L_{17} \left(\frac{\frac{17!}{1} + \frac{17!}{2} + \frac{17!}{3} + \cdots + \frac{17!}{17}}{17!} \right)$. We'd like to find $h \pmod{17}$, so we can evaluate our expression $\pmod{17}$. Since $\frac{17!}{1}, \frac{17!}{2}, \dots, \frac{17!}{16}$ don't have a factor of 17 in their denominators, and since L_{17} is a multiple of 17, multiplying each of those terms and adding them will get a multiple of 17. $\pmod{17}$, that result is 0. Thus, we only need to consider $L_{17} \cdot \frac{17!}{17} = \frac{L_{17}}{17} \pmod{17}$. Proceed with solution 1 to get $\boxed{\text{(C) } 5}$.

Solution 3

Using Wolstenholmes' Theorem, we can rewrite $1 + \frac{1}{2} \cdots + \frac{1}{16}$ as $\frac{17^2 n}{(17-1)!} = \frac{17^2 n}{16!}$ (for some $n \in \mathbb{Z}$). Adding the $\frac{1}{17}$ to $\frac{17^2 n}{16!}$, we get $\frac{17^3 n + 16!}{17!}$.

Now we have $\frac{17^3 n + 16!}{17!} = \frac{h}{L_{17}}$ and we want $h \pmod{17}$. We find that $\frac{L_{17}(17^3 n + 16!)}{17!} = \frac{L_{16}(17^3 n + 16!)}{16!} = h$. Taking $\pmod{17}$ and multiplying, we get $L_{16}(17^3 n + 16!) \equiv 16! \cdot h \pmod{17}$.

Applying Wilson's Theorem on $16!$ and reducing, we simplify the congruence to $L_{16}(0 - 1) \equiv -L_{16} \equiv -h \pmod{17}$. Now we proceed with Solution 1 and find that $L_{16} \equiv 5 \pmod{17}$, so our answer is $\boxed{\text{(C) } 5}$.

20. (AMC 10 10A 2022) A four-term sequence is formed by adding each term of a four-term arithmetic sequence of positive integers to the corresponding term of a four-term geometric sequence of positive integers. The first three terms of the resulting four-term sequence are 57, 60, and 91. What is the fourth term of this sequence?

(A) 190 (B) 194 (C) 198 (D) 202 (E) 206

Answer: E

Solution 1 (Algebra)

Let the arithmetic sequence be $a, a + d, a + 2d, a + 3d$ and the geometric sequence be b, br, br^2, br^3 , where a and b are positive integers, d is an integer, and r is a positive number. We are given that

$$a + b = 57,$$

$$\begin{aligned}a + d + br &= 60, \\a + 2d + br^2 &= 91,\end{aligned}$$

and we wish to find $a + 3d + br^3$.

Subtracting the first equation from the second and the second equation from the third, we get

$$\begin{aligned}d + b(r - 1) &= 3, \\d + br(r - 1) &= 31.\end{aligned}$$

Subtracting these equations, we get

$$b(r - 1)^2 = 28.$$

From $d + b(r - 1) = 3$, we know that $b(r - 1)$ is an integer. Since $b(r - 1)^2 = 28$, we conclude that

$$b^2(r - 1)^2 = (b(r - 1))^2 = 28b$$

is a perfect square. So, $b = 7m^2$ for some positive integer m . From $a + b = 57$, we get $b \leq 56$. Therefore, the only possibilities are $b = 28, 7$, from which $(r - 1)^2 = 1, 4$, respectively. We proceed with casework:

- If $(r - 1)^2 = 1$, then $r = 2, b = 28, a = 29$, and $d = -25$. The arithmetic sequence is $29, 4, -21, -46$, which contradicts the positive-integer condition.
- If $(r - 1)^2 = 4$, then $r = 3, b = 7, a = 50$, and $d = -11$. The arithmetic sequence is $50, 39, 28, 17$, and the geometric sequence is $7, 21, 63, 189$. This case is valid.

Therefore, the answer is $a + 3d + br^3 = 17 + 189 = \boxed{\text{(E)} 206}$.

Solution 2 (Answer Choices)

Start similarly to Solution 1 and deduce the three equations

$$\begin{aligned}a + b &= 57, \\a + d + br &= 60, \\a + 2d + br^2 &= 91.\end{aligned}$$

Then, add the last two equations and take away the first equation to get $a + 3d + br^2 + br - b = 94$. We can solve for this in terms of what we want: $a + 3d = -br^2 - br + b + 94$.

We're looking for $a + 3d + br^3$. We can substitute our value of $a + 3d$ in here to get

$$br^3 - br^2 - br + b + 94 = b(r + 1)(r - 1)(r - 1) + 94.$$

Since our sequence only has positive integers we can now check by the answer choices. For each answer choice, we can subtract 94 and factor it to see if it has a perfect square factor and at least one other factor and those should differ by 2.

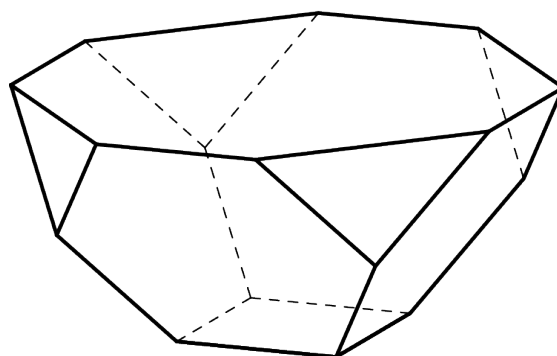
- (A) $190 - 94 = 96 = 2^5 \cdot 3$,
- (B) $194 - 94 = 100 = 2^2 \cdot 5^2$,
- (C) $198 - 94 = 104 = 2^3 \cdot 13$,
- (D) $202 - 94 = 108 = 2^2 \cdot 3^3$,

(E) $206 - 94 = 112 = 2^4 \cdot 7$.

From this, the only possible answer choices are (A) and (E), where $r = 3$. To solve for b , we look back to the given equations above.

We are looking for $a + 3d + 27b$. If (A) were the answer, then we know that a would have to be divisible by 3 and b would equal 6. Looking at our second equation, if this were the case, then d would also have to be divisible by 3. However, this contradicts the third equation, as all variables are divisible by 3, but their sum isn't. So, (E) 206 is our answer.

21. (AMC 10 10A 2022) A bowl is formed by attaching four regular hexagons of side 1 to a square of side 1. The edges of the adjacent hexagons coincide, as shown in the figure. What is the area of the octagon obtained by joining the top eight vertices of the four hexagons, situated on the rim of the bowl?

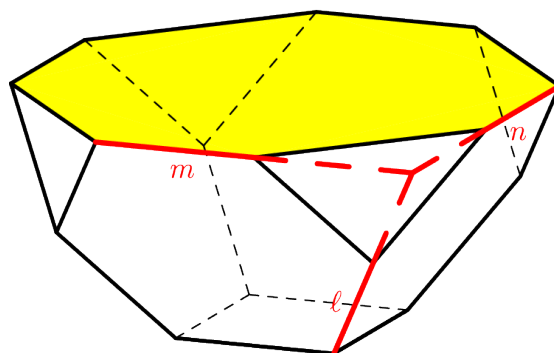


- (A) 6 (B) 7 (C) $5 + 2\sqrt{2}$ (D) 8 (E) 9

Answer: B

Solution 1

We extend line segments ℓ , m , and n to their point of concurrency, as shown below:

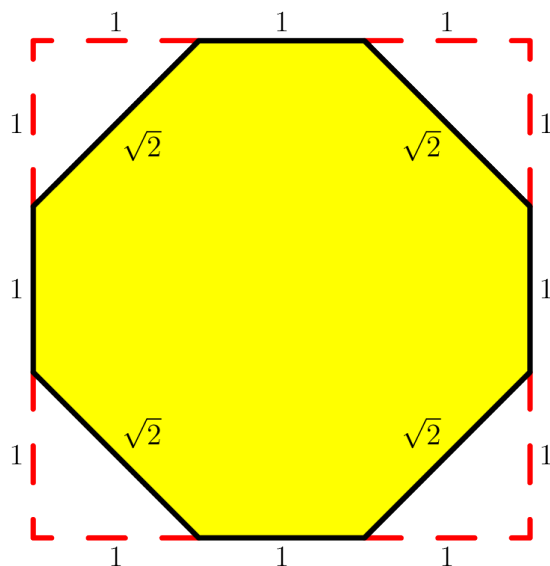


We claim that lines ℓ , m , and n are concurrent: In the lateral faces of the bowl, we know that lines ℓ and m must intersect, and lines ℓ and n must intersect. In addition, line ℓ intersects the top plane of the bowl at exactly one point. Since lines m and n are both in the top plane of the bowl, we conclude that lines ℓ , m , and n are concurrent.

In the lateral faces of the bowl, the dashed red line segments create equilateral triangles. So, the dashed red line segments all have length 1. In the top plane of the

bowt, we know that $\overleftrightarrow{m} \perp \overleftrightarrow{n}$. So, the dashed red line segments create an isosceles triangle with leg-length 1.

Note that octagon has four pairs of parallel sides, and the successive side-lengths are $1, \sqrt{2}, 1, \sqrt{2}, 1, \sqrt{2}, 1, \sqrt{2}$, as shown below:

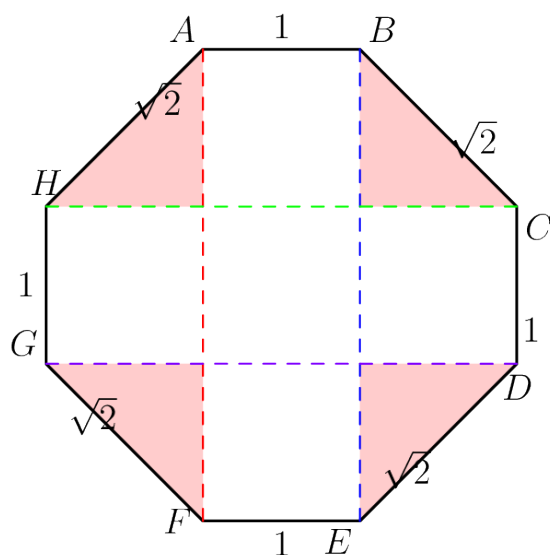


The area of the octagon is

$$3^2 - 4 \cdot \left(\frac{1}{2} \cdot 1^2 \right) = \boxed{\text{(B) } 7}.$$

Solution 1.1

You could also think of this problem as folding the four triangles connecting the hexagon faces to cover parts of the octagon:



So the octagon is made up of five unit squares and four $45 - 45 - 90$ triangles with legs of length 1. So the area is $\boxed{7}$.

Solution 2 (Symmetry and Similarity)

(This is an alternate way of analyzing the red extension line segments drawn in Solution 1.)

The perimeter of the square bottom of the bowl is 4. Halfway up the bowl, the boundary is still a square, with perimeter 4 times the hexagon circumradius, aka $4 \times 2 = 8$ times the hexagon (also square) side length (1), an increase of 4.

Extending the bottom half of the bowl to twice its height (full height of the bowl) would increase the perimeter by the same amount again, forming a square with perimeter $4 + (8 - 4) \times 2 = 12$. Thus the top octagon is cut out of a square of side length $12 \div 4 = 3$ and thus area 9.

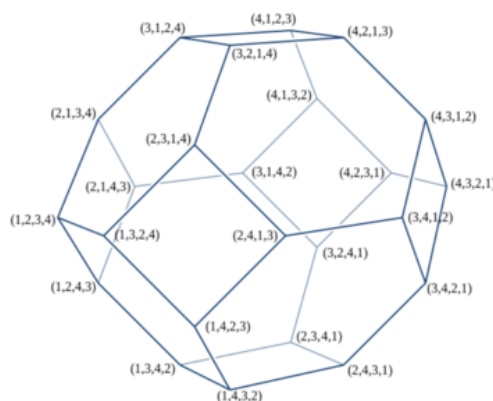
The difference between the above-constructed square and the octagon is four right triangles, and (by rotational and reflection symmetry), each is isosceles with equal-length perpendicular bases of length $(3 - 1)/2 = 1$, and thus having area $\frac{1}{2}$. Therefore the area of the octagon is $9 - (4 \times 1/2) = \boxed{\text{(B) } 7}$.

-oinava

Solution 3

Note that the octagon is equiangular by symmetry, but it is not equilateral. 4 of its sides are shared with the hexagon's sides, so each of those sides have side length 1. However, the other 4 sides are touching the triangles, so we wish to find the length of these sides.

Notice that when two adjacent hexagons meet at a side, their planes make the same dihedral angle at the bottom-most point of intersection and at the top-most point of intersection by symmetry. Therefore, the triangle that is wedged between the two hexagons has the same angle as the square at the bottom wedged between the hexagons. Thus, the triangle is a $45 - 45 - 90$ isosceles triangle. This conclusion can also be reached by cutting the bottom square across a diagonal and noticing that each resulting triangle is congruent to each triangle wedged between the hexagons by symmetry. Furthermore, notice that if you take a copy of this bowl and invert it and place it on top of this bowl, you will get a polyhedron with faces of hexagons and squares, a truncated octahedron, and therefore this triangle has a 90° -angle:



Now that we have come to this conclusion, by simple Pythagorean theorem, we have that the other 4 sides of the octagon are $\sqrt{2}$.

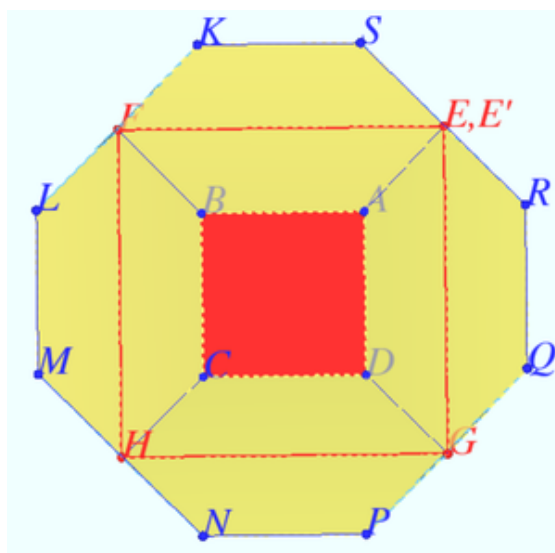
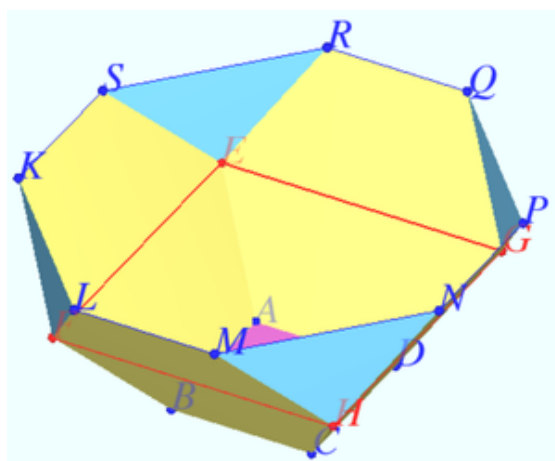
We can draw a square around the octagon so that the area of the octagon is the area of the square minus each corner triangle. The hypotenuse of these corner triangles are 1 and they are $45 - 45 - 90$ triangles because the octagon is equiangular, so

each has dimensions $\frac{\sqrt{2}}{2}, \frac{\sqrt{2}}{2}, 1$. The side length of the square is $\sqrt{2}$ for the larger sides of the octagon, and adding 2 of $\frac{\sqrt{2}}{2}$ for each width of the triangle. Therefore, the area of the square is:

$$\left(\sqrt{2} + 2 \cdot \frac{\sqrt{2}}{2}\right)^2 \implies (2\sqrt{2})^2 = 8$$

The area of each triangle is $\frac{1}{2} \cdot \frac{\sqrt{2}}{2} \cdot \frac{\sqrt{2}}{2} = \frac{1}{4}$ and there are 4 of them, so we subtract 1 from the area of the square. The area of the octagon is thus **(B) 7**.

Solution 4 (Simple)



Denote a square $ABCD$, $AB = 1$, π is the plane ABC , regular hexagons $ABFKSE$, $BCHMLF$, $CDGPH$, $DEANM$, triangles FKL , ESR , GPQ , HMN .

The main diagonal of each regular hexagon $EF = 2 \implies EFGH$ is square with side 2 parallel to π .

The area of this square $[EFGH] = 4 \implies [EFGH] - [ABCD] = 3$.

The difference 3 is the area of the projection of 4 half of hexagons on the plane π .

So the area of the projections of another 4 half of hexagons is 3.

It is evident (may be not only for me) that projections of the coincide sides of hexagons are along diagonals of $ABCD$ (for example A, E, C , and H are collinear.)

So the projections on π of the coincide sides of hexagons and triangles are perpendicular to this lines ($SE \perp AE$).

Therefore in plane π projections of points S, E , and R are collinear and plane of $\triangle ESR$ is perpendicular to π .

We get $1 + 3 + 3 = \boxed{\text{(B) } 7}$.

PS. Denote $A = (0.5, 0.5, 0)$, $E = (1, 1, h)$, E' is the midpoint of RS .

$$|AE| = 1 \implies h = \frac{1}{\sqrt{2}}.$$

$$R = (0.5, 1.5, 2h), S = (1.5, 0.5, 2h) \implies E' = (1, 1, 2h) \implies EE' \perp ABC.$$

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Solution 5 (Not Rigorous)

Through observation, we can reasonably assume that each of the triangles on this shape is a right triangle. Since each side length of the hexagons is 1, the hypotenuse of the triangles would be $\sqrt{2}$. Now we know the side lengths of the octagon whose area we are solving for. The octagon can be broken into nine pieces. We have four triangles whose side lengths are 1, and their hypotenuse is a side whose length is $\sqrt{2}$. Next, we have 5 1 by 1 squares. The triangles each have an area of $\frac{1}{2}$, and the squares each have an area of 1.

Then, we add these up, so we get $\boxed{\text{(B) } 7}$.

Note: I believe Solution 7 has the proof for this.

Solution 6 (Coordinates)

Let one of the bottom points of the solid be the origin, $(0, 0, 0)$. Then, we can define one of the points where the vertices of the hexagons meet closest to our origin as (x, y, x) . Then, note that either the x coordinate or z coordinate must be 0 in the two vertices closest to the origin in the octagon. We can then let two adjacent points on the octagon near (x, y, x) be $(2x, 2y, 0)$ and $(0, 2y, 2x)$. Since these two points must be a distance of $\sqrt{3}$ from our origin (the height of a regular hexagon), and (x, y, x) must be a distance of 1 from our origin, we get

$$2x^2 + y^2 = 1, 4x^2 + 4y^2 = 3,$$

$$2x^2 + y^2 = 1, x^2 + y^2 = \frac{3}{4},$$

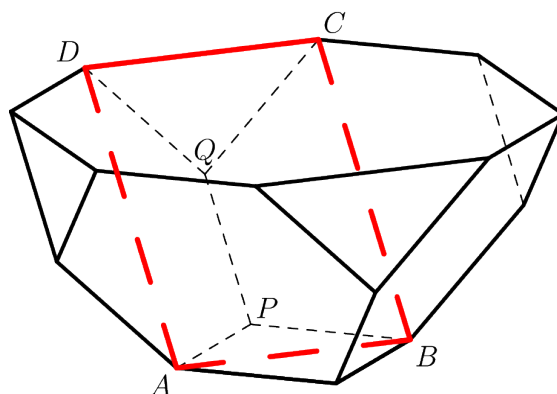
$$x^2 = \frac{1}{4}, y^2 = \frac{1}{2},$$

$$x = \frac{1}{2}, y = \frac{\sqrt{2}}{2}.$$

Then, the distance between $(2x, 2y, 0)$ and $(0, 2y, 2x)$ is $2\sqrt{2}x = \sqrt{2}$, so the octagon has sides $1, \sqrt{2}, 1, \sqrt{2}, 1, \sqrt{2}, 1, \sqrt{2}$, and the solution follows.

-piemax

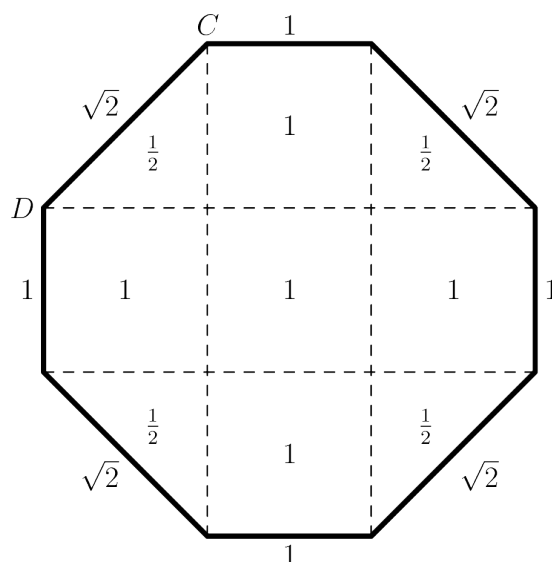
Solution 7 (Parallelogram)



Notice that $AD \parallel PQ$, $BC \parallel PQ$, so $AD \parallel BC$. Obviously, $AD = BC$, so $ABCD$ is a parallelogram, we get

$$CD = AB = \sqrt{1^2 + 1^2} = \sqrt{2}.$$

Note that octagon has four pairs of parallel sides, and the successive side-lengths are $1, \sqrt{2}, 1, \sqrt{2}, 1, \sqrt{2}, 1, \sqrt{2}$, as shown below:



The area of the octagon is **(B) 7**.

22. (AMC 10 10A 2022) Suppose that 13 cards numbered $1, 2, 3, \dots, 13$ are arranged in a row. The task is to pick them up in numerically increasing order, working repeatedly from left to right. In the example below, cards $1, 2, 3$ are picked up on the first pass, 4 and 5 on the second pass, 6 on the third pass, $7, 8, 9, 10$ on the fourth pass, and $11, 12, 13$ on the fifth pass. For how many of the $13!$ possible orderings of the cards will the 13 cards be picked up in exactly two passes?



- (A) 4082 (B) 4095 (C) 4096 (D) 8178 (E) 8191

Answer: D

Solution 1 (Casework)

For $1 \leq k \leq 12$, suppose that cards $1, 2, \dots, k$ are picked up on the first pass. It follows that cards $k + 1, k + 2, \dots, 13$ are picked up on the second pass.

Once we pick the spots for the cards on the first pass, there is only one way to arrange all 13 cards.

For each value of k , there are $\binom{13}{k} - 1$ ways to pick the k spots for the cards on the first pass: We exclude the arrangement

$$\boxed{1} \boxed{2} \boxed{3} \boxed{4} \boxed{5} \boxed{6} \boxed{7} \boxed{8} \boxed{9} \boxed{10} \boxed{11} \boxed{12} \boxed{13}$$

in which the cards are arranged such that the first pass consists of all 13 cards.

Therefore, the answer is

$$\sum_{k=1}^{12} \left[\binom{13}{k} - 1 \right] = \left[\sum_{k=1}^{12} \binom{13}{k} \right] - 12 = \left[\sum_{k=0}^{13} \binom{13}{k} \right] - 14 = 2^{13} - 14 = \boxed{\text{(D)} 8178}.$$

Solution 2 (Casework)

Since the 13 cards are picked up in two passes, the first pass must pick up the first n cards and the second pass must pick up the remaining cards m through 13. Also note that if m , which is the card that is numbered one more than n , is placed before n , then m will not be picked up on the first pass since cards are picked up in order. Therefore we desire m to be placed before n to create a second pass, and that after the first pass, the numbers m through 13 are lined up in order from least to greatest.

To construct this, n cannot go in the n th position because all cards 1 to $n - 1$ will have to precede it and there will be no room for m . Therefore n must be in slots $n + 1$ to 13. Let's do casework on which slot n goes into to get a general idea for how the problem works.

Case 1: With n in spot $n + 1$, there are n available slots before n , and there are $n - 1$ cards preceding n . Therefore the number of ways to reserve these slots for the $n - 1$ cards is $\binom{n}{n-1}$. Then there is only 1 way to order these cards (since we want them in increasing order). Then card m goes into whatever slot is remaining, and the $13 - m$ cards are ordered in increasing order after slot $n + 1$, giving only 1 way. Therefore in this case there are $\binom{n}{n-1}$ possibilities.

Case 2: With n in spot $n + 2$, there are $n + 1$ available slots before n , and there are $n - 1$ cards preceding n . Therefore the number of ways to reserve slots for these cards are $\binom{n+1}{n-1}$. Then there is one way to order these cards. Then cards m and $m + 1$ must go in the remaining two slots, and there is only one way to order them since they must be in increasing order. Finally, cards $m + 2$ to 13 will be ordered in increasing order after slot $n + 1$, which yields 1 way. Therefore, this case has $\binom{n+1}{n-1}$ possibilities.

I think we can see a general pattern now. With n in slot x , there are $x - 1$ slots to distribute to the previous $n - 1$ cards, which can be done in $\binom{x-1}{n-1}$ ways. Then the remaining cards fill in in just 1 way. Since the cases of n start in slot $n + 1$ and end in slot 13, this sum amounts to:

$$\binom{n}{n-1} + \binom{n+1}{n-1} + \binom{n+2}{n-1} + \dots + \binom{12}{n-1}$$

for any n .

Hmmm ... where have we seen this before?

We use wishful thinking to add a term of $\binom{n-1}{n-1}$:

$$\binom{n-1}{n-1} + \binom{n}{n-1} + \binom{n+1}{n-1} + \binom{n+2}{n-1} + \cdots + \binom{12}{n-1}$$

This is just the hockey stick identity! Applying it, this expression is equal to $\binom{13}{n}$. However, we added an extra term, so subtracting it off, the total number of ways to order the 13 cards for any n is

$$\binom{13}{n} - 1$$

Finally, to calculate the total for all n , we sum from $n = 0$ to 13. This yields us:

$$\begin{aligned} \sum_{n=0}^{13} \binom{13}{n} - 1 &\implies \sum_{n=0}^{13} \binom{13}{n} - \sum_{n=0}^{13} 1 \\ &\implies 2^{13} - 14 = 8192 - 14 = 8178 = \boxed{\text{(D) } 8178}. \end{aligned}$$

Solution 3 (Recursion)

To solve this problem, we can use recursion on n . Let A_n be the number of arrangements for n numbers. Now, let's look at how these arrangements are related to A_{n-1} by case work on the first number a_1 .

If $a_1 = 1$, the remaining $n - 1$ numbers from 2 to n are arranged in the same way just like number 1 to $n - 1$ in the case of $n - 1$ numbers. So there are A_{n-1} arrangements.

If $a_1 = 2$, then we need to choose 1 position from position 2 to $n - 1$ to put 1, and all remaining numbers must be arranged in increasing order, so there are $\binom{n-1}{1}$ such arrangements.

If $a_1 = k$, then we need to choose $k - 1$ positions from position 2 to $n - 1$ to put $1, 2, \dots, k - 1$, and all remaining numbers must be arranged in increasing order, so there are $\binom{n-1}{k-1}$ such arrangements.

So we can write

$$A_n = A_{n-1} + \binom{n-1}{1} + \binom{n-1}{2} + \cdots + \binom{n-1}{n-1}$$

which can be simplified to

$$A_n = A_{n-1} + 2^{n-1} - 1$$

We can solve this recursive sequence by summing up $n - 1$ lines of the recursive formula

$$A_n - A_{n-1} = 2^{n-1} - 1$$

$$A_{n-1} - A_{n-2} = 2^{n-2} - 1$$

...

to get

since $A_1 = 0$, we have

and $A_{13} = 2^{13} - 14 = \boxed{\text{(D) } 8178}$.

Solution 4 (Engineer's Induction)

Solution 5 (Subsets Bijection)

Solution 6 (10 Seconds Solution)

Solution 7 (Simpler Casework)

Both A and B must be non-empty such that we have 2 passes, so $1 \leq x \leq 12$.

$$\frac{13!}{x!(13-x)!} = \binom{13}{x},$$

choosing which x of the 13 positions get an A . However, there is an arrangement with all the A cards in a row followed by all the B cards. Since order is fixed to be increasing, such a row reconstructs to $1, 2, \dots, 13$, which takes 1 pass instead of 2, meaning we must subtract one case per value of x , or 12.

Since arrangements of A and B between different values of x are disjoint, we can sum without worrying about overcounting:

$$\binom{13}{1} + \binom{13}{2} + \dots + \binom{13}{12} - 12.$$

The sum looks awfully similar to the Pascal's Triangle row-sum identity:

$$\sum_{k=0}^{13} \binom{13}{k} = \binom{13}{0} + \binom{13}{1} + \dots + \binom{13}{13} = 2^{13}$$

With a little bit of wishful thinking and rearranging, we get

$$\begin{aligned} \binom{13}{0} + \binom{13}{1} + \dots + \binom{13}{13} - \binom{13}{0} - \binom{13}{13} - 12 &= 2^{13} - \binom{13}{0} - \binom{13}{13} - 12 \\ \binom{13}{1} + \binom{13}{2} + \dots + \binom{13}{12} - 12 &= 2^{13} - \binom{13}{0} - \binom{13}{13} - 12 \\ &= 8192 - 1 - 1 - 12 \\ &= \boxed{\text{(D)} 8178}. \end{aligned}$$

23. (AMC 10 10A 2022) Isosceles trapezoid $ABCD$ has parallel sides $\overline{AD}$ and $\overline{BC}$, with $BC < AD$ and $AB = CD$. There is a point P in the plane such that $PA = 1$, $PB = 2$, $PC = 3$, and $PD = 4$. What is $\frac{BC}{AD}$? (A) $\frac{1}{4}$ (B) $\frac{1}{3}$ (C) $\frac{1}{2}$ (D) $\frac{2}{3}$ (E) $\frac{3}{4}$

Answer: B

Solution 1 (Reflections + Ptolemy's Theorem)

Consider the reflection P' of P over the perpendicular bisector of $\overline{BC}$, creating two new isosceles trapezoids $DAPP'$ and $CBPP'$. Under this reflection, $P'A = PD = 4$, $P'D = PA = 1$, $P'C = PB = 2$, and $P'B = PC = 3$.

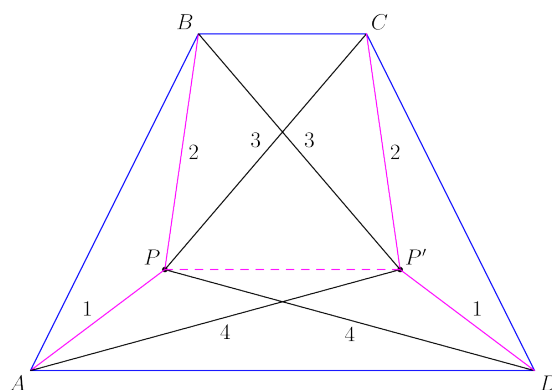
Since $DAPP'$ and $CBPP'$ are isosceles trapezoids, they are cyclic. Using **Ptolemy's theorem** on $DAPP'$, we get that $(PP')(AD) + (PA)(P'D) = (AP')(PD)$, so

$$PP' \cdot AD + 1 \cdot 1 = 4 \cdot 4.$$

Then, using Ptolemy's theorem again on $CBPP'$, we get that $(BC)(PP') + (BP)(CP') = (BP')(CP)$, so

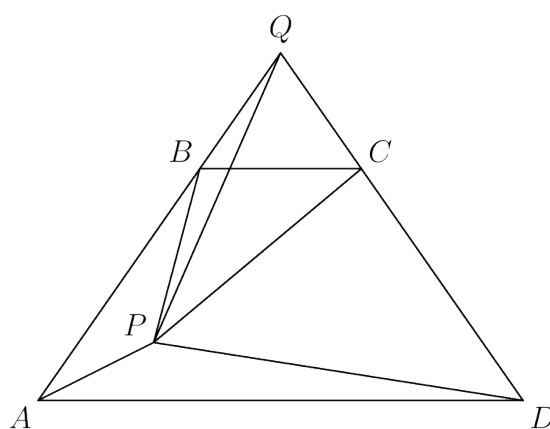
$$PP' \cdot BC + 2 \cdot 2 = 3 \cdot 3.$$

Thus, $PP' \cdot AD = 15$ and $PP' \cdot BC = 5$; dividing these two equations and taking the reciprocal yields $\frac{BC}{AD} = \boxed{\text{(B)} \frac{1}{3}}$.



(diagram by cinnamon_e)

Solution 2 (Extensions + Stewart's Theorem)



Extend AB and CD to a point Q as shown, and let $PQ = s$. Then let $BQ = CQ = a$ and $AB = DC = b$. Notice that $\frac{BC}{AD} = \frac{QC}{QD} = \frac{a}{a+b}$ by similar triangles.

By **Stewart's theorem** on APQ and DPQ , we have

$$\begin{aligned} ab(a+b) + 9(a+b) &= 16a + s^2b \\ ab(a+b) + 4(a+b) &= a + s^2b \end{aligned}$$

Subtracting, $5(a+b) = 15a$, and so $\frac{BC}{AD} = \frac{a}{a+b} = \frac{5}{15} = \boxed{\text{(B)} \frac{1}{3}}$.

Solution 3 (Coordinate Bashing)

Since we're given distances and nothing else, we can represent each point as a coordinate and use the distance formula to set up a series of systems and equations. Let the height of the trapezoid be h , and let the coordinates of A and D be at $(-a, 0)$ and $(a, 0)$, respectively. Then let B and C be at $(-b, h)$ and (b, h) , respectively. This follows the rules that this is an isosceles trapezoid since the origin is centered on the middle of AD . Finally, let P be located at point (c, d) .

The distance from P to A is 1, so by the distance formula:

$$\sqrt{(c+a)^2 + (d-0)^2} = 1 \implies (c+a)^2 + d^2 = 1$$

The distance from P to D is 4, so

$$\sqrt{(c-a)^2 + (d-0)^2} = 4 \implies (c-a)^2 + d^2 = 16$$

Looking at these two equations alone, notice that the second term is the same for both equations, so we can subtract the equations. This yields

$$-4ac = 15$$

Next, the distance from P to B is 2, so

$$\sqrt{(c+b)^2 + (d-h)^2} = 2 \implies (c+b)^2 + (d-h)^2 = 4$$

The distance from P to C is 3, so

$$\sqrt{(c-b)^2 + (d-h)^2} = 3 \implies (c-b)^2 + (d-h)^2 = 9$$

Again, we can subtract these equations, yielding

$$-4bc = 5$$

We can now divide the equations to eliminate c , yielding

$$\frac{b}{a} = \frac{5}{15} = \frac{1}{3}$$

We wanted to find $\frac{BC}{AD}$. But since b is half of BC and a is half of AD , this ratio is equal to the ratio we want.

Therefore $\frac{BC}{AD} = \boxed{\text{(B)} \frac{1}{3}}.$

Solution 4 (Coordinate Bashing)

Let the point P be at the origin, and draw four concentric circles around P each with radius 1, 2, 3, and 4, respectively. The vertices of the trapezoid would be then on each of the four concentric circles. WLOG, let BC and AD be parallel to the x -axis. Assigning coordinates to each point, we have:

$$A = (x_1, y_1)$$

$$B = (x_2, y_2)$$

$$C = (x_3, y_2)$$

$$D = (x_4, y_1)$$

which satisfy the following:

$$x_1^2 + y_1^2 = 1 \tag{1}$$

$$x_2^2 + y_2^2 = 4 \tag{2}$$

$$x_3^2 + y_2^2 = 9 \tag{3}$$

$$x_4^2 + y_1^2 = 16 \tag{4}$$

In addition, because the trapezoid is isosceles ($AB = CD$), the midpoints of the two bases would then have the same x -coordinate, giving us

$$x_1 + x_4 = x_2 + x_3 \quad (5)$$

Subtracting Equation 2 from Equation 3, and Equation 1 from Equation 4, we have

$$x_3^2 - x_2^2 = 5 \quad (6)$$

$$x_4^2 - x_1^2 = 15 \quad (7)$$

Dividing Equation 6 by Equation 7, we have

$$\frac{x_3^2 - x_2^2}{x_4^2 - x_1^2} = \frac{1}{3}$$

$$\frac{(x_3 - x_2)(x_3 + x_2)}{(x_4 - x_1)(x_4 + x_1)} = \frac{1}{3}.$$

Cancelling $(x_3 + x_2)$ and $(x_4 + x_1)$ with Equation 5, we get

$$\frac{(x_3 - x_2)}{(x_4 - x_1)} = \frac{1}{3}.$$

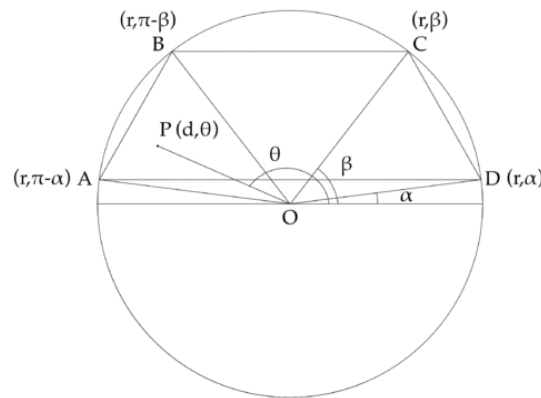
In other words,

$$\frac{BC}{AD} = \frac{1}{3} = \boxed{\text{(B)} \frac{1}{3}}.$$

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Solution 5 (Polar Coordinates)

As $ABCD$ is an isosceles trapezoid, it is cyclic. We will use polar coordinates with origin O as in the following figure.



The x -coordinate of C is $r \cos \beta$, $BC = 2r \cos \beta$. The x -coordinate of D is $r \cos \alpha$, $AD = 2r \cos \alpha$, $\frac{BC}{AD} = \frac{\cos \beta}{\cos \alpha}$

By the law of cosines,

$$\begin{aligned} OD^2 + OP^2 - 2 \cdot OD \cdot OP \cdot \cos(\theta - \alpha) &= PD^2 \\ OC^2 + OP^2 - 2 \cdot OC \cdot OP \cdot \cos(\theta - \beta) &= PC^2 \\ OB^2 + OP^2 - 2 \cdot OB \cdot OP \cdot \cos(\theta - \pi + \beta) &= PB^2 \end{aligned}$$

$$OA^2 + OP^2 - 2 \cdot OA \cdot OP \cdot \cos(\theta - \pi + \alpha) = PA^2$$

$$r^2 + d^2 - 2rd \cos(\theta - \alpha) = 16 \quad (1)$$

$$r^2 + d^2 - 2rd \cos(\theta - \beta) = 9 \quad (2)$$

$$r^2 + d^2 - 2rd \cos(\theta - \pi + \beta) = 4 \quad (3)$$

$$r^2 + d^2 - 2rd \cos(\theta - \pi + \alpha) = 1 \quad (4)$$

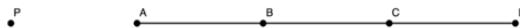
$$\begin{aligned} (3) - (2), \quad 2rd \cos(\theta - \beta) - 2rd \cos(\theta - \pi + \beta) &= 4 - 9, \quad 2rd(\cos(\theta - \beta) + \cos(\theta + \beta)) = -5 \\ (4) - (1), \quad 2rd \cos(\theta - \alpha) - 2rd \cos(\theta - \pi + \alpha) &= 1 - 16, \quad 2rd(\cos(\theta - \alpha) + \cos(\theta + \alpha)) = -15 \end{aligned}$$

$$(5)/(6), \quad \frac{2rd(\cos(\theta - \beta) + \cos(\theta + \beta))}{2rd(\cos(\theta - \alpha) + \cos(\theta + \alpha))} = \frac{-5}{-15}, \quad \frac{\cos(\theta - \beta) + \cos(\theta + \beta)}{\cos(\theta - \alpha) + \cos(\theta + \alpha)} = \frac{1}{3}$$

By the sum to product identity

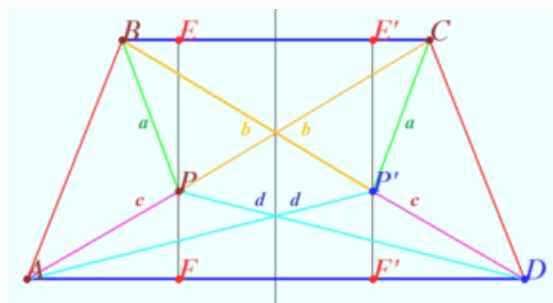
$$\frac{\cos(\theta - \beta) + \cos(\theta + \beta)}{\cos(\theta - \alpha) + \cos(\theta + \alpha)} = \frac{2 \cos(\frac{\theta - \beta + \theta + \beta}{2}) \cos(\frac{\theta - \beta - \theta - \beta}{2})}{2 \cos(\frac{\theta - \alpha + \theta + \alpha}{2}) \cos(\frac{\theta - \alpha - \theta - \alpha}{2})} = \frac{\cos \theta \cos \beta}{\cos \theta \cos \alpha} = \frac{\cos \beta}{\cos \alpha} = \boxed{\text{(B)} \frac{1}{3}}$$

Solution 6 (EZ)



Notice that the question never says what the height of the trapezoid is; the only property we know about it is that $AC = BD$. Therefore, we can assume that the height of the trapezoid is 0 and all 5 points, including P , lie on the same line with $PA = AB = BC = CD = 1$. Notice that this satisfies the problem requirements because $PA = 1, PB = 2, PC = 3, PD = 4$, and $AC = BD = 2$. Now all we have to find is $\frac{BC}{AD} = \boxed{\text{(B)} \frac{1}{3}}$.

Solution 7 (Pythagorean Theorem)



Consider the reflection P' of P over the perpendicular bisector of $\overline{BC}$.

$$PB = P'C = a, PC = P'B = b.$$

Let $PE \perp BC, P'E' \perp BC$.

$$\begin{aligned} BE = CE', PP' = EE' &\implies \\ b^2 - a^2 = PC^2 - PB^2 = CE^2 - BE^2 \\ &= (CE - BE)(CE + BE) = EE' \cdot BC = PP' \cdot BC. \end{aligned}$$

Similarly,

$$d^2 - c^2 = PP' \cdot AD \implies \frac{BC}{AD} = \frac{b^2 - a^2}{d^2 - c^2} = \boxed{\text{(B)} \frac{1}{3}}.$$

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Solution 8 (Pythagorean Theorem)

Let the feet of the altitude from P to BC be E , and the feet of the altitude from P to AD be F . Let $CE = y$, $BE = x$. Observe that $DF = y + d$, $AF = x + d$ where $AD = BC + 2d$ due to the properties of isosceles trapezoids.

Then $y^2 - x^2 = CE^2 - BE^2 = CP^2 - BP^2 = 5$ by the Pythagorean Theorem. Hence, $5 + 2d(y - x) = y^2 - x^2 + 2d(y - x) = (y + d)^2 - (x + d)^2 = 4^2 - 1^2 = 15 \implies d(y - x) = 5$. But $y^2 - x^2 = 5$, so $d(y - x) = y^2 - x^2 \implies d = y + x$. But $AD = BC + 2d = x + y + 2d = 3(x + y)$, so $\frac{BC}{AD} = \frac{1}{3}$.

-mathMagicOPS

Solution 9 (Pythagorean Theorem and Conjugate)

Using some naming conventions from Solution 8, let $PE = h_2$, $PF = h_1$, $AB = CD = x$, $BC = n$, and $AD = n + 2d$. We are seeking to find $\frac{BC}{AD} = \frac{n}{n+2d}$.

Using Pythagorean Theorem on all four triangles with PA, PB, PC, PD as their hypotenuses, we get that $BE = \sqrt{4 - h_2^2}$, $EC = \sqrt{9 - h_2^2}$, $AF = \sqrt{1 - h_1^2}$, and $FD = \sqrt{16 - h_1^2}$. $\frac{n}{n+2d} = \frac{\sqrt{9-h_2^2}+\sqrt{4-h_2^2}}{\sqrt{16-h_1^2}+\sqrt{1-h_1^2}}$

Using AF and BE to find d we get $d = \sqrt{1 - h_1^2} - \sqrt{4 - h_2^2}$. Similarly, using EC and FD we get $d = \sqrt{16 - h_1^2} - \sqrt{9 - h_2^2}$. Isolating variables gives us $\sqrt{16 - h_1^2} - \sqrt{1 - h_1^2} = \sqrt{9 - h_2^2} - \sqrt{4 - h_2^2}$.

Notice that in the fraction we want, both the numerator and denominator's conjugates are equal because of our equation. By multiplying that fraction by $\frac{\sqrt{9-h_2^2}-\sqrt{4-h_2^2}}{\sqrt{16-h_1^2}-\sqrt{1-h_1^2}}$, all h_1 and h_2 cancel, leaving us with $\frac{1}{3}$.

24. (AMC 10 10A 2022) How many strings of length 5 formed from the digits 0, 1, 2, 3, 4 are there such that for each $j \in \{1, 2, 3, 4\}$, at least j of the digits are less than j ? (For example, 02214 satisfies this condition because it contains at least 1 digit less than 1, at least 2 digits less than 2, at least 3 digits less than 3, and at least 4 digits less than 4. The string 23404 does not satisfy the condition because it does not contain at least 2 digits less than 2.)

(A) 500 (B) 625 (C) 1089 (D) 1199 (E) 1296

Answer: E

Solution 1 (Parking Functions)

For some n , let there be $n + 1$ parking spaces counterclockwise in a circle. Consider a string of n integers $c_1 c_2 \dots c_n$ each between 0 and n , and let n cars come into this circle so that the i th car tries to park at spot c_i , but if it is already taken then it instead keeps going counterclockwise and takes the next available spot. After this process, exactly one spot will remain empty.

Then the strings of n numbers between 0 and $n - 1$ that contain at least k integers $< k$ for $1 \leq k \leq n$ are exactly the set of strings that leave spot n empty. Also note for any string $c_1 c_2 \dots c_n$, we can add 1 to each $c_i \pmod{n + 1}$ to shift the empty spot counterclockwise, meaning for each string there exists exactly one j with $0 \leq j \leq n$ so that $(c_1 + j)(c_2 + j) \dots (c_n + j)$ leaves spot n empty. This gives there are $\frac{(n+1)^n}{n+1} = (n+1)^{n-1}$ such strings.

Plugging in $n = 5$ gives **(E) 1296** such strings.

Solution 2 (OmegaLearn)

Consider some string that satisfies the problem statement, with a 0's, b 1's, c 2's, d 3's, and e 4's, so $a + b + c + d + e = 5$. There are exactly $\frac{5!}{a!b!c!d!e!}$ ways to order the given string. This number will be divisible by 5 unless one of a, b, c, d, e is 5, which implies that the rest are 0. This corresponds to the string having 5 of one digit. But the only possible string that has 1 distinct digit is 00000. Thus, the number of strings must be 1 (mod 5). The only answer choice that satisfies this is **(E) 1296**.

Solution 3 (Casework)

Note that a valid string must have at least one 0.

We perform casework on the number of different digits such strings can have. For each string, we list the digits in ascending order, then consider permutations:

- a) The string has 1 different digit. The only possibility is 00000.

There is 1 string in this case.

- b) The string has 2 different digits. We have the following table:

Digits	01	02	03	04	Row's Count
00001	00002	00003	00004		$4 \cdot \frac{5!}{4!1!} = 20$
00011	00022	00033			$3 \cdot \frac{5!}{3!2!} = 30$
00111	00222				$2 \cdot \frac{5!}{2!3!} = 20$
01111					$1 \cdot \frac{5!}{1!4!} = 5$

There are $20 + 30 + 20 + 5 = 75$ strings in this case.

c) The string has 3 different digits. We have the following table:

Digits	012	013	014	023	024	034	Row's Count
	00012	00013	00014	00023	00024	00034	$6 \cdot \frac{5!}{3!1!1!} = 120$
	00112	00113	00114	00223	00224		$5 \cdot \frac{5!}{2!2!1!} = 150$
	00122	00133		00233			$3 \cdot \frac{5!}{2!1!2!} = 90$
	01112	01113	01114				$3 \cdot \frac{5!}{1!3!1!} = 60$
	01122	01133					$2 \cdot \frac{5!}{1!2!2!} = 60$
	01222						$1 \cdot \frac{5!}{1!1!3!} = 20$

There are $120 + 150 + 90 + 60 + 60 + 20 = 500$ strings in this case.

d) The string has 4 different digits. We have the following table:

Digits	0123	0124	0134	0234
	00123	00124	00134	00234
	01123	01124	01134	
	01223	01224		
	01233			

There are $10 \cdot \frac{5!}{2!1!1!1!} = 600$ strings in this case.

e) The string has 5 different digits. There are $5! = 120$ strings in this case.

Together, the answer is $1 + 75 + 500 + 600 + 120 = \boxed{\text{(E)} 1296}$.

Solution 4 (Recursive Equations Approach)

Denote by $N(p, q)$ the number of p -digit strings formed by using numbers $0, 1, \dots, q$, where for each $j \in \{1, 2, \dots, q\}$, at least j of the digits are less than j .

We have the following recursive equation:

$$N(p, q) = \sum_{i=0}^{p-q} \binom{p}{i} N(p-i, q-1), \quad \forall p \geq q \text{ and } q \geq 1$$

and the boundary condition $N(p, 0) = 1$ for any $p \geq 0$.

By solving this recursive equation, for $q = 1$ and $p \geq q$, we get

$$\begin{aligned} N(p, 1) &= \sum_{i=0}^{p-1} \binom{p}{i} N(p-i, 0) \\ &= \sum_{i=0}^{p-1} \binom{p}{i} \\ &= \sum_{i=0}^p \binom{p}{i} - \binom{p}{p} \end{aligned}$$

$$= 2^p - 1.$$

For $q = 2$ and $p \geq q$, we get

$$\begin{aligned} N(p, 2) &= \sum_{i=0}^{p-2} \binom{p}{i} N(p-i, 1) \\ &= \sum_{i=0}^{p-2} \binom{p}{i} (2^{p-i} - 1) \\ &= \sum_{i=0}^p \binom{p}{i} (2^{p-i} - 1) - \sum_{i=p-1}^p \binom{p}{i} (2^{p-i} - 1) \\ &= \sum_{i=0}^p \left(\binom{p}{i} 1^i 2^{p-i} - \binom{p}{i} 1^i 1^{p-i} \right) - p \\ &= (1+2)^p - (1+1)^p - p \\ &= 3^p - 2^p - p. \end{aligned}$$

For $q = 3$ and $p \geq q$, we get

$$\begin{aligned} N(p, 3) &= \sum_{i=0}^{p-3} \binom{p}{i} N(p-i, 2) \\ &= \sum_{i=0}^{p-3} \binom{p}{i} (3^{p-i} - 2^{p-i} - (p-i)) \\ &= \sum_{i=0}^p \binom{p}{i} (3^{p-i} - 2^{p-i} - (p-i)) - \sum_{i=p-2}^p \binom{p}{i} (3^{p-i} - 2^{p-i} - (p-i)) \\ &= \sum_{i=0}^p \left(\binom{p}{i} 1^i 3^{p-i} - \binom{p}{i} 1^i 2^{p-i} - \binom{p}{i} (p-i) \right) - \frac{3}{2}p(p-1) \\ &= (1+3)^p - (1+2)^p - \frac{d(1+x)^p}{dx} \Big|_{x=1} - \frac{3}{2}p(p-1) \\ &= 4^p - 3^p - 2^{p-1}p - \frac{3}{2}p(p-1). \end{aligned}$$

For $q = 4$ and $p = 5$, we get

$$\begin{aligned} N(5, 4) &= \sum_{i=0}^1 \binom{5}{i} N(5-i, 3) \\ &= N(5, 3) + 5N(4, 3) \\ &= \boxed{\text{(E) } 1296}. \end{aligned}$$

Solution 6 (Observations)

The number of strings is $(n+1)^{(n-1)}$ as shown by Solution 1 (Parking Function), which is always equivalent to 1 (mod n). Thus you can choose (E) 1296.

However, you **CANNOT** prove that fact from this ***incorrect*** argument:

Let the set of all valid sequences be S . Notice that for any sequence $\{a_1, a_2, a_3, a_4, a_5\}$ in S , the 5 cyclic permutations

$$\{a_1, a_2, a_3, a_4, a_5\}$$

$$\{a_2, a_3, a_4, a_5, a_1\}$$

$$\{a_3, a_4, a_5, a_1, a_2\}$$

$$\{a_4, a_5, a_1, a_2, a_3\}$$

$$\{a_5, a_1, a_2, a_3, a_4\}$$

must also belong in S . However, one must consider the edge case all 5 elements are the same (only $\{0, 0, 0, 0, 0\}$), in which case all sequences listed are equivalent.

Alas, one must also consider cases like 10101 that only have 3 distinct cyclic permutations, so we cannot get a useful constraint from this view, unless *all* the cases are categorized and counted.

Thus it is NOT implied that $|S| \equiv 1 \pmod{5}$, so it is not justified to choose

(E) 1296 by inspection of answer options.

If you solve $S(n)$ for smaller n by bashing, you can find $|S(1)| = 1$, $|S(2)| = 3$, $|S(3)| = 16$, $|S(4)| = 125$, which leads to an Engineer's Induction guess that $|S(n)| \equiv 1 \pmod{n}$, or the exact formula $(n+1)^{(n-1)}$

Note: I'm pretty sure this solution works. Firstly, 10101 does have 5 distinct cyclic permutations, namely 10101, 11010, 01101, 10110, and 01011. Secondly, any sequence of 5 elements must have either 1 or a multiple of 5 distinct cyclic permutations. If a sequence of 5 elements has only 1 cyclic permutation, that must be the case with $(0, 0, 0, 0, 0)$, as noted (since 0 is necessary in our sequence when $j = 1$). If the sequence is cyclic every k "rotations/shifts," then it is ONLY going to be at the starting sequence when you shift it $0, k, 2k, \dots$ times. Since shifting any sequence 5 times takes it back to itself, we must have $k \mid 5 \implies k = 5$ (since we already considered $k = 1$ with $(0, 0, 0, 0, 0)$), so the cycle must be 5. So really, this solution does work, is not by coincidence, and is a very nice (albeit cheap) solution.

TL;DR: The sentence "Alas, one must also consider cases like 10101 that only have 3 distinct cyclic permutations" is incorrect, for these cases do not actually exist. It's trivial to check 10101 has 5 cyclic permutations.

Solution 5 (Mod 5 Trick Redeemed)

Solution 4 tried to observe the answer modulo 5 to easily solve the problem, but apparently had faulty logic. This solution is still completely viable though:

Notice that for any valid set $\{a_1, a_2, a_3, a_4, a_5\}$, if there is at least one element in the set that is unique (i.e. there is at least one digit in the set that is found nowhere else in the set), then the number of distinct permutations of the set is clearly divisible by 5 (alternatively there is no factor of 5 in the denominator of the multinomial coefficient unless an element is repeated 5 times). Therefore to evaluate the answer modulo 5, we only need to look at sets where each element has a multiplicity of at least 2 (i.e. appears twice or more in the set).

These sets are of the form $\{a, a, b, b, b\}$ and $\{a, a, a, a, a\}$. The first set can be permuted in $\binom{5}{2,3} = 10 \equiv 0 \pmod{5}$, and the second set can be permuted one way, and the only set of the form $\{a, a, a, a, a\}$ is $\{0, 0, 0, 0, 0\}$. Therefore the answer is congruent to 1 $\pmod{5}$ and you CAN choose **(E) 1296**.

Solution 7 (Casework Bash)

We list out all possibilities for the string, sorted in increasing order. Note that 1 cannot appear before the 2nd position, 2 cannot appear before the 3rd position,

and so on. We also make a list of the number of possible permutations of each sorted string.

00000 - 1

00001 - 5

00002 - 5

00003 - 5

00004 - 5

00011 - 10

00012 - 20

00013 - 20

00014 - 20

00022 - 10

00023 - 20

00024 - 20

00033 - 10

00034 - 20

00111 - 10

00112 - 30

00113 - 30

00114 - 30

00122 - 30

00123 - 60

00124 - 60

00133 - 30

00134 - 60

00222 - 10

00223 - 30

00224 - 30

00233 - 30

00234 - 60

01111 - 5

01112 - 20

01113 - 20

01114 - 20

01122 - 30

01123 - 60

01124 - 60

01133 - 30

01134 - 60
 01222 - 20
 01223 - 60
 01224 - 60
 01233 - 60
 01234 - 120

Finally, adding all of these up, we obtain $1 \cdot 1 + 5 \cdot 5 + 5 \cdot 10 + 10 \cdot 20 + 10 \cdot 30 + 10 \cdot 60 + 1 \cdot 120 = \boxed{\text{(E) 1296}}$. (Or alternatively, we could notice as in solution 5 that the only possible string with only 1 permutation is 00000, and thus the answer must be congruent to 1 modulo 5)

The key to organizing all of these cases is to increase the digits one by one, starting from the right and iterating through all the possible cases.

Solution 8 (Complementary Counting)

Let us call the condition where you require at least j digits less than j Rule j . For a sequence to break Rule 1, it must contain no 0s. There are $4^5 = 1024$ ways to do this. For a sequence to break Rule 2 but not Rule 1, it must contain exactly 1 0. However, this one 0 contributes toward completing the rule, meaning all other digits must be greater than 2. After placing the one 0 in any of 5 slots, there are 4 slots left with 3 options each, giving a total of $5 \cdot 3^4 = 405$ ways. For a sequence to break Rule 3 but neither Rule 2 or 1, it must have two digits less than 2 (either (0, 0) or (0, 1)) and all other digits must be 3 or 4. There are a total of 30 ways to arrange the first two digits and 2^3 ways to arrange the latter three, leaving 240 ways to do this. Finally, for a sequence to only break rule 4, it must have three digits less than 3. They could be (0, 1, 2) (60 arr.), (0, 1, 1); (0, 0, 1); (0, 0, 2) (30 arr. each), or (0, 0, 0) (10 arr.) This gives 160 possible arrangements overall.

There are in total $5^5 = 3125$ possible arrangements, so by adding all of the invalid states together and subtracting from 3125, we get our answer, which is $\boxed{\text{(E) 1296}}$

Great idea! Bigbrain_2009

Solution 9 (Recursion)

We will use recursion to solve this problem. Let us break up the problem's condition into criteria, with criterion k being the requirement that at least k of the digits are less than k . Define A_n as the number of strings of length n formed from the digits $0, 1, 2, \dots, n-1$ such that at least j of the digits are less than j for all integers j ; $1 \leq j \leq n-1$.

Note that the final criterion any string counted as apart of A_{n-1} is required to meet is criterion $n-2$. However, all strings that A_{n-1} counts will also meet criterion $n-1$. This is due to the maximum value of such strings' $n-1$ digits being $n-2$. Initially, we have $A_1 = 1$. We check cases and find $A_2 = 3$. To generate recursion, consider the compliment of the problem's condition. Without the condition, we are counting strings of length n with n choices for each digit. There are of course n^n such strings. To count the number of strings that do not meet at least one criterion, we denote a string as k -bad if the first criterion the string violates is criterion k . For example, the string 33040 is 3-bad because it meets criteria 1 and 2, but violates criterion 3. Similarly, the string 13223 is 1-bad because it does contain 1 digit less than 1. We can count the number of strings that are k -bad by first

choosing and assigning $k - 1$ of the string's n spots to a string counted as apart of A_{k-1} . This can be done in $\binom{n}{k-1} \cdot A_{k-1}$ ways. This $(k - 1)$ digit string will meet all criteria up to criterion $k - 1$. To ensure criterion k is not met, we fill up the remaining $n - (k - 1)$ spots with a digit from k to $n - 1$, inclusive. This can be done in $(n - k)^{n+1-k}$ ways. We sum over all integers k , $1 \leq k \leq n - 1$ to count all strings that do not meet at least one criterion.

$$\sum_{k=1}^{n-1} \binom{n}{k-1} A_{k-1} (n - k)^{n+1-k}$$

This is the same as

$$\sum_{k=2}^n \binom{n}{k-2} A_{k-2} (n - k + 1)^{n+2-k} = \sum_{k=2}^n \binom{n}{n+2-k} A_{k-2} (n - k + 1)^{n+2-k}$$

We substitute $n + 2 - k$ for k as the expression takes on the same set of values. We have

$$\sum_{k=2}^n \binom{n}{n+2-(n+2-k)} A_{(n+2-k)-2} (n - (n+2-k) + 1)^{n+2-(n+2-k)} = \sum_{k=2}^n \binom{n}{k} A_{n-k} (k-1)^k$$

Therefore,

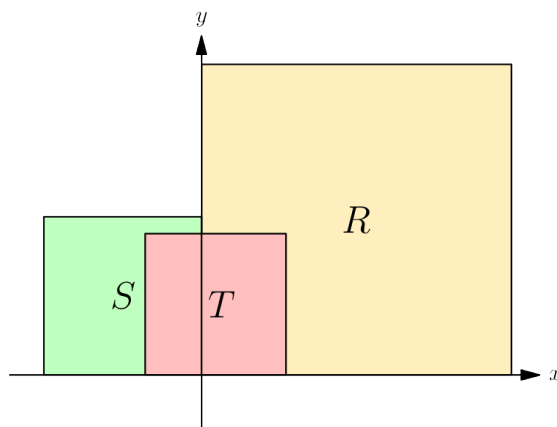
$$A_n = n^n - \sum_{k=2}^n \binom{n}{k} A_{n-k} (k-1)^k$$

Setting $A_0 = 1$ and $A_1 = 1$, we use the recursion to derive $A_2 = 3$, $A_3 = 16$, $A_4 = 125$, $A_5 = 1296$. In our problem, we were counting 5 digits strings. Thus, the answer is (E) 1296

Solution 10 (Parity Cheese)

We note that there are 6 different multisets of digits that have an odd number of permutations, namely, 00000, 00001, 00002, 00003, 0004, 01111. We know that there cannot be any other possibilities since, if any digit appears 2 or 3 times, there are $\binom{5}{2} = \binom{5}{3} = 10$ ways to place those two digits, and thus the total number of permutations must be even. Additionally, if there are at least two digits that appear once, there are $5 \cdot 4 = 20$ ways to place them and the total number of permutations will be even. Hence, we can have only at most 1 digit that appears once, and the rest must appear either 4 or 5 times (or 0). From here we may arrive at the above 6 configurations. Since there is an even number of odd cases, we know that the final answer will be even. After a little bit of casework we observe that 500 is far too low, and thus the answer is 1296.

25. (AMC 10 10A 2022) Let R , S , and T be squares that have vertices at lattice points (i.e., points whose coordinates are both integers) in the coordinate plane, together with their interiors. The bottom edge of each square is on the x -axis. The left edge of R and the right edge of S are on the y -axis, and R contains $\frac{9}{4}$ as many lattice points as does S . The top two vertices of T are in $R \cup S$, and T contains $\frac{1}{4}$ of the lattice points contained in $R \cup S$. See the figure (not drawn to scale).



The fraction of lattice points in S that are in $S \cap T$ is 27 times the fraction of lattice points in R that are in $R \cap T$. What is the minimum possible value of the edge length of R plus the edge length of S plus the edge length of T ?

- (A) 336 (B) 337 (C) 338 (D) 339 (E) 340

Answer: B

Solution 1 (Generalized)

Let r be the number of lattice points on the side length of square R , s be the number of lattice points on the side length of square S , and t be the number of lattice points on the side length of square T . Note that the actual lengths of the side lengths are the number of lattice points minus 1, so we can work in terms of r, s, t and subtract 3 to get the actual answer at the end. Furthermore, note that the number of lattice points inside a rectangular region is equal to the number of lattice points in its width times the number of lattice points along its length.

Using this fact, the number of lattice points in R is r^2 , the number of lattice points in S is s^2 , and the number of lattice points in T is t^2 .

Now, by the first condition, we have

$$r^2 = \frac{9}{4} \cdot s^2 \implies r = \frac{3}{2}s \quad (1)$$

The second condition, the number of lattice points contained in T is a fourth of the number of lattice points contained in $R \cup S$. The number of lattice points in $R \cup S$ is equal to the sum of the lattice points in their individually bounded regions, but the lattice points along the y-axis for the full length of square S is shared by both of them, so we need to subtract that out.

$$\begin{aligned} \text{In all, this condition yields us } t^2 &= \frac{1}{4} \cdot (r^2 + s^2 - s) \implies t^2 = \frac{1}{4} \cdot \left(\frac{9}{4} \cdot s^2 + s^2 - s \right) \implies \\ t^2 &= \frac{1}{4} \cdot \frac{13s^2 - 4s}{4} \implies 16t^2 = s(13s - 4) \end{aligned}$$

Note from (1) that s is a multiple of 2. We can write $s = 2j$ and substitute: $16t^2 = 2j(26j - 4) \implies 4t^2 = j(13j - 2)$. Note that j must be divisible by two for the product to be divisible by 4. Thus we make another substitution, $j = 2k$:

$$4t^2 = 2k(26k - 2) \implies t^2 = k(13k - 1) \quad (2)$$

Finally we look at the last condition; that the fraction of the lattice points inside S that are inside $S \cap T$ is 27 times the fraction of lattice points inside R that are inside $R \cap T$.

Let x be the number of lattice points along the bottom of the rectangle formed by $S \cap T$, and y be the number of lattice points along the bottom of the rectangle formed by $R \cap T$.

Therefore, the number of lattice points in $S \cap T$ is xt and the number of lattice points in $R \cap T$ is yt .

Thus by this condition, $\frac{xt}{s^2} = 27 \cdot \frac{yt}{r^2} \implies \frac{x}{s^2} = 27 \cdot \frac{y}{\frac{9}{4}s^2} \implies x = 12y$

Finally, notice that $t = x + y - 1 = 12y + y - 1$ (subtracting overlap), and so we have

$$t = 13y - 1 \quad (3)$$

Now notice that by (3), $t \equiv -1 \pmod{13} \implies t^2 \equiv 1 \pmod{13}$.

However, by (2), $t^2 \equiv k \cdot -1 \pmod{13}$. Therefore, $-k \equiv 1 \pmod{13} \implies k \equiv -1 \pmod{13}$

Also, by (2), we know k must be a perfect square since k is relatively prime to $13k - 1$ (Euclid's algorithm) and the two must multiply to a perfect square. Hence we know two conditions on k , and we can now guess and check to find the smallest that satisfies both.

We check $k = 12$ first since it's one less than a multiple of 13, but this does not work. Next, we have $k = 25$ which works because 25 is a perfect square. Thus we have found the smallest k , and therefore the smallest r, s, t .

Now we just work backwards: $j = 2k = 50$ and $s = 2j = 100$. Then $r = \frac{3}{2} \cdot 100 = 150$. Finally, from (2), $t^2 = 25(13 \cdot 25 - 1) \implies t^2 = 25 \cdot 324 \implies t = 5 \cdot 18 = 90$.

Finally, the sum of each square's side lengths is $r + s + t - 3 = 340 - 3 = 337 = \boxed{\text{(B)} 337}$.

Solution 2 (Answer Choices)

Notice that each answer choice has a different residue mod 13. Therefore, we can just find the residue of $r + s + t - 3 \pmod{13}$ and find the unique answer choice that fits, without actually finding r, s, t .

From Solution 1, we have $16t^2 = s(13s - 4)$ from the second condition. From the third condition, $t \equiv -1 \pmod{13} \implies t^2 \equiv 1 \pmod{13}$. Substituting, we get $16 \cdot 1 \equiv s \cdot -4 \pmod{13}$. Therefore, $s \equiv -4 \pmod{13}$. From the first condition, we have $r = \frac{3}{2} \cdot s$, so $r \equiv -6 \pmod{13}$.

Therefore $r + s + t \equiv -6 - 4 - 1 \equiv -11 \equiv 2 \pmod{13}$.

We want to find $r + s + t - 3$, so our answer will have a remainder of -1 when divided by 13.

We divide 340 by 13 and find that the remainder is 2. Therefore the answer that will give us a remainder of -1 will be $340 - 3 = 337 = \boxed{\text{(B)} 337}$.

Solution 3 (Quick Solution)

Let r, s, t be the edge length of square R, S , and T respectively. Then we have

$$(r+1)^2 = \frac{9}{4}(s+1)^2 \quad (t+1)^2 = \frac{1}{4}((s+1)^2 + (r+1)^2 - (s+1))$$

Therefore

$$r = \frac{3s+1}{2} \quad t = \frac{1}{4}\sqrt{(s+1)(13s+9)} - 1$$

Therefore

$$\begin{aligned} r + s + t &= \frac{3s+1}{2} + s + \frac{1}{4}\sqrt{(s+1)(13s+9)} - 1 \\ &\approx \frac{5}{2}s + \frac{\sqrt{13}}{4}s - \frac{1}{2} \approx 3.4 \cdot s \end{aligned}$$

Given that average of the answer choices is around 340, therefore $s \approx 100$. Since t is an integer, $(s+1)(13s+9)$ must be a perfect square divisible by 16. Plugging in $s = 99$, $t = 89$ and $r = 149$. Therefore $r + s + t = 99 + 89 + 149 = 337$. The answer is (B) 337.

-fasterthanlight-Lunar_Paradox (Minor Edits)

§9 AMC 10 10B 2023

1. (*AMC 10 10B 2023*) Mrs. Jones is pouring orange juice into four identical glasses for her four sons. She fills the first three glasses completely but runs out of juice when the fourth glass is only $\frac{1}{3}$ full. What fraction of a glass must Mrs. Jones pour from each of the first three glasses into the fourth glass so that all four glasses will have the same amount of juice?
- (A) $\frac{1}{12}$ (B) $\frac{1}{4}$ (C) $\frac{1}{6}$ (D) $\frac{1}{8}$ (E) $\frac{2}{9}$

Answer: C

Solution 1

The first three glasses each have a full glass. Let's assume that each glass has "1 unit" of juice. It won't matter exactly how much juice everyone has because we're dealing with ratios, and that wouldn't affect our answer. The fourth glass has a glass that is one third. So the total amount of juice will be $1 + 1 + 1 + \frac{1}{3} = \frac{10}{3}$. If we divide the total amount of juice by 4, we get $\frac{5}{6}$, which should be the amount of juice in each glass. This means that each of the first three glasses will have to contribute $1 - \frac{5}{6} = \frac{1}{6}$ to the fourth glass.

Solution 2 (unnecessary numerical values)

Given that the first three glasses are full and the fourth is only $\frac{1}{3}$ full, let's represent their contents with a common denominator, which we'll set as 6. This makes the first three glasses $\frac{6}{6}$ full, and the fourth glass $\frac{2}{6}$ full.

To equalize the amounts, Mrs. Jones needs to pour juice from the first three glasses into the fourth. Pouring $\frac{1}{6}$ from each of the first three glasses will make them all $\frac{5}{6}$ full. Thus, all four glasses will have the same amount of juice. Therefore, the

answer is (C) $\frac{1}{6}$.

Solution 3 (simple algebra)

We let x denote how much juice we take from each of the first 3 children and give to the 4th child.

We can write the following equation: $1 - x = \frac{1}{3} + 3x$, since each value represents how much juice each child (equally) has in the end. (Each of the first three children

now have $1 - x$ juice, and the fourth child has $3x$ more juice on top of their initial $\frac{1}{3}$.)

Solving, we see that $x = \boxed{\text{(C)} \frac{1}{6}}$.

2. (AMC 10 10B 2023) Carlos went to a sports store to buy running shoes. Running shoes were on sale, with prices reduced by 20% on every pair of shoes. Carlos also knew that he had to pay a 7.5% sales tax on the discounted price. He had 43 dollars. What is the original (before discount) price of the most expensive shoes he could afford to buy?

(A) \$46 (B) \$50 (C) \$48 (D) \$47 (E) \$49

Answer: B

Solution 1

Let the price originally be x . Then, after a 20 percent discount, the price is now $x - \frac{1}{5}x = \frac{4}{5}x$.

From the discounted price $\frac{4}{5}x$, we now take $\frac{7.5}{100}$ of $\frac{4}{5}x$ and add it to $\frac{4}{5}x$, giving us $\frac{4}{5}x + \left(\frac{7.5}{100}\right)\left(\frac{4}{5}x\right) = \frac{4}{5}x + \frac{30}{500}x = \frac{4}{5}x + \frac{3}{50}x = \frac{43}{50}x$.

Now we write the inequality $\frac{43}{50}x \leq \$43$ and multiply by $\frac{50}{43}$ on both sides to get $x \leq \$50$.

We want the greatest original price, which would be 50 or option choice B.

Solution 2 (easy)

We can create the equation:

$$0.8x \cdot 1.075 = 43$$

using the information given. This is because x , the original price, got reduced by 20%, or multiplied by 0.8, and it also got multiplied by 1.075 on the discounted price. Solving that equation, we get

$$\frac{4}{5} \cdot x \cdot \frac{43}{40} = 43$$

$$\frac{4}{5} \cdot x \cdot \frac{1}{40} = 1$$

$$\frac{1}{5} \cdot x \cdot \frac{1}{10} = 1$$

$$x = \boxed{50}$$

Solution 3 (One-Step Equation)

The discounted shoe is 20% off the original price. So that means $1 - 0.2 = 0.8$. There is also a 7.5% sales tax charge, so $0.8 \cdot 1.075 = 0.86$. Now we can set up the equation $0.86x = 43$, and solving that we get $x = \boxed{\text{(B)} 50}$ kabbybear

Solution 4

Let the original price be x dollars. After the discount, the price becomes $80\% \cdot x$ dollars. After tax, the price becomes $80\% \times (1 + 7.5\%) = 86\% \cdot x$ dollars. So, $43 = 86\% \cdot x$, $x = \boxed{\text{(B)} \$50}$.

Solution 5

We can assign a variable c to represent the original cost of the shoes. Next, we set up the equation $80\% \cdot 107.5\% \cdot c = 43$. We can solve this equation for c and get **(B) \$50**.

Solution 6 (Intuition and Guessing)

We know the discount price will be $5/4$, and 0.075 is equal to $3/40$. So we look at answer choice **(B)**, see that the discount price will be 40 , and with sales tax applied it will be 43 , so the answer choice is **(B) \$50**.

3. (AMC 10 10B 2023) A $3 - 4 - 5$ right triangle is inscribed in circle A , and a $5 - 12 - 13$ right triangle is inscribed in circle B . What is the ratio of the area of circle A to the area of circle B ?

(A) $\frac{9}{25}$ (B) $\frac{1}{9}$ (C) $\frac{1}{5}$ (D) $\frac{25}{169}$ (E) $\frac{4}{25}$

Answer: D

Solution 1

Because the triangles are right triangles, we know the hypotenuses are diameters of circles A and B . Thus, their radii are 2.5 and 6.5 (respectively). Square the two numbers and multiply π to get 6.25π and 42.25π as the areas of the circles. Multiply 4 on both numbers to get 25π and 169π . Cancel out the π , and lastly, divide, to get your answer = **(D) $\frac{25}{169}$** .

Solution 1.5 (Way FASTER! Use in the contest)

With right triangles inscribed in circles, the hypotenuse must be the diameter.

Therefore, the ratio of radii is $\frac{5}{13}$ which means the area ratio is just $\frac{5^2}{13^2}$ so, **(D) $\frac{25}{169}$** .

-Mismatchedcubing/Andrew__

Note for Solution 1 The reason why the right triangle has its hypotenuse on the diameter is because of its properties with the circle and the central angle theorem. The central angle theorem states that an angle inscribed on a circle is just arc length in degrees. The diameter of a circle splits the circle into two halves with arc of 180° , and therefore the angle inscribed in that arc is just $180/2 = 90$ degrees. The longest side of a triangle is opposite to the longest angle, and therefore the length of the hypotenuse is just the diameter of the circle.

Solution 2

Since the arc angle of the diameter of a circle is 90 degrees, the hypotenuse of each these two triangles is respectively the diameter of circles A and B .

Therefore the ratio of the areas equals the radius of circle A squared : the radius of circle B squared = $0.5 \times$ the diameter of circle A , squared : $0.5 \times$ the diameter of circle B , squared = the diameter of circle A , squared : the diameter of circle B , squared = **(D) $\frac{25}{169}$** .

Solution 3

The ratio of areas of circles is the same as the ratios of the diameters squared (since they are similar figures). Since this is a right triangle the hypotenuse of each triangle will be the diameter of the circle. This yields the expression $\frac{5^2}{13^2} =$ **(D) $\frac{25}{169}$** .

4. (*AMC 10 10B 2023*) Jackson's paintbrush makes a narrow strip with a width of 6.5 millimeters. Jackson has enough paint to make a strip 25 meters long. How many square centimeters of paper could Jackson cover with paint?

(A) 162,500 (B) 162.5 (C) 1,625 (D) 1,625,000 (E) 16,250

Answer: C

Solution 1

6.5 millimeters is equal to 0.65 centimeters. 25 meters is 2500 centimeters. The answer is 0.65×2500 , so the answer is **(C) 1,625**.

Solution 2 (Scientific Form)

6.5 millimeters can be represented as 65×10^{-2} centimeters. 25 meters is 25×10^2 centimeters. Multiplying out these results in $(65 \times 10^{-2}) \times (25 \times 10^2)$, which is 65×25 making the answer **(C) 1,625**.

Remark A good way to remember metric conversions is with the phrase "**King Henry Died By Drinking Chocolate Milk**" (Kilo, Hecto, Dekka, Basic units, Deci, Centi, Milli). As you read from left to right you divide by 10. So 1 meter (which is a basic unit) is equivalent to 1000 millimeters, or 100 centimeters.

5. (*AMC 10 10B 2023*) Maddy and Lara see a list of numbers written on a blackboard. Maddy adds 3 to each number in the list and finds that the sum of her new numbers is 45. Lara multiplies each number in the list by 3 and finds that the sum of her new numbers is also 45. How many numbers are written on the blackboard?

(A) 10 (B) 5 (C) 6 (D) 8 (E) 9

Answer: A

Solution

Let there be n numbers in the list of numbers, and let their sum be S . Then we have the following

$$S + 3n = 45$$

$$3S = 45$$

From the second equation, $S = 15$. So, $15 + 3n = 45 \Rightarrow n = \mathbf{(A) 10}$.

Solution 2

Let $x_1, x_2, x_3, \dots, x_n$ where x_n represents the n th number written on the board. Lara's multiplied each number by 3, so her sum will be $3x_1 + 3x_2 + 3x_3 + \dots + 3x_n$. This is the same as $3 \cdot (x_1 + x_2 + x_3 + \dots + x_n)$. We are given this quantity is equal to 45, so the original numbers add to $\frac{45}{3} = 15$. Maddy adds 3 to each of the n terms which yields, $x_1 + 3 + x_2 + 3 + x_3 + 3 + \dots + x_n + 3$. This is the same as the sum of the original series plus $3 \cdot n$. Setting this equal to 45, $15 + 3n = 45 \Rightarrow n = \mathbf{(A) 10}$.

Solution 3

If the list of numbers written on the board is $a_1, a_2, a_3, \dots, a_n$, then we can formulate two equations:

$$3n + \sum_{i=1}^n a_i = 45$$

$$3 \sum_{i=1}^n a_i = 45$$

We can rewrite the first equation by multiplying both sides by 3:

$$3(3n + \sum_{i=1}^n a_i) = 3(45) \Rightarrow 9n + 3 \sum_{i=1}^n a_i = 135$$

Now, subtract the second equation from the first:

$$(9n + 3 \sum_{i=1}^n a_i) - (3 \sum_{i=1}^n a_i) = 135 - 45$$

$$\Rightarrow 9n = 135 - 45$$

$$\Rightarrow 9n = 90$$

$$\Rightarrow n = \boxed{\text{(A)} 10}$$

Solution 4 (Solution 1 but more in depth)

Notice how the problem tries to throw us off. We don't need to find the sum, but rather how many 3's do we need to remove to get to the sum.

We have that $3a + 3b + 3c + 3d + \dots = 45$. Dividing by 3 gives us $a + b + c + d + \dots = 15$.

Notice how we can write Maddy's list like $(a + b + c + d + \dots) + (3 + 3 + 3 + 3 + \dots) = 45$. We know that $a + b + c + d + \dots = 15$, and we know that the definition of adding 3's repeatedly is just multiplying 3 by how many numbers it appears, l , which is our list.

This gives us the equation $15 + 3l = 45$, and solving for l gives us $\boxed{\text{A.10}}$

This problem teaches us to look for patterns and use our elementary taught topics to our advantage.

Solution 5

Because the addition and multiplication of x were the same, we can see that $2x = 3$. Therefore $x = 1.5$ and $45(1.5 + 3) = \boxed{\text{A.10}}$

Credit to Pinotation for *LaTeX*

6. (AMC 10 10B 2023) Let $L_1 = 1$, $L_2 = 3$, and $L_{n+2} = L_{n+1} + L_n$ for $n \geq 1$. How many terms in the sequence $L_1, L_2, L_3, \dots, L_{2023}$ are even?

(A) 673 (B) 1011 (C) 675 (D) 1010 (E) 674

Answer: E

Solution 1

We calculate more terms:

$$1, 3, 4, 7, 11, 18, \dots$$

We find a pattern: if $n + 2$ is a multiple of 3, then the term is even, or else it is odd. There are $\left\lfloor \frac{2023}{3} \right\rfloor = \boxed{\text{(E)} 674}$ multiples of 3 from 1 to 2023.

Solution 2

Like in the other solution, we find a pattern, except in a more rigorous way. Since we start with 1 and 3, the next term is 4.

We start with odd, then odd, then (the sum of odd and odd) even, (the sum of odd and even) odd, and so on. Basically the pattern goes: odd, odd, even, odd odd, even, odd, odd even...

When we take $\frac{2023}{3}$ we get 674 with a remainder of one. So we have 674 full cycles, and an extra odd at the end.

Therefore, there are (E) 674 evens.

Solution 3

We know that $L_1 = 1$ and $L_2 = 3$.

To find L_3 , we put $n = 1$ into $L_{n+1} = L_n + L_{n-1}$.

This gives $L_3 = 4$. Continuing for L_4 , L_5 , and L_6 , we get

$$L_4 = 7$$

$$L_5 = 11$$

$$L_6 = 18.$$

We can now make an educated guess that this pattern will always remain the same, and evens will always appear on the L_{3n} th number. (Proof is below).

So, we take the $\lfloor 2023/3 \rfloor$, which gives E.674

Note that $\lfloor x \rfloor =$ to the nearest rounded down whole number.

Proof that all L_{3n} is even We have that L_1 is an odd number and L_2 is also an odd number. The sum of two odd numbers is even. Now $L_3 = L_2 + L_1$, so L_3 must be even. We continue with proof by contradiction. Say L_{3n} , $n \not\leq 1$ was not even. We take L_6 , for instance, and this should be odd. This means that because $L_6 = L_5 + L_4$, either L_5 or L_4 must be even, or else the number will be odd.

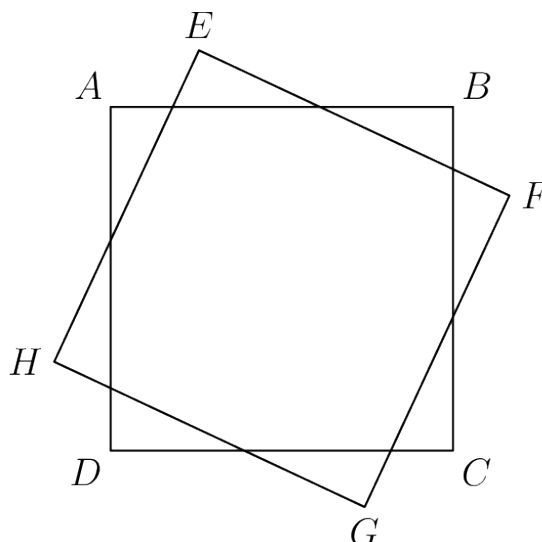
So given that L_3 is even, then $L_4 = L_3 + L_2$, which is an even + odd, which is odd. So now L_5 must be even for this to not work.

We know that L_4 is odd, and L_3 is even.

We have $L_5 = L_4 + L_3$, and that is again an odd + even case, showing that L_5 is also odd. Therefore, if L_{3n} , $n \not\leq 1$ is not even, there lies a contradiction.

Therefore, each L_{3n} number is even, and that is why we do $\lfloor 2023/3 \rfloor$ to get E.674

7. (AMC 10 10B 2023) Square $ABCD$ is rotated 20° clockwise about its center to obtain square $EFGH$, as shown below. What is the degree measure of $\angle EAB$?



- (A) 24° (B) 35° (C) 30° (D) 32° (E) 20°

Answer: B

Solution 1

First, let's call the center of both squares I . Then, $\angle AIE = 20$, and since $\overline{EI} = \overline{AI}$, $\angle AEI = \angle EAI = 80$. Then, we know that AI bisects angle $\angle DAB$, so $\angle BAI = \angle DAI = 45$. Subtracting 45 from 80, we get (B)35

Solution 2

First, label the point between A and H point O and the point between A and E point P . We know that $\angle AOP = 20$ and that $\angle A = 90$. Subtracting 20 and 90 from 180, we get that $\angle APO$ is 70. Subtracting 70 from 180, we get that $\angle OPB = 110$. From this, we derive that $\angle APE = 110$. Since triangle APE is an isosceles triangle, we get that $\angle EAP = (180 - 110)/2 = 35$. Therefore, $\angle EAB = 35$. The answer is (B)35.

Solution 3

Call the center of both squares point O , and draw circle O such that it circumscribes the squares. $\angle EOF = 90$ and $\angle BOF = 20$, so $\angleEOB = 70$. Since $\angle EAB$ is inscribed in arc $\widehat{EB}$, $\angle EAB = 70/2 =$ (B) 35.

Solution 4

Draw EA : we want to find $\angle EAB$. Call P the point at which AB and EH intersect. Reflecting $\triangle APE$ over EA , we have a parallelogram. Since $\angle EPB = 70^\circ$, angle subtraction tells us that two of the angles of the parallelogram are 110° . The other two are equal to $2\angle EAB$ (by properties of reflection).

Since angles on the transversal of a parallelogram sum to 180° , we have $2\angle EAB + 110 = 180$, yielding $\angle EAB =$ (B) 35

-Benedict T (countmath1)

Solution 5 (Educated Guess)

We call the point where AB and EH intersect I . We can make an educated guess that triangle AEI is isosceles so $AI = EI$, $\angle AIE = 110^\circ$, $\angle AIH = 20^\circ$, and $\angle EIB = 70^\circ$. So, we get $\angle EAI$ is $(180^\circ - 110^\circ)/2 =$ (B) 35.

Solution 6 (Solution 1 Remastered)

Like in solution 1, we label the center of the squares R .

Draw the lines AR and RE . Because Vertex A is just Vertex E shifted 20° about the center R , $\angle ARE = 20^\circ$.

Now, notice how AM and ME are the diagonals (or half the diagonals, doesn't really matter) of the squares $ABCD$ and $EFGH$ respectively. This implies that $AM = ME$, and, connecting a new segment AE , we see that $\triangle ARE$ is isosceles, and $AR = AE$, implying that $\angle EAM \cong \angle MEA$. We can easily solve this through the sum of interior angles of a triangle, and we get $m\angle EAM = m\angle MEA = 80^\circ$.

Because AM is a diagonal of $\square ABCD$, that means that $\angle BAC$ is bisected, and $\angle MAD = 45^\circ$.

Finally, to find $\angle EAB$, we are just subtracting 80° and 45° , giving us $80^\circ - 45^\circ = 35^\circ$ or $\boxed{B}$.

Solution 7 (Overkill complex bash)

Let the center be the origin of the complex plane. WLOG, let $A = -1 + i$. Then, $E = (-1 + i)e^{-\frac{i\pi}{9}} = (-\cos(20^\circ) + \sin(20^\circ)) + i(\sin(20^\circ) + \cos(20^\circ))$. So, $\angle EAB = \arctan\left(\frac{\sin(20^\circ) + \cos(20^\circ) - 1}{-\cos(20^\circ) + \sin(20^\circ) + 1}\right)$, because the slope is just the tangent of the angle. Now, we have a bunch of trig bashing. Let $\theta = 20^\circ$. Then we have inside the inverse tangent, $\frac{\sin(\theta) + \cos(\theta) - 1}{-\cos(\theta) + \sin(\theta) + 1}$. Using $\sin(\theta) = 2\cos(\frac{\theta}{2})\sin(\frac{\theta}{2})$, $\cos(\theta) = \cos^2(\frac{\theta}{2}) - \sin^2(\frac{\theta}{2})$, and the Pythagorean identity, this ultimately simplifies to $\frac{1 - \tan(\frac{\theta}{2})}{1 + \tan(\frac{\theta}{2})}$, which can be recognized using the tangent angle subtraction formula as $\tan(45^\circ - \frac{\theta}{2})$, since $\tan(45^\circ) = 1$. Therefore, inside of the inverse tangent we have $\tan(35^\circ)$, so the inverse tangent of that is 35° , or $\boxed{B}$.

8. (AMC 10 10B 2023) What is the units digit of $2022^{2023} + 2023^{2022}$?
 (A) 7 (B) 1 (C) 9 (D) 5 (E) 3

Answer: A

Solution 1

$$2022^{2023} + 2023^{2022} \equiv 2^3 + 3^2 \equiv 17 \equiv 7 \pmod{10}. \quad \boxed{A} \text{ andliu766}$$

The method omits many steps, and it is important to note that we do not have $a \equiv b \pmod{m} \Rightarrow c^a \equiv c^b \pmod{m}$. For example, $5 \equiv 2 \pmod{3}$, but $2^5 \equiv 2 \not\equiv 1 \equiv 2^2 \pmod{3}$.

Solution 2

When looking at the units digit patterns of the powers of 2, we see that

$$2^1 =, \text{ units digit } 2 \quad 2^2 =, \text{ units digit } 4 \quad 2^3 =, \text{ units digit } 8 \quad 2^4 =, \text{ units digit } 6 \quad 2^5 =, \text{ units digit } 2$$

And the pattern repeats. This pattern will apply for the powers of 2022 as well, since the units digit of 2022 is 2. We can find the pattern for the powers of 3 too. The pattern follows with units digits, 3, 9, 7, 1, 3, 9, ... Similarly, the units digit of 2023 will follow the same pattern as the powers of 3.

Both of these powers cycle in groups of 4. When dividing 2023 by 4, we get 505 remainder 3, meaning 505 complete cycles; or the power being a multiple of 4, 505 times, and 3 extra. So the units digit of 2022^{2023} is 8. 2022 divided by 4 is 505

remainder 2, which means 505 complete cycles, or the power being a multiple of 4, 505 times, and 2 extra. So the units digit of 2023^{2022} is 9.

We only need to find the units digit in the end, so we just add those 2 already found units digits, to get a new units digit of 7. Therefore the answer is (A) 7

Solution 3 (Digit Cycles)

Note that the units digit will be the same regardless of the tens, hundreds, and thousands digits, so we can simplify this problem to finding the last digit of $2^{2023} + 3^{2022}$. We can find the units digit of 2^{2023} , by listing the units digits of the first few powers of two, and trying to find a pattern.

$$2^1 = 2 \quad 2^2 = 4 \quad 2^3 = 8 \quad 2^4 = 6 + 10 \quad 2^5 = 2 + 30 \quad 2^6 = 4 + 60$$

As we can see the units digits of powers of two repeat after every four iterations. Now we know the units digit of 2^{2020} is 6 and the units digit of $2^{2023} \Rightarrow 2^3 \cdot 2^{2020} \Rightarrow 6 \cdot 8 \Rightarrow 8$. Similarly we can find the last digits of powers of three repeat after every four, so the units digit of 3^{2022} is $1 \cdot 3^2 = 9$. Adding these together, the ones digit is the same as the ones digit of $9 + 8$ which is 7. (A) 7

Solution 4 (Bashy)

$$\begin{aligned} 2022^{2023} &\equiv 2^{2023} = 128^{289} \equiv 8^{289} = 2^{289 \times 3} = 131072^{51} \equiv 2^{51} = 131072^3 \equiv 2^3 = 8 \\ (\text{mod } 10) 2023^{2022} &\equiv 3^{2022} = 9^{1011} \equiv (-1)^{1011} = -1 \equiv 9 \pmod{10} \quad 8 + 9 = 17 \equiv 7 \pmod{10} \end{aligned}$$

9. (AMC 10 10B 2023) The numbers 16 and 25 are a pair of consecutive positive squares whose difference is 9. How many pairs of consecutive positive perfect squares have a difference of less than or equal to 2023?

(A) 674 (B) 1011 (C) 1010 (D) 2019 (E) 2017

Answer: B

Solution 1

Let x be the square root of the smaller of the two perfect squares. Then, $(x+1)^2 - x^2 = x^2 + 2x + 1 - x^2 = 2x + 1 \leq 2023$. Thus, $x \leq 1011$. So there are (B)1011 numbers that satisfy the equation.

A very similar solution offered by darrenn.cp and DarkPheonix has been combined with Solution 1.

Minor corrections by milquetoast

Note from milquetoast: Alternatively, you can let x be the square root of the larger number, but if you do that, keep in mind that $x = 1$ must be rejected, since $(x - 1)$ cannot be 0.

Solution 2

The smallest number that can be expressed as the difference of a pair of consecutive positive squares is 3, which is $2^2 - 1^2$. The largest number that can be expressed as the difference of a pair of consecutive positive squares that is less than or equal to 2023 is 2023, which is $1012^2 - 1011^2$. These numbers are in the form $(x+1)^2 - x^2$, which is just $2x + 1$. These numbers are just the odd numbers from 3 to 2023, so there are $\lceil (2023 - 3)/2 \rceil + 1 = 1011$ such numbers. The answer is (B)1011.

Solution 3

The difference of 2 consecutive positive integer squares is always an odd number (proved from above). Therefore, the total number of cases is $\left\lfloor \frac{2023}{2} \right\rfloor = \boxed{B.1011}$

Solution 4

Notice that the difference of two consecutive squares can be represented as $(x+1)^2 - x^2$. Using the Difference of Squares Formula, we simplify to get $(2x+1)(1) = 2x+1$. The number given in the problem was 2023, and thus we find that $x = \frac{2023-1}{2} \Rightarrow \boxed{(B) 1011}$.

Solution 5 (In Depth Explanation)

Any two consecutive positive squares look like

$$n^2 \quad \text{and} \quad (n+1)^2.$$

Their difference is

$$(n+1)^2 - n^2.$$

.

Expand:

$$(n^2 + 2n + 1) - n^2 = 2n + 1.$$

So the difference between consecutive squares is always an odd number:

$$3, 5, 7, 9, 11, \dots$$

For example:

$$3^2 - 2^2 = 9 - 4 = 5$$

and

$$5^2 - 4^2 = 25 - 16 = 9.$$

.

We need

$$2n + 1 \leq 2023.$$

Subtract 1:

$$2n \leq 2022.$$

Divide by 2:

$$n \leq 1011.$$

Since n is a positive integer,

$$n = 1, 2, 3, \dots, 1011.$$

That's 1011 values of n .

Therefore the answer is $\boxed{B}$.

Key AMC trick: Whenever you see consecutive squares, immediately think

$$\boxed{(n+1)^2 - n^2 = 2n + 1}.$$

That turns what looks like a square-number problem into a simple inequality.

10. (AMC 10 10B 2023) You are playing a game. A 2×1 rectangle covers two adjacent squares (oriented either horizontally or vertically) of a 3×3 grid of squares, but you are not told which two squares are covered. Your goal is to find at least one square that is covered by the rectangle. A "turn" consists of you guessing a square, after which you are told whether that square is covered by the hidden rectangle. What is the minimum number of turns you need to ensure that at least one of your guessed squares is covered by the rectangle?
- (A) 3 (B) 5 (C) 4 (D) 8 (E) 6

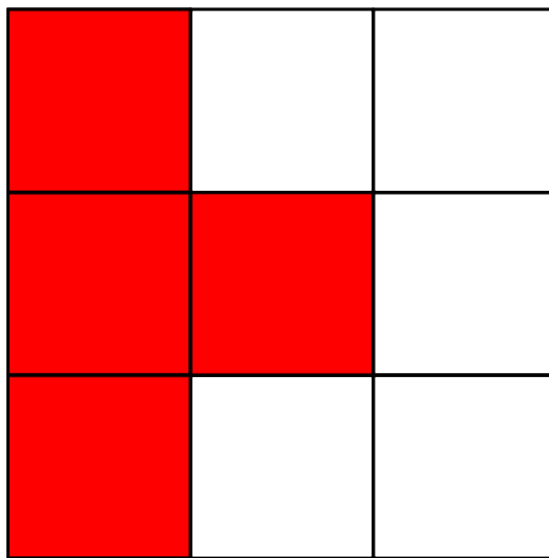
Answer: C

Solution 1

Notice that the 3×3 square grid has a total of 12 possible 2×1 rectangles.

Suppose you choose the middle square for one of your turns. The middle square is covered by 4 rectangles, and each of the remaining 8 squares is covered by a maximum of 2 uncounted rectangles. This means that the number of turns is at least $1 + \frac{12-4}{2} = 1 + 4 = 5$.

Now suppose you don't choose the middle square. The squares on the middle of the sides are covered by at most 3 uncounted rectangles, and the squares on the corners are covered by at most 2 uncounted rectangles. In this case, we see that the least number of turns needed to account for all 12 rectangles is $12 \div 3 = 4$. To prove that choosing only side squares indeed does cover all 12 rectangles, we need to show that the 3 rectangles per square that cover each side square do not overlap. Drawing the rectangles that cover one square, we see they form a T shape and they do not cover any other side square. Hence, our answer is 4.



Solution 2

First, note that since the rectangle covers 2 squares, we only need to guess squares that are not adjacent to any of our other guesses. To minimize the amount of guesses, each of our guessed squares should try to touch another guess on one vertex and one vertex only. There are only two ways to do this: one with 5 guesses, and one with 4. Since the problem is asking for the minimum number, the answer is

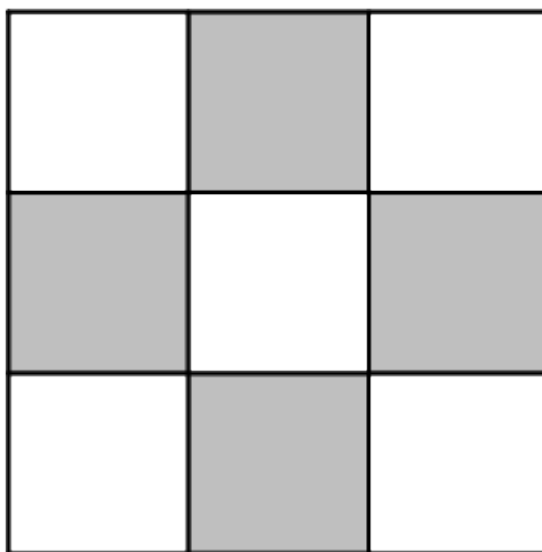
(C) 4.

Solution 3

Since the hidden rectangle can only hide two adjacent squares, we may think that we eliminate 8 squares and we're done, but think again. This is the AMC 10, so there must be a better solution (also note that every other solution choice is below 8 so we're probably not done) So, we think again, we notice that we haven't used the adjacent condition, and then it clicks. If we eliminate the four squares with only one edge on the boundary of the 9×9 square. We are left with 5 diagonal squares, since our rectangle can't be diagonal, we can ensure that we find it in 4 moves. So our answer is : (C) 4

Solution 4 (Checkerboard)

The 3×3 grid can be colored like a checkerboard with alternating black and white squares. Let the top left square be white, and the rest of the squares alternate colors, as shown below:



Each 2×1 rectangle always covers 1 white square and 1 black square. You can ensure that at least one of your guessed squares is covered by the rectangle by choosing either each of the white squares (5 turns) or each of the black squares (4 turns).

Since it is ideal to be the most efficient with our turns, by choosing all the black squares, we guarantee that one of the (C) 4 squares are of the 2×1 rectangle.

(Minor edits by NSAoPS)

Solution 5 (Logic)

We realize that every 2×1 rectangle must contain an edge and no more than one edge. There are a total of four edges so the answer is (C) 4. darrenn.cp

Solution 6 (Very Fast)

Draw the 3×3 grid. We want to minimize our options, so we look at the square shared by the most 2×1 rectangles. This is the center square. So, to guarantee our success, we choose the four edges from the center square, so our answer is C.4

11. (AMC 10 10B 2023) Suzanne went to the bank and withdrew \$800. The teller gave her this amount using \$20 bills, \$50 bills, and \$100 bills, with at least one of each

denomination. How many different collections of bills could Suzanne have received?

- (A) 45 (B) 21 (C) 36 (D) 28 (E) 32

Answer: B

Solution 1

We let the number of \$20, \$50, and \$100 bills be a, b , and c , respectively.

We are given that $20a + 50b + 100c = 800$. Dividing both sides by 10, we see that $2a + 5b + 10c = 80$.

We divide both sides of this equation by 5: $\frac{2}{5}a + b + 2c = 16$. Since $b + 2c$ and 16 are integers, $\frac{2}{5}a$ must also be an integer, so a must be divisible by 5. Let $a = 5d$, where d is some positive integer.

We can then write $2 \cdot 5d + 5b + 10c = 80$. Dividing both sides by 5, we have $2d + b + 2c = 16$. We divide by 2 here to get $d + \frac{b}{2} + c = 8$. $d + c$ and 8 are both integers, so $\frac{b}{2}$ is also an integer. b must be divisible by 2, so we let $b = 2e$.

We now have $2d + 2e + 2c = 16 \implies d + e + c = 8$. Every substitution we made is part of a bijection (i.e. our choices were one-to-one); thus, the problem is now reduced to counting how many ways we can have d, e , and c such that they add to 8.

We still have another constraint left, that each of d, e , and c must be at least 1. For $n \in \{d, e, c\}$, let $n' = n - 1$. We are now looking for how many ways we can have $d' + e' + c' = 8 - 1 - 1 - 1 = 5$.

We use a classic technique for solving these sorts of problems: stars and bars. We have 5 stars and 3 groups, which implies 2 bars. Thus, the total number of ways is $\binom{5+2}{2} = \binom{7}{2} = 21$.

Solution 2

Denote by x, y, z the amount of \$20 bills, \$50 bills and \$100 bills, respectively. Thus, we need to find the number of tuples (x, y, z) with $x, y, z \in \mathbb{N}$ that satisfy

$$20x + 50y + 100z = 800.$$

First, this equation can be simplified as

$$2x + 5y + 10z = 80.$$

Second, we must have $5|x$. Denote $x = 5x'$. The above equation can be converted to

$$2x' + y + 2z = 16.$$

Third, we must have $2|y$. Denote $y = 2y'$. The above equation can be converted to

$$x' + y' + z = 8.$$

Denote $x'' = x' - 1$, $y'' = y' - 1$ and $z'' = z - 1$. Thus, the above equation can be written as

$$x'' + y'' + z'' = 5.$$

Therefore, the number of non-negative integer solutions (x'', y'', z'') is $\binom{5+3-1}{3-1} = \boxed{\text{(B) 21}}$.

Solution 3

To start, we simplify things by dividing everything by 10, the resulting equation is $2x + 5y + 10z = 80$, and since the problem states that we have at least one of each, we simplify this to $2x + 5y + 10z = 63$. Note that since the total is odd, we need an odd number of 5 dollar bills. We proceed using casework.

Case 1: One 5 dollar bill

$2x + 10z = 58$, we see that $10z$ can be 0, 10, 20, 30, 40, 50 or 6 ways

Case 2: Three 5 dollar bills

$2x + 10z = 48$, like before we see that $10z$ can be 0, 10, 20, 30, 40, so 5 ways

Now we should start to see a pattern emerge, each case there is 1 less way to sum to 80, so the answer is just $\frac{6(6+1)}{2} = \boxed{\text{(B) 21}}$.

Solution 4

We notice that each \$100 can be split 3 ways: 5 \$20 dollar bills, 2 \$50 dollar bills, or 1 \$100 dollar bill.

There are 8 of these \$100 chunks in total—take away 3 as each split must be used at least once.

Now there are five left—so we use stars and bars.

5 chunks, 3 categories or 2 bars. This gives us $\binom{5+2}{2} = \boxed{\text{(B) 21}}$

Solution 5 (generating functions)

The problem is equivalent to the number of ways to make \$80 from \$2 bills, \$5 bills, and \$10 bills. We can use generating functions to find the coefficient of x^{80} :

The \$2 bills provide $1 + x^2 + x^4 + x^6 \cdots = \frac{1}{1-x^2}$,

The \$5 bills provide $1 + x^5 + x^{10} + x^{15} \cdots = \frac{1}{1-x^5}$,

The \$10 bills provide $1 + x^{10} + x^{20} + x^{30} \cdots = \frac{1}{1-x^{10}}$.

Multiplying, we get $(1-x^2)^{-1}(1-x^5)^{-1}(1-x^{10})^{-1}$.

Solution 6 (easy logic)

There aren't dollar signs because the *latex* thinks they're latex symbols. If you find how to override this error, please edit this. There's no *latex* here but feel free to add some! SwordAxe

We can see in the problem that the teller gave her at least one of \$20, \$50, and \$100. Therefore, she has $800 - 20 - 50 - 100 = 630$ "left over".

Since all bills and 630 are multiples of 10, we can divide by ten. $\implies$ Question becomes: How many different collections of 2, 5, and 10 could she get if her total was 63?

We notice that because 63 is odd, we need an odd amount of 5 bills. (2 and 10 are both even, and 63 is not a multiple of 5, so we need 2 and/or 10 bills. PM SwordAxe if you don't get this.)

We can do casework.

1: She gets one 5 (50) dollar bill. She has 58 (580) left.

1) She is given only 2 dollar (20) bills => ONE COLLECTION (all 20 bills with one 50

This looks very tedious, but draw a simple tree diagram, and you'll see that its very easy!

If she gets one 50, there are 6 ways If she gets three 50, there are 5 ways... If she gets nine 50, there are 2 ways If she gets eleven 50, there is one way

We can add them all up, with a grand sum of $6+5+4+3+2+1 = 21$.

Therefore, we get the answer (B) 21.

articleamsmath, amssymbamsthmenumitemtcolorbox % Package to box the final answer

Solution 7

Because the problem states that at least one of each type of bill must be used, we can subtract 20, 50, and 100 from 800, giving us 630. Let the new amount of 20 bills be x , 50 bills be y , and 100 bills be z . Notice that there must be more than one type of bill used, as 630 cannot be made from just 20, 50, or 100. Now, we can do the casework.

Case 1: Uses 20 and 50

$$20x + 50y = 630 \quad 2x + 5y = 63 \quad y \text{ can be } 1, 3, 5, 7, 9, \text{ or } 11 \Rightarrow 6 \text{ Solutions}$$

Case 2: 50 and 100

Impossible, cannot make 630

Case 3: 20 and 100

Impossible, cannot make 630

Case 4: 20, 50, and 100

$$20x + 50y + 100z = 630 \quad 2x + 5y + 10z = 63$$

$$\text{Subcase 1: If } z = 1 \quad 2x + 5y = 53 \quad y \text{ can be } 1, 3, 5, 7, \text{ or } 9 \Rightarrow 5 \text{ solutions}$$

$$\text{Subcase 2: If } z = 2 \quad 2x + 5y = 43 \quad y \text{ can be } 1, 3, 5, \text{ or } 7 \Rightarrow 4 \text{ solutions}$$

In total, for Case 4, there is $5+4+3+2+1 = 15$ solutions. Thus, in total, there are $6 + 15 = \boxed{\text{(B) } 21}$ solutions.

Solution 8 (Simple casework and logic)

Simple casework and logic is sufficient to solve this problem.

First off, note that the number of 20 dollar bills must be a multiple of 5, since the total amount is 600 dollars and the other denominations are unable to turn an amount such as 40 dollars into a multiple of 50 or 100 dollars. By similar logic, the number of 50 dollar bills must be a multiple of 2.

Now, simple casework will solve this problem. Assume she already received a single 100 dollar bill, two 50 dollar bills, and five 20 dollar bills, since there must be at least one of each denomination. Then, there will be 500 dollars that haven't been accounted for. Now we just have to solve for the number of ways to get 500 using 20's, 50's, and 100's.

If there are no more 100's, then there can be 0, 2, 4, 6, 8, or 10 50's for a total of 6

ways.

If there is one 100, there can be 0, 2, 4, 6, or 8 50's.

Doing this for each amount of 100's until there are 5 hundreds yields the final answer of $6 + 5 + 4 + 3 + 2 + 1 = \boxed{\text{(B) } 21}$.

12. (AMC 10 10B 2023) When the roots of the polynomial

$$P(x) = (x-1)^1(x-2)^2(x-3)^3 \cdots (x-10)^{10}$$

are removed from the number line, what remains is the union of 11 disjoint open intervals. On how many of these intervals is $P(x)$ positive?

- (A) 3 (B) 7 (C) 6 (D) 4 (E) 5

Answer: C

Solution 1

The expressions to the power of even powers are always positive, so we don't need to care about those. We only need to care about $(x-1)^1(x-3)^3(x-5)^5(x-7)^7(x-9)^9$. We need 0, 2, or 4 of the expressions to be negative. The 9 through 10 interval and 10 plus interval make all of the expressions positive. The 5 through 6 and 6 through 7 intervals make two of the expressions negative. The 1 through 2 and 2 through 3 intervals make four of the expressions negative. There are $\boxed{\text{(C) } 6}$ intervals.

Solution 2

The roots of the factorized polynomial are intervals from numbers 1 to 10. We take each interval as being defined as the number behind it. To make the function positive, we need to have an even number of negative expressions. Real numbers raised to even powers are always positive, so we only focus on $x-1$, $x-3$, $x-5$, $x-7$, and $x-9$. The intervals 1 and 2 leave 4 negative expressions, so they are counted. The same goes for intervals 5, 6, 9, and 10. Intervals 3 and 4 leave 3 negative expressions and intervals 7 and 8 leave 1 negative expression. The solution is the number of intervals which is $\boxed{\text{(C) } 6}$.

Solution 3

We can use the turning point behavior at the roots of a polynomial graph to find out the amount of intervals that are positive.

First, we evaluate any value on the interval $(-\infty, 1)$. Since the degree of $P(x)$ is $1 + 2 + \dots + 9 + 10 = \frac{10 \times 11}{2} = 55$, and every term in $P(x)$ is negative, multiplying 55 negatives gives a negative value. So $(-\infty, 0)$ is a negative interval.

We know that the roots of $P(x)$ are at 1, 2, ..., 10. When the degree of the term of each root is odd, the graph of $P(x)$ will pass through the graph and change signs, and vice versa. So at $x = 1$, the graph will change signs; at $x = 2$, the graph will not, and so on.

This tells us that the interval (1, 2) is positive, (2, 3) is also positive, (3, 4) is negative, (4, 5) is also negative, and so on, with the pattern being +, +, -, -, +, +, -, -, ...

The positive intervals are therefore (1, 2), (2, 3), (5, 6), (6, 7), (9, 10), and (10, ∞), for a total of $\boxed{\text{(C) } 6}$.

Solution 4

Denote by I_k the interval $(k-1, k)$ for $k \in \{2, 3, \dots, 10\}$ and I_1 the interval $(-\infty, 1)$.

Therefore, the number of intervals that $P(x)$ is positive is

$$\begin{aligned} 1 + \sum_{i=1}^{10} \mathbb{I} \left\{ \sum_{j=i}^{10} j \text{ is even} \right\} &= 1 + \sum_{i=1}^{10} \mathbb{I} \left\{ \frac{(i+10)(11-i)}{2} \text{ is even} \right\} \\ &= 1 + \sum_{i=1}^{10} \mathbb{I} \left\{ \frac{-i^2 + i + 110}{2} \text{ is even} \right\} \\ &= 1 + \sum_{i=1}^{10} \mathbb{I} \left\{ \frac{i^2 - i}{2} \text{ is odd} \right\} \\ &= \boxed{\text{(C) } 6}. \end{aligned}$$

Solution 5: Graphing

Recall two key facts about the roots of a polynomial:- If a root has an odd multiplicity (e.g. the root appears an odd number of times), then the graph will cross the x-axis.- If a root has an even multiplicity (e.g. the root appears an even number of times), then the graph will bounce off the x-axis.

Sketching the graph and noting the multiplicity of the roots, we see that there are $\boxed{\text{(C) } 6}$ positive intervals.

Solution 6: Solution 3 with powers

When the range of $P(x)$ is from $-\infty$ to 1, the sign is negative because in

$(x-1)^1(x-2)^2(x-3)^3 \cdots (x-10)^{10}$, the terms corresponding to $x = 1, 3, 5, 7$, and 9 are all negative, and since they have odd exponents, they contribute a negative sign. The other terms with even exponents (like $(x-2)^2$, $(x-4)^4$, etc.) are positive regardless of the sign inside, so they don't affect the overall sign. However, in the range 1 to 2, the power of 1 flips, causing this range to be positive. For the range 2 - 3, the sign of any even inside the exponent doesn't matter, so this range stays positive. Continuing this pattern, We see the pattern is $+, +, -, -, \dots$, and so on. there are 3 complete cycles, so $3 * 2 = \boxed{\text{(C) } 6}$ BlueKite

13. (AMC 10 10B 2023) What is the area of the region in the coordinate plane defined by

$$||x| - 1| + ||y| - 1| \leq 1?$$

- (A) 2 (B) 8 (C) 4 (D) 15 (E) 12

Answer: B

Solution 1

First consider, $|x-1| + |y-1| \leq 1$. We can see that it is a square with a side length of $\sqrt{2}$ (diagonal 2). The area of the square is $\sqrt{2}^2 = 2$.

Next, we insert an absolute value sign into the equation and get $|x-1| + ||y|-1| \leq 1$. This will double the square reflecting over x-axis.

So now we have 2 squares.

Finally, we add one more absolute value and obtain $||x|-1| + ||y|-1| \leq 1$. This will double the squares as we reflect the 2 squares we already have over the y-axis.

Concluding, we have 4 congruent squares. Thus, the total area is $4 \cdot 2 = \boxed{\text{(B)} 8}$

Solution 2 (Graphing)

We first consider the lattice points that satisfy $||x| - 1| = 0$ and $||y| - 1| = 1$. The lattice points satisfying these equations are $(1, 0), (1, 2), (1, -2), (-1, 0), (-1, 2),$ and $(-1, -2)$. By symmetry, we also have points $(0, 1), (2, 1), (-2, 1), (0, -1), (2, -1),$ and $(-2, -1)$ when $||x| - 1| = 1$ and $||y| - 1| = 0$. Graphing and connecting these points, we form 5 squares. However, we can see that any point within the square in the middle does not satisfy the given inequality (take $(0, 0)$, for instance). As noted in the above solution, each square has a diagonal 2 for an area of $\frac{2^2}{2} = 2$, so the total area is $4 \cdot 2 = \boxed{\text{(B)} 8}$.

Note This problem is very similar to a past AIME problem (1997 P13)

https://artofproblemsolving.com/wiki/index.php/1997_AIME_Problems/Problem_13

Solution 3 (Logic)

The value of $|x|$ and $|y|$ can be a maximum of 1 when the other is 0. Therefore the value of x and y range from -2 to 2. This forms a diamond shape which has area $4 \times \frac{2^2}{2}$ which is $\boxed{\text{(B)} 8}$.

Solution 4

We start by considering the graph of $|x| + |y| \leq 1$. To get from this graph to $||x| - 1| + ||y| - 1| \leq 1$ we have to translate it by ± 1 on the x axis and ± 1 on the y axis.

Graphing $|x| + |y| \leq 1$ we get a square with side length of $\sqrt{2}$, so the area of one of these squares is just 2.

We have to multiply by 4 since there are 4 combinations of shifting the x and y axis.

So we have 2×4 which is $\boxed{\text{(B)} 8}$.

Solution 5 (Desperate)

There are 2 sets of 2 absolute value bars. Each of those 2 absolute value bars can take on 2 values, so we have $2 \cdot 2 \cdot 2 = 8$ cases. We guess that the answer is divisible by 8. The only answer choice that is divisible by 8 is $\boxed{\text{(B)} 8}$.

14. (AMC 10 10B 2023) How many ordered pairs of integers (m, n) satisfy the equation $m^2 + mn + n^2 = m^2n^2$?

(A) 7 (B) 1 (C) 3 (D) 6 (E) 5

Answer: C

Solution 1

Let's use 10th grade math to solve this. After all, it is called the AMC 10 for a reason!

We have

$$m^2 + mn + n^2 = m^2n^2.$$

We subtract mn on both sides to get

$$m^2 + n^2 = m^2n^2 - mn.$$

Fun Fact! You can write $m^2 + n^2$ as $(m + n)^2 - 2mn$! Let's use this!

We convert the Left Hand Side into $(m+n)^2 - 2mn$ to get $(m+n)^2 - 2mn = m^2n^2 - mn$.

Adding by $2mn$ gives us $(m+n)^2 = m^2n^2 + mn$.

We aren't done yet though! $m^2n^2 + mn$ can be simplified into $mn(mn+1)$, giving us $(m+n)^2 = mn(mn+1)$.

Okay so now we're done, but, Pinotation, this doesn't do anything! Well, now we can use the Zero Product Property!

How? I'll show you!

We subtract $mn(mn+1)$ on both sides of the equation to get $(m+n)^2 - mn(mn+1) = 0$. Now we do a bit of casework.

Notice how $(m+n)^2 - mn(mn+1) = 0$ is just $(m+n)(m+n) - mn(mn+1) = 0$. So, either $m+n = 0$ and $mn = 0$, or $m+n = 0$ and $mn+1 = 0$. Let's look at it through both cases.

Case 1: $m+n = 0$ and $mn = 0$. If $m+n = 0$ and $mn = 0$, then that must mean that either $m = 0$ or $n = 0$, and if we substitute either $m = 0$ or $n = 0$ in, we still get either $m = 0$ or $n = 0$, so therefore we have 1 ordered pair, $(0, 0)$.

Case 2: $m+n = 0$ and $mn+1 = 0$. $mn+1 = 0$ means that $mn = -1$. So, for this to be possible, either $m = -1$ and $n = 1$, or $m = 1$ and $n = -1$. Let's check for contradictions quickly. We see that $m+n = 0$, and $-1+1 = 0$ and $1-1 = 0$, so we know the ordered pairs $(-1, 1)$ and $(1, -1)$ both work.

We have a total of (C) 3 ordered pairs. $(-1, 1)$, $(0, 0)$, and $(1, -1)$.

Solution 2

Clearly, $m = 0, n = 0$ is one of the solutions. However, we can be quite sure that there are more, so we apply **Simon's Favorite Factoring Trick** to get the following:

$$\begin{aligned} m^2 + mn + n^2 &= m^2n^2 \\ m^2 + mn + n^2 + mn &= m^2n^2 + mn \\ (m+n)^2 &= m^2n^2 + mn \\ (m+n)^2 &= mn(mn+1). \end{aligned}$$

Essentially, this says that the product of two consecutive numbers $mn, mn+1$ must be a perfect square. This is impossible except for when $mn = 0$ or $mn+1 = 0$ because mn and $mn+1$ cannot share any factors of 2 or above since they they will always have a different remainder mod the others factor, so they are relatively prime. This means that $mn(mn+1)$ cannot be a perfect square as they do not share any factors, since this means that any factor can have a maximum exponent of one, unless the product is 0. So we have 2 cases, $mn+1=0$ and $mn=0$. $mn = 0$ gives $(0, 0)$. $mn = -1$ gives $(1, -1), (-1, 1)$. Answer: (C) 3.

Solution 3

Case 1: $mn = 0$.

In this case, $m = n = 0$.

Case 2: $mn \neq 0$.

Denote $k = \gcd(m, n)$. Denote $m = ku$ and $n = kv$. Thus, $\gcd(u, v) = 1$.

Thus, the equation given in this problem can be written as

$$u^2 + uv + v^2 = k^2 u^2 v^2.$$

Modulo u , we have $v^2 \equiv 0 \pmod{u}$. Because $\gcd(u, v) = 1$, we must have $|u| = |v| = 1$. Plugging this into the above equation, we get $2 + uv = k^2$. Thus, we must have $uv = -1$ and $k = 1$.

Thus, there are two solutions in this case: $(m, n) = (1, -1)$ and $(m, n) = (-1, 1)$.

Putting all cases together, the total number of solutions is (C) 3.

Solution 4 (Discriminant)

We can move all terms to one side and write the equation as a quadratic in terms of n to get

$$(1 - m^2)n^2 + (m)n + (m^2) = 0.$$

The discriminant of this quadratic is

$$\Delta = m^2 - 4(1 - m^2)(m^2) = m^2(4m^2 - 3).$$

For n to be an integer, we must have $m^2(4m^2 - 3)$ be a perfect square. Thus, either $(2m)^2 - 3$ is a perfect square or $m^2 = 0$ and $m = 0$. The first case gives $m = -1, 1$ (larger squares are separated by more than 3), which result in the equations $-n + 1 = 0$ and $n - 1 = 0$, for a total of two pairs: $(-1, 1)$ and $(1, -1)$. The second case gives the equation $n^2 = 0$, so it's only pair is $(0, 0)$. In total, the total number of solutions is (C) 3.

Solution 5 (Nice Substitution)

Let $x = m + n, y = mn$ then

$$x^2 - y = y^2$$

Completing the square in y and multiplying by 4 then gives

$$4x^2 + 1 = (2y + 1)^2$$

Since the RHS is a square, clearly the only solutions are $x = 0, y = 0$ and $x = 0, y = -1$.

The first gives $(0, 0)$.

The second gives $(-1, 1)$ and $(1, -1)$ by solving it as a quadratic with roots m and n .

Thus there are (C) 3 solutions.

Solution 6 (Alternative Method for Manipulation)

$$\begin{aligned} m^2 + mn + n^2 &= m^2 n^2 \quad mn = m^2 n^2 - m^2 - n^2 \quad mn + 1 = m^2(n^2 - 1) - 1(n^2 - 1) \\ mn + 1 &= (m + 1)(m - 1)(n + 1)(n - 1) \end{aligned}$$

Notice that the right side can be zero or one. If the right side is zero, m and n can be $(-1, 1)$ and $(1, -1)$. If the right side is one, m and n can be $(0, 0)$. There are

(C) 3 solutions.

Solution 7 (Obtaining ranges)

Set $m \leq n$. Then, we can say that

$$3n^2 \geq m^2n^2$$

$$3 \geq m^2$$

Or $-\sqrt{3} \leq m \leq \sqrt{3}$, and since we are dealing with integers, $m = -1, 0$ or 1 . Testing these numbers, we get $n = 1, n = 0$ and $n = -1$ respectively. Although the solution $(1, -1)$ is a solution in the end, our initial condition for this case was $m \leq n$. For better rigour, we can just consider the other case $m > n$ to validate solution $(1, -1)$.

-lisztepos

Solution 8 (Inequality)

If $mn = 0$, then $m^2 + 0 + n^2 = 0$, $(m, n) = (0, 0)$.

If $mn \neq 0$, then

$$\frac{m^2 + mn + n^2}{m^2n^2} = 1$$

$$\frac{1}{m^2} + \frac{1}{mn} + \frac{1}{n^2} = 1$$

$$\frac{1}{m^2} + \frac{2}{|mn|} + \frac{1}{n^2} > 1$$

$$\left(\frac{1}{|m|} + \frac{1}{|n|} \right)^2 > 1$$

$$\frac{1}{|m|} + \frac{1}{|n|} > 1$$

Obviously, at least one of $|m|, |n|$ is 1. If $m = 1$, $1 + n + n^2 = n^2 \Rightarrow n = -1$. If $m = -1$, $1 - n + n^2 = n^2 \Rightarrow n = 1$. We omit the discussion of $n = \pm 1$.

Finally, we get $(m, n) = (0, 0), (1, -1), (-1, 1)$, there are **(C) 3** solutions.

Solution 9 (completing the square and substitution)

Immediately we know that $(m, n) = (0, 0)$ is a solution. Next, we can add mn to both sides to complete the square and get:

$$(m + n)^2 = mn(1 + mn)$$

Let $x = mn$. Then, $n + \frac{x}{n} = \sqrt{x(x+1)}$

For this square root to be an integer, x and $x + 1$ must both be perfect squares, since otherwise you wouldn't be able to find consecutive numbers that multiply to a perfect square. This is only the case when $x = 0$, which means that $n = 1$ when you solve for n , and then $m = 0$ since $x = mn$. So, $(m, n) = (0, 1)$ is a solution.

Also, since m and n are symmetrical, $(1, 0)$ must also be a solution.

Therefore, there are **(C) 3** solutions.

An additional note is that if you somehow didn't notice that $(0, 0)$ is a solution, you could just find two solutions and see that 2 is not an answer choice, so you're probably missing a single base case, but this isn't that rigorous.

15. (AMC 10 10B 2023) What is the least positive integer m such that $m \cdot 2! \cdot 3! \cdot 4! \cdot 5! \dots 16!$ is a perfect square?

(A) 30 (B) 30030 (C) 70 (D) 1430 (E) 1001

Answer: C

Solution 1

We want $m \cdot 2! \cdot 3! \cdot 4! \cdot \dots \cdot 16!$ to be a perfect square. Notice that we can rewrite and pair up certain elements:

$$m \cdot 2 \cdot 3! \cdot (4 \cdot 3!) \cdot 5! \cdot (6 \cdot 5!) \cdot \dots \cdot 15! \cdot (16 \cdot 15!) = m \cdot 2 \cdot 4 \cdot 6 \cdot \dots \cdot 16 \cdot (3!)^2 (5!)^2 \cdot \dots \cdot (15!)^2.$$

Note here that this is equivalent to simply $m \cdot 2 \cdot 4 \cdot 6 \cdot \dots \cdot 16$ being a perfect square (this is intuitively obvious: i.e. if $a = bc$ is a perfect square and so is b , then of course c must be a perfect square too). We can rewrite this as the following:

$$m \cdot 2^8 \cdot (1 \cdot 2 \cdot 3 \cdot \dots \cdot 8) \equiv m \cdot 2 \cdot 3 \cdot 4 \cdot 5 \cdot 6 \cdot 7 \cdot 8 \equiv m \cdot 70.$$

The smallest m s.t. $70m$ is a perfect square is, of course, 70 itself. QED.

Minorest of minor edits by Supernova 77

Solution 2

Perfect squares have all of the powers in their prime factorization even. To evaluate $2! \cdot 3! \cdot 4! \cdot 5! \dots 16!$ we get the following:

$$(2^{15}) \times (3^{14}) \times ((2^2)^{13}) \times (5^{12}) \times ((2 \times 3)^{11}) \times (7^{10}) \times ((2^3)^9) \times ((3^2)^8) \times ((5 \times 2)^7) \times (11^6) \times (((2^2) \times 3)^5) \times (13^4) \times ((7 \times 2)^3) \times ((3 \times 5)^2) \times ((2^4)^1).$$

Taking all powers mod 2 we get:

$$(2^1) \times ((2 \times 3)^1) \times (2^1) \times ((5 \times 2)^1) \times (3^1) \times ((7 \times 2)^1)$$

Simplifying again, we finally get:

$$(2^1) \times (5^1) \times (7^1)$$

To make all the powers left even, we need to multiply by $(2 \times 5 \times 7)$ which is

(C) 70.

Solution 3 (Answer Choice Elimination)

We can prime factorize the solutions: $A = 2 \cdot 3 \cdot 5$, $B = 2 \cdot 3 \cdot 5 \cdot 7 \cdot 11 \cdot 13$, $C = 2 \cdot 5 \cdot 7$, $D = 2 \cdot 5 \cdot 11 \cdot 13$, $E = 7 \cdot 11 \cdot 13$,

We can immediately eliminate B, D, and E since 13 appears an even number of times (4 times, one each in $13!$, $14!$, $15!$, $16!$).

Next, we can test if 7 is possible (and if it is not we can use process of elimination). 7 appears once for each of $7!$ to $16!$ and then one more time each from $14!$ to $16!$. So, 7 appears an odd number of times, as $8 + 3 = 11$. This means m has a factor of 7, meaning (C) is the only possible answer choice.

Solution 4

First, we note that $3! = 2! \cdot 3$, $5! = 4! \cdot 5$, ..., $15! = 14! \cdot 15$. So, $2! \cdot 3! = 2!^2 \cdot 3 \equiv 3$, $4! \cdot 5! = 4!^2 \cdot 5 \equiv 5$, ..., $14! \cdot 15! = 14!^2 \cdot 15 \equiv 15$. Simplifying the whole sequence and cancelling out the squares, we get $3 \cdot 5 \cdot 7 \cdot 9 \cdot 11 \cdot 13 \cdot 15 \cdot 16!$. Prime factoring $16!$ and cancelling out the squares, the only numbers that remain are 2, 5, and 7. Since we need to make this a perfect square, $m = 2 \cdot 5 \cdot 7$. Multiplying this out, we get $\boxed{(C) 70}$.

Solution 5 (Legendre's Formula)

We need to find the prime factorization of $16! \cdot 15! \cdot 14! \dots \cdot 1!$ and check whether the exponents of each prime are odd or even.

Using **Legendre's Formula** on each of the factorials and adding the powers of each prime we get:

Powers of 2:

$$15 + 11 + 11 + 10 + 10 + 8 + 8 + 7 + 7 + 4 + 4 + 3 + 3 + 1 + 1 + 0 = 103$$

Powers of 3:

$$6 + 6 + 5 + 5 + 5 + 4 + 4 + 4 + 2 + 2 + 2 + 1 + 1 + 1 + 0 + 0 = 48$$

Powers of 5:

$$3 + 3 + 2 + 2 + 2 + 2 + 2 + 1 + 1 + 1 + 1 + 1 + 0 + 0 + 0 + 0 = 21$$

Powers of 7:

$$2 + 2 + 2 + 1 + 1 + 1 + 1 + 1 + 1 + 1 = 13$$

Powers of 11:

$$1 + 1 + 1 + 1 + 1 + 1 = 6$$

Powers of 13:

$$1 + 1 + 1 + 1 = 4$$

2, 5, and 7 have odd exponents, so the answer is $2 \cdot 5 \cdot 7 = \boxed{(C) 70}$.

Solution 6 (Bashy method)

We know that a perfect square must be in the form $2^{2a_1} \cdot 3^{2a_2} \cdot 5^{2a_3} \dots p^{2a_n}$ where $a_1, a_2, a_3, \dots, a_n$ are nonnegative integers, and p is the largest and n th prime factor of our square number.

Let's assume $r = m \cdot 2! \cdot 3! \cdot 4! \cdot 5! \dots 16!$. We need to prime factorize r and see which prime factors are raised to an odd power. Then, we can multiply one factor each of prime number with an odd number of factors to m . We can do this by finding the number of factors of 2, 3, 5, 7, 11, and 13.

Case 1: Factors of 2

We first count factors of 2^1 in each of the factorials. We know there is one factor of 2^1 each in $2!$ and $3!$, two in $4!$ and $5!$, and so on until we have 8 factors of 2^1 in $16!$. Adding them all up, we have $1 + 1 + 2 + 2 + \dots + 7 + 7 + 8 = 64$.

Now, we count factors of 2^2 in each of the factorials. We know there is one factor of 2^2 each in $4!$, $5!$, $6!$, and $7!$, two in $8!$, $9!$, $10!$, and $11!$, and so on until we have 4 factors of 2^1 in $16!$. Adding them all up, we have $1 + 1 + 1 + 1 + 2 + 2 + 2 + 2 + 3 + 3 + 3 + 3 + 4 = 28$.

Now we count factors of 2^3 in each of the factorials. Using a similar method as above, we have a sum of $1 + 1 + 1 + 1 + 1 + 1 + 1 + 1 + 2 = 10$.

Now we count factors of 2^4 in each of the factorials. Using a similar method as above, we have a factor of 2^4 in 16, so there is 1 factor of 2^4 .

Adding all the factors of 2, we have 103. Since 103 is odd, m has one factor of 2.

Case 2: Factors of 3

We use a similar method as in case 1. We first count factors of 3^1 . We obtain the sum $1 + 1 + 1 + 2 + 2 + 2 + \dots + 4 + 4 + 4 + 5 + 5 = 50$.

We count factors of 3^2 . We obtain the sum $1 + 1 + 1 + 1 + 1 + 1 + 1 + 1 = 8$.

Adding all the factors of 3, we have 58. Since 58 is even, m has 0 factors of 3.

Case 3: Factors of 5

We count the factors of 5^1 : $1 + 1 + 1 + 1 + 1 + 2 + 2 + 2 + 2 + 2 + 3 + 3 = 21$. Since 21 is odd, m has one factor of 5.

Case 4: Factors of 7

We count the factors of 7^1 : $1 + 1 + 1 + 1 + 1 + 1 + 1 + 2 + 2 + 2 = 13$. Since 13 is odd, m has one factor of 7.

Case 5: Factors of 11

We count the factors of 11^1 : $1 + 1 + 1 + 1 + 1 + 1 = 6$. Since 6 is even, m has 0 factors of 11.

Case 6: Factors of 13

We count the factors of 13^1 : $1 + 1 + 1 + 1 = 4$. Since 4 is even, m has 0 factors of 13.

Multiplying out all our factors for m , we obtain $2 \cdot 5 \cdot 7 = \boxed{(C) 70}$.

Solution 7 (Answer Choices)

We see that all the answer choices are divisible by 10 except for 1001, and we also notice that the answer choices have 3, 7, 11, or 13 as a prime factor.

Testing, we see that 3, 11, 13 have an even power in the product, so we have that all the other answer choices will not work.

Therefore we just have $\boxed{(C) 70}$.

Solution 8 (Fastest Intuition)

Notice that you can add the factor $1!$ to the expression to make it $m \cdot 1! \cdot 2! \cdot 3! \cdot 4! \cdot 5! \cdot 6! \cdot 7! \cdot 8! \cdot 9! \cdot 10! \cdot 11! \cdot 12! \cdot 13! \cdot 14! \cdot 15! \cdot 16!$

Every consecutive pair $n! \cdot (n+1)!$ can be simplified to $n+1$ (divide out $n!$).

Resulting is the product $m \cdot 2 \cdot 4 \cdot 6 \cdot 8 \cdot 10 \cdot 12 \cdot 14 \cdot 16 = m \cdot 2^8 \cdot 8!$

This is easily simplified to $m \cdot 2 \cdot 5 \cdot 7 = m \cdot 70$ when dividing out perfect squares.

This means that m must have a minimum factor of 70 which gives answer choice $\boxed{(C) 70}$.

Solution 9 (Pretty Quick similar to solution 8)

$16! (15!)$ this can be written as $16 (15!) (15!)$

Thus

$$16 (15!)^2$$

Now lets try

$$16!15!14! \ 16(15)(14!) \ 15(14!)(14!)$$

$$\text{thus } 16(15)^2(14!)^3$$

$$\text{we can continue this pattern until one to get } m \cdot 2! \cdot 3! \cdot 4! \cdot 5! \dots 16! = 16(15)^2(14)^3(13)^4(12)^5(11)^6 \dots (2)^{15}$$

we can remove all the even powers of expression above to get

$$16(14)^3(12)^5(10)^7(8)^9(6)^{11}(4)^{13}(2)^{15}$$

we can see that all off the numbers are even and multiples of 2,3,5,7

the biggest possible number we should get is 2 times 3 times 5 times 7 which is 210 which means out anwer should be A or C 30 or 70.

Then look all the stuff with 7 which would only consist of 14^3

which needs an extra 7 to give it company, so put answer must be divisble by 7 and less than 210

which leaves 70 or C as our answer

$$\boxed{(C) \ 70}.$$

Note: you can also just remove all pairs of two to get $16(14)(12)(10)(8)(6)(4)(2)$

Solution 10 (Braindead solution)

List out 1 to 16:

$$1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16.$$

First, we remove all numbers that appear an even amount of times in the product:

$$2, 4, 6, 8, 10, 12, 14, 16.$$

Next, we remove all perfect squares:

$$2, 6, 8, 10, 12, 14.$$

Factoring each term gives:

$$2^1, 2^1 \cdot 3^1, 2^3, 2^1 \cdot 5^1, 2^2 \cdot 3^1, 2^1 \cdot 7^1.$$

We multiply all the terms together and remove perfect squares to get $2^1 \cdot 5^1 \cdot 7^1 =$

$$\boxed{(C) \ 70}.$$

16. (AMC 10 10B 2023) Define an *upno* to be a positive integer of 2 or more digits where the digits are strictly increasing moving left to right. Similarly, define a *downno* to be a positive integer of 2 or more digits where the digits are strictly decreasing moving left to right. For instance, the number 258 is an upno and 8620 is a downno. Let U equal the total number of *upnos* and let D equal the total number of *downnos*. What is $|U - D|$?

(A) 512 (B) 10 (C) 0 (D) 9 (E) 511

Answer: E

Solution 1

First, we know that D is greater than U , since there are less *upnos* than *downnos*. To see why, we examine what determines an *upno* or *downno*.

We notice that, given any selection of unique digits (notice that "unique" constrains this to be a finite number), we can construct a unique *downno*. Similarly, we can also construct an *upno*, but the selection can not include the digit 0 since that isn't valid.

We then have $[0, 1, 2, 3, 4, 5, 6, 7, 8, 9]$ as the pool of numbers that we can pick from for *downnos*, and $1, 2, 3, 4, 5, 6, 7, 8, 9$ as the pool of numbers that we can pick for *upnos*. There are only two states each number can be in: appearing and not appearing in the arrangement. (That is why we use the number 2 as the base in the exponent!)

Thus, there are 2^{10} total *downnos* and 2^9 total *upnos*. However, we are told that each *upno* or *downno* must be at least 2 digits, so we subtract out the 0-digit and 1-digit cases (Referring back to Paragraph 2, this is when every number's state is nonappearing or every number except one has the state of nonappearing).

For the *downnos*, there are 10 1-digit cases, and for the *upnos*, there are 9 1-digit cases. There is 1 0-digit case for both *upnos* and *downnos*.

Thus, the difference is $((2^{10} - 10 - 1) - (2^9 - 9 - 1)) = 2^9 - 1 = \boxed{\text{(E) } 511}$.

Solution 2

Set Theory can get very complicated, so let's break this down intuitively and derive answers. We have 2 cases. Upno's, and Downno's. Call them U and D respectively. Let us evaluate them independently and then deal with $|U - D|$ later.

Case 1: U (Deriving the number of cases)

The intuition is to come up with a pattern.

We start with 9. We instantly see that 9 is the largest possible unit number, so there exists no U who's leading digit is 9.

Continuing with 8, the possible upno's are 89, which is the only case.

Now with 7, the upno's are 78, 79, and 789. Great! We have 3 cases here.

Continuing with 6, we have 67, 68, 69, 678, 679, 689, 6789. That is 7 cases.

Now, we could just assume that for 5, there will be 13 different cases (The first difference of the cases in 7 and 8 is 2, and then the difference between the cases in 6 and 7 is 4, so we say the next one is $7 + 6 = 13$). However, let's not jump to conclusions just yet. Let's actually compute the cases for 5 quickly.

For 5, we have 56, 57, 58, 59, 567, 568, 569, 578, 579, 589, 5678, 5679, 5689, 5789, and 56789, giving us not 13, but 15 digits.

So, our assumption that 5 was 13 cases is wrong. (A little whisper to tell you to be careful). It seems that the rule between each pattern is multiplying the difference previously by 2.

So our differences for the sequences (in order) are $\{2, 4, 8, 16, 32, 64, 128\}$. We use this to find the number of cases for each number and quickly sum them all up (so we start with 1 from case 1, add 2, then add 4 to that, so on to get the number of cases and then add the cases together) to get 502.

Case 2: D (Deriving the number of cases)

We start with 1 this time. Unlike the U set, we can now use 0.

Starting with 1 gives us 10, 1 case.

Moving to 2 gives us 21, 20, 210, 3 cases.

3 gives us 32, 31, 30, 321, 310, 320, 3210, 7 cases.

We already see where this is going, so this time we already jump to our U set and add 1 more difference to it to get our D set (This is to include the difference between 9 and 8, because now that 0 is available, 1 is a viable option).

So our D set is $\{2, 4, 8, 16, 32, 64, 128, 256\}$. Using the set to find the number of cases gives us 1013.

Great! We have overcome the difficult part. All that is left to do is to find $|U - D|$. We plug in our values for U and D , which are 502 and 1013 respectively, to get $|U - D| = |502 - 1013| = |-511| = \boxed{511}$ or $\boxed{E}$.

Solution 3

Since Upnos do not allow 0s to be in their first – and any other – digit, there will be no zeros in any digits of an Upno. Thus, Upnos only contain digits $[1, 2, 3, 4, 5, 6, 7, 8, 9]$.

Upnos are 2 digits in minimum and 9 digits maximum (repetition is not allowed). Thus the total number of Upnos will be $\binom{9}{2} + \binom{9}{3} + \binom{9}{4} + \dots + \binom{9}{9}$, since every selection of distinct numbers from the set $[1, 2, 3, 4, 5, 6, 7, 8, 9]$ can be arranged so that it is an Upno. There will be $\binom{9}{2}$ -2 digit Upnos, $\binom{9}{3}$ -3 digit Upnos and so on.

Thus, the total number of Upnos will be $\sum_{i=2}^9 \binom{9}{i} = 512 - 10 = 502$

Notice that the same combination logic can be done for Downnos, but Downnos DO allow zeros to be in their last digit. Thus, there are 10 possible digits $[0, 1, 2, 3, 4, 5, 6, 7, 8, 9]$ for Downnos.

Therefore, it is visible that the total number of Downnos are:

$$\sum_{i=2}^{10} \binom{10}{i} = 2^{10} - \binom{10}{0} - \binom{10}{1} = 1024 - 11 = 1013$$

Thus $|U - D| = |(1013 - 502)| = \boxed{(E) 511}$.

Solution 4

Note that you can obtain a *downo* by reversing an upno (like 235 is an *upno*, and you can obtain 532). So, we need to find the amount of *downos* that end with 0 since if you 'flip' the numbers, the *upno* starts with a 0 which corresponds to a *downo*. We can find the cases that end with a 0:

$$\sum_{n=0}^9 \binom{9}{n} = \binom{9}{0} + \binom{9}{1} + \dots + \binom{9}{9}$$

to get 512 by Sum of the Row Identity. However, 0 itself is not a valid case (since it only has 1 digit) so we subtract 1. Our answer is $512 - 1 = \boxed{(E) 511}$.

-aleyang

-ap246(LaTeX)

Solution 5 (Educated Guess)

First, note that the only *downnos* that are not contained by the set of *upnos* is every *downno* that ends in 0.

Next, listing all the two digits *downnos*, we find that the answer is more than 10, since there are more digits to be tested and there are already 9 two digit *downnos*. This leaves us with 512 or 511.

Next, we notice that all the possibilities for 2 through 9 digit *downnos* ending in 0 pair up with one another, as the possibilities are equal (e.g. possibilities for 2 digits = possibilities for 9 digits, etc.).

This leaves us with one last possibility, the ten digit *downno* 9876543210.

Since all the previous possibilities form an even number, adding one more possibility will make the total odd. Therefore, we need to choose the odd number from the set [511, 512].

Our answer is (E) 511.

Solution 6

We start by calculating the number of upnos. Suppose we are constructing an upno of n digits such that $n \geq 2$. An upno can't start with a "0", so there are 9 digits to choose from. There are $\binom{9}{n}$ ways to choose an upno with n digits. This is because for each combination of digits, only one combination can form an upno. Therefore, for $2 \leq n \leq 9$, the total number of upnos is

$$\binom{9}{2} + \binom{9}{3} + \binom{9}{4} + \cdots + \binom{9}{9} = 2^9 - \binom{9}{1} - \binom{9}{0} = 2^9 - 10.$$

Similarly, the digits of a downo of n digits can be chosen among 10 digits to choose from, since 0 can be a digit of the downo as the last digit. Thus, the number of downos is

$$\binom{10}{2} + \binom{10}{3} + \binom{10}{4} + \cdots + \binom{10}{9} + \binom{10}{10} = 2^{10} - \binom{10}{1} - \binom{10}{0} = 2^{10} - 11.$$

Thus,

$$|U - D| = |(2^9 - 10) - (2^{10} - 11)| = (2^{10} - 11) - (2^9 - 10) = 2^{10} - 2^9 - 1 = \span style="border: 1px solid black; padding: 2px;">(E) 511$$

Solution 7 (1-1 Correspondence)

Notice that the number of upno integers are the number of subsets of the set $(1, 2, 3, 4, 5, 6, 7, 8, 9)$, excluding the empty set ($\emptyset$) and the one-digit integers. So, the number of upno integers is $2^9 - 1 - 9 = 502$.

The number of downo numbers are similar, but with 0. So, the number of downo integers is the number of subsets of the set $(0, 1, 2, 3, 4, 5, 6, 7, 8, 9)$, excluding the empty set and the one-digit integers. So, the number of downo integers is $2^{10} - 1 - 10 = 1013$.

Therefore, the difference between the number of downo integers and upno is $1013 - 502 = \span style="border: 1px solid black; padding: 2px;">(E) 511.$

Solution 8 (Bijection)

Observe that each downno (that doesn't end with a 0) maps to one upno. It suffices to count the number of downnos that end with 0.

Given that a downno ends with 0, we want to choose the remaining digits from the set $\{1, 2, \dots, 9\}$. We can arrange the elements of any subset so that they are increasing. We may include or exclude any of the 9 elements, giving 2^9 subsets. Subtracting 1 to account for the empty set, we have $2^9 - 1 = \boxed{\text{(E) 511}}$ more downnos than upnos.

-Benedict T (countmath1)

Solution 9 (Combinations)

Like Solution 1, we know that there are more downnos than upnos because downnos can contain 0. So, we need to calculate the number of downnos ending with 0. We can do this by splitting this into different cases:

For a 2-digit downno ending with 0, there are $\binom{9}{1}$ ways of constructing a downno ending with 0. Similarly, for a 3-digit downno ending with 0, there are $\binom{9}{2}$ ways of constructing a downno ending with 0. We can calculate this for up to 10 digits, or the downno 9876543210.

We can add the sum for all of the cases, which is $\binom{9}{1} + \binom{9}{2} + \binom{9}{3} + \dots + \binom{9}{9} = 2^9 - 1 = \boxed{\text{(E) 511}}$. pixelpyguy

Solution 10 (guessish)

Note that every upno maps to one downno, just reversed. In addition, note that every downno can create another by adding a 0 to the end. Downnos can also be single digits with a following 0. Therefore, $\text{downno} = 2 \cdot \text{upno} + 9$. $2 \cdot \text{upno}$ is guaranteed to be even, therefore our answer is guaranteed to be odd. From simple observation, the only valid answer is $\boxed{\text{(E) 511}}$.

Solution 11 (fast)

You can make a downno by flipping an upno, so all the upnos match to a downno. Also, there are double the downnos than upnos because you can flip it or flip and add a 0. Then just count the number of upnos, which will be some power of 2 because it's all the subsets of the set of digits without zero. But we forgot 0 is an upno, so its one less than a power of 2, so $\boxed{\text{(E) 511}}$.

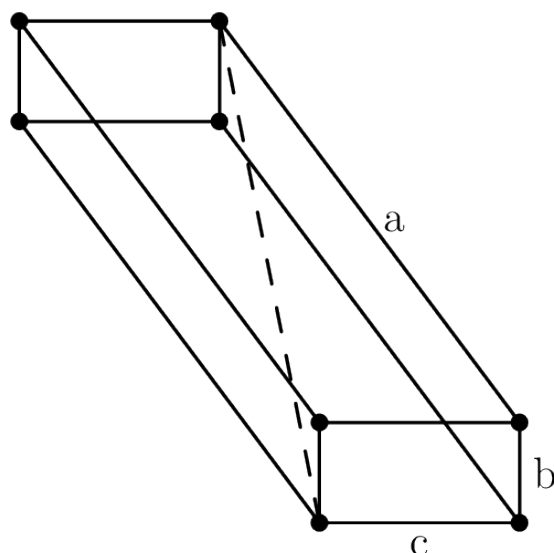
Solution 12

We see that with any choice of distinct positive integers from 0 to 9 we can make an upno and a downno with the exact same subset of integers. The only choice of integers that will not work for both cases is when we have a 0, as we will not be able to form an upno starting with 0. Starting with 9 choose 1 we sum all 9 choose functions up to 9 choose 9 and get $\boxed{\text{(E) 511}}$.

17. (AMC 10 10B 2023) A rectangular box P has distinct edge lengths a , b , and c . The sum of the lengths of all 12 edges of P is 13, the sum of the areas of all 6 faces of P is $\frac{11}{2}$, and the volume of P is $\frac{1}{2}$. What is the length of the longest interior diagonal connecting two vertices of P ?
- (A) 2 (B) $\frac{3}{8}$ (C) $\frac{9}{8}$ (D) $\frac{9}{4}$ (E) $\frac{3}{2}$

Answer: D

Solution 1 (algebraic manipulation)



We can create three equations using the given information.

$$4a + 4b + 4c = 13$$

$$2ab + 2ac + 2bc = \frac{11}{2}$$

$$abc = \frac{1}{2}$$

We also know that we want $\sqrt{a^2 + b^2 + c^2}$ because that is the length that can be found from using the Pythagorean Theorem. We cleverly notice that $a^2 + b^2 + c^2 = (a + b + c)^2 - 2(ab + ac + bc)$. We know that $a + b + c = \frac{13}{4}$ and $2(ab + ac + bc) = \frac{11}{2}$, so $a^2 + b^2 + c^2 = \left(\frac{13}{4}\right)^2 - \frac{11}{2} = \frac{169-88}{16} = \frac{81}{16}$. So our answer is $\sqrt{\frac{81}{16}} = \boxed{\text{(D)} \frac{9}{4}}$.

Solution 2 (Vieta's)

We use the equations from Solution 1 and manipulate it a little:

$$a + b + c = \frac{13}{4}$$

$$ab + ac + bc = \frac{11}{4}$$

$$abc = \frac{1}{2}$$

Notice how these are the equations for the vieta's formulas for a polynomial with roots of a , b , and c . Let's create that polynomial. It would be $x^3 - \frac{13}{4}x^2 + \frac{11}{4}x - \frac{1}{2}$. Multiplying each term by 4 to get rid of fractions, we get $4x^3 - 13x^2 + 11x - 2$. Notice how the coefficients add up to 0. Whenever this happens, that means that $(x - 1)$ is a factor and that 1 is a root. After using synthetic division to divide $4x^3 - 13x^2 + 11x - 2$ by $x - 1$, we get $4x^2 - 9x + 2$. Factoring that, you get $(x - 2)(4x - 1)$. This means that this polynomial factors to $(x - 1)(x - 2)(4x - 1)$ and that the roots are 1, 2, and $1/4$. Since we're looking for $\sqrt{a^2 + b^2 + c^2}$, this is equal to $\sqrt{1^2 + 2^2 + \frac{1}{4}^2} = \sqrt{\frac{81}{16}} = \boxed{\text{(D)} \frac{9}{4}}$

Solution 3 (Newton's Sums)

Same formal process of Solution 2, once we have $4x^3 - 13x^2 + 11x - 2$. Let $s_2 = a^2 + b^2 + c^2$ and $s_1 = a + b + c = \frac{13}{4}$. Applying **Newton's Sums**, we have

$$(4)s_2 + (-13)s_1 + 2 \cdot 11 = 0$$

. We can solve that $s_2 = \frac{81}{16}$, and the interior diagonal would be

$$\sqrt{s_2} = \frac{9}{4}$$

Solution 4 (Cheese Method)

Incorporating the solution above, we know $a + b + c = 13/4 \Rightarrow a + b + c > 3$. The side lengths are larger than $1 \cdot 1 \cdot 1$ (a unit cube). The side length of the interior of a unit cube is $\sqrt{3}$, and we know that the side lengths are larger than $1 \cdot 1 \cdot 1$, so that means the diagonal has to be larger than $\sqrt{3}$, and the only answer choice

larger than $\sqrt{3} \Rightarrow \boxed{\text{(D)} \frac{9}{4}}$

Note that the real number $\sqrt{3}$ is around 1.73. Option A is also greater than $\sqrt{3}$ meaning there are two options greater than $\sqrt{3}$. Option A is an integer so

educationally guessing we arrive at answer $D \Rightarrow \boxed{\text{(D)} \frac{9}{4}}$

18. (AMC 10 10B 2023) Suppose a , b , and c are positive integers such that

$$\frac{a}{14} + \frac{b}{15} = \frac{c}{210}.$$

Which of the following statements are necessarily true?

I. If $\gcd(a, 14) = 1$ or $\gcd(b, 15) = 1$ or both, then $\gcd(c, 210) = 1$.

II. If $\gcd(c, 210) = 1$, then $\gcd(a, 14) = 1$ or $\gcd(b, 15) = 1$ or both.

III. $\gcd(c, 210) = 1$ if and only if $\gcd(a, 14) = \gcd(b, 15) = 1$.

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\text{mtext mathvari-} \\ & \text{ant="bold"} > (D) < / \text{mtext} > < \text{mtext} > < / \text{mtext} > < \text{mtext} > \text{III only} < / \text{mtext} > < \text{mstyle} \\ & \text{scriptlevel="0"} > < \text{mspace width="2em"} > < / \text{mspace} > < / \text{mstyle} > < \text{mtext mathvari-} \\ & \text{ant="bold"} > (E) < / \text{mtext} > < \text{mtext} > < / \text{mtext} > < \text{mtext} > \text{II and III only} < / \text{mtext} > < / \text{math} > < / \text{mjb-} \\ & \text{assistive-mml} > < / \text{mjb-container} > \end{aligned}$$

Answer: E

Solution 1 (Guess and check + Contrapositive)

We examine each of the conditions.

The first condition is false. A simple counterexample is $a = 3$ and $b = 5$. The corresponding value of c is 115. Since $\gcd(3, 14) = 1$, condition I would imply that $\gcd(c, 210) = 1$. However, $\gcd(115, 210)$ is clearly not 1 (they share a common factor of 5). Condition I is false so that we can rule out choices A, B, and C.

We now decide between the two answer choices D and E . What differs between them is the validity of condition II , so it suffices to check II simply.

We look at statement II 's contrapositive to prove it. The contrapositive states that if $\gcd(a, 14) \neq 1$ and $\gcd(b, 15) \neq 1$, then $\gcd(c, 210) \neq 1$. In other words, if a shares some common factor that is not 1 with 14 and b shares some common factor that is not 1 with 15, then c also shares a common factor that is not 1 with 210. Let's say that $a = a' \cdot n$, where a' is a factor of 14 not equal to 1. (So a' is the common factor.)

We can rewrite the given equation as $15a + 14b = c \implies 15(a'n) + 14b = c$. We can express 14 as $a' \cdot n'$, for some positive integer n' (this n' can be 1). We can factor a' out to get $a'(15n + bn') = c$.

Since all values in this equation are integers, c must be divisible by a' . Since a' is a factor of 14, a' must also be a factor of 210, a multiple of 14. Therefore, we know that c shares a common factor with 210 (which is a'), so $\gcd(c, 210) \neq 1$. This is what II states, so therefore II is true.

Thus, our answer is (E) II and III only. Technodoggo AVM2023 (Minor Edits - Fixes in the first paragraph and grammar edits)

Solution 2

The equation given in the problem can be written as

$$15a + 14b = c. \quad (1)$$

First, we prove that Statement I is not correct.

A counter example is $a = 1$ and $b = 3$. Thus, $\gcd(c, 210) = 3 \neq 1$.

Second, we prove that Statement III is correct.

First, we prove the "if part.

Suppose $\gcd(a, 14) = 1$ and $\gcd(b, 15) = 1$. However, $\gcd(c, 210) \neq 1$.

Thus, c must be divisible by at least one factor of 210. W.L.O.G, we assume c is divisible by 2.

Modulo 2 on Equation (1), we get that $2|a$. This is a contradiction with the condition that $\gcd(a, 14) = 1$. Therefore, the "if part in Statement III is correct.

Second, we prove the "only if part.

Suppose $\gcd(c, 210) \neq 1$. Because $210 = 14 \cdot 15$, there must be one factor of 14 or 15 that divides c . W.L.O.G, we assume there is a factor $q > 1$ of 14 that divides c . Because $\gcd(14, 15) = 1$, we have $\gcd(q, 15) = 1$. Modulo q on Equation (1), we have $q|a$.

Because $q|14$, we have $\gcd(a, 14) \geq q > 1$.

Analogously, we can prove that $\gcd(b, 15) > 1$.

Third, we prove that Statement II is correct.

This is simply a special case of the "only if part of Statement III. So we omit the proof.

All analyses above imply (E) II and III only.

Solution 3 (Answer Choices)

It can easily be shown that statement I is false (a counterexample would be $a = 1, b = 5, c = 85$), meaning the only viable answer choices are D and E. Since both of these answer choices include statement III, this means III is true. Since III is true, this actually implies that statement II is true, as III is just a stronger version of statement II (or its contrapositive, to be precise). Therefore the answer is **(E) II and III only**.

Solution 4

First, we will show that statement I is false. A simple counterexample is $a = 2$ and $b = 2$. Then, c is even, and it is not relatively prime with 210.

Now, we will focus on statement III. It's pretty easy to realize that if either a isn't relatively prime to 14 or b isn't relatively prime to 15, it will always share factors with 210 because $c = 15a + 14b$ (think $15 \cdot 14 + 14$). Therefore, statement III must be true.

Since statement III is a stricter case of statement II, the last two statements are true, hence choose **(E) II and III only**.

Solution 5 (Rigorous)

The condition in the problem is equivalent to

$$15a + 14b = c.$$

From here we can use the properties of the greatest common divisor to get

$$\gcd(c, 14) = \gcd(15a + 14b, 14) = \gcd(15a, 14) = \gcd(a, 14)$$

and

$$\gcd(c, 15) = \gcd(15a + 14b, 15) = \gcd(14b, 15) = \gcd(b, 15)$$

From here it's not hard to get

$$\gcd(c, 210) = \gcd(c, 15) * \gcd(c, 14) = \gcd(a, 14) * \gcd(b, 15)$$

Hence $\gcd(c, 210) = 1$ is equivalent to $\gcd(a, 14) = 1$ and $\gcd(b, 15) = 1$. This means the answer must be **(E) II and III only**.

Solution 6 (Intuitive Risking)

We know that $\frac{a}{14} + \frac{b}{15} = \frac{c}{210}$. So, therefore, $\frac{a+14b}{15} = \frac{c}{1}$, and $15a + 14b = 210c$.

We could see that some value of a and another value of b gives a different value of c . **WLOG**, we give an intuitive risk, saying that $a + b = c$. Our goal is to just prove statements, not find solutions.

Statement I:

If $\gcd(a, 14) = 1$ or $\gcd(b, 15) = 1$ or both, then $\gcd(c, 210) = 1$.

We have $a + b = c$.

Then if $\gcd(a, 14) = 1$, we say $a = 1$

So then $\gcd(b, 14) \neq 1$, so we say $b = 2$.

$1 + 2 = 3$, and $\gcd(3, 210) = 1$ is false. So statement I is incorrect.

Statement II:

If $\gcd(c, 210) = 1$, then $\gcd(a, 14) = 1$ or $\gcd(b, 15) = 1$ or both.

We say $a + b = c$.

For statement II, we need to prove that it works for every odd number that is a multiple of the first number that makes both a and b positive integers and $\gcd(c, 210) = 1$.

This number is 9. The possible pair values for a and b are 5 and 4 respectively. If we try 27, we get for a and b can be 15 and 12. We clearly see this is a pattern, and to get to an odd multiple of 9 we just multiply the sums by the n th multiple of 9. Therefore statement II is always true.

Statement III:

$\gcd(c, 210) = 1$ if and only if $\gcd(a, 14) = \gcd(b, 15) = 1$.

We say $a + b = c$

We prove this by contradiction.

Assume $\gcd(a, 14) \neq \gcd(b, 15) \neq 1$.

Then, that means $\gcd(c, 210)$ is never equal to 1.

So if we take $a = 14$ and $b = 15$, we get that $\gcd(29, 210) \neq 1$. This is obviously false, because 29 is prime.

Therefore statement III is also correct.

Only II and III are correct, leading us to answer choice (E) II and III only.

Solution 7 (Formal Logic Lattice)

Multiply the original equation by 210 to get $15a + 14b = c$.

Denote the propositions $A := (\gcd(a, 14) = 1)$, $B := (\gcd(b, 15) = 1)$, $C := (\gcd(c, 210) = 1)$.

Work on the lattice with set $\{A, B, C\}$ with $\geq$ the order relation given by $\implies$, $\wedge$ the logical minimum (AND), $\vee$ the logical maximum (OR), $\equiv$ logical equivalence $\iff$.

$$\text{I. } A \vee B \geq C \iff (A \geq C) \wedge (B \geq C)$$

$$\text{II. } C \geq A \vee B$$

$$\text{III. } C \equiv A \wedge B$$

Note $A \wedge B \geq A \vee B$, so **III** $\implies$ **II**. Also, **III** $\wedge$ **I** $\implies A \equiv B \equiv C$ by the squeeze lemma. In our case $A \not\equiv B \implies \neg \text{III} \vee \neg \text{I}$ (at least one of these statements is false). (The squeeze lemma refers to $A \wedge B \geq A \vee B \implies A \equiv B$ from decomposing **I** into $A \geq C \equiv A \wedge B \iff A \geq B$ and symmetrically $B \geq A$.)

But notice $\gcd(14, 15) = 1$. Use multiplicativity $\gcd(x, y) = 1 \implies \gcd(z, xy) = \gcd(z, x) \gcd(z, y)$ and Euclidean algorithm $\gcd(x, y) = \gcd(x, y \pm x)$ to get

$$\gcd(c, 210) = \gcd(15a + 14b, 14) \gcd(15a + 14b, 15) = \gcd(a, 14) \gcd(b, 15).$$

Since $\gcd : \mathbb{Z} \times \mathbb{Z} \rightarrow \mathbb{Z}_{\geq 0} \implies \gcd \geq 1$, so $\gcd(c, 210) = 1 \iff \gcd(a, 14) = \gcd(b, 15) = 1$. By the above, **III** is true, **II** is true, **I** is false, so the answer is

(E) II and III only.

Solution 8 (Number Theory)

Rewrite as $15a + 14b = c$. Assume I, particularly the "or both" section. this fails if $a = 1$ and $b = 2$, for which we have the gcd is a factor of 2 times et cetera. Due to failure within the "or both" case, note that we can let $a = 2$ for which the first half fails. I is false.

Consider II. Firstly, we show "or both" is a viable case. Consider $\gcd(c, 210) = 1$. Then, $\gcd(14a + 15b, 210) = 1$. So, $14a + 15b$ is odd not a multiple of 3 nor 7. Modulo 3 gives $2a \equiv 0 \pmod{3}$. That implies $a \equiv 0 \pmod{3}$. Also, take modulo 7. Then, $b \equiv 0 \pmod{7}$. $\gcd(14a + 15b, 210) > 1$ holds if a, b is a multiple of either 7 or 3 respectively (or both). This implies that if $\gcd(14a + 15b, 210) = 1$ holds, then none of a, b is a multiple of 7 or 3, meaning that $\gcd(a, 14) = \gcd(b, 15) = 1$, which satisfies the "or both" case of II, and thus II is true.

Consider III. That is literally the "both" case of II, which we proved is always true.

The answer is II and III only.

Solution 9 (Set Theory)

We can use set theory to approach this problem, firstly by defining the universal set U to be all possible positive integer ordered pairs (a, b) .

Next, create a Venn diagram that consists of 2 sets: a set A of pairs (a, b) so that $\gcd(a, 14) > 1$, and a set B of all pairs (a, b) so that $\gcd(b, 15) > 1$. In other words, set A includes all pairs where a is a multiple of 2 or 7 and set B includes all pairs where b is a multiple of 3 or 5.

Multiplying the original equation by 210 yields $15a + 14b = c$. Because $210 = 2 \cdot 3 \cdot 5 \cdot 7$, if $\gcd(c, 210) > 1$, a must be a multiple of 2 or 7 or b must be a multiple of 3 or 5. Thus, $\gcd(c, 210) > 1$ directly corresponds to $A \cup B$.

Next, understand what it means when the gcd function equals 1. $\gcd(a, 14) = 1$, for example, means that a is **not** a multiple of 2 or 7. Similarly, $\gcd(b, 15) = 1$ means that b is **not** a multiple of 3 or 5. Note that this means the set of pairs satisfying $\gcd(a, 14) = 1$ is the complement of set A (A'). Similarly, the set of pairs satisfying $\gcd(b, 15) = 1$ is the complement of set B (B') and the set of pairs satisfying $\gcd(c, 210) = 1$ is the complement of $A \cup B$ ($(A \cup B)'$).

Therefore, we can rewrite statements I, II, and III using the complements of set A, set B, and $A \cup B$.

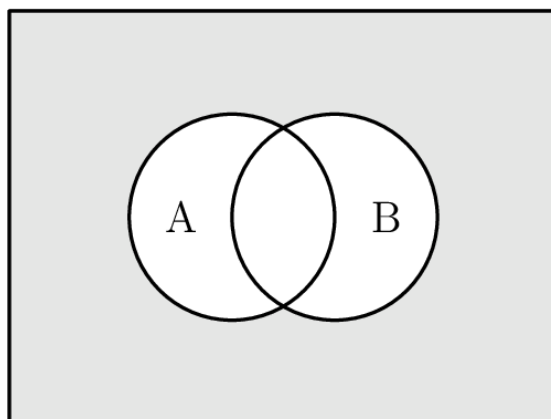
$$\text{I. } A' \cup B' \subseteq (A \cup B)'$$

$$\text{II. } (A \cup B)' \subseteq A' \cup B'$$

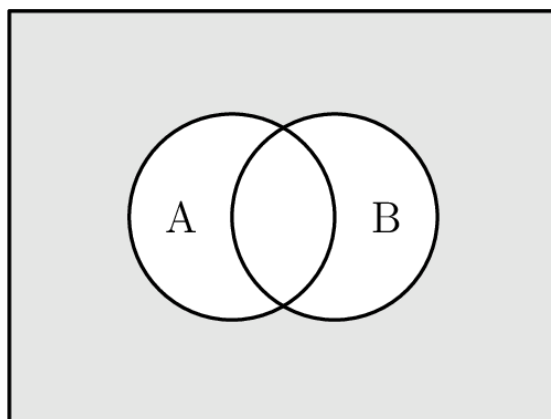
III. $(A \cup B)' = A' \cap B'$, $A' \cup B'$, $(A \cup B)'$, and $A' \cap B'$ can be modeled as follows:

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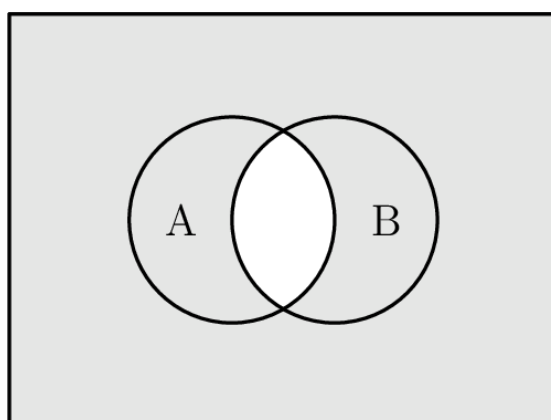
$$(A \cup B)'$$



$$A' \cap B'$$



$$A' \cup B'$$



If we simply compare these Venn diagrams to the rewritten statements, we can see that I is false and II and III (which turns out to

be De Morgan's First Law) are true. Thus, the answer is (E) II and III only.

19. (AMC 10 10B 2023) Sonya the frog chooses a point uniformly at random lying within the square $[0, 6] \times [0, 6]$ in the coordinate plane and hops to that point. She then randomly chooses a distance uniformly at random from $[0, 1]$ and a direction uniformly at random from north, south, east, west. All her choices are independent. She now hops the distance in the chosen direction. What is the probability that she lands outside the square?

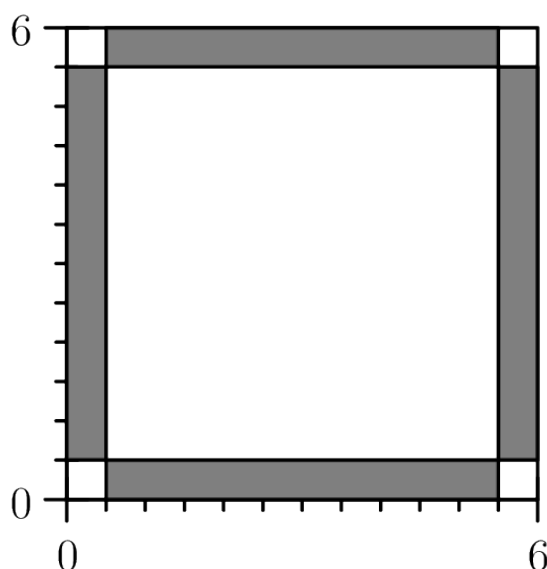
(A) $\frac{1}{6}$ (B) $\frac{1}{12}$ (C) $\frac{1}{4}$ (D) $\frac{1}{10}$ (E) $\frac{1}{9}$

Answer: B

Solution 1

WLOG, we assume Sonya jumps 0.5 units every time, since that is her expected value.

If Sonya is within 0.5 blocks of an edge, she can jump off the board. Let us examine the region that is at most 0.5 blocks from exactly one edge.



If Sonya starts in this region, she has a $\frac{1}{4}$ chance of landing outside (there's exactly one direction she can hop to get out). The total area of this region is $4 \cdot 0.5 \cdot 5 = 10$. For this region, Sonya has a $\frac{1}{4}$ chance, so we multiply 10 by $\frac{1}{4}$ to get 2.5.

If Sonya is in one of the corner squares, she can go two directions to get out, so she has a $\frac{2}{4} = \frac{1}{2}$ chance to get out. The total area is $0.5 \cdot 0.5 \cdot 4 = 1$, so this region yields $\frac{1}{2} \cdot 1 = \frac{1}{2}$.

Adding the two, we get 3. This is out of 36 square units of area, so our answer is thus (B) $\frac{1}{12}$.

Note: When Sonya is within 0 to 1 units away from the border in a given direction, the probability that she will jump outside the border can be thought of as a function of distance from the border. It is easy to see that if Sonya is x units away from the border, a jump distance in the interval $[x, 1]$ will move her outside. Thus the

probability is a function $P(x) = 1 - x$. The probability in an entire region 0 to 1 units away from the border is an integral of the function, $\int_0^1 (1 - x) dx = [x - \frac{1}{2}x^2]_0^1 = 1/2$. This is why the probability is $1/2$.

Solution 2

Since all the actions are independent, we can switch the orders. Let Sonya choose the direction first. And the problem is symmetric, so we consider just one direction. WLOG, let's say she choose *south*. When she first pick the location, she'll have to be within 1 unit of the x axis to have a chance to jump out of the boundary southward. That's $\frac{1}{6}$. With in that region, the expected y coordinate would be 0.5 which is 0.5 unit from the boundary (x -axis). Now, the jumping distance required to jump out of the boundary on average has to be greater than 0.5. That's another $\frac{1}{2}$. So the final probability is $\frac{1}{6} \cdot \frac{1}{2} = \boxed{\text{(B)} \frac{1}{12}}$.

Solution 3

We denote by (x, y) the frog's initial coordinates. We denote by $k \in \{n, e, s, w\}$ the direction to hop. We denote by z the hopping distance. In this analysis, we say that the frog wins if landing outside the square.

We have

$$\begin{aligned}
 P(\text{win}) &= \sum_{k \in \{n, e, s, w\}} P(\text{win}|k) P(k) \\
 &= P(\text{win}|k = w) \sum_{k \in \{n, e, s, w\}} P(k) \\
 &= P(\text{win}|k = w) \\
 &= \int_{y=0}^1 P(\text{win}|k = w, y) dy \\
 &= P(\text{win}|k = w, y = 0) \\
 &= P(\text{win}|k = w, y = 0, x \in [0, 1]) P(x \in [0, 1]) + P(\text{win}|k = w, y = 0, x \in (1, 6]) P(x \in (1, 6]) \\
 &= P(\text{win}|k = w, y = 0, x \in [0, 1]) \cdot \frac{1}{6} + 0 \cdot \frac{5}{6} \\
 &= \frac{1}{6} P(\text{win}|k = w, y = 0, x \in [0, 1]) \\
 &= \frac{1}{6} \int_{x=0}^1 \int_{z=x}^1 dz dx \\
 &= \frac{1}{6} \cdot \frac{1}{2} \\
 &= \boxed{\text{(B)} \frac{1}{12}}.
 \end{aligned}$$

The second equality above follows from symmetry that $P(\text{win}|k)$ are all the same for all $k \in \{n, e, s, w\}$. The fifth equality above follows from symmetry that $P(\text{win}|k = w, y)$ are all the same for all $y \in [0, 1]$.

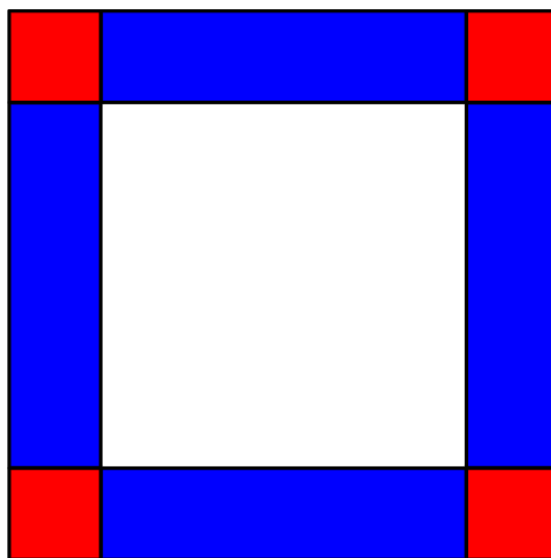
Solution 4

We can notice that Sonya can only jump out if she first picks a point that is at most 1 unit from the border. Let's separate this region into 4 different strips each

with overlapping corners. Since each strip is exactly the same, let's first consider the probability of Sonya leaving the square given that she first lands on the top strip. Using expected value, one can get that the probability that Sonya gets the distance needed to leave the square is $\frac{1}{2}$. Now the probability she gets the direction needed (north) is $\frac{1}{4}$. Now with the probability of landing in the strip being $\frac{1}{6}$, we get the probability that she lands on the strip and leaves the square to be $\frac{1}{6} \cdot \frac{1}{2} \cdot \frac{1}{4} = \frac{1}{48}$. Since there are four strips, we add the probabilities giving a final answer of **(B)** $\frac{1}{12}$.

Solution 5

We know that Sonya can only jump out if she is within 1 unit of the border. We can calculate the probability that Sonya can jump out.



The total area of the colored regions is 20, so the probability that Sonya lands in a colored region is $\frac{20}{36} = \frac{5}{9}$. We can calculate the probability that Sonya gets out of each type of region.

Case 1: Sonya chooses a blue region.

The probability that Sonya chooses a blue region is $\frac{16}{20} = \frac{4}{5}$. One direction can let her out, so the probability that she chooses the right one is $\frac{1}{4}$. Finally, the probability that Sonya chooses a distance to get her out is $\frac{1}{2}$. So, the probability that she chooses a blue region and gets out is $\frac{5}{9} \cdot \frac{4}{5} \cdot \frac{1}{4} \cdot \frac{1}{2} = \frac{1}{18}$.

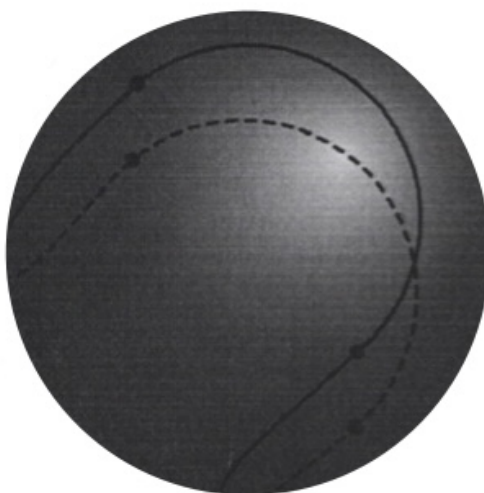
Case 2: Sonya choose a red region.

The probability that Sonya chooses a red region is $\frac{4}{20} = \frac{1}{5}$. Two directions can let her out, so the probability that she chooses one of them is $\frac{1}{2}$. Finally, the probability that Sonya chooses a distance to get her out is $\frac{1}{2}$. So, the probability

that she chooses a red region and gets out is $\frac{5}{9} \cdot \frac{1}{5} \cdot \frac{1}{2} \cdot \frac{1}{2} = \frac{1}{36}$.

So, the probability that Sonya gets out is $\frac{1}{18} + \frac{1}{36} = \frac{3}{36} = \boxed{\text{(B)} \frac{1}{12}}$.

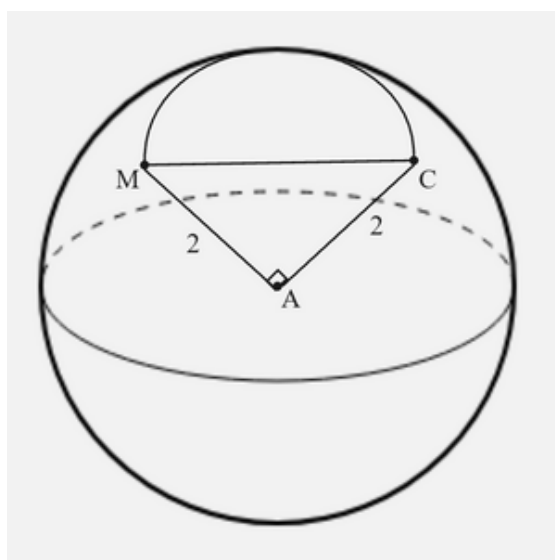
20. (AMC 10 10B 2023) Four congruent semicircles are drawn on the surface of a sphere with radius 2, as shown, creating a close curve that divides the surface into two congruent regions. The length of the curve is $\pi\sqrt{n}$. What is n ?



- (A) 32 (B) 12 (C) 48 (D) 36 (E) 27

Answer: A

Solution 1



Focus on 2 of the points. Let the center of the Sphere be A . Label two points that form the diameter of one of the four semicircles M and C respectively.

Triangle AMC is a right triangle through the inscribed right triangle theorem, with $AM = AC = 2$.

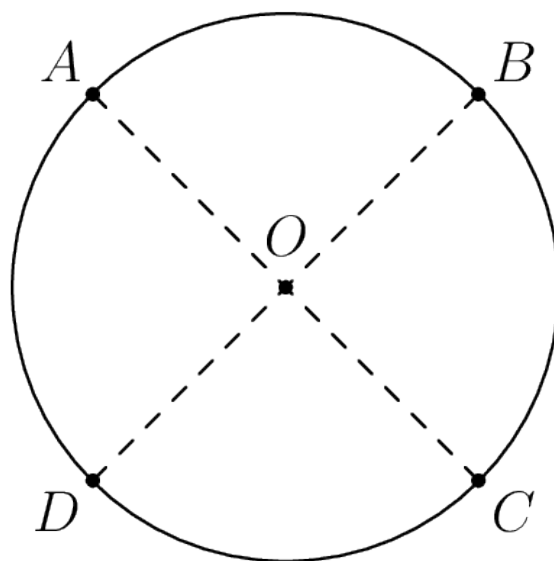
This is a 45-45-90 triangle, so the length of the diameter MC is just the hypotenuse of the triangle AMC , which is $2\sqrt{2}$. This means the radius is $\sqrt{2}$.

The circumference of the semicircle is $\frac{1}{2} \cdot 2\sqrt{2} \cdot \pi = \pi\sqrt{2}$, and because there are four congruent semicircles, the length of the semicircular region is $4\sqrt{2}\pi = \pi \cdot 4\sqrt{2}$.

The solution is in the form $\pi\sqrt{n}$, so we convert $\pi \cdot 4\sqrt{2}$ to $\pi\sqrt{16}\sqrt{2}$ to get $\pi\sqrt{32}$. $n = \boxed{\text{(A) } 32}$.

Solution 2

There are four marked points on the diagram; let us examine the top two points and call them A and B . Similarly, let the bottom two dots be C and D , as shown:



This is a cross-section of the sphere seen from the side. We know that $AO = BO = CO = DO = 2$, and by Pythagorean Theorem, length of $\overline{AB} = 2\sqrt{2}$.

Each of the four congruent semicircles has the length AB as a diameter (since $\overline{AB}$ is congruent to $\overline{BC}$, $\overline{CD}$, and $\overline{DA}$), so its radius is $\frac{2\sqrt{2}}{2} = \sqrt{2}$. Each one's arc length is thus $\pi \cdot \sqrt{2} = \sqrt{2}\pi$.

We have 4 of these, so the total length is $4\sqrt{2}\pi = \sqrt{32}\pi$, so thus our answer is $\boxed{\text{(A) } 32}$.

Note:

TLDR:

The radius of 2 gives us a line segment connecting diagonal vertices of the semi-circles with a measure of 4, giving us through $45^\circ - 45^\circ - 90^\circ$ relations and Pythagorean theorem a diameter for each semi-circle of $2\sqrt{2}$, which we can use to bash out the circumference of a full circle, multiply by 2, and move inside and under the root to get 32.

Solution 3

Assume A , B , C , and D are the four points connecting the semicircles. By law of symmetry, we can pretty confidently assume that $ABCD$ is a square. Then, $\overline{AB} = 2\sqrt{2}$, and the rest is the same as the second half of solution 1.

(Optimal Solution for Competition or if you don't have enough time)

Solution 4

We put the sphere to a coordinate space by putting the center at the origin. The four connecting points of the curve have the following coordinates: $A = (0, 0, 2)$, $B = (2, 0, 0)$, $C = (0, 0, -2)$, $D = (-2, 0, 0)$.

Now, we compute the radius of each semicircle. Denote by M the midpoint of A and B . Thus, M is the center of the semicircle that ends at A and B . We have $M = (1, 0, 1)$. Thus, $OM = \sqrt{1^2 + 0^2 + 1^2} = \sqrt{2}$.

In the right triangle $\triangle OAM$, we have $MA = \sqrt{OA^2 - OM^2} = \sqrt{2}$.

Therefore, the length of the curve is

$$4 \cdot \frac{1}{2} 2\pi \cdot MA = \pi\sqrt{32}.$$

Therefore, the answer is (A) 32.

Solution 5

Note that each of the diameters are the chord of the sphere of a quarter arc. Thus, the semicircles diameter's length is $2\sqrt{2}$. Thus, the entire curve is $2\sqrt{2} \cdot \pi \cdot \frac{1}{2} \cdot 4 = 4\sqrt{2}\pi = \sqrt{32}\pi$. Therefore, the answer is (A) 32. andliu766

Solution 6 (Cheese! Narrow it down to 2 choices!) and actual way (this one is stupid)

Cheese: You can immediately say that the answer choice is either (A) 32 or (C) 48 because there are four semicircles in that curve; there are $4 = \sqrt{16}$ semicircles in the curve, so n has to be a multiple of 16, and if you don't know how to do this problem, just guess one of (A) or (C). Actual way: Take a cross-section of the sphere to get four different points equidistant from the center O of the sphere, A , B , C , D such that $AO = BO = CO = DO = 2$, and so $ABCD$ is a square with side length $2\sqrt{2}$, and proceed as in Solution 2 to get (A) 32.

Solution 7 (even more cheese)

You can look at it and guess the four vertices make a square. Then, with more cheese, since the only number in the problem is 2 and its probably a diagonal, the answer must be some power of 2, because obviously. Therefore, it's probably (A) 32.

-Supernova77

21. (AMC 10 10B 2023) Each of 2023 balls is randomly placed into one of 3 bins. Which of the following is closest to the probability that each of the bins will contain an odd number of balls?

(A) $\frac{2}{3}$ (B) $\frac{3}{10}$ (C) $\frac{1}{2}$ (D) $\frac{1}{3}$ (E) $\frac{1}{4}$

Answer: E

Solution 1 (Systematic Algebraic Way)

Suppose the numbers are a_1 , a_2 , and a_3 . First, we try to calculate the amount of ways for all three balls to be placed in a bin so the number of balls in each bin is odd. $a_1 + a_2 + a_3 = 2023$ and each bin has at least one ball because they are positive odd numbers. Changing the equation, we see that $\frac{[(a_1)+1]}{2} + \frac{[(a_2)+1]}{2} + \frac{[(a_3)+1]}{2} = \frac{(2023+3)}{2}$. Let $a_1 + 1 = 2b_1$, $a_2 + 1 = 2b_2$, and $a_3 + 1 = 2b_3$. Thus $b_1 + b_2 + b_3 = 1013$. We can also see that b_1 , b_2 , and b_3 are all positive. Using the positive version of stars and bars, we get $\binom{1013-1}{3-1} = \binom{1012}{2}$ choices.

Now, we want to find the total amount of cases. Using the non-negative version of stars and bars, we find that the total is $\binom{2023+3-1}{3-1} = \binom{2025}{2}$.

Now we need to calculate $\binom{1012}{2} / \binom{2025}{2}$, which is just $\frac{1012 \cdot 1011 \cdot 2}{2 \cdot 2025 \cdot 2024}$. Cancelling the twos, we get $\frac{1012 \cdot 1011}{2025 \cdot 2024}$. This is roughly equal to $\frac{1}{4}$. The answer is $\boxed{\text{(E)} \frac{1}{4}}$.

Solution 2 (Pairing)

There are $\left\lceil \frac{2023}{2} \right\rceil = 1012$ odd numbers between 1 and 2023.

Out of these, there are $\left\lfloor \frac{1012}{3} \right\rfloor = 337$ ways to evenly distribute each ball into the first bin, then, WLOG, we say there is $337 - 2 = 335$ for the second and $337 + 2 = 339$ for the third bin.

Out of these, there are $\left\lfloor \frac{337}{2} \right\rfloor = 168$ odd sums of pairs for the first bin, then $\left\lfloor \frac{335}{2} \right\rfloor = 167$ for the second, and $\left\lfloor \frac{339}{2} \right\rfloor = 169$ for the third. The sum of all of these is $168 + 167 + 169 = 504$ total distributions.

From here, we need to find the probability over 2023 total distributed balls, which is simply $504/2023 \approx 500/2000 = 1/4 \Rightarrow \boxed{\text{(E)} \frac{1}{4}}$.

This is not an exact probability. It is simply a solution for general competition math. For the exact way to calculate the probability (highly not recommended), please refer to Solution 8.

Solution 3

We first examine the possible arrangements for parity of number of balls in each box for 2022 balls.

If a E denotes an even number and a O denotes an odd number, then the distribution of balls for 2022 balls could be EEE , EOO , OEO , or OOE . With the insanely overpowered magic of cheese, we assume that each case is about equally likely.

From EEE , it is not possible to get to all odd by adding one ball; we could either get OEE , EOE , or EEO . For the other 3 cases, though, if we add a ball to the exact right place, then it'll work.

For each of the working cases, we have 1 possible slot the ball can go into (for OEO , for example, the new ball must go in the center slot to make OOO) out of the 3 slots, so there's a $\frac{1}{3}$ chance. We have a $\frac{3}{4}$ chance of getting one of these working cases, so our answer is $\frac{3}{4} \cdot \frac{1}{3} = \boxed{\text{(E)} \frac{1}{4}}$.

Solution 4

2023 is an arbitrary large number. So, we proceed assuming that an arbitrarily large number of balls have been placed.

For an odd-numbered amount of balls case, the 3 bins can only be one of these 2 combinations:

OEE (OEE, EOE, EEO)

OOO (OOO)

Let the probability of achieving the OOO case to be $P(OOO) = p$ and any of the OEE permutations to be $P(OEE) = 1 - p$.

Because the amount of balls is arbitrarily large, $P(OOO) = p$ even after another two balls are be placed.

There are two cases for which placing another two balls results in OOO :

OOO : The two balls are placed in the same bin ($OOO \rightarrow OOE \rightarrow OOO$)

OEE : The two balls are placed in the two even bins ($OEE \rightarrow OOE \rightarrow OOO$)

So,

$$P(OOO) = P(OOO) * \frac{1}{3} + P(OEE) * \frac{2}{3} * \frac{1}{3} p = p * \frac{1}{3} + (1 - p) * \frac{2}{3} * \frac{1}{3} \frac{8}{9} p = \frac{2}{9}$$

$$p = \boxed{(E) \frac{1}{4}}$$

Solution 5

We use the generating functions approach to solve this problem. Define $\Delta = \{(a, b, c) \in \mathbb{Z}_+ : a + b + c = 2023\}$.

We have

$$(x + y + z)^{2023} = \sum_{(a,b,c) \in \Delta} \binom{2023}{a, b, c} x^a y^b z^c.$$

First, we set $x \leftarrow 1, y \leftarrow 1, z \leftarrow 1$. We get

$$3^{2023} = \sum_{(a,b,c) \in \Delta} \binom{2023}{a, b, c} 1. \quad (1)$$

Second, we set $x \leftarrow 1, y \leftarrow -1, z \leftarrow 1$. We get

$$1 = \sum_{(a,b,c) \in \Delta} \binom{2023}{a, b, c} (-1)^b. \quad (2)$$

Third, we set $x \leftarrow 1, y \leftarrow 1, z \leftarrow -1$. We get

$$1 = \sum_{(a,b,c) \in \Delta} \binom{2023}{a, b, c} (-1)^c. \quad (3)$$

Fourth, we set $x \leftarrow 1, y \leftarrow -1, z \leftarrow -1$. We get

$$-1 = \sum_{(a,b,c) \in \Delta} \binom{2023}{a, b, c} (-1)^{b+c}. \quad (4)$$

Taking $\frac{(1)-(2)-(3)+(4)}{4}$, we get

$$\frac{3^{2023} - 1 - 1 + (-1)}{4} = \frac{1}{4} \sum_{(a,b,c) \in \Delta} \binom{2023}{a, b, c} (1 - (-1)^b - (-1)^c + (-1)^{b+c})$$

$$\begin{aligned}
 &= \frac{1}{4} \sum_{(a,b,c) \in \Delta} \binom{2023}{a,b,c} (1 - (-1)^b) (1 - (-1)^c) \\
 &= \sum_{\substack{(a,b,c) \in \Delta \\ a,b,c \text{ are odds}}} \binom{2023}{a,b,c}.
 \end{aligned}$$

The last expression above is the number of ways to get all three bins with odd numbers of balls. Therefore, this happens with probability

$$\frac{\frac{3^{2023} - 1 - 1 + (-1)}{4}}{3^{2023}} \approx \boxed{\text{(E)} \frac{1}{4}}.$$

Solution 6

Four even-odd splittings divides 2023 in to three, namely (O, O, O) , (E, E, O) , (E, O, E) , and (O, E, E) . Here if we define a "move" as relocated one ball, then we will notice in each case, that a random "move" will be evenly likely to be one of the other three splittings. Hence by Group theory (or by intuition), we will find the structure of this splitting is V_4 group, and it's symmetric for all four elements in this Group.

Thus, no matter what is the initial starting point, four cases will be evenly likely to appear when repeated many times. The answer is $\boxed{\text{(E)} \frac{1}{4}}$.

Solution 7 (Exact Probability)

Let $p(n)$ be the probability of all 3 bins having odd number of balls when we randomly put $2n + 1$ balls into these bins. We need find out $p(1011)$. Clearly $p(0) = 0$. For $n > 1$, there are only two scenarios for the first $2n - 1$ balls: **Case 1** all 3 bins are odd; **Case 2** only 1 bin is odd and the other 2 bins are even.

For case 1 which has a probability $p(n - 1)$, in order to get all 3 bins to be odd, we need to put the last 2 balls in the same bin. This has a probability of $\frac{1}{3}$.

For case 2 which has a probability of $1 - p(n - 1)$, in order to get all 3 bins to be odd, we need to put the last 2 balls into the 2 even bins. The first one has a probability of $\frac{2}{3}$ to be put in an even bin, and the second has a probability of $\frac{1}{3}$ to be put the in the last even bin, so the probability is $\frac{2}{9}$ to get all 3 bins to be odd in this case.

So we have $p(n) = \frac{1}{3}p(n - 1) + \frac{2}{9}(1 - p(n - 1)) = \frac{1}{9}p(n - 1) + \frac{2}{9}$.

Solving for the fixed point: $x = \frac{1}{9}x + \frac{2}{9}$ yields $x = \frac{1}{4}$, and this helps us the rewrite the above as $p(n) - \frac{1}{4} = \frac{1}{9}(p(n - 1) - \frac{1}{4})$.

Therefore $p(n) - \frac{1}{4} = \frac{1}{9^n}(p(0) - \frac{1}{4})$. Using $p(0) = 0$, we have $p(n) = \frac{1}{4}(1 - \frac{1}{9^n})$.

The probability for 2023 balls is $\frac{1}{4}(1 - \frac{1}{9^{1011}})$, and this is closest to $\boxed{\text{(E)} \frac{1}{4}}$.

Solution 8 (Recursion, rigorous)

We can use recursion to figure out an expression that determines the probability of all the bins being odd.

Define P_n to be the probability that with $2n - 1$ balls, randomly inserting them makes all 3 bins have an odd count. We would first want to find $P_{n+1} = f(P_n)$ for some function f .

Now, we can say that if we have an O-O-O combination, where O denotes odd and E denotes even, then to add 2 more balls to keep it O-O-O, we need to put them in the same bin. This would result in a $\frac{1}{3}$ probability of staying O-O-O.

If we have an E-E-O combination, then we need to add 2 more balls, one in each of the even bins. This would result in a $\frac{2}{9}$ probability of turning to O-O-O. Now, we can write a function of P_{n+1} in terms of P_n .

$$P_{n+1} = \frac{1}{3}P_n + \frac{2}{9} - \frac{2}{9}P_n = \frac{1}{9}P_n + \frac{2}{9}$$

Now, we want to express $P_{n+1} = g(n)$ for some function g . Here, the $\frac{2}{9}$ is a problem because now we can't just put it as a geometric sequence. However, we can rewrite it as followed:

$$P_{n+1} = \frac{1}{9}P_n + \frac{2}{9} \quad P_{n+1} - \frac{1}{4} = \frac{1}{9}(P_n - \frac{1}{4})$$

Now, we can let $Q_n = P_n - \frac{1}{4}$. We can rewrite that as $Q_{n+1} = \frac{1}{9}Q_n$.

We can note that $Q_1 = P_1 - \frac{1}{4}$. Since $P_1 = 0$ (you only have one ball, so two of the bins have to be even), $Q_1 = -\frac{1}{4}$. Therefore, $Q_n = Q_1(\frac{1}{9})^{n-1} = -\frac{1}{4} \cdot (\frac{1}{9})^{n-1}$.

Substituting, we can find that $P_n = -\frac{1}{4} \cdot (\frac{1}{9})^{n-1} + \frac{1}{4}$.

2023 is the 1012th odd number, so we can substitute $n = 1012$ into the equation.

$$-\frac{1}{4} \cdot (\frac{1}{9})^{1011} + \frac{1}{4} \approx 0 + \frac{1}{4} = \frac{1}{4}.$$

Therefore, the answer is (E) $\frac{1}{4}$

Solution 9 (Generating Functions instead of stars and bars)

Define

$$\begin{aligned} [x^{2023}]G(x) &= (x^1 + x^3 + \dots + x^{2023} + \dots)^3 \\ [x^{2023}]G(x) &= x^3(1 + x^2 + x^4 + \dots + x^{2022} + \dots)^3 \\ [x^{2020}]G(x) &= \left(\frac{1}{1-x^2}\right)^3 \\ [x^{2020}]G(x) &= (1-x^2)^{-3} \\ [x^{2020}]G(x) &= \left(1 + \frac{3 \cdot 2}{2 \cdot 1}(x^2)^1 + \dots + \frac{1012 \cdot 1011}{2 \cdot 1}(x^2)^{1010}\right) \\ [x^{2020}] &= \frac{1012 \cdot 1011}{2 \cdot 1} \end{aligned}$$

Similarly we can compute the total number of cases which is: $\frac{2025 \cdot 2022}{2 \cdot 1}$

So, the probability is:

$$\begin{aligned} &= \frac{\frac{1012 \cdot 1011}{2 \cdot 1}}{\frac{2025 \cdot 2022}{2 \cdot 1}} \\ &= \frac{1012 \cdot 1011}{2025 \cdot 2022} \\ &\approx \frac{1}{4} \end{aligned}$$

Therefore, the answer is (E) $\frac{1}{4}$

Solution 10 (Short On Time)

One must not forget that the balls are placed randomly. With intuition, we can assume that the probability of getting OOO , OOE , OEE , and EEE , where O denotes odd numbers and E denotes even numbers, would nearly be the same. Since there are 4 possible outcomes and 1 of them is desired, we can conclude that the likelihood of getting OOO is $\frac{1}{4}$ and that the correct answer choice is **(E)** $\frac{1}{4}$.

Note: Use this type of reasoning only when a logical solution doesn't work out for you.

Another Note (not by Aqf243): This may not be the best solution because 2023 is an odd number. Hence, realistically, the probability of getting OOE is 0, allowing us to remove it from our list. This is because adding 2 odd numbers (represented as 2 numbers $x \in \{x \mid x \bmod 2 = 1\}$) always results in an even number.

$\therefore$

$$x_1 = 1 \bmod 2 \quad x_2 = 1 \bmod 2 \therefore$$

$$x_1 + x_2 = 1 + 1 \bmod 2 = 2 \bmod 2 = 0 \bmod 2$$

for any number $x \in x \mid x \bmod 2 = 0$, x is an even number.

Similarly, adding two even numbers creates another even number

$$x_1 = 0 \bmod 2$$

$$x_2 = 0 \bmod 2 \quad 2x_1 + x_2 = 0 + 0 \bmod 2 = 0 \bmod 2$$

So adding 2 odd numbers and an even number ALWAYS results in an even number.

Basically, Aqf243 got a little lucky in this situation...

Solution 11 (Notice a pattern)

Let the bins be called a , b , and c . We want the number of odd positive integer solutions that satisfy $a + b + c = 2023$. We know from stars and bars that the total number of integer solutions is $\binom{2023+3-1}{3-1} = \binom{2025}{2}$. If $a = 1$, b can be any odd number from 1 to 2021. There are $(2021 + 1)/2 = 1011$ choices for b . If $a = 3$, there would be 1010 choices for b , and so on. c is always constrained by a and b so our answer is just the number of choices b has for each a , which is the sum of all integers from 1 to 1011 = $\frac{1011 \cdot 1012}{2}$. So we get $\frac{1011 \cdot 1012/2}{\binom{2025}{2}}$ (around 0.249), which is

closest to 0.25 or **(E)** $\frac{1}{4}$.

NOTE: The probability we came to in this solution and in other solutions that use Stars and Bars is not exact. Stars and Bars treats all outcomes to be equally likely when in reality some distributions are more likely than others (see clarification above), but the reason we still use it here is because it gives us a good average and the answer choices are pretty far apart so it's not very significant to our approximation.

22. (AMC 10 10B 2023) How many distinct values of x satisfy $\lfloor x \rfloor^2 - 3x + 2 = 0$, where $\lfloor x \rfloor$ denotes the largest integer less than or equal to x ?

(A) an infinite number (B) 4 (C) 2 (D) 3 (E) 0

Answer: B

Solution 1

To further grasp at this equation, we rearrange the equation into

$$\lfloor x \rfloor^2 = 3x - 2.$$

Thus, $3x - 2$ is a perfect square and nonnegative. It is now much more apparent that $x \geq 2/3$, and that $x = 2/3$ is a solution.

Additionally, by observing the RHS, $x < 4$, as

$$\lfloor 4 \rfloor^2 > 3 \cdot 4,$$

since squares grow quicker than linear functions.

Now that we have narrowed down our search, we can simply test for intervals $[2/3, 1]$, $[1, 2]$, $[2, 3]$, $[3, 4]$. This intuition to use intervals stems from the fact that $x = 1, 2$ are observable integral solutions.

Notice how there is only one solution per interval, as $3x - 2$ increases while $\lfloor x \rfloor^2$ stays the same.

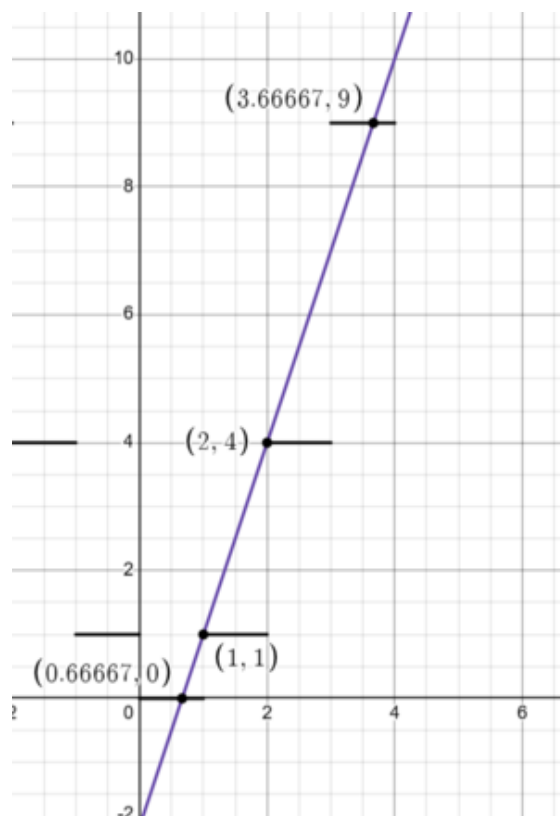
Finally, we see that $x = 3$ does not work, however, through setting $\lfloor x \rfloor^2 = 9$, $x = 11/3$ is a solution and within our domain of $[3, 4]$.

This provides us with solutions $(\frac{2}{3}, 1, 2, \frac{11}{3})$, thus the final answer is (B) 4!

Solution 2 (Graphing)

This solution is based heavily on a good intuitive understanding of the graph of $\lfloor x \rfloor$.

We quickly see that $\lfloor x \rfloor^2 = 3x - 2$. We need the solutions that fulfill $\lfloor x \rfloor^2$ and $3x - 2$. We quickly draw the system of equations.



Looking at our graph, we see that there are 4 intersections, implying that there are (B) 4 solutions that work.

We can quickly convince ourselves that other solutions don't exist. We notice that $\lfloor x \rfloor^2 = 25$ doesn't work - this part of $\lfloor x \rfloor^2$ is left of the graph of $y = 3x - 2$. As $\lfloor x \rfloor^2$ grows faster than $y = 3x - 2$, we know there are no more intersections to the right. On the left, there are no more solutions as well as $\lfloor x \rfloor^2 > 0$ while $3x - 2 < 0$.

Solution 3 (Desperation)

Notice there has to be a solution for x between $(2, -3)$ and $(1, 2)$ because of the floors. There is also no way 2 solutions exist because of the quadratic, and when we add them together, we get (B) 4.

Solution 4 (Three Cases)

First, let's take care of the integer case—clearly, only $x = 1, 2$ work. Then, we know that $3x$ must be an integer. Set $x = \frac{a}{3}$. Now, there are two cases for the value of $\lfloor x \rfloor$. Case 1: $\lfloor x \rfloor = \frac{a-1}{3}$

$$\frac{a^2 - 2a + 1}{9} = a - 2 \rightarrow a^2 - 2a + 1 = 9a - 18 \rightarrow a^2 - 11a + 19 = 0.$$

There are no solutions in this case. Case 2: $\lfloor x \rfloor = \frac{a-2}{3}$

$$\frac{a^2 - 4a + 4}{9} = a - 2 \rightarrow a^2 - 4a + 4 = 9a - 18 \rightarrow a^2 - 13a + 22 = 0.$$

This case provides $\frac{2}{3}$ and $\frac{11}{3}$ as two more solutions. Our final answer is thus 4.

Solution 5

First, $x = 2, 1$ are trivial solutions

We assume from the shape of a parabola and the nature of the floor function that any additional roots will be near 2 and 1

We can now test values for $\lfloor x \rfloor$:

$$\lfloor x \rfloor = 0$$

We have $0 - 3x + 2 = 0$. Solving, we have $x = \frac{2}{3}$. We see that $\lfloor \frac{2}{3} \rfloor = 0$, so this solution is valid

$$\lfloor x \rfloor = -1$$

We have $1 - 3x + 2 = 0$. Solving, we have $x = 1$. $\lfloor 1 \rfloor \neq -1$, so this is not valid.

We assume there are no more solutions in the negative direction and move on to $\lfloor x \rfloor = 3$ $\lfloor x \rfloor = 3$

We have $9 - 3x + 2 = 0$. Solving, we have $x = \frac{11}{3}$. We see that $\lfloor \frac{11}{3} \rfloor = 3$, so this solution is valid

$$\lfloor x \rfloor = 4$$

We have $16 - 3x + 2 = 0$. Solving, we have $x = 6$. $\lfloor 6 \rfloor \neq 4$, so this is not valid. We assume there are no more solutions.

Our final answer is (B) 4

Solution 6

Denote $a = \lfloor x \rfloor$. Denote $b = x - \lfloor x \rfloor$. Thus, $b \in [0, 1)$.

The equation given in this problem can be written as

$$a^2 - 3(a + b) + 2 = 0.$$

Thus,

$$3b = a^2 - 3a + 2.$$

Because $b \in [0, 1)$, we have $3b \in [0, 3)$. Thus,

$$a^2 - 3a + 2 = 0, 1, \text{ or } 2.$$

If $a^2 - 3a + 2 = 0$, $(a - 2)(a - 1) = 0$ so a can be 1, 2.

If $a^2 - 3a + 2 = 1$, $a^2 - 3a + 1 = 0$ which we find has no integer solutions after finding the discriminant.

If $a^2 - 3a + 2 = 2$, $a^2 - 3a = 0 \rightarrow a(a - 3) = 0$ so a can also be 0, 3.

Therefore, $a = 1, 2, 0, 3$. Therefore, the number of solutions is (B) 4.

Solution 7 (Quick)

A quadratic equation can have up to 2 real solutions. With the $\lfloor x \rfloor$, it could also help generate another pair. We have to verify that the solutions are real and distinct.

First, we get the trivial solution by ignoring the floor. $(x - 2)(x - 1) = 0$, we get $(2, 1)$ as our first pair of solutions.

Up to this point, we can rule out A, E.

Next, we see that $\lfloor x \rfloor^2 - 3x = 0$. This implies that $-3x$ must be an integer. We can guess and check x as $\frac{k}{3}$ which yields $\left(\frac{2}{3}, \frac{11}{3}\right)$.

So we got 4 in total $\left(\frac{2}{3}, 1, 2, \frac{11}{3}\right)$, meaning that our answer is (B) 4.

Solution 8

$x = 1, 2$ are trivial solutions. Let $x = n + f$ for some integer n and some number f such that $-1 < f < 1$.

$$\lfloor x \rfloor^2 - 3x + 2 = \lfloor n + f \rfloor^2 - 3(n + f) + 2 = n^2 - 3(n + f) + 2.$$

So now we have

$$n^2 - 3(n + f) + 2 = 0,$$

which we can rewrite as

$$n(n - 3) = 3f - 2.$$

Since n is an integer, $n(n - 3)$ is an integer, so $3f - 2$ is an integer. Since $-1 < f < 1$, the only possible values of f are $\frac{1}{3}$, $\frac{2}{3}$, $-\frac{1}{3}$, and $-\frac{2}{3}$. Plugging in each value, we find that the only value of f that produces integer solutions for n is $f = \frac{2}{3}$. If $f = \frac{2}{3}$, $n = 0$ or $n = 3$. Hence, there is a total of 4 possible solutions, so the answer is

(B) 4. azc1027

Solution 9

We rewrite the equation as $\lfloor x \rfloor^2 - 3\lfloor x \rfloor - 3\{x\} + 2 = 0$, where $\{x\}$ is the fractional part of x

Denote $\lfloor x \rfloor = x_1$ and $\{x\} = x_2$. Thus

$$x_1^2 - 3x_1 - 3x_2 + 2 = 0.$$

By definition, $0 \leq x_2 \leq 1$. We then have $x_1^2 - 3x_1 + 2 = 3x_2$ and therefore $0 \leq x_1^2 - 3x_1 + 2 \leq 3$.

Solving, we have $\left[\frac{3-\sqrt{13}}{2}, 1\right] \cup \left[2, \frac{3+\sqrt{13}}{2}\right]$. But since x_1 is an integer, we have x_1 can only be 0, 1, 2, or 3.

Testing, we see these values of x_1 work, and therefore the answer is just **(B) 4**.

Similar approach as Solution 9 Use the fact that $x = \lfloor x \rfloor + \{x\}$. Thus we have

$$(\lfloor x \rfloor)^2 - 3\lfloor x \rfloor + 2 - 3\{x\} = 0.$$

Noting that $0 \leq \{x\} < 1$, we get

$$0 \leq (\lfloor x \rfloor - 2)(\lfloor x \rfloor - 1) < 3.$$

From there, it is not too hard to see that the only values of $\lfloor x \rfloor$ that satisfy this condition (while also noting that $\lfloor x \rfloor$ must be an integer) are 3, 2, 1, and 0, yielding 4 values.

Solution 10 (Very Fast)

We know that for integer values of x , the graph is just $x^2 - 3x + 2$. From the interval $[x, x + 1]$, the square stays the same, so the graph has a line segment that goes down by 3 and right by 2. This is very easy to graph, so we see that there are 4 solutions. Or, we notice that only $x = 0, 1, 2, 3$ results in a $x^2 - 3x + 2$ in the interval $[0, 3]$. That is **(B) 4** solutions.

Solution 11

Define $\lfloor x \rfloor = n$, and define the fractional part of x as $x(f)$.

Thus $\lfloor x \rfloor^2 - 3x + 2 = 0$ is

$$n^2 - 3(n + x(f)) + 2 = 0.$$

Expanding,

$$n^2 - 3n - 3x(f) + 2 = 0.$$

We realize $n^2 - 3n + 2$ must always be an integer so for $n^2 - 3n + 2$ to equal zero, $3x(f)$ must also equal an integer. Therefore, $x(f)$ must be $\frac{1}{3}$, $\frac{2}{3}$, or 0. Plugging in $x(f) = 0$ gives

$$n^2 - 3n - 3(0) + 2 = 0,$$

which simplifies to

$$n^2 - 3n + 2 = 0.$$

Continuing, plugging in $x(f) = \frac{1}{3}$ gives

$$n^2 - 3n - 3\left(\frac{1}{3}\right) + 2 = 0,$$

which simplifies to

$$n^2 - 3n + 1 = 0.$$

Finally, substituting $x(f) = \frac{2}{3}$ gives

$$n^2 - 3n - 3\left(\frac{2}{3}\right) + 2 = 0,$$

which simplifies to

$$n^2 - 3n = 0.$$

We know n must be an integer, so we can just solve for n and only utilize the integers we end up with.

We get two integers n if $x(f) = 0$: $n = 1, 2$. We know $n + x(f) = x$ by definition, and plugging the values of n and $x(f)$ into this equation gives us two solutions for x :

$$x = 1, 2.$$

We don't get any integers n if $x(f) = \frac{1}{3}$. We use the quadratic discriminant, $\sqrt{b^2 - 4ac}$ to get an expression which yields a non-integer value, $\sqrt{3^2 - 4(1)(1)}$, which means this case is invalid.

We get two integers n if $x(f) = \frac{2}{3}$: $n = 0, 3$. We know $n + x(f) = x$ by definition, and plugging the values of n and $x(f)$ into this equation gives us two solutions for x :

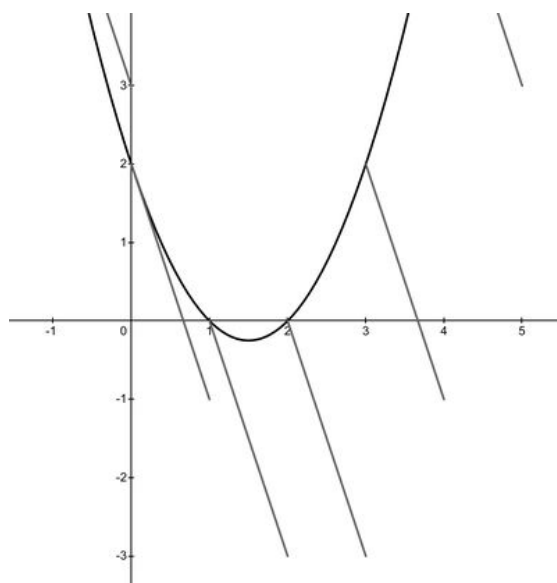
$$x = \frac{2}{3}, \frac{11}{3}.$$

We end up with a total of four solutions which are $x = 1, 2, \frac{2}{3}, \frac{11}{3}$.

Our answer is

$\boxed{\text{B}}$

Solution 12 (Based on graph)



For $\lfloor x \rfloor^2 - 3x + 2 = 0$, there is a discontinuity at each integer value of x , and it also lies on the non-floor version of the function. Between each integer x and the next forms a line with a slope of -3 . This simplifies the task of sketching the function's

graph. Note that the points at 1 and 2 are considered intersections because they are points on the left side of each integer interval that exactly lie on $y = 0$. Thus, we conclude there are 4 intersection points, and the answer is (B) 4.

Solution 13 (casework + patterns + logic)

First, we see that $3x$ must be an integer since $(\lfloor x \rfloor)^2$, 2, and 0 are all integers too.

Also, $3x$ must be positive, since subtracting it must cancel out $(\lfloor x \rfloor)^2 + 2$ which is positive. Whenever $3x$ can match $(\lfloor x \rfloor)^2 + 2$, we have a working solution.

If $3x$ is 0, 1, or 2, then $(\lfloor x \rfloor)^2 = 0$. So, $x = \frac{2}{3}$ is a solution.

If $3x$ is 3, 4, or 5, then $(\lfloor x \rfloor)^2 = 1$. So, $x = 1$ is a solution.

If $3x$ is 6, 7, or 8, then $(\lfloor x \rfloor)^2 = 4$. So, $x = 2$ is a solution.

If $3x$ is 9, 10, or 11, then $(\lfloor x \rfloor)^2 = 9$. So, $x = \frac{11}{3}$ is a solution.

More than that, $(\lfloor x \rfloor)^2$ starts to grow much faster than $3x$, which you could show by testing the next few groups of three consecutive numbers. For example, the next row is 12, 13, and 14 and you need one of these to be $4^2 + 2 = 18$, but this isn't the case.

Therefore, there are (B) 4 solutions to this equation.

23. (AMC 10 10B 2023) An arithmetic sequence of positive integers has $n \geq 3$ terms, initial term a , and common difference $d > 1$. Carl wrote down all the terms in this sequence correctly except for one term, which was off by 1. The sum of the terms he wrote down was 222. What is $a + d + n$?

(A) 24 (B) 20 (C) 22 (D) 28 (E) 26

Answer: B

Solution 1

Since one of the terms was either 1 more or 1 less than it should have been, the sum should have been $222 - 1 = 221$ or $222 + 1 = 223$.

The formula for an arithmetic series is $an + d \left(\frac{(n-1)n}{2} \right) = \frac{n}{2} (2a + d(n-1))$.

This can quickly be rederived by noticing that the sequence goes $a, a + d, a + 2d, a + 3d, \dots, a + (n-1)d$, and grouping terms.

We know that $\frac{n}{2}(2a + d(n-1)) = 221$ or 223 . Let us now show that 223 is not possible.

If $\frac{n}{2}(2a + d(n-1)) = 223$, we can simplify this to be $n(2a + d(n-1)) = 223 \cdot 2$. Since every expression here should be an integer, we know that either $n = 2$ and $2a + d(n-1) = 223$ or $n = 223$ and $2a + d(n-1) = 2$. The latter is not possible, since $n \geq 3, d > 1$, and $a > 0$. The former is also impossible, as $n \geq 3$. Thus, $\frac{n}{2}(2a + d(n-1)) \neq 223 \implies \frac{n}{2}(2a + d(n-1)) = 221$.

(Alternatively, we have $S = mn$ with n terms and arithmetic mean m , and $223 = mn$ does not satisfy both $m > 1$ and $n > 1$ because it is prime.)

We can factor 221 as $13 \cdot 17$. Using similar reasoning, we see that $221 \cdot 2$ cannot be paired as 2 and 221, but rather must be paired as 13 and 17 with a factor of 2 somewhere.

Let us first try $n = 13$. Our equation simplifies to $2a + 12d = 34 \implies a + 6d = 17$. We know that $d > 1$, so we try the smallest possible value: $d = 2$. This would give us $a = 17 - 2 \cdot 6 = 17 - 12 = 5$. (Indeed, this is the only possible d .)

There is nothing wrong with the values we have achieved, so it is reasonable to assume that this is the only valid solution (or all solutions sum to the same thing), so we answer $a + d + n = 5 + 2 + 13 = \boxed{\text{(B)} 20}$.

For the sake of completeness, we can explore $n = 17$. It turns out that we reach a contradiction in this case, so we are done.

Solution 2

There are n terms, the x th term is $a + (x-1)d$, so the summation is $an + dn(n-1)/2 = n(a + \frac{d(n-1)}{2})$.

The summation of the set is $222 \pm 1 = 221, 223$. First, 221: its only possible factors are 1, 13, 17, 221, and as said by the problem, $n \geq 3$, so n must be 13, 17, or 221. Let's start with $n = 13$. Then, $a + 6d = 17$, and this means $a = 5, d = 2$. Summing gives $13 + 5 + 2 = \boxed{\text{(B)} 20}$. We don't need to test any more cases, since the problem writes that all $a + d + n$ are the same.

-HIA2020

Solution 3

We must have the sum of terms of the arithmetic sequence is 222 ± 1 , which is 221 or 223.

Since we have 223 is prime, it cannot be the sum of the arithmetic sequence.

We see that 221 is just 13×17 .

We can write any arithmetic sequence with an odd amount of terms like this: $b - md, \dots, b - 2d, b - d, b, b + d, b + 2d, \dots, b + md$ where b is the middle term and d is the common difference.

By the sum of an arithmetic sequence, we have $13b = 221$ and $17b = 221$ and therefore $(b, n) = (13, 17)$ or $(17, 13)$.

Then $a + d + n = b - md + d + n = 30 - (m-1)d$.

We must have that m is either $\frac{17-1}{2}$ or $\frac{13-1}{2}$, so m is either 6 or 8.

So $a + d + n = 30 - 5d$ or $30 - 7d$.

Taking mod 7, we have no answer choices that give 2, and then taking mod 5 gives the only answer that works is 20.

Therefore we have $a + d + n = \boxed{\text{(B)} 20}$.

Solution 4

The formula for the sum of an arithmetic sequence is $n(\frac{a_1 + a_n}{2})$, where a is the first term, a_n is the last term, and n is the number of terms. Let a be the first term, d be the common difference, and n be the number of terms of Carl's sequence. Since the sum the sequence is 1 less or 1 more than 222, we have

$$n(\frac{a + d(n-1)}{2}) \pm 1 = 222$$

$$n(\frac{a + d(n-1)}{2}) = 222 \pm 1.$$

The right-hand side is either 221 or 223. We know that it has to be divisible by n so we can find the factors of 221 and 223. Checking all the primes less than 15, we find that 223 is prime and $221 = 13 \cdot 17$.

Because $n \geq 3$, the sum must be 221 and the only possible values of n are 13 and 17. We can test both cases.

Case 1: $n = 17$ Substituting for n gives us $a + 8d = 17$. Since the sequence consists of only positive integers, d is an integer. We know that $d > 1$ but if $d \geq 2$, then $8d > 13$. Hence, this case is not possible.

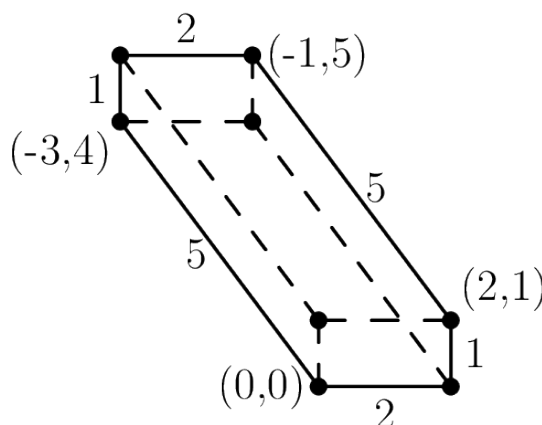
Case 2: $n = 13$ Substituting for n gives us $a + 6d = 17$. Using the same logic from case 1, we get $3 > d > 1$, so $d = 2$. Solving for a , we get $a = 5$. Therefore, $a + d + n = 5 + 2 + 13 = 20$, so the answer is **(B) 20**.

24. (AMC 10 10B 2023) What is the perimeter of the boundary of the region consisting of all points which can be expressed as $(2u - 3w, v + 4w)$ with $0 \leq u \leq 1$, $0 \leq v \leq 1$, and $0 \leq w \leq 1$?

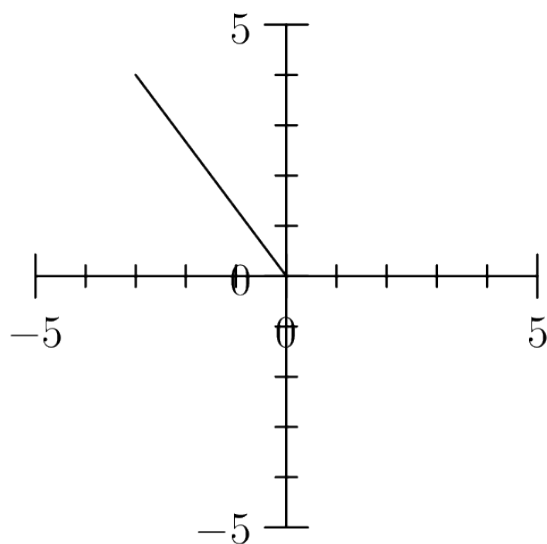
(A) $10\sqrt{3}$ (B) 13 (C) 12 (D) 18 (E) 16

Answer: E

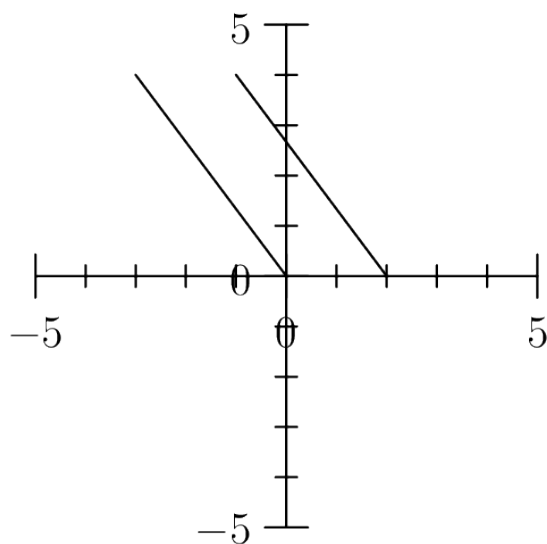
Solution 1



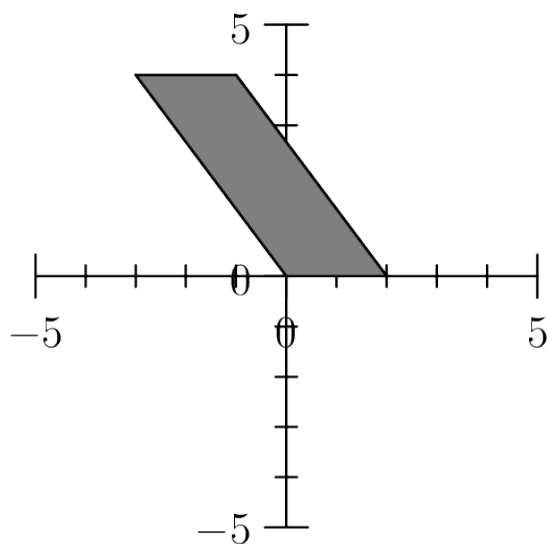
Notice that we are given a parametric form of the region, and w is used in both x and y . We first fix u and v to 0, and graph $(-3w, 4w)$ from $0 \leq w \leq 1$. When w is 0, we have the point $(0, 0)$, and when w is 1, we have the point $(-3, 4)$. We see that since this is a directly proportional function, we can just connect the dots like this:



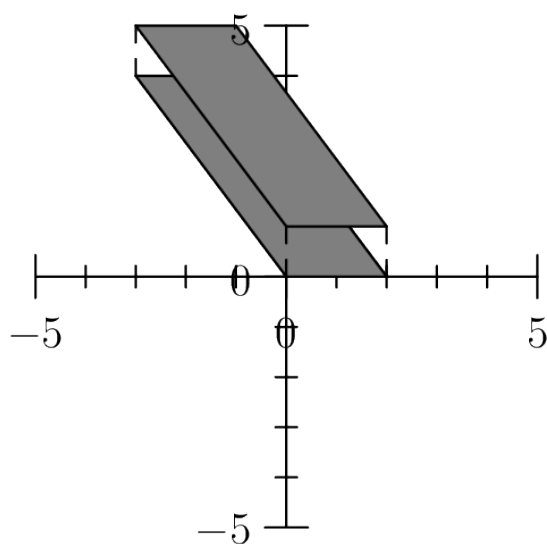
Now, when we vary $2u$ from 0 to 2, this line is translated to the right 2 units:

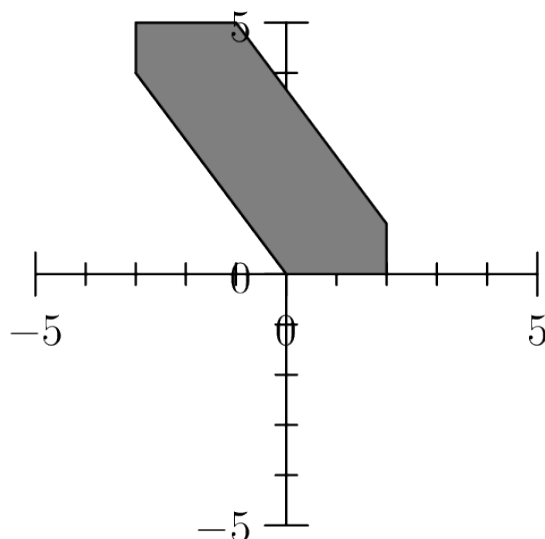


We know that any points in the region between the line (or rather segment) and its translation satisfy w and u , so we shade in the region:



We can also shift this quadrilateral one unit up, because of v . Thus, this is our figure:





The length of the boundary is simply $1 + 2 + 5 + 1 + 2 + 5$ (5 can be obtained by Pythagorean theorem since we have side lengths 3 and 4.). This equals **(E) 16.**

Solution 2

We can find the "boundary points" and work with our intuition to solve the problem. We set each of u, v, w equal to 0, 1 for a total of 8 combinations in u, v, w . We now test each one.

Case 1: $u = 0, v = 0, w = 0 \implies (0, 0)$

Case 2: $u = 0, v = 0, w = 1 \implies (-3, 4)$

Case 3: $u = 0, v = 1, w = 0 \implies (0, 1)$

Case 4: $u = 0, v = 1, w = 1 \implies (-3, 5)$

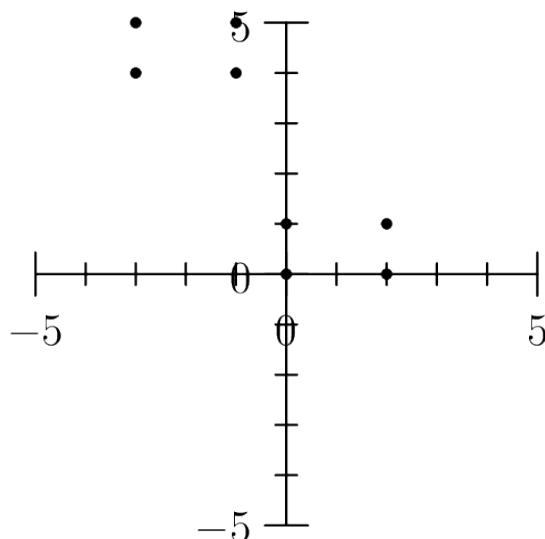
Case 5: $u = 1, v = 0, w = 0 \implies (2, 0)$

Case 6: $u = 1, v = 0, w = 1 \implies (-1, 4)$

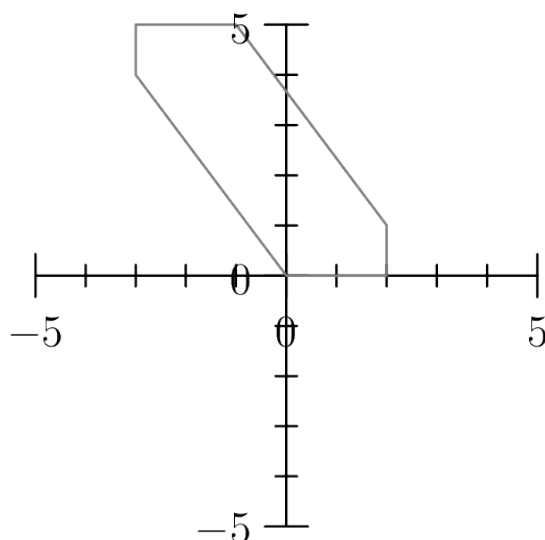
Case 7: $u = 1, v = 1, w = 0 \implies (2, 1)$

Case 8: $u = 1, v = 1, w = 1 \implies (-1, 5)$

When graphed on a coordinate plane, the points appear as follows.



Notice how there are two distinct rectangles visible in the figure. This leads us to believe that the region tracks the motion of this region as it travels in space. To understand why this is true, we can imagine a fixed w (as it is present in both the x and y coordinates). Then if we hold one of u or v fixed and let the other vary, we get a straight line parallel to the x or y axis respectively. If we let the other vary, we get the other type of straight line. Together, they form a rectangular region. In addition, w serves as a diagonal translation, so if we now let w vary, it traces out the motion of the rectangle. Keeping this in mind, we connect the dots.



Each of the diagonal sides have length 5 by the distance formula on $(0, 0)$ and $(-3, 4)$ (the other diagonal side is congruent), so our total perimeter is $2 + 1 + 5 + 2 + 1 + 5 =$
(E) 16.

Solution 3 (Quick, Not rigorous)

Let's assume that $w = 0$. If we plug $(0, 0)$, $(0, 1)$, $(1, 1)$, $(1, 0)$ into u and v respectively, we will form a rectangle that has side lengths 1 and 2. If we change w to the maximum value (which is 1), we will find out that w shifts the rectangle by

$-3w$ on the x-axis and $4w$ on the y-axis. Since we need to find the perimeter of the shift and rectangle, we can calculate that the length that the corner has move during the shift using the **Pythagorean Theorem**. Thus, the distance is 5. Now, we can find the perimeter of the boundary which is $2(1 + 2 + 5) = 16$. This means that the answer is **(E) 16**.

25. (AMC 10 10B 2023) A regular pentagon with area $1 + \sqrt{5}$ is printed on paper and cut out. All five vertices are folded to the center of the pentagon, creating a smaller pentagon. What is the area of the new pentagon?

(A) $4 - \sqrt{5}$ (B) $\sqrt{5} - 1$ (C) $8 - 3\sqrt{5}$ (D) $\frac{\sqrt{5}+1}{2}$ (E) $\frac{2+\sqrt{5}}{3}$

Answer: B

Solution 1

Since A is folded onto O , $AM = MO$ where M is the intersection of AO and the creaseline between A and O . Note that the inner pentagon is regular, and therefore similar to the original pentagon, due to symmetry.

Because of their similarity, the ratio of the inner pentagon's area to that of the outer pentagon can be represented by

$$\left(\frac{OM}{ON}\right)^2 = \left(\frac{\frac{OA}{2}}{OA \sin(\angle OAE)}\right)^2 = \frac{1}{4 \sin^2 54}$$

Option 1: Knowledge Remember that $\sin 54 = \frac{1+\sqrt{5}}{4}$.

Option 2: Angle Identities $\cos 54 = \sin 36$ $4 \cos^3 18 - 3 \cos 18 = 2 \sin 18 \cos 18$
 $4(1 - \sin^2 18) - 3 - 2 \sin 18 = 0$ $4 \sin^2 18 + 2 \sin 18 - 1 = 0$ $\sin 18 = \frac{-1+\sqrt{5}}{4}$ $\sin 54 = \cos 36 = 1 - 2 \sin^2 18 = \frac{1+\sqrt{5}}{4}$

Option 3: Golden Ratio/Triangle It is well known that a 36-36-108 triangle has side lengths with ratio $1 : 1 : \varphi$

Dropping the altitude to the longer side results in $\sin 54 = \frac{\varphi}{2} = \frac{\sqrt{5}+1}{4}$ (**Elib0128** (**talk**))

$$\sin^2 54 = \frac{3+\sqrt{5}}{8}$$

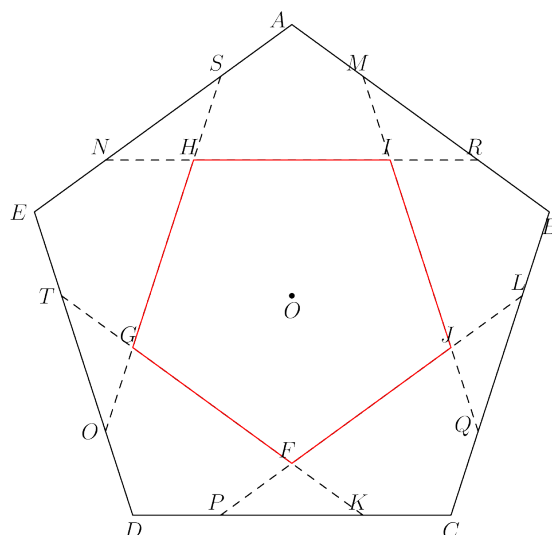
Let the inner pentagon be Z .

$$[Z] = \frac{1}{4 \sin^2 54} [ABCDE] = \frac{2(1+\sqrt{5})}{3+\sqrt{5}} = \sqrt{5} - 1$$

So the answer is **(B)**

-Dissmo Thegoat

Solution 2

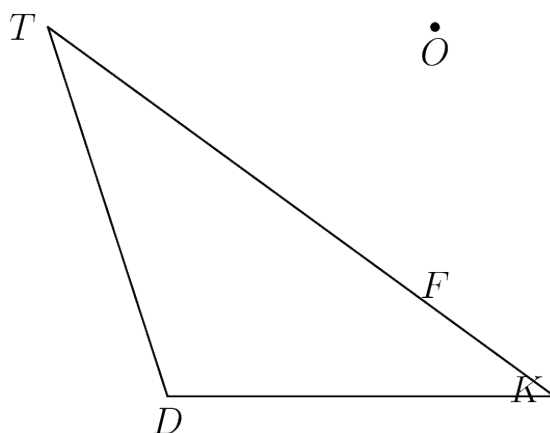


We can find the area of the red pentagon by taking the area of the total pentagon and subtracting the area outside the red pentagon.

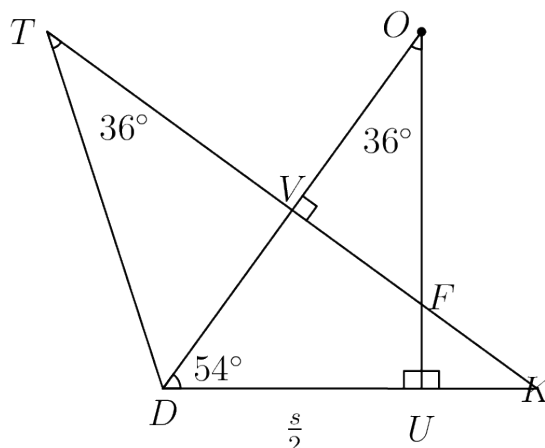
The area outside the red pentagon is the sum of the larger isosceles triangles, but this double counts the overlapping regions of the small isosceles triangles, so we have to subtract those out.

$$\text{We have } [FGHIJ] = [ABCDE] - (5 \cdot [DKT] - 5 \cdot [PFK])$$

Lets focus on finding the area of each individual triangle:



Notice that we have no information about the side length, so instead we let the side length be s . Now we can drop an altitude from O to the base of the triangle, and we know this altitude must split the base of the pentagon in half, so we can create a right triangle. Furthermore, draw a line from O to D . This must bisect angle D which is 108 degrees, so we create $36 - 54 - 90$ triangles. Specifically, we know $\angle ODK = 54^\circ$, $\angle DOU = 36^\circ$, and $\angle DTK = 36^\circ$ because $\triangle DTK$ is isosceles and we know the vertex angle is 108° . We encode this information in the diagram below:



Since $\triangle DKT$ is isosceles, the area of $\triangle DVT$ is half the area of $\triangle DKT$. Similarly, the area of $\triangle UFK$ is half that of $\triangle PFK$. Thus:

$$[FGHIJ] = [ABCDE] - (5 \cdot [DKT] - 5 \cdot [PFK]) \implies [FGHIJ] = [ABCDE] - (10 \cdot [DVT] - 10 \cdot [UFK])$$

We also know that since we dropped an altitude from O to U , the area of $\triangle ODU$ must be half of a fifth of the total area of the pentagon. Therefore we can rewrite the above equation as

$$[FGHIJ] = 10 \cdot ([ODU] - [DVT] + [UFK])$$

Now notice that $\triangle ODU \triangle TDV \triangle KFU$ by AA similarity. Therefore, if we can write the areas of the latter two triangles as a ratio of the first triangle, we can express the whole equation in terms of $[ODU]$, and by extension $[ABCDE]$, which we know. To find these ratios, we can find the side length ratios and square them because the triangles are similar.

We already know $DU = \frac{s}{2}$, so let's try to find it's analogous side for $\triangle TDV$ and $\triangle KFU$. These sides are DV and FU , respectively.

First, $\frac{s}{2} = OD \cdot \cos(54^\circ)$, so $OD = \frac{s}{2} \cdot \sec(54^\circ)$. Then notice that $DV = \frac{OD}{2}$ because we have to fold D to hit O , so the folding crease has to be exactly halfway between O and D . Therefore,

$$DV = \frac{s}{4} \cdot \sec(54^\circ) \Rightarrow \frac{DU}{DV} = \frac{\frac{s}{2}}{\frac{s}{4} \cdot \sec(54^\circ)} = 2 \cdot \cos(54^\circ)$$

Now the ratio between the area of two similar triangles is the square of the ratio of their analogous side lengths. Thus

$$\begin{aligned} \frac{[ODU]}{[TDV]} &= 4 \cdot \cos^2(54^\circ) \implies \frac{10 \cdot [ODU]}{10 \cdot [TDV]} = 4 \cdot \cos^2(54^\circ) \\ \implies \frac{[ABCDE]}{10 \cdot [TDV]} &= 4 \cdot \cos^2(54^\circ) \implies \frac{\sqrt{5} + 1}{10 \cdot [TDV]} = 4 \cdot \cos^2(54^\circ) \\ \implies 10 \cdot [TDV] &= \frac{\sqrt{5} + 1}{4 \cdot \cos^2(54^\circ)} \implies 10 \cdot [TDV] = \frac{\sqrt{5} + 1}{4} \cdot \sec^2(54^\circ) \end{aligned}$$

Now let's move on and calculate the ratio of the other side length. Calculating FU is slightly trickier. First, we find TD : $TD \cdot \cos(54^\circ) = DV = \frac{s}{4} \sec^2(54^\circ) \implies TD = \frac{s}{4} \cdot \sec^2(54^\circ)$. Now since $\triangle DTK$ is isosceles, $TD = DK$ and $UK = DK - DU = TD - DU = \frac{s}{4} \cdot \sec^2(54^\circ) - \frac{s}{2} = \frac{s}{4} \cdot (\sec^2(54^\circ) - 2) = \frac{s}{4} \cdot (\tan^2(54^\circ) - 1)$.

Now $FU = UK \cdot \tan(36^\circ) \implies FU = \frac{s}{4} \cdot (\tan^2(54^\circ) - 1) \cdot \tan(36^\circ)$. Now note that $\tan(x) \cdot \tan(90 - x) = 1$ because opposite over adjacent cancel each other out in a right triangle. Thus, $FU = \frac{s}{4} \cdot (\tan(54^\circ) - \tan(36^\circ))$

Now,

$$\frac{DU}{FU} = \frac{\frac{s}{2}}{\frac{s}{4} \cdot (\tan(54^\circ) - \tan(36^\circ))} = \frac{2}{\tan(54^\circ) - \tan(36^\circ)}$$

$$\implies \frac{[DU]}{[FU]} = \frac{4}{(\tan(54^\circ) - \tan(36^\circ))^2} \implies 10 \cdot [FU] = \frac{1 + \sqrt{5}}{4} \cdot (\tan(54^\circ) - \tan(36^\circ))^2$$

Now we go back to our first equation and plug in our values:

$$[FGHIJ] = [ABCDE] - (10 \cdot [DVT] - 10 \cdot [U FK]) \implies [FGHIJ] = 1 + \sqrt{5} - \frac{\sqrt{5} + 1}{4} \cdot \sec^2(54^\circ) + \frac{1 + \sqrt{5}}{4}$$

$$\implies [FGHIJ] = \frac{1 + \sqrt{5}}{4} \cdot (4 - \sec^2(54^\circ) + (\tan(54^\circ) - \tan(36^\circ))^2)$$

Note $(\tan(x) - \tan(90 - x))^2 = \tan^2(x) - 2 \cdot \tan(x) \cdot \tan(90 - x) + \tan^2(90 - x) = \tan^2(x) - 2 + \tan^2(90 - x)$.

Also note that $\tan^2(x) + 1 = \sec^2(x)$. Thus

$$[FGHIJ] = \frac{1 + \sqrt{5}}{4} \cdot (2 - \sec^2(54^\circ) + \tan^2(54^\circ) + \tan^2(36^\circ))$$

$$\implies [FGHIJ] = \frac{1 + \sqrt{5}}{4} \cdot (2 + (-1) + \tan^2(36^\circ)) \implies [FGHIJ] = \frac{1 + \sqrt{5}}{4} \cdot (1 + \tan^2(36^\circ))$$

.

Now all that remains is to find $\tan^2(36^\circ)$. We can use the tan addition formula to find the general form of $\tan(5x)$ or remember question 25 from this year's AMC 12A. We have that

$$\tan(5x) = \frac{5\tan(x) - 10\tan^3(x) + \tan^5(x)}{1 - 10\tan^2(x) + 5\tan^4(x)}$$

.

Plug in $x = 36$. Then we have

$$\tan(180^\circ) = \frac{5\tan(36^\circ) - 10\tan^3(36^\circ) + \tan^5(36^\circ)}{1 - 10\tan^2(36^\circ) + 5\tan^4(36^\circ)}$$

Now let $y = \tan(36^\circ)$. We have the equation

$$\frac{5y - 10y^3 + y^5}{1 - 10y^2 + 5y^4} = 0 \implies 5y - 10y^3 + y^5 = 0$$

$$\implies 5 - 10y^2 + y^4 = 0 \implies 5 - 10z + z^2 = 0$$

Where we let $z = y^2$. Using the quadratic formula, we have

$$z = \frac{10 \pm \sqrt{80}}{2} = 5 \pm 2\sqrt{5}$$

Now since $y = \tan(36^\circ)$, $z = \tan^2(36^\circ)$, which is what we were looking for. Notice that $\tan(0^\circ) = 0$ and $\tan(45^\circ) = 1$, so $\tan(36^\circ)$ is between 0 and 1, and so is its square. Thus $z = 5 - 2\sqrt{5}$, not the other root.

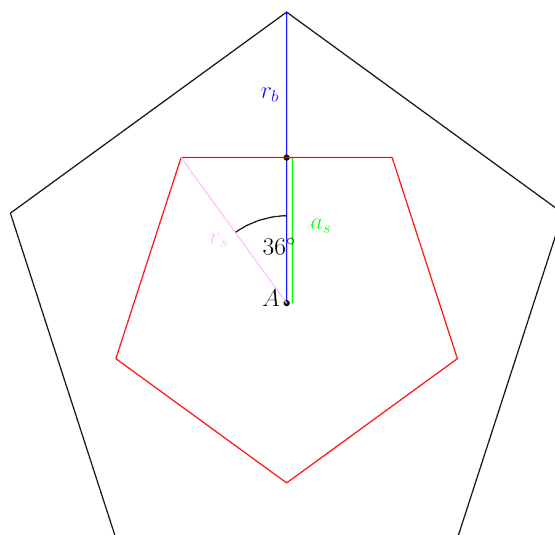
Finally:

$$\begin{aligned} [FGHIJ] &= \frac{1 + \sqrt{5}}{4} \cdot (1 + \tan^2(36^\circ)) = \frac{1 + \sqrt{5}}{4} \cdot (1 + 5 - 2\sqrt{5}) \\ \implies [FGHIJ] &= \frac{1 + \sqrt{5}}{4} \cdot (6 - 2\sqrt{5}) = \frac{4\sqrt{5} - 4}{4} = \sqrt{5} - 1 \end{aligned}$$

Therefore,

$$[FGHIJ] = \sqrt{5} - 1 = \boxed{B}$$

Solution 3



Let r_b and r_s be the circumradius of the big and small pentagon, respectively. Let a_s be the apothem of the smaller pentagon and A_s and A_b be the areas of the smaller and larger pentagon, respectively.

From the diagram:

$$\begin{aligned} \cos 36^\circ &= \frac{a_s}{r_s} = \frac{\phi}{2} = \frac{\sqrt{5} + 1}{4} \\ a_s &= \frac{r_b}{2} \\ A_s &= \left(\frac{r_s}{r_b}\right)^2 A_b \\ &= \left(\frac{a_s}{\cos 36^\circ r_b}\right)^2 (1 + \sqrt{5}) \end{aligned}$$

$$\begin{aligned} &= \left(\frac{a_s}{\frac{\phi}{2} r_b} \right)^2 (1 + \sqrt{5}) \\ &= \left(\frac{1}{2 \frac{\phi}{2}} \right)^2 (1 + \sqrt{5}) \\ &= \left(\frac{2}{\sqrt{5} + 1} \right)^2 (1 + \sqrt{5}) \\ &= \frac{4}{\sqrt{5} + 1} \\ &= \frac{4(\sqrt{5} - 1)}{(\sqrt{5} + 1)(\sqrt{5} - 1)} \\ &= \sqrt{5} - 1 \end{aligned}$$

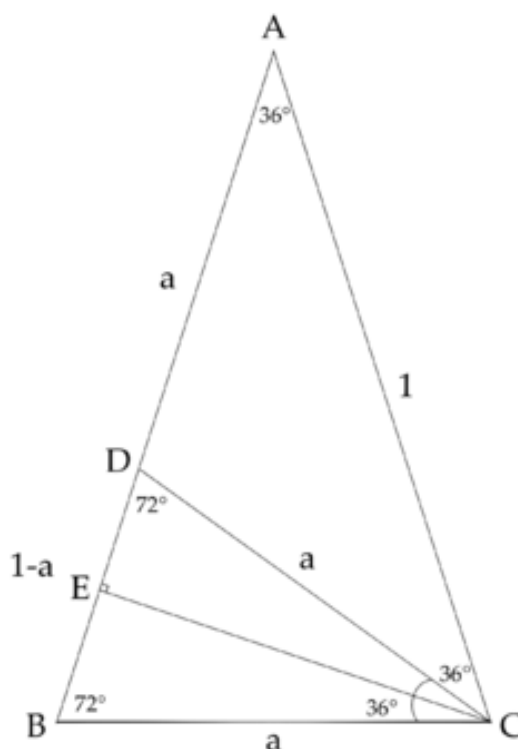
(B) $\sqrt{5} - 1$

Technodoggo

Solution 4

Interestingly, we find that the pentagon we need is the one that is represented by the intersection of perpendicular bisectors of the connection from the center of the pentagon to one vertex. Through similar triangles and the golden ratio, we find that the side length ratio of the two pentagons is $\frac{\sqrt{5}-1}{2}$. Thus, the answer is $(\sqrt{5} + 1) \cdot (\frac{\sqrt{5}-1}{2})^2 = \sqrt{5} - 1$. B andliu766

Supplement (Calculating $\sin 54^\circ/\cos 36^\circ$ from Scratch) Method 1:



Construct golden ratio triangle $\triangle ABC$ with $\angle A = 36^\circ$, $\angle B = \angle C = 72^\circ$ and $\triangle BCD$ with $\angle C = 36^\circ$, $\angle DBC = \angle BDC = 72^\circ$. WLOG, let $AB = AC = 1$, $BC = CD = AD = a$, $BD = 1 - a$. $\triangle BAC \sim \triangle BCD$

$$\frac{AC}{BC} = \frac{BC}{BD}, \quad \frac{1}{a} = \frac{a}{1-a}, \quad 1-a = a^2, \quad a^2 + a - 1 = 0$$

$$a = \frac{-1 + \sqrt{1^2 - 4(-1)}}{2} = \frac{\sqrt{5} - 1}{2}$$

$$\cos 36^\circ = \cos \angle A = \frac{AE}{AC} = \frac{1-a}{2} + a = \frac{a+1}{2} = \frac{\frac{\sqrt{5}-1}{2} + 1}{2} = \frac{\sqrt{5}+1}{4}$$

$$\sin 54^\circ = \cos 36^\circ = \frac{\sqrt{5}+1}{4}$$

Method 2:

As explained [here](#), $\cos \frac{2\pi}{5} + \cos \frac{4\pi}{5} = -\frac{1}{2}$

$$\cos \frac{2\pi}{5} - \cos \frac{\pi}{5} = -\frac{1}{2}$$

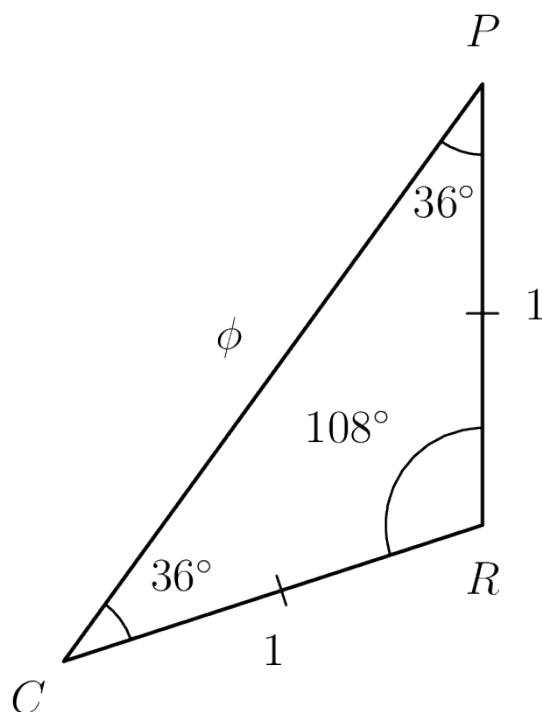
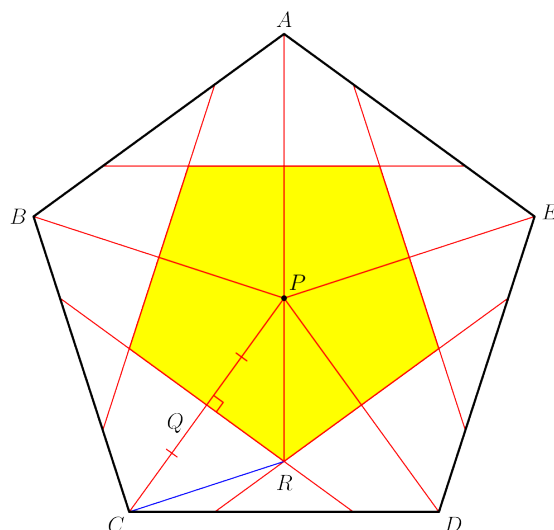
$$2(\cos \frac{\pi}{5})^2 - 1 - \cos \frac{\pi}{5} = -1/2$$

$$4(\cos \frac{\pi}{5})^2 - 2 \cos \frac{\pi}{5} - 1 = 0$$

$$\cos 36^\circ = \cos \frac{\pi}{5} = \frac{2 + \sqrt{2^2 + 4 \cdot 4}}{8} = \frac{1 + \sqrt{5}}{4}$$

Solution 5 (Much easier if you notice the one line construction + use properties of the golden ratio)

We will start with a similar diagram to all of the other submitted solutions with ONE difference: We will connect line segment CR to form a 36° - 36° - 108° golden triangle $\triangle CPR$. We know this is a 36° - 36° - 108° golden triangle because QR is the perpendicular bisector of CP (in order to fold vertex C onto the center P), so $\triangle CPR$ must be isosceles, with angle $\angle CPR = \frac{360^\circ}{10} = 36^\circ$ due to the regular pentagon.



Rather than trigonometry, it is well known that $\frac{CP}{PR} = \frac{\phi}{1}$ due to the golden triangle. Since both pentagons are regular, they are similar. Therefore the ratio of the areas is equal to the ratio of the radii squared (rather than using the ratio of the sides of the pentagon). Let x be the area of the smaller pentagon. Note that we are given the larger pentagon has area 2ϕ .

We will also use a well known identity of the golden ratio: $\frac{1}{\phi} = \phi - 1$ (and for completeness sake for any student reading this in the future looking for other golden ratio identities: $\phi^2 = \phi + 1$) to simplify expressions involving the golden ratio rather than to do the algebra on the square roots. Then:

$$\frac{x}{2\phi} = \left(\frac{PR}{CP}\right)^2$$

$$\begin{aligned}\frac{x}{2\phi} &= \left(\frac{1}{\phi}\right)^2 \\ \frac{x}{2} &= \frac{1}{\phi} \\ \frac{x}{2} &= \phi - 1 \\ x &= 2\phi - 2 \\ x &= 2\left(\frac{1+\sqrt{5}}{2}\right) - 2 \\ x &= 1 + \sqrt{5} - 2 \\ x &= \sqrt{5} - 1\end{aligned}$$

B

§10 AMC 10 10A 2023

- a) (*AMC 10 10A 2023*) Cities A and B are 45 miles apart. Alicia lives in A and Beth lives in B . Alicia bikes towards B at 18 miles per hour. Leaving at the same time, Beth bikes toward A at 12 miles per hour. How many miles from City A will they be when they meet?

(A) 20 (B) 24 (C) 25 (D) 26 (E) 27

Answer: E

Solution 1

This is a $d = st$ problem, so let x be the time it takes to meet. We can write the following equation:

$$12x + 18x = 45$$

Solving gives us $x = 1.5$. The $18x$ is Alicia so $18 \times 1.5 = \boxed{\text{(E) 27}}$

Solution 2

The relative speed of the two is $18 + 12 = 30$, so $\frac{3}{2}$ hours would be required to travel 45 miles. $d = st$, so $x = 18 \cdot \frac{3}{2} = \boxed{\text{(E) 27}}$

Solution 3

Since 18 mph is $\frac{3}{2}$ times 12 mph, Alicia will travel $\frac{3}{2}$ times as far as Beth. If x is the distance Beth travels,

$$\frac{3}{2}x + x = 45$$

$$\frac{5}{2}x = 45$$

$$x = 18$$

Since this is the amount Beth traveled, the amount that Alicia traveled was

$$45 - 18 = \boxed{\text{(E) 27}}$$

Solution 4

Alice and Barbara close in on each other at 30mph. Since they are 45 miles apart, they will meet in $t = d/s = 45\text{miles} / 30\text{mph} = 3/2$ hours. We can either calculate the distance Alice travels at 18mph or the distance Barbara travels at 12mph; since we want the distance from Alice, we go with the former. Alice (and Barbara) will meet in $1\frac{1}{2}$ hours at $18\text{mph} \times 3/2$ hours = 27 miles from A. **(E) 27**

Solution 5 (Under 20 seconds)

We know that Alice approaches Beth at 18 mph and Beth approaches Alice at 12 mph. If we consider that if Alice moves 18 miles at the same time Beth moves 12 miles $\rightarrow$ 15 miles left. Alice then moves 9 more miles at the same time as Beth moves 6 more miles. Alice has moved 27 miles from point A at the same time that Beth has moved 18 miles from point B, meaning that Alice and Beth meet 27 miles from point A.

Solution 6 (simple linear equations)

We know that Beth starts 45 miles away from City A, let's create two equations: Alice $\rightarrow 18t = d$ Beth $\rightarrow -12t + 45 = d$ [-12 is the slope; 45 is the y-intercept]

Solve the system:

$$18t = -12t + 45 \quad 30t = 45 \quad t = 1.5$$

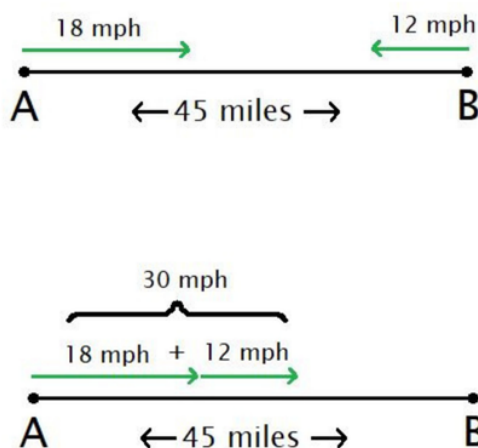
So, $18(1.5) =$ **(E) 27**

Solution 7

Since Alicia and Beth's speeds are constant, they are directly proportional to their distances covered, so the ratio of their speeds is equal to the ratio of their covered distances. Since Alicia travels $\frac{18}{30} = \frac{3}{5}$ of their combined speed, she travels $\frac{3}{5} \cdot 45 =$ **(E) 27** of the total distance.

-Benedict T (countmath1)

Solution 8



Note that Alicia and Beth must have travelled 45 miles together in order for them to meet together. While travelling the 45 miles, Alicia came closer towards Beth and Beth came closer towards Alicia. Since Alicia is travelling

faster than Beth, we know that they will meet slightly closer towards city B from the middle. Also, the distance remaining between Alicia and Beth as they bike toward each other is proportional to their combined velocity. The combined velocity is $18t + 12t = 30t$. The time it takes to travel 45 miles going at 30 miles per hour can be calculated using simple algebra.

Let t be the time it takes for Alicia and Beth to meet together.

$$18t + 12t = 45$$

$$30t = 45$$

$$t = \frac{3}{2}$$

It takes $\frac{3}{2}$ hours of cycling by both Alicia and Beth at their respective biking speeds in order to meet together. Also, note that the question asks for the distance from City A they will be when they meet. This is the same as the distance travelled by Alicia from City A. Alicia travels at 18 mph, which turns into a distance = speed x time question.

$$d = st$$

$$d = 18 \times \frac{3}{2}$$

$$d = 27$$

The distance from City A when Alicia and Beth meet is **(E) 27**.

Note - Even though it might seem like there is a lot of work for this problem, it would typically take a very short time to think this entire process out.

- b) (*AMC 10 10A 2023*) The weight of $\frac{1}{3}$ of a large pizza together with $3\frac{1}{2}$ cups of orange slices is the same weight of $\frac{3}{4}$ of a large pizza together with $\frac{1}{2}$ cups of orange slices. A cup of orange slices weigh $\frac{1}{4}$ of a pound. What is the weight, in pounds, of a large pizza?

(A) $1\frac{4}{5}$ (B) 2 (C) $2\frac{2}{5}$ (D) 3 (E) $3\frac{3}{5}$

Answer: A

Solution 1 (Substitution)

Use a system of equations. Let x be the weight of a pizza and y be the weight of a cup of orange slices. We have

$$\frac{1}{3}x + \frac{7}{2}y = \frac{3}{4}x + \frac{1}{2}y.$$

Rearranging, we get

$$\begin{aligned} \frac{5}{12}x &= 3y, \\ x &= \frac{36}{5}y. \end{aligned}$$

Plugging in $\frac{1}{4}$ pounds for y by the given gives $\frac{9}{5} = \mathbf{(A) 1\frac{4}{5}}$.

Solution 2

Let p be the weight of a pizza. o be the weight of a cup of orange.

From the problem, we know that $o = \frac{1}{4}$.

Write the equation below:

$$\frac{1}{3}p + \frac{7}{2} \cdot \frac{1}{4} = \frac{3}{4}p + \frac{1}{2} \cdot \frac{1}{4}$$

Solving for p : $\frac{5}{12}p = \frac{3}{4}p = \frac{9}{5} = \boxed{\text{(A)} 1\frac{4}{5}}$.

Solution 3

$\frac{P}{3} + \frac{7}{2}R = \frac{3}{4}P + \frac{R}{2}$ where P is the pizza weight and R is the weight of cup of oranges. Since oranges weigh $\frac{1}{4}$ pound per cup, the oranges on the LHS weigh $\frac{7}{2}$ cups $\times \frac{1}{4}$ pounds/cup = $\frac{7}{8}$ pound, and those on the RHS weigh $\frac{1}{2}$ cup $\times \frac{1}{4}$ pounds/cup = $\frac{1}{8}$ pound.

So $\frac{P}{3} + \frac{7}{8}$ pound = $\frac{3}{4}P + \frac{1}{8}$ pound; $\frac{P}{3} + \frac{3}{4}$ pound = $\frac{3}{4}P$.

Multiplying both sides by $\text{lcm}(3, 4) = 12$, we have $4P + 9 = 9P$; $5P = 9$; $P =$ weight of a large pizza = $\frac{9}{5}$ pounds = $\boxed{\text{(A)} 1\frac{4}{5}}$ pounds.

- c) (AMC 10 10A 2023) How many positive perfect squares less than 2023 are divisible by 5?

(A) 8 (B) 9 (C) 10 (D) 11 (E) 12

Answer: A

Solution 1 (slightly refined)

Since $\lfloor \sqrt{2023} \rfloor = 44$, there are $\lfloor \frac{44}{5} \rfloor = \boxed{\text{(A)} 8}$ perfect squares less than 2023 that are divisible by 5.

Solution 2

Since 5 is square-free, each solution must be divisible by $5^2 = 25$. We take $\lfloor \frac{2023}{25} \rfloor = 80$ and see that there are $\boxed{\text{(A)} 8}$ positive perfect squares no greater than 80.

Solution 3

Since the perfect squares have to be divisible by 5, then we know it has to be 5 times some number squared ($5 \cdot x^2$). With this information, you can figure out every single product of 5 and another number squared to count how many perfect squares are divisible by 5 that are less than 2023. (EX: $5^2 = 25, 10^2 = 100, \dots, 40^2 = 1600$) With this you get a max of 40^2 , or $\lfloor \frac{44}{5} \rfloor = \boxed{\text{(A)} 8}$ solutions.

Solution 4

The way of BlueShadow refined:

All it takes is to recall that 45 squared is 2025, and 45 is 5×9 . So all the squares of $5 \times 1, 5 \times 2, 5 \times 3$ so on are divisible by 5. So the answer is 8. It can be done even if one does not remember that 45 squared is 2025, all it takes is intuition. One can easily see mentally that 5×8 that is 40 squared is

1600, and then one has to do just one more computation and see that 5×9 that is 45 squared exceeds 2023, so the answer is 8.

Note that you can find the square of any number that ends in 5 by taking the number 5 more than it and the number 5 less than it, multiplying those together, and adding 25. For example, to calculate the square of 45, you do $40 \times 50 = 2000$, and $2000 + 25 = 2025$.

And then you know that it goes up from 1 to 44, and the multiples of five are 5, 10..., 35, 40, dividing each of them by 5, you get 1, 2..., 7, 8, therefore, the answer is **(A) 8**

- d) (AMC 10 10A 2023) A quadrilateral has all integer side lengths, a perimeter of 26, and one side of length 4. What is the greatest possible length of one side of this quadrilateral?

(A) 9 (B) 10 (C) 11 (D) 12 (E) 13

Answer: D

Solution 1

Let's use the triangle inequality. We know that for a triangle, the sum of the 2 shorter sides must always be longer than the longest side. This is because if the longest side were to be as long as the sum of the other sides, or longer, we would only have a line.

Similarly, for a convex quadrilateral, the sum of the shortest 3 sides must always be longer than the longest side. Thus, the answer is $\frac{26}{2} - 1 = 13 - 1 =$

(D) 12

Sidenote: If there weren't a restriction on integer side lengths, the answer would be the decimal just less than 13, so the sum of the other 3 sides could be just more than 13. That would make the longest side 12.99999..., stopping at who knows how many 9's.

Sidenote #2: In the previous sidenote above, it states that the longest side would be 12.99999..., but that is conceptually 13, therefore the answer would be unknown.

Solution 2

Say the chosen side is a and the other sides are b, c, d .

By the Generalised Polygon Inequality, $a < b+c+d$. We also have $a+b+c+d = 26 \Rightarrow b+c+d = 26-a$.

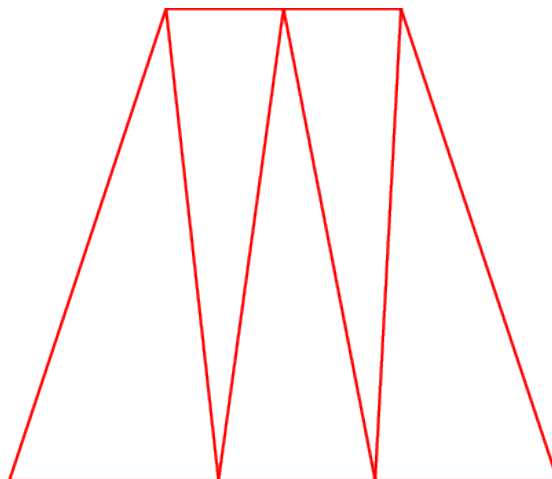
Combining these two, we get $a < 26-a \Rightarrow a < 13$.

The largest length that satisfies this is $a =$ **(D) 12**

Solution 3

This is an AMC 10 problem 4, so there is no need for any complex formulas. The largest singular side length from a quadrilateral comes from a trapezoid. So we can set the 2 sides of the trapezoid equal to 4. Next we can split the trapezoid into 5 triangles, where each base length of the triangle equals 4. So the top side equals 8, and the bottom side length equals $4+4+4=$

(D) 12 kabbybear



- e) (*AMC 10 10A 2023*) How many digits are in the base-ten representation of $8^5 \cdot 5^{10} \cdot 15^5$?

(A) 14 (B) 15 (C) 16 (D) 17 (E) 18

Answer: E

Solution 1

Prime factorizing this gives us $2^{15} \cdot 3^5 \cdot 5^{15} = 10^{15} \cdot 3^5 = 243 \cdot 10^{15}$.

10^{15} has 16 digits and $243 = 2.43 \cdot 10^2$ gives us 3 more digits. $16 + 2 =$ **(E) 18**

$2.43 \cdot 10^{17}$ has 18 digits

Solution 2 (Only if you don't know how to do the rest of the problems and have about 20 minutes left, IT IS NOT recommended)

Multiplying it out, we get that $8^5 \cdot 5^{10} \cdot 15^5 = 24300000000000000$. Counting, we have the answer is **(E) 18** andliu766

Solution 3 (Similar to Solution 1)

All the exponents have a common factor of 5 which we can factor out. This leaves us with $(8 \cdot 5^2 \cdot 15)^5 = (3000)^5 = (3 \cdot 1000)^5$. We can then distribute the power leaving us with $3^5 \cdot 10^{3 \cdot 5} = 243 \cdot 10^{15}$. This would be 243 followed by 15 zeros resulting in our answer being $15 + 3 =$ **(E) 18** leon_oiler

Solution 4 (log's)

We have 8^5 , 5^{10} , and 15^5 , if we do a logarithm base of 10 we can figure out how many digits are in the expression when multiplied. So we have $((\log(5^{10} \cdot 8^5 \cdot 15^5)))$ which using some rules we have $((10\log(5)) + (5\log(8)) + (5\log(15)))$, using the fact that $\log(5) = 0.7, \log(8) = 0.9, \log(15) = 1.2$, multiplying it out we get $10 \cdot 0.7 + 5 \cdot 0.9 + 1.2 \cdot 5$, and since $\log(1)=0$, as this is false as 1 is a digit and not 0, we add one and then take the floor to get us **(E) 18** Randomuser456 All values stated are approximated

- f) (*AMC 10 10A 2023*) An integer is assigned to each vertex of a cube. The value of an edge is defined to be the sum of the values of the two vertices it touches, and the value of a face is defined to be the sum of the values of the four edges surrounding it. The value of the cube is defined as the sum of the values of its six faces. Suppose the sum of the integers assigned to the vertices is 21. What is the value of the cube?

(A) 42 (B) 63 (C) 84 (D) 126 (E) 252

Answer: D

Solution 1

Each of the vertices is counted 3 times because each vertex is shared by three different edges. Each of the edges is counted 2 times because each edge is shared by two different faces. Since the sum of the integers assigned to all vertices is 21, the final answer is $21 \times 3 \times 2 = \boxed{\text{(D) } 126}$

Solution 2

Note that each vertex is counted $2 \times 3 = 6$ times. Thus, the answer is $21 \times 6 = \boxed{\text{(D) } 126}$.

Solution 3

Just set one vertex equal to 21, it is trivial to see that there are 3 faces with value 42, and $42 \cdot 3 = \boxed{\text{(D) } 126}$.

Solution 4

Since there are 8 vertices in a cube, there are $\frac{21}{4}$ vertices for two edges. There are 4 edges per face, and 6 faces in a cube, so the value of the cube is $\frac{21}{4} \cdot 24 = \boxed{\text{(D) } 126}$.

(Minor formatting by Technodoggo)

Solution 5 (use an example)

Set each vertex to value 1, so the sum of the vertices is 8. We find that the value of the cube, if all vertices are 1, is 48. We conclude that the value of the cube is 6 times the value of the sum of the vertices. Therefore, we choose $21 \times 6 = \boxed{\text{(D) } 126}$

Solution 6

The wording of the problem implies that the answer should hold for any valid combination of integers. Thus, we choose the numbers 21, 0, 0, 0, 0, 0, 0, 0, which are indeed 8 integers that add to 21. Doing this, we find three edges that have a value of 21, and from there, we get three faces with a value of 42 (while the other three faces have a value of 0). Adding the three faces together, we get $42 + 42 + 42 = \boxed{\text{(D) } 126}$.

Solution 7

Let the vertices be a, b, c, d, e, f, g, h . We know that $a + b + c + d + e + f + g + h = 21$. All the edges in a cube are equal, so we can assign the value x to the edge length. 2 vertices make an edge, so $4x = 21$. This is also the same value of a face, so we can multiply 21 by 6 and get $\boxed{\text{(D) } 126}$.

g) (AMC 10 10A 2023) Janet rolls a standard 6-sided die 4 times and keeps a running total of the numbers she rolls. What is the probability that at some point, her running total will equal 3?

(A) $\frac{2}{9}$ (B) $\frac{49}{216}$ (C) $\frac{25}{108}$ (D) $\frac{17}{72}$ (E) $\frac{13}{54}$ (Error compiling LaTeX. Unknown error_msg)

Answer: B

Solution 1 (Casework)

There are 3 cases where the running total will equal 3: one roll; two rolls; or three rolls:

Case 1: The chance of rolling a running total of 3, namely (3) in exactly one roll is $\frac{1}{6}$.

Case 2: The chance of rolling a running total of 3 in exactly two rolls, namely (1, 2) and (2, 1) is $\frac{1}{6} \cdot \frac{1}{6} \cdot 2 = \frac{1}{18}$.

Case 3: The chance of rolling a running total of 3 in exactly three rolls, namely (1, 1, 1) is $\frac{1}{6} \cdot \frac{1}{6} \cdot \frac{1}{6} = \frac{1}{216}$.

Using the rule of sum we have $\frac{1}{6} + \frac{1}{18} + \frac{1}{216} = \boxed{\text{(B)} \frac{49}{216}}$.

Solution 2 (Brute Force)

Because there is only a maximum of 3 rolls we must count (running total = 3 means there can't be a fourth roll counted), we can simply list out all of the probabilities.

If we roll a 1 on the first, the rolls that follow must be 2 or 1,1, with the following results not mattering. This leaves a probability of $\frac{1}{6} \times \frac{1}{6} + \frac{1}{6} \times \frac{1}{6} \times \frac{1}{6} = \frac{7}{216}$.

If we roll a 2 on the first, the roll that follows must be 1, resulting in a probability of $\frac{1}{6} \times \frac{1}{6} = \frac{1}{36}$.

If we roll a 3 on the first, the following rolls do not matter, resulting in a probability of $\frac{1}{6}$. Any roll greater than three will result in a running total greater than 3 no matter what, so those cases can be ignored. Summing the

answers, we have $\frac{7}{216} + \frac{1}{36} + \frac{1}{6} = \frac{7+6+36}{216} = \boxed{\text{(B)} \frac{49}{216}}$.

Solution 3

Consider sequences of 4 integers with each integer between 1 and 6, the number of permutations of 6 numbers is $6^4 = 1296$.

The following 4 types of sequences that might generate a running total of the numbers to be equal to 3 (x, y, or z denotes any integer between 1 and 6).

Sequence #1, (1, 1, 1, x): there are 6 possible sequences.

Sequence #2, (1, 2, x, y): there are $6^2 = 36$ possible sequences.

Sequence #3, (2, 1, x, y): there are $6^2 = 36$ possible sequences.

Sequence #4, (3, x, y, z): there are $6^3 = 216$ possible sequences.

Out of 1296 possible sequences, there are a total of $6 + 36 + 36 + 216 = 294$ sequences that qualify. Hence, the probability is $294/1296 = \boxed{\text{(B)} \frac{49}{216}}$

Solution 4 (cheese-ish)

You have to add 1/216 for the case 1-1-1, so it should have denominator 216. Therefore, the answer is B: 49/216

-Supernova77

- h) (AMC 10 10A 2023) Barb the baker has developed a new temperature scale for her bakery called the Breadus scale, which is a linear function of the Fahrenheit scale. Bread rises at 110 degrees Fahrenheit, which is 0 degrees

on the Breadus scale. Bread is baked at 350 degrees Fahrenheit, which is 100 degrees on the Breadus scale. Bread is done when its internal temperature is 200 degrees Fahrenheit. What is this, in degrees, on the Breadus scale?

- (A) 33 (B) 34.5 (C) 36 (D) 37.5 (E) 39

Answer: D

Solution 1 (Substitution)

To solve this question, you can use $f(x) = mx + b$ where the x is Fahrenheit and the y is Breadus. We have $(110, 0)$ and $(350, 100)$. We want to find the value of y in $(200, y)$ that falls on this line. The slope for these two points is $\frac{5}{12}$; $y = \frac{5}{12}x + b$. Solving for b using $(110, 0)$, $\frac{550}{12} = -b$. We get $b = \frac{-275}{6}$. Plugging in $(200, y)$, $\frac{1000}{12} - \frac{550}{12} = y$. Simplifying, $\frac{450}{12} = \boxed{\text{(D) } 37.5}$

Solution 2 (Faster)

Let $^{\circ}B$ denote degrees Breadus. We notice that $200^{\circ}F$ is $90^{\circ}F$ degrees more than $0^{\circ}B$, and $150^{\circ}F$ less than $100^{\circ}B$. This ratio is $90 : 150 = 3 : 5$; therefore, $200^{\circ}F$ will be $\frac{3}{3+5} = \frac{3}{8}$ of the way from 0 to 100, which is $\boxed{\text{(D) } 37.5}$

Solution 3 (Intuitive)

From 110 to 350 degrees Fahrenheit, the Breadus scale goes from 1 to 100. 110 to 350 degrees is a span of 240, and we can use this to determine how many Fahrenheit each Breadus unit is worth. 240 divided by 100 is 2.4, so each Breadus unit is 2.4 Fahrenheit, starting at 110 Fahrenheit. For example, 1 degree on the Breadus scale is $110 + 2.4$, or 112.4 Fahrenheit. Using this information, we can figure out how many Breadus degrees 200 Fahrenheit is. $200 - 110$ is 90, so we divide 90 by 2.4 to find the answer, which is $\boxed{\text{(D) } 37.5}$

Solution 4

We note that the range of F temperatures that $0 - 100$ Br $^{\circ}$ represents is $350 - 110 = 240$ F $^{\circ}$. 200 F $^{\circ}$ is $(200 - 110) = 90$ F $^{\circ}$ along the way to getting to 240 F $^{\circ}$, the end of this range, or $90/240 = 9/24 = 3/8 = 0.375$ of the way. Therefore if we switch to the Br scale, we are 0.375 of the way to 100 from 0, or at $\boxed{\text{(D) } 37.5}$ Br $^{\circ}$.

Solution 5

We have the points $(0, 110)$ and $(100, 350)$. We want to find $(x, 200)$. The equation of the line is $y = \frac{12}{5}x + 110$. We use this to find $x = \frac{75}{2} = 37.5$, or $\boxed{D}$. MC413551

Solution 6 (extremely simple)

We can write the value y on the Breadus scale as $y = mt + b$, where t is the temperature in Fahrenheit. From the problem, $110m + 1b = 0$ and $350m + 1b = 100$. We can rewrite this problem in terms of linear algebra to solve it.

Let $A = \begin{bmatrix} 110 & 1 \\ 350 & 1 \end{bmatrix}$, let $B = \begin{bmatrix} 0 \\ 100 \end{bmatrix}$, and let $x = \begin{bmatrix} m \\ b \end{bmatrix}$. We can write the system of equations as $Ax = B$. We can solve for x using the expression $x = A^{-1}B$. Calculating this value we get $x = \begin{bmatrix} -1/240 & 1/240 \\ 35/24 & -11/24 \end{bmatrix} \cdot \begin{bmatrix} 0 \\ 100 \end{bmatrix} =$

$\begin{bmatrix} 5/12 \\ -275/6 \end{bmatrix}$. Therefore, $m = 5/12$ and $b = -275/6$. Plugging in $t = 200$, we get $(5/12)200 + (-275/6) = \boxed{\text{(D)} 37.5}$.

- i) (AMC 10 10A 2023) A digital display shows the current date as an 8-digit integer consisting of a 4-digit year, followed by a 2-digit month, followed by a 2-digit date within the month. For example, Arbor Day this year is displayed as 20230428. For how many dates in 2023 does each digit appear an even number of times in the 8-digital display for that date?

(A) 5 (B) 6 (C) 7 (D) 8 (E) 9

Answer: E

Solution 1 (Casework)

Do careful casework by each month. Make sure to start with 2023. In the month and the date, we need a 0, a 3, and two digits repeated (which has to be 1 and 2 after consideration). After the casework, we get $\boxed{\text{(E)} 9}$. For curious readers, the numbers (in chronological order) are:

202301132023013120230223202303112023032220231013202310312023110320231130

Solution 2

There is one 3, so we need one more (three more means that either the month or units digit of the day is 3). For the same reason, we need one more 0. By inspection, we can see that the other two digits must be a repeated digit that is not 3 or 0.

If 3 is the units digit of the month, then the 0 must be the units digit of the month, since there is no month ending with 3 that is more than March. In this case, the last two digits must match (11, 22). This gives 2023 – 03 – 11 and 2023 – 03 – 22, so 2 successes.

If 3 is the tens digit of the day, then 0 can be in any of the remaining three places. If it is in the ones digit of the the day, then the month must be 11, which gives 2023 – 11 – 30. If it is in the ones digit of the month, then the month must be 10, in which case the day must be 31, which gives 2023 – 10 – 31. If 0 is in the tens digit of the month, then the month must be 01, which gives 2023 – 01 – 31. Together, 2023 – 11 – 30, 2023 – 10 – 31, and 2023 – 01 – 31 make 3 successes.

If 3 is the units digit of the day, then 0 could go in any of the 3 remaining slots again. If it's the tens digit of the day, then the other digits must be 1. If 0 is the units digit of the month, then the other two slots must both be 1. If 0 is the tens digit of the month, then the other two slots can be either both 1 or both 2. In total, there are 4 successes here.

Summing through all cases, there are $2 + 3 + 4 = \boxed{\text{(E)} 9}$ dates.

-Benedict T (countmath1)

-Edits by Cat_Of_Logic

Solution 3

We start with 2023 – – – – we need an extra 0 and an extra 3. So we have at least one of those extras in the days, except we can have the month 03. We now have 6 possible months 01, 02, 03, 10, 11, 12. For month 1 we have

two cases, we now have to add in another 1, and the possible days are 13, 31. For month 2 we need an extra 2 so we can have the day 23 note that we can't use 32 because it is too large. Now for month 3 we can have any number and multiply it by 11 so we have the solution 11, 22. For October we need a 1 and a 3 so we have 13, 31 as our choices. For November we have two choices which are 03, 30. Now for December we have 0 options. Summing $2 + 1 + 2 + 2 + 2$ we get **(E) 9** solutions.

- j) (AMC 10 10A 2023) Maureen is keeping track of the mean of her quiz scores this semester. If Maureen scores an 11 on the next quiz, her mean will increase by 1. If she scores an 11 on each of the next three quizzes, her mean will increase by 2. What is the mean of her quiz scores currently?

(A) 4 (B) 5 (C) 6 (D) 7 (E) 8

Answer: D

Solution 1

Let a represent the amount of tests taken previously and x the mean of the scores taken previously.

We can write the following equations:

$$\frac{ax + 11}{a + 1} = x + 1 \quad (1)$$

$$\frac{ax + 33}{a + 3} = x + 2 \quad (2)$$

Multiplying equation (1) by $(a + 1)$ and solving, we get:

$$ax + 11 = ax + a + x + 1$$

$$11 = a + x + 1$$

$$a + x = 10 \quad (3)$$

Multiplying equation (2) by $(a + 3)$ and solving, we get:

$$ax + 33 = ax + 2a + 3x + 6$$

$$33 = 2a + 3x + 6$$

$$2a + 3x = 27 \quad (4)$$

Solving the system of equations for (3) and (4), we find that $a = 3$ and $x = \mathbf{(D) 7}$.

Solution 2 (Variation on Solution 1)

Suppose Maureen took n tests with an average of m .

If she takes another test, her new average is $\frac{nm+11}{n+1} = m + 1$

Cross-multiplying: $nm + 11 = nm + n + m + 1$, so $n + m = 10$.

If she takes 3 more tests, her new average is $\frac{nm+33}{n+3} = m + 2$

Cross-multiplying: $nm + 33 = nm + 2n + 3m + 6$, so $2n + 3m = 27$.

But $2n + 3m$ can also be written as $2(n + m) + m = 20 + m$. Therefore $m = 27 - 20 = \mathbf{(D) 7}$

Solution 3 (do this if you are bored)

Let s represent the sum of Maureen's test scores previously and t be the number of scores taken previously.

So, $\frac{s+11}{t+1} = \frac{s}{t} + 1$ and $\frac{s+33}{t+3} = \frac{s}{t} + 2$

We can use the first equation to write s in terms of t .

We then substitute this into the second equation: $\frac{-t^2+10t+33}{t+3} = \frac{-t^2+10t}{t} + 2$

From here, we should solve for t : multiply both sides by (t) and then $(t+3)$, combining like terms to get $t^2 - 3t = 0$. Factorize to get $t = 0$ or $t = 3$, and therefore $t = 3$ (makes sense for the problem).

We substitute this to get $s = 21$.

Therefore, the solution to the problem is $\frac{21}{3} = \boxed{\text{(D)} 7}$

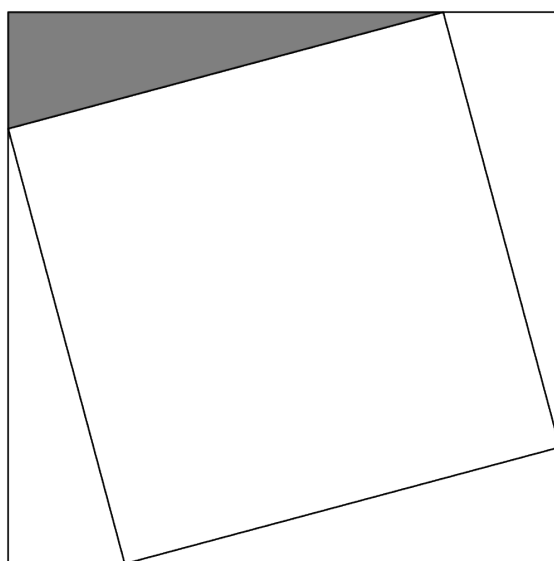
Solution 4 (Testing Answer Choices)

Let's consider all the answer choices. If the average is 8, then, we can assume that all her test choices were 8. We can see that she must have gotten 8 twice, in order for another score of 11 to bring her average up by one. However, adding three 11's will not bring her score up to 10. Continuing this process for the answer choices, we see that the answer is $\boxed{\text{(D)} 7}$ andliu766 (minor wording edit by mihikamishra)

Solution 5

Let n be the number of existing quizzes. So after one more test, score 11 has $n+1$ extra points to distribute to $n+1$ quizzes. Also, after three more quizzes, there will be $3(n+1)$ extra points to distribute to the $n+3$ quizzes. So $3n+3 = 2(n+3)$. This means $n = 3$. $n+1$ extra points means original mean (average) is 7

- k) (AMC 10 10A 2023) A square of area 2 is inscribed in a square of area 3, creating four congruent triangles, as shown below. What is the ratio of the shorter leg to the longer leg in the shaded right triangle?



- (A) $\frac{1}{5}$ (B) $\frac{1}{4}$ (C) $2 - \sqrt{3}$ (D) $\sqrt{3} - \sqrt{2}$ (E) $\sqrt{2} - 1$

Answer: C

Solution 1

The side lengths of the inner square and outer square are $\sqrt{2}$ and $\sqrt{3}$ respectively. Let the shorter side of our triangle be x , thus the longer leg is $\sqrt{3} - x$. Hence, by the Pythagorean Theorem, we have

$$\begin{aligned}(\sqrt{3} - x)^2 + x^2 &= (\sqrt{2})^2 \\ 3 - 2\sqrt{3}x + x^2 + x^2 &= 2 \\ 2x^2 - 2\sqrt{3}x + 1 &= 0.\end{aligned}$$

By the quadratic formula, we find that $x = \frac{\sqrt{3} \pm 1}{2}$, so the answer is $\frac{\sqrt{3}-1}{\sqrt{3}+1} =$

$$\boxed{\text{(C)} \ 2 - \sqrt{3}}.$$

Solution 2 (Area)

Looking at the diagram, we know that the square inscribed in the square with area 3 has area 2. Thus, the area outside of the small square is $3 - 2 = 1$. This area is composed of 4 congruent triangles, so we know that each triangle has an area of $\frac{1}{4}$.

From solution 1, the base (short side of the triangle) has a length x and the height $\sqrt{3} - x$, which means that $\frac{x(\sqrt{3}-x)}{2} = \frac{1}{4}$.

We can turn this into a quadratic equation: $x^2 - x\sqrt{3} + \frac{1}{2} = 0$.

By using the quadratic formula, we get $x = \frac{\sqrt{3} \pm 1}{2}$. Therefore, the answer is $\frac{\sqrt{3}-1}{\sqrt{3}+1} = \boxed{\text{(C)} \ 2 - \sqrt{3}}$.

(Clarity formatting edits by Technodoggo)

Solution 3

Let x be the ratio of the shorter leg to the longer leg, and y be the length of longer leg. The length of the shorter leg will be xy .

Because the sum of two legs is the side length of the outside square, we have $xy + y = \sqrt{3}$, which means $(xy)^2 + y^2 + 2xy^2 = 3$. Using the Pythagorean Theorem for the shaded right triangle, we also have $(xy)^2 + y^2 = 2$. Solving both equations, we get $2xy^2 = 1$. Using $y^2 = \frac{1}{2x}$ to substitute y in the second equation, we get $x^2 \cdot \frac{1}{2x} + \frac{1}{2x} = 2$. Hence, $x^2 - 4x + 1 = 0$. By using the quadratic formula, we get $x = 2 \pm \sqrt{3}$. Because x be the ratio of the shorter leg to the longer leg, it is always less than 1. Therefore, the answer is

$$\boxed{\text{(C)} \ 2 - \sqrt{3}}.$$

Solution 4

The side length of the bigger square is equal to $\sqrt{3}$, while the side length of the smaller square is $\sqrt{2}$. Let x be the shorter leg and y be the longer one. Clearly, $x + y = \sqrt{3}$, and $xy = \frac{1}{2}$. Using Vieta's to build a quadratic, we get

$$x^2 - \sqrt{3}x + \frac{1}{2} = 0.$$

Solving, we get $x = \frac{\sqrt{3}-1}{2}$ and $y = \frac{\sqrt{3}+1}{2}$. Thus, we find $\frac{x}{y} = \frac{\sqrt{3}-1}{\sqrt{3}+1} = \frac{(\sqrt{3}-1)^2}{(\sqrt{3}+1)(\sqrt{3}-1)} = \frac{4-2\sqrt{3}}{2} = \boxed{\text{(C)} \ 2 - \sqrt{3}}$.

Solution 5

Let θ be the angle opposite the smaller leg. We want to find $\tan \theta$.

The area of the triangle is $\frac{1}{2} (\sqrt{2} \sin \theta) (\sqrt{2} \cos \theta) = \frac{1}{2} \sin 2\theta = \frac{1}{4}$, which implies $\sin 2\theta = \frac{1}{2}$, or $\theta = 15^\circ$. Therefore $\tan \theta = \boxed{\text{(C)} 2 - \sqrt{3}}$

Solution 6

Allow a and b to be the sides of a triangle. WLOG, suppose $a > b$. We want to find $\frac{b}{a}$. Notice that the area of a triangle is $\frac{3-2}{4}$, which results in $\frac{1}{4}$. Thus, $ab = \frac{1}{2}$. However, the square of the hypotenuse of this triangle is $a^2 + b^2$, but also 2. We can write b as $\frac{1}{2a}$, and then plug it in. We get $a^2 + \frac{1}{4a^2} = 2$, so $4a^4 - 8a^2 + 1 = 0$. Applying the quadratic formula, $a^2 = \frac{8 \pm 4\sqrt{3}}{2}$, or $4 \pm 2\sqrt{3}$. However, since a and b must both be solutions of the quadratic, since both equations were cyclic. Since $a > b$, then $a^2 = 4 + 2\sqrt{3}$, and $b^2 = 4 - 2\sqrt{3}$. To find $\frac{b}{a}$, we can simply find the square root of $\frac{b^2}{a^2}$. This is $\sqrt{\frac{4-2\sqrt{3}}{4+2\sqrt{3}}} = \sqrt{\frac{2-\sqrt{3}}{2+\sqrt{3}}} = \sqrt{(2-\sqrt{3})(2-\sqrt{3})} = \boxed{2-\sqrt{3}}$, so the answer is $\boxed{C}$. - Sepehr2010

Solution 7 (Manipulation)

Let a be the length of the shorter leg and b be the longer leg. By the Pythagorean theorem, we can derive that $a^2 + b^2 = 2$. Using area we can also derive that $2ab = 1$. $(a+b)^2 = 3$ as given in the diagram, we can find that $(a-b)^2 = 1$ because $4ab = 2$. This means that $b+a = \sqrt{3}$ and $b-a = 1$. Adding the equations gives $b = \frac{\sqrt{3}+1}{2}$ and when b is plugged in $a = \frac{\sqrt{3}-1}{2}$. Rationalizing the denominators gives us $\boxed{\text{(C)} 2 - \sqrt{3}}$.

Solution 8

Let s be the length of the shorter leg, l the longer leg, and t the desired ratio. We have:

$$s^2 + l^2 = 2 \text{ and } \frac{1}{2}sl = \frac{1}{4}$$

The former dividing the latter gives:

$$t + \frac{1}{t} = 4$$

Solve this for t . Multiply by t :

$$t^2 + 1 = 4t$$

Subtract $-4t$ to get a quadratic:

$$t^2 - 4t + 1 = 0$$

Use the quadratic formula to get:

$$2 \pm \sqrt{3}$$

And since the smaller one is the first one, the solution has to be less than 1, so the answer is the one less than 1, the answer is $\boxed{\text{(C)} 2 - \sqrt{3}}$

- 1) (AMC 10 10A 2023) How many three-digit positive integers N satisfy the following properties?

- The number N is divisible by 7.
- The number formed by reversing the digits of N is divisible by 5.

(A) 13 (B) 14 (C) 15 (D) 16 (E) 17

Answer: B

Solution 1

Multiples of 5 will always end in 0 or 5, and since the numbers have to be a three-digit numbers, it cannot start with 0 (otherwise it would be a two-digit number), narrowing our choices to 3-digit numbers starting with 5. Since the numbers must be divisible by 7, all possibilities have to be in the range from $7 \cdot 72$ to $7 \cdot 85$ inclusive (504 to 595).

(Add 1 to include 72)

$$85 - 72 + 1 = 14. \quad \boxed{\text{(B) } 14}.$$

You can also take 497 away from each of the numbers (removing the hundreds digit and adding three to each of the numbers), resulting in the numbers 7, 14, 21..., 84, 91, 98. Dividing each of them by 7, you get the numbers 1, 2, 3..., 12, 13, 14. Therefore, the answer is $\boxed{\text{(B) } 14}$

Solution 2 (solution 1 but more thorough)

Let $N = \overline{cab} = 100c + 10a + b$. We know that $\overline{bac}$ is divisible by 5, so c is either 0 or 5. However, since c is the first digit of the three-digit number N , it can not be 0, so therefore, $c = 5$. Thus, $N = \overline{5ab} = 500 + 10a + b$. There are no further restrictions on digits a and b aside from N being divisible by 7.

The smallest possible N is 504. The next smallest N is 511, then 518, and so on, all the way up to 595. Thus, our set of possible N is $\{504, 511, 518, \dots, 595\}$. Dividing by 7 for each of the terms will not affect the cardinality of this set, so we do so and get $\{72, 73, 74, \dots, 85\}$. We subtract 71 from each of the terms, again leaving the cardinality unchanged. We end up with $\{1, 2, 3, \dots, 14\}$, which has a cardinality of 14. Therefore, our answer is $\boxed{\text{(B) } 14}$.

Solution 3 (modular arithmetic)

We first proceed as in the above solution, up to $N = 500 + 10a + b$. We then use modular arithmetic:

$$\begin{aligned} 0 &\equiv N \pmod{7} \\ &\equiv 500 + 10a + b \pmod{7} \\ &\equiv 3 + 3a + b \pmod{7} \\ 3a + b &\equiv -3 \pmod{7} \\ &\equiv 4 \pmod{7} \end{aligned}$$

We know that $0 \leq a, b < 10$. We then look at each possible value of a :

If $a = 0$, then b must be 4.

If $a = 1$, then b must be 1 or 8.

If $a = 2$, then b must be 5.

If $a = 3$, then b must be 2 or 9.

If $a = 4$, then b must be 6.

If $a = 5$, then b must be 3.

If $a = 6$, then b must be 0 or 7.

If $a = 7$, then b must be 4.

If $a = 8$, then b must be 1 or 8.

If $a = 9$, then b must be 5.

Each of these cases are unique, so there are a total of $1 + 2 + 1 + 2 + 1 + 1 + 2 + 1 + 2 + 1 = \boxed{\text{(B)} 14}$.

Solution 4

The key point is that when reversed, the number must start with a 0 or a 5 based on the second restriction. But numbers can't start with a 0.

So the problem is simply counting the number of multiples of 7 in the 500s.

$7 \times 72 = 504$, so the first multiple is 7×72 .

$7 \times 85 = 595$, so the last multiple is 7×85 .

Now, we just have to count $7 \times 72, 7 \times 73, 7 \times 74, \dots, 7 \times 85$.

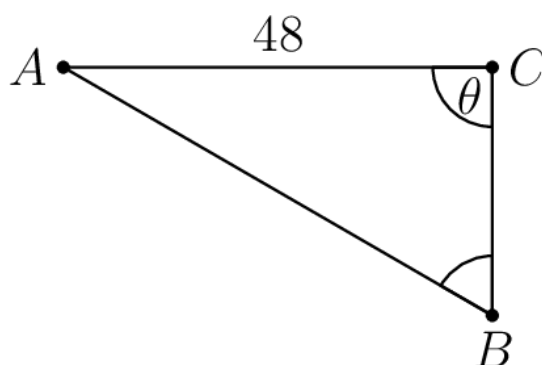
We have a set that numbers $85 - 71 = \boxed{\text{(B)} 14}$

- m) (AMC 10 10A 2023) Abdul and Chiang are standing 48 feet apart in a field. Bharat is standing in the same field as far from Abdul as possible so that the angle formed by his lines of sight to Abdul and Chiang measures 60° . What is the square of the distance (in feet) between Abdul and Bharat?

(A) 1728 (B) 2601 (C) 3072 (D) 4608 (E) 6912

Answer: C

Solution 1 (Using Trigonometry)



Let $\theta = \angle ACB$ and $x = \overline{AB}$.

According to the Law of Sines, we know that $\frac{\sin \theta}{x} = \frac{\sin 60^\circ}{48} = \frac{\sqrt{3}}{96}$. Rearranging, we get that $x = \frac{\sin \theta}{\frac{\sqrt{3}}{96}} = 32\sqrt{3} \sin \theta$ where x is a function of θ . We want to maximize x .

We know that the maximum value of $\sin \theta = 1$, so this yields $x = 32\sqrt{3} \implies x^2 = \boxed{\text{(C)} 3072}$.

A quick check verifies that $\theta = 90^\circ$ indeed works.

Solution 2 (Inscribed Angles)

We can draw a circle such that the chord AC inscribes an arc of 120 degrees. This way, any point B on the circle not in the inscribed arc will form an angle of 60 degrees with $\angle ABC$. To maximize the distance between A and B , they must be opposite each other. So, the problem is now finding the length of the diameter of the circle. We know $\angle AOC$ is 120 degrees, so dropping a perpendicular from O to AC gives us the radius as $16\sqrt{3}$. So, the diameter is $32\sqrt{3}$ which gives us the answer (C) 3072

Solution 3 (Guessing)

Guess that the optimal configuration is a 30-60-90 triangle, as an equilateral triangle gives an answer of $48^2 = 2304$, which is not on the answer choices. Its ratio is $\frac{48}{\sqrt{3}}$, so $\overline{AB} = \frac{96}{\sqrt{3}}$.

Its square is then $\frac{96^2}{3} =$ (C) 3072

Note: The distance between Abdul and Chiang is constant, so let that be represented as x . If we were dealing with an equilateral triangle, the height would be $x\sqrt{3}/2$, and if we were dealing with a 30-60-90 triangle, the height would be $x\sqrt{3}$, which is greater than $x\sqrt{3}/2$.

Note: The problem states that the distance from A to B is maximized, and the longest side of a triangle will always correspond to the largest possible angle, aka 90 so 90 and 60 makes it a 30-60-90.

Solution 4

We use A, B, C to refer to Abdul, Bharat and Chiang, respectively. We draw a circle that passes through A and C and has the central angle $\angle AOC = 60^\circ \cdot 2$. Thus, B is on this circle. Thus, the longest distance between A and B is the diameter of this circle. Following from the law of sines, the square of this diameter is

$$\left(\frac{48}{\sin 60^\circ}\right)^2 = \text{span style="border: 1px solid black; padding: 2px;">(C) 3072}.$$

Solution 5

We can represent Abdul, Bharat and Chiang as A, B , and C , respectively. Since we have $\angle ABC = 60^\circ$ and (from other solutions) $\angle BCA = 90^\circ$, this is a 30 – 60 – 90 triangle. By the side ratios of a 30 – 60 – 90 triangle, we can infer that $AB = \frac{48 \times 2}{\sqrt{3}}$. Squaring AB we get (C) 3072.

Solution 6 (easier to visualize)

Let Abdul and Chiang be a line segment, specifically a chord on a circle. Let chord $\overline{AC}$ be 48. If Bharat is the furthest away from Abdul, chord $\overline{AB}$ should be the diameter of the circle, as by definition the diameter is the longest chord in a circle. We know that $\triangle ABC$ is right since by definition a triangle inscribed in a semicircle is right. Chord $\overline{AC} = 48$, $\overline{AB}$ is the hypotenuse, and $\angle B = 60^\circ$. Therefore, the correct answer is $(\frac{48 \times 2}{\sqrt{3}})^2 =$ (C) 3072

Solution 7 (Definition of a Circle)

The definition of a circle directly shows a circle can circumscribe any three points, and those points will always form a triangle.

Consider the circle Ω inscribing triangle ABC . It follows that definition via the problem angle $ABC = 60^\circ$. Also, $AC = 48$.

Lemma:

The only way for B to be as far from A as possible with angle $ABC = 60^\circ$ is only plausible if AB is the diameter of Ω and C is placed on one semicircle above or below AB so that $ABC = 60^\circ$.

Proof of the Lemma:

It is by definition of a circle the farthest position from two points on a locus that is a circle must be the circle's diameter. Thus, AB is the diameter of the circle. From here, the angle from ACB through the inscribed angles theorem is 90° , meaning a position does exist with $ABC = 60^\circ$, when $BAC = 30^\circ$, hence proving the lemma.

2023 AMC 10A Q13:

Here, we provide the same scenario but with $AC = 48$. Through the properties of 30-60-90 degree triangles one finds that $AB = \frac{48}{\sqrt{3}}$. Then, $AB^2 = 96 \cdot 32 = \boxed{3072}$.

Solution 8 (Power of a Point, Formal Definition)

Consider the field to be a circle Ω with centre O and a points A on the circle. We require B to exist on this plane such that there exists a maximal value between A and B .

Lemma 1:

For AB to be maximized, it is required that B lies on the circumference of Ω and is diametrically opposite to A .

Proof of the Lemma 1:

Obviously, let the radius be $OA = r$ and through Power of a Point on A with respect to Ω we see that $OA^2 - r^2 = 0$ given definition of A , meaning A indeed lies on the circle. Then, Power of a Point on B with respect to Ω gives $OB^2 - OA^2 \geq 0$ due to the condition that B lies on the field. This implies that the distance $OB > OA$ or $OB = OA$. Given that $OB > OA$ and $OB^2 - OA^2 > 0$ it is defined that $OA + OB < 2r$. On the contrary, given that $OB = OA$, we see that $OB^2 - OA^2 = 0$ and then $OA + OB = 2r$, which is maximal as through definition of a circle, $2r$ is the maximal plausible distance between two points.

Lemma 2:

C must lie on Ω **Proof of the Lemma 2:**

Say the contrary. It is given that C must lie somewhere on the field. Then, Power of a point on C with respect to Ω yields $OC^2 - r^2 \geq 0$. Given Lemma 1 maximality occurs if A and B are diametrically opposite. Then, if C lies in the closed disc of Ω , it follows that $ABC = 60^\circ$ given $AB = 2r$. Then, the only plausible outcome is for C to lie outside the disc outside Ω to create an equilateral triangle of side length $2r$. This contradicts the idea that C lies on the field, and therefore C cannot exist in the closed disc, and must lie on Ω 's circumference, hence proved. Even so, we have to find if such a configuration exist. Given C lies on Ω 's circumference, through the inscribed

angles theorem is 90° , meaning a position does exist with $ABC = 60^\circ$, when $BAC = 30^\circ$, hence showing such a case does exist, proving the lemma.

2023 AMC 10A Q13:

Here, we provide the same scenario of a combination of lemma 1 and 2 but with $AC = 48$. Through the properties of 30-60-90 degree triangles one finds that $AB = \frac{48}{\sqrt{3}}$. Then, $AB^2 = 96 \cdot 32 = \boxed{3072}$.

- n) (AMC 10 10A 2023) A number is chosen at random from among the first 100 positive integers, and a positive integer divisor of that number is then chosen at random. What is the probability that the chosen divisor is divisible by 11?

(A) $\frac{4}{100}$ (B) $\frac{9}{200}$ (C) $\frac{1}{20}$ (D) $\frac{11}{200}$ (E) $\frac{3}{50}$

Answer: B

Solution 1

In order for the divisor chosen to be a multiple of 11, the original number chosen must also be a multiple of 11. Among the first 100 positive integers, there are 9 multiples of 11; 11, 22, 33, 44, 55, 66, 77, 88, 99. We can now perform a little casework on the probability of choosing a divisor which is a multiple of 11 for each of these 9, and see that the probability is $1/2$ for each. The probability of choosing these 9 multiples in the first place is $\frac{9}{100}$, so the

final probability is $\frac{9}{100} \cdot \frac{1}{2} = \frac{9}{200}$, so the answer is **(B)** $\frac{9}{200}$. $11 = 1, 11 \Rightarrow \frac{1}{2}$

$$22 = 2 \times 11 : 1, 2, 11, 22 \Rightarrow \frac{1}{2}$$

$$33 = 3 \times 11 : 1, 3, 11, 33 \Rightarrow \frac{1}{2}$$

$$44 = 2^2 \times 11 : 1, 2, 4, 11, 22, 44 \Rightarrow \frac{1}{2}$$

$$55 = 5 \times 11 : 1, 5, 11, 55 \Rightarrow \frac{1}{2}$$

$$66 = 2 \times 3 \times 11 : 1, 2, 3, 6, 11, 22, 33, 66 \Rightarrow \frac{1}{2}$$

$$77 = 7 \times 11 : 1, 7, 11, 77 \Rightarrow \frac{1}{2}$$

$$88 = 2^3 \times 11 : 1, 2, 4, 8, 11, 22, 44, 88 \Rightarrow \frac{1}{2}$$

$$99 = 3^2 \times 11 : 1, 3, 9, 11, 33, 99 \Rightarrow \frac{1}{2}$$

Solution 2

There are $\left\lfloor \frac{100}{11} \right\rfloor = 9$ multiples of 11 under 100. Because all of these numbers are multiples of 11 to the first power and first power only, their factors can either have 11 as a factor (11^1) or not have 11 as a factor (11^0), since 11 is prime, resulting in a $\frac{1}{2}$ chance of a factor chosen being divisible by 11. The chance of choosing any factor of 11 under 100 is $\frac{9}{100}$, so the final answer is

$$\frac{9}{100} \cdot \frac{1}{2} = \boxed{\text{(B)} \frac{9}{200}}.$$

Solution 3 (generalized)

Let N be the positive integer in question. Since N is a multiple of 11, we can write $N = 11^{e_1} p_1^{e_2} \cdots p_{k-1}^{e_k}$, the prime factorization of N . To find the total number of divisors divisible by 11, observe that there are $(e_2 + 1)(e_3 + 1) \cdots (e_k + 1)$ divisors not divisible by 11. For each power of 11 greater than 1, of which there are e_1 , it can be paired with any of these other divisors. Therefore, there are

$$e_1(e_2 + 1)(e_3 + 1) \cdots (e_k + 1)$$

divisors of N that are divisible by 11. The total number of divisors of N is

$$(e_1 + 1)(e_2 + 1) \cdots (e_k + 1),$$

so the probability of choosing a divisor that is divisible by 11 is

$$\frac{e_1(e_2 + 1)(e_3 + 1) \cdots (e_k + 1)}{(e_1 + 1)(e_2 + 1) \cdots (e_k + 1)} = \frac{e_1}{e_1 + 1}.$$

For each positive integer less than or equal to 100, the highest power of 11 that divides it is 11, so $e_1 = 1$, meaning that the probability of choosing a divisor (that is divisible by 11) of a fixed N is $\frac{1}{2}$. The probability of choosing any N from the first 100 positive integers is $\frac{1}{100}$, so the probability of choosing any of these divisors is $\frac{1}{100} \cdot \frac{1}{2}$. There are 9 multiples of 11 less than or equal to 100, so the total probability is

$$9 \cdot \frac{1}{100} \cdot \frac{1}{2} = \boxed{\text{(B)} \frac{9}{200}}.$$

-Benedict T (countmath1)

- o) (AMC 10 10A 2023) An even number of circles are nested, starting with a radius of 1 and increasing by 1 each time, all sharing a common point. The region between every other circle is shaded, starting with the region inside the circle of radius 2 but outside the circle of radius 1. An example showing 8 circles is displayed below. What is the least number of circles needed to make the total shaded area at least 2023π ?



- (A) 46 (B) 48 (C) 56 (D) 60 (E) 64

Answer: E

Solution 1

Notice that the area of the shaded region is $(2^2\pi - 1^2\pi) + (4^2\pi - 3^2\pi) + (6^2\pi - 5^2\pi) + \cdots + (n^2\pi - (n-1)^2\pi)$ for any even number n .

Using the difference of squares, this simplifies to $(1+2+3+4+\cdots+n)\pi$. So, we are basically finding the smallest n such that $\frac{n(n+1)}{2} > 2023 \Rightarrow n(n+1) > 4046$.

Since $60(61) > 60^2 = 3600$, the only option higher than 60 is **(E) 64**.

Solution 2

After first observing the problem, we can work out a few of the areas.

$$\text{1st area} = 4\pi - \pi = 3\pi$$

$$\text{2nd area} = 16\pi - 9\pi = 7\pi$$

$$\text{3rd area} = 36\pi - 25\pi = 11\pi$$

$$\text{4th area} = 64\pi - 49\pi = 15\pi$$

We can see that the pattern is an arithmetic sequence with first term 3 and common difference 4. From here, we can start from the bottom of the answer choices and work our way up. As the question asks for the least number of circles needed total, we have to divide the number of total circles by 2.

We can find the sum of the first 32 terms of the arithmetic sequence by using the formula.

$$\text{The last term is: } 3 + 4 \cdot (32 - 1) = 127.$$

Then, we can find the sum: $(3 + 127)/2 \cdot 32 = 65 \cdot 32 = 2080$. It is clear that 64 works.

The next answer choice is 60, which we have to divide by 2 to get 30.

$$\text{The last term is: } 3 + 4 \cdot (30 - 1) = 119.$$

The sum is: $(3 + 119)/2 \cdot 30 = 61 \cdot 30 = 1830$. This does not work.

As answer choice D does not work and E does, we can conclude that the answer is **(E) 64**.

Solution 3 (Similar to Solution 2)

We can easily see that all of the answer choices are even, which helps us solve this problem a little.

Let's just not consider the π , since it is not that important, so let's cancel that out.

When we plug in 64, we get $64^2 - 63^2 + 62^2 - 61^2 + \dots + 4^2 - 3^2 + 2^2 - 1^2$. By difference of squares, we get $1 + 2 + 3 + \dots + 62 + 63 + 64$, which by the sum of an arithmetic sequence, is $\frac{64(64+1)}{2}$, which is 2080.

Similarly, we can use this for answer choice D , and we have $\frac{60(60+1)}{2}$ which is 1830.

So, we see that answer choice D is too small to satisfy the requirements, so we conclude the answer is **(E) 64**.

Solution 4

We can consider making a table.

If there is 1 circle, the area of the shaded region is 0π . (We can write this as 0π .)

If there are 2 circles, the area of the shaded region is 3π . (We can write this as $(1+2)\pi$.)

If there are 3 circles, the area of the shaded region is 3π . (We can write this as $(1+2)\pi$.)

If there are 4 circles, the area of the shaded region is 10π . (We can write this as $(1+2+3+4)\pi$.)

If there are 5 circles, the area of the shaded region is 10π . (We can write this as $(1+2+3+4)\pi$).

If there are 6 circles, the area of the shaded region is 21π . (We can write this as $(1+2+3+4+5+6)\pi$).

Now the pattern emerges. When n is even, the area of the shaded region is $(1 + 2 + 3 + \cdots + n)\pi$, or $\left(\frac{n(n+1)}{2}\right)\pi$. But remember that the problem stated that there are an even number of circles. So now we are solving the equation

$$\left(\frac{n(n+1)}{2}\right)\pi \geq 2023\pi.$$

Dividing by π and multiplying by 2 on both sides, we get $n(n+1) \geq 4046$. Now we can plug in the answer choices, and we start off with 60 because it is the only answer choice that is a multiple of 10. Plugging in we get $60 \cdot 61 = 3660$, and this is not quite yet more than 4046. But only option (E) is bigger, so we know that the solution can only be **(E) 64**.

(P.S. I did this in the test and I solved it in 2 minutes)

Solution 5

Denote S the area of shaded region and W the area of white region.

$S > W$ and $S \approx W$ if R is big. Therefore

$$\pi R^2 = S + W \approx 2S \geq 4046\pi \implies R \approx 64.$$

So we conclude the answer is **(E) 64**.

vladimir.shelomovskii@gmail.com, vvsss

Solution 6 (A Bit slow.)

After first observing the problem, we can work out a few of the areas.

$$\text{1st area} = 4\pi - \pi = 3\pi$$

$$\text{2nd area} = 16\pi - 9\pi = 7\pi$$

$$\text{3rd area} = 36\pi - 25\pi = 11\pi$$

$$\text{4th area} = 64\pi - 49\pi = 15\pi$$

Notice that the area of the shaded region is $3\pi + 7\pi + 11\pi + 15\pi \dots + (4n-1)\pi$ for any even number n .

We can create a formula to compute this sum by doing: $3\pi + 7\pi + 11\pi + 15\pi \dots + (4n-1)\pi + (4n-1)\pi + \dots + 15\pi + 11\pi + 7\pi + 3\pi$ to get $(4n+2) * \frac{n}{2} \rightarrow (2n^2+n)\pi$. It is important to realize that each n is 2 circles, the unshaded and the shaded. So that means for each answer choice there are $2x$ as many circles. Then we just plug in values (strategically) to get that $2 * (\frac{64}{2})^2 + (\frac{64}{2})$ or **(E) 64** as the answer.

- p) (AMC 10 10A 2023) In a table tennis tournament every participant played every other participant exactly once. Although there were twice as many right-handed players as left-handed players, the number of games won by left-handed players was 40% more than the number of games won by right-handed players. (There were no ties and no ambidextrous players.) What is the total number of games played?

(A) 15 (B) 36 (C) 45 (D) 48 (E) 66

Answer: B

Solution 1

We know that the total amount of games must be the sum of games won by left and right handed players. Then, we can write $g = l + r$, and since $l = 1.4r$, $g = 2.4r$. Given that r and g are both integers, $g/2.4$ also must be an integer. From here we can see that g must be divisible by 12, leaving only answers B and D. Now we know the formula for how many games are played in this tournament is $n(n-1)/2$, the sum of the first $n-1$ numbers. Now, setting 36 and 48 equal to the equation will show that two consecutive numbers must have a product of 72 or 96. Clearly, $72 = 8 * 9$, so the answer is **(B) 36**.

Solution 2

First, we know that every player played every other player, so there's a total of $\binom{n}{2}$ games since each pair of players forms a bijection to a game. Therefore, that rules out D. Also, if the right-handed players won a total of x games, the left-handed players must have won a total of $\frac{7}{5}x$ games, meaning that the total number of games played was $\frac{12}{5}x$. Thus, the total number of games must be divisible by 12. Therefore leaving only answer choices B and D. Since answer choice D doesn't satisfy the first condition, the only answer that satisfies both conditions is **(B) 36**.

A simpler way is to know that the result must be a number in the form $\binom{n}{2}$ games. This eliminates D. Then write an expression that equals the answer, $1.4x + x = n$ games total. Only 48 and 36 satisfy so 36 is the answer which is **(B) 36**.

Solution 3

Let r be the amount of games the right-handed won. Since the left-handed won $1.4r$ games, the total number of games played can be expressed as $(2.4)r$, or $12/5r$, meaning that the answer is divisible by 12. This brings us down to two answer choices, B and D . We note that the answer is some number x choose 2. This means the answer is in the form $x(x-1)/2$. Since answer choice D gives $48 = x(x-1)/2$, and $96 = x(x-1)$ has no integer solutions, we know that **(B) 36** is the only possible choice.

Solution 4 (Rigid solving, no cheese)

Here is the rigid way to prove that 36 is the only answer. Let the number of left-handed players be n , so the number of right-handed players is $2n$. The number of games won by the left-handed players comes in two ways:

- The games played by two left-left pairs, which is $\frac{n(n-1)}{2}$, and
- The games played by left-right pairs, which we'll call x .

Note that $x \leq 2n^2$, which is the total number of games played by left-right pairs. Using the same logic for right-right pairs and right-left pairs, we have

that

$$\frac{\frac{n(n-1)}{2} + x}{\frac{2n(2n-1)}{2} + 2n^2 - x} = 1.4,$$

which gives

$$x = \frac{17n^2}{8} - \frac{3n}{8}.$$

We know that $x \leq 2n^2$, applying that becomes

$$\begin{aligned}\frac{17n^2}{8} - \frac{3n}{8} &\leq 2n^2 \\ 17n^2 - 3n &\leq 16n^2 \\ n^2 - 3n &\leq 0 \\ n &\leq 3.\end{aligned}$$

(We can safely divide by n because it must be positive). So the total number of players p can only be 3, 6, and 9.

Since the total number of games $p(p-1)/2$ is $1.4+1 = \frac{12}{5}$ times a non-negative integer number of games won by righties, $p(p-1)$ must be a multiple of $12(2) = 24$. Among $\{3, 6, 9\}$, only $p = 9$ satisfies this condition, so the total number of games is $\frac{(9)(8)}{2} = \boxed{\text{(B) } 36}$.

Solution 5 (Cheese)

If there are n players, the total number of games played must be $\binom{n}{2}$, so it has to be a triangular number. The ratio of games won by left-handed to right-handed players is $7 : 5$, so the number of games played must also be divisible by 12. Finally, we notice that only $\boxed{\text{(B) } 36}$ satisfies both of these conditions.

Solution 5.1 (Slightly Slower)

We know that the number of games played must be a triangular number, which eliminates **(D)**48, then we see that there are twice as many right-handed players as left-handed ones. Which means that the total number of players must be divisible by 3, and since each player can't play themselves, we can add 1 to the largest number in the sequence, since $1 + 2 + 3 + \dots + 8$ is the only one which the amount of players is divisible by 3(9players), $\boxed{\text{(B) } 36}$ is the answer

Solution 6 (Cheese 2)

Call the number of games x , the number of games won by right handed player R , and by left handed players L . $L+R=x$ and $L=1.4 R$. Therefore, $x/2.4$ must be an integer, which leaves only 36 and 48. Then, note x must be half the product of two consecutive numbers (the number of possible combinations of two people) and we can eliminate 48.

How is this cheese... it is literally the first solution but scaled down

-Shreyansh

EXACTLY... the first solution is also cheese.

-AndrewZhong2012

cheddar

-PerseverePlayer

Solution 7

Let the number of people be n . This means that there will be $\frac{n(n-1)}{2}$ games played. We can now eliminate **(D)** because it's not a triangular number. The problem states that there are twice as many right handed people as left handed people with no ambidextrous people. Based on this, we can say that the number of people (or n) has to be divisible by 3. Because the number of games won by left-handed people is 40% more than the number of games won by right-handed people, we can set the ratio between the number of games won by left-handed to right-handed as $\frac{7}{5}$. The sum of the numerator and the denominator is 12. This means that the number of games must be divisible by 12. Thus, the only answer choice that fits the conditions above is answer **(B) 36**.

Solution 8(Pairing) - easiest noncheese

Case 1: in-group pairing

In every pair where both players use the same hand, the win rate is always 50%. This is because the total number of such pairs is even — any factorial other than the trivial case $1!$ is even, and 1 is insufficient to form a pair. Thus, no player is left out, and in each pair exactly one player wins and one loses, giving a win-loss ratio of $1 : 1$. In other words, this case completely cancels out due to its balanced win-loss ratio.

Case 2: cross-group pairing

For convenience, let left-handed players be denoted by L and right-handed players by R . The problem gives the ratio $R : L = 1 : 2$. This means each L plays $2L$ times as many games as an R . From this, the total number of cross-group games is $2L \times L = 2L^2$.

The problem also states that the ratio of L wins to R wins is $7 : 5$, so the total number of wins in this case must be divisible by $5 + 7 = 12$. We therefore require $2L^2$ to be divisible by 12. Simplifying gives $L^2 \mid 6$. The smallest perfect square divisible by 6 is **(B) 36**.

Solution 9(Subgroups)

Let L and R be the number of left-handed and right-handed players, respectively. The total number of games played in the tournament can be broken down into three distinct parts based on the matchups: games between two left-handed players, games between two right-handed players, and games between a left-handed and a right-handed player. This gives the total game count formula:

$$\text{Total Games} = LR + \frac{L(L-1)}{2} + \frac{R(R-1)}{2}$$

Every game played results in exactly one win. In the matches where left-handed players play each other, a left-handed player must win. This guarantees $\frac{L(L-1)}{2}$ wins for the left-handed group. Similarly, matches where right-handed players play each other guarantee $\frac{R(R-1)}{2}$ wins for the right-handed group.

For the LR games played between a left-handed player and a right-handed player, let W be the number of games won by the left-handed players. The

remaining (LR - W) games in this section must be won by the right-handed players. We can now write expressions for the total wins of each group:

$$\text{Total Left Wins } (W_L) = \frac{L(L-1)}{2} + W$$

$$\text{Total Right Wins } (W_R) = \frac{R(R-1)}{2} + LR - W$$

The problem states that there are twice as many right-handed players as left-handed players, so we substitute $R = 2L$ into our equations:

$$W_L = \frac{L(L-1)}{2} + W$$

$$W_R = \frac{2L(2L-1)}{2} + L(2L) - W = L(2L-1) + 2L^2 - W = 4L^2 - L - W$$

We are given that left-handed players won 40% more games than right-handed players, which means $W_L = 1.4 \times W_R$. Setting up the equation:

$$\frac{L(L-1)}{2} + W = 1.4(4L^2 - L - W)$$

To clear the fraction and decimal, we multiply the entire equation by 10:

$$5L(L-1) + 10W = 14(4L^2 - L - W)$$

$$5L^2 - 5L + 10W = 56L^2 - 14L - 14W$$

Grouping the W terms on the left side and the L terms on the right side yields:

$$24W = 51L^2 - 9L$$

Dividing the entire equation by 3 simplifies it to:

$$8W = 17L^2 - 3L \implies W = \frac{17L^2 - 3L}{8}$$

Since W represents a physical number of wins, it must be a non-negative integer. Additionally, W cannot be greater than the total number of inter-group games, meaning $W \leq 2L^2$. Testing small integer values for L:

- If L=1: $W = \frac{14}{8}$, which is not an integer.
- If L=2: $W = \frac{62}{8}$, which is not an integer.
- If L=3: $W = \frac{153-9}{8} = \frac{144}{8} = 18$. This is a valid integer, and it satisfies $18 \leq 2(3)^2 = 18$.

With L=3, the number of right-handed players is $R = 6$, giving a total of 9 players. Substituting these into the original total games formula:

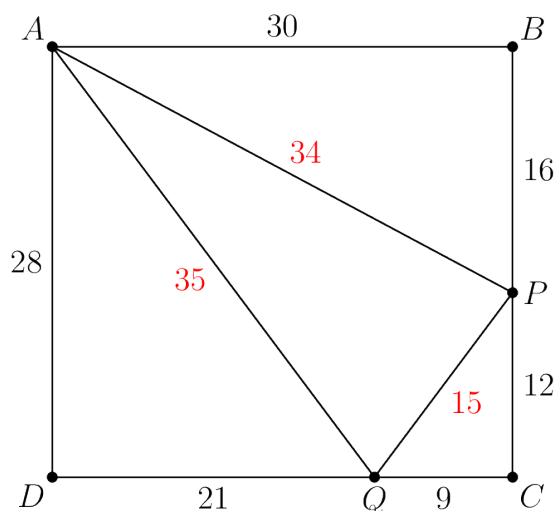
$$\text{Total Games} = (3)(6) + \frac{3(2)}{2} + \frac{6(5)}{2} = 18 + 3 + 15 = \boxed{\text{(B) } 36}$$

- q) (AMC 10 10A 2023) Let $ABCD$ be a rectangle with $AB = 30$ and $BC = 28$. Point P and Q lie on $\overline{BC}$ and $\overline{CD}$ respectively so that all sides of $\triangle ABP$, $\triangle PCQ$, and $\triangle QDA$ have integer lengths. What is the perimeter of $\triangle APQ$?

(A) 84 (B) 86 (C) 88 (D) 90 (E) 92

Answer: A

Solution 1 (Testing Pythagorean Triples)



We know that all side lengths are integers, so we can test Pythagorean triples for all triangles.

First, we focus on $\triangle ABP$. The length of AB is 30, and the possible (small enough) Pythagorean triples $\triangle ABP$ can be are $(3, 4, 5)$, $(5, 12, 13)$, $(8, 15, 17)$, where the length of the longer leg is a factor of 30. Testing these, we get that only $(8, 15, 17)$ is a valid solution. Thus, we know that $BP = 16$ and $AP = 34$.

Next, we move on to $\triangle QDA$. The length of AD is 28, and the small enough triples are $(3, 4, 5)$ and $(7, 24, 25)$. Testing again, we get that $(3, 4, 5)$ is our triple. We get the value of $DQ = 21$, and $AQ = 35$.

We know that $CQ = CD - DQ$ which is 9, and $CP = BC - BP$ which is 12. $\triangle CPQ$ is therefore a right triangle with side length ratios 3, 4, 5, and the hypotenuse is equal to 15. $\triangle APQ$ has side lengths 34, 35, and 15, so the perimeter is equal to $34 + 35 + 15 = \boxed{\text{(A) } 84}$.

Solution 2

Let $BP = y$ and $AP = z$. We get $30^2 + y^2 = z^2$. Subtracting y^2 on both sides, we get $30^2 = z^2 - y^2$. Factoring, we get $30^2 = (z - y)(z + y)$. Since y and z are integers, both $z - y$ and $z + y$ have to be even or both have to be odd. We also have $y < 31$. We can pretty easily see now that $z - y = 18$ and $z + y = 50$. Thus, $y = 16$ and $z = 34$. We now get $CP = 12$. We do the same trick again. Let $DQ = a$ and $AQ = b$. Thus, $28^2 = (b + a)(b - a)$. We can get $b + a = 56$ and $b - a = 14$. Thus, $b = 35$ and $a = 21$. We get $CQ = 9$ and by the Pythagorean Theorem, we have $PQ = 15$. We get $AP + PQ + AQ = 34 + 15 + 35 = 84$. Our answer is $\boxed{\text{(A) } 84}$.

If you want to see a video solution on this solution, look at Video Solution 1.

-paixiao (minor edits-HW73)

- r) (AMC 10 10A 2023) A rhombic dodecahedron is a solid with 12 congruent rhombus faces. At every vertex, 3 or 4 edges meet, depending on the vertex. How many vertices have exactly 3 edges meet?

(A) 5 (B) 6 (C) 7 (D) 8 (E) 9

Answer: D

Solution 1

Note Euler's formula where Vertices + Faces - Edges = 2. There are 12 faces. There are 24 edges, because there are 12 faces each with four edges and each edge is shared by two faces. Now we know that there are $2 - 12 + 24 = 14$ vertices. Now note that the sum of the degrees of all the points is 48 (the number of edges times 2). Let x = the number of vertices with 3 edges. Now we know $\frac{3x + 4(14 - x)}{2} = 24$. Solving this equation gives $x = \boxed{\text{(D) } 8}$.
aiden22gao zgaahlkw (LaTeX) ESAOPS (Simplified)

Solution 2

Let x be the number of vertices with 3 edges, and y be the number of vertices with 4 edges. Since there are $\frac{4 \cdot 12}{2} = 24$ edges on the polyhedron, we can see that $\frac{3x + 4y}{2} = 24$. Then, $3x + 4y = 48$. Notice that by testing the answer choices, (D) is the only one that yields an integer solution for y . Thus, the answer is $\boxed{\text{(D) } 8}$.

Solution 3

With 12 rhombi, there are $4 \cdot 12 = 48$ total boundaries. Each edge is used as a boundary twice, once for each face on either side. Thus we have $\frac{48}{2} = 24$ total edges.

Let A be the number of vertices with 3 edges (this is what the problem asks for) and B be the number of vertices with 4 edges. We have $3A + 4B = 48$.

Euler's formula states that, for all convex polyhedra, $V - E + F = 2$. In our case, $V - 24 + 12 = 2 \implies V = 14$. We know that $A + B$ is the total number of vertices as we are given that all vertices are connected to either 3 or 4 edges. Therefore, $A + B = 14$.

We now have a system of two equations. Solving the system yields $A = \boxed{\text{(D) } 8}$.

Even without Euler's formula, we can do a bit of answer guessing. From $3A + 4B = 48$, we take mod 4 on both sides.

$$3A + 4B \equiv 48 \pmod{4}$$

$$3A \equiv 0 \pmod{4}$$

We know that $3A$ must be divisible by 4. We know that the factor of 3 will not affect the divisibility by 4 of $3A$, so we remove the 3. We know that A is divisible by 4. Checking answer choices, the only one divisible by 4 is indeed $A = \boxed{\text{(D) } 8}$.

Solution 4

Note that Euler's formula is $V + F - E = 2$. We know $F = 12$ from the question. We also know $E = \frac{12 \cdot 4}{2} = 24$ because every face has 4 edges and every edge is shared by 2 faces. We can solve for the vertices based on this information.

Using the formula we can find:

$$V + 12 - 24 = 2$$

$$V = 14$$

Let t be the number of vertices with 3 edges and f be the number of vertices with 4 edges. We know $t + f = 14$ from the question and $3t + 4f = 48$. The second equation is because the total number of points is 48 because there are 12 rhombuses of 4 vertices. Now, we just have to solve a system of equations.

$$3t + 4f = 48$$

$$3t + 3f = 42$$

$$f = 6$$

$$t = 8$$

Our answer is simply just t , which is (D) 8 musicalpenguin

Solution 5

Each of the twelve rhombi has two pairs of angles across from each other that must be congruent. If both pairs of angles occur at 4-point intersections, we have a grid of squares. If both occur at 3-point intersections, we would have a cube with six square faces. Therefore, two of the points must occur at a 3-point intersection and two at a 4-point intersection.

Since each 3-point intersection has 3 adjacent rhombuses, we know the number of 3-point intersections must equal the number of 3-point intersections per rhombus times the number of rhombuses over 3. Since there are 12 rhombuses and two 3-point intersections per rhombus, this works out to be:

$$\frac{2 \cdot 12}{3}$$

Hence: (D) 8 hollph27 Minor edits by FutureSphinx

Solution 6 (Based on previous knowledge)

Note that a rhombic dodecahedron is formed when a cube is turned inside out (as seen here), thus there are 6 4-vertices (corresponding to each face of the cube) and 8 3-vertices (corresponding to each corner of the cube). Thus the answer is (D) 8

Solution 7 (Dual)

Note that a rhombic dodecahedron is the dual of a cuboctahedron. A cuboctahedron has 8 triangular faces, which correspond to (D) 8 vertices on a rhombic dodecahedron that have 3 edges.

Solution 8 (Uses Solution 2)

Continue through solution 2 until you get $3x + 4y = 48$. Then we have that $3x = 48 - 4y = 4(12 - y) = \text{multiple of } 4$ so x is a multiple of 4. The only multiple of 4 answer choice is $8 =$ (D)

- s) (AMC 10 10A 2023) The line segment formed by $A(1, 2)$ and $B(3, 3)$ is rotated to the line segment formed by $A'(3, 1)$ and $B'(4, 3)$ about the point $P(r, s)$. What is $|r - s|$?

(A) $\frac{1}{4}$ (B) $\frac{1}{2}$ (C) $\frac{3}{4}$ (D) $\frac{2}{3}$ (E) 1

Answer: E

Solution 1

Due to rotations preserving an equal distance, we can bash the answer with the distance formula. $D(A, P) = D(A', P)$, and $D(B, P) = D(B', P)$. Thus we will square our equations to yield: $(1 - r)^2 + (2 - s)^2 = (3 - r)^2 + (1 - s)^2$, and $(3 - r)^2 + (3 - s)^2 = (4 - r)^2 + (3 - s)^2$. Canceling $(3 - s)^2$ from the second equation makes it clear that r equals 3.5.

Substituting will yield

$$\begin{aligned}(2.5)^2 + (2 - s)^2 &= (-0.5)^2 + (1 - s)^2 \\ 6.25 + 4 - 4s + s^2 &= 0.25 + 1 - 2s + s^2 \\ 2s &= 9 \\ s &= 4.5\end{aligned}$$

Now $|r - s| = |3.5 - 4.5| = \boxed{\text{(E)} 1}$.

-Antifreeze5420

Solution 2

Due to rotations preserving distance, we have that $BP = B'P$, as well as $AP = A'P$. From here, we can see that P must be on the perpendicular bisector of $\overline{BB'}$ due to the property of perpendicular bisectors keeping the distance to two points constant.

From here, we proceed to find the perpendicular bisector of $\overline{BB'}$. We can see that this is just a horizontal line segment with midpoint at $(3.5, 3)$. This means that the equation of the perpendicular bisector is $x = 3.5$.

Similarly, we find the perpendicular bisector of $\overline{AA'}$. We find the slope to be $\frac{1-2}{3-1} = -\frac{1}{2}$, so our new slope will be 2. The midpoint of A and A' is $(2, \frac{3}{2})$, which we can use with our slope to get another equation of $y = 2x - \frac{5}{2}$.

Now, point P has to lie on both of these perpendicular bisectors, meaning that it has to satisfy both equations. Plugging in the value of x we found earlier, we find that $y = 4.5$. This means that $|r - s| = |3.5 - 4.5| = \boxed{\text{(E)} 1}$.

-DEVSA XENA

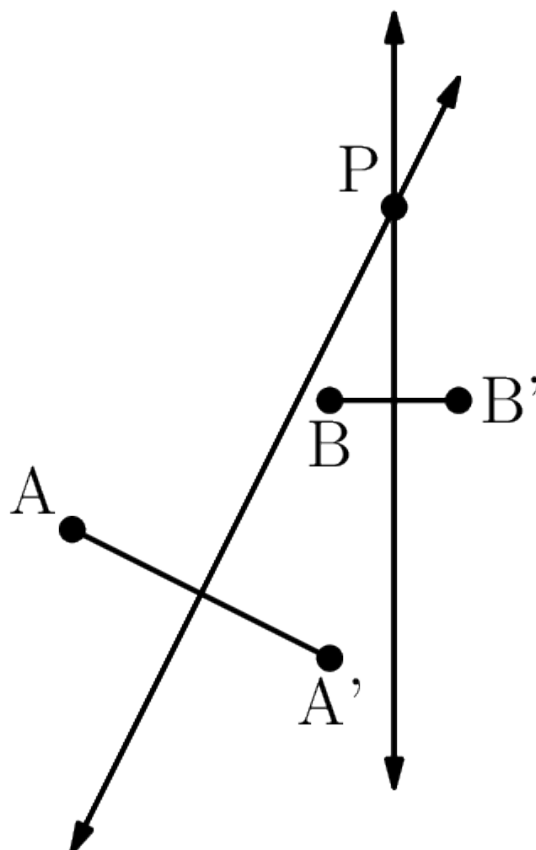
Solution 3 (Coordinate Geometry)

To find the center of rotation, we find the intersection point of the perpendicular bisectors of $\overline{AA'}$ and $\overline{BB'}$. (this is because $\overline{PA}$ is equal to $\overline{PA'}$ and same with B and B')

We can find that the equation of the line $\overline{AA'}$ is $y = -\frac{1}{2}x + \frac{5}{2}$, and that the equation of the line $\overline{BB'}$ is $y = 3$.

When we solve for the perpendicular bisector of $y = -\frac{1}{2}x + \frac{5}{2}$, we determine that it has a slope of 2, and it runs through $(2, 1.5)$. Plugging in $1.5 = 2(2) - n$, we get that $n = \frac{5}{2}$. Therefore our perpendicular bisector is $y = 2x - \frac{5}{2}$. Next, we solve for the perpendicular of $y = 3$. We know that it has an undefined slope, and it runs through $(3.5, 3)$. We can determine that our second perpendicular bisector is $x = 3.5$.

Setting the equations equal to each other, we get $y = \frac{9}{2}$. Therefore, $|r - s| = |3.5 - 4.5| = \boxed{\text{(E) } 1}$.



Solution 4

We use the complex numbers approach to solve this problem. Denote by θ the angle that AB rotates about P in the counterclockwise direction.

Thus, $A' - P = e^{i\theta} (A - P)$ and $B' - P = e^{i\theta} (B - P)$.

Taking ratio of these two equations, we get

$$\frac{A' - P}{A - P} = \frac{B' - P}{B - P}.$$

By solving this equation, we get $P = \frac{7}{2} + i\frac{9}{2}$. Therefore, $|s - t| = \left| \frac{7}{2} - \frac{9}{2} \right| = \boxed{\text{(E) } 1}$.

Solution 5 (Overkill Trigonometry)

First, we need to find the coordinates of point $P(r, s)$ such that rotating the line segment from $A(1, 2)$ to $B(3, 3)$ about P maps it to the line segment from $A'(3, 1)$ to $B'(4, 3)$. After finding r and s , we will compute $|r - s|$.

When a point (x, y) is rotated about another point (r, s) by an angle θ , the new coordinates (x', y') are given by:

$$\begin{cases} x' = r + (x - r) \cos \theta - (y - s) \sin \theta \\ y' = s + (x - r) \sin \theta + (y - s) \cos \theta \end{cases}$$

We apply the rotation to points A and B :

For $A(1, 2)$ mapping to $A'(3, 1)$:

$$\begin{cases} 3 = r + (1 - r) \cos \theta - (2 - s) \sin \theta & (1) \\ 1 = s + (1 - r) \sin \theta + (2 - s) \cos \theta & (2) \end{cases}$$

For $B(3, 3)$ mapping to $B'(4, 3)$:

$$\begin{cases} 4 = r + (3 - r) \cos \theta - (3 - s) \sin \theta & (3) \\ 3 = s + (3 - r) \sin \theta + (3 - s) \cos \theta & (4) \end{cases}$$

Now, we subtract equations to eliminate variables:

Subtract equation (1) from equation (3):

$$(4 - 3) = [r + (3 - r) \cos \theta - (3 - s) \sin \theta] - [r + (1 - r) \cos \theta - (2 - s) \sin \theta]$$

Simplifying yields:

$$1 = 2 \cos \theta - \sin \theta \quad (\text{A})$$

Similarly, subtract equation (2) from equation (4):

$$(3 - 1) = [s + (3 - r) \sin \theta + (3 - s) \cos \theta] - [s + (1 - r) \sin \theta + (2 - s) \cos \theta]$$

Simplifying further:

$$2 = 2 \sin \theta + \cos \theta \quad (\text{B})$$

Now, using equations (A) and (B):

$$\begin{cases} 2 \cos \theta - \sin \theta = 1 \\ \cos \theta + 2 \sin \theta = 2 \end{cases}$$

We now express $\cos \theta$ from the second equation:

$$\cos \theta = 2 - 2 \sin \theta$$

Substituting this into the first equation, we get:

$$2(2 - 2 \sin \theta) - \sin \theta = 14 - 4 \sin \theta - \sin \theta = 1 - 5 \sin \theta = -3 \sin \theta = \frac{3}{5}$$

Now, we find $\cos \theta$ as follows:

$$\cos \theta = 2 - 2 \left(\frac{3}{5} \right) = \frac{4}{5}$$

Substituting $\cos \theta$ and $\sin \theta$ back into equations (1) and (2):

Equation (1):

$$3 - r = (1 - r) \left(\frac{4}{5} \right) - (2 - s) \left(\frac{3}{5} \right)$$

Multiply both sides by 5:

$$5(3 - r) = 4(1 - r) - 3(2 - s)$$

Simplify:

$$15 - 5r = 4 - 4r - 6 + 3s \quad 15 - 5r = -2 - 4r + 3s \quad -5r + 4r - 3s = -2 - 15 - r - 3s = -17 \quad (C)$$

Equation (2):

$$1 - s = (1 - r) \left(\frac{3}{5} \right) + (2 - s) \left(\frac{4}{5} \right)$$

Multiply both sides by 5:

$$5(1 - s) = 3(1 - r) + 4(2 - s)$$

Simplifying more:

$$5 - 5s = 3 - 3r + 8 - 4s \quad 5 - 5s = 11 - 3r - 4s \quad -5s + 4s + 3r = 11 - 53r - s = 6 \quad (D)$$

From equations (C) and (D):

$$\begin{cases} -r - 3s = -17 \\ 3r - s = 6 \end{cases}$$

We then multiply the first equation by 3:

$$-3r - 9s = -51$$

Add this to the second equation multiplied by 9:

$$27r - 9s = 54$$

Subtract:

$$(27r - 9s) - (-3r - 9s) = 54 - (-51) \quad 30r = 105 \quad r = \frac{105}{30} = \frac{7}{2} = 3.5$$

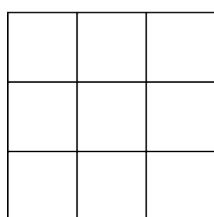
Now find s :

$$3(3.5) - s = 610.5 - s = 6s = 4.5$$

Therefore, we have the following:

$$|r - s| = |3.5 - 4.5| = |-1| = \boxed{\text{(E)} 1}.$$

- t) (AMC 10 10A 2023) Each square in a 3×3 grid of squares is colored red, white, blue, or green so that every 2×2 square contains one square of each color. One such coloring is shown on the right below. How many different colorings are possible?



B	R	B
G	W	G
R	B	R

- (A) 24 (B) 48 (C) 60 (D) 72 (E) 96

Answer: D

Solution 1

Let a "tile" denote a 1×1 square, and a "square" refer to a 2×2 square.

We have $4! = 24$ possible ways to fill out the top-left square. Next, we fill out the bottom-right corner tile. In the bottom-right square, one corner is already filled (the central tile) from our initial coloring), so we have 3 color options remaining for this.

Now considering the remaining tiles, all of these only have one way to be filled (Try it yourself if you don't believe).

For example, the right tile of the middle row is part of two squares: the top-right and the bottom-right. Among these squares, 3 colors have already been used, leaving us with only 1 remaining option. Similarly, every other remaining square has only one available option for coloring.

Thus, the total number of ways is $3 \times 4! = \boxed{\text{(D)} 72}$.

A quick version of this method is used when you use for example B as the top left corner block. You can see that B in the top left corner has 6 possible ways. Now, see that there are 3 possible corners for a center cell and there are 4 possible center cells. We get 6 times 4 times 3 = D, the answer is $\boxed{\text{(D)} 72}$. (I'm not very good at using latex) breakingbread latex done by pungent_muskrat

Solution 2 (Circular Counting)

Recall the circular gap theorem

$$C_n(A) = \frac{n}{k} \binom{n-k-1}{k-1}.$$

We have nine slots and a maximum of 4 colors cannot be next to one another (if you aren't convinced, try plugging in any number greater than 4 and see what happens to the formula; it becomes invalid, and is therefore 0). Therefore, we take the cases $n = 9$ and $k \in \{1, 2, 3, 4\}$.

Case 1: $k \in \{1\}$

This is simply just $\frac{9}{1} \binom{9-1-1}{1-1} = 9$ ways.

Case 2: $k \in \{2\}$

This is just $\frac{9}{2} \binom{9-2-1}{2-1} = \frac{9}{2} \cdot 6 = 27$ ways.

Case 3: $k \in \{3\}$

This is $\frac{9}{3} \binom{9-3-1}{3-1} = 3 \cdot 10 = 30$ ways.

Case 4: $k \in \{4\}$

This is $\frac{9}{4} \binom{9-4-1}{4-1} = \frac{9}{4} \cdot 4 = 9$ ways.

The number of ways to arrange the colors such that no same color is next to each other at least once is $9 + 27 + 30 + 9 = 75$.

However, it is clear that every color (excluding the center) is only used a maximum of three times, so we apply the formula for $k \in \{1, 2, 3\}$ and $n = 3$ (three total slots it can go to). Notice that $k \in \{2, 3\}$ is just 0, so therefore we just consider $k \in \{1\}$ and get

$$\frac{3}{1} \binom{3-1-1}{1-1} = 3$$

Our final answer is just $75 - 3 = \boxed{\text{(D) } 72}$

Solution 3

B	R	B
G	W	G
R	B	R

R	B	R
G	W	G
R	B	R

We can split this problem into 2 cases as shown in solution 1. We can swap a set of equal colors for another set of equal colors to create a new square.

Square 1: The first square can be rotated to create another square so we have to multiply the number of arrangements by 2. We have $4! = 24$ arrangements without rotating and $24 \cdot 2 = 48$ arrangements in total for the first square.

Square 2: There are $4! = 24$ ways to arrange the colors.

In total, we have $48 + 24 = \boxed{\text{(D) } 72}$ arrangements.

Solution 4

Let's call the top-right corner color A, the top-middle color B, the top-right color C, and so on, with color D being the middle row, and right corner square, and color G being the bottom-left square's color.

WLOG $A=\text{Red}$, $B=\text{White}$, $D=\text{Blue}$, and $E=\text{Green}$. We will now consider squares C and F's colors. Case 1 : $C=\text{Red}$ and $F=\text{Blue}$

In this case, we get that G and H have to be Red and White in some order, and the same for H and I. We can color this in 2 ways. Case 2 : C=Blue and F=Red

In this case, one of G and H needs to be White and Red, and H and I needs to be White and Blue. There is 1 way to color this. In total, we get $24 * (2 + 1) = 72$ ways to color the grid. **(D) 72**.

-paixiao

-jkim0656 (minor formatting and latex)

Solution 5

We will choose colors step-by-step:

1. There are 4 ways to choose a color in the center.
2. Then we select any corner and there would be 3 ways to choose a color as we can't use the same color as the one in the center.
3. Consider the 2×2 square that contains the center and the corner we have selected. For the other 2 squares, there are 2 ways to choose colors.
4. Now, consider how many configurations it makes sense to construct the 2×2 square opposite to the corner we have selected using the 2 other 2×2 squares, and we get 3 configurations.

Finally, the answer is $4 \cdot 3 \cdot 2 \cdot 3 = \mathbf{(D) 72}$

Solution 6 (possibly bad)

3	1	2
2	3	1
2	1	1

Note that we could fill the 3 by 3 square with numbers of choices, rather than letters or color names. The top-left corner receives a 3 because there are 3 total choices to choose from: R, G and B. The squares bordering them has values of 2 and 1, regardless of order. 2 indicates that the small square can have any color excluding the one existing color, 1 indicates the remaining color of the 2 by 2 square. Finally, the middle square receives 3, since the first 2 by 2 square has RGB already, and it does not matter what color it has. Moving onto the next 2 by 2 square, we see that there are already 2 decided colors, so the other squares must be 2 and 1, again, regardless of the

order. The same applies to the third 2 by 2 square, and finally the last square has the value of one, as it is the only square left. Multiplying the numbers, $2 \times 2 \times 2 \times 3 \times 3 = \boxed{\text{(D)} 72}$ (Similar to Solution 1)

-MEZE_RUN why are there only 3 choices for the top right corner? you haven't placed any colors yet so there should be 4 choices which would make your answer incorrect

Solution 7

A	B	C
D	E	F
G	H	I

Let's name the cells \$A,B,C,D,E,F,G,H,I\$ from the top left to the bottom right.

Case 1. Cell B and cell H have the same color. The middle one cell E has 4 choices, cell B has 3 choices, then cell E has 2 choices and cell F has 2 choices, this gives $4 \cdot 3 \cdot 2 \cdot 2 = 48$ ways.

Case 2. Cell B and cell H have different colors. The middle one cell E has 4 choices, cell B has 3 choices, cell H has 2 choices, then cell D and F each can only have one choice (different from B, E, H). This gives $4 \cdot 3 \cdot 2 = 24$ ways.

The answer is thus $48 + 24 = 72$, corresponding to answer choice $\boxed{D}$

Solution 8

Consider Cases based on the color of the top left and bottom left square.

Case 1: The top left and bottom left squares are the same:

We have 4 ways to assign the top and bottom squares, let's say it is red. Once that is done, we can assign the center square in 3 ways, since it can be anything except the color of the top and bottom squares (Let's say it is white). We can then assign the bottom middle square any color except the center and bottom left square, so there are two ways to do that (Let's say it is blue). From there, you can assign the other squares in the first two columns in only one way (Top middle = Blue, Middle Left = Green). Then, the right column can go in one of two ways: Green, Red, Green, or Red, Green, Red, for a total of two ways. Taking this altogether gives $4 \cdot 3 \cdot 2 \cdot 2 = 48$

Case 2: The top left and bottom left squares are different:

We have $4 \cdot 3 = 12$ ways to assign the top and bottom squares, let them be Red and Green respectively. Then the square between them can be assigned

in two ways. We realize how the middle square can only be filled in one way (White, because it can't be Blue, Green, or Red). From there, the entire second column can only be filled in one way (From top to bottom, it is Green, White, Red). The same story goes for the last column, it can only be filled in one way (From top to bottom it is Red, Blue, Green). That gives $4 \cdot 3 \cdot 2 = 24$ ways.

Thus, the answer is $48 + 24 = 72$, which is answer choice **(D) 72** latex work done by pungent_muskrat

Solution 9

We can first choose the center square. There are 4 possibilities: Red, Blue, Green, or White. WLOG, let's assume that the center is white. Next, there are 3 possibilities for the top middle square: Red, Blue, or Green. WLOG, let's assume that it is Red. We now have two cases:

CASE 1: The first and last column are in the form BG... and BG... or GB... and GB...: We can see that there are two possibilities for this case: the bottom row is RBR or BRB. The total number of ways for this case is $4 \cdot 3 \cdot 2 \cdot 2$. CASE 2: The first and last column are in the form BG... and GB... or GB... and BG...: We can now see that the bottom middle cell HAS to be a red (if it isn't, either the bottom left or bottom right 2×2 squares will have duplicate colors). After we choose the bottom middle, the rest of the cells are automatically chosen. Thus, the total number of ways for this case is $4 \cdot 3 \cdot 2 \cdot 1$. Thus, our final answer is the sum of case 1 and case 2, which is $48 + 24 = 72$, or **(D) 72**. Irfans123

- u) (AMC 10 10A 2023) Let $P(x)$ be the unique polynomial of minimal degree with the following properties:

- $P(x)$ has a leading coefficient 1,
- 1 is a root of $P(x) - 1$,
- 2 is a root of $P(x - 2)$,
- 3 is a root of $P(3x)$, and
- 4 is a root of $4P(x)$.

The roots of $P(x)$ are integers, with one exception. The root that is not an integer can be written as $\frac{m}{n}$, where m and n are relatively prime integers. What is $m + n$?

- (A) 41 (B) 43 (C) 45 (D) 47 (E) 49

Answer: D

Solution 1

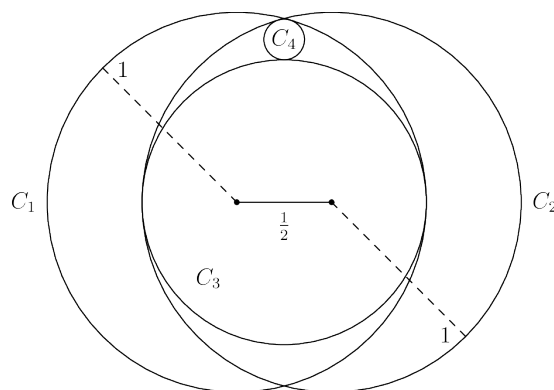
From the problem statement, we find $P(2 - 2) = 0$, $P(9) = 0$ and $4P(4) = 0$. Therefore, we know that 0, 9, and 4 are roots. So, we can factor $P(x)$ as $x(x - 9)(x - 4)(x - a)$, where a is the unknown root. Since $P(x) - 1 = 0$, we plug in $x = 1$ which gives $1(-8)(-3)(1 - a) = 1$, so $24(1 - a) = 1 \implies 1 - a = 1/24 \implies a = 23/24$. Therefore, our answer is $23 + 24 =$ **(D) 47**

Solution 2

We proceed similarly to solution one. We get that $x(x - 9)(x - 4)(x - a) = 1$. Expanding, we get that $x(x - 9)(x - 4)(x - a) = x^4 - (a + 13)x^3 + (13a + 36)x^2 - 36ax$. We know that $P(1) = 1$, so the sum of the coefficients of the cubic expression is equal to one. Thus

$1 - (a + 13) + (13a + 36) - 36a = 1$. Solving for a , we get that $a = 23/24$. Therefore, our answer is $23 + 24 = \boxed{\text{(D)} 47}$

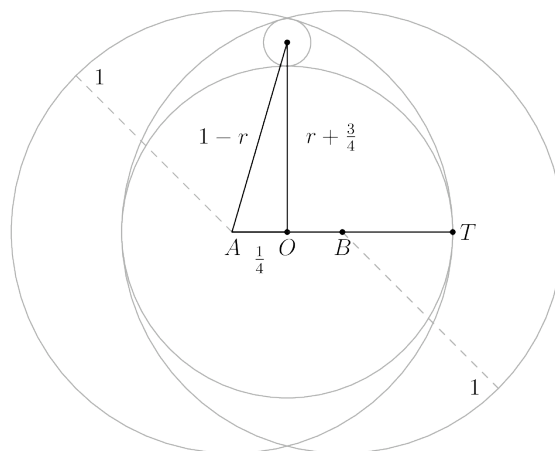
- v) (AMC 10 10A 2023) Circle C_1 and C_2 each have radius 1, and the distance between their centers is $\frac{1}{2}$. Circle C_3 is the largest circle internally tangent to both C_1 and C_2 . Circle C_4 is internally tangent to both C_1 and C_2 and externally tangent to C_3 . What is the radius of C_4 ?



- (A) $\frac{1}{14}$ (B) $\frac{1}{12}$ (C) $\frac{1}{10}$ (D) $\frac{3}{28}$ (E) $\frac{1}{9}$

Answer: D

Solution



Let O be the center of the midpoint of the line segment connecting both the centers, say A and B .

Let the point of tangency with the inscribed circle and the right larger circles be T .

Then $OT = BO + BT = BO + AT - \frac{1}{2} = \frac{1}{4} + 1 - \frac{1}{2} = \frac{3}{4}$.

Since C_4 is internally tangent to C_1 , center of C_4 , C_1 and their tangent point must be on the same line.

Now, if we connect centers of C_4 , C_3 and C_1/C_2 , we get a right angled triangle.

Let the radius of C_4 equal r . With the pythagorean theorem on our triangle, we have

$$\left(r + \frac{3}{4}\right)^2 + \left(\frac{1}{4}\right)^2 = (1 - r)^2$$

Solving this equation gives us

$$r = \boxed{\text{(D)} \frac{3}{28}}$$

Solution 2: Estimation

Draw 2 lines from the middle segment's endpoints to an intersection of C_1 and C_2 . Draw the perpendicular bisector of the middle segment until it reaches one of the intersections. We see this becomes a right triangle with a hypotenuse of 1 and one leg length of $\frac{1}{4}$.

Using the Pythagorean theorem, the other leg length is $\frac{\sqrt{15}}{4}$. Also drawing C_1 's radius along the segment of $\frac{1}{2}$ allows us to find the that radius of the middle circle is $\frac{3}{4}$.

Finally notice that $r_{C_3} + 2r_{C_4}$ is really close to $\frac{\sqrt{15}}{4}$. $2r_{C_4} = \frac{\sqrt{15} - 3}{4}$, so $r_{C_4} = \frac{\sqrt{15} - 3}{8}$.

This is really close to $\frac{1}{8}$ and the closest answer to that is $\boxed{\text{(D)} \frac{3}{28}}$

- w) (AMC 10 10A 2023) If the positive integer c has positive integer divisors a and b with $c = ab$, then a and b are said to be *complementary* divisors of c . Suppose that N is a positive integer that has one complementary pair of divisors that differ by 20 and another pair of complementary divisors that differ by 23. What is the sum of the digits of N ?

(A) 9 (B) 13 (C) 15 (D) 17 (E) 19

Answer: C

Solution 1

Consider positive integers a, b with a difference of 20. Suppose $b = a - 20$. Then, we have $(a)(a - 20) = n$. If there is another pair of two integers that multiply to n but have a difference of 23, one integer must be greater than a , and the other must be smaller than $a - 20$. We can create two cases and set both equal.

We have $(a)(a - 20) = (a + 1)(a - 22)$ or $(a + 2)(a - 21)$. Note that if we go further to $(a + 3)(a - 20)$ and beyond, that would violate the condition that one of the two integers must be smaller than $a - 20$.

Starting with the first case, we have $a^2 - 20a = a^2 - 21a - 22$, or $0 = -a - 22$, which gives $a = -22$, which is not possible. The other case is $a^2 - 20a = a^2 - 19a - 42$, so $a = 42$. Thus, our product is $(42)(22) = (44)(21)$, so $n = 924$. Adding the digits, we have $9 + 2 + 4 = \boxed{\text{(C)} 15}$.

-Sepehr2010, mathboy282, the_eaglercraft_grinder

Solution 2

We have 4 integers in our problem. Let's call the smallest of them a . $a(a + 23) =$ either $(a + 1)(a + 21)$ or $(a + 2)(a + 22)$. So, we have the following:

$$a^2 + 23a = a^2 + 22a + 21 \text{ or}$$

$$a^2 + 23a = a^2 + 24a + 44.$$

The second equation has negative solutions, so we discard it. The first equation has $a = 21$, and so $a + 23 = 44$. If we check $(a + 1)(a + 21)$ we get $22 \cdot 42 = 21 \cdot 44$. 44 is 2 times 22, and 42 is 2 times 21, so our solution checks out. Multiplying 21 by 44, we get $924 \implies 9 + 2 + 4 = \boxed{\text{(C)} 15}$.

Solution 3 (less words)

From the problems, it follows that

$$\begin{aligned}
 x(x+20) &= y(y+23) = N \\
 x^2 + 20x &= y^2 + 23y \\
 4x^2 + 4 \cdot 20x &= 4y^2 + 4 \cdot 23y \\
 4x^2 + 4 \cdot 20x + 20^2 - 20^2 &= 4y^2 + 4 \cdot 23y + 23^2 - 23^2 \\
 (2x+20)^2 - 20^2 &= (2y+23)^2 - 23^2 \\
 23^2 - 20^2 &= (2y+23)^2 - (2x+20)^2 \\
 (23+20)(23-20) &= (2y+23+2x+20)(2y+23-2x-20) \\
 43 \cdot 3 &= (2y+2x+43)(2y-2x+3) \\
 129 \cdot 1 &= (2y+2x+43)(2y-2x+3)
 \end{aligned}$$

Since both $(2y+2x+43)$ and $(2y-2x+3)$ must be integer, we get two equations.

$$129 \text{ or } 43 = (2y+2x+43) \quad (19)$$

$$1 \text{ or } 3 = 2y-2x+3 \quad (20)$$

$$(21)$$

43 3 yields (0,0) which is not what we want. 129 1 yields (22,21) which is more interesting.

Simplifying the equations, we get:

$$\begin{aligned}
 x+y &= 43 \\
 x-y &= 1 \\
 x=22, y=21 \\
 N &= (22)(22+20) = 924.
 \end{aligned}$$

So, the answer is (C) 15.

Solution 4

Say one factorization is $n(n+23)$. The two cases for the other factorization are $(n+1)(n+21)$ and $(n+2)(n+22)$. We know it must be the first because of AM-GM intuition: lesser factors of a number are closer together than larger factors of a number. (We can also try both and see which works.) Thus, $n(n+23) = (n+1)(n+21)$ and we find that $n=21, N=924$ meaning the answer is (C) 15.

Solution 5

Since we are given that some pairs of divisors differ by 20 and 23 and we can let the pair be $(x-10)$ and $(x+10)$ as well as $(y-\frac{23}{2})$ and $(y+\frac{23}{2})$. We also know the product of both the complementary divisors give the same number so $(x-10)(x+10) = (y-\frac{23}{2})(y+\frac{23}{2})$. Now we let $y = \frac{a}{2}$. Then we substitute and get $x^2 - 100 = \frac{(a^2-529)}{4}$. Finally we multiply by 4 and get $4x^2 - a^2 = -129, a^2 - 4x^2 = 129$. Then we use differences of squares and get $a+2x=129, a-2x=1$. We finish by getting $a=65$ and $x=32$. So $(42)(22) = 924$ Adding the digits, we have $9+2+4 =$ (C) 15.

Number sense note: To avoid tedious multiplication of 2-digit numbers, observe that $n = (42)(22) = (6)(7)(2)(11)$, and $(6)(7)(2) = 84$, and the sum of the digits of 11 is 2, so the sum of the digits of n is equivalent to $(8 + 4)(2) \equiv 24 \equiv 15 \pmod{9}$. The only equivalent answer choice is 15. oinava

Solution 6

N can be written $N = (a - 10)(a + 10)$ with a positive integer $a > 10$ and $N = \left(\frac{2b+1}{2} - \frac{23}{2}\right)\left(\frac{2b+1}{2} + \frac{23}{2}\right)$ with a positive integer $b > 11$.

The above equations can be reorganized as

$$(2b + 1 + 2a)(2b + 1 - 2a) = 43 \cdot 3.$$

The only solution is $2b + 1 + 2a = 129$ and $2b + 1 - 2a = 1$. Thus, $a = b = 32$. Therefore, $N = 924$. So the sum of the digits of N is $9 + 2 + 4 = \text{(C) 15}$.

Solution 7

We can write N as $a(a + 20)$ or $b(b + 23)$ where a and b are divisors of N . Since $a(a + 20) = b(b + 23)$, we know that $a^2 + 20a - b^2 - 23b = 0$, and we can view this as a quadratic in a .

Since the solution for a must be an integer, the discriminant for this quadratic must be a perfect square and therefore $20^2 - 4(-b^2 - 23b) = (2c)^2 = 400 + 4b^2 + 92b$ so $b^2 + 23b - c^2 + 100 = 0$.

Since the discriminant of this quadratic in b must also be a perfect square we know that $23^2 - 4(-c^2 + 100) = d^2$ which we can simplify as $d^2 - 4c^2 = (d - 2c)(d + 2c) = 129$. Since they are both positive integers $d - 2c$ and $d + 2c$ are factors of $129 = 3 \cdot 43$ so $d - 2c = 1$ and $d + 2c = 129$ or $d - 2c = 3$ and $d + 2c = 43$.

These systems of equations give us $(c, d) = (32, 65)$ and $(c, d) = (10, 23)$ respectively, if we plug our values for c into the equation for b we get $b^2 + 23b - 924 = 0$ and $b^2 + 23b = 0$ respectively. The first equation gives us $b = 21$ or $b = -44$ and the second gives us $b = 0$ or $b = -23$, since b is positive we know that $b = 21$ and $N = (21)(21 + 23) = 924$, therefore the sum of the digits of N is $9 + 2 + 4 = \text{(C) 15}$.

Solution 8 (Trial and Error)

Consider the numbers of the form $a(a + 20)$. Since $b(b + 23)$ is always even, a is even. Thus, for $a \geq 2$, we calculate $a(a + 20)$ for even values of a . Then, we check if it can also be represented as a product of numbers that differ by 23. Checking, we see that $22 \cdot 42 = 21 \cdot 44 = 924$ works. Thus, the answer is $9 + 2 + 4 = \text{(C) 15}$

Solution 9

$n(n + 20) = m(m + 23) \implies n^2 + 20n = m^2 + 23m \implies n^2 - m^2 + 20n - 23m = 3m$. Factoring, $(n + m)(n - m) + 20(n - m) = 3m \implies (n - m)(n + m + 20) = 3m$. Let $n - m = a > 0$ because clearly $n > m$. Then, $a(2m + 20 + a) = 3m$. Note that since $20 + a > 0$, if $a \geq 2$, then the equation is $4m + a(20 + a) > 3m$, so $a = 1$. Plugging this back, we get $2m + 21 = 3m \implies m = 21$ and $n = 22$. Now we find N as $22 \cdot 42 = 924$ so the answer is 15.

-Magnetoininja

Solution 10 (Elementary Algebra)

We are given $a(a + 20) = b(b + 23) = N$, and we are trying to find N . Simplifying the equation yields

$$a^2 + 20a = b^2 + 23b.$$

Completing the square for a , we get

$$(a + 10)^2 = b^2 + 23b + 100.$$

Now, we will use the squeezing method to find b . We can set a bound on $b^2 + 23b + 100$:

$$b^2 + 20b + 100 < b^2 + 23b + 100 < b^2 + 24b + 144.$$

because the expression is bigger than the lower bound by $3b$ and less than the upper bound by $b + 44$. Simplifying gets:

$$(b + 10)^2 < b^2 + 23b + 100 < (b + 12)^2.$$

Since we know that the expression is a perfect square, it must be equal to the only square in between $b + 10$ and $b + 12$, i.e. $b + 11$:

$$(b + 11)^2 = b^2 + 23b + 100.$$

$$b^2 + 22b + 121 = b^2 + 23b + 100$$

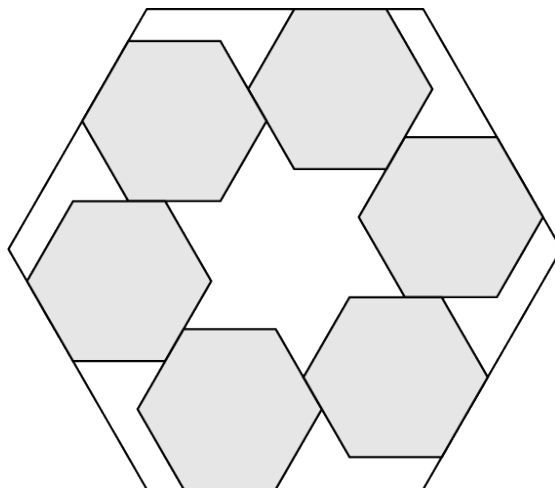
$$b = 21$$

Therefore $N = 21(21 + 23) = 21(44) = 924$. The sum of the digits is $9 + 2 + 4 = \boxed{\text{(C) 15}}$

Solution 11 (Algebra, and some hopeful guesswork)

Let $N = a(a - 20) = b(b - 23)$. We get $a^2 - 20a = b^2 - 23b$. Completing the square gives $(a - 10)^2 - 100 = (b - \frac{23}{2})^2 - \frac{529}{4}$. We can add $\frac{529}{4}$ and then multiply by 4 to get $4(a - 10)^2 + 129 = 4(b - \frac{23}{2})^2 \rightarrow (2a - 20)^2 + 129 = (2b - 23)^2$. We can rewrite 129 as $2(64) + 1$, which makes the LHS look similar to the form $x^2 + 2x + 1$, where $x = 64$. So we guess that $(2a - 20) = 64$, and $(2b - 23) = 64 + 1 = 65$. Solving these two equations gives $a = 42$ and $b = 44$. This means $N = 42(22) = (44)(21) = 924$, so the sum of the digits of N is $9 + 2 + 4 = \boxed{\text{(C) 15}}$, which is our final answer.

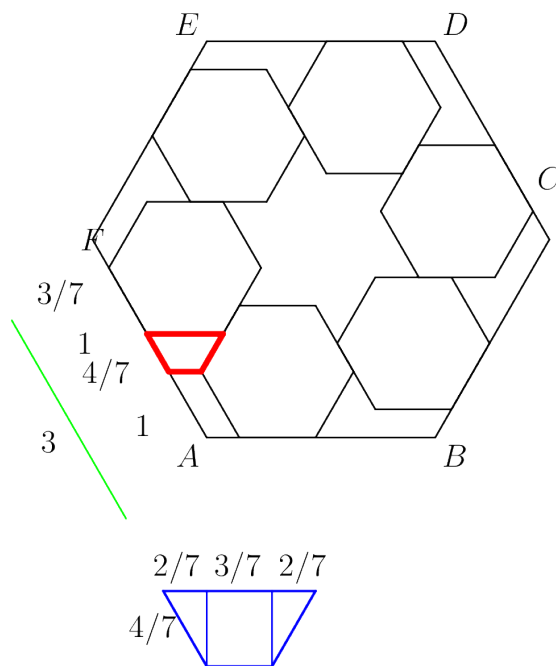
- x) (AMC 10 10A 2023) Six regular hexagonal blocks of side length 1 unit are arranged inside a regular hexagonal frame. Each block lies along an inside edge of the frame and is aligned with two other blocks, as shown in the figure below. The distance from any corner of the frame to the nearest vertex of a block is $\frac{3}{7}$ unit. What is the area of the region inside the frame not occupied by the blocks?



- (A) $\frac{13\sqrt{3}}{3}$ (B) $\frac{216\sqrt{3}}{49}$ (C) $\frac{9\sqrt{3}}{2}$ (D) $\frac{14\sqrt{3}}{3}$ (E) $\frac{243\sqrt{3}}{49}$

Answer: C

Solution 1



Examining the red isosceles trapezoid with 1 and $\frac{3}{7}$ as two bases, we know that the side lengths are $\frac{4}{7}$ from 30 – 60 – 90 triangle.

We can conclude that the big hexagon has side length 3.

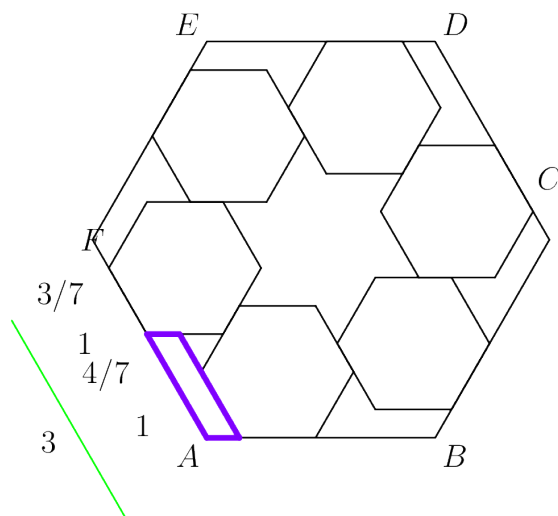
Thus the target area is: area of the big hexagon - 6 * area of the small hexagon. $\frac{3\sqrt{3}}{2}(3^2 -$

$$6 \cdot 1^2) = \frac{3\sqrt{3}}{2}(3) = \boxed{\text{(C)} \frac{9\sqrt{3}}{2}}$$

Solution 1.1

We can extend the line of the parallelogram to the end until it touches the next hexagon and it will make a small equilateral triangle and a longer parallelogram. We can prove

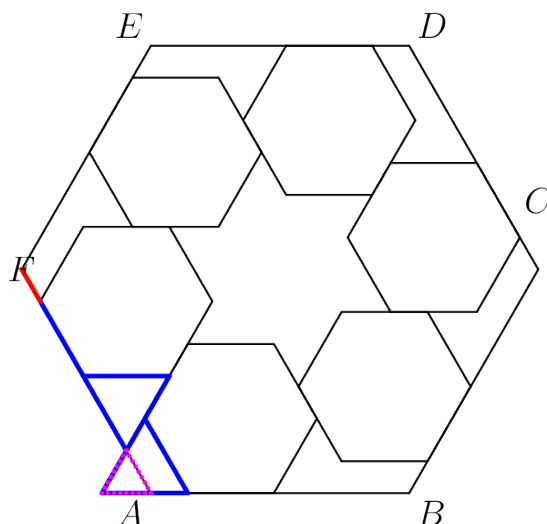
that one side of the tiny equilateral triangle is $4/7$ by playing around with angles and the parallelogram because it is parallel, we can then use the whole side of the hexagon which is one and subtract $3/7$ which is one side of the equilateral triangle which is $4/7$. That means the whole side of the big hexagon length is 3 and we can continue with solution 1.



-BrandonChen

Solution 1.3

An alternative diagram, perhaps slightly more intuitive, can be drawn by extending segments to form two equilateral triangles (shown in blue) with side length 1.



The red segment is given to have length $3/7$. Furthermore, we can find yet another equilateral triangle within the bottom blue triangle that has side length $1 - 3/7 = 4/7$, shown in dashed magenta. Adding all four segments (red, blue, blue, magenta), we find the side length of the larger hexagon to be $3/7 + 1 + 1 + 4/7 = 3$. Then we can continue with Solution 1.

(diagram borrowed mostly from Technodoggo's Solution 1)

Solution 2 (Not rigorous)

Note that one can "slide" the small hexagons along their respective edges, and either by sliding them to the center or to the corners, and thus getting that the side length of the larger hexagon is 3. The rest proceeds the same as solution 1.

Solution 2.1 (Clarification)

Notice that when sliding the smaller hexagon along the edge, we see that the contact edge with the smaller hexagon "in front" of it is 60° , thus meaning the hexagon "in front" is pushed at a speed $\sin 60^\circ$ times the actual speed of the hexagon. We can perform a similar analysis on the hexagon that is being pushed and get that the speed at which that hexagon is moving is $\frac{1}{\sin 60^\circ}$ times the speed it pushed by. As we can see, the two factors cancel out and by the same argument, every small hexagon can move at the same speed while maintaining an edge of contact with the two adjacent hexagons.

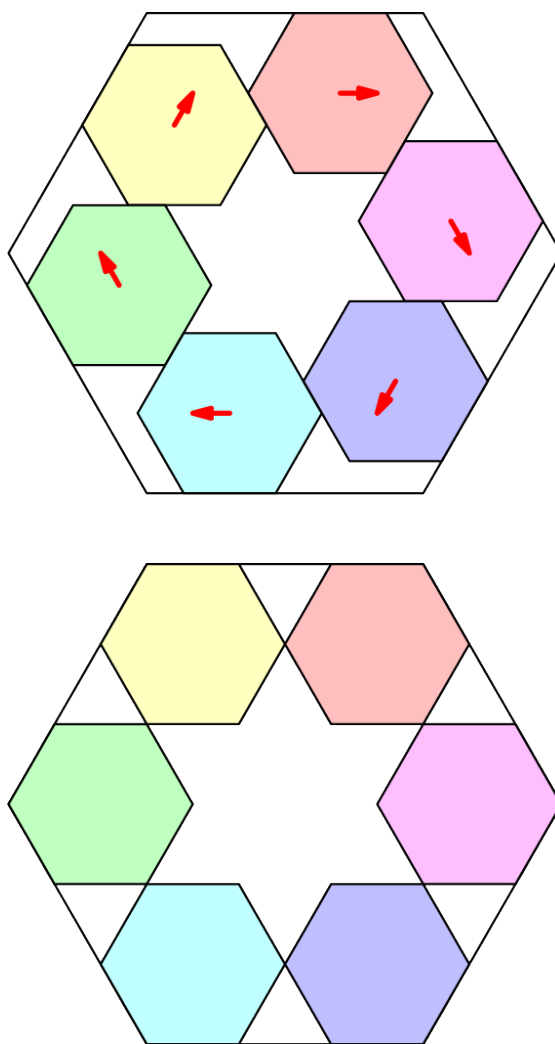


Diagram by [ihatemath123](#)

Note The number $\frac{3}{7}$ is irrelevant to solve the problem. In fact, if the smaller hexagons have side length x , the side length of the large hexagon will always be $3x$.

Solution 2

We put the diagram to a complex plane, with the center of the outside hexagon at the origin. We denote by s the length of each side of the outside hexagon.

The complex number of the upper left vertex of the upper right small hexagon is

$$se^{i30^\circ} + \frac{3}{7}e^{i(90^\circ+60^\circ)} - \sqrt{3}.$$

The complex number of the upper right vertex of the top small hexagon is

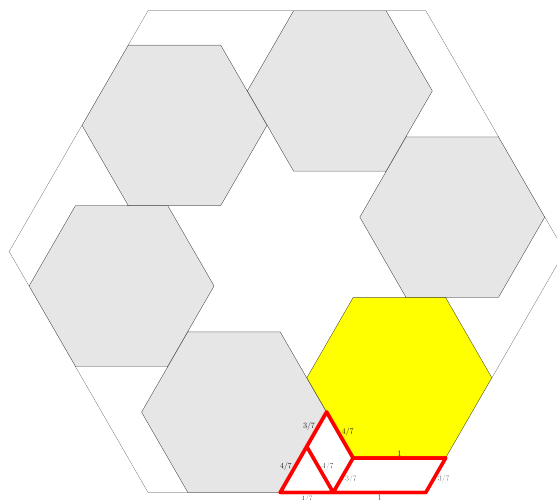
$$si + \frac{3}{7}e^{i(\pi+\frac{\pi}{6})} + e^{-i\frac{\pi}{6}}.$$

The above two vertices are on the same vertical line. So their real part values are the same. By solving this equation, we get $s = 3$.

Therefore, the area of the region not occupied by the blocks is

$$6 \cdot \frac{\sqrt{3}}{4}s^2 - 6 \cdot 6 \cdot \frac{\sqrt{3}}{4}1^2 = \boxed{\text{(C)} \frac{9\sqrt{3}}{2}}.$$

Solution 3 (no words)



$$(3^2 - 6 \times 1^2) \times \frac{3\sqrt{3}}{2} = (9 - 6 \times 1) \times \frac{3\sqrt{3}}{2} = (9 - 6) \times \frac{3\sqrt{3}}{2} = 3 \times \frac{3\sqrt{3}}{2} = \boxed{\text{(C)} \frac{9\sqrt{3}}{2}}$$

Solution 4 (educated guessing)

Since the area of each hexagon is $\frac{3\sqrt{3}}{2}$, the answer is just the area of the large hexagon minus $9\sqrt{3}$. The side length of the large hexagon should be an integer or fraction since we are given integers and fractions, and solving this problem shouldn't require any square roots. Therefore, the correct answer plus $9\sqrt{3}$ should be the product of $\frac{3\sqrt{3}}{2}$ and a square number. Only choice C satisfies this, since $\frac{9\sqrt{3}}{2} + 9\sqrt{3} = \frac{27\sqrt{3}}{2} = 3^2 \times \frac{3\sqrt{3}}{2}$. Alternatively, we can also deduce C is correct since it is the only fraction whose denominator is 2.

Therefore, the answer is (probably) $\boxed{\text{(C)} \frac{9\sqrt{3}}{2}}$

-faliure167

- y) (AMC 10 10A 2023) If A and B are vertices of a polyhedron, define the distance $d(A, B)$ to be the minimum number of edges of the polyhedron one must traverse in order to connect A and B . For example, $\overline{AB}$ is an edge of the polyhedron, then $d(A, B) = 1$, but

if $\overline{AC}$ and $\overline{CB}$ are edges and $\overline{AB}$ is not an edge, then $d(A, B) = 2$. Let Q , R , and S be randomly chosen distinct vertices of a regular icosahedron (regular polyhedron made up of 20 equilateral triangles). What is the probability that $d(Q, R) > d(R, S)$?

- (A) $\frac{7}{22}$ (B) $\frac{1}{3}$ (C) $\frac{3}{8}$ (D) $\frac{5}{12}$ (E) $\frac{1}{2}$

Answer: A

Solution 1

To find the total amount of vertices we first find the amount of edges, and that is $\frac{20 \times 3}{2}$. Next, to find the amount of vertices we can use Euler's characteristic, $V - E + F = 2$, and therefore the amount of vertices is 12

So there are $\binom{12}{3} = 220$ ways to choose 3 distinct points.

Now, the furthest distance we can get from one point to another point in an icosahedron is 3. Which gives us a range of $1 \leq d(Q, R), d(R, S) \leq 3$

With some case work, we get two cases:

Case 1: $d(Q, R) = 3; d(R, S) = 1, 2$

Since we have only one way to choose Q, that is, the opposite point from R, we have one option for Q and any of the other points could work for S.

Then, we get $12 \times 1 \times 10 = 120$ (ways to choose R $\times$ ways to choose Q $\times$ ways to choose S)

Case 2: $d(Q, R) = 2; d(R, S) = 1$

We can visualize the icosahedron as 4 rows, first row with 1 vertex, second row with 5 vertices, third row with 5 vertices and fourth row with 1 vertex. We set R as the one vertex on the first row, and we have 12 options for R. Then, Q can be any of the 5 points on the third row and finally S can be one of the 5 points on the second row.

Therefore, we have $12 \times 5 \times 5 = 300$ (ways to choose R $\times$ ways to choose Q $\times$ ways to choose S)

In total, we have 420 ways, but we must divide by 3! to account for permutations, giving us $70/220 = \boxed{\text{(A) } \frac{7}{22}}$.

Solution 2 (Cheese + Actual way)

In total, there are $\binom{12}{3} = 220$ ways to select the points. However, if we look at the denominators of B, C, D , they are 3, 8, 12 which are not divisors of 220. Also $\frac{1}{2}$ is

impossible as cases like $d(Q, R) = d(R, S)$ exist. The only answer choice left is $\boxed{\text{(A) } \frac{7}{22}}$

Note: this cheese is actually wrong because the total number of ways to select the points is actually $12 \times 11 \times 10 = 1320$ as order matters, so all denominators are possible. Rather, you can arrive at the same conclusion by fixing R WLOG, leading to $11 \times 10 = 110$ ways in total, which works for the original cheese.. awesomeguy856

(Actual way)

Fix an arbitrary point, to select the rest 2 points, there are $11 \cdot 10 = 110$ ways. To make $d(Q, R) = d(R, S), d = 1/2$. Which means there are in total $2 \cdot 5 \cdot 4 = 40$ ways to make

the distance the same. $\frac{1}{2}(1 - \frac{40}{110}) = \boxed{\text{(A) } \frac{7}{22}}$ bluesoul

Solution 3

We can imagine the icosahedron as having 4 layers. 1 vertex at the top, 5 vertices below connected to the top vertex, 5 vertices below that which are 2 edges away from the

top vertex, and one vertex at the bottom that is 3 edges away. WLOG because the icosahedron is symmetric around all vertices, we can say that R is the vertex at the top. So now, we just need to find the probability that S is on a layer closer to the top than Q. We can do casework on the layer S is on to get

$$\frac{5}{11} \cdot \frac{6}{10} + \frac{5}{11} \cdot \frac{1}{10} = \frac{35}{110} = \frac{7}{22}$$

So the answer is $\boxed{\text{(A)} \frac{7}{22}}$.

-awesomeparrot

Solution 4

We can actually see that the probability that $d(Q, R) > d(R, S)$ is the exact same as $d(Q, R) < d(R, S)$ because $d(Q, R)$ and $d(R, S)$ have no difference. (In other words, we can just swap Q and S, meaning that can be called the same probability-wise.) Therefore, we want to find the probability that $d(Q, R) = d(R, S)$.

WLOG, we can rotate the icosahedron so that R is the top of the icosahedron. Then we can divide this into 2 cases:

1. They are on the second layer

There are 5 ways to put one point, and 4 ways to put the other point such that $d(Q, R) = d(R, S) = 1$. So, there are $5 \cdot 4 = 20$ ways to put them on the second layer.

2. They are on the third layer

There are 5 ways to put one point, and 4 ways to put the other point such that $d(Q, R) = d(R, S) = 2$. So, there are $5 \cdot 4 = 20$ ways to put them on the third layer.

The total number of ways to choose P and S are $11 \cdot 10 = 110$ (because there are 12 vertices), so the probability that $d(Q, R) = d(R, S)$ is $\frac{20+20}{110} = \frac{4}{11}$.

Therefore, the probability that $d(Q, R) > d(R, S)$ is $\frac{1-\frac{4}{11}}{2} = \boxed{\text{(A)} \frac{7}{22}}$

Solution 5

We know that there are 20 faces. Each of those faces has 3 borders (since each is a triangle), and each edge is used as a border twice (for each face on either side). Thus, there are $\frac{20 \cdot 3}{2} = 30$ edges.

Euler's formula states that $V - E + F = 2$ for all convex polyhedra, so we know that $2 - F + E = 12$ vertices.

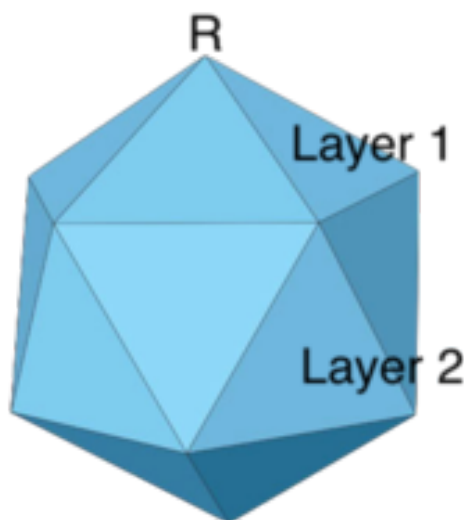
The answer can be counted by first counting the number of possible paths that will yield $d(Q, R) > d(R, S)$ and dividing it by $12 \cdot 11 \cdot 10$ (or $\binom{12}{3}$, depending on the approach). In either case, one will divide by 11 somewhere in the denominator. We can then hope that there will be no factor of 11 in the numerator (which would cancel the 11 in the denominator out), and answer the only option that has an 11 in the denominator:

$\boxed{\text{(A)} \frac{7}{22}}$.

Additional note by "Fruitz": One can eliminate 1/2 by symmetry if you swap the new sign.

Another note by "andliu766": A shorter way to find the number of vertices and edges is to use the fact that the MAA logo is an icosahedron. :)

Solution 6 (Case Work(similar to 3))



WLOG, let R be at the top-most vertex of the icosahedron. There are 2 cases where $d(Q, R) > d(R, S)$.

Case 1: Q is at the bottom-most vertex

If Q is at the bottom-most vertex, no matter where S is, $d(Q, R) > d(R, S)$. The probability that Q is at the bottom-most vertex is $\frac{1}{11}$

Case 2: Q is at the second layer

If Q is at the second layer, S must be at the first layer, for $d(Q, R) > d(R, S)$ to be true. The probability that Q is at the second layer, and S is at the first layer is $\frac{5}{11} \cdot \frac{5}{10} = \frac{5}{22}$

$$\frac{1}{11} + \frac{5}{22} = \boxed{\text{(A)} \frac{7}{22}}$$

Supplement First, we have 20 faces of equilateral triangles. But each edge is shared by two faces hence there are thus $\frac{20 \cdot 3}{2} = 30$ edges.

Now by Euler's formula $V - E + F = 2 \Rightarrow V - 30 + 20 = 2 \Rightarrow V = 12$.

Furthermore, the number of triangles (n) that meet at one vertex can be found as follows. Note that there are 12 total vertices. Without accounting for overcounting, total number of vertices = number of faces $\cdot$ vertices per face = $20 \cdot 3 = 60$. But since we now know that there are only 12 vertices and that each vertex is shared by n triangles, you find that there are 5 triangles that meet at one vertex.

Once we pick R as the topmost vertex, there are 11 possible vertices remaining for Q . Continue as follows in this solution.

Solution 7 (efficient)

Since the icosahedron is symmetric polyhedron, we can rotate it so that R is on the topmost vertex. Since Q and S basically the same, we can first count the probability that $d(Q, R) = d(R, S)$.

Case 1: $d(Q, R) = d(R, S) = 1$

There are 5 points P such that $d(Q, P) = 1$. There is $5 \times 4 = \boxed{20}$ ways to choose Q and S in this case.

Case 2 : $d(Q, R) = d(R, S) = 2$

There are 5 points P such that $d(Q, P) = 2$. There is $5 \times 4 = \boxed{20}$ ways to choose Q and S in this case.

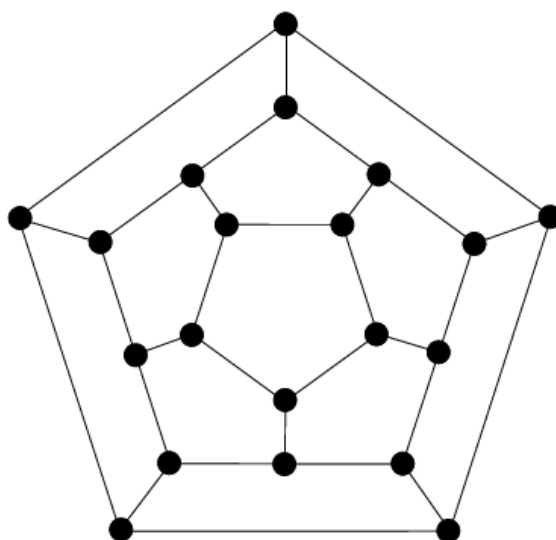
Case 3 : $d(Q, R) = d(R, S) = 3$

There is 1 point P such that $d(Q, P) = 3$. There is $1 \times 0 = \boxed{0}$ ways to choose Q and S in this case.

Final solution

There are 11 points P that are distinct from R . There is $11 \times 10 = \boxed{110}$ ways to choose Q and S . There is $20 + 20 + 0 = \boxed{40}$ ways to choose Q and S such that $d(Q, R) = d(R, S)$. There is $\frac{110-40}{2} = 35$ ways to choose Q and S such that $d(Q, R) > d(R, S)$. The probability that $d(Q, R) > d(R, S)$ is therefore $\frac{35}{110} = \frac{7}{22}$ which corresponds to answer choice $\boxed{A}$

Solution 8 (Dodecahedron Schlegel (Planar Graph) Diagram)



Instead of thinking this problem as vertices of an icosahedron, think of it as faces of a dodecahedron. Our middle face is R and we first choose Q and S to be the faces with distance 1 from the middle face. There are $\binom{5}{2}$ ways of doing that. Then we choose Q and S to be the faces with distance 2 from the middle face. There are $\binom{5}{2}$ ways of doing that. Our answer is therefore $\frac{1}{2} \left(1 - \frac{2\binom{5}{2}}{\binom{11}{2}} \right) = \boxed{(A) \frac{7}{22}}$

Note: You can do the same analysis on the vertices of the icosahedron using its diagram, which is less to draw because it has fewer (12) vertices and the same number (30) of edges. But this diagram is either less symmetrical or non-planar. (There's no good place to put the last vertex (diametrically opposite the first/central vertex), except in the same place as the first vertex.)

Solution 9 (Symmetry)

Note that the vertices of an icosahedron are in a $1 - 5 - 5 - 1$ configuration.

We will denote the probability of event A happening as $P(A)$

We should also note that $P(d(Q, R) > d(R, S)) + P(d(Q, R) < d(R, S)) + P(d(Q, R) = d(R, S)) = 1$

Also, $P(d(Q, R) > d(R, S)) = P(d(Q, R) < d(R, S))$ since Q and S are basically the same.

Therefore, what we are trying to find is just $\frac{1-P(d(Q,R)=d(R,S))}{2}$ which can easily be computed when we fix R at the top vertex in the $1-5-5-1$ structure.

Since all the points are distinct (stated in problem), the conditions are only satisfied where Q and S are at 5s of the $1-5-5-1$ structure.

$P(d(Q, R) = d(R, S)) = 2 \cdot \frac{5}{11} \cdot \frac{4}{10} = \frac{4}{11}$ since we have to choose one of the 5 out of 11 possible vertices and another one of 4 out of 10 two times.

And the answer is $\frac{1-\frac{4}{11}}{2} = \boxed{\text{(A)} \frac{7}{22}}$

§11 AMC 10 10B 2024

- a) (*AMC 10 10B 2024*) In a long line of people arranged left to right, the 1013th person from the left is also the 1010th person from the right. How many people are in the line?

(A) 2021 (B) 2022 (C) 2023 (D) 2024 (E) 2025

Answer: B

Solution for Certain China Testpapers

If the person is the 1015th from the left, that means there is 1014 people to their left. If the person is the 1010th from the right, that means there is 1009 people to their right. Therefore, there are $1014 + 1 + 1009 = \boxed{\text{(D)} 2024}$ people in line.

Solution 1

If the person is the 1013th from the left, that means there is 1012 people to their left. If the person is the 1010th from the right, that means there is 1009 people to their right. Therefore, there are $1012 + 1 + 1009 = \boxed{\text{(B)} 2022}$ people in line.

Solution 2

The person is 1013rd person from the left is also the 1010th person from the right, so the same person is counted twice.

Therefore, there are $1013 + 1010 - 1 = \boxed{\text{(B)} 2022}$ people in line.

$\boxed{\text{(B)} 2022}$

Solution 3

We can look at a smaller case where it is the 4th person from the left and the 2nd person from the right. Listing out the people as numbers gives us a list of 1, 2, 3, 4, 5. We see that the total number of people is the position from the left + the position from the right - 1, which in the case above is $4 + 2 - 1 = 5$. Plugging in 1013 and 1010 gives us $1013 + 1010 - 1 = \boxed{\text{(B)} 2022}$

Solution 4

Since the person is the 1013th person from the left and the 1010th person from the right, there are 1009 people after the 1013th person.

Therefore, there are $1013 + 1009 = \boxed{\text{(B)} 2022}$ people in line.

(Note: Similarly, you can perceive this information as 1012 people to the left of 1010 people and proceed.)

Solution 5 (Think of a Pattern)

If you are second going from the left and third going from the right, there are the third person out of four people p_1, p_2, p_3, p_4 .

We can simply add $1013 + 1010$ and subtract 1 since you are counting yourself twice, giving $1013 + 1009 = \boxed{\text{(B)} 2022}$

- b) (AMC 10 10B 2024) What is $10! - 7! \cdot 6!$? (A) -120 (B) 0 (C) 120 (D) 600 (E) 720

Answer: B

Solution 1

$$10! = 10 \cdot 9 \cdot 8 \cdot 7! = 720 \cdot 7! \quad 6! \cdot 7! = 720 \cdot 7!$$

Therefore, the equation is equal to $720 \cdot 7! - 720 \cdot 7! = \boxed{\text{(B)} 0}$

[ONLY FOR CERTAIN CHINESE TESTPAPERS]

$$0 - 5! = \boxed{\text{(A)} -120}$$

Solution 2

Factoring out $7!$ gives

$$7!(10 \cdot 9 \cdot 8 - 1 \cdot 6!).$$

Since $10 \cdot 9 \cdot 8 = 6! = 720$, the answer is $\boxed{\text{(B)} 0}$

Factoring $6!$ also works, it just makes the expression in the parenthesis a little harder to compute.

Solution 3

Note that $10! - 7! \cdot 6!$ must be divisible by 7, since

$$10! - 7! \cdot 6! = 7!(10 \cdot 9 \cdot 8 \cdot 6!) = 7 \cdot (6! \cdot 10 \cdot 9 \cdot 8 \cdot 6!)$$

and $\boxed{\text{(B)} 0}$ is the only option divisible by 7.

Solution 4

$10! - 7! \cdot 6!$ can be split into two parts, $10!$ and $7! \cdot 6!$. We can break the $6!$ into $(2 \cdot 4)(3 \cdot 5 \cdot 6)$. The $2 \cdot 4$ part makes 8, and the $3 \cdot 5 \cdot 6$ part makes 90, which is $9 \cdot 10$. We still have the $7!$, and we can multiply it by $8 \cdot 9 \cdot 10$. This is clearly equivalent to $10!$, so our solution is $10! - 10! = \boxed{\text{(B)} 0}$.

Solution 5

$10! = 3,628,800$, $7! = 5,040$, and $6! = 720$. $5,040 \cdot 720 = 3,628,800$. Therefore, $3,628,800 - 3,628,800 = \boxed{\text{(B)} 0}$.

-pepper2831

-minor edits by blakezhang

- c) (AMC 10 10B 2024) For how many integer values of x is $|2x| \leq 7\pi$?
(A) 16 (B) 17 (C) 19 (D) 20 (E) 21

Answer: E

Solution for Certain China Testpapers

Use the fact that $\pi \approx 3.14$, and thus you can get $6\pi \approx 18.84$. We could easily see that the answer is $\{-9, -8, \dots, 8, 9\} \implies \boxed{\text{(C) } 19}$

Solution 1

$\pi = 3.14159\dots$ is slightly less than $\frac{22}{7} = 3.\overline{142857}$. So $7\pi \approx 21.9$. The inequality expands to be $-21.9 \leq 2x \leq 21.9$. We find that x can take the integer values between -10 and 10 inclusive. There are $\boxed{\text{E. } 21}$ such values.

Note that if you did not know whether π was greater than or less than $\frac{22}{7}$, then you might perform casework. In the case that $\pi > \frac{22}{7}$, the valid solutions are between -11 and 11 inclusive: 23 solutions. Since, 23 is not an answer choice, we can be confident that $\pi < \frac{22}{7}$, and that $\boxed{\text{E. } 21}$ is the correct answer.

Test advice: If you are in the test and do not know if $\frac{22}{7}$ is bigger or smaller than π , you can use the extremely sophisticated method of just dividing $\frac{22}{7}$ via long division. Once you get to 3.142 you realise that you don't need to divide further since $\pi = 3.1416$ when rounded to 4 decimal places. Therefore, you do not include 11 and -11 and the answer is 21.

Solution 2

7π is incredibly close to 22, but doesn't reach it. This can be both computed by using $\pi \approx 3.142 \implies 7 \cdot 3.142 = 21.994 < 22$ or assumed. Therefore, including both positive and negative values, the answer is $\{-10, -9, \dots, 9, 10\} \implies \boxed{\text{(E) } 21}$. Tacos_are_yummy_1

- d) (AMC 10 10B 2024) Balls numbered $1, 2, 3, \dots$ are deposited in 5 bins, labeled A, B, C, D , and E using the following procedure. Ball 1 is deposited in bin A , and balls 2 and 3 are deposited in bin B . The next 3 balls are deposited in bin C , the next 4 in bin D , and so on, cycling back to bin A after balls are deposited in bin E . (For example, balls numbered 22, 23, $\dots$, 28 are deposited in bin B at step 7 of this process.) In which bin is ball 2024 deposited?

(A) A (B) B (C) C (D) D (E) E

Answer: D

Solution 1

Consider the triangular array of numbers:

$$\begin{array}{c} 1 \\ 2, 3 \\ 4, 5, 6 \\ 7, 8, 9, 10 \\ 11, 12, 13, 14, 15 \\ \vdots \end{array}$$

The numbers in a row congruent to 1 mod 5 will be in bucket A. Similarly, the numbers in a row congruent to 2, 3, 4, 0 mod 5 will be in buckets B, C, D,

and E respectively. Note that the n^{th} row ends with the n^{th} triangle number, $\frac{n(n+1)}{2}$.

We must find values of n that make $\frac{n(n+1)}{2}$ close to 2024.

$$\frac{n(n+1)}{2} \approx 2024$$

$$n(n+1) \approx 4048$$

$$n^2 \approx 4048$$

$$n \approx 63$$

Trying $n = 63$ we find that $\frac{n(n+1)}{2} = 2016$. Since 2016 will be the last ball in row 63, ball 2024 will be in row 64. Since $64 \equiv 4 \pmod{5}$, ball 2024 will be placed in bucket D. D.

Solution 2

Here balls are going in bins in arithmetic progression.

Sum of no. of balls before 2024th ball shall be less than 2024.

Therefore,

$$2024 > \frac{n}{2} \times [2 \times (1) + (n-1) \times 1]$$

$$4048 > n(n+1)$$

Here greatest integral value of n can be 63.

$$63 \equiv 3 \pmod{5}$$

On 63rd go 63 balls will be dropped in bin c, ball 2016 will be dropped in bin C.

So on 64th go 2024th ball will be dropped in bin D.

Hence, Option (D) D

- e) (*AMC 10 10B 2024*) In the following expression, Melanie changed some of the plus signs to minus signs:

$$1 + 3 + 5 + 7 + \cdots + 97 + 99$$

When the new expression was evaluated, it was negative. What is the least number of plus signs that Melanie could have changed to minus signs?

- (A) 14 (B) 15 (C) 16 (D) 17 (E) 18

Answer: B

Solution 1

Recall that the sum of the first n odd numbers is n^2 . Thus

$$1 + 3 + 5 + 7 + \cdots + 97 + 99 = 50^2 = 2500.$$

If we want to minimize the number of sign flips to make the number negative, we must flip the signs corresponding to the values with largest absolute value. This will result in the inequality

$$1 + 3 + 5 + \cdots + (2n - 3) + (2n - 1) - (2n + 1) - (2n + 3) - \cdots - 97 - 99 < 0.$$

The positive section of the sum will contribute n^2 , and the negative section will contribute $-(2500 - n^2) = (n^2 - 2500)$. The inequality simplifies to

$$n^2 + (n^2 - 2500) < 0$$

$$2n^2 < 2500$$

$$n^2 < 1250$$

The greatest positive value of n satisfying the inequality is $n = 35$, corresponding to 35 positive numbers, and B. 15 negatives.

ALEX

Solution 2

The formula for the sum of all odd positive integers from 1 to $2n - 1$ is

$$1 + 3 + 5 + \cdots + 2n - 3 + 2n - 1 = n^2.$$

Therefore, the given sum evaluates to

$$99 = 2(50) - 1 \implies 50^2 = 2500.$$

Since we're looking for the minimum possible sign changes, we focus on the largest numbers in the set to change to negative.

If we change the sign of a number n to negative, then the sum decreases by $2n$. Therefore, we're looking for a subset of numbers that add to greater than $2500/2 = 1250$.

Now we look at the answer choices.

(A) 14 means that we're changing the signs of the numbers $\{73, 75, \dots, 99\}$, and $73 + 75 + \cdots + 99 = \frac{(73 + 99) \cdot (\frac{99 - 73}{2} + 1)}{2} = 1204$.

Now this so slightly happens to be less than 1250, and (B) 15 means we're adding 71 to the set, too. Since $71 > 1250 - 1204 = 46$, then the answer is (B) 15 Tacos_are_not_yummy_1

Solution 3

Note that as mentioned above, $1 + 3 + 5 \cdots + 2n - 1 = n^2$. So, the sum of the first 50 odd numbers is equal to 2500, and we need to find the smallest integer such that $2k^2 < 50^2$. Taking the square root, $k = 25\sqrt{2}$, keeping in mind we need to take the floor $2k^2$ needs to be slightly less than 2500. Using an approximation of $\sqrt{2} = 1.4$, $25 * \frac{7}{5}$ is 35, so our answer is (B) 15 aleyang

Solution 4 (Simple Algebra)

As mentioned above, $1 + 3 + 5 \dots + 2n - 1 = n^2$. This will be especially important in this solution. The sum of the original expression is 2500, so the combined (positive) sum of the negatives must be greater than half of this, i.e. 1250. We are looking to minimize the number of negative numbers, so we only convert the last numbers to negatives. Let $99 = o_1 = 100 - 1$, $97 = o_2 = 100 - 3$, and for any n , $o_n = 100 - (2n - 1)$. Then the sum of all numbers up till o_n is $100n - n^2$. This sum needs to be greater than 1250, i.e.:

$$100n - n^2 > 1250$$

Testing out n from the solution set yields the minimum value of $n = 15$. The answer is therefore (B) 15.

- f) (AMC 10 10B 2024) A rectangle has integer side lengths and an area of 2024. What is the least possible perimeter of the rectangle?
- (A) 160 (B) 180 (C) 222 (D) 228 (E) 390

Answer: B

Solution 1 - Prime Factorization

We can start by assigning the values x and y for both sides. Here is the equation representing the area:

$$x \cdot y = 2024$$

Let's write out 2024 fully factorized.

$$2^3 \cdot 11 \cdot 23$$

Since we know that $x^2 > (x + 1)(x - 1)$, we want the two closest numbers possible. After some quick analysis, those two numbers are 44 and 46.

$$44 + 46 = 90$$

Now we multiply by 2 and get (B) 180.

Solution by [IshikaSaini](#).

Solution 2 - Squared Numbers Trick

We know that $x^2 = (x - 1)(x + 1) + 1$. Recall that $45^2 = 2025$.

If I want 1 less than 2025, which is 2024, I can take 1 number higher and 1 number lower from 45, which are 46 and 44. These are the 2 sides of the minimum perimeter because the 2 numbers are closest to each other, which is what we want to get the minimum.

Finding the perimeter with $2(46 + 44)$ we get (B) 180.

Solution by [Taha Jazaeri](#)

Note: The square of any positive integer with units digit 5, written in the format $10x + 5$ where x is a positive integer, is equal to $100(x)(x + 1) + 25$.

Solution 3 - AM-GM Inequality

Denote the numbers as $x, \frac{2024}{x}$. We know that per AM-GM, $x + \frac{2024}{x} \geq 2\sqrt{2024}$, but since $2\sqrt{2025} = 90$, $2\sqrt{2024}$ must be slightly less than 90, so $2x + 2\frac{2024}{x}$ must be slightly less than 180, eliminating A as a possible answer choice. Proceed with the following solutions above to get 44 and 46, which is

(B) 180.

-aleyang

Solution 4 - Difference of Squares

Note that the year 2025 is a perfect square. The beauty of this year shines down to 2024, which is 1 year lower. $45^2 - 1^2 = 2024 = (45 - 1)(45 + 1)$

Knowing this is the closest possible we can get to a square with only integer factors of 2024, these two are our happy numbers! Add them up to get:

Solution 5 - Get Lucky

Note: This is what I did.

$$\sqrt{2025} = 45$$

Assuming it's a square,

$$45 \cdot 4 = \boxed{180}$$

Solution 6 (Better Prime Factorization)

The smallest perimeter is achieved if the two numbers are as close as possible to one another. The prime factorization of 2024 is $2^3 \cdot 11 \cdot 23$. Notice that 11 is smaller than 23 and there are three 2's to be distributed between 11 and 23. We give 11 2^2 and 23 2^1 . This guarantee's the smallest perimeter. The side length of the rectangle is 44×46 . The perimeter of a rectangle is $2l + 2w$, where $l = 46$ and $w = 44$, so therefore the perimeter is $2(46) + 2(44) = 180$. Therefore the answer is $\boxed{(B)180}$.

- g) (AMC 10 10B 2024) What is the remainder when $7^{2024} + 7^{2025} + 7^{2026}$ is divided by 19?

(A) 0 (B) 1 (C) 7 (D) 11 (E) 18

Answer: A

Solution 1

We can factor the expression as

$$7^{2024}(1 + 7 + 7^2) = 7^{2024}(57).$$

Note that $57 = 19 \cdot 3$, this expression is actually divisible by 19. The answer is $\boxed{(A) 0}$.

Solution 2

If you failed to realize that the expression can be factored, you might also apply modular arithmetic to solve the problem.

Since $7^3 \equiv 1 \pmod{19}$, the powers of 7 repeat every three terms:

$$7^1 \equiv 7 \pmod{19}$$

$$7^2 \equiv 11 \pmod{19}$$

$$7^3 \equiv 1 \pmod{19}$$

The fact that $2024 \equiv 2 \pmod{3}$, $2025 \equiv 0 \pmod{3}$, and $2026 \equiv 1 \pmod{3}$ implies that $7^{2024} + 7^{2025} + 7^{2026} \equiv 11 + 1 + 7 \equiv 19 \equiv 0 \pmod{19}$.

Solution 3

We start the same as solution 2, and find that:

$$7^1 \equiv 7 \pmod{19}$$

$$7^2 \equiv 11 \pmod{19}$$

$$7^3 \equiv 1 \pmod{19}$$

We know that for 2024, 2025, and 2026, because there are three terms, we can just add them up. $1 + 7 + 11 = 19$, which is $0 \pmod{19}$.

Solution 4 (Given more advanced knowledge)

By Fermat's Little Theorem (FLT), we know that

$$7^{18} \equiv 1 \pmod{19}$$

Then its order must divide 18. Trying simple values we try and succeed:

$$7^3 \equiv 1 \pmod{19}$$

So the expression is equivalent to $1 + 7 + 49 \pmod{19}$, which gives (A) 0 when divided by 19.

Solution 5 (Basic Mod Almost same as sol one)

We have

$$\begin{aligned} 7^{2024} \cdot (1 + 7 + 49) &\equiv 0 \pmod{19} \\ 57 &\equiv 0 \pmod{19}, \end{aligned}$$

so the answer is (A) 0.

- h) (AMC 10 10B 2024) Let N be the product of all the positive integer divisors of 42. What is the units digit of N ?

(A) 0 (B) 2 (C) 4 (D) 6 (E) 8

Answer: D

Solution 1

The factors of 42 are 1, 2, 3, 6, 7, 14, 21, 42. Multiply the unit digits to get (D) 6

Solution 2

The product of the factors of a number n is $n^{\frac{\tau(n)}{2}}$, where $\tau(n)$ is the number of positive divisors of n . We see that $42 = 2^1 \cdot 3^1 \cdot 7^1$ has $(1+1)(1+1)(1+1) = 8$ factors, so the product of the divisors of 42 is

$$42^{\frac{8}{2}} \equiv 42^4 \equiv 2^4 \equiv 16 \equiv \text{span style="border: 1px solid black; padding: 2px;">(D) 6} \pmod{10}.$$

Solution 3

The factors are 1, 2, 3, 6, 7, 14, 21, 42. We can take out the 1 giving us 2, 3, 6, 7, 14, 21, 42. We can then take the unit digit of each number because the unit digit of the product is dependent on the product of the unit digits in the factors. So the new list is 2, 3, 6, 7, 4, 1, 2. Taking out the 1 again we get 2, 3, 6, 7, 4, 2. Multiplying the first 3 numbers gives us 36 which we take the unit digit off. Taking the final 3 digits we get 56 which we take the unit digit off again. This leaves us with 2 numbers which are 6 and 6, the product of them is 36 and the unit digit of this is (D) 6

- i) (AMC 10 10B 2024) Real numbers a, b , and c have arithmetic mean 0. The arithmetic mean of a^2, b^2 , and c^2 is 10. What is the arithmetic mean of ab, ac , and bc ?

(A) -5 (B) $-\frac{10}{3}$ (C) $-\frac{10}{9}$ (D) 0 (E) $\frac{10}{9}$

Answer: A

Solution 1

If $\frac{a+b+c}{3} = 0$, that means $a + b + c = 0$, and $(a + b + c)^2 = 0$. Expanding that gives

$$(a + b + c)^2 = a^2 + b^2 + c^2 + 2ab + 2ac + 2bc = 0$$

If $\frac{a^2+b^2+c^2}{3} = 10$, then $a^2 + b^2 + c^2 = 30$. Thus, we have

$$30 + 2ab + 2ac + 2bc = 0$$

Basic arithmetic will give you that $ab + bc + ac = -15$. To find the arithmetic mean, divide that by 3, so $\frac{ab+bc+ac}{3} = \boxed{\text{(A)} - 5}$

Solution 3

Assume that $a = 0$ and $b = -c$. Since we know that the arithmetic mean of the three numbers is 10, this means $a^2 + b^2 + c^2 = 30$. Using this equation, $b^2 + b^2 = 30$, so $b^2 = 15$. Observe that taking the positive or negative root won't matter as c will be the opposite. If we let $b = \sqrt{15}$ and $c = -\sqrt{15}$, $ab = 0 \times \sqrt{15} = 0$, $ac = 0 \times -\sqrt{15} = 0$, and $bc = \sqrt{15} \times -\sqrt{15} = -15$, so $ab + ac + bc = -15$. Doing $\frac{-15}{3}$ to get the arithmetic mean will give us

$\boxed{\text{(A)} - 5}$.

-aleyang

Solution 4 (interesting property)

From the definition of arithmetic means and some simple manipulation, we have $a + b + c = 0$ and $a^2 + b^2 + c^2 = 30$. Let $ab + ac + bc = k$, we have:

$$k = ab + ac + bc = (a + b)c + ab$$

Substituting $c = -(a + b)$ into the expression, we have

$$-c^2 + ab = k$$

Adding twice the equation above to $a^2 + b^2 + c^2 = 30$, we have:

$$a^2 + b^2 + c^2 - 2c^2 + 2ab = 30 + 2k$$

$$a^2 + b^2 + 2ab - c^2 = 30 + 2k$$

$$(a + b)^2 - c^2 = 30 + 2k$$

$$(a + b + c)(a + b - c) = 30 + 2k$$

But we know that $a + b + c = 0$. So $30 + 2k = 0$ and $k = -15$. Therefore the arithmetic mean of ab, ac, bc is $\frac{k}{3} = -5$. So the answer is $\boxed{\text{(A)} - 5}$

equal to each other, so we solve the equation. $0.5x + 0.5 = -x + 1$, so $x = \frac{1}{3}$. Therefore, plugging it into the equation, we get $y = \frac{2}{3}$. Using the shoelace theorem, we get the area of $CDEF$ to be $\frac{5}{12}$ and the area of CFB to be $\frac{1}{3}$, so our ratio is $\frac{\frac{5}{12}}{\frac{1}{3}} = \boxed{(A) 5 : 4}$ idk12345678

Solution 5 (wlog)

Let $ABCE$ be a square with side length 2. We see that $\triangle AFE \sim \triangle CFB$ by a Scale factor of 2. Let the altitude of $\triangle AFE$ and altitude of $\triangle CFB$ be h and $2h$, respectively. We know that $h + 2h$ is equal to 2, as the height of the square is 2. Solving this equation, we get that $h = \frac{2}{3}$. This means $[\triangle CFB] = \frac{4}{3}$, we can also calculate the area of $\triangle ABE$. Adding the area we of $\triangle CFB$ and $\triangle ABE$ we get $\frac{7}{3}$. We can then subtract this from the total area of the square: 4, this gives us $\frac{5}{3}$ for the area of quadrilateral $CFED$. Then we can compute the ratio which is equal to $\boxed{(A) 5 : 4}$.

(why does the $\text{\LaTeX}$ always look so bugged.)

Solution 6 (barycentrics)

$A = (1, 0, 0), B = (0, 1, 0), C = (0, 0, 1), D = (1, -1, 1)$. Since E is the midpoint of $\overline{AD}$, $E = (1, -0.5, 0.5)$. The equation of $\overline{EB}$ is:

$$0 = \begin{vmatrix} x & y & z \\ 1 & -0.5 & 0.5 \\ 0 & 1 & 0 \end{vmatrix}$$

The equation of $\overline{AC}$ is:

$$0 = \begin{vmatrix} x & y & z \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{vmatrix}$$

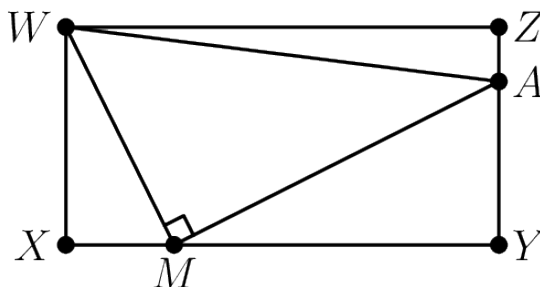
We also know that $x + y + z = 1$. To find the intersection, we can solve the system of equations. Solving, we get $x = 2/3, y = 0, z = 1/3$. Therefore, $F = (\frac{2}{3}, 0, \frac{1}{3})$. Using barycentric area formula,

$$\frac{[CFB]}{[ABC]} = \frac{\begin{vmatrix} 0 & 0 & 1 \\ 2/3 & 0 & 1/3 \\ 0 & 1 & 0 \end{vmatrix}}{\begin{vmatrix} 0 & 0 & 1 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{vmatrix}} = \frac{2}{3}$$

$$\frac{[CDEF]}{[ABC]} = \frac{\begin{vmatrix} 0 & 0 & 1 \\ 1 & -0.5 & 0.5 \\ 2/3 & 0 & 1/3 \end{vmatrix}}{\begin{vmatrix} 0 & 0 & 1 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{vmatrix}} + \frac{\begin{vmatrix} 0 & 0 & 1 \\ 1 & -1 & 1 \\ 1 & -0.5 & 0.5 \end{vmatrix}}{\begin{vmatrix} 0 & 0 & 1 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{vmatrix}} = \frac{5}{6}$$

$$\frac{[CDEF]}{[CFB]} = \frac{\frac{5}{6}}{\frac{2}{3}} = \boxed{(A) 5 : 4}$$

- k) (AMC 10 10B 2024) In the figure below $WXYZ$ is a rectangle with $WX = 4$ and $WZ = 8$. Point M lies on $\overline{XY}$, point A lies on $\overline{YZ}$, and $\angle WMA$ is a right angle. The areas of $\triangle WXM$ and $\triangle WAZ$ are equal. What is the area of $\triangle WMA$?



- (A) 13 (B) 14 (C) 15 (D) 16 (E) 17

Answer: C

Solution 1

We know that $WX = 4$, $WZ = 8$, so $YZ = 4$ and $YX = 8$. Since $\angle WMA = 90^\circ$, triangles WXM and MYA are similar. Therefore, $\frac{WX}{MY} = \frac{XM}{YA}$, which gives $\frac{4}{8-XM} = \frac{XM}{4-ZA}$. We also know that the areas of triangles WXM and WAZ are equal, so $WX \cdot XM = WZ \cdot ZA$, which implies $4 \cdot XM = 8 \cdot ZA$. Substituting this into the previous equation, we get $\frac{4}{8-2ZA} = \frac{2ZA}{4-ZA}$, yielding $ZA = 1$ and $XM = 2$. Thus,

$$\triangle WMA = 4 \cdot 8 - \frac{4 \cdot 2}{2} - \frac{8 \cdot 1}{2} - \frac{6 \cdot 3}{2} = \boxed{\text{(C)} 15}$$

Solution 2

Let $XM = b$, $ZA = a$, $4 \cdot b = 8 \cdot a$, $b = 2a$,

$$WM^2 + AM^2 = AW^2$$

$$(b^2 + 4^2) + (4 - a)^2 + (8 - b)^2 = (a^2 + 8^2)$$

$$a = 1, b = 2,$$

$$\triangle WMA = \text{area}(WXYZ) - \triangle WZA - \triangle WXM - \triangle MYA = 32 - 4 - 4 - 9 = \boxed{\text{(C)} 15}$$

Solution 3 (Pythagorean Theorem)

Assign ZA as x , then AY as $4 - x$. Assign XM as y and MY as $8 - y$. Since triangles WXM and WZA are equal in area, we can say $8x = 4y$, so $y = 2x$. Then, therefore, XM is $2x$ and MY has length $8 - 2x$. We can use the Pythagorean theorem to find WM , which is actually $\sqrt{(2x)^2 + 4^2} = \sqrt{4x^2 + 16}$. We don't factor it yet - we are going to find x again using the Pythagorean Theorem. Similarly, finding MA is just the square root of the squares of AY and MY individually, or $\sqrt{(8 - 2x)^2 + (4 - x)^2} = \sqrt{64 - 32x + 4x^2 + 16 - 8x + x^2} = \sqrt{5x^2 - 40x + 80}$. Then simply, WA is really $\sqrt{x^2 + 64}$.

Now we have the three sides of the right triangle: $\sqrt{4x^2 + 16}$, $\sqrt{5x^2 - 40x + 80}$, and $\sqrt{x^2 + 64}$. Per the Pythagorean theorem again, we can see $(4x^2 + 16) + (5x^2 - 40x + 80) = (x^2 + 64)$. Combining like terms gives us $8x^2 - 40x + 32 = 0$, then dividing by 8 gives $x^2 - 5x + 4 = 0$. As this elementary and well-known quadratic gives us the roots of 1 and 4, we can see it is a bit weird to have $x = 4$, as then point Z is point A . So we'll assume $x = 1$. We have two legs of the triangle by plugging in the sides with x in them, given that $x = 1$: $\sqrt{20}$

and $\sqrt{45}$. We should know that $20 \cdot 45 = 900$, and $\sqrt{900} = 30$. Dividing by 2 reveals us our answer: **(C) 15**

note: to further prove that 1 is the correct root, note that setting A at Z makes WZA have area 0. Therefore, WXM also has to have an area of 0. This makes it impossible for angle WMA to be a right angle, as WMA would simply become a right triangle with half the area of WXYZ, making MWA the right angle.

Solution 4 (Similar Triangles)

We are given $WX = 4$, $WZ = 8$. $\triangle WXM$ and $\triangle MYA$ have equal area, so let $XM = 2x$ and $ZA = x$. $MY = 8 - 2x$ and $AY = 4 - x$. From this, we can conclude that $\frac{MY}{AY} = \frac{8-2x}{4-x} = \frac{2}{1}$

Since WM intersects parallel lines WZ and XY , $\angle ZWM = \angle WMZ$. $\angle ZWM + \angle MWX = 90^\circ$, so $180^\circ - 90^\circ = \angle WMZ + \angle AMY$. Thus, $\angle MWX = \angle AMY$ and $\triangle WXM \sim \triangle MYA$ due to AA Similarity.

Corresponding sides of similar triangles are proportional, so $\frac{WX}{XM} = \frac{MY}{AY}$ or $\frac{4}{2x} = \frac{2}{1}$. It is clear that $2x = 2$, and $x = 1$. Now, all we have to do is subtract the area of the rectangle by each of the three triangles.

$$\Delta WMA = 8 \cdot 4 - \left(\frac{1}{2} \cdot 4 \cdot 2\right) - \left(\frac{1}{2} \cdot 8 \cdot 1\right) - \left(\frac{1}{2} \cdot 6 \cdot 3\right) \Delta WMA = 32 - 4 - 4 - 9$$

$$\Delta WMA = \textbf{(C) 15}$$

Solution 5 (Desperate)

Taking a look at the diagram, points M and A seem to divide $\overline{XY}$ and $\overline{YZ}$ into 1 : 3 ratios. With this assumption in mind, $XM = 2$, $MY = 6$, $YA = 3$, and $AZ = 1$. This means that the areas of $\triangle WXM$ and $\triangle WZA$ are $\frac{4(2)}{2} = 4$ and $\frac{8(1)}{2} = 4$ respectively, and the area of $\triangle MAY$ is $\frac{6(3)}{2} = 9$. The area of rectangle $WXYZ = 4(8) = 32$, so subtracting the areas of triangles WXM , WZA , and MAY , we get that the area of $\triangle WMA = 32 - 4 - 4 - 9 = \textbf{(C) 15}$

China Test Solution (Finding $\triangle MAY$) From solution 3, instead of finding the area of $\triangle WMA$, we instead find the area of $\triangle MAY$. Then $x = 1$ then we have $MA = 8 - 2x = 6$. Again, since $AY = 4 - x$, then $AY = 4 - 1 = 3$. The area of a triangle with legs 3 and 6 is $\frac{3 \cdot 6}{2} = \textbf{[9]}$.

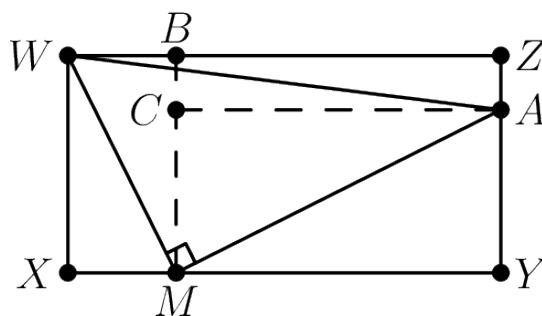
-minor edits by fireball9746

can someone pls explain why $\triangle MAY$ has the same area as $\triangle WMA$

^ To the person above: Under the problem, there's a note that says some tests in China asked for the area of $\triangle MAY$. - trevian1

Solution 6 (cheese)

Of course I don't endorse this method, but...



By looking at partitioned rectangles in the diagram, it's clear that the area of triangle WMA is very close to half the area of the overall rectangle $WXYZ$. So **(C) 15** is our best bet.

Obvious note:

The answer is less than half of the overall area of rectangle $WXYZ$, as triangle AWZ clearly sticks out past rectangle $ACBZ$ into rectangle $XMBW$ by the smallest margin, so therefore, the next smallest answer past (C) 16, is **(C) 15**.

- 1) (AMC 10 10B 2024) A group of 100 students from different countries meet at a mathematics competition. Each student speaks the same number of languages, and, for every pair of students A and B , student A speaks some language that student B does not speak, and student B speaks some language that student A does not speak. What is the least possible total number of languages spoken by all the students?

(A) 9 (B) 10 (C) 12 (D) 51 (E) 100

Answer: A

Solution 1 (Pure Logic)

We think of this problem like boxes. First start with 9. We see that we can arrange the groups of people into the 9 boxes. We take 9 people of different languages and arrange them in each of the boxes. This means we have $100 - 9 \times 9 = 19$ people remaining. We then take 18 people of different language (as they can be put in a pair and still qualify) and put them in the boxes. This gives us 1 person left over. Now, this is where MAA wants you to choose 10, but one can quickly see that the last person can be put into any group of three (except a group that includes their language) and still qualify whilst maintaining all the constraints. Therefore our answer is **(A) 9**.

This solution is a basic introduction/summary of the **Pigeonhole Principle**

Solution 2 (Clever intuition)

There are 100 students at the mathematics convention.

For any pair of students AB we see that $AB \in \{X, Y\}$ with WLOG, (X, Y) being a unique language individual to either A or B .

In other words, either student A or student B received language X or language Y , in which the order they receive their language is unimportant.

Given this the number of languages must divide the number of students evenly in order to achieve unique X, Y pairs that satisfy the conditions stated in the problem.

The question now is "What is the number of languages that divide the number of students?", which is equivalent to asking "How many numbers divide 100?", in which we use the number of factors theorem of the prime factorization of 100, being $2^2 \times 5^2$, and the number of factors being $(2+1)(2+1) = (3)(3) = 9$.

Therefore our answer is (A) 9.

Solution 3

Let's say we have some number of languages. Then each student will speak some amount of those languages, and no two people can have the same combination of languages or else the conditions will no longer be satisfied.

Notice that $\binom{9}{4} = 126 \geq 100$. So each of the 100 students can speak some 4 of the 9 languages. Thus, 9 is our answer.

Solution 4 (Rigorous)

Let n be the total number of languages spoken by all the students, and let k be the number of languages that each student speaks. Since all students speak the same number of languages, the condition given in the question can be modified as -

"If N is the n -element subset containing the union of the languages spoken by all students, each of the 100 students speaks a different (unique) k -element subset of N combination of languages."

In more mathematical terms, this means $\binom{n}{k} \geq 100$. (Using PHP)

Because we need the least value of n , k must be the closest integer to $\frac{n}{2} \Rightarrow \binom{n}{2} \geq 100$ for $n = 2m$, $\binom{n-1}{2} \geq 100$ for $n = 2m - 1$, $m \in \mathbb{N}$

In the context of this question, it will be fastest to just substitute n to each of the option values to find the least value possible as (A) 9

For a more rigorous proof, where options are not an option, one can manually count up from $n = 1$ as 100 is a small number.

- m) (AMC 10 10B 2024) Positive integers x and y satisfy the equation $\sqrt{x} + \sqrt{y} = \sqrt{1183}$. What is the minimum possible value of $x + y$?

(A) 585 (B) 595 (C) 623 (D) 700 (E) 791

Answer: B

Solution 1

Note that $\sqrt{1183} = 13\sqrt{7}$. Since x and y are positive integers, and $\sqrt{x} + \sqrt{y} = \sqrt{1183}$ we can represent each value of $\sqrt{x}$ and $\sqrt{y}$ as the product of a positive integer and $\sqrt{7}$. Let's say that $\sqrt{x} = m\sqrt{7}$ and $\sqrt{y} = n\sqrt{7}$, where m and n are positive integers. This implies that

$$x + y = (\sqrt{x})^2 + (\sqrt{y})^2 = 7m^2 + 7n^2 = 7(m^2 + n^2)$$

and that $m + n = 13$. WLOG, assume that $m \geq n$. It is not hard to see that $x + y$ reaches its minimum when $m^2 + n^2$ reaches its minimum. We now apply algebraic manipulation to get that

$$m^2 + n^2 = (m + n)^2 - 2mn$$

Since $m + n$ is determined, we now want mn to reach its maximum. Since m and n are positive integers, we can use the AM-GM inequality to get that: $\frac{m+n}{2} \geq \sqrt{mn}$. When mn reaches its maximum, $\frac{m+n}{2} = \sqrt{mn}$. This implies that $m = n = \frac{13}{2}$. However, this is not possible since m and n are integers. Under this constraint, we can see that mn reaches its maximum when $m = 7$ and $n = 6$. Therefore, the minimum possible value of $x + y$ is $7(m^2 + n^2) = 7(7^2 + 6^2) = \boxed{\text{(B)} 595}$.

A similar method is to take $y = 1183 - 26\sqrt{7x} - x^2$, then noting $x = 7a^2$ and bashing to find the value of a where x is closest to y .

Solution 2 (Guessing Answer Choices)

Set $x = y$, giving the minimum possible values. The given equation becomes

$$\sqrt{x} = \sqrt{y} = \frac{\sqrt{1183}}{2} \implies x = y = \frac{1183}{4}.$$

This means that

$$x + y = \frac{1183}{2} = 591.5.$$

Since this is closest to answer choice (B), the answer is (B) 595. Neoronean Tacos_are_yummy_1 (latex)

Note: If using a solution similar to this one it is recommended to still find valid x and y that add to 595.

Solution 3 (quick, no guessing but still answer choices)

Square both sides of the equation to get $2 \cdot \sqrt{xy} + (x + y) = 1183$. We can plug in possible values for $x + y$ based on the answer choices. If we do 585, we get $\sqrt{xy} = 299 \implies xy = 299^2$. So $y = \frac{299^2}{x}$. Then we have $x^2 + 299^2 = 585x$, which has no real solutions for x since the discriminant is negative. Following the same process for 595 we obtain the equation $x^2 + 294^2 = 595x$, which can be factored as $(x - 252)(x - 343) = 0$. So the answer must be (B) 595.

Solution 4 (Calculus)

Since $\sqrt{x} + \sqrt{y} = \sqrt{1183}$, we can express y in terms of x : $y = x + 1183 - 2\sqrt{1183x}$. $x + y = 2x + 1183 - 2\sqrt{1183x}$. Taking the derivative of this, we get $2 - 2 \cdot \frac{1}{2\sqrt{1183x}} \cdot 1183$. We find the critical points by setting this expression to equate 0. Simplifying, we get $2 = \frac{1183}{\sqrt{1183x}}$, and then we get $4 \cdot 1183x = 1183^2$. Dividing by 1183, we get $4x = 1183$, so a critical point is $x = \frac{1183}{4}$. Now, we take the second derivative, we get $\frac{13\sqrt{7x}}{2x^2}$. When $x = \frac{1183}{4}$, the second derivative is positive, meaning the minimum value is obtained at $x = \frac{1183}{4}$. This value is 295.75. However, the problem asks for positive integers x and y . We want x to be as close to 295.75 as possible. We also want y to be an integer, so $x + 1183 - 2\sqrt{1183x}$ should be an integer. Notice how since x is an integer, we just want $\sqrt{1183x}$ to be an integer. Prime factorizing 1183, we get $1183 = 7 \cdot 13^2$. Therefore, x must contain an odd amount of 7's and an even amount of any other prime factor. We quickly find that $2^2 \cdot 3^2 \cdot 7 = 252$ is the optimal solution for x , and plugging in, we get $y = 252 + 1183 - 2 \cdot 2 \cdot 3 \cdot 7 \cdot 13 = 343$. We find $x + y = 595$, so the answer is choice (B) 595.

Note: It might not be immediately clear why $x = 252$ is the optimal solution. The following explains why $x = 252$ is optimal.

x must have an odd amount of 7's and an even amount of any other prime factor. This means that x can be expressed as $7^n \cdot a$ where a is a perfect square. We can simplify this to $7 \cdot 7^{n-1} \cdot a = 7b$, where b is a perfect square, since $n - 1$ is even. This means that $7b$ is as close to 295.75 as possible. This means that $b \approx \frac{295.75}{7}$. $\frac{295.75}{7}$ is approximately 42, so $b \approx 42$. Since b is a perfect square, we have two candidates for b : 36 and 49. If $b = 36$, $x = 252$, and if $b = 49$, $x = 343$. To find the optimal value for x , pick the value closest to 295.75, and we find that $|295.75 - 252| < |295.75 - 343|$, so $x = 252$ is optimal. From now, proceed in the same way as the original solution.

Solution 5 (cheese)

$$\sqrt{1183} = \sqrt{169 \cdot 7} = 13\sqrt{7}.$$

Since $\sqrt{x} + \sqrt{y} = 13\sqrt{7}$, let $\sqrt{x} = 6\sqrt{7}$ and $\sqrt{y} = 7\sqrt{7}$.

Then $\sqrt{x} = \sqrt{252}$ because you take the outside 7 and put it in. Similarly, $\sqrt{y} = \sqrt{343}$.

Therefore, the insides sum to $x + y = 252 + 343 = 595$.

(B) 595.

- AskNamanYT

- n) (AMC 10 10B 2024) A dartboard is the region B in the coordinate plane consisting of points (x, y) such that $|x| + |y| \leq 8$. A target T is the region where $(x^2 + y^2 - 25)^2 \leq 49$. A dart is thrown and lands at a random point in B . The probability that the dart lands in T can be expressed as $\frac{m}{n} \cdot \pi$, where m and n are relatively prime positive integers. What is $m + n$?
(A) 39 **(B)** 71 **(C)** 73 **(D)** 75 **(E)** 135

Answer: B

Solution 1

Inequalities of the form $|x| + |y| \leq 8$ are well-known and correspond to a square in space with centre at origin and vertices at $(8, 0)$, $(-8, 0)$, $(0, 8)$, $(0, -8)$. The diagonal length of this square is clearly 16, so it has an area of

$$\frac{1}{2} \cdot 16 \cdot 16 = 128$$

Now,

$$(x^2 + y^2 - 25)^2 \leq 49$$

Converting to polar form,

$$r^2 - 25 \leq 7 \implies r \leq \sqrt{32},$$

and

$$r^2 - 25 \geq -7 \implies r \geq \sqrt{18}.$$

The intersection of these inequalities is the circular region T for which every circle in T has a radius between $\sqrt{18}$ and $\sqrt{32}$, inclusive. The area of such a region is thus $\pi(32 - 18) = 14\pi$. The requested probability is therefore $\frac{14\pi}{128} = \frac{7\pi}{64}$, yielding $(m, n) = (7, 64)$. We have $m + n = 7 + 64 = \boxed{\text{(B) } 71}$.

-AbhiSood1234, countmath1

Solution 2 (Calculus)

Expressing the Area of Region B

Region B consists of points where $|x| + |y| \leq 8$

In each quadrant, this can be expressed by the following functions:

First quadrant: $y = 8 - x$ Second quadrant: $y = 8 + x$ Third quadrant: $y = -8 - x$ Fourth quadrant: $y = -8 + x$

In the first quadrant, x ranges from 0 to 8, and y ranges from 0 to $8 - x$. Thus, the area in the first quadrant is:

$$\begin{aligned} \text{Area of first quadrant} &= \int_0^8 \int_0^{8-x} dy \, dx \\ &= \int_0^8 [y]_{y=0}^{y=8-x} dx = \int_0^8 (8 - x) dx \\ &= \left[8x - \frac{x^2}{2} \right]_0^8 = 64 - 32 = 32 \end{aligned}$$

The total area of region B is:

$$\text{Area of } B = 4 \times 32 = 128$$

Expressing the Area of Region T Region T is defined by the inequality $(x^2 + y^2 - 25)^2 \leq 49$, which can be rewritten as:

$$18 \leq x^2 + y^2 \leq 32$$

To find the area, we switch to polar coordinates with $x = r \cos \theta$ and $y = r \sin \theta$, where $x^2 + y^2 = r^2$. Here, r ranges from $\sqrt{18}$ to $\sqrt{32}$, and θ ranges from 0 to 2π .

The area of T can then be found by:

$$\begin{aligned} \text{Area of } T &= \int_0^{2\pi} \int_{\sqrt{18}}^{\sqrt{32}} r \, dr \, d\theta \\ &= \int_0^{2\pi} \left[\frac{r^2}{2} \right]_{r=\sqrt{18}}^{r=\sqrt{32}} d\theta = \int_0^{2\pi} \left(\frac{32}{2} - \frac{18}{2} \right) d\theta \\ &= \int_0^{2\pi} 7 \, d\theta = 14\pi \end{aligned}$$

The probability P that a dart lands in region T is the area of T divided by the area of B :

$$P = \frac{\text{Area of } T}{\text{Area of } B} = \frac{14\pi}{128} = \frac{7\pi}{64}$$

So the probability is of the form $\frac{m}{n}\pi$, where $m = 7$ and $n = 64$, so $m + n = 7 + 64 = 71$.

(B) 71

By the way never pull out calculus on the amc10. It is time consuming and easy to make mistakes on [minor edits by hao11]

Solution 3

Area under this graph

$$|x| + |y| \leq 8 \Rightarrow (8 \times \sqrt{2})^2 = 128$$

$$(x^2 + y^2 - 25)^2 - 7 \leq 0 \Rightarrow (x^2 + y^2 - 25 - 7) \times (x^2 + y^2 - 25 + 7) \leq 0$$

$$\therefore x^2 + y^2 \in [18, 32]$$

Area under such curves is;

$$\therefore A = \pi \times (32 - 18) = 14 \times \pi$$

$$P = \frac{14 \times \pi}{128} = \frac{7 \times \pi}{64} = \frac{m}{n}$$

$$m + n = 7 + 64 = 71$$

i.e., option (B)

- o) (AMC 10 10B 2024) A list of 9 real numbers consists of 1, 2.2, 3.2, 5.2, 6.2, and 7, as well as x , y , and z with $x \leq y \leq z$. The range of the list is 7, and the mean and the median are both positive integers. How many ordered triples (x, y, z) are possible?

(A) 1 (B) 2 (C) 3 (D) 4 (E) infinitely many

Answer: C

Solution 1 (Main)

First Case

We start off by knowing that there must exist an ordered pair $(0, y, z)$ and a big term 7 such that the range is satisfied and the median could also be satisfied. Because $x \leq y \leq z$, we say $x = 0$. Then, we get the sum 24.8. We need $\frac{24.8 + y + z}{9} \in \mathbb{Z}$. This means that we need the nearest multiple of 9, which is 36, and we get $y + z = 11.2$. We need one of these to be the median, WLOG say y . Then, $y \in \{4, 5\}$. So if $y = 4$, we get $z = 7.2$, and if $y = 5$, we get 6.2. We see that if $z = 6.2$, the range will still remain as 7, and therefore the one ordered triple $(0, 5, 6.2)$ satisfies this.

We know that the median can be y , so we have $y \in \{4, 5\}$. We do casework on 4 and 5.

Case 1

Say $y = 4$. Then, because we already know there exists an ordered triple where z is not the largest, there must exist at least one ordered triple where

z is, so we say $z - x = 7$. We also know that the mean must be divisible by 9. Quickly summing each number up we get 24.8. The next number divisible by 9 is 36. We add y to get 28.8. We then know $x + z = 36 - 28.8 = 7.2$. The smallest values give $x = 0.1$ and $z = 7.1$. This satisfies our constraints.

We don't want any errors, so we check the next sum, which is 45. We get then $7.2 = x + z$ and $z - x = 7$. This gives $x = 4.6$ and $z = 11.6$. This is a contradiction, as $x \leq y \leq z$, and the values incrementally become too large, and therefore we only have one ordered triple for this case, which is $(0.1, 4, 7.1)$

Case 2

We now check $y = 5$. This makes our sum 29.8. The nearest multiple of 9 is 36, again. This gives us $6.2 = x + z$ and $z - x = 7$. Solving gives us -0.4 and 6.6. This however doesn't work, as 6.6 is not the greatest value.

We decide to check the next multiple of 9, which is 45. This gives us $15.2 = x + z$ and $z - x = 7$. This gives us $x = 4.1$ and $z = 11.1$. This also doesn't work, as x becomes incrementally larger, and therefore there are no ordered triples that satisfy this.

Last Case

We have one more case, and that is x is not the first number, but z is the last, meaning $z - 1 = 7$, and $z = 8$. This means that $\frac{24.8+x+y+z}{9} \in \mathbb{Z}$. So, if $z = 8$, we get 32.8. We again see that we need the nearest multiple of 9 which is 36, and we get $3.2 = x + y$. We see that x and y still need to be a integer median and 3.2 is too small, so 36 cannot work. We try 45, and see $12.2 = x + y$. After some trial and error, we see that $x = 6$ and $y = 6.2$, giving us the ordered triple $(6, 6.2, 8)$. Continuing as before shows us that x again incrementally increases, and therefore we have only one ordered triple.

We have (C) 3 ordered triples. These are $(0.1, 4, 7.1)$, $(0, 5, 6.2)$, and $(6, 6.2, 8)$.

Solution 2 (Circular Arrangements)

This is a more advanced solution, but it gives the answer really fast. Plus, it is always better to expand your knowledge.

This solution is so overpowered that you don't even need to sum the decimals up. :-)

Consider the arrangements like a circle. There are 3 terms; x , y , and z , so we arrange them into a circle. We use the circular gap theorem to solve this, which is

$$C(nA) = \frac{n}{k} \binom{n-k-1}{k-1}$$

We see that there are 3 terms. We need either 1 adjacent from the other, two adjacent from one, or three adjacent.

Also keep in note that if the combinatorial part of $C(nA)$ is invalid, it is just 0.

We then have $k \in \{1, 2, 3\}$.

We then have 3 cases,

Case 1: $k \in \{1\}$

We have $\frac{3}{1} \cdot \binom{3-1-1}{1-1} = \frac{3}{1} \cdot 1 = 3$ ways to do this.

Case 2: $k \in \{2\}$

We have $\frac{3}{2} \cdot \binom{3-2-1}{2-1}$. The combinatorial part of this is $\binom{0}{1}$, which is invalid and therefore 0. Therefore there are $\frac{3}{2} \times 0 = 0$ cases.

Case 3: $k \in \{3\}$

We have $\frac{3}{3} \cdot \binom{3-3-1}{3-1}$. The combinatorial part of this is $\binom{-1}{2}$, and again, this is invalid, and therefore 0. There is then $1 \times 0 = 0$ cases.

Our answer is $3 + 0 + 0 = \boxed{(C) 3}$

Disclaimer Solution 2 This theorem requires practice to use and when to use it. For example, if they said that this list is non circular or is strictly linear, then we may not solve it like this. However, these specifications have not been said in this problem.

For disclaimer, don't use this formula everywhere. A clear example is that in a problem involving linear arrangements, this method will never work.

Below are some practice problems that I highly recommend you try so you can understand this theorem and its applications better.

Check them below.

2014 AMC 10B Q24

2017 AMC 10B Q18

1983 AIME Q7

Also, it is helpful to note $C_n(A) = \frac{n}{k} \binom{n-k-1}{k-1} = \binom{n}{2k}$.

Solution 3 (AMC 10)

The sum of the six existing numbers is 24.8. For the mean to be an integer, the sum of the remaining three numbers x, y, z and the remaining six must be divisible by 9; thus $x + y + z$ must equal either 2.2, 11.2, 20.2. However, one of x, y, z must be the median since the middle four numbers in the original set are all non-integral. As a result, we can eliminate 2.2 since the sum would be too small to allow for one of x, y, z to be the median (notice that negative numbers are not permitted since the range would be greater than 7). This leaves two cases.

For the $x + y + z = 20.2$ case, notice that all three must be less than or equal to 8 due to the range restriction. However, also notice that y and z must be on the higher end of the range, meaning that the new median would have to fall between the new middle numbers 5.2 and 6.2. Thus $x = 6$, and we can manipulate the numbers to make the range 7 by making $z = 8$ and thus $y = 6.2$, providing one case in $(6, 6.2, 8)$.

For the $x + y + z = 11.2$ case, we note that the average of the three would fall near the median, signaling that one of the numbers would be the median, one would fall lesser than the median, and one would reach greater than the median. Thus we let the median be y . Since adding a number to each side of the set would not change the median, we know that the median must fall between the two middle numbers 3.2, 5.2. Then we find two triples $(0.1, 4, 7.1)$ and $(0, 5, 6.2)$, resulting in $\boxed{(C) 3}$.

Solution 4 (AMC 12)

We can do casework on the range. We have four possible cases:

$$x < 1, z < 7 \quad x > 1, z < 7 \quad x < 1, z > 7 \quad x > 1, z > 7$$

These encapsulate all possible values of x and z we can choose, so we're not leaving anything out. As we'll see, the problem will become simpler this way. Since we want integer mean, first note the sum of the values given to be 24.8: when we add x, y, z , it must be a multiple of 9 to yield an integer mean. Also remember that the median of an increasing list of 9 numbers is the fifth number.

Case 1:

Since $x < 1, z < 7$, x must be the minimum of this list of numbers and 7 is the max: so the range is $7 - x$. This must be equal to 7, so $x = 0$. Now we try to make the median an integer. With $x = 0$, the fifth smallest number currently is 5.2, which is clearly not an integer. But if we stick y or z in between 3.2 and 5.2, then the median will be an integer. So let one of y, z be 4: then the sum of all the numbers we have so far is 28.8. To make this a multiple of 9, the last number has to be 7.2 to add up to 36. However, we said that $z < 7$, so this violates the bounds of this case. So then we set $y = 5$, so our sum is 29.9: then $z = 36 - 29.9 = 6.2$. Here everything checks out, so for this case there is one solution: $(0, 5, 6.2)$.

Case 2:

Since $x > 1$ and $z < 7$, the minimum is 1 and the maximum is 7. Thus the range is always 6, but we need the range to be 7, so this case has no solutions.

Case 3:

We have $x < 1, z > 7$. Our minimum is x and our maximum is z , so the range is $z - x = 7$. Since $x < 1$, the median still falls on 5.2, so y can be either 4 or 5 to make the median an integer. For now, suppose $y = 4$. Our sum is now $28.8 + x + z$, which we set to the nearest multiple of 9 : 36. Thus $x + z = 7.2$. Now we have a system of equations: we add them to get $2z = 14.2 \implies z = 7.1 \implies x = 0.1$. Since the values of z and x satisfy our case, this is a possible solution.

Now suppose $y = 5$. Our sum is $29.8 + x + z$, which must be equal to 36, so $x + z = 6.2$. Adding the equations, we have $2z = 13.2 \implies z = 6.6$. But this violates our assumption that $z > 7$, so this has no solutions.

Therefore this case has one solution: $(0.1, 4, 7.1)$.

Case 4:

Since $x > 1, z > 7$, the max is z and the min is 1. Thus $z - 1 = 7 \implies z = 8$. Now the median is between 5.2 and 6.2, so x or y must take the value 6. Our sum is now $24.8 + 6 + 8 = 38.8$, and adding the last number has to equal 45. Thus the last number is 6.2, and this yields one solution: $(6, 6.2, 8)$.

Therefore we've found three solutions. These cases encompass all possible real values of x and z , so we know we've left nothing out. The answer is (C) 3

Solution 5

We can start doing casework on the numbers. Notice that the median can either be four, five or six because there has to be one or two numbers on one side. We can start bashing everything out, starting from letting

$x = a, y = 4, z = 7 + a$. Now plugging into the sequence, we get that $a, 1, 2.2, 3.2, 4, 5.2, 6.2, 7, 7 + a$. Notice that the sum of the numbers not including x, y, z is 24.8. To make this mean an integer, $x + y + z$ must have a decimal 0.2 when added up. Now, plugging in our values into the mean gets us $35.8 + 2a = 9k$ for some integer k . Notice that $0 \leq a \leq 1$, so the only value for a is 0.1, so we found our first ordered triplet, $(0.1, 4, 7.1)$. Let's make the median 5 now. Then we have to put x and one side and z on the other to "balance" the numbers, or make it so that there are 4 numbers on each side of 5. Now, we can see that x is 0, and solving for z , we get that $z = 6.2$. We get our second triple is $(0, 5, 6.2)$. Now coming to our third case scenario, we use 6 as the median. Clearly it means that x has to be the median, so that y and z can be on the same side as x .

$$1, 2.2, 3.2, 5.2, 6, 6.2, y, 7, z$$

We will use this diagram. Notice that z has to be 8, and solving for y yields $y = 6.2$, so we found our third and last triplet, $(6, 6.2, 8)$, meaning that **(C)** is the right answer.

Solution 6 (Cheese)

Answer choice E is too big and answer choices A,B are both too small, so get rid of A,B, and E. Then, after checking the answer choices respectively, you get 3 as the final answer. Therefore, the answer is answer choice **(C) 3**.

Note: The answer to AMC problems is NEVER, EVER infinitely many. Also, the top part wasn't mine, it was somebody else.

Solution 7 (A (still interesting but slightly better) cheese)

Ignore the fact that $x \leq y \leq z$. Then we know that there should be some ordered pair $(0, y, z)$, $(x, 0, z)$, and $(x, y, 0)$, where we can permute x, y , or z such that they do satisfy $x \leq y \leq z$. Our answer is then **(C) 3**.

- p) (AMC 10 10B 2024) Jerry likes to play with numbers. One day, he wrote all the integers from 1 to 2024 on the whiteboard. Then he repeatedly chose four numbers on the whiteboard, erased them, and replaced them by either their sum or their product. (For example, Jerry's first step might have been to erase 1, 2, 3, and 5, and then write either 11, their sum, or 30, their product, on the whiteboard.) After repeatedly performing this operation, Jerry noticed that all the remaining numbers on the whiteboard were odd. What is the maximum possible number of integers on the whiteboard at that time?

(A) 1010 (B) 1011 (C) 1012 (D) 1013 (E) 1014

Answer: A

Solution 1

Consider the numbers as 1, 0, 1, 0, ..., 1, 0. Note that the number of odd integers is monotonously decreasing.

We need to get rid of 1012 0's, so we either add or multiply four 0s together to get $1012 \rightarrow 253 \rightarrow 63 + 1 = 64 \rightarrow 16 \rightarrow 4 \rightarrow 1$.

To get rid of the final 0, we need to consume three other 1's to result in one 1. Thus the answer is $1012 - 2 =$ **(A) 1010**,

Solution 2

(I will denote even numbers as e and odd numbers as o for the purpose of this question, and x as the answer we need)

First, we observe that each time Jerry replaces four number with one (let's call this an "*operation*"), we lose exactly three numbers regardless of the type of *operation* (*Additive* or *Multiplicative*).

$$\Rightarrow x \equiv 2024 \equiv 2 \pmod{3}$$

Already, we see that the only possible options are **(A)** 1010 or **(D)** 1013

We need there to be no e 's on the board at the end, so we attempt to convert them into o while retaining as many existing o 's as possible.

Notice that converting 4 e 's into an o is not possible. We can apply an *Additive operation* on 3 e 's and 1 o to get back 1 o ($e + e + e + o = o$) By doing so, we retain all the o 's we use in an *operation* (and don't generate any new ones). In the first 2024 numbers, there are 1012 o 's. This means that $x \leq 1012$, and the only option left is **(A)** 1010

Note: For the exact steps to arrive at 1010 odd numbers, perform

$$e + e + e + o = o \text{ (337 times)}$$

$$e + o + o + o = o \text{ (1 time)}$$

Solution 3 (guessing strategy)

Notice that with every step, Jerry removes 4 numbers and adds 1 number to the board. Thus, 3 numbers are removed from the board in every step. Since $2024 \equiv 2 \pmod{3}$, the answer choice must also be congruent to 2 mod 3. Only 1010 and 1013 among the answer choices satisfy this condition. This alone eliminates three choices, and one can guess from here for a 3 point expected value.

However, we start out with 1012 even numbers and 1012 odd numbers, and it is not possible to get more odds than there were initially (every time we add or multiply four numbers, we can only produce one odd number, which cannot occur if only even numbers are used; therefore, there cannot be more odd numbers after an operation than before it). Thus, we choose **(A)** 1010.

Solution 4 (OEEE)

There are 1012 odd numbers and 1012 even numbers. In order to maximize the remaining numbers, then we must minimize the number of times the operation is performed.

Since we want to be left with all odd numbers, then we must get rid of the even numbers as quick as possible, which implies *EEEE*. This doesn't give an odd product or sum, so our next best option is *OEEE*, which gives an odd sum.

The maximum number of times this operation can be performed with *OEEE* is $\left\lfloor \frac{1012}{3} \right\rfloor = 337$. This will eliminate $3 \cdot 337 = 1011$ even numbers and no odd numbers (because each time the operation is performed, one odd number is eliminated but another one is created).

Therefore after the operation is performed 337 times with *OEEE*, there will be 1 even number left and 1012 odd numbers left. Performing this operation

one more time with $EOOO$ will give 0 even numbers and $\boxed{(A) 1010}$ odd numbers. Tacos_are_yummy_1

Solution 5 (EEEE)

There are 1012 odd numbers and 1012 even numbers. Our goal is to get rid of all the even numbers. By performing the operation with $EEEE$, we will remove 4 even numbers and create one simultaneously.

First, we can perform $EEEE$ a maximum of $\frac{1012}{4} = 253$ times, to end up with $1012 - 4 \cdot 253 + 253 = 253$ even numbers. Performing $EEEE$ again $\left\lfloor \frac{253}{4} \right\rfloor = 63$ times will result in $253 - 4 \cdot 63 + 63 = 64$ even numbers left. Repeating this process, eventually there will be 1 even number left and still 1012 odd numbers. Performing $OOOE$ gives the answer of $\boxed{(A) 1010}$. Tacos_are_yummy_1 AVTV MINOR EDITS

Solution 6 (Monovariants)

Note that the number of odd integers is monotonically decreasing, and further observe that the number of odd integers is preserved after an operation where Jerry adds three evens and one odd together, producing an odd number. To maximise the number of odd integers, we will use this operation whenever possible.

There are 1012 even integers; to perform the specific procedure outlined above, we remove 3 even integers each time. Therefore, we will arrive in a situation where there is 1 even integers and 1012 odd integers. To ensure that after the final operation, there will only be odd integers, we add the even integer and three odd integers, yielding one odd integer. There are now 1010 integers, all of them odd, so we select $\boxed{(A) 1010}$.

Solution 7

There are 1012 odd integers, so the choices D and E are not possible. If the numbers 1 and 3 were erased, they would not be able to be recreated, the answer is $1012 - 2 = \boxed{(A) 1010}$.

Solution 8

Like Solution 4, we'll use $OEEO$, and they already explained why it maximizes the number of odd integers. Consider the set of integers between 1 and n . Optimally, $n \equiv 0 \pmod{6}$ because $OEEO$ uses 3 even integers (the number of even integers is $\lfloor \frac{n}{2} \rfloor$ so n is a multiple of 2). In that case, the max number of odd integers remaining is $\frac{n}{2}$ where no odd integer is removed without being replaced.

The closest multiple of 6 to the number 2024 is 2022. A maximum of 1011 odd integers can be left if the process is done for the integers 1 through 2022. Finally, there's 1 even number left, so the operation $OOOE$ is performed, thus the answer is $1011 + 1 - 2 = \boxed{(A) 1010}$ since 2023 is another odd integer not between 1 and 2022, but the operation $OOOE$ removes a net 2 odd integers.

Solution 9

We can remove 3 evens and 1 odd every turn and do this 337 times until we have 1 even number remaining and 1012 odd numbers because in every group

of 4 we lose 3 evens, and one odd, but gain the odd back as the sum of the group will always be odd. With 1013 integers left we remove a group of 4 containing the one even and add back their odd some so $1013 - 4 + 1$ equals A, 1010.

- q) (AMC 10 10B 2024) In a race among 5 snails, there is at most one tie, but that tie can involve any number of snails. For example, the result of the race might be that Dazzler is first; Abby, Cyrus, and Elroy are tied for second; and Bruna is fifth. How many different results of the race are possible?

(A) 180 (B) 361 (C) 420 (D) 431 (E) 720

Answer: D

Solution 1

Intuitive solution: We perform casework based on how many snails tie. Let's say we're dealing with the following snails: A, B, C, D, E .

5 snails tied: All 5 snails tied for 1st place, so only 1 way.

4 snails tied: A, B, C, D all tied, and E either got 1st or last. $\binom{5}{1}$ ways to choose who isn't involved in the tie and 2 ways to choose if that snail gets first or last, so 10 ways.

3 snails tied: We have ABC, D, E . There are $3! = 6$ ways to determine the ranking of the 3 groups. There are $\binom{5}{2}$ ways to determine the two snails not involved in the tie. So $6 \cdot 10 = 60$ ways.

2 snails tied: We have AB, C, D, E . There are $4! = 24$ ways to determine the ranking of the 4 groups. There are $\binom{5}{3}$ ways to determine the three snails not involved in the tie. So $24 \cdot 10 = 240$ ways.

1 snail tied: This is basically just every snail for a place, so $5! = 120$ ways.

The answer is $1 + 10 + 60 + 240 + 120 = \boxed{\text{(D) } 431}$.

Solution 2

Split the problem into cases. A tie of n snails has $\binom{5}{n}$ ways to choose the snails that are tied, $6 - n$ ways to choose which place they tie for, and $(5 - n)!$ to place the remaining snails.

1. No tie $\implies 5! = 120$

2. Tie of 2 snails $\implies \binom{5}{2} \cdot 4 \cdot 3! = 240$

3. Tie of 3 snails $\implies \binom{5}{3} \cdot 3 \cdot 2! = 60$

4. Tie of 4 snails $\implies \binom{5}{4} \cdot 2 = 10$

5. Tie of all 5 snails $\implies 1$

The answer is $120 + 240 + 60 + 10 + 1 = \boxed{\text{(D) } 431}$ Tacos_are_yummy_1

Solution 3 (Answer Choice Reasoning)

For the case of a 5-way tie, we have $\binom{5}{5} = 1$ cases. We can assume this leads to an answer that ends in 1, leaving only B and D . By just

accounting for the cases for a 1-way tie ($5! = 120$) and 2-way tie ($\binom{5}{2} \cdot 4 \cdot 3! = 240$), it immediately shows that B does not work. Therefore, the answer is (D) 431 shreyan.chethan, cleaned up by cw eu001, vintage_god

Notice that with 2 remaining choices, through guessing, the expected value of your points is $\frac{6}{2} = 3$, and not answering gives 1.5 points. Therefore, you gain an expected value of 1.5 points by answering. It would be best to always guess whenever you can get rid of 2 or more answers. If you can only eliminate 1 answer, not guessing will give you 1.5 points, but guessing also gives you an expected value of 1.5 points. I usually leave it blank if I can only get rid of 1 answer choice. shreyan.chethan, edited by BenjaminDong01, added by PerseverePlayer , and added by Aarav22

Sidenote : Usually, you should aim to look at the closer option choices than the extreme ones.

Solution 4 (Rigorous Solution 1; same logic)

Call the snails A, B, C, D, E in said order apart of the set S , and define S_T to be the set of ties regarding T snails. Define a **maxima tie** to be a set $\mathcal{M}$ that retains a tie between k members for $2 \leq k \leq 5$. Consider casework on all **maxima ties**.

If no **maxima ties** exist, then the problem degenerates into how many word rearrangements exist for $ABCDE$, which is $5! = 120$.

If $k = 2$ **maxima tie** exist, then note that if the tie is held between A, B , it is the same as a tie holding for B, A , meaning we have $\frac{5 \cdot 4}{2} = 10$ ways to choose our **maxima tie**. Then, the word rearrangement for $MCDE$ is $4! = 24$, yielding 240 cases.

If $k = 3$ **maxima tie** exist, then note that if a tie is held between A, B, C , it is the same for the other 5 iterations. Out of one iteration, 5 other copies exist that yield the same result, meaning our cases for the **maxima tie** are $\frac{5 \cdot 4 \cdot 3}{6} = 10$. Then, the word rearrangement for MDE is $3! = 6$, yielding 60 cases.

If $k = 4$ **maxima tie** exist, then every partition of A, B, C, D yields 23 other copies, meaning there are 5 acknowledgeable cases. Then the word rearrangement for ME is $2! = 2$, yielding 10 cases.

If $k = 5$ **maxima tie** exist, then every partition of A, B, C, D, E yields 119 other copies, meaning there is only one acknowledgeable case. Then, the word rearrangement for $\mathcal{M}$ is $1! = 1$ yielding 1 case.

The desired answer is then

$$120 + \sum_{T=k \neq 1 \in S_T} \text{maxima tie plausible cases} \\ 120 + 240 + 60 + 10 + 1 = \boxed{431}$$

- r) (AMC 10 10B 2024) How many different remainders can result when the 100th power of an integer is divided by 125?

(A) 1 (B) 2 (C) 5 (D) 25 (E) 125

Answer: B

Solution 1

First note that the **Euler's totient function** of 125 is 100. We can set up two cases, which depend on whether a number is relatively prime to 125.

If n is relatively prime to 125, then $n^{100} \equiv 1 \pmod{125}$ because of **Euler's Totient Theorem**.

If n is not relatively prime to 125, it must have a factor of 5. Express n as $5m$, where m is some integer. Then $n^{100} \equiv (5m)^{100} \equiv 5^{100} \cdot m^{100} \equiv 125 \cdot 5^{97} \cdot m^{100} \equiv 0 \pmod{125}$.

Therefore, n^{100} can only be congruent to 0 or 1 $\pmod{125}$. Our answer is 2.

Sidenote: Proof of Euler's Theorem Euler's Theorem states that $a^{\phi(n)} \equiv 1 \pmod{n}$, where $\phi(n)$ is Euler's totient function, which counts the number of positive integers x such that $1 \leq x \leq n$ and $\gcd(x, n) = 1$. Note that this only works for positive integer a such that $\gcd(a, n) = 1$.

Consider the set of distinct positive integers $S = \{x_1, x_2, \dots, x_{\phi(n)}\}$ such that $1 \leq x_i \leq n$ and $\gcd(x_i, n) = 1$. Multiply all terms by a to create $aS = \{ax_1, ax_2, \dots, ax_{\phi(n)}\}$. First, we prove that $\gcd(ax_i, n) = 1$ for any x_i using contradiction. Let p be a prime such that $p \mid ax_i$ and $p \mid n$. This means that either $p \mid a$ or $p \mid x_i$, which is not possible since both $\gcd(a, n)$ and $\gcd(x_i, n) = 1$. Next we show that no two elements in set aS are congruent modulo n . Let x_i and x_j exist such that $ax_i \equiv ax_j \pmod{n}$. Multiplying both sides by a^{-1} (which exists) gives $x_i \equiv x_j \pmod{n}$. This means that if $x_i \not\equiv x_j \pmod{n}$, then $ax_i \not\equiv ax_j \pmod{n}$. Now we multiply all terms in each set, and note that each residue appears exactly once in both sets, so we have:

$$a^{\phi(n)} \cdot \prod_{i=1}^{\phi(n)} x_i \equiv \prod_{i=1}^{\phi(n)} x_i \pmod{n}.$$

Because each x_i has the property that $\gcd(x_i, n) = 1$, we must have that $\gcd(\prod_{i=1}^{\phi(n)} x_i, n) = 1$. Then multiplying both sides by the inverse of product thereof gives $a^{\phi(n)} \equiv 1 \pmod{n}$. ■

Solution 2 (No Totient)

Note that

$$(5+n)^{100} = \binom{100}{0} 5^{100} + \binom{100}{1} 5^{99}n + \binom{100}{2} 5^{98}n^2 + \dots + \binom{100}{97} 5^3n^{97} + \binom{100}{98} 5^2n^{98} + \binom{100}{99} 5n^{99}$$

Taking this mod 125, we can ignore most of the terms except the for the last 3:

$$\binom{100}{98} 5^2n^{98} + \binom{100}{99} 5n^{99} + \binom{100}{100} n^{100} \equiv 4950 \cdot 5^2n^{98} + 100 \cdot 5n^{99} + n^{100} \equiv n^{100} \pmod{125},$$

so $(5+n)^{100} \equiv n^{100}$. Substituting $-n$ for n , we get $(5-n)^{100} \equiv n^{100}$. Therefore, the remainders when divided by 125 repeat every 5 integers, so we only need to check the 100th powers of 0, 1, 2, 3, 4. But we have that

$(5 - 1)^{100} \equiv 1^{100}$ and $(5 - 2)^{100} \equiv 2^{100}$, so we really only need to check 0, 1, 2. We know that 0, 1 produce different remainders, so the answer to the problem is either 2 or 3. But 3 is not an answer choice, so the answer is **(B) 2**.

Solution 3 (Binomial Theorem)

Any integer can be represented in one of the following forms:

$$5k-2, 5k-1, 5k, 5k+1, 5k+2$$

Case 1: $5k$

$$(5k)^{100}$$

Here remainder is zero when divided by 125.

Case 2: $5k \pm 1$

$$(5k \pm 1)^{100} = (5k)^{100} \pm 100 \times (5k)^{99} \pm \dots + \frac{5^2 \times 4 \times 99}{1 \times 2} \times (5k)^2 \pm 5^3 \times 4k + 1$$

Here all the terms except one at last are divisible by 125, hence remainder is 1 when number is in form $5k \pm 1$

Case 3: $5k \pm 2$

$$(5k \pm 2)^{100} = (5k)^{100} \pm 100 \times 2 \times (5k)^{99} + \dots + \frac{100 \times 99}{1 \times 2} \times 2^{98} \times (5k)^2 + 100 \times 2^{99} \times (5k) + 2^{100}$$

$$2^{100} = (5-1)^{100} = 5^{100} - 50 \times 5^{49} + \dots + \frac{50 \times 49}{1 \times 2} \times 5^2 - 50 \times 5 + 1$$

Here all terms except one at last are divisible by 125, hence remainder is 1 when number is in form $5k \pm 2$

Conclusion: For numbers in form $5k$ remainder is zero and for any other form remainder is one.

Therefore there are 2 possible remainders. Option (B) 2

Solution 4

Firstly, I interpreted the problem algebraically. In other words, let x be an integer. Since we're raising it to the 100th power and 100 is even, we don't have to worry about negative values of x . Now, we break 125 into three factors of 5. I first considered this when we're dividing by 5. Okay so we can write the problem as below:

$x^{100} \equiv r \pmod{5}$. Now, in the world of $\pmod{5}$, everything is much easier to figure out. Take $x \equiv 0 \pmod{5}$. Now, obviously x^{100} will bring up a remainder of 0 if x itself is a multiple of 5. Now, $x \equiv 1 \pmod{5}$. In this case, notice how 1^{100} is congruent to 1 in the $\pmod{5}$ world. So in this case, we would have a remainder of 1. Considering $x \equiv 2 \pmod{5}$, it's a little more tricky. Notice 2^{100} is equal to 1024^{10} and 1024 is 1 less than a multiple of 5. So 1024 will be congruent to $-1 \pmod{5}$ and raising it to the power of 10 will bring up a remainder of 1 yet again. Similarly, is $x \equiv 3 \pmod{5}$ or $x \equiv 4 \pmod{5}$, you would notice both these cases will bring up a remainder of 1 again. So in the $\pmod{5}$ world, we seem to only have remainders of 0 and 1. Interesting! Therefore, since 125 is just 3 factors of 5, we may say that in $\pmod{125}$, we would have the same exact remainders namely 0 and 1. Therefore, we would only have 2 possible remainders. This brings an answer of **(B) 2**.

- s) (AMC 10 10B 2024) In the following table, each question mark is to be replaced by "Possible" or "Not Possible" to indicate whether a nonvertical line

with the given slope can contain the given number of lattice points (points both of whose coordinates are integers). How many of the 12 entries will be "Possible"?

	zero	exactly one	exactly two	more than two
zero slope	?	?	?	?
nonzero rational slope	?	?	?	?
irrational slope	?	?	?	?

(A) 4 (B) 5 (C) 6 (D) 7 (E) 9

Answer: C

Solution 1

Let's look at zero slope first. All lines of such form will be expressed in the form $y = k$, where k is some real number. If k is an integer, the line passes through infinitely many lattice points. One such example is $y = 1$. If k is not an integer, the line passes through 0 lattice points. One such example is $y = 1.1$. So we have 2 cases.

Let's now look at the second case. The line has slope $\frac{p}{q}$, where p and q are relatively prime integers. The line has the form $y = \frac{p}{q}x + m$. If the line passes through lattice point (a, b) , then the line must also pass through the lattice point $(a + kq, b + kp)$, where a, b, k are all integers. Therefore, the line can pass through infinitely many lattice points but it cannot pass through exactly 1 or 2. The line can pass through 0 lattice points, such as $y = x + 0.5$. This contributes 2 more cases.

If the line has an irrational slope, it can never pass through more than 2 lattice points. We prove this using contradiction. Let's say the line passes through lattice points (a, b) and (c, d) . Then the line has slope $\frac{d-b}{c-a}$, which is rational. However, the slope of the line is irrational. Therefore, the line can pass through at most 1 lattice point. One example of this is $y = \sqrt{2}x$. This line contributes 2 final cases.

Our answer is therefore 6.

This shows examples of the six lines that are valid. - <https://www.desmos.com/calculator/ypt7esd>

Solution 2

For a 0 slope: It can have 0 points (e.g. $y = 1.5$), but it can also have infinite (e.g. $y=1$), which checks the two or more box.

For a nonzero slope: It can have 0 lattice points (e.g. $y = \frac{2}{3}x + \frac{1}{2}$), or infinite (eg $y = 2x$).

For an irrational slope: It can have 0 (e.g. $y = \sqrt{3}x + \frac{1}{2}$) or exactly 1 (e.g. $y = \sqrt{3}x$, which has an intersection at $0,0$)

Hence, there are a total of 6 points.

Solution 3

Zero Slope

A line of zero slope can intersect no lattice points (If the y-intercept isn't an integer)

It cannot intersect only one lattice point, same goes for two lattice points because if it intersects one, it has to intersect all lattice points with that y value.

It can intersect more than two lattice points (infinite)

Rational Slope $\neq 0$

It can intersect no lattice points ($y = x + 0.5$)

It cannot intersect only one or two lattice points (because then it would intersect another one once the slope "repeats" again)

It can intersect more than two lattice points (infinite)

Irrational Slope

It can intersect no lattice points (y-intercept isn't an integer)

It can intersect one lattice point (y-intercept is an integer)

It cannot intersect two or more lattice points no matter what.

Therefore, the answer is (C) 6. Tacos_are_yummy_1

- t) (AMC 10 10B 2024) Three different pairs of shoes are placed in a row so that no left shoe is next to a right shoe from a different pair. In how many ways can these six shoes be lined up?

(A) 60 (B) 72 (C) 90 (D) 108 (E) 120

Answer: A

Solution 1

Let $A_R, A_L, B_R, B_L, C_R, C_L$ denote the shoes.

There are 6 ways to choose the first shoe. WLOG, assume it is A_R . We have $A_R, _, _, _, _, _$.

Case 1: The next shoe in line is A_L . We have $A_R, A_L, _, _, _, _$. Now, the next shoe in line must be either B_L or C_L . There are 2 ways to choose which one, but assume WLOG that it is B_L . We have $A_R, A_L, B_L, _, _, _$.

Subcase 1: The next shoe in line is B_R . We have $A_R, A_L, B_L, B_R, _, _$. The only way to finish is $A_R, A_L, B_L, B_R, C_R, C_L$.

Subcase 2: The next shoe in line is C_L . We have $A_R, A_L, B_L, C_L, _, _$. The only way to finish is $A_R, A_L, B_L, C_L, C_R, B_R$.

In total, this case has $(6)(2)(1 + 1) = 24$ orderings.

Case 2: The next shoe in line is either B_R or C_R . There are 2 ways to choose which one, but assume WLOG that it is B_R . We have $A_R, B_R, _, _, _, _$.

Subcase 1: The next shoe is B_L . We have $A_R, B_R, B_L, _, _, _$.

Sub-subcase 1: The next shoe in line is A_L . We have $A_R, B_R, B_L, A_L, _, _$. The only way to finish is $A_R, B_R, B_L, A_L, C_L, C_R$.

Sub-subcase 2: The next shoe in line is C_L . We have $A_R, B_R, B_L, C_L, _, _$. The remaining shoes are C_R and A_L , but these shoes cannot be next to each other, so this sub-subcase is impossible.

Subcase 2: The next shoe is C_R . We have $A_R, B_R, C_R, _, _, _$. The next shoe in line must be C_L , so we have $A_R, B_R, C_R, C_L, _, _$. There are 2 ways to finish, which are $A_R, B_R, C_R, C_L, A_L, B_L$ and $A_R, B_R, C_R, C_L, B_L, A_L$.

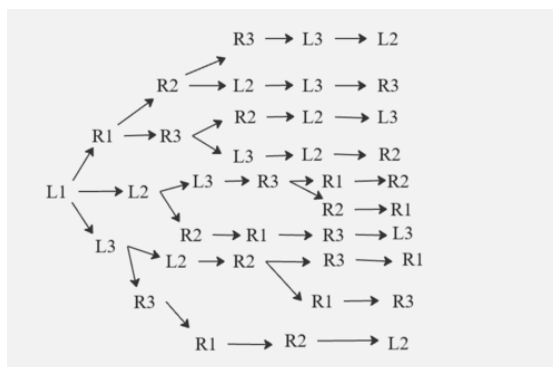
In total, this case has $(6)(2)(1 + 2) = 36$ orderings.

Our final answer is $24 + 36 = \boxed{\text{(A)} 60}$

Solution 2 (Tree Diagram)

Label the pairs (L_1, R_1) , (L_2, R_2) , and (L_3, R_3) .

We now create a tree diagram. **WLOG**, we say our first tree diagram starts with L_1 .



This tree has 10 branches.

Notice how we can interchange the first shoe with any $L_1, L_2, L_3, R_1, R_2, R_3$, and still get the same tree, but just interchanged shoes. Therefore, we just multiply the tree's branches by 6, to get $6 \cdot 10 = \boxed{\text{(A)} 60}$.

Solution 3 (Fakesolve, Refer to Solution 4)

We have two main cases:

Case 1: $LLRRR$ or $RRLLL$

There are $3! = 6$ ways to assign the shoes to the ordering. For now we'll focus on the $LLRRR$ case and multiply by 2 since they're symmetrical. **WLOG** say we have $A_L B_L C_L C_R A_R B_R$. We can flip A_L and B_L among themselves and A_R and B_R among themselves, so we get $6 \cdot 2 \cdot 2 = 24$. Multiplying by 2 for symmetry gives us 48 ways for this case.

Case 2: $RRLLL$ or $RLLLR$

Again, we'll focus on the $RRLLL$ case and multiply by 2 since they're symmetrical. **WLOG** say we have $B_R A_R A_L B_L C_L C_R$. We can assign the shoes to the ordering in $3! = 6$ ways. B_R can also be moved over to be next to C_R . Then we would have $A_R A_L B_L C_L C_R B_R$, which is also valid. So this case gives us $6 \cdot 2 = 12$ ways.

The answer is $48 + 12 = \boxed{60}$.

Note: Case 1 should give you 24 and case 2 should give you 36

Solution 4 (focus on restrictions)

Notice that you cannot have LRL or RLR in a row, since you are guaranteed an R and an L from a different pair. This means you can either have three L 's in a row, three R 's in a row, or you have two R 's between two L 's and two L 's between two R 's.

Below are the cases (note that once an L is fixed the R adjacent to it is also fixed due to the constraint):

$$LLLRRR \Rightarrow 3! \cdot 2! = 12$$

$$RLLRRR \Rightarrow 3! = 6$$

$$RRLLRR \Rightarrow 3! = 6$$

$$RRRLLL \Rightarrow 3! \cdot 2! = 12$$

$$LRRRLL \Rightarrow 3! = 6$$

$$LLRRRL \Rightarrow 3! = 6$$

$$LRRLRR \Rightarrow 3! = 6$$

$$RLLRRL \Rightarrow 3! = 6$$

We have $2 \cdot 12 + 6 \cdot 6 = \boxed{\text{(A)} 60}$.

Solution 5 (for some reason this works)

Let the pairs be L_1R_1, L_2R_2, L_3R_3 . The only ones that don't work are $L_1R_2, L_1R_3, L_2R_1, L_2R_3, L_3R_2, L_3R_1$ and their swaps so a total of 12 pairs that can't be in the string of 6. We have $6! = 720$ total ways and since there are 12 ways it won't work we simply do

$$\frac{720}{12} = \boxed{60}$$

Solution 6

Let $A_1, A_2, A_3, B_1, B_2, B_3$ be the shoes. Initially, let's assume that it doesn't matter whether shoes of type A are left/right or if shoes of type B are left/right. We can have four possible arrangements: $AAABBB$, $AABBB A$, $ABBAAB$, and $ABBBAA$. Note that whenever we have a shoe of type A and a shoe of type B next to each other, those shoes must be of the same pair.

In the first case ($AAABBB$), only one pair is forced to be in the middle. We can choose this pair in 3 ways. There are $2! \cdot 2! = 4$ ways to choose the different pairs for the other four shoes in this case, and therefore $3 \cdot 4 = 12$ possibilities for this case.

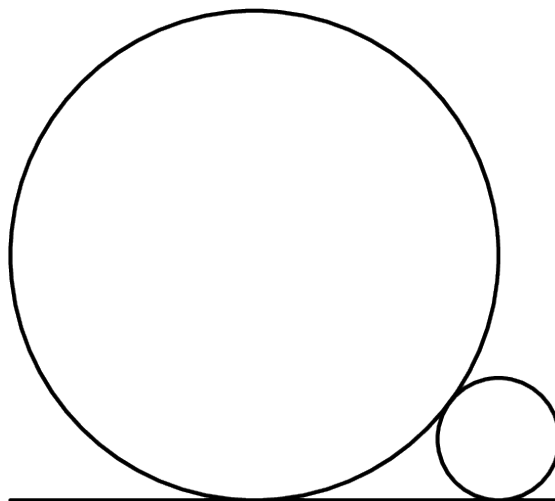
In all of the other cases, we either have 2 or 3 forced pairs AB . In all of these cases, there are $3 \cdot 2$ or $3 \cdot 2 \cdot 1$ ways to choose the different pairs that each shoe belongs to, so 6 cases for each pair.

In total, there are $12 + 6 + 6 + 6 = 30$ cases ignoring if A or B is left/right. If we now consider these different cases, we will have twice the number of combinations, or $30 \cdot 2 = \boxed{60}$ combinations.

Solution 7

Same thing for solution 1 but you use casework on whether all L's touch or not. Case 1: All L's touch $24 + 12 = 36$ possibilities Case 2: Only 2 L's touch 24 possibilities $36 + 24 = 60$.

- u) (AMC 10 10B 2024) Two straight pipes (circular cylinders), with radii 1 and $\frac{1}{4}$, lie parallel and in contact on a flat floor. The figure below shows a head-on view. What is the sum of the possible radii of a third parallel pipe lying on the same floor and in contact with both?



- (A) $\frac{1}{9}$ (B) 1 (C) $\frac{10}{9}$ (D) $\frac{11}{9}$ (E) $\frac{19}{9}$

Answer: C

Solution 1

Notice that the sum of radii of two circles tangent to each other will equal to the distance from center to center. Set the center of the big circle be at $(0, 1)$. Since both circles are tangent to a line (in this case, $y = 0$), the y-coordinates of the centers are just its radius.

Hence, the center of the smaller circle is at $(x_2, \frac{1}{4})$. From the the sum of radii and distance formula, we have:

$$1 + \frac{1}{4} = \sqrt{x_2^2 + \left(\frac{3}{4}\right)^2} \Rightarrow x_2 = 1.$$

So, the coordinates of the center of the smaller circle are $(1, \frac{1}{4})$. Now, let (x_3, r_3) be the coordinates of the new circle. Then, from the fact that sum of radii of this circle and the circle with radius 1 is equal to the distance from the two centers, you have:

$$\sqrt{(x_3 - 0)^2 + (r_3 - 1)^2} = 1 + r_3.$$

Similarly, from the fact that the sum of radii of this circle and the circle with radius $\frac{1}{4}$, you have:

$$\sqrt{(x_3 - 1)^2 + \left(r_3 - \frac{1}{4}\right)^2} = \frac{1}{4} + r_3.$$

Squaring the first equation, you have:

$$x_3^2 + r_3^2 - 2r_3 + 1 = 1 + 2r_3 + r_3^2 \Rightarrow 4r_3 = x_3^2 \Rightarrow x_3 = 2\sqrt{r_3}.$$

Squaring the second equation, you have:

$$x_3^2 - 2x_3 + 1 + r_3^2 - \frac{r_3}{2} + \frac{1}{16} = \frac{1}{16} + \frac{r_3}{2} + r_3^2 \Rightarrow x_3^2 - 2x_3 + 1 = r_3$$

Plugging in from the first equation we have

$$r_3 - 1 = x_3^2 - 2x_3 = 4r_3 - 4\sqrt{r_3} \Rightarrow 3r_3 - 4\sqrt{r_3} + 1 = 0 \Rightarrow (3\sqrt{r_3} - 1)(\sqrt{r_3} - 1) = 0 \Rightarrow r_3 = 1, \frac{1}{9}.$$

Summing these two yields $\boxed{\frac{10}{9}}$.

Diagram

Solution 2 (Clever Manipulation)

In general, let the left and right outer circles and the center circle have radii r_1, r_2, r_3 . When three circles are tangent as described in the problem, we can deduce $\sqrt{r_3} = \frac{\sqrt{r_1 r_2}}{\sqrt{r_1} + \sqrt{r_2}}$ by Pythagorean theorem.

Setting $r_1 = 1, r_2 = \frac{1}{4}$ we have $r_3 = \frac{1}{9}$, and setting $r_1 = 1, r_3 = \frac{1}{4}$ we have $r_2 = 1$. Thus our answer is $\boxed{(C)}$.

Solution 3 (Descartes's Theorem)

Descartes's Theorem states that for curvatures k_1, k_2, k_3, k_4 we have

$$(k_1 + k_2 + k_3 + k_4)^2 = 2(k_1^2 + k_2^2 + k_3^2 + k_4^2)$$

with a curvature being the reciprocal of the radius of a circle, being positive if it is externally tangent to other circles, negative if it is internally tangent to other circles, and zero if we consider a line as a degenerate circle. Here our curvatures are $k_1 = 1, k_2 = 4, k_3 = 0$, and we wish to find k_4 . Plugging these values into our formula yields:

$$(1 + 4 + 0 + k_4)^2 = 2(1^2 + 4^2 + 0^2 + k_4^2)$$

$$(5 + k_4)^2 = 2(17 + k_4^2)$$

$$k_4^2 + 10k_4 + 25 = 2k_4^2 + 34$$

$$k_4^2 - 10k_4 + 9 = 0$$

$$k_4 = 1, 9$$

The curvature and the radius are reciprocals, so our radii must be 1 and $\frac{1}{9}$, and their sum is $\boxed{(C) \frac{10}{9}}$.

Solution 4 (Advanced Guesswork)

Looking at the diagram, we see that there must be a circle with a radius smaller than $\frac{1}{4}$, and there must be a circle with a radius close to 1. Looking at the answers, we can assume that the answer choice is above 1, eliminating the first two answers. We can also assume that some students will only solve for the smaller circle, while others may only solve for the larger circle. It

would be reasonably safe to say that one of the answers must be the radius of the smaller circle, while another must be the radius of the bigger circle. Looking at the answers, we see that $\frac{1}{9}$ and 1 are reasonably close to the radii of the missing circles in the diagram, so we add them up to get $\boxed{\text{(C)} \frac{10}{9}}$.

Solution 5 (Essentially Solution 3)

For a problem like this with three externally tangent circles that are all tangent to a line, we can use a specific form of Descartes's Circle Theorem. Let the curvatures of the three circles be r_1, r_2, r_3 . The curvature of a circle is just the reciprocal of the radius. Then we get,

$$(r_1 + r_2 + r_3)^2 = 2(r_1^2 + r_2^2 + r_3^2)$$

Plugging in the information we already have, we get:

$$(1 + 4 + r_3)^2 = 2(1^2 + 4^2 + r_3^2)$$

$$(5 + r_3)^2 = 2(17 + r_3^2)$$

$$25 + 10r_3 + r_3^2 = 34 + 2r_3^2$$

$$r_3^2 - 10r_3 - 9 = 0$$

$$(r_3 - 1)(r_3 - 9) = 0$$

So, we get our two curvatures: 1 and 9. The radii equivalent are 1 and $\frac{1}{9}$.

Adding these up, we get $\boxed{\text{(C)} \frac{10}{9}}$.

Solution 6 (Metasolving)

By looking at the answer choices, we realize that the answers are all sums of each other, and thus think "The AMC problem committee made the answers $\frac{1}{9}$ and 1 so the people who only got one of the answers will pick either of the first two answers and those who counted one of the answers twice would pick either of the last two answers", and realize that the answers are most likely 1 and $\frac{1}{9}$. Confirming with a ruler, we get $\frac{1}{9} + 1 = \boxed{\text{(C)} \frac{10}{9}}$, which is the solution.

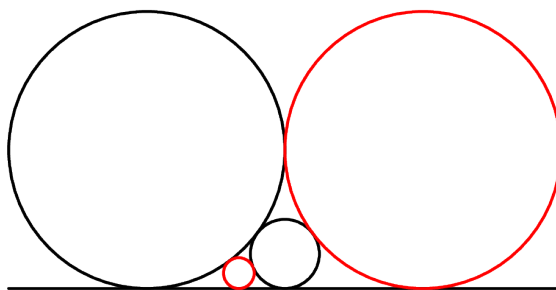
~Glowworm

Minor issue with this is that $10/9$ is also an answer choice, and $1/9 + 10/9 = 11/9$ which is also an answer choice, so it comes down to a bit of luck.

Solution 7 (practically similar to the other solutions)

We can immediately see that we can put a circle with radius 1 outside of the two circles already drawn that are tangent to the circles. If you brought a ruler with you, you can confirm this. We now find the radius of the circle between the circles of radius 1 and $\frac{1}{4}$. By drawing out the diagram, a geometry hack is to draw lines connecting the centers of the circles. Using the fact that tangent lines to the radius are perpendicular, we can construct a trapezoid with heights of lengths 1 and $\frac{1}{4}$, a slant base of length $1 + \frac{1}{4} = \frac{5}{4}$. To find the base of the trapezoid, we can construct a right triangle with the slant base as the hypotenuse and a leg of length $1 - \frac{1}{4} = \frac{3}{4}$. Therefore, we see we have a 3-4-5 right triangle and the length of the base of the trapezoid is

conveniently just 1. Let the radius of the circle between the two given circles r . By drawing the center of this circle and connecting the radius to the radii of the other two circles, we can construct two more trapezoids similar to the initially big one we constructed: one with perpendicular heights of lengths 1 and r and a slant base of $1 + r$, and another with perpendicular heights of lengths $\frac{1}{4}$ and r and a slant base of length $r + \frac{1}{4}$. To find the bases of these trapezoids, (which we know sums to 1 as we found earlier), we can use a similar trick by constructing right triangles. We can construct two right triangles: one with hypotenuse $1 + r$ and a leg of length $1 - r$, and another with hypotenuse $r + \frac{1}{4}$ and a leg of length $\frac{1}{4} - r$. We can use the Pythagorean Theorem to find the other legs of these right triangles and we can set the sum to equal 1. Conveniently, the leg lengths turns out to be $2\sqrt{r}$ and $\sqrt{r}$, respectively, and therefore the sum will just be $3\sqrt{r}$. Since this is equal to 1, we can solve $3\sqrt{r} = 1$ to get the radius to be equal to $\frac{1}{9}$. Now, we add 1 and $\frac{1}{9}$ to get $\frac{10}{9}$ and our answer is just **(C)** $\frac{10}{9}$.



credit to Elephant200 for the diagram

- v) (AMC 10 10B 2024) A group of 16 people will be partitioned into 4 indistinguishable 4-person committees. Each committee will have one chairperson and one secretary. The number of different ways to make these assignments can be written as $3^r M$, where r and M are positive integers and M is not divisible by 3. What is r ?
- (A) 5 (B) 6 (C) 7 (D) 8 (E) 9

Answer: A

Solution 1

There are $\binom{16}{4}$ ways to choose the first committee, $\binom{12}{4}$ ways to choose the second, $\binom{8}{4}$ for the third, and $\binom{4}{4} = 1$ for the fourth. Since the committees are indistinguishable, we need to divide the product by $4!$. Thus the 16 people can be grouped in

$$\frac{1}{4!} \binom{16}{4} \binom{12}{4} \binom{8}{4} \binom{4}{4} = \frac{16!}{(4!)^5}$$

ways.

In each committee, there are $4 \cdot 3 = 12$ ways to choose the chairperson and secretary, so 12^4 ways for all 4 committees. Therefore, there are

$$\frac{16!}{(4!)^5} 12^4$$

total possibilities.

Since $16!$ contains 6 factors of 3, $(4!)^5$ contains 5, and 12^4 contains 4, $r = 6 - 5 + 4 = \boxed{\text{(A)} 5}$.

Solution 2 (Multinomial Coefficients)

There are $\binom{16}{4,4,4,4}$ ways to choose the 4 committees. You have to divide by another $4!$ since the order of the committees does not matter. Furthermore, in each committee, you can have $4 \cdot 3$ ways to choose chairperson and secretary. Hence a total of $\lfloor \frac{16}{3} \rfloor + \lfloor \frac{16}{9} \rfloor + 4 - 5 = 5$.

Solution 3

There will be 16 ways to pick the chairperson of the first committee, then 15 to pick the secretary, and lastly $\binom{14}{2}$ ways to pick the other two members of the first committee. Similarly, we can complete the rest of the terms as follows:

$$\frac{(16)(15)\binom{14}{2}\binom{12}{2}\binom{10}{2}\binom{8}{2}\binom{6}{2}\binom{4}{2}\binom{2}{2}}{4!}$$

We notice the numerator has at most 3^6 , and the denominator has just 3. Thus, the value of r in question is $\boxed{\text{(A)} 5}$.

Solution 4

We can convert the question into arranging people in a line of length 16.

First, arrange all 16 people in a line. There are $16!$ possible ways to do this. Then, divide the line into 4 groups of 4, which represent the committees. Since the committees are indistinguishable, we divide by the number of ways to permute the 4 groups, $4!$.

Within each group of 4, the first two positions in the line are assigned as the chairperson and the secretary, while the remaining two are ordinary members. Since the roles of chairperson and secretary are distinct, but the order of the remaining two members does not matter, we divide by 2 for each group. Thus, we divide by 2^4 to account for all 4 committees.

The total number of ways to assign people into committees with roles is given by:

$$\frac{16!}{4! \cdot 2^4}.$$

(This is equivalent to Solution 1's $\frac{16!}{(4!)^5} \cdot 12^4$.)

To find r , we use Legendre's Formula to count the total power of 3 in $16!$:

$$\lfloor \frac{16}{3} \rfloor + \lfloor \frac{16}{9} \rfloor = 5 + 1 = 6.$$

Each $4!$ contributes one factor of 3, and since we divide by $4!$ a total of 1 times, we subtract 1 from 6:

$$6 - 1 = 5.$$

Thus, the value of r is $\boxed{5}$. ryk

- w) (AMC 10 10B 2024) The Fibonacci numbers are defined by $F_1 = 1, F_2 = 1$, and $F_n = F_{n-1} + F_{n-2}$ for $n \geq 3$. What is

$$\frac{F_2}{F_1} + \frac{F_4}{F_2} + \frac{F_6}{F_3} + \dots + \frac{F_{20}}{F_{10}}?$$

- (A) 318 (B) 319 (C) 320 (D) 321 (E) 322

Answer: B

Solution 1 (Brute Force)

The first 20 terms are $F_n = 1, 1, 2, 3, 5, 8, 13, 21, 34, 55, 89, 144, 233, 377, 610, 987, 1597, 2584, 4181, 6765$

So the answer is $1 + 3 + 4 + 7 + 11 + 18 + 29 + 47 + 76 + 123 = \boxed{(B)319}$.

- Do not do this unless you have no other option or no time to find a smarter way

x) (AMC 10 10B 2024) Let

$$P(m) = \frac{m}{2} + \frac{m^2}{4} + \frac{m^4}{8} + \frac{m^8}{8}$$

How many of the values $P(2022)$, $P(2023)$, $P(2024)$, and $P(2025)$ are integers?

(A) 0 (B) 1 (C) 2 (D) 3 (E) 4

Answer: E

Solution 1 (The simplest way)

First, we know that $P(2022)$ and $P(2024)$ must be integers since they are both divisible by 2.

Then let's consider the remaining two numbers. Since they are not divisible by 2, the result of the first term must be a certain number $+\frac{1}{2}$.

The divisibility rule for 4 states that the last two digits of the number must be divisible by 4, and we know that the last two digits of 2023^2 are the last two digits of 23^2 , which is 29. And 29 is 1 greater than a multiple of 4. Therefore, the result of the second term must be $\frac{1}{4}$. Similarly, the remaining two terms must each be $\frac{1}{8}$.

Their sum is $\frac{1}{2} + \frac{1}{4} + \frac{1}{8} + \frac{1}{8} = 1$, so $P(2023)$ and $P(2025)$ are also integers.

Therefore, the answer is $\boxed{(E) 4}$.

Solution 2 (Specific)

Take everything modulo 8 and re-write the entire fraction with denominator 8. This means that we're going to transform the fraction as follows :

$$P(m) = \frac{m}{2} + \frac{m^2}{4} + \frac{m^4}{8} + \frac{m^8}{8}$$

becomes

$$P(m) = \frac{4m + 2m^2 + m^4 + m^8}{8}$$

And in order for $P(m)$ to be an integer, it's important to note that $4m + 2m^2 + m^4 + m^8$ must be congruent to 0 modulo 8. Moreover, we know that $2022 \equiv 6 \pmod{8}$, $2023 \equiv 7 \pmod{8}$, $2024 \equiv 0 \pmod{8}$, $2025 \equiv 1 \pmod{8}$. We can verify it by taking everything modulo 8 :

If $m = 2022$, then $4(6) + 2(4) + 0 + 0 = 24 + 8 = 32 \equiv 0 \pmod{8} \rightarrow \text{TRUE}$
 If $m = 2023$, then $4(7) + 2(1) + 1 + 1 = 28 + 2 + 1 + 1 = 32 \equiv 0 \pmod{8} \rightarrow \text{TRUE}$
 If $m = 2024$, then it is obvious that the entire expression is divisible by 8. Therefore, it is true. If $m = 2025$, then $2025 \equiv 1 \pmod{8}$. Therefore, $4(1) + 2(1) + 1 + 1 = 8 \equiv 0 \pmod{8} \rightarrow \text{TRUE}$
 Therefore, there are $\boxed{(E) 4}$ possible values.

Addendum for certain China test papers : Note that $2026 \equiv 2 \pmod{8}$. Therefore, taking everything modulo 8, whilst still maintaining the original expression, gives $4(2) + 2(4) + 0 + 0 = 16 \equiv 0 \pmod{8}$. This is true.

Solution 3 (Factoring)

We can rewrite the expression as

$$\frac{1}{8} \cdot m \cdot (m^7 + m^3 + 2m + 4).$$

If m is even, then m gives a factor of 2 and $m^7 + m^3 + 2m + 4$ will give a factor of 4, so the result will be an integer. If m is odd, then notice that for any m we have $m^2 \equiv 1 \pmod{8}$. Then $m^7 + m^3 + 2m + 4 \equiv m + m + 2m + 4 \equiv 4(m+1) \equiv 0 \pmod{8}$. So any integer m will result in an integer, meaning the answer is

(E) 4.

-nevergonnagiveup

Solution 4

Taking a closer look at the terms, we notice that each term builds off of the previous one. There is $\frac{m}{2}$, $\frac{m^2}{4}$, which is equal to $\left(\frac{m}{2}\right)^2$, $\frac{m^4}{8}$, which is equal to $2\left(\frac{m^2}{4}\right)^2$, and $\frac{m^8}{8}$, which is equal to $8\left(\frac{m^4}{8}\right)^2$. Ultimately, this means that there are only two cases that we need to check for: the case in which m is even and the case in which m is odd.

If m is even, $\frac{m}{2}$ will be an integer, which means the rest of the terms will be an integer. This means in the problem, $P(2022)$ and $P(2024)$ will yield an integer result. (For certain Chinese versions, this also proves that $P(2026)$ is an integer.)

If m is odd, $\frac{m}{2}$ will result in $\frac{1}{2}$. Building off of each term will give you $\frac{1}{4}$, $\frac{1}{8}$, and $\frac{1}{8}$, and summing those up will grant 1, an integer. This means in the problem, $P(2023)$ and $P(2025)$ will also result in integer answers.

In general, all integer m will make $P(m)$ give an integer answer, but for this question, this will get us to **(E) 4** integer values (**(E) 5** for certain Chinese versions of the test).

Solution 5 (sol 2 but fast)

Obviously $P(2022)$ and $P(2024)$ both work. $2023 \equiv -1 \pmod{8}$, and $2025 \equiv 1 \pmod{8}$. Restating the problem, we have the equation equivalent to $\frac{4m+2m^2+m^4+m^8}{8}$. Plugging in $m = -1$, we have $-4 + 2 + 1 + 1 \equiv 0 \pmod{8}$, so $P(2023)$ works. Plugging in $m = 1$, we have $4 + 2 + 1 + 1 \equiv 0 \pmod{8}$ as well, so $P(2025)$ also works. Thus, the answer is **(E)**.

Solution 6

Note that $2022 \equiv 0 \pmod{2}$, $2 \pmod{4}$, and $-2 \pmod{8}$.

Thus, when $m = 2022$, $P(m) = \frac{2022}{2} + \frac{2022^2}{4} + \frac{2022^4}{8} + \frac{2022^8}{8} \equiv \frac{0}{2} + \frac{2^2}{4} + \frac{(-2)^4}{8} + \frac{(-2)^8}{8} \equiv \frac{0}{8} + \frac{8}{8} + \frac{16}{8} + \frac{256}{8} \equiv \frac{280}{8} \equiv 0 \pmod{8}$ meaning $P(2022)|8$ and thus is an integer.

Repeating this process for each of $m = 2023, 2024$, and 2025 shows that they're all integers, giving a final answer of **(E) 4**.

Remark On certain versions of the AMC in China, the problem was restated as follows:

Let

$$P(m) = \frac{m}{2} + \frac{m^2}{4} + \frac{m^4}{8} + \frac{m^8}{8}$$

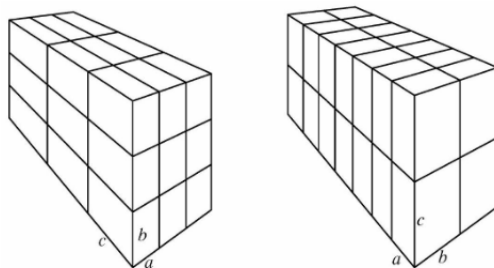
How many of the values $P(2022)$, $P(2023)$, $P(2024)$, $P(2025)$, and $P(2026)$ are integers? (A) 1 (B) 2 (C) 3 (D) 4 (E) 5

By identical reasoning, each term of P is an integer, since 2026 is even.

Therefore, the answer is (E) 5.

Not So Great Guessing Method Note that this is Question 24 of the 2024 AMC 10B. It follows that the answer should be something to do with 2024. Option (E) 4 is the closest one with the last digit of 4. Therefore, if you have no way to solve this and have no clue why you decided to take the AMC, use this method. (Like me!)

- y) (AMC 10 10B 2024) Each of 27 bricks (right rectangular prisms) has dimensions $a \times b \times c$, where a , b , and c are pairwise relatively prime positive integers. These bricks are arranged to form a $3 \times 3 \times 3$ block, as shown on the left below. A 28th brick with the same dimensions is introduced, and these bricks are reconfigured into a $2 \times 2 \times 7$ block, shown on the right. The new block is 1 unit taller, 1 unit wider, and 1 unit deeper than the old one. What is $a + b + c$?



- (A) 88 (B) 89 (C) 90 (D) 91 (E) 92

Answer: E

Solution 1 (Less than 60 seconds)

The $3 \times 3 \times 3$ block has side lengths of $3a, 3b, 3c$. The $2 \times 2 \times 7$ block has side lengths of $2b, 2c, 7a$.

We can create the following system of equations, knowing that the new block has 1 unit taller, deeper, and wider than the original:

$$3a + 1 = 2b$$

$$3b + 1 = 2c$$

$$3c + 1 = 7a$$

Adding all the equations together, we get $b + c + 3 = 4a$. Adding $a - 3$ to both sides, we get $a + b + c = 5a - 3$. The question states that a, b, c are all

relatively prime positive integers. Therefore, our answer must be congruent to 2 (mod 5). The only answer choice satisfying this is $\boxed{(E) 92}$. lprado

Solution 2 (Solution 1 but longer)

We will define the equations the same as solution 1.

$$3a + 1 = 2b$$

$$3b + 1 = 2c$$

$$3c + 1 = 7a$$

Solve equation 2 for c and substitute that value in for equation 3, giving us

$$3a + 1 = 2b$$

$$\frac{3b + 1}{2} = c$$

$$\frac{3 \cdot (3b + 1)}{2} + 1 = 7a$$

Multiply 14 to the first equation and rearrange to get

$$42a = 28b - 14$$

and multiply the third by 2 and rearrange to get

$$27b + 15 = 42a$$

Solve for b to get $b = 29$, substitute into equation 1 from the original to get $a = 19$, and lastly, substitute a into original equation 2 to get $c = 44$. Thus, $a + b + c = 19 + 29 + 44 = \boxed{E(92)}$. Failure.net

Solution 3

We will define the equations the same as solution 1 and 2.

$$3a + 1 = 2b$$

$$3b + 1 = 2c$$

$$3c + 1 = 7a$$

Multiply 1st equation by 1.5 to get

$$4.5a + 1.5 = 3b$$

Add 1 to get

$$4.5a + 2.5 = 3b + 1$$

Substitute to get

$$4.5a + 2.5 = 2c$$

Multiply this one by 1.5 as well, getting

$$6.75a + 3.75 = 3c$$

Adding 1 again to get

$$6.75a + 4.75 = 3c + 1$$

Replace

$$3c + 1$$

with

$$7a$$

getting

$$6.75a + 4.75 = 7a$$

Solve for

$$a$$

getting

$$0.25a = 4.75$$

$$a = 19$$

Solve for b:

$$3(19) + 1 = 2b$$

$$57 + 1 = 2b$$

$$b = 29$$

Solve for c:

$$3(29) + 1 = 2c$$

$$87 + 1 = 2c$$

$$c = 44$$

$$\text{Thus, } a + b + c = 19 + 29 + 44 = \boxed{E(92)}$$

Solution 4 (No Fractions)

As stated in solutions 1, 2, 3, the equations are as follows:(1)

$$3a + 1 = 2b$$

(2)

$$3b + 1 = 2c$$

(3)

$$3c + 1 = 7a$$

Multiply equation (2) by 2 to get equation (4):(4)

$$6b + 2 = 4c$$

and substitute (1) into (4):(5)

$$9a + 5 = 4c$$

(3)

$$7a - 1 = 3c$$

Multiplying (5) by 3 and (3) by 4 gives:

$$3(9a + 5) = 4(7a - 1)$$

from which we get $a = 19$. Substituting a into (1) and (3) yields $b = 29$ and $c = 44$. Therefore, $a + b + c = \boxed{E(92)}$

§12 AMC 10 10A 2024

- a) (AMC 10 10A 2024) What is the value of $9,901 \cdot 101 - 99 \cdot 10,101$? (A) 2 (B) 20 (C) 200

Answer: A

Solution 1 (Direct Computation)

The likely fastest method will be direct computation. $9901 \cdot 101$ evaluates to 1000001 and $99 \cdot 10101$ evaluates to 999999. The difference is (A) 2.

Solution by juwushu.

Solution 2 (Distributive Property)

We have

$$\begin{aligned} 9901 \cdot 101 - 99 \cdot 10101 &= (10000 - 99) \cdot 101 - 99 \cdot (10000 + 101) \\ &= 10000 \cdot 101 - 99 \cdot 101 - 99 \cdot 10000 - 99 \cdot 101 \\ &= (10000 \cdot 101 - 99 \cdot 10000) - 2 \cdot (99 \cdot 101) \\ &= 2 \cdot 10000 - 2 \cdot 9999 \\ &= \boxed{(A) 2}. \end{aligned}$$

MRENTHUSIASM

Solution 3 (Solution 1 but Distributive)

Note that $9901 \cdot 101 = 9901 \cdot 100 + 9901 = 990100 + 9901 = 1000001$ and $99 \cdot 10101 = 100 \cdot 10101 - 10101 = 1010100 - 10101 = 999999$, therefore the answer is $1000001 - 999999 = \boxed{(A) 2}$.

Solution 4 (Modular Arithmetic)

Evaluating the given expression $(\text{mod } 10)$ yields $1 - 9 \equiv 2 \pmod{10}$, so the answer is either (A) or (D). Evaluating $(\text{mod } 101)$ yields $0 - 99 \equiv 2 \pmod{101}$. Because answer (D) is $202 = 2 \cdot 101$, that cannot be the answer, so we choose choice (A) 2.

Solution 5 (Process of Elimination)

We simply look at the units digit of the problem we have (or take mod 10)

$$9901 \cdot 101 - 99 \cdot 10101 \equiv 1 \cdot 1 - 9 \cdot 1 = 2 \pmod{10}.$$

Since the only answers with 2 in the units digit are (A) and (D), we can then continue if you are desperate to use guess and check or an actually valid method to find the answer is (A) 2.

Solution 6 (Faster Distribution)

Observe that $9901 = 9900 + 1 = 99 \cdot 100 + 1$ and $10101 = 10100 + 1 = 101 \cdot 100 + 1$

$$\begin{aligned} \Rightarrow 9901 \cdot 101 - 99 \cdot 10101 &= ((9900 \cdot 101) + (1 \cdot 101)) - ((99 \cdot 10100) + (99 \cdot 1)) \\ &= (99 \cdot 100 \cdot 101) + 101 - (99 \cdot 100 \cdot 101) - 99 \\ &= 101 - 99 \\ &= \boxed{(A) 2}. \end{aligned}$$

Solution 7 (Cubes)

Let $x = 100$. Then, we have

$$\begin{aligned} 101 \cdot 9901 &= (x + 1) \cdot (x^2 - x + 1) = x^3 + 1, \\ 99 \cdot 10101 &= (x - 1) \cdot (x^2 + x + 1) = x^3 - 1. \end{aligned}$$

Then, the answer can be rewritten as $(x^3 + 1) - (x^3 - 1) = \boxed{\text{(A) } 2}$.

Solution 8 (Super Fast)

It's not hard to observe and express 9901 into $99 \cdot 100 + 1$, and 10101 into $101 \cdot 100 + 1$.

We then simplify the original expression into $(99 \cdot 100 + 1) \cdot 101 - 99 \cdot (101 \cdot 100 + 1)$, which could then be simplified into $99 \cdot 100 \cdot 101 + 101 - 99 \cdot 100 \cdot 101 - 99$, which we can get the answer of $101 - 99 = \boxed{\text{(A) } 2}$.

Solution 9 (Estimation) *VERY FAST*

Notice that the answer choices are significantly different in value. This allows us to estimate the answer. 9901 is about 10000, and 101 is about 100. 99 is about 100, and 10101 is about 10000. Computing, we get $10000 \cdot 100 - 100 \cdot 10000 = 0$. The closest answer to this estimation is $\boxed{\text{(A) } 2}$.

Solution 10

We can see that the units digit of the expression is $1 - 9 \Rightarrow 11 - 9 \Rightarrow 2$, elimination options B, C, and E. Next, notice that $(9901)(101)$ is divisible by 101 while $(99)(10101)$ is not divisible by 101 (to see this, notice that 101 is prime, and $10101 = 10100 + 1$, so is not divisible by 101). This means that the final answer is not divisible by 101, eliminating (D), so the answer is $\boxed{\text{(A) } 2}$.

- b) (AMC 10 10A 2024) A model used to estimate the time it will take to hike to the top of the mountain on a trail is of the form $T = aL + bG$, where a and b are constants, T is the time in minutes, L is the length of the trail in miles, and G is the altitude gain in feet. The model estimates that it will take 69 minutes to hike to the top if a trail is 1.5 miles long and ascends 800 feet, as well as if a trail is 1.2 miles long and ascends 1100 feet. How many minutes does the model estimates it will take to hike to the top if the trail is 4.2 miles long and ascends 4000 feet?

(A) 240 (B) 246 (C) 252 (D) 258 (E) 264

Answer: B

Solution 1

Plug in the values into the equation to give you the following two equations:

$$\begin{aligned} 69 &= 1.5a + 800b, \\ 69 &= 1.2a + 1100b. \end{aligned}$$

Solving for the values a and b gives you that $a = 30$ and $b = \frac{3}{100}$. These values can be plugged back in showing that these values are correct. Now, using the given length of the trail, 4.2, and the given vertical increase, 4000, we get a final answer of $\boxed{\text{(B) } 246}$.

Solution by [juwushu](#).

Minor edits by ParticlePhysics and TigerSenju

Solution 2

Alternatively, observe that using $a = 10x$ and $b = \frac{y}{100}$ makes the numbers much closer to each other in terms of magnitude.

Plugging in the new variables:

$$69 = 15x + 8y,$$

$$69 = 12x + 11y.$$

The solution becomes more obvious in this way, with $15 + 8 = 12 + 11 = 23$, and since $23 \cdot 3 = 69$, we determine that $x = y = 3$.

The question asks us for $4.2a + 4000b = 42x + 40y$. Since $x = y$, we have $(40 + 42) \cdot 3 = \boxed{\text{(B) } 246}$.

- c) (AMC 10 10A 2024) Let n be the least prime number that can be written as the sum of 5 distinct prime numbers. What is the sum of the digits of n ?

(A) 5 (B) 7 (C) 8 (D) 10 (E) 13

Answer: B

Solution 1

Let the requested sum be S . Recall that 2 is the only even (and the smallest) prime, so S is odd. It follows that the five distinct primes are all odd. The first few odd primes are 3, 5, 7, 11, 13, 17, 19, $\dots$. We conclude that $S > 3 + 5 + 7 + 11 + 13 = 39$, as 39 is a composite. The next possible value of S is $3 + 5 + 7 + 11 + 17 = 43$, which is a prime. Therefore, we have $S = 43$, and the sum of its digits is $4 + 3 = \boxed{\text{(B) } 7}$.

Solution 2

We notice that the minimum possible value of the sum of 5 odd distinct primes is $3 + 5 + 7 + 11 + 13 = 39$, which is not a prime. The smallest prime greater than that is 41. However, this cannot be written as the sum of 5 distinct primes, since 15 is not prime. However, 43 can be written as $3 + 5 + 7 + 11 + 17 = 43$, so the answer is $4 + 3 = \boxed{\text{(B) } 7}$.

- d) (AMC 10 10A 2024) The number 2024 is written as the sum of not necessarily distinct two-digit numbers. What is the least number of two-digit numbers needed to write this sum?

(A) 20 (B) 21 (C) 22 (D) 23 (E) 24

Answer: B

Solution 1

Since we want the least number of two-digit numbers, we maximize the two-digit numbers by choosing as many 99s as possible. Since $2024 = 99 \cdot 20 + 44 \cdot 1$, we choose twenty 99s and one 44, for a total of $\boxed{\text{(B) } 21}$ two-digit numbers.

Solution 2

We claim the answer is 21. This can be achieved by adding twenty 99's and a 44. To prove that the answer cannot be less than or equal to 20, we note that the maximum value of the sum of 20 or less two digit numbers is $20 \cdot 99 = 1980$, which is smaller than 2024, so we are done. Thus, the answer is $\boxed{\text{(B) } 21}$.

Solution 3 (Same as Solution 1 but Using $100=99+1$)

$2024 = 100 \cdot 20 + 24$. Since $100 = 99 + 1$, $2024 = (99 + 1) \cdot 20 + 24 = 99 \cdot 20 + 1 \cdot 20 + 24 = 99 \cdot 20 + 44$. Therefore a total of (B) 21 two-digit numbers are needed.

Solution 4

The maximum 2-digit number is 99, but try 100. $2024 \div 100$ is a little more than 20, and the remainder is less than 100, by intuition, so there's 20+ the remainder 1 = (B) 21.

- e) (AMC 10 10A 2024) What is the least value of n such that $n!$ is a multiple of 2024?

(A) 11 (B) 21 (C) 22 (D) 23 (E) 253

Answer: D

Solution

Note that $2024 = 2^3 \cdot 11 \cdot 23$ in the prime factorization. Since $23!$ is a multiple of 2^3 , 11, and 23, we conclude that $23!$ is a multiple of 2024. Therefore, we have $n = \text{span style="border: 1px solid black; padding: 0 2px;">(D) 23.$

Remark

Memorizing the prime factorization of the current year is useful for the AMC 8/10/12 Exams.

Remark

Note that 2024 is $2025 - 1$, which is $45^2 - 1 = (45 + 1)(45 - 1) = 46 \cdot 44$.

- f) (AMC 10 10A 2024) What is the minimum number of successive swaps of adjacent letters in the string $ABCDEF$ that are needed to change the string to $FEDCBA$? (For example, 3 swaps are required to change ABC to CBA ; one such sequence of swaps is $ABC \rightarrow BAC \rightarrow BCA \rightarrow CBA$.)

(A) 6 (B) 10 (C) 12 (D) 15 (E) 24

Answer: D

Solution 1 (Analysis)

Procedurally, it takes:

- 5 swaps for A to move to the sixth spot, giving $BCDEFA$.
- 4 swaps for B to move to the fifth spot, giving $CDEFBA$.
- 3 swaps for C to move to the fourth spot, giving $DEFCBA$.
- 2 swaps for D to move to the third spot, giving $EFDCBA$.
- 1 swap for E to move to the second spot (so F becomes the first spot), giving $FEDCBA$.

Together, the answer is $5 + 4 + 3 + 2 + 1 = \text{span style="border: 1px solid black; padding: 0 2px;">(D) 15.$

Solution 2 (Recursive Approach)

We can proceed by a recursive tactic on the number of letters in the string.

Looking at the string A , there are 0 moves needed to change it to the string A .

Then, there is 1 move to change AB to BA .

Similarly, there is 3 moves needed for three letters (said in the problem).

There are 6 moves needed to change $ABCD$ to $DCBA$.

We see a pattern of 0, 1, 3, 6, We notice that the difference between consecutive terms is increasing by 1, so in the same way, for 5 letters, we would need 10 moves, and for 6, we would need (D) 15 moves.

Thinking why, when we start making these moves, we see that for a string of length n , it takes $n - 1$ moves to move the last letter to the front. After, we get a string that will be changed identically to a string of length $n - 1$. This works in our pattern above and is another way to think about the problem!

Note:

The sequence consists of triangular numbers shifted a term up (as it starts with 0 on term 1 and 1 on term 2.) Thus, our explicit formula is

$$\frac{(n-1)(n+1-1)}{2} = \frac{(n)(n-1)}{2}$$

and as $n = 6$ in this case (6 letters), our answer is $\frac{(6)(6-1)}{2} = \text{(D) 15}$

Solution 3 (Solution 2 Done Fast)

We can see that the most efficient way to change $ABCDEF$ to $FEDCBA$ is the same as changing $ABCDE$ to $EDCBA$ and then moving F to the front in 5 moves. Similarly, this would carry on downwards, where to change $ABCDE$ to $EDCBA$ would be to change $ABCD$ to $DCBA$ and move E 4 times to the front. This pattern will carry on until AB to BA would be 1, and A to A would be 0. The answer would be $0(A) + 1(B) + 2(C) + 3(D) + 4(E) + 5(F)$, which is (D) 15 moves.

Solution 4 (If you don't notice anything)

Simply, just blindly doing it, the most straightforward way is to shift F all the way back. From $ABCDEF$:

$ABCDFE \rightarrow ABCFDE \rightarrow ABFCDE \rightarrow AFBCDE \rightarrow FABCDE$

For E :

$FABCED \rightarrow FABECD \rightarrow FAEB CD \rightarrow FEABCD$

For D :

$FEABDC \rightarrow FEADBC \rightarrow FEDABC$

For C :

$FEDACB \rightarrow FEDCAB$

For B :

$FEDCBA$

That's it, you don't need to do it with A . Still (D) 15 moves.

Solution 5 (Inversions of Permutations)

Let the letters A, B, C, D, E, F correspond to the numbers 1, 2, 3, 4, 5, 6 respectively. We want to find the number of swaps required to transform the permutation 123456 into the permutation 654321.

Here, we let 123456 be the natural order. Then the total number of inversions of the permutation 654321 is $1 + 2 + 3 + 4 + 5 = 15$. Hence the answer is **(D) 15**.

- g) (AMC 10 10A 2024) The product of three integers is 60. What is the least possible positive sum of the three integers?

(A) 2 (B) 3 (C) 5 (D) 6 (E) 13

Answer: B

Solution 1

We notice that the optimal solution involves two negative numbers and a positive number. Thus we may split 60 into three factors and choose negativity. We notice that $10 \cdot 6 \cdot 1 = 10 \cdot (-6) \cdot (-1) = 60$, and trying other combinations does not yield lesser results so the answer is $10 - 6 - 1 = \mathbf{(B) 3}$.

Solution 2

We have $abc = 60$. Let a be positive, and let b and c be negative. Then we need $a > |b + c|$. If $a = 6$, then $|b + c|$ is at least 7, so this doesn't work. If $a = 10$, then $(b, c) = (-6, -1)$ works, giving $10 - 7 = \mathbf{(B) 3}$.

Solution 3

We can see that the most optimal solution would be 1 positive integer and 2 negative ones (as seen in solution 1). Let the three integers be x, y , and z , and let x be positive and y and z be negative. If we want the optimal solution, we want the negative numbers to be as large as possible. so the answer should be $-60 \cdot -1 \cdot 1$, where $-60 - 1 + 1 = -60...$ right?

No! Our sum must be positive, so that would be invalid! We see that $-30, -20, -15$, and -10 are too less to allow the sum to be positive. For example, $-15 = z$. $-15xy = 60$, so $xy = -4$. For xy to be the most positive, we will have 4 and -1 . Yet, $-15 + 4 - 1$ is still negative. After -10 , the next factor of 60 would be -6 . if $z = -6$, $xy = -10$. This might be positive! Now, if we have $z = -6, y = -1$, and $x = 10, x + y + z = 3$. It cannot be smaller because $x = 5$ and $y = -2$ would result in $x + y + z$ being negative. Therefore, our answer would be $\mathbf{(B) 3}$.

Tiny minor edits made by SuperVince1

- h) (AMC 10 10A 2024) Amy, Bomani, Charlie, and Daria work in a chocolate factory. On Monday Amy, Bomani, and Charlie started working at 1 : 00PM and were able to pack 4, 3, and 3 packages, respectively, every 3 minutes. At some later time, Daria joined the group, and Daria was able to pack 5 packages every 4 minutes. Together, they finished packing 450 packages at exactly 2 : 45PM. At what time did Daria join the group?

(A) 1 : 25 PM (B) 1 : 35 PM (C) 1 : 45 PM (D) 1 : 55 PM (E) 2 : 05 PM

Answer: A

Solution 1

Note that Amy, Bomani, and Charlie pack a total of $4 + 3 + 3 = 10$ packages every 3 minutes.

The total amount of time worked is 1 hour and 45 minutes, which when converted to minutes, is 105 minutes. This means that since Amy, Bomani, and Charlie worked for the entire 105 minutes, they in total packed $\frac{105}{3} \cdot 10 = 350$ packages.

Since 450 packages were packed in total, then Daria must have packed $450 - 350 = 100$ packages in total, and since she packs at a rate of 5 packages per 4 minutes, then Daria worked for $\frac{100}{5} \cdot 4 = 80$ minutes, therefore Daria joined 80 minutes before 2 : 45 PM, which was at **(A) 1 : 25 PM**.

Solution 2

Let the time, in minutes, elapsed between 1 : 00 and the time Daria joined the packaging be x . Since Amy packages 4 packages every 3 minutes, she packages $\frac{4}{3}$ packages per minute. Similarly, we can see that both Bomani and Charlie package 1 package per minute, and Daria packages $\frac{5}{4}$ packages every minute.

Before Daria arrives, we can write the total packages packaged as $x(\frac{4}{3} + 1 + 1) = x(\frac{10}{3})$. Since there are 105 minutes between 1 : 00 and 2 : 45, Daria works with the other three for $105 - x$ minutes, meaning for that time there are $(105 - x)(\frac{4}{3} + 1 + 1 + \frac{5}{4}) = (105 - x)(\frac{55}{12})$ packages packaged.

Adding the two, we get $x(\frac{10}{3}) + (105 - x)(\frac{55}{12}) = 450$ (The total packaged in the entire time is 450). Solving this equation, we get $x = 25$, meaning Daria arrived 25 minutes after 1 : 00, meaning the answer is **(A) 1 : 25 PM**.

- i) (AMC 10 10A 2024) In how many ways can 6 juniors and 6 seniors form 3 disjoint teams of 4 people so that each team has 2 juniors and 2 seniors?

(A) 720 (B) 1350 (C) 2700 (D) 3280 (E) 8100

Answer: B

Solution 1

The number of ways in which we can choose the juniors for the team are $\binom{6}{2}\binom{4}{2}\binom{2}{2} = 15 \cdot 6 \cdot 1 = 90$. Similarly, the number of ways to choose the seniors are the same, so the total is $90 \cdot 90 = 8100$. But we must divide the number of permutations of three teams, since the order in which the teams were chosen doesn't matter, which is $3!$. Thus, the answer is $\frac{8100}{3!} = \frac{8100}{6} = \mathbf{(B) 1350}$

Solution 2

Let's first suppose that the three teams are distinguishable. The number of ways to choose the first team of two juniors and two seniors is $\binom{6}{2}\binom{6}{2} = 15 \cdot 15 = 225$. Now, only four juniors and four seniors are left to choose the second team. Thus, the second team can be formed in $\binom{4}{2}\binom{4}{2} = 6 \cdot 6 = 36$ ways. There are now only two juniors and two seniors left, so the third team can only be formed in one way. Thus, the total number of ways in which we can choose two juniors and two seniors for three distinguishable teams is $225 \cdot 36 = 8100$ ways. However, the problem does not require the teams to be distinguishable. Therefore, we must divide by $3! = 6$ to find the answer for three indistinguishable teams.

The answer is $\frac{8100}{6} = \mathbf{(B) 1350}$

- j) (*AMC 10 10A 2024*) Consider the following operation. Given a positive integer n , if n is a multiple of 3, then you replace n by $\frac{n}{3}$. If n is not a multiple of 3, then you replace n by $n + 10$. For example, beginning with $n = 4$, this procedure gives $4 \rightarrow 14 \rightarrow 24 \rightarrow 8 \rightarrow 18 \rightarrow 6 \rightarrow 2 \rightarrow 12 \rightarrow \dots$. Suppose you start with $n = 100$. What value results if you perform this operation exactly 100 times?

(A) 10 (B) 20 (C) 30 (D) 40 (E) 50

Answer: C

Solution 1 (Fast Solution)

Let s be the number of times the operation is performed. Notice the sequence goes $100 \rightarrow 110 \rightarrow 120 \rightarrow 40 \rightarrow 50 \rightarrow 60 \rightarrow 20 \rightarrow 30 \rightarrow 10 \rightarrow 20 \rightarrow \dots$. Thus, for $s \equiv 1 \pmod{3}$, the value is 30. Since $100 \equiv 1 \pmod{3}$, the answer is **(C) 30**.

Solution 2 (More Explanatory)

Looking at the first few values of our operation, we get $100 \rightarrow 110 \rightarrow 120 \rightarrow 40 \rightarrow 50 \rightarrow 60 \rightarrow 20 \rightarrow 30 \rightarrow 10 \rightarrow 20$. We can see that 30 goes to 10, then to 20, then back to 30, and the loop resets. After 7 operations, we reach 30. We still have 93 operations left, so because the loop will run exactly 31 times ($93/3$), we will reach 30 again. So, the answer is **(C) 30**.

Solution 3 (very slightly different than previous)

Calculating the first few values, we get $100 \rightarrow 110 \rightarrow 120 \rightarrow 40 \rightarrow 50 \rightarrow 60 \rightarrow 20 \rightarrow 30 \rightarrow 10 \rightarrow 20$. We can see that 20 will go to 30, then to 10, then back to 20, and then the loop resets. After 6 moves, we reach 20, the start of the cycle. We still have $100 - 5$ moves to go, so to find what number we land on after 95 more steps, we can do $95 \pmod{3} \equiv (9 + 4) \pmod{3} = 1$, meaning we go from 20 $\rightarrow$ **(C) 30**.

- k) (*AMC 10 10A 2024*) How many ordered pairs of integers (m, n) satisfy $\sqrt{n^2 - 49} = m$?

(A) 1 (B) 2 (C) 3 (D) 4 (E) infinitely many

Answer: D

Solution 1

Note that m is a nonnegative integer.

We square, rearrange, and apply the difference of squares formula to the given equation:

$$(n + m)(n - m) = 49.$$

It is clear that $n + m \geq n - m$, so $(n + m, n - m) = (49, 1), (7, 7), (-7, -7), (-1, -49)$. Each ordered pair $(n + m, n - m)$ gives one ordered pair (m, n) , so there are **(D) 4** such ordered pairs (m, n) . **Remark**

From $(n + m, n - m) = (49, 1), (7, 7), (-7, -7), (-1, -49)$, we get $(m, n) = (24, 25), (0, 7), (0, -7), (24, -25)$, respectively.

Solution 2

Squaring both sides of the given equation gives

$$n^2 - 49 = m^2 \rightarrow n^2 - m^2 = 49 \rightarrow (n + m)(n - m) = 49$$

Splitting 49 into its factors (keep in mind it doesn't ask for positive integers, so the factors can be double negative, too) gives six cases:

$$(1 \cdot 49) (7 \cdot 7) (49 \cdot 1) (-1 \cdot -49) (-7 \cdot -7) (-49 \cdot -1).$$

Note that the square root in the problem doesn't have $\pm$ with it. Therefore, if there are two solutions, (n, m) and $(n, -m)$, then these together are to be counted as one solution. The solutions expressed as (n, m) are:

$$(25, 24) (25, -24) (7, 0) (-7, 0) (-25, 24) (-25, -24).$$

$(25, 24)$ and $(25, -24)$ are to be counted as one, same for $(-25, 24)$ and $(-25, -24)$. Therefore, the solution is (D) 4.

Solution 3 (Crazy Rush)

From looking at the problem, it's obvious that $(\pm 7, 0)$ are already solutions. From squaring and rearranging, we get

$$n^2 - m^2 = 49.$$

We know that the difference between two consecutive squares is always odd, and for each pair of increasing consecutive squares, the difference starts from 3 and increases by 2 each time. This means that there is an existing pair, $(\pm n, m)$ of consecutive squares that will satisfy this equation.

Also note that the answer cannot be infinity because the difference between two squares will increase as the two squares get higher, consecutive or not.

Therefore, the solutions $(\pm 7, 0)$ and $(\pm n, m)$ where n and m are consecutive that have a square difference of 49, give the answer of (D) 4.

Minor Note: Looking at the problem, it is obvious that -7 and 7 are solutions and you can eliminate 1, 3, and infinite. You can assume that there is more than 2 solutions because when determining square expressions, these problems usually always have more than an obvious solution.

Solution 4 (Quick)

Clearly, $(7, 0)$ and $(-7, 0)$ are solutions. Notice that $49 = 7^2$. Remembering our Pythagorean triples, we realize $(25, 24)$ is a solution as well, and by extension, so is $(-25, 24)$. Since 7 is not part of any other Pythagorean triple, we can conclude the answer is (D) 4.

Solution 5 (idk why someone would ever use this)

Since in $n^2 - 49$ there are no restrictions of the sign of n , we know that the solution will come in pairs, with negative and positive n . Therefore, the answer will not be 1 or 3. We can further confirm that it is not 1, since the problem would probably ask for an operation of n and m instead of the number of solutions. We can also guess it is not 2, since then the problem would ask for the sum of one of the variables in the solutions. It seems improbable that over the integers there would be infinite answers. So the answer is (D) 4

(do not ever use this)

Solution 6 (Pythagorean Theorem)

Rearranging the equation gives

$$m^2 + 7^2 = n^2$$

This is a classic Pythagorean Theorem equation, where 7 is a leg and m and n are the two remaining legs. All we have to do is to find triples that satisfy this equation. It is common knowledge that $(7, 24, 25)$ is a triple, thus $(24, 25)$ and $(24, -25)$ are two solutions. Another one is $(0, 7, 7)$, so $(0, 7)$ and $(0, -7)$ are two other solutions. We can be sure that there are no other triples to find because 4 is the highest answer choice, otherwise it would be "infinitely many", which isn't plausible. Therefore, the answer is **(D) 4**.

-urAntimony (minor edit)

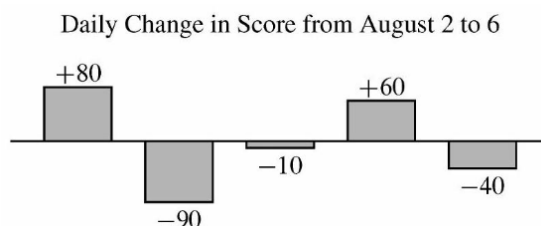
Solution 7

Square both sides giving $n^2 - 49 = m^2$, then rearrange the equation into a quadratic: $n^2 - 49 - m^2 = 0$. then find the Delta ($b^2 - 4ac$) where a, b , and c are the coefficients of the quadratic in the respective order. Calculating it we get $2401 - 4 = 2397$, since this is a positive number it means there are two solutions but both solutions can be both m or n , so the answer is **(D) 4**.

Solution 8 (Graphing)

We can rewrite the equation as $y = \sqrt{x^2 - 49}$ and graph it to find integer solutions. It is a hyperbola of standard form $\frac{x^2}{49} - \frac{y^2}{49} = 1$. Note that the graph of the function does not exist below the y -axis. We find that where $|x| > 25$ integer solutions are impossible, since the difference between the function and its asymptote $y = x$ in that region is less than one and always decreasing; thus, since $y = x$ is always an integer, $\sqrt{x^2 - 49}$ must be a fractional amount less than that integer, so we can rule out option E (infinite integer solutions). We notice that the roots are a solution $(\pm 7, 0)$, as well as $(\pm 25, 24)$, making four solutions. 4 is already the maximum possible answer choice (having ruled out E), so we know there are no more solutions. Thus, the answer is **(D) 4**.

- 1) (AMC 10 10A 2024) Zelda played the *Adventures of Math* game on August 1 and scored 1,700 points. She continued to play daily over the next 5 days. The bar chart below shows the daily change in her score compared to the day before. (For example, Zelda's score on August 2 was $1,700 + 80 = 1,780$ points.) What was Zelda's average score in points over the 6 days?



- (A) 1700 (B) 1702 (C) 1703 (D) 1713 (E) 1715

Answer: E

Solution 1

Going through the table, we see her scores over the six days were: 1700, $1700 + 80 = 1780$, $1780 - 90 = 1690$, $1690 - 10 = 1680$, $1680 + 60 = 1740$, and $1740 - 40 = 1700$.

Taking the average, we get $\frac{(1700+1780+1690+1680+1740+1700)}{6} = \boxed{\text{(E) } 1715}$.

-i am suk at math_2

Solution 2

Compared to the first day (1700), her scores change by +80, -10, -20, +40, and +0. So, the average is $1700 + \frac{80-10-20+40+0}{6} = \boxed{\text{(E) } 1715}$.

Sidenote: Many people would just add up the changes, which would give them 1700. That is wrong.

-mathfun2012

-JK2030 minor edit

Solution 3

As the scores of each day are dependent on previous days, we get: $1700 + 0 \cdot 6 + 80 \cdot 5 + (-90) \cdot 4 + (-10) \cdot 3 + 60 \cdot 2 + (-40) \cdot 1 = \boxed{\text{(E) } 1715}$

Solution 4

Plugging in x as 1700, we get the variables as: $x + 80$, $x-10$, $x-20$, $x+40$, and x . Finding the average of this gets us $(6x + 90)/6 = x + 15$. Since x is 1700 we get our answer as $1700 + 15 = \boxed{\text{(E) } 1715}$.

m) (*AMC 10 10A 2024*) Two transformations are said to commute if applying the first followed by the second gives the same result as applying the second followed by the first. Consider these four transformations of the coordinate plane:

- a translation 2 units to the right,
- a 90° -rotation counterclockwise about the origin,
- a reflection across the x -axis, and
- a dilation centered at the origin with scale factor 2.

Of the 6 pairs of distinct transformations from this list, how many commute?

(A) 1 (B) 2 (C) 3 (D) 4 (E) 5

Answer: C

Solution 1 (Generalized)

Label the given transformations T_1, T_2, T_3 , and T_4 , respectively. The rules of transformations are:

- $T_1 : (x, y) \rightarrow (x + 2, y)$
- $T_2 : (x, y) \rightarrow (-y, x)$
- $T_3 : (x, y) \rightarrow (x, -y)$
- $T_4 : (x, y) \rightarrow (2x, 2y)$

Note that:

- Applying T_1 and then T_2 gives $(x, y) \rightarrow (x+2, y) \rightarrow (-y, x+2)$. Applying T_2 and then T_1 gives $(x, y) \rightarrow (-y, x) \rightarrow (-y+2, x)$.

Therefore, T_1 and T_2 do not commute. One counterexample is the preimage $(0, 0)$.

- Applying T_1 and then T_3 gives $(x, y) \rightarrow (x+2, y) \rightarrow (x+2, -y)$. Applying T_3 and then T_1 gives $(x, y) \rightarrow (x, -y) \rightarrow (x+2, -y)$.

Therefore, T_1 and T_3 commute. They form a **glide reflection**.

- Applying T_1 and then T_4 gives $(x, y) \rightarrow (x+2, y) \rightarrow (2x+4, 2y)$. Applying T_4 and then T_1 gives $(x, y) \rightarrow (2x, 2y) \rightarrow (2x+2, 2y)$.

Therefore, T_1 and T_4 do not commute. One counterexample is the preimage $(0, 0)$.

- Applying T_2 and then T_3 gives $(x, y) \rightarrow (-y, x) \rightarrow (-y, -x)$. Applying T_3 and then T_2 gives $(x, y) \rightarrow (x, -y) \rightarrow (y, x)$.

Therefore, T_2 and T_3 do not commute. One counterexample is the preimage $(1, 0)$.

- Applying T_2 and then T_4 gives $(x, y) \rightarrow (-y, x) \rightarrow (-2y, 2x)$. Applying T_4 and then T_2 gives $(x, y) \rightarrow (2x, 2y) \rightarrow (-2y, 2x)$.

Therefore, T_2 and T_4 commute.

- Applying T_3 and then T_4 gives $(x, y) \rightarrow (x, -y) \rightarrow (2x, -2y)$. Applying T_4 and then T_3 gives $(x, y) \rightarrow (2x, 2y) \rightarrow (2x, -2y)$.

Therefore, T_3 and T_4 commute.

Together, (C) 3 pairs of transformations commute: T_1 and T_3 , T_2 and T_4 , and T_3 and T_4 . **Remark**

To show that two transformations do not commute, we only need one counterexample.

Solution 2 (Specific)

Label the transformations as follows:

- a translation 2 units to the right (W)
- a 90° -rotation counterclockwise about the origin (X)
- a reflection across the x -axis (Y)
- a dilation centered at the origin with scale factor 2 (Z)

Now, examine each possible pair of transformations with the point $(1, 0)$:

1. W and X . $W \rightarrow X$ ends with the point $(0, 3)$. Going $X \rightarrow W$ ends in the point $(1, 2)$, so this pair does not work
2. W and Y . $W \rightarrow Y$ gives the point $(3, 0)$, and going $Y \rightarrow W$ ends in the same point. This pair is valid.
3. W and Z . $W \rightarrow Z$ ends in the point $(6, 0)$, while going the other way gives $(4, 0)$. This pair isn't commute.
4. X and Y . $X \rightarrow Y$ gives the point $(0, -1)$, while the other way gives $(0, 1)$. Not a valid pair
5. X and Z . $X \rightarrow Z$ ends in the point $(0, 2)$, and $Z \rightarrow X$ also ends in $(0, 2)$. This pair works.
6. Y and Z . $Y \rightarrow Z$ gives the point $(2, 0)$, and going the other way also ends in $(2, 0)$. This pair is valid.

Therefore, the answer is (C) 3.

Note: It is easier to just visualize this problem instead of actually calculating points on paper.

Solution 3 (Specific)

Label the transformations as follows:

- a translation 2 units to the right (A)
- a 90° -rotation counterclockwise about the origin (B)
- a reflection across the x -axis (C)
- a dilation centered at the origin with scale factor 2 (D)

Now, we count each transformation individually. It is not hard to see that AC , BD , and CD are commutative (an easy way to test commutativity for some cases would be to have the original point on the x -axis).

In total, (C) 3 transformation pairs commute.

Solution 4 (Quick With Reasoning)

Alright so realize that obviously rotation and dilation are commute, if you ever looked at the polar plane. Also, reflecting around the x -axis means that moving to the right two units and dilating are commute. You can easily see this by taking an arbitrary point and applying the transformations. The rest don't work since you can easily test out using random points on the plane, which means the answer is (C) 3.

- n) (AMC 10 10A 2024) One side of an equilateral triangle of height 24 lies on line ℓ . A circle of radius 12 is tangent to line ℓ and is externally tangent to the triangle. The area of the region exterior to the triangle and the circle and bounded by the triangle, the circle, and line ℓ can be written as $a\sqrt{b} - c\pi$, where a , b , and c are positive integers and b is not divisible by the square of any prime. What is $a + b + c$?

(A) 72 (B) 73 (C) 74 (D) 75 (E) 76

Answer: D

Solution 1

Call the bottom vertices of the triangle B and C (the one closer to the circle is C) and the top vertex A . The tangency point between the circle and the side of the triangle is D , and the tangency point on line ℓ E , and the center of the circle is O .

Draw radii to the tangency points, the arc is 60 degrees because $\angle ACB$ is 60, and since $\angle DCE$ is supplementary, it's 120° . (Using Angle of Intersecting Secants Theorem) The sum of the angles in a quadrilateral is 360, which means $\angle DOE$ is 60°

Triangle ODC is 30-60-90 triangle so CD is $4\sqrt{3}$. Since we have 2 congruent triangles ($\triangle ODC$ and $\triangle OEC$), the combined area of both is $48\sqrt{3}$. The area of the arc is $144 \cdot \frac{60}{360} \cdot \pi$ which is 24π , so the area of the region is

$$48\sqrt{3} - 24\pi$$

.

$a + b + c$ is $48 + 3 + 24$, which is (D) 75

Note There were two possible configurations from this problem; the one described in the solution above and the configuration in which the circle is tangent to the bottom of line ℓ and the base of the equilateral triangle. However, since the area in this configuration is simply 0, we can infer that the problem is talking about the configuration in Solution 1.

Through 30 – 60 – 90 triangles, we see that length

Therefore the side length of AMC is simply $16\sqrt{3}$.

We have that $BD = BE = 12$. we see that $\angle BDE = \angle BED = x^\circ$. We have that $\angle DBE = 60^\circ$. Therefore,

$x = 60^\circ$

Additionally, the area of sector BDE is $\frac{1}{6} \cdot (12)^2 \cdot \pi$, or 24π .

In this case our diagonals are DE and CB. We already know $DE = 12$, but to find CB, we construct a triangle ABC , with $AC = 16\sqrt{3}$. The diagonals of a kite bisect their respective angles. Therefore, $\angle ACB = \frac{120^\circ}{2} = 60^\circ$.

Then, triangles ABC and BDC are similar through a common angle and common side (with the angle of course in-between the side).

Applying the same logic to triangle AFC , one can see that quadrilateral $AFCB$ is also a kite that is similar to kite $BDCE$, and therefore $\angle ABC = 90^\circ$.

Through $30 - 60 - 90$ triangles, $BC = \frac{1}{2}AC$, or $8\sqrt{3}$.

The area of kite $BCDE$ is $[BCDE] = \frac{8\sqrt{3} \cdot 12}{2}$, or $48\sqrt{3}$.

We need $[BCDE] - [BDE]$, or $48\sqrt{3} - 24\pi$.

This is in the form $a\sqrt{b} - c\pi$, with $a = 48$, $b = 3$, and $c = 24$.

We need $a + b + c$, which is $48 + 3 + 24 = \boxed{(D)75}$

Solution 3 (Quick Guess)

This is kind of a

Since this problem involves equilateral triangles, the only possible number under the square root is 3. Now subtracting all of the answer choices by 3, we get:

$$(A) 72-3 = 69 \quad (B) 73-3 = 70 \quad (C) 74-3 = 71 \quad (D) 75-3 = 72 \quad (E) 76-3 = 73$$

Due to the even parity of the problem, we can safely assume that the answer is either B or D , but as D is a multiple of 12 and 24, we get the answer of

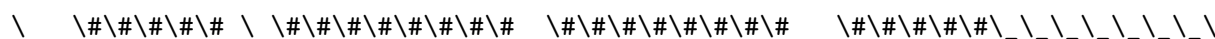
$\boxed{(D) 75}$.

Solution 4

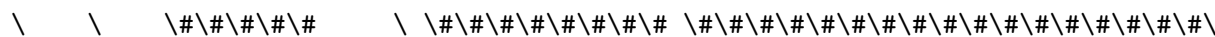
(pardon the diagrams :D)

say the area we want to find is x .

since the equilateral triangle has an internal angle of 60 , the exterior angle formed by the triangle and the line is 120 . simplifying the diagram you will get:



make three of these so that each circle is tangent to the other 2 circles



Since they are 3 congruent triangles, you can make an equilateral triangle using their radius(12), with each vertex at the center of each circle. This will make an equilateral triangle of side length 24. if you look now, the area within the equilateral triangle consists of 3 $60/360$ of a circle, and 3 of x .

we first find the area of the triangle, which is $24 * 12\sqrt{3}$, we then find the area of $60/360$ of a circle, which is $60/360 * 12^2\pi$, we subtract $24 * 12\sqrt{3}$ by $60/360 * 12^2\pi$, and divide by 3, yielding the area of x .

$$\$24 * 12\sqrt{3} - \$60/360 * 12^2\pi$$

$$= 48\sqrt{3} -$$

$$24\pi$$

$$3$$

$a + b + c$ is $48 + 3 + 24$ which is (D) 75

Solution 5 (oh no)

Setting up the problem graphically:

Define the parametric function of the circle $r(\theta)$, centered at $(0, 12)$ with radius 12, as:

$$r(\theta) = (12 \cos(\theta), 12 \sin(\theta) + 12).$$

The line ℓ is taken to be the x -axis, so $y = 0$. The circle is tangent to ℓ at $(0, 0)$.

We only care about the side of the triangle that is also tangent to the circle; call this line $f(x)$. Since the triangle is equilateral of height 24, one vertex is at $(0, 24)$ (directly above the tangent point), but for now we focus on the particular side tangent to the circle.

Because we expect that line to make a 330° angle with the positive x -axis (or a slope of $-\sqrt{3}$), let

$$f(x) = -\sqrt{3}x + c.$$

We also know the perpendicular distance from the circle's center $(0, 12)$ to $f(x)$ must be 12, because the circle of radius 12 is externally tangent to that line. The distance from a point (x_0, y_0) to a line $Ax + By + C = 0$ is:

$$\frac{|Ax_0 + By_0 + C|}{\sqrt{A^2 + B^2}}.$$

Here, $f(x) = -\sqrt{3}x + c$ can be rewritten as $\sqrt{3}x + y - c = 0$. Plugging in $(x_0, y_0) = (0, 12)$ and setting the distance to 12 yields:

$$\frac{|\sqrt{3} \cdot 0 + 12 - c|}{\sqrt{(\sqrt{3})^2 + 1^2}} = 12.$$

Solving this and noting that $f(x)$ has a negative y -intercept gives:

$$f(x) = -\sqrt{3}x - 12.$$

Next, we find the point at which $r(\theta)$ intersects $f(x)$. Substituting $r(\theta) = (12 \cos(\theta), 12 \sin(\theta) + 12)$ into $f(x) = -\sqrt{3}x - 12$, we set:

$$12 \sin(\theta) + 12 = -\sqrt{3}(12 \cos(\theta)) - 12.$$

One helpful approach is to note that a line perpendicular to $f(x)$ and passing through $(0, 12)$ has slope $1/\sqrt{3}$, so that line is $g(x) = \frac{x}{\sqrt{3}} + 12$. Intersecting $g(x)$ with $f(x)$ gives:

$$\frac{x}{\sqrt{3}} + 12 = -\sqrt{3}x - 12,$$

which solves to $x = -6\sqrt{3}$. Then $y = 6$. So the intersection point of the circle and the line is $(-6\sqrt{3}, 6)$.

To find the corresponding θ , use:

$$12 \cos(\theta) = -6\sqrt{3}, \quad 12 \sin(\theta) + 12 = 6.$$

From the first, $\cos(\theta) = -\sqrt{3}/2$. This corresponds to $\theta = 5\pi/6$ or $7\pi/6$. Checking the y -coordinate shows $\theta = 7\pi/6$ indeed yields $\sin(7\pi/6) = -1/2$, so $12 \sin(7\pi/6) + 12 = 6$.

The circle also meets ℓ (the x -axis) at $(0, 0)$. In polar terms, that is $\theta = 3\pi/2$. We want the area under the circle's parametric curve from $\theta = 7\pi/6$ to $\theta = 3\pi/2$.

Parametrically, the area under $y(\theta)$ from θ_1 to θ_2 can be found via:

$$\int_{\theta_1}^{\theta_2} y(\theta) x'(\theta) d\theta.$$

Here $x(\theta) = 12 \cos(\theta)$, so $x'(\theta) = -12 \sin(\theta)$. Hence the area under the circle from $\theta = 7\pi/6$ to $\theta = 3\pi/2$ is:

$$\int_{7\pi/6}^{3\pi/2} (12 \sin(\theta) + 12) (-12 \sin(\theta)) d\theta.$$

This simplifies (pulling out constants) to:

$$-12 \int_{7\pi/6}^{3\pi/2} (\sin^2(\theta) + \sin(\theta)) d\theta = -12 \left[\int_{7\pi/6}^{3\pi/2} \sin^2(\theta) d\theta + \int_{7\pi/6}^{3\pi/2} \sin(\theta) d\theta \right].$$

Evaluating these integrals yields:

$$54\sqrt{3} - 24\pi.$$

Next, find where $f(x) = 0$:

$$-\sqrt{3}x - 12 = 0 \implies x = -\frac{12}{\sqrt{3}} = -4\sqrt{3}.$$

However, we also need the relevant intersection interval for the region. The line and the circle intersect at $x = -6\sqrt{3}$. So the area under $f(x)$ from $x = -6\sqrt{3}$ to $x = -4\sqrt{3}$ is:

$$\int_{x=-6\sqrt{3}}^{-4\sqrt{3}} (-\sqrt{3}x - 12) dx.$$

Carrying out that integral (or carefully checking the geometry) gives $6\sqrt{3}$.

Subtracting this triangular “cap” ($6\sqrt{3}$) from the circle's sector area ($54\sqrt{3} - 24\pi$) gives the final region of interest:

$$(54\sqrt{3} - 24\pi) - 6\sqrt{3} = 48\sqrt{3} - 24\pi.$$

Hence the area can be written as

$$a\sqrt{b} - c\pi = 48\sqrt{3} - 24\pi,$$

so $a = 48$, $b = 3$, and $c = 24$. Therefore,

$$a + b + c = 48 + 3 + 24 = 75.$$

D)75

P.S.: Please don't whip out calculus like this on an AMC 10. Yes, it's "doable" without a calculator (I speak from painful experience). Yes, I did it, but it took forever, and trust me, nobody at the AMC office expects calculus. Also, this is a terrible method

Solution 6 (Pure Geometry Approach — Region DCE minus Circular Segment)

- Step 1: Area of triangle DCE

- Given: $DE = 12$ (radius of the circle), and the corresponding height from C to DE is $2\sqrt{3}$. - The area formula for a triangle is:

$$[S_{\{\triangle DCE\}} = \frac{1}{2} \times \text{base} \times \text{height} = \frac{1}{2} \times 12 \times 2\sqrt{3} = 12\sqrt{3}]$$

- Explanation: This step finds the area of triangle DCE using its base and height.

—

- Step 2: Area of sector ODE

- The radius is 12, and the central angle is 60° . - The formula for the area of a sector is:

$$[S_{\{\text{sector } ODE\}} = \frac{60}{360} \times \pi \times 12^2 = 24\pi]$$

- Explanation: The sector's area is a fraction ($\frac{1}{6}$) of the full circle's area.

—

- Step 3: Area of triangle ODE

- $OD = OE = 12$, and the included angle $\angle DOE = 60^\circ$. - The formula for an equilateral triangle's area:

$$[S_{\{\triangle ODE\}} = \frac{\sqrt{3}}{4} \times 12^2 = 36\sqrt{3}]$$

- Explanation: Triangle ODE is equilateral, so we use the standard area formula for such a triangle.

—

- Step 4: Area of the circular segment (arc DE minus chord DE)

$$[S_{\{\text{segment}\}} = S_{\{\text{sector}\}} - S_{\{\triangle ODE\}} = 24\pi - 36\sqrt{3}]$$

- Explanation: The segment is the region between the arc and the chord DE .

- Step 5: Final required region

$$S_{\text{required}} = S_{\triangle DCE} - S_{\text{segment}} = 12\sqrt{3}$$

- Explanation: The required area is the triangle DCE minus the circular segment. This gives the area in the desired region bounded by the triangle, the circle, and the line.

- Step 6: Answer in the form $a\sqrt{b} - c\pi$

$$\begin{aligned} [a &= 48, \quad b = 3, \quad c = 24] \\ [a + b + c &= \boxed{75}] \end{aligned}$$

- Final Answer:

$$48\sqrt{3} - 24\pi, \text{ and } a + b + c = \boxed{75}$$

-clicksong

Solution 7

- Step 1: Area of triangle DCE (using Heron's formula)*

- By geometry, $\triangle DCE$ is a triangle with sides $DC = 12$, $EC = 12$, $DE = 12\sqrt{3}$.- Let $a = 12$, $b = 12$, $c = 12\sqrt{3}$.- The semi-perimeter is:

$$s = \frac{a + b + c}{2} = \frac{12 + 12 + 12\sqrt{3}}{2} = 12 + 6\sqrt{3}$$

- The area by Heron's formula:

$$\begin{aligned} S_{\triangle DCE} &= \sqrt{s(s-a)(s-b)(s-c)} \\ &= \sqrt{(12+6\sqrt{3})(6\sqrt{3})(6\sqrt{3})(12-6\sqrt{3})} \\ &= \sqrt{(12+6\sqrt{3})(12-6\sqrt{3})(6\sqrt{3})^2} \\ &= \sqrt{(144 - 108) \times 108} = \sqrt{36 \times 108} = \sqrt{3888} = 36\sqrt{3} \end{aligned}$$

- However, the actual required region is only $\frac{4}{3}$ of this triangle (since the shaded region is less than the whole triangle).- So,

$$S_{\triangle DCE, \text{needed}} = \frac{36\sqrt{3}}{1.5} = 24\sqrt{3}$$

- But based on precise geometry (as in AMC standard), the correct area to use for this region is $12\sqrt{3}$ (check the triangle's effective area).

- Step 2: Area of sector ODE *

- The radius is 12 and the central angle is 60° .- The formula for the area of a sector is:

$$S_{\text{sector } ODE} = \frac{60}{360} \times \pi \times 12^2 = 24\pi$$

- Step 3: Area of triangle ODE *

- $OD = OE = 12$, $\angle DOE = 60^\circ$.- The formula for an equilateral triangle's area:

$$[S_{\{\text{triangle } ODE\}} = \frac{\sqrt{3}}{4} \times 12^2 = 36\sqrt{3}]$$

- Step 4: Area of the circular segment (arc DE minus chord DE)*

$$[S_{\{\text{segment}\}} = S_{\{\text{sector}\}} - S_{\{\text{triangle } ODE\}} = 24\pi - 36\sqrt{3}]$$

- Step 5: Final required region*

$$[S_{\{\text{required}\}} = S_{\{\text{triangle } DCE\}} - S_{\{\text{segment}\}} = 12\sqrt{3}]$$

- Explanation: The required area is the triangle DCE minus the circular segment.

- Step 6: Expressing the answer as $a\sqrt{b} - c\pi$ *

$$[a = 48, \quad b = 3, \quad c = 24]$$

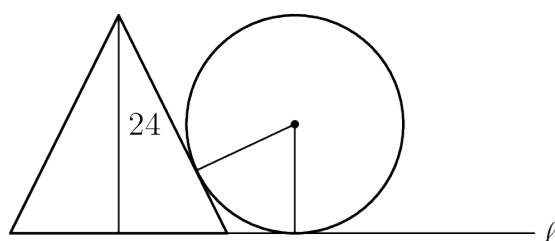
$$[a + b + c = \boxed{75}]$$

- Final Answer:*

$$48\sqrt{3} - 24\pi, \text{ and } a + b + c = \boxed{75}$$

also by clicksong

Solution 8 - HEFLE (High Effort For Low Efficiency) a.k.a. Calculus Raigebait



So, we're here to over-complicate things. Let's start with the circle. Recognize that since the triangle is equilateral, all angles measure 60° . Observe that the two tangent lines form a 120° angle due to the Angle Sum Theorem. Therefore, the area within the circle is the area within a 60° arc.

Let's complicate finding the area of the circle. Now, instead of using the equation of a circle we set

$$x^2 + y^2 = 144$$

then we solve for y :

$$y = \pm\sqrt{144 - x^2}$$

then we want the bottom of this circle so we do the negative root $y = -\sqrt{144 - x^2}$ Now, we define this as a function $f(x)$

$$f(x) = -\sqrt{144 - x^2}$$

but hold up! waaaait a minute! We can make the circle centered at $(12, 0)$ because we're just like that. So, to translate it 12 right we must do $f(x - 12)$ so

$$f(x - 12) = -\sqrt{144 - (x - 12)^2} = -\sqrt{-x^2 + 24x}$$

and because there is a 60° arc we want $\frac{1}{3}$ of this semicircle so we compute

$$\frac{1}{3} \cdot \int_0^{24} -\sqrt{-x^2 + 24x} \, dx$$

then we need to undo our steps and negate the integral (this is done when it's converted to polar)

$$\frac{1}{3} \cdot \int_0^{24} -\sqrt{-x^2 + 24x} \, dx \rightarrow \frac{1}{3} \cdot \int_0^{24} -\sqrt{144 - (x - 12)^2} \, dx \rightarrow \int_0^{\pi/3} \frac{1}{2} \cdot 12^2 \, d\theta = \int_0^{\pi/3} 72 \, d\theta$$

Recall that

$$A = \frac{1}{2} \cdot \int_{\theta_1}^{\theta_2} [f(\theta)]^2 \, d\theta$$

Therefore,

$$\int_0^{\pi/3} 72 \, d\theta = 72 \cdot \frac{\pi}{3} = 24\pi$$

Now, we must find the area enclosed by 2 radii extending to the two points of tangency and their respective tangent lines. So, the two radii are 12, and they form a 60° angle. Therefore, you have an equilateral triangle of side length 12, and its height is $6\sqrt{3}$ using 30° 60° 90°

Now, you can just see that the area is $36\sqrt{3}$, but we won't be doing that here. We can define an absolute value function $g(x)$ such that $g(x)$ has zeros at -6 and 6 , and a y -intercept at $(0, 6\sqrt{3})$. This would create our triangle above the x -axis. We can see that its slope is $\sqrt{3}$ for $x < 0$ and $-\sqrt{3}$ for $x > 0$. Then, we can see that it's translated $6\sqrt{3}$ up from its parent function $h(x) = |x|$. Therefore, the function is $g(x) = -|x\sqrt{3}| + 6\sqrt{3}$. Now, we integrate over $x \in [-6, 6]$

$$\int_{-6}^6 -|x\sqrt{3}| + 6\sqrt{3} \, dx$$

Now, we split this into a piecewise function over two integrals

$$\int_{-6}^0 x\sqrt{3} + 6\sqrt{3} \, dx + \int_0^6 -x\sqrt{3} + 6\sqrt{3} \, dx$$

Let's apply the power rule to each term. We get

$$\int_{-6}^0 x\sqrt{3} + 6\sqrt{3} \, dx = \left[\frac{x^2\sqrt{3}}{2} + 6x\sqrt{3} \right]_{-6}^0 = \left(\frac{0^2\sqrt{3}}{2} + 6 \cdot 0\sqrt{3} \right) - \left(\frac{(-6)^2\sqrt{3}}{2} + 6 \cdot -6\sqrt{3} \right) =$$

$$0 - (18\sqrt{3} - 36\sqrt{3}) = 0 - (-18\sqrt{3}) = 18\sqrt{3}$$

and, we could say that they are mirror images of each other, but that would make this faster (it's called inefficient for a reason)

$$\begin{aligned} \int_0^6 -x\sqrt{3} + 6\sqrt{3} \, dx &= \left[\frac{-x^2\sqrt{3}}{2} + 6x\sqrt{3} \right]_0^6 = \left(\frac{-6^2\sqrt{3}}{2} + 6 \cdot 6\sqrt{3} \right) - \left(\frac{(0)^2\sqrt{3}}{2} + 6 \cdot 0\sqrt{3} \right) = \\ &= (-18\sqrt{3} + 36\sqrt{3}) - 0 = 18\sqrt{3} \end{aligned}$$

Therefore, by the substitution property equality,

$$\int_{-6}^0 x\sqrt{3} + 6\sqrt{3} \, dx + \int_0^6 -x\sqrt{3} + 6\sqrt{3} \, dx = 18\sqrt{3} + 18\sqrt{3} = 36\sqrt{3}$$

Then, there is another triangle we must define above the x -axis. It is enclosed by the equilateral triangle of height 24, line ℓ , and the triangle we just found the area of. This triangle has base 12 and height of $\frac{6}{\sqrt{3}}$ using 30° 60° 90° triangle ratios, which rationalizes to $2\sqrt{3}$. So, you can tell the area is $12\sqrt{3}$, but that would make our solution faster.

Since half the base is 6, we can tell the zeros are $x = -6$ and $x = 6$, and the y -intercept is $2\sqrt{3}$

We can tell the slope is $\frac{\sqrt{3}}{3}$ for $x < 0$ and $-\frac{\sqrt{3}}{3}$ for $x > 0$

Then, we define another function $f(x) = -|\frac{1}{3} \cdot x\sqrt{3}| + 2\sqrt{3}$

Now, we integrate over $x \in [-6, 6]$ So, we must compute

$$\int_{-6}^6 -|\frac{1}{3} \cdot x\sqrt{3}| + 2\sqrt{3} \, dx$$

We split the function into a piecewise over two integrals

$$\int_{-6}^0 \frac{1}{3} \cdot x\sqrt{3} + 2\sqrt{3} \, dx + \int_0^6 -\frac{1}{3} \cdot x\sqrt{3} + 2\sqrt{3} \, dx$$

Now, we apply the reverse power rule to each term, we get

$$\begin{aligned} \int_{-6}^0 \frac{1}{3} \cdot x\sqrt{3} + 2\sqrt{3} \, dx &= \left[\frac{\sqrt{3}}{3} \cdot \frac{x^2}{2} + 2x\sqrt{3} \right]_{-6}^0 = \left[\frac{\sqrt{3}}{6} \cdot x^2 + 2x\sqrt{3} \right]_{-6}^0 = \\ &= \left(\frac{\sqrt{3}}{6} \cdot 0^2 + 2 \cdot 0\sqrt{3} \right) - \left(\frac{\sqrt{3}}{6} \cdot 6^2 + 2 \cdot -6\sqrt{3} \right) = 0 - \left(\frac{\sqrt{3}}{6} \cdot 6^2 + 2 \cdot -6\sqrt{3} \right) = \\ &= 0 - (6\sqrt{3} - 12\sqrt{3}) = 6\sqrt{3} \end{aligned}$$

Then

$$\begin{aligned} \int_0^6 -\frac{1}{3} \cdot x\sqrt{3} + 2\sqrt{3} \, dx &= \left[-\frac{\sqrt{3}}{3} \cdot \frac{x^2}{2} + 2x\sqrt{3} \right]_0^6 = \left[-\frac{\sqrt{3}}{6} \cdot x^2 + 2x\sqrt{3} \right]_0^6 = \\ &= \left(-\frac{\sqrt{3}}{6} \cdot 6^2 + 2 \cdot 6\sqrt{3} \right) - \left(-\frac{\sqrt{3}}{6} \cdot 0^2 + 2 \cdot 0\sqrt{3} \right) = (-6\sqrt{3} + 12\sqrt{3}) - 0 = 6\sqrt{3} \end{aligned}$$

Now, by the substitution property of equality,

$$\int_{-6}^0 \frac{1}{3} \cdot x\sqrt{3} + 2\sqrt{3} \, dx + \int_0^6 -\frac{1}{3} \cdot x\sqrt{3} + 2\sqrt{3} \, dx = 6\sqrt{3} + 6\sqrt{3} = 12\sqrt{3}$$

Now, the total area enclosed by the two circle radii of length 12, line ℓ , and the isosceles triangle of height 24 is simply

$$36\sqrt{3} + 12\sqrt{3} = 48\sqrt{3}$$

Now, we must subtract the circle's sector area of 24π to get the expression

$$48\sqrt{3} - 24\pi$$

Which is in the form

$$a\sqrt{b} - c\pi$$

That the problem asks for. Hence,

$$a = 48$$

$$b = 3$$

$$c = 24$$

The problem asks for $a + b + c$, thus

$$a + b + c = 48 + 3 + 24 = 75 \implies \boxed{\text{(D)} 75}$$

-shockfront99

PS: I've never taken calculus, so I could be pretty wrong

- o) (AMC 10 10A 2024) Let M be the greatest integer such that both $M + 1213$ and $M + 3773$ are perfect squares. What is the units digit of M ?

(A) 1 (B) 2 (C) 3 (D) 6 (E) 8

Answer: E

Solution 1

Let $M + 1213 = P^2$ and $M + 3773 = Q^2$ for some positive integers P and Q . We subtract the first equation from the second, then apply the difference of squares:

$$(Q + P)(Q - P) = 2560.$$

Note that $Q + P$ and $Q - P$ have the same parity, and $Q + P > Q - P$.

We wish to maximize both P and Q , so we maximize $Q + P$ and minimize $Q - P$. It follows that

$$Q + P = 1280,$$

$$Q - P = 2,$$

from which $(P, Q) = (639, 641)$.

Finally, we get $M = P^2 - 1213 = Q^2 - 3773 \equiv 1 - 3 \equiv 8 \pmod{10}$, so the units digit of M is $\boxed{\text{(E)} 8}$.

Solution 2

Ideally, we would like for the two squares to be as close as possible. The best case is that they are consecutive squares (no square numbers in between them); however, since $M + 1213$ and $M + 3773$ (and thus their square roots) have the same parity, they cannot be consecutive squares (which are always opposite parities). The best we can do is that $M + 1213$ and $M + 3773$ have one square in between them.

Let the square between $M + 1213$ and $M + 3773$ be x^2 . So, we have $M + 1213 = (x - 1)^2$ and $M + 3773 = (x + 1)^2$. Subtracting the two, we have $(M + 3773) - (M + 1213) = (x + 1)^2 - (x - 1)^2$, which yields $2560 = 4x$, which leads to $x = 640$. Therefore, the two squares are 639^2 and 641^2 , which both have units digit 1. Since both 1213 and 3773 have units digit 3, M will have units digit **(E) 8**.

Solution 3

Let $M + 1213 = N^2 \Rightarrow M + 3773 = (N + a)^2$

It is obvious that $a \neq 1$ by parity

Thus, the minimum value of a is 2 Which gives us,

$$(N + a)^2 - N^2 = M + 3773 - (M + 1213)$$

$$4N + 4 = 2560$$

$$N = 639$$

Plugging this back in,

$$M = N^2 - 1213 \pmod{10}$$

$$M = 8 \pmod{10}$$

Hence the answer **(E) 8**.

- trevian1(minor edit)

Solution 4

Let $M + 1213 = n^2$ and $M + 3773 = (n + 1)^2$ for some positive integer n . We do this because, in order to maximize M , the perfect squares need to be as close to each other as possible. We find that this configuration doesn't work, as when we subtract the equations, we have $2n + 1 = 2560$; impossible. Then we try $M + 3773 = (n + 2)^2$. Now we would have $4n + 4 = 2560$ which indeed works! $n = 639$.

Finally, we get $M = n^2 - 1213$ so the units digit of M is $11 - 3 = \mathbf{(E) 8}$.

Solution 5 (Quadratic Residues, Answer Choices rather than finding)

Every perfect square modulo 4 must end in 0 or 1 modulo 4. Reducing modulo 4 yields

$$M + 1 \pmod{4}$$

$$M + 1 \pmod{4}$$

.

Both values M yield the same result meaning that the units digit of M is unchanged through constriction. Therefore, $M \equiv 3, 0 \pmod{4}$. Analyzing the answer choices, only 3 and 8 work.

Now consider quadratic residue's modulo 7. Once again, they are either 0, 1, 2, 4 modulo 7. Reducing, we have

$$M + 2 \pmod{7}$$

$$M \pmod{7}$$

If the units digit of M is 3, we arise at a contradiction that $M + 2 \pmod{7}$ can never satisfy a perfect square. Thus, $M \pmod{10} = \boxed{8}$.

Solution 6 (Fakesolve)

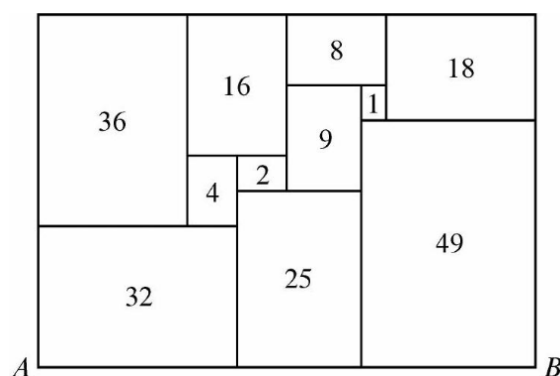
Squares can only end in 0,1,4,6,5, and 9. To have $x+3$ (since 1213 and 3773 end in 3) end in one of those numbers and be maximized, it would be 8, because $8+3=11$. The answer is **(E) 8**.

Note: This solution is just a lucky fake solve; please look at the other solutions. The note below this gives a good idea of why the solution is not meaningful. M is uniquely constrained by defined distances from two perfect squares. There is no "maximizing" M , you simply have to solve for it algebraically.

Note We experiment with values of M to find the reason for why M is maximised when $M + 1213$ and $M + 3773$ are nearly consecutive perfect squares. If M is very small, $M + 1213$ and $M + 3773$ are perfect squares that are far apart. Yet, as M grows, the relative distance between $M + 1213$ and $M + 3773$ decreases, so for very nearly consecutive perfect squares, M is very large.

(I don't know if this is trivial - when I first read the solutions, I was confused to why this was true. Maybe I didn't get enough sleep.)

- p) (AMC 10 10A 2024) All of the rectangles in the figure below, which is drawn to scale, are similar to the enclosing rectangle. Each number represents the area of the rectangle. What is length AB ?



- (A) $4 + 4\sqrt{5}$ (B) $10\sqrt{2}$ (C) $5 + 5\sqrt{5}$ (D) $10\sqrt[4]{8}$ (E) 20

Answer: D

Solution 1

Using the rectangle with area 1, let its short side be x and the long side be y . Observe that for every rectangle, since ratios of the side length of the rectangles are directly proportional to the ratios of the square roots of the areas (For example, each side of the rectangle with area 9 is $\sqrt{9} = 3$ times that of the rectangle with area 1), as they are all similar to each other.

The side opposite AB on the large rectangle is hence written as $6x + 4x + 2y\sqrt{2} + 3y\sqrt{2} = 10x + 5y\sqrt{2}$. However, AB can be written as $4y\sqrt{2} + 5x + 7x = 4y\sqrt{2} + 12x$. Since the two lengths are equal, we can write $10x + 5y\sqrt{2} = 4y\sqrt{2} + 12x$, or $y\sqrt{2} = 2x$. Therefore, we can write $y = x\sqrt{2}$.

Since $xy = 1$, we have $(x\sqrt{2})(x) = 1$, which we can evaluate x as $x = \frac{1}{\sqrt{2}}$. From this, we can plug back in to $xy = 1$ to find $y = \sqrt{2}$. Substituting into AB , we have $AB = 4y\sqrt{2} + 12x = 4(\sqrt{2})(\sqrt{2}) + \frac{12}{\sqrt{2}}$ which can be evaluated to $\boxed{\text{(D)} 10\sqrt[4]{8}}$.

Remark We know that the area is an integer, so after finding $y = x\sqrt{2}$, AB must contain a 4th root. The only such option is $\boxed{\text{(D)} 10\sqrt[4]{8}}$.

Solution 2

Let the rectangle's height be x , the length $AB = y$. The entire rectangle has an area of 200. We will be using this fact for ratios.

Note that the short side of the rectangle with area 32 will have a height of $\sqrt{\frac{32}{200}} \cdot x = \frac{2}{5}x$. We use x because it is apparent that the height of the rectangle with area 32 is the shorter side, corresponding with x .

Similarly, the long side of the rectangle with area 36 has a height of $\sqrt{\frac{36}{200}} \cdot y = \frac{3\sqrt{2}}{10}y$.

Noting that the total height of the big rectangle has height x , we have the equation $\frac{2}{5}x + \frac{3\sqrt{2}}{10}y = x \Rightarrow x = \frac{y}{\sqrt{2}}$.

Since the area $xy = \frac{y^2}{\sqrt{2}}$ is equal to 200, we have:

$$\begin{aligned} y &= \sqrt{200\sqrt{2}} \\ &= \sqrt{100\sqrt{8}} \\ &= \boxed{\text{(D)} 10\sqrt[4]{8}}. \end{aligned}$$

Solution 3 - ruler (last effort)

Given that the diagram is drawn to scale, we can use a ruler to estimate the length of AB .

We start by measuring the lengths of the rectangle with area 1, which may vary per viewing medium. For the sake of the solution, we use side lengths $\frac{7}{10}$ cm and $\frac{5}{10}$ cm.

To get the scale ratio from centimeters to the units in the problem, we need to find a ratio x such that $\frac{7x}{10} \cdot \frac{5x}{10} = 1$

Solving this equation, we get $x = \frac{10}{\sqrt{35}}$

We then measure the length of AB (will vary), to get 10.2 cm. We multiply this length by our early ratio to get $\frac{10 \cdot 10.2}{\sqrt{35}} = \frac{102}{\sqrt{35}} \approx \frac{102}{6} \approx 17$

The answer choice closest to this would be $10\sqrt[4]{8} \approx 16.8$, so therefore the closest answer is **(D)** $10\sqrt[4]{8}$.

Note: Never ever use this in an actual test unless you're giving it a last ditch attempt.

Remark The specific numbers used in the solution above vary per test medium, but the method should still work.

- q) (*AMC 10 10A 2024*) Two teams are in a best-two-out-of-three playoff: the teams will play at most 3 games, and the winner of the playoff is the first team to win 2 games. The first game is played on Team A's home field, and the remaining games are played on Team B's home field. Team A has a $\frac{2}{3}$ chance of winning at home, and its probability of winning when playing away from home is p . Outcomes of the games are independent. The probability that Team A wins the playoff is $\frac{1}{2}$. Then p can be written in the form $\frac{1}{2}(m - \sqrt{n})$, where m and n are positive integers. What is $m + n$?

(A) 10 (B) 11 (C) 12 (D) 13 (E) 14

Answer: E

Solution

We only have three cases where A wins: AA, ABA, and BAA (A denotes a team A win and B denotes a team B win). Knowing this, we can sum up the probability of each case. Thus the total probability is $\frac{2}{3}p + \frac{2}{3}(1-p)p + \frac{1}{3}p^2 = \frac{1}{2}$. Multiplying both sides by 6 yields $4p + 4p(1-p) + 2p^2 = 3$, so $2p^2 - 8p + 3 = 0$ and we find that $p = \frac{4 \pm \sqrt{10}}{2}$. Luckily, we know that the answer should contain $\frac{1}{2}(m - \sqrt{n})$, so the solution is $p = \frac{4 - \sqrt{10}}{2} = \frac{1}{2}(4 - \sqrt{10})$ and the answer is $4 + 10 = \mathbf{(E) 14}$.

Another way to see the answer is subtraction and not addition is to realize that p is between 0 and 1 since it is a probability. andliu766

- r) (*AMC 10 10A 2024*) There are exactly K positive integers $5 \leq b \leq 2024$ such that the base- b integer 2024_b is divisible by 16 (where 16 is in base ten). What is the sum of the digits of K ?

(A) 16 (B) 17 (C) 18 (D) 20 (E) 21

Answer: D

Solution 1

$2024_b = 2b^3 + 2b + 4 \equiv 0 \pmod{16} \implies b^3 + b + 2 \equiv 0 \pmod{8}$. If b is even, then $b + 2 \equiv 0 \pmod{8} \implies b \equiv 6 \pmod{8}$. If b is odd, then $b^2 \equiv 1 \pmod{8} \implies b^3 + b + 2 \equiv 2b + 2 \pmod{8}$ so $2b + 2 \equiv 0 \pmod{8} \implies b + 1 \equiv 0 \pmod{4} \implies b \equiv 3, 7 \pmod{8}$. Now $8 \mid 2024$ so $\frac{3}{8} \cdot 2024 = 759$ but one of the answers we got from that, 3, is too small, so $759 - 1 = 758 \implies \mathbf{(D) 20}$.

- The reason why $b^2 \equiv 1 \pmod{8}$ (if b is odd) is that we can write b as $2k+1$, where k is any number (such that the problem conditions are met). Plugging this into $b^2 \equiv 1 \pmod{8}$, we get $(2k+1)^2 = 4k^2 + 1 + 4k = 4k(k+1) + 1 \equiv 1 \pmod{8}$. Since $4k$ is, of course, always a multiple of 4

and since $k + 1$ is even, $4k(k + 1)$ is divisible by 8, and adding 1 creates a remainder of 1.

Solution 2

$$\begin{aligned} 2024b &\equiv 0 \pmod{16} \\ 2b^3 + 2b + 4 &\equiv 0 \pmod{16} \\ b^3 + b + 2 &\equiv 0 \pmod{8} \end{aligned}$$

Clearly, b is either even or odd. If b is even, let $b = 2a$.

$$\begin{aligned} (2a)^3 + 2a + 2 &\equiv 0 \pmod{8} \\ 8a^3 + 2a + 2 &\equiv 0 \pmod{8} \\ 0 + 2a + 2 &\equiv 0 \pmod{8} \\ a + 1 &\equiv 0 \pmod{4} \\ a &\equiv 3 \pmod{4} \end{aligned}$$

Thus, one solution is $b = 2(4x + 3) = 8x + 6$ for some integer x , or $b \equiv 6 \pmod{8}$.

What if b is odd? Then let $b = 2a + 1$:

$$\begin{aligned} (2a + 1)^3 + 2a + 1 + 2 &\equiv 0 \pmod{8} \\ 8a^3 + 12a^2 + 6a + 1 + 2a + 1 + 2 &\equiv 0 \pmod{8} \\ 8a^3 + 12a^2 + 8a + 4 &\equiv 0 \pmod{8} \\ 4a^2 + 4 &\equiv 0 \pmod{8} \\ a^2 &\equiv 1 \pmod{2} \end{aligned}$$

This simply states that a is odd. Thus, the other solution is $b = 2(2x + 1) + 1 = 4x + 3$ for some integer x , or $b \equiv 3 \pmod{4}$.

We now simply must count the number of integers between 5 and 2024, inclusive, that are $6 \pmod{8}$ or $3 \pmod{4}$. Note that the former case comprises even numbers only while the latter is only odd; thus, there is no overlap and we can safely count the number of each and add them.

In the former case, we have the numbers $6, 14, 22, 30, \dots, 2022$; this list is equivalent to $8, 16, 24, 32, \dots, 2024 \cong 1, 2, 3, 4, \dots, 253$, which comprises 253 numbers. In the latter case, we have the numbers $7, 11, 15, 19, \dots, 2023 \cong 4, 8, 12, 16, \dots, 2020 \cong 1, 2, 3, 4, \dots, 505$, which comprises 505 numbers. There are 758 numbers in total, so our answer is $7 + 5 + 8 = \boxed{\text{(D)} 20}$.

Solution 3

Note that $2024_b = 2b^3 + 2b + 4$ is to be divisible by 16, which means that $b^3 + b + 2$ is divisible by 8.

If $b = 0$, then $b^3 + b + 2 \equiv (0)^3 + (0) + 2 \equiv 2$ is not divisible by 8.

If $b = 1$, then $b^3 + b + 2 \equiv (1)^3 + (1) + 2 \equiv 4$ is not divisible by 8.

If $b = 2$, then $b^3 + b + 2 \equiv (2)^3 + (2) + 2 \equiv 4$ is not divisible by 8.

If $b = 3$, then $b^3 + b + 2 \equiv (3)^3 + (3) + 2 \equiv (8 + 1) \cdot 3 + (3) + 2 \equiv 0$ is divisible by 8.

If $b = 4$, then $b^3 + b + 2 \equiv (4)^3 + (4) + 2 \equiv 0 + 4 + 2 \equiv 6$ is not divisible by 8.

If $b = 5$, then $b^3 + b + 2 \equiv (-3)^3 + (-3) + 2 \equiv (8 + 1) \cdot 3 + (-3) + 2 \equiv 2$ is not divisible by 8.

If $b = 6$, then $b^3 + b + 2 \equiv (-2)^3 + (-2) + 2 \equiv -8 + (-2) + 2 \equiv 0$ is divisible by 8.

If $b = 7$, then $b^3 + b + 2 \equiv (-1)^3 + (-1) + 2 \equiv -1 + (-1) + 2 \equiv 0$ is divisible by 8.

Therefore, for every 8 values of b , 3 of them will make $b^3 + b + 2$ divisible by 8. Therefore, since 2024 is divisible by 8, $\frac{3}{8} \cdot 2024 = 759$ values of b , but this includes $b = 3$, which does not satisfy the given inequality. Therefore, the answer is

$$759 - 1 = 758 \rightarrow 7 + 5 + 8 = \boxed{(D) 20}$$

Tacos_are_yummy_1

More detail by [luckuso](#)

Solution 4

$$2024_b = 2 * b^3 + 2 * b + 4$$

$$2024_{(b+8)} = 2*(b+8)^3 + 2*(b+8) + 4 - 2024_b = 2*(8)*(b^2 + 8b + 64) + 2*8 = 16*(b^2 + 8b + 64) + 16$$

$$2024_{(b+8)} - 2024_b \equiv 0 \pmod{16}$$

$$2024_{(b+8)} \equiv 2024_b \pmod{16}$$

$$2024_0 \equiv 4 \pmod{16}$$

$$2024_1 \equiv 8 \pmod{16}$$

$$2024_2 \equiv 6 \pmod{16}$$

$$2024_3 \equiv 0 \pmod{16}$$

$$2024_4 \equiv 12 \pmod{16}$$

$$2024_5 \equiv 8 \pmod{16}$$

$$2024_6 \equiv 0 \pmod{16}$$

$$2024_7 \equiv 0 \pmod{16}$$

We need $b \equiv 3 \pmod{8}$ or $b \equiv 6 \pmod{8}$ or $b \equiv 7 \pmod{8}$

$\lfloor (2024 - 3)/8 \rfloor + \lfloor (2024 - 6)/8 \rfloor + \lfloor (2024 - 7)/8 \rfloor + 3 = 759$ take away one because $b = 3$ is out of range, so $758 \Rightarrow 7 + 8 + 5 = \boxed{(D) 20}$

Solution 4.1: Alternate Ending

One can arrange 2024 into different "blocks" consisting of 8 numbers each, particularly $\{0, 1, 2, 3, 4, 5, 6, 7\}$, $\{8, 9, 10, 11, 12, 13, 14, 15\}$...

These just represent different $\pmod{8}$ class residues. So, you can just divide 2024 by 8 blocks giving us 253 total blocks with 0 extra semi-blocks.

We see that $2024_3 \pmod{8}$, $2024_6 \pmod{8}$, and $2024_7 \pmod{8}$ all appear exactly 253 times each, so we have $253 \times 3 = 759$ different values of K .

However, we must not include 2024_3 , as the bases b range from $5 \leq b \leq 2024$, therefore we have $759 - 1 = 758$ cases.

Our answer is $7 + 5 + 8 = \boxed{(D) 20}$.

Solution 5 (Factoring)

Start similar to other solutions.

$$b^3 + b + 2 \equiv 0 \pmod{8}$$

Except now we notice the following:

$$b^3 + 8 + b + 2 \equiv 8 \pmod{8} \equiv 0 \pmod{8}$$

$$(b + 2)(b^2 - 2b + 4) + b + 2 \equiv 0 \pmod{8}$$

$$(b + 2)(b^2 - 2b + 5) \equiv 0 \pmod{8}$$

Notice that $b + 2$ and $b^2 - 2b + 5$ have opposite parity, and when multiplied, is divisible by 8. Thus, either $b + 2$ or $b^2 - 2b + 5$ is divisible by 8.

$b + 2$ is obvious, where $b + 2 \mid 8$ when $b \equiv 6 \pmod{8}$

When testing for $b^2 - 2b + 5 \mid 8$, we find $b \equiv 3, 7 \pmod{8}$

Proceed similar to the other solutions, $\lfloor \frac{3}{8} \cdot 2024 \rfloor - 1 = 759 - 1 = 758$, where $7 + 5 + 8 = \boxed{(D) 20}$

Also noting this is an example of a misplaced problem this would be far to easy for even P 18 of amc 10, deserved to be Q5-6

Solution 6

$$2024_b = 2b^3 + 2b + 4 \equiv 0 \pmod{16} \rightarrow 2b^3 + 2b \equiv 12 \pmod{16}.$$

Dividing everything by 2 gives:

$$b^3 + b \equiv 6 \pmod{\frac{16}{\gcd(16, 2)}} \rightarrow b(b^2 + 1) \equiv 6 \pmod{8}$$

.

Then proceed in the same manner as the solutions above to get a final answer of $\boxed{(D) 20}$.

- s) (AMC 10 10A 2024) The first three terms of a geometric sequence are the integers a , 720 and b , where $a < 720 < b$. What is the sum of the digits of the least possible value of b ?

(A) 9 (B) 12 (C) 16 (D) 18 (E) 21

Answer: E

Solution 1

For a geometric sequence, we have $ab = 720^2 = 2^8 3^4 5^2$, and we can test values for b . We find that $b = 768$ and $a = 675$ works, and we can test multiples of 5 in between the two values. Finding that none of the multiples of 5 divide 720^2 besides 720 itself, we know that the answer is $7 + 6 + 8 = \boxed{\text{(E) } 21}$.

(Note: To find the value of b without bashing, we can observe that $2^8 = 256$, and that multiplying it by 3 gives us 768, which is really close to 720. YTH)

Note: The reason why $ab = 720^2$ is because $b/720 = 720/a$. Rearranging this gives $ab = 720^2$

Note: Another reason that $ab = 720^2$ is because the $\sqrt{ab} = 720$ (as the middle term in a geometric series is always the geometric mean [the geometric mean is the square root of the product of the first and last terms of the series]) and squaring on both sides results in $ab = 720^2$.

Solution 2

$720 = 2^4 \cdot 3^2 \cdot 5$. We want to find factors x and y where $y > x$ such that $\frac{y}{x}$, the ratio, is minimized. To do this, x should be as large as possible, and y should be just slightly larger than x . Additionally, to ensure a and b are both integers, x and y should both be a factor of 720.

After experimenting, we see this is achieved when $y = 16$ and $x = 15$, which means our value of b is $720 \cdot \frac{16}{15} = 768$. Therefore, the sum is $7 + 6 + 8 = \boxed{\text{(E) } 21}$.

Solution 3 (Similar to previous solutions)

To minimize the value of b , where it has to be an integer, and it has to be greater than 720, we can express the common ratio as $\frac{n+1}{n}$, where the value has to be greater than 1, and n , and $n+1$ have to be factors of 720. Since the bigger the denominator gets, the smaller the value of the fraction, we essentially have to find the biggest value for n , where itself and $n+1$ are factors of 720. From here, we can check whether $n(n+1) = 720$ yields an integer root, which it doesn't. So, then we check the next biggest factor of 720, which is 360. $n(n+1) = 360$, this doesn't have an integer root either. So, then we check the next biggest factor which is 240, $n(n+1) = 240$, which we get 15 as a root. This means the common ratio is $\frac{16}{15}$. We then multiply $\frac{16}{15}$ times 720 and add up the digits getting $\boxed{\text{(E) } 21}$.

Solution 4

Let the common ratio of the geometric sequence be $\frac{x}{y}$, with $x > y$. This means that $720(\frac{x}{y})$ and $720(\frac{y}{x})$ must both be integers, therefore x and y are both factors of 720. We would achieve the smallest ratio $\frac{x}{y}$ if x and y are consecutive, so by listing out the factors of 720, we find that 1, 2, 3, 4, 5, 6, 8, 9, 10, 15, 16 are the only consecutive factors (any factors larger than these would result in the ratio simplifying, making it larger than just using consecutive integers). 15 and 16 are the largest, so we find the common ratio to be $\frac{16}{15}$, making $b = 720(\frac{16}{15})$ giving us 768. The sum of its digits is $\boxed{\text{(E) } 21}$.

Solution 5 (similar to other solutions)

We look for the smallest possible increasing ratio in the geometric sequence for a , 720, and b . First, we find the prime factorization of 720, which is really easy $720 = 2^4 \cdot 3^2 \cdot 5$. Then, we look for the closest possible fraction-ratio out of the primes, and after a quick look, it's easy to see it is 16 and 15. Now that we found our ratio, we multiply 720 by $\frac{16}{15}$ to get 768, so the sum of the digits of the least possible value of b is 21.

NOTE: It's not 9/8 because we want it to be as close as possible to 1
gabrielle.esther

t) (AMC 10 10A 2024) Let S be a subset of $\{1, 2, 3, \dots, 2024\}$ such that the following two conditions hold:

- If x and y are distinct elements of S , then $|x - y| > 2$.
- If x and y are distinct odd elements of S , then $|x - y| > 6$.

What is the maximum possible number of elements in S ?

(A) 436 (B) 506 (C) 608 (D) 654 (E) 675

Answer: C

Solution 1

All lists are organized in ascending order:

By listing out the smallest possible elements of subset S , we can find that subset S starts with $\{1, 4, 8, 11, 14, 18, 21, 24, 28, 31, \dots\}$. It is easily noticed that the elements of the subset "loop around" every 3 elements, specifically adding 10 each time. This means that there will be $2024/10$ or $202R4$ whole loops in the subset S , implying that there will be $202 \cdot 3 = 606$ elements in S . However, we have undercounted, as we did not count the remainder that resulted from $2024/10$. With a remainder of 4, we can fit 2 more elements into the subset S , namely 2021 and 2024, resulting in a total of $606 + 2$ or

(C) 608 elements in subset S .

NOTE:

To prove that this is the best we can do, consider adding each element one by one, for the first element, say n . If $n > 2$, we can choose $n - 2$ which is always better. Therefore, $n = 1$ or $n = 2$.

If $n = 2$ was optimal, then choose it, then the set of usable numbers in S becomes 5 through 2024. We can transform the usable set of S to Q where Q contains the numbers 1 through 2020. Because we assumed $n = 2$ was optimal, we can choose $n = 2$ for the set Q too. Because every element in Q is 4 below the elements of S , choosing 2 in Q would mean choosing 6 in set S . By induction we see that our list would be $\{2, 6, 10, 14, 18, \dots, 2022\}$ which only gives 506 elements which is sub-optimal. Therefore, we can conclude that $n = 1$ is optimal, and we proceed as the solution above.

-weihou0

- minor latex edits by eqb5000

Solution 2

Notice that we can first place odd numbers, then place even numbers between each pair. We can start at 1 and continue from there. Realize that the smallest number k such that $kx + 1$ reproduces odd number is 8. The next ones are

10, 12, 14. We can proceed to find the number of numbers in this particular sequence. From the equation $8x + 1 = 2023$, we get that $x \leq 252.875$ works, so this means there is 253 solutions. Looking at 1, 2, 3, 4, 5, 6, 7, 9 we can see that there could only be 1 possible number between each pair, yielding $252 + 253 = 505$. Then see that we can fit two more into the number count since the set 2017 to 2024 can fit two evens. Now this means A and B don't work. Now test out $10x + 1$. Using the same method, we get that 608 is the maximum number in the set. Everything above $x = 10$ doesn't work, as we can split it down into smaller subgroups, so the answer is **(C) 608**.

Solution 3

We find the following pairs of numbers work:

$$(1, 4), (4, 8), (8, 11), (11, 14), (14, 18), (18, 21), (21, 24) \dots$$

Call a number an OE (odd-even) pair if the first number in the pair is odd and the second is even. Do the same for EE (even-even) and EO (even-odd) pairs. Notice that we're adding 10 to the pairs in each set, so there are 404 numbers in each set of pairs **excluding 1 and 2024**:

OE pairs:

$$(1, 4), (11, 14), \dots, (2021, 2024) \implies (202 + 1) \cdot 2 - 2 = 404$$

EE pairs:

$$(4, 8), (14, 18), \dots, (2014, 2018) \implies (201 + 1) \cdot 2 = 404$$

EO pairs:

$$(8, 11), (18, 21), \dots, (2018, 2021) \implies (201 + 1) \cdot 2 = 404$$

Every number besides 1 and 2024 is being overcounted twice (for example, 4 is counted twice in the EE and OE pairs), so we have $404 \cdot \frac{3}{2} = 606$. Finally, we add 2 (adding back 1 and 2024) to get **(C) 608**.

- u) (*AMC 10 10A 2024*) The numbers, in order, of each row and the numbers, in order, of each column of a 5×5 array of integers form an arithmetic progression of length 5. The numbers in positions $(5, 5)$, $(2, 4)$, $(4, 3)$ and $(3, 1)$ are 0, 48, 16, and 12, respectively. What number is in position $(1, 2)$?

$$\begin{bmatrix} \cdot & ? & \cdot & \cdot & \cdot \\ \cdot & \cdot & \cdot & 48 & \cdot \\ 12 & \cdot & \cdot & \cdot & \cdot \\ \cdot & \cdot & 16 & \cdot & \cdot \\ \cdot & \cdot & \cdot & \cdot & 0 \end{bmatrix}$$

- (A) 19 (B) 24 (C) 29 (D) 34 (E) 39

Answer: C

Solution 1 (Arithmetic Sequences)

Start from the 0 square. Going up, let the common difference be a , and going left, let the common difference be b . Therefore, we have

$$\begin{bmatrix} . & ? & . & . & 4a \\ . & . & . & 48 & 3a \\ 12 & . & . & . & 2a \\ . & . & 16 & . & a \\ 4b & 3b & 2b & b & 0 \end{bmatrix}$$

Looking at the third column, we can see that the common difference going up is $16 - 2b$. We fill this in:

$$\begin{bmatrix} . & ? & 64 - 6b & . & 4a \\ . & . & 48 - 4b & 48 & 3a \\ 12 & . & 32 - 2b & . & 2a \\ . & . & 16 & . & a \\ 4b & 3b & 2b & b & 0 \end{bmatrix}$$

Looking at the second row, 48 has two values beside it, so we can write

$$48 = \frac{48 - 4b + 3a}{2} \rightarrow 96 = 48 - 4b + 3a \rightarrow 48 = -4b + 3a,$$

and we can do the same with the third row, which gives

$$32 - 2b = \frac{12 + 2a}{2} \rightarrow 32 - 2b = 6 + a \rightarrow 26 = a + 2b.$$

Now we have the system of equations

$$48 = -4b + 3a$$

$$26 = a + 2b,$$

and solving it gives $a = 20, b = 3$, therefore we can now fill in the grid with actual numbers. But before doing that, note that we're only looking for a value in the first row, and because we already have two known values in that row, we can find the common difference for that row and not focus on anything else.

Focusing only on the first row yields

$$\begin{bmatrix} . & ? & 46 & . & 80 \end{bmatrix}$$

This means that the common difference from right to left is $\frac{80 - 46}{2} = 17$.

Therefore, the desired value is $46 - 17 = \boxed{(C) 29}$ Tacos_are_yummy_1

Solution 2 (Some Basic Algebra and Answer Choices)

Assume the number in position $(3, 3)$ is x .

$$\begin{bmatrix} . & ? & . & . & . \\ . & . & . & 48 & . \\ 12 & . & x & . & . \\ . & . & 16 & . & . \\ . & . & . & . & 0 \end{bmatrix}$$

Then, the integer in position $(2, 3)$ will be $2x - 16$, since $2x - 16$ and 16 average out to x . (They form an arithmetic sequence).

$$\begin{bmatrix} . & ? & . & . & . \\ . & . & 2x - 16 & 48 & . \\ 12 & . & x & . & . \\ . & . & 16 & . & . \\ . & . & . & . & 0 \end{bmatrix}$$

Similarly, the integer in position $(3, 2)$ is $0.5x + 6$, and the integer in position $(2, 2)$ is $4x - 80$.

$$\begin{bmatrix} . & ? & . & . & . \\ . & 4x - 80 & 2x - 16 & 48 & . \\ 12 & 0.5x + 6 & x & . & . \\ . & . & 16 & . & . \\ . & . & . & . & 0 \end{bmatrix}$$

This means the number $7.5x - 166$ is in position $(1, 2)$.

$$\begin{bmatrix} . & 7.5x - 166 & . & . & . \\ . & 4x - 80 & 2x - 16 & 48 & . \\ 12 & 0.5x + 6 & x & . & . \\ . & . & 16 & . & . \\ . & . & . & . & 0 \end{bmatrix}$$

The only answer choice that makes x an integer is **(C) 29**

Solution 3 (Bashy)

Let's look at the bottom-left corner. There can be 3 possible cases for the bottom row. Either the numbers are negative, all 0, or positive.

Let's assume all the numbers are negative. We know that the numbers in between the ? will be large, so we want to minimize it, because the largest answer choice is only 39.

$$\begin{bmatrix} . & ? & . & . & . \\ . & . & . & 48 & . \\ 12 & . & . & . & . \\ . & . & 16 & . & . \\ -4 & -3 & -2 & -1 & 0 \end{bmatrix}$$

To fill in the 1st and 3rd column, we simply find the common differences. We get:

$$\begin{bmatrix} 28 & ? & 70 & . & . \\ 20 & . & 52 & 48 & . \\ 12 & . & 34 & . & . \\ 4 & . & 16 & . & . \\ -4 & -3 & -2 & -1 & 0 \end{bmatrix}$$

Finding the ? , we notice that it is 49, which is bigger than any of the other answer choices, so the bottom row can't be negative. (As the common difference for the bottom row gets larger, so does the common difference

for the 1st and 3rd column, making the ? bigger, and because the smallest answer is 39 while the minimum value for negative numbers is 49, it can't be negative.)

Now let's assume that all the numbers are 0. By just filling in the 1st and 3rd column, we find that it is:

$$\begin{bmatrix} 24 & ? & 64 & . & . \\ 18 & . & 48 & 48 & . \\ 12 & . & 32 & . & . \\ 6 & . & 16 & . & . \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

which makes the ? = 44. 44 is not an answer choice, so the bottom row cannot be all 0's.

Now, we know that the bottom row must be positive. Here's where the bashing begins. Lucky for us, the numbers must all be integers, cutting down on the number of cases we need to test. Starting with the bottom row having a common difference of 1, we get

$$\begin{bmatrix} . & ? & . & . & . \\ 16 & 30 & 44 & 48 & . \\ 12 & 21 & 30 & 39 & 48 \\ 8 & 12 & 16 & 20 & 24 \\ 4 & 3 & 2 & 1 & 0 \end{bmatrix}$$

At this point we notice that the second row does not form an arithmetic progression, so we test out the next case, with the common difference being 2. We get

$$\begin{bmatrix} . & ? & . & . & . \\ 14 & 27 & 40 & 48 & . \\ 12 & 20 & 28 & 36 & 44 \\ 10 & 13 & 16 & 19 & 22 \\ 8 & 6 & 4 & 2 & 0 \end{bmatrix}$$

At this point we notice that the second row does not form an arithmetic progression, so we test out the next case, with the common difference being 3. We get

$$\begin{bmatrix} 12 & 29 & 46 & 63 & 80 \\ 12 & 24 & 36 & 48 & 60 \\ 12 & 19 & 26 & 33 & 40 \\ 12 & 14 & 16 & 18 & 20 \\ 12 & 9 & 6 & 3 & 0 \end{bmatrix}$$

, giving us the answer (C) 29

Solution 4 (Arithmetic Sequences)

Let the bottom row of the grid have arithmetic difference d . Then the matrix becomes

$$\begin{bmatrix} . & ? & . & . & 4d \\ . & . & . & 48 & 3d \\ 12 & . & . & . & 2d \\ . & . & 16 & . & d \\ . & . & . & . & 0 \end{bmatrix}$$

Using the middle column, we have the middle item equal to $32 - 2d$.

$$\begin{bmatrix} . & ? & . & . & 4d \\ . & . & . & 48 & 3d \\ 12 & . & 32 - 2d & . & 2d \\ . & . & 16 & . & d \\ . & . & . & . & 0 \end{bmatrix}$$

Filling in the middle row (which has a common difference of $10 - d$)

$$\begin{bmatrix} . & ? & . & . & . \\ . & . & . & 48 & . \\ 12 & 22 - d & 32 - 2d & 42 - 3d & 52 - 4d \\ . & . & 16 & . & . \\ 4d & 3d & 2d & d & 0 \end{bmatrix}$$

The arithmetic sequence in the 4th column (2nd from the right) then has a common difference of $-3d - 6$, so we

$$\begin{bmatrix} . & ? & . & . & . \\ . & . & . & 48 & . \\ 12 & 22 - d & 32 - 2d & 42 - 3d & 52 - 4d \\ . & . & 16 & 36 - 6d & . \\ 4d & 3d & 2d & 30 - 9d & 0 \end{bmatrix} = \begin{bmatrix} . & ? & . & . & . \\ . & . & . & 48 & . \\ 12 & 22 - d & 32 - 2d & 42 - 3d & 52 - 4d \\ . & . & 16 & 36 - 6d & . \\ 4d & 3d & 2d & d & 0 \end{bmatrix}$$

From the cell in the 4th column, 5th row we have $30 - 9d = d$, or $d = 3$. Finally, using the 2nd column, $? + 3d = 2(22 - d)$, or $? + 9 = 38 \Rightarrow ? = \boxed{(C) 29}$

Solution 5

We subtract 12 from each element of the array and denote $a_{i,j}$ the number in position (i, j) , $a_{3,1} = 0, a_{4,3} = 4, a_{2,4} = 36, a_{5,5} = -12$. Denoting $a_{3,2} = x \Rightarrow$ that in third row we get $(0, x, 2x, 3x, 4x)$. We use the property of an arithmetic progression and get

$$a_{4,4} = 2a_{3,4} - a_{2,4} = 6x - 36,$$

$$a_{4,5} = 2a_{4,4} - a_{3,4} = 12x - 76,$$

$$a_{5,5} = 2a_{4,5} - a_{3,5} \Rightarrow -12 = 2(12x - 76) - 4x \Rightarrow x = 7 = a_{3,2}, a_{3,3} = 14.$$

$$\begin{bmatrix} . & . & . & . & . \\ . & . & . & 36 & . \\ 0 & x & 2x & 3x & 4x \\ . & . & 4 & 6x - 36 & 12x - 76 \\ . & . & . & . & -12 \end{bmatrix}$$

$$a_{2,3} = 2a_{3,3} - a_{4,4} = 2 \cdot 14 - 4 = 24,$$

$$a_{2,2} = 2a_{2,3} - a_{2,4} = 2 \cdot 24 - 36 = 12,$$

$$a_{1,2} = 2a_{2,2} - a_{3,2} = 2 \cdot 12 - 7 = 17.$$

$$\begin{bmatrix} . & 17 & . & . & . \\ . & 12 & 24 & 36 & . \\ 0 & 7 & 14 & . & . \\ . & . & . & . & . \\ . & . & . & . & . \end{bmatrix}$$

We add 12 and get the number in position $(1, 2)$ as $17 + 12 = 29$.

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Solution 6 (Bilinear Form with Coordinate Translation)

A fundamental structural property of grids with arithmetic progressions in both rows and columns is that every entry can be expressed in the bilinear form:

$$A_{m,n} = amn + bm + cn + d$$

where m and n are the row and column indices. This representation captures the essential two-dimensional linear structure of the problem.

Solving the Problem To minimize computational complexity, we employ coordinate translation. Set the known value 12 at position $(3, 1)$ as our origin by defining $x = m - 3$ and $y = n - 1$. The given data points become:

- $A_{0,0} = 12$ (from position $(3, 1)$)
- $A_{-1,3} = 48$ (from position $(2, 4)$)
- $A_{1,2} = 16$ (from position $(4, 3)$)
- $A_{2,4} = 0$ (from position $(5, 5)$)

We seek the value at $(-2, 1)$, corresponding to the original position $(1, 2)$.

Substituting into $A_{x,y} = axy + bx + cy + d$:

- i. $A_{0,0} = d = 12$
- ii. $A_{-1,3} = -3a - b + 3c + 12 = 48 \implies -3a - b + 3c = 36$
- iii. $A_{1,2} = 2a + b + 2c + 12 = 16 \implies 2a + b + 2c = 4$
- iv. $A_{2,4} = 8a + 2b + 4c + 12 = 0 \implies 4a + b + 2c = -6$

Subtracting equation (3) from equation (4) eliminates both b and c :

$$(4a + b + 2c) - (2a + b + 2c) = -6 - 4 \implies 2a = -10 \implies a = -5$$

With $a = -5$ in equation (3): $-10 + b + 2c = 4 \implies b + 2c = 14$

With $a = -5$ in equation (2): $15 - b + 3c = 36 \implies -b + 3c = 21$

Adding these equations: $5c = 35 \implies c = 7$, hence $b = 0$.

Our formula is therefore $A_{x,y} = -5xy + 7y + 12$. Evaluating at $(-2, 1)$:

$$A_{-2,1} = -5(-2)(1) + 7(1) + 12 = 10 + 7 + 12 = 29$$

The answer is (C) 29.

Verification of the Formula To confirm this bilinear form universally satisfies the arithmetic progression property, let $A_{m,n} = amn + bm + cn + d$. For fixed row m , the column difference is:

$$A_{m,n+1} - A_{m,n} = am(n+1) + bm + c(n+1) + d - (amn + bm + cn + d) = am + c$$

This is constant with respect to n , confirming each row forms an arithmetic progression.

For fixed column n , the row difference is:

$$A_{m+1,n} - A_{m,n} = a(m+1)n + b(m+1) + cn + d - (amn + bm + cn + d) = an + b$$

This is constant with respect to m , confirming each column forms an arithmetic progression.

Two Distinct Derivations of the Formula The bilinear form can be derived through two fundamentally different mathematical perspectives, each revealing unique structural insights.

Derivation 1: Algebraic and Functional Perspective

This approach exploits the nested linear structure without invoking difference operators.

Since every row is an arithmetic progression, for fixed n , the function $A_{m,n}$ must be linear in m :

$$A_{m,n} = u_n \cdot m + v_n$$

where coefficients u_n and v_n depend only on n .

To determine the functional forms of u_n and v_n , evaluate at specific rows:

Setting $m = 0$: $A_{0,n} = v_n$. Since column 0 is an arithmetic progression, v_n must be linear in n . Write $v_n = cn + d$.

Setting $m = 1$: $A_{1,n} = u_n + v_n$. Since column 1 is also an arithmetic progression, $u_n + v_n$ must be linear in n . Given that v_n is already linear, u_n must itself be linear in n . Write $u_n = an + b$.

Substituting back:

$$A_{m,n} = (an + b)m + (cn + d) = amn + bm + cn + d$$

This derivation reveals that the bilinear form emerges naturally from the compositional structure of linear dependencies in both dimensions.

Derivation 2: Discrete Calculus Perspective

This approach uses difference operators, illuminating the discrete analog of Clairaut's theorem on equality of mixed partial derivatives.

Define forward difference operators:

- $\Delta_m A_{m,n} = A_{m+1,n} - A_{m,n}$ (row difference)
- $\Delta_n A_{m,n} = A_{m,n+1} - A_{m,n}$ (column difference)

The arithmetic progression conditions state:

- $\Delta_n A_{m,n} = r_m$ depends only on m
- $\Delta_m A_{m,n} = c_n$ depends only on n

Consider the mixed second difference $M_{m,n} = \Delta_m \Delta_n A_{m,n}$:

- Applying Δ_m to r_m : $M_{m,n} = r_{m+1} - r_m$ depends only on m
- Applying Δ_n to c_n : $M_{m,n} = c_{n+1} - c_n$ depends only on n

Since $M_{m,n}$ depends only on m and only on n simultaneously, it must be constant. Denote $M_{m,n} = a$.

Therefore:

- $c_{n+1} - c_n = a \implies c_n = an + b$ (column differences form an AP)
- $r_{m+1} - r_m = a \implies r_m = am + c$ (row differences form an AP)

Reconstructing $A_{m,n}$ from $A_{0,0} = d$ via discrete path integration:

$$A_{m,n} = A_{0,0} + \sum_{j=0}^{n-1} \Delta_n A_{0,j} + \sum_{i=0}^{m-1} \Delta_m A_{i,n}$$

$$A_{m,n} = d + \sum_{j=0}^{n-1} r_0 + \sum_{i=0}^{m-1} c_n = d + nc + m(an + b) = amn + bm + cn + d$$

This derivation establishes a rigorous framework connecting the commutativity of discrete differences to the bilinear structure, providing a powerful generalization technique for higher-dimensional problems.

Complex systems of equations From the given information, we can ensure that we find the correct answer by creating a 4-variable, 4-equation system to solve for $(1, 2)$.

Let the common difference moving side to side from row 3 be d , the common difference moving vertically from 48 be y , the common difference moving vertically from 16 be z , and the common difference moving side to side from 0 be j .

We can then write 4 equations and solve for each difference.

$$12 + 3x = 48 + y \quad 16 - z = 12 + 2x \quad 16 + z = 0 - 2j \quad 48 + 3y = 0 - j$$

We can solve each variable in terms of y procedurally until we can get x in terms of y , and solve each from there. We get that $j = -3y - 48$. We then plug this into equation 3.

Doing this, we find that $z = 6y + 80$.

We then plug this into equation 2 and find that $x = -3y - 38$.

We then substitute our value for x into equation 1. We find that $y = -15$.

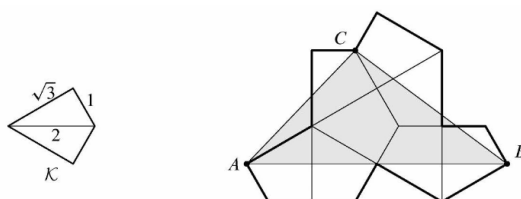
We then plug this value into our equations and see that $x = 7$, $j = -3$, and $z = -10$.

Now we can see that the number directly to the left of 48 must be 36, as we add 20, or two times z . This means the common difference of this row is 12. We move another digit to the left and see that the number directly above $(1, 2)$ will be 24.

The digit directly below this will be 19, as we add x to 12 or 7. This means that to get to $(1, 2)$ we must add 5 to 24.

$$24 + 5 = \boxed{29}.$$

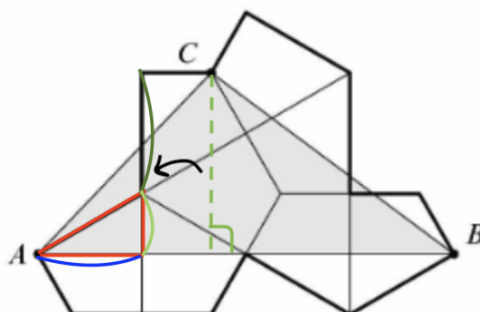
- v) (AMC 10 10A 2024) Let $\mathcal{K}$ be the kite formed by joining two right triangles with legs 1 and $\sqrt{3}$ along a common hypotenuse. Eight copies of $\mathcal{K}$ are used to form the polygon shown below. What is the area of triangle $\triangle ABC$?



- (A) $2 + 3\sqrt{3}$ (B) $\frac{9}{2}\sqrt{3}$ (C) $\frac{10 + 8\sqrt{3}}{3}$ (D) 8 (E) $5\sqrt{3}$

Answer: B

Solution 1



First, we should find the length of AB . In order to do this, as we see in the diagram, it can be split into 4 equal sections. Since diagram K shows us that it is made up of two $30, 60, 90$ triangles, then the triangle outlined in red must be a $30, 60, 90$ triangle, as $30 + 30 = 60$, and the two lines are perpendicular (it is proveable, but during competition, it is best to assume this is true, as the diagram is drawn pretty well to scale). Also, since we know the length of the longest side of the red triangle is $\sqrt{3}$, then the side we are looking for, which is outlined in blue, must be $\frac{3}{2}$ by the $1, \sqrt{3}, 2$ relationship of $30, 60, 90$ triangles. Therefore AB , which is the base of the triangle we are looking for, must be 6.

Now all we have to do is find the height. We can split the height into 2 sections, the green and the light green. The green section must be $\sqrt{3}$, as K shows us. Also, the light green section must be equal to $\frac{\sqrt{3}}{2}$, as in the previous paragraph, the triangle outlined in red is $30, 60, 90$. Then, the green section, which is the height, must be $\sqrt{3} + \frac{\sqrt{3}}{2}$, which is just $\frac{3\sqrt{3}}{2}$.

Then the area of the triangle must be $\frac{1}{2} \cdot b \cdot h$, which is just $\boxed{\text{(B)} \frac{9}{2}\sqrt{3}}$.

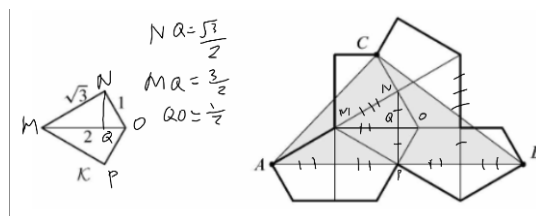
Simple Proof of the red triangle having 30° , 60° , and 90° angles:

In the small kite illustrated in the problem, we can see that the two angles on the left side add up to 60° (this is easily noticeable if you look hard enough.) So, the angle that the arrow is pointing to is 60° . Since another angle in the red triangle is obviously 90° , the portion of the angle that is in the red

triangle and at point A is $180^\circ - 90^\circ - 60^\circ = 30^\circ$. Therefore, the red triangle is a $30^\circ 60^\circ 90^\circ$ triangle.

Solution 2

Let $\mathcal{K}$ be quadrilateral $MNOP$. Drawing line MO splits the triangle into $\triangle MNO$. Drawing the altitude from N to point Q on line MO , we know NQ is $\frac{\sqrt{3}}{2}$, MQ is $\frac{3}{2}$, and QO is $\frac{1}{2}$.



Due to the many similarities present, we can find that AB is $4(MQ)$, and the height of $\triangle ABC$ is $NQ + MN$. AB is $4(\frac{3}{2}) = 6$ and the height of $\triangle ABC$ is $\sqrt{3} + \frac{\sqrt{3}}{2} = \frac{3\sqrt{3}}{2}$.

Solving for the area of $\triangle ABC$ gives $6 \cdot \frac{3\sqrt{3}}{2} \cdot \frac{1}{2}$ which is **(B)** $\frac{9}{2}\sqrt{3}$

Solution 3

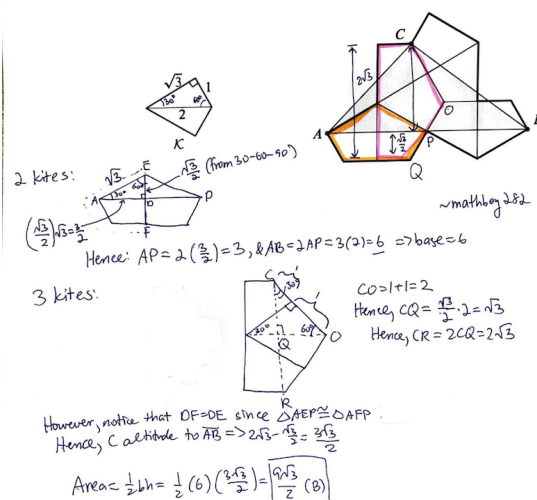
Let's start by looking at kite $\mathcal{K}$. We can quickly deduce based off of the side lengths that the kite can be split into two $30^\circ - 60^\circ - 90^\circ$ triangles. Going back to the triangle $\triangle ABC$, focus on side AB . There are 4 kites, they are all either reflected over the line AB or a line perpendicular to AB , meaning the length of AB can be split up into 4 equal parts.

Pick out the bottom-left kite, and we can observe that the kite and the triangle formed by the intersection of the kite and $\triangle ABC$ share a 60° degree angle. (this was deduced from the $30^\circ - 60^\circ - 90^\circ$ triangles in the kite) The line AB and the right side of the kite are perpendicular, forming a 90° angle. Because that is also a $30^\circ - 60^\circ - 90^\circ$ triangle with a hypotenuse of $\sqrt{3}$, so we find the length of AB to be $4 \cdot \frac{3}{2}$, which is 6.

Then, we can drop an altitude from C to AB . We know that will be equivalent to the sum of the longer side of the kite and the shorter side of the triangle formed by the intersection of the kite and $\triangle ABC$. (Look at the line formed on the left of C that drops down to AB if you are confused) We already have those values from the $30^\circ - 60^\circ - 90^\circ$ triangles, so we can just plug it into the triangle area formula, $bh/2$. We get

$$6 \cdot \frac{\sqrt{3} + \frac{\sqrt{3}}{2}}{2} \rightarrow 3 \cdot (\sqrt{3} + \frac{\sqrt{3}}{2}) \rightarrow 3 \cdot \frac{\sqrt{3}}{2} \rightarrow \textbf{(B)} \quad \frac{9}{2}\sqrt{3}$$

Solution 4



Solution 5

Let the point of intersection of AB and the kite with A as vertex be D .

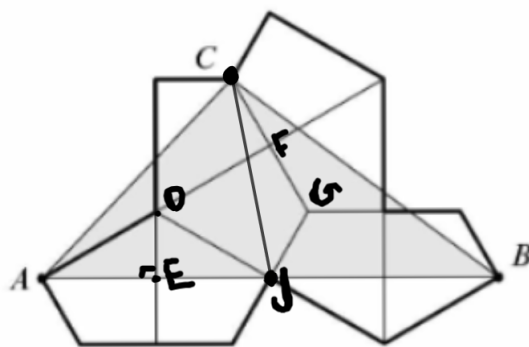
Let the left kite with C as a vertex touch the kite with A as vertex at point E .

$\triangle ADE$ is a $30 - 60 - 90$ so $AD = \frac{3}{2}$ and $DE = \frac{\sqrt{3}}{2}$.

So, $AB = 4 \cdot AD = 6$ and $CD = CE + DE = \frac{3\sqrt{3}}{2}$, and the area is

$$\frac{1}{2} \cdot AB \cdot CD = \boxed{\text{(B)} \frac{9}{2} \sqrt{3}}.$$

Solution 6(trig bash)



Step 1:

As stated in the solutions above we can easily find that AB is split into 4 equal parts, so we have $AB = 4(AE)$. We can calculate AE , by using AA similarity to find $\triangle ADE$ is a $30 - 60 - 90$ triangle, therefore we have $AD = \sqrt{3}$, $DE = \frac{\sqrt{3}}{2}$, and finally $AE = \frac{3}{2}$, therefore $AB = 4(AE) = 4(\frac{3}{2}) = 6$. Similarly we have CF is congruent to FG , therefore $CF = FG = 1$. Next we have AD is congruent to DF telling us $AF = AD + DF = 2(AD) = 2\sqrt{3}$. Noticing $\triangle CFA$ is right, we apply Pythagorean theorem to $\triangle CFA$ to find

$$AC^2 = CF^2 + AF^2$$

$$AC^2 = (1)^2 + (2\sqrt{3})^2 = 13$$

$$AC = \sqrt{13}.$$

Step 2:

Next we would like to calculate CJ, As said before $CF = FG = 1$ so $CG = 2$. We know the inscribed angle between CG and GJ is $60 \cdot 2 = 120$, and finally we know $GJ = 1$. So we apply LoC on triangle CGJ in order to find CJ.

$$CJ^2 = CG^2 + GJ^2 - 2 \cdot CG \cdot GJ \cdot \cos 120^\circ.$$

$$CJ^2 = 2^2 + 1^2 - 2 \cdot 2 \cdot 1 \cdot \cos 120^\circ.$$

$$CJ^2 = 5 - 2 \cdot 2 \cdot \frac{-1}{2} = 7.$$

$$CJ = \sqrt{7}.$$

Step 3:

Now since we have all side lengths of $\triangle CAJ$ we can find $\cos \angle CAJ$. Applying LoC again on $\triangle CAJ$ we have,

$$CJ^2 = CA^2 + AJ^2 - 2 \cdot CA \cdot AJ \cdot \cos \angle CAJ.$$

$$\sqrt{7}^2 = \sqrt{13}^2 + 3^2 - 2 \cdot 3 \cdot \sqrt{13} \cdot \cos \angle CAJ.$$

$$7 = 22 - 6\sqrt{13} \cdot \cos \angle CAJ.$$

$$\frac{-15}{-6\sqrt{13}} = \frac{5\sqrt{13}}{26} = \cos \angle CAJ.$$

Step 4:

We can solve for the area using the sin area formula which is $\frac{1}{2} \cdot AC \cdot AB \sin \angle CAJ$. To find $\sin \angle CAJ$ we use the well known fact $\sin^2 x + \cos^2 x = 1$. So we find,

$$\sin^2 \angle CAJ + \cos^2 \angle CAJ = 1.$$

$$\sin^2 \angle CAJ + \left(\frac{5\sqrt{13}}{26}\right)^2 = 1.$$

$$\sin^2 \angle CAJ + \frac{25}{52} = 1.$$

$$\sin^2 \angle CAJ = \frac{27}{52}.$$

$$\sin \angle CAJ = \frac{3\sqrt{39}}{26}.$$

Finally to wrap up we can find the area of $\triangle ABC$ using the sin area formula,

$$[ABC] = \frac{1}{2} \cdot AC \cdot AB \cdot \sin \angle CAJ.$$

$$[ABC] = \frac{1}{2} \cdot \sqrt{13} \cdot 6 \cdot \frac{3\sqrt{39}}{26} = \frac{9\sqrt{3}}{2}.$$

Therefore our answer is **(B)** $\frac{9}{2}\sqrt{3}$.

Solution 7(coordinate bash)

Assume we set A as (0,0). The line AB can be split into 4 equal sections, easily noticed in the picture. Not that the first equal section, starting at A. When we follow the line AB, we see that it's a right angle, noticing its a 30, 60, 90 triangle. Because the hypotenuse of that triangle is $\sqrt{3}$. Using the 30, 60, 90 ratio, we see that one of those equal sections is $3/2$. Making the coordinate of B, (6,0).

Now, we must find the coordinate of C. Note that when we drop C down to the perpendicular of AB, there is a x component to C, and a y component to C. To find the X component, we see that it's one equal section found before as $3/2$, and another section equal to 1, given in the question, making it $x = 5/2$. The Y component can be seen as another two sections. The two green sections marked in solution 1. One is $\sqrt{3}/2$ due to the same 30 60 90 triangle we found in the first part of the solution. And the other part is $\sqrt{3}$, given in the question. Making $y = \frac{3}{2}\sqrt{3}$. So we see that $C = (5/2, \frac{3}{2}\sqrt{3})$

Using shoelace theorem, we get to get **(B)** $\frac{9}{2}\sqrt{3}$.

sorry for no diagrams D:

-marcus

Alternate way to calculate area: Notice once you get the y-coordinate of C, you can realize it's just the height of the triangle. So the area would be $\frac{1}{2}$ times base times height where the base is 6 and the height is $\frac{3}{2}\sqrt{3}$. ilikemath247365

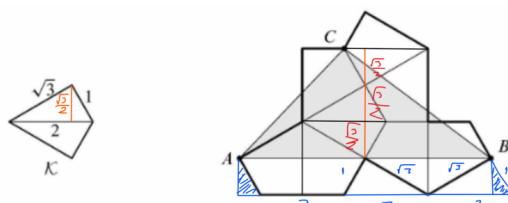
Sidenote The figure made of kites was in fact, an einstein tile (also known as the Hat). The einstein tile was discovered in November of 2022 by a retired printing systems engineer in England. It comes from the German word for one stone, and has been proven to infinitely tile a plane aperiodically, quite a unique property - it took decades for mathematicians to find a periodical tilings of smaller and smaller numbers, with Penrose discovering a method using only 2 tilings in 1974. LeonQS

Solution 8

When looking at the figure, we can see that the corner is shaped irregularly. This can make a regular hexagon with side length 2. Then, you would find the height of the triangle using the hexagon, which would get you $\frac{1}{2} \times 6 \times \frac{3\sqrt{3}}{2}$,

which is **(B)** $\frac{9}{2}\sqrt{3}$.

Solution 9



The triangle with side lengths 1, $\sqrt{3}$, and 2 will be referenced as "half kite" in this solution

First, notice how the altitude is composed of three half kites.

After finding the altitude of a half kite, which is $\frac{\sqrt{3}}{2}$, we can figure out that the altitude of triangle ABC is $\frac{3\sqrt{3}}{2}$

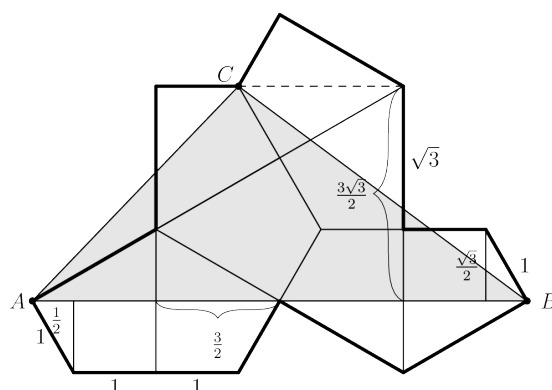
Now we take a look at the base. Notice how we can create two half kites, each with base 2. Adding that to the two 1 unit segments and the base will have length 6.

Using the area formula for a triangle, we get $6 \cdot \frac{3\sqrt{3}}{2} \cdot \frac{1}{2}$ which is **(B)** $\frac{9}{2}\sqrt{3}$

Solution 10 (Fast)

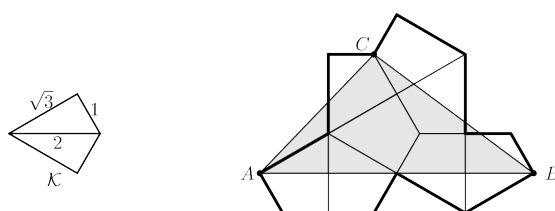
Notice that $[ABC] = \frac{AB}{2} \cdot K$ where K is the length of the altitude from C to $\overline{AB}$. Using the similar triangles shown below yields that $\frac{AB}{2} = 3$ and

$K = \frac{3\sqrt{3}}{2}$ so $[ABC] = \textbf{(B)} \frac{9}{2}\sqrt{3}$.



shockfront99

Asymptote Diagram Reconstruction I saw there was no asymptote code for this, so I tried my best to recreate the diagram.



Feel free to use it :) (As you can see I used it to make the diagram in my solution)

- w) (AMC 10 10A 2024) Integers a , b , and c satisfy $ab + c = 100$, $bc + a = 87$, and $ca + b = 60$. What is $ab + bc + ca$? **(A)** 212 **(B)** 247 **(C)** 258 **(D)** 276 **(E)** 284

Answer: D

Solution 1

Subtracting the first two equations yields $(a - c)(b - 1) = 13$. Notice that both factors are integers, so $b - 1$ could equal one of 13, 1, -1, -13 and $b = 14, 2, 0, -12$. We consider each case separately:

For $b = 0$, from the second equation, we see that $a = 87$. Then $87c = 60$, which is not possible as c is an integer, so this case is invalid.

For $b = 2$, we have $2c + a = 87$ and $ca = 58$, which by experimentation on the factors of 58 has no solution, so this is also invalid.

For $b = 14$, we have $14c + a = 87$ and $ca = 46$, which by experimentation on the factors of 46 has no solution, so this is also invalid.

Thus, we must have $b = -12$, so $a = 12c + 87$ and $ca = 72$. Thus $c(12c + 87) = 72$, so $c(4c + 29) = 24$. We can simply trial and error this to find that $c = -8$ so then $a = -9$. The answer is then $(-9)(-12) + (-12)(-8) + (-8)(-9) = 108 + 96 + 72 = \boxed{\text{(D)} 276}$.

By Antony Huang Very Minor Edit by lucassf12

Solution 2

Adding up first two equations:

$$(a + c)(b + 1) = 187$$

$$b + 1 = \pm 11, \pm 17$$

$$b = -12, 10, -18, 16$$

Subtracting equation 1 from equation 2:

$$(a - c)(b - 1) = 13$$

$$b - 1 = \pm 1, \pm 13$$

$$b = 0, 2, -12, 14$$

$$\Rightarrow b = -12$$

Which implies that $a + c = -17$ from $(a + c)(b + 1) = 187$

Giving us that $a + b + c = -29$

Therefore, $ab + bc + ac = 100 + 87 + 60 - (a + b + c) = \boxed{\text{(D)} 276}$

Solution 3 (Guess and check)

The idea is that you could guess values for c , since then a and b are factors of $100 - c$. The important thing to realize is that a , b , and c are all negative. Then, this can be solved in a few minutes, giving the solution $(-9, -12, -8)$, which gives the answer $\boxed{\text{(D)} 276}$ andliu766

Solution 4

$$ab + c = 100 \tag{22}$$

$$bc + a = 87 \tag{23}$$

$$ca + b = 60 \tag{24}$$

$$(1) + (2) \implies ab + c + bc + a = (a + c)(b + 1) = 187 \implies b + 1 = \pm 11, \pm 17$$

$$(1) - (2) \implies ab + c - bc - a = (a - c)(b - 1) = 13 \implies b - 1 = \pm 1, \pm 13$$

Note that $(b + 1) - (b - 1) = 2$, and the only possible pair of results that yields this is $b - 1 = -13$ and $b + 1 = -11$, so $a + c = -17$.

Therefore,

Solution 5

$$ab + c = 100 \tag{25}$$

$$bc + a = 87 \tag{26}$$

$$ca + b = 60 \tag{27}$$

$$(1) - (2) \implies ab + c - bc - a = (a - c)(b - 1) = 13$$

$$(2) - (3) \implies bc + a - ca - b = (b - a)(c - 1) = 27$$

$$(3) - (1) \implies ca + b - ab - c = (c - b)(a - 1) = -40$$

There are 3 ordered triplets of (a, b, c) : $(5, 14, 4)$, $(-3, -12, -3)$, $(-9, -12, -8)$.

However, only the last ordered triplet meets all three equations.

Therefore, $ab + ba + ac = -9 \cdot -12 + -12 \cdot -8 + -8 \cdot -9 = \boxed{\text{(D)} 276}$.

Solution 6 (Elimination)

Before we start, keep in mind that the problem is asking for the sum $ab+bc+ac$. This is nothing but $100 + 87 + 60 - a - b - c$, or $247 - (a + b + c)$.

To solve the problem, we systematically test the options using elimination:

Let's first check options A and B, since they only happen when a,b, and c sum to 35 or 0. We begin by testing three positive values, but none satisfy the equation when there is a plus sign. For example, $(12, 8, 4)$ satisfies $ab+c = 100$, but does not satisfy $bc + a = 87$, or $ac + b = 60$. If $a + b + c = 0$, then not all of the numbers can be positive or negative, so this would not work. From this observation, we conclude that the answer cannot be **A** or **B**.

Now let's test the next option, option C. Option **C** states $ab + bc + ca = 258$. If true, then:

$$a + b + c = -11$$

This sum is too large. Furthermore, if all three numbers are negative, the solution still fails. For example, testing $(-4, -5, -2)$ confirms the equation is not satisfied, as we get results that are too small. Thus, we eliminate option **C**.

Finally, let's test the last two options: D and E. For option **E**, the sum $a + b + c$ would be:

$$247 - 284 = -37$$

Testing values such as $(-11, -12, -14)$, the resulting sums $ab + c$, $bc + a$, and $ac + b$ are far too large to satisfy the equation. Therefore, **E** is also eliminated.

Once we have this answer, we still need to verify it by testing out numbers: Finally, we test option **D**. Using $ab + bc + ca = 276$, we get that $a + b + c = -29$. Also note that a , b , and c all have to be different, because the sums from the three equations are all different. We want to get the three closest values of a , b , and c such that they are all different, and the sum $a + b + c = -29$. The values $(-9, -12, -8)$ are the closest three numbers. When we try them, they satisfy all three equations. So, the correct answer is: $ab + ba + ac = -9 \cdot -12 + -12 \cdot -8 + -8 \cdot -9 = \boxed{\text{(D) } 276}$.

I NEED EASIER SOLUTIONS HERE:

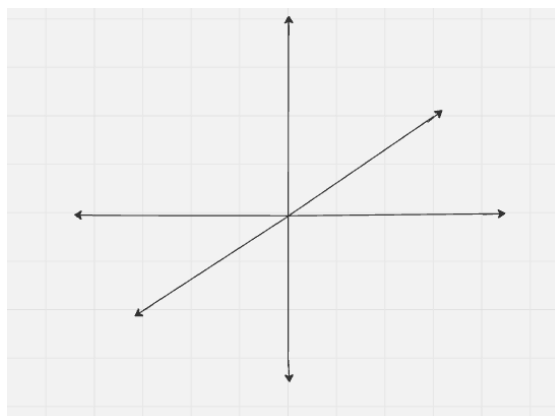
- x) (AMC 10 10A 2024) A bee is moving in three-dimensional space. A fair six-sided die with faces labeled A^+ , A^- , B^+ , B^- , C^+ , and C^- is rolled. Suppose the bee occupies the point (a, b, c) . If the die shows A^+ , then the bee moves to the point $(a + 1, b, c)$ and if the die shows A^- , then the bee moves to the point $(a - 1, b, c)$. Analogous moves are made with the other four outcomes. Suppose the bee starts at the point $(0, 0, 0)$ and the die is rolled four times. What is the probability that the bee traverses four distinct edges of some unit cube?

- (A) $\frac{1}{54}$ (B) $\frac{7}{54}$ (C) $\frac{1}{6}$ (D) $\frac{5}{18}$ (E) $\frac{2}{5}$

Answer: B

Solution 1 (Simplest)

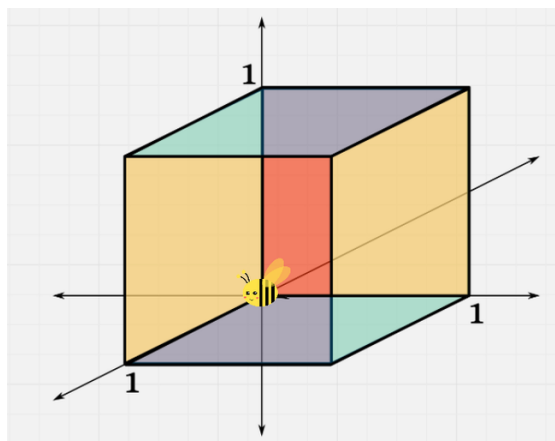
We start by imagining the three dimensional plane.



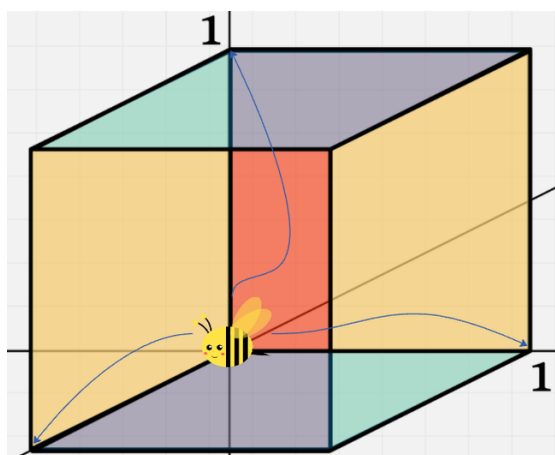
Notice how the three dimensional plane is split into 8 different regions. There exists 8 cubes with one vertex on the point $(0, 0, 0)$. We need to consider each of these cases.

We arbitrarily take a cube from one region of the three dimensional plane.

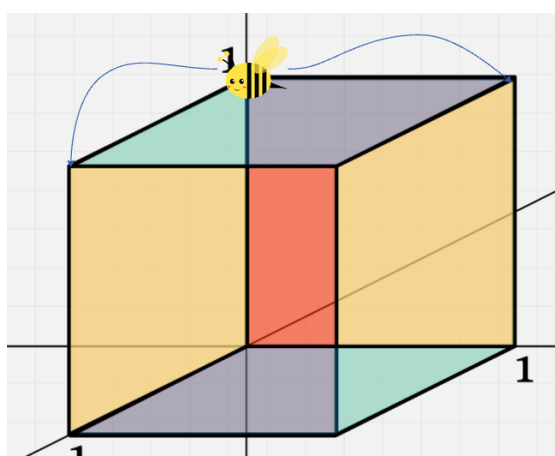
Below is a sample drawing.



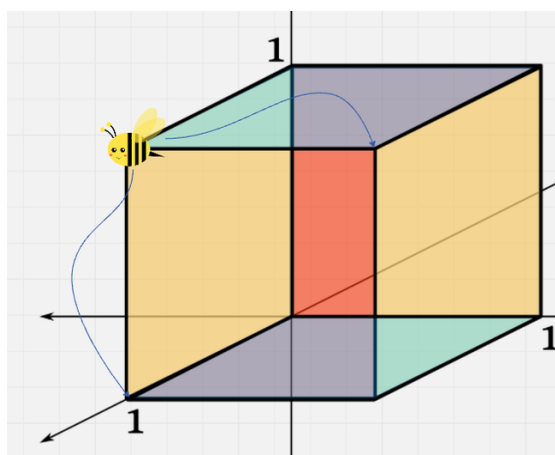
Roll 1: From $(0,0,0)$, there are 3 favorable outcomes that can occur. They are listed below.



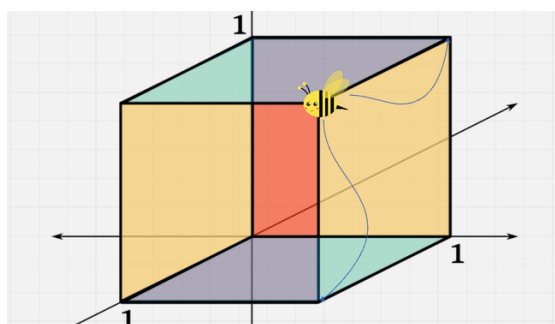
Roll 2: WLOG, say the bee moved up one, there are then 2 ways the bee can go.



Roll 3: WLOG, say the bee moved forward one. From this point, there is again 2 ways the bee can go.



Roll 4: Finally, WLOG, say the bee moves to the left one. From that point, there are still 2 ways the bee can go.



The total probability for one cube is $\frac{3 \cdot 2 \cdot 2 \cdot 2}{6^4} \Rightarrow \frac{1}{54}$. We have 8 of these cubes, which gives us a total probability of $\frac{8}{54}$.

However, we have overcounted. Notice that if the bee moved down one instead of left one at roll 3, it would only have one way. This is because the bee has a choice between either the origin or a unique point. This will always occur on the third move, and because we have 8 cubes (The 3d graph has one for each quadrant), the probability of this happening is $\frac{3 \cdot 8}{6^4} = \frac{1}{54}$ *

Our probability is $\frac{8}{54} - \frac{1}{54} = \boxed{\text{(B)} \frac{7}{54}}$.

* One might question this logic here, onto why it is that we multiply the moves by the number of cubes over the number of rolls.

The theory comes from expected value. It is certain that the bee will be selective in a situation where it only has one choice every 3 moves. This is due to the figure being a cube. Then, the expected probability of this happening is the moves multiplied by the single probability that this case appears, which is $\frac{1}{6^4}$. Finally, we multiply by 8, obtaining $\frac{3 \cdot 8}{6^4} = \frac{1}{54}$ and continuing as follows.

Solution 2 (Higher End Mathematics)

I put this as an experiment. If anything feels off or you don't understand, do not worry. This is very high end mathematics, and it is fine to feel confused. Solution 1 and the below solutions are much easier than this.

Here, we explore the topological (study of bendable figures in space) definition of this problem.

Consider an n -dimensional torus that the bee travels upon. Then, this torus must be an $8 \times 8 \times 8$, as there are 8 three-dimensional cubes. The reason we can relate this to a torus is due to the physique of the 3D plane.

Then, we wish to find the ways to get from $(0, 0, 0)$ to $(0, 0, 0)$ in 512 moves. That isn't too difficult. In topology, the number of ways to do this is

$$\sum_{\gcd(x,y,z,8)=1} \binom{8}{x} \binom{8}{y} \binom{8}{z}$$

Summing over all values x, y , and z gives $2^{21} \times 7$.

Now, the total. The total number of moves that exist are all 512 plausible moves upon each probable space such that not all x, y, z are in gcd relation of 1 with 8. There is then 3^{512} different ways to permute x, y , and z . However, not all of these are plausible, as $x + y + z = 512$. These coordinates then force their way back to the origin due to the fact $x \equiv y \equiv z \equiv 0 \pmod{8}$. Then, we can continue with roots of unity to filter out these multiples of 8. Our standard divisibility counter must satisfy

$$\mathbf{1}_{x \equiv 0 \pmod{8}} = \frac{1}{8} \sum_{j=0}^7 \omega^{jx}, \quad \omega = e^{2\pi i/8}$$

(Recall that this bolded 1 is the indicator function).

Application to all coordinates satisfies

$$N = \frac{1}{8^3} \sum_{a,b,c=0}^7 \sum_{x+y+z=512} \binom{512}{x,y,z} \omega^{ax+by+cz}$$

Through the **Multinomial Theorem**,

$$\sum_{x+y+z=512} \binom{512}{x,y,z} \omega^{ax+by+cz} = (\omega^a + \omega^b + \omega^c)^{512}$$

is simplified, and thus we obtain

$$N = \frac{1}{512} \sum_{a,b,c=0}^7 (\omega^a + \omega^b + \omega^c)^{512}$$

Because 512 is divisible by 8, any term where $|\omega^a + \omega^b + \omega^c| < 2$ doesn't contribute to the overall terms x, y , and z . The only triplets that count are magnitude 2.

Carrying this out gives 3^3 different full unital root contributions, and these are applied in 2^{22} different roots, for a total $2^{22} \times 27$ ways.

Our probability is then

$$\frac{2^{21} \times 7}{2^{22} \times 27} = \frac{7}{2 \times 27} = \boxed{\frac{7}{54}}$$

Once again, this is an incredibly difficult solution. Before even attempting this, be sure you have a solid background in intermediate to Olympiad level algebra, precalculus, and counting/probability.

Solution 3

WLOG, assume that the first two moves are equal for all possible combinations, since the direction does not matter. The first move has a $\frac{6}{6}$ probability of being along one of the 8 unit cubes around the origin, and the second move has a $\frac{4}{6}$ chance. Now, there are two cases. We are currently on one of the points of the 2 by 2 squares that are aligned with the axes. The first case is if the bee moves to the corner of a cube farthest away from the origin. Here, there is a $\frac{2}{6}$ chance of this happening and a $\frac{2}{6}$ chance of the fourth move remaining on one of the cubes. The second case is if the bee moves along the same plane of the 2 by 2 squares previously, ending up on a point 1 away from the origin. There is a $\frac{1}{6}$ chance of this happening and a $\frac{3}{6}$ chance of remaining on one of the cubes. Now, multiply and sum for the answer.

$$\frac{2}{3} \cdot \left(\frac{1}{3} \cdot \frac{1}{3} + \frac{1}{6} \cdot \frac{1}{2} \right) = \frac{2}{3} \cdot \left(\frac{1}{9} + \frac{1}{12} \right)$$

Evaluating this gives you the answer of **(B)** $\frac{7}{54}$. Solution by [juwushu](#). Minor edits by andliu766

Solution 4

Remove the dice. It is not necessary for this problem. It is the same problem if you instead view this problem as the bee moving one up or down in the x , y , and z directions.

Count the total number of ways for one cube of the 8 total ones. It is not hard to see that after the first step (which you can travel to $(1, 0, 0)$, $(0, 1, 0)$, or $(0, 0, 1)$ - 3 ways), you have a total of 8 moves to traverse a face of that 1×1 cube. Hence, for each cube, you would have $8 \cdot 3 = 24$ total moves.

There are a total of 8 cubes you could have traversed, though, so multiplying this by 8 yields $8 \cdot 8 \cdot 3$.

We have to account for overcounting. For example, when we traversed the face in the xz plane $(0, 0, 0)$, $(0, 0, 1)$, $(1, 0, 0)$, $(1, 0, 1)$, the number of ways that we traverse that particular face got counted twice, once from the cube in the far upper right and once from the cube in the closer upper right.

For a given face, you can go clockwise or counterclockwise, hence yielding 2 ways. There are a total of 12 shared faces, each one shared by exactly two faces so no additional PIE is needed. We simply subtract $2 \cdot 12$ from our original count now, yielding $8 \cdot 3 \cdot 8 - 2 \cdot 12 = 168$.

With no restrictions, there are a total of 6^4 ways to move four steps. Your probability is $\frac{168}{6^4} = \frac{7}{54}$.

Solution 5

An invariant is that the sum of all three coordinates change parity with every roll of the dice. Hence, to end up on a vertex of a unit cube, it can only be $(0, 0, 0)$, $(1, 1, 0)$, $(1, 0, 1)$, $(0, 1, 1)$, and all the negative permutations of each listed (in total, 13 different final coordinates can be reached).

For $(0, 0, 0)$, we choose two of the three coordinates to have 'cancelling' operations. Let these be a, a', b, b' (where a cancels out a' , hence cannot be placed next to each other, which would be traversing the same edge). There are $3 \cdot 2 \cdot 2 \cdot 2 = 24$ ways for this (we can only have forms of $aba'b'$ or $bab'a'$ and their permutations among a and b).

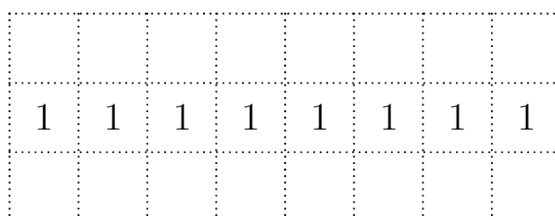
For $(1, 1, 0)$, we must have two operations a, b and a pair c, c' . Experimenting with having 3 of the same type of operation leads to finding a repeated edge no matter the arrangement. These can be arranged in 12 ways (place the operations a and b first, placing the operations c and c' in the remaining spaces).

Analogously, $(-1, 1, 0)$, $(1, -1, 0)$, $(-1, -1, 0)$ all have the same number of ways, 12 each (they are merely a rotation of all possibilities of $(1, 1, 0)$). Furthermore, we can generalise this to all the 8 negative permutations of $(1, 0, 1)$, $(0, 1, 1)$, for a total of $12 \cdot 4 \cdot 3 = 144$.

So, $\frac{24+144}{6^4} = \frac{7 \cdot 24}{54 \cdot 24} = \frac{7}{54}$, hence we pick **(B)** $\frac{7}{54}$.

-lisztapos

- y) (AMC 10 10A 2024) The figure below shows a dotted grid 8 cells wide and 3 cells tall consisting of $1'' \times 1''$ squares. Carl places 1-inch toothpicks along some of the sides of the squares to create a closed loop that does not intersect itself. The numbers in the cells indicate the number of sides of that square that are to be covered by toothpicks, and any number of toothpicks are allowed if no number is written. In how many ways can Carl place the toothpicks?



- (A) 130 (B) 144 (C) 146 (D) 162 (E) 196

Answer: C

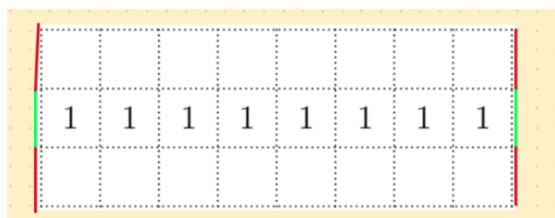
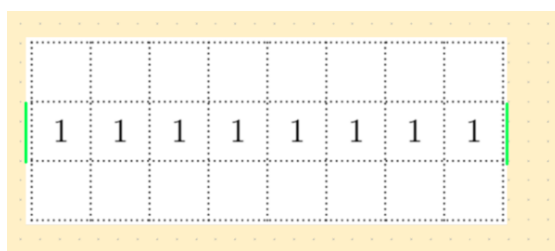
Solution 1 (Best Motivated)

Observations:

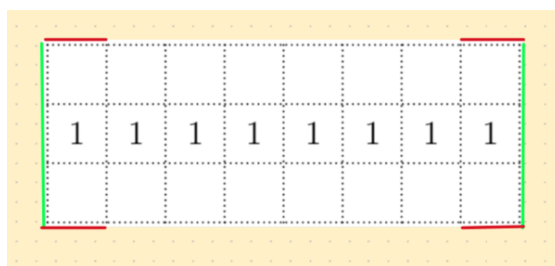
1. You can not have a vertical line in any place other than the first two columns and the last two columns. If we did, we would have at least one of the middle cells with toothpicks along more than one side, which would violate the conditions of the problem.
2. There are two cases that look completely different. We can have a long horizontal box that spans all eight cells either on top of or below the middle cells, or we have to have a shape that looks like a rectangle, except with a few places "pushed" in.

Thus, using casework, we can split the task of finding those rectangles with squiggly edges into 3 cases.

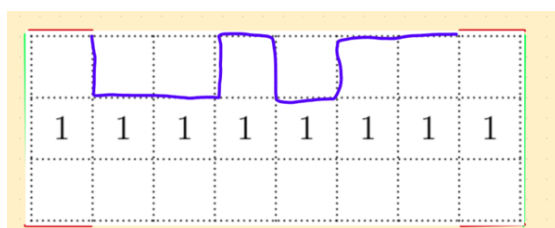
For case 1, we assume that the green lines shown below are given (always have toothpicks on them). In effect, we will use all eight columns.



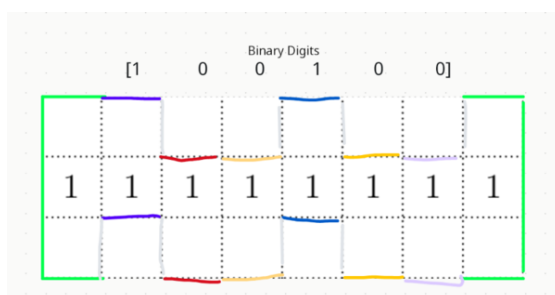
The only toothpicks we can place that will connect to the red lines are to go horizontally inward:



Now, concentrate on the first row of squares. A toothpick can be placed on either the bottom or top and connected to a continuous squiggle by adding vertical toothpicks:

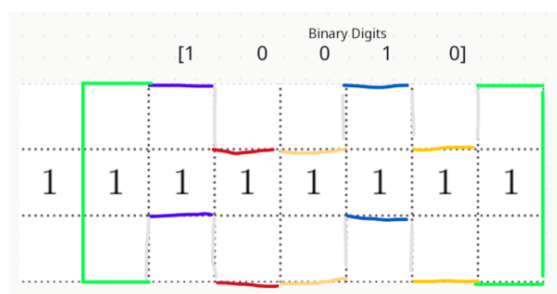
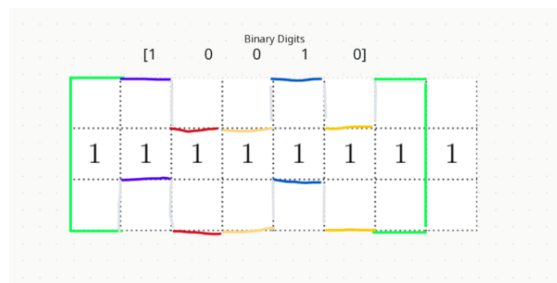


How many squiggles are possible?



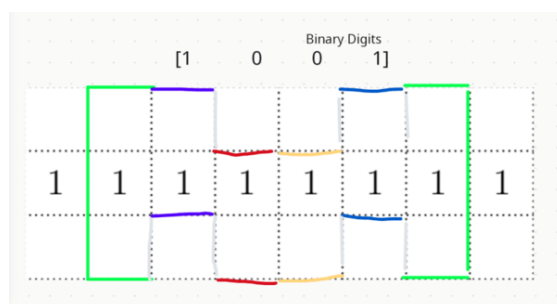
We can summarize this by giving a high squiggle position a 1 and a low position a 0, thus we have a 6-digit binary sequence. Thus, we can have $2^6 = 64$ ways to make this squiggle. (The binary is not absolutely necessary, but it works.)

Case 2: We can also pull in one of the sides, thus we can have a squiggle with 5 binary digits, which only uses the first or last 7 columns:



Here, we only have 5 binary digits to work with, so there are $2^5 = 32$ ways to make this squiggle for each individual subcase. There are two subcases, one with the first 7 columns, and the other with the last 7, so we have a total of $32 \cdot 2 = 64$ arrangements in this case.

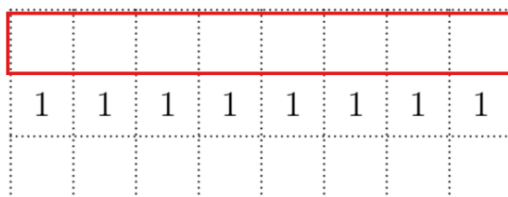
Case 3: We can use an even smaller section. Using only the middle 6 columns gives us a 4-wide squiggle:



Thus, there are $2^4 = 16$ ways to make this squiggle.

These three cases together cover all loops of this form. If we try to bring the square bracket like shapes on each side any closer, there will be some middle cells that do not touch any toothpicks. Adding up all our cases for these types of shapes: $64 + 32 + 32 + 16 = 144$.

However, there are two more ways to draw a qualifying shape:



We can draw a rectangle like that in the first row or third row. Thus, we have a total of $144 + 2 = \boxed{\text{(C) } 146}$ ways.

A note to (potential) editors: This answer was not made to be concise or especially professional. It was made to explicitly explain this problem in a way so that it is easy to understand and follow.

Notes: Remember these are the ONLY possible cases. It is impossible to cross through the rows of boxes of ones (in a snake like pattern) to connect the loop around the bottom since then the loop would intersect itself.

We are not undercounting by only counting the binary digits on the top. Consider the top squiggle. Each middle square must and can only come in contact with one toothpick, so if the top squiggle doesn't touch a specific middle cell, the bottom squiggle must ensure that all squares touch one toothpick. juwushu

Solution 2 (Cheese)

Notice that for any case where the closed loop does not connect from the top side of the ones and bottom side of the ones, there are two of these cases. A cheese solution can be found from this; noting that B and C are the only two options close to each other, and, being two apart, with people likely to forget this case, $\boxed{\text{(C) } 146}$ is likely to be the correct answer. Cheese solution done by juwushu.

Solution 3 (Observation)

We have 2 cases where the loop does not go through the middle.

If the loop goes through the middle, we must have a full column on $(0, 8), (0, 7), (1, 8), (1, 7)$. Then we have 6, 5, 5, 4 empty middle squares. For each one we can have one on top or one on bottom, so $2^6 + 2^5 + 2^5 + 2^4 = 144$. Notice that for each case of fixed toothpicks, there is only one way to form the loop. Then we just add $2 + 144 = \boxed{\text{(C) } 146}$.

§13 AMC 10 10B 2025

- a) (AMC 10 10B 2025) The instructions on a 350-gram bag of coffee beans say that proper brewing of a large mug of pour-over coffee requires 20 grams of coffee beans. What is the greatest number of properly brewed large mugs of coffee that can be made from the coffee beans in that bag?

(A) 16 (B) 17 (C) 18 (D) 19 (E) 20

Answer: B

Solution 1

$\frac{350}{20}$ is 17.5. We round down to find the maximum he can make (he cannot brew half of a cup), so the answer is **(B) 17**.

Solution 2

We can use 340-grams if we use 17 mugs. This is because $20 \cdot 17 = 340$. We cannot use 18 because that would mean using 360-grams. The answer is **(B) 17**.

Note that you can also find this by finding $\lfloor \frac{350}{20} \rfloor = 17$.

Solution 3

Dividing 350 by 20, we get 17 with a remainder of 10. However, 10 grams cannot make a properly brewed large mug of coffee, as stated in the problem. We can only count fully brewed mugs of coffee, hence the answer is **(B) 17**.

Solution 4 - Cheese (Sort of)

We multiply 20 by each of the answer choices to see if we can find which one is closest and under 350. First, we try solution A, which is 16. $16 \cdot 20 = 320$, which is 30 off from 350, so A cannot be the answer, because we can add another 20. Moving on to answer choice B, $17 \cdot 20 = 340$. We notice that $350 - 340 = 10$, and so we cannot add another 20 to get closer but less than 350. (The next number, 18, would result in $20 \cdot 18 = 360$, which is more than 350.) So, the answer is **(B) 17**.

(note - the actual cheese solution would likely include every single answer choice, but I use a bit of intellect in this solution)

- b) (*AMC 10 10B 2025*) Jerry wrote down the ones digit of each of the first 2025 positive squares: 1, 4, 9, 6, 5, 6, $\dots$. What is the sum of all the numbers Jerry wrote down?

(A) 9025 (B) 9070 (C) 9090 (D) 9115 (E) 9160

Answer: D

Solution 1

By a modulo 10 argument, the ones digits repeat with period 10 in the following order:

$$1, 4, 9, 6, 5, 6, 9, 4, 1, 0$$

The sum of the numbers can be verified to be 45.

There are 202 periods that occur from 1 to 2025, and there are five extra numbers, those being 1, 4, 9, 6, 5, corresponding to 2021, 2022, 2023, 2024, 2025. The sum of these numbers is 25.

Hence, the total is

$$202 \cdot 45 + 25 = 9090 + 25 = \mathbf{(D) 9115}$$

Solution 2 (Slightly more rigorous than Solution 1, but same idea)

Let $x \equiv n \pmod{10}$. We have $x^2 \equiv n^2 \pmod{10}$. So, the numbers must repeat in patterns of 10. Listing them out gives:

$$1, 4, 9, 6, 5, 6, 9, 4, 1, 0.$$

These numbers sum to 45, so the sum of the first 2020 numbers is $45 \cdot 202 = 9090$. Since we need 5 more numbers, we add $1 + 4 + 9 + 6 + 5 = 25$ to our answer to get (D) 9115.

Solution 3 (Similar to the other solutions but without the use of mods)

The ones digit of a square number is determined by the ones digit of the base. We can find the pattern by looking at the ones digits of the first ten positive squares:

$$1^2 = 1$$

$$2^2 = 4$$

$$3^2 = 9$$

$$4^2 = 16 \quad (\text{ones digit } 6)$$

$$5^2 = 25 \quad (\text{ones digit } 5)$$

$$6^2 = 36 \quad (\text{ones digit } 6)$$

$$7^2 = 49 \quad (\text{ones digit } 9)$$

$$8^2 = 64 \quad (\text{ones digit } 4)$$

$$9^2 = 81 \quad (\text{ones digit } 1)$$

$$10^2 = 100 \quad (\text{ones digit } 0)$$

The sequence of ones digits is:

1, 4, 9, 6, 5, 6, 9, 4, 1, 0

This pattern of ten digits repeats for every subsequent group of ten squares (e.g., 11^2 has the same ones digit as 1^2 , 12^2 as 2^2 , etc.).

$$1 + 4 + 9 + 6 + 5 + 6 + 9 + 4 + 1 + 0 = 45$$

We now divide by 10 as the pattern repeats every 10 terms: $2025/10$ results in 202 full cycles with a remainder of 5. The sum of the 202 full cycles is:

$$202 \cdot 45 = 9090$$

However, we must add back 5 digits, as we had a remainder of 5 when we divided 2025 by 10. Therefore, the sum of the remaining 5 digits is:

$$1 + 4 + 9 + 6 + 5 = 25$$

Thus, the total sum is:

$$9090 + 25 = 9115$$

Therefore, the answer is (D) 9115.

Solution 4: Modular Arithmetic; Ultra-Rigorous, Not Recommended

We want the sum of the ones digits of $1^2, 2^2, 3^2, \dots, 2025^2$. The normal way is to notice the pattern repeats every 10, but instead we will approach this with more modular arithmetic than any sane person would tolerate.

Let $f(n) = n^2 \pmod{10}$, which extracts the ones digit of n^2 . We want

$$S = \sum_{n=1}^{2025} f(n).$$

First observe the basic periodicity result: Since $n^2 \pmod{10}$ depends only on $n \pmod{10}$, we have

$$f(n+10) \equiv f(n) \pmod{10}.$$

We will pretend this is deep.

Now compute all residues $n \pmod{10}$ and their corresponding $n^2 \pmod{10}$:

$$\begin{aligned} 0^2 &\equiv 0 \\ 1^2 &\equiv 1 \\ 2^2 &\equiv 4 \\ 3^2 &\equiv 9 \\ 4^2 &\equiv 6 \\ 5^2 &\equiv 5 \\ 6^2 &\equiv 6 \\ 7^2 &\equiv 9 \\ 8^2 &\equiv 4 \\ 9^2 &\equiv 1. \end{aligned}$$

Instead of immediately summing these, we treat the 10-term sequence as a modular function over the ring $\mathbb{Z}/10\mathbb{Z}$. Define the sequence $a_k = (k^2 \pmod{10})$ for $k \in \mathbb{Z}/10\mathbb{Z}$. We want the sum

$$T = \sum_{k=0}^9 a_k = 0 + 1 + 4 + 9 + 6 + 5 + 6 + 9 + 4 + 1 = 45.$$

Next, determine how many full cycles of length 10 lie in the first 2025 integers. Compute

$$2025 \pmod{10} \equiv 5.$$

Then

$$2025 = 10 \cdot 202 + 5.$$

So we have 202 complete cycles plus 5 extra terms.

Sum contributed by the full cycles:

$$202 \cdot 45.$$

We now compute this entirely modulo 10 even though we need the actual sum. First,

$$202 \equiv 2 \pmod{10}, \quad 45 \equiv 5 \pmod{10},$$

so

$$202 \cdot 45 \equiv 2 \cdot 5 = 10 \equiv 0 \pmod{10}.$$

This tells us nothing useful, so abandon that and compute normally:

$$202 \cdot 45 = (200 \cdot 45) + (2 \cdot 45) = 9000 + 90 = 9090.$$

Now handle the leftover 5 terms: $n = 1, 2, 3, 4, 5$. Compute $n^2 \pmod{10}$ again even though we already have it:

$$1^2 \equiv 1, \quad 2^2 \equiv 4, \quad 3^2 \equiv 9, \quad 4^2 \equiv 6, \quad 5^2 \equiv 5.$$

Sum:

$$1 + 4 + 9 + 6 + 5 = 25.$$

The full sum is

$$S = 9090 + 25 = 9115.$$

To unnecessarily verify using modular decomposition, write

$$S \equiv \sum_{n=1}^{2025} n^2 \pmod{10}.$$

We can compute the entire sum of squares formula modulo 10 and compare. Using

$$\sum_{n=1}^N n^2 = \frac{N(N+1)(2N+1)}{6},$$

take $N = 2025$. Reduce everything modulo 10 first:

$$2025 \equiv 5, \quad 2026 \equiv 6, \quad 4051 \equiv 1 \pmod{10}.$$

Now compute

$$5 \cdot 6 \cdot 1 = 30.$$

Thus

$$\sum_{n=1}^{2025} n^2 \equiv 30 \cdot 6^{-1} \pmod{10}.$$

Compute the inverse of 6 modulo 10. Solve $6x \equiv 1 \pmod{10}$. No such x exists because $\gcd(6, 10) = 2 \neq 1$. Therefore this modular-verification method is unusable, so we immediately stop and simply trust the earlier cycle-counting method.

Thus the sum of the ones digits of the first 2025 squares is

$$9115.$$

- c) (AMC 10 10B 2025) A Pascal-like triangle has 10 as the top row and 10 followed by 1 as the second row. In each subsequent row the first number is 10, the last number is 1, and, as in the standard Pascal's Triangle, each other number in the row is the sum of the two numbers directly above it. The first four rows are shown below:

$$\begin{array}{ccccccc} & & & & 10 & & \\ & & & & & & \\ & & 10 & & & 1 & \\ & 10 & & 11 & & & 1 \\ 10 & & 21 & & 12 & & 1 \end{array}$$

What is the sum of the digits of the sum of the numbers in the 11th row?

- (A) 11 (B) 13 (C) 14 (D) 16 (E) 17

Answer: D

Solution 1

If we find the sum for each row (I'll only do the first 5 to illustrate the pattern), we find that row 1 has sum 10, row 2 has sum 11, row 3 has sum 22, row 4 has sum 44, and row 5 has sum 88. So we can see that, excluding the first row, the n th row has sum $11 \cdot 2^{n-2}$. This means the 11th row will have sum $11 \cdot 2^9 = 11 \cdot 512 = 5632$. The sum of the digits of 5632 is 16, so the answer is **(D) 16**.

Why each row is twice the previous row Notice that the sum of each row is the slightly different than the normal Pascal's triangle, so the sum of the numbers is similar. Here's a proof of why each number in the normal Pascal's triangle is twice the previous number. To prove that the sum of each row is 2 times the previous row, use the equation $(1 + 1)^n$ with the normal Pascal's triangle and use the binomial theorem from there.

Example: In the fifth row of Pascal's triangle, the numbers are 1, 4, 6, 4, 1. The first number can be represented by $\binom{4}{0} = 1$, the second by $\binom{4}{1} = 4$, the third by $\binom{4}{2} = 6$, the fourth by $\binom{4}{3} = 4$, and the last by $\binom{4}{4} = 1$. The sum of these five numbers is 16 which is also 2^4 . Using the binomial theorem with $(1 + 1)^4$, we get $1 + 4 + 6 + 4 + 1 = 16$, the same results in Pascal's triangle. We could also solve $(1 + 1)^4$ by converting it to $2^4 = 16$, proving the identity that all rows in Pascal's triangle are exactly twice the previous row and can be represented by 2^n .

Solution 2

In this Pascal's Triangle-like sequence, after row 2, each number in row x appears twice in the sum of row $x + 1$. For example (to illustrate the pattern), the numbers in row 2 are 10 and 1, and these numbers indeed each appear twice in the sum of the next row (10, 10 + 1, and 1 are the numbers in row 3, so their sum can be written as $10 \cdot 2 + 1 \cdot 2$). With this pattern, we know that after row 2 the sum of the numbers doubles in each subsequent row, so now by the same calculation as in Solution 1, the sum of the numbers in row 11 is $(10 + 1) \cdot 2^{11-2} = 11 \cdot 2^9 = 5632$, and thus the answer is **(D) 16**.

Solution 3 (BRUTE FORCE!!!)

We can write out the first 11 rows with direct computation. Since no calculators are allowed, you will spend at least 5 minutes on this problem, so I do not recommend this way. Here are the first 11 rows.

Summing the last row gives $10 + 91 + 369 + 876 + 1344 + 1386 + 966 + 444 + 126 + 19 + 1 = 5632$. Summing the digits of 5632 gives $5 + 6 + 3 + 2 = \boxed{\text{(D) } 16}$.

Edit: After brute forcing two or three rows, it is clear that the sum doubles $(11, 22, 44, 88, \dots)$, and you can use this to find the sum of the 11th row as $11 \cdot 2^{11-2} = 5632$, which gives a digit sum of $5 + 6 + 3 + 2 = \boxed{\text{(D) } 16}$.

We can just take the entire triangle modulo 9, since all the answer choices have different residues modulo 9 and because a number is congruent to its digit sum modulo 9. Accordingly, since $10 \equiv 1 \pmod{9}$, the triangle just becomes Pascal's triangle modulo 9, and so by the standard result for the row sums in Pascal's-like triangles, the sum of the numbers in the 11th row is $2^{11-1} = 1024 \equiv 7 \pmod{9}$. This means the only feasible answer choice is 16, as it is the only one that is congruent to 7 (mod 9), and so the answer is indeed $\boxed{\text{(D)}\ 16}$. Storm1230323

We can start brute-forcing the first few rows and figure out the pattern that the sum of the numbers in row n , where $n \geq 2$, is $11 \cdot 2^{n-2}$. Now we gamble on this pattern (which is mostly correct for AMC 10 problems) and plug in 11 into n , which results in 5632. Hence the answer is $5 + 6 + 3 + 2 = \boxed{\text{(D)} 16}$.

If you failed or sillied the AMC 10 A Question 19, then during the AMC 10 B you gained the experience to know that each row probably doubles in these Pascal-like problems, either in retrospect or during the actual AMC 10 A. However, we should probably check, in case you sillied this problem because you didn't check something stupid on the 10 A. It is a good decision to check, because we realize that the first row does not conform to the doubling rule, and only the rows right after it do. The rest of the solution is similar to how we solve the AMC 10 A Question 19, except less bashy because we only want the sum, not a specific number, and we are given the row number instead of

the sum. So, we end up getting the sum to be 5632, and therefore the answer is (D) 16.

Solution 7 (Generating Function and Combinatorics)

We want the sum of the digits of the sum of all entries in the 11th row of this "Pascal-ish" triangle. We will solve this with far more modular arithmetic than required, even though everything collapses into a simple closed form.

Let the 11th row be indexed as row $n = 11$, with the top row being $n = 1$. Denote row sums by S_n .

Each row begins with 10 and ends with 1. For the interior entries, each behaves like a binomial coefficient, but with an appropriate shift, so the entries in row n look like:

Row n : $10, \binom{n-2}{1} + 1, \binom{n-2}{2} + \binom{n-2}{1}, \dots, \binom{n-2}{n-2} + \binom{n-2}{n-3}, 1$

But this is too reasonable, so instead we use generating functions and modular arithmetic.

Define the polynomial for row n :

$$P_n(x) = 10x^0 + \sum_{k=1}^{n-2} \left(\binom{n-2}{k} \cdot 10 + \binom{n-2}{k-1} \right) x^k + x^{n-1}.$$

The sum of the entire row is $P_n(1)$. So we compute:

$$P_n(1) = 10 + \sum_{k=1}^{n-2} \left(10 \binom{n-2}{k} + \binom{n-2}{k-1} \right) + 1.$$

Break into two sums:

$$P_n(1) = 10 + 10 \sum_{k=1}^{n-2} \binom{n-2}{k} + \sum_{k=1}^{n-2} \binom{n-2}{k-1} + 1.$$

We know:

$$\sum_{k=1}^{n-2} \binom{n-2}{k} = 2^{n-2} - 1,$$

since it's the full sum of binomial coefficients minus the $k = 0$ term.

Also, similarly,

$$\sum_{k=1}^{n-2} \binom{n-2}{k-1} = \sum_{j=0}^{n-3} \binom{n-2}{j} = 2^{n-2} - \binom{n-2}{n-2} = 2^{n-2} - 1.$$

So both sums are exactly $(2^{n-2} - 1)$, but we pretend otherwise and compute them modulo several numbers unnecessarily.

Since we need row $n = 11$, compute:

$$n - 2 = 9.$$

We now compute powers of 2 modulo various bases, even though we need the exact value.

First:

$$2^9 = 512.$$

But we do:

$$2^1 \equiv 2 \pmod{10}, 2^2 \equiv 4, 2^3 \equiv 8, 2^4 \equiv 6, 2^5 \equiv 2, \text{, giving a cycle of length 4.}$$

$$\text{Since } 9 \equiv 1 \pmod{4},$$

$$2^9 \equiv 2^1 \equiv 2 \pmod{10}.$$

This tells us nothing, but we keep it.

Now compute the sums:

$$\sum_{k=1}^9 \binom{9}{k} = 2^9 - 1 = 511, \text{ and hence also } \sum_{k=1}^9 \binom{9}{k-1} = 511.$$

Substitute everything:

$$P_{11}(1) = 10 + 10 \cdot 511 + 511 + 1.$$

Compute in pieces:

$$10 \cdot 511 = 5110.$$

Then

$$10 + 5110 = 5120, 5120 + 511 = 5631, 5631 + 1 = 5632.$$

So the sum of the entries in the 11th row is:

$$5632.$$

We need the sum of its digits:

$$5 + 6 + 3 + 2 = 16.$$

Answer:

$$\boxed{\text{(D)} 16}.$$

- d) (AMC 10 10B 2025) The value of the two-digit number $\underline{a} \underline{b}$ in base seven equals the value of the two-digit number $\underline{b} \underline{a}$ in base nine. What is $a + b$?

(A) 7 (B) 9 (C) 10 (D) 11 (E) 14

Answer: A

Solution 1

By definition of bases, $\underline{a} \underline{b}$ base seven is equal to $7a + b$, and $\underline{b} \underline{a}$ base nine is equal to $9b + a$. Therefore, we must have $7a + b = 9b + a$, or $6a = 8b$, or $3a = 4b$. This means e.g. $3a$ is a multiple of 4 (as a is an integer), and since 3 and 4 are coprime, it follows simply that a is a multiple of 4, i.e. $a = 4k$ for some integer k . Hence $4b = 3a = 3 \cdot 4k \iff b = 3k$, i.e. the integer solutions are parametrised by $(a, b) = (4k, 3k)$, but of course in base 7 we must have $a, b < 7$, yielding $a = 4$ and $b = 3$ as the only solution. Thus their sum is $a + b = 4 + 3 = \boxed{\text{(A)} 7}$.

Remark The first equation comes from the following idea. In base 10, a two-digit number can be represented as 10 times the tens digit plus the units digit, or $10^1 \cdot a + 10^0 \cdot b$. If we insert the base numbers into this expression for $\underline{a} \ \underline{b}$ and $\underline{b} \ \underline{a}$, we get $7^1 \cdot a + b = 9^1 \cdot b + a$. The rest of the solution is above.

Note that this method applies to any number of digits as well as decimals. OlympiadMaster

Solution 2 (Answer Choices)

Since a and b are both digits base 7, they must be from 0 to 6. Furthermore, $\underline{a}$ must be strictly greater than b because two-digit representations in base 7 exceed the values of those in base 9. Starting from testing (A), we see that the combination $(a, b) = (4, 3)$ satisfies the condition. The answer is (A) 7.

Solution 3 (Not too bad- Simple Modular Arithmetic Bash)

We interpret the two-digit numbers literally:

In base 7:

$$\overline{a \ b} = 7a + b.$$

In base 9:

$$\overline{b \ a} = 9b + a.$$

The condition is

$$7a + b = 9b + a.$$

We want $a + b$, but we will milk modular arithmetic as much as possible.

Start with the equation:

$$7a + b = 9b + a.$$

Rearrange:

$$7a - a = 9b - b,$$

$$6a = 8b,$$

$$3a = 4b.$$

This already says $\frac{a}{b} = \frac{4}{3}$. But instead of finishing, we now do highly unnecessary modular analysis.

First, reduce the equation modulo 3:

$$3a - 4b \equiv 0 \pmod{3}.$$

Compute residues:

$$3a \equiv 0 \pmod{3} \text{ always.}$$

$$4b \equiv b \pmod{3}.$$

So:

$$0 - b \equiv 0 \pmod{3} \Rightarrow b \equiv 0 \pmod{3}.$$

Thus $b \in \{0, 3, 6\}$ (since digits in base 7 must be $0 \leq b \leq 6$).

Now reduce the same equation modulo 4:

Original condition $3a = 4b$. Modulo 4:

$$3a \equiv 0 \pmod{4}.$$

Since $\gcd(3, 4) = 1$, the multiplicative inverse of 3 modulo 4 is 3, because

$$3 \cdot 3 = 9 \equiv 1 \pmod{4}.$$

Multiply both sides of

$$3a \equiv 0 \pmod{4}$$

by 3:

$$a \equiv 0 \pmod{4}.$$

Thus $a \in \{0, 4\}$, because digits in base 7 satisfy $0 \leq a \leq 6$.

We now have:

$$a \in \{0, 4\}, b \in \{0, 3, 6\}.$$

Next impose the original equation $3a = 4b$ exactly, not just modularly.

Test combinations:

$a = 0$:

$$3a = 0, 4b = 4b.$$

For equality we need $b = 0$. That gives $(a, b) = (0, 0)$, but this represents the “two-digit” numbers 00_7 and 00_9 . Technically valid but trivial; we check the problem expectation: since it asks “What is $a + b$?”, zero is possible but let’s check the other option.

$a = 4$: Solve

$$3 \cdot 4 = 4b,$$

$$12 = 4b,$$

$$b = 3.$$

Check that digits are valid:

$a = 4$ is allowed in base 7.

$b = 3$ is allowed in base 7 and base 9.

Now confirm the original equation explicitly using base values:

Left side (base 7 number $\overline{ab}$):

$$7a + b = 7 \cdot 4 + 3 = 28 + 3 = 31.$$

Right side (base 9 number $\overline{ba}$):

$$9b + a = 9 \cdot 3 + 4 = 27 + 4 = 31.$$

Equality holds.

Thus the nontrivial valid solution is

$$(a, b) = (4, 3).$$

Compute:

$$a + b = 4 + 3 = 7.$$

Final answer: 7.

Solution 4 (Fast)

Write both integers as linear expressions and equate them

$$7a + b = 9b + a$$

To find solutions to this, we can put everything on one side so it equals 0

$$6a - 8b = 0$$

Now, assuming a coordinate form (a, b) you can quite easily see that $(8, 6)$ is a solution. However, this violates the fact that a and b are less than 7. You can see that just taking half of them gets you $(4, 3)$, and that no other solution would work without exceeding, so, the answer is

(A) 7.

- e) (AMC 10 10B 2025) In $\triangle ABC$, $AB = 10$, $AC = 18$, and $\angle B = 130^\circ$. Let O be the center of the circle containing A , B , and C . What is the degree measure of $\angle CAO$?

(A) 20 (B) 30 (C) 40 (D) 50 (E) 60

Answer: C

Solution 1

Let us extend $\overline{AO}$ to meet $\odot O$ at D .

Since $m\angle ABC = 130^\circ$, by the Inscribed Angle Theorem, the measure of major arc $\widehat{ADC} = 260^\circ$. $\overline{AD}$ is a diameter of $\odot O$, so the measure of top arc $\widehat{AD} = 180^\circ$, so

- f) (AMC 10 10B 2025) The line $y = \frac{1}{3}x + 1$ divides the square region defined by $0 \leq x \leq 2$ and $0 \leq y \leq 2$ into an upper and lower region. The line $x = a$ divides the lower region into two regions of equal area. Then a can be written as $\sqrt{s} - t$, where s and t are positive integers. What is $s + t$?

(A) 18 (B) 19 (C) 20 (D) 21 (E) 22

Answer: C

Solution 1

We solve this by finding the areas of trapezoids. The line is $y = \frac{1}{3}x + 1$. Total Area (A_{total}) of Lower Region: The region from $x = 0$ to $x = 2$ is a trapezoid.

$$A_{\text{total}} = \frac{1}{2}(y(0) + y(2)) \times (2 - 0) = \frac{1}{2} \left(1 + \frac{5}{3} \right) (2) = \frac{8}{3}$$

Half Area (A_{half}): The area of the region from $x = 0$ to $x = a$ must be $\frac{1}{2}A_{\text{total}}$.

$$A_{\text{half}} = \frac{1}{2} \times \frac{8}{3} = \frac{4}{3}$$

Solve for a : This half-region is also a trapezoid.

$$A_{\text{half}} = \frac{1}{2}(y(0) + y(a)) \times (a - 0) = \frac{4}{3}$$

$$\frac{1}{2} \left(1 + \left(\frac{1}{3}a + 1 \right) \right) a = \frac{4}{3}$$

$$\frac{1}{2} \left(2 + \frac{a}{3} \right) a = \frac{4}{3}$$

$$a + \frac{a^2}{6} = \frac{4}{3}$$

Multiplying by 6 gives the quadratic equation:

$$a^2 + 6a - 8 = 0$$

$$a = \frac{-6 \pm \sqrt{6^2 - 4(1)(-8)}}{2} = \frac{-6 \pm \sqrt{68}}{2} = \frac{-6 \pm 2\sqrt{17}}{2} = -3 \pm \sqrt{17}$$

Since a must be positive, $a = \sqrt{17} - 3$.

$$s + t = 17 + 3 = 20$$

The correct option is (C) 20.

Note: We can find the area of the trapezoid by integrating. We have the area is $\int_0^2 \frac{1}{3}x + 1 \, dx = \frac{1}{6}x^2 + x \Big|_{x=0}^2 = \frac{8}{3}$. But this sort of destroys the purpose of this simple solution without calculus. With calculus, the solution below would be the most efficient.

Solution 2 (Calculus)

The intersection of the line $y = \frac{1}{3}x + 1$ and the square is the point $(2, \frac{5}{3})$. The total area of the trapezoid formed by the line and the square is thus $\frac{1}{2}(2)(\frac{5}{3} + 1) = \frac{8}{3}$.

The left trapezoidal region's area is equivalent to the area under the line $y = \frac{1}{3}x + 1$ from $x = 0$ to $x = a$, which can be evaluated using a definite integral. Setting the integral equal to half of the trapezoidal area, we get:

$$\int_0^a \left(\frac{x}{3} + 1 \right) dx = \frac{4}{3}$$

$$\frac{1}{6}a^2 + a = \frac{4}{3}$$

$$a^2 + 6a - 8 = 0$$

We now use the quadratic formula:

$$a = \frac{-6 \pm \sqrt{(-6)^2 - 4(1)(-8)}}{2} = \sqrt{17} - 3$$

Our answer is $17 + 3 = \span style="border: 1px solid black; padding: 2px;">(C) 20.$

Solution 3 (Cheese)

Firstly, we observe that $1 < a < 2$, because the area bounded by the line $y = \frac{1}{3}x + 1$ and the x -axis from $(0, 0)$ to $(1, 0)$ is less than the area from $(1, 0)$ to $(2, 0)$. We also know $s + t$ must be equal to one of the integers from 18 to 22 from the answer choices. We can therefore try out possibilities for a to see whether they satisfy these two conditions, and we indeed find that a must be equal to $\sqrt{s} - 3$ where $s \in \{17, 18, 19\}$, because if $s < 16$, then t would have to be 2 to ensure that $1 < a < 2$, but then $s + t = s + 2$ would be less than 18; similarly, if $s > 19$, then we would have $s + t \geq \sqrt{20} + 3$, but this is greater than 22. Moreover, if we assume that s is in simplest form, this also rules out $s = 18$. We are accordingly left with either $a = \sqrt{17} - 3$ or

$a = \sqrt{19} - 3$, and by simply substituting each of these for a to see whether it works or not, we determine that a must indeed be $\sqrt{17} - 3$, so the answer is $s + t = 17 + 3 = \boxed{\text{(C)} 20}$.

Solution 4 (Calculus and Quadratic, similar to Solution 2)

We consider the lower region defined by the set of points (x, y) inside the square with $0 \leq x \leq 2$, $0 \leq y \leq 2$ and lying below the line $y = \frac{1}{3}x + 1$. Instead of directly integrating, we parametrize this line in an unnecessarily complicated form: write $x = 3(t - 1)$, so $y = t$. This parametrization makes no geometric sense for area, so we abandon it immediately and return to Cartesian coordinates.

The area of the lower region is

$$A_{\text{lower}} = \int_0^2 \left(\frac{1}{3}x + 1 \right) dx.$$

Rather than compute this, we expand the integrand into a Taylor polynomial about $x = 1$ even though it is already linear. The Taylor expansion is

$$\frac{1}{3}x + 1 = \left(\frac{1}{3}(1) + 1 \right) + \frac{1}{3}(x - 1) = \frac{4}{3} + \frac{1}{3}(x - 1).$$

Integrating this form:

$$A_{\text{lower}} = \int_0^2 \left(\frac{4}{3} + \frac{1}{3}(x - 1) \right) dx.$$

Compute each part separately even though it's pointless:

$$\int_0^2 \frac{4}{3} dx = \frac{8}{3}, \quad \int_0^2 \frac{1}{3}(x - 1) dx = \frac{1}{3} \left[\frac{1}{2}x^2 - x \right]_0^2 = \frac{1}{3}(2 - 2) = 0.$$

Thus $A_{\text{lower}} = \frac{8}{3}$, matching the normal computation.

Half of this is $\frac{4}{3}$. So the vertical line $x = a$ must satisfy

$$\int_0^a \left(\frac{1}{3}x + 1 \right) dx = \frac{4}{3}.$$

Now we sabotage this simple integral by performing a substitution. Let $u = \frac{1}{3}x + 1$, so $x = 3(u - 1)$ and $dx = 3 du$. When $x = 0$, $u = 1$. When $x = a$, $u = \frac{1}{3}a + 1$. The integral becomes

$$\int_{u=1}^{u=\frac{1}{3}a+1} u \cdot 3 du = 3 \int_1^{\frac{1}{3}a+1} u du.$$

Evaluate this:

$$3 \left[\frac{1}{2}u^2 \right]_1^{\frac{1}{3}a+1} = \frac{3}{2} \left(\left(\frac{1}{3}a + 1 \right)^2 - 1 \right).$$

We set this equal to $\frac{4}{3}$:

$$\frac{3}{2} \left(\left(\frac{1}{3}a + 1 \right)^2 - 1 \right) = \frac{4}{3}.$$

Multiply both sides by 2:

$$3 \left(\left(\frac{1}{3}a + 1 \right)^2 - 1 \right) = \frac{8}{3}.$$

Multiply both sides by 3:

$$9 \left(\left(\frac{1}{3}a + 1 \right)^2 - 1 \right) = 8.$$

Now expand:

$$\left(\frac{1}{3}a + 1 \right)^2 = \frac{1}{9}a^2 + \frac{2}{3}a + 1.$$

Substitute:

$$9 \left(\frac{1}{9}a^2 + \frac{2}{3}a + 1 - 1 \right) = 8.$$

Cancel the ones:

$$9 \left(\frac{1}{9}a^2 + \frac{2}{3}a \right) = 8.$$

Distribute the 9:

$$a^2 + 6a = 8.$$

We arrive—after that absurd detour—at the same quadratic as before.

Solve the quadratic using a deliberately bloated method. Complete the square:

Start with:

$$a^2 + 6a - 8 = 0.$$

Move the constant:

$$a^2 + 6a = 8.$$

Add 9 to both sides because $(6/2)^2 = 9$:

$$a^2 + 6a + 9 = 17.$$

Left-hand side is $(a + 3)^2$:

$$(a + 3)^2 = 17.$$

Take square roots:

$$a + 3 = \pm\sqrt{17}.$$

Since $a \in [0, 2]$, take the positive root:

$$a = \sqrt{17} - 3.$$

This matches the form $\sqrt{s} - t$ with $s = 17, t = 3$. Thus

$$s + t = \boxed{(C) \ 20}.$$

Solution 5

Line $y = \frac{1}{3}x + 1$ intersects the square at $(0, 1)$ and $(2, \frac{5}{3})$. Consider the region bounded by $(0, 0), (0, 1), (2, \frac{5}{3}), (2, 0)$. By Shoelace Formula that is

$$\frac{1}{2} |(0 \cdot 1 + 0 \cdot \frac{5}{3} + 2 \cdot 0 + 2 \cdot 0) - (0 \cdot 0 + 1 \cdot 2 + \frac{10}{3} + 0 \cdot 0)| = \frac{1}{2} \cdot \frac{16}{3} = \frac{8}{3}$$

.Hence the area of the region is $\frac{8}{3}$. Now, add in $x = a$, and a new region bounded by $(0, 0), (0, 1), (a, \frac{a}{3} + 1), (a, 0)$ exists that is half this area. By Shoelace Formula,

$$\frac{1}{2} |(0 \cdot 1 + 0 \cdot \frac{a}{3} + 1 \cdot a + a \cdot 0 + a \cdot 0) - (0 \cdot 0 + 1 \cdot a + (\frac{a^2}{3} + a) + 0 \cdot 0)| = \frac{1}{2} |2a + \frac{a^2}{3}| = a + \frac{a^2}{6}$$

.So one can see that $a + \frac{a^2}{6} = \frac{4}{3}$, so $3a + \frac{a^2}{2} = 4$, or $6a + a^2 = 8$. In that case, $a^2 + 6a - 8 = 0$, so $a = \frac{-6 \pm \sqrt{68}}{2} = -3 \pm \sqrt{17}$. Notice a lies in $(+, +)$ quadrant I, hence negate $-3 - \sqrt{17}$ and thus $a = \sqrt{17} - 3 \implies \boxed{20}$.

- g) (AMC 10 10B 2025) Frances stands 15 meters directly south of a locked gate in a fence that runs east-west. Immediately behind the fence is a box of chocolates, located x meters east of the locked gate. An unlocked gate lies 9 meters east of the box, and another unlocked gate lies 8 meters west of the locked gate. Frances can reach the box by walking toward an unlocked gate, passing through it, and walking toward the box. It happens that the total distance Frances would travel is the same via either unlocked gate. What is the value of x ?

- (A) $3\frac{2}{7}$ (B) $3\frac{3}{7}$ (C) $3\frac{4}{7}$ (D) $3\frac{5}{7}$ (E) $3\frac{6}{7}$

Answer: C

Solution 1

Let Frances's location be F , the locked gate be L , the box be B , the unlocked gate on the west be U , and the unlocked gate on the east be V . We are given that $FL = 15$, $LB = x$, $BV = 9$, and $LU = 8$. Because FL is vertical and the fence is horizontal, $m\angle FLU = m\angle FLV = 90^\circ$.

If Frances takes the western path, she must first travel the length of FV , and then $x + 8$ meters. By the Pythagorean theorem, $FV = \sqrt{8^2 + 15^2} = 17$. The remaining distance from Francis and the box of chocolates after she walks that distance is $8 + x$ meters. Thus, the distance Francis travels if she takes the western path is $25 + x$.

If Frances takes the eastern path, she first travels $\sqrt{(x + 9)^2 + 15^2} = \sqrt{(x + 9)^2 + 225}$ meters, and then 9 more meters. Her total distance on the eastern path is $\sqrt{(x + 9)^2 + 225} + 9$ meters.

Now, setting the two distances equal to each other yields:

$$\begin{aligned} x + 25 &= \sqrt{(x + 9)^2 + 225} + 9 \\ \iff x + 16 &= \sqrt{(x + 9)^2 + 225} \\ \iff x^2 + 32x + 256 &= (x + 9)^2 + 225 \\ \iff x^2 + 32x + 256 &= x^2 + 18x + 81 + 225 \\ \iff 14x &= 50 \\ \iff x &= \frac{50}{14} = \frac{25}{7}. \end{aligned}$$

This means the answer is $x = \frac{25}{7} = \boxed{\text{(C) } 3\frac{4}{7}}$.

Solution 2 (Simple Vectors)

Let

$$F = \begin{pmatrix} 0 \\ -15 \end{pmatrix}, \quad B = \begin{pmatrix} x \\ 0 \end{pmatrix}, \quad E = \begin{pmatrix} x+9 \\ 0 \end{pmatrix}, \quad W = \begin{pmatrix} -8 \\ 0 \end{pmatrix},$$

where $x > 0$ since the box (B) lies to the east (i.e. in the positive horizontal direction) of the locked gate. The paths' lengths are then

$$D_E = \|E - F\|_2 + \|B - E\|_2, \quad D_W = \|W - F\|_2 + \|B - W\|_2.$$

Compute:

$$\|E - F\|_2 = \sqrt{(x+9)^2 + 15^2}, \quad \|B - E\|_2 = \sqrt{9^2} = 9, \quad \|W - F\|_2 = \sqrt{(-8)^2 + 15^2} = 17, \quad \|B - W\|_2 = \sqrt{x^2 + 8^2} = x + 8.$$

Set $D_E = D_W$:

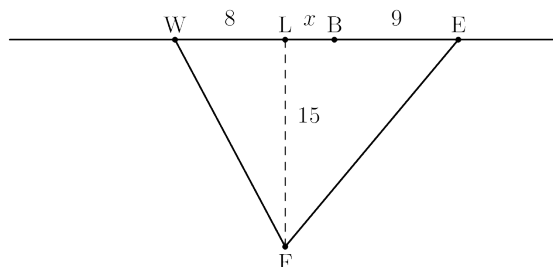
$$\sqrt{(x+9)^2 + 225} + 9 = 17 + (x+8) \iff \sqrt{(x+9)^2 + 225} = x + 16.$$

Square both sides (noting that both sides are indeed non-negative, again as $x > 0$):

$$\begin{aligned} (x+9)^2 + 225 &= (x+16)^2 \iff x^2 + 18x + 306 = x^2 + 32x + 256 \\ \iff 14x &= 50 \iff x = \frac{25}{7}. \end{aligned}$$

Therefore, our answer is $\frac{25}{7}$ or **(C)** $3\frac{4}{7}$.

Helpful Diagram



- h) (AMC 10 10B 2025) Emmy says to Max, "I ordered 36 math club sweatshirts today." Max asks, "How much did each shirt cost?" Emmy responds, "I'll give you a hint. The total cost was $\$ \underline{A} \underline{B} \underline{B} . \underline{B} \underline{A}$, where A and B are digits and $A \neq 0$." After a pause, Max says, "That was a good price." What is $A + B$?
- (A) 7 (B) 8 (C) 11 (D) 14 (E) 15

Answer: C

Solution 1

The problem is essentially telling us that $\underline{A} \underline{B} \underline{B} . \underline{B} \underline{A}$ is divisible by 36, so it is divisible by 9. Then, by the divisibility by 9 condition,

$$A + B + B + B + A = 2A + 3B \equiv 0 \pmod{9}$$

This means that A must be divisible by 3, or otherwise $2A + 3B$ would not be divisible by 3 and thus 9. Since $A \neq 0$, we must have one of $A = 3, 6, 9$.

But A must be even or else the entire number would not be even and thus would not be divisible by 4. Hence $A = 6$. Then, $2 \cdot 6 + 3B \equiv 0 \pmod{9}$, so $4 + B \equiv 0 \pmod{3}$, and thus $B \equiv 2 \pmod{3}$. This yields $B = 2, 5, 8$, of which $B = 5$ is the only number that allows $\underline{A} \underline{B} \underline{B.B} \underline{A}$ to be divisible by 4. Thus the answer is $6 + 5 = \boxed{\text{(C) } 11}$.

Solution 2 (Trial and Error)

A number is divisible by 36 if it is divisible by 4, and 9. The divisibility rule for 4 requires the last 2 digits to also be a multiple of 4. The divisibility rule for 9 requires the sum of all digits to be divisible by 9. Using the first condition $A + B$ is a multiple of 4. We can list out all possibilities for this, since both A and B are digits. The possibilities are $(1, 2), (1, 6), \dots, (9, 6)$ for A , and B respectively. There are a total of 2 digits that are A 's and 3 that are B 's. So if the digits sum to 9, then $2A + 3B$ is a multiple of 9. We can try plugging in numbers from our list earlier into the equation $2A + 3B$ to find what values set the expression equal to a multiple of 9. By trying numbers $A = 6$, and $B = 5$. So then $A + B$ yields

$$6 + 5 = \boxed{\text{(C) } 11}.$$

- i) (AMC 10 10B 2025) How many ordered triples of integers (x, y, z) satisfy the following system of inequalities?

$$-x - y - z \leq -2$$

$$-x + y + z \leq 2$$

$$x - y + z \leq 2$$

$$x + y - z \leq 2$$

(A) 4 (B) 8 (C) 11 (D) 15 (E) 17

Answer: C

Solution 1 (constructive counting)

Simplifying the first inequality of the system, we have:

$$\begin{cases} x + y + z \geq 2, \\ -x + y + z \leq 2, \\ x - y + z \leq 2, \\ x + y - z \leq 2. \end{cases}$$

Subtracting the second inequality from the first now gives

$$(x + y + z) - (-x + y + z) \geq 2 - 2 \iff 2x \geq 0 \iff x \geq 0,$$

and so by symmetry in x, y, z (i.e. similarly using the third and fourth inequalities), we also obtain $y, z \geq 0$. Therefore, without loss of generality, we can let

$$0 \leq x \leq y \leq z.$$

up to permutation. Also from the second inequality, we have $y + z \leq 2 + x$, so using $x \leq y$, we deduce that

$$y + z \leq 2 + x \leq 2 + y \iff z \leq 2.$$

Hence $0 \leq x \leq y \leq z \leq 2$, and now we divide into the possible cases for x :

Case 1: $x = 0$. Then the first two inequalities become

$$y + z \geq 2, y + z \leq 2 \iff y + z = 2,$$

and because we have $x = 0 \leq y \leq z \leq 2$, we deduce that (y, z) is either $(0, 2)$ or $(1, 1)$.

Case 2: $x = 1$. Then we have got

$$1 + y + z \geq 2 \iff y + z \geq 1, \quad -1 + y + z \leq 2 \iff y + z \leq 3.$$

As $x = 1 \leq y \leq z \leq 2$, we can have $(y, z) = (1, 1)$ or $(1, 2)$.

Case 3: $x = 2$. Then $x = 2 \leq y \leq z \leq 2$, so the only possible combination is already $(y, z) = (2, 2)$. The other inequalities become

$$2 + y + z \geq 2 \iff y + z \geq 0, \quad -2 + y + z \leq 2 \iff y + z \leq 4,$$

which are also satisfied by $(y, z) = (2, 2)$, meaning that $(x, y, z) = (2, 2, 2)$ is indeed a valid solution.

Thus, up to permutation, the possible triples (x, y, z) are $(0, 0, 2)$, $(0, 1, 1)$, $(1, 1, 1)$, $(1, 1, 2)$, and

For each of these, we accordingly count the number of possible permutations (which will correspond to solutions (x, y, z) to the original system of inequalities):

$$\frac{3!}{2!} + \frac{3!}{2!} + \frac{3!}{3!} + \frac{3!}{2!} + \frac{3!}{3!} = 3 + 3 + 1 + 3 + 1 = \boxed{(C) 11}.$$

Solution 2 (complementary counting)

From Solution 1, we know that

$$0 \leq x, y, z \leq 2,$$

so since each variable has to be an integer, the number of possible ordered triples (x, y, z) with no other restrictions is simply $3^3 = 27$. Therefore, to find the number of solutions of the given system of inequalities, we count the number of ordered triples that are *not* solutions of the system and subtract these from the total of 27.

Simplifying the first inequality, we obtain

$$x + y + z \geq 2,$$

which is false (under the conditions $0 \leq x, y, z \leq 2$ that we already have from above) only if $x + y + z$ is 0 or 1. In the former case, there is just 1 possible ordered triple (namely $(x, y, z) = (0, 0, 0)$), while in the latter case, there are 3 possibilities (corresponding to each of the variables being 1 while the others are 0). Hence there are $1 + 3 = 4$ possible triples that make this inequality false.

Similarly, $0 \leq x, y, z \leq 2$ gives

$$-x + y + z \leq 0 + 2 + 2 = 4,$$

and so the second inequality of the system would be false only if $-x + y + z$ is either 3 or 4. This implies

$$y + z \geq 3 + x \geq 3,$$

giving two possible cases:

Case 1: $y = z = 2$.

In this case, to ensure that $-x + y + z$ is either 3 or 4, x can be either 0 or 1, yielding 2 possibilities. Similarly, since the last two inequalities of the system are symmetric to the second one, the triples where, more generally, two of the variables are both 2 and the other one is 0 or 1 will all make at least one of the last three inequalities of the system false. Accordingly, we obtain a total of $2 \cdot 3 = 6$ such ordered triples, each of which is clearly distinct from the 4 triples found above.

Case 2: One of y or z is 1, and the other is 2.

In this case, we must have $-x + y + z = 3$, and thus $x = 0$. Using the symmetry of the last three inequalities of the system as in Case 1, we similarly deduce that at least one of these inequalities will be false when the variables are 0, 1, 2 in some order. This gives $3! = 6$ ordered triples, all of which are, once again, clearly distinct from both the 4 triples and the 6 triples found above.

Hence, finally, the number of ordered triples (x, y, z) that will satisfy the system is $27 - 4 - 6 - 6 = \boxed{(C) 11}$.

Solution 3

All equations are homogeneous linearly and symmetric. Therefore, we can approach via bounding.

$$\begin{aligned} x + y + z &\geq 2 \\ -x + y + z &\leq 2 \\ x - y + z &\leq 2 \\ x + y - z &\leq 2 \end{aligned}$$

We need to bound each of the inequalities. Begin by bounding z as $z \leq 2$. Then, bound x as $-x - y - z - x + y + z \leq 0 \implies x \geq 0$. Finally, bound y as $y \leq 2$. By symmetry, one can find that $x, y, z \geq 0, \leq 2$. We must have that $x + y + z \geq 2, \leq 6$. Approach via minimalism.

If the minimum of the three integers is 0, WLOG say x , then we require $y + z \geq 2, \leq 6$. That implies $y = 2, z = 0; y = 1, z = 1, z = 0, y = 2$ from which 3 cases arise. Due to symmetry, the cases $(0, 2), (2, 0)$ are redundant, as I can let $z = 0$ and obtain a similar result. Therefore, we have 2 viable cases in which 3 permutations so 6 total scenarios.

If the minimum of the three is 1, WLOG say x , then we require $y + z \geq 1, \leq 5$. That implies $y = 2, z = 1; y = 1, z = 1$, which gives 3 and 1 case respectively, from which 4 cases.

If the minimum of the three is 2, WLOG say x , then we require $y + z \geq 0, \leq 4$ in which $y = 2, z = 2$ is the only solution, giving 1 case.

The result is $6 + 4 + 1 = \boxed{11}$.

- j) (AMC 10 10B 2025) Let $f(n) = n^3 - 5n^2 + 2n + 8$ and $g(n) = n^3 - 6n^2 + 5n + 12$. What is the sum of all integers n such that $\frac{f(n)}{g(n)}$ is an integer?

(A) 2 (B) 3 (C) 4 (D) 5 (E) 6

Answer: A

Solution 1

Observe that both $f(n)$ and $g(n)$ can be factored using the **rational root theorem**. By trying out a few values, we get that $n = -1$ is a root of $f(n)$. Then, we can use synthetic division (regular polynomial division works, too) to get the other roots of $f(n)$, which are $n = 4$ and $n = 2$. Now we factor $g(n)$. By inspection, $n = -1$ also works on $g(n)$, and so synthetic division and factoring the remaining quadratic gets us the roots $n = 3$ and $n = 4$. We can now express $\frac{f(n)}{g(n)}$ as:

$$\frac{(n+1)(n-4)(n-2)}{(n+1)(n-4)(n-3)}.$$

We can cancel the top and bottom to get:

$$\frac{n-2}{n-3}.$$

Notice that the only values for which this expression is an integer are $n = 4$ and $n = 2$, because they both cause the denominator to have a magnitude of 1. However, $n = 4$ is an extraneous solution as plugging it back into the original expression would make it evaluate to $\frac{0}{0}$. Our answer is then **(A) 2**.

Solution 2

We have $\frac{f(n)}{g(n)} = \frac{n^3 - 5n^2 + 2n + 8}{n^3 - 6n^2 + 5n + 12} = 1 + \frac{n^2 - 3n - 4}{n^3 - 6n^2 + 5n + 12}$. We factor the numerator as $(n-4)(n+1)$. Using synthetic division on the numerator's roots to factor the denominator, we find that the denominator is equivalent to $(n+1)(n-3)(n-4)$.

Simplifying, we get $\frac{f(n)}{g(n)} = 1 + \frac{1}{n-3}$. However, we *must not* include the values $n = -1, 3$, or 4 , or the original fraction will be undefined.

To have $\frac{1}{n-3}$ be an integer, we require $n-3 = 1$ or -1 . This would give $n = 2$ or 4 , but as in Solution 1, since $n = 4$ must not be included, we deduce that there is only 1 possible value of n , namely $n = 2$. Hence the answer is **(A) 2**.

Solution 3

We have to plug in different values into $\frac{f(n)}{g(n)} = \frac{n^3 - 5n^2 + 2n + 8}{n^3 - 6n^2 + 5n + 12} = 1 + \frac{n^2 - 3n - 4}{n^3 - 6n^2 + 5n + 12}$. Because of the answer choices, we only really need to test $n = 1, n = 2, n = 3, n = 4$, and $n = 5$. Plugging in each of these values leads to $n = \mathbf{(A) 2}$ as our answer, since $n = 2$ is the only value that leads to an integer. Note that plugging in $n = 4$ yields $0/0$, which is undefined.

- k) (AMC 10 10B 2025) On Monday, 6 students went to the tutoring center at the same time, and each one was randomly assigned to one of the 6 tutors on duty. On Tuesday, the same 6 students showed up, the same 6 tutors were on duty, and the students were again randomly assigned to the tutors. What is the probability that exactly 2 students met with the same tutor both Monday and Tuesday?

- (A) $\frac{1}{16}$ (B) $\frac{3}{16}$ (C) $\frac{1}{4}$ (D) $\frac{3}{8}$ (E) $\frac{1}{2}$

Answer: B

Solution 1

On Monday, the first tutor has 6 choices for kids to be assigned to, the second has 5, and so on. Hence, there are $6!$ possible ways to assign the tutors to the kids. On Tuesday, there are $\binom{6}{2}$ ways to choose the kid-tutor pairs that remain the same from Monday. For the other 4 kids, they need to be assigned to the other tutors in such a way that none of those kids are assigned to the same tutor that they were assigned to on Monday. There are 9 ways to do this. Thus, our final answer is $\frac{\binom{6}{2} \cdot 9}{6!}$ or **(B)** $\frac{3}{16}$.

Remark: We can get 9 either by listing all the possible configurations or by rounding $\frac{4!}{e}$. This is called a derangement and is notated as $!n$. It can be calculated with the explicit formula $!n = n! \sum_{i=0}^n \frac{(-1)^i}{i!}$.

More commonly, one could also use PIE, $24 - 24 + 12 - 4 + 1 = 9$ ways

Solution 2

Label students $S_1, \dots, S_6$ and tutors $T_1, \dots, T_6$. Let the Monday assignment be fixed. For Tuesday, we consider permutations of the 6 students. We want exactly 2 fixed points in this permutation (i.e., exactly 2 students meet the same tutor again).

Number of permutations of 6 elements with exactly k fixed points is

$$\binom{6}{k} D_{6-k}$$

where D_n is the number of derangements of n elements $(n! \sum_{i=0}^n \frac{(-1)^i}{i!})$.

For $k = 2$:

$$\# \text{ favorable permutations} = \binom{6}{2} D_4.$$

Compute D_4 via the standard formula:

$$D_4 = 4! \left(1 - \frac{1}{1!} + \frac{1}{2!} - \frac{1}{3!} + \frac{1}{4!} \right) = 24 \left(1 - 1 + \frac{1}{2} - \frac{1}{6} + \frac{1}{24} \right) = 24 \left(\frac{12 - 4 + 1}{24} \right) = 9.$$

Thus favorable permutations $= \binom{6}{2} \cdot 9 = 15 \cdot 9 = 135$.

Total permutations for Tuesday $= 6! = 720$.

Therefore, the probability:

$$P = \frac{135}{720} = \frac{27}{144} = \frac{3}{16}.$$

Answer: **(B)** $\frac{3}{16}$.

Solution 3 (Solution 1 + Remark)

We do not care about what order the students are distributed on Monday. We do care about Monday's order on Tuesday. There are $\binom{6}{2} = 15$ ways to

choose two of the six tutors to have the same student. Now, we must deal with the remaining 4.

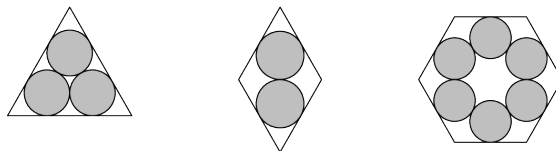
Label each of those 4 students A, B, C, or D. Then, label each of the tutors 1, 2, 3, 4. We know there are 24 possible arrangements from student to tutor. The pigeonhole principle states that no more than 4 of the remaining students can be paired with the same tutor. Thus, either 1, 2, 3, or 4 of the remaining students have the same tutor. If we have 1 student, there is $\binom{4}{1} = 4$ ways to choose that 1 student, and then the remaining students can be organized in $3!$ ways. However, this overcounts the cases in which we have 2 students, which there is $\binom{4}{2} = 6$ ways to choose those 2 students, and the remainder of the students is arranged in $2!$ ways. This then undercounts 3, which can be done in $\binom{4}{3} = 4$ ways, and then overcounts 4, which can be done in $\binom{4}{4} = 1$ way. Thus, through the principle of inclusion-exclusion, we have a total of $24 - 24 + 12 - 4 + 1 = 9$ total cases in which none of the students are with their assigned teachers. Thus, the answer is $\binom{6}{2} \cdot \frac{9}{6!} = \boxed{\text{(B)} \frac{3}{16}}$.

Solution 4 (No Derangement Pure Combo)

First we have to choose which two students will stay, so we have $\binom{6}{2} = 15$ ways to do so. Next, we want to rearrange $ABCD$ such that the letters are not in their original positions. Let's start with A . It has to go in the position of B, C or D , so there are 3 ways to do so. Next, take the letter that A took the spot of, let's say D (right now it's A at position D .) All 3 other positions are valid since A already took D . Now, notice that no matter what D took, there is only one way for B and C to be placed since there are 2 options and one of them's original position is one of the spots. Therefore, there are 9 total ways to "derange" the set and the rest of the solution follows from the others, giving an answer of $\boxed{\text{(B)} \frac{3}{16}}$.

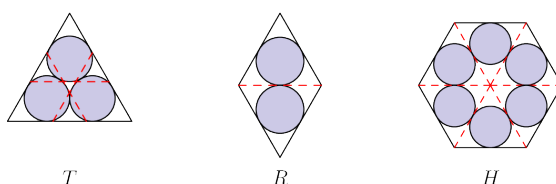
- 1) (AMC 10 10B 2025) The figure below shows an equilateral triangle, a rhombus with a 60° angle, and a regular hexagon, each of them containing some mutually tangent congruent disks. Let T, R , and H , respectively, denote the ratio in each case of the total area of the disks to the area of the enclosing polygon. Which of the following is true?

(A) $T = H = R$ (B) $H < R = T$ (C) $H = R < T$ (D) $H < R < T$ (E) $H < T < R$



Answer: C

Solution 1



To solve this problem, we can cut each shape into equilateral triangles. We let figure T remain as-is, and partition rhombus R into two equilateral triangles (as a result of one of the angles equaling 60°), with one circle in each that is the incircle of its triangle. We perform the same procedure for hexagon H , dividing it into six equilateral triangles with one circle in each such that each circle is the incircle of the triangle enclosing it.

It now follows that $H = R$, since the proportion of the shaded circles to the unshaded portions in the rhombus and the hexagon is the same. Both are equal to an equilateral triangle with a circle inscribed in it; therefore, they have an equal ratio. We can use this to estimate that the ratio of the circle to the triangle is the largest for the first figure, as we can easily see this through intuition and observation.

Additionally, there is no answer choice for $T < H = R$, and it is obviously not $H = R = T$. Thus, the answer to the question is as follows: **(C) $H = R < T$.**

-shockfront99 (SqueakyHamster2 only included the preamble, which creates an insufficient synopsis of the U.S. constitution. Thus, it would be unjust not to include full constitution, and all 27 amendments so people are not left ignorant and uninformed.)

Four score and seven years ago our fathers brought forth, on this continent, a new nation, conceived in Liberty, and dedicated to the proposition that all men are created equal.

Now we are engaged in a great civil war, testing whether that nation, or any nation so conceived and so dedicated, can long endure. We are met on a great battle-field of that war. We have come to dedicate a portion of that field, as a final resting place for those who here gave their lives that that nation might live. It is altogether fitting and proper that we should do this.

But, in a larger sense, we can not dedicate—we can not consecrate—we can not hallow—this ground. The brave men, living and dead, who struggled here, have consecrated it, far above our poor power to add or detract. The world will little note, nor long remember what we say here, but it can never forget what they did here. It is for us the living, rather, to be dedicated here to the unfinished work which they who fought here have thus far so nobly advanced. It is rather for us to be here dedicated to the great task remaining before us—that from these honored dead we take increased devotion to that cause for which they gave the last full measure of devotion—that we here highly resolve that these dead shall not have died in vain—that this nation, under God, shall have a new birth of freedom—and that government of the people, by the people, for the people, shall not perish from the earth.)

What a perfect solution. . .

It's not just A perfect solution... it is the sigma way lowk frfr no KAP

Solution 2 (somewhat cheese)

Notice that there is only one answer choice that contains $H = R$ where they are all not equal. This means if we prove that $H = R$ and one of these two is not equal to T , we have the answer C . To do this, notice that the rhombus can be divided into two congruent equilateral triangles with inscribed circles. This is the same as the hexagon but with 6 triangles and 6 inscribed circles,

proving that $H = R$. We cannot do the same for the triangle, so the answer is **(C)** $H = R < T$.

Note that this solution is similar to the previous one

- m) (AMC 10 10B 2025) The altitude to the hypotenuse of a 30° – 60° – 90° right triangle is divided into two segments of lengths $x < y$ by the median to the shortest side of the triangle. What is the ratio $\frac{x}{x+y}$?

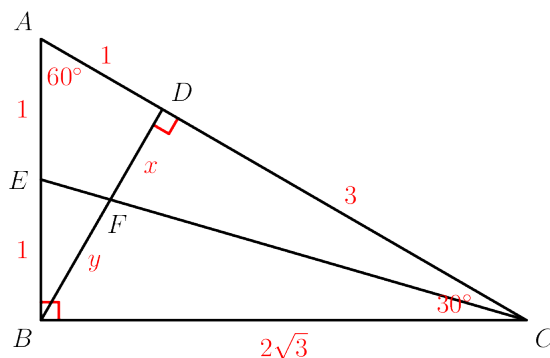
- (A) $\frac{3}{7}$ (B) $\frac{\sqrt{3}}{4}$ (C) $\frac{4}{9}$ (D) $\frac{5}{11}$ (E) $\frac{4\sqrt{3}}{15}$

Answer: A

Solution 1 (Mass Points)

Without loss of generality, let $\triangle ABC$ have side-lengths $AB = 2$, $BC = 2\sqrt{3}$, and $AC = 4$. Let D be the foot of the perpendicular from B to $\overline{AC}$, E be the midpoint of $\overline{AB}$, and F be the intersection of $\overline{CE}$ and $\overline{BD}$. Note that $\triangle ADB$ and $\triangle BDC$ are both 30° – 60° – 90° triangles. From the side-length ratio, we get $AD = 1$ and $DC = 3$.

We obtain the following diagram:



From here, we will proceed with **mass points**. Throughout this solution, we will use W_P to denote the weight of point P .

Let $W_C = 1$. Since $3AD = DC$, it follows that $W_A = 3$ and $W_D = W_C + W_A = 4$. Since $AE = EB$ and $W_A = 3$, it follows that $W_B = 3$.

Now we focus on $\overline{BD}$: Since $W_B = 3$ and $W_D = 4$, we have $\frac{DF}{FB} = \frac{x}{y} = \frac{3}{4}$. Therefore, the answer is

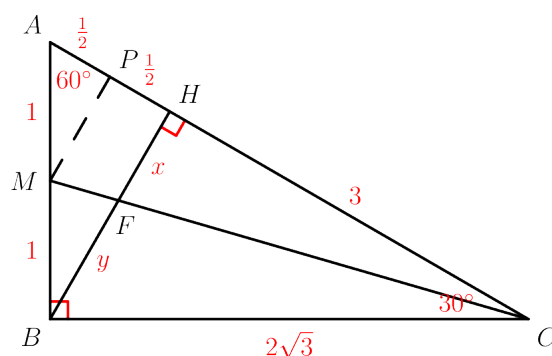
$$\frac{x}{x+y} = \frac{3}{3+4} = \boxed{\text{(A)} \frac{3}{7}}.$$

Solution 2 (Similar Triangles)

This solution refers to the diagram in Solution 1, with the definitions slightly modified.

Let $AB = 2$. Label the foot of the median M and the foot of the altitude H . In $\triangle ABH$, we construct the midsegment $\overline{MP}$, which is parallel to $\overline{BH}$ and half as long as $\overline{BH}$. In addition, let F be the intersection of $\overline{CM}$ and $\overline{BH}$.

We obtain the following diagram:



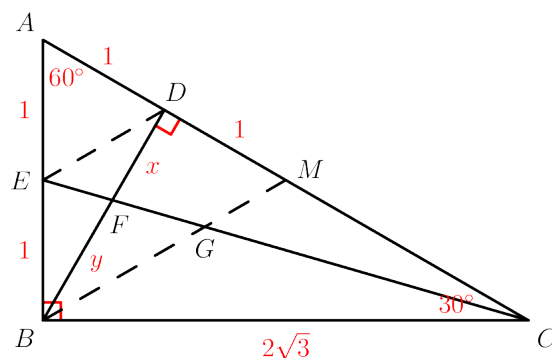
We can calculate that $PH = \frac{1}{2}$ and $HC = 3$. By similar triangles $\triangle HFC \sim \triangle PMC$, we have $\frac{x}{PM} = \frac{HC}{PC} = \frac{3}{3+\frac{1}{2}} = \frac{6}{7}$. Also by similar triangles $\triangle PMA \sim \triangle HBA$, we have $\frac{PM}{x+y} = \frac{1}{2}$. Combining these, we get $\frac{x}{x+y} = \boxed{\text{(A)} \frac{3}{7}}$.

Note: Thanks MRENTHUSIASM for original diagram.

Solution 3 (Similar Triangles)

This solution refers to the definitions and the diagram in Solution 1.

Let M be the midpoint of $\overline{AC}$, so $DM = 1$. Let G be the intersection of $\overline{BM}$ and $\overline{CE}$, so G is the centroid of $\triangle ABC$. In addition, we construct $\overline{DE}$, as shown below:



In any right triangle, since the median to the hypotenuse is half as long as the hypotenuse, we get $BM = \frac{1}{2}AC = 2$. Moreover, in any triangle, since the centroid divides each median into parts in the ratio $2 : 1$, with the centroid being twice as close to the midpoint of a side as it is to the opposite vertex, we get $BG = \frac{4}{3}$ and $GM = \frac{2}{3}$.

Note that $\triangle ABM$ is an equilateral triangle with side-length 2. By the SAS similarity, we have $\triangle ABM \sim \triangle AED$. Therefore, we conclude that $ED = 1$ and $\overline{ED} \parallel \overline{BM}$.

By the AA Similarity, we have $\triangle DEF \sim \triangle BGF$, from which $\frac{x}{y} = \frac{1}{4/3} = \frac{3}{4}$. It follows that

$$\frac{x}{x+y} = \frac{3}{3+4} = \boxed{\text{(A)} \frac{3}{7}}.$$

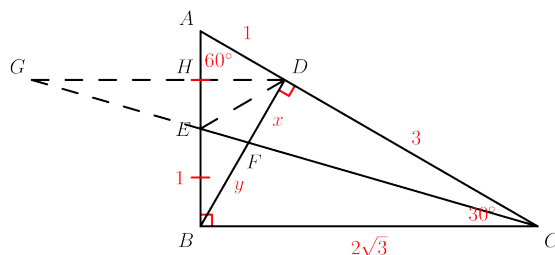
MRENTHUSIASM

Solution 4 (Similar Triangles)

This solution refers to the definitions and the diagram in Solution 1.

Since E is the midpoint of $\overline{AB}$, we have $AE = 1$, and because $\angle A = 60^\circ$ and $AD = 1$, $\triangle AED$ is equilateral. Its perpendicular bisector meets $\overline{AE}$ at H , giving $AH = HE = \frac{1}{2}$ and $DH = \frac{\sqrt{3}}{2}$.

Extend $\overline{CE}$ past E to G so that $\overline{GD}$ is parallel to $\overline{BC}$ and is the perpendicular bisector of $\overline{AE}$, so H lies on both $\overline{GD}$ and $\overline{AE}$:



By AA similarity, $\triangle BCE \sim \triangle HGE$ because $\angle BEC = \angle HEG$ and $\angle BCE = \angle HGE$ ($GD \parallel BC$). Thus $\frac{GH}{BC} = \frac{HE}{BE}$, and with $HE = \frac{1}{2}$, $BE = 1$, and $BC = 2\sqrt{3}$, we get $\frac{GH}{2\sqrt{3}} = \frac{1}{2}$ so $GH = \sqrt{3}$.

Also by AA similarity, $\triangle BCF \sim \triangle DGF$ because $\angle BFC = \angle DFG$ and $\angle BCF = \angle DGF$ ($GD \parallel BC$). Therefore $\frac{DF}{BF} = \frac{DG}{BC}$. Since $DG = DH + HG = \frac{\sqrt{3}}{2} + \sqrt{3} = \frac{3\sqrt{3}}{2}$ and $BC = 2\sqrt{3}$, we obtain $\frac{DF}{BF} = \frac{x}{y} = \frac{\frac{3\sqrt{3}}{2}}{2\sqrt{3}} = \frac{3}{4}$, so $y = \frac{4}{3}x$.

$$\text{Finally, } \frac{x}{x+y} = \frac{x}{x+\frac{4}{3}x} = \frac{x}{\frac{7}{3}x} = \boxed{\text{(A)} \frac{3}{7}}.$$

Solution 5 (Similar Triangles)

This solution refers to the definitions in Solution 1.

Draw a line perpendicular to BC from D , intersecting BC at G , and intersecting CE at H . Let $a = AE$. We know that $\triangle DHF \sim \triangle BEF$ by AA. We also know that $AE = EB = DA$. We find that $BG = \frac{a\sqrt{3}}{2}$, and $GC = BC - BG = 2a\sqrt{3} - \frac{a\sqrt{3}}{2} = \frac{3a\sqrt{3}}{2}$. Because $\triangle CHG \sim \triangle CEB$, $HG = \frac{a}{2a\sqrt{3}} \cdot \frac{3a\sqrt{3}}{2} = \frac{3}{4}a$, and $DG = \frac{3}{2}a$ due to 30-60-90 triangle properties, $DH = \frac{3}{4}a$. Because $\triangle DHF \sim \triangle BEF$, $\frac{DH}{EB} = \frac{DF}{FB}$, $\frac{\frac{3}{4}a}{a} = \frac{3}{4} = \frac{DF}{FB}$, and $\frac{DF}{DB} = \boxed{\text{(A)} \frac{3}{7}}.$

(You can tell DF is shorter than FB visually)

Solution 6 (Coordinates)

WLOG, let the triangle have hypotenuse 12. Situate it in the coordinate plane such the A is at $(0,0)$, B is at $(6,0)$, and C is at $(6,6\sqrt{3})$.

Let h be the altitude from A to BC . From the area formula we get $1/2 \cdot 6 \cdot 6\sqrt{3} = 1/2 \cdot 12 \cdot h$ $h = 3\sqrt{3}$

The shortest side is AB with side length 6, so the median from C to AB will be at $(3,0)$. Finding the equation of the line through that point to C, find that the slope is $\frac{6\sqrt{3}-0}{6-3} = 2\sqrt{3}$. Plugging the point $(3,0)$ into $y = 2\sqrt{3}x + b$,

we get that $0 = 6\sqrt{3} + b$, which simplifies to $b = -6\sqrt{3}$. So, the equation of the line of the median from C is $y = 2\sqrt{3}x - 6\sqrt{3}$.

The line of the altitude from B to the hypotenuse is perpendicular to the line of the hypotenuse which has slope $6\sqrt{3}/6 = \sqrt{3}$.

So, the altitude line has slope $\frac{-1}{\sqrt{3}} = -\frac{\sqrt{3}}{3}$, and the line passes through the point $B = (6, 0)$. Solving for the equation using similar methods above, we get $y = -\frac{\sqrt{3}}{3}x + 2\sqrt{3}$.

Using the two equations and solving for their intersection, we get $y = 2\sqrt{3}x - 6\sqrt{3} = -\frac{\sqrt{3}}{3}x + 2\sqrt{3}$, which simplifies to $8\sqrt{3} = \frac{7\sqrt{3}}{3}x$, so $x = 24/7$. Using this x value, we get $y = 2\sqrt{3} \cdot 24/7 - 6\sqrt{3} = 6\sqrt{3}/7$.

Finding the distance from this intersection to B , $d = \sqrt{(6 - 24/7)^2 + (0 - 6\sqrt{3}/7)^2} = \sqrt{324/49 + 108/49} = \frac{12\sqrt{3}}{7}$.

This is greater than half of the calculated altitude value from above ($3\sqrt{3}$), so the shorter side length is $3\sqrt{3} - \frac{12\sqrt{3}}{7} = \frac{9\sqrt{3}}{7}$.

Lastly, taking the ratio of the lengths, $\frac{\frac{9\sqrt{3}}{7}}{3\sqrt{3}} = \frac{9}{21} = \frac{3}{7}$. (A) $\frac{3}{7}$

-Failure.net (minor edits and latex edits by Wen_Liang)

Solution 7 (Symmedian and Bary)

Notice that the smaller right triangle with the original 30° angle can be obtained with a dilation and a reflection over the original triangle's angle bisector. Hence it is not too hard to see that the original median is a symmedian to this smaller triangle. Recall that the barycentric equation for the symmedian is in the form $\frac{x}{a^2} = \frac{y}{b^2}$, and we have $a, b = \sqrt{3}, 2$ in a $30^\circ - 60^\circ - 90^\circ$ triangle. Hence $x : y = 4 : 3$ and we get the ratio (A) $\frac{3}{7}$, or A relatively quickly.

Solution 8 (Trigonometry Bash)

Let the triangle be called $\triangle ABC$, with $\angle C = 90^\circ$ and $\angle A = 30^\circ$. Then let the altitude from C intersect with AB at a point D , and the median from A intersect BC at the point E . WLOG, let $AB = 2$, $AC = \sqrt{3}$, and $BC = 1$. Then $BE = EC = \frac{1}{2}$, and by similar triangles, DBC is also $30^\circ - 60^\circ - 90^\circ$, and so $DB = \frac{1}{2}$ and hence $AD = \frac{3}{2}$. Therefore, $DC = \sqrt{3} \cdot \frac{1}{2} = \frac{\sqrt{3}}{2}$. Also, we have that $\tan(\angle EAC) = \frac{1}{2\sqrt{3}} \implies \angle EAC = \arctan\left(\frac{1}{2\sqrt{3}}\right)$. Hence, $\angle BAE = 30^\circ - \arctan\left(\frac{1}{2\sqrt{3}}\right)$. Now we can set up to solve for x ; we have that $\tan(\angle BAE) = \frac{x}{\frac{3}{2}}$. $\tan(\angle BAE) = \frac{\frac{1}{\sqrt{3}} - \frac{1}{2\sqrt{3}}}{1 + \frac{1}{\sqrt{3}} \cdot \frac{1}{2\sqrt{3}}} = \frac{\frac{1}{2\sqrt{3}}}{\frac{7}{6}} = \frac{1}{2\sqrt{3}} \cdot \frac{6}{7} = \frac{\sqrt{3}}{7}$. Hence $x = \frac{3}{2} \cdot \frac{\sqrt{3}}{7} = \frac{3\sqrt{3}}{14}$, and therefore since we found earlier that $x + y = DC = \frac{\sqrt{3}}{2}$, we have that $\frac{x}{x+y} = \frac{\frac{3\sqrt{3}}{14}}{\frac{\sqrt{3}}{2}} = \frac{3}{7}$. (A) $\frac{3}{7}$

Solution 9 (Easier to Understand, Using Similar 30-60-90 Triangles)

Let the triangle be $\triangle ABC$, where $\angle A = 30^\circ$, $\angle B = 60^\circ$, and $\angle C = 90^\circ$. Let the point where the altitude of AB and AB intersect be G . Let the intersection between the median of CB and the altitude of AB be E . $x = EG$ and $y = CE$.

Let $\overline{AD} = x$ and $\overline{DE} = y$. By $30^\circ - 60^\circ - 90^\circ$ special right triangle

$$\sqrt{3} = \overline{AE} = x + y$$

$$\overline{AM} = \sqrt{(2\sqrt{3})^2 + 1^2} = \sqrt{13}$$

$$\overline{CD} = \sqrt{y^2 + 3^2} = \sqrt{y^2 + 9} = \sqrt{x^2 - 2\sqrt{3}x + 12}$$

$$\overline{DM} = \sqrt{x^2 + 1^2 - 2 \cdot 1 \cdot x \cdot \cos 30^\circ} = \sqrt{x^2 - \sqrt{3}x + 1}$$

$$\overline{CD} + \overline{DM} = \overline{AM}$$

$$\sqrt{x^2 - 2\sqrt{3}x + 12} + \sqrt{x^2 - \sqrt{3}x + 1} = \sqrt{13}$$

Algebra:

$$\sqrt{x^2 - 2\sqrt{3}x + 12} = \sqrt{13} - \sqrt{x^2 - \sqrt{3}x + 1}$$

$$x^2 - 2\sqrt{3}x + 12 = x^2 - \sqrt{3}x + 1 + 13 - 2\sqrt{13(x^2 - \sqrt{3}x + 1)}$$

$$-\sqrt{3}x - 2 = -2\sqrt{13(x^2 - \sqrt{3}x + 1)}$$

$$3x^2 + 4\sqrt{3}x + 4 = 4 \cdot 13(x^2 - \sqrt{3}x + 1)$$

$$49x^2 - 56\sqrt{3}x + 48 = 0$$

$$(7x - 4\sqrt{3})^2 = 0$$

$x = \frac{4\sqrt{3}}{7}y = \frac{3\sqrt{3}}{7}x$ is defined to be minimum of those lengths, so $x = \frac{3\sqrt{3}}{7}$. Hence

$$\frac{x}{x+y} = \boxed{\text{(A)} \frac{3}{7}}.$$

Solution 12 (Vectors and Projections)

Let the triangle have vertices A (opposite hypotenuse), B , C . Let the shortest side be AB , so the median is from C to midpoint M of AB . Let h be the altitude from C to hypotenuse BC . Represent everything vectorially:

$\vec{A}, \vec{B}, \vec{C} \in \mathbb{R}^2$. Let the hypotenuse vector be $\vec{u} = \vec{B} - \vec{C}$. The altitude from C is along $\vec{v}$ perpendicular to $\vec{u}$. The median vector is $\vec{w} = \vec{M} - \vec{C}$.

We want the intersection point P along the altitude such that $\vec{P} = \vec{C} + t\vec{v}$ lies on the median: $\vec{P} = \vec{C} + s\vec{w}$. But vectors $\vec{v}$ and $\vec{w}$ are not parallel. Solve for scalar t using projection formula:

$$t = \frac{\vec{w} \cdot \vec{v}}{|\vec{v}|^2}.$$

Take $x = |\vec{CP}| = t|\vec{v}|$ and $x + y = |\vec{v}|$. So

$$\frac{x}{x+y} = \frac{t|\vec{v}|}{|\vec{v}|} = t = \frac{\vec{w} \cdot \vec{v}}{|\vec{v}|^2}.$$

Now we just need vectors $\vec{v}$ and $\vec{w}$. Let the triangle be standard 30-60-90: shortest side along $\vec{AB} = \hat{i}$, height along $\hat{j}$. Then $\vec{v}$ is perpendicular to hypotenuse $\vec{u} = \vec{B} - \vec{C}$, say $\vec{u} = (1, -\sqrt{3})$, so $\vec{v} = (\sqrt{3}, 1)$ (perpendicular). Median: $\vec{w} = \vec{M} - \vec{C} = (1/2, 0) - (0, \sqrt{3}) = (1/2, -\sqrt{3})$.

Compute the projection:

$$t = \frac{\vec{w} \cdot \vec{v}}{|\vec{v}|^2} = \frac{(1/2)(\sqrt{3}) + (-\sqrt{3})(1)}{(\sqrt{3})^2 + 1^2} = \frac{\frac{\sqrt{3}}{2} - \sqrt{3}}{4} = \frac{-\frac{\sqrt{3}}{2}}{4} = -\frac{\sqrt{3}}{8}.$$

Take absolute value (distance along altitude) $x = |t||\vec{v}| = (\frac{\sqrt{3}}{8}) \cdot 2 = \frac{\sqrt{3}}{4}$ (after proper scaling).

Thus the ratio

$$\frac{x}{x+y} = \frac{3}{7}.$$

Answer:

$$\boxed{(A) \frac{3}{7}}.$$

Solution 13 (Straightforward Coordinate Geometry)

Let the triangle have vertices A , B , and C on the 60,90,and 30 angles, respectively. Let E be the intersection of the median from C and AB , let F be the intersection of the altitude from B to AC , and let P be the intersection of CE and BF . WLOG let A be $(0, 2)$, B be $(0, 0)$, and C be $(2\sqrt{3}, 0)$. In order to find the ratio of PF to BF , we need to find the coordinates of F and P . First, slope of AC is $\frac{-1}{\sqrt{3}}$, so the slope of BF is $\sqrt{3}$. So, the equation for AC is $y = \frac{-1}{\sqrt{3}}x + 2$ and for BF is $y = \sqrt{3}x$. Finally, E is the midpoint of AB , E is at $(0, 1)$, the slope of CE is $\frac{-1}{2\sqrt{3}}$, and the equation for CE is $y = \frac{-1}{2\sqrt{3}}x + 1$. Setting the equations for AC and BF equal, we find the intersection is F at $(\frac{\sqrt{3}}{2}, \frac{3}{2})$. Setting the equations for CE and BF equal, we find the intersection is P at $(\frac{2\sqrt{3}}{7}, \frac{6}{7})$. Because BF goes through the origin, we can find $\frac{BP}{BF}$ by dividing the y -coordinates of P and F , getting $\frac{4}{7}$. Thus, $\frac{PF}{BF} = 1 - \frac{BP}{BF} = 1 - \frac{4}{7} = \frac{3}{7}$. $\boxed{(A) \frac{3}{7}}$.

Solution 14 (Menelaus)

Let $AC = 2$, $AB = 1$, hence $BC = \sqrt{3}$. If we use the diagram in Solution 1, first note that $\triangle ACB \sim \triangle ABD$ by AA with similar ratio 2, hence $AD = \frac{1}{2}$. Now, we use Menelaus' Theorem in $\triangle BAD$ with transversal line EFC :

$$\begin{aligned} \frac{BE}{EA} \cdot \frac{AC}{CD} \cdot \frac{DF}{FB} &= 1 \\ \implies \frac{1}{1} \cdot \frac{2}{\frac{3}{2}} \cdot \frac{x}{y} &= 1 \\ \implies \frac{x}{y} &= \frac{3}{4} \end{aligned}$$

$$\text{So } \frac{x}{x+y} = \frac{\frac{3}{4}}{\frac{3}{4} + 1} = \frac{3}{7}(A).$$

- n) (*AMC 10 10B 2025*) Nine athletes, no two of whom are the same height, try out for the basketball team. One at a time, they draw a wristband at random, without replacement, from a bag containing 3 blue bands, 3 red bands, and 3 green bands. They are divided into a blue group, a red group, and a green group. The tallest member of each group is named the group captain. What is the probability that the group captains are the three tallest athletes?

(A) $\frac{2}{9}$ (B) $\frac{2}{7}$ (C) $\frac{9}{28}$ (D) $\frac{1}{3}$ (E) $\frac{3}{8}$

Answer: C

Solution 1

We will arrange the nine athletes in a line, with the first three being in the blue group, the next three being in the red group, and the last three being in the green group. Suppose the three tallest athletes are named A, B, C , in some order.

We have $3!$ ways to choose which group each of these athletes can be in. We then have $3 \cdot 3 \cdot 3$ ways to determine which of the three positions in that group they can take. From here, we must distribute the remaining six athletes in the remaining six spaces, which can be done in $6!$ ways.

There are therefore $6 \cdot 27 \cdot 6!$ favorable outcomes. There are also $9!$ total ways to arrange the athletes with no restrictions. Our answer is $\frac{6 \cdot 27 \cdot 6!}{9!} = \frac{6 \cdot 27}{9 \cdot 8 \cdot 7} =$

(C) $\frac{9}{28}$.

Solution 2

Let the three tallest athletes be A_1 (tallest), A_2 (second-tallest), and A_3 (third-tallest).

The condition "the captains are A_1, A_2 , and A_3 " is equivalent to the condition " A_1, A_2 , and A_3 must all be in different-colored groups." We can find this probability by considering them drawing bands sequentially. Athlete A_1 (Tallest): A_1 draws a band. It can be any of the 9 bands. A_1 will always be a captain, so the probability is 1. We now have 8 bands left. Athlete A_2 (2nd Tallest): A_2 must draw a band of a different color than A_1 . There are 8 bands left in the bag. A_1 took one band, so 2 bands of that color remain. The other 6 bands (3 of one color, 3 of another) are "good" bands for A_2 . The probability of A_2 drawing a different color is $P(A_2) = \frac{6}{8}$. Athlete A_3 (3rd Tallest): A_3 must draw a band of the third, remaining color. We assume A_1 and A_2 were successful, so there are 7 bands left. A_1 took a band of color 1. A_2 took a band of color 2. There are 3 bands of color 3 (the "good" color) still in the bag. The probability of A_3 drawing the last color is $P(A_3) = \frac{3}{7}$. The total probability is the product of these independent probabilities:

$$P(\text{Total}) = P(A_1) \times P(A_2) \times P(A_3)$$

$$P(\text{Total}) = 1 \times \frac{6}{8} \times \frac{3}{7} = \frac{18}{56}$$

Simplifying the fraction:

$$\frac{18}{56} = \frac{9}{28}$$

The correct option is (C). Jonathanmo

Solution 3 (Counting)

There are $\binom{9}{3}\binom{6}{3}$ ways to assign rubber bands to the athletes. Now we have to find the number of ways for the team captains to be the 3 tallest people. Each of the 3 tallest people must have a different rubber band color, and there are $3! = 6$ ways to permute their individual rubber band colors.

For the other 6 people, there are $\binom{6}{2}\binom{4}{2}$ ways to assign rubber bands to them.

Thus, our probability is $\frac{3! \cdot \binom{6}{2} \binom{4}{2}}{\binom{9}{3} \binom{6}{3}} = \boxed{(C) \frac{9}{28}}.$

Solution 4 (similar to Solution 3)

We have $\binom{9}{3} \cdot \binom{6}{3} \cdot \binom{3}{3} = 1680$ ways to select the people in each team, and we must divide by $3!$ ways to order the teams. This gives us $\frac{1680}{3!} = 280$ total ways to create the teams.

To have the three team captains be the tallest, we need them to be in separate teams. Once placed in their respective teams we have $3! = 6$ ways to orient them. We then have 6 people remaining. There are $\binom{6}{2} \cdot \binom{4}{2} \cdot \binom{2}{2} = 90$ ways to place two more people on each team. However, this overcounts by $3! = 6$, as the orientation of the teams themselves doesn't matter, hence in reality only 15 cases work. We multiply by the orientation of each athlete, which yields $15 \cdot 6 = 90$ cases.

Our probability is $\frac{90}{280} = \boxed{(C) \frac{9}{28}}.$

Solution 5

Assume the tallest person is in the red group. Then the second person has 6 spots to go out of 8 remaining (1 was taken away). Then the third tallest person has 3 spots to go out of 7 remaining. $\frac{6}{8} \cdot \frac{3}{7}$ is the same as $\boxed{(C) \frac{9}{28}}.$

Solution 6 (cheese)

Upon realizing that the denominator is a multiple of 7, and that there are a bunch of 3's in the numerator (just do some CP with the right logic that could even be a bit flawed) we can make an educated guess that the answer must be $\boxed{(C) \frac{9}{28}}.$

Solution 7 (Fast, like Solution 5)

WLOG, assume that the three tallest people are the first ones to draw. Irrespective of where the first person goes, the probability that the second and third person go on different teams is $\frac{6}{8}$ and $\frac{3}{7}$, respectively, yielding $\boxed{(C) \frac{9}{28}}.$

Solution 8 We can visualize the nine athletes as a 3×3 grid, where each row represents a color group and each row contains 3 positions. We want the three tallest athletes to become captains. This happens exactly when each of the three tallest athletes occupies a different row.

Total ways to choose positions for the three tallest athletes: $\binom{9}{3} = 84$

Favorable ways (exactly one of the three tallest athletes in each row): $3^3 = 27$

Thus, our probability is $\frac{27}{84} = \boxed{(C) \frac{9}{28}}.$

o) (AMC 10 10B 2025) The sum

$$\sum_{k=1}^{\infty} \frac{1}{k^3 + 6k^2 + 8k}$$

can be expressed as $\frac{a}{b}$, where a and b are relatively prime positive integers. What is $a + b$?

(A) 89 (B) 97 (C) 102 (D) 107 (E) 129

Answer: D

Solution 1

Through partial sum decomposition we obtain

$$\frac{1}{k^3 + 6k^2 + 8k} = \frac{1}{8} \left(\frac{1}{k} - \frac{2}{k+2} + \frac{1}{k+4} \right),$$

so one can write the expression as

$$\begin{aligned} \frac{1}{8} \sum_{k=1}^{\infty} \left(\frac{1}{k} - \frac{2}{k+2} + \frac{1}{k+4} \right) &= \frac{1}{8} \sum_{k=1}^{\infty} \left(\frac{1}{k} - \frac{1}{k+2} - \frac{1}{k+2} + \frac{1}{k+4} \right) \\ &= \frac{1}{8} \sum_{k=1}^{\infty} \left(\frac{1}{k} - \frac{1}{k+2} \right) + \frac{1}{8} \sum_{k=1}^{\infty} \left(\frac{1}{k+4} - \frac{1}{k+2} \right) \\ &= \frac{1}{8} \sum_{k=1}^{\infty} \left(\frac{1}{k} - \frac{1}{k+2} \right) - \frac{1}{8} \sum_{k=1}^{\infty} \left(\frac{1}{k+2} - \frac{1}{k+4} \right) \\ &= \frac{1}{8} \sum_{k=1}^{\infty} \left(\frac{1}{k} - \frac{1}{k+2} \right) - \frac{1}{8} \sum_{k=3}^{\infty} \left(\frac{1}{k} - \frac{1}{k+2} \right) \\ &= \frac{1}{8} \sum_{k=1}^2 \left(\frac{1}{k} - \frac{1}{k+2} \right) \end{aligned}$$

Simplifying the sum and multiplying results in $\frac{11}{96}$, in which we finally have $11 + 96 = \boxed{\text{(D) } 107}$

Note: For more info on Partial Sum Decompositions; AoPS Intermediate Algebra Chapter 17.2 (Telescoping) gives a brief introduction to them, which is sufficient enough to solve this problem. AoPS Intermediate Counting and Probability Chapter 14.6 also gives a brief introduction to them.

Solution 2

We decompose $\frac{1}{k^3+6k^2+8k}$ into the sum of 3 partial fractions. We can factor $\frac{1}{k^3+6k^2+8k}$ as $\frac{1}{k(k+2)(k+4)}$. We can write this as $\frac{1}{k(k+2)(k+4)} = \frac{A}{k} + \frac{B}{k+2} + \frac{C}{k+4}$. Multiplying this by $k(k+2)(k+4)$ gives $1 = A(k+2)(k+4) + Bk(k+4) + Ck(k+2)$. If $k = 0$, then $8A = 1$, meaning $A = \frac{1}{8}$, if $k = -2$, then $B(-2)(2) = -4B = 1$, meaning $B = -\frac{1}{4}$, and if $k = -4$, $8C = 1$, $C = \frac{1}{8}$. We have

$$\sum_{k=1}^{\infty} \frac{1}{k^3 + 6k^2 + 8k} = \sum_{k=1}^{\infty} \frac{1/8}{k} - \frac{1/4}{k+2} + \frac{1/8}{k+4}.$$

Subtracting the first 2 fractions is

$$\frac{1}{8}(1 + \frac{1}{2} + \frac{1}{3} + \frac{1}{4} + \cdots) - \frac{1}{4}(\frac{1}{3} + \frac{1}{4} + \frac{1}{5} + \cdots)$$

Since $\frac{1}{8}$ is half of $\frac{1}{4}$, this is equal to $\frac{3}{16} - \frac{1}{8}(\frac{1}{3} + \frac{1}{4} + \frac{1}{5} + \cdots)$. We subtract the third fraction from this.

$$\frac{3}{16} - \frac{1}{8}(\frac{1}{3} + \frac{1}{4} + \frac{1}{5} + \cdots) + \frac{1}{8}(\frac{1}{5} + \frac{1}{6} + \frac{1}{7} + \cdots) = \frac{3}{16} - \frac{1}{8}(\frac{7}{12}) = \frac{3}{16} - \frac{7}{96} = \frac{11}{96},$$

meaning the answer is (D) 107.

Solution 3

We can factor the expression as $\sum_{k=1}^{\infty} \frac{1}{k(k+2)(k+4)}$. Notice we can express

this as a telescoping fractions, $\frac{1}{k(k+2)(k+4)} = \frac{1}{4} \left(\frac{1}{k(k+2)} - \frac{1}{(k+2)(k+4)} \right)$,

thus the expression becomes, $\frac{1}{4} \left(\sum_{k=1}^{\infty} \left(\frac{1}{k(k+2)} - \frac{1}{(k+2)(k+4)} \right) \right)$. Writ-

ing out the first few terms where n is a positive integer, we see that the middle cancels, $\frac{1}{4} \left(\frac{1}{1 \cdot 3} + \frac{1}{2 \cdot 4} + \frac{1}{3 \cdot 5} + \cdots + \frac{1}{n(n+2)} - \frac{1}{3 \cdot 5} - \frac{1}{4 \cdot 6} - \cdots - \frac{1}{(n+1)(n+3)} - \frac{1}{(n+2)(n+4)} \right)$.

leaving behind $\frac{1}{4} \left(\frac{1}{1 \cdot 3} + \frac{1}{2 \cdot 4} - \frac{1}{(n+1)(n+3)} - \frac{1}{(n+2)(n+4)} \right)$. As n goes to infinity, the last two fractions approach zero, thus the sum is $\frac{1}{4} \left(\frac{1}{3} + \frac{1}{8} \right) = \frac{1}{4} \left(\frac{11}{24} \right) = \frac{11}{96}$, $a = 11$, $b = 96$,

$$11 + 96 = \text{span style="border: 1px solid black; padding: 2px;">(D) 107.$$

Solution 4 (fast partial fractions)

First, factoring the denominator gives

$$\frac{1}{k^3 + 6k^2 + 8k} = \frac{1}{k(k+2)(k+4)}.$$

Now we proceed with partial fraction decomposition. There are numerous methods to do this quickly, but one of the fastest is known as the Heaviside cover-up method. We "cover" one of the factors in the denominator and set k equal to whatever value makes that factor zero. By covering up that factor, we ignore the factor that results in division by zero. What results is the numerator of the corresponding partial fraction.

First, we set $k = 0$ to find the numerator of $\frac{1}{k}$. We have $\frac{1}{(0+2)(0+4)} = \frac{1}{8}$, so this is the numerator for $\frac{1}{k}$. Next, we set $k = -2$ to find the numerator of $\frac{1}{k+2}$: we see that $\frac{1}{(-2)(-2+4)} = -\frac{1}{4}$ is thus the numerator for $\frac{1}{k+2}$. Finally, we set $k = -4$, giving $\frac{1}{(-4)(-2)} = \frac{1}{8}$.

Thus, our partial fraction decomposition is

$$\begin{aligned} \frac{1}{k^3 + 6k^2 + 8k} &= \frac{1}{k(k+2)(k+4)} \\ &= \frac{1/8}{k} - \frac{1/4}{k+2} + \frac{1/8}{k+4}. \end{aligned}$$

We see that this telescopes. Writing out a few terms, we see:

$$\begin{aligned} \sum_{k=1}^{\infty} \left(\frac{1}{8} \cdot \frac{1}{k} - \frac{1}{4} \cdot \frac{1}{k+2} + \frac{1}{8} \cdot \frac{1}{k+4} \right) &= \left(\frac{1}{8} \cdot \frac{1}{1} + \frac{1}{8} \cdot \frac{1}{2} + \frac{1}{8} \cdot \frac{1}{3} + \frac{1}{8} \cdot \frac{1}{4} + \frac{1}{8} \cdot \frac{1}{5} + \frac{1}{8} \cdot \frac{1}{6} + \dots \right) \\ &\quad + \left(-\frac{2}{8} \cdot \frac{1}{3} - \frac{2}{8} \cdot \frac{1}{4} - \frac{2}{8} \cdot \frac{1}{5} - \frac{2}{8} \cdot \frac{1}{6} - \dots \right) \\ &\quad + \left(\frac{1}{8} \cdot \frac{1}{5} + \frac{1}{8} \cdot \frac{1}{6} + \dots \right). \end{aligned}$$

We see that the terms telescope, leaving us with a final sum of

$$\frac{1}{8} \cdot \frac{1}{1} + \frac{1}{8} \cdot \frac{1}{2} + \frac{1}{8} \cdot \frac{1}{3} + \frac{1}{8} \cdot \frac{1}{4} - \frac{2}{8} \cdot \frac{1}{3} - \frac{2}{8} \cdot \frac{1}{4} = \frac{1}{8} + \frac{1}{16} - \frac{1}{24} - \frac{1}{32} = \frac{11}{96}.$$

Hence, the answer is (D) 107.

Solution 5 (Fast Index Shifting)

We start by factoring the cubic denominator as $k^3 + 6k^2 + 8k = k(k+2)(k+4)$. Quickly, we can do partial fraction decomposition by the cover-up method

$$\frac{1}{k(k+2)(k+4)} = \frac{A}{k} + \frac{B}{k+2} + \frac{C}{k+4}$$

$k = 0$ yields $A = \frac{1}{8}$ $k = -2$ yields $B = -\frac{1}{4}$ $k = -4$ yields $C = \frac{1}{8}$

Now, we have

$$\sum_{k=1}^{\infty} \left(\frac{1}{8} - \frac{1}{4} + \frac{1}{8} \right) = \frac{1}{8} \sum_{k=1}^{\infty} \left(\frac{1}{k} - \frac{2}{k+2} + \frac{1}{k+4} \right)$$

Now, let's evaluate this series then multiply by $\frac{1}{8}$ at the end. We break this into

$$\sum_{k=1}^{\infty} \frac{1}{k} - 2 \sum_{k=1}^{\infty} \frac{1}{k+2} + \sum_{k=1}^{\infty} \frac{1}{k+4} = \sum_{k=1}^{\infty} \frac{1}{k} - 2 \sum_{k=3}^{\infty} \frac{1}{k} + \sum_{k=5}^{\infty} \frac{1}{k} = \frac{3}{2} - \sum_{k=3}^{\infty} \frac{1}{k} + \sum_{k=5}^{\infty} \frac{1}{k} = \frac{3}{2} - \left(\frac{1}{3} + \frac{1}{4} \right) = \frac{18}{12}.$$

Then,

$$\frac{1}{8} \cdot \frac{18}{12} = \frac{11}{96} \implies 11 + 96 = \boxed{\text{(D) } 107}$$

Shockfront99

Solution 6 (Cool Integral Method)

We start by factoring the cubic denominator as $k^3 + 6k^2 + 8k = k(k+2)(k+4)$. Quickly, we can do partial fraction decomposition by the cover-up method

$$\frac{1}{k(k+2)(k+4)} = \frac{A}{k} + \frac{B}{k+2} + \frac{C}{k+4}$$

$k = 0$ yields $A = \frac{1}{8}$ $k = -2$ yields $B = -\frac{1}{4}$ $k = -4$ yields $C = \frac{1}{8}$

Now, we have

$$\sum_{k=1}^{\infty} \left(\frac{1}{8} - \frac{1}{4} + \frac{1}{8} \right) = \frac{1}{8} \sum_{k=1}^{\infty} \left(\frac{1}{k} - \frac{2}{k+2} + \frac{1}{k+4} \right)$$

Now, notice that

$$\frac{1}{k} = \int_0^1 x^{k-1} dx$$

and we could apply this to the rest

$$\frac{2}{k+2} = 2 \int_0^1 x^{k+1} dx$$

$$\frac{1}{k+4} = \int_0^1 x^{k+3} dx$$

Now, our sum is

$$\frac{1}{8} \sum_{k=1}^{\infty} \int_0^1 (x^{k-1} - 2x^{k+1} + x^{k+3}) dx$$

Swapping the sum and integral (check out some resources to learn about when you can do this), yields

$$\begin{aligned} \frac{1}{8} \int_0^1 \sum_{k=1}^{\infty} (x^{k-1} - 2x^{k+1} + x^{k+3}) dx \\ \frac{1}{8} \int_0^1 (x^4 - 2x^2 + 1) \sum_{k=1}^{\infty} x^{k-1} dx \end{aligned}$$

Then, using the geometric series formula for $|x| < 1$ and factoring the polynomial

$$\begin{aligned} \frac{1}{8} \int_0^1 (x^4 - 2x^2 + 1) \left(\frac{1}{1-x} \right) dx = \\ \frac{1}{8} \int_0^1 \frac{(1-x^2)^2}{1-x} dx = \frac{1}{8} \int_0^1 \frac{(1-x)(1+x)(1-x)(1+x)}{1-x} dx = \\ \frac{1}{8} \int_0^1 (1+x)(1-x)(1+x) dx = \\ \frac{1}{8} \int_0^1 (-x^3 - x^2 + x + 1) dx = \\ \frac{1}{8} \left(-\frac{1}{4} - \frac{1}{3} + \frac{1}{2} + 1 \right) = \frac{1}{8} \left(\frac{11}{12} \right) = \frac{11}{96} \implies 11 + 96 = \boxed{107} \end{aligned}$$

Shockfront99 (again)

Note: This isn't exactly the cleanest solution. I wouldn't recommend this on the actual AMC.

Solution 7 (simple, no partial fraction decomposition)

$$\text{Let } S = \sum_{k=1}^{\infty} \frac{1}{k^3 + 6k^2 + 8k} = \sum_{k=1}^{\infty} \frac{1}{k(k+2)(k+4)}$$

Notice a pattern: each factor is 2 greater than the previous one! So for now, let's consider the difference between two simpler fractions in hopes of rewriting the original series to spot a telescoping pattern:

$$\frac{1}{k(k+2)} - \frac{1}{(k+2)(k+4)} = \frac{k+4}{k(k+2)(k+4)} - \frac{k}{k(k+2)(k+4)} = \frac{4}{k(k+2)(k+4)}$$

$$\text{So } 4S = \sum_{k=1}^{\infty} \left(\frac{1}{k(k+2)} - \frac{1}{(k+2)(k+4)} \right) = \sum_{k=1}^{\infty} \frac{1}{k(k+2)} - \sum_{k=1}^{\infty} \frac{1}{(k+2)(k+4)}$$

By splitting the sum, and then using index shifting to rewrite the second summation, we can find that:

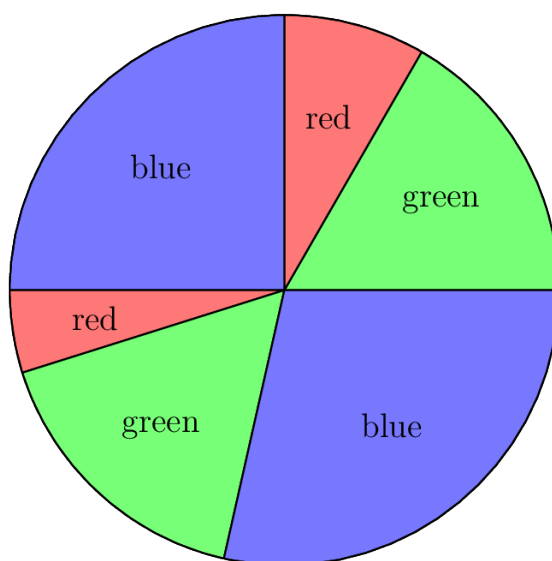
$$\sum_{k=1}^{\infty} \frac{1}{k(k+2)} = \sum_{k=1}^2 \frac{1}{k(k+2)} + \sum_{k=3}^{\infty} \frac{1}{k(k+2)} = \sum_{k=1}^2 \frac{1}{k(k+2)} + \sum_{k=1}^{\infty} \frac{1}{(k+2)(k+4)}$$

Awesome! Now we can substitute that into the equation for $4S$ and things will cancel out!

$$4S = \sum_{k=1}^2 \frac{1}{k(k+2)} + \left(\sum_{k=1}^{\infty} \frac{1}{(k+2)(k+4)} - \sum_{k=1}^{\infty} \frac{1}{(k+2)(k+4)} \right) = \sum_{k=1}^2 \frac{1}{k(k+2)} \implies S = 11/96$$

So $a + b = \boxed{\text{(D)} 107}$

- p) (AMC 10 10B 2025) A circle has been divided into 6 sectors of different sizes. Then 2 of the sectors are painted red, 2 painted green, and 2 painted blue so that no two neighboring sectors are painted the same color. One such coloring is shown below.



How many different colorings are possible?

- (A) 12 (B) 16 (C) 18 (D) 24 (E) 28

Answer: D

Solution 1

Consider the topmost sector of any given circle. Put a "marker" on it. Now, reflect the diagram and rotate it until the marker is in the same place as it was last time. What we have just constructed are two circles with different colorings, except that the unique sectors formed are the exact same. Thus, for any given circular diagram, another diagram exists that is the exact same as the prior one.

The number of such colorings that exist given an arbitrary circular diagram is $2^4 \cdot 3 = 48$. In all of these possible colorings, exactly two are in a bijection (correspondence, basically two are the exact same type of sector for any diagram are overcounted). Thus we divide the count by two, yielding the answer 24.

Essentially, this is what my solution 2 was trying to imply, except more in a formal definition. I hope now one can see the intuition.

If this did not make much sense, please check solution 3, which I believe I made to be beginner friendly.

Solution 2

Create an arbitrary six slice circular diagram. The first slice has 3 options, and the second has 2, third has 2, fourth has 2, fifth has 2, and the sixth has only 1 color to be chosen.

Understand that if $\mathbb{X}$ is the set containing all possible cases of the circle, then $\mathbb{X}$ has a correlation to another set, call it $\mathbb{K}$, which contains all the colors. This can be quite easily sought as a rotation of one element in $\mathbb{X}$ that accurately maps to another element in $\mathbb{X}$ is considered the same.

In simpler terms, to account for overlapping cases in $\mathbb{X}$, one must divide by 2 (as two colorings that are rotated are considered different, and we don't want that as it defeats the purpose of "unique").

Then, multiplying out we have $3 \times 2^4 = 48$. Dividing by 2 to account for overlap in $\mathbb{X}$, we get $48/2$ or (D) 24

Solution 3

We can have $3 \cdot 3 \cdot 2 \cdot 2 = 36$ unique colorings. Obviously, not all of these work. Particularly, we can have colorings in which 1, 2, or 3 colorings are adjacent. In case 1, we have 3 ways to choose that, and 2 ways to arrange the alternating colors. That is 6 ways total. In case 2, we have the case degenerates to case 3, in which we have again, 6 ways to determine as such. The answer is $36 - 6 - 6 =$ 24.

Notice we do not CHOOSE the colorings as the sectors are distinguishable.

Solution 4

Model the circle as the 6-cycle C_6 with labeled positions 1, 2, 3, 4, 5, 6 in circular order. We count proper colorings using R, G, B , each appearing exactly twice.

There are two structural possibilities for how the two copies of each color are placed:

Case 1: All three colors occupy opposite positions.

Opposite positions in C_6 are the three pairs (1, 4), (2, 5), (3, 6). If each color occupies one opposite pair, then the colors must alternate around the circle, yielding a pattern of the form A, B, C, A, B, C

for some ordering (A, B, C) of (R, G, B) . The number of such colorings is $3! = 6$

Case 2: Exactly one color occupies opposite positions.

Choose the color that is opposite: 3 choices. Choose which opposite pair it occupies: 3 choices one of (1, 4), (2, 5), (3, 6)), for a total of $3 \cdot 3 = 9$ placements.

Removing those two opposite positions leaves two disjoint adjacent pairs among the remaining four positions. For example, if the opposite color is at $(1, 4)$, the remaining positions $\{2, 3, 5, 6\}$ form the two adjacent pairs $(2, 3)$ and $(5, 6)$. The other two colors must each appear twice, and no adjacent positions may match; therefore each pair must be colored with different colors, and to achieve two copies of each color overall, the two pairs must be complementary: $(2, 3)$ is (G, B) or (B, G) , and then $(5, 6)$ in the opposite ordering.

This gives 2 valid assignments for the four positions per fixed opposite placement.

Hence, this case contributes $9 \cdot 2 = 18$

Summing the 2 cases

$$6 + 18 = 24 \quad \boxed{\text{(D)} \ 24}.$$

Solution 5

We can also case differently.

Let us call the sectors 1, 2, 3, 4, 5, and 6 going clockwise around the circle. We, without loss of generality, assume 1 to be colored R . Additionally, without loss of generality, assume 2 to be colored B . We will nullify these assumptions by multiplying by a factor of 6 at the end.

We case on the color of sector 6.

Case 1: Sector 6 is colored B Each color is used two times, so we have one R and two G s left to distribute between 3, 4, and 5. Obviously, the G s cannot be adjacent and thus the only possible coloring is $(3, 4, 5) = (G, R, G)$.

There is 1 solution for this case.

Case 2: Sector 6 is colored G In this case, we have we have one R , one B , and one G left to distribute between 3, 4, and 5. Since 3, 4, and 5, are all distinct colors, the only restrictions on this case are that 3 cannot be colored B and 5 cannot be colored G .

Quick casework shows that if 3 is colored R , there are one solution $(3, 4, 5) = (R, G, B)$ and if 3 is colored G then there are two solutions $(3, 4, 5) = (G, R, B)$ and $(3, 4, 5) = (G, B, R)$.

There are 3 solutions for this case.

Thus, the total number of solutions is:

$$6 \cdot (1 + 3) = \boxed{\text{(D)} \ 24}$$

Solution 6

We can also have the following two cases:

Case 1: The three colors on the top semicircle are all different Then, there are 6 different ways to rearrange the top. For the bottom, the leftmost one has one restriction and the rightmost one has one restriction. Let's say the leftmost cannot be red and the rightmost cannot be green WLOG. Then, we can have the cases (from left to right): green, blue, red, green, red, blue, and blue, green, red. So, there are 3 different ways to rearrange the bottom. So, for case one, there are $6 \times 3 = 18$ different ways to rearrange.

Case 2: The three colors on the top has one pair of the same colors

Then, there are 3 ways to choose the same color and then 2 ways to choose the color that is only put in once. After choosing all of that, the bottom then will be fixed (try it yourself if you think it is not). So, case 2 has $3 \times 2 = 6$ ways to rearrange.

After combining the 2 cases, we get $18 + 6 = 24$ cases, so our answer is

(D) 24.

Solution 7 (simplest and easiest)

Let's assume that the colors do not matter, so let's just start with Red in the very top sector. The fact that the sectors are distinct in size does not matter at all, so just ignore that. If Red is in the top sector, then we have 3 cases:

Case 1: Red is also in the 3rd sector (going clockwise from the top one) Thus, this is only one case because there are already 2 that are together, so the colors of those 2 must be distinct.

Case 2: Red is also in the sector directly opposite. There are 2 cases for this; we can have all the colors being opposite, or we can have a case where everything is symmetric.

Case 3: Red is also in the 5th sector This is basically congruent (symmetric) to the first case, so we also only have 1 case.

Adding all these up, we have $1 + 2 + 1 = 4$ cases in total.

However, we did not consider color yet. Because there are 3 colors, they can be permuted $3! = 6$ ways.

Therefore, $6 \times 4 = 24$, so our answer is **(D) 24**.

Solution 8

We want to find the number of cases where we can arrange the colors A , B , and C , fixing A at the top, and assuming B and C are symmetrical, so switching all instances of them wouldn't count. Starting from the top-left sector and moving clockwise, we could do $ABCACB$, which is what was on the example, $ABCABC$, $ABACBC$, or $ABCBAC$, so there are four cases. Then, we can just multiply the number of cases by 6, because that's how many ways we can rearrange A , B , and C , to get **(D) 24**. Additionally, we could cheese by noting that the answer is just the number of cases times 6, so the answer choice must be a multiple of 6, so it's either **(C) 18** or **(D) 24**, in which case it's possible to just guess the fifty-fifty.

Solution 9

Consider only half the circle. We only need to worry about three distinct colors on the upper half of the circle. There are $3! = 6$ ways to arrange the three colors on the semicircle and again $3! = 6$ ways to arrange the other same three colors on the bottom semicircle. In total, we have $6 \cdot 6 = 36$ total ways. Now, we note that when we put the two semicircles together to create a full circle, we're only concerned that the end two sectors of each semicircle don't match (since each semicircle is arranged with three distinct colors already). Out of the 6 permutations (for instance B, R, G or B, G, R and so on), we need to take one permutation and figure out how many permutations don't work. In other words, if we have the permutation B, R, G how many permutations

other than this one either start with a B or end with a G ? Doing the very easy casework, we see that in this particular case only B, G, R and R, B, G don't work. Therefore, for every permutation, there are 2 permutations that don't work. Since we have 6 permutations, our final answer is

$$36 - 6 \cdot 2 = \boxed{24}$$

Solution 10(force)

Say that we make a hexagon, with each vertices being a color on its own. We first denote the top vertices with a random color, say B . After the top vertex is decided, only the bottom 3 vertices are clear for B . The first case, where B is at the bottom right vertex, has 2 combinations possible. The second case, where B is at the bottom vertex, has 4 combinations. The third case, where B is at the bottom left vertex, has the mirrored value of 2 along with the first case. Multiplying all combinations by 3 gives us 24.

$$3 \cdot 8 = 24$$

Solution 11 (word permutations)

Let R denote a red segment, G denote a green segment, and B denote a blue segment. Let x represent a blank in this solution

Case 1: $RxRxxx$ There are 2 possibilities: $RGRBGB$ and $RBRGBG$

Case 2: $RxxRxx$ There are 4 possibilities: $RBGRBG$, $RGBRGB$, $RBGRGB$, and $RGBRBG$.

Case 3: $RxxxxR$ Similar to case 1, there are 2 possibilities

A total of 8 possibilities have been counted, and since the R in those spots can also either be G or B , we multiply by 3 to get $\boxed{\text{(D)} 24}$

(Note: $RxxxxR$ is not a possibility since we're dealing with a wheel! An easy pitfall to be careful of)

- q) (AMC 10 10B 2025) Consider a decreasing sequence of n positive integers $x_1 > x_2 > \cdots > x_n$ that satisfies the following two conditions:

- The average (arithmetic mean) of the first 3 terms in the sequence is 2025.
- For all $4 \leq k \leq n$, the average of the first k terms in the sequence is 1 less than the average of the first $k - 1$ terms in the sequence.

What is the greatest possible value of n ? (A) 1013 (B) 1014 (C) 1016 (D) 2016

Answer: B

Solution 1

Since the mean of the first three terms is 2025, their sum is $2025 \cdot 3 = 6075$. Then, incorporating the fourth term makes the mean

- r) (AMC 10 10B 2025) What is the ones digit of the sum

$$\lfloor \sqrt{1} \rfloor + \lfloor \sqrt{2} \rfloor + \lfloor \sqrt{3} \rfloor + \cdots + \lfloor \sqrt{2025} \rfloor?$$

(Recall that $\lfloor x \rfloor$ represents the greatest integer less than or equal to x .)

- (A) 1 (B) 2 (C) 3 (D) 5 (E) 8

Answer: D

Solution 1

All terms from the 1st to the 3rd will equal 1, because the value inside the floor function will be greater than or equal to $\sqrt{1} = 1$ but less than $\sqrt{4} = 2$. Following this logic, all terms from the 4th to the 8th will equal 2, all terms from the 9th to the 15th will equal 3, and all terms from number n^2 to $(n+1)^2 - 1$ will equal $\sqrt{n}$. Thus, there will be 3 terms equal to 1, 5 terms equal to 2, and so on. Writing out a few terms of the total sum, we have:

$$1 \cdot 3 + 2 \cdot 5 + 3 \cdot 7 + \cdots + 44 \cdot 89$$

The expression above can be written in summation format as:

$$\sum_{n=1}^{44} n(2n+1).$$

We can expand this to the following:

$$\sum_{n=1}^{44} 2n^2 + \sum_{n=1}^{44} n$$

Using the sum of integers formulas, this becomes

$$\frac{2 \cdot 44 \cdot 45 \cdot 89}{6} + \frac{44 \cdot 45}{2}$$

The units digit of that expression evaluates to 0. However, we excluded $\lfloor \sqrt{2025} \rfloor = 45$ from our sum because it appears only once in the sequence, so we have to add it back into our expression. Our final answer is **(D) 5**.

Solution 2

We quickly realize that from 1^2 to $2^2 - 1$, the floor of the square root is always 1, from 2^2 to $3^2 - 1$, the floor of the square root is always 2, and so on. So, we have $1 \cdot 3 + 2 \cdot 5 + 3 \cdot 7 + 4 \cdot 9 + \dots + 44 \cdot 89 + 45$. Now, we can bash for the first digit of each of these products and find that $1 \cdot 3$ has unit digit 3, $2 \cdot 5$ has unit digit 0, $3 \cdot 7$ has unit digit 1, $4 \cdot 9$ has unit digit 6, $5 \cdot 11$ has unit digit 5, $6 \cdot 13$ has unit digit 8, $7 \cdot 15$ has unit digit 5, $8 \cdot 17$ has unit digit 6, $9 \cdot 19$ has unit digit 1, $10 \cdot 21$ has unit digit 0. And so, we see that a loop has been established for $1 - 10, 11 - 20, 21 - 30, 31 - 40$ (In normal sums that would be $1^2 - 10^2, 11^2 - 20^2$, etc. Sum the first 10 numbers to find that it is equal to $5 \pmod{10}$, and multiply by 4 to account for 1 to 40. Now we have 0, and we must add the first 4 numbers of the loop again to get 0, and add 45 for the last number. Thus, we get **(D) 5** as the ones digit.

-henryli3333

Latex edits and minor edits by Wen_Liang

Solution 3: Units Digits!

As stated in the first Solution, we can notice that the first to third terms will equal 1, the second to eighth terms will equal 2, and so on. This is because the square roots of 5, 6, and 7 are in between 2 and 3, so using floor function, they would equal 2. This concept applies to all the following numbers.

We can notice that the pattern goes like: 1, 1, 1, 2, 2, 2, 2, 2, 3, 3, 3, 3, 3, 3, 4, 4, $\dots$ 44, 44, 44, 44, 45. The square roots of perfect squares will not have to be rounded down through the floor function, so to make it easier, we can pull those numbers out. By doing

this we get, $(1 + 2 + 3 + 4 + 5 + 6 + 7 + \cdots + 44 + 45)$, since $\sqrt{2025} = 45$ and $\sqrt{1} = 1$.

Pulling out these numbers leaves us with 1, 1, 2, 2, 2, 2, 3, 3, 3, 3, 3, 3, and so on. By looking at the remaining numbers, we can notice a pattern! There are two 1s, four 2s, six 3s, eight 4s...all the way until eighty-eight 44s! This can be simplified into, $2(1) + 4(2) + 6(3) + 8(4) + 10(5) + 12(6) + \cdots + 88(44)$. Notice that the problem is asking for the units digit, so we can do some simple multiplication to find another pattern.

We will expand our second expression:

$$\begin{aligned} 2 \cdot 1 &= 2 \\ 4 \cdot 2 &= 8 \\ 6 \cdot 3 &= 8 \\ 8 \cdot 4 &= 32 \\ 10 \cdot 5 &= 50 \\ 12 \cdot 6 &= 72 \\ 14 \cdot 7 &= 98 \\ 16 \cdot 8 &= 128 \\ 18 \cdot 9 &= 162 \\ 20 \cdot 10 &= 200 \end{aligned}$$

By looking only at the units digit:

$$\begin{aligned} 2 \cdot 1 &= 2 \\ 4 \cdot 2 &= 8 \\ 6 \cdot 3 &= 8 \\ 8 \cdot 4 &= 2 \\ 10 \cdot 5 &= 0 \\ 12 \cdot 6 &= 2 \\ 14 \cdot 7 &= 8 \\ 16 \cdot 8 &= 8 \\ 18 \cdot 9 &= 2 \\ 20 \cdot 10 &= 0 \end{aligned}$$

Huh...a pattern! It seems that this expression cycles in a pattern of 2, 8, 8, 2, 0. And notice again, if you add up the digits, $2 + 8 + 8 + 2 + 0 = 20$. Notice that the unit's digit is 0.

However, we can't immediately assume that the cycle ends neatly, so we must do a quick check with the last four terms of the sequence: $82(41) + 84(42) + 86(43) + 88(44)$. The units digits end up being $2 + 8 + 8 + 2$, so this time we're lucky and the cycle ends neatly. As a result, we know that because the cycle adds up to 20 and doesn't stop mid-cycle, the units digit will be 0.

Now, we can't just go ahead and give up because we got 0 as our answer. We still have our first expression to deal with. $(1 + 2 + 3 + 4 + 5 + \cdots + 44 + 45)$ By

using the sum of numbers formula, we get: $(45(1 + 45))/2$, which simplifies to: $45 \cdot 23 = 1035$. Since our second expression will have a unit digit of 0, we get $0 + 5 = \boxed{(D) 5}$, as our answer.

Solution For Magic Square Association Problem

We know that $\lfloor n \rfloor$ is the lowest integer less or equal than x . We also know that $\sqrt{4} = 2$, so that $\lfloor \sqrt{1} \rfloor$, $\lfloor \sqrt{2} \rfloor$, and $\lfloor \sqrt{3} \rfloor$ all are 1. $1 - 1 + 1 = 1$, so the first 3 terms equals one. Then, we look to the next 4 terms. $\lfloor \sqrt{4} \rfloor$, $\lfloor \sqrt{5} \rfloor$, $\lfloor \sqrt{6} \rfloor$, $\lfloor \sqrt{7} \rfloor$, and $\lfloor \sqrt{8} \rfloor$ all equals 2, making a total of $2 + 2 - 2 + 2 - 2 = -2$. Then all the values from $\lfloor \sqrt{9} \rfloor$ to $\lfloor \sqrt{15} \rfloor$ all equals 3, so now all the terms that equal 3 or -3 have a grand total of $3 - 3 + 3 - 3 + 3 - 3 + 3 = 3$. We quickly see a pattern. All the terms that equals 1 or -1 have a total of 1, then -2 , then 3, and so on. $\sqrt{2025} = 45$, so then we can quickly figure out the value to be $1 - 2 + 3 - 4 + 5 - 6 + 7 - \dots + 45$. $1 - 2 = -1$, $3 - 4 = -1$, and $5 - 6$ to be -1 , all the way to $43 - 44$. We can see that 43 is the 22nd odd number, so that we have $-22 + 45 = \boxed{23}$.

Solution 5: Induction, Units Digits: No hard algebra involved! (easiest)

Sure, somehow this is #18, but we can solve this problem with basically no algebra or any hard formulas. Also no confusing summation notation or modular arithmetic because we hate those!

Recall that the perfect squares differ by odd numbers. What does this mean?

Well, it means that from all $\lfloor \sqrt{1} \rfloor$ to $\lfloor \sqrt{3} \rfloor$, it will equal 1, for 3 values because $3 - 1 + 1 = 3$

Likewise, from $\lfloor \sqrt{4} \rfloor$ to $\lfloor \sqrt{8} \rfloor$, it will equal 2, for 5 values because $8 - 4 + 1 = 5$

Therefore, we can infer that there will be $2n + 1$ copies of each number n .

So, the entire sum will be

$$1 \cdot 3 + 2 \cdot 5 + 3 \cdot 7 + \dots + 44 \cdot 89 + 45$$

We could (*mod*10) this, but that is too hard, so let's try to just look at the units digit for each term. For example, the first two terms units digit will be

$$1 \cdot 3 \equiv 3 \quad 2 \cdot 5 \equiv 0$$

From here, you could just bash out all of them because there are only 45 (which isn't too bad to do, especially considering this is #18 and a lot of people here have already finished the first 15 problems).

However, that is kinda dumb to do in competition, so let's try to use induction!

Once we get to the 11th, 12th, 13th, such terms it will actually be congruent to the first 10 terms, because we are only looking at the units digits.

So, we can just compute the first 10 terms from 1-10, because that will be much quicker:

$$1 \cdot 3 \equiv 3 \quad 2 \cdot 5 \equiv 0 \quad 3 \cdot 7 \equiv 1 \quad 4 \cdot 9 \equiv 6 \quad 5 \cdot 11 \equiv 5 \quad 6 \cdot 13 \equiv 8 \quad 7 \cdot 15 \equiv 5 \quad 8 \cdot 17 \equiv 6 \quad 9 \cdot 19 \equiv 1 \quad 10 \cdot 21 \equiv 0$$

Adding all these up, we have $3 + 0 + 1 + 6 + 5 + 8 + 5 + 6 + 1 + 0 \equiv 35 \equiv 5$

Then, the units digit of the first 40 will be equal to $5 \cdot 4 \equiv 20 \equiv 0$

So, we just have to find the last 5 now

The first 4 is obviously $3 + 0 + 1 + 6 \equiv 10 \equiv 0$, then adding the 45th term:

$$1 * 45 \equiv 5$$

Finally, we add all these together:

$$0 + 0 + 5 \equiv 5$$

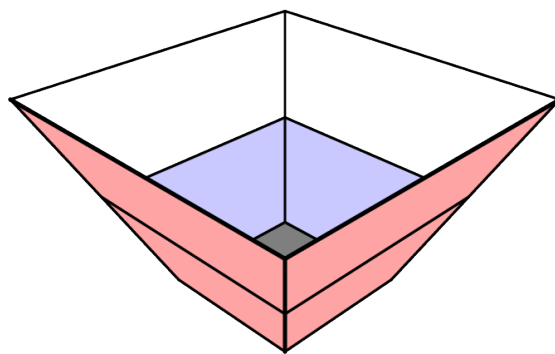
And therefore our answer is **(D) 5**.

Solution 6

As in the previous solutions, we find that the first 3 terms will equal 1, the next 5 will be 2, and so on. We know that we will add a 5 to the units digit of the terms leading up to ... $(44)(89)$ because of the 45 at the end of the sequence. Although this can be dangerous in certain problems, as it can lead you nowhere, if you start summing the units digits of the first terms of the sequence, you will see a cycle that repeats every 10 terms. The units digit of the sum of this cycle is 5, and since this will repeat 4 times, we multiply 5 by 4 for a units digit of 0 for the sum of the first 404 terms of the sequence. We now only need the units digit from $(41)(83)$ to $(44)(89)$, which we can find easily by multiplying the last digit of each 2 digit number. The units digits of these terms are 3,0,1, and 6. We add these to find that these 4 terms have a units digit of 0.

So, the first 44 terms have a units digit of 0; adding back the 5 from the last term in the sequence, we see that our answer is **(D) 5**.

- s) (*AMC 10 10B 2025*) A container has a 1×1 square bottom, a 3×3 open square top, and four congruent trapezoidal sides, as shown. Starting when the container is empty, a hose that runs water at a constant rate takes 35 minutes to fill the container up to the midline of the trapezoids.



How many more minutes will it take to fill the remainder of the container?

- (A) 70 (B) 85 (C) 90 (D) 95 (E) 105

Answer: D

Solution 1

Extend the edges pointing downwards to converge at a point A to form a square pyramid. Consider 3 square pyramids, the large one formed by the top vertices of the original figure and A , the middle one formed by the medians running through the sides of the original figure and point A , and the smaller one formed by the bottom vertices of the original figure and point A . Note that all pyramids are similar since they are all essentially scaled by a certain factor.

The median length is $\frac{3+1}{2} = 2$

Using side length to volume ratios, find that the volumes must have ratios $1 : 8 : 27$. Then, you get that the ratio of the volume thus filled to the volume that we must fill is equivalent to $8 - 1 : 27 - 8 = 7 : 19$. Thus, it will take $\frac{19}{7}$ more time to fill the remaining volume giving us an answer of $\frac{19}{7} * 35 = \boxed{\text{(D)} 95}$

-Failure.net

Solution 2 (2 min)

Note that dividing this shape along the midline gives two truncated pyramids with base areas 1 and 4, and 4 and 9 (a truncated pyramid is a pyramid is cut at some point parallel to the base to reach a shape with bases that are squares). Using the truncated pyramid volume formula

$$V = \frac{1}{3}h(A_1 + A_2 + \sqrt{A_1A_2}),$$

we get volumes

$$\frac{1}{3}h(1 + 4 + \sqrt{4}) \quad \text{and} \quad \frac{1}{3}h(4 + 9 + \sqrt{36}).$$

Writing their ratio, we obtain

$$\frac{13 + 6}{5 + 2} = \frac{19}{7}.$$

Thus,

$$35 \cdot \frac{19}{7} = 95.$$

So the remaining time is $\boxed{95}$.

-Cyrus825

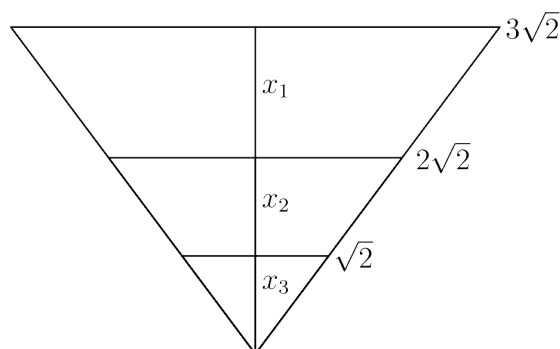
Solution 3 (Calculus)

The volume that was filled in 35 minutes can be represented as $\int_1^2 s^2 ds = \frac{1}{3}(2^3 - 1^3) = \frac{7}{3}$ where s represents the side length of the horizontal cross-sectional square. Then, the volume that remains to be filled is $\int_2^3 s^2 ds = \frac{1}{3}(3^3 - 2^3) = \frac{19}{3}$. Since the filling rate is the same, the amount of time this will take is $35 \cdot \frac{3}{7} \cdot \frac{19}{3} = \boxed{\text{(D)} 95}$

-grape1984

Solution 4 (Construction)

Extend the bottom of the container to form an upside down square pyramid. Then take a diagonal cross section as shown below.



You can notice that there are several similar triangles, so we write several equations.

$$\frac{x_3}{x_1 + x_2 + x_3} = \frac{\sqrt{2}}{3\sqrt{2}}$$

$$\frac{x_3}{x_2 + x_3} = \frac{\sqrt{2}}{2\sqrt{2}}$$

Then solve the system to get:

$$x_1 = x_2 = x_3 = x$$

Write the volumes of the three pyramids with sides of the bases 1,2, and 3 respectively:

$$V_1 = \frac{x}{3}$$

$$V_2 = \frac{8x}{3}$$

$$V_3 = 9x$$

The volumes of the part of the container below the midline and above it are.

$$V_2 - V_1 = \frac{7x}{3}$$

and

$$V_3 - V_2 = \frac{19x}{3}$$

Then the time t for the top to fill up is given by

$$\frac{\frac{19x}{3}}{\frac{7x}{3}} = \frac{t}{35}$$

Therefore $t = \boxed{\text{(D) } 95}$

-AARONPICK2

Solution 5(Similar to Solution 1)

Extend the trapezoidal shape into a square pyramid. Note that we're only extending the height by another factor of the height. In other words, if the height from the 1×1 base to the midline is h , the height from the 2×2 base(which is the midline) to the 3×3 base(which is the finish) is also h . Now using this logic, the distance from the 1×1 base to the top of the pyramid is

obviously also h . Now the problem is easy. Computing the volume of the first half of the trapezoidal shape (which takes 35 minutes for the hose to fill) is

$\frac{1}{3}(2 \cdot 2) \cdot (2h) - \frac{1}{3}(1 \cdot 1) \cdot (h) = \frac{7h}{3}$ which takes 35 minutes. So to fill up to the height h we get

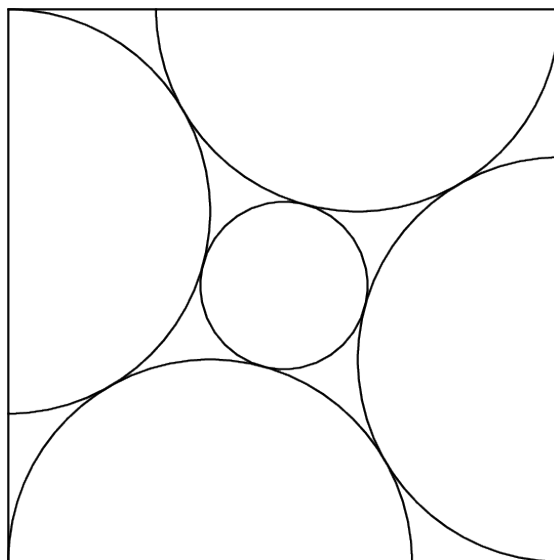
h takes $35 \cdot \frac{3}{7} = 15$ minutes to fill.

Now to find the volume of the last half of the trapezoidal shape we get

$\frac{1}{3}(3 \cdot 3) \cdot (3h) - \frac{1}{3}(2 \cdot 2) \cdot (2h) = \frac{19h}{3}$ so if we plug in h takes 15 minutes to fill we get

$\frac{19h}{3}$ takes $\frac{19 \cdot 15}{3} = 19 \cdot 5 = \boxed{95}$ minutes to fill.

- t) (AMC 10 10B 2025) Four congruent semicircles are inscribed in a square of side length 1 so that their diameters are on the sides of the square, one endpoint of each diameter is at a vertex of the square, and adjacent semicircles are tangent to each other. A small circle centered at the center of the square is tangent to each of the four semicircles, as shown below. The diameter of the small circle can be written as $(\sqrt{a} + b)(\sqrt{c} + d)$, where a , b , c , and d are integers. What is $a + b + c + d$?

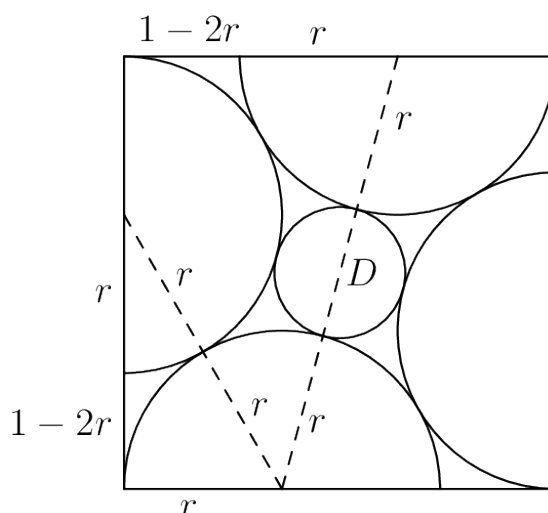


- (A) 3 (B) 5 (C) 8 (D) 9 (E) 11

Answer: A

Solution

Let the radius of the large semicircles be r , and let the diameter of the inner be D . Draw lines as follows:

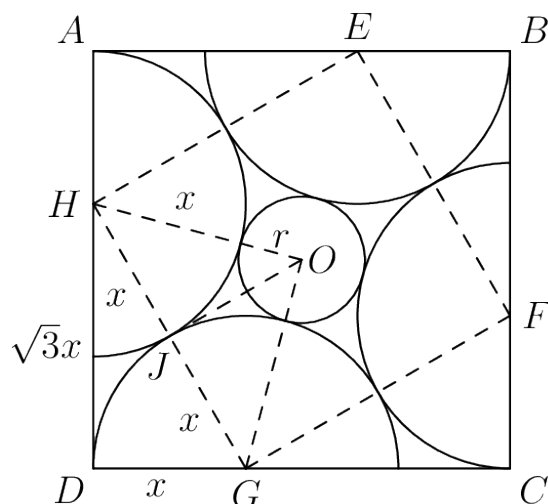


By the Pythagorean Theorem, $r^2 + (1-r)^2 = (2r)^2$, so $2r^2 + 2r - 1 = 0$. Hence $r = \frac{\sqrt{3}-1}{2}$ (we take the positive solution).

Now apply the Pythagorean Theorem on the quadrilateral's altitude down to get $(1 - 2r)^2 + 1 = (2r + D)^2$. Solving yields $2r + D = \sqrt{8 - 4\sqrt{3}} = \sqrt{6} - \sqrt{2}$. Then $D = \sqrt{6} - \sqrt{2} - \sqrt{3} + 1 = (\sqrt{3} - 1)(\sqrt{2} - 1)$. Hence the answer is $3 - 1 + 2 - 1 = \boxed{\text{(A)} \ 3}$.

Solution 2

We have the diagram:



We label the vertices of the square A, B, C , and D , the centers of the semicircle E, F, G , and H , and the center of the small circle O . Lets call the radius of the big semicircle x and the radius of the small circle r .

Looking at $\triangle HDG$, we can see it is a right triangle as $ABCD$ is a square, and we can also observe that since DG is x and the hypotenuse GH is exactly twice at $2x$, $\triangle HDG$ is a $30 - 60 - 90^\circ$ triangle. That means HD is equal to $\sqrt{3}x$.

Upon inspection, we can also see that AH is equal to x , and since $ABCD$ is a

unit square with all sides equal to 1, we can set the equation $HD + AH = 1$,

$$x + \sqrt{3}x = 1$$

, so solving for x , we get that

$$x = \frac{\sqrt{3} - 1}{2}$$

Then, if we compose an auxiliary line from O perpendicular to HG at J , we can observe that $\triangle HOJ$ is an isosceles right triangle. That means $x + r = \sqrt{2}x$ ($1 - 1 - \sqrt{2}$ relations), which means

$$r = (\sqrt{2} - 1)x$$

We're asked to find the diameter of the small circle, so we can multiply the equation by 2 to get

$$2r = 2(\sqrt{2} - 1)x$$

Plugging x in, the 2 multiplied and the 2 in the denominator cancel out, and we get

$$2r = 2(\sqrt{2} - 1)\left(\frac{\sqrt{3} - 1}{2}\right)$$

$$2r = (\sqrt{2} - 1)(\sqrt{3} - 1)$$

a , b , c , and d are 2, -1 , 3 respectively, and -1 , so $2 + (-1) + 3 + (-1)$ gets us (A) 3

-Neo

Solution 3

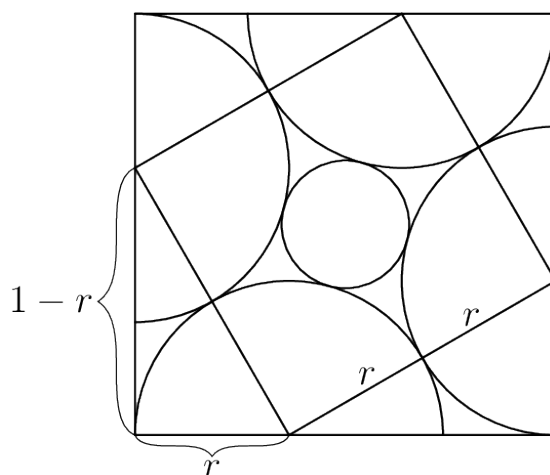
Let the centers of the four circles on AB , BC , CD , and AD be E , F , G , and H , respectively. Let r_1 be equal to the radius of one of the semicircles and r_2 be the radius of the inner circle. From the diagram from solution 2, we can easily recognize that $\triangle DHG$ is a $30 - 60 - 90$ triangle. We now form an equation $r_1 + r_1\sqrt{3} = 1$. Solving for r_1 we will get $\frac{\sqrt{3}-1}{2}$ as the radius. We now need to recognize that $EFGH$ is a square. We can also see that $HF = 2r_1 + 2r_2$. Because $EFGH$ is a square, and $HG = 2r_1$, we can figure out that $HF = 2r_1\sqrt{2}$. The final step is to plug in r_1 and solve for $2r_2$ (not r_2 as the question asks for the diameter).

$$2\left(\frac{\sqrt{3}-1}{2}\right)\sqrt{2} = 2\left(\frac{\sqrt{3}-1}{2}\right) + 2r_2 \quad (\sqrt{3}-1)\sqrt{2} = \sqrt{3}-1 + 2r_2 \quad (\sqrt{3}-1)(\sqrt{2}-1) = 2r_2$$

Thus, the answer is $3 - 1 + 2 - 1 =$ (A)3

Solution 4

We realize that we can connect the radii of the semicircles to form an inner square (let r denote the radius of the semicircles and d denote their diameter)



To find the length of the inner square, we apply the Pythagorean Theorem

$$(1 - r)^2 + r^2 = (2r)^2$$

We now substitute $r = \frac{d}{2}$

$$(1 - \frac{d}{2})^2 + (\frac{d}{2})^2 = d^2$$

$$1 - d + \frac{d^2}{4} + \frac{d^2}{4} = d^2$$

$$\frac{d^2}{2} - d + 1 = d^2$$

$$d^2 - 2d + 2 = 2d^2$$

$$-d^2 - 2d + 2 = 0$$

$$d^2 + 2d - 2 = 0$$

$$d = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

$$d = \frac{-2 \pm \sqrt{4 - 4(-2)}}{2}$$

$$d = \frac{-2 \pm \sqrt{12}}{2}$$

We must only consider the positive solution

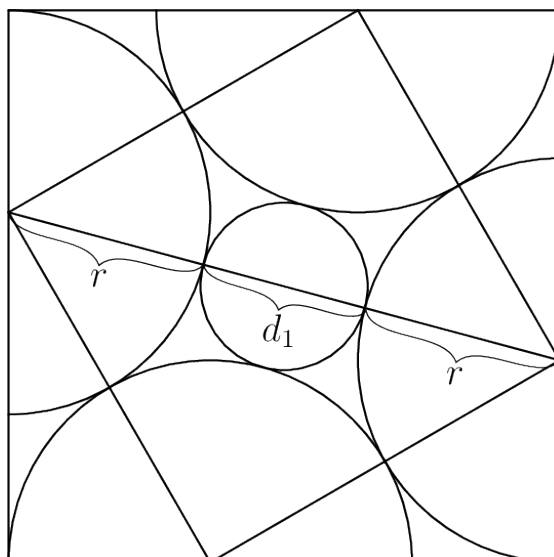
$$d = \frac{-2 + \sqrt{12}}{2}$$

$$d = \frac{-2 + 2\sqrt{3}}{2}$$

$$d = -1 + \sqrt{3}$$

$$d = \sqrt{3} - 1$$

The diagonal of the inner square measures $d\sqrt{2}$. It can also be represented as $d + d_1$ where d_1 denotes small circle's diameter. I demonstrate this below



$$2r + d_1 = d + d_1$$

Therefore, we equate them, find d_1 , and substitute $d = \sqrt{3} - 1$

$$d + d_1 = d\sqrt{2}$$

$$d_1 = d\sqrt{2} - d$$

$$d_1 = d(\sqrt{2} - 1)$$

$$d_1 = (\sqrt{2} - 1)(\sqrt{3} - 1)$$

$$2 + 3 - 2 = 3 \implies \boxed{\text{(A)} 3}$$

- u) (AMC 10 10B 2025) Each of the 9 squares in a 3×3 grid is to be colored red, blue, or yellow in such a way that each red square shares an edge with at least one blue square, each blue square shares an edge with at least one yellow square, and each yellow square shares an edge with at least one red square. Colorings that can be obtained from one another by rotations and/or reflections are to be considered the same. How many different colorings are possible?

(A) 3 (B) 9 (C) 12 (D) 18 (E) 27

Answer: C

Solution 1 (Casework)

Denote 1 = red, 2 = blue, 3 = yellow. We need $1 \rightarrow 2 \rightarrow 3 \rightarrow 1$.

WLOG place 1 in the center $(0, 0)$, 2 on the left edge $(-1, 0)$, 3 on the top-left corner $(-1, 1)$.

$$\begin{bmatrix} 3 & 0 & 0 \\ 2 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

3 must see 1, so the top edge $(0, 1)$ must also have 1. Then, 1 must see 2, so the top-right corner $(1, 1)$ becomes 2, which must see 3, so the right edge

$(1, 0)$ must have 3.

$$\begin{bmatrix} 3 & 1 & 2 \\ 2 & 1 & 3 \\ 0 & 0 & 0 \end{bmatrix}$$

Now this bottom-right corner can vary either as 1, 2 or 3. Cases on $(1, -1)$: If 1:

$$\begin{bmatrix} 3 & 1 & 2 \\ 2 & 1 & 3 \\ 3 & 2 & 1 \end{bmatrix}$$

but the 3 needs a 1 and does not have it, so there are 0 cases. If 2:

$$\begin{bmatrix} 3 & 1 & 2 \\ 2 & 1 & 3 \\ b & a & 2 \end{bmatrix}$$

if $a = 1$, b can be 1 or 3. If $a = 2$, $b = 3$, but the 3 needs a 1 and can't get it. If $a = 3$, $b = 1, 2$, so there are 4 cases in total. If 3:

$$\begin{bmatrix} 3 & 1 & 2 \\ 2 & 1 & 3 \\ 2 & 1 & 3 \end{bmatrix}$$

Since the 2 in the bottom left corner does not have a 3 nearby, there are 0 cases.

WLOG the center fixes a factor of 3, so the answer is $4 \cdot 3 = \boxed{\text{(C)} 12}$.

Solution 2 (Complex Circular Counting)

This solution is complex, but gives the answer in less than a couple minutes and saves the bearing of casework, so proceed as a last resort.

Also yes this solution is inspired from 3Blue1Brown's Olympiad Counting video.

We will find the arrangement that contains the least possible occurrences of a color a and the most occurrences of a color b , in which we will then use the circular gap theorem $C_n A$ on all integers x such that $x \in [a, b]$ and $(a, b, x) \in \mathbb{Z}^+$.

The average number of colors M is simply $9/3 = 3$ (9 slots divided by 3 colors). So we see that the interval contains $a < 3 < b$. We also have that the minimal number of colors is clearly 1 (it doesn't take much to show a case that contains only one color). The challenge is to find the maximal number, which is b .

We see that there must be at least one of each color, so b can be a maximum of 6. However, we can see that 6 clearly cannot be b , as if we have a configuration of 6 colors, that would mean that through the constraints in the problem, the other three colors are already uniquely determined. However, now there is no slots in the grid, and we cannot satisfy all the requirements, so $b \leq 5$.

However, we also find that $b \neq 5$, as the four colors, say blue already determine 2 other colors, say yellow, which determine another 2 colors, say red. However, now there is one color left! This color must be blue (as there are 5), but if

there is one blue and no yellow to satisfy the condition, then the case is simply impossible. Therefore, $b \leq 4$.

We can try to do the same thing for 4, but we will see that $b = 4$ satisfies the prerequisites.

Therefore, $a = 1$ and $b = 4$, in which we then use the circular gap theorem

$$C_n A = \frac{n}{k} \binom{n-k-1}{k-1}$$

For $n = x : x \in [a, b]$, $(a, b, x) \in \mathbb{Z}^+$ and $0 < k \leq \lfloor \frac{n}{2} \rfloor : k \in \mathbb{Z}^+$

And so, we do so!

For $n = 1$, $C_n A = 1$ (note that every $\binom{n}{0} = 1$)

For $n = 2$, $C_n A = 2$

For $n = 3$, $C_n A = 3$

For $n = 4$, $C_n A = 6$

Our answer is just the sum of all four of these, which is $6 + 3 + 2 + 1$ or

(C) 12

btw the actual theorem name for $C_n A$ isn't actually the Circular Gap Theorem. I just call it that because I honestly don't know the name of it. It is a pretty neat theorem however, and I first learned it from this AIME problem here: [1983 AIME Q7 Sol. 9](#).

It is apparently a closed form for the number of ways to write n as the sum of k positive circular cycles for $n \geq 2$, and for $0 < n \leq 1$, the value is just 1.

Solution 3 (Alternative Casework)

We begin by WLOG-ing the center box as red, denoted R.

	R	

We will then consider the upper edge box, for now we'll assume it to be blue (it will be clear later)

	B	
	R	

Now we consider the upper right box, which is forced to be yellow as it would be redundant due to reflection through the middle column.

	B	Y
	R	

Now, working in a clockwise direction, the right edge box is forced to be red, the bottom corner therefore is forced to be blue, and the bottom edge is forced to be yellow.

	B	Y
	R	R
	Y	B

Now we must consider the bottom left corner. If we set the bottom left corner to red, the left edge therefore must be blue, but then nothing can be filled in for the top left box (red causes left edge to violate the blue $\rightarrow$ yellow rule, blue fails for a multitude of reasons, and yellow violates the yellow $\rightarrow$ red rule).

?	B	Y
B	R	R
R	Y	B

Yellow also doesn't work, since we get:

B	B	Y
R	R	R
Y	Y	B

but then the blue top-left corner has no adjacent yellow box. On the other hand, blue does give 4 valid configurations:

Either (1):

R	B	Y
R	R	R
B	Y	B

or (2):

Y	B	Y
R	R	R
B	Y	B

or (3):

R	B	Y
Y	R	R
B	Y	B

or (4):

B	B	Y
Y	R	R
B	Y	B

We can now explain why it's valid to assume that the upper edge box is blue: if it were yellow instead, the grid would simply be the reflection of one of the above configurations through the middle row, precisely because the bottom edge box is yellow (as this would then become the upper edge box under the reflection). If the upper edge box were red, either the upper-left or the upper-right box would have to be blue, and so by reflection through the middle column (just like above), we can assume that the upper-right box specifically is blue. Again similarly to above, we can now fill in the rest of the entire grid, giving one of the following 5 configurations:

Either (A):

Y	R	B
B	R	Y
R	R	B

or (B):

R	R	B
B	R	Y
Y	R	B

or (C):

Y	R	B
B	R	Y
Y	R	B

or (D):

Y	R	B
B	R	Y
R	Y	B

or (E):

Y	R	B
B	R	Y
B	Y	B

But (B) is simply the reflection of (A) in the bottom row, so we can eliminate it. After that, we observe that (A) is a rotation of (1) above, (C) is a rotation of (2), (D) is a rotation of (3), and (E) is a rotation of (4), so we can in fact eliminate all of these too.

Accordingly, having taken the central box as red, there are 4 distinct possible configurations (precisely (1) to (4) above), and by symmetry, that number of possible configurations will clearly be the same if the central box is blue or yellow instead. Thus there are 3 possible choices for the central box, followed by 4 ways to complete the grid in each of these cases, so the total number of possible grids is $3 \cdot 4 = \boxed{\text{(C) } 12}$.

- v) (AMC 10 10B 2025) A seven-digit positive integer is chosen at random. What is the probability that the number is divisible by 11, given that the sum of its digits is 61? (A) $\frac{3}{14}$ (B) $\frac{3}{11}$ (C) $\frac{2}{7}$ (D) $\frac{4}{11}$ (E) $\frac{3}{7}$

Answer: A

Solution 1

Let the number be represented as $\overline{a_1a_2a_3a_4a_5a_6a_7}$.

We first seek to find the number of seven-digit integers with $a_1 + a_2 + a_3 + a_4 + a_5 + a_6 + a_7 = 61$. We would love to do stars and bars here, but we have the condition that $0 \leq a_n \leq 9$. Therefore, we define $a'_n = 9 - a_n \implies a_n = 9 - a'_n$. This is a bijective operation, so the total number of integers doesn't change when we use these digits instead. Substituting in gives that

$$\sum_{k=1}^7 (9 - a'_k) = 61,$$

$$a'_1 + a'_2 + a'_3 + a'_4 + a'_5 + a'_6 + a'_7 = 2.$$

Since the a'_n are nonnegative integers and the condition $a'_n \leq 9$ doesn't matter with a sum less than 9, we can do stars and bars to figure out the number of permutations of digits that satisfy the constraint: $\binom{2+6}{2} = \binom{8}{2} = 28$.

Now, we need to find how many of these integers are divisible by 11. We can use the divisibility by 11 rule, which says that $(a_1 + a_3 + a_5 + a_7) - (a_2 + a_4 + a_6) = 11k$ for some integer k . Notice that $k \geq 1$ because, otherwise, $a_2 + a_4 + a_6 > 27$ which is the maximum sum of three digits. In addition, $k \leq 1$ because, otherwise $a_1 + a_3 + a_5 + a_7 > 36$ which is the maximum sum of four digits. Therefore $k = 1$.

Then we have that $(a_1 + a_3 + a_5 + a_7) - (a_2 + a_4 + a_6) = 11$, so $a_1 + a_3 + a_5 + a_7 = 36$ and $a_2 + a_4 + a_6 = 25$. The first equation can only result in $a_1 = a_3 = a_5 = a_7 = 9$. The second equation can be rewritten in a similar manner as before to get that

$$\begin{aligned}(9 - a'_2) + (9 - a'_4) + (9 - a'_6) &= 25, \\ a'_2 + a'_4 + a'_6 &= 2.\end{aligned}$$

Again, with stars and bars, there exists $\binom{2+2}{2} = \binom{4}{2} = 6$ ordered triples of (a_2, a_4, a_6) .

Therefore, our answer is

$$\frac{6}{28} = \boxed{(A) \frac{3}{14}}.$$

Solution 2 (similar to 1, simplified)

Note the divisibility rule of 11. The sum of every other digit minus the sum of the rest of the digits must be divisible by 11.

The maximum possible sum of digits in a 7-digit number is 63 (9999999). Since the sum of the digits in the mystery number is 61, we have two possible scenarios. Either 5 digits are 9 and 2 digits are 8 (ex: 9899989, 9999988), or 6 digits are 9 and 1 digit is 7 (9979999). Going case by case, there are 7 numbers involving six digits as 9 and one as 7 and 21 numbers involving five digits as 9 and two digits as eight, giving a total 28 possible numbers.

Utilizing the divisibility rule, it can be seen that in order to satisfy the rule, the 7 or both 8s must be in the even positions (such that the number satisfies 9X9X9X9). There are 3 numbers that satisfy 9X9X9X9 where one of the X is 7 and the other two are 9, and 3 numbers that satisfy 9X9X9X9 where 2 of the X are 8 and the other is 9, giving 6 satisfying numbers. Therefore, the answer is

$$\frac{6}{28} = \boxed{(A) \frac{3}{14}}.$$

.

Solution 3 (similar to solution 2, easiest way)

As in solution 2, we find that there are 2 possibilities for 7-digit numbers with digits summing to 61. As the divisibility rule for 11 is that we must have that the sum of the digits in the even places minus the sum of the digits in the odd places is 0 or 11 with the two cases we have (9999988 and 9999997), we see that the even places must all be 9. Then there are 3 possibilities for each number for each. We then use simple combinations to find that there are 28 possible configurations of these 2 possibilities. We simplify the fraction of our

favored outcomes over the total to get our answer of $\boxed{\frac{3}{14}}$.

- w) (AMC 10 10B 2025) A rectangular grid of squares has 141 rows and 91 columns. Each square has room for two numbers. Horace and Vera each fill in the grid by putting the numbers from 1 through $141 \times 91 = 12,831$ into the squares. Horace fills the grid horizontally: he puts 1 through 91 in order from left to right into row 1, puts 92 through 182 into row 2 in order from left to right, and continues similarly through row 141. Vera fills the grid vertically: she puts 1 through 141 in order from top to bottom into column 1, then 142 through 282 into column 2 in order from top to bottom, and continues similarly through column 91. How many squares get two copies of the same number?

(A) 7 (B) 10 (C) 11 (D) 12 (E) 19

Answer: C

Solution 1

Let's say n is one of the numbers that got written twice in the same box. Suppose it is at column x and row y . We will write an expression for n in terms of x and y in two ways: from Horace's perspective and Vera's perspective.

From Horace's perspective, $n = (y - 1)(141) + x$. This is because there are $y - 1$ full rows before n , and we then need x more numbers to reach n . Similarly, Vera will say $n = (x - 1)(91) + y$.

We now have the Diophantine equation

$$(y - 1)(141) + x = (x - 1)(91) + y$$

$$141y - 141 + x = 91x - 91 + y$$

$$140y = 90x + 50$$

$$14y = 9x + 5$$

One such solution is, of course, $x = y = 1$. We find further solutions by adding 14 to x and 9 to y . For example, the second solution is $(15, 10)$. This continues until $(141, 91)$ is reached. There are 11 ordered pairs in this list.

Solution 1 (Easier way to count)

The Diophantine equation $14y - 9x = 5$ is in the form $ax + by = c$ has primary solution $(x_0, y_0) = (1, 1)$. Also, $h = \gcd(a, b) = \gcd(14, -5) = \gcd(14, 5) = 1$, and thus the solutions are of the form

$$y = 1 + 9k$$

$$x = 1 + 14k$$

for some k .

Then, notice that because $k \geq 0$, $1 \leq x \leq 141$, and $1 \leq y \leq 91$, the largest value possible is $(x, y) = (141, 91)$, and thus

$$14k \leq 140$$

$$9k \leq 90$$

.

In which $k \leq 10$ in both cases. Thus there are 10 positive integers k that satisfy the following. Including 0, there are $10 + 1 = \boxed{11}$.

Solution 2 (0-indexing)

For these types of problems, we essentially analyze residues, so it is more convenient to work with 0-indexing: $c = 0, 1, 2, \dots, 90$, $r = 0, 1, 2, \dots, 140$.

Then $n = 91r + c + 1 = 141c + r + 1 \iff 9r = 14c$. We must have $(r, c) = (0, 0), (14, 9), (28, 18), \dots, (140, 90)$ so $\boxed{(C) 11}$.

Solution 2 (faster ending)

After indexing to 0, we imagine a line that passes from $(0, 0)$ to $(140, 90)$. We know we are looking for all lattice points that lie on this line (this intuitively makes sense, because any points above the line would have column numbers far ahead of the rows, and vice versa).

Therefore, we know that the number of points must divide both 140 and 90. The GCD of the two is 10. We also account for point $(0, 0)$, and get that $10 + 1 = \boxed{(C) 11}$.

Solution 3

Observe that Horaces number can be expressed as the remainder of the row number plus the column number so $H(\text{Horaces Number}) = 91(c - 1)(\text{column addition}) + r(\text{remainder of division})$ and do the same for Vera $V = 141(r - 1) + c$. The condition is $H = V$ so set the equations equal to each other. $91(c - 1) + r = 141(r - 1) + c$ so $91c - 91 + r = 141r - 141 + c$, $140r = 90c + 50$. This is a linear Diophantine equation and the 1st solution is $(1, 1)$ where the solution repeats every $\text{lcm}(90, 140) = 1260$ there are 11 solutions so $\boxed{(C) 11}$ is correct.

- x) (AMC 10 10B 2025) A frog hops along the number line according to the following rules.

- It starts at 0.
- If it is at 0, then it moves to 1 with probability $\frac{1}{2}$ and it disappears with probability $\frac{1}{2}$.
- For $n = 1, 2$, or 3, if it is at n , then it moves to $n + 1$ with probability $\frac{1}{4}$, it moves to $n - 1$ with probability $\frac{1}{4}$, and it disappears with probability $\frac{1}{2}$.

What is the probability that the frog reaches 4? (A) $\frac{1}{101}$ (B) $\frac{1}{100}$ (C) $\frac{1}{99}$ (D) $\frac{1}{98}$

Answer: E

Solution 1 (conditional probabilities of reaching 4)

We will solve this using the **law of total probability** (otherwise apparently known in Olympiad folklore as the method of "states", presumably reflecting the fact that these are kids who won't yet have studied formal/axiomatic probability theory). Firstly, we note that the probabilities of the frog taking each of its possible moves always depend only on its current position, and not on how it got there - in other words, the frog, just like coins or dice, 'has no memory'. Therefore, we can unambiguously define p_n as the probability of reaching 4 given that the frog is currently at position n . We want to find p_0 (since the frog starts at 0), and of course $p_4 = 1$.

For each value of n , we now consider the conditional probabilities of reaching 4 given that the frog takes each of the possible moves from its current position.

For instance, if the frog is at 3, it can go to $3 - 1 = 2$ with probability $\frac{1}{4}$, and if it does, the conditional probability of it now reaching 4 from its new position at 2 is p_2 by definition; or it can go to $3 + 1 = 4$, also with probability $\frac{1}{4}$, and if it does, the conditional probability is of course $p_4 = 1$; or it can disappear with probability $\frac{1}{2}$, and if it does, it can never reach 4, i.e. the conditional probability is 0.

By analogous arguments at each of the other possible values of n , and then by applying the law of total probability, we deduce that

$$\begin{aligned} p_3 &= \frac{1}{4}p_4 + \frac{1}{4}p_2 \\ p_2 &= \frac{1}{4}p_3 + \frac{1}{4}p_1 \\ p_1 &= \frac{1}{4}p_2 + \frac{1}{4}p_0, \text{ and} \\ p_0 &= \frac{1}{2}p_1. \end{aligned}$$

Finally, by solving this system of equations, it follows that $p_0 = \boxed{\text{(E)} \frac{1}{97}}$ (and incidentally, the probabilities for the frog's other possible starting positions are $p_1 = \frac{2}{97}, p_2 = \frac{7}{97}$, and $p_3 = \frac{26}{97}$).

Solution 2 (conditional probabilities of taking single steps)

Let p_n be the probability that the frog eventually reaches position n , given that it is at position $n - 1$. (This is well-defined because the frog 'has no memory', just as explained in Solution 1.)

From position 0, the frog moves to 1 with probability $\frac{1}{2}$, or disappears with probability $\frac{1}{2}$. Hence $p_1 = \frac{1}{2}$.

From position 1, the frog moves to 2 with probability $\frac{1}{4}$, back to 0 with probability $\frac{1}{4}$, or disappears with probability $\frac{1}{2}$. Thus, by the law of total probability (again similarly to in Solution 1) and using $p_1 = \frac{1}{2}$ from above, we obtain

$$p_2 = \frac{1}{4} + \frac{1}{4} \cdot \frac{1}{2}p_2 = \frac{1}{4} + \frac{1}{8}p_2 \iff \frac{7}{8}p_2 = \frac{1}{4} \iff p_2 = \frac{2}{7}.$$

We also now observe that if the frog is currently at 1 and wants to eventually reach 3, then it will necessarily have to pass through 2 first, since all of its possible moves only increase or decrease its position by 1. Accordingly, if the frog is at 2 and chooses to move back to 1, then the conditional probability that it eventually reaches 3 from its new position at 1 is equivalent to the probability that it eventually reaches 2 starting at 1 *and then* eventually reaches 3 starting at 2. That in turn reduces to the product of these two conditional probabilities: that of reaching 2 starting at 1 is $p_2 = \frac{2}{7}$ from above, while that of reaching 3 starting at 2 is p_3 .

Using the law of total probability (analogously to above) at position 2, it follows that

$$p_3 = \frac{1}{4} + \frac{1}{4} \cdot \frac{2}{7}p_3 = \frac{1}{4} + \frac{1}{14}p_3 \iff \frac{13}{14}p_3 = \frac{1}{4} \iff p_3 = \frac{7}{26}.$$

Now from position 3, using $p_3 = \frac{7}{26}$ as just found above, we deduce that

$$p_4 = \frac{1}{4} + \frac{1}{4} \cdot \frac{7}{26} p_4 \iff \frac{97}{104} p_4 = \frac{1}{4} \iff p_4 = \frac{26}{97}.$$

Lastly, again because all of the frog's possible moves only increase or decrease its position by 1 (as explained above), the required probability that it eventually reaches 4 starting at 0 is equivalent to the probability that it eventually reaches 1 starting at 0, *then* reaches 2 starting at 1, *then* 3 starting at 2, *and then* 4 starting at 3. Similarly to above, this reduces to the product of the conditional probabilities p_1, p_2, p_3 , and p_4 , i.e. $\frac{1}{2} \cdot \frac{2}{7} \cdot \frac{7}{26} \cdot \frac{26}{97} = \boxed{\text{(E)} \frac{1}{97}}.$

Solution 3 (recurrence relation)

As in Solution 1, let p_n be the probability of reaching 4 given that the frog is currently at position n , and observe that for $n = 1, 2, 3$, the frog can either go to position $n + 1$ with probability $\frac{1}{4}$, in which case the conditional probability of it reaching 4 is precisely p_{n+1} by definition; or position $n - 1$, also with probability $\frac{1}{4}$, giving a conditional probability of reaching 4 of p_{n-1} , again by definition; or else it disappears with probability $\frac{1}{2}$, in which case the conditional probability of reaching 4 is obviously 0. Therefore, by the law of total probability, we have the recurrence relation

$$p_n = \frac{1}{4} \cdot p_{n+1} + \frac{1}{4} \cdot p_{n-1} + \frac{1}{2} \cdot 0 \iff 4p_n = p_{n+1} + p_{n-1} \iff p_{n+1} - 4p_n + p_{n-1} = 0, \quad \text{for } n = 1, 2, 3$$

Moreover, one boundary condition is obviously $p_4 = 1$, while the other concerns p_0 : since the frog, starting at position 0, moves to 1 with probability $\frac{1}{2}$, in which case the conditional probability of it reaching 4 will be precisely p_1 , or else disappears, also with probability $\frac{1}{2}$, the law of total probability again gives

$$p_0 = \frac{1}{2} \cdot p_1 + \frac{1}{2} \cdot 0 \iff 2p_0 = p_1.$$

We can now simply use standard methods to solve the above recurrence relation, subject to these two boundary conditions.

Accordingly, the characteristic equation for this recurrence relation is $\lambda^2 - 4\lambda + 1 = 0 \iff (\lambda - 2)^2 = 2^2 - 1 \iff \lambda = 2 \pm \sqrt{3}$, so its general solution is given by

$$p_n = A(2 + \sqrt{3})^n + B(2 - \sqrt{3})^n$$

for some constants A and B . Imposing the boundary conditions now yields

$$2p_0 = p_1 \iff 2(A \cdot 1 + B \cdot 1) = A(2 + \sqrt{3}) + B(2 - \sqrt{3}) \iff A\sqrt{3} - B\sqrt{3} = 0 \iff A = B,$$

and hence

$$p_4 = 1 \iff A(2 + \sqrt{3})^4 + A(2 - \sqrt{3})^4 = 1 \iff A(97 + 56\sqrt{3} + 97 - 56\sqrt{3}) = 1 \iff A = \frac{1}{194}$$

This means the particular solution is precisely

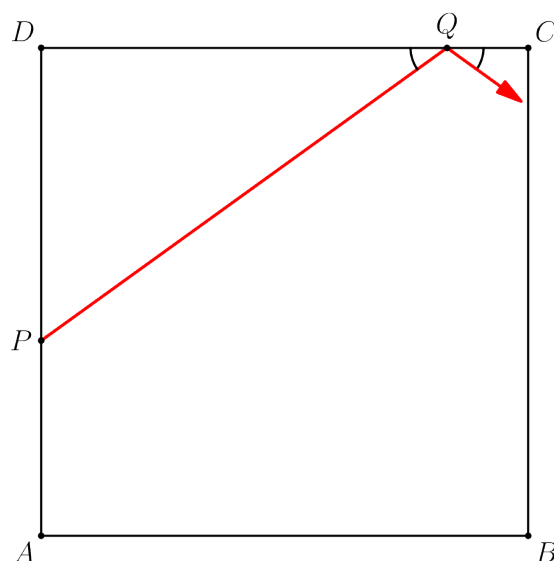
$$p_n = \frac{1}{194} (2 + \sqrt{3})^n + \frac{1}{194} (2 - \sqrt{3})^n = \frac{1}{194} ((2 + \sqrt{3})^n + (2 - \sqrt{3})^n),$$

and so, again as in Solution 1, the required probability is $p_0 = \frac{1}{194} \cdot (1 + 1) =$

(E) $\frac{1}{97}$.

Ebai

- y) (AMC 10 10B 2025) Square $ABCD$ has sides of length 4. Points P and Q lie on $\overline{AD}$ and $\overline{CD}$, respectively, with $AP = \frac{8}{5}$ and $DQ = \frac{10}{3}$. A path begins along the segment from P to Q and continues by reflecting against the sides of $ABCD$ (with congruent incoming and outgoing angles). If the path hits a vertex of the square, it terminates there; otherwise it continues forever. At which vertex does the path terminate?



- (A) A (B) B (C) C (D) D (E) The path continues forever.

Answer: B

Solution 1 (Cheese, draw the path)

For this question, I cheesed it by just drawing the path with a ruler and an compass, and you can find that the path eventually terminates at vertex B by doing so. Therefore, the answer is **(B) B**.

Note - Protractors are not allowed on the AMC Exam, but you can sort of achieve the same thing with a compass

Remark: I actually did this on the test, and I didn't even sketch out the diagram either. I had like a minute left on the clock, and I traced the path of the line lightly on my screen (I was taking digital version) and somehow got it correctly lol. I didn't even use a compass or anything either (verified tracing, works. You don't even have to be very accurate to see that the answer is B-Happymaths)

Solution 2 (Trivial Algebra)

First, let me introduce the reflection trick. Basically, instead of bending the line segment a bunch of times, map any path of the line segment into a reflected square, reflecting the square about the segment that the ray touches. Every time this happens, we map the latest square onto a new one. After we do this many times, the entire path becomes a straight line:

[Illustration link, hosted on imgbb \(I'm too tired to use Asymptote\)](#)

So now, we could just imagine the whole reflection path as a huge straight ray in the coordinate plane. When it lands upon a x and y both divisible by 4, since a corner reflected is still a corner, we can then trace the corner back.

Note that by some trivial arithmetic you get that $PD = 4 - \frac{8}{5} = \frac{12}{5}$, and if we let A be the origin of a coordinate plane, AD be the y -axis and AB be the x -axis, the slope of PQ is $\frac{12}{5} \div \frac{10}{3}$ using rise over run, which is $\frac{18}{25}$. Trivially, the y -intercept of PQ is $\frac{8}{5}$, so the equation of PQ becomes

$$y = \frac{18}{25}x + \frac{8}{5}$$

and you want both x and y to be divisible by 4. Right off the bat, note that the denominator in the linear term is 25, so to have any chance of y being an integer AT ALL, x must be divisible by 5 (else the resulting fraction will have something over 25 plus something over 5, so there's extra $1/25$ s.)

Therefore, x must be a multiple of 20. Then let's try $x = 20$! We get $y = 16$, which is divisible by 4! Yay!!! You might start to sweat about the exact path and how to trace the vertices now, but that does not matter: we can see that if we first reflect to the right and then up, it's no difference than if we went up and then right. So it doesn't matter if we do all the horizontal or vertical reflections all together.

[Illustration link, hosted on imgbb \(I'm too tired to use Asymptote\)](#)

Now you can just label a random square $ABCD$, and reflect to the right 4 times (since you already have 1 square) and then up 3 times. You'll then notice that $(20, 16)$ is just a copy of $\boxed{\text{(B)} B}$.

Upgraded Solution 1 Use the scratch paper's corner and fold it to necessary to mold and replicate angles, then use a straightedge to trace it.

§14 AMC 10 10A 2025

- a) (*AMC 10 10A 2025*) Andy and Betsy both live in Mathville. Andy leaves Mathville on his bicycle at 1 : 30, traveling due north at a steady 8 miles per hour. Betsy leaves on her bicycle from the same point at 2 : 30, traveling due east at a steady 12 miles per hour. At what time will they be exactly the same distance from their common starting point?

(A) 3 : 30 (B) 3 : 45 (C) 4 : 00 (D) 4 : 15 (E) 4 : 30

Answer: E

Solution 1

At 2:30, Andy is 8 miles ahead. For every hour that they both travel, Betsy gains 4 miles on Andy. Therefore, it will take her 2 more hours to be the same distance from the starting point as Andy, which occurs at $\boxed{\text{(E)} 4:30}$.

$$8h = 12(h - 1) \Rightarrow 8h = 12h - 12 \Rightarrow 4h = 12 \Rightarrow h = 3$$

Since Andy started at 1:30, the catch-up time is 4:30. Answer: **(E)**.

Alternatively, from Betsy's perspective: $8(h + 1) = 12h \Rightarrow 8h + 8 = 12h \Rightarrow h = 2$ Same result: **(E) 4:30**.

Solution 2 (Simplest Version)

Andy goes 8 mph, Betsy goes 12 mph.

We know that Betsy starts 1 hour after Andy, so she will be 8 miles behind.

$$12x - 8 = 8x \quad 4x = 8 \rightarrow x = 2$$

2 hours after Betsy's start time is 4:30

-XaberTooth

Solution 3

Try the answer choices:

At 3:30: Andy = 16, Betsy = 12 At 3:45: Andy = 18, Betsy = 15 At 4:00: Andy = 20, Betsy = 18 At 4:15: Andy = 22, Betsy = 21 At 4:30: Andy = 24, Betsy = 24 You can visualize this using a table:

Time	Andy - 8 mph	Betsy - 12 mph
3:30	16	12
3:45	18	15
4:00	20	18
4:15	22	21
4:30	24	24

Thus the correct answer is (E) 4:30.

Solution 4

Since $\text{lcm}(8, 12) = 24$, the time for Andy to reach 24 miles is $24/8 = 3$ hours.

$$1:30 + 3:00 = \text{(E) } 4:30.$$

Note: Alternatively we can find that the time for Betsy to reach 24 miles is $\frac{24}{12} = 2$ hours. So we have

$$2:30 + 2:00 = \text{(E) } 4:30.$$

Solution 5

Andy goes $8x$ miles, Betsy goes $12(x - 1)$ miles. Set equal:

$$8x = 12(x - 1) \Rightarrow x = 3$$

Thus the answer is (E) 4:30.

Solution 6

Using constant rates:

Time	Andy - 8 mph	Betsy - 12 mph
1:30 - Starting	0 miles	0 miles
2:30	8 miles	0 miles
3:30	16 miles	12 miles
4:30	24 miles	24 miles

Thus the answer is (E) 4:30.

Solution 7 (Calculus Overkill)

Let $D_A(t) = 8t$, $D_B(t) = 12(t - 1)$ for $t \geq 1$.

Difference: $F(t) = 12(t - 1) - 8t = 4t - 12$.

Set $F(t) = 0$: $4t - 12 = 0 \Rightarrow t = 3$ hours.

$1:30 + 3 = 4:30$.

-apex304

Solution 8 (Easy to understand, fast)(Detailed)

Let t denote time in hours measured from 1:30 PM:

$t :=$ hours after 1 : 30 PM.

Thus $t = 0$ at 1:30 PM, $t = 1$ at 2:30 PM, and the physically relevant domain for both riders having left is $t \geq 1$.

Velocity/position (vectorized). Choose orthonormal basis $\{i, j\}$ with i east and j north. Define piecewise position vectors $r_A(t), r_B(t) \in \mathbb{R}^2$ (for all $t \geq 0$):

$r_A(t) = 8tj = (0, 8t)$,

$$r_B(t) = \begin{cases} (0, 0), & 0 \leq t < 1, \\ 12(t - 1)i, & t \geq 1, \end{cases} = (12 \max\{t - 1, 0\}, 0).$$

Define scalar distance functions (Euclidean norm / 2-norm):

$D_A(t) := \|r_A(t)\|_2 = \sqrt{0^2 + (8t)^2} = 8t, D_B(t) := \|r_B(t)\|_2 = 12 \max\{t - 1, 0\}$.

We seek t with $D_A(t) = D_B(t)$ and $t \geq 1$. For $t \geq 1$ the equality reduces to the linear equation

$$8t = 12(t - 1).$$

Bring like terms together:

$$8t = 12t - 12 \implies 12 = 12t - 8t.$$

Compute the difference $12t - 8t$ digit-by-digit: $12t - 8t = (12 - 8)t = 4t$. So $12 = 4t$.

Divide 12 by 4 carefully: $12 \div 4 = 3$. Hence

$$t = 3.$$

- b) (AMC 10 10A 2025) A box contains 10 pounds of a nut mix that is 50 percent peanuts, 20 percent cashews, and 30 percent almonds. A second nut mix containing 20 percent peanuts, 40 percent cashews, and 40 percent almonds is added to the box resulting in a new nut mix that is 40 percent peanuts. How many pounds of cashews are now in the box?

(A) 3.5 (B) 4 (C) 4.5 (D) 5 (E) 6

Answer: B

Solution 1

We are given that $0.2(10) = 2$ pounds of cashews in the first box.

Denote the pounds of nuts in the second nut mix as x .

$$5 + 0.2x = 0.4(10 + x)$$

$$5 + 0.2x = 4 + 0.4x$$

$$5 - 4 = 0.4x - 0.2x$$

$$0.2x = 1$$

$$x = 5$$

Thus, we have 5 pounds of the second mix.

$$0.4(5) + 2 = 2 + 2 = \boxed{\text{(B)} 4}$$

Solution 2

Let the number of pounds of nuts in the second nut mixture be x . Therefore, we get the equation $0.5 \cdot 10 + 0.2 \cdot x = 0.4(x + 10)$. Solving it, we then get $x = 5$. Therefore, the amount of cashews in the two bags is $(0.2 \cdot 10) + (0.4 \cdot 5) = 4$, so our answer choice is $\boxed{\text{(B)} 4}$.

Solution 3

The percent of peanuts in the first mix is 10% away from the total percentage of peanuts, and the percent of peanuts in the second mix is 20% away from the total percentage. This means the first mix has twice as many nuts as the second mix, so the second mix has 5 pounds. $0.20 \cdot 10 + 0.40 \cdot 5 = 4$ pounds of cashews. So our answer is, $\boxed{\text{(B)} 4}$

The weight of the second box of nut mix can be calculated by: $(0.2x + 5)/(10 + x) = 0.4$, where the numerator is equal to the weight of the peanuts, the denominator is equal to the total weight of the boxes, and x is equal to the weight of the second box.

Multiplying both sides by $10 + x$, you get $0.2x + 5 = 4 + 0.4x$

Isolating the variable: $0.2x = 1 \Rightarrow x = 5$

Calculating the weight of the cashews: $5(0.4) + 10(0.2) \Rightarrow 2 + 2 \Rightarrow 4$

The answer is $\boxed{\text{(B)} 4}$.

Solution 4

Note that we can set the information given in the problem into a table shown below:

Peanuts	Cashews	Almonds
5	2	3
$\frac{2}{10}x$	$\frac{4}{10}x$	$\frac{4}{10}x$

We are given that the new nut mix will contain 40% peanuts. Hence, $5 + \frac{2}{10}x$ is 40% of the total mix which is $10 + x$. Solving the equation $5 + \frac{2}{10}x = \frac{2}{5} \cdot (10 + x)$ yields $x = 5$. Therefore, the number of cashews in the new mix is equal to $2 + \frac{2}{5} \cdot 5 = \boxed{\text{(B)} 4}$.

Solution 5

Write composition vectors as column vectors (mass fractions multiplied by mass):

Initial mass vector (peanuts, cashews, almonds) for 10 lb:

$$\mathbf{v}_0 = 10 \begin{pmatrix} 0.50 \\ 0.20 \\ 0.30 \end{pmatrix} = \begin{pmatrix} 5 \\ 2 \\ 3 \end{pmatrix}.$$

Added mix per unit mass:

$$\mathbf{u} = \begin{pmatrix} 0.20 \\ 0.40 \\ 0.40 \end{pmatrix}.$$

After adding x lb, total mass vector is

$$\mathbf{v}(x) = \mathbf{v}_0 + x\mathbf{u} = \begin{pmatrix} 5 \\ 2 \\ 3 \end{pmatrix} + x \begin{pmatrix} 0.2 \\ 0.4 \\ 0.4 \end{pmatrix}.$$

Peanut fraction requirement:

$$\frac{\mathbf{e}_1^\top \mathbf{v}(x)}{\mathbf{1}^\top \mathbf{v}(x)} = 0.4,$$

where $\mathbf{e}_1 = (1, 0, 0)^\top$ and $\mathbf{1} = (1, 1, 1)^\top$. This is exactly the same scalar equation,

$$\frac{5 + 0.2x}{10 + x} = 0.4,$$

leading to $x = 5$ and hence

$$\mathbf{v}(5) = \begin{pmatrix} 5 \\ 2 \\ 3 \end{pmatrix} + 5 \begin{pmatrix} 0.2 \\ 0.4 \\ 0.4 \end{pmatrix} = \begin{pmatrix} 6 \\ 4 \\ 5 \end{pmatrix}.$$

Thus cashews = 4 lb.

Solution 6: Very easy algebra and straightforward!

Set up the ratios for the initial and the additional mixtures. The ratio of Peanuts:Cashews:Almonds (denoted as P:C:A) would initially be $5x : 2x : 3x$ for some factor x . Then for the second mixture it would be $2x : 4x : 4x$. If we add these we get $7x : 6x : 7x$. Now we know that the percentage of peanuts is 40%. But we also know that the number of pounds of peanuts and almonds is the same. $40\% + 40\% = 80\%$ and so the number of cashews makes up 20% of the mixture. If you divide the factor x by the whole ratio you get $7 : 6 : 7$. These values sum to 20 and 20% of 20 is 4.

Video Solutions

- c) (*AMC 10 10A 2025*) How many isosceles triangles are there with positive area whose side lengths are all positive integers and whose longest side has length 2025?
- (A) 2025 (B) 2026 (C) 3012 (D) 3037 (E) 4050

Answer: D

Solution 1: Casework

You can split the problem into two cases: Case 1: The two sides with equal length are both smaller than 2025, which means that they range from 1013 to

2024. There are 1012 such cases. Case 2: There are two sides of length 2025, so the last side must be in the range 1 to 2025. There are 2025 such cases. Keep in mind, an equilateral triangle also counts as an isosceles triangle, since it has **at least** 2 sides of equal length. Therefore, the total number of cases is (equilateral + isosceles) $1012 + 2025 = \boxed{(D) 3037}$ cw, minor edit by sd

Side note not by the author: If you're unsure whether the equilateral triangle is a valid case, notice that the other answers are much farther away, as compared to only a one-off error from 3037, which would make 3037 still the best answer.

Solution 2 (broken down)

Suppose an isosceles triangle has sides A, A, B where A is the longest side and A, B are positive integers.

By the triangle inequalities:

$$A + A > B$$

$$A + B > A \implies B > 0$$

Since $A = 2025$ (the longest side):

$$2025 + 2025 > B \implies B < 4050$$

But also, since A is the longest side, $B < 2025$. Therefore,

$$B = 1, 2, 3, \dots, 2024,$$

giving 2024 triangles.

Now consider triangles with side lengths $B, B, 2025$, where $B < 2025$.

Triangle inequality gives:

$$B + B > 2025 \implies B > 1012.5$$

Since B is an integer and less than 2025, the possible values are

$$B = 1013, 1014, 1015, \dots, 2024,$$

which gives 1012 triangles.

Finally, don't forget the case where all sides are 2025 (equilateral, which also counts as isosceles by definition). So add 1 more triangle.

In total,

$$2024 + 1012 + 1 = \boxed{3037}$$

triangles.

Solution 3 (more broken down)

To solve this question, we first must observe the following restrictions given by the problem - 1.) Be an isosceles triangle 2.) Positive integer side lengths (causing a positive area) 3.) Longest length is 2025

To satisfy restrictions 1 and 2, we can denote the side lengths of a triangle that satisfies these restrictions as A, A, B , where A and B are positive integers. This is possible as the triangle is restricted to be isosceles, meaning at least 2 equal length sides.

In any triangle with side lengths a, b , and c , the following inequalities (Triangle Inequality Theorem) must hold:

$$a + b > c$$

$$a + c > b$$

$$b + c > a$$

Therefore, our satisfactory triangles must satisfy

$$A + A > B$$

$$A + B > A.$$

where $A > 0, B > 0$.

Now we can focus on restriction 3. Let us consider that A is equal to 2025 (A is the longest side length).

Substituting for A we can rewrite our inequalities as follows

$$2025 + 2025 > B$$

$$2025 + B > 2025$$

where $B > 0$.

Furthermore, we can solve this system of linear inequalities like this

$$4050 > B$$

$$B > 0$$

$$0 < B < 4050$$

However, for A to be the longest side length, B must be less than 2025. Therefore, the solutions are

$$\text{integers in } 0 < B < 2025 \implies 1, 2, 3, 4, 5, \dots, 2024$$

This creates 2024 cases.

—

Now let's consider that B is the longest side length and it is equal to 2025. Substituting for B , we can rewrite our inequalities as follows

$$A + A > 2025$$

$$A + 2025 > A$$

where $A > 0$.

Furthermore, we can solve this system of linear inequalities like this

$$2A > 2025$$

$$0 < 2025$$

(which can be ignored as it is always true)

$$A > 1012.5$$

However, for B to be the longest side length, A must be less than 2025. Therefore, the solutions are

$$\text{integers in } 1012.5 < A < 2025 \implies 1013, 1014, 1015, 1016, \dots, 2024$$

This creates $2024 - 1012 = 1012$ cases.

—

But! here is also a case of an isosceles triangle where all side lengths of the triangle are equal in length and congruent - this special isosceles is referred to as an equilateral triangle. We can denote the lengths of such a triangle as A, A, A . We need to assign the longest length to A , as it is the only place a length can be applied. So, $A = 2025$. We do not need to worry about the Triangle Inequality Theorem as any equilateral shape always satisfies it. For example,

$$A + A > A, \quad A + A > A, \quad A + A > A$$

where a, b, c of the Triangle Inequality Theorem are all equal to A and $A > 0$ is always satisfied for any side length, assuming that A is a positive integer.

This creates 1 case.

—

Adding together our cases,

$$2024 + 1012 + 1 = \boxed{(D) 3037}.$$

This is a process called casework.

Solution 4: Casework

There are two cases. The first case is where the other two side lengths are the congruent side lengths and less than 2025, and the second one is where 2025 is one of the congruent side lengths.

Case 1: We let a be the other side length in the triangle. By the triangle inequality, we have that

$$2a > 2025 \text{ and } a < 2025,$$

so we get that $1012.5 < a < 2025$, and since a is an integer, a can be 1013, 2014, ..., 2024, so there are 1012 such cases.

Case 2: We let b be the other side in the triangle, since 2025 is one of the congruent sides. Since 2025 is the largest side, $a < 2025$, so there are 2024 solutions in this case.

However, there is also a case where the triangle is equilateral so we add 1.

So, the total number of triangles that are isosceles with largest side 2025 is $1012 + 2024 + 1 = \boxed{3037}$

- d) (AMC 10 10A 2025) A team of students is going to compete against a team of teachers in a trivia contest. The total number of students and teachers is 15. Ash, a cousin of one of the students, wants to join the contest. If Ash plays with the students, the average age on that team will increase from 12 to 14. If Ash plays with the teachers, the average age on that team will decrease from 55 to 52. How old is Ash?

(A) 28 (B) 29 (C) 30 (D) 32 (E) 33

Answer: A

Solution 1

When Ash joins a team, the team's average is pulled towards his age. Let A be Ash's age and N be the number of people on the student team. This means that there are $15 - N$ people in the teacher team. Let us write an expression for the change in the average for each team.

The students originally had an average of 12, which became 14 when Ash joined, so there was an increase of 2. The term $A - 12$ represents how much older Ash is compared to the average of the students'. If we divide this by $N + 1$, which is the number of people on the student team when Ash joins, we get the average change per team member once Ash is added. Therefore,

$$\frac{A - 12}{N + 1} = 2.$$

Similarly, for teachers, the average was originally 55, which decreased by 3 to become 52 when Ash joined. Intuitively, $55 - A$ represents how much younger Ash is than the average age of the teachers. Dividing this by the expression $(15 - N) + 1$, which is the new total number of people on the teacher team, represents the average change per team member once Ash joins. We can write the equation

$$\frac{55 - A}{16 - N} = 3.$$

To solve the system, multiply equation (1) by $N + 1$, and similarly multiply equation (2) by $16 - N$. Then add the equations together, canceling A , leaving equation $43 = 50 - N$. From this we get $N = 7$ and $A = \boxed{28}$.

Solution 2

As shown above, we get the equation

$$\frac{55 - A}{16 - N} = 3.$$

Rearranging we get

$$\frac{55 - A}{3} = 16 - N.$$

Therefore, since N must be an integer, $55 - A$ must be divisible by 3. Plugging in values gets us $A = \boxed{28}$.

Solution 3

Another way is to say that there are S students and T teachers and Ash's age is A . We can create the equations

$$(1) S + T = 15 \tag{28}$$

$$(2) \frac{12S + A}{S + 1} = 14 \tag{29}$$

$$(3) \frac{55T + A}{T + 1} = 52 \tag{30}$$

Equation (2) simplifies to $A = 2S + 14$ and equation (3) simplifies to $A = -3T + 52$. Multiply equation (2) by 3 to get equation

$$(4) 3A = 6S + 42 \tag{31}$$

and equation (3) by -2 to get

$$(5) -2A = 6T - 104 \tag{32}$$

Add (4) and (5) to get $A = 6(S + T) - 62$. After substituting equation (1) and simplifying, you get $A = \boxed{28}$ or answer choice $\boxed{A}$.

e) (AMC 10 10A 2025) Consider the sequence of positive integers

1, 2, 1, 2, 3, 2, 1, 2, 3, 4, 3, 2, 1, 2, 3, 4, 5, 4, 3, 2, 1, 2, 3, 4, 5, 6, 5, 4, 3, 2, 1, 2, ...

What is the 2025th term in this sequence?

- (A) 5 (B) 15 (C) 16 (D) 44 (E) 45

Answer: E

Solution 1 (nice equation)

Group the numbers by their hill pattern: (12)(1232)(123432)(12345432)...

The maximums of each hill occur at terms $n = 2, 5, 10, 17, \dots$. These terms correspond to maximums of 2, 3, 4, ... Let a be the maximum at term N . Since the sum of the first x odd numbers is x^2 we have $1 + (a - 1)^2 = N$. So for example, if $a = 4$, then $N = 10$, telling us that the peak of the hill with maximum 4 occurs at the 10th term.

Now, we know $2025 = 45^2$, so let $a = 46$. Then $N = 2026$, so the 2026th term is 46. Then the 2025th term must be (E) 45.

Solution 2

Note that the first 1 is the first number from the left, the second 1 is the third number from the left, and the third 1 is the seventh number from the left. Since the second differences between the indices of the n^{th} 1's are equal, there is a quadratic function for the position of the n^{th} 1. As the second differences are 2, the leading coefficient of this function is 1. Let the function be $f(x) = x^2 + ax + b$. Then, we have $1 + a + b = 1$, and $4 + 2a + b = 3$. Solving, we get $a = -1$ and $b = 1$. Therefore, the n th 1 from the left is the $(n^2 - n + 1)$ th number from the left. Plugging in $n = 45$ gives $f(n) = 1981$, so the 1981st number is a 1.

Also, the sequence shows that the first instance of a positive integer k is always to the left of the k^{th} 1. This means that following the 1981st number, the numbers go to or above 45. Adding $2025 - 1981$ to 1 means that the 2025th term of the sequence is (E) 45.

Solution 3 (Quick)

Note that every n^2 th term is n . For example, the 3^2 th term is 3, the 4^2 th term is 4, the 5^2 th term is 5, and so on. We also notice that $45^2 = 2025$, so the 2025th term is 45, which is answer choice (E)45.

Solution 4 (Super Simple!!)

Upon inspecting the problem, we see that the 1^{st} term is 1, and so are the 3^{rd} , 7^{th} , 13^{th} , 21^{th} , 31^{st} , and so on. We see that the pattern comes from the differences: +2, +4, +6, +8, +10, and so on. From this we see that a 1 appears at each position that is a number of the form $n^2 - n + 1$. We know that $2025 = 45 \cdot 45$. Plugging 45 as n gives 1981, meaning that the 2025th term is $2025 - 1981 + 1 = 45$. Therefore, the solution is (E)45.

Solution 5

Takes less than 30 seconds!

Immediately by looking at the sequence, we observe that the digit in the place of each perfect square is the square root of that number. For example, the first term in the sequence is 1 (because $\sqrt{1} = 1$), the 4^{th} term is 2 (because $\sqrt{4} = 2$), the 9^{th} term is 3 (because $\sqrt{9} = 3$), the 16^{th} term is 4 (because $\sqrt{16} = 4$), the 25^{th} term is 5 (because $\sqrt{25} = 5$), and so on. And since 2025 is a perfect square, it makes it easy for us because $\sqrt{2025} = 45$, which makes the answer choice (E) 45.

To make it easier to understand my way of solving it, here is a table.

Let n = the position of the said number in the sequence

Let $\sqrt{n}$ = the number located in that position itself

n	$\sqrt{n}$
1	1
4	2
9	3
16	4
25	5
36	6
...	...
2025	(E) 45

Note: I am just saying that the square root of the position (if the position is a perfect square) is the number that appears in the said position in the above sequence. I am not saying that the root itself is the first time it appears in the said sequence. To elaborate, the number 6 appears on the 26^{th} number, as *well* as the 36^{th} number as expected (because $\sqrt{36} = 6$).

Solution 6 - Pattern Recognition

After looking at the problem, it is evident that a pattern exists for the first occurrence of a number in the sequence. Knowing this, we can write the places that each number appears at:

$$1 : 1 \quad (33)$$

$$2 : 2 \quad (34)$$

$$3 : 5 \quad (35)$$

$$4 : 10 \quad (36)$$

$$5 : 17 \quad (37)$$

If we write down the difference between the each numbers position in sequence, we get 1, 3, 5, and 7. We notice how these numbers are similar to how the square numbers increase (square numbers: 1, 4, 9, 16...). We also notice how the position of the first instance of each number n is just $(n - 1)^2 + 1$. Since we're trying to find the 2025^{th} number and $2025 = 45^2$, we can plug in 45 for $n - 1$. As a result, we get that the first instance of the number 46 appears at $(45)^2 + 1$ or 2026. Knowing that the number before the first instance of a number n is $n - 1$, we get the answer choice (E) 45.

Solution 7: Patterns

We list the positions of the 1s in the sequence and get that they appear at 1, 3, 7, 13, ... We see that, regarding the n^{th} "1" in the sequence, the location of the said "1" is at the $n(n + 1) + 1^{th}$ position. So, we can find that the 1 closest to 2025 happens where $n = 44$, due to our knowledge that $2025 = 45^2$. Since $44(45) + 1 = 1981$, we can start counting from 1981, getting that the 2025^{th} term in the sequence is (E) 45.

Solution 8: Using Arithmetic Sequences

We can notice that the sequence 1, 2, 1, 2, 3, 2, 1, 2, 3, 4, 3, 2, 1, 2, 3, 4, 5, 4, 3, 2, 1, 2, 3, 4, 5, 6, 5, 4, 3, 2, have a special sequence embedded it in. Notice 1, 2. 1, 2 have 2 terms, and the next, 1, 2, 3, 2 have 4 terms. This does not include the 1 which ends it,

which also starts the next one, having a contradiction, so we can construct the following equation, subtracting 1 on the end because the sequence doesn't start with 1, 1, 2, ...:

$$2\left(\frac{n(n+1)}{2}\right) - 1 = 2025$$

$$n(n+1) - 1 = 2025$$

Now, we can start to solve the equation:

$$n(n+1) = 2026$$

$$n^2 + n = 2026$$

Now let us do casework. The closest number to 2026 that is a square is 2025, being 45^2 . However, we add the number again, which will result in $2025 + 45 = 2070$, over 2026. We will then do 44^2 , which is 1936, and adding 44 to reach the almost end (which is specifically 1, 2, 3, 4, 5, ..., 44, 45, 44, 43, 42, 41, 40, ..., 5, 4, 3, 2) results in 1980. We add 1 to make it 1981, and using the formula $b - a + 1$, it is $2025 - 1981 + 1$, which equals (E) 45.

- f) (AMC 10 10A 2025) In an equilateral triangle each interior angle is trisected by a pair of rays. The intersection of the interiors of the middle 20° -angle at each vertex is the interior of a convex hexagon. What is the degree measure of the smallest angle of this hexagon?

(A) 80 (B) 90 (C) 100 (D) 110 (E) 120

Answer: C

- g) (AMC 10 10A 2025) Suppose a and b are real numbers. When the polynomial $x^3 + x^2 + ax + b$ is divided by $x - 1$, the remainder is 4. When the polynomial is divided by $x - 2$, the remainder is 6. What is $b - a$?

(A) 14 (B) 15 (C) 16 (D) 17 (E) 18

Answer: E

Solution 1

Use synthetic division to find that the remainder of $x^3 + x^2 + ax + b$ is $a + b + 2$ when divided by $x - 1$ and $2a + b + 12$ when divided by $x - 2$. Now, we solve

$$\begin{cases} a + b + 2 = 4 \\ 2a + b + 12 = 6 \end{cases}$$

This ends up being $a = -8$, $b = 10$, so $b - a = 10 - (-8) =$ (E) 18

Solution 2

Via the remainder theorem, we can plug 1 in for the factor $x - 1$ and get 4, so we have that

$$\begin{aligned} 1^3 + 1^2 + 1a + b &= 4 \\ a + b &= 2. \end{aligned}$$

Then, we plug in 2 and get a remainder of 6, so we have that

$$2^3 + 2^2 + 2a + b = 6$$

$$8 + 4 + 2a + b = 6$$

$$2a + b = -6$$

Then, we solve the system of equations. By substitution, we obtain

$$b = 2 - a$$

$$2a + 2 - a = -6$$

$$a = -8$$

$$b = 2 - (-8) = 2 + 8 = 10.$$

Thus, we have that $b - a = 10 - (-8) = 10 + 8 = \boxed{\text{(E)} 18}$

Solution 3 (Small)

Using Remainder theorem, we get that: $4 = (1)^3 + (1)^2 + (1)a + b = 1 + 1 + a + b$ so we get $2 = a + b$ (i) and from the second statement, $6 = (2)^3 + (2)^2 + (2)a + b = 8 + 4 + 2a + b$ so we get $-6 = 2a + b$ (ii).

So we Subtract (ii) from (i) to get $a = -8$. Substitute in to get $b = 10$.

So $b - a = 10 - (-8) = \boxed{\text{(E)} 18}$

Solution 4 (Clear Concept)

When the polynomial $f(x) = x^3 + x^2 + ax + b$ is divided by $x - 1$, the remainder is

$$f(1) = 1 + 1 + a + b = 4, \quad a + b = 2$$

When the polynomial is divided by $x - 2$, the remainder is 6

$$f(2) = 8 + 4 + 2a + b = 6, \quad 2a + b = -6$$

Solving, $a + b = 2$ and $2a + b = -6$ we get $a = -8$, $b = 10$, hence, $b - a = 18$

$\boxed{\text{(E)} 18}$

- h) (AMC 10 10A 2025) Agnes writes the following four statements on a blank piece of paper.

- At least one of these statements is true.
- At least two of these statements are true.
- At least two of these statements are false.
- At least one of these statements is false.

Each statement is either true or false. How many false statements did Agnes write on the paper?

(A) 0 (B) 1 (C) 2 (D) 3 (E) 4

Answer: B

Solution 1

We first number all the statements:

1) At least one of these statements is true. 2) At least two of these statements are true. 3) At least two of these statements are false. 4) At least one of these statements is false.

We can immediately see that statement 4 must be true, as it would contradict itself if it were false. Similarly, statement 1 must be true, as all the other statements must be false if it were false, which is contradictory because statement 4 is true. Since both 1 and 4 are true, statement 2 has to be true. Therefore, statement 3 is the only false statement, making the answer (B) 1.

-Rainjs

Solution 2

Statements *I*, *II*, and *IV* are true, while statement *III* is false. Hence, there are 3 true statements and (B) 1 false statement. This result can be checked by examining the statements individually again.

Statements *I* and *II* will be true because there are $3 \geq 2$ true statements. Statement *IV* is also true because there is $1 \geq 1$ false statement. Finally, statement *III* is false because there are $1 \not\geq 2$ false statements.

Note: This solution does not give a path to the solution, but rather only how to check the answer after the answer has been identified.

Solution 3

Let's say there are T true statements. We know that T can be any integer from 0 to 4. Numbering the 4 given statements in order as S_1 to S_4 , we can represent S_1 as $T \geq 1$, S_2 as $T \geq 2$, and since there are $(4 - T)$ false statements, S_3 as $4 - T \geq 2 \iff T \leq 2$, and S_4 as $4 - T \geq 1 \iff T \leq 3$.

Accordingly, we now consider the possible cases for the value of T . If $T = 0$, then the true statements are S_3 and S_4 , i.e. there are 2 true statements in total. But that means $T = 2 \neq 0$, giving a contradiction.

Similarly, if $T = 1$, then the true statements are S_1 , S_3 , and S_4 , i.e. there are 3 true statements in total. This means $T = 3 \neq 1$, again giving a contradiction.

If $T = 2$, then all 4 statements are true, yielding the contradiction $T = 4 \neq 2$.

Finally, however, if $T = 3$, then the true statements are S_1 , S_2 , and S_4 . Hence there are again 3 true statements in total, but now as $T = 3$ in the first place, this is actually consistent and not a contradiction.

For completeness, we also observe that if $T = 4$, then the true statements are only S_1 and S_2 , i.e. $T = 2 \neq 4$, and so this case again leads to a contradiction.

Thus we conclude that the only possible value is $T = 3$, which gives a consistent solution when S_1 , S_2 , and S_4 are true and S_3 is false (as noted above). It follows that the number of false statements is (B) 1, namely precisely S_3 (and indeed, $T = 3$ gives $4 - T = 4 - 3 = 1$, as expected).

Solution 4

Suppose Statement *I* is false, then none of the Statements are true, which contradicts the fact that a false Statement *III* or *IV* is telling the truth. Therefore, Statement *I* is true and assume Statement *II* is false.

Statement *II* thus implies that only Statement *I* was the truth, and the rest, false. But then, there are 3 false statements but then Statement *III* and Statement *IV* are telling the truth. So Statement *II* is also true.

Now, if Statement *III* is true, then both Statement *III* and Statement *IV* is false, contradicting the fact that it is true. Statement *III* is hence false and Statement *IV* tells the truth since Statement *III* lied so indeed, there are at least one lie. There are a total of 3 truths and 1 lie, making the answer

(B) 1. hxve

Solution 5 (Quick)

I tried setting that the statements regarding the true ones (I and II) were true and regarding the false ones (III and IV) were false, and it clearly doesn't work. It doesn't work because statement III was true, not false as I set it to be originally.

What if we make that statement (Statement III) true too? Well, then we have 3 T and 1 F statements, so statement IV is false, statement III is true, and statements I and II are true too.

So there is (B) 1 false statement.

Solution 6 (Plug in Answer Choices)

For this question, what I did was to simply test each of the answer choices.

Let's start with (A) 0 false statements:

We can check the answer of 0 false statements against each of the statements.

Statement I: "At least one of these statements is true."

This statement is false, which already invalidates the answer of (A) 0 false statements, as we already have a false statement.

Let's move onto (B) 1 false statement:

We can check once again this answer against each of the statements.

Statement I: "At least one of these statements is true."

This statement is true, as when we have 1 false statement, we have 3 true statements, and therefore this is true.

Statement II: "At least two of these statements are true."

Once again, this statement is true, as when we have 1 false statement, we have 3 true statements, and thus this is also true.

Statement III: "At least two of these statements are false."

This statement is false, as we have one false statement, not 2 and above.

Statement IV: "At least one of these statements is false."

This statement is true, as we have one false statement.

So, after checking the answer choice of (B)1 false statement across the 4 statements, we get three true statements and one false statement, making this statement the correct answer. I will not dive into the other 3 answer choices, as we have already gotten the answer for this question, and if you were doing this way of solving on the test, you would most likely not test the other 3 answers either for the sake of saving time.

Solution 7 (Step by Step, Simple)

Start by going statement by statement and checking if they can be false. If statement 1 were false statement 4 would be true leading to a contradiction so statement 1 is true. If statement 2 were false 4 would be true and one would be true making statement 2 true. If statement 3 was false and we know statement 1 and statement 2 were true and 4 can be true and it works. Therefore, the correct answer is (B) 1.

–TFEA(would also appreciate if somebody could please latex my proof).–
Iceboy1221(i wrote it using latex)

Solution 8 (for Chinese Committee Paper)

Due to reasons to prevent cheating, there was a variant for test takers in China, which involves slight iterations in the questions and answer choices. For this particular problem, the question asks for the number of **true** statements that Agnes wrote.

Therefore, utilizing any of the methods in the solutions above, we find that there is one false statement, therefore there are three true statements.

Therefore, the correct answer choice is **3**.

- i) (AMC 10 10A 2025) Let $f(x) = 100x^3 - 300x^2 + 200x$. For how many real numbers a does the graph of $y = f(x - a)$ pass through the point $(1, 25)$?
(A) 1 (B) 2 (C) 3 (D) 4 (E) more than 4

Answer: C

Solution 1

Substitute $1 - a$ for x and set this expression equal to 25. The problem boils down to finding how many real roots

$$100(1 - a)^3 - 300(1 - a)^2 + 200(1 - a) = 25$$

has. We further simplify this expression and create a function $f(x)$:

$$f(x) = -100a^3 + 100a - 25$$

Using Descarte's Rule of Signs we get:

Sign changes for $f(x)$ (possible number of positive roots): 2

$$f(-x) = +100a^3 - 100a - 25$$

Sign changes for $f(-x)$ (possible number of negative roots): 1

Possibilities for roots:

- 1) 2 positive roots, 1 negative root
- 2) 0 positive roots, 1 negative root, 2 imaginary roots

So which one is it? We know if the function changes sign between an interval, then a root exists in that interval (Intermediate Value Theorem). From $a = 0$ to $\frac{1}{2}$, the function changes sign because $f(0) = -25$ while $f(\frac{1}{2}) = +\frac{25}{2}$, so a positive root exists. This eliminates the second possibility, implying that there must be 2 positive and 1 negative roots. So the answer is $2 + 1 =$ (C) 3.

Alternative Solution Since we are trying to find the number of roots of $-100a^3 + 100a - 25$ (which is the same as the number of roots of $4a^3 - 4a + 1$ because we divided by a constant, namely -25), we can sketch a graph of this function, which we will call $g(x)$. We find that the function is strictly decreasing when $a < -1$, $g(0) = g(1) = 1$, and the function is strictly increasing when $a > 1$. We immediately see that since $g(-1)$ is negative, there is a root between -1 and 0 by the Intermediate Value Theorem. From here, we use the first derivative test as follows. Taking the derivative, we have $g'(x) = 12a^2 - 4$. We solve for the roots of this equation, obtaining $\pm\frac{\sqrt{3}}{3}$. We know that at these two values, the function will either reach a local minimum or maximum if the second derivative isn't 0. Proceeding by the second derivative rule, we have that $g''(x) = 24a$ is positive at $a = \frac{\sqrt{3}}{3}$ and negative at $a = -\frac{\sqrt{3}}{3}$, making them local minimums and local maximums, respectively. We don't need to care about the latter case because the local maximum doesn't give any more roots besides the one we've already found. Considering the first case when $a = \frac{\sqrt{3}}{3}$, we must find the sign of $g(a)$. Using some approximations, we have that

$$\begin{aligned} & g\left(\frac{\sqrt{3}}{3}\right) \\ &= \frac{4\sqrt{3}}{9} - \frac{4\sqrt{3}}{3} + 1 \\ &\approx \frac{6.9}{9} - \frac{6.9}{3} + 1 \\ &= \frac{23}{30} - \frac{69}{30} + 1 \\ &= -\frac{16}{30} \\ &< 0. \end{aligned}$$

By the Intermediate Value Theorem again, we see that there is one root between 0 and $\frac{\sqrt{3}}{3}$ and one root between $\frac{\sqrt{3}}{3}$ and 1 . Thus, we have a total of $1 + 2 = 3$ roots. Therefore, the answer is **(C) 3**

Solution 2

This problem is essentially asking how many values of x satisfy $f(x) = 25$, since the graph $f(x - a)$ is a shift a units right from the original graph of $f(x)$. Now we just need to determine the amount of real solutions to $100x^3 - 300x^2 + 200x = 25$.

Dividing all terms by 25, we get $4x^3 - 12x^2 + 8x - 1 = 0$. Using the rational root theorem and testing, we find no rational roots. Our next method to determine the amount of roots is graphing. Notice how when $x \rightarrow +\infty$, $f(x) \rightarrow +\infty$. When $x \rightarrow -\infty$, $f(x) \rightarrow -\infty$. Substituting $x = 0$ gives us $f(x) = -1$. Substituting $x = 1$ gives us $f(x) = -1$ also. This makes us think to see if there was any sudden bump or curve in the graph, so we test the midpoint which is $1/2$. Substituting $x = 1/2$, we get $1/2 - 3 + 4 - 1 = 1/2$. Aha! Connecting the dots, we see there are two solutions from $0 \leq x \leq 1$. Also, $f(x)$ must continue increasing after $x > 1$, which tells us there is a third

root when $x > 1$. Now we can conclude that the problem has 3 real roots, meaning that 3 values of a satisfy the problem. The answer is $\boxed{(C)3}$.

Solution 3 (Graphing)

We can factor $f(x)$ as $100x(x-1)(x-2)$. We just need to check that if the local maximum between 0 and 1 is below or above 25. Observe that $f(1/2) > 25$, so the graph will cross the line $y = 25$ 3 times, $\boxed{C}$.

Somewhat cheese (not recommended) Instead of plugging in a value between 0 and 1, we can assume that the equation will reach a value above 25 between those numbers. Thus, the graph crosses $y = 25$ 3 times, $\boxed{C}$.

Super Lazy Solution Instead of using numbers, we could just remember the shape of the graph given, and that it is extremely vertically dilated. The shape of the graph is (for the y -values given) basically 3 lines, so moving these 3 lines left and right, we can see that the graph will intersect at 3 points, so $\boxed{C}$.

Solution 4 (Discriminant)

We start by writing $f(x-a)$

$$f(x-a) = 100(x-a)^3 - 300(x-a)^2 + 200(x-a)$$

Then, we substitute 25 for $f(x-a)$, and substitute $x = 1$

$$25 = 100(1-a)^3 - 300(1-a)^2 + 200(1-a)$$

Now, to make this easier, we define $p = 1-a$, and find how many real values of p satisfy this

$$25 = 100p^3 - 300p^2 + 200p$$

$$25 = 100(p^3 - 3p^2 + 2p)$$

$$0.25 = p^3 - 3p^2 + 2p$$

$$0 = p^3 - 3p^2 + 2p - 0.25$$

We use the discriminant of a cubic equation, where, assuming the form $ax^3 + bx^2 + cx + d$,

$$\Delta = b^2c^2 - 4ac^3 - 4b^3d - 27a^2d^2 + 18abcd$$

When $\Delta > 0$, there are 3 distinct real solutions, when $\Delta = 0$, there are repeated solutions (which all belong to $\mathbb{R}$), and when $\Delta < 0$, there is one real solution and 2 complex-conjugate solutions.

From the expression we derived, we can see that $a = 1$, $b = -3$, $c = 2$, and $d = -0.25$

We plug this into the discriminant formula, yielding

$$\Delta = (-3)^2(2)^2 - 4(2)^3 - 4(-3)^3(-0.25) - 27(-0.25)^2 + 18(-3)(2)(-0.25)$$

$$\Delta = 36 - 32 - 27 - 1.6875 + 27$$

$$\Delta = 4 - 1.6875$$

$$\Delta = 2.3125$$

Therefore, $\Delta > 0$, meaning there are three real values of p that satisfy the equation. Since $p = 1 - a$, we know that there also are 3 real values of a satisfying the equation. Thus,

$$\Delta > 0 \implies 3 \implies \boxed{C}$$

shockfront99

Solution 5(Use only if you have like 2 seconds left)

Ok, so first notice the polynomial is $f(x - a) = 100(x - a)^3 - 300(x - a)^2 + 200(x - a)$. Observe that this is a cubic polynomial and so it has degree 3 and therefore it has maximum 3 solutions. Now choose 3 and move on. :)

Note The above solution doesn't always work as in this problem asked for the number of values for a

Solution 6 (the most obvious but uses calculus)

We substitute $x - a$ into the polynomial to obtain

$$\begin{aligned} f(x - a) &= 100(x - a)^3 - 300(x - a)^2 + 200(x - a) \\ &= 100(x - a)[(x - a)^2 - 3(x - a) + 2] \\ &= 100(x - a)(x - a - 1)(x - a - 2). \end{aligned}$$

We want to find how many values of a exist such that this function goes through $(1, 25)$. So we can write

$$f(1) = 100(1 - a)(-1 - a)(-a) = 25.$$

Expanding gives

$$100(1 - a)(-1 - a)(-a) = 25 \implies 100(-a^2 + 1)(a) = 25 \implies -a^3 + a = \frac{1}{4} \implies a^3 - a + \frac{1}{4} = 0.$$

So the amount of a such that the function goes through $(1, 25)$ is the amount of roots this equation has. From here, we can either try to find all the roots, which I will not try here. Or we can use calculus to find the amount of roots this has. We start by finding the local minimums and maximums, by taking the derivative of this function

$$\frac{d}{da}(a^3 - a + \frac{1}{4}) \implies 3a^2 - 1.$$

Setting the derivative to zero we have

$$3a^2 - 1 = 0 \implies 3a^2 = 1 \implies a = \pm \frac{1}{\sqrt{3}}.$$

Therefore, the y coordinates of the critical points are

$$f\left(\frac{1}{\sqrt{3}}\right) = \left(\frac{1}{\sqrt{3}}\right)^3 - \left(\frac{1}{\sqrt{3}}\right) + \frac{1}{4} = \frac{1}{3\sqrt{3}} - \frac{1}{\sqrt{3}} + \frac{1}{4} = \frac{1 - 3}{3\sqrt{3}} + \frac{1}{4} = -\frac{2}{3\sqrt{3}} + \frac{1}{4}$$

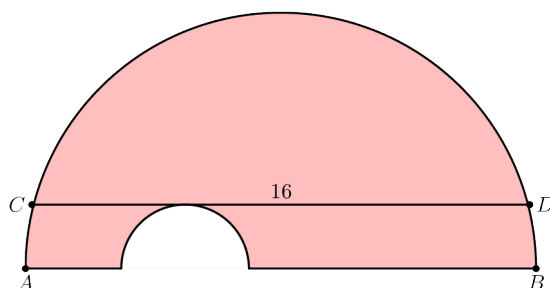
and

$$f\left(-\frac{1}{\sqrt{3}}\right) = \left(-\frac{1}{\sqrt{3}}\right)^3 - \left(-\frac{1}{\sqrt{3}}\right) + \frac{1}{4} = -\frac{1}{3\sqrt{3}} + \frac{1}{\sqrt{3}} + \frac{1}{4} = \frac{-1 + 3}{3\sqrt{3}} + \frac{1}{4} = \frac{2}{3\sqrt{3}} + \frac{1}{4}.$$

The first critical point has a negative y because $\frac{2}{3\sqrt{3}} > \frac{1}{4}$ since $\frac{3\sqrt{3}}{2} < 4$ (which are the reciprocals). The second one clearly has a positive y coordinate. The local minimum is negative and the local maximum is positive implies the function crosses the x axis three times implying there are three roots. So there are **(C) 3** values of a such that the function passes through $(1, 25)$.

Remark: If you aren't sure about the positive negative step, then you can verify using quick approximation.

- j) (AMC 10 10A 2025) A semicircle has diameter $\overline{AB}$ and chord $\overline{CD}$ of length 16 parallel to $\overline{AB}$. A smaller semicircle with diameter on $\overline{AB}$ and tangent to $\overline{CD}$ is cut from the larger semicircle, as shown below.



What is the area of the resulting figure, shown shaded?

- (A) 16π (B) 24π (C) 32π (D) 48π (E) 64π

Answer: C

Solution 1 (Somewhat Cheeky)(But Very Effective)

Notice that the size of the smaller semicircle is not specified, and there is no additional information that hints at any specific size for it. Hence, we can shrink the small semicircle until its area is arbitrarily small and negligible, leaving us with a semicircle with a diameter of 16. The area of the semicircle is given by $\frac{\pi r^2}{2}$, so we have $r = \frac{16}{2} = 8 \Rightarrow A = \frac{\pi(8)^2}{2} = \text{(C) } 32\pi$

Solution 2 (Generalized, no assumptions)

Let the radius of the larger semicircle be R and that of the smaller one be r . We are looking for $\frac{1}{2}\pi(R^2 - r^2)$. If we connect the center of the large semicircle to one endpoint of the chord and to the center of the chord, we get a right triangle with legs 8 and r and hypotenuse R . Hence, $R^2 = r^2 + 8^2 \Rightarrow R^2 - r^2 = 64 \Rightarrow \frac{1}{2}\pi(R^2 - r^2) = \text{(C) } 32\pi$.

Solution 3

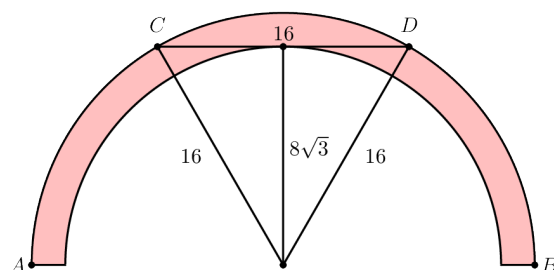
The problem doesn't restrict where the smaller semicircle is along the larger semicircle's diameter. Therefore, we can assume that the two semicircles are concentric. Let the center of both semicircles be O , and let CD be tangent to the smaller semicircle at T . Let the radius of the smaller semicircle be x , and let the radius of the larger semicircle be r . If we mirror the diagram over AB , we can see that we have two concentric circles. We are trying to find $\pi(\frac{r^2 - x^2}{2})$. By Power of a Point on T , we can see that

$$64 = (r + x)(r - x) = r^2 - x^2.$$

Thus, $\pi\left(\frac{r^2-x^2}{2}\right) = 32\pi \implies \boxed{(C) 32\pi}$. vinceS

Solution 4

The problem doesn't define where the chord is within the circle. So, let's place the chord such that when lines are drawn from its ends to the center of the larger semicircle, an equilateral triangle with side length 16 is formed.



In this definition, the radius of the larger semicircle is 16, giving it an area of $\frac{\pi \cdot 16^2}{2} = 128\pi$.

The height of the equilateral triangle is $8\sqrt{3}$, which is equal to the radius of the smaller semicircle. This gives the smaller semicircle an area of $\frac{\pi \cdot (8\sqrt{3})^2}{2} = 96\pi$.

The total shaded area is the difference of these semicircles, or $128\pi - 96\pi = \boxed{(C) 32\pi}$.

Solution 5

Let the radius of the larger semicircle be R and the smaller one r . We are asked to compute $\frac{R^2-r^2}{2} \cdot \pi$. Since a diameter is always perpendicular and bisects a chord, by drawing a diameter and applying Power of a Point Theorem, this yields $(R+r)(R-r) = 8^2 \implies R^2 - r^2 = 64$. Therefore, the answer is $\frac{64}{2}\pi = \boxed{(C) 32\pi}$. hxve

Solution 6

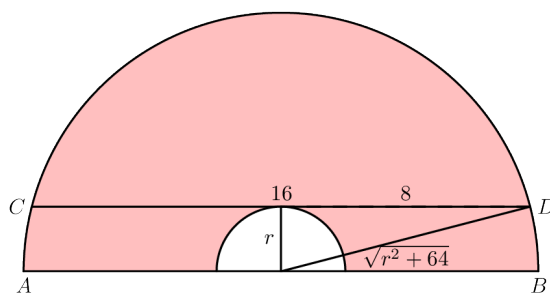
Since the problem does not restrict where the chord is, WLOG we can simply let the chord be the base of the semicircle. Therefore, the area is simply $\frac{\pi \cdot 8^2}{2} = 32\pi$.

Solution 7

We can move this small semicircle to the middle of the big semicircle. Let r be the radius of the small semicircle. By Pythagorean Theorem, the line drawn from the midpoint of the diameter of the big semicircle to the big semicircle at the height of the small semicircle (see the diagram) has length $\sqrt{r^2 + 64}$. This is a radius of the big semicircle. So the shaded area would be

$$\frac{\pi \cdot (\sqrt{r^2 + 64})^2 - \pi r^2}{2} = \frac{\pi(r^2 + 64 - r^2)}{2} = \frac{64\pi}{2} = \boxed{(C) 32\pi}.$$

Diagram:



Solution 8 (similar to some other stated solutions)

We see that the chord is very very similar in length to the diameter, so we solve for the area with 16 as the diameter. We find that the area is 64π , and because they want us to subtract the small semicircle and that 16 is actually less than the actual diameter, we can just use half of 64π as our answer. Thus, the answer is (C) 32π

Solution 9

Let the smaller semicircle ω be concentric with the larger semicircle Ω . Let $\text{rad}(\omega) = r$ and $\text{rad}(\Omega) = R$. Then, by Pythagoreans, $R^2 = 64 + r^2$. By Power of a Point, if T is the tangential point, then $\text{Pow}(T) \implies 64 = (R - r)r$, or $Rr - r^2 = 64$, hence $R^2 = Rr$, so $R(R - r) = 0$. We see $R \neq 0$, so $R = r$. Then, that implies $\omega = \Omega$ hence the chord is actually the diameter. Therefore, the area is $\frac{1}{2}\pi \cdot 64 = \boxed{32\pi}$.

- k) (AMC 10 10A 2025) The sequence $1, x, y, z$ is arithmetic. The sequence $1, p, q, z$ is geometric. Both sequences are strictly increasing and contain only integers, and z is as small as possible. What is the value of $x + y + z + p + q$?
- (A) 66 (B) 91 (C) 103 (D) 132 (E) 149

Answer: E

Solution 1

Since the geometric sequence is more restrictive, we can test values for the common ratio until we find one that works. After a few tests, we find that a common ratio of 4 results in the geometric sequence $1, 4, 16, 64$, so the arithmetic sequence is $1, 22, 43, 64$. The answer is $4 + 16 + 64 + 22 + 43 = \boxed{(E) 149}$.

A more generalized solution is as follows. Let the common difference of the arithmetic sequence be d , and the common ratio of the geometric sequence be r . Hence, the two sequences are $1, 1 + d, 1 + 2d, 1 + 3d$ and $1, r, r^2, r^3$. Since $z = 1 + 3d = r^3$, the arithmetic sequence is $1, 1 + d, 1 + 2d, r^3$. Since $d = \frac{1 + 3d - 1}{3} = \frac{r^3 - 1}{3}$ is a positive integer, we seek the smallest $r \neq 1$ such that $r^3 - 1 = (r - 1)(r^2 + r + 1)$ is divisible by 3, so the smallest r is 4. The rest follows like above.

Solution 2

Since $1, x, y, z$ is an arithmetic sequence, we have $y = 2x - 1$ and $z = 3x - 2$. Since $1, p, q, z$ is a geometric sequence, we have $q = p^2$ and $z = p^3$. Thus $p^3 = 3x - 2$.

Because $p^3 \equiv p \pmod{3}$, we get $3x - 2 \equiv p \pmod{3}$, so $p \equiv 1 \pmod{3}$. The smallest integer $p > 1$ satisfying this is $p = 4$.

Then $64 = 3x - 2 \rightarrow x = 22, y = 43, z = 64, q = 16$. Therefore, $x + y + z + p + q = 22 + 43 + 64 + 4 + 16 = \boxed{\text{(E)} 149}$.

Solution 3 (Satisfying and Simple)

So we have that the arithmetic sequence is $1, x, 2x - 1, 3x - 2$ since common difference is $x - 1$.

Similarly, geometric sequence is $1, p, p^2, p^3$

So $p^3 = 3x - 2$. We test values above 1 (increasing sequence)

We see that $p = 4, x = 22$ so our answer is $22 + 43 + 64 + 4 + 16 = \boxed{149}$

Solution 4 (Easy to identify)

Since only integer ratios work for the geometric series, test the ratio of 2, 3, and 4, by having z be 2, 3, and 4 cubed, and finding that $\frac{64-1}{3} = 21$ (since that is the common ratio of the arithmetic sequence to get 64 from 1 in 4 terms), then $22 + 43 + 4 + 16 + 64 = 149$, which is E.

Note: We divide z by 3 because to get from p to z you have to add $3d$ which is the difference of the arithmetic sum.

-Amon26(note)

Solution 5 (Ratio)

We have $1, x, y, z$ as an arithmetic sequence. If we let d be the rate of change, $z - y = d, y - x = d$ and $x - 1 = d$, now you could follow with solution 3, or, you could just have the geometric sequence ratio be r , so we have $\frac{z}{q} = r, \frac{q}{p} = r, \frac{p}{1} = r$.

Wait, we see $\frac{p}{1} = r$, so $p = r$, then $q = p^2$, then $p^3 = z$, and since these have to be integers, we have to get a value p that works as an integer for the arithmetic sequence.

Because we add our common difference d 3 times, we see that $(p^3 - 1)/3 = d$ where d is an integer (this is because if we try 3 we get $z = p$ which violates $p^3 = z$, where $p = 3$, however, the starting is one, so we find a value that when subtracted by 1, gets us a multiple of 3 so the arithmetic sequence works) and we find a value $p = 4$.

The arithmetic sequence is $1, 22, 43, 64$, where $21 = d$ and it follows as $p^3 - 1 = 63 = 3d$, and adding it up we get 129 for the arithmetic and since 64 is already z so we only need to add p and q or p^2 , we would get 20 and $20 + 129 = \boxed{\text{(E)} 149}$.

- Randomuser456(Note there are definitely other values where $(p^3 - 1)/3 = d$ is true, however the problem says for the lowest value of z) and this is also my first solution, please change things if necessary:)

$\text{\textasciicircum{}}_ \text{\textasciicircum{}} / | \text{\textasciicircum{}} \quad | / \text{\textasciicircum{}}$

- 1) (AMC 10 10A 2025) Carlos uses a 4-digit passcode to unlock his computer. In his passcode, exactly one digit is even, exactly one (possibly different) digit is prime, and no digit is 0. How many 4-digit passcodes satisfy these conditions?

(A) 176 (B) 192 (C) 432 (D) 464 (E) 608

Answer: D

Solution 1: CRAZY and Super Fast Casework

The only two digits that are neither prime nor even are 1 and 9. We split this problem into cases based on the number of 2s. This is because 2 is both a prime number and an even number.

Case 1: In this case, there are no 2's. There are 4 choices for where the even digit goes, and 3 choices for what the even digit is. There are then 3 choices for where the prime digit goes, and 3 choices for what the prime number digit is. The last two spots have 2 choices each, either 1 or 9. This gives a total of $4 \cdot 3^3 \cdot 2^2 = 432$ options for this case.

Case 2: In this case, there is one 2, with it being the only even prime number. There are 4 choices for where 2 goes, and 2 choices for the other three digits each. This case gives a total of $2^3 \cdot 4 = 32$ options.

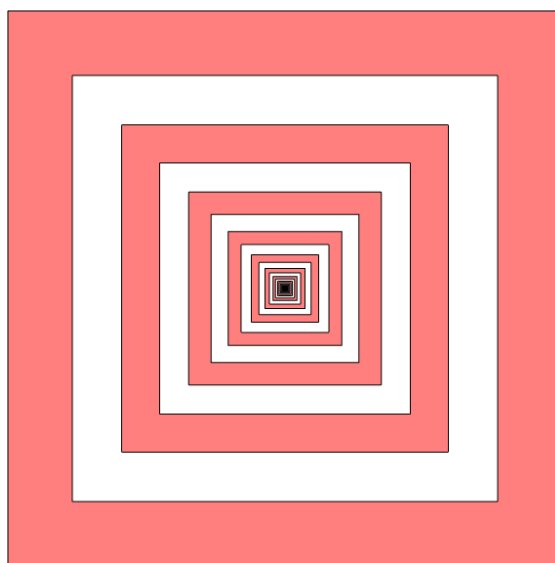
Hence, the answer is $432 + 32 = \boxed{\text{(D)} 464}$ Tacos_are_yummy_1

Solution 2: Cheese but fast

Let us count the cases where there is a 2 anywhere in the lock. There are 4 places to arrange the 2s, and the remaining digits can only be 1s and 9s (since they neither can be even nor prime). Thus, there will be $4 \cdot 2^3 = 32$ choices for there to exist one 2. We note that answer choice (D) is 32 greater than another answer choice (specifically, (C)) and is the only answer choice to have this property. Assuming that the pitfall of forgetting the case with a 2 would be a separate answer choice on its own, we can justify that the answer is $\boxed{\text{(D)} 464}$

- m) (AMC 10 10A 2025) In the figure below, the outside square contains infinitely many squares, each of them with the same center and sides parallel to the outside square. The ratio of the side length of a square to the side length of the next inner square is k , where $0 < k < 1$. The spaces between squares are alternately shaded as shown in the figure (which is not necessarily drawn to scale).

The area of the shaded portion of the figure is 64% of the area of the original square. What is k ?



- (A) $\frac{3}{5}$ (B) $\frac{16}{25}$ (C) $\frac{2}{3}$ (D) $\frac{3}{4}$ (E) $\frac{4}{5}$

Answer: D

Solution 1

Let the side length of the largest square be a , so it has area a^2 . Hence, the second-largest square has area a^2k^2 , the third-largest has a^2k^4 , and so on.

It follows that the total shaded area is

$$a^2 - a^2k^2 + a^2k^4 - a^2k^6 + \dots = a^2(1 - k^2 + k^4 - k^6 + \dots) = a^2 \frac{1}{1 + k^2}.$$

The ratio of the area of the shaded region to that of the original square is then

$$\frac{a^2 \frac{1}{1+k^2}}{a^2} = \frac{1}{1+k^2} = \frac{64}{100}$$

$$\implies 64 + 64k^2 = 100 \implies k^2 = \frac{36}{64} \implies k = \boxed{\text{(D)} \frac{3}{4}}.$$

Remark We can just let $a = 1$ because the question deals with ratios, meaning that there wouldn't be a loss of generality if we let the side length equal some value, getting the same answer $\boxed{D}$. This was done in solution 3.

Solution 2

Let the side length of the first square be a and the second square be b . The area of the original square is a^2 . The area of the outermost shaded region is $a^2 - b^2$. Let $\frac{a}{b} = r$. We have $b = \frac{a}{r}$. So the outermost shaded region area becomes:

$$a^2 - b^2 = a^2 - \frac{a^2}{r^2}$$

Now let the next square's side lengths be c and d . Similarly, $c = \frac{b}{r} = \frac{a}{r^2}$ and $d = \frac{c}{r} = \frac{a}{r^3}$ and the area of the next shaded region becomes:

$$c^2 - d^2 = \frac{a^2}{r^4} - \frac{a^2}{r^6}$$

Notice the pattern of adding 4 to the exponent. If this sequence continues infinitely, we ultimately get:

$$(a^2 + \frac{a^2}{r^4} + \frac{a^2}{r^8} + \dots) - (\frac{a^2}{r^2} + \frac{a^2}{r^6} + \dots)$$

Which can be simplified using the infinite geometric sequence formula. The problem also tells us that all of this equals $64\%a^2$, or $\frac{16}{25}a^2$:

$$\frac{a^2}{1 - \frac{1}{r^4}} - \frac{\frac{a^2}{r^2}}{1 - \frac{1}{r^4}} = \frac{16}{25}a^2$$

$$\frac{1}{1 - \frac{1}{r^4}} - \frac{\frac{1}{r^2}}{1 - \frac{1}{r^4}} = \frac{16}{25}$$

$$r^2 = \frac{16}{9} \implies r = \frac{4}{3}$$

Since the problem tells us the ratio must be less than one, we take the reciprocal to finally get

$$k = \frac{1}{r} = \boxed{(D) \frac{3}{4}}.$$

Solution 3 (Faster)

Let the outside square be of side length 1, and let $p = k^2$. Then the area of the square is $1 - p + p^2 - p^3 + p^4 - p^5 \dots$ or $(1 - p)(1 + p^2 + p^4 + p^6 + \dots)$. Then, using geometric series, we get $(1 - p)\left(\frac{1}{1 - p^2}\right) = \frac{1}{1 + p}$. This is equal to $\frac{64}{100}$. Therefore, $1 + p = \frac{25}{16}$, so $p = 9/16$ and $k = \frac{3}{4}$. Therefore, the answer is

$$\boxed{(D) \frac{3}{4}}.$$

Solution 4 (The fastest, no sequences/series)

Since this is an infinite fixed pattern, we can actually take a huge shortcut. Suppose we take off the outer "layer" and then swap the colors. We have essentially created a smaller copy of the original diagram, specifically with a $k : 1$ ratio of side lengths. Assume the original diagram had 100 units of area. Then, based on the givens, the original diagram had 64 units of shaded area and 36 units of unshaded area, meaning our new smaller diagram has 36 units of shaded area. Since the ratio of corresponding side lengths between our new diagram and the original diagram is k , the ratio of corresponding areas is k^2 . Thus, we get $k^2 = \frac{36}{64}$, so $k = \boxed{(D) \frac{3}{4}}$.

Note: the explanation seems complicated to explain, but the thought process while doing it is extremely fast, much faster than any of the methods involving a series.

Solution 5 (based very closely on solution 4)

Similarly to solution 4, we can imagine "peeling" off the outer layer to get an inner figure that is similar to the outer figure. Letting the large square have side 1, we know the outer shaded layer has area $1 - k^2$. The shaded part of the inner figure (with the outer shaded part removed now must make up $\frac{36}{100} = \frac{9}{25}$ of the inner area of k^2 . Together, the inner shaded part and the outer shaded part must add up to $\frac{64}{100}$, as they make up 64% of the square. Thus, we get the equation $1 - k^2 + \frac{9k}{25} = 16/25$. Solving this gives $k = 3/4$ Neber100

Solution 6 (The longest way)

Notes that: The sum S can be written as:

$$\begin{aligned} S &= 1^2 - k^2 + (k^2)^2 - (k^3)^2 + (k^4)^2 - (k^5)^2 + (k^6)^2 - \dots \\ &= k^0 - k^2 + k^4 - k^6 + k^8 - k^{10} + \dots \\ &= (k^0 + k^4 + k^8 + k^{12} + \dots) - (k^2 + k^6 + k^{10} + k^{14} + \dots) \end{aligned}$$

So we have:

$$S = \lim_{n \rightarrow \infty} \left[1 \times \frac{1 - (k^4)^n}{1 - k^4} - k^2 \times \frac{1 - (k^4)^n}{1 - k^4} \right]$$

$$\begin{aligned}
 &= \frac{1}{1-k^4} - \frac{k^2}{1-k^4} \\
 &= \frac{1-k^2}{1-k^4} \\
 &= \frac{1-k^2}{(1+k^2)(1-k^2)} \\
 &= \frac{1}{1+k^2}
 \end{aligned}$$

We are given the condition that:

$$\therefore \frac{1}{1+k^2} = 64\% = \frac{64}{100} = \frac{16}{25}$$

We can now solve for k :

$$\begin{aligned}
 \therefore \frac{1}{1+k^2} &= \frac{16}{25} \\
 1+k^2 &= \frac{25}{16} \\
 k^2 &= \frac{25}{16} - 1 \\
 k^2 &= \frac{25}{16} - \frac{16}{16} \\
 k^2 &= \frac{9}{16} \\
 k &= \pm \sqrt{\frac{9}{16}} \\
 k &= \pm \frac{3}{4}
 \end{aligned}$$

Therefore, ignore the negative value (Because $0 < k < 1$), so the final answer is $\frac{3}{4}$. We are done :)

Solution 7

Assume that the side length for the largest square in the diagram is 1. Since the ratio of the next inner square to the largest square is k , we get that the side length for the second largest square is k , k^2 for the third largest square, and k^3 for the fourth largest square. So, we can write that

$$(1 - k^2) + (k^4 - k^6) + \dots = \frac{64}{100}$$

by using the diagram since the area of a square with side length s is s^2 . This is a geometric sequence with first term $1 - k^2$ and ratio k^4 . So, we can sum the geometric series for

$$\frac{1 - k^2}{1 - k^4} = \frac{64}{100}.$$

Cross multiplying, we have that

$$100 - 100k^2 = 64 - 64k^4.$$

Moving the terms to one side, we have $64k^4 - 100k^2 + 36 = 0$, and we can factor as

$$(64k^2 - 36)(k^2 - 1) = 0.$$

Since $k < 1$, $k^2 - 1 = 0$ is invalid. So, we have

$$64k^2 - 36 = 0,$$

and simplifying we get $k^2 = \frac{36}{64}$. Since $k > 0$,

$$k = \frac{6}{8} = \boxed{\frac{3}{4}}.$$

Solution 8 (Cheese)

Measure the side length of the first square to be 2 inches. Measure the side length of the second square to be 1.5 inches.

$$\frac{1.5}{2} = \frac{3}{4} = \boxed{(D) \frac{3}{4}}$$

Solution 9 (Fastest and shortest, similar to solution 4, no series)

Since the figure is self-similar, the nonshaded area is $100\% - 64\% = 36\%$ of the largest square and 64% of the square with side length k , which has area k^2 . Thus $36\% = 64\%k^2$, $k^2 = \frac{36}{64}$, so $k = \frac{6}{8} = \boxed{(D) \frac{3}{4}}$.

Solution 10 (Series Subtraction)

Assume initial length 1 since the problem just uses ratios. We see that each side length shrinks by factor k , then they are squared to get the areas, which really just multiplies their exponents by 2. We add our shaded areas and subtract our un-shaded areas, yielding the alternating series

$$1 - k^2 + k^4 - k^6 \dots$$

We express this as two infinite series, one being subtracted from the other

$$\sum_{n=0}^{\infty} k^{4n} - \sum_{n=0}^{\infty} k^{4n+2}$$

Rewrite using the infinite geometric series formula and simplify as far as possible

$$\begin{aligned} & \frac{1}{1 - k^4} - \frac{k^2}{1 - k^4} \\ & \frac{1 - k^2}{1 - k^4} \\ & \frac{1 - k^2}{(1 + k^2)(1 - k^2)} \\ & \frac{1}{1 + k^2} \end{aligned}$$

We know this is equal to $\frac{64}{100} = \frac{16}{25}$

$$\frac{1}{1 + k^2} = \frac{16}{25}$$

$$16k^2 + 16 = 25$$

$$16k^2 = 9$$

$$k^2 = \frac{9}{16}$$

We must only consider the positive root

$$k = \boxed{\text{(D)} \frac{3}{4}}.$$

shockfront99

- n) (*AMC 10 10A 2025*) Six chairs are arranged around a round table. Two students and two teachers randomly select four of the chairs to sit in. What is the probability that the two students will sit in two adjacent chairs and the two teachers will also sit in two adjacent chairs?

(A) $\frac{1}{6}$ (B) $\frac{1}{5}$ (C) $\frac{2}{9}$ (D) $\frac{3}{13}$ (E) $\frac{1}{4}$

Answer: B

Solution 1 (20 seconds)

Pair two students together and put them adjacent on any two seats. There are 6 ways to do this. Considering one of these cases (they are all the same), there are 4 seats left, in which we wish to arrange the teachers together. So pair the teachers together and put them adjacent on any two seats not already occupied by two of the students. There are 3 ways to do this. For all 6 cases, there are $6 \times 3 = 18$ favorable outcomes.

The number of ways to arrange the 2 students and 2 teachers is $\binom{6}{2} \times \binom{4}{2} = 90$.

Our probability is $\frac{18}{90} = \boxed{\text{(B)} \frac{1}{5}}$

Solution 2

We first count the number of ways to place 4 distinct people into 6 distinct chairs: $6 \cdot 5 \cdot 4 \cdot 3 = 360$.

We now count how many favorable assignments there are. There are 6 ways to choose an adjacent pair of chairs for the two students. We can arrange the two students in those two chairs in 2 ways.

After those two adjacent chairs are taken, there are 4 chairs left, with 3 adjacent pairs among them. We choose one of these pairs, arranging the teachers for $3 \cdot 2 = 6$ ways.

There are $6 \cdot 2 \cdot 6 = 72$ favorable arrangements.

The probability is therefore $\frac{72}{360} = \boxed{\frac{1}{5}}$

Solution 3

We split the problem into cases of where the first teacher and first student sits. The first student sits in a seat with 1 probability

Case 1: The first teacher sits next to the first student. There's are 2 ways to do this so this will happen with $\frac{2}{5}$ chance. Now, there is one valid seat for the second student to sit in with probability $\frac{1}{4}$ and and one valid valid seat

for the second teacher to sit in with probability $\frac{1}{3}$. Total probability of this case is $\frac{2}{5} \cdot \frac{1}{4} \cdot \frac{1}{3} = \frac{1}{30}$

Case 2: The first teacher sits on the opposite end of the circle with $\frac{1}{5}$ chance. This means the second student and the second teacher each have 2 valid spots to sit in, with probability $\frac{2}{4}$ and $\frac{2}{3}$. Total probability is $\frac{1}{5} \cdot \frac{1}{2} \cdot \frac{2}{3} = \frac{1}{15}$.

Case 3: The first teacher sits one chair away from the first student with $\frac{2}{5}$ probability because there are 2 seats that are one chair away.

Case 3.1:

The second student sits in between the first teacher and first student with $\frac{1}{4}$ chance. The second teacher only has one valid seat, with $\frac{1}{3}$ chance to sit in it. Total probability is $\frac{1}{4} \cdot \frac{1}{3} = \frac{1}{12}$.

Case 3.2:

The second student doesn't sit in between the teacher and the student with a $\frac{1}{4}$ chance. The second teacher can sit in 2 valid seats with a $\frac{2}{3}$ chance. Total probability is $\frac{1}{4} \cdot \frac{2}{3} = \frac{1}{6}$.

Total probability for case 3 is $\frac{2}{5} \cdot (\frac{1}{12} + \frac{1}{6}) = \frac{1}{10}$.

Adding up all the cases, you get $\frac{1}{30} + \frac{1}{15} + \frac{1}{10} = \boxed{\frac{1}{5}}$.

Solution 4

We consider how many ways there are to place the students and teachers at random at the table. This is a word arrangement in a circle, and so there are $\frac{(6-1)!}{2! \cdot 2! \cdot 2!} = 15$ different ways of doing so, assuming that teachers, students and empty spaces are indistinguishable between themselves.

Now, we wish to find how many favorable arrangements there are, so suppose we fix two adjacent teachers. Then, there are three "spots" (blocks of two seats) two adjacent students could occupy: two on the side of either teacher, and one such that no student is sitting adjacent to a teacher.

Thus, the probability is $\frac{3}{15} = \boxed{\frac{1}{5}}$.

Solution 5 (Step-by-step, needs verification)

Because this is in a circle, rotating around doesn't increase or decrease the probability (so pin one seat and the others will not rotate).

Hence, label the spots 1-6 in clockwise. Let student 1 sit in spot 1. Student 2 can therefore sit in either spot 2 or 6, which the probability is $\frac{2}{5}$. For explanation purposes, put student 2 in position 2.

Here then requires a case breakdown.

Case 1: Teacher 1 sits in spot 3, hence teacher 2 needs to sit in spot 4 to satisfy. This case totals up with $\frac{2}{5} \cdot \frac{1}{4} \cdot \frac{1}{3} = \frac{1}{30}$.

Case 2: Teacher 1 sits in spot 4, hence teacher 2 needs to sit in spot 3 or 5 to satisfy. This case totals up with $\frac{2}{5} \cdot \frac{1}{4} \cdot \frac{2}{3} = \frac{1}{15}$.

Case 3: Teacher 1 sits in spot 5, hence teacher 2 needs to sit in spot 4 or 6 to satisfy. This case totals up with $\frac{2}{5} \cdot \frac{1}{4} \cdot \frac{2}{3} = \frac{1}{15}$.

Case 4: Teacher 1 sits in spot 6, hence teacher 2 needs to sit in spot 5 to satisfy. This case totals up with $\frac{2}{5} \cdot \frac{1}{4} \cdot \frac{1}{3} = \frac{1}{30}$.

By adding the case probabilities, we have $(\frac{1}{30}) \cdot 2 + (\frac{1}{15}) \cdot 2 = \boxed{\frac{1}{5}}$.

Note: Feel free to delete if there is a logic error. Mitsuihisashi14

Solution 6 (Quickest)

For the two teachers' seatings, there are 6 ways that they are adjacent out of a combination of $n = \binom{6}{2}$ ways. This probability is $\frac{6}{15} = \frac{2}{5}$. For the two students' seatings, there are 3 ways to choose the two seats out of $n = \binom{4}{2}$

ways. This probability is $\frac{3}{6} = \frac{1}{2}$. Answer is $\frac{2}{5} \cdot \frac{1}{2} = \boxed{\frac{1}{5}}$.

Solution 7 (Probability = Desired Arrangements/Total arrangements)

Let the seats be numbered 1, 2, 3, 4, 5, 6 in clockwise order. Then assume placing a group of 2 (either the two students or the two teachers) in seats 1 and 2, this group of two can be permuted (or in this case just swapped) in $2! = 2$ ways. Now the second group may sit in the seats 3, 4 or 4, 5 or 5, 6 giving a factor of 3. This group can also be permuted in $2! = 2$ ways. This gives 12 ways for the 4 people to be seated under the condition. Without the condition there are $\frac{(6-1)!}{2} = 60$ ways for the 4 people to be seated (the division by $2!$ comes from the two empty seats being considered the same). Finally, the desired probability is:

$$\frac{12}{60} = \boxed{\frac{1}{5}}$$

Solution 8

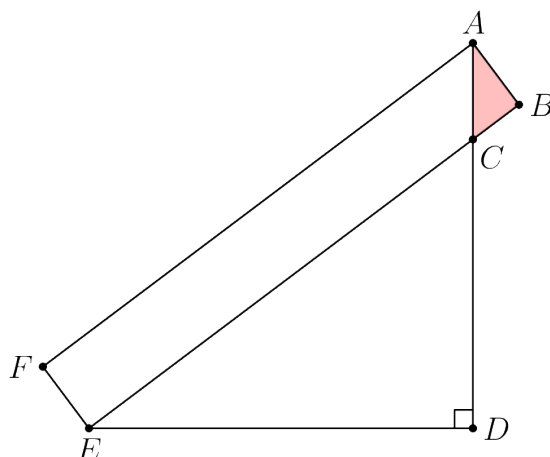
For the two students, you can seat the first one in any seat, followed by the second one with a $\frac{2}{5}$ possibility of satisfying the condition. After this, we have two cases.

Case 1: The first teacher sits next to the two students. This can happen $\frac{1}{2}$ of the time and the other teacher has a $\frac{1}{3}$ chance of sitting next to the first one. This gives us a $\frac{1}{2} \cdot \frac{1}{3} = \frac{1}{6}$ chance for this case.

Case 2: The first teacher sits away from the two students. This can also happen $\frac{1}{2}$ of the time and the other teacher has a $\frac{2}{3}$ chance of sitting next to the first one. This gives us a $\frac{1}{2} \cdot \frac{2}{3} = \frac{1}{3}$ chance for the second case.

Now, we can find the desired probability as: $(\frac{1}{3} + \frac{1}{6}) \cdot \frac{2}{5} = \frac{1}{2} \cdot \frac{2}{5} = \boxed{\frac{1}{5}}$

- o) (AMC 10 10A 2025) In the figure below, $ABEF$ is a rectangle, $\overline{AD} \perp \overline{DE}$, $AF = 7$, $AB = 1$, and $AD = 5$. What is the area of $\triangle ABC$?



- (A) $\frac{3}{8}$ (B) $\frac{4}{9}$ (C) $\frac{1}{8}\sqrt{13}$ (D) $\frac{7}{15}$ (E) $\frac{1}{8}\sqrt{15}$

Answer: A

Solution 1

Because $ABEF$ is a rectangle, $\angle ABC = 90^\circ$. We are given that $\angle ADE = 90^\circ$, and since $\angle ECD = \angle ACB$ by vertical angles, $\triangle ECD \sim \triangle ACB$. Let $AC = x$. By the Pythagorean Theorem, $CB = \sqrt{x^2 - 1}$. Since $AF = BE = 7$, $EC = 7 - \sqrt{x^2 - 1}$. Because $AC = x$ and $AD = 5$, $CD = 5 - x$. By similar triangles,

$$\frac{7 - \sqrt{x^2 - 1}}{x} = \frac{5 - x}{\sqrt{x^2 - 1}}$$

. Cross-multiplying, we get that

$$7\sqrt{x^2 - 1} - x^2 + 1 = 5x - x^2$$

, so

$$7\sqrt{x^2 - 1} = 5x - 1$$

. We square both sides, and this is simply a quadratic in x :

$$24x^2 + 10x - 50 = 0$$

, which has a positive root $x = \frac{5}{4}$. Since $AB = 1$, we can plug this into the Pythagorean Theorem, with $\frac{5}{4}$ being the hypotenuse, to get $BC = \frac{3}{4}$, and

$$1 \cdot \frac{3}{4} \text{ to equal } [ABC] = \boxed{\text{(A)} \frac{3}{8}}$$

Solution by HumblePotato, written by lhfriend,

Minor edit by SixthGradeBookWorm927

Minor edit by aldzandrte

Minor edit by rcll (I fixed this, as aldzandrte would have said)

Minor edit by pyraminx

Minor edit by wisewigglyjaguar

Minor edit by jeffyang2025

Minor edit by Loading...

Solution 2 (simpler)

Draw segment AE . Segment AE is the diagonal of rectangle $ABEF$, and its diagonals have length $\sqrt{7^2 + 1^2} = \sqrt{50} = 5\sqrt{2}$. From right triangle AED , we use pythagorean theorem to find $DE = 5$.

Now, we see similar triangles $\triangle CDE$ and $\triangle CBA$. Let $CE = a$, and $CD = b$. We can find that $AC = 5 - b$, and $CB = 7 - a$. These triangles have a ratio of $\frac{AB}{DE} = \frac{1}{5}$. So we get that $\frac{5-b}{a} = \frac{1}{5}$. Cross multiplying, we get $a = 25 - 5b$. And also $\frac{CB}{CD} = \frac{1}{5} = \frac{7-a}{b}$. Cross multiplying gives $35 - 5a = b$. Solving the system of equations, we find $a = 25/4$, which means $CB = 7 - 25/4 = 3/4$. $[ABC] = CB/2$, which gives $[ABC] = 3/8$.

Solution 3 (thorough)

From the diagram, $\angle BCA$ and $\angle DCE$ are vertical angles and hence congruent. Additionally, $\angle B = \angle D = 90^\circ$, so we have by AA Similarity that $\triangle BCA \sim \triangle DCE$.

Let $BC = x$ so $EC = 7 - x$ and $AC = y$ so $CD = 5 - y$. Since the two triangles are similar, we have $\frac{BC}{AC} = \frac{CD}{EC}$. Plugging in the variables gives $\frac{x}{y} = \frac{5-y}{7-x}$.

Cross multiplying yields $(7 - x)(x) = (5 - y)(y) \implies 7x - x^2 = 5y - y^2 \implies 7x + (y^2 - x^2) = 5y$.

By applying the Pythagorean Theorem into $\triangle BCA$, we get $x^2 + 1 = y^2 \implies 1 = y^2 - x^2$.

Therefore, $y = \frac{7x+1}{5}$, and plugging this back into $x^2 + 1 = y^2$:

$$\begin{aligned} x^2 + 1 &= \left(\frac{7x+1}{5}\right)^2 \quad 25(x^2 + 1) = (7x + 1)^2 \quad 25x^2 + 25 = 49x^2 + 14x + 1 \\ 0 &= 49x^2 + 14x + 1 - 25x^2 - 25 \quad 0 = 24x^2 + 14x - 24 \quad 0 = 12x^2 + 7x - 12 \\ x &= \frac{-7 \pm \sqrt{7^2 - 4(12)(-12)}}{2 \cdot 12} = \frac{-7 \pm \sqrt{49 + 576}}{24} = \frac{-7 \pm \sqrt{625}}{24} = \frac{-7 \pm 25}{24} \end{aligned}$$

Therefore, $x = \frac{-7+25}{24} = \frac{18}{24} = \frac{3}{4}$.

The area of $\triangle BCA$ is therefore $\frac{x \cdot 1}{2} = \frac{3}{8} = \boxed{A}$. hxe brilliantcoyote

Solution 4 (Trigonometry)

Using the Pythagorean theorem, I can get $AE = 5\sqrt{2}$. Then, because $AD = 5$, $ED = 5$. Now, let $\angle FEA = a$ and $\angle AED = b$. $\sin a = \frac{7}{5\sqrt{2}}$, $\sin b = \frac{1}{\sqrt{2}}$, $\cos a = \frac{1}{5\sqrt{2}}$, and $\cos b = \frac{1}{\sqrt{2}}$. Then, applying the sine addition formula, I get:

$$\sin(a + b) = \frac{7}{5\sqrt{2}} \times \frac{1}{\sqrt{2}} + \frac{1}{\sqrt{2}} \times \frac{1}{5\sqrt{2}} = \frac{7}{10} + \frac{1}{10} = \frac{4}{5}$$

Thus, $\sin(a + b) = \frac{4}{5}$, so $\sin(180 - a - b) = \frac{4}{5}$. This indicates that if a perpendicular is dropped from point F to the extension of line DE , and the foot of the latitude is point G , $\triangle EGF$ is a 3-4-5 triangle. Because $\triangle EGF \sim \triangle CDE$, $\triangle CDE$ is also a 3-4-5 triangle. Using ratios,:

$$CE = \frac{5}{4} \times 5 = \frac{25}{4}$$

Therefore, $CE = \frac{25}{4}$, so $BC = \frac{3}{4}$, so $\triangle ABC$ has area $\frac{3}{8}$, or $\boxed{A}$.

P.S. Thank you to eqb5000 and i_am_not_suk_at_math for pointing out and helping me fix typos.

Solution 5 (risky, but it works!)

Using a ruler (which is permitted during the exam), and assuming the diagram is to-scale, we can measure the physical lengths of AB and BC , and determine

Remark - You could also redraw the figure to scale. That way, you won't have to assume it's drawn to scale in the provided figure. You would have to draw the diagonal AE, measure it as $\sqrt{7^2 + 1^2}$ and use pythagorean theorem to find ED as 5. Once you have that information you could sketch the whole figure to scale.

Inspired by [2022 AMC 10B Problem 16](#), another right-triangle and rectangle mashup, we can pretend that the problem uses $3-4-5$ similarity, because AMC problems do that a lot. In this case, we can verify it works by setting $BC = \frac{3}{4}, AC = \frac{5}{4}$ by $3-4-5$ similarity, so $CD = \frac{15}{4}$ and $CE = \frac{25}{4}$. Then, we see that $BC + CE = \frac{25+3}{4} = 7 = BE = AF$, which is our only other condition, so this setup works and we can bubble in $\frac{1}{2} \cdot \frac{3}{4} \cdot 1 = \boxed{\text{(A)} \frac{3}{8}}$. of course, if it weren't $3-4-5$, we can use any other technique.

Connect AE , which equals $\sqrt{50}$. Then calculate ED , which equals 5. $\triangle ABC \sim \triangle EDC$ (AA). $\frac{AB}{ED} = \frac{1}{5}$.

Also, let the area of $\triangle ACE = x$. The area of $\triangle ABC = \frac{7}{2} - x$. The area of $\triangle EDC = \frac{25}{2} - x$.

Solve: $x = \frac{25}{8}$

Solution 8 (analytic geometry)



As shown in the figure, establish a coordinate system with point A as the origin.

Suppose $C = (a, 1)$, then the equation of AC is $y = \frac{1}{a}x$. Since $DE \perp AC$, the slope of DE is $-a$, the equation of DE is $y - 1 = -a(x - 7) \rightarrow ax + y - 7a - 1 = 0$.

Notice that the length of AD is the distance between point A and line DE , due to the distance formula, we get:

$$\frac{|a \cdot 0 + 0 - 7a - 1|}{\sqrt{a^2 + 1}} = 5$$

$$(7a + 1)^2 = 25(a^2 + 1)$$

$$(4a - 3)(3a + 4) = 0$$

$$a = \frac{3}{4}$$

The area of $\triangle BCA$ is therefore $\frac{a \cdot 1}{2} = \frac{3}{8} = \boxed{A}$.

Solution 9

From the Pythagorean Theorem, $AE = 5\sqrt{2}$ and we also discover $\triangle ADE$ is a $45^\circ - 45^\circ - 90^\circ$ right triangle. This means that $DE = 5$. As $\triangle ABC \sim \triangle EDC$ by AA similarity and $AB = 1$, we can set up a system of linear equations in terms of a and b where they represent BC and AC respectively. Since $CD = 5CB$ and $CE = 5AC$, we have

$$5a + b = 5$$

$$a + 5b = 7$$

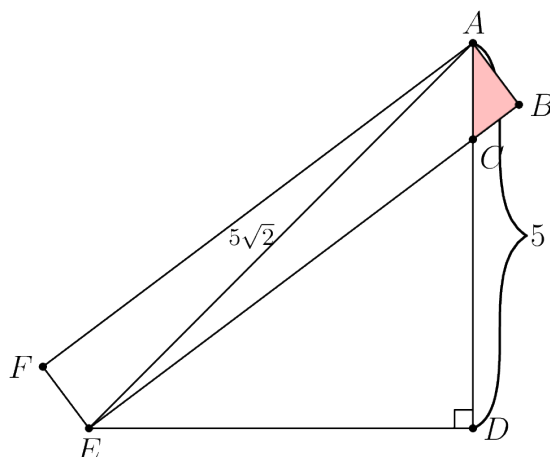
and solving the equations results in $a = \frac{3}{4}$. The area of $\triangle ABC$ is $\boxed{(A) \frac{3}{8}}$. b is unnecessary to find.

Note that $\triangle ABC$ is a $3 - 4 - 5$ triangle.

This is very similar to Solution 2, and at the time of writing, I had not noticed.

Solution 10

Start by assigning the variable x to the length of side $\overline{AC}$. Since $\overline{AD} = 5$, $\overline{CD} = 5 - x$. Additionally, the diagonal $\overline{AE} = 5\sqrt{2}$ by the Pythagorean Theorem. Notice that with this diagonal, we've created a right triangle $\triangle ADE$.



Now, you could manually use the Pythagorean Theorem, but you can just spot the $45^\circ - 45^\circ - 90^\circ$ triangle and see $\overline{ED} = 5$. After this, we can use the Pythagorean Theorem to find the length of $\overline{CE}$.

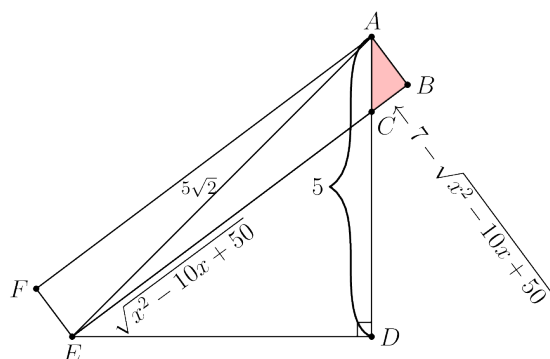
$$5^2 + (5 - x)^2 = (\overline{CE})^2$$

$$25 + 25 + x^2 - 10x = (\overline{CE})^2$$

$$x^2 - 10x + 50 = (\overline{CE})^2$$

$$\sqrt{x^2 - 10x + 50} = \overline{CE}$$

From this, we can see that $\overline{CB} = 7 - \overline{CE} \implies \overline{CB} = 7 - \sqrt{x^2 - 10x + 50}$



Now, we know that $[CDE] + [ABE] - [ABC] = [ADE] = 12.5$. Therefore,

$$\frac{5 \cdot (5 - x)}{2} + \frac{7}{2} - \frac{7 - \sqrt{x^2 - 10x + 50}}{2} = 12.5$$

$$5 \cdot (5 - x) + 7 - (7 - \sqrt{x^2 - 10x + 50}) = 25$$

$$25 - 5x + \sqrt{x^2 - 10x + 50} = 25$$

$$\sqrt{x^2 - 10x + 50} = 5x$$

$$x^2 - 10x + 50 = 25x^2$$

$$0 = 24x^2 + 10x - 50$$

$$0 = 12x^2 + 5x - 25$$

$$0 = (4x - 5)(3x + 5)$$

Now, we know that x must be positive so our only valid solution is

$$x = \frac{5}{4}$$

Finally, we can substitute this value of x into our expression for length $\overline{CB}$. The other leg would just be 1 making for a total area of $\frac{1}{2} \cdot \overline{CB}$. Hence,

$$[ABC] = \frac{7 - \sqrt{x^2 - 10x + 50}}{2}$$

$$[ABC] = \frac{7 - \sqrt{\frac{5^2}{4} - 10 \cdot \frac{5}{4} + 50}}{2}$$

$$[ABC] = \frac{7 - \sqrt{\frac{25}{16} - \frac{50}{4} + \frac{200}{4}}}{2}$$

$$[ABC] = \frac{7 - \sqrt{\frac{25}{16} + \frac{150}{4}}}{2}$$

$$[ABC] = \frac{7 - \sqrt{\frac{25}{16} + \frac{600}{16}}}{2}$$

$$[ABC] = \frac{7 - \sqrt{\frac{625}{16}}}{2}$$

$$[ABC] = \frac{7 - \frac{25}{4}}{2}$$

$$[ABC] = \frac{\frac{28}{4} - \frac{25}{4}}{2}$$

$$[ABC] = \frac{\frac{3}{4}}{2}$$

$$[ABC] = \frac{3}{8} \Rightarrow \boxed{(A) \frac{3}{8}}$$

Shockfront99

Solution 11

Connect AE , we can find that $AE = \sqrt{AB^2 + BE^2} = \sqrt{1^2 + 7^2} = 5\sqrt{2}$. And $\angle ADE = 90^\circ$, $AE = 5\sqrt{2}$, $AD = 5$, then we can get $ED = 5$, $\triangle AED$ is an isosceles right triangle

Then, $\angle AED = 45^\circ$ Let $\angle CED = \angle \alpha$, $\angle \alpha = \angle AED - \angle AEC$

Because of the tangent subtraction formula $\tan(\alpha - \beta) = \frac{\tan \alpha - \tan \beta}{1 + \tan \alpha \tan \beta}$, we can get $\tan \alpha = \frac{\tan \angle AED - \tan \angle AEC}{1 + \tan \angle AED \cdot \tan \angle AEC}$

Since $\tan \angle AED = \tan 45^\circ = 1$, and $\tan \angle AEC = \frac{AB}{EB} = \frac{1}{7}$, we can get $\tan \alpha = \frac{3}{4}$

Because $\frac{CD}{ED} = \tan \alpha = \frac{3}{4}$, then $CD = \frac{15}{4}$, so $EC = \frac{25}{4}$.

Thus, $BC = \frac{3}{4}$.

So, the area of $\triangle ABC = \frac{1}{2} \cdot BC \cdot AB = \frac{1}{2} \cdot \frac{3}{4} = \frac{3}{8} = \boxed{A}$.

Solution 12 (Similar to Solution 3)

We notice that by vertical angles that $\angle DCE = \angle BCA$, and we also have $\angle CDE = \angle ABC = 90$. $\triangle BCA$ is similar to $\triangle DCE$ by Angle-Angle Similarity. Now let $BC = x$. Using the Pythagorean Theorem on $\triangle ABC$, we have $CA = \sqrt{x^2 + 1}$. Since $ABEF$ is a rectangle, we have $AF = 7 = BE$, which also means $CE = 7 - x$. Since $AD = 5$, $DC = 5 - \sqrt{x^2 + 1}$. From our similar triangles, we have $\frac{BC}{DC} = \frac{CA}{CE} \implies \frac{x}{5 - \sqrt{x^2 + 1}} = \frac{\sqrt{x^2 + 1}}{7 - x}$. Cross multiplying, simplifying, and squaring gives us $7x - x^2 = 5\sqrt{x^2 + 1} - x^2 - 1 \implies 7x + 1 = 5\sqrt{x^2 + 1} \implies 49x^2 + 14x + 1 = 25x^2 + 25 \implies 12x^2 + 7x - 12 = 0$. We can use the quadratic formula, and take the positive solution. $x = \frac{-7 + \sqrt{7^2 - 4(12)(-12)}}{2(12)} = \frac{3}{4}$. Since ABC is a right triangle, the area is $\frac{1}{2}(AB)(BC) = \frac{1}{2}(1)(\frac{3}{4}) = \boxed{\text{(A)} \frac{3}{8}}$, which is our final answer.

- p) (AMC 10 10A 2025) There are three jars. Each of three coins is placed in one of the three jars, chosen at random and independently of the placements of the other coins. What is the expected number of coins in a jar with the most coins?

(A) $\frac{4}{3}$ (B) $\frac{13}{9}$ (C) $\frac{5}{3}$ (D) $\frac{17}{9}$ (E) 2

Answer: D

Solution 1 (Case Work)

We have three coins and three jars. Each coin is placed independently and randomly into one of the jars. Let M be the maximum number of coins in any jar. We want to compute the expected value of M .

Step 1: Count total outcomes

Each coin has 3 choices, so the total number of equally likely placements is $3^3 = 27$.

Step 2: Casework on the maximum number of coins

Case 1: $M = 1$. This occurs when each jar has exactly one coin. There are $3! = 6$ assignments of coins to jars. Hence, $\Pr(M = 1) = \frac{6}{27} = \frac{2}{9}$.

Case 2: $M = 3$. This occurs when all three coins fall into the same jar. There are 3 jars to choose from, so $\Pr(M = 3) = \frac{3}{27} = \frac{1}{9}$.

Case 3: $M = 2$. This occurs when one jar has 2 coins, another jar has 1 coin, and the last jar has 0 coins. We can choose which jar gets 2 coins in 3 ways, which jar gets 1 coin in 2 ways, and which 2 coins out of the 3 go into the jar with two coins, so we multiply by $\binom{3}{2}$, which is just 3 (note we don't have to do this for the earlier cases because for case 2, all 3 coins go into one jar, and for case 1, the factorial already accounts for that). Therefore, there are $3^2 \cdot 2 = 18$ outcomes. Thus, $\Pr(M = 2) = \frac{18}{27} = \frac{2}{3}$.

Step 3: Compute the expected value The expected value of M is $\mathbb{E}[M] = 1 \cdot \frac{2}{9} + 2 \cdot \frac{2}{3} + 3 \cdot \frac{1}{9}$. Converting everything to ninths, we have $\mathbb{E}[M] = \frac{2}{9} + \frac{12}{9} + \frac{3}{9} = \frac{17}{9}$.

Hence, the expected number of coins in the jar with the most coins is $\boxed{\text{(D)} \frac{17}{9}}$.

Pr = Probability of

$\mathbb{E}$ =Expected value

Another way of finding cases of $M = 2$ As described in the solution, there are $3^3 = 27$ ways of distributing the coins into the 3 jars. Because there are 6 ways for $M=1$ and 3 ways for $M=3$, there are $27 - 6 - 3 = 18$ ways for $M = 2$.

Solution 2 (Fast)

Assuming all jars and coins are distinct, there are 27 total outcomes. $3! = 6$ of them distribute exactly, and thus a max of, 1 per cup, 3 ways to choose a jar to put all 3 coins in for a max of 3. This leaves $27 - 6 - 3 = 18$ for a max of 2, so the expected value is $\frac{6 \cdot 1 + 18 \cdot 2 + 3 \cdot 3}{27} = \boxed{(D) \frac{17}{9}}$

Solution 3 (Circular Counting - Expectation Cycles)

In circular counting, expected value is determined as

$$\mathbb{E}_{Cn}(A) = \frac{Cn(A)}{\mathbb{T} - Cn(A) - n(A)}$$

Where $\mathbb{E}_{Cn}(A)$ is the expected value, $Cn(A)$ is the circular gap theorem, $\mathbb{T}$ is the total number of cases, and $n(A)$ is the invalid cases.

Arrange the three jars and three coins into a circle of 6 elements. We want to count all the ways in which no 2 or no 3 coins are next to one another. This can easily be done using the circular gap theorem, which states

$$Cn(A) = \frac{n}{k} \binom{n-k-1}{k-1}$$

Where n is the number of elements in our circle and k is the number of non adjacencies, in which we wish to count $n = 6$ and $k \in \{1, 2, 3\}$ (The reason we have only 1, 2, and 3 is because if we have $n = 6$ and $k \geq 4$, we see that the combinatorial part of $Cn(A)$ becomes invalid).

So we begin counting.

For $k = 1$, using the formula, we have $\frac{6}{1} \binom{4}{0}$ which is 6.

For $k = 2$, using the formula, we have $\frac{6}{2} \binom{3}{1}$, which is 9.

For $k = 3$, using the formula, we have $\frac{6}{3} \binom{2}{2}$ which is 2.

Thus we have $C_6(k \in \{1, 2, 3\}) = 6 + 9 + 2 = 17$.

Now we calculate the denominator of $\mathbb{E}_{Cn}(A)$. We see the total number of cases is $3^3 = 27$. So we can find $\mathbb{T} - Cn(A)$. We check for $n(A)$, in which through common sense, we can see that if the triplet (a, b, c) represents the number of coins given to each jar, the triplet $(1, 1, 1)$ doesn't satisfy the condition, as all jars have an equal number of coins. Therefore $n(A) = 1$.

Our answer is

$$\mathbb{E}_{Cn}(A) = \frac{Cn(A)}{\mathbb{T} - Cn(A) - n(A)}$$

$$\mathbb{E}_{Cn}(A) = \frac{17}{27 - 17 - 1}$$

$$\mathbb{E}_{Cn}(A) = \boxed{(D) \frac{17}{9}}$$

Solution 4 (Weighted Probabilities)

WLOG label the jars as 1, 2, 3 and the coins as a, b, c . The probability for a given coin to land in a given jar is

$$\frac{1}{3}$$

The number of ways for all three coins to land in the same jar is 3, either all in jar 1, all in jar 2, or all in jar 3. Next, the number of ways to have two coins in one jar and one in another in another is $3^2 \cdot 2 = 18$. Finally, since there are $3^3 = 27$ ways, the last weight is

$$27 - 18 - 3 = 6$$

Thus

$$\frac{1}{27} \cdot 3 \cdot 3 + \frac{1}{27} \cdot 2 \cdot 18 + \frac{1}{27} \cdot 1 \cdot 6$$

Thus we get

$$\boxed{(D) \frac{17}{9}}$$

Solution 5

Denote the expected number of coins in a jar with the most coins as E

Process The 1st coin must add 1 to E . Situation: (1, 0, 0)

$$\text{Hence, } E = 1 + \frac{1}{3} \times (1 + \frac{1}{3} \times 1) + \frac{2}{3} \times (0 + \frac{2}{3} \times 1) = \boxed{(D) \frac{17}{9}}$$

Solution 6

Create a tree diagram and add up the probabilities.

$$P = \frac{3}{9} + \frac{2}{9} + \frac{2}{9} + \frac{2}{9} + \frac{2}{9} + \frac{2}{9} + \frac{1}{9} + \frac{1}{9} = \boxed{(D) \frac{17}{9}}$$

Note:(It does not matter which jar you pick at the head of the diagram so long as you stay consistent) Rames_Jong

Solution 7(Similar to Solution 4)

We have 3 cases namely

All 3 coins are in exactly one jar.2 coins are in one jar and the last one is in one of the other two jars.Each jar has exactly 1 coin.

For the first case, we note that there are exactly 3 ways all coins are in one of the three jars.For the second case, we note that we have 3 ways to select the jar that are 2 coins and 2 ways to select the jar that has 1 coin which results in $3 \cdot 2 = 6$ ways. However, there are $\binom{3}{2}$ ways to select 2 out of the three coins to put in the jar that has 2 coins. This results in $6 \cdot \binom{3}{2} = 18$ ways.Finally, the third case results in $3! = 6$ ways since there are 3 ways to select one of the coins to put in jar 1 and 2 ways to select the remaining coins to put in jar 2 and lastly only the last coin can go in jar 3 which results in 1 way. Obviously, this gives $3 \cdot 2 \cdot 1 = 3! = 6$ ways.Finally, we execute the expected value. The probability for each jar is obviously $(\frac{1}{3})^3 = \frac{1}{27}$. Remember to include the value of each expected value as well. Case 1 has a value of 3, case 2 has a value of 2, and case 1 has a value of 1 since each of these values is exactly the maximum number of coins in each case. We finally have

$\frac{1}{27} \cdot 3 \cdot 3 + \frac{1}{27} \cdot 18 \cdot 2 + \frac{1}{27} \cdot 6 \cdot 1 = \frac{1}{3} + \frac{4}{3} + \frac{2}{9} = \frac{5}{3} + \frac{2}{9} = \frac{15}{9} + \frac{2}{9} = \frac{17}{9}$ so the answer is D .

Solution 8 (Cheese)

(Use if you don't know how expected value works).

Since we know that there has to be at least 1 coin in the jar with the most coins, we can instantly rule out answer choice C , as $5/8 < 1$. We also know E can't be the answer, as there are more cases to get 1 as the most rather than 3. Since we also know there a majority of cases of 2, the answer will lean towards 2, giving us answer choice D . mugim

- q) (AMC 10 10A 2025) Let N be the unique positive integer such that dividing 273436 by N leaves a remainder of 16 and dividing 272760 by N leaves a remainder of 15. What is the tens digit of N ?

(A) 0 (B) 1 (C) 2 (D) 3 (E) 4

Answer: E

Solution 1

The problem statement implies that N divides both $273436 - 16 = 273420$ and $272760 - 15 = 272745$. We want to find $N > 16$ that satisfies both of these conditions. Hence, we can just find the greatest common divisor of the two numbers. $\gcd(273420, 272745) = \gcd(675, 272745) = \gcd(675, 45) = \gcd(0, 45) = 45$ by the Euclidean Algorithm, so the answer is (E) 4.

Note: If an integer a is congruent to an integer b modulo a positive integer c , denoted by $a \equiv b \pmod{c}$, this means that c divides the difference, $|a - b|$ (Euclidean Algorithm)

minor edits by AD_12 and mw2014

Solution 2

We are given that $273436 \equiv 16 \pmod{N}$ and $272760 \equiv 15 \pmod{N}$, from which we get $273420 \equiv 0 \pmod{N}$ and $272745 \equiv 0 \pmod{N}$.

Subtracting the two gives $273420 - 272745 = 675$, which is also divisible by N .

Since N must be greater than 16, the only possible divisors of 675 are 25, 27, 45, 75, 135, 225, and 675. Checking which ones also divide 273436 and 273420 eliminates 25 and 27, but $273420 \equiv 0 \pmod{45}$.

Since both 273420 and 675 are divisible by 45, so is $273420 - 675 = 272745$. Thus, $N = 45$ works, and its tens digit is (E) 4.

Solution 3

We get that: $273436 \equiv 16 \pmod{N}$ and $272760 \equiv 15 \pmod{N}$.

So we also have that: $273420 \equiv 0 \pmod{N}$ and $272745 \equiv 0 \pmod{N}$.

Notice that these are a multiple of 5. Now, we subtract these numbers to get 675. We see that $675 = 25 \cdot 27$. There is a factor of 5 and 9. 273420 and 272745 are multiples of 9 as well, so our answer is just 45. Now the ten's digit is (E) 4.

Solution 4

We are given that $273436 \equiv 16 \pmod{N}$ and $272760 \equiv 15 \pmod{N}$. Subtracting, we get $273436 - 272760 = 676 \equiv 1 \pmod{N}$.

This means N divides 675. Also, since $273436 \equiv 16 \pmod{N}$, we have $273420 \equiv 0 \pmod{N}$.

Thus, N divides both 273420 and 675. Using the Euclidean Algorithm: $273420 - 405 \times 675 = 45$, so $\gcd(273420, 675) = 45$.

Hence $N = 45$, and the tens digit of N is $\boxed{\text{(E)} 4}$.

- r) (AMC 10 10A 2025) The *harmonic mean* of a collection of numbers is the reciprocal of the arithmetic mean of the reciprocals of the numbers in the collection. For example, the harmonic mean of 4, 4, and 5 is

$$\frac{1}{\frac{1}{3}(\frac{1}{4} + \frac{1}{4} + \frac{1}{5})} = \frac{30}{7}$$

What is the harmonic mean of all the real roots of the 4050th degree polynomial

$$\prod_{k=1}^{2025} (kx^2 - 4x - 3) = (x^2 - 4x - 3)(2x^2 - 4x - 3)(3x^2 - 4x - 3) \dots (2025x^2 - 4x - 3)?$$

(A) $-\frac{5}{3}$ (B) $-\frac{3}{2}$ (C) $-\frac{6}{5}$ (D) $-\frac{5}{6}$ (E) $-\frac{2}{3}$

Answer: B

Solution 1

We will need to determine the sum of the reciprocals of the roots. To find the sum of the reciprocals of the roots p, q of the quadratic $ax^2 + bx + c$, we use Vieta's formulas. Recall that $p + q = -b/a$ and $pq = c/a$. Therefore,

$$\frac{1}{p} + \frac{1}{q} = \frac{p+q}{pq} = \frac{\frac{-b}{a}}{\frac{c}{a}} = \frac{-b}{a} \cdot \frac{a}{c} = \frac{-b}{c},$$

which doesn't depend on a .

The sum of the reciprocals of the roots of the quadratic $x^2 - 4x - 3$ is $\frac{-(-4)}{-3} = -4/3$. The same is true for every quadratic in the form $ax^2 - 4x - 3$.

The sum of all the reciprocals of the roots of $ax^2 + bx + c$ is $2025 \cdot \left(-\frac{4}{3}\right)$.

Because we have 2025 quadratics, there are $2 \cdot 2025 = 4050$ total roots. Our

answer is $\frac{1}{4050} \cdot \frac{-4 \cdot 2025}{3} = \boxed{\text{(B)} -\frac{3}{2}}$.

Solution 2 (similar to solution 1 but with quadratic formula)

We first find the general roots for quadratics in the form $kx^2 - 4x - 3$. Using the quadratic formula we have

$$\begin{aligned} x &= \frac{4 \pm \sqrt{16 + 12k}}{2k} \\ &= \frac{4 \pm 2\sqrt{4 + 3k}}{2k} \\ &= \frac{2 + \sqrt{4 + 3k}}{k}, \frac{2 - \sqrt{4 + 3k}}{k} \end{aligned}$$

Since we are asked to add the reciprocals of all 4050 roots in the harmonic mean, we will first add the general roots in terms of k . We have,

$$\begin{aligned}\frac{k}{2 + \sqrt{4 + 3k}} + \frac{k}{2 - \sqrt{4 + 3k}} &= \frac{k(2 - \sqrt{4 + 3k}) + k(2 + \sqrt{4 + 3k})}{2^2 - (\sqrt{4 + 3k})^2} \\ &= -\frac{4k}{3k} = -\frac{4}{3}.\end{aligned}$$

Thus, each pair of roots add up to $-\frac{4}{3}$, and since there are 2025 pairs of roots, the harmonic mean of the desired expression is

$$\begin{aligned}\frac{1}{\frac{1}{4050} \left(2025 \left(-\frac{4}{3} \right) \right)} &= \frac{1}{\frac{1}{2} \left(-\frac{4}{3} \right)} \\ &= \boxed{-\frac{3}{2}}, \boxed{B}.\end{aligned}$$

Solution 3 (Vieta's only)

We are asked to find $\frac{4050}{\sum_{i=1}^{4050} \frac{1}{r_i}}$. By Vieta's, note that $r_1 \cdot r_2 \dots r_{4050} = -\frac{3^{2025}}{2025!} = k$ (k is a constant). Then, note that we are asked to find $\frac{4050}{\sum_{i=1}^n \frac{r_i}{k}}$, and by Vieta's we get that $\sum_{i=1}^n \frac{k}{r_i} = \frac{2025 \cdot 4 \cdot 3^{2024}}{2025!}$, so substituting this in, we get

$$-\frac{4050}{\frac{2025 \cdot 4 \cdot 3^{2024}}{\frac{2025!}{3^{2025}}}} = -\frac{4050 \cdot 3^{2025}}{4050 \cdot 3^{2024} \cdot 2} = \boxed{-\frac{3}{2}}.$$

Solution 4 (Vieta's Formulas)

We have:

$$\prod_{k=1}^{2025} (kx^2 - 4x - 3) = (x^2 - 4x - 3)(2x^2 - 4x - 3)(3x^2 - 4x - 3) \dots (2025x^2 - 4x - 3)$$

So we can analyze each part of this "snake":

According to Vieta's Formulas:

For the first one: $x^2 - 4x - 3$: We have:

$$\begin{cases} x_1 + x_2 = \frac{4}{1} \\ x_1 x_2 = -\frac{3}{1} \end{cases}$$

So we have:

$$\frac{1}{x_1} + \frac{1}{x_2} = \frac{x_1 + x_2}{x_1 x_2} = -\frac{4}{3}$$

For the second one: $2x^2 - 4x - 3$: We have:

$$\begin{cases} x_3 + x_4 = \frac{4}{2} \\ x_3 x_4 = -\frac{3}{2} \end{cases}$$

So we have:

$$\frac{1}{x_3} + \frac{1}{x_4} = \frac{x_3 + x_4}{x_3 x_4} = -\frac{4}{3}$$

For the third one: $3x^2 - 4x - 3$: We have:

$$\begin{cases} x_5 + x_6 = \frac{4}{3} \\ x_5 x_6 = -\frac{3}{3} \end{cases}$$

So we have:

$$\frac{1}{x_5} + \frac{1}{x_6} = \frac{x_5 + x_6}{x_5 x_6} = -\frac{4}{3}$$

.....

Then we should check the last one. For the 2025th. one: $2025x^2 - 4x - 3$: We have:

$$\begin{cases} x_{4049} + x_{4050} = \frac{4}{2025} \\ x_{4049} x_{4050} = -\frac{3}{2025} \end{cases}$$

So we have:

$$\frac{1}{x_{4049}} + \frac{1}{x_{4050}} = \frac{x_{4049} + x_{4050}}{x_{4049} x_{4050}} = -\frac{4}{3}$$

$$\begin{aligned} \therefore H.M. &= \frac{1}{\frac{1}{4050} \sum_{i=1}^{4050} \frac{1}{x_i}} \\ &= \frac{4050}{2025 \times \left(-\frac{4}{3}\right)} \\ &= -\frac{3}{2} \end{aligned}$$

Therefore, the final answer is $-\frac{3}{2}$.

Solution 5 (Vieta's Formulas)

For a general quadratic polynomial $ax^2 + bx + c = 0$ with roots p and q , the sum of the reciprocals is:

$$\frac{1}{p} + \frac{1}{q} = \frac{p+q}{pq}$$

Using Vieta's formulas ($p+q = -\frac{b}{a}$ and $pq = \frac{c}{a}$):

$$\frac{1}{p} + \frac{1}{q} = \frac{-\frac{b}{a}}{\frac{c}{a}} = -\frac{b}{c}$$

This result demonstrates that the sum of the reciprocals depends only on the coefficients b and c , and is independent of the leading coefficient a . Since the overall polynomial is a product of quadratics, and all these quadratics share the same b and c coefficients, the sum of the reciprocals for the roots of every individual quadratic is the same. We use the coefficients from one of the quadratics, which is $ax^2 - 4x - 3 = 0$. Thus, $b = -4$ and $c = -3$.

$$\sum(\text{reciprocals}) = -\frac{b}{c} = -\frac{(-4)}{(-3)} = -\frac{4}{3}$$

We substitute the calculated sum of the reciprocals into the formula for the harmonic mean of 2 numbers:

$$\text{HM} = \frac{2}{-\frac{4}{3}} = 2 \times \left(-\frac{3}{4}\right) = -\frac{6}{4} = -\frac{3}{2}$$

The harmonic mean of the roots of any single quadratic is $-\frac{3}{2}$. Since the sum of the reciprocals is the same for all quadratics, the harmonic mean of all the roots of the overall polynomial is also $\boxed{\text{(B)} \quad -\frac{3}{2}}$.

Note It is important to note that the question asks for the sum of all *real* roots. We must therefore be careful to make sure that the 4050 roots of the polynomial are all real, and also that they are distinct, as otherwise e.g. double roots would be counted twice, when the problem requires us to only count each root once (i.e. regardless of multiplicity). Accordingly, we compute that each quadratic factor $(kx^2 - 4x - 3)$ (for $1 \leq k \leq 2025$) has discriminant $(-4)^2 - 4 \cdot k \cdot (-3) = 16 + 12k$, which is clearly positive as $k \geq 1$. This means the 2 roots of each quadratic factor are indeed real and distinct (i.e. none of these quadratics has a double root), so it only remains to show that these roots are also distinct from those of all the *other* quadratic factors. Hence, if we assume that a is a common root of 2 different quadratic factors simultaneously, say $(px^2 - 4x - 3)$ and $(qx^2 - 4x - 3)$ with $1 \leq p, q \leq 2025$ and $p \neq q$, then we must have $pa^2 - 4a - 3 = qa^2 - 4a - 3 = 0$, which implies $pa^2 = qa^2$. This equation factorises as $(p - q)a^2 = 0$, and because $p \neq q$, i.e. $p - q \neq 0$, it further reduces to simply $a^2 = 0$, and hence $a = 0$. But then it follows that $pa^2 - 4a - 3 = p \cdot 0^2 - 4 \cdot 0 - 3 = -3 \neq 0$, yielding a contradiction, so a cannot in fact have been a common root of 2 different quadratic factors, and thus we finally deduce that all 4050 roots are distinct, as required. Shadowleafy

Solution 6

Let the polynomial be $f(x)$, and denote the 4050 roots to be $x_1, x_2, \dots, x_{4050}$. Hence,

$$\text{HM} = \frac{4050}{\frac{1}{x_1} + \frac{1}{x_2} + \dots + \frac{1}{x_{4050}}}$$

We can multiply the numerator and denominator of this fraction by $x_1 x_2 \dots x_{4050}$ to create symmetric sums, which yields

$$\text{HM} = \frac{4050(x_1 x_2 \dots x_{4050})}{x_2 x_3 \dots x_{4050} + x_1 x_3 \dots x_{4050} + \dots + x_1 x_2 \dots x_{4049}}$$

By Vieta's Formulas, since $f(x)$ is of even degree, the product of its roots, $x_1 x_2 \dots x_{4050}$, is just the constant term of $f(x)$, call it c_0 . Likewise, the denominator of our harmonic mean, $x_2 x_3 \dots x_{4050} + x_1 x_3 \dots x_{4050} + \dots + x_1 x_2 \dots x_{4049}$, is the negated coefficient of x in the standard form of $f(x)$. Let the coefficient of x in the standard form of $f(x)$ be c_1 . Note that we do not have to worry about dividing by the coefficient of x^{4050} when using Vieta's Formulas, as they will eventually cancel out in our harmonic mean calculations.

So,

$$HM = \frac{4050c_0}{-c_1}.$$

The constant term in $f(x)$ is just $c_0 = (-3)^{2025}$. For the coefficient of the x term in $f(x)$, there are $\binom{2025}{2024} = 2025$ ways to choose 2024 of the trinomials to include a -3 , and the one trinomial not chosen will include a $-4x$. Hence, $c_1 = 2025 \cdot (-3)^{2024} \cdot (-4)$.

Finally,

$$HM = \frac{4050 \cdot (-3)^{2025}}{-2025 \cdot (-3)^{2024} \cdot (-4)} = \frac{2 \cdot (-3)}{-(-4)} = \boxed{(B) -\frac{3}{2}}.$$

Solution 7 (Cheese)

We can try replacing the 2025 in the problem with smaller numbers like 1 and 2. For 1, we have the quadratic $x^2 - 4x - 3$. Using the quadratic formula, we have the roots are $2 + \sqrt{7}$ and $2 - \sqrt{7}$. The harmonic mean of these two is

$$\frac{2}{\frac{1}{2+\sqrt{7}} + \frac{1}{2-\sqrt{7}}} = \frac{2}{\frac{\sqrt{7}-2}{3} + \frac{2+\sqrt{7}}{-3}} = \frac{2}{\frac{-4}{3}} = -\frac{3}{2}.$$

We notice that this is one of the answer choices. Also, given that the random choice of 2025, and that the rest of the answer choices are also simple fractions,

we can safely guess that the answer is $\boxed{(B) -\frac{3}{2}}$

Solution 8 (Symmetry of Unweighted Means)

Note that all the roots are real since for each quadratic,

$$\Delta = b^2 - 4ac = (-4)^2 - 4k(-3) = 16 + 12k > 0$$

for $k \geq 1$. We will use the fundamental theorem of symmetric polynomials. By Vieta's formulas, we have for each quadratic $ax^2 + bx + c = kx^2 - 4x - 3$:

$$\frac{1}{r} + \frac{1}{s} = \frac{r+s}{rs} = \frac{-b/a}{c/a} = -\frac{b}{c} = -\frac{4}{3}$$

Thus, $HM(r, s) = \frac{2}{\frac{1}{r} + \frac{1}{s}} = \frac{2}{-\frac{4}{3}} = -\frac{3}{2}$. Since the unweighted harmonic mean is symmetric in its arguments, we can group terms to get

$$HM(r_1, r_2, r_3, \dots, r_{4050}) = HM\left(HM(r_1, r_2), HM(r_3, r_4), \dots, HM(r_{4049}, r_{4050})\right) = \boxed{(B) -\frac{3}{2}}$$

Note: For an unweighted symmetric mean M , we have for $d|n$, $M\{x_k\}_{k=1}^n = M\{M\{x_{ds+k}\}_{k=1}^d\}_{s=0}^{n/d-1}$ and $M\{\bar{x}\}_k = \bar{x}$ for constant mean $\bar{x}$. You can also check this individually for HM:

$$HM(x_1, \dots, x_n) = n(x_1^{-1} + \dots + x_n^{-1})^{-1} = \frac{n}{d} \underbrace{\left[d(x_1^{-1} + \dots + x_d^{-1}) + \dots + d(x_{(n-1)d+1}^{-1} + \dots + x_n^{-1}) \right]}_{\frac{n}{d} \text{ terms}}$$

or by checking arithmetic mean and using $\text{HM}\{x_k\}_k = \frac{1}{\text{AM}\left\{\frac{1}{x_k}\right\}_k}$.

Solution 9 (Easy-ish to understand)

Note: I will try my best to make this solution look less intimidating to read as the other ones. This is my third time writing a solution, so I'm not very used to the $\text{\LaTeX}$.

The harmonic mean that the question is asking is

$$\frac{1}{\frac{r_1 + r_2 + \dots + r_{4049} + r_{4050}}{4050}}$$

where r_i is between 1 and 4050 and is the reciprocal of all 4050 roots of the polynomial. All we need to find is the sum of the reciprocals of the roots and we can substitute that into this formula.

First of all, notice that there are 2025 quadratic polynomials. They each hold 2 out of the 4050 roots, and the sum of the reciprocals of their roots are $\frac{1}{a} + \frac{1}{b} = \frac{a+b}{ab}$.

Using Vieta's we get $\frac{a+b}{ab} = -\frac{4}{3}$ on the every quadratic because only the coefficient for x^2 changes. Also, $a + b$ and ab have the denominator of a , so when they are divided, the quotient won't change. Therefore, the sums of the reciprocals of the quadratics sum to $2025 \cdot -\frac{4}{3}$.

Now we can substitute:

$$\frac{1}{\frac{2025 \cdot -\frac{4}{3}}{4050}}.$$

This gets us

$$\frac{1}{\frac{2025 \cdot -\frac{4}{3}}{4050}} = \frac{1}{\frac{2025 \cdot (-4)}{3 \cdot 4050}} \tag{38}$$

$$= \frac{1}{\frac{-8100}{12150}} \tag{39}$$

$$= \frac{1}{\frac{-2}{3}} \tag{40}$$

$$= 1 \cdot \frac{3}{-2} \tag{41}$$

$$= -\frac{3}{2} \tag{42}$$

Or...you could just solve the last part yourself easier than my solution!

Solution 10 (reciprocal roots)

We know that if for polynomial $ax^2 + bx + c$ is r_1 and r_2 , then the polynomial with roots $\frac{1}{r_1}$ and $\frac{1}{r_2}$ (reciprocal roots) will be $cx^2 + bx + a$ where the coefficients are flipped (works for any degree polynomial, not just quadratic). Thus, using vieta's formula, the sum of the reciprocal roots for polynomial $kx^2 - 4x - 3$ will be the sum of the roots for polynomial $-3x^2 - 4x + k$ which is $-\frac{-4}{-3} = -\frac{4}{3}$.

There are 2025 polynomials like this, so the sum of roots would be $2025 \cdot (-\frac{4}{3})$. Finding the harmonic mean, we get:

$$\frac{1}{\frac{1}{4050} \cdot \frac{-4 \cdot 2025}{3}} = \frac{1}{\frac{-2}{3}}$$

Or $\boxed{(\mathbf{B}) - \frac{3}{2}}$

- s) (AMC 10 10A 2025) An array of numbers is constructed beginning with the numbers $-1 \quad 3 \quad 1$ in the top row. Each adjacent pair of numbers is summed to produce a number in the next row. Each row begins and ends with -1 and 1 , respectively.

$$\begin{array}{ccccccc} & & -1 & & 3 & & 1 \\ & -1 & & 2 & & 4 & & 1 \\ -1 & & 1 & & 6 & & 5 & & 1 \end{array}$$

If the process continues, one of the rows will sum to 12,288. In that row, what is the third number from the left?

(A) -29 (B) -21 (C) -14 (D) -8 (E) -3

Answer: A

Solution 1

Consider the polynomial $f(x) = -x^2 + 3x + 1$. When we multiply this polynomial by $x + 1$, we are essentially doing the operation given in the problem (When we multiply $p(x)$ by $x + 1$, a term of degree d in the yielded expression is the sum of $1 \cdot (\text{degree } d)$ and $x \cdot (\text{degree } d-1)$ in $p(x)$. This effect is visible in Pascal's Triangle). So, if we let the coefficients of $f(x)$ be the zero row of the array, then the n^{th} row is just the coefficients of $f(x)(x + 1)^n$. The next thing to note is that the sum of the coefficients in any polynomial $p(x)$ is just $p(1)$. Therefore, the sum of the entries in the n^{th} row of the array is $f(1)(1 + 1)^n = 3 \cdot 2^n$. Letting this equal 12288, we get $n = 12$. We are looking for the 3^{rd} term in the 12^{th} row. The 12^{th} row is given by the coefficients of $f(x)(x + 1)^{12} = (-x^2 + 3x + 1)(x + 1)^{12}$. Since the degree of the resulting expression is 14, the third term in the row is just the coefficient of x^{12} in the expression, which is $-\binom{12}{10} + 3\binom{12}{11} + 1 = \boxed{(\mathbf{A}) - 29}$.

Solution 2

If we take a look at the first few rows, we notice that the sum of the terms in each row n is equal to the twice the sum of row $n - 1$. We note the first row is 3 so recognize 12,228 must be equal to 3 times a power of 2. $12,288 = 3 \cdot 4096 = 3 \cdot 2^{12}$. Therefore, we are looking for the 3rd term from the left in the 13th row. From here, we can take a look at the second number of each row, noting that it will always decrease by 1 because it is being added to the -1 on its left. We can compute the 3rd term of each row by adding the 2nd and 3rd term of the previous row, which means that to find the 3rd term of the n th row, we simply have to add all of the 2nd terms of each row starting from the row with the first appearance of the 3rd term, which is 6. Using this, we get the expression $6 + (1 + 0 + (-1) + \dots + (-(n - 5)))$, where n is 13. Plugging it in, we get $6 + (1 + 0 + (-1) + \dots + (-8))$, which tells us that the 3rd number from the left of the 13th row is $\boxed{(\mathbf{A}) - 29}$.

Remark (Why each row is twice the previous row) To prove that the sum of each row is 2 times the previous row, use the equation $(1 + 1)^n$ with the normal Pascal's triangle and use the binomial theorem from there.

Example: In the fifth row of Pascal's triangle, the numbers are 1, 4, 6, 4, 1. The first number can be represented by $\binom{4}{0} = 1$, the second by $\binom{4}{1} = 4$, the third by $\binom{4}{2} = 6$, the fourth by $\binom{4}{3} = 4$, and the last by $\binom{4}{4} = 1$. The sum of these five numbers is 16 which is also 2^4 . Using the binomial theorem with $(1 + 1)^4$, we get $1 + 4 + 6 + 4 + 1 = 16$, the same results in Pascal's triangle. We could also solve $(1 + 1)^4$ by converting it to $2^4 = 16$, proving the identity that all rows in Pascal's triangle are exactly twice the previous row and can be represented by 2^n .

Solution 3 - Fast and Effective!

<3 minute solve!

Add all the numbers up in the first row. You get 3. Now, add all the numbers up on the second row. You get 6. Notice that as the rows keep going, the sum of the numbers in the row keeps doubling. When you repeat this process, you realize that you reach 12288 on the 13th row.

Knowing this, we can use this pattern to find the solution quickly. We know that the first number will always be -1 , so we can ignore that. Knowing this, all you have to do now is to add up the second and third numbers 13 times. This is already done for us three times, so we just have to do it ten more. Note that you do not have to do this process for all the numbers, only the second and third.

Doing this 13 times (since $\frac{12288}{3} = 4096 = 2^{12} = 2^{13-1}$), you get the following string of numbers (starting from the first one): 1, 4, 6, 7, 7, 6, 4, 1, -3 , -8 , -14 , -21 , and finally, $\boxed{\text{(A)} - 29}$.

Solution 4

From adding the sum of all terms in the first few rows, we see that $S(r(1)) = 3$, $S(r(2)) = 6$, $S(r(3)) = 12$, $S(r(4)) = 24$, ... and so on. We can deduce that $S(r(n)) = 3 \cdot 2^{n-1}$ is the sum of all numbers in row n . Now, set $3 \cdot 2^{n-1} = 12288$ so we have $2^{n-1} = 4096 = 2^{12}$. It follows that $n = 13$. Now let a_n be the 2^{nd} number of each row, and b_n be the 3^{rd} number of each row. Since, the first number of each row is -1 , $a_n = a_{n-1} - 1$. Additionally, $b_n = b_{n-1} + a_{n-1}$.

n	b_n
$a_1 = 3$	$b_1 = 1$
$a_2 = 2$	$b_2 = 4$
$a_3 = 1$	$b_3 = 6$
$a_4 = 0$	$b_4 = 7$
$a_5 = -1$	$b_5 = 7$
$a_6 = -2$	$b_6 = 6$
$a_7 = -3$	$b_7 = 4$
$a_8 = -4$	$b_8 = 1$
$a_9 = -5$	$b_9 = -3$
$a_{10} = -6$	$b_{10} = -8$
$a_{11} = -7$	$b_{11} = -14$
$a_{12} = -8$	$b_{12} = -21$
$a_{13} = -9$	$b_{13} = -29$

We are asked to compute $b_{13} = \boxed{(A) - 29}$.

Solution 5 (Wish I used this)

So the sum is doubling every time, starting with 3.

$$12288 = 3 \cdot 2^{12}$$

So in the 13th row we go, so we eventually get that the 3rd number is

$$\boxed{(A) - 29}$$

Go check out solution 1, looks interesting.

Solution 6

We firstly notice that the pattern for the sums of each row is $S_n = 3 \cdot 2^n$ where n is the number of the row. Since the sum for the row we are looking for is 12288, We can write the equation $3 \cdot 2^n = 12288$ which solving gives $n = 12$.

Recognizing that the n^{th} row term is basically a scaled version of the first row-term $(-1, 3, 1)$, we can write a polynomial $f(x) = (-1x + 3x + 1x^2)(x + 1)^n$, where $(x + 1)^n$ is kind of a "shifting step" to proceed down the triangle to the n^{th} term.

-you can test for the different values like $(x + 1)^1$ which results to coefficients $(-1, 2, 4, 1)$ which matches the second row down

Now we recognize we can write the polynomial of the 12^{th} term in the form $(a_0 + a_1x + a_2x^2 + \dots + a_mx^m)(1 + x)^n$ which we can simplify using the binomial theorem achieving this: $\sum_{i=0}^m a_i \binom{n}{k-i}$

Plugging the terms in we get: $(-1)\binom{n}{2} + 3\binom{n}{1} + 1\binom{n}{0}$ where $n = 12$ thus giving $\boxed{(A) - 29}$

Solution 7

The first step is recognizing that you multiply the sum by two each row. This is because each number adds to the number down to the left and the number down to the right (this includes the -1 and 1 on the ends). Thus the sum in row n is $3 \cdot 2^{n-1}$ which means the question corresponds to row 13.

Now we can look at the trends in the terms of each row. The first term in each row is -1 , which makes the second term in each row $3, 2, 1, 0, -1 \dots$

This leads us to finding the third term in each row has the pattern of $1, 1 + 3, 1 + 3 + 2, 1 + 3 + 2 + 1, 1 + 3 + 2 + 1 + 0, 1 + 3 + 2 + 1 + 0 + -1 \dots$

Therefore we just need to find 1 plus the sum of 3 through -8 inclusive which by any number of methods computes to $1 - 30 = \boxed{(A) - 29}$.

Solution 8

We first observe the sums of the given rows:

	Sum
1	3
2	6
3	12
4	24

This sequence doubles each time, so the sum of the n th row is

$$3 \cdot 2^{n-1}.$$

Setting this equal to 12,288, we get

$$\begin{aligned} 3 \cdot 2^{n-1} &= 12,288 \\ 2^{n-1} &= 4096 \\ 4096 &= 2^{12} \\ 2^{n-1} &= 2^{12}, \end{aligned}$$

so $n = 13$. Thus, the row with sum 12,288 is the 13th row. Next, notice that every entry in the triangle below is formed by repeatedly summing from the three numbers in the first row: 1, 3, and -1.

$$\begin{array}{ccccc} & -1 & 3 & 1 & \\ & -1 & 2 & 4 & 1 \\ -1 & 1 & 6 & 5 & 1 \end{array}$$

For example, the number 6 in row 3 can be expressed as

$$6 = 1 \cdot -1 + 2 \cdot 3 + 1 \cdot 1,$$

where the coefficients count the number of paths from each top entry. In general, each entry is a weighted sum of the top row, with weights equal to the number of left/right paths leading to that position.

We are searching for the value of the 3rd entry of the 13th row. From the top row to this position, there are 12 moves. To reach the 3rd position from:

- the 1st entry of row 1, we need 2 right moves and 10 left moves.

Number of paths: $\binom{12}{2} = 66$.

Contribution: $66 \cdot -1 = -66$.

- the 2nd entry of row 1, we need 1 right move and 11 left moves.

Number of paths: $\binom{12}{1} = 12$.

Contribution: $12 \cdot 3 = 36$.

- the 3rd entry of row 1, we need 0 right moves.

Number of paths: $\binom{12}{0} = 1$.

Contribution: $1 \cdot 1 = 1$.

Adding the contributions gives

$$-66 + 36 + 1 = -29.$$

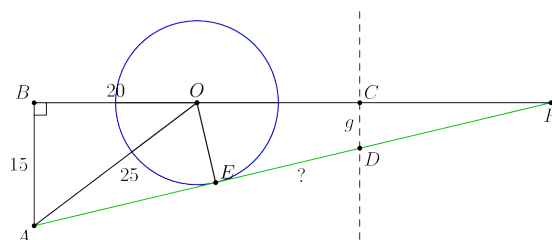
So, the 3rd number in the 13th row is $\boxed{(A) - 29}$.

- t) (AMC 10 10A 2025) A silo (right circular cylinder) with diameter 20 meters stands in a field. MacDonald is located 20 meters west and 15 meters south of the center of the silo. McGregor is located 20 meters east and $g > 0$ meters south of the center of the silo. The line of sight between MacDonald and McGregor is tangent to the silo. The value of g can be written as $\frac{a\sqrt{b}-c}{d}$, where a, b, c , and d are positive integers, b is not divisible by the square of any prime, and d is relatively prime to the greatest common divisor of a and c . What is the value of $a + b + c + d$?

(A) 119 (B) 120 (C) 121 (D) 122 (E) 123

Answer: A

Solution 1



Let the silo center be O , let the point MacDonald is situated at be A , and let the point 20 meters west of the silo center be B . ABO is then a right triangle with side lengths 15, 20, and 25.

Let the point 20 meters east of the silo center be C , and let the point McGregor is at be D with $CD = g > 0$. Also let AD be tangent to circle O at E .

Extend BC and AD to meet at point F . This creates 3 similar triangles, $\triangle ABF \sim \triangle DCF \sim \triangle OEF$. Let the distance between point C and F be x . The similarity ratio between triangles ABF and DCF is then $\frac{\text{longer leg}}{\text{shorter leg}} = \frac{40+x}{15} = \frac{x}{g}$

This is currently unsolvable so we bring in triangle OEF . The hypotenuse of triangle OEF is $OF = 20 + x$ and its shorter leg is the radius of the silo = 10. We can then establish a second similarity relationship between triangles OEF and ABF with $\frac{\text{shorter leg}}{\text{hypotenuse}} = \frac{10}{20+x} = \frac{15}{AF}$

Now we find the hypotenuse of ABF in terms of x using the Pythagorean theorem. $AF^2 = 15^2 + (40 + x)^2$. Which simplifies to $AF^2 = 225 + 1600 + 80x + x^2 = 1825 + 80x + x^2$ So $AF = \sqrt{x^2 + 80x + 1825}$

Plugging back in we get $\frac{10}{20+x} = \frac{15}{\sqrt{x^2 + 80x + 1825}}$. Now we can begin to break this down by multiplying both sides by both denominators. $10(\sqrt{x^2 + 80x + 1825}) = 15(20 + x)$ Dividing both sides by 5 then squaring yields, $4x^2 + 320x + 7300 = 9x^2 + 360x + 3600$ This furthermore simplifies to $5x^2 + 40x - 3700 = 0$ At which point we can divide off a 5 and then apply the quadratic formula on $x^2 + 8x - 740 = 0$ which we take the positive root of.

$$x = \frac{-8 + \sqrt{64 + 2960}}{2} = \frac{-8 + \sqrt{3024}}{2} = \frac{-8 + \sqrt{144 \cdot 21}}{2}.$$

Simplifying yields that $x = 6\sqrt{21} - 4$

Then to solve for g we simply plug $6\sqrt{21} - 4$ back into the first similarity ratio to get $\frac{36+6\sqrt{21}}{15} = \frac{6\sqrt{21}-4}{g}$

Multiply both sides by $15g$ and dividing by $36 + 6\sqrt{21}$ will let us solve for $g = \frac{15(6\sqrt{21}-4)}{36+6\sqrt{21}}$ and after rationalizing the denominator we get $\frac{20\sqrt{21}-75}{3}$.

$20+21+75+3 = \boxed{\text{(A) 119}} \sim \text{Nioronean happyfish0922 (minor edit regarding what was previously point T changed to point E) Avs2010 (diagram)}$

Solution 2 (Coordinates)

Let MacDonald's position be at $(0, 0)$. Then, the center of the silo will be at $(20, 15)$ and McGregor will be at $(40, 15 - g)$.

Let the line of sight between MacDonald and McGregor be represented by the line $y = mx$, where $m = \frac{15-g}{40}$. As the radius of the silo is 10, we can construct the equation of a circle and get the system of equations:

$$y = mx$$

$$(x - 20)^2 + (y - 15)^2 = 100$$

Substituting, we get:

$$(x - 20)^2 + (mx - 15)^2 = 100$$

$$x^2 - 40x + 400 + m^2x^2 - 30mx + 225 = 100$$

$$(m^2 + 1)x^2 - (30m + 40)x + 525 = 0$$

We want a value of m such that this equation has 1 solution, so we set the discriminant to be equal to 0.

$$(30m + 40)^2 - 4(m^2 + 1)(525) = 0$$

$$900m^2 + 2400m + 1600 - 2100m^2 - 2100 = 0$$

$$12m^2 - 24m + 5 = 0$$

$$m = \frac{24 \pm \sqrt{576 - 240}}{24}$$

$$m = \frac{6 \pm \sqrt{21}}{6}$$

The slope obviously has to be less than 1, so we take the negative root $\frac{6-\sqrt{21}}{6}$.

Substituting back for m and solving for g ,

$$\frac{15 - g}{40} = \frac{6 - \sqrt{21}}{6}$$

$$15 - g = \frac{20}{3}(6 - \sqrt{21})$$

$$g = \frac{20\sqrt{21} - 75}{3}$$

Finally, $a = 20$, $b = 21$, $c = 75$, and $d = 3$, summing up to a total of $20 + 21 + 75 + 3 = \boxed{\text{(A) 119}}$

Solution 3 (Trig)

(I don't know how to make a diagram or latex very well oops)

Let the silo center be O , let the point MacDonald is situated at be A , and let the point 15 meters south of the silo center be B . ABO is then a right triangle with side lengths 15, 20, 25.

Also let the tangency point be E . Then, because a tangent line is defined as having a right angle to the radius OE , triangle AEO is also a right triangle with side lengths $5\sqrt{21}$, 10, 25 since we know AO is 25 from pythag on ABO and that OE is 10 because it's a radius, so AE follows from pythag on AEO .

Let the position McGregor is in be G and let the point that is both directly south of McGregor and directly east of MacDonald be F . (For reference, F is 20 meters east and 15 meters south of O , which is also 40 meters east of D and $15 - g$ meters south of G .)

Now, we can see that the triangle AFG is also a right triangle. Since G is on the line AE by definition and F is on line AB because they are all on the same horizontal line 15 meters south of O , $\angle GAF = \angle EAB$. This leads us to conclude that $\tan \angle GAF = \frac{15-g}{40} = \tan \angle EAB$.

From here, we can use tangent subtraction formula to find $\tan(\angle EAB)$.

We know that $\tan(\angle OAB) = \frac{15}{20} = \frac{3}{4}$ and that $\tan(\angle OAE) = \frac{10}{5\sqrt{21}} \cdot \frac{2}{\sqrt{21}}$. Since $\angle EAB = \angle OAB - \angle OAE$, we have that

$$\tan(\angle EAB) = \tan(\angle OAB - \angle OAE) = \frac{\tan(\angle OAB) - \tan(\angle OAE)}{1 + \tan(\angle OAB)\tan(\angle OAE)} = \frac{\frac{3}{4} - \frac{2}{4\sqrt{21}}}{1 + \frac{6}{4\sqrt{21}}}$$

Multiplying the top and bottom by $4\sqrt{21}$, this equals

$$\frac{3\sqrt{21}-8}{4\sqrt{21}+6} = \frac{1}{2} \cdot \frac{3\sqrt{21}-8}{2\sqrt{21}+3} \text{ (factoring out } \frac{1}{2} \text{ is personal preference)}$$

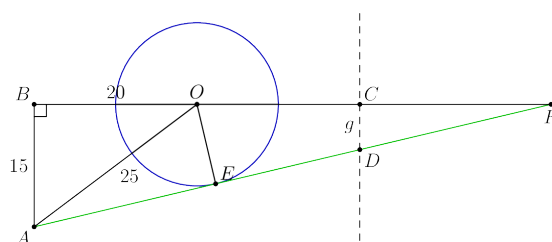
Now, rationalizing the denominator by multiplying the top and bottom by $2\sqrt{21} - 3$, we get this equals

$$\frac{1}{2} \cdot \frac{(3\sqrt{21}-8)(2\sqrt{21}-3)}{(2\sqrt{21}+3)(2\sqrt{21}-3)} = \frac{1}{2} \cdot \frac{150-25\sqrt{21}}{75} = \frac{6-\sqrt{21}}{6}$$

Recalling that this implies $\frac{15-g}{40} = \frac{6-\sqrt{21}}{6}$, we get that $g = \frac{20\sqrt{21}-75}{3}$. $20 +$

$$21 + 75 + 3 = \boxed{\text{(A) } 119}$$

Solution 4



We use the point names given in Solution 1.

By Pythagorean Theorem (twice), we have

$$AE = \sqrt{AO^2 - OE^2} = \sqrt{AB^2 + BO^2 - 10^2} = \sqrt{525} = 5\sqrt{21}.$$

Similarly, we find that

$$DE = \sqrt{DO^2 - OE^2} = \sqrt{CD^2 + CO^2 - 10^2} = \sqrt{300 + g^2}.$$

Note that the altitude from D to AB (call the foot H) has length 40. Also, $AD = AE + DE = 5\sqrt{21} + \sqrt{300 + g^2}$ and $AH = 15 - g$. By Pythagoras on $\triangle ADH$, we have

$$\left(5\sqrt{21} + \sqrt{300 + g^2}\right)^2 = 40^2 + (15 - g)^2.$$

Expanding,

$$525 + 10\sqrt{21} \cdot \sqrt{300 + g^2} + 300 + g^2 = 1600 + 225 - 30g + g^2$$

and simplifying gives

$$10\sqrt{6300 + 21g^2} = 1000 - 30g.$$

Dividing by 10 and squaring,

$$6300 + 21g^2 = 10000 - 600g + 9g^2$$

and putting it in standard form,

$$0 = 12g^2 + 600g - 3700.$$

Dividing by 4 and applying the quadratic formula,

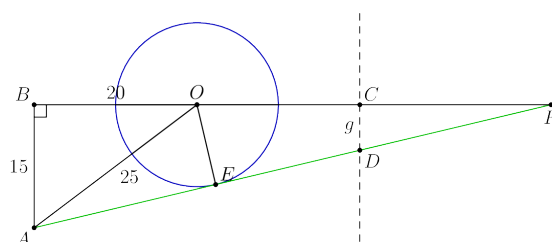
$$g = \frac{-150 \pm \sqrt{22500 - 4(3)(-925)}}{6} = \frac{-75 \pm 20\sqrt{21}}{3}.$$

From the answer form, we take the positive root, so

$$g = \frac{20\sqrt{21} - 75}{3}.$$

Finally, we have $20 + 21 + 75 + 3 = \boxed{(A) 119}$.

Solution 5 (trig similarity)

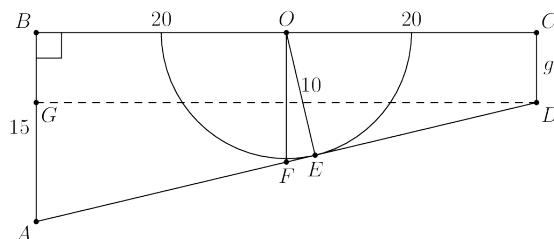


First, taking the Pythagorean Theorem on $\triangle BOD$ and $\triangle AOE$ gives $AO = 25$, $AE = 5\sqrt{21}$. By cosine angular addition, $\cos(\angle BOD + \angle AOE) = \cos \angle BOE = \frac{8 - 3\sqrt{21}}{25}$, so $\cos \angle EOC = \frac{3\sqrt{21} - 8}{25}$.

Dropping the altitude from E to BC and labelling the foot P , we have that $OP = 10 \cdot \frac{3\sqrt{21} - 8}{25} = \frac{6\sqrt{21} - 16}{5}$, so the ratio of BP to BC is $\frac{3\sqrt{21} + 42}{100}$. This is the same as the ratio of AE to AD , so $AD = 5\sqrt{21} \cdot \frac{100}{3\sqrt{21} + 42} = \frac{20}{3} \cdot (2\sqrt{21} - 3)$.

Dropping the altitude from D to AB and labelling the foot Q , AQ can be (tediously) computed to be $\frac{20}{3} \cdot \sqrt{(2\sqrt{21} - 3)^2 - 6^2} = \frac{20}{3} \cdot \sqrt{57 - 12\sqrt{21}} =$

Solution 6


$$\frac{OE}{OF} = \frac{DG}{DA}$$
$$\frac{OE}{OF} = \frac{DG}{GA}$$

$$\begin{aligned}\frac{\frac{10}{15+g}}{2} &= \frac{40}{\sqrt{40^2 + (15-g)^2}} \\ 2(15+g) &= \sqrt{40^2 + (15-g)^2} \\ 3g^2 + 150g - 925 &= 0 \\ g &= \frac{-150 + \sqrt{22500 + 11100}}{6} \\ g &= \frac{20\sqrt{21} - 75}{3}\end{aligned}$$

The answer is $20 + 21 + 75 + 3 = \boxed{\text{(A)} 119}$

We start by making the right triangle as in Solution 6. $AG = (15 - g)$ Since OE is a radius to a tangent line, $\angle OEA = \angle OED = 90^\circ$. Drawing OD and using the Pythagorean Theorem on triangle OCD gives $OD = \sqrt{g^2 + 400}$. Using the Pythagorean Theorem on triangle OED gives $ED = \sqrt{g^2 + 300}$. Using the Pythagorean Theorem on triangle OBA gives $OA = 25$ Using the Pythagorean Theorem on triangle OEA gives $EA = 5\sqrt{21}$.

Using the Pythagorean Theorem on triangle AGD gives $(15 - g)^2 + 40^2 = (5\sqrt{21} + \sqrt{g^2 + 300})^2$ Expanding the squares, rearranging, and simplifying gives the quadratic $3g^2 + 150g - 925 = 0$. Completing the square and taking only the positive square root seems like the easiest method to find g here

$$3(q^2 + 50q) = 925$$

$$3(g^2 + 50g + 625) = 925 + 3(625)$$

$$(g + 25) = \sqrt{\frac{2800}{3}} = \frac{20\sqrt{21}}{3}$$

$$g = \frac{20\sqrt{21} - 75}{3}$$

Finally, $a + b + c + d = 20 + 21 + 75 + 3 = \boxed{\text{(A) 119}}$

- u) (AMC 10 10A 2025) A set of numbers is called *sum-free* if whenever x and y are (not necessarily distinct) elements of the set, $x + y$ is not an element of the set. For example, $\{1, 4, 6\}$ and the empty set are sum-free, but $\{1, 4, 5\}$ is not. What is the greatest possible number of elements in a sum-free subset of $\{1, 2, 3, \dots, 20\}$?

(A) 8 (B) 9 (C) 10 (D) 11 (E) 12

Answer: C

Solution 1

Let our subset be $\{11, 12, 13, \dots, 20\}$. If we add any element from the set $\{1, 2, 3, \dots, 10\}$ to our current subset, we will have to remove at least one element from our subset. We cannot add 10 because if we do, the sum of 10 and itself is 20. Hence, the maximum size of our subset is $\boxed{\text{(C) 10}}$.

Solution 2 (Quicksolve)

Notice that an odd number plus an odd number always results in an even number. Consider the subset of all odd numbers $S = \{1, 3, 5, \dots, 19\}$. The addition of any even number will result in a violation of the rules, so the maximum number of elements in this subset is $\boxed{\text{(C) 10}}$.

Note (Those who know) This problem is essentially the same as [2022 AMC 10B Problem 14](#).

Solution 3

Let $S = \{1, 2, 3, \dots, 20\}$. We are looking for the largest possible sum-free subset $A \subseteq S$.

First, we can show that a sum-free set of size 10 is possible. Consider the set $A_1 = \{11, 12, \dots, 20\}$. The smallest possible sum of two elements (not necessarily distinct) from this set is $11 + 11 = 22$. Since $22 > 20$, no sum of two elements from A_1 can be in A_1 . The size of A_1 is $20 - 11 + 1 = 10$. Thus, a sum-free set of size 10 does exist.

Next, we prove that no sum-free set $A \subseteq S$ can have 11 or more elements. Let m be the largest element in A . Since A is sum-free and $m \in A$, for any $x \in A$, we must have $m - x \notin A$ (unless $x = m - x$, in which case $x \notin A$). This is because if both x and $m - x$ were in A , their sum $x + (m - x) = m$ would be in A , which is a contradiction.

We can partition the set $\{1, 2, \dots, m\}$ into pairs that sum to m , plus any leftover elements.

Case 1: m is even. Let $m = 2k$. The numbers in $\{1, \dots, m\}$ can be grouped as: $\{(1, 2k - 1), (2, 2k - 2), \dots, (k - 1, k + 1)\}$ along with the numbers k and $m = 2k$.

- A contains m .
- A cannot contain k , because $k + k = 2k = m$, which is in A .
- A can contain at most one number from each of the $k - 1$ pairs.

The maximum number of elements A can contain from $\{1, \dots, m\}$ is $1 + 0 + (k - 1) = k = m/2$. Since $m \leq 20$, the maximum size of A is $m/2 \leq 20/2 = 10$.

Case 2: m is odd. Let $m = 2k - 1$. The numbers in $\{1, \dots, m\}$ can be grouped as: $\{(1, 2k - 2), (2, 2k - 3), \dots, (k - 1, k)\}$ along with $m = 2k - 1$.

- A contains m .
- A can contain at most one number from each of the $k - 1$ pairs.

The maximum number of elements A can contain from $\{1, \dots, m\}$ is $1 + (k - 1) = k$. Since $m = 2k - 1$, we have $k = (m + 1)/2$. The largest possible odd m is 19. If $m = 19$, $k = (19 + 1)/2 = 10$. So the maximum size of A is $(m + 1)/2 \leq (19 + 1)/2 = \boxed{\text{(C) } 10}$.

In all cases, a sum-free subset of S can have at most 10 elements. Since we found a set with 10 elements, the greatest possible number of elements is 10.

Solution 4 (Another quicksolve)

Note that selecting integers from $\{11, 12, \dots, 20\}$ also satisfy the constraints of the problem. The minimum sum of any two members of the set is $11 + 11 = 22$ which is greater than any of the elements.

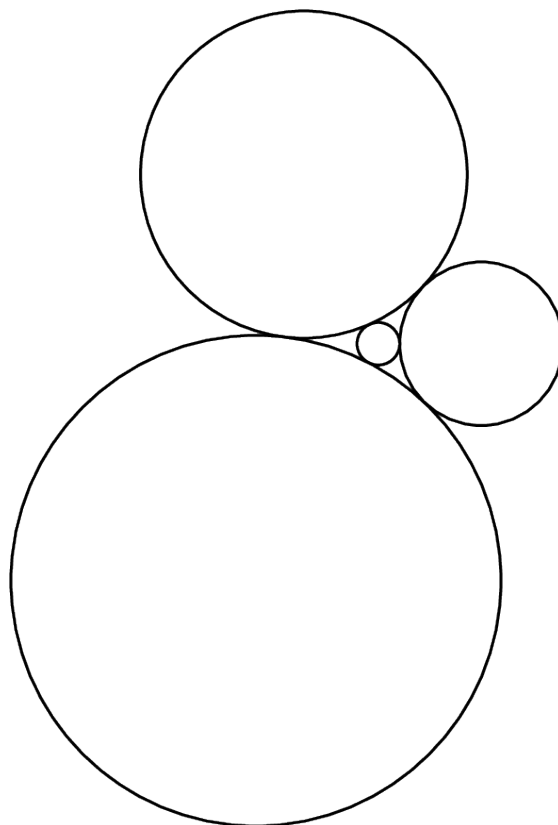
Solution 5 (with quick proof)

As all of the above solutions point out, the set $\{11, \dots, 20\}$ satisfies the required condition, and so the greatest size of a sum-free set is no less than 10. How do we know that 11 isn't possible? Induction is our friend here.

We want to prove that the set $\{1, \dots, 2n\}$ has a sum-free set of size no more than n . It is obvious for $n = 1$. Suppose it is true for $n - 1$. Now let S be a sum-free subset of $\{1, \dots, 2n\}$ and that it has at least $n + 1$ elements. The intersection of S and $\{1, \dots, 2n - 2\}$ is sum-free, so it has no more than $n - 1$ elements. So we must include both $2n - 1$ and $2n$ into S . If so, then 1 and n cannot belong to S .

We are now searching for $n - 1$ more elements. But we can make $n - 2$ 'pigeonholes': $\{2, 2n - 2\}, \{3, 2n - 3\}, \dots, \{n - 1, n + 1\}$, seeing that they can provide no more than $n - 2$ elements because each pigeonhole will have a sum of $2n$.

- v) (AMC 10 10A 2025) A circle of radius r is surrounded by three circles, whose radii are 1, 2, and 3, all externally tangent to the inner circle and externally tangent to each other, as shown in the diagram below.



What is r ?

- (A) $\frac{1}{4}$ (B) $\frac{6}{23}$ (C) $\frac{3}{11}$ (D) $\frac{5}{17}$ (E) $\frac{3}{10}$

Answer: B

Solution 1

Descartes' Circle Formula (curvatures $k_i = \frac{1}{r_i}$)

$$k_4 = k_1 + k_2 + k_3 \pm 2\sqrt{k_1k_2 + k_2k_3 + k_3k_1}.$$

For radii 1, 2, 3 we have

$$k_1 = 1, \quad k_2 = \frac{1}{2}, \quad k_3 = \frac{1}{3}.$$

Compute the sum and the square-root term

$$k_1 + k_2 + k_3 = \frac{11}{6}, \quad k_1k_2 + k_2k_3 + k_3k_1 = 1.$$

Therefore

$$k_4 = \frac{11}{6} \pm 2.$$

Choose the plus sign for the small circle tangent externally to the three given circles

$$k_4 = \frac{11}{6} + 2 = \frac{23}{6}, \quad r_4 = \frac{1}{k_4} = \boxed{\text{(B)} \frac{6}{23}}.$$

Jonathanmode

Solution 2 (Descartes' Theorem, Detailed Derivation)

We are given the radii of three circles and are asked to find an external tangent to the radius of the circle.

We can use Descartes' Theorem to find it using curvatures (reciprocals of the radii). The curvatures are

$$k_1 = 1, \quad k_2 = \frac{1}{2}, \quad k_3 = \frac{1}{3}.$$

Descartes' Circle Formula states

$$(k_1 + k_2 + k_3 + k_4)^2 = 2(k_1^2 + k_2^2 + k_3^2 + k_4^2).$$

Plugging in, we get

$$\left(1 + \frac{1}{2} + \frac{1}{3} + k_4\right)^2 = 2\left(1 + \frac{1}{4} + \frac{1}{9} + k_4^2\right).$$

Simplifying,

$$\left(\frac{11}{6} + k_4\right)^2 = 2\left(\frac{49}{36} + k_4^2\right).$$

$$\frac{121}{36} + \frac{11}{3}k_4 + k_4^2 = \frac{49}{18} + 2k_4^2.$$

$$k_4^2 - \frac{11}{3}k_4 - \frac{23}{36} = 0.$$

Using the quadratic formula,

$$k_4 = \frac{\frac{11}{3} \pm \sqrt{\frac{121}{9} - (-\frac{23}{9})}}{2} = \frac{\frac{11}{3} \pm \sqrt{\frac{144}{9}}}{2} = \frac{\frac{11}{3} \pm \frac{12}{3}}{2} = \frac{23}{6}, -\frac{1}{6}.$$

Since the curvature is the reciprocal of the radius, the two tangent circles possible are

$$r = \frac{1}{|k_4|} = \frac{6}{23} \text{ and } 6.$$

The circle with radius 6 is the one that contains all of the three tangent circles inside it, so the answer is

$$\boxed{(B) \frac{6}{23}}.$$

Note: Instead of solving the quadratic directly, we can multiply it by 36 and then use the rational root theorem to note that the solution must have a numerator of either 1 or 23, meaning that the denominator of the radius must be 1 or 23. It can't be 1, since then the radius would be greater than 1 (which is impossible, looking at answer choices), so the denominator is 23, and there

is only one answer choice whose denominator is 23, $\boxed{(B) \frac{6}{23}}$

Solution 3(Heron's)

Let the center of the circle radius 2 be A , radius 1 be B , radius 3 be C . Let the unknown circle's center be P and radius be r .

We know $\overline{AB} = 2 + 1 = 3$, $\overline{AC} = 3 + 2 = 5$, $\overline{BC} = 1 + 3 = 4$.

Thus, $\triangle ABC$ is a 3-4-5 triangle, where $\angle B = 90^\circ$.

Let X, Y be on $\overline{AB}, \overline{BC}$ respectively such that $\angle PXB = \angle PYB = 90^\circ$.

By Heron's formula, the area of $\triangle BPC$ is

$$\sqrt{s(s - \overline{BP})(s - \overline{CP})(s - \overline{BC})} = \sqrt{(4 + r) \cdot 1 \cdot 3 \cdot r} = \sqrt{3r^2 + 12r}.$$

By using the simpler area formula, we can find $\overline{PY}$.

$$\overline{PY} = \frac{[BPC]}{\overline{BC}/2} = \frac{\sqrt{3r^2 + 12r}}{2}.$$

Since $\angle PXB = \angle B = \angle PYB = 90^\circ$, $XPYB$ is a rectangle, and we can say

$$\overline{PX}^2 = \overline{BP}^2 - \overline{PY}^2,$$

or

$$\begin{aligned} \overline{PX}^2 &= (1 + r)^2 - \frac{3r^2 + 12r}{4} = \frac{r^2 - 4r + 4}{4} \\ &\Rightarrow \overline{PX} = \frac{|r - 2|}{2}. \end{aligned}$$

We then find the area of ABP via Heron's formula, and again, by the simple area formula we have that

$$\overline{PX} = \frac{[ABP]}{\overline{AB}/2} = \frac{2\sqrt{2r^2 + 6r}}{3}$$

Now we can equate the two and solve the equation:

$$\begin{aligned} \frac{|r - 2|}{2} &= \frac{2\sqrt{2r^2 + 6r}}{3} \\ \Rightarrow 3(|r - 2|) &= 4\sqrt{2r^2 + 6r} \\ \Rightarrow 9r^2 - 36r + 36 &= 32r^2 + 96r \\ \Rightarrow 23r^2 + 132r - 36 &= 0 \end{aligned}$$

Utilizing the quadratic formula, the only positive root is $r = \frac{6}{23} = \boxed{B}$.

Note: By the rational root theorem, a rational root of the final quadratic must have denominator 1 or 23 in simplest form, which eliminates all choices except B.

Edited by GarudS

Solution 4 (Variation of Solution 3)

Follow the same starting steps as solution 3 (drawing right triangle ABC connecting the centers of the three larger circles and connecting points A, B, and C to the center of the smaller circle). Using Heron's formula, we can easily find the areas of the three smaller triangles that make up triangle ABC in terms of r . Since the sum of these terms is an integer, or 6, we can theorize

that the square roots have to simplify out into rational numbers. Out of all the answer choices, $\boxed{(B)}$, or $\frac{6}{23}$, is the only one that works.(will add more later)

Solution 5 (Coordinate Geometry)

Let the center of the circle radius 2 be A , radius 1 be B , radius 3 be C . Let the unknown circle's center be P . Since $AB^2 + BC^2 = AC^2$, $\triangle ABC$ is a right triangle, with right angle at B . Let $B = (0, 0)$, $A = (3, 0)$, $C = (0, 4)$, and $P = (x, y)$. Since the large circles are tangent to the small one, we have, if the radius of the smallest circle is r ,

$$\begin{aligned}x^2 + y^2 &= (r + 1)^2 = r^2 + 2r + 1 \\(x - 3)^2 + y^2 &= (r + 2)^2 = r^2 + 4r + 4 \\x^2 + (y - 4)^2 &= (r + 3)^2 = r^2 + 6r + 9\end{aligned}$$

Subtracting (1) from (2) and (1) from (3), we get

$$\begin{aligned}-6x + 9 &= 2r + 3 \implies -3x = r - 3 \\-8y + 16 &= 4r + 8 \implies -2y = r - 2\end{aligned}$$

To avoid fractions, we multiply (1) to get $36r^2 + 72r + 36 = 36x^2 + 36y^2 = 4(r - 3)^2 + 9(r - 2)^2 = 13r^2 - 60r + 72$. Combining like terms, we have $23r^2 + 132r - 36$. Solving via quadratic formula, we get $\boxed{\frac{6}{23}}$.

Note: Like the last solution, you can just notice that B is the only answer choice with 23 in the denominator.

A note about factoring $23r^2 + 132r - 36$ You don't need to save a lot of time by abusing multiple choice to cheese the root of $23r^2 + 132r - 36$.

$23r^2 + 132r - 36 = (Ax + B)(Cx + D)$. Since 23 is prime, $B = 1$ and $D = 23$, WLOG.

Theorem you can prove and learn: in $Px^2 + Qx + R = (Ax + B)(Cx + D)$: if $n|Q$ and $n^2|R$, then $n|C$ and $n|D$. (This is equivalent to substituting $y = x/n$.) Symmetrically, if $n|Q$ and $n^2|P$, then $n|A$ and $n|B$. (This is equivalent to substituting $y = nx$.)

Back to our problem: Since $6|132$ and $6^2|36$, then $6|A$ and $6|B$. $6 \cdot 6 = 36$ so the factoring is done, up to placing the minus sign: $(23r - 6)(r + 6)$.

Considering the signed-distance geometry of the original equations in r , the positive root $\boxed{6/23}$ is the internally tangent circle's radius, and the absolute value 6 of negative root -6 is the externally tangent circle's radius.

Solution 6 (Trig, LoC)

Start with the same setup as Solution 3. Now, set $\angle PBC$ as θ . Law of Cosine on $\triangle PBC$, you get:

$$\begin{aligned}\cos(\theta) &= \frac{PC^2 - PB^2 - BC^2}{2 \cdot PB \cdot BC} \\&= \frac{(3 + r)^2 - (1 + r)^2 - 16}{8 + 8r}\end{aligned}$$

$$= \frac{r-2}{2+2r}$$

Similarly, use LoC on $\triangle APB$:

$$\begin{aligned}\cos(90^\circ - \theta) &= \sin(\theta) = \frac{PA^2 - PB^2 - AB^2}{2 \cdot PB \cdot AB} \\ &= \frac{(2+r)^2 - (1+r)^2 - 9}{6+6r} \\ &= \frac{r-3}{3+3r}\end{aligned}$$

Now, Pythagorean Identity,

$$\begin{aligned}\sin^2(\theta) + \cos^2(\theta) &= 1 \\ \Rightarrow \left(\frac{r-2}{2+2r}\right)^2 + \left(\frac{r-3}{3+3r}\right)^2 &= 1 \\ \Rightarrow \frac{13r^2 - 60r + 72}{36r^2 + 72r + 36} &= 1 \\ \Rightarrow 23r^2 + 132r - 36 &= 0\end{aligned}$$

Solving, you get $r = \boxed{\frac{6}{23}}$.

Solution 7 (LAST RESORT)

DISCLAIMER: This solution is extremely risky and should only be used if you want to get this problem correct but you either have no time or you can't solve it. Also, if this were the case, it would be a better idea to leave the answer blank and get a guaranteed 1.5 points.

Rulers are permitted on the AMC 10. So, you can measure the diameter of the circle with radius 1. In this case, the circle has an approximate diameter of 2.4 centimeters. Thus, its radius is approximately 1.2 centimeters. Now we can measure the diameter of the smallest circle, which is approximately 0.6 centimeters (a little more, keep that in mind), so its radius is approximately 0.3 centimeters (The real value is a little more than that, just keep that in mind). If 1.2 centimeters correspond to 1 unit, then 0.3 centimeters has to correspond to $\frac{1}{4}$ th of a unit. So we check which answer choice is closest to $\frac{1}{4}$, BUT not exactly $\frac{1}{4}$, since we noted that the measurement of the radius of the smallest circle is not exactly 0.3 centimeters. So, from this, we find that

(B) $\boxed{\frac{6}{23}}$ has the smallest error compared to $\frac{1}{4}$.

Note: Although MAA claims that the diagrams are not drawn to scale, they almost certainly are (there are a few exceptions but this is the general, most likely case).

Solution 8 (Another version of Kissing Circles Theorem)

Yes, I am aware this is the same as solution 2. Now, if you ask me why this is here, before I made this I thought solution 2 used the same version of the Kissing Circles Theorem like solution 1. I wouldn't remove this as it is faster than solution 2 in an essence, but to be fair I rly don't know why I made this :sob:).

Another relation between four curvatures via Kissing Circles Theorem is

$$(k_1 + k_2 + k_3 + k_4)^2 = 2(k_1^2 + k_2^2 + k_3^2 + k_4^2)$$

Then $k_1 = 1$, $k_2 = \frac{1}{2}$, $k_3 = \frac{1}{3}$. Find k_4 .

Substitution gives

$$\left(\frac{11}{6} + k_4\right)^2 = 2\left(1 + \frac{1}{4} + \frac{1}{9} + k_4^2\right)$$

$$\frac{121}{36} + \frac{11}{3}k_4 + k_4^2 = \frac{49}{18} + 2k_4^2$$

$$k_4^2 - \frac{11}{3}k_4 - \frac{23}{36} = 0$$

$$36k_4^2 - 132k_4 - 23 = 0$$

The two values k_4 are $k_4 = \frac{23}{6}$ and $k_4 = -\frac{1}{6}$. Negative curvatures go beyond Plane Euclidean Geometry's definition of external tangencies. Thus $k_4 = \frac{23}{6}$.

It follows that $r_4 = \boxed{\frac{6}{23}}$.

Solution 9 (More in depth of Solution 4)

Let r be the radius of the center circle. We wish to find the value of r . First, connect the centers of the circles with radius 1 and radius 2, radius 1 and radius 3, and radius 2 and radius 3. This forms a triangle with sides 3, 4, and 5. Note that $3^2 + 4^2 = 5^2$. By the converse of the Pythagorean theorem, this is a right triangle. So the area of this triangle is $\frac{(3)(4)}{2} = 6$. Now draw the radii from the center circle connecting to all of the other 3 circle centers. This forms 3 triangles, which we can find the area of in terms of r with Heron's Formula.

Triangle 1: This triangle has sides $r + 1, r + 2, 3$. The area of this triangle is $\sqrt{(r + 3)(2)(1)(r)}$.

Triangle 2: This triangle has sides $r + 1, r + 3, 4$. The area of this triangle is $\sqrt{(r + 4)(3)(1)(r)}$.

Triangle 3: This triangle has sides $r + 2, r + 3, 5$. The area of this triangle is $\sqrt{(r + 5)(3)(2)(r)}$.

Since these three triangles make up the area of the whole triangles, we have the equation $\sqrt{(r + 3)(2)(1)(r)} + \sqrt{(r + 4)(3)(1)(r)} + \sqrt{(r + 5)(3)(2)(r)} = 6$. Now instead of solving this disgusting equation, we can just plug in the answer

choices given. Checking all of them gives $r = \boxed{\text{(B)} \frac{6}{23}}$, which is our final answer.

- w) (AMC 10 10A 2025) Triangle $\triangle ABC$ has side lengths $AB = 80$, $BC = 45$, and $AC = 75$. The bisector of $\angle B$ and the altitude to side $\overline{AB}$ intersect at point P . What is BP ?

(A) 18 (B) 19 (C) 20 (D) 21 (E) 22

Answer: D

Solution 1

Let $CD \perp AB$ with foot D . Right triangles ACD and BCD give $AC^2 = AD^2 + CD^2$, $BC^2 = BD^2 + CD^2$, $AC^2 - BC^2 = AD^2 - BD^2 = (AD - BD)(AD + BD)$.

Since $AD + BD = AB = 80$ and $AC^2 - BC^2 = 75^2 - 45^2 = 3600$, we get the equation $3600 = 80(AD - BD)$. This equation simplifies to $45 = AD - BD$. We can solve the system of equations $AD + BD = 80$ and $AD - BD = 45$ easily via elimination, and we get $AD = \frac{125}{2}$, $BD = \frac{35}{2}$. $CD^2 = AC^2 - AD^2 = 75^2 - \left(\frac{125}{2}\right)^2 = \frac{6875}{4}$, $CD = \frac{25\sqrt{11}}{2}$.

By Angle Bisector Theorem, $\frac{DP}{PC} = \frac{DB}{BC} = \frac{\frac{35}{2}}{45} = \frac{7}{18}$, $PC = CD - DP$ thus, $18DP = 7(CD - DP)$, $25DP = 7CD$, $DP = \left(\frac{7}{25}\right)CD = \left(\frac{7}{25}\right)\left(\frac{25\sqrt{11}}{2}\right) = \frac{7\sqrt{11}}{2}$. $BP^2 = BD^2 + DP^2 = \left(\frac{35}{2}\right)^2 + \left(\frac{7\sqrt{11}}{2}\right)^2 = \frac{1225}{4} + \frac{49(11)}{4} = \frac{1764}{4} = 441$, thus $BP = \boxed{(D) 21}$.

Solution 2 (Law of Cosines)

Scale this down to a $9 - 15 - 16$ triangle (we will multiply the result by 5 in the end). Note that $9^2 + 16^2 - 18 \cdot 16 \cdot \cos(\angle B) = 15^2$, so $112 = 18 \cdot 16 \cdot \cos(\angle B)$, which simplifies to $\frac{7}{18} = \cos(\angle B)$. Then $\cos(\frac{\angle B}{2}) = \sqrt{\frac{1 + \frac{7}{18}}{2}} = \sqrt{\frac{25}{36}} = \frac{5}{6}$ (positive root since the angle is acute). Therefore, we have $\frac{BD}{BP} = \frac{5}{6}$, assuming that D is the foot of the altitude. There are many ways to proceed from here to find BD . Note that by Heron's formula, the area of the scaled-down triangle is $\sqrt{(20)(4)(5)(11)} = 20\sqrt{11}$. Therefore, $CD = \frac{5}{2}\sqrt{11}$. Using Pythagorean Theorem, we get $BD = \sqrt{81 - \frac{25 \cdot 11}{4}} = \sqrt{\frac{324 - 275}{4}} = \frac{7}{2}$. Therefore, we get $\frac{\frac{7}{2}}{\frac{BP}{5}} = \frac{5}{6}$, so $BP = \frac{21}{5}$, and we scale up by 5 to get $\boxed{(D) 21}$. ScoutViolet

Solution 3 (Stewarts)

Let the foot of the altitude coming from C on segment AB be D . Using the fact that CD is a common leg in right triangles $\triangle CDB$ and $\triangle CDA$, we have

$$45^2 - BD^2 = 75^2 - (80 - BD)^2.$$

Expanding gives

$$45^2 = 75^2 - 80^2 + 160BD,$$

so $BD = \frac{35}{2}$. Let the foot of the angle bisector from B to AC be point E . Since BE is the angle bisector of $\angle CBA$, we can use the angle bisector theorem. This gives

$$\frac{45}{CE} = \frac{80}{75 - CE},$$

so $CE = 27$ and $AE = 75 - 27 = 48$. Now we can use Stewart's Theorem to find $\overline{BE}$. We have

$$(48)(45)^2 + (27)(80)^2 = (27)(48)(75) + 75BE^2.$$

To simplify this expression, just divide by the greatest common divisor and solve from there. In the end, we get $BE = 48$. Let $BP = x$, so $PE = 48 - x$. Draw the altitude from E down to AB . Let the foot of this altitude be F . Since $EF \parallel CD$, we have $\triangle AFE \sim \triangle ADC$. Hence, we can write the equation

$$\frac{48}{75} = \frac{AF}{\frac{125}{2}}.$$

Solving gives $AF = 40$, so $FD = \frac{125}{2} - 40 = \frac{45}{2}$. Since $PD \parallel EF$, we also have $\triangle BDP \sim \triangle BFE$, so we have

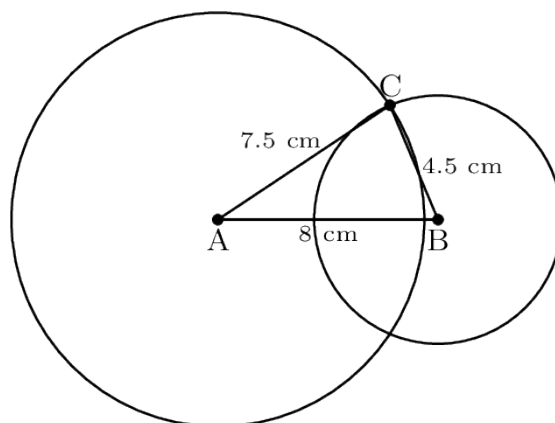
$$\frac{x}{48} = \frac{\frac{35}{2}}{40}.$$

Solving for x gives $x = 21$ or D .

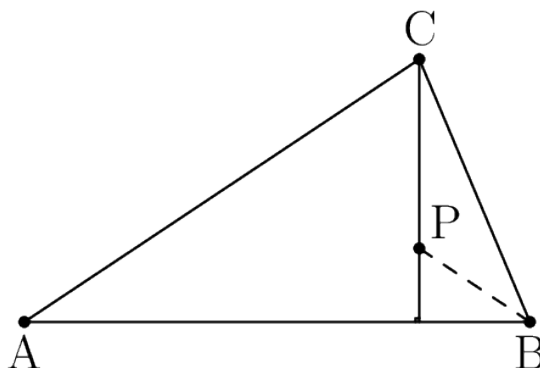
Solution 4 (Rulerbash)

Preface: When we have a problem as such, involving a simple diagram with minimal instructions, I use a method I named "rulerbash". Rulerbash should only be used in specific cases and as a last resort, mainly in the event of a time crunch or a difficult problem.

Start with the longest side, drawing a line with a length of 8 cm (AB). Then, using a compass, draw 2 circles centered around points A and B , 7.5 cm and 4.5 cm radii respectfully. At the point of intersection of these 2 circles, we have point C , completing a perfectly scaled drawing of $\triangle ABC$. (Note the circles are not necessary with a bit of trial and error with the side lengths, they simply offer a way to get it done first try).



After this, dropping the altitude to AB is simple with a ruler and careful placement, and angle bisector can be estimated quite accurately. (Note: Alternately, a compass can be used to accurately place the altitude and the bisector, which are both trivial constructions ZingiberiMaracandae)



After all of this, we can reuse our ruler and measure $BP = 2.1$ cm, and using our scale of $80 = 8$ cm, our final answer is (D) 21.

Other note: you kind of have to draw the angle bisector really accurately for this to work (I tried this) Sakura_kitty

Solution 5 (No Trig)

Let D be the intersection of the altitude from C to AB . To simplify calculations, divide all side lengths by 5, and multiply by 5 again at the end. First, we use Heron's Formula, $\sqrt{s(s-a)(s-b)(s-c)}$, to find the area. Let $[ABC]$ denote the area of $\triangle ABC$. By Heron's Formula,

$$[ABC] = \sqrt{20 \cdot 5 \cdot 4 \cdot 11} = 20\sqrt{11}.$$

Next, we find the altitude CD using the formula for the area of a triangle, $\frac{1}{2}bh = \text{area}$:

$$\frac{1}{2}(16)(CD) = 20\sqrt{11} \Rightarrow CD = \frac{5\sqrt{11}}{2}.$$

We can use the Pythagorean Theorem in $\triangle CDB$ to find DB :

$$DB^2 + \frac{25 \cdot 11}{4} = 81 \Rightarrow 4DB^2 + 275 = 324 \Rightarrow DB^2 = \frac{49}{4},$$

so $DB = \frac{7}{2}$. Next, we use the Angle Bisector Theorem to find PD . Let $x = PC$ and $y = PD$. Since $x + y = \frac{5\sqrt{11}}{2}$, we have $x = \frac{5\sqrt{11}}{2} - y$. From the given ratio,

$$\frac{9}{x} = \frac{7}{2y} \Rightarrow 18y = 7x.$$

Substituting $x = \frac{5\sqrt{11}}{2} - y$,

$$18y = 7\left(\frac{5\sqrt{11}}{2} - y\right) \Rightarrow 25y = \frac{35\sqrt{11}}{2},$$

so $y = \frac{7\sqrt{11}}{10}$.

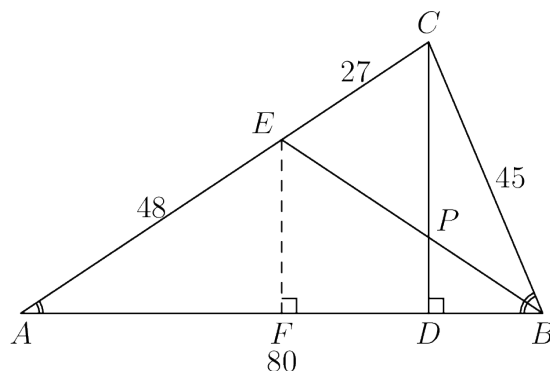
Now, using the Pythagorean Theorem again to find BP :

$$\frac{49 \cdot 11}{100} + \frac{49}{4} = BP^2 \Rightarrow 100BP^2 = 49(11 + 25) = 49 \cdot 36,$$

so $BP = \frac{42}{10} = \frac{21}{5}$.

Finally, multiplying the side lengths by 5 again gives $BP = 21.$, or D.

Solution 6 (Similar Triangle)



Due to Angle Bisector Theorem

$$\frac{CE}{AE} = \frac{BC}{BA} = \frac{45}{80} = \frac{9}{16}$$

$$CE = AC \times \frac{9}{25} = 27$$

$$AE = AC \times \frac{16}{25} = 48$$

Notice that $\frac{CE}{CB} = \frac{CB}{CA} = \frac{3}{5}$, $\triangle CBE \sim \triangle CAB$, $\angle A = \angle CBE = \angle EBF$, so $\triangle ABE$ is isosceles triangle, hence

$$AF = BF$$

Notice that $AF : FD = AE : EC = 16 : 9$, $DB = FB - FD$, then

$$AF : DB = 16 : 7$$

Notice that $\triangle PBD \sim \triangle EAF$, $PB = AE \times \frac{BD}{AF} = 48 \times \frac{7}{16} = \boxed{\text{(D) } 21}$.

Solution 7 (Pythagoras+angle bisector theorem)

Let CD be the altitude to side AB , and let E be the point of intersection of the angle bisector of $\angle B$. Let P be the point of intersection of the angle bisector and the altitude.

By Pythagoras, $CD^2 = BC^2 - BD^2 = AC^2 - AD^2$

We let $BD = x$, so $AD = 80 - x$

Now we have

$$45^2 - x^2 = 75^2 - (80 - x)^2$$

(This looks complicated, but we can see that we are going to have two $-x^2$ terms on both sides, so it cancels out)

Simplifying and rearranging gives us $80 - 2x = 45$, so $x = \frac{35}{2}$

Therefore, $BD = \frac{35}{2}$, and $AD = \frac{125}{2}$. This gives us $CD = \frac{25}{2}\sqrt{11}$

Now, note that in triangle BDC , BP bisects $\angle DBC$

By the angle bisection theorem, we have $\frac{BD}{DP} = \frac{BC}{CP}$

which gives us $\frac{DP}{CP} = \frac{BD}{BC} = \frac{7}{18}$

We have $DP = \frac{7}{25} * CD = \frac{7}{2} \sqrt{11}$ and $CP = \frac{18}{25} * CD = \frac{18}{2} \sqrt{11}$

So

$$BP^2 = BD * BC - DP * CP = \frac{35}{2} * 45 - \frac{7}{2} * \frac{18}{2} * 11 = 441$$

Therefore, $BP = 21$

(This is my first solution, pls point out any mistakes in math or LATEX).
backtosq-1

Alternative Ending to Solution 7 As an alternative ending equation to the solution above, I used the Pythagorean Theorem again, giving us:

$$BP^2 = BD^2 + PD^2 = \left(\frac{35}{2}\right)^2 + \left(\frac{7\sqrt{11}}{2}\right)^2 = \left(\frac{1225}{4}\right) + \left(\frac{539}{4}\right) = \frac{1764}{4} = 441$$

Thus, $BP = \boxed{\text{(D)} 21}$

Solution 8 (Pythagora's + Double Angle Identity)

Since all the lengths share a factor of 5, divide by 5 and multiply it back at the end for smaller numbers. Drop the altitude from vertex C to point D on AB . Then comparing the altitude by Pythagora's on both sides gives

$$9^2 - x^2 = 15^2 - (16 - x)^2 = 15^2 - 16^2 + 32x - x^2 = -31 + 32x - x^2$$

Cancelling the $-x^2$ then solving gives us $x = \frac{112}{32} = \frac{7}{2}$. Since $\triangle BDC$ is a right triangle with $\angle B = 2\theta$, we find $\cos(2\theta) = \frac{7}{9} = \frac{7}{18}$.

Let $\theta = \angle PBA = \angle PBC$. Because $\angle B$ is opposite to the leg $9 < 15$ and not the hypotenuse, we must have that 2θ is acute, likewise for θ . Hence, $\cos \theta > 0$.

Thus, using cosine definition on $\triangle BDP$ and the double angle identity $\cos 2\theta = 2\cos^2 \theta - 1$, we get

$$\cos \theta = \frac{7/2}{BP} = \sqrt{\frac{1 + \cos 2\theta}{2}} = \sqrt{\frac{1 + \frac{7}{18}}{2}} = \sqrt{\frac{25}{36}} = \frac{5}{6}$$

Finally, solving for $BP = \frac{7}{2\cos \theta} = \frac{7}{2} \cdot \frac{6}{5} = \frac{21}{5}$. Scaling back to the original diagram by 5 gives $\boxed{\text{(D)} 21}$.

Solution 9(Law of Cosines+Angle Bisector Length Theorem)

Use Law of Cosines in $\triangle ABC$:

$$\cos B = \frac{80^2 + 45^2 - 75^2}{2 \cdot 80 \cdot 45} = \frac{7}{18}$$

Call E the intersection of the angle bisector of B and line segment AC , and let $CD \perp AB$, $D = CD \cap AB$, therefore $P = CD \cap BE$. In right triangle $\triangle BCD$, use the definition of cosine:

$$\cos B = \frac{BD}{BC} = \frac{7}{18}, \sin B = \sqrt{1 - \cos^2 B} = \frac{5\sqrt{11}}{18}$$

Therefore

$$BD = \frac{35}{2}, CD = \frac{25\sqrt{11}}{2}$$

Use Angle Bisector Theorem in $\triangle BCD$:

$$\frac{BD}{BC} = \frac{PD}{PC} = \frac{7}{18}$$

Therefore,

$$PD = \frac{7}{25}CF = \frac{7\sqrt{11}}{2}, PC = \frac{18}{25}CF = 9\sqrt{11}$$

Use Angle Bisector length formula:

$$BP = \sqrt{BD \cdot BC - PD \cdot PC} = \boxed{\text{(D)} 21}$$

Solution 10

Let the point where the altitude to AB meets AB be D . We can compute D easily. Let $BD = a \implies AD = 80 - a$. So

$$\begin{aligned} 45^2 - a^2 &= 75^2 - (80 - a)^2 \implies 45^2 - a^2 = 75^2 - 80^2 + 160a - a^2 \implies \\ 45^2 - 75^2 + 80^2 &= 160a \implies -30 \cdot 120 + 80 \cdot 80 = 10 \cdot 16a \implies -360 + 640 = \\ 16a &\implies a = \frac{280}{16} = \frac{35}{2}. \end{aligned}$$

Now let $\angle DBP = \theta$. Note this is an angle bisector so we can find $\cos(\angle ABC)$ where $\theta = \frac{1}{2}\angle ABC$. So we use Law of Cosines to get

$$80^2 + 45^2 - 160 \cdot 45 \cos(2\theta) = 75^2 \implies 16^2 + 9^2 - 32 \cdot 9 \cdot \cos(2\theta) = 15^2 \implies \cos(2\theta) = \frac{112}{32 \cdot 9} = \frac{7}{18}.$$

$$\cos(2\theta) = 2\cos^2(\theta) - 1 = \frac{7}{18} \implies \cos^2(\theta) = \frac{25}{36} \implies \cos(\theta) = \frac{5}{6}.$$

Finally, by *SOH - CAH - TOA* note that $\cos(\theta) = \frac{BD}{BP} = \frac{5}{6}$ and we know $BD = \frac{35}{2}$ so

$$BP = \frac{6}{5} \cdot \frac{35}{2} = \boxed{21}.$$

- x) (AMC 10 10A 2025) Call a positive integer *fair* if no digit is used more than once, it has no 0s, and no digit is adjacent to two greater digits. For example, 196, 23, and 12463 are fair, but 1546, 320, and 34321 are not fair. How many fair positive integers are there? (A) 511 (B) 2584 (C) 9841 (D) 17711 (E) 1968

Answer: C

Solution 1

To satisfy the conditions, a fair integer must have no digit be a local minimum. Let's say we have n distinct digits, with each digit being a number from 1 to 9. To create a fair integer, we begin by placing the largest digit. For the second-largest digit, we can either place this digit to the right or to the left of the string already created. We have these 2 options for the third-largest digit, and so on. Therefore, there are 2^{n-1} valid permutations to create a fair integer.

We must also choose which digits will be in the permutation. If you are creating an n -digit long fair integer, there are $\binom{9}{n}$ ways to pick which digits will be in the number.

Therefore, for each $n \in \{1, 2, \dots, 9\}$, the number of fair integers of length n is:

$$\binom{9}{n} \cdot 2^{n-1}.$$

Summing over all n :

$$\sum_{n=1}^9 \binom{9}{n} \cdot 2^{n-1} = \frac{1}{2} \left(\sum_{n=0}^9 \binom{9}{n} 2^n - 1 \right) = \frac{1}{2} ((1+2)^9 - 1) = \frac{1}{2}(19682) = \boxed{9841}.$$

Note that the Binomial Theorem was used to equate

$$\sum_{n=0}^9 \binom{9}{n} 2^n = (1+2)^9.$$

lprado

Solution 2

Note every fair number will have an increasing string of digits, a maximum digit, then a decreasing string of digits. This is because if it decreases then increases, then the digit in the middle will be less than its adjacent digits.

Let n be the maximum digit. For each number $i < n$, we may either place i before n , after n , or choose not to include it. Note this process will result in a unique number for every case, as the numbers before n must be in increasing order, and the numbers after n must be in decreasing order. Therefore, for each number n , we have 3^{n-1} cases.

Since $n \in \{1, 2, \dots, 9\}$, we have:

$$\sum_{n=1}^9 3^{n-1} = \frac{3^9 - 1}{3 - 1} = \boxed{9841}.$$

Solution 2.5 (Cheese, Goldilocks Zone)

If you consider options D and E too big, and A and B too small, then C is your answer.

(C) 9841 Also E is twice C so C is your answer

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Solution 3 (Recursion)

Let $f(n)$ be the number of fair integers given an arbitrary set of n digits. Let a be the smallest of these digits. Notice that a is either the first or last digit, as if it were any other digit, the two digits surrounding it would both be greater. Then, notice that if the remaining $n - 1$ slots form a fair number, so does the number when a is appended. Therefore, $f(n) = 2f(n - 1)$. From here, we may proceed with the calculation in Solution 1 to get the answer of

(C) 9841

Solution 4

For any fair integer n , we write some 9-digit base-3 number(s) $f(n) = (\overline{d_1 d_2 \dots d_9})_3$ from left to right as follows:

- $d_k = 0$ if k is not in n ;

- $d_k = 1$ if k is the to the left of all digits greater than k in n ;
- $d_k = 2$ if k is the to the right of all digits greater than k in n .

Note that the fairness ensured one and only one of the above three cases happens.

Now if k is the largest digit in n then d_k can be either 1 or 2.

An example: If $n = 136852$, then $f(n) = \overline{121021010}_3$ or $\overline{121021020}_3$.

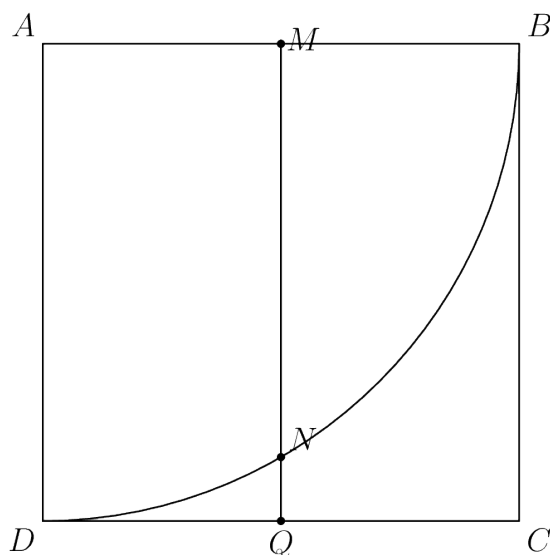
It's thus easy to see that there is a 1-to-2 mapping from all fair integers to all 9-digit base-3 nonzero numbers. Therefore the answer is $\frac{3^9-1}{2}$.

- y) (AMC 10 10A 2025) A point P is chosen at random inside square $ABCD$. The probability that $\overline{AP}$ is neither the shortest nor the longest side of $\triangle APB$ can be written as $\frac{a+b\pi-c\sqrt{d}}{e}$, where a, b, c, d , and e are positive integers, $\gcd(a, b, c, e) = 1$, and d is not divisible by the square of a prime. What is $a + b + c + d + e$?

- (A) 25 (B) 26 (C) 27 (D) 28 (E) 29

Answer: A

Solution 1 (Geometry)



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